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UPDATED
EDITION



Quantitative Aptitude for CAT

& Other MBA Entrance Exams

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Shortcut Techniques**

- ✓ 23 Chapters with Illustrations
- ✓ 4 levels of Exercise
- ✓ **5 CAT Mock Tests**

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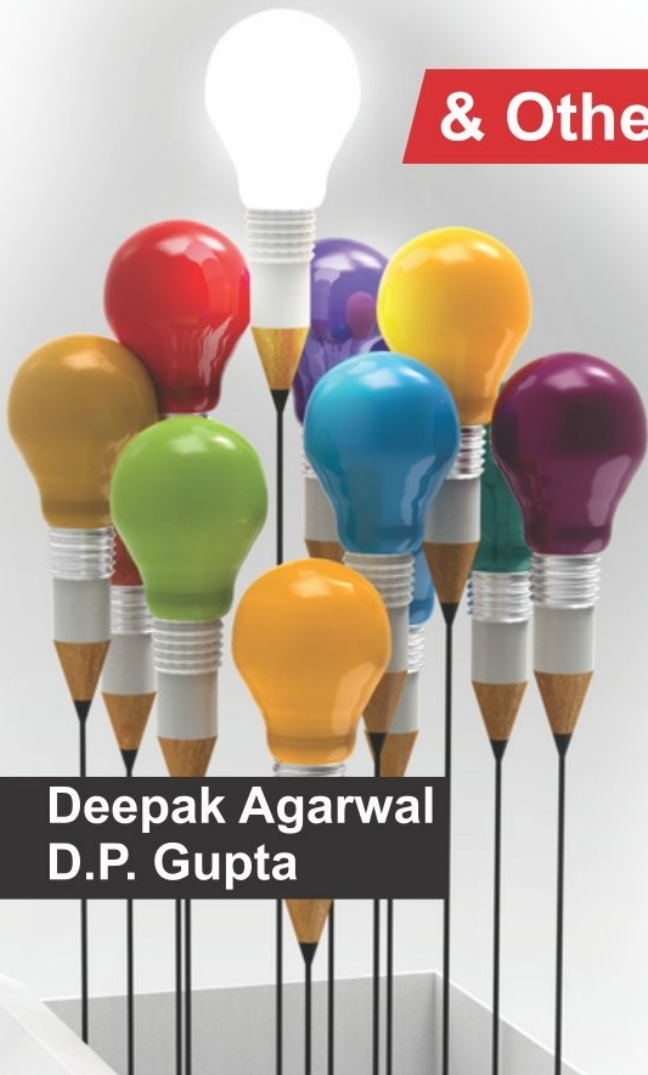
**Fully Solved
Questions**

5 Years
(2012-16)

**Topic-wise CAT
Solved Paper**

**Deepak Agarwal
D.P. Gupta**

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- **Head Office** : B-32, Shivalik Main Road, Malviya Nagar, New Delhi-110017
 - **Sales Office** : B-48, Shivalik Main Road, Malviya Nagar, New Delhi-110017
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Preface

Quantitative Aptitude forms a very important part of preparation of MBA aspirants. Not just the Quant section but it forms the backbone of the Data Interpretation, Data Sufficiency and Reasoning. Disha's Quantitative Aptitude for CAT/ XAT/ IIFT/ CMAT/ MAT is a book focused on mastering techniques to crack these examinations. The book starts from a foundation level and moves to an expert level.

Structure of the book: *The book comprises of 5 Units (Numbers, Arithmetic, Algebra, Geometry and Counting Principles) which have been further divided into 23 chapters followed by 5 Mock Tests. Each chapter consists of*

- Theory with Illustrations
- Foundation Level Exercise
- Standard Level Exercise
- Expert Level Exercise
- Solutions to the 3 levels of exercises
- Test Yourself
- Solutions to Test Yourself

Salient Features of the Book:

- Each chapter includes detailed review of all the concepts involved with exhaustive number of well discussed Illustrations.
- The theory is followed by 3 levels of exercises – Foundation Level, Standard Level and Expert Level. The detailed solution to each and every question has been provided immediately at the end of the 3 exercises.
- **Foundation Level :** Here the focus is to expose the students to solve problems based on the concepts they have learned in theory part. The student develops a good foundation and is ready for the Standard level.
- **Standard Level :** The Standard level is a collection of excellent quality problems which will test a student on the application of the concepts learned in various real-life situations. The problems provide a good platform to develop a very good problem solving aptitude so as to take up the expert level confidently.
- **Expert Level :** This is the toughest part of the book and involves the trickiest questions on the concepts involved. Here most of the problems will pose good challenge to the students.
- The book contains 23 Chapter-wise Tests – Test Yourself - on the basis of latest CAT pattern after the exercises in each chapter. The students must attempt these tests in specified time limits and conditions. A new chapter introducing Trigonometry has been added in the book.
- At the end of the book 5 Mock Tests are provided based on the pattern of latest CAT exams. The solutions to the test are provided at the end of the tests.
- The book contains questions of past exams of CAT/ XAT/ IIFT/ SNAP/ NMAT/ ATMA/ FMS in the various exercises and Illustrations.

We would like to thank the DTP team at Disha, especially Mr. Amit Kumar Jha, who have worked really hard to bring the book to the present shape. Although we have taken utmost care while preparing the book but errors might have crept in. We would like to request our readers to highlight these errors.

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UNIT-I

Numbers

Chapter 1

Fundamentals

Chapter 2

Number System

1

FUNDAMENTALS

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- Shortcuts For Addition and Subtraction
- Shortcuts for Multiplication
- Rounding off and Its Uses
- 'BODMAS' Rule
- Brackets
- Factorial
- Roman Numbers
- Important Conversion
- Absolute Value or Modulus of a Number
- Properties of a Modulus
- Powers or Exponents
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- Properties of Squares
- Square Roots
- Cubes

INTRODUCTION

In the CAT and the likes of competitions, 25% to 35% questions are based on numeracy. So, to save the time for other questions in competitions, it is essential to command over shortcuts of addition, subtraction and multiplications given in this chapter.

SHORTCUTS FOR ADDITION AND SUBTRACTION

Addition is the mother of all calculations, which gives you an extra edge that makes your calculations faster. Subtraction is the extension of addition.

- I. Addition of smaller number to larger number is easier than addition of larger number to smaller number. For example addition in the order $5817 + 809 + 67 + 8$ is easier than the addition in the order $8 + 67 + 809 + 5817$. Hence to add the numbers, it is better to first arrange them in decreasing order and then add them.
- II. To find the sum like $6345 + 2476 + 802$, first add the thousands and then hundreds, tens and once in order. Thus

$$6345 + 2476 + 802 = 6000 + 2000 (= 8000) + 300 (= 8300) + 400 (= 8700) + 800 (= 9500) + 40 (= 9540) + 70 (= 9610) + 5 (= 9615) + 6 (= 9621) + 2 = 9623$$
- III. To find the sum of large numbers like $64083 + 43102 + 94320 + 8915 + 7042$

First add the thousands like

$$64 + 43 (= 107) + 94 (= 201) + 8 (= 209) + 7 = 216$$

At this stage you know the answer would be 216000 + (a maximum of 5000), as there are five numbers whose last 3 digits numbers are not added. If the range from 216000 to 221000 is sufficient to choose the correct option, then no need to add further otherwise add the hundredth digits of given numbers $1 + 3 + 9 = 13$.

At this stage you know the answer would be

$$217300 + (\text{a maximum of } 500)$$

If the range from 217300 to 217800 is sufficient to choose the correct option, then no need to add further otherwise add the last two digits of numbers

$$83 + 2 + 20 + 15 + 42 = 162.$$

Hence the correct sum will be $217300 + 162 = 217462$.

There are two advantages of process of addition (i) No need to get final sum as in this process of addition, you could choose the correct option at earlier stage also. (ii) In the entire calculation, you have not gone above two digits additions.

- IV. Sometimes you have to add so many large numbers. In that case you can find the required sum using the following methods.

(A) Column Form

Write the given numbers one below the other with right align if the given numbers are whole numbers and with decimal point align if the given numbers are decimal numbers as we write in conventional method of addition.

$$\begin{array}{r}
 6580125 \\
 8924708 \\
 608907 \\
 895 \\
 2130 \\
 85704 \\
 730956 \\
 9547684 \\
 4675 \\
 532689
 \end{array}
 \left. \vphantom{\begin{array}{r} 6580125 \\ 8924708 \\ 608907 \\ 895 \\ 2130 \\ 85704 \\ 730956 \\ 9547684 \\ 4675 \\ 532689 \end{array}} \right\} \leftarrow \text{Right align}$$

$$\begin{array}{r}
 85406.487 \\
 672028.32 \\
 4927.052 \\
 531486.2 \\
 564.8 \\
 62089.204 \\
 701438.909
 \end{array}
 \left. \vphantom{\begin{array}{r} 85406.487 \\ 672028.32 \\ 4927.052 \\ 531486.2 \\ 564.8 \\ 62089.204 \\ 701438.909 \end{array}} \right\} \leftarrow \text{Decimal point align}$$

(i) Addition of Whole Numbers

To add the whole numbers with right align, we start adding the digits in the right most column by going down but when the running total becomes 10 or higher than 10, then we reduce it by 10 and go ahead with reduced number. As we do so, we make a small dash ‘ ’ at the right top corner of the digit that makes our total 10 or higher than 10 as given below for right most column.

$8 \rightarrow 5 + 8 = 13$, which is more than 10, so we subtract 10 from 13 and mark a dash at the right top corner of the digit 8 and start adding again.

$7 \rightarrow 3 + 7 = 10$, so we subtract 10 from it and mark a dash at the right top corner of the digit 7 and start adding again.

$$5 \rightarrow 0 + 5 = 5$$

$$0 \rightarrow 5 + 0 = 5$$

$$4 \rightarrow 5 + 4 = 9$$

$6 \rightarrow 9 + 6 = 15$, which is more than 10, so we subtract 10 to from 15 and mark a dash at the right top corner of the digit 6 and start adding again.

$$4 \rightarrow 5 + 4 = 9$$

$5 \rightarrow 9 + 5 = 14$, which is greater than 10, so we subtract from it and mark a dash at the right top corner of the digit 5 and start adding again.

$9 \rightarrow 4 + 9 = 13$, which is greater than 10, so we subtract from it and mark a dash at the right top corner of the digit 9.

The dashes and the final figure 3 will be written under the first column from right as

$$\begin{array}{r}
 5 \\
 8' \\
 7' \\
 5 \\
 0 \\
 4 \\
 6'
 \end{array}$$

$$\begin{array}{r}
 4 \\
 5' \\
 9' \\
 \hline
 3
 \end{array}$$

Now we count the dashes marked in the first column from right. Number of dashes in this column is 5. Now add the number of dashes 5 in the top digits 2 of the second column from right, then start adding this column as we add the first column from right.

In the same way, we add the other columns one by one from right. After adding the left most column, write the number of dashes in this column in the left of the total of this column as given below.

$$\begin{array}{cccccccc}
 \textcircled{4} & \textcircled{3} & \textcircled{3} & \textcircled{6} & \textcircled{4} & \textcircled{5} & \text{←} & \text{Number of dashes in the just right column} \\
 6' & 5 & 8' & 0 & 1 & 2 & 5 & \\
 8 & 9' & 2 & 4' & 7' & 0 & 8' & \\
 & 6' & 0 & 8 & 9' & 0 & 7' & \\
 & & & & 8 & 9' & 5 & \\
 & & & 2' & 1' & 3 & 0 & \\
 & & 8' & 5 & 7 & 0 & 4 & \\
 & 7' & 3 & 0 & 9' & 5' & 6' & \\
 9' & 5 & 4 & 7' & 6' & 8' & 4 & \\
 & & & 4 & 6 & 7 & 5' & \\
 & 5' & 3' & 2 & 6' & 8' & 9' & \\
 \hline
 2 & 7 & 0 & 1 & 8 & 4 & 7 & 3
 \end{array}$$

The advantage of this process is that the entire calculation is done only by adding one digit numbers.

(ii) Addition of Decimal Numbers

Addition of decimal numbers with decimal point align is the same as addition of whole numbers with right align.

In addition of decimal numbers, we put a decimal point in the sum total align with decimal in the given numbers as given below.

$$\begin{array}{cccccccc}
 \textcircled{2} & \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{5} & \textcircled{2} & \textcircled{1} & \textcircled{2} & \text{←} & \text{Number of dashes in just right column} \\
 8 & 5 & 4 & 0 & 6 & . & 4 & 8' & 7 & \\
 6 & 7' & 2 & 0 & 2 & 8' & . & 3 & 2 & \\
 & 4' & 9' & 2 & 7' & . & 0 & 5 & 2 & \\
 5' & 3 & 1 & 4' & 8' & 6 & . & 2' & & \\
 & & & 5 & 6' & 4' & . & 8 & & \\
 & 6' & 2 & 0 & 8' & 9' & . & 2' & 0 & 4' \\
 7' & 0 & 1 & 4 & 3 & 8' & . & 9 & 0 & 9' \\
 \hline
 2 & 0 & 5 & 7 & 9 & 4 & 0 & . & 9 & 7 & 2
 \end{array}$$

Illustration 1: Find the sum of the following numbers using column form.

564.39, 4237.8, 4.213, 56.8, 9423.41 and 46.98

Solution:

$$\begin{array}{cccccc}
 \textcircled{1} & \textcircled{2} & \textcircled{3} & \textcircled{3} & \textcircled{1} & \\
 & 5 & 6 & 4 & . & 3 & 9 \\
 & 4 & 2 & 3' & 7' & . & 8' \\
 & & & & 4 & . & 2 & 1' & 3 \\
 & & & & 5 & 6' & . & 8' \\
 9' & 4' & 2 & 3 & . & 4 & 1 \\
 & & 4' & 6' & . & 9' & 8 \\
 \hline
 14 & 3 & 3 & 3 & . & 5 & 9 & 3
 \end{array}$$

(B) Row Form

To find the sum of numbers, it is not necessary to write them one below the other with align i.e., column form. You can find the sum of numbers written in a row form using the same method discussed above for column form but there is a problem of alignment. To overcome this problem of alignment, we use the method of column form in slightly different way as discussed below.

This method of addition is very important. If you get command over it, you can stop wasting time in writing the numbers in column form.

(i) Addition of Decimal Numbers

Suppose you have to find the sum

$$707.325 + 1923.82 + 58.009 + 564.943 + 65.6$$

- (a) Put zeros to the right of the last digit after decimal to make the number of digits after decimal equal in each number.

For example, the above addition may be written as

$$707.325 + 1923.820 + 58.009 + 564.943 + 65.600$$

- (b) Start adding the last digit from right of all the numbers. During running total, don't exceed 10. That is, when you exceed 10, mark a tick with pencil anywhere near about your calculation and go ahead with the number exceeding 10.

$$707.325 + 1923.820 + 58.009 + 564.943 + 65.600$$

$$= \text{-----} 7$$

5 plus 0 is 5; 5 plus 9 is 14, mark a tick in rough area and carry over 4; 4 plus 3 is 7; 7 plus 0 is 7, so write down 7. During addition we strike off all the digits which are added. It saves us from confusion and duplication.

- (c) Add the number of ticks (marked near by calculation in rough) with the digits at 2nd places from right and erase that tick from rough.

$$707.325 + 1923.820 + 58.009 + 564.943 + 65.600$$

$$= \text{-----} 97$$

1 (number of tick) plus 2 is 3; 3 plus 2 is 5; 5 plus 0 is 5; 5 plus 4 is 9 and 9 plus 0 is 9; so write down 9 at the second place from right in the sum.

$$707.325 + 1923.820 + 58.009 + 564.943 + 65.600$$

$$= \text{-----} 697$$

3 plus 8 is 11; mark a tick in rough and carry over 1; 1 plus 0 is 1; 1 plus 9 is 10, mark another tick in rough and carry over zero; 0 plus 6 is 6, so put down 6 at the third place from right in the sum.

- (e) Following the same way get the result:

$$707.325 + 1923.820 + 58.009 + 564.943 + 65.600$$

$$+ 65.600$$

$$= 3319.697$$

Illustration 2: Find the sum of the following numbers using row form.

$$564.39, 4237.8, 4.213, 56.8, 9423.41 \text{ and } 46.98$$

Solution:

$$564.390 + 4237.800 + 4.213 + 56.800$$

$$+ 9423.410 + 46.980$$

$$= 14333.593$$

Carry over		
/		= 1
/ / /		= 3
/ / /		= 3
/ /		= 2
/		= 1

(ii) Addition of Whole Numbers

Suppose you have to find the sum

$$707325 + 192382 + 58009 + 564943 + 656$$

Follow the steps mentioned in steps (b), (c), (d) and (e) of section (B) (i) (addition of decimal numbers in row form) above (without considering the decimal). Thus,

$$707325 + 192382 + 58009 + 564943 + 656$$

$$= \text{-----} 5$$

$$707325 + 192382 + 58009 + 564943 + 656$$

$$= \text{-----} 15$$

$$707325 + 192382 + 58009 + 564943 + 656$$

$$= \text{-----} 315$$

$$707325 + 192382 + 58009 + 564943 + 656$$

$$= \text{-----} 3315$$

$$707325 + 192382 + 58009 + 564943 + 656$$

$$= \text{-----} 23315$$

$$707325 + 192382 + 58009 + 564943 + 656$$

$$= \text{-----} 1523315$$

Illustration 3: Find the sum of the following numbers using row form.

$$5834, 96182, 459, 2128, 87582 \text{ and } 735$$

Solution:

$$5834 + 96182 + 459 + 2128 + 87582 + 735$$

$$= 192920$$

Carry over		
/ / /		= 3
/ / /		= 3
/ /		= 2
/ /		= 2
/		= 1

V. Single Step Solution for Addition and Subtraction in a Single Row: Digit-Sum Method

To understand this method, let us find the value of

$$6531 - 468 + 8901 - 3210$$

First of all, we check that the required value or number will +ve or -ve by just looking at the given numbers with signs. In the case of +ve required number, the digits of the required number will be zero or +ve integer and in the case of -ve required number, the digits of the required number will be zero or -ve integer. Clearly the required number will be +ve. Hence, digits of required number will be zero or +ve.

Now to find the unit digit of the required number, add and subtract the digits at units places of these given numbers according to the sign attached with these numbers as

$$1 - 8 + 1 - 0 = -6$$

Since required number will be +ve, therefore its unit digit can not be -ve.

To make (-6) positive, we borrow from tens of largest given positive number. You should remember that we can't borrow from negative given number if required number is +ve. So, we borrow 1 from tens digit 0 of 8901. Now, we add 10 to (-6), this can be shown as

$$\begin{array}{r} (-1)(-1) \\ 6531 - 468 + 8 \quad 9 \quad 0 \quad 1 - 3210 = \text{-----} 4 \end{array}$$

Now, we add and subtract the digits at tens places of given numbers according to the sign attached with these numbers.

$$3 - 6 + 9 - 1 = 5$$

Since 5 is positive, hence 5 is the tens digit of the required number.

This can be shown as

$$\begin{array}{r} (-1)(-1) \\ 6531 - 468 + 8 \quad 9 \quad 0 \quad 1 - 3210 = \text{-----} 54 \end{array}$$

Now add the digit at hundredth places as

$$5 - 4 + (9 - 1) - 2 = 7, \text{ which is positive.}$$

Hence hundredth digit of required number will be 7.

Now add the digit at thousand places as

$$6 + 8 - 3 = 11$$

Thus the last two digits of the required number are 11. Hence

$$\begin{array}{r} (-1)(-1) \\ 6531 - 468 + 8 \quad 9 \quad 0 \quad 1 - 3210 = \text{-----} 11754 \end{array}$$

The same above method is used for decimal numbers also after making the equal number of digits after decimal in all the given numbers by putting zero(s) at the end of the number after decimals.

This method requires some practice. But after some practice, you will find it is faster method.

Illustration 4: $6598 - 2401 + 2281 - 516 = ?$

Solution: (-1) (+1)

$$6 \quad 5 \quad 9 \quad 8 - 2401 + 2281 - 516 = 5962$$

After adding and subtracting the digits at tens places according to the sign attached with the respective numbers, we get 16, which has two digits. So, 6 is written at tenth place in the required number and 1 is added to 5 (hundredth digit) of 6598.

Illustration 5: $5603 - 2281 + 210 - 1472 = ?$

Solution: (-2)

$$5 \quad 6 \quad 0 \quad 3 - 2281 + 210 - 1472 = 2060$$

After adding and subtracting the digits at tens places according to the sign attached with the respective numbers, we get -14. Since 14 is more than 10 but not more than 20. Therefore to make -14 as a single positive digit we have to borrow 2 from hundredth digit i.e., 6 of 5603.

Now $-14 + 20 = 6$, therefore tens digit of the required number is 6.

Illustration 6: $3584 - 1502 + 2191 - 213 = ?$

Solution: (+1)

$$3 \quad 5 \quad 8 \quad 4 - 1502 + 2191 - 213 = 4060$$

After adding and subtracting the digits at tens places according to the sign attached with the respective numbers, we get 16. So we take 6 as tenth digit of the required number and add 1 to the hundredth digit i.e., 5 of 3584.

Illustration 7: $125 - 2827 + 5163 - 2131 = ?$

Solution: (-1)

$$125 - 2827 + 5163 - 2131 = 330$$

Illustration 8: $2513 - 6718 + 1231 - 3414 = ?$

Solution: (-1)

$$2513 - 6 \quad 7 \quad 1 \quad 8 + 1231 - 3414 = -6388$$

By observing the given numbers with signs, it is clear that the required number or value will be -ve. Hence digits of the required number will be zero or negative integer.

Now $3 - 8 + 1 - 4 = -8$, so unit digit of required number is 8 (without the sign).

$$1 - 1 + 3 - 1 = 2, \text{ which is +ve.}$$

To make 2 negative, borrow 1 from hundredth digit of the largest given -ve number i.e. borrow 1 from 7 of 6718.

Now subtract 10 from 2, which gives -8. So 8 is the tens digit of required number. Similarly, we find the hundredth and thousand digit of the required number as 3 and 6 respectively. Since required number will be -ve, therefore we put a -ve sign before 6388 which gives -6388 as required number.

Illustration 9: $765.819 - 89.003 + 12.038 - 86.89 = ?$

Solution: First equate the number of digits after decimals by putting zero(s) at the end. So, $765.819 - 89.003 + 12.038 - 86.89$

$$\begin{array}{r} (-1) \quad (-1) \quad (-1) \quad (-1) \quad (+1) \\ = 7 \quad 6 \quad 5 \quad 8 \quad 1 \quad 9 - 89.003 + 12.038 - 86.890 \\ = 601.964 \end{array}$$

SHORTCUTS FOR MULTIPLICATION

1. Line Segment Method of Multiplications of Two Whole Numbers of any Number of Digits

To clearly understand this method, we will discuss some examples.

(i) Consider the multiplication of two digit numbers,

$$\begin{array}{r} 7 \quad 6 \\ \times 4 \quad 9 \\ \hline \end{array}$$

The digit of the different places of the required product will be found out as follows.

(a) Finding the Units Place Digit

To Find the unit's digit of the product of any two numbers, we always find the product their unit's digits.

$$\text{Here product of unit digits} = 6 \times 9 = 54$$

Unit's digit 4 of 54 is the unit's digit of the required product. Tenth digit 5 of 54 will be carry over to the tens place.

Thus

$$\begin{array}{r} 7 \quad 6 \\ \times 4 \quad 9 \\ \hline 4 \end{array}$$

5 carry over to the tens place.

(b) Finding the Tens Place Digit

$$\begin{array}{r} 7 \quad 6 \\ \times 4 \quad 9 \\ \hline 4 \end{array}$$

$$7 \times 9 + 6 \times 4 = 63 + 24 = 87$$

$$87 + 5 \text{ (from carry over)} = 92$$

Here unit's digit 2 of 92 is the tens place digit of the required product. Tens digit 9 of 92 will be carry over to the hundred's place digit.

Thus

$$\begin{array}{r} 7 \quad 6 \\ \times 4 \quad 9 \\ \hline 2 \quad 4 \end{array}$$

9 carry over to the hundred's place.

(c) Finding the Hundred's Place Digit

$$\begin{array}{r} 7 \quad 6 \\ \times 4 \quad 9 \\ \hline 2 \quad 4 \end{array}$$

$$7 \times 4 = 28$$

$$28 + 9 \text{ (from carry over)} = 37$$

Since 7 and 4 are the last digits on the left in both the given numbers, so this is the last calculation in this multiplication and hence we can write 37 for the remaining 2 digits in the required product.

Thus

$$\begin{array}{r} 7 \quad 6 \\ \times 4 \quad 9 \\ \hline 3 \quad 7 \quad 2 \quad 4 \end{array}$$

(ii) Consider the multiplication of more than 2 digits numbers,

$$\begin{array}{r} 5 \quad 4 \quad 0 \quad 2 \quad 3 \\ \times 3 \quad 1 \quad 5 \quad 6 \\ \hline \end{array}$$

Study the following table which explains the process of finding the digit of different places of the required product.

Finding the digit	Diagram showing the calculation process	Calculation	Required digit(s)	Carry on to the next place digit	Explanation of the diagram showing the calculation process
Unit digit	$\begin{array}{r} 5 \quad 4 \quad 0 \quad 2 \\ \times 3 \quad 1 \quad 5 \\ \hline 0 \end{array}$	$5 \times 2 = 10$	0	1	Multiplication between unit's digit of both the number shows by line segment between 2 and 5.
Tens digit	$\begin{array}{r} 5 \quad 4 \quad 0 \rightarrow 2 \\ \times 3 \quad 1 \leftarrow 5 \\ \hline 3 \quad 0 \end{array}$	$5 \times 0 + 1 \times 2 = 2$ $2 + 1 \text{ (carry over)} = 3$	3	0	Multiplication of tens digit 0 of 5402 by unit's digit 5 of 315 shows by line segment between 0 and 5, then rotate this line segment in clockwise direction about their midpoint to find the next pair of digits to be multiplied
Hundred digit	$\begin{array}{r} 5 \quad 4 \rightarrow 0 \rightarrow 2 \\ \times 3 \leftarrow 1 \leftarrow 5 \\ \hline 6 \quad 3 \quad 0 \end{array}$	$5 \times 4 + 1 \times 0 + 3 \times 2 = 26$	6	2	Multiplication of hundred's digit 4 of 5402 by unit's digit 5 of 315 shows by line segment between 4 and 5, then rotate this line segment in clockwise direction about their mid-point to find the next pair of digits to be multiplied.
Thousand digit	$\begin{array}{r} 5 \rightarrow 4 \rightarrow 0 \rightarrow 2 \\ \times 3 \leftarrow 1 \leftarrow 5 \\ \hline 1 \quad 6 \quad 3 \quad 0 \end{array}$	$5 \times 5 + 1 \times 4 + 3 \times 0 = 29$ $29 + 2 \text{ (carry over)} = 31$	1	3	Similar explanation as for given above for hundred digit but there is no digit in the left of 3 in 315, so the unit digit 2 of 5402 will not be multiplied by any digit.
Ten thousand digit	$\begin{array}{r} 5 \rightarrow 4 \quad 0 \quad 2 \\ \times 3 \leftarrow 1 \quad 5 \\ \hline 0 \quad 1 \quad 6 \quad 3 \quad 0 \end{array}$	$1 \times 5 + 3 \times 4 = 17$ $17 + 3 \text{ (carry out)} = 20$	0	2	Since unit digit 5 of 315 is multiplied by left most digit 5 of 5402 in finding the thousand digit. Hence tens digit 1 of 315 multiplies the left most digit 5 of 5402 and rotate the line segment in clockwise direction between 1 and 5 about their mid-point to find the next pair of digits to be multiplied but there is no digit in the left of 3 in 315, so further rotation of line segment between 3 and 4 in clockwise direction will not find any two digits to be multiplied and hence the ten's and unit's digit of 5402 will not be multiplied by any digit.

Finding the digit	Diagram showing the calculation process	Calculation	Required digit(s)	Carry on to the next place digit	Explanation of the diagram showing the calculation process
Last digit(s)	$ \begin{array}{r} 5 \quad 4 \quad 0 \quad 2 \\ \times 3 \quad 1 \quad 5 \\ \hline 1 \quad 7 \quad 0 \quad 1 \quad 6 \quad 3 \quad 0 \end{array} $	$3 \times 5 = 15$ $15 + 2 \text{ (carry over)} = 17$	17	0	Since ten's digit 1 of 315 is multiplied by left most digit 5 of 5402 in finding the ten thousand's digit, so hundred digit 3 of 315 multiplies the left most digit 5 of 5402. Since there is no digit in the left of 3 in 315, so rotation of line segment between 3 and 5 about their mid-point in anticlockwise direction will not find any two digits to be multiplied further and hence hundred, tens and unit digits of 5402 will not be multiplied by any digit.
Hence required product = 1701630					

In CAT and CAT like competitions large multiplications might not be required but it might be required to find any specific digit of the product of large multiplication, then the above method of multiplication is quite useful.

2. Multiplication of Two Numbers Using Formulae $(a - b)(a + b) = a^2 - b^2$

If the difference between two numbers x and y is a small even number, then the smaller is expressed as $(a - b)$ whereas larger is expressed as $(a + b)$, then the product of x and y is found out by the formulae $x \cdot y$ i.e., $(a - b)(a + b) = a^2 - b^2$

Here a should be such that a^2 is very easily calculated.

For example:

$$(i) \quad 38 \times 42 = (40 - 2) \times (40 + 2) = (40)^2 - (2)^2 = 1600 - 4 = 1596$$

$$(ii) \quad 66 \times 74 = (70 - 4) \times (70 + 4) = (70)^2 - (4)^2 = 4900 - 16 = 4884$$

$$(iii) \quad 2094 \times 2106 = (2100 - 6) \times (2100 + 6) = (2100)^2 - (6)^2 = 4410000 - 36 = 4409964$$

If the difference between the two numbers is not even, still this method is used by modify as

$$\begin{aligned}
 47 \times 54 &= 47 \times 53 + 47 \\
 &= (50 - 3) \times (50 + 3) + 47 \\
 &= (50)^2 - (3)^2 + 47 \\
 &= 2500 - 9 + 47 = 2538
 \end{aligned}$$

3. Multiplying Two Numbers Close to 100, 1000, 10000, 100000, etc

To multiply two numbers close to 100, 1000, 10000 or 100000; we can use a specific method which is discussed in the following illustrations.

(i) Let us multiply 92 and 97.

Step (a): Calculate the difference from 100 of both the numbers and write them as follows:

$$\begin{array}{r}
 \text{Difference} \\
 \text{from 100} \\
 92 \quad - 8 \\
 \times 97 \quad - 3 \\
 \hline
 \hline
 \end{array}$$

Step (b):

$$\begin{array}{r}
 92 \quad - 8 \\
 \times 97 \quad - 3 \\
 \hline
 89 \quad \quad 24 \\
 \hline
 \end{array}$$

Initial digits of the required product is found out by cross addition as
 $92 + (-3)$ or
 $97 + (-8) = 89$

Last two digits of the required product
 $(-8) \times (-3) = 24$

Thus, $92 \times 97 = 8924$

(ii) Let us multiply 1008 and 994.

$$\begin{array}{r}
 \text{Difference} \\
 \text{from 1000} \\
 1008 \quad + 8 \\
 \times 994 \quad - 6 \\
 \hline
 1002 \quad \quad 000 \\
 \hline
 - \quad \quad 48 \\
 \hline
 1001 \quad \quad 952 \\
 \hline
 \text{Initial} \quad \quad \text{Last three} \\
 \text{digits} \quad \quad \text{digits}
 \end{array}$$

Here we first find the initial digits by cross addition as $1008 + (-6)$ or $994 + 8 = 1002$

Now write 1002 as initial digits and write last three digits as 000, (i.e., last three zeroes of 1000) which means numbers' value is 1002000. Now in 1002000 add the product $8 \times (-6) = -48$, which gives the required product i.e., 1001952.

Illustration 10: Find the product 108×104 .

Solution:

$$\begin{array}{r}
 \text{Difference} \\
 \text{from 100} \\
 108 \quad + 8 \\
 \times 104 \quad + 4 \\
 \hline
 112 \quad \quad 32 \\
 \hline
 \text{Last} \quad \quad \text{First two} \\
 \text{digits} \quad \quad \text{digits}
 \end{array}$$

Hence $108 \times 104 = 11232$

ROUNDING OFF AND ITS USES

Sometimes we need to calculate approximate value of an expression. We find the approximate value of an expression like

$$71 \times 86 \text{ can be taken as } 70 \times 90 = 6300$$

$$63 + 47 + 24 + 69 \text{ can be taken as } 60 + 50 + 20 + 70 = 200$$

$$42 \times 33 + 72 - 33 \times 18 + 27 \text{ can be taken as}$$

$$40 \times 30 + 70 - 30 \times 20 + 30 = 1200 + 70 - 600 + 30 = 700$$

'BODMAS' RULE

A given series of calculations or operations is done in a specific order as each letter of BODMAS in order represent.

B → Brackets and order of operation of brackets is (), { }, []

O → Of (Calculation is done the same as multiplication)

D → Division

M → Multiplication

A → Addition

S → Subtraction

So, first of all we solve the inner most brackets moving outwards. Then we perform 'of' which means multiplication, then division, addition and subtraction.

- Addition and subtraction can be done together or separately as required.

- Between any two brackets if there is not any sign of addition, subtraction and division it means we have to do multiplication
 $(20 \div 5) (7 + 3 \times 2) + 8 = 4 (7 + 6) + 8$

$$= 4 \times 13 + 8 = 52 + 8 = 60$$

BRACKETS

They are used for the grouping of things or entities. The various kind of brackets are:

- '-' is known as line (or bar) bracket or vinculum.
- () is known as parenthesis, common bracket or small bracket.
- { } is known as curly bracket, brace or middle bracket.
- [] is known as rectangular bracket or big bracket.

The order of eliminating brackets is:

- line bracket
- small bracket (i.e., common bracket)
- middle bracket (i.e., curly bracket)
- big bracket (i.e., rectangular bracket)

Illustration 11: Find the value of

$$\left[5 - \left\{ 6 - (5 - \overline{4 - 3}) \right\} \right] \text{ of } \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} \div \frac{\frac{1}{2} + \frac{1}{3}}{\frac{1}{2} - \frac{1}{3}}$$

$$\text{Solution: } \left[5 - \left\{ 6 - (5 - \overline{4 - 3}) \right\} \right] \text{ of } \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} \div \frac{\frac{1}{2} + \frac{1}{3}}{\frac{1}{2} - \frac{1}{3}}$$

$$= [5 - \{6 - (5 - 1)\}] \text{ of } \frac{\frac{3}{2}}{\frac{1}{2}} \sqrt{\frac{\frac{5}{6}}{\frac{1}{6}}}$$

$$= \{5 - (6 - 4)\} \text{ of } \left(\frac{3}{2} \times \frac{2}{1} \right) \div \left(\frac{5}{6} \times \frac{6}{1} \right)$$

$$= (5 - 2) \text{ of } 3 \div 5$$

$$= 3 \text{ of } 3 \div 5 = 3 \times \frac{3}{5} = \frac{9}{5}$$

FACTORIAL

The product of n consecutive natural numbers (or positive integers) from 1 to n is called as the factorial ' n '. Factorial n is denoted by $n!$.

$$\text{i.e., } n! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 \dots (n-2)(n-1)n$$

$$4! = 1 \times 2 \times 3 \times 4 = 4 \times 3 \times 2 \times 1$$

$$5! = 1 \times 2 \times 3 \times 4 \times 5 = 5 \times 4 \times 3 \times 2 \times 1$$

$$6! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 = 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

Note: $0! = 1$ and $1! = 1$

Properties

- $n!$ is always an even number if $n \geq 2$.
- $n!$ always ends with zero if $n \geq 5$.

ROMAN NUMBERS

In this system there are basically seven symbols used to represent the whole Roman number system. The symbols and their respective values are given below.

$$I = 1, V = 5, X = 10, L = 50,$$

$$C = 100, D = 500 \text{ and } M = 1000$$

In general, the symbols in the numeral system are read from left to right, starting with the symbol representing the largest value; the same symbol cannot occur continuously more than three times; the value of the numeral is the sum of the values of the symbols.

For example LXVII = $50 + 10 + 5 + 1 + 1 = 67$.

An exception to the left to the right reading occurs when a symbol of smaller value is followed immediately by a symbol of greater value, then the smaller value is subtracted from the larger. For example.

$$CDXLVIII = (500 - 100) + (50 - 10) + 5 + 1 + 1 + 1 = 448.$$

Illustration 12: The value of the numeral MCDLXIV is:

- (a) 1666 (b) 664 (c) 1464 (d) 656

$$\text{Solution: } MCDLXIV = 1000 + (500 - 100) + 50 + 10 + (5 - 1) = 1464$$

Hence (c) is the correct option.

Illustration 13: Which of the following represents the numeral for 2949

- (a) MMMIX (b) MMXMIX
(c) MMCMIL (d) MMCMXLIX

$$\begin{aligned} \text{Solution: } 2949 &= 2000 + 900 + 40 + 9 \\ &= (1000 + 1000) + (1000 - 100) \\ &\quad + (50 - 10) + (10 - 1) \\ &= MMCMXLIX \end{aligned}$$

Hence (d) is the correct option.

IMPORTANT CONVERSION

$$1 \text{ trillion} = 10^{12} = 1000000000000$$

$$1 \text{ billion} = 10^9 = 1000000000$$

$$1 \text{ million} = 10^6 = 1000000$$

$$1 \text{ crore} = 10^7 = 100 \text{ lakh}$$

$$10 \text{ lakh} = 10^6 = 1 \text{ million}$$

$$1 \text{ lakh} = 10^5 = 100000 = 100 \text{ thousand}$$

$$1 \text{ thousand} = 10^3 = 1000$$

ABSOLUTE VALUE OR MODULUS OF A NUMBER

Absolute value of a number is its numerical value irrespective of its sign.

If x be a real number N then $|N|$ indicates the absolute value of N .

Thus $|6| = 6$, $|-6| = 6$, $|0| = 0$, $|1| = 1$, $|3.4| = 3.4$, $|-6.8| = 6.8$, etc.

$|-6| = 6$ can also be written as $|-6| = -(-6) = 6$. Thus, if x is a negative number, then $|x| = -x$ and if x is non-negative number, then $|x| = x$

$$\text{Hence } |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

PROPERTIES OF A MODULUS

$$(i) |a| = |-a| \quad (ii) |ab| = |a| |b|$$

$$(iii) \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad (iv) |a+b| \leq |a| + |b|$$

(The sign of equality holds only when the sign of a and b are same)

$$(v) \text{ If } |a| \leq k \Rightarrow -k \leq a \leq k$$

$$(vi) \text{ If } |a-b| \leq k \Rightarrow -k \leq a-b \leq k \Rightarrow b-k \leq a \leq b+k$$

Illustration 14: Solution of the equation $|x-2| = 5$ is

- (a) 3, -7 (b) -3, 7
(c) 3, 6 (d) None of these

$$\text{Solution: } |x-2| = 5 \Rightarrow x-2 = 5 \text{ or } x-2 = -5 \\ \Rightarrow x = 7 \text{ or } x = -3$$

Hence (b) is the correct option.

Illustration 15: The minimum value of the expression $|17x-8|-9$ is

- (a) 0 (b) -9
(c) $\frac{8}{17}$ (d) none of these

Solution: The value of expression $|17x-8|-9$ is minimum only when $|17x-8|$ is minimum. But the minimum value of $|k|$ is zero.

$$\text{Hence minimum value of } |17x-8|-9 = 0-9 = -9$$

Hence (b) is the correct answer.

POWERS OR EXPONENTS

When a number is multiplied by itself, it gives the square of the number. i.e., $a \times a = a^2$ (Example $5 \times 5 = 5^2$)

If the same number is multiplied by itself twice we get the cube of the number i.e., $a \times a \times a = a^3$ (Example $4 \times 4 \times 4 = 4^3$)

In the same way $a \times a \times a \times a \times a = a^5$

and $a \times a \times a \times \dots$ upto n times $= a^n$

There are five basic rules of powers which you should know:

If a and b are any two real numbers and m and n are positive integers, then

$$(i) a^m \times a^n = a^{m+n} \quad (\text{Example: } 5^3 \times 5^4 = 5^{3+4} = 5^7)$$

$$(ii) \frac{a^m}{a^n} = a^{m-n}, \text{ if } m > n \quad \left(\text{Example: } \frac{6^5}{6^2} = 6^{5-2} = 6^3 \right)$$

$$\frac{a^m}{a^n} = \frac{1}{a^{n-m}}, \text{ if } m < n \quad \left(\text{Example: } \frac{4^3}{4^8} = \frac{1}{4^{8-3}} = \frac{1}{4^5} \right)$$

$$\text{and } \frac{a^m}{a^n} = a^0 = 1, \text{ if } m = n \quad \left(\text{Example: } \frac{3^4}{3^4} = 3^{4-4} = 3^0 = 1 \right)$$

$$(iii) (a^m)^n = a^{mn} = (a^n)^m \quad (\text{Example: } (6^2)^4 = 6^{2 \times 4} = 6^8 = (6^4)^2)$$

$$(iv) (a)(ab)^n = a^n \cdot b^n \quad (\text{Example: } (6 \times 4)^3 = 6^3 \times 4^3)$$

$$(b) \left(\frac{a}{b} \right)^n = \frac{a^n}{b^n}, b \neq 0 \quad \left(\text{Example: } \left(\frac{5}{3} \right)^4 = \frac{5^4}{3^4} \right)$$

$$(v) a^{-n} = \frac{1}{a^n} \quad \left(\text{Example: } 5^{-3} = \frac{1}{5^3} \right)$$

$$(vi) \text{ For any real number } a, a^0 = 1$$

$$\text{Illustration 16: } \frac{5^{n+3} - 6 \times 5^{n+1}}{9 \times 5^n - 5^n \times 2^2} = ?$$

$$\text{Solution: } \frac{5^n \times 5^3 - 6 \times 5^n \times 5}{5^n (9 - 2^2)} \\ = \frac{5^n (5^3 - 6 \times 5)}{5^n \times 5} \\ = \frac{125 - 30}{5} \\ = \frac{95}{5} = 19$$

$$\text{Illustration 17: } \left\{ \left(\sqrt[3]{(81)^2} \right)^{3/2} \right\}^{1/4} = ?$$

$$\text{Solution: } \left\{ \left(\sqrt[3]{(81)^2} \right)^{3/2} \right\}^{1/4} \\ = \left\{ (81^2)^{1/3 \times \frac{3}{2}} \right\}^{1/4} \\ = (81)^{2 \times \frac{1}{3} \times \frac{3}{2} \times \frac{1}{4}} = (3^4)^{\frac{1}{4}} = 3$$

ALGEBRAIC IDENTITIES

Consider the equality $(x+2)(x+3) = x^2 + 5x + 6$

Let us evaluate both sides of this equality for some value of variable x say $x = 4$

$$\text{LHS} = (x+2)(x+3) = (4+2)(4+3) = 6 \times 7 = 42$$

$$\text{RHS} = (4)^2 + 5 \times 4 + 6 = 16 + 20 + 6 = 42$$

$$\text{So for } x = 4, \text{ LHS} = \text{RHS}$$

Let us calculate LHS and RHS for $x = -3$

$$\text{LHS} = (-3+2)(-3+3) = 0$$

$$\text{RHS} = (-3)^2 + -(-3) + 6 = 9 - 15 + 6 = 0$$

$$\therefore \text{ for } x = -3, \text{ LHS} = \text{RHS}$$

If we take any value of variable x , we can find that LHS = RHS

Such an equality which is true for every value of the variable present in it is called an identity. Thus $(x+2)(x+3) = x^2 + 5x + 6$, is an identity.

Identities differ from equations in the following manners.

An equation is a statement of equality of two algebraic expression involving one or more variables and it is true for certain values of the variable.

For example:

$$4x + 3 = x - 3 \quad \dots (1)$$

$$\Rightarrow 3x = -6 \Rightarrow x = -2$$

Thus equality (1) is true only for $x = -2$, no other value of x satisfy equation (1).

Standard Identities

- (i) $(a + b)^2 = a^2 + 2ab + b^2$
- (ii) $(a - b)^2 = a^2 - 2ab + b^2$
- (iii) $a^2 - b^2 = (a + b)(a - b)$
- (iv) $(x + a)(x + b) = x^2 + (a + b)x + ab$
- (v) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

Some More Identities

We have dealt with identities involving squares. Now we will see how to handle identities involving cubes.

- (i) $(a + b)^3 = a^3 + b^3 + 3a^2b + 3ab^2$
 $\Rightarrow (a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- (ii) $(a - b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$
 $\Rightarrow (a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- (iii) $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- (iv) $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- (v) $a^3 + b^3 + c^3 - 3abc$
 $= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
 If $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$

SQUARES

When a number is multiplied by itself, then we get the square of the number.

For example, square of 5 = 5×5 (or 5^2) = 25

Square of 2 and 3 digits numbers and cube of 2 digits numbers are very useful in CAT and CAT like competitions.

For this it is advised to learn the square of 1 to 30 as given in the table:

Number	Square	Number	Square
1	1	16	256
2	4	17	289
3	9	18	324
4	16	19	361
5	25	20	400
6	36	21	441
7	49	22	484
8	64	23	529

Number	Square	Number	Square
9	81	24	576
10	100	25	625
11	121	26	676
12	144	27	729
13	169	28	784
14	196	29	841
15	225	30	900

Shortcuts to Find the Squares of Numbers From 31 to 49

- (i) The numbers from 31 to 49 is written in the form $(50 - x)$.
 Thus to find the square of 38, 38 can be written as $(50 - 12)$.
- (ii) The last two digits of the square of 38 is the last two digits of $(12)^2 = 144$
 Thus last two digits of $(38)^2$ is 44 and 1 is carry over.
- (iii) The first two digits of the square of 38 = $25 - 12 + 1 = 14$
 Here 25 is the standard number used for finding square of any number from 31 to 49.
 12 is the value of x , when the number from 31 to 49 written in the form $(50 - x)$ and 1 is the carry over.
 Thus square of 38 = $38^2 = 1444$

The whole process can be shown in a single line as
 $38 \rightarrow 50 - 12 \rightarrow (12)^2 = 1\text{ }44 \rightarrow 25 - 12 + 1 = 14 \rightarrow 1444$

Illustration 18: Find the square of 31.

- Solution:** (i) $31 = 50 - 19$
 (ii) $(19)^2 = 361$
 Thus last two digits of the square of 31 is 61 and carry over 3.
 (iii) $25 - 19 + 3 = 9$
 Thus $(31)^2 = 961$
 The whole process can be shown in a single line as
 $31 \rightarrow 50 - 19 \rightarrow 19^2 = 3\text{ }61 \rightarrow 25 - 19 + 3 = 9 \rightarrow 961$

Shortcuts to Find the Squares of Numbers From 51 to 79

- (i) The numbers from 51 to 79 is written in the form $(50 + x)$.
 Thus to find square of 78, 78 can be written as $(50 + 28)$.
- (ii) The last two digits of the square of 78 is the last two digits of $(28)^2 = 784$.
 Thus last two digits of $(28)^2$ is 84 and carry over 2.
- (iii) The first two digits of the square of 78 = $25 + 28 + 7 = 60$.
 Here 25 is the standard number used for finding square of any number from 51 to 79. 28 is the value of x , when 78 is written in the form $(50 + x)$ and 2 is the carry over.
 Thus square of 78 = $78^2 = 6084$.
 The whole process can be shown in a single line as
 $78 \rightarrow 50 + 28 \rightarrow (28)^2 = 7\text{ }84 \rightarrow 25 + 28 + 7 = 60 \rightarrow 6084$

Shortcuts to Find the Squares of Numbers From 81 to 99

- (i) The numbers from 81 to 99 is written in the form $(100 - x)$.
 Thus to find square of 83, 83 can be written as $(100 - 17)$.

(ii) The last two digits of the square of 83 is the last two digits of $(17)^2 = 289$.

Thus last two digits of $(83)^2$ is 89 and carry over 2.

(iii) The first two digits of the square of 83
 $= 83 - 17 + 2 = 68$

Here 83 is the number whose square is to be found out. 17 is the value of x when 83 is written as $(100 - x)$ and 2 is carry over. Thus square of 83 $= (83)^2 = 6889$.

The whole process can be shown in a single line as

$$83 \rightarrow 100 - 17 \rightarrow (17)^2 = 289 \rightarrow 83 - 17 + 2 \rightarrow 68 \rightarrow 8968$$

Another Shortcut Method to Find the Square of any Two and Three Digits Numbers

This method of squaring is directly connected with a process known as 'Duplex Combination (D)'.

See the duplex (D) of some two and three digits numbers

- (i) Duplex (D) of 7 = square of 7 = $7^2 = 49$
- (ii) Duplex (D) of 5 = $(5)^2 = 25$
- (iii) Duplex (D) of 27 = Twice the product of the digits 2 and 7.
 $= 2 \times (2 \times 7) = 28$
- (iv) Duplex (D) of 54 = $2 \times (5 \times 4) = 40$
- (v) Duplex (D) of 69 = $2 \times (6 \times 9) = 108$
- (vi) Duplex (D) of 83 = $2 \times (8 \times 3) = 48$
- (vii) Duplex (D) of 97 = $2 \times (9 \times 7) = 126$
- (viii) Duplex (D) of 238 = Twice the product of extreme digits + Square of the central digit
 $= 2 \times (2 \times 8) + (3)^2 = 32 + 9 = 41$

(ix) Duplex (D) of 789 = $2 \times (7 \times 9) + (8)^2 = 126 + 64 = 190$

In the same way, we can find the Duplex (D) of any two and three digits number.

Now see the squaring of some two and three digits numbers.

(i) $(57)^2 = D \text{ of } 5 / D \text{ of } 57 / D \text{ of } 7$
 $= (5)^2 / 2 \times (5 \times 7) / (7)^2$
 $= 25 / 70 / 49$
 $= 25 + 7 (=32) / 70 + 4 (=74) / 49$
↖ carry over ↗
 $= 3249$

(ii) $(78)^2 = D \text{ of } 7 / D \text{ of } 78 / D \text{ of } 8$
 $= 49 / 112 / 64$
 $= 49 + 11 (=60) / 112 + 6 (=118) / 64$
↖ carry over ↗
 $= 6084$

(iii) $(83)^2 = 64 / 48 / 9 = 6889$

(iv) $(96)^2 = 81 / 108 / 36 = 9216$

(v) $(769)^2 = D \text{ of } 7 / D \text{ of } 76 / D \text{ of } 769 / D \text{ of } 69 / D \text{ of } 9$
 $= (7)^2 / 2 \times (7 \times 6) / 2 \times (7 \times 9) + (6)^2 / 2$
 $\quad \quad \quad \times (6 \times 9) / (9)^2$
 $= 49 / 84 / 162 / 108 / 81$
 $= 49 + 10 (=59) / 84 + 17 (=101) / 162 + 11 (=173) / 108 + 8 (=116) / 81$
↖ carry over ↗
 $= 591361$

(vi) $(483)^2 = D \text{ of } 4 / D \text{ of } 48 / D \text{ of } 483 / D \text{ of } 83 / D \text{ of } 3$
 $= 16 / 64 / 88 / 48 / 9 = 233289$

(vii) $(238)^2 = 4 / 12 / 41 / 48 / 64 = 56644$

In the same way, we can find the square of two and three digits numbers.

Properties of Squares

- (i) The difference between the squares of two consecutive natural numbers is always equal to the sum of the natural numbers. Thus

$$(87)^2 - (86)^2 = 87 + 86 = 173$$

This property is very useful in opposite direction. For example, if difference between squares of two consecutive numbers is given say 85, then you can immediately find the two numbers are 43 and 42.

- (ii) First two digits of the square of any number say 65, ending in 5 is always 25. The remaining digits of the square will be found out by the product of the given number leaving the units digit 5 i.e., 6 and the number 1 more than 6 i.e., 7.
 Now $6 \times 7 = 42$

Hence $(65)^2 = 4225$

Similarly first two digits of two 235 is 25 and the remaining digit is $23 \times 24 = 552$

$\therefore (235)^2 = 55225$

- (iii) The square of a number is always non-negative i.e., $a^2 \geq 0$, where a is any real number.

- (iv) The sum of square of first n natural numbers,

$$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- (v) (a) Square of 0 and 1 are the number itself i.e. $(0)^2 = 0$, $(1)^2 = 1$.

- (b) Square of any number between 0 and 1 is less than the number i.e., $(0.2)^2 < 0.2$.

- (vi) Number of zeroes at the end in the square of a given number is equal to twice the number of zeroes at the end of the given number.

SQUARE ROOTS

If $b = a \times a$ or a^2 , then a is called square root of b and it is represented as $\sqrt{b} = a$ or $(b)^{1/2} = a$.

Since, $16 = 4 \times 4$ or 4^2 , therefore $\sqrt{16} = 4$

And $25 = 5 \times 5$ or 5^2 , therefore $\sqrt{25} = 5$

There are two methods for finding the square root of a number.

(i) Prime Factorisation Method

To find the square root by this method, we first factorise the given number into prime numbers as given below for the number 3136.

$$3136 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 7 \times 7$$

Now pair the same prime factor like

$$3136 = \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{7 \times 7}$$

Now product of prime numbers taken one number from each pair of prime factors is the square root of the given number

2	3136
2	1568
2	784
2	392
2	196
2	98
7	49
7	7
	1

$$\therefore \sqrt{3136} = 2 \times 2 \times 2 \times 7 = 56$$

$$\text{If we write, } 3136 = (2)^6 \times (7)^2$$

Then square root of 3136 is the product of prime factors 2 and 7 with the powers half of the powers raised on 2 and 7 respectively.

$$\text{i.e., } \sqrt{3136} = (2)^3 \times 7 = 56$$

(ii) Division Method

In this method first of all pair the digits of the given number from right side. But there may be left a single digit at the left end of the number. Further process is shown below for the number 2304.

$$\sqrt{2304} = 48$$

$$\begin{array}{r} 48 \\ 4 \overline{) 2304} \\ \underline{4} \\ 88 \\ \underline{8} \\ 8 \\ \underline{8} \\ 0 \\ \times \times \times \end{array}$$

Illustration 19: Find the square root of 15625.

$$\text{Solution: } \sqrt{15625} = 125$$

$$\begin{array}{r} 125 \\ 1 \overline{) 15625} \\ \underline{1} \\ 22 \\ \underline{2} \\ 245 \\ \underline{2} \\ 25 \\ \underline{25} \\ 0 \\ \times \times \times \end{array}$$

CUBES

When a number multiplies itself three times, we get the cube of the number.

$$\text{Cube of } 4 = 4 \times 4 \times 4 = 64$$

Cubes of large numbers are rarely used. It is advised to you to learn the cube of the integers from 1 to 10.

Number	1	2	3	4	5	6	7	8	9	10
Cube	1	8	27	64	125	216	343	512	729	1000

To find the cube of a two digit number, first write the square of tens digit of the number and then write three numbers separately right of the square of tens digit, which form a G.P. (geometrical progression) of four terms whose first term is the square of tens digit and common ratio is equal to the ratio of unit's digit to tens digit of the given numbers.

Note that the sequence 3, 6, 12, 24, ... is a G.P. whose first term is 3. In this sequence we get each number after multiplying just previous number by 2.

$$\therefore \frac{6}{3} = \frac{12}{6} = \frac{24}{12} = \dots = 2$$

Thus 2 is called common ratio.

Let us find the cube of a two digits number 26. For this we find the cube of tens digit i.e. $(2)^3 = 8$, then we find the ratio of its unit's digit to tens digit $= \frac{6}{2} = 3$.

Now we write square of tens digit i.e. 8 and the three numbers right of 8 separately in such a way that 8 and the three number are in G.P. whose common ratio is 3 as

$$8 \quad 24 \quad 72 \quad 216$$

(we get each of the numbers 24, 72 and 216 by multiplying just previous number of it by the common ratio 3.)

Now write twice the two middle number 24 and 72 just below to 24 and 72 respectively and add the numbers one below the other in two rows with carry over the digits except units digit of each sum from right to left as shown:

$$\begin{array}{r} 8 \quad 24 \quad 72 \quad 216 \\ 48 \quad 144 \\ \hline \begin{array}{c} \text{⑨} \text{ carry over} \quad \text{②3} \text{ carry over} \quad \text{②1} \text{ carry over} \\ 17 \quad \text{⑨} 5 \quad \text{②3} 7 \quad \text{②1} 6 \end{array} \end{array}$$

The final number 17576 obtained is the cube of 26 i.e., $(26)^3 = 17576$.

Illustration 20: Find the cube of 42.

Solution: Cube of tens digit 4 = 64.

$$\text{Ratio of unit's digit to tens digit} = \frac{2}{4} = \frac{1}{2}$$

The next three numbers are

$$64 \times \frac{1}{2}, \left(64 \times \frac{1}{2}\right) \times \frac{1}{2}, \left(64 \times \frac{1}{2} \times \frac{1}{2}\right) \times \frac{1}{2} \text{ i.e. } 32, 16, 8.$$

$$\begin{array}{r} 64 \quad 32 \quad 16 \quad 8 \\ 64 \quad 32 \\ \hline \begin{array}{c} \text{⑩} \text{ carry over} \quad \text{④} \text{ carry over} \\ 74 \quad \text{⑩} 0 \quad \text{④} 8 \quad 8 \end{array} \end{array}$$

$$\therefore (42)^3 = 74088$$

Illustration 21: Find the cube of 14.

Solution: $(1)^3 = 1$,

$$\text{Ratio of unit digit to tens digit} = \frac{4}{1} = 4$$

$$\begin{array}{r} 1 \quad 4 \quad 16 \quad 64 \\ 8 \quad 32 \\ \hline \begin{array}{c} \text{①} \quad \text{⑤} \quad \text{⑥} \\ 2 \quad \text{①} 7 \quad \text{⑤} 4 \quad \text{⑥} 4 \end{array} \end{array}$$

$$\therefore (14)^3 = 2744$$

Illustration 22: Find the cube of 33.

Solution: $(3)^3 = 27$, Ratio $= \frac{3}{3} = 1$

$$\begin{array}{r} 27 \quad 27 \quad 27 \quad 27 \\ 54 \quad 54 \\ \hline \begin{array}{c} \text{⑧} \quad \text{⑧} \quad \text{②} \\ 35 \quad \text{⑧} 9 \quad \text{⑧} 3 \quad \text{②} 7 \end{array} \end{array}$$

$$\therefore (33)^3 = 35937$$

Illustration 23: Find the cube of 39.

Solution:

27	81	243	729
	162	486	
32	80	72	
59	323	801	729

$$\therefore (39)^3 = 59319$$

Illustration 24: Find the cube of 28.

Solution:

8	32	128	512
	64	256	
13	43	51	
21	9	5	2

$$\therefore (28)^3 = 21952.$$

Practice Exercises



Foundation Level

- $287 \times 287 + 269 \times 269 - 2 \times 287 \times 269 = ?$
(a) 534 (b) 446
(c) 354 (d) 324
- If $(64)^2 - (36)^2 = 20 \times x$, then $x = ?$
(a) 70 (b) 120
(c) 180 (d) 140
- If $\sqrt{3} = 1.732$ and $\sqrt{2} = 1.414$, the value of $\frac{1}{\sqrt{3} + \sqrt{2}}$ is
(a) 0.064 (b) 0.308
(c) 0.318 (d) 2.146
- $\sqrt{0.01 + \sqrt{0.0064}} = ?$
(a) 0.3 (b) 0.03
(c) $\sqrt{0.18}$ (d) None of these
- $356 \times 936 - 356 \times 836 = ?$
(a) 35600 (b) 34500
(c) 9630 (d) 93600
- The value of $\frac{\frac{1}{2} \div \frac{1}{2} \text{ of } \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \text{ of } \frac{1}{2}}$ is
(a) $\frac{2}{3}$ (b) 2
(c) $\frac{4}{3}$ (d) 3
- The simplified value of
$$\frac{\left(1 + \frac{1}{1 + \frac{1}{100}}\right) \left(1 + \frac{1}{1 + \frac{1}{100}}\right) - \left(1 - \frac{1}{1 + \frac{1}{100}}\right) \left(1 - \frac{1}{1 + \frac{1}{100}}\right)}{\left(1 + \frac{1}{1 + \frac{1}{100}}\right) + \left(1 - \frac{1}{1 + \frac{1}{100}}\right)}$$
 is
(a) 100 (b) $\frac{200}{101}$
(c) 200 (d) $\frac{202}{100}$
- If $5^a = 3125$, then the value of $5^{(a-3)}$ is
(a) 25 (b) 125
(c) 625 (d) 1625
- In a group of buffaloes and ducks, the number of legs are 24 more than twice the number of heads. What is the number of buffaloes in the group?
(a) 6 (b) 8
(c) 10 (d) 12
- $\sqrt{\frac{0.081 \times 0.324 \times 4.624}{1.5625 \times 0.0289 \times 72.9 \times 64}}$ is equal to
(a) 0.024 (b) 0.24
(c) 2.4 (d) 24
- $\sqrt[3]{4 \frac{12}{125}}$?
(a) $1\frac{2}{5}$ (b) $1\frac{3}{5}$
(c) $1\frac{4}{5}$ (d) $2\frac{2}{5}$
- If $3^{4X-2} = 729$, then find the value of X.
(a) 4 (b) 3
(c) 2 (d) 5
- What number must be added to the expression $16a^2 - 12a$ to make it a perfect square?
(a) $\frac{9}{4}$ (b) $\frac{11}{2}$
(c) $\frac{13}{2}$ (d) 16
- The value of $\left[\frac{1}{\sqrt{9}-\sqrt{8}}\right] - \left[\frac{1}{\sqrt{8}-\sqrt{7}}\right] + \left[\frac{1}{\sqrt{7}-\sqrt{6}}\right] - \left[\frac{1}{\sqrt{6}-\sqrt{5}}\right] + \left[\frac{1}{\sqrt{5}-\sqrt{4}}\right]$ is
(a) 6 (b) 5
(c) -7 (d) -6

15. Simplify: $5\sqrt[3]{250} + 7\sqrt[3]{16} - 14\sqrt[3]{54}$
- (a) $-2\sqrt[3]{2}$ (b) $-3\sqrt[3]{2}$
(c) $3\sqrt[3]{2}$ (d) $2\sqrt[3]{3}$
16. The no. plate of a bus had peculiarity. The bus number was a perfect square. It was also a perfect square when the plate was turned upside down. The bus company had only five hundred buses numbered from 1 to 500. What was the number?
- (a) 169 (b) 36
(c) 196 (d) Cannot say
17. If * means adding six times of second number into first number, then find the value of $(1*2)*3$.
- (a) 121 (b) 31
(c) 93 (d) 91
18. If a and b are positive integers, such that $a^b = 125$, then $(a - b)^{a+b-4} = ?$
- (a) 16 (b) 25
(c) 28 (d) 30
19. If $p \times q = p + q + \frac{p}{q}$, then value of $8 \times 2 = ?$
- (a) 2 (b) 10
(c) 14 (d) 16
20. If $x*y = x^2 + y^2 - xy$, then value of $9*11$ is
- (a) 93 (b) 103
(c) 60.5 (d) 121
21. The least number by which we multiply to the 11760, so that we can get a perfect square number
- (a) 2 (b) 3
(c) 5 (d) None of these
22. $\sqrt{\frac{0.081 \times 0.484}{0.0064 \times 6.25}}$?
- (a) 0.99 (b) 0.88
(c) 0.77 (d) 0.98
23. If $5\sqrt{5} \times 5^3 \div 5^{-3/2} = 5^{(a+2)}$, then value of a is
- (a) 5 (b) 4
(c) 6 (d) 7
24. If difference between the $\frac{4}{5}$ of $\frac{3}{4}$ of a number and $\frac{2}{5}$ of $\frac{1}{6}$ of the same number is 648, then number is
- (a) 1110 (b) 1215
(c) 1325 (d) 1440
25. If sum of two numbers is 42 and their product is 437, then find their difference.
- (a) 3 (b) 4
(c) 5 (d) 7
26. $54.327 \times 357.2 \times 0.0057$ is the same as:
- (a) $5.4327 \times 3.572 \times 5.7$
(b) $5.4327 \times 3.572 \times 0.57$
(c) $54327 \times 3572 \times 0.0000057$
(d) None of these
27. Write the 44000 in Roman numerals
- (a) XLI (b) XLVI
(c) XLIV (d) XLVIC
28. Write LXXIX in Hindu-Arabic numerals
- (a) 70000 (b) 70009
(c) 7009 (d) 700009
29. If $\frac{a+b}{b+c} = \frac{c+d}{d+a}$, then
- (a) a must equal c
(b) $a + b + c + d$ must equal zero
(c) either $a = c$ or $a + b + c + d = 0$, or both
(d) $a(b + c + d) = c(a + b + d)$
30. A number lies between 300 and 400. If the number is added to the number formed by reversing the digits, the sum is 888 and if the unit's digit and the ten's digit change places, the new number exceeds the original number by 9. Find the number.
- (a) 339 (b) 341
(c) 378 (d) 345
31. x and y are 2 different digits. If the sum of the two digit numbers formed by using both the digits is a perfect square, then find $x + y$.
- (a) 10 (b) 11
(c) 12 (d) 13
32. If $a > 1$, then arrange the following in ascending order.
- I. $\sqrt[3]{4\sqrt{a^3}}$ II. $\sqrt[3]{5\sqrt{a^4}}$
III. $\sqrt[3]{a}$ IV. $\sqrt[5]{a^3}$
- (a) I, II, III, IV (b) I, II, IV, III
(c) IV, I, III, II (d) III, I, II, IV
33. Arrange the following in the descending order:
 $5^{1/4}, 4^{1/3}, 6^{1/5}$.
- (a) $4^{1/3}, 5^{1/4}, 6^{1/5}$ (b) $5^{1/4}, 4^{1/3}, 6^{1/5}$
(c) $6^{1/5}, 4^{1/3}, 5^{1/4}$ (d) $5^{1/4}, 4^{1/3}, 6^{1/5}$

34. If $a + b + c = 13$, $a^2 + b^2 + c^2 = 69$, then find $ab + bc + ca$.
 (a) -50 (b) 50
 (c) 69 (d) 75
35. Which of the following is correct if $A = 3^{3^3}$, $B = 3^{3^{3^3}}$, $C = 3^{3^{3^3}}$ and $D = 3^{3^{3^3}}$?
 (a) $A > B = C > D$ (b) $C > A > B > D$
 (c) $A > C > D > B$ (d) $C > B > D > A$
36. If $a - 8 = b$, then determine the value of $|a - b| - |b - a|$.
 (a) 16 (b) 0
 (c) 4 (d) 2
37. Find the value of x in $\sqrt{x + 2\sqrt{x + 2\sqrt{x + 2\sqrt{3x}}}} = x$.
 (a) 1 (b) 3
 (c) 6 (d) 12
38. Find the possible integral value of x , if $x^2 + |x - 1| = 1$.
 (a) 1 (b) -1
 (c) 0 (d) 1 and 0
39. Find two numbers such that their sum, their product and the differences of their squares are equal.
 (a) $\left(\frac{3+\sqrt{3}}{2}\right)$ and $\left(\frac{1+\sqrt{2}}{2}\right)$ or $\left(\frac{3+\sqrt{2}}{2}\right)$ and $\left(\frac{1+\sqrt{2}}{2}\right)$
 (b) $\left(\frac{3+\sqrt{7}}{2}\right)$ and $\left(\frac{1+\sqrt{7}}{2}\right)$ or $\left(\frac{3+\sqrt{6}}{2}\right)$ and $\left(\frac{1-\sqrt{6}}{2}\right)$
 (c) $\left(\frac{3-\sqrt{5}}{2}\right)$ and $\left(\frac{1-\sqrt{5}}{2}\right)$ or $\left(\frac{3+\sqrt{5}}{2}\right)$ and $\left(\frac{1+\sqrt{5}}{2}\right)$
 (d) None of these

Standard Level

- Value of $999\frac{995}{999} \times 999 = ?$
 (a) 990809 (b) 998996
 (c) 153.6003 (d) 213.0003
- $7892.35 \times 99.9 = ?$
 (a) 753445.765 (b) 764455.765
 (c) 788445.765 (d) None of these
- How many $\frac{1}{12}$ in $18\frac{3}{4}$
 (a) 522 (b) 252
 (c) 225 (d) 253
- The least possible positive number which should be added to 575 to make a perfect square number is
 (a) 0 (b) 1
 (c) 4 (d) None of these
- If $a * b * c = \sqrt{\frac{(a+2)(b+3)}{(c+1)}}$, then the value of $(6 * 15 * 3)$ is
 (a) 6 (b) 3
 (c) 4 (d) can't be determined
- If $x = 3 + \sqrt{8}$, then $\left(x^2 + \frac{1}{x^2}\right) = ?$
 (a) 34 (b) 24
 (c) 38 (d) 36
- If $x^a = y^b = z^c$ and $y^2 = zx$ then the value of $\frac{1}{a} + \frac{1}{c}$ is
 (a) $\frac{b}{2}$ (b) $\frac{c}{2}$
 (c) $\frac{2}{b}$ (d) $2a$
- If $\frac{9^n \times 3^2 \times (3^{-n/2})^{-2} - (27)^n}{3^{3m} \times 2^3} = \frac{1}{27}$ then the value of $(m - n)$ is
 (a) -1 (b) 1
 (c) 2 (d) -2
- If $\frac{2x}{1 + \frac{1}{1 + \frac{x}{1-x}}}$ 1, then find the value of x .
 (a) $\frac{2}{3}$ (b) $\frac{3}{2}$
 (c) 2 (d) $\frac{1}{2}$
- Find the value of $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \frac{1}{5 \times 6} + \dots + \frac{1}{9 \times 10}$
 (a) $\frac{3}{2}$ (b) $\frac{2}{5}$
 (c) $\frac{2}{3}$ (d) $\frac{3}{5}$
- Find the square root of $7 - 2\sqrt{10}$.
 (a) $\sqrt{5} + \sqrt{2}$ (b) $-\sqrt{5} - \sqrt{2}$
 (c) $\pm(\sqrt{5} - \sqrt{2})$ (d) $\pm(\sqrt{5} + \sqrt{2})$
- The product of two 2-digit numbers is 1938. If the product of their unit's digits is 28 and that of ten's digits is 15, find the larger number.
 (a) 34 (b) 57
 (c) 43 (d) 75
- If $P + P! = P^3$, then the value of P is
 (a) 4 (b) 6
 (c) 0 (d) 5
- For any real value of x the maximum value of $8x - 3x^2$ is
 (a) $\frac{8}{3}$ (b) 4
 (c) 5 (d) $\frac{16}{3}$
- If x is a number satisfying the equation $\sqrt[3]{x+9} - \sqrt[3]{x-9} = 3$, then x^2 is between
 (a) 55 and 65 (b) 65 and 75
 (c) 75 and 85 (d) 85 and 95
- The value of $\left[35.7 - \left(3 + \frac{1}{3 + \frac{1}{3}}\right) - \left(2 + \frac{1}{2 + \frac{1}{2}}\right)\right]$ is
 (a) 30 (b) 34.8
 (c) 36.6 (d) 41.4
- Which one of the following sets of surds is in correct sequence of ascending order of their values?
 (a) $\sqrt[4]{10}, \sqrt[3]{6}, \sqrt{3}$ (b) $\sqrt{3}, \sqrt[4]{10}, \sqrt[3]{6}$
 (c) $\sqrt{3}, \sqrt[3]{6}, \sqrt[4]{10}$ (d) $\sqrt[4]{10}, \sqrt{3}, \sqrt[3]{6}$
- Rohan is asked to figure out the marks scored by Sunil in three different subjects with the help of certain clues. He is told that the product of the marks obtained by Sunil is 72 and the sum of the marks obtained by Sunil is equal to the Rohan's current age (in completed years). Rohan could not answer the question with this information. When he was also told that Sunil got the highest marks in Physics among the three subjects, he immediately answered the question

correctly. What is the sum of the marks scored by Sunil in the two subjects other than Physics?

- (a) 6 (b) 8
(c) 10 (d) cannot be determined

19. The last three-digits of the multiplication 12345×54321 will be

- (a) 865 (b) 745
(c) 845 (d) 945

20. The sum of the two numbers is 12 and their product is 35. What is the sum of the reciprocals of these numbers ?

- (a) $\frac{12}{35}$ (b) $\frac{1}{35}$
(c) $\frac{35}{8}$ (d) $\frac{7}{32}$

21. Find the value of $\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{5}\right) \dots \left(1 - \frac{1}{100}\right)$.

- (a) $\frac{1}{5}$ (b) $\frac{1}{10}$
(c) $\frac{1}{50}$ (d) $\frac{2}{5}$

22. An employer pays `20 for each day a works, and forfeits `3 for each day he is idle. At the end of 60 days, a worker gets `280. For how many days did the worker remain idle?

- (a) 28 (b) 40
(c) 52 (d) 60

23. Simplify :
$$1 + \frac{\frac{1}{2}}{\frac{2}{3}} \div \frac{1 + \frac{2}{3} + \frac{9}{2}}{1 - \frac{2}{3}}$$

- (a) $\frac{11}{13}$ (b) $\frac{13}{15}$
(c) $\frac{13}{11}$ (d) $\frac{15}{13}$

24. If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, where x, y, z are natural numbers,

then x, y, z are

- (a) 1, 2, 5 (b) 1, 5, 2
(c) 5, 2, 11 (d) 11, 2, 5

25. The value of $\sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{(0.003)^2 + (0.021)^2 + (0.0065)^2}}$ is

- (a) 0.1 (b) 10
(c) 10^2 (d) 10^3

26. If $\frac{x^2 + y^2 + z^2 - 64}{xy - yz - zx} = -2$ and $x + y = 3z$, then the value

of z is

- (a) 2 (b) 3
(c) 4 (d) None of these

27. If $\sqrt{24} = 4.899$, the value of $\sqrt{\frac{8}{3}}$ is

- (a) 0.544 (b) 1.333
(c) 1.633 (d) 2.666

28. If $(X + (1/X)) = 4$, then the value of $X^4 + 1/X^4$ is

- (a) 124 (b) 64
(c) 194 (d) Can't be determined

29. If $\sqrt{15625} = 125$, then the value of

$\sqrt{15625} + \sqrt{156.25} + \sqrt{1.5625}$ is

- (a) 1.3875 (b) 13.875
(c) 138.75 (d) 156.25

30. A hostel has provisions for 250 students for 35 days. After 5 days, a fresh batch of 25 students was admitted to the hostel. Again after 10 days, a batch of 25 students left the hostel. How long will the remaining provisions survive?

- (a) 18 days (b) 19 days
(c) 20 days (d) 17 days

31. If $\frac{97}{19} = a + \frac{1}{b + \frac{1}{c}}$ where a, b and c are positive integers,

then what is the sum of a, b and c ?

- (a) 16 (b) 20
(c) 9 (d) Cannot be determined

Expert Level

- Let a, b, c, d and e be integers such that $a = 6b = 12c$, and $2b = 9d = 12e$. Then which of the following pairs contains a number that is not an integer?
 - $\left(\frac{a}{27}, \frac{b}{e}\right)$
 - $\left(\frac{a}{36}, \frac{c}{e}\right)$
 - $\left(\frac{a}{12}, \frac{bd}{18}\right)$
 - $\left(\frac{a}{6}, \frac{c}{d}\right)$
- Which one among $2^{1/2}$, $3^{1/3}$, $4^{1/4}$, $6^{1/6}$ and $12^{1/12}$ is the largest?
 - $2^{1/2}$
 - $3^{1/3}$
 - $4^{1/4}$
 - $6^{1/6}$
- If x and y are any natural numbers and $x \square y = x \times y$, if $x + y$ is an even, $= y^2$, if $x + y$ is an odd. $x \Delta y = x^2$, if x^y is an even $= x^2 - y^2$, if x^y is an odd $(9 \square 11) \Delta 4$ equals to –
 - 297
 - 9785
 - 9801
 - None of these
- Number S is obtained by squaring the sum of digits of a two digit number D . If difference between S and D is 27, then the two digit number D is
 - 24
 - 54
 - 34
 - 45
- The value of $(2^{3/4} + 2^{1/2} + 2^{1/4} + 1)(2^{1/4} - 1)$ is
 - 5
 - 7
 - 1
 - 1
- Each of the series $S_1 = 2 + 4 + 6 + \dots$ and $S_2 = 3 + 6 + 9 + \dots$ is continued to 100 terms. Find how many terms are identical.
 - 34
 - 33
 - 32
 - None of the these
- $\sqrt{x}\sqrt{y} = \sqrt{xy}$ is true only when
 - $x > 0, y > 0$
 - $x > 0$ and $y < 0$
 - $x < 0$ and $y > 0$
 - All of these
- $\sqrt{(0.798)^2 + 0.404 \times 0.798 + (0.202)^2} + 1 = ?$
 - 0
 - 2
 - 1.596
 - 10.404
- Square root of $\frac{(0.75)^3}{1-0.75} + (0.75 + 0.75^2 + 1) = ?$
 - 4
 - 3
 - 2
 - 1
- If $2 * 3 = \sqrt{13}$ and $3 * 4 = 5$, then the value of $5 * 12$ is
 - 17
 - $\sqrt{29}$
 - 21
 - 13
- The value of $\left[35.7 - \left(3 + \frac{1}{3 + \frac{1}{3}} \right) - \left(2 + \frac{1}{2 + \frac{1}{2}} \right) \right]$ is
 - 30
 - 34.8
 - 36.6
 - 41.4
- The value of the expression $\frac{4^n \times 20^{m-1} \times 12^{m-n} \times 15^{m+n-2}}{16^m \times 5^{2m+n} \times 9^{m-1}}$ is
 - 500
 - 1
 - 200
 - $\frac{1}{500}$
- If $3a = 4b = 6c$ and $a + b + c = 27\sqrt{29}$, then $\sqrt{a^2 + b^2 + c^2}$ is
 - $3\sqrt{29}$
 - 81
 - 87
 - None of these
- Find the number of zeros at the end of the product of $2^{222} \times 5^{555}$.
 - 1
 - 22
 - 222
 - 555
- If both x and y are integers, how many solutions are there of the equation $(x - 8)(x - 10) = 2^y$?
 - 1
 - 2
 - 3
 - more than 3
- p and q are positive numbers such that $p^q = q^p$, and $q = 9p$. The value of p is
 - $\sqrt{9}$
 - $\sqrt[9]{9}$
 - $\sqrt[2]{9}$
 - $\sqrt[8]{9}$
- $M = \sqrt{3 - \sqrt{5} + \sqrt{9 - 4\sqrt{5}}}$ and $N = \sqrt{\sqrt{7} - 1 - \sqrt{11 - 4\sqrt{7}}}$. What is the value of $\frac{M - N}{M + N}$?
 - 0
 - 1
 - 1
 - None of these
- If a and b are real numbers such that $a^{a^b} = b$ and $a \neq b$, then what is the value of $a^b - b$?
 - 1
 - 0
 - 1
 - 2

Test Yourself

- Find the sum of the following numbers using row form.
86324, 698, 4366, 32517, 10651
(a) 134566 (b) 134666
(c) 134556 (d) 143556
- Find the product of 100008×100004 .
(a) 1000120032 (b) 10001200032
(c) 1100120032 (d) 12001200332
- Represent 3949 in Roman numbers.
(a) MMMCMLXIX (b) MMMCMLXIX
(c) MMMCMLX (d) MMMCLIX
- Find 16^3
(a) 4036 (b) 4056
(c) 4096 (d) 4076
- $9998 \times 999 = ?$
(a) 9997001 (b) 9988002
(c) 9987012 (d) 9898012
- $\sqrt{\frac{0.324 \times 0.081 \times 4.624}{1.5625 \times 0.0289 \times 72.9 \times 64}}$?
(a) 24 (b) 2.40
(c) 0.024 (d) None of these
- If $Z + \frac{1}{Z} = 1$, $Z^{64} + \frac{1}{Z^{64}}$ is equal to
(a) 0 (b) 1
(c) -1 (d) -2
- Find the square root of $7 + \sqrt{48}$.
(a) $2 - \sqrt{3}$ (b) $2 + \sqrt{3}$
(c) $3 + \sqrt{2}$ (d) $3 - \sqrt{2}$
- Find the square of 112.
(a) 12444 (b) 12504
(c) 12584 (d) 12544
- Find the square of 679.
(a) 461041 (b) 461141
(c) 460041 (d) 451011
- If $x^{1/p} = y^{1/q} = z^{1/r}$ and $xyz = 1$, then the value of $p + q + r$ would be
(a) 0 (b) 1
(c) 2 (d) $x + y + z$
- $\left(\frac{1}{64}\right)^0 + (64)^{-1/2} - (32)^{4/5}$ is equal to
(a) $-15\frac{7}{8}$ (b) $16\frac{1}{8}$
(c) $-14\frac{7}{8}$ (d) $17\frac{1}{8}$
- Find the cube root of $38 - 17\sqrt{5}$
(a) $2 + \sqrt{5}$ (b) $2 - \sqrt{5}$
(c) $3 - \sqrt{5}$ (d) $4 - \sqrt{5}$
- If $|x - 4| + |y - 4| = 4$, then how many integer values can the set (x, y) have?
(a) Infinite (b) 5
(c) 16 (d) 9
- $\frac{3}{2}$ of $\left(\frac{4}{3} \div \frac{5}{7}\right) + \frac{1}{2} \left[3\frac{4}{5} - \left\{ \frac{2}{5} \div \left(\frac{1}{3} + \frac{1}{2} + \frac{1}{5} - \frac{1}{6} \right) \right\} \right]$
(a) $\frac{130}{581}$ (b) $\frac{581}{130}$
(c) $\frac{641}{170}$ (d) $\frac{541}{170}$

Hints & Solutions

Foundation Level

1. (d) Given Exp. = $a^2 + b^2 - 2ab$, where $a = 287$ and $b = 269 = (a - b)^2 = (287 - 269)^2 = (18)^2 = 324$.

2. (d) $20 \times x = (64 + 36)(64 - 36) = 100 \times 28$

$$\Rightarrow x = \frac{100 \times 28}{20} = 140.$$

3. (c) $\frac{1}{\sqrt{3} + \sqrt{2}} \cdot \frac{1}{(\sqrt{3} + \sqrt{2})} \times \frac{(\sqrt{3} - \sqrt{2})}{(\sqrt{3} + \sqrt{2})} = \left(\frac{\sqrt{3} - \sqrt{2}}{3 - 2} \right)$
 $= (\sqrt{3} - \sqrt{2}) = (1.732 - 1.414) = 0.318$

4. (a) Given expression = $\sqrt{0.01 + 0.08} = \sqrt{0.09} = 0.3$

5. (a) $356 \times 936 - 356 \times 836 = 356 \times (936 - 836)$
 $= 356 \times 100 = 35600$

6. (a) $\frac{\frac{1}{2} \div \frac{1}{2} \times \frac{1}{2}}{\frac{1}{2} + \frac{1}{2} \times \frac{1}{2}} \cdot \frac{\frac{1}{2} \times 2 \times \frac{1}{2}}{\frac{3}{4}} = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}$

7. (b) Given exp.

$$= \frac{a^2}{a+b} \cdot a \cdot b \cdot \left(1 + \frac{1}{1 + \frac{1}{100}} \right) \left(1 + \frac{1}{1 + \frac{1}{100}} \right)$$

$$= 2 \times \frac{1}{(101/100)} = 2 \times \frac{100}{101} = \frac{200}{101}$$

8. (a) $5^a = 3125 \Rightarrow 5^a = 5^5 \Rightarrow a = 5$
 $\Rightarrow 5^{(a-3)} = 5^{(5-3)} = 5^2 = 25$

9. (d) Let the number of buffaloes be x and the number of ducks be y .
 Then, $4x + 2y = 2(x + y) + 24 \Leftrightarrow 2x = 24 \Leftrightarrow x = 12$.

10. (a) Given exp. = $\sqrt{\frac{81 \times 324 \times 4624}{15625 \times 289 \times 729 \times 64}}$
 $= \frac{9 \times 18 \times 68}{125 \times 17 \times 27 \times 8} \cdot \frac{3}{125} = 0.024.$

11. (b) $\sqrt[3]{4 \frac{12}{125}} = \sqrt[3]{\frac{512}{125}} = \left(\frac{8 \times 8 \times 8}{5 \times 5 \times 5} \right)^{1/3} = \frac{8}{5} \cdot \frac{1}{5}$

12. (c) $729 = 9^3 = 3^6$, Now $4X - 2 = 6$ or $X = 2$.

13. (a) $16a^2 - 12a = (4a)^2 - 2(4a)(3/2)$
 $\therefore$ The number is $(3/2)^2 = (9/4)$.

14. (b) By rationalization we have

$$\left[\frac{1}{\sqrt{9} - \sqrt{8}} \right] \left[\frac{1}{\sqrt{9} - \sqrt{8}} \right] \times \frac{\sqrt{9} + \sqrt{8}}{\sqrt{9} + \sqrt{8}} = \frac{\sqrt{9} + \sqrt{8}}{9 - 8}$$

$$= \sqrt{9} + \sqrt{8}$$

Similarly, $\left[\frac{1}{\sqrt{8} - \sqrt{7}} \right] = \sqrt{8} + \sqrt{7}$ and $\frac{1}{\sqrt{7} - \sqrt{6}} = \sqrt{7} + \sqrt{6}$

and so on. The given expression

$$= (\sqrt{9} + \sqrt{8}) - (\sqrt{8} + \sqrt{7}) + (\sqrt{7} + \sqrt{6}) - (\sqrt{6} + \sqrt{5}) + (\sqrt{5} + \sqrt{4})$$

$$= \sqrt{9} + \sqrt{4} = 3 + 2 = 5.$$

15. (b) $5\sqrt[3]{250} + 7\sqrt[3]{16} - 14\sqrt[3]{54}$

$$= 5\sqrt[3]{125 \times 2} + 7\sqrt[3]{8 \times 2} - 14\sqrt[3]{27 \times 2}$$

$$= 5 \times 5\sqrt[3]{2} + 7 \times 2\sqrt[3]{2} - 14 \times 3 \times \sqrt[3]{2}$$

$$= (25 + 14 - 42)\sqrt[3]{2} = -3\sqrt[3]{2}$$

16. (a) Work from the choices: only 169 when reversed becomes 961 and both numbers are squares.

17. (b) $1*2 = 1 + 2 \times 6 = 13$
 $13*3 = 13 + 3 \times 6 = 31$

18. (a) $a^b = 125 \Rightarrow a^b = 5^3$
 $\therefore a = 5, b = 3$
 $(a - b)^{a+b-4} = (5 - 3)^{5+3-4} = 2^4 = 16$

19. (c) $p \times q = p + q + \frac{p}{q} \Rightarrow 8 \times 2 = 8 + 2 + \frac{8}{2} = 14$

20. (b) $x*y = x^2 + y^2 = xy$
 $9*11 = 9^2 + 11^2 - 9 \times 11$
 $= 81 + 121 - 99 = 103$

21. (d) Since the factors of 11760 are $2 \times 2 \times 2 \times 2 \times 3 \times 5 \times 7 \times 7$ so we need to multiply it with 3×5 because all the factors are paired but 3 and 5 are unpaired, hence (d) is the correct choice.

22. (a) $\sqrt{\frac{0.081 \times 0.484}{0.0064 \times 6.25}} = \sqrt{\frac{81 \times 484}{64 \times 625}}$
 $= \frac{9 \times 22}{8 \times 25} \cdot \frac{99}{100} = 0.99$

23. (b) $5\sqrt{5} \times 5^3 \div 5^{-3/2} = 5^{(a+2)}$

$$5^1 \times 5^{\frac{1}{2}} \times 5^3 \times 5^{3/2} = 5^{a+2}$$

$$5^{1 + \frac{1}{2} + 3 + \frac{3}{2}} = 5^{a+2}, \frac{12}{5^{a+2}}, 5^{a+2}, a + 2 = 6$$

$$\therefore a = 4$$

24. (b) Let number be x

$$x \times \frac{4}{5} \times \frac{3}{4} - x \times \frac{2}{5} \times \frac{1}{6} = 648$$

$$\frac{3x}{5} - \frac{x}{15} = 648$$

$$\frac{9x - x}{15} = 648$$

$$8x = 648 \times 15 \quad x = \frac{648 \times 15}{8} = 81 \times 15 = 1215$$

25. (b) If sum of two is even, their difference is always even, So (b) is right answer.

26. (a) Number of decimal places in the given expression = 8

Number of decimal places in (a) = 8

Number of decimal places in (b) = 9

Number of decimal places in (c) = 7.

Clearly, the expression in (a) is the same as the given expression.

27. (c)

28. (b)

29. (c) $\frac{a+b}{b+c} \quad \frac{c+d}{d+a}$

$$\Rightarrow ad + a^2 + bd + ab = bc + c^2 + bd + cd$$

$$\Rightarrow (a^2 - c^2) + (ad - cd) + (ab - bc) = 0$$

$$\Rightarrow (a - c)(a + c + d + b) = 0$$

$$\Rightarrow a = c \text{ or } a + b + c + d = 0 \text{ or both}$$

30. (d) Sum is 888 $\Rightarrow$ unit's digit should add up to 8. This is possible only for 4th option as "3" + "5" = "8".

31. (b) The numbers that can be formed are xy and yx . Hence $(10x + y) + (10y + x) = 11(x + y)$. If this is a perfect square then $x + y = 11$.

32. (d) $I = \sqrt[3]{\sqrt[4]{a^3}} \quad \left((a^3)^{1/4}\right)^{1/3} \quad a^{1/4}$

$$II = \sqrt[3]{\sqrt[5]{a^4}} = \left((a^4)^{1/5}\right)^{1/3} \quad a^{4/15}$$

$$III = \sqrt{\sqrt[3]{a}} = \left(a^{1/3}\right)^{1/2} = a^{1/6}$$

$$IV = \sqrt{\sqrt[5]{a^3}} = \left((a^3)^{1/5}\right)^{1/2} \quad a^{3/10}$$

Now again, to compare these numbers, we need to bring the indices to a common denominator.

$$\therefore I = a^{1/4} = a^{15/60}, \quad II = a^{4/15} = a^{16/60},$$

$$III = a^{1/6} = a^{10/60}, \quad IV = a^{3/10} = a^{18/60}.$$

$\therefore$ The ascending order is III, I, II, IV.

33. (a) Comparing $4^{1/3}$ and $5^{1/4}$
 $(4^{1/3})^{12}$ and $(5^{1/4})^{12}$ i.e., 4^4 and 5^3
 $= 256 > 125$

$$\therefore 4^{1/3} > 5^{1/4}$$

Similarly, comparing $5^{1/4}$ and $6^{1/5}$

$$(5^{1/4})^{20} \text{ and } (6^{1/5})^{20} \text{ i.e., } 5^5 \text{ and } 6^4 = 3125 > 1296$$

$$\therefore 5^{1/4} > 6^{1/5}$$

34. (b) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
 $\Rightarrow 2(ab + bc + ca) = (a + b + c)^2 - (a^2 + b^2 + c^2)$
 $= 169 - 69 = 100$
 $ab + bc + ca = 50$

35. (b) $A^{3^{33}} = 3^{3^{27}}$

$$\text{and } C^{3^{33}} = 3^{3^{33}}$$

Hence $C > A$.

Hence either (b) and (d) option is correct.

$$\text{Now } A = 3^{3^3} = 3^{3^{27}}$$

$$\text{and } D = 3^{3^{33}}$$

$$\text{Hence } A > D \quad (\text{Since } 3^{27} > 333)$$

Thus the correct relation is $C > A > B > D$.

Hence, option (b) is correct.

36. (b) $|a - b| = |8| = 8 \Rightarrow |b - a| = |-8| = 8$
 $\Rightarrow |a - b| - |b - a| = 8 - 8 = 0$

37. (b) If we try to put x as 12, we get the square root of $3x$ as 6. Then the next point at which we need to remove the square root sign would be $12 + 2(6) = 24$ whose square root would be an irrational number. This leaves us with only 1 possible value ($x = 3$). Checking for this value of x we can see that the expression is satisfied as LHS = RHS.

38. (d) At a value of $x = 0$ we can see that the expression $x^2 + |x - 1| = 1 \rightarrow 0 + 1 = 1$. Hence, $x = 0$ satisfies the given expression. Also at $x = 1$, we get $1 + 0 = 1$.

39. (d) Solve this question through options. Also realize that $a \times b = a + b$ only occurs for the situation $2 \times 2 = 2 + 2$.

Hence, clearly the answer has to be none of these.

Standard Level

1. (b) $999 \frac{995}{999} \times 999 = \left(999 + \frac{995}{999}\right) \times 999$

$$= 999 \times 999 + \frac{995}{999} \times 999 = 999^2 + 995$$

$$= 998001 + 995 = 998996$$

$$\begin{aligned}
 2. \quad (c) \quad & 7892.35 \times 99.9 \\
 &= \frac{789235 \times 999}{1000} = \frac{789235 \times (1000 - 1)}{1000} \\
 &= \frac{789235000 - 789235}{1000} = 788445.765
 \end{aligned}$$

$$\begin{aligned}
 3. \quad (c) \quad & \text{Total number of } \frac{1}{12} \quad \frac{18\frac{3}{4}}{\frac{1}{12}} \\
 &= \frac{75}{4} \times \frac{12}{1} = 225
 \end{aligned}$$

4. (b) This problem can't be solved by factorisation because we need not factor. So we have to solve it by division method as follows

	23.9
2	575
2	4
43	175
3	129
469	4600
9	4221

(If the number is not a perfect square then by putting decimal we can increase the zeros in pairs for further calculation.)

The result obtained is ≈ 23.9 .

So by adding some number we can make it the perfect square of 24. Now since we know that $(24)^2 = 576$. So we need to add 1 ($\because 576 - 575 = 1$)

Thus (b) is the correct option.

Alternatively : Using options we can solve this problem as if we consider option (a) then 575 itself be a perfect square but its not a perfect square. Again if we add 1 (i.e., using option (b) we get the number 576 and then check it, we find that 576 is a perfect square. Hence (b) is correct.

Alternatively : Since we know that $(20)^2 = 400$ and $(25)^2 = 625$. It means the value of perfect square must lies in the range of 400 and 625. So we can try it manually and get that $(23)^2 = 529$ and $(24)^2 = 576$. So simply we need to add 1 to make a perfect square number.

$$5. \quad (a) \quad 6 \times 15 \times 3 \sqrt{\frac{(6+2)(15+3)}{(3+1)}} = \sqrt{\frac{8 \times 18}{4}} = 6$$

$$\begin{aligned}
 6. \quad (a) \quad & x + \frac{1}{x} = 3 + \sqrt{8} + \frac{1}{3 + \sqrt{8}} \\
 &= \frac{(3 + \sqrt{8})^2 + 1}{(3 + \sqrt{8})} = \frac{9 + 8 + 6\sqrt{8} + 1}{(3 + \sqrt{8})}
 \end{aligned}$$

$$= \frac{18 + 6\sqrt{8}}{(3 + \sqrt{8})} \cdot \frac{6(3 + \sqrt{8})}{(3 + \sqrt{8})} = 6, \left(x + \frac{1}{x}\right)^2 = 6^2$$

$$x^2 + \frac{1}{x^2} + 2 \cdot x \cdot \frac{1}{x} = 36, x^2 + \frac{1}{x^2} = 36 - 2 = 34$$

$$\begin{aligned}
 7. \quad (c) \quad & \text{If } x^a = y^b = z^c \text{ and } y^2 = zx \\
 & \text{Let } x^a = y^b = z^c = k \\
 & \Rightarrow x = k^{1/a}, y = k^{1/b}, z = k^{1/c} \\
 & \text{Now, } \because y^2 = zx \\
 & \therefore (k^{1/b})^2 = (k^{1/c}) \cdot (k^{1/a})
 \end{aligned}$$

$$\Rightarrow k^{2/b} = k^{1/c + 1/a}$$

$$\Rightarrow \frac{1}{a} + \frac{1}{c} = \frac{2}{b}$$

Hence (c) is the correct option.

$$8. \quad (b) \quad \frac{9^n \times 3^2 \times (3^{-n/2})^{-2} - (27)^n}{3^{3m} \times 2^3} = \frac{1}{27}$$

$$\Rightarrow \frac{3^{2n} \times 3^2 \times 3^n - 3^{3n}}{3^{3m} \times 2^3} = \frac{1}{3^3}$$

$$\Rightarrow \frac{3^{2n+2+n} - 3^{3n}}{8 \times 3^{3m}} = \frac{1}{3^3}$$

$$\Rightarrow \frac{3^{3n+2} - 3^{3n}}{3^{3m} \times 8} = 3^{-3}$$

$$\Rightarrow \frac{3^{3n}(3^2 - 1)}{3^{3m} \times 8} = 3^{-3}$$

$$\Rightarrow 3^{3n-3m} = 3^{-3}$$

$$\Rightarrow 3n - 3m = -3$$

$$\Rightarrow m - n = 1$$

Hence (b) is the correct option.

HCF : The HCF of $(a^m - 1)$ and $(a^n - 1)$ is equal to the $(a^{\text{HCF of } m, n} - 1)$

$$9. \quad (a) \quad \text{We have : } \frac{2x}{1 + \frac{1}{(1-x)+x}} = 1 \Leftrightarrow \frac{2x}{1 + \frac{1}{[1/(1-x)]}} = 1$$

$$\Leftrightarrow \frac{2x}{1 + (1-x)} = 1$$

$$\Leftrightarrow 2x = 2 - x \Leftrightarrow 3x = 2 \Leftrightarrow x = \frac{2}{3}$$

10. (b) Given expression

$$= \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{5}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \dots + \left(\frac{1}{9} - \frac{1}{10}\right)$$

$$= \left(\frac{1}{2} - \frac{1}{10}\right) = \frac{4}{10} = \frac{2}{5}$$

11. (c) $7 - 2\sqrt{10} = 5 + 2 - 2\sqrt{5 \times 2}$

$$\Rightarrow 7 - 2\sqrt{10} = (\sqrt{5})^2 + (\sqrt{2})^2 - 2\sqrt{5} \cdot \sqrt{2}$$

$$\Rightarrow 7 - 2\sqrt{10} = (\sqrt{5} - \sqrt{2})^2$$

Thus the $\sqrt{7 - 2\sqrt{10}} = \pm(\sqrt{5} - \sqrt{2})$

12. (b) We have,
Product of unit's digits = 28
 $\Rightarrow$ Product of units digits = 4×7
[$\because$ Unit's digits are one digit numbers]
 $\Rightarrow$ Unit's digits are 4 and 7.
 $\Rightarrow$ Product of ten's digits = 15
 $\Rightarrow$ Product of ten's digits = 3×5
[Ten's digit are one digit numbers]
 $\Rightarrow$ Ten's digits are 3 and 5.
Thus, the two numbers either 34 and 57 or 37 and 54.
Now, $34 \times 57 = 34 \times (50 + 7)$ [$\because 57 = 50 + 7$]
 $= 34 \times 50 + 34 \times 7$ [$\because a \times (b + c) = a \times b + a \times c$]
 $= 1700 + 238 = 1938$
and, $37 \times 54 = 37 \times (50 + 4)$ [$\because 54 = 50 + 4$]
[$\because a \times (b + c) = a \times b + a \times c$]
 $= 1850 + 148 = 1998$

13. (d) Consider $P = 5$, then
 $5 + 5! = 5^3$
 $5 + 120 = 125$
 $125 = 125$
Thus (d) is correct option.

14. (d) Let $Z = 8x - 3x^2$
- $$\Rightarrow Z = -3 \left[x^2 - \frac{8}{3}x \right]$$
- $$\Rightarrow Z = -3 \left[x^2 - 2 \times x \times \frac{4}{3} + \left(\frac{4}{3}\right)^2 - \left(\frac{4}{3}\right)^2 \right]$$
- $$\Rightarrow Z = -3 \left(x - \frac{4}{3} \right)^2 + 3 \times \left(\frac{4}{3} \right)^2$$
- So the maximum value occurs when $x = \frac{4}{3}$

Maximum value $-3 \times 0 + 3 \times \frac{16}{9} = \frac{16}{3}$

15. (c) $\sqrt[3]{x+9} - \sqrt[3]{x-9} = 3$,
- $$\Rightarrow (x+9) - (x-9) - 3 \sqrt[3]{x+9} \cdot \sqrt[3]{x-9} = 3$$
- $$(\sqrt[3]{x+9} - \sqrt[3]{x-9}) = 3^3$$
- $$(\because (a-b)^3 = a^3 - b^3 - 3ab(a-b))$$
- $$\Rightarrow 18 - 3 \sqrt[3]{x^2 - 81} \times 3$$
- $$\Rightarrow \frac{18 - 27}{9} = \sqrt[3]{x^2 - 81}$$
- $$\Rightarrow -1 = x^2 - 81 \Rightarrow x^2 = 80$$

16. (a) Given expression
- $$= 35.7 - \left(3 + \frac{1}{10} \right) - \left(2 + \frac{1}{5} \right)$$
- $$= 35.7 - \left(3 + \frac{3}{10} \right) - \left(2 + \frac{2}{5} \right)$$
- $$= 35.7 - \frac{33}{10} - \frac{12}{5} = 35.7 - \left(\frac{33}{10} + \frac{12}{5} \right)$$
- $$= 35.7 - \frac{57}{10} = 35.7 - 5.7 = 30.$$

17. (b) $\sqrt[4]{10} (10)^{1/4} = (10)^{3/4} = (1000)^{1/4}$
 $\sqrt[3]{6} (6)^{1/3} = (6)^{4/3} = (1296)^{1/3}$
 $\sqrt{3} (3)^{1/2} = (3)^{3/2} = (729)^{1/2}$
 $\therefore \sqrt{3} < \sqrt[4]{10} < \sqrt[3]{6}$ is the correct order and hence (b) is correct.
18. (a) The product of the marks obtained = 72
As Rohan was not able to figure out the marks obtained by Sunil initially, there must be at least two possible ways of getting that same sum. The two possible cases are 2, 6, 6 and 3, 3, 8 (Sum = 14).
When Rohan got to know that Sunil got the highest in Physics among the three subjects, he could answer correctly as this is possible only with 3, 3 and 8.
Therefore, the sum of the marks obtained by Sunil in the other two subjects is 3 + 3 i.e. 6.
19. (b) The unit's digit will be $1 \times 5 = 5$ (no carry over). The tens digit will be $(4 \times 1 + 5 \times 2) = 4$ (carry over 1). The hundreds digit will be $(3 \times 1 + 4 \times 2 + 5 \times 1) = 6 + 1$ (carried over) = 7. Hence, answer is 745.

20. (a) Let the numbers be a and b . Then, $a + b = 12$ and $ab = 35$.

$$\therefore \frac{a+b}{ab} = \frac{12}{35} \Rightarrow \left(\frac{1}{b} + \frac{1}{a}\right) = \frac{12}{35}$$

$$\therefore \text{Sum of reciprocals of given numbers} = \frac{12}{35}$$

21. (c) Given expression

$$= \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \dots \times \frac{99}{100} = \frac{2}{100} = \frac{1}{50}$$

22. (b) Suppose the worker remained idle for x days. Then, he worked for $(60 - x)$ days.

$$\therefore 20(60 - x) - 3x = 280 \Leftrightarrow 1200 - 23x = 280 \Leftrightarrow 23x = 920 \Leftrightarrow x = 40.$$

So, the worker remained idle for 40 days.

$$23. (b) \text{ Given exp.} = \frac{1}{2} \div \frac{1}{1 + \frac{2/3}{1 + \frac{3}{5 + \frac{8}{3 + \frac{9}{(1/3)}}}}}$$

$$= \frac{1}{1 + \frac{2/3}{(13/3)}} = \frac{1}{1 + \frac{2}{13}} = \frac{13}{15}$$

$$24. (b) 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}} = \frac{37}{13} = 2\frac{11}{13} = 2 + \frac{11}{13}$$

$$\Rightarrow \frac{1}{x + \frac{1}{y + \frac{1}{z}}} = \frac{11}{13} \Rightarrow x + \frac{1}{y + \frac{1}{z}} = \frac{13}{11}$$

$$\Rightarrow x + \frac{1}{y + \frac{1}{z}} = 1 + \frac{2}{11} \Rightarrow x = 1,$$

$$y + \frac{1}{z} = \frac{11}{2} = 5\frac{1}{2} = 5 + \frac{1}{2} \Rightarrow x = 1, y = 5, z = 2.$$

$$25. (b) \text{ Given exp. } \sqrt{\frac{(0.03)^2 + (0.21)^2 + (0.065)^2}{\left(\frac{0.03}{10}\right)^2 + \left(\frac{0.21}{10}\right)^2 + \left(\frac{0.065}{10}\right)^2}} \\ = \sqrt{\frac{100[(0.03)^2 + (0.21)^2 + (0.065)^2]}{(0.03)^2 + (0.21)^2 + (0.065)^2}} \\ = \sqrt{100} = 10.$$

26. (c) Given : $x^2 + y^2 + z^2 - 64 = -2(xy - yz - zx) \dots(i)$

$$\text{Now, } [x + y + (-z)]^2 = x^2 + y^2 + z^2 + 2(xy - yz - zx)$$

$$\Rightarrow (3z - z)^2 = x^2 + y^2 + z^2 + 2(xy - yz - zx)$$

$$\Rightarrow -2(xy - yz - zx) = (x^2 + y^2 + z^2) - (2z)^2 \dots(ii)$$

From (i) and (ii), we get:

$$(2z)^2 = 64 \Leftrightarrow 4z^2 = 64 \Leftrightarrow z^2 = 16 \Leftrightarrow z = 4.$$

$$27. (c) \sqrt{\frac{8}{3}} \sqrt{\frac{8 \times 3}{3 \times 3}} = \frac{\sqrt{24}}{3} = \frac{4.899}{3} = 1.633.$$

$$28. (c) \left[X + \frac{1}{X}\right]^2 = X^2 + \frac{1}{X^2} + 2 = 16 \text{ or } X^2 + \frac{1}{X^2} = 14$$

$$\text{Now, } X^4 + \frac{1}{X^4} + 2 = 196 \text{ or } X^4 + \frac{1}{X^4} = 194.$$

$$29. (c) \text{ Given expression} = \sqrt{15625} + \sqrt{\frac{15625}{100}} + \sqrt{\frac{15625}{10000}} \\ = \left(125 + \frac{125}{10} + \frac{125}{100}\right) = (125 + 12.5 + 1.25) = 138.75$$

30. (b) Provisions for one student = $250 \times 35 = 8750$
250 students used provisions for 5 days.
Total provisions used by 250 students in 5 days = $250 \times 5 = 1250$
Remaining provision = $8750 - 1250 = 7500$
After 5 days total number of student = $250 + 25 = 275$
Total provisions used by 275 student in 10 days = $275 \times 10 = 2750$
Now remaining = $7500 - 2750 = 4750$
After 15 days no. of student = $275 - 25 = 250$
 $4750 = 250 \times \text{no. of extra days}$

$$\text{No. of extra days} = \frac{4750}{250} = 19 \text{ days}$$

31. (a) $\frac{97}{19} = 5 + \frac{2}{19}$. Also, $\frac{19}{2}$ can be written as $9 + \frac{1}{2}$. So the values of a , b and c are 5, 9 and 2 respectively. Hence, the sum of a , b and c is 16.

Expert Level

1. (d) Given $a = 6b = 12c = 27d = 36e$
Multiplied and Divide by 108 in whole expression

$$\frac{108a}{108} = \frac{108b}{18} = \frac{108c}{9} = \frac{108d}{4} = \frac{108e}{3}$$

$$\frac{1}{108}a = \frac{1}{18}b = \frac{1}{9}c = \frac{1}{4}d = \frac{1}{3}e = 1 \text{ (say)}$$

$$\Rightarrow a = 108, b = 18, c = 9, d = 4, e = 3$$

$$\text{So it is clear that } \left(\frac{a}{6}, \frac{c}{d}\right) \text{ contains a number } \frac{c}{d} \left(\frac{9}{4}\right)$$

which is not an integer

2. (b) In this question it is advisable to raise all the numbers to the power of 12, so the numbers become,

$$(2^{1/2})^{12}, (3^{1/3})^{12}, (4^{1/4})^{12}, (6^{1/6})^{12}$$

$$\text{or } 2^6, 3^4, 4^3, 6^2 \quad \text{or } 64, 81, 64, 36$$

So, $3^{1/3}$ is the largest.

3. (b) $(9 \square 11) = 9 \times 11$, if $9 + 11$ is even = 99

$99 \Delta 4 = 99^2 - 4^2$, if 99^4 is odd.

$$= (99 - 4)(99 + 4) = 95 \times 103 = 9785$$

4. (b) Suppose $D = 24 \therefore S = (2 + 4)^2 = 36$

According to the Question

$$S - D = 27 \Rightarrow 36 - 24 = 12 \neq 27 \therefore D \neq 24$$

$$\text{If } D = 54 \text{ then } (5 + 4)^2 - 54 \quad 81 - 54 = 27$$

therefore D is 54.

5. (d) The given expression is

$$(2^{3/4} + 2^{1/2} + 2^{1/4} + 1)(2^{1/4} - 1)$$

$$= [2^{1/2}(2^{1/4} + 1) + 1(2^{1/4} + 1)](2^{1/4} - 1)$$

$$= (2^{1/2} + 1)(2^{1/4} + 1)(2^{1/4} - 1)$$

$$[\because (a - b)(a + b) = a^2 - b^2]$$

$$= (2^{1/2} + 1)(2^{1/2} - 1) = 2 - 1 = 1$$

6. (b) $S_1 = 2 + 4 + 6 + \dots + 200$

$$S_2 = 3 + 6 + 9 + \dots + 300$$

In S_2 , 6 and multiples of 6 upto 200 will be identical to terms in S_1 .

$$\text{Identical terms} = 6 + 12 + 18 + \dots + 198$$

$$\text{No. of terms} = 198 \div 6 = 33$$

7. (a) For the expression to hold true, x and y should both be positive.

8. (b) $\sqrt{(0.798)^2 + 0.404 \times 0.798 + (0.202)^2} + 1$

$$= \sqrt{(0.798)^2 + 2 \times 0.202 \times 0.798 + (0.202)^2} + 1$$

$$= \sqrt{(0.798 + 0.202)^2} + 1$$

$$= 1.000 + 1 = 1 + 1 = 2$$

9. (c) $\sqrt{\frac{(0.75)^3}{1-0.75} + (0.75 + 0.75^2 + 1)}$

$$= \sqrt{\frac{(0.75)^3 + (1-0.75)(1^2 + 0.75 + 0.75^2)}{(1-0.75)}}$$

$$= \sqrt{\frac{(0.75)^3 + (1)^3 - (0.75)^3}{(1-0.75)}} = \sqrt{\frac{1}{0.25}} = \frac{1}{0.5} = 2$$

10. (d) By observaiton we get to know that

$$2 * 3 = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$\text{and } 3 * 4 = \sqrt{3^2 + 4^2} = 5$$

$$\text{then } 5 * 12 = \sqrt{5^2 + 12^2} = 13$$

11. (a) Given expression = $35.7 - \left(3 + \frac{1}{10}\right) - 2 + \frac{1}{5}$

$$35.7 - \left(3 + \frac{3}{10}\right) - \left(2 + \frac{2}{5}\right)$$

$$= 35.7 - \frac{33}{10} - \frac{12}{5} = 35.7 - \left(\frac{33}{10} + \frac{12}{5}\right)$$

$$= 35.7 - \frac{57}{10} = 35.7 - 5.7 = 30.$$

12. (d) $\frac{4^n \times 20^{m-1} \times 12^{m-n} \times 15^{m+n-2}}{16^m \times 5^{2m+n} \times 9^{m-1}}$

$$2^{2n} \times 2^{2m-2} \times 5^{m-1} \times 2^{2m-2n} \times 3^{m-n}$$

$$\frac{\times 3^{m+0-2} \times 5^{m+n-2}}{2^{4m} \times 5^{2m+n} \times 3^{2m-2}}$$

$$= 2^{2n+2m-2+2m-2n-4m} \times 3^{m-n+m+n-2-2m+2} \times 5^{m-1+m+n-2-2m-n}$$

$$= 2^{-2} \times 3^0 \times 5^{-3}$$

$$\frac{1}{4} \times \frac{1}{125} = \frac{1}{500}$$

Hence (d) is the correct option.

13. (c) $4b = 6c \Rightarrow b = \frac{3}{2}c$ and $3a = 4b \Rightarrow a = \frac{4}{3}b$

$$= \frac{4}{3} \left(\frac{3}{2}c\right) = 2c.$$

$$a + b + c = 27\sqrt{29} \Rightarrow 2c + \frac{3}{2}c + c = 27\sqrt{29}$$

$$\frac{9}{2}c = 27\sqrt{29} \Rightarrow c = 6\sqrt{29}.$$

$$\therefore \sqrt{a^2 + b^2 + c^2}$$

$$= \sqrt{(a+b+c)^2 - 2(ab+bc+ca)}$$

$$= \sqrt{(27\sqrt{29})^2 - 2\left(2c \times \frac{3}{2}c + \frac{3}{2}c \times c + c \times 2c\right)}$$

$$= \sqrt{(729 \times 29) - 2\left(3c^2 + \frac{3}{2}c^2 + 2c^2\right)}$$

$$= \sqrt{(729 \times 29) - 2 \times \frac{13}{2}c^2}$$

$$= \sqrt{(729 \times 29) - 13 \times (6\sqrt{29})^2} = \sqrt{29(729 - 468)}$$

$$= \sqrt{29 \times 261} = \sqrt{29 \times 29 \times 9} = 29 \times 3 = 87.$$

14. (c) Since the number of 2's are less than the number of 5's hence the restriction is imposed by the number of 2's.

Thus there can be only 222 pairs of (5×2) . Hence the number of zeros at the end of the product of the given expression will be 222.

15. (a) $(x-8)(x-10) = 2^y \Rightarrow (x-10)(x-10+2) = 2^y$

But LHS can be of the form 2^y only if $x-10 = 2$
 $\Rightarrow x = 12$ is the only solution.

16. (d) Given, $p^q = q^p$

$$(p^q)^{1/p} = (q^p)^{1/p}$$

$$p^{q/p} = q$$

$$\frac{9p}{p^9} = q \quad [\because q = 9p]$$

$$\text{or } p^9 = q \quad \dots (1)$$

$$\text{Also } p = \frac{q}{9} \quad \dots (2)$$

Dividing equation (1) by (2), we get
 $p^8 = 9$

$$\text{or } p = \sqrt[8]{9}$$

17. (a) $M = \sqrt{3-\sqrt{5}+\sqrt{9-4\sqrt{5}}}$

$$\sqrt{3-\sqrt{5}+\sqrt{((\sqrt{5})^2-2 \times 2 \times \sqrt{5}+2^2)}}$$

$$\sqrt{3-\sqrt{5}+\sqrt{(\sqrt{5}-2)^2}}$$

$$\sqrt{3-\sqrt{5}+\sqrt{5}-2} = \sqrt{1} = 1$$

$$N = \sqrt{\sqrt{7}-1-\sqrt{11-4\sqrt{7}}}$$

$$\sqrt{\sqrt{7}-1-\sqrt{(\sqrt{7})^2-(2 \times 2 \times \sqrt{7})+(2)^2}}$$

$$\sqrt{\sqrt{7}-1-\sqrt{(\sqrt{7}-2)^2}}$$

$$\sqrt{\sqrt{7}-1-\sqrt{7}+2} = \sqrt{1} = 1$$

$$\text{Hence, } \frac{M-N}{M+N} = \frac{1-1}{1+1} = 0.$$

18. (b) It is given that $a^{a^b} = b$

Putting the value of b in left-hand side, we get

$$a^{a^{a^b}} = b$$

On repeating the same step n times, we get

$$a^{a^{a^{\dots^b}}} = b$$

when n tends to infinity, we get

$$a^{a^{a^{\dots^b}}} = a^b = b$$

$$\text{Hence } a^b - b = 0$$

Explanation of Test Yourself

1. (c) = 134556

Carry	Over
Carry	=2
Carry	=2
Carry	=2
Carry	=1
Carry	=1

2. (b) Difference from 100000

$$\begin{array}{r}
 100008 \quad +8 \\
 \times 100004 \quad +4 \\
 \hline
 100012 \quad 00000 \\
 + \quad \quad \quad 32 \\
 \hline
 100012 \quad 00032
 \end{array}$$

Hence $100008 \times 100004 = 10001200032$

3. (a) $3949 = 3000 + 900 + 40 + 9$
= MMMCMLXIX

4. (c) 1 6 36 216

$$\begin{array}{r}
 12 \quad 72 \\
 \textcircled{3} \quad \textcircled{12} \quad \textcircled{21} \\
 \swarrow \quad \searrow \quad \swarrow \\
 4 \quad \textcircled{30} \quad \textcircled{129} \quad \textcircled{216}
 \end{array}$$

= 4096

5. (b) $9998 \times 999 = 9998 \times (1000 - 1)$
= $9998 \times 1000 - 9998 \times 1 = 9998000 - 9998 = 9998002$

6. (c) Give expression = $\sqrt{\frac{324 \times 81 \times 4624}{15625 \times 289 \times 729 \times 64}}$
(sum of decimal places being equal in nume. and deno.)

$$= \frac{18 \times 9 \times 68}{125 \times 17 \times 27 \times 8} = \frac{3}{125} = 0.024$$

7. (c) $Z + \frac{1}{Z} = 1$

$$\Rightarrow Z^2 - Z + 1 = 0$$

$$\Rightarrow (Z + 1)(Z^2 - Z + 1) = 0 \text{ (if } Z \neq -1)$$

$$\Rightarrow (Z^3 + 1) = 0$$

$$\therefore Z^3 = -1 \text{ and } Z \neq -1$$

$$\text{Now, } Z^{64} + \frac{1}{Z^{64}} = Z^{63} \cdot Z + \frac{1}{Z^{63} \cdot Z}$$

$$\Rightarrow (Z^3)^{21} \cdot Z + \frac{1}{(Z^3)^{21} \cdot Z} = -Z - \frac{1}{Z} = -1$$

8. (b) Given $7 + \sqrt{48} = 7 + 4\sqrt{3}$ let its square root is $p + \sqrt{q}$

$$\text{Then } (p + \sqrt{q})^2 = p^2 + q + 2p\sqrt{q}$$

$$= 7 + 4\sqrt{3} \text{ equating rational and irrational part.}$$

$$p^2 + q = 7 \text{ and } 2p\sqrt{q} = 4\sqrt{3}$$

$$\text{or } p\sqrt{q} = 2\sqrt{3} \text{ or } p = 2 \text{ and } q = 3$$

$$\text{Hence required square root is } 2 + \sqrt{3}$$

9. (d) Since this number is closer to 100, we will take 100 as the base.

$$(112)^2 = (100 + 12)^2 = (100)^2 + 2 \times 100 \times 12 + (12)^2$$

$$= 10000 + 2 \times 1200 + 144 = 12544$$

10. (a) $(679)^2 = (700 - 21)^2$

$$= 490000 - 2 \times 700 \times 21 + 441$$

$$[\text{using } (a - b)^2 = a^2 - 2ab + b^2]$$

$$= 461041$$

11. (a) $x^{1/p} = y^{1/q} = z^{1/r}$ and $xyz = 1$ (given)

$$x^{1/p} = y^{1/q} = z^{1/r} = k \text{ [constant]}$$

$$x^{1/p} = k \Rightarrow x = k^p, \text{ same as others two terms}$$

$$= y = k^q, z = k^r.$$

$$\text{multiply } x, y, z = xyz = k^p k^q k^r = k^{p+q+r}$$

$$\Rightarrow 1 = k^0 = k^{p+q+r}$$

$$\Rightarrow p + q + r = 0$$

12. (c) $\left(\frac{1}{64}\right)^0 + (64)^{-1/2} - (32)^5$

$$= 1 + \frac{1}{8} - 16 = \frac{-119}{8} = -14\frac{7}{8}$$

13. (b) Let the required cube root is $p - \sqrt{q}$

$$\text{Then } \sqrt[3]{38 - 17\sqrt{5}} = p - \sqrt{q} \text{ and}$$

$$\sqrt[3]{38 + 17\sqrt{5}} = p + \sqrt{q}$$

$$\text{Multiplying these two relations, we will get}$$

$$p^2 - q = \sqrt[3]{1444 - 1445} = \sqrt[3]{-1} = -1$$

$$\text{or } p^2 = q - 1$$

$$38 - 17\sqrt{5} = (p - \sqrt{q})^3 = p^3 - 3p^2\sqrt{q} + 3pq - q\sqrt{q}$$

$$\text{Equating rational and irrational part we will get}$$

$$p^3 + 3pq = 38 \text{ and } \sqrt{q}(3p^2 + q) = 17\sqrt{5}$$

$$\text{From little observation and trial we will get } q = 5 \text{ and } p = 2$$

$$\text{Hence required cube root is } 2 - \sqrt{5}$$

14. (c) The expression would have solutions based on a structure of:
 $4 + 0; 3 + 1; 2 + 2; 1 + 3$ or $0 + 4$.

There will be $2 \times 1 = 2$ solutions for $4 + 0$ as in this case x can take the values of 8 and 0, while y can take a value of 4;

Similarly, there would be $2 \times 2 = 4$ solutions for $3 + 1$ as in this case x can take the values 7 or 1, while y can take a value of 5 or 3;

Thus, the total number of solutions can be visualised as:

2 (for $4 + 0$) + 4 (for $3 + 1$) + 4 (for $2 + 2$) + 4 (for $1 + 3$) + 2 (for $0 + 4$) = 16 solutions for the set (x, y) where both x and y are integers.

$$\begin{aligned}
 15. \quad (b) \quad & \frac{3}{2} \text{ of } \left(\frac{4}{3} \div \frac{5}{7} \right) + \frac{1}{2} \left[3 \frac{4}{5} - \left\{ \frac{2}{5} \div \left(\frac{1}{3} + \frac{1}{2} + \frac{1}{5} - \frac{1}{6} \right) \right\} \right] \\
 &= \frac{3}{2} \text{ of } \left(\frac{4}{3} \times \frac{7}{5} \right) + \frac{1}{2} \left[\frac{19}{5} - \left\{ \frac{2}{5} \div \left(\frac{1}{3} + \frac{1}{2} + \frac{1}{5} - \frac{1}{6} \right) \right\} \right] \\
 &= \frac{3}{2} \text{ of } \left(\frac{28}{15} \right) + \frac{1}{2} \left[\frac{19}{5} - \left\{ \frac{2}{5} \div \left(\frac{13}{15} \right) \right\} \right] \\
 &= \frac{3}{2} \times \frac{28}{15} + \frac{1}{2} \left\{ \frac{19}{5} - \left(\frac{2}{5} \times \frac{15}{13} \right) \right\} \\
 &= \frac{14}{5} + \frac{1}{2} \left(\frac{19}{5} - \frac{6}{13} \right) \\
 &= \frac{14}{5} + \frac{1}{2} \times \frac{217}{65} \\
 &= \frac{14}{5} + \frac{217}{130} = \frac{581}{130}
 \end{aligned}$$

2

NUMBER SYSTEM

- Introduction
- Tabular Classification of Numbers
- Concept of Number Line (or Real Number Line)
- Conversion of Rational number of the Form $\frac{p}{q}$ Non-terminating Recurring Decimal into the Rational Number of the Form $\frac{p}{q}$
- Division
- Tests of Divisibility
- Prime Numbers
- Complex Numbers, Real Numbers and Imaginary Numbers
- General or Expanded Form of 2 and 3 Digits Numbers
- Sum of Numbers formed with given different digits
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- Number of Factors of a Composite Number
- Number of ways of Expressing a Composite Number as a Product of two Factors
- Sum of Factors (or Divisors) of A Composite Number
- Sum of Unit Digits
- The Last Digit From Left (i.e., unit digit) of Any Power of a Number
- Concept of Remainders
- To Find the Last Digits of the Expression Like $a_1 \times a_2 \times a_3 \times \dots \times a_n$
- Last Two Digits of a Number with Large Power
- Number of zeroes in an expression like $a \times b \times c \times \dots$, where $a, b, c, \dots$ are natural numbers
- Powers of a Number Contained in a Factorial
- Base System
- Successive Division
- Factors and Multiples
- Highest Common Factor (HCF) or Greatest Common Divisor (GCD)
- Least Common Multiple (LCM)
- Greatest Integral Value

INTRODUCTION

This chapter is most important chapter of quantitative aptitude for CAT and the likes of competitions. 20% to 35% questions of CAT and CAT like competitions are based on number system. So, you are advised to go through each and every concept, example and question of this chapter.

TABULAR CLASSIFICATION OF NUMBERS

First read the chart classification of numbers given on the next page carefully.

From the chart of classification of real numbers given on the next page, it is clear that both rational and irrational numbers combined together are called real numbers i.e., each rational number is a real number as well as each irrational number is a real number.

Note that sum, difference, product or quotient [provided denominator not equal to 0 (zero)] of two rationals, two irrationals or one rational and one irrational number is also a real number.

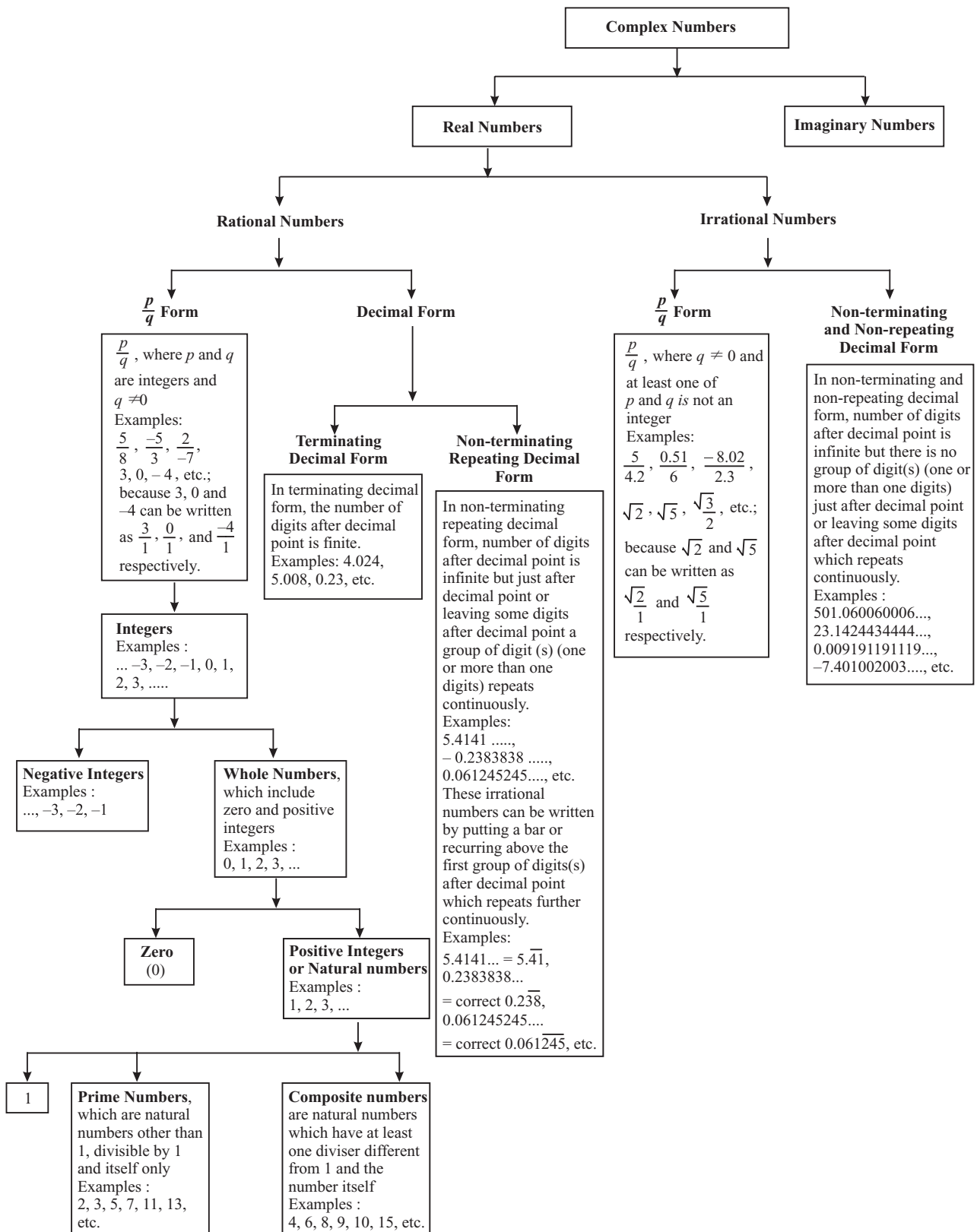
Note that π is an irrational number, which is actually the ratio of circumference of the diameter of a circle i.e. $\pi = \frac{c}{d}$, where c and d are the circumference and diameter of a circle. Approximate value of π is taken as $\frac{22}{7}$ or 3.14.

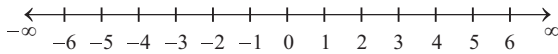
CONCEPT OF NUMBER LINE (OR NUMBER LINE)

A number line is a straight line from negative infinite $(-\infty)$ in left hand side to positive infinite $(+\infty)$ in right hand side as given:

Topic continue on Page 31

Chart: Classification of Numbers





Each point on the number line represents a unique real number and each real number is denoted by a unique point on the number line.

Symbols of some special sets are:

N : the set of all natural numbers

Z : the set of all integers

Q : the set of all rational numbers

R : the set of all real numbers

Z^+ : the set of positive integers

Q^+ : the set of positive rational numbers, and

R^+ : the set of positive real numbers

The symbols for the special sets given above will be referred to throughout the text.

Even Integers

An integer divisible by 2 is called an even integer. Thus, ..., -6, -4, -2, 0, 2, 4, 6, 8, 10, 12, ..., etc. are all even integers. $2n$ always represents an even number, where n is an integer.

For example, by putting $n = 5$ and 8 in $2n$, we get even integer $2n$ as 10 and 16 respectively.

Odd Integers

An integer not divisible by 2 is called an odd integer.

Thus, ..., -5, -3, -1, 1, 3, 5, 7, 9, 11, 13, 15, ..., etc. are all odd integers.

$(2n - 1)$ or $(2n + 1)$ always represents an odd number, where n is an integer.

For example by putting $n = 0, 1$ and 5 in $(2n - 1)$, we get odd integer $(2n - 1)$ as -1, 1 and 9 respectively.

Properties of Positive and Negative Numbers

If n is a natural number then

(A positive number)^{natural number} = A positive number

(A negative number)^{even positive number} = A positive number

(A negative number)^{odd positive number} = A negative number

CONVERSION OF RATIONAL NUMBER OF THE FORM NON-TERMINATING RECURRING DECIMAL INTO THE RATIONAL NUMBER OF THE FORM $\frac{p}{q}$

First write the non-terminating repeating decimal number in recurring form i.e., write

64.20132132132..... as $64.\overline{20132}$

Then using formula given below we find the required $\frac{p}{q}$ form

Rational number in the form $\frac{p}{q}$

$$= \frac{\left[\begin{array}{l} \text{Complete number neglecting} \\ \text{the decimal and bar over} \\ \text{repeating digit(s)} \end{array} \right] - \left[\begin{array}{l} \text{Non-recurring part of} \\ \text{the number neglecting} \\ \text{the decimal} \end{array} \right]}{m \text{ times } 9 \text{ followed by } n \text{ times } 0}$$

where m = number of recurring digits in decimal part

and n = number of non-recurring digits in decimals part

$$\begin{aligned} \text{Thus, } \frac{p}{q} \text{ form of } 64.\overline{20132} &= \frac{6420132 - 6420}{99900} \\ &= \frac{6413712}{99900} = \frac{534476}{8325} \end{aligned}$$

$$\text{In short; } 0.\overline{a} = \frac{a}{9}, 0.\overline{ab} = \frac{ab}{99}, 0.\overline{abc} = \frac{abc}{999}, \text{ etc. and}$$

$$0.a\overline{b} = \frac{ab - a}{90}, 0.a\overline{bc} = \frac{abc - a}{990}, 0.a\overline{bcd} = \frac{abcd - a}{9900}, \text{ etc.}$$

$$0.ab\overline{cd} = \frac{abcd - ab}{9900}, ab.\overline{cde} = \frac{abcde - abc}{990}, \text{ etc.}$$

Illustration 1: Convert $2.\overline{46102}$ in the $\frac{p}{q}$ form of rational number.

$$\text{Solution: Required } \frac{p}{q} \text{ form} = \frac{246102 - 2}{99999} = \frac{246100}{99999}$$

Illustration 2: Convert $0.167320\overline{6}$ in the $\frac{p}{q}$ form of rational number.

$$\text{Solution: Required } \frac{p}{q} \text{ form} = \frac{1673206 - 167}{9999000} = \frac{1673039}{9999000}$$

Illustration 3: Convert $31.026415555 \dots$ into $\frac{p}{q}$ form of rational number.

Solution: First write $31.026415555 \dots$ as $31.02641\overline{5}$

$$\begin{aligned} \text{Now required } \frac{p}{q} \text{ form} &= \frac{31026415 - 3102641}{900000} = \frac{27923774}{900000} \\ &= \frac{13961887}{450000} \end{aligned}$$

DIVISION

$$\begin{array}{r} 4 \overline{) 275} 68 \\ \underline{24} \\ 35 \\ \underline{32} \\ 3 \end{array}$$

Here 4 is the divisor, 275 is the dividend, 68 is the quotient and 3 is the remainder. Remainder is always less than divisor.

$$\text{Thus, Divisor} \overline{) \text{Dividend}} \left(\begin{array}{l} \text{Quotient} \\ \hline \text{Remainder} \end{array} \right)$$

Thus,

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

For example, $275 = 4 \times 68 + 3$

When quotient is a whole number and remainder is zero, then dividend is divisible by divisor.

TESTS OF DIVISIBILITY

I. Divisibility by 2:

A number is divisible by 2 if its unit digit is any of 0, 2, 4, 6, 8.

Ex. 58694 is divisible by 2, while 86945 is not divisible by 2.

II. Divisible by 3:

A number is divisible by 3 only when the sum of its digits is divisible by 3.

- Ex.** (i) Sum of digits of the number 695421 = 27, which is divisible by 3.
 $\therefore$ 695421 is divisible by 3.
 (ii) Sum of digits of the number 948653 = 35, which is not divisible by 3.
 $\therefore$ 948653 is not divisible by 3.

III. Divisible by 4:

A number is divisible by 4 if the number formed by its last two digits i.e. ten's and unit's digit of the given number is divisible by 4.

- Ex.** (i) 6879376 is divisible by 4, since 76 is divisible by 4.
 (ii) 496138 is not divisible by 4, since 38 is not divisible by 4.

IV. Divisible by 5:

A number is divisible by 5 only when its unit digit is 0 or 5.

Ex. Each of the numbers 76895 and 68790 is divisible by 5.

V. Divisible by 6:

A number is divisible by 6 if it is simultaneously divisible by both 2 and 3.

Ex. 90 is divisible by 6 because it is divisible by both 2 and 3 simultaneously.

VI. Divisible by 7:

A number is divisible by 7 if and only if the difference of the number of its thousands and the remaining part of the given number is divisible by 7 respectively.

Ex. 473312 is divisible by 7, because the difference between 473 and 312 is 161, which is divisible by 7.

VII. Divisible by 8:

A number is divisible by 8 if the number formed by its last three digits i.e. hundred's, ten's and unit's digit of the given number is divisible by 8.

- Ex.** (i) In the number 16789352, the number formed by last 3 digits, namely 352 is divisible by 8.
 $\therefore$ 16789352 is divisible by 8.
 (ii) In the number 576484, the number formed by last 3 digits, namely 484 is not divisible by 8.
 $\therefore$ 576484 is not divisible by 8.

VIII. Divisible by 9:

A number is divisible by 9 only when the sum of its digits is divisible by 9.

- Ex.** (i) Sum of digits of the number 246591 = 27, which is divisible by 9.
 $\therefore$ 246591 is divisible by 9.
 (ii) Sum of digits of the number 734519 = 29, which is not divisible by 9.
 $\therefore$ 734519 is not divisible by 9.

IX. Divisible by 10:

A number is divisible by 10 only when its unit digit is 0.

- Ex.** (i) 7849320 is divisible by 10, since its unit digit is 0.
 (ii) 678405 is not divisible by 10, since its unit digit is not 0.

X. Divisible by 11:

A number is divisible by 11 if the difference between the sum of its digits at odd places from right and the sum of its digits at even places also from right is either 0 or a number divisible by 11.

Ex. (i) Consider the number 29435417.

(Sum of its digits at odd places from right) –
 (Sum of its digits at even places from right)
 $(7 + 4 + 3 + 9) - (1 + 5 + 4 + 2) = (23 - 12) = 11$,
 which is divisible by 11.

$\therefore$ 29435417 is divisible by 11.

(ii) Consider the number 57463822.

(Sum of its digits at odd places) –
 (Sum of its digits at even places)
 $= (2 + 8 + 6 + 7) - (2 + 3 + 4 + 5) = (23 - 14)$
 $= 9$, which is neither 0 nor divisible by 11.

$\therefore$ 57463822 is not divisible by 11.

XI. Divisible by 12:

A number is divisible by 12, if it is simultaneously divisible by both 3 and 4.

Properties of Divisibility

- If a is divisible by b then ac is also divisible by b .
- If a is divisible by b , and c is divisible by d then ac is divisible by bd .
- If m and n both are divisible by d then $(m + n)$ and $(m - n)$ are both divisible by d .
- Out of n consecutive whole numbers, one and only one is divisible by n .
 For example, out of the five consecutive whole numbers 8, 9, 10, 11, 12 only one i.e., 10 is divisible by 5.
- The square of an odd integer when divided by 8 will always leave a remainder of 1.
- The product of 3 consecutive natural numbers is divisible by 6.
- The product of 3 consecutive natural numbers, the first of which is even, is divisible by 24.
- Difference between any number and the number obtained by writing the digits in reverse order is divisible by 9.
- Any number written in the form $(10^n - 1)$ is divisible by 3 and 9.
- Any six-digits, twelve-digits, eighteen-digits or any such number with number of digits equal to multiple of 6, is divisible by each of 7, 11 and 13 if all of its digits are the same.
 For example 666666, 888888, 333333333333 are all divisible by 7, 11 and 13.

As 666666 can be written as $666 \times 1000 + 666$
 $= 666 (1000 + 1) = 666 \times (1001) = 666 \times (7 \times 11 \times 13)$
 Hence, 666666 is divisible by all of 7, 11 and 13.

Illustration 4: Find the least value of * for which $7*5462$ is divisible by 9.

Solution: Let the required value be x . Then,

$$(7 + x + 5 + 4 + 6 + 2) = (24 + x) \text{ should be divisible by 9.}$$

$$\Rightarrow x = 3$$

Illustration 5: Find the least value of * for which 4832*18 is divisible by 11.

Solution: Let the digit in place of * be x .

(Sum of digits at odd places from right) –

(Sum of digits at even places from right)

$$= (8 + x + 3 + 4) - (1 + 2 + 8) = (4 + x),$$

which should be divisible by 11.

$$\therefore x = 7.$$

Illustration 6: The number 523 abc is divisible by 7, 8 and 9. Then find the value of $a \times b \times c$

Solution: The LCM of 7, 8, and 9 is 504. Therefore, 523 abc should be divisible by 504. Now $523\ abc = 504000 + 19\ abc$.

Therefore, $19abc$ should be divisible by 504.

$19abc = 19000 + abc = 18648 + 352 + abc$. Now 18648 is divisible by 504 $\Rightarrow 352 + abc$ should be divisible by 504. Therefore, $abc = 504 - 352 = 152$. Therefore, $a \times b \times c = 1 \times 5 \times 2 = 10$.

Note:

- | | |
|--------------------------------|--------------------------|
| (i) Even + Even = Even | (ii) Odd + Odd = Even |
| Even – Even = Even | Odd – Odd = Even |
| Even $\times$ Even = Even | Odd $\times$ Odd = Odd |
| Even $\div$ Even = Even or odd | Odd $\div$ Odd = Odd |
| (iii) Even + Odd = Odd | (iv) Odd + Even = Odd |
| Even – Odd = Odd | Odd – Even = Odd |
| Even $\times$ Odd = Even | Odd $\times$ Even = Even |
| Even $\div$ Odd = Even | Odd $\div$ Even |
- = (never divisible)

PRIME NUMBERS

A number other than 1 is called a prime number if it is divisible by only 1 and itself.

All prime numbers less than 100 are:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

Note that 2 is the smallest prime number. 2 is the only even prime number.

Smallest odd prime number is 3.

Twin Primes: A pair of prime numbers are said to be twin prime when they differ by 2. For example 3 and 5 are twin primes.

Co-primes or Relative primes: A pair of numbers are said to be co-primes or relative primes to each other if they do not have any common factor other than 1. For example 13 and 21.

Some Properties which Help in Finding Two Co-prime Numbers

- Two consecutive natural numbers are always co-prime.
Ex. 8 and 9 are co-prime.
Also 12 and 13 are co-prime.
- Two consecutive odd integers are always co-prime.
Ex. 7, 9; 15, 17; 21, 23; etc.
- Two prime numbers are always co-prime.
Ex. 19 and 23 are co-prime.
Also 29 and 41 are co-prime.
- A prime number and a composite number such that the composite number is not a multiple of the prime number are always co-prime.
Ex. 7 and 15 are co-prime.

- Square of two co-prime numbers are always co-prime numbers.

Some Properties which Help in Finding Three Co-prime Numbers

3 numbers are co-prime to each other means all the possible pair of numbers out of these three numbers are co-prime. For example from three numbers 7, 8, 13 three pairs (7, 8), (7, 13) and (8, 13) are formed and each of these pair is a pair of co-prime. Hence, 7, 8, 13 are three co-prime numbers.

Following are some properties helping in finding three co-prime numbers:

- Three consecutive odd integers are always co-prime.
Ex. 9, 11, 13 are co-prime.
- Three consecutive natural numbers with first one being odd are always co-primes.
Ex. 7, 8, 9 are co-prime.
- Two consecutive natural numbers along with the next odd numbers are always co-primes.
Ex. 12, 13, 15 are co-prime. Also 17, 18, 19 are co-prime.
- Three prime numbers are always co-prime.
Ex. 3, 11, 13 are co-prime.

To Test Whether a Given Number is Prime Number or Not

In CAT and CAT like competitions you are required to check whether a given number maximum upto 400 is prime number or not.

If you want to test whether any number is a prime number or not, take an integer equal to the square root of the given number but if square root is not an integer then take an integer just larger than the approximate square root of that number. Let it be 'x'. Test the divisibility of the given number by every prime number less than 'x'. If the given number is not divisible by any prime number less than, then the given number is prime number; otherwise it is a composite number.

Square root of 361 is 19. Prime numbers less than 19 are clearly 2, 3, 5, 7, 11, 13 and 17. Since, 361 is not divisible by any of the numbers 2, 3, 5, 7, 11, 13 and 17. Hence, 361 is a prime number.

It is advisable to learn the squared numbers of all integers from 1 to 20, which are very useful to find whether a given number is a prime or not.

From the table it is clear that if any number, say 271 lies between 256 and 289, then its square root lies between 16 and 17, because $16^2 = 256$ and $17^2 = 289$. Thus square root of the given number is not an integer. So, we take 17 as an integer just greater than the square root of the given number. Now all the prime numbers less than 17 are 2, 3, 5, 7, 11 and 13. Since 271 is not divisible by any of the numbers 2, 3, 5, 7, 11 and 13. Hence 361 is a prime number.

$1^2 = 1$
$2^2 = 4$
$3^2 = 9$
$4^2 = 16$
$5^2 = 25$
$6^2 = 36$
$7^2 = 49$
$8^2 = 64$
$9^2 = 81$
$10^2 = 100$
$11^2 = 121$
$12^2 = 144$
$13^2 = 169$
$14^2 = 196$
$15^2 = 225$
$16^2 = 256$
$17^2 = 289$
$18^2 = 324$
$19^2 = 361$
$20^2 = 400$

Illustration 7: Is 171 is a prime number ?

Solution: Square root of 171 lies between 13 and 14, because $13^2 = 169$ and $14^2 = 196$. Therefore, the integer just greater than the square root of 171 is 14.

Now prime numbers less than 14 are 2, 3, 5, 7, 11 and 13.

Since 171 is divisible by 3, therefore 171 is not a prime number.

Illustration 8: Is 167 is a prime number ?

Solution: Square root of 167 lies between 12 and 13, because $12^2 = 144$ and $13^2 = 169$. Therefore the integer just greater than the square root of 167 is 13.

Now prime numbers less than 13 are 2, 3, 5, 7 and 11.

Since 167 is not divisible by any of the prime numbers 2, 3, 5, 7 and 11; therefore 167 is a prime number.

Illustration 9: Find the number of positive integers n in the range $12 \leq n \leq 40$ such that the product $(n-1)(n-2)(n-3) \dots 3.2.1$ is not divisible by n .

Solution: The product $(n-1)(n-2)(n-3) \dots 3.2.1$ will not be divisible by n only when this product does not contain factors of n , i.e., n is a prime number. The prime numbers that satisfy the above conditions are 13, 17, 19, 23, 29, 31, and 37.

Hence there are 7 required prime numbers.

Note:

In solving linear equations related to word problems and in solving many other problems it is required to know whether the value of any of the expressions like $\frac{M}{P_1} + \frac{N}{P_2}, \frac{M}{P_1} - \frac{N}{P_2}$, etc., is an integer or not. If P_1 and P_2 do not divide M and N respectively and P_1, P_2 are co-prime then any of the expressions like $\frac{M}{P_1} + \frac{N}{P_2}, \frac{M}{P_1} - \frac{N}{P_2}$, etc., will never be an integer on simplify.

Perfect Numbers

A number n is said to be a perfect number if the sum of all the divisors of n (including n) is equal to $2n$.

Ex. Divisor of 6 are 1, 2, 3 and 6. Sum of the divisors
 $= 1 + 2 + 3 + 6 = 12 = 2 \times 6$

Hence, 6 is a perfect number.

COMPLEX NUMBERS, REAL NUMBERS AND IMAGINARY NUMBERS

The numbers in the form $a + ib$, where a and b are real numbers and $i = \sqrt{-1}$; are called complex numbers, 'i' is read as 'iota'.

For example $5 + 3i, \sqrt{2} - 5i$, and $7 + 2\sqrt{3}i$ are complex numbers, A complex number is denoted by C . Thus $C = a + ib$,

If $b = 0$, then the complex number a is purely real number and if $a = 0$, then the complex number ib is purely imaginary number. Thus in complex number $a + ib$, a is called real part and b is called imaginary part.

Ex. $5, \sqrt{2}, 3.4, \frac{5}{7}, 2.1102$, etc. are real numbers.

$2i, -2\sqrt{3}i, -5i$, etc. are imaginary numbers.

All real numbers are complex numbers.

All imaginary numbers are complex numbers.

Properties of 'iota' i.e. 'i'

$i = \sqrt{-1}$, hence $(i)^2 = -1, (i)^3 = -i, (i)^4 = 1,$

$(i)^5 = \sqrt{-1}, (i)^6 = -1$ and so on.

In short, $(i)^{4n} = 1, (i)^{4n+1} = \sqrt{-1}, (i)^{4n+2} = -1, (i)^{4n+3} = -i$, where n is any natural number.

Conjugate of a Complex Number

By changing the sign, +ve to -ve or -ve to +ve of imaginary part of a complex number, we get the conjugate of the given complex number.

Thus conjugate of the complex number $a + ib = a - ib$

Conjugate of the complex number $a - ib = a + ib$

Operations on Complex Numbers

Let z_1, z_2 be two complex numbers such that $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, then

(i) $z_1 \pm z_2 = (a_1 \pm a_2) + i(b_1 \pm b_2)$

(ii) $z_1 \cdot z_2 = (a_1a_2 - b_1b_2) + i(a_1b_2 + b_1a_2)$

(iii) $z_1 = z_2 \Leftrightarrow a_1 = a_2$ and $b_1 = b_2$

GENERAL OR EXPANDED FORM OF 2 AND 3 DIGITS NUMBERS

(i) In a two digits number AB , A is the digit of tenth place and B is the digit of unit place, therefore AB is written using place value in expanded form as

$AB = 10A + B$

Ex. $35 = 10 \times 3 + 5$

(ii) In a three digits number ABC , A is the digit of hundred place, B is the digit of tenth place and C is the digit of unit place, therefore ABC is written using place value in expanded form as

$ABC = 100A + 10B + C$

Ex. $247 = 100 \times 2 + 10 \times 4 + 7$

These expanded forms are used in forming equations related to 2 and 3 digits numbers.

Illustration 10: A two-digit number pq is added to the number formed by reversing its original digits. If their sum is divisible by 11, 9, and 2. Find the number pq .

Solution: Let the original number be pq . The value of the number
 $= 10p + q$.

The number formed by reversing the digits $= qp$. Value of this number $= 10q + p$.

Sum of the two numbers $= 11p + 11q = 11(p + q)$

Now, if the sum is divisible by 11, 9, 2, it means that $(p + q)$ must be divisible by both 9 and 2. Hence, $p + q = 18$. So, it means $p = q = 9$. The original number is 99.

Illustration 11: In a two digit prime number, if 18 is added, we get another prime number with reversed digits. How many such numbers are possible ?

Solution: Let a two-digit number be pq .

$\therefore 10p + q + 18 = 10q + p$

$\Rightarrow -9p + 9q = 18 \Rightarrow q - p = 2$

Satisfying this condition and also the condition of being a prime number (pq and qp both), there are 2 numbers 13 and 79.

SUM OF NUMBERS FORMED WITH GIVEN DIFFERENT DIGITS

Sum of the total numbers which can be formed with given n different digits $a_1, a_2, \dots, a_n$ is $(a_1 + a_2 + a_3 + \dots + a_n) (n-1)!$. (111 ... n times).

Illustration 12: Find the sum of all 4 digit numbers formed with the digits 1, 2, 4 and 6.

Solution: Sum = $(a_1 + a_2 + a_3 + \dots + a_n) (n-1)!$
(111 ... n times)
 $= (1 + 2 + 4 + 6) \cdot 3! \cdot (1111) = 13 \times 6 \times 1111 = 86658$.

FACTORISATION

It is a process of representing a given number as a product of two or more prime numbers.

Here each prime number which is present in the product is called a factor of the given number.

For example, 12 is expressed in the factorised form in terms of its prime factors as $12 = 2^2 \times 3$.

Illustration 13: If $N = 2^3 \times 3^7$, then

- What is the smallest number that you need to multiply with N in order to make it a perfect square?
- What is the smallest number that you need to divide by N in order to make it a perfect square?

Solution:

- Any perfect square number in its factorised form has prime factors with even powers. So in order to make $2^3 \times 3^7$ a perfect square, the smallest number that we need to multiply it with would be 2×3 i.e. 6. The resulting perfect square will be $2^4 \times 3^8$.
- Similarly, in order to arrive at a perfect square by dividing the smallest number, we need to divide the number by 2×3 i.e., 6. The resulting perfect square will be $2^2 \times 3^6$.

NUMBER OF FACTORS OF A COMPOSITE NUMBER

It is possible to find the number of factors of a composite number without listing all those factors.

Take 12 for instance, it can be expressed as $12 = 2^2 \times 3^1$.

The factors of 12 are $(2^0 \times 3^0)$, $(2^0 \times 3^1)$, $(2^1 \times 3^0)$, $(2^1 \times 3^1)$, $(2^2 \times 3^0)$ and $(2^2 \times 3^1)$.

Here the powers of 2 can be one of 0, 1, 2 and the powers of 3 can be one of 0, 1. So number of combinations of a power of 2 and a power of 3 is $3 \times 2 = 6$. All the combinations of power of 2 and a power of 3 are 0, 0; 0, 1; 1, 0; 1, 1; 2, 0; 2, 1. Each combination of the powers of 2 and 3 gives a distinctly different factor. Since there are 6 different combinations of the powers of 2 and 3, hence there are 6 distinctly different factors of 12.

Let N be a composite number such that $N = (x)^a (y)^b (z)^c \dots$ where $x, y, z, \dots$ are different prime numbers. Then the number of divisors (or factors) of $N = (a+1)(b+1)(c+1) \dots$

Here factors and divisors means the same.

Illustration 14: Find the total number of factors of 576.

Solution: The factorised form of $576 = 2^6 \times 3^2$
So the total number of factors = $(6+1)(2+1) = 21$

Illustration 15: Find the number of divisors of 21600.

Solution: $21600 = 2^5 \times 3^3 \times 5^2$
 $\Rightarrow$ Number of divisors = $(5+1) \times (3+1) \times (2+1)$
 $= 6 \times 4 \times 3 = 72$.

Illustration 16: How many divisors of 21600 are odd numbers?

Solution: $21600 = 2^5 \times 3^3 \times 5^2$
An odd number does not have a factor of 2 in it. Therefore, we will consider all the divisors having powers of 3 and 5 but not 2. Therefore, ignoring the powers of 2, the number of odd divisors = $(3+1) \times (2+1) = 4 \times 3 = 12$.

Illustration 17: How many divisors of 21600 are even numbers?

Solution: $21600 = 2^5 \times 3^3 \times 5^2$
Total number of divisors of 21600 = $6 \times 4 \times 3 = 72$
Number of odd divisors of 21600 = $4 \times 3 = 12$
 $\therefore$ Number of even divisors of 21600 = $72 - 12 = 60$

Illustration 18: How many divisors of 21600 are perfect squares?

Solution: $21600 = 2^5 \times 3^3 \times 5^2$
In a perfect square, all the prime factors have even powers. Therefore, all the divisors made by even powers of 2, 3 and 5 will be perfect squares.
The even powers of 2 are $2^0, 2^2, 2^4$, even powers of 3 are 3^0 and 3^2 , and even powers of 5 are 5^0 and 5^2 . We can select and even power of 2 in 3 ways, even power of 3 in 2 ways and even power of 5 in 2 ways. Therefore, the number of combinations of even powers of 2, 3 and 5 = $3 \times 2 \times 2 = 12$.
Hence there are 12 perfect square divisors of 21600.

Number of Factors Satisfying Special Condition(s) for any Composite Number

To understand the concept, consider a composite number 1080. Suppose we want to find the number of factors of 1080 which are divisible by 6. For this, we factorise 1080 and 6 into its prime factors as $1080 = 2^3 \times 3^3 \times 5$; $6 = 2 \times 3$

Since each factor of 1080 should be divisible by 6, therefore each factor must contain at least one 2 and one 3.

Now the remaining factors (except one 2 and one 3) of 1080 = $2^2 \times 3^2 \times 5$

Hence number of factors of 1080 divisible by 6
= Number of factors of $(2^2 \times 3^2 \times 5)$
 $= (2+1) \times (2+1) \times (1+1) = 18$

NUMBER OF WAYS OF EXPRESSING A COMPOSITE NUMBER AS A PRODUCT OF TWO FACTORS

- Number of ways of expressing a composite number N which is not a perfect square as a product of two factors
 $= \frac{1}{2} \times (\text{Number of prime factors of the } N)$

- Number of ways of expressing a perfect square number M as a product of two factors = $\frac{1}{2} [(\text{Number of prime factors of } M + 1)]$

Illustration 19: Find the number of ways of expressing 180 as a product of two factors.

Solution: $180 = 2^2 \times 3^2 \times 5^1$

Number of factors = $(2 + 1)(2 + 1)(1 + 1) = 18$

Since 180 is not a perfect square, hence there are total $\frac{18}{2} = 9$

ways in which 180 can be expressed as a product of two factors.

Illustration 20: Find the number of ways expressing 36 as a product of two factors.

Solution: $36 = 2^2 \times 3^2$

Number of factors = $(2 + 1)(2 + 1) = 9$

Since 36 is a perfect square, hence the number of ways of expressing 36 as a product of two factors

$$= \frac{9+1}{2} = 5, \text{ as } 36 = 1 \times 36, 2 \times 18, 3 \times 12, 4 \times 9 \text{ and } 6 \times 6.$$

SUM OF FACTORS (OR DIVISORS) OF A COMPOSITE NUMBER

Let N be a composite number in such a way that $N = (x)^a (y)^b (z)^c \dots$ where $x, y, z, \dots$ are prime numbers. Then, the sum of factors

$$(\text{or divisors}) \text{ of } N = \frac{x^{a+1} - 1}{x - 1} \times \frac{y^{b+1} - 1}{y - 1} \times \frac{z^{c+1} - 1}{z - 1} \dots$$

Illustration 21: What is the sum of the divisors of 60 ?

Solution: $60 = 2^2 \times 3 \times 5$

$$\Rightarrow \text{Sum of the divisors} = \frac{2^3 - 1}{2 - 1} \times \frac{3^2 - 1}{3 - 1} \times \frac{5^2 - 1}{5 - 1} = 168.$$

SUM OF UNIT DIGITS

For given n different digits $a_1, a_2, a_3, \dots, a_n$; the sum of the digits at unit place of all different numbers formed is

$$(a_1 + a_2 + a_3 + \dots + a_n) (n - 1)! \text{ i.e., (Sum of the digits) } (n - 1)!$$

Illustration 22: Find the sum of unit digits of all different numbers formed from digits 4, 6, 7 and 9.

Solution: Required sum = $(4 + 6 + 7 + 9) - (4 - 1)!$
 $= 26 - 3! = 26 - 6 = 20.$

THE LAST DIGIT FROM LEFT (i.e., UNIT DIGIT) OF ANY POWER OF A NUMBER

The last digits (from left) of the powers of any number follow a cyclic pattern i.e., they repeat after certain number of steps. If we find out after how many steps the last digit of the powers of a number repeat, then we can find out the last digit of any power of any number.

Let us look at the powers of 2:

Last digit of 2^1 is 2 .	Last digit of 2^6 is 4 .
Last digit of 2^2 is 4 .	Last digit of 2^7 is 8 .
Last digit of 2^3 is 8 .	Last digit of 2^8 is 6 .
Last digit of 2^4 is 6 .	Last digit of 2^9 is 2 .
Last digit of 2^5 is 2 .	

Since last digit of 2^5 is the same as the last digit of 2^1 , then onwards the last digit will start repeating, i.e., digits of $2^5, 2^6, 2^7, 2^8$ will be the same as those of $2^1, 2^2, 2^3, 2^4$. Then the last digit of 2^9 is again the same as the last digit of 2^1 and so on. Thus, we see that when power of 2 increases, the last digits repeat after every 4 steps.

In above pattern, we can see that whenever the power of 2 is a multiple of 4, the last digit of that number will be the same as the last digit of 2^4 .

Suppose we want to find out the last digit of 2^{66} , we should look at a multiple of 4 which is just less than or equal to the power 66 of 2. Since 64 is a multiple of 4, the last digit of 2^{64} will be the same as the last digit of 2^4 .

Then the last digits of $2^{65}, 2^{66}$ will be the same as the last digits of $2^1, 2^2$ respectively. Hence the last digit of 2^{66} is the same as the last digit of 2^2 i.e., 4.

Similarly, we can find out the last digit of 3^{75} by writing down the pattern of the powers of 3.

Last digit of 3^1 is 3.	Last digit of 3^4 is 1.
Last digit of 3^2 is 9.	Last digit of 3^5 is 3.
Last digit of 3^3 is 7.	Last digit of 3^6 is 9
	Last digit of 3^7 = 7
	Last digit of 3^8 = 1
	Last digit of 3^9 = 3

The last digit repeats after 4 steps (like in the case of powers of 2).

Whenever the powers of 3 is a multiple of 4, the last digit of that number will be the same as the last digit of 3^4 .

To find the last digit of 3^{75} , we look for a multiple of 4 which is just less than or equal to the power 75 of 3. Since, 72 is multiple of 4, the last digit of 3^{72} will be the same as that of 3^4 . Hence the last digit of 3^{75} will be the same as the last digit of 3^3 i.e., 7.

Last Digit (i.e., Unit Digit) of a Product

Last digit of the product $a \times b \times c \dots$ is the last digit of the product of last digits of $a, b, c, \dots$

Illustration 23: Find the last digit of $2^{416} \times 4^{430}$.

Solution: Writing down the powers of 2 and 4 to check the pattern of the last digits, we have

We have seen that whenever the power of 2 is a multiple of 4, the last digit of that number will be the same as the last digit of 2^4 .

Now, Last digit of $4^1 = 4.$
Last digit of $4^2 = 6.$
Last digit of $4^3 = 4.$
Last digit of $4^4 = 6.$

Thus last digit of any power of 4 is 4 for an odd power and 6 for an even power. The last digit of 2^{416} will be the same as 2^4 because 416 is a multiple of 4. So the last digit of 2^{416} is 6.

Last digit of 4^{430} is 6, since the power of 4 is even.

Hence the last digit of $2^{416} \times 4^{430}$ will be equal to the last digit of $6 \times 6 = 6$.

CONCEPT OF REMAINDERS

(I) Suppose the numbers $N_1, N_2, N_3, \dots$ give quotients $Q_1, Q_2, Q_3, \dots$ and remainder $R_1, R_2, R_3, \dots$ when divided by a common divisor D .

Let S be the sum of $N_1, N_2, N_3, \dots$

Therefore, $S = N_1 + N_2 + N_3 + \dots$

$$= (D \times Q_1 + R_1) + (D \times Q_2 + R_2) + (D \times Q_3 + R_3) + \dots$$

$$= D \times K + (R_1 + R_2 + R_3 + \dots), \dots (1)$$

where K is some number

Hence the remainder when S is divided by D is the remainder when $(R_1 + R_2 + R_3 \dots)$ is divided by D .

- (II) Suppose the numbers, $N_1, N_2, N_3, \dots$ give quotients $Q_1, Q_2, Q_3, \dots$ and remainders $R_1, R_2, R_3, \dots$ respectively, when divided by a common divisor D .

Therefore $N_1 = D \times Q_1 + R_1, N_2 = D \times Q_2 + R_2,$

$N_3 = D \times Q_3 + R_3 \dots$ and so on.

Let P be the product of $N_1, N_2, N_3, \dots$

Therefore,

$$\begin{aligned} P &= N_1 N_2 N_3 \dots \\ &= (D \times Q_1 + R_1) (D \times Q_2 + R_2) (D \times Q_3 + R_3) \dots \\ &= D \times K + (R_1 R_2 R_3 \dots), \end{aligned} \quad \dots (2)$$

where K is some number

In the above equation, since only the product $(R_1 R_2 R_3 \dots)$ is free of D , therefore the remainder when P is divided by D is the remainder when the product $(R_1 R_2 R_3 \dots)$ is divided by D .

Illustration 24: What is the remainder when the product $1991 \times 1992 \times 2000$ is divided by 7?

Solution: The remainder when 1991, 1992 and 2000 are divided by 7 are 3, 4 and 5 respectively.

Hence the final remainder is the remainder when the product $3 \times 4 \times 5 = 60$ is divided by 7. Therefore, remainder = 4.

Illustration 25: What is the remainder when 2^{2010} is divided by 7?

Solution: 2^{2010} is a product $(2 \times 2 \times 2 \dots (2010 \text{ times}))$. Since, 2 is a number less than 7, we try to convert the product into product of numbers higher than 7. Notice that $8 = 2 \times 2 \times 2$. Therefore, we convert the product in the following manner

$$2^{2010} = 8^{670} = 8 \times 8 \times 8 \dots (670 \text{ times}).$$

The remainder when 8 is divided by 7 is 1. Hence the remainder when 8^{670} is divided by 7 is the remainder obtained when the product $1 \times 1 \times 1 \dots (670 \text{ times})$ is divided by 7. Therefore, remainder = 1.

Illustration 26: What is the remainder when 2^{2012} is divided by 7?

Solution: This problem is like the previous one, except that 2012 is not an exact multiple of 3, so we cannot convert it completely into the form 8^x . We will write it in following manner $2^{2012} = 8^{670} \times 4$. Now, 8^{670} gives the remainder 1 when divided by 7 as we have seen in the previous problem. And 4 gives a remainder of 4 only when divided by 7. Hence the remainder when 2^{2012} is divided by 7 is the remainder when the product 1×4 is divided by 7. Therefore, remainder = 4.

Illustration 27: What is the remainder when 25^{24} is divided by 9?

Solution: Again $25^{24} = (18 + 7)^{24} = (18 + 7) (18 + 7) \dots 24 \text{ times} = 18K + 7^{24}$.

Hence, remainder when 25^{24} is divided by 9 is the remainder when 7^{24} is divided by 9.

Now, $7^{24} = 7^3 \times 7^3 \times 7^3 \dots (8 \text{ times}) = 343 \times 343 \times 343 \dots (8 \text{ times})$

Now when 343 is divided by 9 the remainder is 1

So, the remainder when dividing $(343)^8$ by 9 means remainder when dividing $(1)^8$ by 9. So the required remainder is 1.

Illustration 28: What is the remainder when $39^{32^{32}}$ is divided by 7?

Solution: Steps for finding remainder when X^Y is divided by D .

- Divide X by D . Let the remainder be R . Therefore, you have to find the remainder when R^Z is divided by D . 39 gives a remainder 4 when divided by 7. Therefore, you have to find the remainder when $4^{32^{32}}$ is divided by 7.
- Find a power of R that gives a remainder +1 when divided by D . If you find a power that gives a remainder -1, twice of that power will give a remainder of +1. Now we know that $4^3 = 64$ gives a remainder 1 when divided by 7.
- Find the remainder when Y is divided by the power R . Here, find the remainder when 32^{32} is divided by 3. The remainder is 1. Therefore, when 32^{32} is divided by 3. The remainder is 1. Therefore, 32^{32} can be written as $3k + 1$ and $4^{32^{32}}$ can be written as $4^{3k+1} = (4^3)^k \times 4$.
- Now 4^3 gives a remainder 1 when divided by 7. Therefore, the required remainder is the remainder when 4 is divided by 7. Hence, the required remainder is 4.

(III) Some Special Cases

(A) When Both the Dividend and the Divisor have a Factor in Common

To find the remainder,

- Divide both dividend and divisor by the common factor (K) i.e. HCF of dividend and divisor.
- Divide the resulting dividend (A) by resulting divisor (B) and find the remainder (R_1).
- The real remainder R is the remainder R_1 multiplied by the common factor (K).

Illustration 29: What is the remainder when 2^{96} is divided by 96?

Solution: The common factor between 2^{96} and 96 is $32 = 2^5$.

Divide both dividend and divisor by 32 i.e., 2^5

You will get the resulting dividend and the divisor as the numbers 2^{91} and 3 respectively.

$$\text{Now, } 2^{91} = (2^4)^{22} \cdot 2^3 = (16)^{22} \cdot 8$$

When 16 and 8 are divided by 3, remainder are 1 and 2 respectively.

Hence when $(16)^{22}$ when divided by 3, we get the remainder $(1)^{22}$ i.e. 1.

Hence the remainder when $(2)^{91}$ is divided by 3 is the remainder when 1×2 is divided by 3. Hence the remainder when $(2)^{91}$ is divided by 3 is 2.

Hence the real remainder will be 2 multiplied by common factor 32.

$$\text{i.e. real remainder} = 64$$

(B) The Concept of Negative Remainder

$$15 = 16 \times 0 + 15 \quad \text{or} \quad 15 = 16 \times 1 - 1.$$

The remainder when 15 is divided by 16 is 15 in the first case and -1 in the second case. Hence, the remainder when 15 is divided by 16 is 15 or -1.

$$\text{Also } 23 = 7 \times 3 + 2 \quad \text{or} \quad 23 = 7 \times 4 - 5.$$

Thus, when 23 is divided by 7, then remainder is 2 or -5.

Thus, when a number is divided by A gives a negative remainder B , then positive remainder = $A + B$.

For example, when a number gives a negative remainder of -2 when divided by 23, it means that the number gives a positive remainder of $23 - 2 = 21$ when divided by 23.

Using the concept of positive and negative remainders, you can find the remainder more easily by reducing calculations. You must remember that we always take the positive remainder in final answer.

- (i) To find the remainder when $76 \times 55 \times 67 \times 51$ is divided by 8, we can follow as

$$\begin{aligned} \frac{76 \times 55 \times 67 \times 51}{8} &\xrightarrow{\text{Remainder}} \frac{4 \times (-1) \times 3 \times 3}{8} \xrightarrow{\text{Remainder}} \\ &\frac{-36}{8} \xrightarrow{\text{Remainder}} -4 \xrightarrow{\text{Remainder}} -4 + 8 = 4 \end{aligned}$$

Thus required remainder = 4

To convert the negative remainder into positive remainder, we add the divisor to the negative remainder.

Note that if you transform $\frac{-36}{8}$ into $\frac{-9}{2}$ by dividing both numerator and denominator by common factor 4 of -36 and 8, then original remainder -4 is also divided by 4 giving -1 as remainder. So to find the correct answer, we multiply the incorrect remainder by 4.

- (ii) When $69 \times 68 \times 71 \times 66$ is divided by 72 to find the remainder, we can follow as

$$\begin{aligned} \frac{69 \times 68 \times 71 \times 66}{72} &\xrightarrow{\text{Remainder}} \frac{(-3) \times (-4) \times (-1) \times (-6)}{72} \\ &\xrightarrow{\text{Remainder}} \frac{72}{72} \xrightarrow{\text{Remainder}} 1. \end{aligned}$$

Thus required remainder = 1.

- (iii) When $53 \times 55 \times 57 \times 61$ is divided by 60, then to find the remainder we can follow as

$$\begin{aligned} \frac{53 \times 55 \times 57 \times 61}{60} &\xrightarrow{\text{Remainder}} \frac{(-7) \times (-5) \times (-3) \times 1}{60} \\ &\xrightarrow{\text{Remainder}} \frac{-105}{60} \xrightarrow{\text{Remainder}} -45 \end{aligned}$$

Thus Remainder = $-45 + 60 = 15$

(C) Remainder in Case of a Number with Large Power

- (i) When remainder becomes 1 directly. In such a case, no matter how large the value of the power n is, the remainder is 1.

For example,

$$\begin{aligned} \frac{(33)^{24139}}{8} &\xrightarrow{\text{Remainder}} \frac{(8-4+1)^{24139}}{8} \xrightarrow{\text{Remainder}} \frac{(1)^{24139}}{8} \xrightarrow{\text{Remainder}} 1 \end{aligned}$$

If a is the divisor and dividend can be expressed as $(ax - 1)^n$, where x and n are two natural numbers, then

- (ii) Remainder will be +1, if n is even and -1, if n is odd and hence positive remainder is $(a - 1)$, when n is odd.

For example,

$$\frac{41^{127}}{7} \xrightarrow{\text{Remainder}} \frac{(7 \times 6 - 1)^{127}}{7} \xrightarrow{\text{Remainder}} \frac{(-1)^{127}}{7}$$

$$\xrightarrow{\text{Remainder}} (-1) \xrightarrow{\text{Remainder}} 7 - 1 = 6.$$

Illustration 30: Find the remainder when 7^{52} is divided by 2402.

Solution:

$$\begin{aligned} \frac{7^{52}}{2402} &\xrightarrow{\text{Remainder}} \frac{(7^4)^{13}}{2402} \xrightarrow{\text{Remainder}} \frac{(2401)^{13}}{2402} \\ &\xrightarrow{\text{Remainder}} \frac{(2402 - 1)^{13}}{2402} \xrightarrow{\text{Remainder}} (-1)^{13} = -1. \end{aligned}$$

Hence, the remainder when 7^{52} is divided by 2402 is equal to -1 or $2402 - 1 = 2401$.

TO FIND THE LAST DIGITS OF THE EXPRESSION LIKE $a_1 \times a_2 \times a_3 \times \dots \times a_n$

Last r digits (from right) of the product $a_1 \times a_2 \times a_3 \times \dots \times a_n$ is the remainder when $a_1 \times a_2 \times a_3 \times \dots \times a_n$ is divided by $(10)^r$.

Let us find the last two digits of $29 \times 47 \times 53 \times 76 \times 89$.

$$\begin{aligned} \text{Now } \frac{29 \times 47 \times 53 \times 76 \times 89}{100} &\xrightarrow{\text{Remainder}} \\ \frac{29 \times 47 \times 53 \times 19 \times 89}{25} &\xrightarrow{\text{Remainder}} \frac{4 \times (-3) \times 3 \times (-6) \times (-11)}{25} \\ \xrightarrow{\text{Remainder}} \frac{-2376}{25} &\xrightarrow{\text{Remainder}} -1 \end{aligned}$$

(on dividing by 4 in both numerator and denominator)

Thus remainder is -1 (after dividing by 4)

Hence actual remainder = $-1 \times 4 = -4$, which is negative

Now actual positive remainder = $-4 + 100 = 96$

Hence required last two digits = 96

Similarly you can find any number of last digits of a product.

Theorem 1: $(a^n + b^n)$ is divisible by $(a + b)$ when n is odd.

Theorem 2: $(a^n - b^n)$ is divisible by $(a + b)$ when n is even.

Theorem 3: $(a^n - b^n)$ is always divisible by $(a - b)$ when n is an integer.

Hence $(a^n - b^n)$ is divisible by both $(a + b)$ and $(a - b)$ when n is even and $(a^n - b^n)$ is divisible by only $(a - b)$ when n is odd.

Illustration 31: What is the remainder when $3^{444} + 4^{333}$ is divided by 5?

Solution: The dividend is in the form $a^x + b^y$. We need to change it into the form $a^n + b^n$.

$3^{444} + 4^{333} = (3^4)^{111} + (4^3)^{111}$. Now $(3^4)^{111} + (4^3)^{111}$ will be divisible by $3^4 + 4^3 = 81 + 64 = 145$.

Since the number is divisible by 145, it will certainly be divisible by 5. Hence, the remainder is 0.

Illustration 32: What is the remainder when $(5555)^{2222} + (2222)^{5555}$ is divided by 7?

Solution: The remainder when 5555 and 2222 are divided by 7 are 4 and 3 respectively. Hence, the problem reduces to finding the remainder when $(4)^{2222} + (3)^{5555}$ is divided by 7.

Now $(4)^{2222} + (3)^{5555} = (4^2)^{1111} + (3^5)^{1111} = (16)^{1111} + (243)^{1111}$. Now $(16)^{1111} + (243)^{1111}$ is divisible by $16 + 243$ or it is divisible by 259, which is a multiple of 7. Hence the remainder when $(5555)^{2222} + (2222)^{5555}$ is divided by 7 is zero.

Illustration 33: $20^{2008} + 16^{2008} - 3^{2008} - 1$ is divisible by:

- (a) 314 (b) 323
(c) 253 (d) 91

Solution: (b) $20^{2008} + 16^{2008} - 3^{2008} - 1 = (20^{2008} - 3^{2008}) + (16^{2008} - 1^{2008})$. Now $20^{2008} - 3^{2008}$ is divisible by 17 (Theorem 3) and $16^{2008} - 1^{2008}$ is divisible by 17 (Theorem 2). Hence the complete expression is divisible by 17.

$20^{2008} + 16^{2008} - 3^{2008} - 1 = (20^{2008} - 1^{2008}) + (16^{2008} - 3^{2008})$. Now $20^{2008} - 1^{2008}$ is divisible by 19 (Theorem 3) and $16^{2008} - 3^{2008}$ is divisible by 19 (Theorem 2). Hence the complete expression is also divisible by 19.

Hence the complete expression is divisible by $17 \times 19 = 323$.

Illustration 34: If $p = 1! + (2 \times 2!) + (3 \times 3!) + \dots + (10 \times 10!)$, where $n! = 1 \times 2 \times 3 \times \dots \times n$ for integer $n \geq 1$, then $p + 2$ when divided by $11!$, leaves a remainder

- (a) 10 (b) 0 (c) 7 (d) 1

Solution: (b) n th term of series

$$= n \times n! = (n + 1 - 1) \times n! = (n + 1)! - n!$$

Therefore, $p = 2! - 1! + 3! - 2! + 4! - 3! + \dots + 11! - 10!$
 $= 11! - 1! \Rightarrow p + 2 = 11! + 1$

Hence when $(p + 2)$ is divided by $11!$, then remainder = 1.

Illustration 35: Find the remainder when

$1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + 99 \times 100$ is divided by 101.

Solution: n th term of the series $= n \times (n + 1) = n^2 + n$.

Therefore, sum of the series,

$$\begin{aligned} \sum (n^2 + n) &= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \\ &= \frac{n(n+1)(n+2)}{3} = \frac{99 \times 100 \times 101}{3} \\ &= 33 \times 100 \times 101 \end{aligned}$$

$\Rightarrow$ Remainder on dividing by 101 = 0.

Note:

$$\begin{aligned} \text{(i)} \quad 1 + 2 + 3 + \dots + n &= \frac{n(n+1)}{2} \Rightarrow \sum n = \frac{n(n+1)}{2} \\ \text{(ii)} \quad 1^2 + 2^2 + 3^2 + \dots + n^2 &= \frac{n(n+1)(2n+1)}{6} \Rightarrow \sum n^2 = \frac{n(n+1)(2n+1)}{6} \\ \text{(iii)} \quad 1^3 + 2^3 + 3^3 + \dots + n^3 &= \left(\frac{n(n+1)}{2} \right)^2 \Rightarrow \sum n^3 = \left(\frac{n(n+1)}{2} \right)^2 \end{aligned}$$

LAST TWO DIGITS OF A NUMBER WITH LARGE POWER

Last Two Digits of Numbers Ending in 1

Let's start with an example $(31)^{786}$.

Multiply the tens digit of the number (3 here) with the last digit of the exponent (6 here) to get the tens digit $3 \times 6 = 18$.

The unit digit 8 of the product 18 is tens digit of the required number.

Unit digits of the required number is equal to 1.

Last two digits of 41^{2789} is 61

(Since $4 \times 9 = 36$. Therefore, 6 is the tens digit and 1 is the units digit).

Last Two Digits of Numbers Ending in 3, 7 or 9

Convert the number till the base of the number ends in 1 and then find the last two digits according to the previous method.

To find the last two digits of 19^{266} :

$19^{266} = (19^2)^{133}$. Now, 19^2 ends in 61 ($19^2 = 361$) therefore, we need to find the last two digits of $(61)^{133}$.

Last two digits of $(61)^{133}$ and hence $(19)^{266}$ is 81 by the previous method (tens digit = unit digit of $(6 \times 3 = 18)$ and unit digit = 1).

To find the last two digits of 33^{288} :

$33^{288} = (33^4)^{72}$. Now 33^4 ends in 21 ($33^4 = 33^2 \times 33^2 = 1089 \times 1089 = \text{xxxxx}21$), therefore we need to find the last two digits of 21^{72} . By the previous method, the last two digits of $21^{72} = 41$ (tens digit = $2 \times 2 = 4$, unit digit = 1).

Now try the method with a number ending in 7:

$$87^{474} = 87^{472} \times 87^2 = (87^4)^{118} \times 87^2 \rightarrow (69 \times 69)^{118} \times 69$$

(The last two digits of 87^2 are 69) $\rightarrow 61^{118} \times 69 \rightarrow 81 \times 69 \rightarrow 89$.

Last Two Digits of Numbers Ending in 2, 4, 6 or 8

(i) There is only one even two-digit number 76, raised to any power gives the last two digits as 76.

(ii) 2^{10} ends in 24.

(iii) 24 raised to an even power always ends with 76 and 24 raised to an odd power always ends with 24. Therefore, 24^{34} will end in 76 and 24^{53} will end in 24.

Now apply this concept in the following examples.

(iv) When 76 is multiplied with 2^n for $n \geq 2$, the last two digits of the product is the same as the last two digits of 2^n . Therefore, the last two digits of 76×2^7 will be the last two digits of $2^7 = 28$.

Illustration 36: Find the last two digits of 2^{543} .

Solution: $2^{543} = (2^{10})^{54} \times 2^3$

$$= (24)^{54} \text{ (i.e. 24 raised to an even power)} \times 2^3 \rightarrow 76 \times 8 \rightarrow 08.$$

Illustration 37: Find the last two digits of 64^{236} .

Solution: $64^{236} \rightarrow (2^6)^{236} \rightarrow 2^{1416} \rightarrow (2^{10})^{141} \times 2^6$

$$\rightarrow (24)^{141} \text{ (24 raised to odd power)} \times 64 \rightarrow 24 \times 64 \rightarrow 36.$$

Illustration 38: Find the last two digits of 62^{586} .

Solution: $62^{586} = (2 \times 31)^{586} = 2^{586} \times 31^{586}$

$$= (2^{10})^{58} \times 2^6 \times 31^{586} = 76 \times 64 \times 81 = 84$$

Illustration 39: Find the last two digits of 54^{380} .

Solution: $54^{380} = (2 \times 3^3)^{380} = 2^{380} \times 3^{1140} = (2^{10})^{38} \times (3^4)^{285}$
 $= 76 \times 81^{285} = 76 \times 01 = 76.$

Illustration 40: Find the last two digits of 56^{283} .

Solution: $56^{283} = (2^3 \times 7)^{283} = 2^{849} \times 7^{283}$
 $= (2^{10})^{84} \times 2^9 \times (7^4)^{70} \times 7^3$
 $= 76 \times 12 \times (01)^{70} \times 43 = 16$

Illustration 41: Find the last two digits of 78^{379} .

Solution: $78^{379} = (2 \times 39)^{379} = 2^{379} \times 39^{379}$
 $= (2^{10})^{37} \times 2^9 \times (39^2)^{189} \times 39 \rightarrow 24 \times 12 \times$
 $(21)^{189} \times 39 \rightarrow 24 \times 12 \times 81 \times 39 = 92.$

NUMBER OF ZEROES IN AN EXPRESSION LIKE $a \times b \times c \times \dots$, WHERE $a, b, c, \dots$ ARE NATURAL NUMBERS

Consider an expression $8 \times 15 \times 20 \times 30 \times 40$.

The expression can be written in the standard form as :

$$8 \times 15 \times 20 \times 30 \times 40$$

$$= (2^3) \times (3 \times 5) \times (2^2 \times 5) \times (2 \times 3 \times 5) \times (2^3 \times 5)$$

$$= 2^9 \times 3^2 \times 5^4, \text{ in which base of each factor is a prime number.}$$

A zero is formed by the product of 2 and 5 i.e. 2×5 . Hence number of zeroes is equal to the number of pair(s) of 2's and 5's formed.

In the above standard form of the product there are 9 twos and 4 fives. Hence number of pairs of 2 and 5 i.e. (2×5) is 4. Hence, there will be 4 zeroes at the end of the final product.

In the same above way, we can find the number of zeroes at the end of any product given in the form of an expression like $a \times b \times c \times \dots$, where $a, b, c, \dots$ are natural numbers.

If there is no pair of 2 and 5 i.e. 2×5 , then there is no zero at the end of the product. For example, consider the expression $9 \times 21 \times 39 \times 49$.

The given expression in standard form,

$$9 \times 21 \times 39 \times 49 = (3^2) \times (3 \times 7) \times (3 \times 13) \times (7^2) \\ = 3^4 \times 7^3 \times 13$$

There is no pair of 2 and 5 in the standard form of expression given as product, therefore there will be no zero at the end of the final product.

Illustration 42: Find the number of zeroes in the product

$$1^1 \times 2^2 \times 3^3 \times 4^4 \times 5^5 \times 6^6 \times \dots \times 49^{49}$$

Solution: Clearly the fives will be less than the twos. Hence, we need to count only the fives.

$$\text{Now, } 5^5 \times 10^{10} \times 15^{15} \times 20^{20} \times 25^{25} \times 30^{30} \times 35^{35} \times 40^{40} \times 45^{45} \\ = (5)^5 \times (5 \times 2)^{10} \times (5 \times 3)^{15} \times (5 \times 4)^{20} \times (5 \times 5)^{25} \times \\ (5 \times 6)^{30} \times (5 \times 7)^{35} \times (5 \times 8)^{40} \times (5 \times 9)^{45}$$

It gives us $5 + 10 + 15 + 20 + 25 + 30 + 35 + 40 + 45$ fives i.e., 825 fives

Thus the product has 825 zeroes.

Illustration 43: Find the number of zeroes in:

$$100^1 \times 99^2 \times 98^3 \times 97^4 \times \dots \times 1^{100}$$

Solution: Clearly the fives will be less than the twos. Hence we need to count the number of fives only. This can get done by :

$$100^1 \times 95^6 \times 90^{11} \times 85^{16} \times 80^{21} \times 75^{26} \times \dots \times 5^{96} \\ = (5^2 \times 4)^1 \times (5 \times 19)^6 \times (5 \times 18)^{11} \times (5 \times 17)^{16} \times (5 \times 16)^{21} \times \\ (5^2 \times 3)^{26} \times (5 \times 14)^{31} \times \dots \times (5 \times 11)^{56} \times (5^2 \times 2)^{51} \times (5 \times 9)^{56} \\ \times \dots \times (5 \times 6)^{71} \times (5^2)^{76} \times (5 \times 4)^{81} \times \dots \times (5 \times 2)^{91} \times (5)^{96} \\ \rightarrow (1 + 6 + 11 + 16 + 21 + 26 + 31 + 36 + 41 + 46 + \dots + 96) \\ + (1 + 26 + 51 + 76)$$

In $1 + 6 + 11 + 16 + \dots + 96$, if any term is subtract from just next term, we get the same number

$5 (= 6 - 1 = 11 - 6 = \dots = 96 - 91)$. This difference 5 is called common difference and any series having a common difference is called an A.P. (Arithmetic Progression).

Hence $1 + 6 + 11 + 16 + \dots + 96$ is an A.P.

The formula to find the number of terms in an A.P. is

$$l = a + (n - 1) d$$

where, l is the last term (here $l = 96$)

a is the first term (here $a = 1$)

d is the common difference (here $d = 5$)

and n is the number of terms.

$$\therefore 96 = 1 + (n - 1) \times 5 \Rightarrow n = 20$$

The formula to find the sum of all terms of an A.P. is

$$S = \frac{n}{2} (a + l), \text{ where } S \text{ is the sum of all terms}$$

$$\therefore S = \frac{20}{2} (1 + 96) = 970$$

$$\text{i.e. } 1 + 6 + 11 + 16 + \dots + 96 = 970$$

We will study A.P. in detail in the chapter: Progressions.

$$\text{Now } 1 + 26 + 51 + 76 = 154$$

$$\therefore (1 + 6 + 11 + 16 + \dots + 96) + (1 + 26 + 51 + 76) \\ = 970 + 1124.$$

POWERS OF A NUMBER CONTAINED IN A FACTORIAL

(I) First Method

Highest power of prime number p in $n!$

$$= \left[\frac{n}{p} \right] + \left[\frac{n}{p^2} \right] + \left[\frac{n}{p^3} \right] + \left[\frac{n}{p^4} \right] + \dots + \left[\frac{n}{p^r} \right], \text{ where } [x]$$

denotes the greatest integer less than or equal to x and r is a natural number such that $p^r < n$.

Illustration 44: Find the highest power of 2 in 50!

Solution: The highest power of 2 in 50!

$$= \left[\frac{50}{2} \right] + \left[\frac{50}{4} \right] + \left[\frac{50}{8} \right] + \left[\frac{50}{16} \right] + \left[\frac{50}{32} \right] \\ = 25 + 12 + 6 + 3 + 1 = 47$$

Illustration 45: Find the highest power of 6 in 60!.

Solution: Here given number 6 is not a prime number so first convert 6 as a product of primes. $6 = 2 \times 3$, therefore we will find the highest power of 2 and 3 in 60!.

Highest power of 2 in 60!

$$= \left[\frac{60}{2} \right] + \left[\frac{60}{4} \right] + \left[\frac{60}{8} \right] + \left[\frac{60}{16} \right] + \left[\frac{60}{32} \right] \\ = 30 + 15 + 7 + 3 + 1 = 56$$

Highest power of 3 in 60!

$$= \left[\frac{60}{3} \right] + \left[\frac{60}{9} \right] + \left[\frac{60}{27} \right] = 20 + 6 + 2 = 28$$

So 60! contains $(2)^{56} \times (3)^{28}$. Hence it contains 28 pairs of 2 and 3. Therefore, required power of 6 is 28, which is actually the power of the largest prime factor 3 of 6, because power of largest prime factor is always equal or less than the other prime factors of any number.

Illustration 46: Find the highest power of 30 in 50!

Solution: $30 = 2 \times 3 \times 5$. Now 5 is the largest prime factor of 30, therefore, the powers of 5 in 50! will be less than those of 2 and 3. Therefore, power of 30 is equal to the power of 5 in 50! So we find the highest power of 5 in 50!

The highest power of 5 in 50!

$$= \left[\frac{50}{5} \right] + \left[\frac{50}{25} \right] = 10 + 2 = 12$$

Hence the highest power of 30 in 50! = 12.

(II) Second Method

We will discuss this method through an example,

Let us find the highest power of prime number 7 in 400! to find the highest power of prime number 7 in 400! we divide 400! by

7 and find the quotient. Divide this quotient by 7 again and find the next quotient and proceed as given below. In this way, we will find the last quotient, which is less than the divisor 7 as follows :

$$\begin{array}{r} 7 \overline{) 400} \\ 7 \overline{) 57} \leftarrow \text{First quotient} \\ 7 \overline{) 8} \leftarrow \text{Second quotient} \\ 1 \leftarrow \text{Third quotient} \end{array}$$

Here the last quotient is 1, which is less than 7.

In this method, highest power of 7 in 400!

$$\begin{aligned} &= \text{Sum of all the quotients} \\ &= \text{First Quotient} + \text{Second Quotient} + \text{Third Quotient} \\ &= 57 + 8 + 1 = 66 \end{aligned}$$

Above rule is valid only for prime numbers not for composite numbers. If we need to find the highest power of composite number in a given factorial, then first convert the composite number as product of primes and then find the highest power of largest prime factor of the given composite number, which is the required highest power of the given composite number.

Illustration 47: Find the number of zeroes present at the end of 100!

Solution: We get a zero at the end of a number when we multiply that number by 10. So, to calculate the number of zeroes at the end of 100!, we have to find highest power of 10 present in the number. Since, 10 (a composite number) = 2×5 and hence 5 is the largest prime factor of 10, therefore we have to find the highest power of 5 in 100!

$$\begin{array}{r} 5 \overline{) 100} \\ 5 \overline{) 20} \\ 4 \end{array}$$

The highest power of 5 in 100!

$$\begin{aligned} &= \text{First Quotient} + \text{Second Quotient} = 20 + 4 = 24 \\ \text{Therefore, the number of zeroes at the end of } 100! &= 24. \end{aligned}$$

BASE SYSTEM

The number system in which we carry out all calculation is decimal (base 10) system. It is called decimal system because there are 10 digits 0 to 9.

There are other number systems also depending on the number of digits contained in the base system. Some of the most common systems are Binary system, Octal system, and Hexadecimal system. A number system containing two digits 0 and 1 is called binary (base 2) system. Number system containing eight digits 0, 1, 2, 3, ..., 7 is called Octal (base 8) system.

Hexadecimal (base 16) system has 16 digits 0, 1, 2, 3, ..., 9, A, B, C, D, E, F; where A has a value 10, B has a value 11 and so on.

Let a number $abcde$ be written in base p , where a, b, c, d and e are single digits less than p . The value of the number $abcde$ in base 10 = $e \times p^0 + d \times p^1 + c \times p^2 + b \times p^3 + a \times p^4$

For example, The number 7368 can be written as

$$8 \times 10^0 + 6 \times 10^1 + 3 \times (10)^2 + 7 \times (10)^3 = 7368 \text{ in decimal (base 10) number system.}$$

The number 7368 in base 9 is written in decimal number system as

$$8 \times 9^0 + 6 \times 9^1 + 3 \times 9^2 + 7 \times 9^3 = 5408$$

There are mainly two types of operations associated with conversion of bases: First conversion from any base to base ten and second conversion from base 10 to any base.

(i) Conversion From Any Base to Base Ten

The number $(pqrst)_{10}$ (i.e., the number $pqrst$ on base a) is converted to base 10 by finding the value of the number.

$$(pqrst)_{10} = u + ta + sa^2 + ra^3 + qa^4 + pa^5.$$

Here subscript ' a ' in $(pqrst)_a$ denotes the base of the number $pqrst$.

Illustration 48: Convert $(21344)_5$ to base 10.

Solution:

$$\begin{aligned} (21344)_5 &= 4 \times 5^0 + 4 \times 5^1 + 3 \times 5^2 + 1 \times 5^3 + 2 \times 5^4 \\ &= 4 + 4 \times 5 + 3 \times 25 + 1 \times 125 + 2 \times 625 = 1474. \end{aligned}$$

(ii) Conversion From Base 10 to Any Base

A number written in base 10 can be converted to any base ' a ' by first dividing the number by ' a ' and then successively dividing the quotients by ' a '. The remainders, written in reverse order, give the equivalent number in base ' a '.

For example the number 238 in base 3 is found as

$$\begin{array}{r} 3 \overline{) 238} \\ 79 \\ 26 \\ 8 \\ 2 \end{array} \begin{array}{l} \leftarrow 1 \\ \leftarrow 1 \\ \leftarrow 2 \\ \leftarrow 2 \end{array} \begin{array}{l} \leftarrow \\ \leftarrow \\ \leftarrow \\ \leftarrow \end{array} \begin{array}{l} \text{Remainders} \end{array}$$

The remainders in reverse order is 22211.

Hence, 22211 is the required number in base 3.

Note: Value of a single digit number to all bases are the same;

For example,

$$5_4 = 5_7 = 5_8 = 5_{10}$$

Addition, Subtraction and Multiplication in the Same Bases

Illustration 49: Add the numbers $(4235)_7$ and $(2354)_7$.

Solution: The numbers are written as

$$\begin{array}{r} 4 \ 2 \ 3 \ 5 \\ 2 \ 3 \ 5 \ 4 \end{array}$$

The addition of 5 and 4 (first digit from right of both numbers) is 9 which being more than 7 would be written as $9 = 7 \times 1 + 2$. Here 1 is the quotient and 2 is the remainder when 9 is divided by 7. The remainder 2 is placed at the first place from right of the answer and the quotient 1 gets carried over to the second place from the right.

At the second place from the right $3 + 5 + 1$ (carry) = $9 = 7 \times 1 + 2$

$$\begin{array}{r} +1 \ +1 \\ 4 \ 2 \ 3 \ 5 \\ 2 \ 3 \ 5 \ 4 \\ \hline 6 \ 6 \ 2 \ 2 \end{array}$$

The remainder 2 is placed at the second place from right of the answer and the quotient 1 carry over to the third place from right.

In the same way, we can find the other digits of the answer.

Illustration 50: $(52)_7 + 46_8 = (?)_{10}$

- (a) $(75)_{10}$ (b) $(50)_{10}$
(c) $(39)_{39}$ (d) $(28)_{10}$

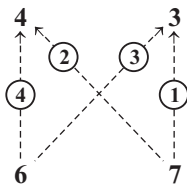
Solution: (a) $(52)_7 = (5 \times 7^1 + 2 \times 7^0)_{10} = (37)_{10}$
Also, $(46)_8 = (4 \times 8^1 + 6 \times 8^0)_{10} = (38)_{10}$
Sum $= (37)_{10} + (38)_{10} = (75)_{10}$

Illustration 51: $(11)_2 + (22)_3 + (33)_4 + (44)_5 + (55)_6 + (66)_7 + (77)_8 + (88)_9 = (?)_{10}$

- (a) 396 (b) 276
(c) 250 (d) 342

Solution: (b) $(11)_2 = (1 \times 2^1 + 1 \times 2^0)_{10} = (3)_{10}$
 $(22)_3 = (2 \times 3^1 + 2 \times 3^0)_{10} = (8)_{10}$
 $(33)_4 = (3 \times 4^1 + 3 \times 4^0)_{10} = (15)_{10}$
 $(44)_5 = (4 \times 5^1 + 4 \times 5^0)_{10} = (24)_{10}$
 $(55)_6 = (5 \times 6^1 + 5 \times 6^0)_{10} = (35)_{10}$
 $(66)_7 = (6 \times 7^1 + 6 \times 7^0)_{10} = (48)_{10}$
 $(77)_8 = (7 \times 8^1 + 7 \times 8^0)_{10} = (63)_{10}$
 $(88)_9 = (8 \times 9^1 + 8 \times 9^0)_{10} = (80)_{10}$
Sum $= (3)_{10} + (8)_{10} + (15)_{10} + (24)_{10}$
 $+ (35)_{10} + (48)_{10} + (63)_{10} + (80)_{10}$
 $= (276)_{10}$

Illustration 52: Multiply $(43)_8 \times (67)_8$



Solution:

$7 \times 3 = 21 = 8 \times 2 + 5 \Rightarrow$ we write 5 and carry 2 forward
 $7 \times 4 + 2 \text{ (carry)} = 30 = 8 \times 3 + 6 \Rightarrow$ we write 6 and carry 3 forward
 $6 \times 3 = 18 = 8 \times 2 + 2 \Rightarrow$ we write 2 and carry 2 forward
 $6 \times 4 + 2 \text{ (carry)} = 26 = 8 \times 3 + 2 \Rightarrow$ we write 2 and carry 3 forward. Thus

$$\begin{array}{r} (4 \ 3)_8 \\ \times (6 \ 7)_8 \\ \hline 3 \ 6 \ 5 \\ 3 \ 2 \ 2 \\ \hline 3 \ 6 \ 0 \ 5 \end{array}$$

Illustration 53: Subtract $(247)_8$ from $(345)_8$.

Solution:

- (i) 5 is less than 7. So borrow 1 from the previous digit 4. Since, we are working in octal system, so 5 become $5 + 8 = 13$. Subtract 7 from 13, you will get 6.

$$\begin{array}{r} 3 \ 4 \ 5 \\ -2 \ 4 \ 7 \\ \hline 6 \end{array}$$

- (ii) Since, we have borrowed 1, the 4 in the first row has now become 3, which is less than the digit (4), just below it in the second row, So borrow 1 from 3 of first row. So, the 4

in first row is now becomes $3 + 8 = 11$. Subtracting 4 of second row from 11, we get 7. Hence,

$$\begin{array}{r} 3 \ 4 \ 5 \\ -2 \ 4 \ 7 \\ \hline 0 \ 7 \ 6 \end{array}$$

SUCCESSIVE DIVISION

If the quotient of a division is taken as the dividend in the next division, such a division is called successive division. A successive division process can continue upto any number of step- until the quotient in a division becomes zero for the first time i.e., the quotient in the first division is taken as dividend in the second division, the quotient in the second division is taken as the dividend in the third division, the quotient in the third division is taken as the dividend in the fourth division and so on.

If we say that 3305 is divided successively by 4, 5, 7 and 2; then the quotients and remainder are as follows in the successive division:

Dividend	Divisor	Quotient	Remainder
3305	4	826	1
826	5	165	1
165	7	23	4
23	2	11	1

Here we see that when 3305 is successively divided by 4, 5, 7 and 2 then the respective remainders are 1, 1, 4, and 1.

Illustration 54: A number when divided successively by 13 and 3 gives respective remainders of 5 and 1. What will be the remainder when the largest such two-digit number is divided by 19.

Solution: We write down the divisors one after the other and their respective remainders below them.

Divisors : 13 3
Remainders : 5 1

We know that in successive division first quotient is taken as second dividend and second quotient is taken as third dividend.

Here we are given second divisor = 3

Second remainder = 1

Assume second quotient = k

So as per division equation,

Second dividend = Divisor $\times$ Quotient + Remainder

$$= 3 \times k + 1 = 3k + 1$$

Now this second dividend is quotient of the first division.

Actual No. (N) = First divisor $\times$ First quotient (or second dividend) + First remainder

$$= 13 \times (3k + 1) + 5 = 39k + 18$$

So number (N) = $39k + 18$.

Now largest such 2 digit number is obtained when $k = 2$. Now when $k = 2$, then $N = 39 \times 2 + 18 = 96$

Now when 96 is divided by 19, the remainder is 1.

Illustration 55: A number when successively divided by 9, 5 and 4 leaves remainders 2, 1 and 3 respectively. What will be the remainders when the number is divided successively by 7, 3 and 5?

Solution: Here we will find the smallest number which will satisfy the given conditions.

Therefore, $12x \times 12y = 31104 \rightarrow xy = 216$.

Now we need to find co-prime pairs whose product is 216.

$216 = 2^3 \times 3^3$. Therefore, the co-prime pairs will be (1, 216) and (8, 27). Therefore, (12, 12×216) and (8×12 , 27×12) are two possible numbers.

Illustration 62: Find the HCF of $3^{100} - 1$ and $3^{120} - 1$.

Solution: $3^{100} - 1 = (3^{20})^5 - 1^5 \Rightarrow$ divisible by $3^{20} - 1$ (Since $a^n - b^n$ is always divisible by $a - b$).

Similarly, $3^{120} - 1 = (3^{20})^6 - 1^6 \Rightarrow$ divisible $3^{20} - 1$.

$\therefore$ HCF = $3^{20} - 1$.

(ii) Division Method

To find the HCF of two numbers by division method, we divide the larger number by the smaller number. Then we divide the smaller number by the first remainder, then first remainder by the second remainder.. and so on, till the remainder becomes 0. The last divisor is the required HCF.

Illustration 63: Find the HCF of 288 and 1080 by the division method.

Solution:

$$\begin{array}{r} 288 \overline{) 1080} \quad | 3 \\ \underline{864} \\ 216 \overline{) 288} \quad | 1 \\ \underline{216} \\ 72 \overline{) 216} \quad | 3 \\ \underline{216} \\ 0 \end{array}$$

The last divisor 72 is the HCF of 288 and 1080.

Shortcut for Finding HCF or GCD

To find the HCF of any number of given numbers, first find the difference between two nearest given numbers. Then find all factors (or divisors) of this difference. Highest factor which divides all the given numbers is the HCF.

Illustration 64: Find the HCF of 12, 20 and 32.

Solution: Difference of nearest two numbers 12 and 20 = $20 - 12 = 8$

All factors (or divisor) of 8 are 1, 2, 4 and 8.

1, 2 and 4 divides each of the three given numbers 12, 20 and 32. Out of 1, 2 and 4; 4 is the highest number. Hence, HCF = 4.

LEAST COMMON MULTIPLE (LCM)

The least common multiple (LCM) of two or more numbers is the lowest number which is divisible by all the given numbers.

Consider two numbers 12 and 15.

Multiples of 12 are 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132,...

While the multiples of 15 are 15, 30, 45, 60, 75, 90, 105, 120, 135, 150,.....

Out of these series of multiples, we have some common multiples like 60, 120, 180, ..., etc. Out of these common multiples, 60 is the lowest, so 60 is called the Lowest Common Multiple (LCM) of 12 and 15.

Methods to Find The LCM

There are two methods to find the LCM.

(i) Prime Factorization Method

After performing the prime factorization of all the given numbers, we find the highest index of all the prime numbers among the given numbers. The LCM is the product of all these prime numbers with their respective highest indices because LCM must be divisible by all of the given numbers.

Illustration 65: Find the LCM of 72, 288 and 1080.

Solution: $72 = 2^3 \times 3^2$

$288 = 2^5 \times 3^2$

$1080 = 2^3 \times 3^3 \times 5$

Hence, LCM = $2^5 \times 3^3 \times 5^1 = 4320$

(ii) Division Method

To find the LCM of 5, 72, 196 and 240, we use the division method in the following way:

Check whether any prime number that divides at least two of all the given numbers. If there is no such prime number, then the product of all these numbers is the required LCM, otherwise find the smallest prime number that divides at least two of the given numbers. Here, we see that smallest prime number that divides at least two given numbers is 2.

Divide those numbers out of the given numbers by 2 which are divisible by 2 and write the quotient below it. The given number(s) that are not divisible by 2 write as it is below it and repeat this step till you do not find at least two numbers that are not divisible by any prime number.

2	5, 72, 196, 240
2	5, 36, 98, 120
2	5, 18, 49, 60
3	5, 9, 49, 30
5	5, 3, 49, 10
	1, 3, 49, 2

After that find the product of all divisors and the quotient left at the end of the division. This product is the required LCM.

Hence, LCM of the given numbers = product of all divisors and the quotient left at the end.

$$= 2 \times 2 \times 2 \times 3 \times 5 \times 3 \times 49 \times 2 = 35280$$

Illustration 66: On a traffic signal, traffic light changes its colour after every 24, 30 and 36 seconds in green, red and orange light. How many times in an hour only green and red light will change simultaneously.

Solution: LCM. of 24 and 30 = 120

So in 1 hr both green and red light will change simultaneously $3600/120$ times = 30 times

LCM of 24, 30 and 36 is 360

Hence in 1 hr all three lights will change simultaneously 3600/360 times = 10 times

So in 1 hr only red and green lights will change $30 - 10 = 20$ times simultaneously.

Shortcut For Finding LCM

Using idea of co-prime, you can find the LCM by the following shortcut method:

LCM of 9, 10, 15 and 36 can be written directly as $9 \times 10 \times 2$.

The logical thinking that behind it is as follows:

Step 1: If you can see a set of 2 or more co-prime numbers in the set of numbers of which you are finding the LCM, write them down by multiply them.

In the above situation, since we see that 9 and 10 are co-prime to each other, we start off writing the LCM by writing 9×10 as the first step.

Step 2: For each of the other numbers, consider what prime factor(s) of it is/are not present in the LCM (if factorised into primes) taken in step 1. In case you see some prime factors of each of the other given numbers separately are not present in the LCM (if factorised into primes) taken in step 1, such prime factors will be multiplied in the LCM taken in step 1.

Prime factorisation of $9 \times 10 = 3 \times 3 \times 2 \times 5$

Prime factorisation of $15 = 3 \times 5$

Prime factorisation of $36 = 2 \times 2 \times 3 \times 3$

Here we see that both prime factors of 15 are present in the prime factorisation of 9×10 but one prime factor 2 of 36 is not present in the LCM taken in step 1. So to find the LCM of 9, 10, 15 and 36; we multiply the LCM taken in step 1 by 2.

Thus required LCM = $9 \times 10 \times 2 = 180$

Rule For Finding HCF and LCM of Fractions

(I) HCF of two or more fractions

$$= \frac{\text{HCF of numerator of all fractions}}{\text{LCM of denominator of all fractions}}$$

(II) LCM of two or more fractions

$$= \frac{\text{LCM of numerator of all fractions}}{\text{HCF of denominator of all fractions}}$$

Illustration 67: Find the HCF and LCM of $\frac{4}{5}, \frac{6}{11}, \frac{3}{5}$.

Solution: $\text{HCF} = \frac{\text{HCF of } 4, 6, 3}{\text{LCM of } 5, 11, 5} = \frac{1}{55}$

$$\text{LCM} = \frac{\text{LCM of } 4, 6, 3}{\text{HCF of } 5, 11, 5} = \frac{12}{1} = 12$$

For any two numbers, HCF $\times$ LCM = product of the two numbers

This formula is applicable only for two numbers.

For example, HCF of 288 and 1080 is 72 and LCM of these two numbers is 4320.

We can see that $72 \times 4320 = 311040 = 288 \times 1080$.

Properties of HCF and LCM

- The HCF of two or more numbers is smaller than or equal to the smallest of those numbers.
- The LCM of two or more numbers is greater than or equal to the largest of those numbers.
- If numbers $N_1, N_2, N_3, N_4, \dots$, etc. give an equal remainder when divided by the same number P , then P is a factor of $(N_1 - N_2), (N_2 - N_3), (N_3 - N_4), \dots$

As $N_1 = P \times Q_1 + R, N_2 = P \times Q_2 + R,$

$$N_3 = P \times Q_3 + R$$

Now $N_1 - N_2 = P(Q_1 - Q_2)$

$$N_2 - N_3 = P(Q_2 - Q_3)$$

$$N_3 - N_4 = P(Q_3 - Q_4)$$

$$\Rightarrow P \text{ is the common factor of } N_1 - N_2, N_2 - N_3, N_3 - N_4.$$

- If L is the LCM of N_1, N_2, N_3, N_4 ; then all the multiples of L are divisible by these numbers.
- If a number P always leaves a remainder R when divided by the numbers $N_1, N_2, N_3, N_4, \dots$, etc., then $P = \text{LCM}$ (or a multiple of LCM) of $N_1, N_2, N_3, N_4, \dots + R$.
- If HCF of x and y is G , then HCF of
 - (i) x and $(x + y)$ is also G .
 - (ii) x and $(x - y)$ is also G .
 - (iii) $(x + y)$ and $(x - y)$ is also G .
- HCF is always a factor of LCM.
- For two numbers A and B if HCF is h , then we can assume $A = hx, B = hy$ and LCM of A and B is given by ' hxy '. Here x and y are co-prime.

Illustration 68: Find the highest four-digit number that is divisible by each of the numbers 24, 36, 45 and 60.

Solution: $24 = 2^3 \times 3, 36 = 2^2 \times 3^2, 45 = 3^2 \times 5$ and $60 = 2^2 \times 3 \times 5$. Hence, the LCM of 24, 36, 45 and $60 = 2^3 \times 3^2 \times 5 = 360$. Now all the multiples of the LCM 360 will be divisible by each of 24, 36, 45 and 60. The highest four-digit number is 9999, which when divided by 360 gives the remainder 279. Hence the highest four-digit number divisible by 24, 36, 45 and $60 = 9999 - 279 = 9720$.

Illustration 69: Find the highest number less than 1900 that is divisible by each of the numbers 2, 3, 4, 5, 6 and 7.

Solution: The LCM of 2, 3, 4, 5, 6 and 7 is 420. Hence 420 and every multiple of 420, is divisible by each of these numbers. Hence, the number 420, 840, 1260 and 1680 are all divisible by each of these numbers. We can see that 1680 is the highest number less than 1900 which is a multiple of 420. Hence, the highest number less than 1900 divisible by each one of 2, 3, 4, 5, 6 and 7 is 1680.

Illustration 70: Find the lowest number which gives a remainder 3 when divided by any of the numbers 6, 7 and 8.

Solution: The LCM of 6, 7 and 8 is 168. Hence 168 is divisible by 6, 7 and 8. Therefore, $168 + 3 = 171$ is the lowest

number, which will give a remainder 3, when divided by these numbers.

Illustration 71: What is the smallest number which when divided by 6, 18, 24 leaves a remainder of 2, 14 and 20 respectively ?

Solution: The common difference between the divisor and the remainder is 4 ($6 - 2 = 4$, $18 - 14 = 4$, $24 - 20 = 4$). Now the LCM of 6, 18 and 24 is 72. Now $72 - 4 = 72 - 6 + 2 = 72 - 18 + 14 = 72 - 24 + 20$. Therefore, if we subtract 4 from 72, the resulting number will give remainders of 2, 14 and 20 when divided by 6, 18 and 24.

Hence, the required number = $72 - 4 = 68$.

Illustration 72: A number when divided by 3, 4, 5 and 6 always leaves a remainder of 2, but leaves no remainder when divided by 7. What is the lowest such number possible ?

Solution: The LCM of 3, 4, 5 and 6 is 60. Therefore, the number is of the form $60k + 2$, i.e., 62, 122, 182, 242, etc. We can see that 62 and 122 are not divisible by 7 but 182 is divisible by 7. Therefore, the lowest such number possible = 182.

Illustration 73: How many pairs of number exist such that their HCF is 4 and LCM is 48 ?

Solution: Since, HCF of two numbers is 4. Hence we can assume two numbers as $4a$ and $4b$. Here a and b are co-prime to each other.

We know for two numbers:

H.C.F. $\times$ L.C.M. = Product of two numbers

Hence $4 \times 48 = 4a \times 4b \Rightarrow ab = 12 = 3 \times 2^2$

Now the number of ways, we can express 12 as a product of two of its co-prime factors = $2^{2-1} = 2^1 = 2$. So total 2 pairs of number that satisfy the above condition.

Illustration 74: How many pairs of two digit numbers exist such that their HCF is 8 and LCM is 120 ?

Solution: Since the HCF of two numbers is 8, hence the two numbers will be $8a$ and $8b$, where a and b are co-prime. Therefore $8a \times 8b = 8 \times 120$

or $ab = 15$

So pair of a and b is 3 and 5 that will give numbers as 24 and 40.

If we take a and b as 1 and 15 then we will get the numbers as 8 and 120 but here one of them is not a two digit number. Hence only one such pair exists.

GREATEST INTEGRAL VALUE

If x be a real number, then $[x]$ indicates greatest integer equal or less than x .

If the given number is an integer, then the greatest integer gives the number itself, otherwise it gives the first integer towards the left of the number x on the number line.

For example $[4] = 4$, $[3.4] = 3$, $[6.8] = 6$, $[-2.3] = -3$, $[-5.6] = -6$ and so on.

Note that -3 is less than -2.3 and -6 is less than -5.6 , etc.

Illustration 75: What is the value of

$$[\sqrt{1}] + [\sqrt{2}] + [\sqrt{3}] + \dots + [\sqrt{49}] + [\sqrt{50}]$$

where $[x]$ denotes greatest integer function?

Solution: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$, $6^2 = 36$, $7^2 = 49$, $8^2 = 64$

Therefore, from $[\sqrt{1}]$ to $[\sqrt{3}]$, the value will be 1, from $[\sqrt{4}]$ to $[\sqrt{8}]$ the value will be 2, from $[\sqrt{9}]$ to $[\sqrt{15}]$ the value will be 3 and so on.

Therefore, the total value

$$= 1 \times 3 + 2 \times 5 + 3 \times 7 + 4 \times 9 + 5 \times 11 + 6 \times 13 + 7 \times 2 \\ = 3 + 10 + 21 + 36 + 55 + 78 + 14 = 217.$$

Illustration 76: What is the value of x for which $x[x] = 32$?

Solution: If the value of x is 5, $x[x] = 25$, and if the value of x is 6, then $x[x] = 36$

Therefore, the value of x lies between 5 and 6.

If x lies between 5 and 6, then $[x] = 5$.

$$\Rightarrow x = \frac{28}{[x]} = \frac{32}{5} \quad 6.4.$$

Practice Exercises



Foundation Level

- The greatest number which will divide 116, 221, 356 leaving the same remainder in each case is
(a) 15 (b) 5
(c) 10 (d) 20
- What number has to be added to 345670 in order to make it divisible by 6?
(a) 2 (b) 4
(c) 5 (d) 6
- The least number which when divided by 35 leaves a remainder 25, when divided by 45 leaves the remainder 35 and when divided by 55 leaves 45 is
(a) 3465 (b) 3645
(c) 3655 (d) 3455
- If n is any even number, then $n(n^2 + 20)$ is always divisible by
(a) 15 (b) 20
(c) 24 (d) 32
- When 2^{256} is divided by 17 the remainder would be
(a) 1 (b) 16
(c) 14 (d) None of these
- The last digit of 2137^{753} is
(a) 9 (b) 7
(c) 3 (d) 1
- Find the least square number which is divisible by 3, 5, 6, and 9.
(a) 900 (b) 90
(c) 8100 (d) 81
- In order that the number $1y3y6$ be divisible by 11, the digit y should be
(a) 1 (b) 2
(c) 5 (d) 6
- If n is an even natural number, then the largest natural number by which $n(n+1)(n+2)$ is divisible is
(a) 6 (b) 8
(c) 12 (d) 24
- Which number should be added to 459045 to make it exactly divisible by 27?
(a) 3 (b) 9
(c) 0 (d) None of these
- Find the last digit of the sum $19^{81} + 4^{9k}$, $K \in N$.
(a) 4 (b) 9
(c) 3 (d) Cannot be determined
- The sum of prime numbers that are greater than 60, but less than 70 is
(a) 128 (b) 191
(c) 197 (d) 260
- The number 311311311311311311311 is
(a) divisible by 3 but not by 11
(b) divisible by 11 but not by 3
(c) divisible by both 3 and 11
(d) neither divisible by 3 nor by 11
- A difference between two numbers is 1365, when larger number is divided by the smaller one, the quotient is 6 and the remainder is 15. What is the smaller number?
(a) 240 (b) 360
(c) 270 (d) 295
- If the number $517 * 324$ is completely divisible by 3, then the smallest whole number in place of $*$ will be:
(a) 0 (b) 1
(c) 2 (d) None of these
- If the product $4864 \times 9P2$ is divisible by 12, the value of P is
(a) 2 (b) 5
(c) 6 (d) None of these
- The largest 4-digit number exactly divisible by 88 is
(a) 9944 (b) 9768
(c) 9988 (d) 8888
- $(x^n - a^n)$ is completely divisible by $(x + a)$, when
(a) n is any natural number
(b) n is an even natural number
(c) n is an odd natural number
(d) n is prime
- When $0.\overline{47}$ is converted into a fraction the result is
(a) $\frac{46}{90}$ (b) $\frac{46}{99}$
(c) $\frac{47}{90}$ (d) $\frac{47}{99}$
- Which of the following statements are true:
(i) The rational number $\frac{29}{23}$ lies to the left of zero on the number line.
(ii) The rational number $\frac{-12}{-17}$ lies to the right of zero on the number line.

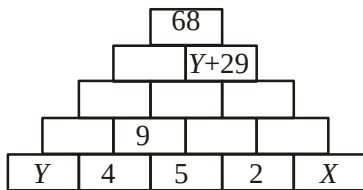
- (iii) The rational numbers $\frac{-12}{5}$ and $\frac{-7}{17}$ are on the opposite side of zero on the number line.
- (v) The rational numbers $\frac{-21}{5}$ and $\frac{7}{-31}$ are on the opposite side of zero on the number line.
- (a) Only (i) (b) (i) & (ii)
(c) Only (iii) (d) (i), (ii) & (iv)
21. I have a certain number of beads which lie between 600 and 900. If 2 beads are taken away the remainder can be equally divided among 3, 4, 5, 6, 7 or 12 boys. The number of beads I have
- (a) 729 (b) 842
(c) 576 (d) 961
22. Find the digit at the unit's place of $(377)^{59} \times (793)^{87} \times (578)^{129} \times (99)^{99}$
- (a) 1 (b) 2
(c) 7 (d) 9
23. Four different electronic devices make a beep after every 30 minutes, 1 hour, $1\frac{1}{2}$ hour and 1 hour 45 minutes respectively. All the devices beeped together at 12 noon. They will again beep together at:
- (a) 12 midnight (b) 3 a.m.
(c) 6 a.m. (d) 9 a.m.
24. If N is the sum of first 13,986 prime numbers, then N is always divisible by
- (a) 6 (b) 4
(c) 8 (d) None of these
25. If two numbers when divided by a certain divisor give remainder 35 and 30 respectively and when their sum is divided by the same divisor, the remainder is 20, then the divisor is
- (a) 40 (b) 45
(c) 50 (d) 55
26. Find the least number which when divided by 12, leaves a remainder 7, when divided by 15, leaves a remainder 10 and when divided by 16, leaves a remainder 11
- (a) 115 (b) 235
(c) 247 (d) 475
27. How many even integers n , where $100 \leq n \leq 200$, are divisible neither by seven nor by nine?
- (a) 40 (b) 37
(c) 39 (d) 38
28. A number is *interesting* if on adding the sum of the digits of the number and the product of the digits of the number, the result is equal to the number. What fraction of numbers between 10 and 100 (both 10 and 100 included) is *interesting*?
- (a) 0.1 (b) 0.11
(c) 0.16 (d) 0.22
29. In a cricket match, Team A scored 232 runs without losing a wicket. The score consisted of byes, wides and runs scored by two opening batsmen : Ram and Shyam. The runs scored by the two batsman are 26 times wides. There are 8 more byes than wides. If the ratio of the runs scored by Ram and Shyam is 6 : 7, then the runs scored by Ram is
- (a) 88 (b) 96
(c) 102 (d) 112
30. If $x + y + z = 1$ and x, y, z are positive real numbers, then the least value of $\left(\frac{1}{x}-1\right)\left(\frac{1}{y}-1\right)\left(\frac{1}{z}-1\right)$ is
- (a) 4 (b) 8
(c) 16 (d) None of these
31. The last digit of $3^{3^{4n}} + 1$, is
- (a) 0 (b) 4
(c) 8 (d) 2
32. The last digit in $(25_)^{32}$ and $(25_)^{33}$ both is 6. The missing digit is :
- (a) 4 (b) 8
(c) 6 (d) 5
33. Which digits should come in place of * and \$ if the number 62684*\$ is divisible by both 8 and 5?
- (a) 4, 0 (b) 0, 4
(c) 0, 0 (d) 4, 4
34. At a college football game, $\frac{4}{5}$ of the seats in the lower deck of the stadium were sold. If $\frac{1}{4}$ of all the seating in the stadium is located in the lower deck, and if $\frac{2}{3}$ of all the seats in the stadium were sold, then what fraction of the unsold seats in the stadium was in the lower deck?
- (a) $\frac{3}{20}$ (b) $\frac{1}{6}$
(c) $\frac{1}{5}$ (d) $\frac{1}{3}$
35. The integers 1, 2, ..., 40 are written on a blackboard. The following operation is then repeated 39 times; In each repetition, any two numbers, say a and b , currently on the blackboard are erased and a new number $a + b - 1$ is written. What will be the number left on the board at the end?
- (a) 820 (b) 821
(c) 781 (d) 819
36. If $653xy$ is divisible by 80 then the value of $x + y$ is
- (a) 2 (b) 3
(c) 4 (d) 6
37. How many numbers are there between 200 and 800 which are divisible by both 5 and 7?
- (a) 35 (b) 16
(c) 17 (d) can't be determined
38. How many numbers are there in the set $S = \{200, 201, 202, \dots, 800\}$ which are divisible by neither of 5 or 7?
- (a) 411 (b) 412
(c) 410 (d) None of these
39. When a number divided by 9235, we get the quotient 888 and the remainder 222, such a least possible number is
- (a) 820090 (b) 8200920
(c) 8200680 (d) None of these

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Standard Level

- What is the remainder obtained on dividing $34^{43} + 43^{34}$ by 7 ?
(a) 4 (b) 3
(c) 1 (d) 0
- Two different prime numbers X and Y , both are greater than 2, then which of the following must be true ?
(a) $X - Y = 23$ (b) $X + Y \neq 87$
(c) Both (a) and (b) (d) None of these
- What is the remainder when $1! + 2! + 3! + \dots + 100!$ is divided by 7 ?
(a) 0 (b) 5
(c) 6 (d) 3
- On dividing 2272 as well as 875 by 3-digit number N , we get the same remainder. The sum of the digits of N is:
(a) 10 (b) 11
(c) 12 (d) 13
- Which one of the following numbers will completely divide $(3^{25} + 3^{26} + 3^{27} + 3^{28})$?
(a) 11 (b) 16
(c) 25 (d) 30
- There are two integers 34041 and 32506, when divided by a three-digit integer n , leave the same remainder. What is the value of n ?
(a) 298 (b) 307
(c) 461 (d) can't be determined
- After distributing the sweets equally among 25 children, 8 sweets remain. Had the number of children been 28, 22 sweets would have been left after equally distributing. What was the total number of sweets ?
(a) 328 (b) 348
(c) 358 (d) Data inadequate
- Find the remainder when 7^{99} is divided by 2400.
(a) 1 (b) 343
(c) 49 (d) 7
- A number N when factorized can be written as $N = p_1^4 \times p_2^3 \times p_3^7$. Find the number of perfect squares which are factors of N . (The 3 prime numbers $p_1, p_2, p_3 > 2$)
(a) 12 (b) 24
(c) 36 (d) 6
- The number $\log_2 7$ is
(a) An integer (b) A rational number
(c) An irrational number (d) A prime number
- Which of the following is true ?
(a) The cube of an odd integer is of the form $8q + 1$, where q is an integer.
(b) the square of an odd integer is of the form $8q + 1$, where q is an integer.
(c) The fourth power of any integer is of the form $10q + 1$, where q is an integer
(d) None of these
- $94^3 - 23^3 - 71^3$ is atleast divisible by
(a) 71 and 23 (b) 23 and 74
(c) 71 and 94 (d) 23, 71 and 94
- Find the smallest natural number n that satisfies the following statement : $981^n, 982^n, 983^n, 984^n, 985^n, 986^n$ leave the same remainder when divided. If $n > 0$, then $n =$?
(a) 1 (b) 3
(c) 5 (d) 6
- How many whole numbers between 100 and 800 contain the digit 2?
(a) 200 (b) 214
(c) 220 (d) 240
- p, q and r are three non-negative integers such that $p + q + r = 10$. The maximum value of $pq + qr + pr + pqr$ is
(a) ≥ 40 and < 50 (b) ≥ 50 and < 60
(c) ≥ 60 and < 70 (d) ≥ 70 and < 80
- Let a, b, c, d and e be integers such that $a = 6b = 12c$, and $2b = 9d = 12e$. Then which of the following pairs contains a number that is not an integer?
(a) $\left(\frac{a}{27}, \frac{b}{e}\right)$ (b) $\left(\frac{a}{36}, \frac{c}{e}\right)$
(c) $\left(\frac{a}{12}, \frac{bd}{18}\right)$ (d) $\left(\frac{a}{6}, \frac{c}{d}\right)$
- If $x = (16^3 + 17^3 + 18^3 + 19^3)$, then x divided by 70 leaves a remainder of
(a) 0 (b) 1
(c) 69 (d) 35
- Find the total number of prime factors in $2^{17} \times 6^{31} \times 7^5 \times 10^{11} \times 11^{10} \times (323)^{23}$
(a) 162 (b) 161
(c) 346 (d) 97
- The digits of a three-digit number A are written in the reverse order to form another three-digit number B . If $B > A$ and $B - A$ is perfectly divisible by 7, then which of the following is necessarily true?
(a) $100 < A < 299$ (b) $106 < A < 305$
(c) $112 < A < 311$ (d) $118 < A < 317$
- If $N = 1! - 2! + 3! - 4! + \dots + 47! - 48! + 49!$, then what is the unit digit of N^N ?
(a) 0 (b) 9
(c) 7 (d) 1
- The digits of a 3-digit number in Base 4 get reversed when it is converted into Base 3. How many such numbers exist?
(a) 0 (b) 1
(c) 2 (d) 3
- Find the remainder when $73 \times 75 \times 78 \times 57 \times 197$ is divided by 34.
(a) 22 (b) 30
(c) 15 (d) 28
- What is the ten's place digit of 12^{42} ?
(a) 2 (b) 4
(c) 6 (d) 8
- Find the HCF of $(3^{125} - 1)$ and $(3^{35} - 1)$.
(a) 5 (b) 3
(c) $(3^5 - 1)$ (d) $(3^{35} - 1)$

25. A computer program was tested 300 times before its release. The testing was done in three stages of 100 tests each. The software failed 15 times in Stage I, 12 times in Stage II, 8 times in Stage III, 6 times in both Stage I and Stage II, 7 times in both Stage II and Stage III, 4 times in both Stage I and Stage III, and 4 times in all the three stages. How many times the software failed in a single stage only?
- (a) 10 (b) 13
(c) 15 (d) 17
26. In the figure, number in any cell is obtained by adding two numbers in the cells directly below it. For example, 9 in the second row is obtained by adding the two numbers 4 and 5 directly below it. The value of $X - Y$ is



- (a) 2 (b) 4
(c) 3 (d) 5
27. A natural number when increased by 50% has its number of factors unchanged. However, when the value of the number is reduced by 75%, the number of factors is reduced by 66.66%. One such number could be:
- (a) 32 (b) 84
(c) 126 (d) None of these
28. Let x denote the greatest 4-digit number which when divided by 6, 7, 8, 9 and 10 leaves a remainder of 4, 5, 6, 7 and 8 respectively. Then, the sum of the four-digits of x is 8 respectively. Then, the sum of the four-digits of x is
- (a) 25 (b) 18
(c) 20 (d) 22
29. $(x-1)(x-2)(x-3) = 6y$
How many integer solutions exist for the given equation?
- (a) 0 (b) 1
(c) 2 (d) More than 2
30. A is the set of the first 100 natural numbers. What is the minimum number of elements that should be picked from A to ensure that atleast one pair of numbers whose difference is 10 is picked?
- (a) 51 (b) 55
(c) 20 (d) 11
31. The power of 45 that will exactly divide $123!$ is
- (a) 28 (b) 30
(c) 31 (d) 59
32. What is the remainder when 32^{32} is divided by 7?
- (a) 2 (b) 3
(c) 4 (d) 6
33. Two different two-digit natural numbers are written beside each other such that the larger number is written on the left. When the absolute difference of the two numbers is subtracted from the four-digit number so formed, the number obtained is 5481. What is the sum of the two two-digit numbers?
- (a) 70 (b) 71
(c) 72 (d) 73
34. In a three-digit number, the unit digit is twice the tens digit and the tens digit is twice the hundreds digit. The same

number is written as $1XY$ and $1YX$ in base 8 and base 9 respectively. Find the sum of X and Y in the decimal system.

(a) 15 (b) 7
(c) 11 (d) Cannot be determined

$$35. a + \frac{1}{b + \frac{1}{c + \frac{1}{d + \dots}}}$$

- If a, b, c, d etc. are positive integers, then what is the value of ' b '?
- (a) 2 (b) 4
(c) 3 (d) 5
36. If m and n are positive integers such that $(m-n)^2 = \frac{4mn}{(m+n-1)}$, then how many pairs (m, n) are possible?
- (a) 4 (b) 10
(c) 16 (d) Infinite
37. $x^2 - 3y^2 = 1376$
How many integer solutions exist for the given equation?
- (a) One (b) Two
(c) Four (d) Zero
38. The number of zeros at the end of the product of $222^{111} \times 35^{53} + (7!)^{61} \times (10!)^{51} + 42^{42} \times 25^{25}$ is
- (a) 42 (b) 53
(c) 1055 (d) None of these
39. The highest power of 17 which can divide exactly the following expression : $(18^2 - 1)(18^4 - 1)(18^6 - 1)(18^8 - 1)(18^{10} - 1) \times \dots (18^{16} - 1)(18^{18} - 1)$ is :
- (a) 1 (b) 17
(c) 9 (d) can't be determined
40. The remainder when $2^2 + 22^2 + 222^2 + 2222^2 + \dots (222 \dots 49 \text{ twos})^2$ is divided by 9 is:
- (a) 2 (b) 5
(c) 6 (d) 7
41. Find the last non-zero digit of $96!$.
- (a) 2 (b) 4
(c) 6 (d) 8
42. When 96 is added to a N^2 , it gives another perfect square. If N is a natural no., how many distinct values of N are possible?
- (a) 3 (b) 4
(c) 5 (d) None of these
43. The numbers 1 to 29 are written side by side as follows 1234567891011..... 28 29
If the number is divided by 9, then what is the remainder?
- (a) 3 (b) 1
(c) 0 (d) None of these
44. The remainder when the number 123456789101112..... 484950 is divided by 16 is
- (a) 3 (b) 4
(c) 5 (d) 6
45. How many zeroes will be there at the end of the expression $(2!)^{2!} + (4!)^{4!} + (8!)^{8!} + (9!)^{9!} + (10!)^{10!} + (11!)^{11!}$?
- (a) $(8!)^{8!} + (9!)^{9!} + (10!)^{10!} + (11!)^{11!}$
(b) $10^{10!}$
(c) $4! + 6! + 8! + 2(10!)$
(d) $(0!)^{0!}$

Expert Level

- Number of prime factors of $(216)^{3/5} \times (2500)^{2/5} \times (300)^{1/5}$ is
(a) 3.5 (b) 4.5
(c) 6 (d) 7
- If p is divided by q , then the maximum possible difference between the minimum possible and maximum possible remainder can be?
(a) $p - q$ (b) $p - 1$
(c) $q - 1$ (d) None of these
- ' x ' stands for a number. If the sum of all the three digits of $(x! - x)$ is divisible by ' x ', what is ' x '?
(a) 2 (b) 6
(c) 4 (d) 12
- A number which when divided by 3, 4, 5, 6, & 7 leaves respectively, the remainder 2, 3, 4, 5 and 6. Such smallest 6 digit number is
(a) 100379 (b) 1000379
(c) 100019 (d) None of these
- Find the remainder when $(2222^{5555} + 5555^{2222})$ is divided by 7.
(a) 1 (b) 2
(c) 3 (d) 0
- The last digit of the expression $4 + 9^2 + 4^3 + 9^4 + 4^5 + 9^6 + \dots + 4^{99} + 9^{100}$ is
(a) 0 (b) 3
(c) 5 (d) None of these
- If a , b , and c are positive integers such that $(a - b + c)(b - c + a)(c - a + b) = 15$, then what is the product of a , b and c ?
(a) 24 (b) 64
(c) 42 (d) Cannot be determined
- Find the maximum value of n such that $570 \times 60 \times 30 \times 90 \times 100 \times 500 \times 700 \times 343 \times 720 \times 81$ is perfectly divisible by 30^n .
(a) 12 (b) 11
(c) 14 (d) 13
- What will be the value of x for $\frac{(100^{17} - 1) + (10^{34} + x)}{9}$; the remainder = 0
(a) 3 (b) 6
(c) 9 (d) 8

Directions (Qs. 10-11) : Consider the information given below:

In the diagram below, the seven letters correspond to seven unique digits chosen from 0 to 9. The relationship among the digits is such that:

$$P.Q.R = X.Y.Z = Q.A.Y$$

P		X
Q	A	Y
R		Z

- The value of A is
(a) 0 (b) 2
(c) 3 (d) 6
- The sum of digits which are not used is:
(a) 8 (b) 10
(c) 14 (d) 12
- Find the number of zeros in the product:
 $1^1 \times 2^2 \times 3^3 \times 4^4 \times \dots \times 98^{98} \times 99^{99} \times 100^{100}$
(a) 1200 (b) 1300
(c) 1050 (d) 1225
- Let X be a four-digit number with exactly three consecutive digits being same and is a multiple of 9. How many such X 's are possible?
(a) 12 (b) 16
(c) 19 (d) 20
- If N is a natural number less than 100, then for how many values of N are the numbers $6N + 1$ and $15N + 2$ relatively prime?
(a) 16 (b) 10
(c) 33 (d) All of these
- What is the remainder when $2(8!) - 21(6!)$ divides $14(7!) + 14(13!)$?
(a) 1 (b) 7!
(c) 8! (d) 9!
- For how many natural number values of N , $N^4 + 4$ will be a prime number?
(a) 0 (b) 1
(c) 2 (d) None of these
- $N = 7777 \dots 7777$, where the digit 7 repeats itself 429 times. What is the remainder left when N is divided by 1144?
(a) 913 (b) 1129
(c) 777 (d) None of these
- If $x \Delta (y + 1) = y \Delta (x + 1)$, $x \Delta x = 1$ and $(x - y) \Delta (x + y) = (x - y) \Delta (x + y) = x \Delta y$, then what is the value of $1001 \Delta 1$?
(a) 1000 (b) 100
(c) 10 (d) 1
- The question given below is followed by two statements, A and B . Mark the answer using the following instructions: Mark (a) if the question can be answered by using one of the statements alone, but cannot be answered by using the other statement alone.

Mark (b) if the question can be answered by using either statement alone.

Mark (c) if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

Mark (d) if the question cannot be answered even by using both the statements together.

- Q. N is a natural number that has exactly 24 factors. What is the number of factors of N^3 ?
- A. When N is multiplied by 3, the resultant number has 32 factors.
- B. When N is multiplied by 5, the resultant number has 30 factors.
20. What is the remainder obtained when the sum of the squares of any thirty consecutive natural numbers is divided by 12?
- (a) 0 (b) 3
(c) 11 (d) Cannot be determined
21. The sum of the digits of a four-digit number is 31. What fraction of such numbers are divisible by 11?
- (a) $\frac{1}{4}$ (b) $\frac{1}{5}$
(c) $\frac{1}{6}$ (d) None of these
22. $A = k^2 - 1$ and $B = (k + 1)^2 - 1$, where k is a natural number greater than 1. How many prime numbers are there by which both A and B are divisible for at least one value of k ?
- (a) 0 (b) 1
(c) 2 (d) More than 2
23. $A = 2812$, $B = 188$ and $C = 216$. How many natural numbers are there by which at least one among A , B and C is divisible?
- (a) 499 (b) 501
(c) 504 (d) 505
24. The H.C.F. of a , b and c is 8. If $a - b = b - c = 8$ and the L.C.M. of a , b and c is a four-digit number, then what is the maximum possible value of c ?
- (a) 80 (b) 88
(c) 96 (d) 100
25. $P = b^2c^2 - ac - bd$, where a , b , c and d , in that order, are four consecutive natural numbers ($a < b$). Which of the following statements is correct?
- (a) $\sqrt{P}$ is always a prime number.
- (b) $\sqrt{P}$ is always a rational number, though not necessarily prime.
- (c) P can be rational or irrational depending on the numbers.
- (d) P is always an irrational number.
26. What digit does "a" represent, if $35! = 10333147966386144929a66651337523200000000$?
- (a) 4 (b) 6
(c) 2 (d) 1
27. If p is a prime number and w, x, y, z are four natural numbers whose sum is less than p , then $(w + x + y + z)^p - (w^p + x^p + y^p + z^p)$ is always divisible by
- (a) $p - 1$ (b) p^2
(c) p (d) $p + 1$
28. $N = a^4 + b^4 + c^4 + d^4 \dots$ till 31 terms, where a, b, c, d etc. are distinct prime numbers. If N is divisible by 30, then which of the following statements is/are definitely true?
- I. One of the numbers is 2.
II. One of the numbers is 3.
III. One of the numbers is 5.
- (a) Only I (b) I and III
(c) II and III (d) I, II and III
29. $N = 70! \times 69! \times 68! \times \dots 3! \times 2! \times 1!$ Which of the following represents the 147th digit from the right end of N ?
- (a) 2 (b) 0
(c) 5 (d) 7
30. If the integers m and n are chosen at random between 1 and 100, then atmost distinct numbers of the form $7^m + 7^n$ is divisible by 5 equals to
- (a) 1250 (b) 10000
(c) 2500 (d) None of these
31. What will be the value of x for $\frac{(100^{17} - 1) + (10^{34} + x)}{9}$; the remainder = 0
- (a) 3 (b) 6
(c) 9 (d) 8
32. P, Q, R, S and T are five prime numbers, where $P < Q < R < S < T$. It is also given that $P + Q + R + S + T = 452$. What is the value of P^5 ?
- (a) 243 (b) 32
(c) 16807 (d) More than one value

Test Yourself

1. $(11)_2 + (22)_3 + (33)_4 + (44)_5 + (55)_6 + (66)_7 + (77)_8 + (88)_9 = (?)_{10}$
 (a) 396 (b) 276
 (c) 250 (d) 342
2. What is the remainder when $1! + 2! + 3! + \dots + 100!$ is divided by 7?
 (a) 0 (b) 5
 (c) 6 (d) 3
3. Convert $0.234\bar{5}$ in $\frac{p}{q}$ form of rational number
 (a) $\frac{129}{550}$ (b) $\frac{469}{1980}$
 (c) $\frac{2368}{9900}$ (d) None of these
4. The remainder when 7^{84} is divided by 342 is
 (a) 0 (b) 1
 (c) 21 (d) 340
5. Find the number of factors of 1200
 (a) 30 (b) 24
 (c) 32 (d) 36
6. If $x959y$ is divisible by 44 and $y > 5$, then what are values of the digit x and y ?
 (a) $x = 7, y = 6$ (b) $x = 4, y = 8$
 (c) $x = 6, y = 7$ (d) None of these
7. Find the unit's digit in the product $(2467)^{153} \times (341)^{72}$.
 (a) 6 (b) 7
 (c) 8 (d) 9
8. There is one number which is formed by writing one digit 6 times (e.g. 111111, 444444 etc.). Such a number is always divisible by:
 (a) 7 and 11 (b) 11 and 13
 (c) 7, 11 and 13 (d) All of these
9. A number when divided successively by 4 and 5 leaves remainder 1 and 4 respectively. When it is successively divided by 5 and 4, then the respective remainders will be:
 (a) 1, 2 (b) 2, 3
 (c) 3, 2 (d) 4, 1
10. Find the last two digit of $(545454)^{380}$
 (a) 01 (b) 67
 (c) 76 (d) 34
11. Find the highest power of 5 in $100!$.
 (a) 19 (b) 22
 (c) 25 (d) None of these
12. If HCF of two numbers A and B is 12 while that of C and D is 15 and that of E and F is 18, then what is the HCF of A, B, C, D, E & F ?
 (a) 6 (b) 12
 (c) 90 (d) 80
13. Three bells chime at an interval of 18, 24 and 32 minutes respectively. At a certain time they begin to chime together. What length of time will elapse before they chime together again?
 (a) 2 hours 24 minutes (b) 4 hours 48 minutes
 (c) 1 hour 36 minutes (d) 5 hours
14. Find the number of zeroes in the following multiplication: $5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$.
 (a) 6 (b) 7
 (c) 8 (d) 9
15. Find the number of even factors of 60060.
 (a) 128 (b) 64
 (c) 32 (d) 80

Hints & Solutions

Foundation Level

1. (a) Let 'r' be the remainder $\Rightarrow 221 - r, 116 - r, 356 - r$ are exactly divisible by that number. Now, if two numbers are divisible by a number, then so is their difference
 $\Rightarrow [(221 - r) - (116 - r)], [(356 - r) - (116 - r)].$
 and $[(356 - r) - (221 - r)]$ are divisible by that number
 $\Rightarrow 105, 135, 240$ are divisible by that number
 $= \text{HCF of } 105, 135, 140 = 15.$
2. (a) On dividing the given number 345670 by 6, we get 4 as the remainder.
 So 2 must be added to the given number.
3. (d) Since $(35 - 25) = 10, (45 - 35) = 10, (55 - 45) = 10.$
 Now take the LCM of 35, 45, 55 and subtract 10 from it
 $\Rightarrow 3465 - 10 = 3455.$
4. (c) $n(n^2 + 20)$ is always divisible by 24, if n is even number.
5. (a) When 2^{256} is divided by 17 then

$$\Rightarrow \frac{2^{256}}{2^4 + 1} \Rightarrow \frac{(2^2)^{64}}{(2^4 + 1)}$$
 By remainder theorem when $f(x)$ is divided by $x + a$ the remainder $= f(-a)$
 Here $f(1) = (2^2)^{64}$ and $x = 2^4$ and $a = 1$
 $\therefore \text{Remainder} = f(-1) = (-1)^{64} = 1$
6. (b) The last digit of 2137^1 is 7.
 Last digit of 2137^2 is 9. Last digit of 2137^3 is 3, the last digit of 2137^4 is 1, last digit 2137^5 is 7 and the last digit of 2137^6 is 9 and so on. Hence it form a pattern and the last digit repeats for every 5th. $753 = 4 \times 188 + 1$. Thus the last digit of 2137^{753} is the same as that of 2137^1 i. e., 7.
7. (a) We have to find the least number which is divisible by 3, 5, 6 and 9 and is also a perfect square. The LCM of 3, 5, 6 and 9 is $3 \times 3 \times 2 \times 5 = 90$. Hence, the required number is $90 \times 2 \times 5 = 900$.
8. (c) Use test of 11 after putting $y = 5$.
9. (d) Out of n and $n + 2$, one is divisible by 2 and the other by 4, hence $n(n + 2)$ is divisible by 8. Also $n, n + 1, n + 2$ are three consecutive numbers, hence one of them is divisible by 3. Hence $n(n + 1)(n + 2)$ must be divisible by 24. This will be true for any even number n .
10. (b) Check the number for divisibility by 3.
 So, $4 + 5 + 9 + 0 + 4 + 5 = 27$. Hence it is divisible by 3 and the quotient is 153015.
 Now, check the quotient for divisibility by 9.
 $1 + 5 + 3 + 0 + 1 + 5 = 15$
 So, the number is not divisible by 9.
11. (c) Last digit in $19 - 9$
 $18^2 - 1$
 $19^3 - 1$
 for odd powers of 19
 Last digit is 9 and for even it is 1
 Last digit in 19^{81} is 9
 Last digit in 41 is 4
 42 is 6
 43 is 4
 for odd powers of 4
 $3^9 k$ is odd irrespective of the value of k
 $\therefore$ last digit in $4^9 k$ is 4. Last digit in $19^{81} + 4^9 k$ is last digit in $9 + 4$ i.e., in $13 = 3$
12. (a) Sum of prime numbers that are greater than 60, but less than 70 is
 $61 + 67 = 128$
13. (d) 311 is repeated seven times in the number, 311 is not divisible by 3 but 311 repeated twice is not divisible by 3, but divisible by 11.
 Similarly 311 repeated thrice is divisible by 3, but not by 11.
 As 311 is repeated seven times, which is neither multiple of 2 nor 3.
 So, number is not divisible by 3 or 11.
14. (c) $\frac{1365 - 15}{5} = 270$
15. (c) Sum of digits $= (5 + 1 + 7 + x + 3 + 2 + 4)$
 $= (22 + x)$, which must be divisible by 3.
 $\therefore x = 2.$
16. (d) Clearly, 4864 is divisible by 4.
 So, $9P2$ must be divisible by 3, so, $(9 + P + 2)$ must be divisible by 3.
 $\therefore P = 1.$
17. (a) Largest 4-digit number = 9999

$$\begin{array}{r} 88 \overline{) 9999} \\ 88 \\ \hline 119 \\ 88 \\ \hline 319 \\ 264 \\ \hline 55 \end{array}$$
 Required number $= (9999 - 55) = 9944$
18. (a) $(x^n - a^n)$ is always divisible by $(x + a)$, when n is even natural number.

19. (d) $0.\overline{47} \frac{47}{99}$.
20. (c)
21. (b) LCM of the numbers = 420. Hence there must be $(420 \times 2) + 2 = 842$ beads.
22. (b) Since $59 = 4 \times 14 + 3 \Rightarrow$ last digit of $(377)^{59} = 3$
 $87 = 4 \times 21 + 3 \Rightarrow$ last digit of $(793)^{87} = 7$
 $129 = 4 \times 32 + 1 \Rightarrow$ last digit of $(578)^{129} = 8$
 $99 = 2 \times 49 + 1 \Rightarrow$ last digit of $(99)^{59} = 9$
Hence the last digit of the result is equal to the last digit of
 $3 \times 7 \times 8 \times 9$, i.e., 2.
 $\therefore$ digit at unit's place = 2
23. (d) Interval after which the devices will beep together = (L.C.M. of 30, 60, 90, 105) min = 1260 min. = 21 hrs. So, the devices will again beep together 21 hrs. after 12 noon i.e., at 9 a.m.
24. (d) N will be an odd number because N is sum of one even number (b) and 13985 odd numbers. Hence, N will not be divisible by an even number.
25. (b) Divisor = $r_1 + r_2 - r_3 = 35 + 30 - 20 = 45$
26. (b) $12 - 7 = 5$, $15 - 10 = 5$ and $16 - 11 = 5$
Hence the desired number is 5 short for divisibility by 12, 15 and 16.
L.C.M. of 12, 15, 16 is 240
Hence the least number = $240 - 5 = 235$
27. (c) We have to find numbers between 100 and 200 which are even and are neither divisible by 7 nor by 9.
 $\therefore$ No. that are even and are divisible by 7 are 7 and no. which are even and divisible by 9 are 6.
Nos. even and divisible by 7 and 9 both are (e.g., 63) is only 126 :
 $\therefore$ Required answer = $7 + 6 - 1 = 12$
 $\therefore 51 - 12 = 39$.
28. (a) Let the numbers be the form $10x + y$
According to question
 $10x + y = x + y + xy$
 $9x = xy$
 $\therefore y = 9$
The numbers are 19, 29, 39, 49, 59, 69, 79, 89 and 99 total of 9 numbers
Hence the required fraction = $\frac{9}{91}$
 $= 0.099 \approx 0.1$
29. (b) Let there be w wide runs.
Byes = $w + 8$
Runs scored by batsmen = $26w$
Total run = 232
or $w + w + 8 + 26W = 323$
 $\Rightarrow w = \frac{224}{28} = 8$
 $\therefore$ Run scored by Ram $\frac{6}{13} \times 208 = 96$
30. (b) The value of the expression will be least when $x = y = z = 1/3$.
Hence, the least value = $\left(\frac{1}{1/3} - 1\right)^3$
 $= 2 \times 2 \times 2 = 8$.
31. (b) Consider $3^{4n} = (81)^n = (1 + 80)^n = 1 + 80q$, $q \in N$
 $\therefore 3^{4n} = 3^{80q+1} = (81)^{20q} \cdot 3$
Since the last digit of $(81)^{20q}$ is 1, so the last digit of $3^{4n} + 1$ is $1 \times 3 + 1 = 4$
32. (c) The last digit in the number must be 6: for only numbers ending in 6, when raised to any power, result in another no. ending in 6.
33. (a) Since the given number is divisible by 5, so 0 or 5 must come in place of \$. But, a number ending with 5 is never divisible by 8. So, 0 will replace \$. Now, the number formed by the last three digits is $4*0$, which becomes divisible by 8, if * is replaced by 4. Hence, digits in place of * and \$ are 4 and 0 respectively.
34. (a) Let total number of seats in the stadium be p ; number of seats in the lower deck be x and number of seats in upper deck be y .
 $\therefore p = x + y$, $x = p/4$, $y = 3p/4$
Now in the lower deck, $4x/5$ seats were sold and $x/5$ seats were unsold.
No. of total seats sold in the stadium = $2p/3$.
No. of unsold seats in the lower deck = $x/5 = p/20$
No. of unsold seats in the stadium = $p/3$
 $\therefore$ Required fraction = $\frac{p/20}{p/3} = \frac{3}{20}$
35. (c) $1 + 2 + 3 + \dots + 40 = \frac{40 \times 41}{2} = 820$
Since at each time any two numbers a and b are erased and a single new number $(a + b - 1)$ is written. Hence, each one is subtracted and this process is repeated 39 times. Therefore, number left on the board at the end = $820 - 39 = 781$.
36. (d) Since $80 = 8 \times 10$ or $80 = 16 \times 5$
Thus y (i.e., unit digit) must be zero.
 $\therefore 653xy = 653x0$, where $653x0$ must be divisible by 16 or $653x$ is divisible by 8.
Thus the last 3-digit number $53x$ will be divisible by 8. Hence, at $x = 6$, we get the required result.
 $\therefore x + y = 6 + 0 = 6$
37. (c) In the given range, the last number which is divisible by both 5 and 7. i.e., 35 is 210 and the highest number is 770. So the total number of numbers between 200 and 800 which are divisible by both 5 and 7 is
 $\left(\frac{770 - 210}{35}\right) + 1 = 17$
Hence option (c) is correct.

38. (a) Total numbers in the set = $(800 - 200) + 1 = 601$
Number of numbers which are divisible by 5

$$\frac{(800-200)}{5} + 1 = 121$$

Number of numbers which are divisible by 7

$$\frac{(798-203)}{7} + 1 = 86$$

Number of numbers which are divisible by both 5 & 7

$$\frac{(770-210)}{35} + 1 = 17$$

$\therefore$ Number of numbers which are either divisible by 5 or 7 or both
= $(121 + 86) - 17 = 190$

39. (d) Since Dividend = Divisor $\times$ Quotient + Remainder
 $\therefore$ Dividend = $9235 \times 888 + 222$
Thus the number = 8200902
Hence (d) is the correct choice.

40. (c) Let this number be N then
 $N = 32 \times Q_1 + 29$... (1)
Again $N = 8 \times Q_2 + R$... (2)
From (1) and (2)
 $32Q_1 + 29 = 8Q_2 + R$ (where R is the remainder)
 $8Q_2 - 32Q_1 = 29 - R$
 $8(Q_2 - 4Q_1) = 29 - R$
or $(Q_2 - 4Q_1) = \frac{29 - R}{8}$
Since Q_1, Q_2, R are integers also $Q_2 - 4Q_1$ is an integer.
Therefore $29 - R$ must be divisible by 8.

41. (d) $(0.\bar{1})^2 [1 - 9(0.\bar{16})^2]$
 $= \left(\frac{1}{9}\right)^2 \left[1 - 9 \times \left(\frac{16}{99}\right)^2\right]$
 $= \frac{1}{81} \left[1 - 9 \times \frac{256}{9801}\right]$
 $= \frac{1}{81} \left[1 - \frac{256}{1089}\right] = \frac{1}{81} \times \frac{833}{1089} = \frac{833}{88209}$

42. (d) Since the 7, 11 and 13 all are the factors of such a number so (d) is the correct answer.
43. (c) $\therefore 7056 = 2^4 \times 3^2 \times 7^2$
 $\therefore$ Number of factors/divisors of 7056
Product of factors = $(7056)^{45/2} = (84)^{45}$
Hence (c) is the correct option.
44. (d) The sum of digits of the number will be 114, which leaves a remainder of 6 when divided by 9. So when divided by 18 it would leave either 6 or $6 + 9 = 15$ as the remainder.
Since the number is odd, it will leave an odd remainder only when divided by 18. So the remainder will be 15.

45. (a) $\frac{a+220}{a+4} - \frac{a+4+216}{a+4} = 1 + \frac{216}{a+4}$

Therefore, $(a + 4)$ must be a factor of 216.

The number of factors of 216 = 16

But $(a + 4)$ cannot be equal to 1, 2, 3 and 4 as 'a' has to be a positive integer.

Total possible values = $16 - 4 = 12$

46. (a) Sum of all even factors:
 $(2^1)(5^0 + 5^1 + 5^2)(7^0 + 7^1 + 7^2) = 3534$
Number of even factors = $1 \times 3 \times 3 = 9$
47. (d) Sum of divisors of 544 which are perfect square is:
 $(2^0 + 2^2 + 2^4)(17^0) = 21$.
48. (c) Count the number of fives. This can get done by:
 $100^1 \times 95^6 \times 90^{11} \times 85^{16} \times 80^{21} \times 75^{26} \times \dots 5^{96}$
 $(1 + 6 + 11 + 16 + 21 + 26 + 31 + 36 + 41 + 46 + \dots + 96) + (1 + 26 + 51 + 76)$
= $20 \times 48.5 + 4 \times 38.5$ (Using sum of A.P. explained in the next chapter.)
= $970 + 154 = 1124$.
49. (a) $(23)_5 = (2 \times 5^1 + 3 \times 5^0)_{10} = (13)_{10} = (1 \times 8^1 + 5 \times 8^0)_8 = (15)_8$
also, $(47)_9 = (4 \times 9^1 + 7 \times 9^0)_{10} = (43)_{10}$
= $(5 \times 8^1 + 3 \times 8^0) = (53)_8$
sum = $(13)_{10} + (43)_{10} = (56)_{10} \rightarrow (70)_8$
50. (b) If we look at the numbers $100 < N \leq 105$, we see only 101 and 103 do not have their factors in N (because these are primes). So, obviously the new LCM will be $101 \times 103 \times N$.
51. (c) The number needs to be less than $13 \times 52 = 676$. The highest power of 13 in 676! is 56.
The power of 13 in the smallest such number needs to be exactly 52. If we subtract $13 \times 3 = 39$ from 676, we get 637. The number 637! will be the smallest number of type N! that is completely divisible by 1352.
52. (c) The maximum possible remainder must be less than 18 as the sum of any two digits can not be greater than 18. So we check when the sum of digits is 18. If the sum of digits is 18 the only possible remainder is 9 in case of 99. Similarly, if the sum of digits is 17 the maximum possible remainder is 14 in case of 98. Similarly, if the sum of digits is 16 the maximum possible remainder is 15 in case of 79. The remainder we have already got is 15 and all other sums of digits will be 15 or less than 15. So 15 has to be the answer.
53. (d) $12^{55}/3^{11} = 3^{44} \cdot 4^{55} \rightarrow 4$ as units place.
Similarly, $8^{48}/16^{18} = 2^{72} \rightarrow 6$ as the units place.
Hence, 0 is the answer.
54. (c) It can be seen that the first expression is larger than the second one. Hence, the required answer would be given by the (units digit of the first expression - units digit of the second expression) = $6 - 0 = 6$.

55. (c) The given numbers are two consecutive even numbers, so their HCF = 2
Now, using LCM $\times$ HCF = Product of two numbers
LCM $\times$ 2 = (...6) $\times$ (...8)
It can be seen now that the unit digit of LCM = 4
56. (a) Answer is LCM of 40, 42, 45 = $2^3 \times 3^2 \times 5^1 \times 7^1$
= 2520 cm = 25.2 m.
57. (b) Go through option
 $S_n = 1 + 3 + 5 + 7 + \dots + 22221$
 $S_{11111} = (11111)^2$
Hence it is divisible by 11111. Thus option (b) is correct.
58. (b) $(50^2 - 1) = (50 + 1)(50 - 1) = (17 \times 3) \times (7 \times 7)$
hence divisible by 17.
and $(729)^5 - 729 = 729(729^4 - 1)$
= $729(729^2 - 1)(729^2 + 1)$
= $(729)(729 - 1)(729 + 1)(729^2 + 1)$
= $729 \times 728 \times 730 \times (729^2 + 1)$
Hence it is divisible by 5.
59. (b) Since $\frac{6}{10} \rightarrow$ Remainder is 6
 $\frac{6^6}{10} \rightarrow$ Remainder is 6
 $\frac{6^6}{10} \rightarrow$ Remainder is 6
60. (b) The answer will be 50 since, 125×122 will give 50 as the last two digits.
61. (a) $(2^{100} - 1)$ and $(2^{120} - 1)$ will yield the GCD as $2^{20} - 1$.
62. (c) Let us assume that the quotient is Q and divisor is D.
Using the condition given in question, $1997 = QD + 41$
 $\Rightarrow QD = 1956$. Now we will factorize 1956 in two parts such that D (divisor) is more than 41.
63. (a) 6^n (where n is a natural number) will always leaves the remainder 6 when divide by 10.
64. (c) For any n, 199^{2n} has last digit as 1,
But the last digit of 144^{3n} is 4 for odd values of n and 6 for even values of n.
Therefore, last digit of the given expression is either 5 or 7.
65. (d) For looking at the zeroes in the expression we should be able to see that the number of zeroes in the third term onwards is going to be very high. Thus, the number of zeroes in the expression would be given by the zeroes in the expression would be given by the number of zeroes in: $4 + 24^{24}$. 24^{24} has a unit digit 6. Hence, the number of zeroes in the expression would be 1. Option (d) is correct.
- will be -1. Similarly, last term of $(42 + 1)^{34}$ will be +1. Therefore, $34^{43} + 43^{34}$ will leave remainder $[(-1) + (+1)] = 0$, when divided by 7.
2. (b) Two prime numbers greater than 2 must be odd. Sum of two odd numbers must always be even, thus, $X + Y = 87$ is not possible.
3. (b) $7! + 8! + 9! + 10! + \dots + 100 = 7.6! + 8.7.6! + 9.8.7.6! + \dots + 100!$ is completely divisible by 7 as each of the terms contain at least one 7 in it.
Now, $1! + 2! + 3! + 4! + 5! + 6!$
= $1 + 2 + 6 + 24 + 120 + 720$
which leaves a remainder of 6 when divided by 7.
4. (a) Clearly, $(2272 - 875) = 1397$, is exactly divisible by N. Now, $1397 = 11 \times 127$
 $\therefore$ The required 3-digit number is 127, the sum of whose digits is 10.
5. (d) $(3^{25} + 3^{26} + 3^{27} + 3^{28}) = 3^{25} \times (1 + 3 + 3^2 + 3^3)$
= $3^{25} \times 40 = 3^{24} \times 3 \times 4 \times 10 = (3^{24} \times 4 \times 30)$, which is divisible by 30.
6. (b) Let the common remainder be x. Then numbers $(34041 - x)$ and $(32506 - x)$ would be completely divisible by n.
Hence the difference of the numbers $(34041 - x)$ and $(32506 - x)$ will also be divisible by n
or $(34041 - x - 32506 + x) = 1535$ will also be divisible by n.
Now, using options we find that 1535 is divisible by 307.
7. (c) Let the total number of sweets be $(25x + 8)$.
Then, $(25x + 8) - 22$ is divisible by 28
 $\Leftrightarrow (25x - 14)$ is divisible by 28 $\Leftrightarrow 28x - (3x + 14)$ is divisible by 28
 $\Leftrightarrow (3x + 14)$ is divisible by 28 $\Leftrightarrow x = 14$.
 $\therefore$ Total number of sweets = $(25 \times 14 + 8) = 358$.
8. (b) $7^4/2400$ gives us a remainder of 1. Thus, the remainder of $7^{99}/2400$ would depend on the remainder of $7^{2400}/2400$ which is 1. The powers of p_1 can be 0, 2 i.e., 2, Powers of p_2 can be 0, 2, 4 or 6 i.e., 4. Hence, a combination of these powers gives $3 \times 2 \times 4$ i.e., 24 numbers. So, there are 24 perfect squares that divide N.
9. (b) The powers of p_1 can be 0, 2, 4 or 6 i.e., 4. Hence, a combination of these powers gives $3 \times 2 \times 4$ i.e., 24 numbers. So, there are 24 perfect squares that divide N.
10. (c) Suppose, possible, $\log_2 7$ is rational, say p/q where p and q are integers, prime to each other.
Then, $\frac{p}{q} \log_2 7 \Rightarrow 7 = 2^{p/q} \Rightarrow 2^p = 7^q$
which is false since L.H.S. is even and R.H.S. is odd. Obviously $\log_2 7$ is not an integer and hence not a prime number.
11. (b) Square of the odd numbers can be written as $1^n = 8 \times 0 + 1$, $3^2 = 8 \times 1 + 1$, $5^2 = 8 \times 3 + 1$ & so on i. e., square of the odd number is of the form $8q + 1$.

Standard Level

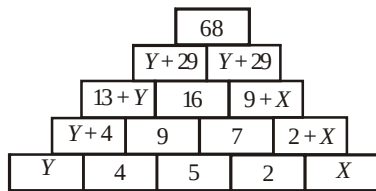
1. (d) $(34^{43} + 43^{34})/7 = [(35 - 1)^{43} + (42 + 1)^{34}/7]$.
Applying binomial theorem to $(35 - 1)^{43}$, all terms will be divisible by 35 (i.e. 7) except the last term which

12. (d) $94^3 - 23^3$ is divisible by $94 - 23 = 71$
 $94^3 - 71^3$ is divisible by $94 - 71 = 23$
 $23^3 - 71^3$ is divisible by $23 + 71 = 94$
 $\therefore 94^3 - 23^3 - 71^3$ is divisible by 23, 71 and 94
13. (d) The remainder when 981^n is divided by 7 can be determined as follows: $981^n = (980 + 1)^n$ binomial expansion of the expression every term will be divisible by 7, except for the last term i.e., 1^n for any value of n , the remainder is 1. Similarly, when 982^n is divided by 7, the remainder is equal to the remainder obtained when 2^n is divided by 7. Thus for the remainder to be equal to 1, n will be 3 or 6 Applying a similar logic to the other number, we see that the remainder is equal to 1 for all numbers when $n = 6$.
14. (b) Between 100 and 199, there will be 19 numbers which contain '2'. They are as follows.
 102, 112, 120 – 129 (10 numbers), 132, 142, 152, 162, 172, 182, 192.
 Similar would be the case for 300 – 399, 400 – 499, 500 – 599, 600 – 699.
 For 200 – 299, all 100 numbers will have 2.
 $\therefore$ Total number of numbers containing '2' = $19 \times 6 + 100 = 114 + 100 = 214$.
15. (c) As p, q, r are non-negative integers, the maximum will be achieved when the value of each variable is closed to each other.
 i.e., p, q, r are 3, 3, 4 (not necessarily in the same order).
 Hence the value of
 $pq + qr + pr + pqr = 3 \times 3 + 3 \times 4 + 3 \times 4 + 3 \times 3 \times 4$
 $= 9 + 12 + 12 + 36 = 69$
16. (d) Given $a = 6b = 12c = 27d = 36e$
 Multiplied and Divide by 108 in whole expression

$$\frac{108a}{108} \quad \frac{108b}{18} \quad \frac{108c}{9} = \frac{108d}{4} = \frac{108e}{3}$$

$$\frac{1}{108}a \quad \frac{1}{18}b \quad \frac{1}{9}c \quad \frac{1}{4}d = \frac{1}{3}e = 1 \text{ (say)}$$
 $\Rightarrow a = 108, b = 18, c = 9, d = 4, e = 3$
 So it is clear that $\left(\frac{a}{6}, \frac{c}{d}\right)$ contains a number $\frac{c}{d} \left(\frac{9}{4}\right)$
 which is not an integer
17. (a) Remember that, $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$
 $x = (16^3 + 17^3 + 18^3 + 19^3)$
 $x = (16^3 + 19^3) + (17^3 + 18^3)$
 $x = (16 + 19)(16^2 + 19^2 - 16 \times 19) + (17 + 18)(17^2 + 18^2 - 17 \times 18)$
 $x = 35[16^2 + 19^2 - 16 \times 19 + 17^2 + 18^2 - 17 \times 18]$
 $x = 35 \times (\text{Even number})$
 Hence, x is divisible by 70 and leaves remainder as zero.
18. (a) $2^{17} \times 6^{31} \times 7^5 \times 10^{11} \times 11^{10} \times (323)^{23}$
 $2^{17} \times 2^{31} \times 3^{31} \times 7^5 \times 2^{11} \times 5^{11} \times 11^{10} \times 17^{23} \times 19^{23}$
 $2^{59} \times 3^{31} \times 5^{11} \times 7^5 \times 11^{10} \times 17^{23} \times 19^{23}$
 $\therefore$ Total number of prime factors
 $= 59 + 31 + 11 + 5 + 10 + 23 + 23 = 162$
19. (b) Let the 3 digits of number A be x, y and z
 Hence $A = 100x + 10y + z$
 On reversing the digits of number A , we get the number B i.e., zyx .
 $\therefore B = 100z + 10y + x$
 As $B > A \Rightarrow z > x$... (1)
 $B - A = 99z - 99x = 99(z - x)$
 As 99 is not divisible by 7
 so $(z - x)$ has to be divisible by 7. ... (2)
 Using (1) & (2), the only possible values of z and x are (8, 1) and (9, 2)
 So the minimum and maximum range of A are 108 and 299, which $\in 106 < A < 305$
20. (d) The unit digit of every term from $5!$ to $49!$ is 0.
 Also, $1! - 2! + 3! - 4! = 1 - 2 + 6 - 24 = -19$.
 Hence, the unit digit of N will be $10 - 9 = 1$.
 The unit digit of N^N will also be 1.
21. (b) Let the 3-digit number be abc . Now according to the given condition, $(abc)_4 = (cba)_3$.
 $16a + 4b + c = 9c + 3b + a$
 $\Rightarrow 15a + b = 8c$
 The only set of numbers which satisfies the relation given above is $a = 1, b = 1$ and $c = 2$.
22. (a) The remainder would be given by: $(5 \times 7 \times 10 \times 23 \times 27)/34 \rightarrow 35 \times 230 \times 27/34 \rightarrow 1 \times 26 \times 27/34 = 702/34 \rightarrow \text{remainder} = 22$.
23. (b) For this, we need to break 12^{42} first by using binomial theorem as $(10 + 2)^{42}$. Obviously this expression will have 43 terms, and out of these 43 terms first 41 terms will have both of their tens and units place digit as 0.
 Last two terms will be
 $\rightarrow {}^{42}C_{41} \times 10^1 \times 2^{41} + {}^{42}C_{42} \times 10^0 \times 2^{42}$
 Now, we will find the ten's place digit of all these terms individually.
 Tens digit of ${}^{42}C_{41} \times 10^1 \times 2^{41} = 42 \times 10 \times (02)$
 [Cyclicity of 2 is 20, so 2^{41} will have same tens digits as 2^1] = 840, so 40 are the last two digits.
 Similarly, ${}^{42}C_{42} \times 10^0 \times 2^{42} = 1 \times 1 \times 04 = 04$
 So, finally last two digits are $\rightarrow 40 + 04 = 44$, so 4 is the ten's place digit.
24. (a) The solution of this question is based on the rule that: The HCF of $(a^m - 1)$ and $(a^n - 1)$ is given by $(a^{\text{HCF of } m, n} - 1)$
 Thus, in this question the answer is : $(3^5 - 1)$. Since 5 is the HCF of 35 and 125.]
25. (b) Assume that the software fails a, b , and c times in a single stage, in two stage, and in all stages respectively.
 $\therefore b + 3c = 6 + 7 + 4 = 17$
 but $c = 4$, hence $b = 5$
 Similarly, we have
 $a + 2b + 3c = 15 + 12 + 8 = 35$
 $a = 35 - 12 - 10 = 35 - 22 = 13$
 Hence option (b)

26. (b)



$$Y + 29 + Y + 29 = 68$$

or $2Y = 68 - 58 = 10$

$$\therefore Y = \frac{10}{2} = 5$$

$$Y + 29 = 16 + 9 + X$$

$$Y = 25 + X - 29$$

$$Y = X - 4$$

$$5 + 4 = X \quad (\because Y = 5)$$

$$\Rightarrow X = 9$$

27. (b) You need to solve this question using trial and error. For 32 (option 1):

$32 = 2^5$, Hence 6 factors. On increasing by 50%, $48 = 2^4 \times 3^1$ has 10 factors. Thus the number of factors is increasing when the number is increased by 50% which is not what the question is defining for the number. Hence, 32 is not the correct answer.

Checking for option (b) 84.

$$84 = 2^2 \times 3^1 \times 7^1 \rightarrow (2 + 1)(1 + 1)(1 + 1) = 12 \text{ factors}$$

On increasing by 50% $\rightarrow 126 = 2^1 \times 3^2 \times 7^1 \rightarrow (1 + 1)(2 + 1)(1 + 1) = 12$ factors. (no change in number of factors).

Second Condition: When the value of the number is reduced by 75% $\rightarrow 84$ would become 21 ($3^1 \times 7^1$) and the number of factors would be $2 \times 2 = 4$ - a reduction of 66.66% in the number of factors.

28. (a) The number will be a multiple of 6, 7, 8, 9, 10

LCM of 6, 7, 8, 9, 10 = 2520

$$\therefore \text{Largest 4-digit number divided by this} = 7560$$

$$\therefore \text{Required number} = 7558$$

$$\text{Sum of the digits of this number} = 25$$

29. (b) In the given equation the right hand side contains the powers of 2 and 3 only; therefore the left hand side should contain the powers of 2 and 3 only.

Since $(x - 1)(x - 2)(x - 3)$ is a product of three consecutive numbers, it will always contain either one or two multiples of 2 and one multiple of 3. Let's make two cases:

(a) If $(x - 1)$ and $(x - 3)$ are multiples of 2:

Let $(x - 1)$ be equal to $2k$; then $(x - 3)$ is equal to $2(k + 1)$. Now k and $(k + 1)$ should both contain powers of 2 or 3 only. This is possible with $k = 1, 2$ or 3 . Also if any of k or $(k + 1)$ is a multiple of 3, $(x - 2)$ will not be a multiple of 3 or 2. So again it will not satisfy.

(b) If $(x - 2)$ is a multiple of 2:

Here $(x - 1)$ and $(x - 3)$ will both be odd, out of which only one will be a multiple of 3. Hence

the other number will be a multiple of an odd number other than 3. So the equation can be satisfied only if that other odd number is 1. Hence taking one odd number as 1 we get $1 \times 2 \times 3$ which is equal to 6.

Hence the equation is satisfied for $x = 4$ only

30. (a) Let's divide the first 100 natural numbers in five sets of 20 numbers each:

$\{1, 2, 3, \dots, 20\}$, $\{21, 22, 23, \dots, 40\}$, $\dots, \{81, 82, 83, \dots, 100\}$. If we pick the first ten numbers from each set we will not get any pair of two numbers whose difference is 10.

However, if we pick just one more number from any of the sets, it would have a difference of 10 with one of the numbers which has already been picked.

So the answer is $10 \times 5 + 1 = 51$.

31. (a) $45 = 3^2 \times 5$. Hence, we need to count the number of 3^2 's and 5 's that can be made out of $123!$.

$$\text{Number of } 3\text{'s} = 41 + 13 + 4 + 1 = 59 \rightarrow \text{Number of } 3^2\text{'s} = 29$$

$$\text{Number of } 5\text{'s} = 24 + 4 = 28.$$

The required answer is the lower of the two (viz. 28 and 29). Hence, option (a) 28 is correct.

32. (c) Remainder of $(_{32}32^{32} \text{ divided by } 7) = \text{Remainder of } (_432^{32} \text{ divided by } 7)$

Now find cyclicity of remainder of $(_432^n \text{ divided by } 7)$.

$$\text{Remainder when } _432^1 \text{ divided by } 7 = 2$$

$$\text{Remainder when } _432^2 \text{ divided by } 7 = 4$$

$$\text{Remainder when } _432^3 \text{ divided by } 7 = 2$$

So, the cyclicity is 2, 4, 2, 4 and so on.

For every even value of n , remainder = 4

33. (d) As the larger number is written on the left, the larger number is either 54 or 55.

Let the smaller number be x .

Case I: The larger number is 54.

$$5400 + x = 5481 + 54 - x$$

$$2x = 5535 - 5400 = 135$$

(In this case x will not be a natural number.)

Case II: The larger number is 55.

$$5500 + x = 5481 + 55 - x$$

$$2x = 5536 - 5500 = 36$$

$$\Rightarrow x = 18$$

Hence, the required sum = 73.

34. (c) Let the hundreds digit be n .

The tens digit will be $2n$.

The unit digit will be $4n$.

The possible values of ' n ' are 1 and 2 and hence the possible numbers are 124 and 248 respectively.

On converting 248 in base 8 and base 9, the given condition gets violated.

On converting 124 in base 8 and base 9, we get

$$(174)_8 = (147)_9.$$

$$\text{Required sum} = 4 + 7 = 11.$$

35. (c) The expression can be written as $a + \frac{1}{b+x}$, where x lies in the interval $(0, 1)$. Since $(1.25)^3 = 1.953125$ and $(1.3)^3 = 2.197$, it can be concluded that $2^{\frac{1}{3}}$ belongs to the interval $(1.25, 1.3)$. Hence, $a = 1$. This implies that $\frac{1}{b+x}$ lies in the interval $(0.25, 0.3)$. The only possible value of $b = 3$.

36. (d) $(m-n)^2 = \frac{4}{(m+n-1)}$
 $\Rightarrow (m-n)^2 (m+n-1) = 4mn$
 $\Rightarrow (m-n)^2 (m+n-1) = (m+n)^2 \cdot (m-n)^2$
 $\Rightarrow (m-n)^2 (m+n) = (m+n)^2$
 $\Rightarrow (m-n)^2 = (m+n)$
 (Since, $m+n \neq 0$)
 The above equation has infinitely many solutions where m and n are positive integers. We can put $m+n = v$ and $m \cdot n = u$, and re-write the equation as $u^2 = v$ and then plug in different values of u and v to get different pairs of (m, n) .
37. (d) $3y^2 = x^2 - 1376$
 As we can see L.H.S. is definitely a multiple of 3 and in R.H.S. 1376 leaves a remainder of 2 when divided by 3. There are three possibilities for x in R.H.S.:
 (i) If x is multiple of 3, so is x^2 , and R.H.S. will leave a remainder of 1 when divided by 3.
 (ii) If x is of the form $3m+1$, x^2 will be of the form $3n+1$ and R.H.S. will leave a remainder of 2. $m, n \in \mathbb{N}$
 (iii) If x is of the form $3m+2$, x^2 will be of the form $3n+1$ and R.H.S. will leave a remainder of 2. $m, n \in \mathbb{N}$ So R.H.S. can never be a multiple of 3, while L.H.S. is always a multiple of 3. Hence no real solution exists.

38. (a) The number of zeros at the end of $222^{111} \times 35^{53}$ is 53. The number of zeros at the end of $(7!)^{61} \times (10!)^{51}$ is 960. The number of zeros at the end of $42^{42} \times 25^{25}$ is 42. Thus the number of zeros at the end of the whole expression is 42.
39. (c) $(18^2 - 1) = (17)(18+1)$
 $(18^4 - 1) = (18^2 + 1)(18^2 - 1)$
 $= (18^2 + 1)(18+1)(18-1)$
 $(18^6 - 1) = (18^3)^2 - 1$
 $= (18^3 + 1)(18^3 - 1) = 17 \times k$ etc.
 Hence there will only 9 times 17 in the whole expression.
40. (c) $2^2 + 22^2 + 222^2 + 2222^2 + \dots + (2222 \dots 49 \text{ twos})^2$
 $= 2^2 + (2)^2 + 2^2 + \dots + 2^2 (49 \text{ twos})$
 $= 4 + 4 + 4 + \dots + 4 (49 \text{ twos})$
 $= 4 \times 49 = \text{last digit is } 6.$

41. (d) Factorize 96! into prime factors. Find the unit digit of all the factors individually and multiply to get the unit digit of 96!.

$$96! = 2^{92} \times 3^{46} \times 5^{22} \times \dots$$

Now 5^{22} and 2^{22} can be eliminated, since these will result in zeroes. Find the unit digit of all the remaining.

42. (b) $N^2 + 96 = P^2$, or, $96 = P^2 - N^2$. Now factorize 96 and equate it with $(P+N)(P-N)$.

43. (a) Sum of the digits of the 'super' number
 $= 1 + 2 + 3 + \dots + 29$

$$\frac{29}{2} \cdot \{2 \times 1 + (29-1) \cdot 1\}$$

$$= \frac{29}{2} \cdot (2+28) = \frac{29 \times 30}{2} \quad 29 \times 15 = 435$$

Now, sum of digits in the number 435 = 4 + 3 + 5 = 12 which gives a remainder of 3 when divided by 9.

44. (d) The remainder when a number is divided by 16 is given by the remainder of the last 4 digits divided by 16 (because 10000 is a multiple of 16. This principle is very similar in logic to why we look at last 2 digits for divisibility by 4 and the last 3 digits for divisibility by 8.) Thus, the required answer would be the remainder of 4950/16 which is 6.

45. (d) For looking at the zeroes in the expression we should be able to see that the number of zeroes in the third term onwards is going to be very high. Thus, the number of zeroes in the expression would be given by the number of zeroes in:

$4 + 24^{24}$. 24^{24} has a unit digit 6. Hence, the number of zeroes in the expression would be 1. Option (d) is correct.

Expert Level

1. (d) $(216)^{3/5} \times (2500)^{2/5} \times (300)^{1/5}$
 $= (3^3 \times 2^3)^{3/5} \times (5^4 \times 2^2)^{2/5} \times (5^2 \times 2^2 \times 3)^{1/5}$
 $= 3^{\left(\frac{3 \times 3}{5}\right)} \times 2^{\left(\frac{3 \times 3}{5}\right)} \times 5^{\left(\frac{4 \times 2}{5}\right)} \times 2^{\left(\frac{2 \times 2}{5}\right)} \times 5^{\left(\frac{2 \times 1}{5}\right)} \times 2^{\left(\frac{2 \times 1}{5}\right)} \times 3^{\frac{1}{5}}$
 $= 3^{\frac{9}{5}} \times 2^{\frac{9}{5}} \times 5^{\frac{8}{5}} \times 2^{\frac{4}{5}} \times 5^{\frac{2}{5}} \times 2^{\frac{2}{5}} \times 3^{\frac{1}{5}}$
 $= 3^{\left(\frac{9}{5} + \frac{1}{5}\right)} \times 2^{\left(\frac{9}{5} + \frac{4}{5} + \frac{2}{5}\right)} \times 5^{\left(\frac{8}{5} + \frac{2}{5}\right)} = 3^2 \times 2^3 \times 5^2$
 Hence, the number of prime factors = $(2 + 3 + 2) = 7$
2. (c) $\frac{p}{q}$ minimum possible remainder = 0 (when q exactly divides p)
 Maximum possible remainder = $q - 1$
 So, required maximum possible difference
 $= (q - 1) - 0 = (q - 1)$

3. (b) $(x! - x)$ is a 3-digit number.
 $\therefore x!$ is a 3-digit number.
 So, 'x' is either 5 or 6.
 If $x = 5$, then $x! - x = 120 - 5 = 115$.
 Sum of 3-digits is $1 + 1 + 5 = 7$ which is not divisible by 5.
 Hence, x is not 5. If $x = 6$, then $x! - x = 720 - 6 = 714$.
 Sum of 3-digits is $7 + 1 + 4 = 12$ which is divisible by 6.
 Hence, $x = 6$.
4. (a) Looking at the choice, (b) is seven digits number, examining the remaining two 6 digit numbers, we find: 100019 does not give remainder 6 with divisor 7. By looking at the last digits 7 & 9, we can say it suits 4 & 5. All digits totaling to 20 and the number being odd, we can also say it suits requirements of 3 & 6. Actually, dividing it by 7. Hence (a) viz 100379.
5. (d) This is of the form : $[(2222^{5555})/7 + (5555^{2222})/7]$
 We now proceed to find the individual remainder of : $(2222^{5555})/7$.
 Let the remainder be R_1 .
 When 2222 is divided by 7, it leaves a remainder of 3.
 Hence, for remainder purpose $(2222^{5555})/7 \approx (3^{5555})/7$
 $= (3 \cdot 3^{5554})/7 = [3(3^2)^{2777}]/7$
 $= [3 \cdot (7 + 2)^{2777}]/7 \approx (3 \cdot 2^{2777})/7$
 $= (3 \cdot 2^2 \cdot 2^{2775})/7 = [3 \cdot 2^2 \cdot (2^3)^{925}]/7$
 $= [3 \cdot 2^2 \cdot (8)^{925}]/7 \approx (12)/7$
 Remainder = 5.
 Similarly, $(5555^{2222})/7 \approx (4^{2222})/7 = [(2^2)^{2222}]/7$
 $= (b)^{4444}/7 = (2 \cdot 2^{4443})/7$
 $= [2 \cdot (2^3)^{1481}]/7 \approx [2 \cdot (a)^{1481}]/7$ 2 (remainder).
 Hence, $(2222^{5555})/7 + (5555^{2222})/7 \approx (5 + 2)/7$
 $\Rightarrow$ Remainder = 0
6. (a) The unit digit of each pair is 5 and there are total 50 such pairs.
- $$\begin{array}{ccccccc} 4+9^2 & + & 4^3+9^4 & + & 4^5+9^6 & + & \dots + & 4^{99}+9^{100} \\ \downarrow & & \downarrow & & \downarrow & & & \downarrow \\ \text{Unit digit} & & 5 & & 5 & & 5 & & 5 \end{array}$$
- Thus $5 + 5 + 5 + \dots 5$ (50 times)
 Hence the unit digit = 0
 [Since $5 \times 50 = 250 \rightarrow$ unit digit is zero.]
7. (d) We can break 15 into factors (5, 3, 1) and (15, 1, 1). Solving for a, b, c we get (4, 3, 2) and (8, 8, 1) as two possible triplets for (a, b, c) .
 Thus $abc = 24$ or 64 . So a unique answer cannot be determined.
Note: We should also check for (5, -3, -1) and (15, -1, -1) as other possible triplets because here $(a + b + c) > 0$. However, since not all the individual values of a, b and c will come out to be positive, the cases will get rejected.

8. (b) Checking for the number of 2's 3's and 5's in the given expression you can see that the minimum is for the number of 3's (there are 11 of them while there are 125's and more than 11 2's) Hence, option (b) is correct.
9. (d) $\frac{(100^{17} - 1) + (10^{34} + x)}{9}$
- $$100^{17} - 1 = \frac{1000\dots00 - 1}{17 \text{ zeroes}} = \frac{9999\dots99}{16 \text{ nines}}$$
- $\Rightarrow$ divisible by 9 $\Rightarrow R = 0$
 Since the first part of the expression is giving a remainder of 0, the second part should also 0 as a remainder if the entire remainder of the expression has to be 0. Hence, we now evaluate the second part of the numerator.
- $$10^{34} + x = \frac{1000\dots00 + x}{34 \text{ zeroes}} = \frac{1000\dots00x}{33 \text{ zeroes}}$$
- with x at the right most place, In order for this number to be divisible by 9, the sum of digits should be divisible by 9.
 $\Rightarrow 1 + 0 + 0 \dots + 0 + x$ should be divisible by 9.
 $\Rightarrow 1 + x$ should be divisible by 9 $\Rightarrow x = 8$
10. (b) $P.Q.R. = X.Y.Z. = Q.A.Y.$
 $\Rightarrow A = \frac{P.Q.R}{Q.Y} \cdot \frac{P.R}{Y}$
- Also, $A = \frac{X.Y.Z.}{Q.Y} \cdot \frac{XZ}{Q}$
- As A, P, Q, R, X, Y and Z are seven unique digits chosen from 0 to 9, it is clear that none of them can be 0, 5 or 7.
 Hence the required numbers are 1, 2, 3, 4, 6, 8, 9
 The only possibility for the following expression is :
- $$A = \frac{P.R}{Y} \cdot \frac{XZ}{Q} = \frac{1 \times 8}{4} = \frac{3 \times 6}{9} = 2$$
- Hence, $A = 2$
- Note :** No other combination results in a unique value of A .
11. (d) The digits which are not used are 0, 5 and 7.
 Required sum = $0 + 5 + 7 = 12$
12. (b) The number zeroes would be given by counting the number of 5's. The relevant numbers for counting the number of 5's in the product would be given by: $5^5; 10^{10}, 15^{15}, 20^{20}, 25^{25} \dots$ and so on till 100^{100}
 The number of 5's in these values would be given by: $(5 + 10 + 15 + 20 + 50 + 30 + 35 + 40 + 45 + 100 + 55 + 60 + 65 + 70 + 150 + 80 + 85 + 90 + 95 + 200)$

This can also be written as:

$$(5 + 10 + 15 + 20 + 25 + 30 + 35 + 40 + 45 + 50 + 55 + 60 + 65 + 70 + 75 + 80 + 85 + 90 + 95 + 100) + (25 + 50 + 75 + 100) = 1050 + 250 = 1300$$

13. (d) A number is divisible by 9, if the sum of its digits = 9. As three consecutive digits of the 4 digits remain together so any such 4 digit number can be written in 2 ways aaab or baaa. Hence the possible cases are

A's value	B's value	Possible Nos.
1	6	2
2	3	2
3	0, 9	1 + 2 = 3 (0333 is 3 digit no.)
4	6	2
5	3	2
6	0, 9	1 + 2 (0666 is 3 digit no.)
7	6	2
8	3	2
9	0	1 (0999 is 3 digit no.)
0	9	1 (0009 is 1 digit no.)

Hence, total numbers possible

$$= 2 + 2 + 3 + 2 + 2 + 3 + 2 + 2 + 1 + 1 = 20$$

14. (d) If $6N + 1$ and $15N + 2$ are divisible by x , then their difference i.e., $9N + 1$ will always be divisible by x . Similarly $(9N + 1) - (6N + 1)$ i.e., $3N$ will also be divisible by x . If x divides $3N$ then it can also divide $6N$. So, x divides $6N$ and $6N + 1$ both i.e., two consecutive numbers. Hence x cannot be anything but 1. So for all the values of N , the given two numbers will be co-prime.
15. (a) $[7! (14 + 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8)] / [7! (16 - 3)] = [(14 + 14 \times 13 \times 12 \times 11 \times 10 \times 9 \times 8)] / [(13)] \rightarrow$ remainder 1.
16. (b) First of all we should understand that we cannot solve this question by taking different values of N and checking if its prime or not. We need to devise some alternative method.
 $N^4 + 4 = (N^2 + 2)^2 - (2N)^2 = (N^2 + 2 + 2N)(N^2 + 2 - 2N)$
 Since N is a natural number, each of the above factors will also be a natural number.
 We can conclude that $N^4 + 4 = (N^2 + 2 + 2N)(N^2 + 2 - 2N)$ is product of two natural numbers. Hence it cannot be a prime number except for the value $N = 1$ for which the values are as follows:
 $(N^2 + 2 + 2N) = 5$ and $(N^2 + 2 - 2N) = 1$, So $(N^2 + 2 + 2N)(N^2 + 2 - 2N) = 5 \times 1 = 5$
 (For $N = 1$, value can be obtained using $N^4 + 4$. Though for higher values of N , calculating the value and checking if its prime or not is difficult).
17. (c) A natural number formed by using the same digit written 6 times (e.g., 777777) is always divisible by 1001 i.e., $7 \times 11 \times 13$.

(Since, $1001 \times 111 = 111111$)

Also, N can be written as: 7777.

$$\dots\dots\dots 7777 \times 1000 + 777$$

$$\longleftarrow 426 \text{ times } \longrightarrow$$

$\therefore 7777 \dots\dots\dots 7777 \times 1000$ is always divisible

$$\longleftarrow 426 \text{ times } \longrightarrow$$

by 1144 i.e., $11 \times 13 \times 8$.

Required remainder = 777

18. (d) Three operations have been given:

$$(i) \quad x \Delta (y + 1) = y \Delta (x + 1)$$

$$(ii) \quad x \Delta x = 1$$

$$(iii) \quad (x - y) \Delta (x + y) = x \Delta y$$

Putting $x = 1000$ and $y = 1001$ in operation (i), we get

$$1000 \Delta 1002 = 1001 \Delta 1001$$

From operation (ii), $1001 \Delta 1001 = 1$

$$\therefore 1000 \Delta 1002 = 1$$

Putting $x = 1001$ and $y = 1$ in operation (iii), we get

$$\therefore 1000 \Delta 1002 = 1001 \Delta 1$$

$$\therefore 1001 \Delta 1 = 1$$

19. (c) As N has exactly 24 factors, N can be of the form p^{23} , pq^{11} , p^2q^7 , p^3q^5 , pqr^5 , pq^2r^3 or $pqrs^2$, where p, q, r and s represent different prime numbers.

From Statement A:

As the number of factors of the resultant number is less than twice the number of factors of N , 3 must be a factor of N . Thus N can be of the form p^2q^7 , pq^2r^3 or $pqrs^2$, where the prime factor raised to the power 2 represents 3. But we cannot determine the number of factors of N^3 with certainty and hence this statement alone is not sufficient.

From Statement B:

As the number of factors of the resultant number is less than twice the number of factors of N , 5 must be a factor of N . Thus N can be of the form p^3q^5 or pq^2r^3 , where the prime factor raised to the power 3 represents 5. But we cannot determine the number of factors of N^3 with certainty and hence this statement alone is not sufficient.

From Statements A and B:

The only possibility is that N is of the form pq^2r^3 .

$$\therefore N^3 = p^3q^6r^9 \text{ and the number of factors of } N^3 = 4 \times 7 = 280.$$

20. (c) Let $a, a + 1, a + 2, \dots, a + 29$ be thirty consecutive natural numbers and N be the sum of their squares.
 $N = a^2 + (a + 1)^2 + (a + 2)^2 + \dots + (a + 29)^2$
 $N = 30a^2 + 2a(1 + 2 + \dots + 29) + (1^2 + 2^2 + \dots + 29^2)$

$$N = 30a^2 + 2a \left(\frac{29 \times 30}{2} \right) + \frac{29 \times 30 \times 59}{6}$$

Remainder when $\frac{29 \times 30 \times 59}{6}$ is divided by 12 = 11

$$\text{Now, } 30a^2 + 2a \left(\frac{29 \times 30}{2} \right) = 30(a^2 + 29a)$$

As $(a^2 + 29a)$ is always even, remainder obtained when $30(a^2 + 29a)$ is divided by $12 = 0$.

So the remainder obtained when the sum of the squares of any thirty consecutive natural numbers is divided by $12 = 11$.

21. (d) Let the four-digit number be $abcd$.
For the number to be divisible by 11, $(a + c) - (b + d) = 0$ or $\pm 11k$, where k is a natural number.
Let us assume that $a + c = x$ and $b + d = y$.
 $x + y = 31$ and $x - y = 11$
(For x and y to be integers, $x - y$ can neither be 0 nor an even multiple of 11.)
Solving the above equations, we get
 $x = 21$ and $y = 10$.

As x is the sum of two single digit numbers, the maximum possible value of x is 18.

Therefore, no such number is possible.

22. (b) $A = (k - 1)(k + 1)$
 $B = k(k + 2)$
For all values of k greater than or equal to 2, the natural numbers ' $k - 1$ ' and ' $k + 1$ ' are coprime with both ' k ' and ' $k + 2$ ' except when ' $k - 1$ ' and ' $k + 2$ ' are both multiples of 3.
Note that $(k + 2) - (k - 1) = 3$.
Here, the common factor of A and B , which is also a prime number, is 3.
E.g. when $k - 1 = 3$ or $k = 4$, $A = 15$ and $B = 24$.
The only common factor of A and B in this case is 3.

23. (d) $A = 28^{12} = 2^{24} \times 7^{12}$
 $B = 18^8 = 2^8 \times 3^{16}$
 $C = 21^6 = 3^6 \times 7^6$
The total number of factors of A will be
 $(24 + 1)(12 + 1) = 325$.
Similarly, the total number of factors of B and C will be 153 and 49 respectively.
Case I: Factors common to A , B and C .
HCF of $(A, B \text{ and } C) = 1$.
The only factor common to A , B and C is 1.
Case II: Factors common to exactly two among A , B and C .
Factors common to A and $B = 20, 21, \dots, 27, 28$
Factors common to B and $C = 30, 31, \dots, 35, 36$
Factors common to A and $C = 70, 71, \dots, 75, 76$
Hence, the answer
 $= (325 + 153 + 49) - (9 + 7 + 7) + 1 = 505$.

24. (c) Let $c = 8k$. Hence, $b = 8(k + 1)$ and $a = 8(k + 2)$, where k is a natural number.
As the H.C.F. of ' k ', ' $k + 1$ ' and ' $k + 2$ ' is always 1, we can say that the L.C.M. of ' k ', ' $k + 1$ ' and ' $k + 2$ ' will be
 $k(k + 1)(k + 2)$, when k is odd, and
 $\frac{k(k + 1)(k + 2)}{2}$, when k is even.
The L.C.M. of a , b and c is either $8k(k + 1)(k + 2)$ or $4k(k + 1)(k + 2)$.

L.C.M. < 10000 (given)

For highest value of k we'll take $4k(k + 1)(k + 2) < 10000$,

where k is even.

Hence, $k(k + 1)(k + 2) < 2500$.

The maximum possible value of $k = 12$.

The maximum possible value of $c = 96$.

25. (b) Since it is given that the four numbers are consecutive natural numbers and $a < b$,
 $b = a + 1$, $c = a + 2$ and $d = a + 3$.
 $P = b^2c^2 - ac - bd$
 $P = (a + 1)^2(a + 2)^2 - a(a + 2) - (a + 1)(a + 3)$
 $P = a^4 + 6a^3 + 11a^2 + 6a + 1$
 $P = (a^2 + 3a + 1)^2$

$\sqrt{P} = a^2 + 3a + 1$, which is always a rational number, though not necessarily prime. E.g. at $a = 6$.

26. (b) This can be done using the Divisibility rule of a number like 9 or 11. E.g., Since $35!$ is divisible by 11, either the sum of the digits at the odd places must be equal to the sum of the digits at the even places or their difference should be a multiple of 11. In this case the latter is not possible. Hence, $66 + a = 72$ and $a = 6$.
27. (c) If p is prime and m is not a multiple of p , then m^{p-1} when divided by p leaves remainder 1.
So $(w + x + y + z)^p$ when divided by p will leave remainder $(w + x + y + z)$ and so will $(w^p + x^p + y^p + z^p)$.
Hence, $(w + x + y + z)^p - (w^p + x^p + y^p + z^p)$ will always be divisible by p .
28. (d) As $30 = 2 \times 3 \times 5$, any natural number which is divisible by 30 must also be divisible by 2, 3 and 5.
The 4th power of any prime number is an odd natural number except when the prime number is 2. Hence if none of the 31 prime numbers is 2, we can say that N must be an odd number. This is not true as N is divisible by 2. Hence, one of the numbers is definitely 2.
All prime numbers, except 2 and 5, have 1, 3, 7 or 9 as the unit digit. Thus the 4th power of each of these prime numbers must end with 1. The remainder obtained when 5 divides the fourth power of each of these numbers is 1. The same is true for 24, which is definitely one of the terms as concluded above.
Hence, if none of the prime numbers is 5, the overall remainder when N is divided by 5, will be $1 + 1 + 1 + \dots + 31 \text{ times} \equiv 31 \equiv 1$. This is not true as N is divisible by 5. Hence, one of the numbers is definitely 5.
Similarly, the 4th power of any prime number except 34 leaves a remainder of 1 on division by 3. Further analysis (as done in the case above) will lead to the conclusion that N won't be divisible by 3 if none of the 31 numbers is 3. Hence, one of the numbers is definitely 3.

29. (b) We have to calculate the number of zeroes starting from the right end of the number N .

The number of zeroes from:

$$1! \text{ to } 4! = 0$$

$$5! \text{ to } 9! = 1 \times 5 = 5$$

$$10! \text{ to } 14! = 2 \times 5 = 10$$

$$15! \text{ to } 19! = 3 \times 5 = 15$$

$$20! \text{ to } 24! = 4 \times 5 = 20$$

$$25! \text{ to } 29! = 6 \times 5 = 30$$

$$30! \text{ to } 34! = 7 \times 5 = 35$$

$$35! \text{ to } 39! = 8 \times 5 = 40$$

So we get 155 zeroes till 39! only. From this we can easily conclude that the 147th digit from the right end of N will be zero.

30. (a) We know that if a number is divisible by 5, then its unit digit must be either 0 or 5.

Again, the unit digits of $7^1 = 7$

$$7^2 = 9$$

$$7^3 = 3 \text{ and } 7^4 = 1$$

i.e., we can get only 7, 9, 3, 1 as unit digits.

Now, the combinations of 7, 9, 3, 1 never gives unit digit, but $7 + 3 = 9 + 1 = 10$, these two combinations gives us unit digit zero.

So $7^m + 7^n$ is divisible where

case (i) $m = 1, 5, 9, \dots, 97$ correspondingly $n = 3, 7, 11, \dots, 99$

case (ii) $m = 2, 6, 10, \dots, 98$ correspondingly $n = 4, 8, 12, \dots, 100$

or the values of m and n can be reversed in each case mutually but then we get the same values.

Hence total distinct numbers so formed, divisible by 5, are

$$1250 = 25 \times 25 + 25 \times 25$$

Hence option (a) is correct.

$$31. (d) \frac{(100^{17} - 1) + (10^{34} + x)}{9}$$

$$100^{17} - 1 = \frac{1000 \dots 00 - 1}{17 \text{ zeroes}} = \frac{9999 \dots 99}{16 \text{ nines}}$$

$\Rightarrow$ divisible by 9

$\Rightarrow R = 0$

Since the first part of the expression is giving a remainder of 0, the second part should also give 0 as a remainder if the entire remainder of the expression has to be 0. Hence, we now evaluate the second part of the numerator.

$$10^{34} + x = \frac{1000 \dots 00 + x}{34 \text{ zeroes}} + \frac{1000 \dots 00x}{33 \text{ zeroes}}$$

with x at the right most place. In order for this number to be divisible by 9, the sum of digits should be divisible by 9.

$\Rightarrow 1 + 0 + 0 \dots + 0 + x$ should be divisible by 9.

$\Rightarrow 1 + x$ should be divisible by 9 $\Rightarrow x = 8$

32. (b) $P + Q + R + S + T = 482$. Sum of five prime numbers is even possible only if four of these are odd and one is even (all these cannot be odd). So, $P = 2$.

$$\therefore P^5 = 2^5 = 32$$

Explanation of Test Yourself

1. (b) $(11)_2 \left(1 \times 2^1 + 1 \times 2^0\right)_{10} = (3)_{10}$
 $(22)_2 \left(2 \times 2^1 + 2 \times 2^0\right)_{10} = (8)_{10}$
 $(33)_4 \left(3 \times 4^1 + 3 \times 4^0\right)_{10} = (15)_{10}$
 $(44)_5 \left(4 \times 5^1 + 4 \times 5^0\right)_{10} = (24)_{10}$
 $(55)_6 = \left(5 \times 6^1 + 5 \times 6^0\right)_{10} = (35)_{10}$
 $(66)_7 = \left(6 \times 7^1 + 6 \times 7^0\right)_{10} = (48)_{10}$
 $(77)_8 \left(7 \times 8^1 + 7 \times 8^0\right)_{10} = (63)_{10}$
 $(88)_9 \left(8 \times 9^1 + 8 \times 9^0\right)_{10} = (80)_{10}$
 sum $(276)_{10}$
2. (b) $7! + 8! + 9! + 10! + \dots + 100! = 7.6! + 8.7.6! + 9.8.7.6! + \dots + 100!$ is completely divisible by 7 as each of the terms contain at least one 7 in it.
 Now, $1! + 2! + 3! + 4! + 5! + 6!$
 $1 + 2 + 6 + 24 + 120 + 720 = 873$
 which leaves a remainder of 5 when divided by 7.
3. (a) $\frac{p}{q}$ form = $\frac{2345 - 23}{9900} = \frac{2322}{9900} = \frac{129}{550}$
4. (b) $\frac{7^{84}}{342} = \frac{(7^3)^{28}}{342} = \frac{(343)^{28}}{342}$
 By remainder theorem, $\frac{(343)^{28}}{342}$ will have the same remainder as $\frac{1^{28}}{342}$ i.e., the remainder is 1.
5. (a) Since $1200 = 12 \times 100 = 4 \times 3 \times 4 \times 25 = 2^4 3^1 5^2$ hence its number of factors is $(4 + 1)(1 + 1)(2 + 1) = 5 \times 2 \times 3 = 30$
6. (a) Here $44 = 11 \times 4$
 $\therefore$ the number must be divisible by 4 and 11 respectively. Test of 4 says that 9y must be divisible by 4 and since $y > 5$, so $y = 6$
 Again, $x9596$ is divisible by 11, so $x + 5 + 6 = 9 + 9 \Rightarrow x = 7$
 Thus $x = 7, y = 6$
7. (b) Clearly, unit's digit in the given product = unit's digit in $7^{153} \times 1^{72}$.
 Now, 7^4 gives unit digit 1.
 $\therefore 7^{153}$ gives unit digit $(1 \times 7) = 7$. Also 1^{72} gives unit digit 1.
 Hence, unit's digit in the product is $= (7 \times 1) = 7$.
8. (d) Since 111111 is divisible by each one of 7, 11 and 13, so each one of given type of numbers is divisible by each one of 7, 11, and 13. as we may write, $222222 = 2 \times 111111$, $333333 = 3 \times 111111$, etc.

9. (b) Complete remainder $= d_1 r_2 + r_1$
 $= 4 \times 4 + 1 = 17$
 Now, 17 when divided successively by 5 and 4
 $\therefore$ The remainders are 2, 3.
10. (c) Since last two digit of $(545454)^{380}$ is same as $(54)^{380} = (2 \times 3)^{380} = 2^{380} \times 3^{1140}$
 Now the 1st term is $2^{380} = (2^{20})^{19}$ which has last two digits as 76.
 While 2nd term is $3^{1140} = (3^4)^{285} = (81)^{285}$ whose last two digits is 01.
 Hence required last two digits is $76 \times 01 = 76$
11. (d) Divide 100 successively by 5 and keep on writing the quotient and then find the summation of all the quotient this summation will give us the highest power of 5 in 100!.

	100
5	20
5	4

Sum of all the quotient is $20 + 4 = 24$, hence highest power of 5 in 100! Is 24.

Alternately

Required highest power of 5 is

$$\left\lfloor \frac{100}{5^1} \right\rfloor + \left\lfloor \frac{100}{5^2} \right\rfloor = 20 + 4 = 24 \text{ hence highest power of 5 in } 100! \text{ Is } 24.$$

12. (c) The numbers are $A = 12a, b = 12b, c = 15c, d = 15d, E = 18e$ and $F = 18f$ where $(a, b), (c, d)$ and (e, f) in pairs co prime to each other hence HCF of these 6 numbers are same as HCF of 12, 15 and 18 that is equal to 3.
13. (b) L.C.M of 18, 24 and 32 = 288
 Hence they would chime together after every 288 min. or 4 hrs. 48 min.
14. (c) The number of zeroes depends on the number of fives and the number of twos. The expression can be written as
 $5 \times (5 \times 2) \times (5 \times 3) \times (5 \times 2 \times 2) \times (5 \times 5) \times (5 \times 2 \times 3) \times (5 \times 7) \times (5 \times 2 \times 2 \times 2) \times (5 \times 3 \times 3) \times (5 \times 5 \times 2)$
 Number of 5s – 12, Number of 2s – 8.
 Hence: 8 zeroes
15. (b) Since $60060 = 2^2 \times 3 \times 5 \times 7 \times 11 \times 13$ here we need to find the number of even factors hence we can write the number as $60060 = 2(2 \times 3 \times 5 \times 7 \times 11 \times 13)$.
 Require number of even factor is same as number of factors of $2 \times 3 \times 5 \times 7 \times 11 \times 13 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$.



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UNIT-II

Arithmetic

Chapter 3	Averages
Chapter 4	Alligations
Chapter 5	Percentages
Chapter 6	Profit, Loss and Discount
Chapter 7	Interest
Chapter 8	Ratio, Proportion and Variation
Chapter 9	Time and Work
Chapter 10	Time, Speed and Distance

3

AVERAGES

- Average
- Position of the Average on the Number Line
- Weighted Average
- Properties of Average (Arithmetic Mean)

AVERAGE

An average is a simple concept of mathematics but its uses are very common in day-to-day life. In CAT and CAT like aptitude test exams at least one question is always asked, the nature of the question asked in CAT is applied and blended with logical reasoning.

An average of a group of numbers is a number that is the best representative of the group of numbers because it tells a lot about the entire numbers of the group.

In other words an average is a measure of central tendency called arithmetic mean of a group of numbers,

The formula for finding the average is

$$\text{Average} = \frac{\text{Sum of all numbers}}{\text{Number of numbers}}$$

Thus if A_v be the average of n numbers $x_1, x_2, x_3, \dots, x_n$ then

$$A_v = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

$$\Rightarrow x_1 + x_2 + x_3 + \dots + x_n = n \cdot A_v$$

$$\Rightarrow \text{Sum of } n \text{ numbers} = (\text{Number of numbers}) \times (\text{Average})$$

Illustration 1: The average of the first nine prime numbers is:

- (a) 9 (b) 11 (c) $11\frac{1}{9}$ (d) $11\frac{2}{9}$

Solution: (c) Average

$$= \frac{2+3+5+7+11+13+17+19+23}{9}$$

$$= \frac{100}{9} = 11\frac{1}{9}$$

Illustration 2: The average of five consecutive odd numbers is 61. What is the difference between the highest and lowest numbers?

- (a) 2 (b) 5
(c) 8 (d) Cannot be determined

Solution: (c) Let the numbers be $x, x+2, x+4, x+6$ and $x+8$.

$$\text{Then, } \frac{x+(x+2)+(x+4)+(x+6)+(x+8)}{5} = 61$$

$$\text{or } 5x + 20 = 305 \text{ or } x = 57.$$

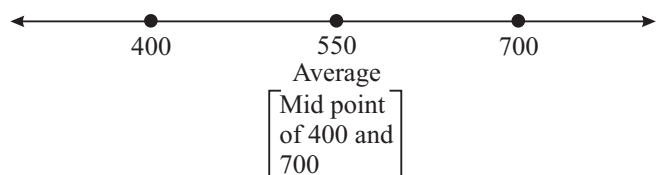
$$\text{So, required difference} = (57+8) - 57 = 8.$$

POSITION OF THE AVERAGE ON THE NUMBER LINE

Let us see the position of an average on the number line:

Suppose you purchased one book of Quant for ₹ 700 and one book of reasoning for ₹ 400. The average cost of a quant's book and a reasonings book is $\frac{700+400}{2} = 550$.

See the position of the average on the number line:

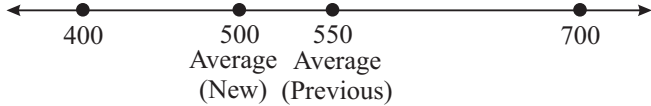


You can see that the average is the mid-point of the 400 and 700 on the number line.

Now you suppose that you purchased one book of quant and two books of reasoning. One of reasoning books for yourself and

other book of reasoning for your friend then average cost price of the three books = $\frac{700 + 400 + 400}{3} = 500$.

Now see the position of the two averages 500 and 550 on the number line:



Clearly the position of the average (new) is shifted towards 400 i.e. the average (new) is closer to 400 than to 700. This has happened because we again add a number 400 in the group of two numbers 400 and 700 to find the average and the new added number 400 is less than the mean (previous) 550.

WEIGHTED AVERAGE

If we have two or more groups of numbers whose individual averages are known, then combined average of all the numbers of all the groups is known as Weighted Average. Thus if there are k groups having number of numbers $n_1, n_2, n_3, \dots, n_k$ with averages $A_1, A_2, A_3, \dots, A_k$ respectively; then weighted average,

$$A_w = \frac{n_1 A_1 + n_2 A_2 + n_3 A_3 + \dots + n_k A_k}{n_1 + n_2 + n_3 + \dots + n_k}$$

Illustration 3: The average score of a cricketer in two matches is 27 and in three other matches is 32. Then find the average score in all the five matches.

Solution:

$$\text{Average in 5 matches} = \frac{2 \times 27 + 3 \times 32}{2 + 3} = \frac{54 + 96}{5} = 30.$$

Illustration 4: The average age of students of a class is 15.8 years. The average age of boys in the class is 16.4 years and that of the girls is 15.4 years. The ratio of the number of boys to the number of girls in the class is

- (a) 1 : 2 (b) 2 : 3 (c) 3 : 4 (d) 3 : 5

Solution: (b) Let the number of boys in a class be x .

Let the number of girls in a class be y .

$\therefore$ Sum of the ages of the boys = $16.4x$

Sum of the ages of the girls = $15.4y$

$\therefore 15.8(x + y) = 16.4x + 15.4y$

$$\Rightarrow 0.6x = 0.4y \Rightarrow \frac{x}{y} = \frac{2}{3}$$

$\therefore$ Required ratio = 2 : 3

Illustration 5: The average of six numbers is 3.95. The average of two of them is 3.4, while the average of the other two is 3.85. What is the average of the remaining two numbers?

- (a) 4.5 (b) 4.6 (c) 4.7 (d) 4.8

Solution: (b) Sum of the remaining two numbers

$$= (3.95 \times 6) - [(3.4 \times 2) + (3.85 \times 2)] \\ = 23.70 - (6.8 + 7.7) = 23.70 - 14.5 = 9.20$$

$$\therefore \text{Required average} = \left(\frac{9.2}{2} \right) = 4.6.$$

PROPERTIES OF AVERAGE (ARITHMETIC MEAN)

We know that average, $A_v = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$

$$\Rightarrow x_1 + x_2 + x_3 + \dots + x_n = n \cdot A_v \\ = A_v + A_v + A_v + \dots + \text{upto } n^{\text{th}} \text{ term}$$

Thus average of a group of numbers is such a number by which we can replace each and every number of the group without changing the total of the group of numbers.

Consider five numbers 16, 22, 25, 19 and 38.

$$\text{Its average} = \frac{16 + 22 + 25 + 19 + 38}{5} = \frac{120}{5} = 24$$

This means that if each of the five numbers 16, 22, 25, 19 and 38 were replaced by 24, there would be no change in the total.

See how it will happen:

Number	→	Increase/Decrease	→	Mean
16	→	+ 8	→	24
22	→	+ 2	→	24
25	→	- 1	→	24
19	→	+ 5	→	24
38	→	- 14	→	24
120		0		120

Thus sum of all increase and sum of all decrease in the group of numbers are equal i.e., the net deficit due to the numbers below the average always equals the net surplus due to the numbers above the average. Therefore there is no change in the sum if each number were replaced by the average of the group of numbers. This is an important way to look at the average.

Illustration 6: Find a number which when added in a group of five numbers, increases the average 20 by 2.

Solution: Let us replace each number of the group of five given numbers by its average i.e., 20. Now a new number when added in the group, the average is increased by 2. This can be understood as follows:

Due to the new number 2 is added to each of the five averages which replaced each number of the group. Apart from this, a number 22 (= 20 + 2) is also added as the sixth number in the group. Thus the new number increases the sum of all numbers of the group in two ways. First by adding 2 in each of the five averages which replaced each of the five numbers of the group and second by adding a number 22.

This can be visualised as

$$\begin{array}{rclcl} 20 & + & 2 & = & 22 \\ 20 & + & 2 & = & 22 \\ 20 & + & 2 & = & 22 \\ 20 & + & 2 & = & 22 \\ 20 & + & 2 & = & 22 \\ \hline 22 \text{ (Sixth number)} & & & = & 22 \end{array}$$

↑
Contributed by new number

Hence the new number = $(2 + 2 + 2 + 2 + 2) + 22$
 $= 5 \times 2 + 22 = 32.$

We can find the solution by using the formula,

New number = (No. of original numbers) $\times$
 (Increase in average) + (New average)
 $= 5 \times 2 + 22 = 32$

Illustration 7: The average age of 30 boys of a class is equal to 14 yrs. When the age of the class teacher is included the average becomes 15 yrs. Find the age of the class teacher.

Solution: Total ages of 30 boys = $14 \times 30 = 420$ yrs.

Total ages when class teacher is included = $15 \times 31 = 465$ yrs.

$\therefore$ Age of class teacher = $465 - 420 = 45$ yrs.

By direct formula,

Age of new entrant

= (No. of old members) $\times$ (Increase in average) + (New average)
 $= 30 (15 - 14) + 15 = 45$ yrs.

Illustration 8: The average weight of 29 students in a class is 48 kg. If the weight of the teacher is included, the average weight rises by 500 g. Find the weight of the teacher.

Solution: Here, weight of the teacher is added and final average of the group increases.

$\Rightarrow$ weight of the teacher = $48 + (0.5 \times 30) = 63$ kg.

Illustration 9: The average weight of 45 students in a class is 52 kg. 5 of them whose average weight is 48 kg leave the class and other 5 students whose average weight is 54 kg join the class. What is the new average weight (in kg) of the class?

- (a) 52.6 (b) $52\frac{2}{3}$
 (c) $52\frac{1}{3}$ (d) None of these

Solution: (b) Total weight of 45 students = $45 \times 52 = 2340$ kg

Total weight of 5 students who leave
 $= 5 \times 48 = 240$ kg

Total weight of 5 students who join
 $= 5 \times 54 = 270$ kg

Therefore, new total weight of 45 students
 $= 2340 - 240 + 270 = 2370$

$\Rightarrow$ New average weight = $\frac{2370}{45} = 52\frac{2}{3}$ kg.

Remember

(i) Ages and Averages

If the average age of a group of persons is x years today then after n years their average age will be $(x + n)$ years because for a group of people, 1 year is added to each person's age every year.

Similarly, n years ago their average age would have been $(x - n)$ years, because 1 year is subtracted from each person's age before every year.

(ii) Average Speed if Equal Distances are Travelled by Two Different Speeds

If a car travels at a speed S_1 from A to B and at a speed S_2 from B to A . Then

$$\text{Average speed} = \frac{2S_1 \cdot S_2}{S_1 + S_2}$$

The above formula can be found out as follows:

If distance between A and B is d , then

$$\begin{aligned} \text{Average speed} &= \frac{\text{Total distance}}{\text{Total time}} = \frac{2d}{\frac{d}{S_1} + \frac{d}{S_2}} \\ &= \frac{2}{\frac{1}{S_1} + \frac{1}{S_2}} = \frac{2S_1 \cdot S_2}{S_2 + S_1} \end{aligned}$$

Similarly for three equal distances travelled by three different speeds S_1, S_2 and S_3 ;

$$\text{Average speed} = \frac{3S_1 \cdot S_2 \cdot S_3}{S_1 \cdot S_2 + S_2 \cdot S_3 + S_3 \cdot S_1}$$

Let's find the average speed of a car which goes from Delhi to Panipat at a speed of 60 kmph and returns at a speed of 90 kmph.

$$\begin{aligned} \text{Average speed} &= \frac{2S_1 \cdot S_2}{S_1 + S_2} = \frac{2 \times 60 \times 90}{60 + 90} = \frac{2 \times 60 \times 90}{150} \\ &= 72 \text{ kmph.} \end{aligned}$$

You can find the average speed by an another way also as illustrated in the following examples.

First find the difference of the speeds, which is equal to $90 - 60 = 30$ kmph.

Now find the ratio of the speeds, which is equal to
 $60 : 90 = 2 : 3$

Now find the sum $2 + 3 = 5$

Now find $\frac{1}{5}$ of 30, which is equal to $\frac{1}{5} \times 30 = 6$.

Now add $2 \times \left(\frac{1}{5} \times 30\right)$ i.e. 12 to 60 or subtract $3 \times \left(\frac{1}{5} \times 30\right)$ i.e. 18 from 90, to get the average speed.

Hence average speed = $60 + 12 = 72$ kmph
 or average speed = $90 - 18 = 72$ kmph

The whole process can be shown in a single line as

$$90 - 60 = 30 \rightarrow 60 : 90 = 2 : 3 \rightarrow 60 + \frac{2}{5} \times 30 = 72$$

$$\left(\text{or } 90 - \frac{3}{5} \times 30 = 72 \right)$$

Illustration 10: The average age of a family of 6 members is 22 yrs. If the age of the youngest member be 7 yrs, then what was the average age of the family at the birth of the youngest member?

Solution: Total ages of all members = $6 \times 22 = 132$ yrs.

7 yrs. ago, total sum of ages = $132 - (6 \times 7) = 90$ yrs.

But at that time there were 5 members in the family.

$\therefore$ Average at that time = $90 \div 5 = 18$ yrs.

Illustration 11: The average of marks obtained by 120 candidates in a certain examination is 35. If the average marks of passed candidates is 39 and that of the failed candidates is 15, what is the number of candidates who passed the examination?

Sol. Let the number of passed candidates be x .

Then total marks = $120 \times 35 = 39x + (120 - x) \times 15$

or, $4200 = 39x + 1800 - 15x$ or $24x = 2400$

$\therefore x = 100$

$\therefore$ number of passed candidates = 100.

Illustration 12: The average of 11 results is 50. If the average of first six results is 49 and that of last six is 52, find the sixth result.

Solution: The total of 11 results = $11 \times 50 = 550$

The total of first 6 results = $6 \times 49 = 294$

The total of last 6 results = $6 \times 52 = 312$

The 6th result is common to both;

$\therefore$ Sixth result = $294 + 312 - 550 = 56$

Illustration 13: A batsman in his 17th innings makes a score of 85, and thereby increases his average by 3. What is his average after 17 innings?

Solution: Let the average after 16th innings be x , then

$16x + 85 = 17(x + 3)$ (= Total score after 17th innings)

$\therefore x = 85 - 51 = 34$

$\therefore$ average after 17 innings = $x + 3 = 34 + 3 = 37$.

Illustration 14: A cricketer has completed 10 innings and his average is 21.5 runs. How many runs must he make in his next innings so as to raise his average to 24?

Solution: Total of 10 innings = $21.5 \times 10 = 215$

Suppose he needs a score of x in 11th innings; then
average in 11 innings = $\frac{215 + x}{11} = 24$

or, $x = 264 - 215 = 49$.

Illustration 15: An aeroplane flies along the four sides of a square at the speeds of 200, 400, 600 and 800 km/h. Find the average speed of the plane around the field.

(a) 384 km/h

(b) 370 km/h

(c) 368 km/h

(d) None of these

Solution: (a) Let each side of the square be x km and let the average speed of the plane around the field be y km/h.

Then, $\frac{x}{200} + \frac{x}{400} + \frac{x}{600} + \frac{x}{800} = \frac{4x}{y}$

$\Rightarrow \frac{25x}{2400} = \frac{4x}{y} \Rightarrow \left(\frac{2400 \times 4}{25} \right) = 384$.

$\therefore$ Average speed = 384 km/h.

Practice Exercises



Foundation Level

- The average age of 24 students and the class teacher is 16 years. If the class teacher's age is excluded, the average reduces by one year. What is the age of the class teacher?
(a) 50 years (b) 45 years
(c) 40 years (d) Data inadequate
- The average age of 36 students in a group is 14 years. When teacher's age is included in it, the average increases by one. What is the teacher's age in years?
(a) 31 (b) 36
(c) 51 (d) cannot be determined
- A school has 4 section of Chemistry in Class X having 40, 35, 45 and 42 students. The mean marks obtained in Chemistry test are 50, 60, 55 and 45 respectively for the 4 sections. Determine the overall average of marks per student
(a) 50.25 (b) 52.25
(c) 51.25 (d) 53.25
- The average of six numbers is 3.95. The average of two of them is 3.4, while the average of the other two is 3.85. What is the average of the remaining two numbers?
(a) 4.5 (b) 4.6
(c) 4.7 (d) 4.8
- The average of 5 consecutive numbers is n . If the next two numbers are also included the average will
(a) remain the same (b) increase by 1
(c) increase by 1.4 (d) increase by 2
- The average of 11 numbers is 10.9. If the average of the first six numbers is 10.5 and that of the last six numbers is 11.4, then the middle number is :
(a) 11.5 (b) 11.4
(c) 11.3 (d) 11.0
- The average temperature for the first four days of a week is 40.2°C and that of the last four days is 41.3°C . If the average temperature for the whole week is 40.6°C , then the temperature on the fourth day is
(a) 40.8°C (b) 38.5°C
(c) 41.3°C (d) 41.8°C
- A person covers half his journey by train at 60 kmph, the remainder half by bus at 30 kmph and the rest by cycle at 10 kmph. Find his average speed during the entire journey.
(a) 36 kmph (b) 24 kmph
(c) 48 kmph (d) None of these
- The marks obtained by Hare Rama in Mathematics, English and Biology are respectively 93 out of 100, 78 out of 150 and 177 out of 200. Find his average score in percent.
(a) 87.83 (b) 86.83
(c) 76.33 (d) 77.33
- P is going to Delhi from Gurgaon by his car at a speed of 40 km/h. While coming back, he returns with a speed of x km/h. What should be the value of x so that his average speed during the entire journey is 80 km/h?
(a) 160 km/h (b) 40 km/h
(c) 120 km/h (d) It is not possible
- The average age of Mr. and Mrs Sinha at the time of their marriage in 1972 was 23 years. On the occasion of their anniversary in 1976, they observed that the average age of their family had come down by 4 years compared to their average age at the time of their marriage. This was due to the fact that their son Vicky was born during that period. What was the age of Vicky in 1980?
(a) 6 (b) 7
(c) 8 (d) 5
- A library has an average of 510 visitors of Sunday and 240 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is
(a) 250 (b) 276
(c) 280 (d) 285
- The mean of 30 values was 150. It was detected on rechecking that one value 165 was wrongly copied as 135 for the computation of the mean. Find the correct mean.
(a) 151 (b) 149
(c) 152 (d) None of these
- The average of 10 numbers is 40.2. Later it is found that two numbers have been wrongly copied. The first is 18 greater than the actual number and the second number added is 13 instead of 31. Find the correct average.
(a) 40.2 (b) 40.4
(c) 40.6 (d) 40.8
- In 1919, W. Rhodes, the Yorkshire cricketer, scored 891 runs for his county at an average of 34.27; in 1920, he scored 949 runs at an average of 28.75; in 1921, 1329 runs at an average of 36.70. What was his county batting average for the four years?
(a) 36.23 (b) 37.81
(c) 35.88 (d) 28.72

16. The average salary of all the workers in a workshop is ₹ 8,000. The average salary of 7 technicians is ₹ 12,000 and the average salary of the rest is ₹ 6,000. The total number of workers in the workshop is :
 (a) 21 (b) 20
 (c) 23 (d) 22
17. The average monthly salary of employees, consisting of officers and workers, of an organisation is ₹ 3000. The average salary of an officer is ₹ 10,000 while that of a worker is ₹ 2000 per month. If there are total 400 employees in the organisation, find the number of officers.
 (a) 60 (b) 50
 (c) 80 (d) 40
18. Of the three numbers, the first is twice the second and the second is twice the third. The average of the reciprocal of the numbers is $\frac{7}{72}$. The numbers are
 (a) 16, 8, 4 (b) 20, 10, 5
 (c) 24, 12, 6 (d) 36, 18, 9
19. In a bag, there are 150 coins of ₹ 1, 50 p and 25 p denominations. If the total value of coins is ₹ 150, then find how many rupees can be constituted by 50 coins.
 (a) ₹ 16 (b) ₹ 20
 (c) ₹ 28 (d) None of these
20. The average age of a group of persons going for picnic is 16 years. Twenty new persons with an average age of 15 years join the group on the spot due to which their average age becomes 15.5 years. The number of persons initially going for picnic is
 (a) 5 (b) 10
 (c) 20 (d) 30
21. The average weight of 47 balls is 4 gm. If the weight of the bag (in which the balls are kept) be included, the calculated average weight per ball increases by 0.3 gm. What is the weight of the bag?
 (a) 14.8 gm (b) 15.0 gm
 (c) 18.6 gm (d) None of these
22. On an average 300 people watch the movie in Sahu cinema hall on Monday, Tuesday and Wednesday and the average number of visitors on Thursday and Friday is 250. If the average number of visitors per day in the week be 400, then the average number of people who watch the movie in weekends (i.e., on Saturday and Sunday) is
 (a) 500 (b) 600
 (c) 700 (d) None of these
23. A train travels with a speed of 20 m/s in the first 10 minutes, goes 8.5 km in the next 10 minutes, 11 km in the next 10, 8.5 km in the next 10 and 6 km in the next 10 minutes. What is the average speed of the train in kilometer per hour for the journey described?
 (a) 42 kmph (b) 35.8 kmph
 (c) 55.2 kmph (d) 46 kmph

24. Find the average increase rate if increase in the population in the first year is 30% and that in the second year is 40%.
 (a) 41 (b) 56
 (c) 40 (d) 38
25. Find the average weight of four containers, if it is known that the weight of the first container is 100 kg and the total of the second, third and fourth containers' weight is defined by $f(x) = x^2 - 3/4(x^2)$ where $x = 100$
 (a) 650 kg (b) 900 kg
 (c) 750 kg (d) 450 kg

Directions for Qs. 26–28 : Read the information given below and answer the questions that follow :

There are 60 students in a class. These students are divided into three groups A, B and C of 15, 20 and 25 students each. The groups A and C are combined to form group D.

26. What is the average weight of the students in group D?
 (a) More than the average weight of A
 (b) More than the average weight of C
 (c) Less than the average weight of C
 (d) Cannot be determined
27. If one student from Group A is shifted to group B, which of the following will be true?
 (a) The average weight of both groups increases
 (b) The average weight of both the groups decreases
 (c) The average weight of the class remains the same
 (d) Cannot be determined
28. If all the students of the class have the same weight, then which of the following is false?
 (a) The average weight of all the four groups is the same.
 (b) The total weight of A and C is twice the total weight of B.
 (c) The average weight of D is greater than the average weight of A.
 (d) The average weight of all the groups remains the same even if the number of students are shifted from one group to another.

Directions for Qs. 29–31: Eight years ago there were 5 members in the Arthur's family and then the average age of the family was 36 years. Mean while Arthur got married and gave birth to a child. Still the average age of his family is same now.

29. The present age of his wife is
 (a) 25 years (b) 26 years
 (c) 32 years (d) Data insufficient
30. The age of his wife at the time of his child's birth was. If the difference between the age of her child and herself was 26 years
 (a) 25 years (b) 26 years
 (c) 20 years (d) Can't be determined

31. The age of Arthur at the time of his marriage was
 (a) 22 years (b) 23 years
 (c) 26 years (d) Can't be determined
32. The average age of a group of 14 persons is 27 years and 9 months. Two persons, each 42 years old, left the group. What will be the average age of the remaining persons in the group?
 (a) 26.875 years (b) 26.25 years
 (c) 25.375 years (d) 25 years
33. A school has only four classes that contain 10, 20, 30 and 40 students respectively. The pass percentage of these classes are 20%, 30%, 60% and 100% respectively. Find the pass % of the entire school.
 (a) 56% (b) 76%
 (c) 34% (d) 66%
34. Find the average of $f(x)$, $g(x)$, $h(x)$, $d(x)$ at $x = 10$. $f(x)$ is equal to $x^2 + 2$, $g(x) = 5x^2 - 3$, $h(x) = \log x^2$ and $d(x) = (4/5)x^2$.
 (a) 170 (b) 170.25
 (c) 70.25 (d) 70
35. The average of 'n' numbers is z. If the number x is replaced by the number x^1 , then the average becomes z^1 . Find the relation between n, z, z^1 , x and x^1 .
 (a) $\left[\frac{z^1 - 2}{x^1 - x} \quad \frac{1}{n} \right]$ (b) $\left[\frac{x^1 - x}{z^1} \quad \frac{1}{n} \right]$
 (c) $\left[\frac{z - z^1}{x - x^1} \quad \frac{1}{n} \right]$ (d) $\left[\frac{x - x^1}{z - z^1} \quad \frac{1}{n} \right]$
36. A man's average expenditure for the first 4 months of the year was ₹ 251.25. For the next 5 months the average monthly expenditure was ₹ 26.27 more than what it was during the first 4 months. If the person spent ₹ 760 in all during the remaining 3 months of the year, find what percentage of his annual income of ₹ 3000 he saved in the year.
 (a) 14% (b) -5.0866%
 (c) 12.5% (d) None of these
37. A curious student of Statistics calculated the average height of all the students of his class as A. He also calculated the average of the average heights of all the possible pairs of students (two students taken at a time) as B. Further, he calculated the average of the average heights of all the possible triplets of students (three students taken at a time) as C. Which of the following is true of the relationship among A, B and C?
 (a) $A + 2B = C$ (b) $A + B = 2C$
 (c) $A = B = 3C$ (d) None of these
38. We write down all the digits from 1-9 side by side. Now we put '+' between as many digits as we wish to, so that the sum of numbers become 666. It is explained below
 $1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9 = 666$
 Now suppose we put plus signs at following places.
 $12 + 345 + 67 + 89 = 513$
 Since there are four numbers, so the average can be calculated by dividing the sum by 4. What is the average if the sum is 666?
 (a) 166.5 (b) 111
 (c) 133.2 (d) Cannot be determined

Standard Level

1. The average weight of 3 men A, B and C is 84 kg. Another man D joins the group and the average now becomes 80 kg. If another man E, whose weight is 3 kg more than that D, replaces A then the average weight of B, C, D and E becomes 75 kg. The weight of A is
 (a) 70 kg (b) 72 kg
 (c) 79 kg (d) 78 kg
2. In Arun's opinion, his weight is greater than 65 kg but less than 72 kg. His brother does not agree with Arun and he thinks that Arun's weight is greater than 60 kg but less than 70 kg. His mother's view is that his weight cannot be greater than 68 kg. If all of them are correct in their estimation, what is the average of different probable weights of Arun?
 (a) 67 kg (b) 68 kg
 (c) 69 kg (d) None of these
3. In the month of July of a certain year, the average daily expenditure of an organisation was ₹68. For the first 15 days of the month, the average daily expenditure was ₹85 and for the last 17 days, ₹51. Find the amount spent by the organisation on the 15th of the month.
 (a) ₹42 (b) ₹36
 (c) ₹34 (d) ₹52
4. There are five boxes in a cargo hold. The weight of the first box is 200 kg and the weight of the second box is 20% higher than the weight of the third box, whose weight is 25% higher than the first box's weight. The fourth box at 350 kg is 30% lighter than the fifth box. Find the difference in the average weight of the four heaviest boxes and the four lightest boxes.
 (a) 51.5 kg (b) 75 kg
 (c) 37.5 kg (d) 112.5 kg
5. Of the three numbers, the average of the first and the second is greater than the average of the second and the third by 15. What is the difference between the first and the third of the three numbers?
 (a) 15 (b) 45
 (c) 60 (d) None of these
6. The average monthly expenditure of Ravi was ₹1100 during the first 3 months, ₹2200 during the next 4 months and ₹4620 during the subsequent five months of the year. If the total saving during the year was ₹2100, find Ravi's average monthly income.
 (a) ₹1858 (b) ₹3108.33
 (c) ₹3100 (d) None of these
7. Rajeev earns $\frac{3}{2}$ times in January, April, July and October than his average earning of ₹600 per month in the rest of the month. So his savings in the January, April, July and October goes to $\frac{5}{4}$ times that of the rest months saving of ₹400 per month in the year. The average expenditure of per month is:
 (a) ₹266.66 (b) ₹250
 (c) ₹233.33 (d) ₹433.33
8. There were five sections in MAT paper. The average score of Pooja in first 3 sections was 83 and the average in the last 3 sections was 97 and the average of all the sections (i.e., whole paper) was 92, then her score in the third section was
 (a) 85 (b) 92
 (c) 88 (d) None of these
9. Mr. Anant Roy, the renowned author, recently got his new novel released. To his utter dismay he found that for the 1,007 pages on an average there were 2 mistakes every page. While, in the first 612 pages there were only 434 mistakes, they seemed to increase for the latter pages. Find the average number of mistakes per page for the remaining pages.
 (a) 6 (b) 4
 (c) 2 (d) None of these
10. If the average marks of 17 students in a class is A. The marks of the students when arranged in either an ascending or a descending order was found to be in arithmetic progression. The class teacher found that the students who were ranked 3rd, 7th, 8th, 11th, 15th had copied in the exam and hence got all of them rusticated. The average of the remainder of the class was B. Then
 (a) $A = B$ (b) $A > B$
 (c) $A < B$ (d) Data Insufficient
11. In hotel Trident, the rooms are numbered from 101 to 130 on the first floor, 221 to 260 on the second floor and 306 to 345 on the third floor. In the month of June 2012, the room occupancy was 60% on the first floor, 40% on the second floor and 75% on the third floor. If it is also known that the room charges are ₹200, ₹100 and ₹150 on each of the floors, then find the average income per room for the month of June 2012.
 (a) ₹151.5 (b) ₹88.18
 (c) ₹78.3 (d) ₹65.7
12. The average age of a couple is 25 years. The average age of the family just after the birth of the first child was 18 years. The average age of the family just after the second child was born was 15 years. The average age of the family after the third and the fourth children (who are twins) were born was 12 years. If the present average age of the family of six persons is 16 years, how old is the eldest child?
 (a) 6 years (b) 7 years
 (c) 8 years (d) 9 years

13. The average monthly rainfall for a year in Guntur district is 2.7 inches, the average for the first 7 months is 1.1 inches less than the annual average. If the total rainfall for the next 4 months is 20.8 inches, then the rainfall in the last month will be

(a) 0.1 inch (b) 0.2 inch
(c) 0.4 inch (d) 0.6 inch

Directions for Questions 14–17

Read the following passage and answer the questions the follow. In a family of five persons A, B, C, D and E, each and everyone loves one another very much. Their birthdays are in different months and on different dates. A remembers that his birthday is between 25th and 30th, of B it is between 20th and 25th, of C it is between 10th and 20th, of D it is between 5th and 10th and of E it is between 1st to 5th of the month, the sum of the date of birth is defined as the addition of the date and the month, for example 12th January will be written as 12/1 and will add to a sum of the date of 13. (Between 25th and 30th includes both 25 and 30).

14. What may be the maximum average of their sum of the dates of birth?
(a) 24.6 (b) 15.2
(c) 28 (d) 32
15. What may be the minimum average of their sum of the dates of births?
(a) 24.6 (b) 15.2
(c) 28 (d) 32
16. If it is known that the dates of birth of three of them are even numbers then find maximum average of their sum of the dates of birth.
(a) 24.6 (b) 15.2
(c) 27.6 (d) 28
17. If the date of birth of four of them are prime numbers, then find the maximum average of the sum of their dates of birth.
(a) 27.2 (b) 26.4
(c) 28 (d) None of these
18. Eleven years earlier the average age of a family of 4 members was 28 years. Now the age of the same family with six members is yet the same, even when 2 children were born in this period. If they belong to the same parents and the age of the first child at the time of the birth of the younger child was same as there were total family members just after the birth of the youngest members of this family, then the present age of the youngest member of the family is
(a) 3 years (b) 5 years
(c) 6 years (d) None of these
19. The average earning of a group of persons is ₹ 50 per day. The difference between the highest earning and lowest earning of any two persons of the group is ₹ 45. If these

two people are excluded the average earning of the group decreased by ₹ 1. If the minimum earning of the person in the group lies between 42 and 47 and the number of persons initially in the group was equal to a prime number, with both its digits prime. The number of persons in the group initially was:

(a) 29 (b) 53
(c) 31 (d) None of these

20. The class X of a Vidhyalaya has four sections: A, B, C and D. The average weight of the students of A, B, C together and A, C, D together are 45 kg and 55 kg respectively, while the average weight of the students of A, B, D together and B, C, D together are 50kg and 60kg respectively. Which of the following could be the average weight of the students of all the four sections together?
(a) 47.6 kg (b) 52.5 kg
(c) 53.7 kg (d) 56.5 kg

21. The average market price of three shares x, y and z is ₹ m. Shares x and y lose ₹ n each and z gains ₹ $\frac{n}{2}$. As a result, the average market price of the three shares decrease by ₹ 1. The value of n is
(a) 2 (b) 3
(c) 4 (d) dependent of x

Directions Qs. 22–25: Read the following and answer the questions that follows.

During a cricket match. India playing against NZ scored in the following manner:

Partnership	Runs scored
1st wicket	112
2nd wicket	58
3rd wicket	72
4th wicket	92
5th wicket	46
6th wicket	23

22. Find the average runs scored by the first four batsmen
(a) 83.5 (b) 60.5
(c) 66.8 (d) Cannot be determined.
23. The maximum average runs scored by the first five batsmen could be
(a) 80.6 (b) 66.8
(c) 76 (d) Cannot be determined
24. The minimum average runs scored by the last five batsmen to get out could be
(a) 53.6 (b) 44.4
(c) 66.8 (d) 0
25. If the fifth down batsman gets out for a duck, then find the average runs scored by the first six batsmen.
(a) 67.1 (b) 63.3
(c) 48.5 (d) Cannot be determined

Expert Level

- The average price of 3 diamonds of same weights is ₹5 crore, where the average price of the two costliest diamonds is double the price of the cheapest diamond. The price of the cheapest diamond is
(a) 3 crore (b) 5 crore
(c) 1.66 crore (d) can't be determined
 - Three Maths classes: X, Y and Z take an algebra test. The average score in class X is 83. The average score in class Y is 76. The average score in class Z is 85. The average score of all students in classes X and Y together is 79. The average score of all students in classes Y and Z together is 81. What is the average for all the three classes?
(a) 81 (b) 81.5
(c) 82 (d) 84.5
 - A set of consecutive positive integers beginning with 1 is written on the blackboard. A student came along and erased one number. The average of the remaining numbers is $35\frac{7}{17}$. What was the number erased?
(a) 7 (b) 8
(c) 9 (d) None of these
 - A salesman gets a bonus according to the following structure: If he sells articles worth ₹ x then he gets a bonus of $(x/100 - 1)$. In the month of January, his sales value was ₹ 100, in February it was ₹ 200, from March to November it was ₹ 300 for every month and in December it was ₹ 30 per month from his employer. Find his average income per month during the year.
(a) ₹ 31.25 (b) ₹ 30.34
(c) ₹ 32.5 (d) ₹ 34.5
 - There were 42 students in a hostel. Due to the admission of 13 new students, the expenses of the mess increase by ₹ 31 per day while the average expenditure per head diminished by ₹ 3. What was the original expenditure of the mess?
(a) ₹ 633.23 (b) ₹ 583.3
(c) ₹ 623.3 (d) ₹ 632
- Directions (Qs. 6–7) :** Refer to the data below and answer the questions that follow.
- Five heavy weight boxers measure their weights. Following results were obtained :
- P is heavier than R by 14 lb.
 - B is lighter than S by 10 lb.
 - M 's weight is equal to the average weight of the four other boxers.
 - P 's weight and B 's weight taken together equals the weight of M and S .
- What is the average of the weights of M and R ?
(a) 102 lb (b) 104 lb
(c) 100 lb (d) None of these
 - What is the average of the weights of P , S and B ?
(a) 102.3 lb (b) 107.3 lb
(c) 105.3 lb (d) None of these
 - The weight of a body as calculated by the average of 7 different experiments is 53.735 gm. The average of the first three experiments is 54.005 gm, of the fourth is 0.004 gm greater than the fifth, while the average of the sixth and seventh experiment was 0.010 gm less than the average of the first three. Find the weight of the body obtained by the fourth experiment.
(a) 49.353 gm (b) 51.712 gm
(c) 53.072 gm (d) 54.512 gm
 - In a set of prime and composite numbers, the composite numbers are twice the number of prime numbers and the average of all the numbers of the set is 9. If the number of prime numbers and composite numbers are exchanged then the average of the set of numbers is increased by 2. If during the exchange of the numbers the average of the prime numbers and composite numbers individually remained constant, then the ratio of the average of composite numbers to the average of prime numbers (initially) was
(a) $\frac{7}{13}$ (b) $\frac{13}{7}$
(c) $\frac{9}{11}$ (d) None of these
 - The average marks of Sameer decreased by 1, when he replaced the subject in which he has scored 40 marks by the other two subjects in which he has just scored 23 and 25 marks respectively. Later he has also included 57 marks of Computer Science, then the average marks increased by 2. How many subjects were there initially?
(a) 6 (b) 12
(c) 15 (d) can't be determined
 - There are three categories of jobs A, B and C. The average salary of the students who got the job of A and B categories is 26 lakh per annum. The average salary of the students who got the job of B and C category is 44 lakh per annum and the average salary of those students who got the job of

A and C categories is 34 lakh per annum. The most appropriate (or closest) range of average salary of all the three categories (if it is known that each student gets only of jobs i.e., A, B and C) :

- (a) lies between 30 and 44 (b) lies between 28 and 34
(c) lies between 34 and 43 (d) lies between 29 and 48
12. There are only five people in the Aman Verma's family. Aman, his wife, a son and two daughters. The younger daughter's age is $\frac{4}{5}$ th of the elder daughter's age. The age of eldest daughter is $\frac{3}{8}$ times that of her father Aman and the age of the son is $\frac{1}{5}$ th that of his father Aman. 4 years ago the age of her wife was 8 times that of his son and now the sum of the ages of the younger daughter and wife is same as the sum of the ages of Aman and his son. The average age of the family is:
(a) 22.22 years (b) 22.4 years
(c) 21.2 years (d) None of these
13. Sachin Tendulkar has a certain batting average N (a whole number) in his career of 86 innings. In the 87th inning, he gets out after scoring 270 runs which increases his batting average by a whole number. The batting average is calculated by dividing the total number of runs scored by the total number of innings played by the player. How many values of his new average is/are possible?
(a) 0 (b) 1
(c) 2 (d) None of these

Directions (Qs. 14–15): It is possible to arrange eight of the nine numbers 2, 3, 4, 5, 7, 10, 11, 12, 13 in the vacant squares of the 3

by 4 array shown below so that the arithmetic average of the numbers in each row and column is the same integer.

1			15
	9		
		14	

14. The arithmetic average is
(a) 6 (b) 7
(c) 8 (d) 9
15. Which one of the nine numbers must be left out when completing the array ?
(a) 4 (b) 5
(c) 7 (d) 10
- Direction for questions :** In the entrance examination of IIMs, there were 200 questions, each of which carried the same marks. A correct answer gets 2 marks and there is 100% negative marking. A total of 70 candidates took the exam and it was later found that the average marks obtained by these 70 candidates was 240. The candidates were not required to attempt all the questions. None of the candidates got more number of incorrect answers than correct answers.
16. When the scores of the top four students are deleted, the average score of the remaining 66 students falls by 6 marks. Assume that it is possible for two or more students to have the same net score. What is the minimum score possible for the fourth ranking student if no student got a net score of more than 352?
(a) 280 (b) 300
(c) 308 (d) 320

Test Yourself

- One-fourth of a certain journey is covered at the rate of 25 km/h, one-third at the rate of 30 km/h and the rest at 50 km/h. Find the average speed for the whole journey.
(a) 600/53 km/h (b) 1200/53 km/h
(c) 1800/53 km/h (d) 1600/53 km/h
- The average of 20 numbers is zero. Of them, at the most, how many may be greater than zero?
(a) 0 (b) 1
(c) 10 (d) 19
- The average age of A and B is 20 years. If C were to replace A, the average would be 19 and if C were to replace B, the average would be 21. What are the age of A, B and C?
(a) 22, 18, 20 (b) 20, 20, 18
(c) 18, 22, 20 (d) None of these
- A batsman in his 12th innings makes a score of 65 and thereby increases his average by 2 runs. What is his average after the 12th innings if he had never been 'not out'?
(a) 42 (b) 43
(c) 44 (d) 45
- The average of a batsman for 40 innings is 50 runs. His highest score exceeds his lowest score by 172 runs. If these two innings are excluded, his average drops by 2 runs. Find his highest score.
(a) 172 (b) 173
(c) 174 (d) 175
- The mean of 30 values was 150. It was detected on rechecking that one value 165 was wrongly copied as 135 for the computation of the mean. Find the correct mean.
(a) 151 (b) 149
(c) 152 (d) None of these
- A car owner buys petrol at ₹ 7.50, ₹ 8.00 and ₹ 8.50 per litre for three successive years. What approximately is his average cost per litre of petrol if he spends ₹ 4000 each year?
(a) ₹ 8 (b) ₹ 9
(c) ₹ 7.98 (d) ₹ 8.50
- In Arun's opinion, his weight is greater than 65 kg but less than 72 kg. His brother does not agree with Arun and he thinks that Arun's weight is greater than 60 kg but less than 70 kg. His mother's view is that his weight cannot be greater than 68 kg. If all of them are correct in their estimation, what is the average of different probable weights of Arun?
(a) 67 kg (b) 68 kg
(c) 69 kg (d) None of these
- The average weight of 5 men is decreased by 3 kg when one of them weighing 150 kg is replaced by another person. This new person is again replaced by another person whose weight is 30 kg lower than the person he replaced. What is the overall change in the average due to this dual change?
(a) 6 kg (b) 9 kg
(c) 12 kg (d) 15 kg
- 19 persons went to a hotel for a combined dinner party. 13 of them spent ₹ 79 each on their dinner and the rest spent ₹ 4 more than the average expenditure of all the 19. What was the total money spent by them?
(a) 1628.4 (b) 1536
(c) 1492 (d) None of these
- There are a certain number of pages in a book. Arjun tore a certain page out of the book and later found that the average of the remaining page numbers is $46\frac{10}{13}$. Which of the following were the page number of the page that Arjun had torn?
(a) 57 and 58 (b) 59 and 60
(c) 45 and 46 (d) 47 and 48
- In an exam, the average was found to be x marks. After deducting computational error, the average marks of 94 candidates got reduced from 84 to 64. The average thus came down by 18.8 marks. The numbers of candidates who took the exam were:
(a) 100 (b) 90
(c) 110 (d) 105
- Ten years ago, the ages of the members of a joint family of eight people added up to 231 years. Three years later, one member died at the age of 60 years and a child was born during the same year. After another three years, one more member died, again at 60, and a child was born during the same year. The current average age of this eight-member joint family is nearest to
(a) 24 years (b) 23 years
(c) 22 years (d) 21 years
- I was born 30 years after my father was born. My sister was born 25 years after my mother was born. The average age of my family is 26.25 years right now. My sister will get married 4 years from now and will leave the family. Then the average age of the family will be $\frac{107}{3}$ years. What is the age of my father?
(a) 30 year (b) 35 year
(c) 40 year (d) 45 year
- The average weight of A, B and C is x kg. A and C lose y kg each after dieting and B gains $\frac{y}{2}$ kg. After this their average weight decreases by 1 kg. Find y.
(a) 1 kg (b) 2 kg
(c) 3 kg (d) Cannot be determined

Hints & Solutions

Foundation Level

1. (c) Age of the class teacher = $25 \times 16 - 24 \times 15$
 $= 400 - 360 = 40$ yrs.
2. (c) Age of the teacher = $(37 \times 15 - 36 \times 14)$ years
 $= 51$ years.
3. (b) Required average marks

$$\frac{40 \times 50 + 35 \times 60 + 45 \times 55 + 42 \times 45}{40 + 35 + 45 + 42}$$

$$\frac{2000 + 2100 + 2475 + 1890}{162} = \frac{8464.9997}{162}$$
 $= 52.25$
4. (b) Sum of the remaining two numbers
 $= (3.95 \times 6) - [(3.4 \times 2) + (3.85 \times 2)]$
 $= 23.70 - (6.8 + 7.7) = 23.70 - 14.5 = 9.20$
 $\therefore$ Required average = $\left(\frac{9.2}{2}\right) = 4.6$.
5. (b) Check as follows,

$$\frac{1+2+3+4+5}{5} = 3$$

$$\frac{1+2+3+4+5+6+7}{7} = 4$$
6. (a) The middle number = Sum of the first six + Sum of the last six – Sum of all the 11
 $= 6 \times 10.5 + 6 \times 10.5 - 11 \times 10.9$
 $= 63 + 68.4 - 119.9$
 $= 11.5$
7. (d) Temperature on the fourth day
 $= 40.2 \times 4 + 41.3 \times 4 - 40.6 \times 7$
 $= 160.8 + 165.2 - 284.2 = 41.8^\circ\text{C}$
8. (b) Recognise that the journey by bus and that by cycle are of equal distance. Hence, we can use the short cut illustrated earlier to solve this part of the problem.
 Using the process explained above, we get average speed of the second half of the journey as
 $10 + 1 \times 5 = 15$ kmph
 Then we employ the same technique for the first part and get $15 + 1 \times 9 = 24$ kmph
9. (d) His total score is $93 + 78 + 177 = 348$ out of 450% score = 77.33.
10. (d) $80 \quad \frac{2 \cdot x \cdot 40}{40 + x}$
 $40 + x = x$
 Hence, not possible
11. (b) Sum of ages of Mr. and Mrs. Sinha in 1972 = 46 years
 Sum of age of their family in 1976 = $19 \times 3 = 57$ years
 Sum of ages of Mr. and Mrs. Sinha in 1976 = $(46 + 8)$ years
 $= 54$ years
 $\therefore$ Age of Vicky in 1980 = $57 - 54 + 4 = 7$ years.
12. (d) Since the month begins with Sunday, so there will be five Sundays in the month
 $\therefore$ Required average = $\left(\frac{510 \times 5 + 240 \times 25}{30}\right)$
 $= \frac{8550}{30} = 285$
13. (a) Corrected mean $\frac{150 \times 30 - 135 + 165}{30}$

$$\frac{4500 - 135 + 165}{30} = \frac{4530}{30} = 151$$
14. (a) Sum of 10 numbers = 402
 Corrected sum of 10 numbers
 $= 402 - 13 + 31 - 18 = 402$
 Hence, correct average $\frac{402}{10} = 40.2$
15. (c) Find out the number of innings in each year. Then the answer will be given by:

$$\frac{\text{Total runs in 4 years}}{\text{Total innings in 4 years}} \quad (4270/119 = 35.88)$$
16. (a) Let the total no. of workers be x .
 Now, $8000x = 7 \times 12000 + (x - 7) \times 6000$
 $\Rightarrow x = \frac{42000}{2000} = 21$
17. (b) Let the number of officers be x .
 Number of workers = $400 - x$
 $\Rightarrow 10000 \times x + 2000(400 - x) = 3000(400)$
 $\Rightarrow 10000x + 800000 - 2000x = 12,00,000$
 $\Rightarrow 4x = 600 - 400 = 200 \Rightarrow x = 50$
 $\therefore$ Number of officers = 50
Shortcut Method:

$$\begin{array}{ccccc} 10000 & & & & 2000 \\ & \searrow & & \swarrow & \\ & 3000 & & 7000 & \\ & \swarrow & & \searrow & \\ 1000 & & & & \end{array}$$
 $1 : 7$
 Ratio of officers to workers = $1 : 7$
 Number of officers = $\frac{1}{8} \times 400 = 50$

18. (c) Let the third number be x . Then, second number = $2x$. First number = $4x$.

$$\therefore \frac{1}{x} + \frac{1}{2x} + \frac{1}{4x} = \left(\frac{7}{72} \times 3 \right) \text{ or } \frac{7}{4x} = \frac{7}{24} \text{ or } 4x = 24$$

$$\text{or } x = 6$$

So, the numbers are 24, 12 and 6.

19. (d) For 150 coins to be of a value of ₹150, using only 25 paise, 50 paise and ₹1 coins, we cannot have any coins lower than the value of ₹1. Thus, the number of 50 paise coins would be 0. Option (d) is correct.

20. (c) Let the initial number of persons be x . Then,
 $16x + 20 \times 15 = 15.5(x + 20) \Leftrightarrow 0.5x = 10$
 $\Leftrightarrow x = 20$.

21. (d) The average weight per ball is asked. Hence the bag does not have to be counted as the 48th item.

22. (c) $400 \times 7 = (300 \times 3) + (250 \times 2) + (n \times 2) \Rightarrow 700$.

23. (c) Find the total distance covered in each segment of 10 minutes. You will get total distance = 46 kilometers in 50 mins.

$$\therefore \text{Average speed} = \frac{46 \times 60}{50} = 55.2 \text{ kmph}$$

24. (a) $100 \rightarrow 130 \rightarrow 182$. Hence, $82/2 = 41$.

25. (a) Put $x = 100$ to get the weight of the containers. Use these weights of find average weight as $2600/4 = 650$.

26. (d) Average weight of the students in group D cannot be determined since we do not know the average weight of each student. The given data is insufficient to compare its average with other groups.

27. (c) If one student from group A is shifted to group B, still there is no effect on the whole class. In any case, the no. of students inside the class is same. Hence the average weight of the class remains same.

28. (c) Since all the students of the class have the same weight, then the average of weight of any group of any no. of students will be the same as that of each students weight. Hence, the average weight of D cannot be greater than average weight of A.

Solutions for question numbers 29–31:

	No. of family members	Average	Total
8 years ago →	5	36	180
Presently →	(if) 5	$(36+8) = 44$	220
	7	36	252

29. (d) From the above explanation we have no any clue about his wife's age.

30. (b) Since we know that the difference between the age of any two persons remains always constant, while the

ratio of their ages gets changed as the time changes.

So, if the age of his child be x (presently)

Then the age of wife be $x + 26$ (presently)

Thus, the total age = $x + (x + 26) = 32$

$$[\because 252 - 220 = 32]$$

$$\Rightarrow x = 3$$

Therefore the age of her child is 3 years and herself is 29 years. Hence her age at the time of the birth of her child was 26 years.

Alternatively: As we have mentioned above that the age difference remains always constant. Therefore her age at the time of her child's birth was 26 years.

31. (d) Since there is no clue. So we can't determine.

32. (c) $(14 \times 27.75 - 20 + 42)/12 = 25.375$

33. (d) The number of pass candidates are $2 + 6 + 18 + 40 = 66$ out of a total of 100. Hence, 66%

34. (b) Put $x = 10$ in the given equations and find the average of the resultant values.

35. (c) $nz - x + x^1 = nz^1$ Simplify to get option (c) correct.

36. (b) $251.25 \times 4 + 277.52 \times 5 + 760 = 3152.6$

37. (d) Let the height of four students be 150, 160, 170, 180 cm then

$$A = \frac{150 + 160 + 170 + 180}{4} = \frac{660}{4} = 165 \text{ cm}$$

$$B = \frac{\frac{150+160}{2} + \frac{160+170}{2} + \frac{170+180}{2} + \frac{150+180}{2} + \frac{150+170}{2} + \frac{160+180}{2}}{6}$$

$$= \frac{155 + 165 + 175 + 165 + 160 + 170}{6} = \frac{990}{6} = 175 \text{ cm}$$

$$= \frac{670}{4} = 167.5$$

Similarly, $C = 167.5$

Now going through the options.

38. (d) We can get the sum 666 in two ways:

$$1 + 2 + 3 + 4 + 567 + 89 = 666$$

$$\text{or } 123 + 456 + 78 + 9 = 666$$

So, average cannot be uniquely determined.

Standard Level

1. (c) D's weight = $4 \times 80 - 3 \times 84 = 320 - 252 = 68$.
 E's weight = $68 + 3 = 71$.
 Now, we know that $A + B + C + D = 4 \times 80 = 320$
 and $B + C + D + E = 78 \times 4 = 312$.
 Hence, A's weight is 8 kg more than E's weight.
 $A = 71 + 8 = 79$.

2. (d) Let Arun's weight be x kg.
According to Arun, $65 < x < 72$.
According to Arun's brother, $60 < x < 70$.
According to Arun's mother, $x < 68$
The values satisfying all the above conditions are 66 and 67
 $\therefore$ Required average
$$= \left(\frac{66+67}{2} \right) = \left(\frac{133}{2} \right) = 66.5 \text{ kg}$$
3. (c) Standard question requiring good calculation speed. Obviously, the 15th day is being double counted. Calculations can be reduced by thinking as:
Surplus in first 15 days – Deficit in last 17 days
 $= 255 - 289 \rightarrow$ Net deficit of 34. This means that the average is reducing by 34 due to the double counting of the 15th day. This can only mean that the 15th day's expenditure is $68 - 34 = 34$.
(Lengthy calculations would have yielded the following calculations:
 $85 \times 15 + 51 \times 17 - 68 \times 31 = 34$)
4. (b) The weight of the boxes are 1st box $\rightarrow 200$, 3rd box $\rightarrow 250$ kg, 2nd box $\rightarrow 300$ kg, 4th box $\rightarrow 350$ and 5th box $\rightarrow 500$ kg. Hence difference between the heavier 4 and the lighter 4 is 300. Hence, difference in the averages is 75.
5. (d) Let the numbers are x , y and z .
Then, $\left(\frac{x+y}{2} \right) - \left(\frac{y+z}{2} \right) = 15$ or $(x+y) - (y+z)$
 $= 30$ or $x - z = 30$
6. (b) Required average income = (Total expenditure + total savings)/12
 $= [(1100 \times 3 + 2200 \times 4 + 4620 \times 5) + 2100]/12$
 $= 37300/12 = 3108.333$
7. (a) Earning in the 8 months = $600 \times 8 = 4800$
Earning in the 4 months = $\left(600 \times \frac{3}{2} \right) \times 4 = 3600$
Total earning = 8400
Saving in 8 months = $400 \times 8 = 3200$
Saving in 4 months = $\left(400 \times \frac{5}{4} \right) \times 4 = 2000$
Total savings = 5200
Total expenditure for 12 months = $8400 - 5200 = 3200$
Therefore average saving per month = $\frac{3200}{12} = 266.66$
8. (d) $a + b + c + d + e = 5 \times 92 = 460$
 $a + b + c = 3 \times 83 = 249$
 $c + d + e = 3 \times 97 = 291$
 $\therefore c = (a + b + c) + (c + d + e) - (a + b + c + d + e)$
or $c = 540 - 460$ or $c = 80$
9. (b) Total mistakes = $1007 \times 2 = 2014$
Let x be average mistake per page for the remaining pages
 $434 + 395x = 2014$
 $395x = 1580$
 $x = 4$
10. (a) The average (in this case, it is A), surplus generated by the marks of 3rd student will be same as deficit incurred due to 15th student. So, rustinating both of them is not going to create any difference on average marks of the class (remember marks are in AP). And similar will be the impact of rustinating 7th student and 11th student and then finally 9th student. So, $A = B$
11. (a) The number of rooms is $18 + 16 + 30$ on the three floors respectively.
Total revenues are: $18 \times 200 + 16 \times 100 + 30 \times 150 = 9700$ required average = $9700/110 = 88.18$.
Note here that if you could visualize here that since the number of rooms is 110 the decimal values cannot be. (c) or (d) which effectively means that options 3 and 4 are rejected.
12. (d) The total age of the family at the birth of first child
 $= 18 \times 3 = 54$
While the total age of the couple at marriage
 $= 25 \times 2 = 50$.
 $\Rightarrow$ The years from marriage till the first child's birth
 $= \frac{54-50}{2} = 2$ years.
The total age of family at the birth of the second child.
 $= 15 \times 4 = 60$ years.
 $\Rightarrow$ Second child was born = $\frac{60-54}{3} = 2$ years after the first.
Similarly the twins were born = $\frac{(12 \times 6) - 60}{4} = 3$ years.
After the second child and today the twins are 4 years old.
($\therefore$ average age of the family became 16 years from 12 years)
 $\therefore$ Age of eldest son = $4 + 3 + 2 = 9$ years.
13. (c) Total annual rainfall = $2.7 \times 12 = 32.4$ inches
Rainfall for first seven months = $(2.7 - 1.1) \times 7 = 11.2$
Total for first 11 months = $11.2 + 20.8 = 32$ inches
Rainfall for last month = $32.4 - 32 = 0.4$ inches
14-17. You have to take between 25th and 30th to mean that both these dates are also included.
14. (c) The maximum average will occur when the maximum possible values are used. Thus:
A should have been born on 30th,
B on 25th,
C on 20th

D on 10th and E on 5th.

Further, the months of births in random order will have to be between August to December to maximize the average.

Hence total will be $30 + 25 + 20 + 10 + 5 + 12$

$+ 11 + 10 + 9 + 8 = 140$. Hence average is 28.

15. (b) The minimum average will be when we have $1 + 5 + 10 + 20 + 25 + 1 + 2 + 3 + 4 + 5 = 76$. Hence average is 15.2

16. (d) This does not change anything. Hence the answer is the 28.

17. (a) The prime dates must be 29th, 23rd, 19th and 5th. Hence, the maximum possible average will reduce by $4/5 = 0.8$. Hence, answer will be 27.2

18. (a)

	No. of family members	Average	Total
Eleven years earlier	4	28	112
Presently	if 4	39	156
	6	28	168

Since it is obvious that just after the birth of the youngest member (i.e., child) was 6 family members in the family. Therefore at the time of the birth of the youngest child the elder child's age was 6 years. Now the sum of their ages $= x + (x + 6) = 12 = (168 - 156)$
 $\Rightarrow x = 3$

19. (d) Let there be n people (initially) in the group, then the total earning of the group $= n \times 50$
 Again $n \times 50 = (n - 2) \times 49 + (2x + 45)$
 $\Rightarrow n = 2x - 53$; where x is the lowest earning of any person.

Now, since $42 < x < 47$ and $n \in$ prime numbers
 Then the only possible value of $n = 37$ for $x = 45$.

20. (b) Let a, b, c, d , the number of students in section A, B, C, D respectively
 then,

$$45(a + b + c) + 55(a + c + d) + 50(a + b + d) + 60(b + c + d)$$

$$= \frac{3(a + b + c + d)}{3(a + b + c + d)}$$

$$50 + \frac{5b + 10c + 15d}{3(a + b + c + d)}$$

Clearly, a, b, c, d are natural no. put $a = b = c = d = 1$

$$\text{Then, required average} = 50 + \frac{30}{12} = 50 + 2.5 = 52.5$$

21. (a) The net decrease in the average can be expressed

$$\frac{y + y - \left(\frac{y}{3}\right)}{3} = 1 \Rightarrow y = 2$$

Q. 22-25:

22. (d) You don't know who got out when. Hence, cannot be determined.
23. (a) Since possibilities are asked about, you will have to consider all possibilities. Assume, the sixth and seventh batsmen have scored zero. Only then will the possibility of the first 5 batsmen scoring the highest possible average arise. In this case the maximum possible average for the first 5 batsmen could be $403/5 = 80.6$.
24. (d) Again it is possible that only the first batsman has scored runs.
25. (d) We cannot find out the number of runs scored by the 7th batsman. Hence answer is (d).

Expert Level

1. (a) Let the price of $A > B > C$

$$\text{then } \frac{A + B}{2} = 2C$$

$$A + B = 4C$$

$$\text{Now, } A + B + C = 5 \times 3 = 15 \text{ crore}$$

$$\Rightarrow 5C = 15 \text{ crore}$$

$$\Rightarrow C = 3 \text{ crore}$$

2. (b) Let no. of students in classes X, Y and Z = x, y and z respectively.

$$\frac{83x + 76y}{x + y} = 79 \quad \dots(1)$$

$$\text{and } \frac{76y + 85z}{y + z} = 81 \quad \dots(2)$$

$$\text{We have to find } \frac{83x + 76y + 85z}{x + y + z}$$

$$\text{From (1) } 4x = 3y \text{ or } y = \frac{4x}{3};$$

$$\text{From (2) } 5y = 4z \text{ or } z = \frac{5}{4}y = \frac{5}{4} \times \frac{4x}{3} = \frac{5}{3}x$$

$$\therefore \text{ Required average } = \frac{83x + 76 \times \frac{4x}{3} + 85 \times \frac{5}{3}x}{\frac{3x}{3} + \frac{4x}{3} + \frac{5x}{3}}$$

$$\frac{83 + \frac{729}{3}}{4} = 81.5$$

3. (a) Let number erased be x
The average of the remaining no.

$$= \frac{\left(\frac{n(n+1)}{2} - x\right)}{n-1} = 35 \frac{7}{17} = \frac{602}{17}$$

Here $n = 69$ and $x = 7$ satisfy the above equation.

4. (c) Replace x with the sales value to calculate the bonus in a month.
Detailed method

$$\text{Bonus for Jan.} = \frac{100}{100} - 1 = -1$$

$$\text{Bonus for Feb.} = \frac{200}{100} - 1 = 1$$

$$\text{Bonus from March to November} = 9 \left(\frac{300}{100} - 1 \right) = 18$$

$$\text{Bonus for December} = \frac{1200}{100} - 1 = 11$$

Total Bonus = 29

$$\therefore \text{Total salary} = 12 \times 30 + 29 = 389$$

$$\text{Hence, average income per month} = \frac{365}{12} = 32.5$$

5. (a) $42A + 31 = 55(A - 3) \rightarrow 13A = 196 \rightarrow A = 196/13 = 15.07$. Total expenditure original = $15.07 \times 42 = 633.23$

6. (a), 7. (c)

Let p, m, r, s and b be the weights of boxers P, M, R, S and B respectively.

From data :

$$p = r + 14 \quad \dots(1)$$

$$b = s - 10 \quad \dots(2)$$

$$4m = p + b + r + s \quad \dots(3)$$

$$p + b = m + s \quad \dots(4)$$

$$p + b + m + r + s = 520 \quad \dots(5)$$

$$\text{From (3) and (5); } 5m = 520 \Rightarrow m = 104 \text{ lb}$$

$$\text{From (1) and (2); } p + b = r + s + 4 \quad \dots(6)$$

$$\text{From (4) and (6); } r + s + 4 = m + s$$

$$\Rightarrow r = m - 4 = 100 \text{ lb}$$

$$\text{From (1); } p = r + 14 = 114 \text{ lb}$$

$$\text{From (5); } 114 + b + 104 + s + 100 = 520$$

$$\Rightarrow b + s = 202 \quad \dots(7)$$

From (7) and (2); $b - s = 10$

$$\text{and } b + s = 202 \Rightarrow b = \frac{192}{2} = 96 \text{ lb and } s = 106 \text{ lb}$$

$\therefore$ Average of the weights of M and R

$$= \frac{104 + 100}{2} = 102 \text{ lb}$$

Average of the weights of P, S and B

$$= \frac{114 + 106 + 96}{3} = 105.3 \text{ lb.}$$

8. (c) Detailed method:- $54.005 - 53.735 = 0.27$

$$\text{Avg of Last 4} = 53.735 - \frac{0.27 \times 3}{4} = 53.53$$

$$\text{avg. 6th and 7th} = 54.005 - 0.010 = 53.995$$

$$6\text{th} + 7\text{th} = 53.995 \times 2 = 107.99$$

Let 4th be x

$$x + x - 0.004 + 107.99 = 53.53 \times 2$$

$$\therefore x = 53.072$$

9. (a) Let the average of prime numbers be P and average of composite numbers be C .

Again the number of prime numbers be x , then the number of composite numbers be $2x$.

$$\text{Then } \frac{Px + 2Cx}{3x} \Rightarrow P + 2C = 27 \quad \dots(1)$$

$$\text{and } \frac{2Px + Cx}{3x} = 11$$

$$\Rightarrow 2P + C = 33 \quad \dots(2)$$

On adding eq. (1) and (2)

$$\text{we get } P + C = 20$$

and on subtracting eq. (1) from (2)

$$\text{we get } P - C = 6$$

$$\text{Therefore } P = 13 \text{ and } C = 7$$

$$\text{Thus } \frac{C}{P} = \frac{7}{13}$$

10. (c) Let the number of subjects be n and average marks be x , then, total marks = nx

$$\text{Again } (n+1)(x-1) = (nx-40) + (23+25)$$

$$\Rightarrow x - n = 9 \quad \dots(1)$$

$$\text{Further } (n+2)(x+1) = (nx-40) + (23+25) + 57$$

$$\Rightarrow nx + 2x + n + 2 = nx + 63$$

$$\text{Solving equations (1) and (2), we get } n = 15 \quad \dots(2)$$

11. (a) Let the number of students who got the jobs of A, B and C categories is a , b and c respectively, then the total salary

$$= \frac{26(a+b) + 44(b+c) + 34(c+a)}{2(a+b+c)}$$

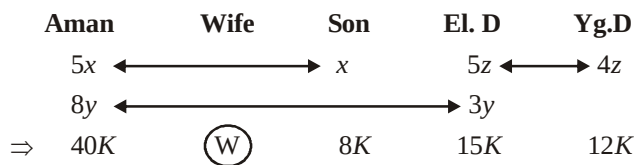
$$= \frac{60a + 70b + 78c}{2(a+b+c)}$$

$$= \frac{30(a+b+c) + (5b+9c)}{a+b+c}$$

$$= 30 + \text{some positive value}$$

So the minimum salary must be ` 30 lakh and the maximum salary can not exceed 44, which is the highest of the three.

12. (a)



Again since $Yg. D + W = A + S$ $(K = x \cdot y)$

$$\Rightarrow 12K + W = 40K + 8K \quad W \rightarrow \text{Age of wife}$$

$$\Rightarrow W = 36K$$

Thus 4 years ago

$$(36K - 4) = 8(8K - 4)$$

$$\Rightarrow 28K = 28 \Rightarrow K = 1$$

Therefore, the age of Aman = 40

$$\text{Wife} = 36$$

$$\text{Son} = 8$$

$$\text{Elder daughter} = 15$$

$$\text{Younger daughter} = 12$$

$$\text{Hence, the average age of the family} = \frac{111}{5} = 22.22 \text{ years}$$

13. (d) Total number of runs scored till 86th inning = $86N$
Now, $86N + 270 = 87(N + S)$, Where S is the increase in batting average.

Different values of S possible now $S = 0, 1, 2$ and 3 .

14. (c) Let us add all the 13 numbers
 $\Rightarrow 1 + 9 + 14 + 15 + [2 + 3 + 4 + 5 + 7 + 10 + 11 + 12 + 13]$
 $= 106$

As there are 4 columns and 3 rows so the sum of the 12 numbers has to be divisible by 12, i.e. the sum should be $96 (12 \times 8)$.

$$\text{So the sum of all the numbers in a row} = \frac{96}{3} = 32$$

$$\text{and in a column} = \frac{96}{4} = 24$$

Further the arithmetic mean of the numbers in a row

$$\text{or column} = \frac{32}{4} \text{ or } \frac{24}{3} = 8$$

15. (d) Clearly 10 has to be left out.
16. (b) The sum of scores of top four students = 1356
Maximum possible score possible for top three students = $352 \times 3 = 1056$
Hence, the minimum possible score for the 4th student = 300

Explanation of Test Yourself

1. (c) Assume that the distance is 120 km. Hence, 30 km is covered @ 25 kmph, 40 km @ 30 kmph and rest 50 km @ 50 kmph.
Then average speed = $\frac{\text{Total distance}}{\text{Total time}}$
2. (d) Average of 20 numbers = 0.
 $\therefore$ Sum of 20 numbers = $(0 \times 20) = 0$.
It is quite possible that 19 of these numbers may be positive and if their sum is a , then 20th number is $(-a)$.
3. (a) Given $A + B = 40$... (1)
 $C + B = 38$... (2)
 $A + C = 42$... (3)
 $(1) + (2) + (3) \Rightarrow A + B + C = 60$... (4)
From (1) and (4), we get
 $C = 20$ years
 $\therefore B = 18$ years and $A = 22$ years
4. (b) Let 'x' be the average score after 12 th innings
 $\Rightarrow 12x = 11 \times (x - 2) + 65$
 $\therefore x = 43$
5. (c) Total runs = $40 \times 50 = 2000$
Let his highest score be = x
Then his lowest score = $x - 172$
Now $\frac{2000 - x - (x - 172)}{38} = 48$
 $\Rightarrow 2x = 2172 - 1824$
 $\Rightarrow x = 174$
6. (a) Corrected mean $\frac{150 \times 30 - 135 + 165}{30}$
 $\frac{4500 - 135 + 165}{30} = \frac{4530}{30} = 151$
7. (c) Let average cost of petrol per litre be 'x'
Then $x = \frac{12000}{\frac{4000}{7.5} + \frac{4000}{8} + \frac{4000}{8.5}}$
 $\frac{3}{\frac{2}{15} + \frac{1}{8} + \frac{2}{17}} = \frac{6120}{767} = \text{` 7.98 per litre}$
8. (d) Let Arun's weight be X kg.
According to Arun, $65 < X < 72$.
According to Arun's brother, $60 < X < 70$.
According to Arun's mother, $X < 68$.
The values satisfying all the above conditions are 66 and 67.
 $\therefore$ Required average $\left(\frac{66 + 67}{2} \right) = \left(\frac{133}{2} \right) = 66.5$ kg.
9. (a) The weight of the second man is 135 and that of the third is 105. Hence, net result is a drop of 45 for 5 people. Hence, 9 kg is the drop.
10. (b) Assume x is the average expenditure of 19 people.
Then, $19x = 13 \times 79 + 6(x + 4)$
11. (a) Since only two page numbers are missing, the average would not change considerably and hence the number of pages remaining (after tearing away two page numbers)
 $= 45 \times 2 = 90$.
 $\therefore$ Since the average (of the remaining pages) was found by dividing with the remaining number of pages and as we have 13 as the denominator, the number of pages remaining 'n' must be 13 or a multiple of it close to 90. i.e., $13 \times 7 = 91$.
 $\therefore$ Total pages = $91 + 2 = 93$.
 $\therefore$ Sum of all pages (initially)
 $= \Sigma 93 = 93 \times \frac{(93+1)}{2} = 4371$ and sum after two pages missing
 $= 46 \times \frac{10}{13} \times 91 = 4256$.
 $\Rightarrow$ missing pages = m and $(m + 1)$, say then
 $4371 - 4256 = m + (m + 1)$
 $\Rightarrow 115 = 2m + 1 \Rightarrow m = 57$, and then $m + 1 = 58$.
 $\therefore$ The missing page numbers are 57 and 58.
12. (a) $\frac{(84 - 64) \times 94}{18.8} = 100$
13. (a) Ten years ago, sum of the ages of 8 people of the family = 231 years
Age of the members before 8 years who died after 3 years = $60 - 3 = 57$ years.

And age of the member before 8 years, who died after 6 years = $60 - 6 = 54$ years

Sum of ages of the two children in the current year = $7 + 4 = 11$ years

Sum of the ages of 8 members in the current year = (Sum of ages of 6 members before 10 years) + 6×10 + (Sum of ages of two children in the current year) = $231 - (57 + 54) + 60 + 11 = 191$

Hence, average age $\frac{191}{8} = 24$ years (Approx).

14. (d) Let present age of father = x year
and present age of mother = y year
therefore present age of son = $(x - 30)$
and present age of daughter = $(y - 25)$
Sum of their ages = $4 \times 26.25 = 105$ years

$$\text{i.e., } x + y + x - 30 + y - 25 = 105$$

$$2x + 2y = 160$$

$$x + y = 80 \quad \dots(1)$$

After 4 years, their total ages will be (excluding the daughter)

$$x + 4 + y + 4 + x - 30 + 4 = 107$$

$$2x + y - 18 = 107$$

$$2x + y = 125 \quad \dots(2)$$

Solving (1) and (2)

$$x = 45$$

$$15. \quad (b) \quad \frac{3x - y - y + \frac{y}{2}}{3} = x - 1$$

$$3x - \frac{3y}{2} = 3x - 3$$

$$\frac{3y}{2} = 3, \quad y = 2$$

4

ALLIGATIONS

- Alligation
- Solving the Problems of Alligations Using Alligation Formula
- Graphical Representation of Alligation– Cross Method
- The Straight Line Approach to Solve the Problems Related to Alligations
- Recognition of Different Situations Where Alligation can be Used
- A Typical Problem

ALLIGATION

Alligation is the simplified, faster technique to solve the problems based on weighted average. This method plays a vital role in saving the time in solving the problems related to weighted average situation.

We know that

$$\text{Weighted Average} = \frac{\text{Sum total of all numbers of all groups}}{\text{Total number of numbers in all groups together}}$$

Therefore weighted average A_w of two groups having n_1 and n_2 numbers with averages A_1 and A_2 respectively is

$$A_w = \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}$$

$$\Rightarrow (n_1 + n_2) A_w = n_1 A_1 + n_2 A_2$$

$$\Rightarrow n_1 (A_w - A_1) = n_2 (A_2 - A_w) \Rightarrow \frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1}$$

Equation $\frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1}$ is called Alligation Formula.

For convenient, we take $A_1 < A_2$. Hence $A_1 < A_w < A_2$.

SOLVING THE PROBLEMS OF ALLIGATIONS USING ALLIGATION FORMULA

Illustration 1: 10 kg of wheat costing ₹ 12 per kg and 15 kg of wheat costing ₹ 20 per kg are mixed. Find the average cost of the mixture per kg.

$$\text{Solution: } \frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1} \Rightarrow \frac{10}{15} = \frac{20 - A_w}{A_w - 12}$$

$$\Rightarrow \frac{2}{3} = \frac{20 - A_w}{A_w - 12} \Rightarrow 5A_w = 84$$

$$\Rightarrow A_w = \frac{84}{5} = 16.8$$

Hence average cost of the mixture = ₹ 16.8 per kg.

Illustration 2: A mixture worth ₹ 3.25 per kg is formed by mixing two types of salts, one costing ₹ 3.10 per kg while the other ₹ 3.60 per kg. In what ratio must they have been mixed?

$$\text{Solution: } \frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1} \Rightarrow \frac{n_1}{n_2} = \frac{3.60 - 3.25}{3.25 - 3.10} = \frac{35}{15}$$

$$\Rightarrow n_1 : n_2 = 7 : 3$$

Hence required ratio = 7 : 3.

GRAPHICAL REPRESENTATION OF ALLIGATION– CROSS METHOD

The alligation formula $\frac{n_1}{n_2} = \frac{A_2 - A_w}{A_w - A_1}$ is graphically represented by the following cross diagram:

$$\begin{array}{ccc} A_1 & & A_2 \\ & \diagdown & \diagup \\ & A_w & \\ & \diagup & \diagdown \\ [A_2 - A_w : A_w - A_1] \\ [n_1 : n_2] \end{array}$$

The ratios in the bracket [] are equal i.e.

$$n_1 : n_2 = A_2 - A_w : A_w - A_1$$

In the above graphical representation five variables A_1, A_2, A_w, n_1 and n_2 are involved.

Based on the problem situation, one of the following three cases may occur with respect to the known and the unknown out of the five variables A_1, A_2, A_w, n_1 and n_2 involved in the problem.

Case	Known	Unknown
I	(a) A_1, A_2, A_w	(a) $n_1 : n_2$
	(b) A_1, A_2, A_w, n_1	(b) n_2 and $n_1 : n_2$
II	A_1, A_2, n_1, n_2	A_w
III	A_1, A_w, n_1, n_2	A_2

Solving the problem using graphical representation of alligation is called cross method.

Let us solve some problems in each of the three cases using cross method.

Case I: When A_1, A_2, A_w are known and one of n_1 and n_2 may be also known then to find $n_1 : n_2$ and n_2 if n_1 is known OR n_1 if n_2 is known.

Illustration 3: If the average weight of the students of a class is 15kg, the average weight of the students of another class is 30kg and average weight of the students of both the classes is 25kg, then find the ratio of the number of students of the first class to the another class.

Solution:

$$\begin{array}{ccc} 15 & & 30 \\ & \searrow \quad \nearrow & \\ & 25 & \\ & \nearrow \quad \searrow & \\ \left[\begin{array}{ccc} 5 & : & 10 \\ n_1 & : & n_2 \end{array} \right] \end{array}$$

$$\therefore n_1 : n_2 = 5 : 10 = 1 : 2$$

Hence required ratio = 1 : 2

Case II: When A_1, A_2, n_1, n_2 are known and A_w is unknown then to find A_w .

Illustration 4: 5 kg of superior quality of sugar is mixed with 25 kg of inferior quality sugar. The price of superior quality and inferior quality sugar is ₹ 35 and ₹ 23 respectively. Find the average price per kg of the mixture.

Solution:

$$\begin{array}{ccc} 23 & & 35 \\ & \searrow \quad \nearrow & \\ & x & \\ & \nearrow \quad \searrow & \\ \left[\begin{array}{ccc} 35 - x & : & x - 23 \\ 25 & : & 5 \end{array} \right] \end{array}$$

$$\therefore \frac{35 - x}{x - 23} = \frac{25}{5}$$

$$\Rightarrow 30x = 175 + 575 \Rightarrow x = \frac{750}{30} = 25$$

Hence average price per kg of the mixture = ₹ 25

Case-III: When A_1, A_w, n_1, n_2 are known and A_2 is unknown, then to find the value of A_2 .

Illustration 5: The ratio of number of girls to number of boys is 1 : 2. If the average weight of the boys is 30 kg and the average weight of both the boys and girls is 25 kg, then find the average weight of the girls.

Solution:

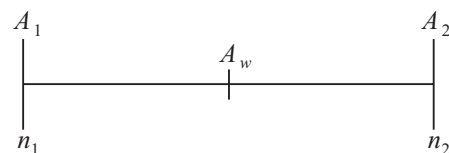
$$\begin{array}{ccc} x & & 30 \\ & \searrow \quad \nearrow & \\ & 25 & \\ & \nearrow \quad \searrow & \\ \left[\begin{array}{ccc} 5 & : & 25 - x \\ 1 & : & 2 \end{array} \right] \end{array}$$

$$\frac{5}{25 - x} = \frac{1}{2} \Rightarrow x = 15$$

Hence average weight of the girls = 15 kg.

THE STRAIGHT LINE APPROACH TO SOLVE THE PROBLEMS RELATED TO ALLIGATIONS

The straight line approach is actually the cross method.



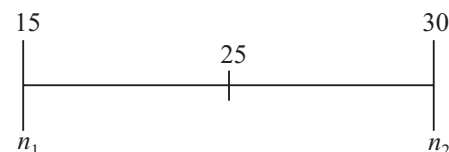
The above diagram is the straight line diagram in which the symbols A_1, A_2, A_w, n_1 and n_2 denote the same quantity as shown in cross method. Here $A_1 < A_w < A_2$.

In the above diagram,

- (a) n_1 corresponds to $(A_2 - A_w)$
- (b) n_2 corresponds to $(A_w - A_1)$
- (c) $(n_1 + n_2)$ corresponds to $(A_2 - A_1)$

Now, we again solve the examples 3, 4 and 5 given in case-I, II and III respectively of cross method using straight line approach.

Sol. of illustration 3 by straight line approach.



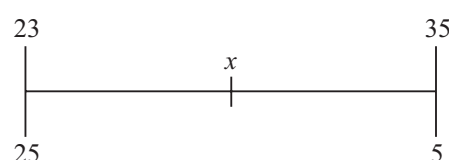
n_1 corresponds to 5 ($= 30 - 25$)

and n_2 corresponds to 10 ($= 25 - 15$)

$$\therefore n_1 : n_2 = 5 : 10 = 1 : 2$$

Hence required ratio = 1 : 2

Sol. of illustration 4 by straight line approach.



25 corresponds to $(35 - x)$

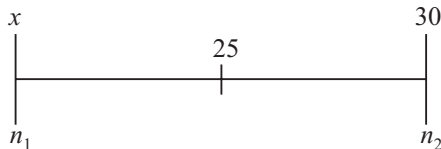
Also 5 corresponds to $(x - 23)$

$$\frac{25}{5} = \frac{35 - x}{x - 23}$$

$$\Rightarrow x = 25$$

Hence average price per kg of mixture = ₹ 25

Sol. of illustration 5 by straight line approach.



Here $n_1 : n_2 = 1 : 2$

Now, n_1 corresponds to 5 $(= 30 - 25)$

and n_2 corresponds to $(25 - x)$

$$\therefore \frac{n_1}{n_2} = \frac{5}{25 - x} \Rightarrow \frac{1}{2} = \frac{5}{25 - x} \Rightarrow x = 15$$

Hence average weight of the girls = 15 kg

RECOGNITION OF DIFFERENT SITUATIONS WHERE ALLIGATION CAN BE USED

There are many types of situations where alligation can be used, which must be recognised by the students. Here you are given some situations (or problems) which help you to recognise different alligation situations and identify A_1, A_2, n_1, n_2 and A_w in each alligation situation.

In each of the following problems

$$A_1 = 20, A_2 = 35, n_1 = 20, n_2 = 40$$

and answer as $A_w = 30$.

1. An average weight of students of a class of 40 students is 35 kg and an average weight of students of a class of 20 students is 20 kg. Find the average weight of the students of both the combined classes. (30 kg)
2. 20 litres of one variety of soda water is mixed with 40 litres of other variety of soda water. The price of first variety of soda water is ₹ 20 per litre and price of other variety of soda water is ₹ 35 per litre. Find the cost of the mixture per litre. (₹ 30)
3. A car travels at 20 km/h for 20 minutes and at 35 km/h for 40 minutes. Find the average speed of the car for the whole journey. (30 km/hr)
4. A car agency sold 20 cars at 20% profit and 40 cars at 35% profit. Find the gain percent on the sale of all these cars. (30%)
5. A trader earns a profit of 20% on 20% of his goods sold while he earns a profit of 35% on 40% of his goods sold. Find the percentage profit on whole. (30%)
6. A 40 litres mixture of water and milk contains 35% of milk and in another 20 litres of mixture of water and milk contains 20% of milk. If a new mixture is formed by mixing the both mixtures, then find the percentage of milk in new mixture. (30%)

7. A shopkeeper sold the 40% hardware at the profit of 35% and 20% software at a profit of 20%. Find the average profit% on the whole goods sold, if he sells only these two kind of things. (30%)

Some Keys to Identify A_1, A_2 & A_w and Differentiate These from n_1 and n_2

1. Normally, there are 3 averages mentioned in the problem, while there are only 2 quantities. This is not foolproof. Sometimes the question might confuse the students by giving 3 values for quantities representing n_1, n_2 and $n_1 + n_2$ respectively.
2. A_1, A_2 and A_w are always rate units, while n_1 and n_2 are quantity units.
Rate units are like ₹ x/kg, y km/hour, etc. and corresponding quantity units are kg, hour etc.
3. The denominator of the average unit corresponds to the quantity unit (i.e., unit for n_1 and n_2).
For example, denominator kg and hour of rate units ₹ x/kg and y km/hour are the units of quantity corresponding to rates.

A TYPICAL PROBLEM

Let's discuss the solution of a typical problem given below:

Illustration 6: A person used to draw out 20% of the honey from a jar containing 10 kg honey and replaced it with sugar solution. He has repeated the same process three times.

Find the final amount of honey left in the jar and the final ratio of honey to sugar solution finally left in the jar.

Solution: In first step: Honey drawn out 20% of 10 kg from the jar and then 2 kg sugar solution is put in the jar.

Hence after first step,

Honey remains in the jar = $10 - 20\% \text{ of } 10 = 10 - 2 = 8 \text{ kg}$
and sugar solution remains in the jar = 2 kg

In second step: 20% of (8 kg honey and 2 kg sugar solution) is drawn out from the jar and then 2 kg of sugar solution is put in the jar.
 $\Rightarrow 20\% \text{ of } 8 \text{ kg honey and } 20\% \text{ of } 2 \text{ kg sugar solution is drawn out from the jar and then } 2 \text{ kg of sugar solution is put in the container.}$

Thus in each step of drawing, 20% of remaining honey is drawn out.

Hence honey left in the container after second draw

$$= 8 - 20\% \text{ of } 8 = 8 - 1.6 = 6.4 \text{ kg}$$

Honey left in the container after third draw

$$= 6.4 - 20\% \text{ of } 6.4$$

$$= 6.40 - 1.28 = 5.12 \text{ kg}$$

Hence the final amount of the honey left in the jar = 5.12 kg

The above whole process can be shown in a single line as
 $10 - 20\% \text{ of } 10 \rightarrow 8 - 20\% \text{ of } 8 \rightarrow 6.4 - 20\% \text{ of } 6.4 \rightarrow 5.12 \text{ kg}$

Now the final amount of sugar solution left in the jar

$$= 10 - 5.12 \text{ kg} = 4.88 \text{ kg}$$

Hence final ratio of honey to the sugar solution left in the jar

$$= \frac{5.12}{4.88} = 64 : 61.$$

Practice Exercises



Foundation Level

1. A mixture of certain quantity of milk with 16 litres of water is worth 90 P per litre. If pure milk be worth ` 1.08 per litre, how much milk is there in the mixture?
(a) 60 (b) 70
(c) 80 (d) 90
2. In my pocket there are ` 25 consisting of only the denominations of 20 paise and 50 paise. Thus there are total 80 coins in my pocket. The no. of coins of the denomination of 50 paise is
(a) 30 (b) 70
(c) 50 (d) 25
3. There are some shepherds and their sheep in a grazing field. The no. of total heads are 60 and total legs are 168 including both men and sheep. The no. of sheep is
(a) 18 (b) 26
(c) 24 (d) 36
4. If 5 kg of salt costing ` 5/kg and 3 kg of salt costing ` 4/ kg are mixed, find the average cost of the mixture per kilogram.
(a) ` 4.5 (b) ` 4.625
(c) ` 4.75 (d) ` 4.125
5. In what ratio should two qualities of coffee powder having rates of ` 47 per kg and ` 32 per kg be mixed in order to get a mixture that would have a rate of ` 37 per kg?
(a) 1 : 2 (b) 2 : 1
(c) 1 : 3 (d) 3 : 1
6. In what ratio should milk and water be mixed so that after selling the mixture at the cost price a profit of $16\frac{2}{3}\%$ is made?
(a) 1 : 2 (b) 1 : 6
(c) 2 : 3 (d) 2 : 5
7. Gold is 19 times as heavy as water and copper 9 times. In what ratio should these metals be mixed so that the mixture may be 15 times as heavy as water?
(a) 1 : 2 (b) 3 : 2
(c) 2 : 3 (d) 4 : 5
8. In a mixture of 60 litres, the ratio of milk to water is 2 : 1. If the ratio of milk to water is to be 1 : 2, then amount of water to be further added is _____.
(a) 20 (b) 40
(c) 60 (d) 80
9. In a mixture of 45 litres, the ratio of milk and water is 4 : 1. How much water must be added to make the mixture ratio 3 : 2 ?
(a) 72 litres (b) 24 litres
(c) 15 litres (d) 1.5 litres
10. In a class of 30 students, the average weight of boys is 20 kg and the average weight of the girls is 25 kg. The fraction of boys out of the total students of the class is
(a) $\frac{4}{5}$ (b) $\frac{5}{6}$
(c) $\frac{3}{4}$ (d) Data insufficient
11. Milk and water are mixed in a vessel A in the proportion 5 : 2, and in vessel B in the proportion 8 : 5. In what proportion should quantities be taken from the two vessels so as to form a mixture in which milk and water will be in the proportion of 9 : 4?
(a) 4 : 5 (b) 5 : 7
(c) 7 : 2 (d) 7 : 9
12. A container has a capacity of 20 gallons and is full of spirit. 4 gallons of spirit is drawn out and the container is again filled with water. This process is repeated 5 times. Find out how much spirit is left in the resulting mixture finally?
(a) $6\frac{257}{525}$ gallons (b) $6\frac{346}{625}$ gallons
(c) 6.5 gallons (d) 6.25 gallons
13. A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is:
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{2}{5}$ (d) $\frac{3}{5}$
14. A dishonest grocer professes to sell pure butter at cost price, but he mixes it with adulterated fat and thereby gains 25%. Find the percentage of adulterated fat in the mixture assuming that adulterated fat is freely available.
(a) 20% (b) 25%
(c) 33.33% (d) 40%

15. A merchant purchased two qualities of pulses at the rate of ₹ 200 per quintal and ₹ 260 per quintal. In 52 quintals of the second quality, how much pulse of the first quality should be mixed so that by selling the resulting mixture at ₹ 300 per quintal, he gains a profit of 25%?
 (a) 100 quintals (b) 104 quintals
 (c) 26 quintals (d) None of these
16. There are two mixtures of honey and water, the quantity of honey in them being 25% and 75% of the mixture. If 2 gallons of the first are mixed with three gallons of the second, what will be the ratio of honey to water in the new mixture?
 (a) 11 : 2 (b) 11 : 9
 (c) 9 : 11 (d) 2 : 11
17. Two solutions of 90% and 97% purity are mixed resulting in 21 litres of mixture of 94% purity. How much is the quantity of the first solution in the resulting mixture?
 (a) 15 litres (b) 12 litres
 (c) 9 litres (d) 6 litres
18. A 20 percent gain is made by selling the mixture of two types of ghee at ₹ 480 per kg. If the type costing 610 per kg was mixed with 126 kg of the other, how many kilograms of the former was mixed ?
 (a) 138 kg (b) 34.5 kg
 (c) 69 kg (d) Cannot be determined
19. A man makes 60 articles in the 1st hour. His efficiency decreases by 25% in the 2nd hour, increases by 40% in the 3rd hour, decreases by 33% in the 4th hour and increases by 50% in the 5th hour. If he has to work for more than 1 hour, then in which hour the average number of articles produced per hour then would be minimum ?
 (a) 2nd hour (b) After 5th hour
 (c) 3rd hour (d) None of these
20. There are two solutions of Sulphuric acid (acid + water) with concentration of 50% and 80% respectively. They are mixed in a certain ratio to get a 62% sulphuric acid solution. This solution is mixed with 6 liters of water to get back 50% solution. How much of the 80% solution has been used in the entire process?
 (a) 15 liters (b) 12 liters
 (c) 10 litres (d) None of these

Standard Level

1. 300 gm of sugar solution has 40% sugar in it. How much sugar should be added to make it 50% in the solution?
(a) 40 gm (b) 50 gm
(c) 60 gm (d) 70 gm
2. There are 65 students in a class. 39 rupees are distributed among them so that each boy gets 80 P and each girl gets 30 P. Find the number of boys and girls in that class.
(a) 45, 20 (b) 40, 25
(c) 39, 26 (d) 29, 36
3. How much water must be added to a cask which contains 40 litres of milk at cost price ₹ 3.5/litre so that the cost of milk reduces to ₹ 2/litre?
(a) 20 (b) 35
(c) 45 (d) None of these
4. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is _____.
(a) 10% (b) 15%
(c) 20% (d) 25%
5. Jayashree purchased 150 kg of wheat of the rate of ₹ 7 per kg. She sold 50 kg at a profit of 10%. At what rate per kg should she sell the remaining to get a profit of 20% on the total deal?
(a) 6.50 (b) 8.75
(c) 7.50 (d) 9.75
6. The ratio of milk and water in 55 litres of adulterated milk is 7 : 4. How much water must be added to make the mixture's ratio 7 : 6?
(a) 5 l (b) 10 l
(c) 15 l (d) 25 l
7. From a cask full of milk, 10 litres are taken out of 50 litres and is filled with water. This was done twice. What is the quantity of milk now left in the cask?
(a) 20 litres (b) 32 litres
(c) 25 litres (d) 30 litres
8. The average weight of boys in a class is 30 kg and the average weight of girls in the same class is 20 kg. If the average weight of the whole class is 23.25 kg, what could be the possible strength of boys and girls respectively in the same class?
(a) 14 and 26 (b) 13 and 27
(c) 17 and 27 (d) None of these
9. In what ratio should water be mixed with soda costing ₹ 12 per litre so as to make a profit of 25% by selling the diluted liquid at ₹ 13.75 per litre?
(a) 10 : 1 (b) 11 : 1
(c) 1 : 11 (d) 12 : 1
10. Two vessels A and B of equal capacities contain mixtures of milk and water in the ratio 4 : 1 and 3 : 1, respectively. 25% of the mixture from A is taken out and added to B. After mixing it thoroughly, an equal amount is taken out from B and added back to A. The ratio of milk to water in vessel A after the second operation is
(a) 79 : 21 (b) 83 : 17
(c) 77 : 23 (d) 81 : 19
11. Two alloys composed of gold and silver together weight 20 kg. One lump contains 75% gold and 31.25 gm per kg silver. Another alloy contains 85% gold and 30 gm per kg silver. The total quantity of silver in two lumps is 617.5 gm. If the two lumps are melted and formed into one, what percentage of gold will it contain?
(a) 50% (b) 89%
(c) 78% (d) 67%
12. Two vessels A and B contain spirit and water mixed in the ratio 5 : 2 and 7 : 6 respectively. Find the ratio in which these mixture be mixed to obtain a new mixture in vessel C containing spirit and water in the ratio 8 : 5?
(a) 4 : 3 (b) 3 : 4
(c) 5 : 6 (d) 7 : 9
13. Two vessels A and B contain milk and water mixed in the ratio 8 : 5 and 5 : 2 respectively. The ratio in which these two mixtures be mixed to get a new mixture containing $69\frac{3}{13}\%$ milk, is :
(a) 2 : 7 (b) 3 : 5
(c) 5 : 2 (d) 5 : 7
14. A can contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and the can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the can initially?
(a) 10 (b) 20
(c) 21 (d) 25

15. Ram prepares solutions of alcohol in water according to customers' needs. This morning Ram has prepared 27 litres of a 12% alcohol solution and kept it ready in a 27 litre delivery container to be shipped to the customer. Just before delivery, he finds out that the customer had asked for 27 litres of 21% alcohol solution. To prepare what the customer wants, Ram replaces a portion of 12% solution by 39% solution. How many litres of 12% solution are replaced?
- (a) 5 (b) 9
(c) 10 (d) 12
16. There are two containers : the first contains 500 ml of alcohol, while the second contains 500 ml of water. Three cups of alcohol from the first container is removed and is mixed well in the second container. Then three cups of this mixture is removed and is mixed in the first container. Let 'A' denote the proportion of water in the first container and 'B' denote the proportion of alcohol in the second container. Then,
- (a) $A > B$ (b) $A < B$
(c) $A = B$ (d) Cannot be determined
17. An industrial solvent of 90% strength is prepared and stored in a 150 litres capacity container. The container is filled to 80% of its capacity. It is required to decrease the concentration of the solvent in the container to less than 60%. For doing this, either pure water can be added to it and/or solution can be removed. But removal can only be done if there is no place for adding water. At a given time, only 10 litres of the solution can be added or removed. How many times should water be added to the solution?
- (a) 4 (b) 5
(c) 9 (d) 6

Expert Level

1. A person has a chemical of ₹ 25 per litre. In what ratio should water be mixed in that chemical so that after selling the mixture at ₹ 20/litre he may get a profit of 25%?
(a) 14 (b) 15
(c) 16 (d) 17
2. A trader has 50 kg of rice, a part of which he sells at 14% profit and the rest at 6% loss. On the whole his loss is 4%. What is the quantity sold at 14% profit and that at 6% loss?
(a) 2, 48 (b) 4, 46
(c) 5, 45 (d) 7, 43
3. A vessel of 80 litre is filled with milk and water. 70% of milk and 30% of water is taken out of the vessel. It is found that the vessel is vacated by 55%. Find the initial quantity of milk and water.
(a) 20, 60 (b) 30, 50
(c) 50, 30 (d) 60, 20
4. A container contained 80 kg of milk. From this container, 8 kg of milk was taken out and replaced by water. This process was further repeated two times. How much milk is now contained by the container?
(a) 48 kg (b) 56 kg
(c) 58.32 kg (d) 59.46 kg
5. In what ratio must a person mix three kinds of wheat costing him ₹ 1.20, ₹ 1.44 and ₹ 1.74 per kg, so that the mixture may be worth ₹ 1.41 per kg?
(a) 1 : 2 : 3 (b) 4 : 5 : 7
(c) 12 : 7 : 7 (d) 13 : 7 : 9
6. In what ratio must a grocer mix two varieties of tea worth ₹ 60 a kg and ₹ 65 a kg so that by selling the mixture at ₹ 68.20 a kg he may gain 10%?
(a) 3 : 2 (b) 3 : 4
(c) 3 : 5 (d) 4 : 5
7. A vessel is full of refined oil. $\frac{1}{4}$ of the refined oil is taken out and the vessel is filled with mustard oil. If the process is repeated 4 times and 10 liters of refined oil is finally left in the vessel, what is the capacity of the vessel?
(a) 33 litres (b) $\frac{2460}{81}$ litres
(c) $\frac{2560}{81}$ litres (d) 30 litres
8. There are two mixtures of honey and water, the quantity of honey in them being 25% and 75% of the mixture. If 2 gallons of the first are mixed with three gallons of the second, what will be the ratio of honey to water in the new mixture?
(a) 11 : 2 (b) 11 : 9
(c) 9 : 11 (d) 2 : 11
9. A bartender stole champagne from a bottle that contained 50% of spirit and he replaced what he had stolen with champagne having 20% spirit. The bottle then contained only 25% spirit. How much of the bottle did he steal?
(a) 80% (b) 83.33%
(c) 85.71% (d) 88.88%
10. There is a vessel holding 40 litres of milk. 4 litres of milk is initially taken out from the vessel and 4 litres of water is then poured in. After this, 5 litres of mixture is replaced with 6 litres of water. And finally 6 litres of the mixture is replaced with 6 litres of water. How much milk (in liters) is there in the vessel?
(a) 26.775 (b) 29.16
(c) 24.72 (d) 27.42
11. In three vessels each of 10 litres capacity, mixture of milk and water is filled. The ratios of milk and water are 2 : 1, 3 : 1 and 3 : 2 in the three respective vessels. If all the three vessels are emptied into a single large vessel, find the proportion of milk and water in the mixture.
(a) 181 : 49 (b) 101 : 49
(c) 121 : 59 (d) 131 : 69
12. A butler stores wine from a butt of sherry which contained 30% of spirit and he replaced what he had stolen by wine containing only 12% of spirit. The butt was then 18% strong only. How much of the butt did he steal?
(a) $\frac{1}{3}$ (b) $\frac{2}{5}$
(c) $\frac{2}{3}$ (d) $\frac{4}{7}$
13. A thief steals four gallons of liquid soap kept in a train compartment's bathroom from a container that is full of liquid soap. He then fills it with water to avoid detection. Unable to resist the temptation he steals 4 gallons of the mixture again, and fills it with water. When the liquid soap is checked at a station it is found that the ratio of the liquid soap now left in the container to the water in it is 36 : 13. What was the initial amount of the liquid soap in the container if it is known that the liquid soap is neither used nor augmented by anybody else during the entire period?
(a) 7 gallons (b) 14 gallons
(c) 21 gallons (d) 28 gallons
14. Gopal sells fruit juice mixture using orange juice and pineapple juice. Gopal prepares this mixture by drawing out a jug of orange juice from a 10 litre container filled with orange juice, and replacing it with pineapple juice. If Gopal draws out another jug of the resultant mixture and replaces it with pineapple juice, the container will have equal volumes of orange juice and pineapple juice. The volume of the jug, in litres, is
(a) 2 (b) > 2 and ≤ 2.5
(c) 2.5 (d) > 2.5 and ≤ 3
15. There is a vessel holding 40 liters of milk. 4 liters of milk is initially taken out from the vessel and 5 liters of water is poured in. After this, 5 liters of the mixture from this vessel is replaced with 6 liters of water. And finally 6 liters of mixture from this vessel is replaced with 7 liters of water. How much of the milk (in litres) is there in the vessel now?
(a) 22.42 (b) 27.09
(c) 24.72 (d) 29.42

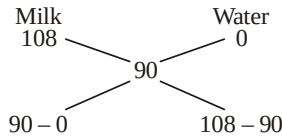
Test Yourself

1. A mixture of 125 gallons of wine and water contains 20% water. How much water must be added to the mixture in order to increase the percentage of water to 25% of the new mixture?
(a) 10 gals (b) 8.5 gals
(c) 8 gals (d) 8.33 gals
2. A dishonest milkman purchased milk at ₹ 10 per litre and mixed 5 litres of water in it. By selling the mixture at the rate of ₹ 10 per litre he earns a profit of 25%. The quantity of the amount of the mixture that he had was:
(a) 15 litres (b) 20 litres
(c) 25 litres (d) 30 litres
3. A cistern contains 50 litres of water, 5 litres of water is taken out of it and replaced by wine. The process is repeated again. Find the proportion of wine and water in the resulting mixture.
(a) 1 : 4 (b) 41 : 50
(c) 19 : 81 (d) 81 : 19
4. There are two kinds of alloys of tin and copper. The first alloy contains tin and copper such that 93.33% of it is tin. In the second alloy there is 86.66% tin. What weight of the first alloy should be mixed with some weight of the second alloy so as to make a 50 kg mass containing 90% of tin?
(a) 15 kg (b) 30 kg
(c) 20 kg (d) 25 kg
5. Two solutions of 90% and 97% purity are mixed resulting in 21 litres of mixture of 94% purity. How much is the quantity of the first solution in the resulting mixture?
(a) 15 litres (b) 12 litres
(c) 9 litres (d) 6 litres
6. What will be the ratio of petrol and kerosene in the final solution formed by mixing petrol and kerosene that are present in three vessels of equal capacity in the ratios 4 : 1, 5 : 2 and 6 : 1 respectively?
(a) 166 : 22 (b) 83 : 22
(c) 83 : 44 (d) None of these
7. Two vessels contain a mixture of spirit and water. In the first vessel the ratio of spirit to water is 8 : 3 and in the second vessel the ratio is 5 : 1. A 35 litre cask is filled from these vessels so as to contain a mixture of spirit and water in the ratio of 4 : 1. How many litres are taken from the first vessel?
(a) 11 litres (b) 22 litres
(c) 16.5 litres (d) 17.5 litres
8. A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is:
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{2}{5}$ (d) $\frac{3}{5}$
9. Baniya sells two types of tea viz. Desi Chai and Videshi Chai. He sells Desi Chai at ₹ 18 per kg and incurs a loss of 10% whereas on selling the Videshi Chai at ₹ 30 per kg, he gains 20%. In what proportion should the Desi Chai and Videshi Chai be mixed such that he can gain a profit of 25% by selling the mixture at ₹ 27.5 per kg?
(a) 3 : 2 (b) 2 : 3
(c) 2 : 5 (d) 3 : 5
10. A butler stole wine from a butt of sherry which contained 32% of spirit and then replaced what he stole, by wine containing only 18% spirit. The butt was then of 24% strength only. How much of the butt had he stolen?
(a) $\frac{3}{8}$ (b) $\frac{5}{7}$
(c) $\frac{4}{7}$ (d) $\frac{7}{11}$
11. The average age of boys in class is 16.66, while the average age of girls is 18.75. Thus the average age of all the 40 students of the class is 17.5. If the difference between the no. of boys and girls is 8, then the no. of girls in the class is:
(a) 12 (b) 16
(c) 18 (d) Data insufficient
12. In two alloys, the ratio of zinc to tin is 5 : 2 and 3 : 4. If 7 kg of the first alloy and 21 kg of the second alloy are mixed together to form a new alloy, then what will be the ratio of zinc and tin in the new alloy?
(a) 1 : 1 (b) 2 : 1
(c) 1 : 2 (d) 2 : 2
13. The average marks of the students in our sections A, B, C and D together is 60%. The average marks of the students of A, B, C and D individually are 45%, 50%, 72% and 80% respectively. If the average marks of the students of sections A and B together is 48% and that of the students of B and C together is 60%. What is the ratio of number of students in sections A and D?
(a) 2 : 3 (b) 4 : 3
(c) 5 : 3 (d) 3 : 5
14. 4 kg of a metal contains $\frac{1}{5}$ copper and rest is zinc. Another 5 kg of metal contains $\frac{1}{6}$ copper and rest is zinc. The ratio of copper and zinc into the mixture of these two metals:
(a) 49 : 221 (b) 39 : 231
(c) 97 : 181 (d) None of these
15. ₹ 1500 is invested in two such part that if one invested at 6%, and the other at 5% the total interest in one year from both investments is ₹ 85. How much invested at 5%?
(a) ₹ 500 (b) ₹ 1000
(c) ₹ 1500 (d) ₹ 4500

Hints & Solutions

Foundation Level

1. (c) The mean value is 90 P and the price of water is 0 P.



By the Alligation Rule, Milk and water are in the ratio of 5 : 1.

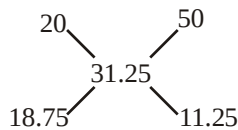
∴ quantity of milk in the mixture = $5 \times 16 = 80$ litres.

2. (a) Go through options :

$$30 \times 50 + 50 \times 20 = 2500 \text{ paise}$$

Alternatively : Since the average price of a coin

$$= \frac{2500}{80} = 31.25 \text{ paise}$$



So the ratio of no. of 20 paise coins to the no. of 50 paise coins

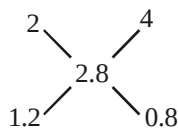
$$= 18.75 : 11.25 \\ = 75 : 45 = 5 : 3$$

Therefore, the no. of coins of the denominations of 50 paise is 30.

3. (c) Go through options :

$$24 \times 4 + 36 \times 2 = 168$$

Alternatively :



$$\Rightarrow 3 : 2$$

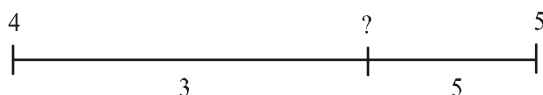
Therefore, the ratio of men and sheep is 3 : 2

Alternatively : Suppose there are only men, then the no. of legs = $60 \times 2 = 120$.

Now since there are $48 = (168 - 120)$ legs extra, it

means there are $24 \left(\frac{48}{2} \right)$ sheep, since a sheep has 2 extra legs than a man has.

4. (b) Solving the following alligation figure:



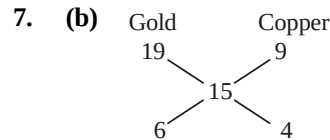
The answer would be 4.625/kg

5. (a) The ratio would be 1 : 2 as seen from the figure:



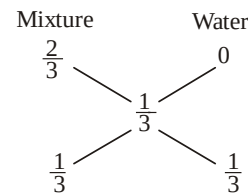
6. (b) **Short-Cut-Method :** In such questions the ratio is

$$\text{water : milk} = 16 \frac{2}{3} : 100 = 1 : 6$$



$$\therefore \text{Gold : Copper} = 6 : 4 = 3 : 2$$

8. (c) Apply the alligation on fraction of milk in each mixture.



Ratio of mixture to water = 1 : 1

Therefore, if there is 60 litre of solution, 60 litres of water should be added.

9. (c) Quantity of milk = $45 \times \frac{4}{5} = 36$ litres

$$\text{Quantity of water} = 45 \times \frac{1}{5} = 9 \text{ litres}$$

Let x litres of water be added to make the ratio 3 : 2

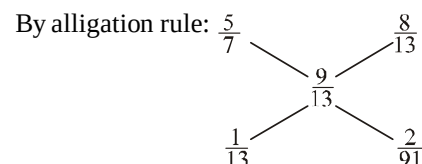
$$\text{Then, } \frac{36}{9+x} = \frac{3}{2} \Rightarrow 72 = 27 + 3x \Rightarrow x = 15 \text{ litres}$$

10. (d) Since we do not know either the average weight of the whole class or the ratio of no. of boys to girls.

11. (c) In vessel A, milk = $\frac{5}{7}$ of the weight of mixture

$$\text{In vessel B, milk} = \frac{8}{13} \text{ of the weight of mixture. Now,}$$

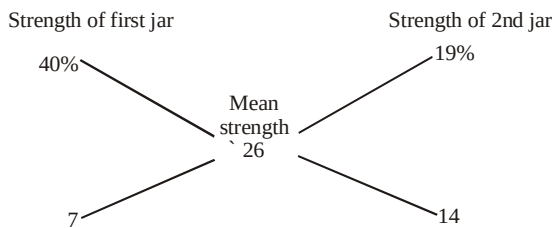
we want to form a mixture in which milk will be $\frac{9}{13}$ of the weight of this mixture.



$$\therefore \text{required proportion is } \frac{1}{13} : \frac{2}{91} = 7 : 2$$

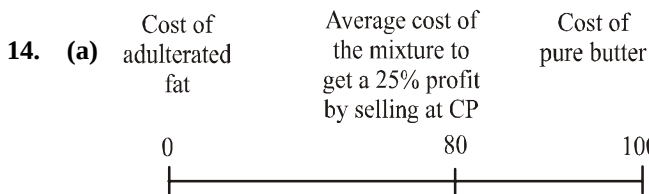
12. (b) The amount of spirit left
 $= 20 \times \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} \times \frac{4}{5}$
 $= 4096/625 = 6 \frac{346}{625}$.

13. (b) By the rule of alligation, we have:



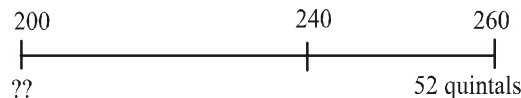
∴ So, Ratio of 1st and 2nd quantities = 7 : 14 = 1 : 2

∴ Required quantity replaced = $\frac{2}{3}$



The ratio of mixing required would be 1 : 4 which means that the percentage of adulterated fat would be 20%.

15. (c) By selling at 300 if we need to get a profit of 25% it means that the cost price would be $300/1.25 = 240$.



Ratio of mixing required to get an average of 240 per quintal = 1 : 2

Thus, in 52 quintals of the second we need to mix 26 quintals of the first.

16. (b) The percentage of honey in the new mixture would be:

$(2 \times 25 + 3 \times 75)/5 = 275/5 = 55\%$. The ratio of honey to water in the new mixture would be 55 : 45 = 11 : 9

17. (c) 90% and 97% mixed to form 94% means that the mixing ratio is 3 : 4. The first solution would be $3 \times 21/7 = 9$ litres.

18. (d) We cannot determine the answer to this question as we do not know the price per kg of the other type of ghee. Hence, we cannot find the ratio of mixing which would be required in order to move further in this question.

19. (d) Number of articles made in 1st hour = 60
Number of articles made in 2nd hour = 45
Number of articles made in 3rd hour = 63
Number of articles made in 4th hour = 42
Number of articles made in 5th hour = 63

So, obviously articles made in 4th hour is minimum.

20. (c) Let x liters of 50% solution and y litres of 80% solutions are used

$$\frac{x}{y} = \frac{80-62}{62-50} = \frac{18}{12} = \frac{x}{y} = \frac{3}{2}$$

Solution get mixed in the ratio 3 : 2.

Now, suppose the value of acid is Z litres

$$\Rightarrow \frac{0.62z}{z+6} = \frac{1}{2}$$

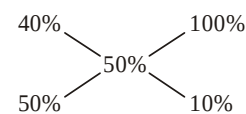
$$\Rightarrow 1.24 Z = Z + 6 \Rightarrow 0.24 Z = 6$$

$$Z = 25$$

Hence, required rate = $\frac{2}{5} \times 25 = 10$ litres

Standard Level

1. (c) The existing solution has 40% sugar. And sugar is to be mixed; so the other solution has 100% sugar. So, by alligation method:

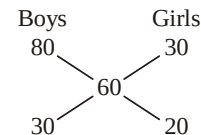


∴ The two mixtures should be added in the ratio 5 : 1.

Therefore, required sugar = $\frac{300}{5} \times 1 = 60$ gm

2. (c) Here, alligation is applicable for 'money per boy or girl'.

Mean value of money per student = $\frac{3900}{65} = 60$ P

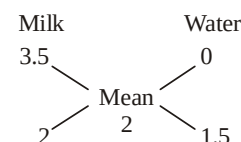


∴ Boys : Girls = 3 : 2

∴ Number of boys = $\frac{65}{3+2} \times 3 = 39$

and number of girls = $65 - 39 = 26$

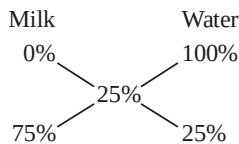
3. (d) This question can be solved in so many different ways. But the method of alligation method is the simplest of all the methods. We will apply the alligation on price of milk, water and mixture.



∴ ratio of milk and water should be 2 : 15 = 4 : 3

∴ added water = $\frac{40}{4} \times 3 = 30$ litres

4. (d) We will apply alligation on % profit. If he sells the milk at CP, he gains 0%. But if he sells water at CP, he gains 100%.



Ratio of milk to water in the mixture should be 3 : 1

$$\therefore \% \text{ of water in mixture} = \frac{1}{3+1} \times 100 = 25\%$$

5. (b) Selling price of 150 kg wheat at 20% profit

$$= 150 \times 7 \left(\frac{120}{100} \right) = ₹ 1260$$

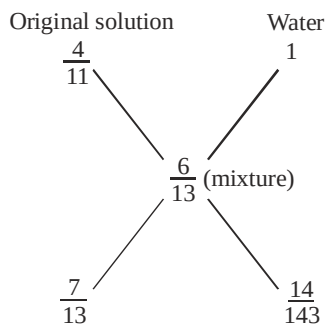
Selling price of 50 kg wheat at 10% profit

$$= 50 \times 7 \left(\frac{110}{100} \right) = ₹ 385$$

$\therefore$ Selling price per kg of remaining 100 kg wheat

$$= \frac{1260 - 385}{100} = ₹ 8.75$$

6. (b) By the rule of alligation, water concentration,



$\therefore$ water must be added to the mixture in the ratio

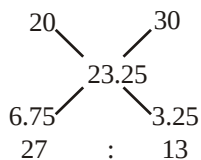
$$\frac{14}{143} : \frac{7}{13} \text{ i.e., } 2 : 11$$

$$\text{Quantity of water to be added} = \frac{2}{11} \times 55 = 10 \text{ litres}$$

7. (b) $\therefore$ 10 litres are withdrawn $= \frac{10}{50} = \frac{1}{5}$ of the whole
Quantity of milk after 2nd operation

$$= 50 \left(1 - \frac{1}{5} \right)^2 = \frac{16}{25} \times 50 = 32 \text{ litres}$$

8. (b) Therefore no. of boys : Number of girls = 13 : 27



9. (c) In order to sell at a 25% profit by selling at 13.75 the cost price should be $13.75/1.25 = 11$. Also since water is freely available, we can say that the ratio of water and soda must be 1:11.

10. (a) Assume there is 20 liters of the mixture in both the vessels.

In vessel A, milk = 16 liters and water = 4 litres

25% from A to B = milk in B = 15 + 4 = 19 litres

= water in B = 5 + 1 = 6 liters ratio = 19 : 6

Equal amount from vessel B to vessel A

$$= \text{milk in A} = 12 + \frac{19}{5} = \frac{79}{5}$$

$$= \text{water in A} = 3 + \frac{6}{5} = \frac{21}{5}$$

Hence, the ratio is 79 : 21

11. (c) Eliminating the option, we get (c) as answer because average always lies between greatest and lowest.

12. (d) Let the C.P. of spirit be ₹ 1 per litre.

Spirit in 1 litre mix. of A = $\frac{5}{7}$ litre; C.p. of a litre mix. in

$$A = ₹ \frac{5}{7}.$$

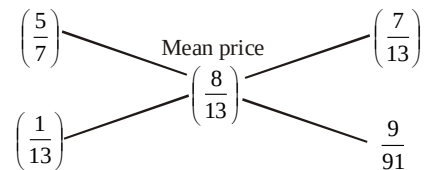
Spirit in 1 litre mix. of B = $\frac{7}{13}$ litre; C.P. of 1 litre mix. in

$$B = ₹ \frac{5}{13}.$$

Spirit in 1 litre mix. of C = $\frac{8}{13}$ litre; Mean price = ₹ $\frac{8}{13}$.

By the rule of alligation, we have :

C.P. of 1 litre mixture in A C.P. of 1 litre mixture in B



$$\therefore \text{Required ratio} = \frac{1}{13} : \frac{9}{91} = 7 : 9.$$

13. (a) Let cost of 1 litre milk be ₹ 1.

Milk in 1 litre mix. in B = $\frac{8}{13}$ litre, C.P. of a litre

$$\text{mix. in B} = ₹ \frac{5}{7}.$$

Milk in 1 litre mix. in B = $\frac{5}{7}$ litre, C.P. of 1 litre mix. in

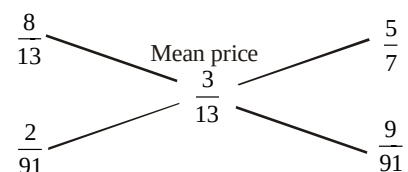
$$B = ₹ \frac{5}{7}$$

Milk in 1 litre of final mix. $\left(\frac{900}{13} \times \frac{1}{100} \times 1 \right) = \frac{9}{13}$ litre;

$$\text{mean price} = ₹ \frac{9}{13}.$$

By the rule of alligation, we have:

C.P. of 1 litre mixture in A C.P. of 1 litre mixture in B



$$\therefore \text{Required ratio} = \frac{2}{91} : \frac{1}{13} = 2 : 7.$$

14. (c) Suppose the can initially contains $7x$ and $5x$ litres of mixtures A and B respectively.

Quantity of A mixture left $= \left(7x - \frac{7}{12} \times 9\right)$ litres

$$\left(7x - \frac{21}{4}\right) \text{ litres.}$$

Quantity of B in mixture left $= \left(5x - \frac{5}{12} \times 9\right)$ litres

$$\left(5x - \frac{15}{4}\right) \text{ litres.}$$

$$\therefore \frac{\left(7x - \frac{21}{4}\right)}{\left(5x - \frac{15}{4}\right) + 9} = \frac{7}{9} \Rightarrow \frac{28x - 21}{20x + 21} = \frac{7}{9}$$

$$\Rightarrow 252x - 189 = 140x + 147$$

$$\Rightarrow 112x = 336 \Rightarrow x = 3.$$

So, the can contained 21 litre.

15. (b) Let Ram replaces x litres of 12% sol. with 39% solution.

Now, quality of 12% sol. in 27 litre $= \frac{27 \times 12}{100}$

$\therefore$ After replacing we have volume of 12% sol.

$$= \frac{27 \times 12}{100} - \frac{12x}{100} + \frac{39x}{100} = \frac{324 + 27x}{100}$$

This will be equal to 27 litre of 21% sol.

$$\therefore \frac{324 + 27x}{100} = \frac{21 \times 27}{100}$$

$$\therefore x = \frac{567 - 324}{27} = \frac{243}{27} = 9$$

Hence option (b)

16. (c) Let capacity of each cup be 100 ml
After first operation, first container will have 200 ml of alcohol and second container will have 300 l alcohol and 500 ml water.

Ratio of water to alcohol in the second container $= 5 : 3$.

After second operation, the quantity of water and alcohol left would be $\left(300 \times \frac{5}{8}\right) = 187.5$ ml and

$\left(300 \times \frac{3}{8}\right) = 112.5$ ml respectively in the first container.

and the quantity of water and alcohol in the first container is 187.5 ml and $(200 + 112.5)$ ml = 312.5 ml

hence, ratio of water and alcohol $= 187.5 : 312.5 = 3 : 5$ and the ratio of alcohol to water $= 5 : 3$.

Hence, on comparing ratio of water and alcohol in both the containers we find that $A = B$.

17. (d) Value of solvent in container $= \left(\frac{4}{5} \times 150\right) l = 12l$

Concentration of solvent $= \left(\frac{9}{10} \times 120\right) l = 108l$

When first time water is added the concentration of

$$\text{solvent} = \frac{108}{130} \times 100 = 83\%$$

When 2nd time water is added the concentration of solvent

$$= \frac{108}{140} \times 100 = 77\%$$

when 3rd the water is added the concentration of

$$\text{solvent} = \frac{108}{150} \times 100 = 72\%$$

Now, 10l solution should be removed

Solvent present in container $= (108 - 7.2) l = 100.8 l$

Again 10 l water is added and 10 l solvent is removed.

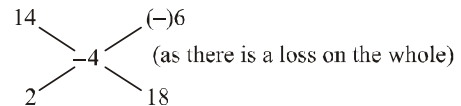
To get the required solution, water is added for 6 times.

Expert Level

1. (c) In this question, the alligation method is applicable on prices, so we should get the average price of mixture.
SP of mixture $= ₹ 20/\text{litre}$; profit $= 25\%$

$$\therefore \text{average price} = 20 \times \frac{100}{125} = ₹ 16/\text{litre}$$

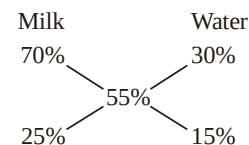
2. (c) **I Part** **II Part**



$\therefore$ ratio of quantities sold at 14% profit and 6% loss $= 2 : 18 = 1 : 9$.

$\therefore$ quantity sold at 14% profit $= \frac{50}{1+9} \times 1 = 5$ kg and sold at 6% loss $= 50 - 5 = 45$ kg.

3. (c) Here, the % values of milk and water that is taken from the vessel should be taken into consideration.



$$\Rightarrow 5 : 3$$

Ratio of milk to water $= 5 : 3$

$$\therefore \text{quantity of milk} = \frac{80}{5+3} \times 5 = 50 \text{ litres}$$

$$\text{and quantity of milk} = \frac{80}{5+3} \times 3 = 30 \text{ litres}$$

4. (c) Amount of liquid left after n operations, when the container originally contains x units of liquid from which y units are taken out each time is

$$x \left(\frac{x-y}{x} \right)^n \text{ units.}$$

Thus, in the above case, amount of milk left

$$= 80 \left[\frac{80-8}{80} \right]^3 \text{ kg} = 58.32 \text{ kg}$$

5. (c) 1st wheat 120 2nd wheat 144 3rd wheat 174
following the above rule, we have,

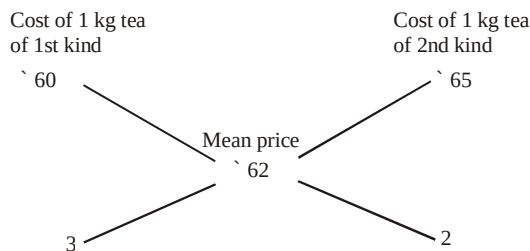
$$[(144 - 141) + (174 - 141)]$$

$$[= 141 - 120]$$

$$[= 141 - 120]$$

Therefore, the required ratio = 36 : 21 : 21 = 12 : 7 : 7

6. (a) S.P. of 1 kg of the mixture = ₹ 68.20, Gain = 10%
C.P. of 1 kg of the mixture = ₹ $\left(\frac{100}{110} \times 68.20 \right)$ = ₹ 62.
By the rule of alligation, we have



∴ Required ratio = 3 : 2

7. (c) Let the quantity of refined oil initially be Q . Then we have $Q \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = 10 \rightarrow Q = 2560/81$ litres.
8. (b) The percentage of honey in the new mixture would be:
 $(2 \times 25 + 3 \times 75)/5 = 275/5 = 55\%$. The ratio of honey to water in the new mixture would be 55 : 45 = 11 : 9.
9. (b) 20% spirit is mixed with 50% spirit to get 25% spirit. The ratio of mixing would be 5:1. This means he stole 5/6th of the bottle or 83.33% of the bottle.
10. (a) After 1st operation milk left

$$= 40 \left(1 - \frac{4}{40} \right) = 40 \left(1 - \frac{1}{10} \right) = 40 \cdot \frac{9}{10} = 36L$$

$$\text{After 2nd operation milk left} = 36 \left[1 - \left(\frac{5}{40} \right) \right] = 31.5$$

$$\text{After 3rd operation milk left} \left(36.5 - 6 \times \frac{31.5}{40} \right)$$

$$= (26.775 L)$$

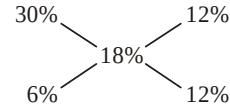
11. (c) By the above theorem the required ratio is

$$\left(\frac{2}{2+1} + \frac{3}{3+1} + \frac{3}{3+2} \right) : \left(\frac{1}{2+1} + \frac{1}{3+1} + \frac{2}{3+2} \right)$$

$$= \left(\frac{2}{3} + \frac{3}{4} + \frac{3}{5} \right) : \left(\frac{1}{3} + \frac{1}{4} + \frac{2}{5} \right)$$

$$= \frac{40+45+36}{3 \times 4 \times 5} : \frac{20+15+24}{3 \times 4 \times 5} = 121 : 59$$

12. (c) By the alligation rule, we find that wine containing 30% of spirit and wine containing 12% of spirit should be mixed in the ratio 1 : 2 to produce a mixture containing 18% of spirit.



$$\text{Ratio} = 6 : 12 = 1 : 2$$

This means that $\frac{1}{3}$ rd of the butt of sherry was left, i.e.

to say, the butler drew out $\frac{2}{3}$ rd of the butt.

∴ $\frac{2}{3}$ rd of the butt was stolen.

13. (d) It can be seen from the ratio 36 : 13 that the proportion of liquid soap to water is 36/49 after two mixings. This means that 6/7th of the liquid soap must have been allowed to remain in the container and hence 1/7th of the container's original liquid soap would have been drawn out by the thief. Since he takes out 4 gallons every time, there must have been 28 gallons in the container. (as 4 should be 1/7th of 28)

14. (d) Let vol. of jug = v litre
After first replacement, volume of orange juice = $(10-v)$
Volume of pineapple juice = v
After second replacement, volume of orange juice remaining

$$= (10-v) - \frac{(10-v)v}{10}$$

$$= \frac{(10-v)^2}{10}$$

Volume of pineapple juice remaining

$$= v - \frac{v^2}{10} = \frac{v(10-v)}{10}$$

Total volume of pineapple juice

$$= v \frac{(10-v)}{10} + v = \frac{20v - v^2}{10}$$

$$\text{Given } \frac{10-v}{10} = \frac{20v - v^2}{10} \Rightarrow v^2 - 20v + 50 = 0$$

$$\therefore v = 17.07 \text{ and } 2.93$$

since $v = 17.07$ is mentioned now here in the question/option

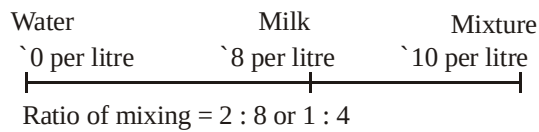
$$\therefore v = 2.93 \text{ Litre}$$

15. (b) When 4 litres of milk is taken out, volume of milk left in the vessel = 36 liters $[40 \{ 1 - (4/40) \}]$
When 5 litres of mixture is taken out, volume of milk left in the vessel = $[40 \{ 1 - (4/40) \} \{ 1 - (5/41) \}]$
When 6 litres of mixture is taken out, volume of milk left in the vessel = $[40 \{ 1 - (4/40) \} \{ 1 - (5/41) \} \{ 1 - (6/42) \}]$
= (27.09 L)

Explanation of Test Yourself

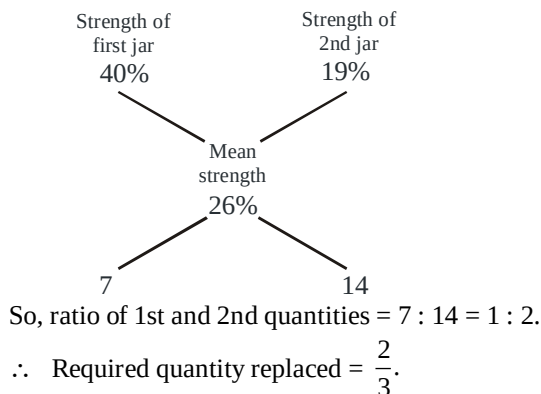
1. (d) In 125 gallons we have 25 gallons water and 100 gallons wine. To increase the percentage of water to 25%, we need to reduce the percentage of wine to 75%. This means that 100 gallons of wine = 75% of the new mixture. Thus the total mixture = 133.33 gallons. Thus, we need to mix $133.33 - 125 = 8.33$ gallons of water in order to make the water equivalent to 25% of the mixture.

2. (c) The cost price of the mixture would have been ₹ 8 per litre for him to get a profit of 25% by selling at ₹ 10 per litre. The ratio of mixing would have been 1: 4 (water is to milk) as can be seen in the figure:



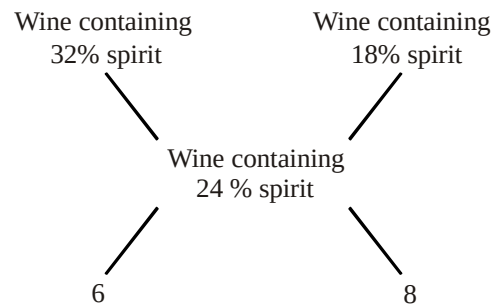
Since we are putting in 5 litres of water, the amount of milk must be 20 litres. The amount of mixture then would become 25 litres.

3. (c) Amount of water left = $50 \times 9/10 \times 9/10 = 40.5$ litres. Hence, wine = 9.5 litres. Ratio of wine and water = 19 : 81.
4. (d) In order to mix two tin alloys containing 86.66% tin and 93.33% tin to get 90% tin, the ratio of mixing should be 1 : 1. Thus, each variety should be 25 kg each.
5. (c) 90% and 97% mixed to form 94% means that the mixing ratio is 3 : 4. The first solution would be $3 \times 21/7 = 9$ litres.
6. (b) In order to solve this we need to assume a value for the amounts in the vessels. If we assume 35 litres as the quantities in all the three vessels we will get: 28 litres + 25 litres + 30 litres = 83 litres of petrol and 22 litres of kerosene in 105 litres of the mixture. The required ratio is 83 : 22.
7. (a) Fits the conditions of the problem as if there are 11 litres in the first vessel, there would be 8 litres of spirit. Also it means that we would be taking 24 litres from the second vessel out of which there would be 20 litres of spirit. Thus, total spirit would be 28 out of 35 litres giving us 7 litres of water.
8. (b) By the rule of alligation, we have :



9. (b) The S.P. of Desi Chai = ₹ 18
The S.P. of Videshi Chai = ₹ 30
The C.P. of Desi Chai = ₹ 20
The C.P. of Videshi Chai = ₹ 25
The S.P. of mixture = ₹ 27.5
The C.P. of mixture = ₹ 22

10. (c) By the rule of alligation,



$$\frac{\text{Quantity of 32\% spirit}}{\text{Quantity of 18\% spirit}} = \frac{6}{8} = \frac{3}{4}$$

Now, wine of 32% spirit = $\frac{3}{7}$ of the butt

∴ The rest part i.e., $1 - \frac{3}{7} = \frac{4}{7}$ of the butt has been stolen.

11. (d)
- | | |
|-------|-------|
| 16.66 | 18.75 |
| | |

$$\Rightarrow \frac{50}{3} \times \frac{4}{4} \quad \frac{75}{4} \times \frac{3}{3}$$

$$\Rightarrow \frac{35}{2} \times \frac{6}{6}$$

x	y
---	---

$$\frac{200}{12} \quad \frac{225}{12}$$

$$\Rightarrow \frac{210}{12}$$

$\frac{15}{12}$	:	$\frac{10}{12}$
3		2

⇒ Thus the number of girls = 16 and number of boys = 24

12. (a) Quantity of X = Quantity of Zinc.

Quantity of Y = Quantity of Tin.

$$M = 7 \text{ kg}$$

$$N = 21 \text{ kg}$$

$$a = 5$$

$$b = 2$$

$$x = 3$$

$$y = 4$$

Putting these values in the formula, we get

$$\frac{\text{Quantity of Zinc}}{\text{Quantity of Tin}} = \frac{7 \times \left(\frac{5}{5+2} \right) + 21 \times \left(\frac{3}{3+4} \right)}{7 \times \left(\frac{2}{5+2} \right) + 21 \times \left(\frac{4}{3+4} \right)}$$

$$= \frac{14}{14} = 1 : 1$$

13. (b) Since the average marks of sections B and C together are equal the average marks of all the four sections (i.e., A, B, C and D), therefore the average marks of the remaining two sections A and D together will also be equal i.e., 60%.

$$\begin{array}{ccc} 45 & & 80 \\ & \searrow \quad \nearrow & \\ & 60 & \\ & \nearrow \quad \searrow & \\ 20 & & 15 \\ \Rightarrow & 4 & : \quad 3 \end{array}$$

Hence, the required ratio is 4 : 3.

14. (a) Copper in 4 kg = $\frac{4}{5}$ kg

$$\text{and zinc in 4 kg} = 4 \times \frac{4}{5} = \frac{16}{5} \text{ kg}$$

$$\text{Copper in 5 kg} = 5 \times \frac{1}{6} = \frac{5}{6} \text{ kg}$$

$$\text{Zinc in 5 kg} = 5 \times \frac{5}{6} = \frac{25}{6} \text{ kg}$$

Therefore, Copper in mixture

$$= \frac{4}{5} + \frac{5}{6} = \frac{49}{30} \text{ kg}$$

$$\text{and zinc in the mixture} = \frac{16}{5} + \frac{25}{6} = \frac{221}{30} \text{ kg}$$

Therefore, the required ratio = 49 : 221.

15. (a) If the whole money is invested at 6% the annual income is 6% of ₹ 1,500 = ₹ 90. If the whole money is invested at 5%, the annual income is 5% of ₹ 1,500 = ₹ 75. But real income = ₹ 85.

∴ Applying the alligation rule, we have

$$\begin{array}{ccc} 6\% & & 5\% \\ \text{₹ 90} & & \text{₹ 75} \\ & \searrow \quad \nearrow & \\ & 85 & \\ & \nearrow \quad \searrow & \\ \text{₹ 10} & & \text{₹ 5} \\ 2 & & 1 \end{array}$$

$$\therefore \text{Money invested at 5\%} = \frac{1}{3} \times ₹ 1,500 = ₹ 500$$

5

PERCENTAGES

- Introduction
- Basic Definition of Percentage
- Percentage Increase, Percentage Decrease and Percentage Change
- Percentage Point Change and Percentage Change
- Calculation of Percentage Value Through Addition
- Effect of Percentage Change in the Numerator on the Value of a Ratio
- Percentage Change Graphic
- Application of Percentage Change Graphic (PCG)
- Calculation of Multiplication by Numbers Like 2.14, 1.04, 0.35, 0.94 and so on Using Percentage

INTRODUCTION

The chapter 'Percentages' is one of the most important chapters. Concept of this chapter is the backbone of Commercial Arithmetic, because this chapter plays a vital role in developing the calculation skills, which are used in chapters like Profit and Loss; Ratio and Proportion; Time and Work; Time, Speed and Distance; Data of Interpretation, etc.

On an average, two questions are asked from this chapter in CAT every year.

BASIC DEFINITION OF PERCENTAGE

The word per cent means per hundred or for every hundred. The symbol '%' is used for the term percent. Thus, 20 per cent is written as 20% and it means 20 out of 100.

This is written in ratio form as $\frac{20}{100}$.

The percentage value of a ratio is obtained when we multiplying the ratio by 100.

Thus percentage value of the ratio $\frac{3}{5}$ will be $\frac{3}{5} \times 100\% = 60\%$.

Illustration 1: A person saves ₹ 5,000 per month from his monthly salary ₹ 30,000. Find the percentage monthly saving of the person.

Solution: Out of monthly salary ₹ 30,000, saving is Rs 5,000

⇒ Out of monthly salary ₹ 1, saving is ₹ $\frac{5,000}{30,000}$

⇒ Out of monthly salary ₹ 100, saving is ₹ $\frac{5,000}{30,000} \times 100$

$$= ₹ \frac{50}{3} = ₹ 16.67 \text{ (approx.)}$$

Hence percentage monthly saving = 16.67% (approx.)

Illustration 2: 250 students of ABC school and 350 students of XYZ school appeared in secondary board examination conducted by CBSE in 2013. 20 students of ABC school and 25 students of XYZ school did not pass in this board examination. Students of which of the two schools ABC and XYZ have shown poor performance?

Solution: We cannot compare the performance of the students of the two schools in secondary board examination by just looking the number of students 20 of ABC school and 25 of XYZ school who did not pass in secondary board examination.

To compare the performance, you have to find the percentage of the students who did not pass the secondary board examination of each school out of those students of each school who appeared in the secondary board examination.

Percentage of the students of ABC school who did not pass

$$= \frac{20}{250} \times 100\% = 8\%$$

Percentage of the student of XYZ school who did not pass

$$= \frac{25}{350} \times 100\% = 7.1\% \text{ (approximately)}$$

Hence students of the XYZ school have shown poor performance.

Illustration 3: In a survey, voters of a national party A are increase by 2.5 lakhs and voters of national party B are increase by 4 lakhs in 2012. Which party A or B has grown more in 2012 ?

Solution: In first shot the answer to the question seems to be national party *B*. But actually the question can not be answered, because we don't know the just previous year's voters of each of the national party *A* and *B*.

If we had further information that in 2011, voters of national party *A* were 5 lakhs and voters of national party *B* were 10 lakhs, we can compare growth rates of two national parties.

Percentage growth rate of national party *A* in 2012

$$= \frac{250000}{500000} \times 100\% = 50\%$$

Percentage growth rate of national party *B* in 2012

$$= \frac{400000}{1000000} \times 100\% = 40\%$$

Hence, national party *A* has higher growth rate in 2012. Thus national party *A* has grown more than *B* in 2012.

In the illustrations 2 and 3, you have seen that percentage is the most powerful tool for comparing the data. 500000 and 1000000 in illustration 3 are called base values of percentage growth rate of party *A* and party *B* respectively.

Without knowing these base values, percentage growth rate of party *A* and party *B* could not be determined.

$$\text{Thus percentage of anything (let } X) = \frac{\text{Value of } X}{\text{Base value of } X} \times 100$$

In illustration 1, ₹ 30000 is the base value of percentage monthly saving. In illustration 2, 250 is the base value of the percentage of students of *ABC* school who did not pass and 350 is the base value of the percentage of student of *XYZ* school who did not pass.

Illustration 4: Express the following as fraction

- (a) 25% (b) $33\frac{1}{3}\%$

Solution :

$$(a) 25\% = \frac{25}{100} \left(\text{Since } \% \text{ means } \frac{1}{100} \right) = \frac{1}{4}$$

$$(b) 33\frac{1}{3}\% = \frac{100}{3}\% = \frac{100}{3 \times 100} = \frac{1}{3}$$

Illustration 5: 25% of a number is 80. What is the number ?

Solution:

Let the number be *X*. According to the given condition

$$\frac{25}{100} \times X = 80 \Rightarrow X = \frac{80 \times 100}{25} = 320.$$

Illustration 6: Express $\frac{1}{8}$ as a percentage.

$$\text{Solution: } \frac{1}{8} = \frac{1}{8} \times 100\%$$

$$= \frac{100}{8}\% = \frac{25}{2}\% = 12\frac{1}{2}\%$$

Illustration 7: Two third of three fifth of one eighth of a certain number is 268.50. What is 30% of the number?

- (a) 1611 (b) 1616
(c) 1343 (d) 594.60

Solution: (a) Let the number be *x*.

According to the question $\frac{2}{3}$ of $\frac{3}{5}$ of $\frac{1}{8} \times x = 268.50$

$$\Rightarrow \frac{2}{3} \times \frac{3}{5} \times \frac{1}{8} \times x = 268.50$$

$$x = \frac{268.50 \times 3 \times 5 \times 8}{2 \times 3} = 5370$$

$$30\% \text{ of } x = \frac{30}{100} \times 5370 = 1611.00$$

Illustration 8: 4598 is 95% of ?

- (a) 4800 (b) 4840
(c) 4850 (d) 4880

Solution: (b) Let 95% of *x* = 4598.

$$\text{Then, } \frac{95}{100} \times x = 4598 \Rightarrow x = \left(4598 \times \frac{100}{95} \right) = 4840.$$

PERCENTAGE INCREASE, PERCENTAGE DECREASE AND PERCENTAGE CHANGE

$$\text{Percentage increase} = \frac{\text{Increase}}{\text{Initial value (i.e., Base value)}} \times 100$$

$$\text{Percentage decrease} = \frac{\text{Decrease}}{\text{Initial value (i.e., Base value)}} \times 100$$

$$\text{Percentage change} = \frac{\text{Change}}{\text{Initial value (i.e., Base value)}} \times 100$$

Let income of a family in the years 2010, 2011 and 2012 are ₹ 50000, ₹ 80000 and ₹ 60000 respectively.

Here income of the family increases in 2011 but decreases in 2012.

Increase in family income in 2011 from 2010

$$\begin{aligned} &= (\text{Higher Income}) - (\text{Lower Income}) \\ &= (\text{Income in 2011}) - (\text{Income in 2010}) \\ &= ₹ 80000 - ₹ 50000 = ₹ 30000 \end{aligned}$$

Decrease in family income in 2012 from 2011

$$\begin{aligned} &= (\text{Higher Income}) - (\text{Lower Income}) \\ &= (\text{Income in 2011}) - (\text{Income in 2012}) \\ &= ₹ 80000 - ₹ 60000 = ₹ 20000 \end{aligned}$$

Percentage increase in family income in 2011 from 2010

$$\begin{aligned} &= \frac{(\text{Increase in income in 2011 from 2010})}{(\text{Income in 2010})} \times 100 \\ &= \frac{30000}{50000} \times 100 = 60\% \end{aligned}$$

Percentage decrease in family income in 2012 from 2011

$$\begin{aligned} &= \frac{(\text{Decrease in income in 2012 from 2011})}{(\text{Income in 2011})} \times 100 \\ &= \frac{20000}{80000} \times 100 = 25\% \end{aligned}$$

Illustration 9: Rent of the house is increased from ₹ 7000 to ₹ 7700. Express the increase in price as a percentage of the original rent.

Solution:

$$\text{Increase value} = ₹ 7700 - ₹ 7000 = ₹ 700$$

$$\text{Increase \%} = \frac{\text{Increase value}}{\text{Base value}} \times 100 = \frac{700}{7000} \times 100 = 10$$

$$\therefore \text{Percentage rise} = 10\%$$

Illustration 10: The cost of a bike last year was ₹ 19000. Its cost this year is ₹ 17000. Find the percent decrease in its cost.

Solution:

$$\begin{aligned} \text{\% decrease} &= \frac{19000 - 17000}{19000} \times 100 \\ &= \frac{2000}{19000} \times 100 = 10.5\% \end{aligned}$$

$$\therefore \text{Percent decrease} = 10.5\%$$

If the value of any thing increases, then percentage change is the percentage increase and if the value of any thing decreases, then percentage change is the percentage decrease. Thus,

Percentage change = Percentage increase, if value of any thing increases

and Percentage change = Percentage decrease, if value of anything decreases.

PERCENTAGE POINT CHANGE AND PERCENTAGE CHANGE

Percentage point change and percentage change are easily understood through the following example:

Let monthly salary of a person be ₹ 60,000 in 2012. His monthly expenditure increases from 60% to 80% of his monthly salary in November, 2012.

Here percentage point change in monthly expenditure in November, 2012.

$$\begin{aligned} &= (80\% \text{ of monthly salary}) - (60\% \text{ of monthly salary}) \\ &= 20\% \text{ of monthly salary} \end{aligned}$$

And percentage change in monthly expenditure in November, 2012

$$\begin{aligned} &= \text{Percentage change in initial percentage expenditure} \\ &= \frac{80 - 60}{60} \times 100\% \end{aligned}$$

$$= 33\frac{1}{3}\% \text{ of initial expenditure in November 2012.}$$

Here percentage point change and percentage change look different, as they are 20% and $33\frac{1}{3}\%$ respectively but actually their values the same as shown below.

Percentage point change = 20% of monthly salary

$$= \frac{20}{100} \times 60,000 = ₹ 12,000$$

$$\text{Percentage change} = 33\frac{1}{3}\% \text{ of initial expenditure}$$

$$= 33\frac{1}{3}\% \text{ of } \left(60,000 \times \frac{60}{100}\right)$$

$$= \frac{100}{3}\% \text{ of } (36000)$$

$$= \frac{100}{3 \times 100} \times 36,000 = ₹ 12,000$$

$$\text{Note that } x\% \text{ of } y = \frac{x}{100} \times y$$

CALCULATION OF PERCENTAGE VALUE THROUGH ADDITION

Calculation of percentage values through addition is easily understood through the following example:

Suppose, we want to calculate 123 is how much per cent of 35

i.e. the percentage value of $\frac{123}{35}$, which is $\frac{123}{35} \times 100 = 351.43$.

you can find this percentage value through addition also.

To understand this process, note that 100% of 35

$$= \frac{100}{100} \times 35 = 35, 10\% \text{ of } 35 = \frac{10}{100} \times 35 = 3.5,$$

$$1\% \text{ of } 35 = \frac{1}{100} \times 35 = 0.35.$$

In this process of required calculation, we remove the multiple of 100%, multiple of 10%, multiple of 1% of the denominator from the numerator in order.

$$\text{Let percentage value of } \frac{123}{35} \text{ i.e. } \frac{123}{35} \times 100\% = x . abcd \dots,$$

where x is a natural number and $a, b, c, d, \dots$ are any digits from 0 to 9.

$$35 \times 3 (= 105, R_1 = 18) + 3.5 \times 5 (= 17.5, R_2 = 0.5) + 0.35 \times 1 (= 0.35, R_3 = 0.15)$$

$$\text{Then } x = \frac{35}{35}$$

Here, $R_1 (= 18)$ is the remaining part of 123 after subtracting 105 from it,

$R_2 (= 0.5)$ is the remaining part of 18 after subtracting 17.5 from it,

and $R_3 (= 0.15)$ is the remaining part of 0.5 after subtracting 0.35 from it.

Leaving the remainder R_3 , which is less than 1% (i.e., 0.35) of 35, we get

$$x = \frac{35 \times 3 + 3.5 \times 5 + 0.35 \times 1}{35}$$

$$= 3 \times \frac{35}{35} + 5 \times \frac{3.5}{35} + 1 \times \frac{0.35}{35}$$

$$= 3 \times 100\% + 5 \times 10\% + 1 \times 1\%$$

$$= 300\% + 50\% + 1\% = 351\%$$

From the last remainder $R_3 (= 0.15)$, we calculate the value of first digit 'a' after decimal in $x . abcd\dots$ after multiplying the numerator of $\frac{0.15}{35}$ by 10 and proceed as follow to calculate the value of a .

$$a = \frac{0.15 \times 10}{35} = \frac{1.5}{35} = \frac{0.35 \times 4 (= 1.4, R_4 = 0.1)}{35} = 4 \times \frac{0.35}{35},$$

leaving the last remainder R_4 , which is less than 1% of $\frac{123}{35}$.

$$= 4 \times 1\% = 4\%$$

From R_4 , we proceed as follow to calculate the value of b .

$$b = \frac{0.1 \times 10}{35} = \frac{1}{35} = \frac{0.35 \times 2 (= 0.70, R_5 = 0.30)}{35} = 2 \times \frac{0.35}{35} = 2 \times 1\% = 2\%$$

Similarly, we can find the value of $c, d, \dots$

$$\text{Hence, } \frac{123}{35} = x \cdot abcd \dots = 351.42 \dots$$

We can summarise the whole process as:

$$\begin{aligned} \frac{123}{25} &= x \cdot abcd \dots \\ 35 \times 3 (= 105, R_1 = 18) + 3.5 \times 5 (= 17.5, R_2 = 0.5) \\ &+ 0.35 \times 1 (= 0.35, R_3 = 0.15) \\ x &= \frac{35}{35} \\ &= 3 \times 100\% + 5 \times 10\% + 1 \times 1\% = 351\% \\ a &= \frac{0.15 \times 10}{35} = \frac{1.5}{35} = \frac{0.35 \times 4 (= 1.4, R_4 = 0.1)}{35} = 4 \times 1\% = 4\% \\ b &= \frac{0.1 \times 10}{35} = \frac{1}{35} = \frac{0.35 \times 2 (= 0.7, R_5 = 0.3)}{35} = 2 \times 1\% = 2\% \\ \therefore \frac{123}{25} &= 351.42 \dots \end{aligned}$$

The advantage of this process is that you only calculate as long as you need so that you guess the right option.

For example, if you have given four options (a) 233.47%, (b) 243.47%, (c) 255.60% and (d) 260.23% for a question (single correct MCQ): 56 is how much per cent of 23? Then you need to calculate the percentage value of $\frac{56}{23}$ upto the multiple of 10 % of $\frac{56}{23}$ only to guess the correct option.

$$\begin{aligned} \frac{56}{23} &= \frac{23 \times 2 (= 46, R_1 = 10) + 2.3 \times 4 (= 9.2, R_2 = 0.8)}{23} \\ &= 2 \times \frac{23}{23} + 4 \times \frac{2.3}{23} \\ &= 2 \times 100\% + 4 \times 10\% = 200\% + 40\% = 240\% \end{aligned}$$

The correct value is lies between 240% and 250%.

Hence the correct option is (b) 243.47%

Illustration 11: Find the percentage value of the ratio $\frac{442}{75}$.

$$\begin{aligned} \text{Solution: } \frac{442}{75} &= x \cdot abcd \dots (\text{let}) \\ x &= \frac{442}{75} = \frac{75 \times 5 (= 375, R_1 = 67)}{75} + \frac{7.5 \times 8 (= 60, R_2 = 7)}{75} \\ &\quad + \frac{0.75 \times 9 (= 6.75, R_3 = 0.25)}{75} \\ &= 5 \times 100\% + 8 \times 10\% + 9 \times 1\% \\ &= 500\% + 80\% + 9\% = 589\% \end{aligned}$$

$$a = \frac{0.25 \times 10}{75} = \frac{2.5}{75} = \frac{0.75 \times 3 (= 2.25, R_4 = 0.25)}{75}$$

$$= 3 \times \frac{0.75}{75} = 3 \times 1\% = 3\%$$

Since remainder R_3 and R_4 are the same 0.25, therefore

$$a = b = c = d = \dots$$

$$\therefore \text{Percentage value of } \frac{442}{75} = 589.333\dots$$

Illustration 12: 45 is how many per cent of 56?

- (a) 78.43 (b) 80.25
(c) 80.35 (d) 81.45

Solution: Percentage value of $\frac{45}{56} = x \cdot abcd \dots (\text{let})$

$$\begin{aligned} x &= \frac{45}{56} = \frac{5.6 \times 8 (= 44.8, R_1 = 0.2)}{56}, R_1 \text{ is less than 1\% of 56.} \\ &= 8 \times 10\% = 80\% \end{aligned}$$

$$\begin{aligned} a &= \frac{0.2 \times 10}{56} = \frac{2}{56} = \frac{0.56 \times 3 (= 1.68, R_2 = 0.32)}{56} \\ &= 3 \times 1\% = 3\% \end{aligned}$$

$$\therefore \frac{45}{56} = (80.3 \dots)\%$$

Hence correct option is (c).

EFFECT OF PERCENTAGE CHANGE IN THE NUMERATOR ON THE VALUE OF A RATIO

Numerator of a ratio has direct relationship with the ratio, if it's, denominator remains constant. That is if denominator of a ratio remains constant, then the value of ratio increases if numerator increases and the value of the ratio decreases if numerator decreases. The percentage increase or decrease in a value of ratio is the same as the percentage increases or decreases in the numerator respectively, if denominator remains constant.

For example, $\frac{60}{82}$ is 20% more than $\frac{50}{82}$ (in terms of percentage change) as numerator 60 is 20% more than the numerator 50.

PERCENTAGE CHANGE GRAPHIC

Increase or decrease in a number by a certain per cent can be represented in many ways. A very simple and common way to represent it is given below.

$$30\% \text{ increase in 50 is represented as } 50 \xrightarrow{+15} 65$$

This means that 30% of 50 is 15, which when added to 50, we get 65.

30% $\uparrow$ means 30% increase

$$20\% \text{ decrease in 60 is represented as } 60 \xrightarrow{-12} 48$$

This means that 20% of 60 is 12, which when decreases from 60, we get 48.

20% $\downarrow$ means 20% decrease

These common representation is known as Percentage Change Graphic (PCG).

APPLICATION OF PERCENTAGE CHANGE GRAPHIC (PCG)

Percentage change graphic (PCG) has many applications. Six major applications of PCG are given below:

- Successive percentage changes
- Product change
- Product constancy
- $A \rightarrow B \rightarrow A$ (Reverse Relation)
- Effect of change in denominator on the value of the ratio
- Calculation of ratio change

Application-1: Successive Percentage Changes

Successive percentage changes are very common in many questions. Suppose you have to solve a problem, in which a number 50 increases successively by two percentages 20% and 10% then decreases by 25%.

These successive changes are handled by using PCG as:

$$50 \xrightarrow[+10]{20\% \uparrow} 60 \xrightarrow[+6]{10\% \uparrow} 66 \xrightarrow[-16.5]{25\% \downarrow} 49.5$$

Here another method is also used. In this method, if value of a quantity is first increased by $a\%$ and then decreased by $b\%$, then the original quantity is increased or decreased by $\left(a - b - \frac{ab}{100}\right)\%$ according to the +ve or -ve sign respectively of the value of $\left(a - b - \frac{ab}{100}\right)$. But PCG method is based on common sense and no need to learn formula, so we preferred PCG method.

Also through PCG, we can find the result after any number of successive changes whereas using formula, we can find the result only when first once increases and then once decreases only.

Illustration 13: A shopkeeper marked selling price of a particular item by 25% more than his cost price. Due to off season, he has given a discount of 10% on his marked price. What percentage of profit he makes?

Solution: $100 \xrightarrow[+25]{25\% \uparrow} 125 \xrightarrow[-12.5]{10\% \downarrow} 112.50$

Percentage profit = 12.5%

Illustration 14: A trader gives successive discounts of 40% and 20% respectively. Find the percentage of the original cost price he will recover.

Solution: $100 \xrightarrow[-40]{40\% \downarrow} 60 \xrightarrow[-12]{20\% \downarrow} 48$

Hence the trader will recover 48% of his original cost price.

Application-2: Product Change

Suppose $20 \times 20 \times 20$ increases to $22 \times 24 \times 30$

Here first factor, second factor and third factors are increased by 10%, 20% and 50% respectively.

Here, we can find the percentage increase in the product $20 \times 20 \times 20$ using PCG as follows:

$$100 \xrightarrow[+10]{10\% \uparrow} 110 \xrightarrow[+22]{20\% \uparrow} 132 \xrightarrow[+66]{50\% \uparrow} 198$$

Hence the product $20 \times 20 \times 20$ increases 98%.

Note that: You will get the same result irrespective of the order in which you use the respective percentage changes as

$$100 \xrightarrow[+20]{20\% \uparrow} 120 \xrightarrow[+60]{50\% \uparrow} 180 \xrightarrow[+18]{10\% \uparrow} 198$$

This method is used for any number of the same factors. Also note that this method is similar to the successive percentage change.

The following two formulae are also used to find the result of product change.

For the product of two same factors like $a \times a$, if one of the two factors is increased by $x\%$ and other factor is increased by $y\%$, then the product $a \times a$ is increased by $\left(x + y + \frac{xy}{100}\right)\%$.

For the product $a \times a$, if one of the two factors is increased by $x\%$ and other factor is decreased by $y\%$, then the product $a \times a$ increased or decreased by $\left(x - y - \frac{xy}{100}\right)\%$ according to the +ve or -ve sign of the value of $\left(x - y - \frac{xy}{100}\right)\%$

Note that if length of a side 'a' of a square is increased or decreased by $x\%$ then its original area $a \times a$ is increased or decreased respectively by $\left(2x \pm \frac{x^2}{100}\right)\%$.

Here we study two methods for product change but PCG method is based on common sense and no need to learn formula. So, we preferred PCG method. Also PCG is used for any number of the same factors.

Illustration 15: If length of each side of a square is increased by 20%, then find the percentage increase in its area.

Solution: Area of a square = side $\times$ side

Method-I: $100 \xrightarrow[+20]{20\% \uparrow} 120 \xrightarrow[+24]{20\% \uparrow} 144$

Hence, percentage increase in area = 44%

Method-II: Percentage increase in area
 $= \left(20 + 20 + \frac{20 \times 20}{100}\right)\% = (40 + 4)\% = 44\%.$

Illustration 16: A number is increased by 10%. and then it is decreased by 10%. Find the net increase or decrease per cent.

Solution: Method I: $100 \xrightarrow[+10]{10\% \uparrow} 110 \xrightarrow[-11]{10\% \downarrow} 99$

Hence 1% decrease.

Method II: % change = $10 - 10 - \frac{10 \times 10}{100} = -1\%$

Hence 1% decrease.

Illustration 17: The price of a car is decreased by 10 % and 20 % in two successive years. What per cent of price of the car is decreased after two years ?

Solution: Method I: $100 \xrightarrow[-10]{10\% \downarrow} 90 \xrightarrow[-18]{20\% \downarrow} 72$

Hence percentage decrease = $100 - 72 = 28\%$.

Method II: $-10 - 20 + \frac{(-10)(-20)}{100} = -28\%$

$\therefore$ The price of the car decreases by 28%.

Illustration 18: If the radius of a circle is diminished by 10%, the area is diminished by

- (a) 36% (b) 20% (c) 19% (d) 10%

Solution: (c) If the radius is diminished by $r\%$, then

area is diminished by $\left(2r - \frac{r^2}{100}\right)\%$

Hence required percentage decrease

$$= 2 \times 10 - \frac{10^2}{100} = 19\%$$

Application-3: Product Constancy

This application is easily understood by the following illustration.

Suppose price of a commodity has gone up by 100% and you want to keep the total expenditure on the commodity the same. Obviously, you need to reduce the consumption of the commodity but by what percent? By use of PCG, you can easily find the answer as follows:

$$100 \xrightarrow[\substack{+100 \\ \text{Increase} \\ \text{in price}}]{100\% \uparrow} 200 \xrightarrow[\substack{-100 \\ \text{Reduction in} \\ \text{consumption}}]{50\% \downarrow} 100$$

Hence required percentage of reduction in consumption in commodity to keep the expenditure the same = 50%

Application-4: A → B → A (Reverse Relation)

Many times, a relationship from A to B is given and you have to find its reverse relationship i.e., a relationship from B to A. This reverse relationship can be determined by using PCG like product consistency as discussed in the following illustration.

Illustration 19: B's salary is 25% more than A's salary. By what per cent is A's salary less than B's salary?

Solution: $100 (A) \xrightarrow[\substack{+25 \\ +25}]{25\% \uparrow} 125 (B) \xrightarrow[\substack{-25 \\ -25}]{20\% \downarrow} 100 (A)$

Hence A's salary is 20% less than B's salary.

Application-5: Effect of Change in Denominator on the Value of the Ratio

When denominator of a ratio increases then value of the ratio decreases and when denominator of the ratio decreases, then value of the ratio increases. Hence there is a reverse relation between change in denominator of a ratio and the value of a ratio.

Hence, if numerator of two ratios are the same then for any change in denominator, we can find the change in the value of ratio using product constancy as given below:

Consider the two ratios,

Ratio -1: $\frac{16}{20}$ Ratio -2: $\frac{16}{25}$

Clearly to make both the ratios equal, we have to drop the denominator of second ratio by 20%. Therefore, by reverse relationship between denominators and ratios, the first ratio is 20% more than second ratio.

This can be shown by PCG,

$$100 \xrightarrow[\substack{+25 \\ \text{Denominator} \\ \text{of first ratio}}]{25\% \uparrow} 125 \xrightarrow[\substack{-25 \\ \text{Denominator} \\ \text{of second ratio}}]{20\% \downarrow} 100$$

∴ 20% drop in denominator ⇒ First ratio is 20% more than the second ratio.

Application-6: Calculation of Ratio Change

When we have given two ratios of which both numerators and also both denominators are different, then we calculate percentage change between the two ratios through PCG as discussed in the following illustration.

Illustration 20: Find the percentage change between the ratios

$\frac{20}{60}$ and $\frac{30}{75}$.

Solution: Ratio -1: $\frac{20}{60}$, Ratio -2: $\frac{30}{75}$

Here the change in ratios occurs due to two reasons:

- (i) Percentage change in numerator (Numerator Effect)
- (ii) Percentage change in denominator (Denominator Effect)

By finding the numerator effect and denominator effect and then segregating the two effects, we get the percentage change between the two given ratios as follows:

Numerator Effect: Numerator changes from 20 to 30. Since, there is a direct relationship between the percentage change in numerator and the percentage change in the value of a ratio. Hence numerator effect is 50% increase.

Denominator Effect: Denominator changes from 60 to 75. Since, there is a reverse relationship between the percentage change in denominator and percentage change in the value of the ratio.

Hence, the denominator effect will be seen by going reverse from 75 to 60 i.e. 20% drop.

Thus overall percentage change in ratio is seen as

$$100 \xrightarrow[\substack{+50 \\ \text{Numerator} \\ \text{Effect}}]{50\% \uparrow} 150 \xrightarrow[\substack{-30 \\ \text{Denominator} \\ \text{Effect}}]{20\% \downarrow} 120$$

Hence second ratio is $(120 - 100 = 20)\%$ more than the first ratio.

CALCULATION OF MULTIPLICATION BY NUMBERS LIKE 2.14, 1.04, 0.35, 0.94 AND SO ON USING PERCENTAGE

We will discuss this method of multiplication in the following illustration.

Illustration 21: Calculate 2.46×532 .

Solution: To find the value of the product 2.46×532 , you can multiply 2.46 and 532 directly but it is a time taking process, which will yield the final value at the end of the process but nothing before that. So you will have no clue about the answer's range till you reach the end of the calculation.

To find the product above using percentages, you should view the given multiplication 2.46×532 as 46% more than twice the 532.

$$\begin{aligned} \text{i.e. } 2.46 \times 532 &= 2 \times 532 + 46\% \text{ of } 532 \\ &= 1064 + 40\% \text{ of } 532 + 6\% \text{ of } 532 \\ &= 1064 + 4 \times (10\% \text{ of } 532) + 6 \times (1\% \text{ of } 532) \\ &= 1064 + 4 \times 53.2 + 6 \times 5.32 \\ &= 1064 + 212.80 + 31.92 = 1308.72 \end{aligned}$$

You can express the whole process in a single line as

$$\begin{aligned} 2.46 \times 532 &= 2 \times 532 + 4 \times 53.2 + 6 \times 5.32 \\ &= 1064 + 212.80 + 31.92 = 1308.72 \end{aligned}$$

Similarly, you should view:

Multiply a number by 1.04 as 4% more than the number.

Multiply a number by 0.35 as 35% of the number.

Multiply a number by 0.94 as 94% of the number or 6% less than the number.

Practice Exercises



Foundation Level

- In a public library there are 110,000 books, 40% of which are science books. It was decided to add 20,000 new books to the library. How many of the new books should be science books in order to bring the percentage of science books in the library up to 45%?
 - 15000
 - 1500
 - 1450
 - 14500
- A batsman scored 110 runs which included 3 boundaries and 8 sixes. What percent of his total score did he make by running between the wickets?
 - 45%
 - $45\frac{5}{11}\%$
 - $54\frac{6}{11}\%$
 - 55%
- A student secures 90%, 60% and 54% marks in test papers with 100, 150 and 200 respectively as maximum marks. The percentage of his aggregate is:
 - 64
 - 68
 - 70
 - None of these
- If two numbers are respectively 20% and 50% of a third number, what is the percentage of the first number to the second?
 - 10
 - 20
 - 30
 - 40
- In an examination, 5% of the applicants were found ineligible and 85% of the eligible candidates belonged to the general category. If 4275 eligible candidates belonged to other categories, then how many candidates applied for the examination?
 - 30,000
 - 35,000
 - 37,000
 - None of these
- Deepa decided to donate 8% of her salary to an orphanage, On the day of donation she changed her mind and donated ₹ 2240 which was 80% of what she had decided earlier. How much is Deepa's salary?
 - ₹ 36000
 - ₹ 42000
 - ₹ 35000
 - ₹ 45000
- When the price of a radio was reduced by 20%, its sale increased by 80%. What was the net effect on the sale?
 - 44% increase
 - 44% decrease
 - 66% increase
 - 75% increase
- If the price of sugar is increased by 7%, then by how much per cent should a housewife reduce her consumption of sugar, to have no extra expenditure?
 - 7 over 107%
 - 107 over 100%
 - 100 over 107%
 - 7%
- A salesman's terms were changed from a flat commission of 5% on all his sales to a fixed salary of ₹ 1,000 plus 2.5% commission on all sales exceeding ₹ 4,000. If his remuneration as per the new scheme was ₹ 600 more than by the first scheme, what were his sales worth?
 - 10,000/-
 - 11,000/-
 - 12,000/-
 - 14,000/-
- An inspector rejects 0.08% of the metres as defective. How many metres will he examine to reject 2 metres?
 - 200m
 - 250m
 - 2500m
 - 3000m
- A invested 10% more than B. B invested 10% less than C. If the total sum of their investment is ₹ 14450. how much did C get?
 - ₹ 5000
 - ₹ 4800
 - ₹ 5100
 - None of these
- A sum of ₹ 4558 is divided among A, B and C such that A receives 20% more than C, and C receives 25% less than B. What is A's share in the amount?
 - ₹ 1548
 - ₹ 1720
 - ₹ 1290
 - ₹ 1345
- In an election between two candidates, 75% of the voters cast their votes, out of which 2% of the votes were declared invalid. A candidate got 9261 votes which were 75% of total valid votes. Find the total number of votes enrolled in that election.
 - 16080
 - 16800
 - 18600
 - 16008
- A spider climbed $62\frac{1}{2}\%$ of the height of the pole in one hour and in the next hour it covered $12\frac{1}{2}\%$ of the remaining height. If the height of the pole is 192 m, then distance climbed in second hour is:
 - 3 m
 - 5 m
 - 7 m
 - 9 m
- A number is increased by 10% and then reduced by 10%. After these operations, the number:
 - does not change
 - decreases by 1%
 - increases by 1%
 - increases by 0.1%
- The difference between the value of a number increased by 25% and the value of the original number decreased by 30% is 22. What is the original number?
 - 70
 - 65
 - 40
 - 90

17. A salesman is allowed $5\frac{1}{2}\%$ discount on the total sales made by him plus a bonus of $\frac{1}{2}\%$ on the sales over ₹ 10,000. If his total earnings were ₹ 1990, then his total sales (in ₹) were:
 (a) 30,000 (b) 32,000
 (c) 34,000 (d) 35,000
18. If 12% of 75% is greater than 5% of a number by 75, the number is
 (a) 1875 (b) 1890
 (c) 1845 (d) 1860
19. Mr Yadav spends 60% of his monthly salary on consumable items and 50% of the remaining on clothes and transport. He saves the remaining amount. If his savings at the end of the year were ₹ 48456, how much amount per month would he have spent on clothes and transport?
 (a) ₹ 4038 (b) ₹ 8076
 (c) ₹ 9691.20 (d) ₹ 4845.60
20. In a certain year, the population of a certain town was 9000. If in the next year the population of males increases by 5% and that of the females by 8% and the total population increases to 9600, then what was the ratio of population of males and females in that given year?
 (a) 4 : 5 (b) 5 : 4
 (c) 2 : 3 (d) None of these
21. The salary of Raju and Ram is 20% and 30% less than the salary of Saroj respectively. By what % is the salary of Raju is more than the salary of Ram?
 (a) 33.33% (b) 50%
 (c) 15.18% (d) 14.28%
22. Wheat is now being sold at ₹ 27 per kg. During last month its cost was ₹ 24 per kg. Find by how much per cent a family reduces its consumption so as to keep the expenditure fixed.
 (a) 10.2% (b) 12.1%
 (c) 12.3% (d) 11.1%
23. $33\frac{1}{3}\%$ of a man's daily output is equal to 50% of a second man's daily output. If the second man turns out 1500 screws daily, then the first man's output in terms of making screws is:
 (a) 500 (b) 1000
 (c) 2000 (d) 2250
24. A fraction is such that if the double of the numerator and the triple of the denominator is changed by + 10% and -30% respectively then we get $\frac{11}{16}$ of $\frac{16}{21}$. Find the fraction.
 (a) $\frac{4}{25}$ (b) $\frac{2}{25}$
 (c) $\frac{3}{25}$ (d) None of these
25. The entry fee in an exhibition was ₹ 10. Later this was reduced by 25%, which increased the sale of tickets by 20%. Find the percentage increase in the number of visitors.
 (a) 54 (b) 57
 (c) 60 (d) 66
26. An inspector rejects 0.08% of the metres as defective. How many metres will he examine to reject 2 metres?
 (a) 200m (b) 250m
 (c) 2500m (d) 3000m
27. If the numerator of a fraction be increased by 15% and its denominator be diminished by 8%, the value of the fraction is $\frac{15}{16}$. Find the original fraction.
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) $\frac{3}{4}$
28. In a class, 65% of the students are boys. On a particular day 80% of girl students were present. What was the fraction of boys who were present that day if the total number of students present that day was 70%?
 (a) $\frac{2}{3}$ (b) $\frac{28}{65}$
 (c) $\frac{5}{6}$ (d) $\frac{42}{65}$
29. A's income is 60% of B's income, and A's expenditure is 70% of B's expenditure. If A's income is 75% of B's expenditure, find the ratio of A's saving to B's saving.
 (a) 5 : 1 (b) 1 : 5
 (c) 3.5 : 1 (d) 2 : 7
30. Due to fall in manpower, the production in a factory decreases by 25%. By what per cent should the working hour be increased to restore the original production?
 (a) $33\frac{1}{3}\%$ (b) 20%
 (c) 25% (d) 4%
31. A number is increased by 20% and then again by 20%. By what percent should the increased number be reduced so as to get back the original number ?
 (a) $19\frac{11}{31}\%$ (b) $30\frac{5}{9}\%$
 (c) 40% (d) 44%
32. In the month of January, the Railway Police caught 4000 ticketless travellers. In February, the number rose by 5%. However, due to constant vigil by the Police and the total number of ticketless travellers caught in the month of April was :
 (a) 3125 (b) 3255
 (c) 3575 (d) 3591
33. In a mixture of milk and water the proportion of water by weight was 75%. If in the 60 gm mixture, 15 gm water was added, what would be the percentage of water ?
 (a) 75% (b) 88%
 (c) 90% (d) None of these
34. A shopkeeper employed a servant at a monthly salary of ₹ 1500. In addition to it, he agreed to pay him a commission of 15% on the monthly sale. How much sale in Rupees should the servant do if he wants his monthly income as ₹ 6000?
 (a) ₹ 30000 (b) ₹ 415000
 (c) ₹ 31500 (d) ₹ 50000

35. How many litres of a 30% alcohol solution should be added to 40 litres of a 60% alcohol solution to prepare a 50% solution ?
 (a) 30 (b) 20
 (c) 24 (d) 32
36. 720 sweets were distributed equally among children in such a way that number of sweets received by each child is 20% of the total number of children. How many sweets did each child receive ?
 (a) 8 (b) 10
 (c) 9 (d) 12
37. A company bought a total of 60 computers and 20 printers to modernise billing operations. If the price of each computer was three times the price of each printer then what per cent of the total cost of the purchase was the total cost of the printers ?
 (a) 10% (b) 11%
 (c) 15% (d) 20%
38. p% of the students of a class passed the exam. g% of the passed students are girls and b% of the fail students are boys. The percentage of passed boys over the failed girls is :
 (a) $\left(\frac{bg}{p} \times 100\right)$ (b) $\frac{100(100-g) \cdot p}{(100-p)(100-b)}$
 (c) $\frac{(100-g)(100-b)}{(100-p)}$ (d) None of these
39. A city had a population of 30,00,000 in the beginning of 1999. Its average growth rate is 4% per year, but due to a massive earthquake in 2001, its population is reduced by 8% in that year. But it again maintained the same growth rate of 4% in following years. What will be the approx. population of the city at the end of 2003?
 (a) 32,06,460 (b) 34,68,420
 (c) 31,52,360 (d) 32,28,810
40. In a factory there are three types of machines M_1 , M_2 and M_3 which produces 25%, 35% and 40% of the total products respectively. M_1 , M_2 and M_3 produces 2%, 4% and 5% defective products, respectively. What is the percentage of non-defective products?
 (a) 89% (b) 97.1%
 (c) 96.1% (d) 86.1%
41. A person saves 6% of his income. Two years later, his income shoots up by 15% but his savings remain the same. Find the hike in his expenditure.
 (a) 15.95% (b) 15%
 (c) 14.8% (d) 15.5%
42. In the half yearly exam only 70% of the students were passed. Out of these (passed in half yearly) only 60% student are passed in annual exam. Out of those who did not pass the half yearly exam, 80% passed in annual exam. What percent of the students passed the annual exam?
 (a) 42% (b) 56%
 (c) 66% (d) None of these
43. Two vessels contain equal quantities of 40% alcohol. Anil changed the concentration of the first vessels to 50% by adding extra quantity of pure alcohol. Balu changed the concentration of the second vessels to 50% replacing a certain quantity of the solution with pure alcohol. By what percentage is the quantity of alcohol added by Anil more than that replaced by Balu?
 (a) 20% (b) 25%
 (c) 40% (d) Cannot be determined
44. Lagaan is levied on the 60% of the cultivated land. The revenue department collected total ₹ 3,84,4000 through the lagaan from the village of Sukhiya. Sukhiya, a very rich farmer, paid only ₹ 480 as lagaan. The percentage of total land of Sukhiya over the total taxable land of the village is:
 (a) 0.15% (b) 15%
 (c) 0.125% (d) None of these
45. Mr. Abhimanyu Banerjee is worried about the balance of his monthly budget. The price of petrol has increased by 40%. By what percent should he reduce the consumption of petrol so that he is able to balance his budget?
 (a) 33.33 (b) 28.56
 (c) 25 (d) 14.28
46. In an election between 2 candidates, Bhiku gets 65% of the total valid votes. If the total votes were 6000, what is the number of valid votes that the other candidate Mhatre gets if 25% of the total votes were declared invalid?
 (a) 1625 (b) 1575
 (c) 1675 (d) 1525
47. A machine depreciates in value each year at the rate of 10% of its previous value. However, every second year there is some maintenance work so that in that particular year, depreciation is only 5% of its previous value. If at the end of the fourth year, the value of the machine stands at ₹ 1, 46, 205, then find the value of machine at the start of the first year.
 (a) ₹ 1, 90, 000 (b) ₹ 2, 00, 000
 (c) ₹ 1, 95, 000 (d) ₹ 2, 10, 000
48. After three successive equal percentage rise in the salary the sum of 100 rupees turned into 140 rupees and 49 paise. Find the percentage rise in the salary.
 (a) 12% (b) 22%
 (c) 66% (d) 82%
49. The salary of Anil and Vinit is 20% and 30% less than the salary of Dheeraj respectively. By what percentage is the salary of Anil more than the salary of Vinit?
 (a) 33.33% (b) 50%
 (c) 10% (d) 14.28%
50. In a certain town, at least 50% of the people read a newspaper. Among those who read a newspaper, at most 25% read more than one paper. Only one of the following statements follows from the statements given below. Which one is it?
 (a) At the most 25% read exactly one newspaper
 (b) At least 25% read all the newspaper
 (c) At the most 37½% read exactly one newspaper
 (d) At least 37½% read exactly one newspaper

Standard Level

- A man buys a house for ₹ 100000 and rents it. He puts 12.5% of each month's rent aside for upkeep & repairs, pays ₹ 325 per year as taxes and realizes 5.5% annually on his investment. Find the monthly rent.
(a) 550 (b) 554.76
(c) 654.76 (d) 1620.45
- When the cost of petroleum increases by 40%, a man reduces his annual consumption by 20%. Find the percentage change in his annual expenditure on petroleum.
(a) 20% (b) 16%
(c) 12% (d) 40%
- A obtains $33\frac{1}{3}\%$ of the marks in a paper for which the maximum was 300. B is ahead of A by 40% of A's marks, while C is ahead of B by two-ninths of his own marks. How many marks does C get?
(a) 180 (b) 140
(c) 150 (d) 210
- In a city, 35% of the population is composed of migrants, 20% of whom are from rural areas. Of the local population, 48% is female while this figure for rural and urban migrants is 30% and 40% respectively. If the total population of the city is 728400, what is its female population ?
(a) 324138 (b) 349680
(c) 509940 (d) None of these
- Madan pays income tax at the rate of 10%. If his income increased by 10% and his tax rate increases to 15%, his net income would increase by ₹ 350. What is Madan's income ?
(a) ₹ 8000 (b) ₹ 10,000
(c) ₹ 12,000 (d) ₹ 14,000
- The digit at unit place of a two-digit number is increased by 100% and the digit at ten places of the same number is increased by 50%. The new number thus formed is 19 more than the original number. What is the original number?
(a) 22 (b) 63
(c) 24 (d) None of these
- Chunilal invests 65% in machinery, 20% in raw material and still has ₹ 1,305 cash with him. Find his total investment.
(a) ₹ 6,500 (b) ₹ 7,225
(c) ₹ 8,500 (d) None of these
- The price of oil is increased by 25%. If the expenditure is not allowed to increase, the ratio between the reduction in consumption and the original consumption is
(a) 1 : 3 (b) 1 : 4
(c) 1 : 5 (d) 1 : 6
- A scooter costs ₹ 25,000 when it is brand new. At the end of each year, its value is only 80% of what it was at the beginning of the year. What is the value of the scooter at the end of 3 years?
(a) ₹ 10,000 (b) ₹ 12,500
(c) ₹ 12,800 (d) ₹ 12,000
- If the price of sugar rises from ₹ 6 per kg to ₹ 7.50 per kg, a person, to have no increase in his expenditure on sugar, will have to reduce his consumption of sugar by
(a) 15% (b) 20%
(c) 25% (d) 30%
- The sum of two numbers is $\frac{28}{25}$ of the first number. The second number is what percent of the first?
(a) 12% (b) 14%
(c) 16% (d) 18%
- A positive number is by mistake divided by 6 instead of being multiplied by 6. What is the % error on the basis of correct answer?
(a) 3 (b) 97
(c) 17 (d) 83
- p is six times as large as q . The percent that q is less than p , is :
(a) $16\frac{2}{3}$ (b) 60
(c) $83\frac{1}{3}$ (d) 90
- 5% of income of A is equal to 15% of income of B and 10% of income of B is equal to 20% of income of C. If C's income is ₹ 2000, then the total income of A, B and C is :
(a) ₹ 6000 (b) ₹ 14,000
(c) ₹ 18,000 (d) ₹ 20,000
- By reduction of 20% in the price of oranges, one can purchase 5 oranges more for ₹ 2.50. Find the reduced price of the oranges per dozen. Find also the original price.
(a) 120 paise, 140 paise (b) ₹ 0.8, ₹ 1.5
(c) ₹ 1.0, ₹ 1.5 (d) ₹ 1.2, ₹ 1.5
- In a factory, producing parts of an automobile, the parts manufactured on the shop floor are required to go through quality checks, each conducted after a specific part of the processing on the raw material is completed. Only parts that are not rejected at one stage are put through subsequent stages of production and testing. If average rejection rates at these three testing stages during a month are 10%, 5% and 2% respectively, then what is the effective rejection rate for the whole plant ?
(a) 17% (b) 15.20%
(c) 84.80% (d) 16.21%

17. In some quantity of ghee, 60% is pure ghee and 40% is vanaspati. If 10 kg of pure ghee is added, then the strength of vanaspati ghee becomes 20%. The original quantity was :
 (a) 10 kg (b) 15 kg
 (c) 20 kg (d) 25 kg
18. The strength of a school increases and decreases every alternate year. It starts with increase by 10% and there-after the percentage of increase/decrease is the same. Which of the following is definitely true about the strength of the school in 2001 as compared to that in 1996?
 (a) Increase approximately by 8%
 (b) Decrease approximately by 8%
 (c) Increase approximately by 20%
 (d) Decrease approximately by 20%
19. $A = 10\%$ of x , $B = 10\%$ of y , $C = 10\%$ of $x + 10\%$ of y . On the basis of the above equalities, what is true in the following?
 (a) A is equal to B
 (b) A is greater than B
 (c) B is greater than A
 (d) Relation cannot be established between A and B
20. Sumitra has an average of 56% on her first 7 examinations. How much should she make on her eighth examination to obtain an average of 60% on 8 examinations?
 (a) 88% (b) 78%
 (c) 98% (d) Cannot be determined
21. Due to an increase of 30% in the price of eggs, 3 eggs less are available for ₹ 7.80. The present rate of eggs per dozen is :
 (a) ₹ 8.64 (b) ₹ 8.88
 (c) ₹ 9.36 (d) ₹ 10.40
22. A speaks truth in 75% and B in 80% cases. In what percentage of cases are they likely to contradict each other when narrating the same incident?
 (a) 35 (b) 30
 (c) 25 (d) 20
23. A's income is 60% of B's income, and A's expenditure is 70% of B's expenditure. If A's income is 75% of B's expenditure, find the ratio of A's savings to B's savings.
 (a) 5 : 1 (b) 1 : 5
 (c) 3.5 : 1 (d) 2 : 7
24. In a market survey, 20% opted for product A whereas 60% opted for product B. The remaining individuals were not certain. If the difference between those who opted for product B and those who were uncertain was 720, how many individuals were covered in the survey ?
 (a) 1440 (b) 1800
 (c) 3600 (d) Data inadequate
25. The income of A is 150% of the income of B and the income of C is 120% of the income of A. If the total income of A, B and C together is ₹ 86000, what is C's income ?
 (a) ₹ 30000 (b) ₹ 32000
 (c) ₹ 20000 (d) ₹ 36000
26. Of the 1000 inhabitants of a town, 60% are males of whom 20% are literate. If, of all the inhabitants, 25% are literate, then what percent of the females of the town are literate ?
 (a) 22.5 (b) 27.5
 (c) 32.5 (d) 37.5
27. When 60% of a number A is added to another number B, B becomes 175% of its previous value. Then which of the following is true regarding the values of A and B?
 (a) $A > B$
 (b) $B > A$
 (c) $B \geq A$
 (d) Either (a) or (b) can be true depending upon the values of A and B
28. In the month of January, the Railway Police caught 4000 ticketless travellers. In February, the number rose by 5%. However, due to constant vigil by the Police and the Railway staff, the number reduced by 5% and in April it further reduced by 10%. The total number of ticketless travellers caught in the month of April was:
 (a) 3125 (b) 3255
 (c) 3575 (d) 3591
29. The ratio of Jim's salary for October to his salary for November was 1.5 : 1.333 and the ratio of the salary for November to that for December was 2 : 2.6666. The worker got 40 rupees more for December than for October and received a bonus constituting 40 per cent of the salary for three months. Find the bonus. (Assume that the number of workdays is the same in every month.)
 (a) 368.888 rupees (b) 152.5555 rupees
 (c) 222.22 rupees (d) 265.6 rupees
30. King Dashratha, at his eleventh hour, called his three queens and distributed his gold in the following way: He gave 50% of his wealth to his first wife, 50% of the rest to his second wife and again 50% of the rest to his third wife. If their combined share is worth 1,30,900 kilograms of gold, find the quantity of gold King Dashratha was having initially?
 (a) 1,50,000 kg (b) 1,49,600 kg
 (c) 1,51,600 kg (d) 1,52,600 kg
31. After three successive raises, Aftab's salary became equal to $\frac{378}{125}$ of his initial salary. By what percentage was the salary raised the first time if the third rise was twice as high (in percentage) as the second rise was twice as high (in percentage) as the first rise?
 (a) 10% (b) 15%
 (c) 20% (d) 25%
32. In an assembly election at Surat, the total turnout was 80% out of which 16% of the total voters on the voting list were declared invalid. Find which of the following can be the percentage votes got by the winner of the election if the candidate who came second got 20% of the total voters on the voting list. (There were only three contestants, only one winner and the total number of voters on the voters' list was 20000.)
 (a) 44.8% (b) 46.6%
 (c) 48% (d) None of these

33. Find the value of x in $\sqrt{x + 2\sqrt{x + 2\sqrt{x + 2\sqrt{3x}}}} = x$.
- (a) 1 (b) 3
(c) 6 (d) 12
34. In the university examination last year, Rajesh scored 65% in English and 82% in History. What is the minimum percent he should score in Sociology. Which is out of 50 marks (if English and History were for 100 marks each), if he aims at getting 78% overall?
- (a) 94% (b) 92%
(c) 98% (d) 96%
35. In an election of 3 candidates A, B and C, A gets 50% more votes than B. A also beats C by 1,80,00 votes. If it is known that B gets 5 percentage more votes than C, find the number of voters on the voting list (given 90% of the voters on the voting list voted and no votes were illegal)
- (a) 72,000 (b) 81,000
(c) 90,000 (d) 1,00,000
36. A number is mistakenly divided by 5 instead of being multiplied by 5. Find the percentage change in the result due to this mistake.
- (a) 96% (b) 95%
(c) 2400% (d) 200%
37. Ambani, a businessman, started off a business with very little capital. In the first year, he earned a profit of 50% and donated 50% of the total capital (initial capital + profit) to a charitable organisation. The same course was followed in the 2nd and 3rd years also. If at the end of three years he is left with ₹ 16,875, then find the amount donated by him at the end of the 2nd year.
- (a) ₹ 45,000 (b) ₹ 12,500
(c) ₹ 22,500 (d) ₹ 20,000
38. Arushi's project report consists of 25 pages each of 60 lines with 75 characters on each line. In case the number of lines is reduced to 55 but the number of characters is increased to 90 per lines, what is the percentage change in the number of pages. (Assume the number of pages to be a whole number.)
- (a) +10% (b) +5%
(c) -8% (d) -10%
39. Recently I had gone to a locality called Shadigarh for conducting a survey about the number of married persons in the locality. The population of the locality is 7,200 and $\frac{11}{18}$ th of those are males and the rest females. If 40% of the males are married, find percentage of married females in the locality
- (a) $48\frac{1}{7}\%$ (b) $52\frac{4}{7}\%$
(c) $62\frac{6}{7}\%$ (d) $71\frac{1}{7}\%$
40. Chintu is given a quadratic equation $ax^2 + bx + c = 0$ and is asked to make another quadratic equation from this with $a = 1$. Also one root of the second quadratic equation is same as one of the roots of the first equation but opposite in sign and the other root of the second equation is two times the second root of the first equation. Find the percentage change in the constant term of the second equation as compared to the first equation?
- (a) 200% increase
(b) 300% decrease
(c) 400% increase
(d) 100% decrease

Expert Level

- Lucknow bound Shatabdi Express has a capacity of 500 seats of which 10% are in the Executive class and the rest chair cars. During one journey, the train was booked to 85% of its capacity. If Executive class was booked to 96% of its capacity, then how many chair car seats were empty during that journey?
(a) 78 (b) 73
(c) 72 (d) None of these
- Shobha's Mathematics test had 75 problem i.e., 10 Arithmetic, 30 Algebra and 35 geometry problems. Although she answered 70% of the Arithmetic, 40% of the Algebra and 60% of the geometry problems correctly. She did not pass the test because she got less than 60% of the problems right. How many more questions she would have needed to answer correctly to earn a 60% passing grade?
(a) 4 (b) 5
(c) 6 (d) 7
- 50% of $a\%$ of b is 75% of $b\%$ of c . Which of the following is c ?
(a) $1.5a$ (b) $0.667a$
(c) $0.5a$ (d) $1.25a$
- 18% of A plus 15% of B plus 19% of C is equal to 17% of the sum of A, B, and C. If $A - B = 500$ and $A - C = 3400$, what is the value of $A + B + C$?
(a) 12,400 (b) 11,600
(c) 13,500 (d) None of these
- In the Chidambaram's family the ratio of expenses to the savings is 5 : 3. But his expenses is increased by 60% and income increases by only 25% thus there is a deficit of ₹ 3500 in the savings. The increased income of Mr. Chidambaram's family is :
(a) ₹ 35,000 (b) ₹ 28,000
(c) ₹ 25,000 (d) ₹ 18,500
- At IIM Bangalore, 60% of the students are boys and the rest are girls. Further 15% of the boys and 7.5% of the girls are getting a fee waiver. If the number of those getting a fee waiver is 90, find the total number of students getting 50% concession if it is given that 50% of those not getting a fee waiver are eligible to get half fee concession ?
(a) 360 (b) 280
(c) 320 (d) 330
- A and B have some guavas divided among themselves. A says to B "If I give you 25% of the guavas I have, I will still have 2 more guavas than you have." To this, B says "If you give me guavas equal to 70% of what I have now, I will have 4 more guavas than you have." What is the total number of guavas that they have?
(a) 80 (b) 64
(c) 36 (d) 88
- The annual earning of Mr. Sikkawala is ₹ 4 lakhs per annum for the first year of his job and his expenditure was 50%. Later on for the next 3 years his average income increases by ₹ 40,000 per annum and the saving was 40%, 30% and 20% of the income. What is the percentage of his total savings over the total expenditure if there is no any interest is applied on the savings for these four years :
(a) $49\frac{37}{87}\%$ (b) $41\frac{73}{83}\%$
(c) 53% (d) None of these
- A, B, C and D purchased a cine-multiplex for ₹ 56 lakhs. The contribution of B, C and D together is 460% that of A, alone. The contribution of A, C and D together is 366.66% that of B's contribution and the contribution of C is 40% that of A, B and D together. The amount contributed by D is :
(a) 10 lakh (b) 12 lakh
(c) 16 lakh (d) 18 lakh
- Ram prepares solutions of alcohol in water according to customers' needs. This morning Ram has prepared 27 litres of a 12% alcohol solution and kept it ready in a 27 litre delivery container to be shipped to the customer. Just before delivery, he finds out that the customer had asked for 27 litres of 21% alcohol solution. To prepare what the customer wants, Ram replaces a portion of 12% solution by 39% solution. How many litres of 12% solution are replaced?
(a) 5 (b) 9
(c) 10 (d) 12
- In every month Ravindra consumes 25 kg rice and 9 kg wheat. The price of rice is 20% of the price of wheat and thus he spends total ₹ 350 on the rice and wheat per month. If the price of wheat is increased by 20% then what is the percentage reduction of rice consumption for the same expenditure of ₹ 350? Given that the price of rice and consumption of wheat is constant :
(a) 36% (b) 40%
(c) 25% (d) 24%
- My friend Siddhartha Ghosh is working in the Life Insurance Corporation of India (LIC). He was hired on the basis of commission and he got the bonus only on the first years commission. He got the policies of ₹ 2 lakh having maturity period of 10 year. His commission in the first, second, third, fourth and for the rest of the years is 20%, 16%, 12%, 10% and 4% respectively. The bonus is 25% of the commission. If the annual premium is ₹ 20,000 then what is his total commission if the completion of the maturity of all the policies is mandatory :
(a) ₹ 174,00 (b) ₹ 23,600
(c) ₹ 15,000 (d) ₹ 15,500

13. 800 people were supposed to vote on a resolution, but 1/3rd of the people who had decided to vote for the motion were abducted. However, the opponents of the motion, through some means managed to increase their strength by 100%. The motion was then rejected by a majority, which was 50% of that by which it would have been passed if none of these changes would have occurred. How many people finally voted for the motion and against the motion?
- (a) 200 (for), 400 (against)
 (b) 100 (for) and 200 (against)
 (c) 150 (for), 300 (against)
 (d) 200 (for) and 300 (against)
14. There are five contestants A, B, C and D in the assembly election from the Bihar Sharif constituency. It is given that none of the contestants got less than 1% of the valid vote. Consider the following statements in isolation to each other :
- A got 49% of the total valid votes
 - B got 55% of the total votes
 - C got 46% of the total valid votes
 - D got 48% of the total valid votes
- With how many of the following statements when used independently, it is possible to find out the winner of the election.
- (a) 0 (b) 1
 (c) 2 (d) 3
15. Mr. A is a computer programmer. He is assigned three jobs for which time allotted is in the ratio of 5 : 4 : 2 (job are needed to be done individually). But due to some technical snag, 10% of the time allotted for each job gets wasted. Thereafter, owing to the lack of interest, he invests only 40%, 30%, 20% of the hours of what was actually allotted to do the three jobs individually. Find how much percentage of the total time allotted is the time invested by A.
- (a) 38.33% (b) 39.4545%
 (c) 32.72% (d) 36.66%
16. An index of 12 shares contains, among others, the shares of Reliance, HLL and Infosys with weightage of 7%, 13% and 15% respectively. What is the increase in the prices of other shares, if these three rise by 9%, 10% and 4% respectively, while the index rises by 6%?
- (a) 5.34% (b) 4.91%
 (c) 4.58% (d) Cannot be determined
17. 10% of salty sea water contained in a flask was poured out into a beaker. After this, a part of the water contained in the beaker was vapourised by heating and due to this, the percentage of salt in the beaker increased M times. If it is known that after the content of the beaker was

poured into the flask, the percentage of salt in the flask increased by x%, find the original quantity of sea water in the flask.

- (a) $\frac{9M + 1\%}{M - 1}$ (b) $\frac{(9M + 1)x\%}{M - 1}$
 (c) $\frac{9M - 1x\%}{M + 1}$ (d) $\frac{9M + x\%}{M + 1}$

18. A 14.4 kg gas cylinder runs for 104 hours when the smaller burner on the gas stove is fully opened while it runs for 80 hours when the larger burner on the gas stove is fully opened. Which of these values are the closest to the percentage difference in the usage of gas per hour, between the smaller and the larger burner?
- (a) 26.23% (b) 30%
 (c) 32.23% (d) 23.07%
19. Hursh Sarma has a salary of `10,800 per month. In the first month of the year, he spends 40% of his income on food, 50% on clothing and saves 11.11% of what he has spent. In the next two months, he saves 9.09% of what he has spent (spending 38.33% of his income of food). In the fourth month, he gets an increment of 11.11% on his salary and spends every single paise on celebrating his raise. But from the fifth month onwards good sense prevails on him and he saves 12.5%, 15%, 20%, 10%, 8.33%, 12.5%, 15% and 20% on his new income per month. The ratio between the sum of the savings for the two months having the highest savings to the sum of the savings for the two months having the lowest savings is
- (a) 2.6666 (b) 5.3333
 (c) 8 (d) None of these
20. Krishan Iyer, a motorist user 24% of his fuel in covering the first 20% of his total journey (in city driving conditions). If he knows that he has to cover another 25% of his total journey in city driving conditions, what should be the minimum percentage increase in the fuel efficiency for non-city driving over the city driving fuel efficiency, so that he is just able to cover his entire journey without having to refuel?
- (Approximately)
- (a) 39.2% (b) 43.5%
 (c) 45.6% (d) 41.2%
21. An empty fuel tank of a car was filled with A type petrol. When the tank was half-empty, it was filled with B type petrol. Again when the tank was half-empty, it was filled with A type petrol. When the tank was half-empty again, it was filled with B type petrol. What is the percentage of A type petrol at present in the tank?
- (a) 33.5% (b) 37.5%
 (c) 40% (d) 50%

22. One type of liquid contains 20% water and the second type of liquid contains 35% of water. A glass is filled with 10 parts of first liquid and 4 parts of second liquid. The percentage of water in the new mixture in the glass is:
- (a) 20% (b) $24\frac{2}{7}\%$
(c) 37% (d) 40%
23. An empty fuel tank of a car was filled with A type petrol. When the tank was half empty, it was filled with B type petrol. Again when the tank was half-empty, it was filled with A type petrol. When the tank was half-empty again, it was filled with B type petrol. What is the percentage of A type petrol at present in the tank ?
- (a) 33.5% (b) 37.5%
(c) 40% (d) 50%
24. A bag contains 600 coins of 25 p denomination and 1200 coins of 50 p denomination. If 12% of 25p coins and 24% of 50p coins are removed, the percentage of money removed from the bag is nearly :
- (a) 15.6% (b) 17.8%
(c) 21.6% (d) 30%
25. If $3x + 7 = x^2 + M = 7x + 5$, what is the value of 120% of M ?
- (a) 8.90 (b) 9.90
(c) 9.98 (d) None of these
26. There are three galleries in a coal mine. On the first day, two galleries are operative and after some time, the third gallery is made operative. With this, the output of the mine became half as large again. What is the capacity of the second gallery as a percentage of the first, if it is given that a four-month output of the first and the third galleries was the same as the annual output of the second gallery?
- (a) 70% (b) 64%
(c) 60% (d) 65%
27. A clock is set right at 12 noon on Monday. It loses $1\frac{1}{2}\%$ on the correct time in the first week but gains $1\frac{1}{4}\%$ on the true time during the second week. The time shown on Monday after two weeks will be
- (a) 12:25:12 (b) 11:34:48
(c) 12:50:24 (d) 12:24:16
28. Harish Sharma has a salary of ₹ 10,800 per month. In the first month of the year, he spends 40% of his income on food, 50% on clothing and saves 11.11% of what he has spent. In the next two months, he saves 9.09% of what he has spent (spending 38.33% of his income on food). In the fourth month, he gets an increment of 11.11% on his salary and spends every single paise on celebrating his raise. But from the fifth month onwards good sense prevails on him and he saves 12.5%, 15%, 20%, 10%, 8.33%, 12.5%, 15% and 20% on his new income per month. The ratio between the sum of the savings for the two months having the highest savings to the sum of the savings for the two months having the lowest savings is
- (a) 8:3 (b) 16:3
(c) 8:1 (d) None of these
29. An index of 12 shares contains, among others, the shares of Reliance, HLL and Infosys with weightage of 7%, 13% and 15% respectively. What is the increase in the prices of other shares, if these three rise by 9%, 10% and 4% respectively, while the index rises by 6%?
- (a) 5.34% (b) 4.91%
(c) 4.58% (d) Cannot be determined

Test Yourself

- When any number is divided by 12, then dividend becomes $\frac{1}{4}$ th of the other number. By how much percent first number is greater than the second number ?
(a) 150 (b) 200
(c) 300 (d) Data inadequate
- The price of a jewel, passing through three hands, rises on the whole by 65%. If the first and the second sellers earned 20% and 25% profit respectively, find the percentage profit earned by the third seller.
(a) 10% (b) 20%
(c) 30% (d) 40%
- In the MOCK CAT paper at AMS, questions were asked in five sections. Out of the total students, 5% candidates cleared the cut-off in all the sections and 5% cleared none. Of the rest, 25% cleared only one section and 20% cleared four sections. If 24.5% of the entire candidates cleared two sections and 300 candidates cleared three sections, find out how many candidates appeared at the MOCK CAT at AMS?
(a) 1000 (b) 1200
(c) 1500 (d) 2000
- A's income is $6\frac{1}{4}\%$ more than B's. How much % is B's less than A's ?
(a) 5.89% (b) 4.78%
(c) 2.39% (d) None of these
- The petrol prices shot up by 7% as a result of the hike in the price of crudes. The price of petrol before the hike was ₹ 28 per litre. Vawal travels 2400 kilometres every month and his car gives a mileage of 18 kilometres to a litre. Find the increase in the expenditure that Vawal has to incur due to the increase in the price of petrol (to the nearest rupee)?
(a) ₹ 270 (b) ₹ 262
(c) ₹ 276 (d) ₹ 272
- Due to a reduction of $6\frac{1}{4}\%$ in the price of sugar, a man is able to buy 1 kg more for ₹ 120. Find the original rate of sugar.
(a) ₹ 5.00 per kg (b) ₹ 5.50 per kg
(c) ₹ 7.00 per kg (d) ₹ 7.50 per kg
- What percentage of numbers from 1 to 70 have squares that end in the digit 1 ?
(a) 1 (b) 14
(c) 20 (d) 21
- Which of the following statement/s is/are true?
I. If two numbers are such that one is 25% more than the second then the sum of the two numbers is nine times that of the difference of the two numbers.
II. If 56% of a number A is 24% of another number B then 49% of A is 21% of B.
III. If $\frac{7}{11}x$ is same as $\frac{2}{9}y$ then 81% of x is same as $26\frac{6}{7}$ of y.
(a) Only I (b) Only III
(c) Only I and II (d) Only II and III
- The population of a town is 8000. If the males increase by 6% and the females by 10%, the population will be 8600. Find the number of females in the town.
(a) 1,000 (b) 2,000
(c) 3,000 (d) 5,000
- Ram purchased a flat at ₹ 1 lakh and Prem purchase a plot of land worth ₹ 1.1 lakh. The respective annual rates at which the prices of the flat and the plot increased were 10% and 5%. After two years they exchanged their belongings and one paid the other the difference. Then
(a) Ram paid ₹ 275 to Prem (b) Ram paid ₹ 475 to Prem
(c) Ram paid ₹ 2750 to Prem (d) Prem paid ₹ 475 to Ram
- $\frac{2}{5}$ of the voters promise to vote for P and the rest promised to vote for Q. Of these, on the last day 15% of the voters went back of their promise to vote for P and 25% of voters went back of their promise to vote for Q and P lost by 2 votes. Then the total number of voters is
(a) 100 (b) 110
(c) 90 (d) 95
- On a train journey, there are 5 kinds of tickets AC I, AC II, AC III, 3-tier, and general. The relationship between the rates of the tickets for the Eurail is:
AC II is 20% higher than AC III and AC I is 70% of AC III's value higher than the AC II ticket's value. The 3-tier ticket is 25% of the AC I's ticket cost and the general ticket is $\frac{1}{3}$ the price of the AC II ticket. The AC II ticket costs 780 euros between London and Paris. The difference in the rates of 3 tier and general ticket is
(a) 41.25 euros (b) 55.8 euros
(c) 48.75 euros (d) 52.75 euros
- A student gets an aggregate of 60% marks in five subjects in the ratio 10 : 9 : 8 : 7 : 6. If the passing marks are 50% of the maximum marks and each subject has the same maximum marks, in how many subjects did he pass the exam?
(a) 2 (b) 3
(c) 4 (d) 5
- Fresh grapes contain 90% water while dry grapes contain 20% water. What is the weight of dry grapes obtained from 20 kg of fresh grapes?
(a) 2 kg (b) 2.5 kg
(c) 2.4 kg (d) None of these
- In a company 40% are male, out of which 75% earn a salary of 25,000 plus. If 45% of the employees 25,000 plus salaries, what is the fraction of female employees earning less than or equal to 25,000?
(a) $\frac{1}{4}$ (b) $\frac{3}{7}$
(c) $\frac{3}{4}$ (d) $\frac{5}{9}$

Hints & Solutions

Foundation Level

1. (d) Let X be the number of new science books. Then,
Total Science books / Total books = 45%.

$$\Rightarrow \frac{\left(X + 110000 \times \frac{40}{100}\right)}{(20000 + 10000)} = \frac{45}{100} \Rightarrow X = 14500.$$

2. (b) Number of runs made by running
= $110 - (3 \times 4 + 8 \times 6) = 50$.

$$\therefore \text{Required percentage} = \left(\frac{50}{110} \times 100\right)\% \\ = 45\frac{5}{11}\%$$

3. (a) Total marks secured = (90% of 100 + 60% of 150 + 54% of 200)

$$= \left(\frac{90}{100} \times 100 + \frac{60}{100} \times 150 + \frac{54}{100} \times 200\right)$$

$$= (90 + 90 + 108) = 288.$$

$$\text{Total maximum marks} = (100 + 150 + 200) = 450.$$

$$\therefore \text{Aggregate Percentage} = \left(\frac{288}{450} \times 100\right)\% = 64\%$$

4. (d) Let the third number be 100. Then, the first and second numbers will be 20 and 50, respectively.

$$\text{Required \%} = \frac{20}{50} \times 100 = 40\%$$

5. (a) Let the total number of applicants be x . Number of eligible candidates = 95% of x . Eligible candidates of other categories = 15% of (95% of x)

$$= \left(\frac{15}{100} \times \frac{95}{100}\right) \times x = \frac{57}{400}x$$

$$\therefore \frac{57}{400}x = 4275 \Leftrightarrow x = \left(\frac{4275 \times 400}{57}\right) = 30000$$

6. (c) Let the salary of Deepa be ` x .
Then, 80% of 8% of $x = 2240$

$$\Rightarrow \frac{80}{100} \times \frac{8}{100} \times x = 2240$$

$$\Rightarrow x \times \frac{2240 \times 100 \times 100}{80 \times 8} = 35000$$

$$\text{Hence, the salary of Deepa} = \text{` } 35000$$

7. (a) Let the original price be x and sale be of y units.

Then, the revenue collected initially = $x \times y$

Now, new price = $0.8x$, new sale = $1.8y$

Then, new revenue collected = $1.44xy$

$$\% \text{ increase in revenue} = \frac{0.44xy}{xy} \times 100 \\ = 44\% \text{ increase}$$

8. (a) % reduction in consumption

$$= \frac{\% \text{ change in price}}{100 + \% \text{ change in price}} \times 100$$

$$= \frac{7}{100 + 7} \% = \frac{7}{107} \%$$

9. (c) Let his sales be worth ` x . Then,

$$1000 + 2.5\% \text{ of } (x - 4000) = 5\% \text{ of } x + 600$$

$$\Rightarrow \frac{5x}{100} - \frac{2.5(x - 4000)}{100} = 1000 - 600$$

$$\Rightarrow 2.5x + 10000 = 40,000$$

$$\Rightarrow x \times \frac{30,000}{2.5} = 12,000 / -$$

10. (c) Let the inspector examined x metres,
then 0.08% of $x = 2$

$$\Rightarrow \frac{x \times 0.08}{100} = 2$$

$$\text{or } x = \frac{200}{0.08} = 2500 \text{ metres}$$

11. (a) Let the investment of $C =$ ` 100

Then B 's investment = ` 90 and  $A$ 's investment = ` 99

Sum of investment = `  $(100 + 90 + 99) =$  ` 289

$$\text{Hence, } C\text{'s actual investment} = \left(\frac{14450 \times 100}{289}\right)$$

$$= \text{` } 5000$$

12. (a) Let B get ` x . Then C gets = 75% of $x = \frac{3x}{4}$

$$\text{and } A \text{ gets} = 120\% \text{ of } \frac{3x}{4} = \frac{120}{100} \times \frac{3x}{4} = \frac{9x}{10}$$

$$\text{Now, } \frac{9x}{10} + \frac{3x}{4} + x = 4558$$

$$\Rightarrow \frac{53x}{20} = 4558 \Rightarrow x = \frac{4558 \times 20}{53} = 1720$$

$$\text{Hence, } A\text{'s share} = \frac{9x}{10} = \frac{9 \times 1720}{10} = \text{` } 1548$$

13. (b) Let the total number of votes enrolled be x . Then,
Number of votes cast = 75% of x . Valid votes = 98% of (75% of x).

$$\therefore 75\% \text{ of } [98\% \text{ of } (75\% \text{ of } x)] = 9261$$

$$\Rightarrow \left(\frac{75}{100} \times \frac{98}{100} \times \frac{75}{100} \times x \right) = 9261$$

$$\Rightarrow x = \left(\frac{9261 \times 100 \times 100 \times 100}{75 \times 98 \times 75} \right) = 16800.$$

14. (d) Height climbed in second hour

$$= 12\frac{1}{2}\% \text{ of } \left(100 - 62\frac{1}{2} \right) \% \text{ of } 192 \text{ m}$$

$$= \left(\frac{25}{2} \times \frac{1}{100} \times \frac{75}{2} \times \frac{1}{100} \times 192 \right) \text{ m} = 9 \text{ m}.$$

15. (b) Let the original number be 100.
Then, the new number = $100 \times 1.1 \times 0.9 = 99$
i.e. the number decreases by 1%.

16. (c) Work with option, $\left(\frac{5}{4} \right) x - \left(\frac{7}{10} \right) x = 22$

Only $x = 40$ fulfil the above equation.

17. (c) Let the total sales be x . Then, $5\frac{1}{2}\%$ of $x + \frac{1}{2}\%$ of $(x - 10000) = 1990$

$$\Leftrightarrow \frac{11}{2} \times \frac{1}{100} \times x + \frac{1}{2} \times \frac{1}{100} \times (x - 10000) = 1990$$

$$\Leftrightarrow 12x - 10000 = 398000 \Leftrightarrow 12x = 408000$$

$$\Leftrightarrow x = 34000$$

18. (a) Let the number be x ,

$$\text{Then, } \frac{12}{100} \times \frac{75}{100} \times x - \frac{5}{100} \times x = 75$$

$$\Rightarrow \frac{9x}{100} - \frac{5x}{100} = 75 \Rightarrow \frac{4x}{100} = 75$$

$$\Rightarrow x = \frac{75 \times 100}{4} = 1875$$

19. (a) $\therefore$ Amount, he have spent in 1 month on clothes transport = Amount spent on saving per month
Amount, spent on clothes and transport

$$= \frac{48456}{12} = \text{₹ } 4038$$

20. (a) Let the population of males = x ; then the population of females = $9000 - x$

$$\text{Now, } 5\% \text{ of } x + 8\% \text{ of } (9000 - x)$$

$$= (9600 - 9000) = 600$$

$$\text{or } 0.05x + 720 - 0.08x = 600$$

$$\text{or } 720 - 600 = 0.08x - 0.05x$$

$$\text{or, } 120 = 0.03x$$

$$\therefore x = 4000$$

$\therefore$ Reqd. ratio of population of males and females

$$\frac{4000}{9000 - 4000} = \frac{4000}{5000} = 4:5$$

21. (d) Let salary of Saroj be x .

$$\therefore \text{Salary of Raju} = \frac{80}{100} x$$

$$\text{Salary of Ram} = \frac{70}{100} x$$

$$\text{Required percentage} = \left(\frac{\frac{80x}{100} - \frac{70x}{100}}{\frac{70x}{100}} \right) \times 100$$

$$= \frac{10x}{76x} \times 100 = \frac{100}{7} = 14.28\%$$

22. (d) Let the family consumes 1 kg wheat
To keep expenditure at Rs. 24, its new consumption

$$\text{should be } \frac{24}{27} \times \frac{8}{9} \text{ kg}$$

$\therefore$ Percentage decrease in consumption

$$= \frac{\left(1 - \frac{8}{9} \right)}{1} \times 100 = 11.1\%$$

Alternative method :

$$\text{Required \%} = \frac{27 - 24}{27} \times 100 = 11.1\%$$

23. (d) Let the first man's output be x .

$$\text{Then, } 33\frac{1}{3}\% \text{ of } x = 50\% \text{ of } 1500 \Leftrightarrow \left(\frac{100}{3} \times \frac{1}{100} \times x \right)$$

$$= 750 \Leftrightarrow x = 750 \times 3 = 2250.$$

24. (b) Solve using options. $2/25$ fits the requirement.

25. (c) $10 \times 100 = 1000$, 100 = no. of visitors

$$\text{Now, } 7.5 \times \text{No. of visitors} = 1200$$

$$\text{No. of visitors} = 160$$

$$\text{Increase \%} = \frac{160 - 100}{100} \times 100 = 60\%$$

26. (c) Let the inspector examined x metres,
then 0.08% of $x = 2$

$$\Rightarrow \frac{x \times 0.08}{100} = 2$$

$$\text{or } x = \frac{200}{0.08} = 2500 \text{ metres}$$

27. (d) Let the original fraction be $\frac{x}{y}$

$$\text{Then, } \frac{115\% \text{ of } x}{92\% \text{ of } y} = \frac{15}{16} \Rightarrow \frac{115x}{92y} = \frac{15}{16}$$

$$\Rightarrow \frac{x}{y} \left(\frac{15}{16} \times \frac{92}{115} \right) = \frac{3}{4}$$

28. (d) Let the class has 100 students.

$\Rightarrow$ Number of girls = 35 and number of boys = 65.

Since total number of present students = 70 and number of girls present = 80% of 35 = 28, so number of boys present = 70 - 28 = 42.

$\Rightarrow$ Required fraction = 42/65.

29. (b) Let 100 units be B's income and X units be B's expenditure

$\Rightarrow$ A's income = 60 units.

A's expenditure = 70X/100 units.

But 60 = 75/100 $\times$ X $\Rightarrow$ X = 80.

i.e., B's saving = (100 - 80) units = 20 units.

$$\text{Hence A's saving} = 60 - \frac{70}{100} \times 80 = 4 \text{ units.}$$

i.e., A's saving : B's saving = 4 : 20 = 1 : 5.

30. (a) Decrease in production is only due to decrease in manpower. Hence, manpower is decreased by 25%

Now, suppose that to restore the same production, working hours are increased by x%

$$\text{Production} = \text{Manpower} \times \text{Working hours} \\ = M \times W \text{ (say)}$$

$$\text{Now, } M \times W = (M - 25\% \text{ of } M) \times (W + x\% \text{ of } W)$$

$$\text{or, } M \times W = \frac{75}{100} M \times \frac{100+x}{100} W$$

$$\text{or, } 100 \times 100 = 75 (100 + x)$$

$$\text{or, } \frac{400}{3} = 100 + x \quad \therefore x = \frac{100}{3} = 33\frac{1}{3}\%$$

31. (b) Let original number = 100

New number = 120% of 120% of 100

$$= \left(\frac{120}{100} \times \frac{120}{100} \times 100 \right) = 144.$$

Decrease on 144 = 44. Decrease on 100

$$= \left(\frac{44}{144} \times 100 \right) \% = 30\frac{5}{9}\%$$

32. (d) Number of ticketless travellers in April

$$= 4000 \times \left(1 + \frac{5}{100} \right) \left(1 - \frac{5}{100} \right) \left(1 - \frac{10}{100} \right)$$

$$= \left(4000 \times \frac{21}{20} \times \frac{19}{20} \times \frac{9}{10} \right) = 3591.$$

33. (d) Weight of water in the mixture of 60 g water

$$= 60 \times \frac{75}{100} = 45\text{g}$$

weight of water in the mixture of 45 g water
= 45 + 15 = 60 g

$$\therefore \text{Percentage of water} = \frac{60 \times 100}{75} = 80\%$$

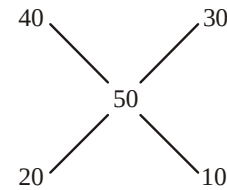
34. (a) Servant's commission amount

$$= 6000 - 1500 = ₹ 4500$$

i.e., 15% = 4500

$$\text{or, } 100\% = \frac{4500}{15} \times 100 = ₹ 30000$$

35. (b) Required litres of solution



$$= \frac{40}{2} = 20 \text{ litres}$$

36. (d) Let the total number of children = x

$$\text{Then, } \frac{720}{x} = 20\% \text{ of } x = \frac{20}{100} \times x = \frac{x}{5}$$

$$\Rightarrow x^2 = 720 \times 5 = 3600$$

$$\Rightarrow x = 60$$

$$\therefore \text{Each child receive} = \frac{720}{60} = 12 \text{ sweets}$$

37. (a) Suppose price of the printer = P

$\therefore$ Price of a computer = 3P

Total cost of 60 computers = 180P

Total cost of 20 printers = 20P

$\therefore$ Total cost of the purchase = 200P

Thus total cost of the printers is 10% of the total cost.

38. (b)

39. (d) Population after 2000 = 3244800

Population after 2001 = 2985216

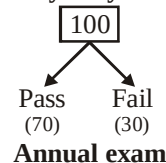
Population at the end of 2003 = 3228810

40. (c) Non-defective products

$$\frac{25 \times 0.98 + 35 \times 0.96 + 40 \times 0.95}{100} \times 100 = 96.1\%$$

41. (a) On ₹ 100 he saves ₹ 6. On 115 he still saves ₹ 6.
percentage increase of 15 on 94 = 15.95%

42. (c) Half yearly exam



$$\frac{70 \times 0.6}{42} + \frac{30 \times 0.8}{24}$$

$$\therefore \text{Total pass in annual exam} = 42 + 24 = 66$$

43. (a) Solution = 100 ml and Alcohol = 40 ml

For first vessel

$$\frac{40+x}{100+x} = \frac{1}{2},$$

so, $x = 20$ ml

For second vessel

$$\frac{40+\frac{3}{5}y}{100+\frac{2}{5}y} = \frac{1}{2},$$

so, $y = 25$ ml

$$\text{Required percentage} = \frac{5}{25} \times 100 = 20\%$$

44. (d) Total land of Sukhiya = $\frac{480x}{0.6} = 800x$

$\therefore$ Cultivated land of village = 384000 x

$$\therefore \text{Required percentage} = \frac{800x}{384000} \times 100 = 0.20833.$$

45. (b) The following PCG will give the answer:

$$100 \xrightarrow[\text{Price effect}]{40\% \uparrow} 140 \xrightarrow[\text{Consumption effect}]{28.56\% \downarrow} 100$$

Hence, the percentage reduction required is 28.56% (40/140)

46. (b) Total votes = 6000. Valid votes = 75% of 6000 = 4500. Bhiku gets 65% of 4500 votes and Mhatre gets 35% of 4500. Hence, Mhatre gets: $0.35 \times 4500 = 1575$ votes.

47. (b) Solve using options. Checking for option (b), gives us:

200000 $\rightarrow$ 180000 $\rightarrow$ 171000 $\rightarrow$ 153900 $\rightarrow$ 146205
(by consecutively decreasing 200000 by 10% and 5% alternately)

48. (a) Solve through trial and error using the options. 12% (option (a)) is the only value that fits the situation.

49. (d) Salary of Dheeraj = ₹ 100

Salary of Anil = ₹ 80

Salary of Vinit = ₹ 70

$$\text{Required percent} = \frac{10}{70} \times 100 = 14.28\%$$

50. (d) Let population = 100

At least 50 people read a newspaper

At most 12.5 people read more than a newspaper

Hence, at least 37.5 people read only one newspaper.

Standard Level

1. (b) We have 5.5% of 100000

= Rent - 12.5% of Rent - 325.

$$\Rightarrow \frac{5500}{12} - \text{Rent} - \frac{\text{Rent}}{8} - \frac{325}{12}$$

$$\Rightarrow \frac{5500}{12} + \frac{325}{12} = \frac{7}{8} \times \text{Rent}.$$

$$\Rightarrow \text{Rent} = \frac{5500}{12} \times \frac{8}{7} = ₹ 554.76 \text{ per month.}$$

2. (c) **First expenditure:** Suppose 100 litres of petroleum at 100 units of money per litre, then total expenditure = 100×100 units of money = 10000 units of money.

Second expenditure: Now 80 litres of petroleum at 140 units of money per litre, total expenditure = 80×140 units of money = 11200 units.

$\Rightarrow$ Expenditure increases by

$$\frac{11200 - 10000}{10000} \times 100 = 12\%$$

Short-cut: $\text{Exp}_1 = PX$, $\text{Exp}_2 = 1.4P (0.8X) = 1.12 PX$.

$\Rightarrow$ Directly we see, answer = 12%.

3. (a) A's marks = $300 \times \frac{33\frac{1}{3}}{100} = 100.$

B's marks = $100 \times (1 + 40/100) = 140.$

$\Rightarrow$ C is ahead of B by $2/9$ of his own marks i.e.

$7/9$ of C's marks = 140

$\Rightarrow$ C's marks = $140 \times 9/7 = 180.$

4. (d) Migrants = 35% of 728400 = $\left(\frac{35x}{100} \times 728400 \right)$

= 254940.

Local population = $(728400 - 254940) = 473460$

Rural population = 20% of 473460 = 94692.

Urban population = $(254940 - 94692) = 160248.$

$\therefore$ Female population = 48% of 473460 + 30% of 94692 + 40% of 160248

$$= \left(\frac{48}{100} \times 473460 + \frac{30}{100} \times 94692 + \frac{40}{100} \times 160248 \right)$$

= 227260.8 + 28407.6 + 64099.2 = 896660.

5. (b) Let Madan's income be ₹ x.

Then, Net income = $(100 - 10)\%$ of ₹ x

$$= 90\% \text{ of } x = \frac{9x}{10}.$$

New net income = 85% of 110% of ₹ x

$$= \left(\frac{85}{100} \times \frac{110}{100} \times x \right) = \frac{187}{200} x$$

$$\therefore \frac{187x}{200} - \frac{9x}{10} = 350 \Leftrightarrow \frac{7x}{200} = 350$$

$$\Leftrightarrow x = \left(\frac{350 \times 200}{7} \right) = 10000.$$

6. (d) Working with options, we have

	Original number	New number	Difference
(a)	22	34	12
(b)	63	96	33
(c)	24	38	14

Obviously, (d) is the correct option.

7. (d) Let he had originally ₹ x . Then
 $65\% \text{ of } x + 20\% \text{ of } x + 1305 = x$
 $0.65x + 0.2x + 1305 = x$
 $\Rightarrow 0.15x = 1305 \Rightarrow x = ₹ 8700$
 $\therefore$ His total investment = $65\% \text{ of } 8700 + 20\% \text{ of } 8700$
 $= 85\% \text{ of } 8700 = ₹ 7395$

8. (c) Let original consumption be 1 unit costing ₹ 100
 New cost = ₹ 125. New consumption

$$= \left(\frac{1}{125} \times 100 \right) = \frac{4}{5} \text{ unit.}$$

$$\therefore \frac{\text{Reduction in consumption}}{\text{Original consumption}} = \frac{\left(1 - \frac{4}{5}\right)}{1} = \frac{1}{5}$$

i.e., 1 : 5.

9. (c) After first year, the value of the scooter = ₹ 20,000
 After second year, the value of scooter = ₹ 16,000
 After third year, the value of scooter = ₹ 12,800
10. (b) Let original consumption = 100 kg and new consumption = x kg
 So, $100 \times 6 = x \times 7.50 \Leftrightarrow x = 80$ kg
 $\therefore$ Reduction in consumption = 20%
11. (a) Let the numbers be x and y . Then,
 $x + y = \frac{28}{25}x \Rightarrow y = \frac{28}{25}x - x \Rightarrow y = \frac{3}{25}x$
 $\Rightarrow \frac{y}{x} = \left(\frac{3}{25} \times 100 \right) \% = 12\%.$
12. (b) Let the number be x . Then,
 $\% \text{ error} = \frac{6x - x/6}{6x} \times 100 = \frac{35}{36} \times 100 = 97.2\%$
13. (c) $p = 6q$. So, q is less than p by $5q$.
 $\therefore$ Required percentage = $\left(\frac{5q}{p} \times 100 \right) \%$
 $= \left(\frac{5q}{6q} \times 100 \right) \% = 83\frac{1}{3} \%$
14. (c) $\frac{5}{100}A + \frac{15}{100}B + \frac{10}{100}B + \frac{20}{100}C \Rightarrow A = 3B$ and
 $B = 2C = 2 \times 2000 = 4000.$
 $\therefore A = 3 \times 4000 = 12000.$
 Hence, $A + B + C = (12000 + 4000 + 2000) = 18000.$

15. (d) Let original price be ₹ x per orange. Then,
 Reduced rate = $(1 - 0.2)x = ₹ 0.8x$

$$\therefore \frac{2.50}{0.8x} \times \frac{2.50}{x} = 5$$

$$\Rightarrow \frac{25}{8x} \times \frac{2.5}{x} = 5 \Rightarrow x = \frac{1}{8}$$

$$\therefore \text{Original price of oranges per dozen} = \frac{1}{8} \times 12 = ₹ 1.5$$

$$\text{and Reduced price} = ₹ (0.8)(1.5) = ₹ 1.2$$

16. (d) Let the total no. of parts produced at initial stage be 100. Then after three successive percentage rejections of 10%, 5% and 2%, we have
 $100 \times 0.9 \times 0.95 \times 0.98 = 83.79$
 Therefore, a single effective rejection
 $= 100 - 83.79 = 16.21$

17. (a) Let the original quantity be x kg. Vanaspati ghee in

$$x \text{ kg} = \left(\frac{40}{100}x \right) \text{ kg} = \left(\frac{2x}{5} \right) \text{ kg.}$$

$$\text{Now, } \frac{\frac{2x}{5}}{x+10} = \frac{20}{100} \Leftrightarrow \frac{2x}{5x+50} = \frac{1}{5} \Leftrightarrow 5x = 50$$

$$\Leftrightarrow x = 10$$

18. (a) Let the strength of school was x in 1998
 $\therefore$ strength in 2001 will be

$$x \times \frac{110}{100} \times \frac{90}{100} \times \frac{110}{100} \times \frac{90}{100} \times \frac{110}{100} = 1.07811x$$

$$\therefore \text{increment} = 1.07811x - x = 0.07811x$$

$$\therefore \% \text{ increase} = 7.811 \approx 8\%$$

19. (d) The given information gives no indication regarding the comparison of x and y .
20. (d) Since the weightage of eighth examination is not known, hence can not be determined.
21. (c) Let the original price per egg be ₹ x . Then, increased

$$\text{price} = ₹ \left(\frac{130}{100}x \right)$$

$$\therefore \frac{7.80}{x} - \frac{7.80}{\frac{130}{100}x} = 3 \Leftrightarrow \frac{7.80}{x} - \frac{780}{130x} = 3$$

$$\Leftrightarrow 1014 - 780 = 3 \times 130x \Leftrightarrow 390x = 234$$

$$\Leftrightarrow x = 0.6$$

$$\text{So, present price per dozen} = ₹ \left(12 \times \frac{130}{100} \times 0.6 \right)$$

$$= ₹ 9.36.$$

22. (a) Let the truth spoken by A and B be p_1 and p_2

$$\text{respectively, i.e., } p_1 = \frac{3}{4} \text{ and } p_2 = \frac{4}{5}$$

They will contradict each other only when one speaks truth and the other is lying.

$$\text{i.e., } \frac{3}{4} \cdot \frac{1}{5} + \frac{4}{5} \cdot \frac{1}{4} = \frac{3}{20} + \frac{4}{20} = \frac{7}{20} = \frac{35}{100} \text{ i.e., } 35\%$$

23. (b) Let B's Income = `x

$$A's \text{ Income} = \frac{3}{5}x$$

$$\text{And B's expenditure} = \frac{7}{10}y$$

$$A's \text{ expenditure} = \frac{7}{10}y$$

$$\text{Also, } \frac{3}{5}x = \frac{3}{4} \cdot \frac{7}{10}y$$

$$\frac{A's \text{ savings}}{B's \text{ savings}} = \frac{x - \frac{7}{10}y}{\frac{3}{5}x - \frac{7}{10}y} = \frac{\frac{7}{8}y - y}{\frac{3}{5} \cdot \frac{7}{10}y - \frac{7}{10}y} = \frac{-y/8}{\frac{21y}{40} - \frac{7}{10}y} = \frac{-y/8}{\frac{21y - 28y}{40}} = \frac{-y/8}{-7y/40} = \frac{5}{28} \approx 1:5$$

24. (b) Percentage of uncertain individuals

$$= [100 - (20 + 60)] \% = 20\%$$

$$\therefore 60\% \text{ of } x - 20\% \text{ of } x = 720 \Leftrightarrow 40\% \text{ of } x = 720$$

$$\Leftrightarrow \frac{40}{100}x = 720 \Leftrightarrow x = \left(\frac{720 \times 100}{40} \right) = 1800.$$

25. (d) Suppose Income of B = `x

$$\text{Income of A} = \frac{150}{100} \times x = \frac{3x}{2}$$

$$\text{Income of C} = \frac{120}{100} \times \frac{3x}{2}$$

$$\frac{6}{5} \times \frac{3x}{2} = \frac{9x}{5}$$

$$\therefore x + \frac{3x}{2} + \frac{9x}{5} = 86000$$

$$\frac{10x + 15x + 18x}{10} = 86000$$

$$43x = 860000$$

$$x = 20000$$

$$\text{So, income of C} = \frac{9}{5} \times 20000 = \text{`} 36000$$

26. (c) Number of males = 60% of 1000 = 600.

$$\text{Number of females} = (1000 - 600) = 400.$$

$$\text{Number of literates} = 25\% \text{ of } 1000 = 250.$$

$$\text{Number of literate males} = 20\% \text{ of } 600 = 120$$

$$\text{Number of literate females} = (250 - 120) = 130$$

$$\therefore \text{Required percentage} = \left(\frac{130}{400} \times 100 \right) \% = 32.5\%$$

27. (d) B + 60% of A = 175% of B $\rightarrow$ 60% of A = 75% of B.

$$\text{i.e. } 0.6A = 0.75B$$

$$A/B = 5/4$$

Apparently it seems that A is bigger, but if you consider A and B to be negative the opposite would be true.

Hence, option (d) is correct

28. (d) Number of ticketless travellers in April

$$4000 \times \left(1 + \frac{5}{100} \right) \left(1 - \frac{5}{100} \right) \left(1 - \frac{10}{100} \right)$$

$$\left(4000 \times \frac{21}{20} \times \frac{19}{20} \times \frac{9}{10} \right) = 3591.$$

29. (d) October : November : December = 9 : 8 : 10.666 since, he got `40 more in December than October, we can conclude that 1.666 = 40 $\rightarrow$ 1 = 24.

Thus, total Bonus for the three months is:

$$0.4 \times 27.666 \times 24 = 265.6$$

30. (b) The total wealth given would be 50% + 25% (which is got by 50% of the remaining 50%) + 12.5% (which is got by 50% of the remaining 25%). Thus, the total wealth given by him would be equivalent to 87.5% of the total. Since, this is equal to 130900 kilograms of gold, the total gold would be:
130900 $\times$ 8/7 = 149600.

31. (c) $\frac{378}{125} \times 3 + \frac{3}{125} = 302.4\%$

Let original salary be `100

And now going through option, we get (c) as answer.

32. (d) Out of a total of 100% votes; 80% voted. 16% were invalid and 20% went to the second placed candidate. This means that the maximum the winner can get is 44%. Options a, b and c are greater than 44% and hence cannot be correct. Hence, none of these.

33. (b) The value of x should be such that the left hand side after completely removing the square root signs should be an integer. For this to happen, first of all the square root of 3x should be an integer. Only 3 and 12 from the options satisfy this requirement. If we try to put x as 12, we get the square root of 3x as 6. Then the next point at which we need to remove the square root sign would be 12 + 2(6) = 24 whose square root would be an irrational number. This leaves us with only 1 possible value (x = 3). Checking for this value of x we can see that the expression is satisfied as LHS = RHS.

34. (d) Rajesh's scores in each area is 65 and 82 respectively out of 100 each. Since, the exam is of a total of 250 marks (100 + 100 + 50) he needs a total of 195 marks in order to get his target of 78% overall. Thus, he should score 195 - 65 - 82 = 195 - 147 = 48 marks in Sociology which would mean 96%

35. (d) The only values that fit this situation are C 25%, B 30%, and A 45%. These are the percentage of votes polled. (Note: these values can be got either through trial and error or through solving $c + c + 5 + 1.5(c + 5) = 100\%$

Then, 20% is 18000 (the difference between A & C.)

Hence, 90000 people must have voted and 100000 people must have been on the voter's list.

36. (a) Let the number be N . Then, $5N$ should be the correct outcome. But instead the value got is $0.2N$. Change in value = $5N - 0.2N = 4.8N$. The percentage change in the value = $4.8N \times 100/5N = 96\%$
37. (c) $100 \rightarrow 150 \rightarrow 75$ (yr. 1) $\rightarrow 112.5 \rightarrow 56.25$ (yr. 2)
 $\rightarrow 84.375 \rightarrow 42.1875$
 Now, $42.1875 = \frac{1}{16} \times 100$
 Hence, $1 \rightarrow 400$
 Also year 2 donation is $56.25 \times 400 = 22500$
38. (c) Total characters in her report = $25 \times 60 \times 75$
 Let the new no. of pages be n
 Then: $n \times 55 \times 90 = 25 \times 60 \times 75$
 $n = 22.72$
 This means that her report would require 23 pages. A drop of 8% in terms of the pages.
39. (c) No. of males = $\frac{11}{18} \times 7200 = 4400$
 No. of males married = $\frac{40}{100} \times 4400 = 1760$
 No. of females married = 1760
 Required percentage = $\frac{1760}{2800} \times 100 = 62\frac{6}{7}\%$
40. (b) Let the equation be
 $x^2 - 2x + 1 = 0$... (1)
 and $x^2 - x - 2 = 0$... (2)
 Required percentage = $\frac{1 - (-2)}{1} \times 100 = 300\%$

Expert Level

1. (b) Seats in executive class = 50
 Seats for chair car = 450
 Booked seats in total = 425
 Booked in executive class = 48
 Therefore, seats booked in chair class = $(425 - 48) = 377$
 Empty seats for chair class = $450 - 377 = 73$
2. (b) Number of questions attempted correctly
 = $(70\% \text{ of } 10 + 40\% \text{ of } 30 + 60\% \text{ of } 35)$
 = $(7 + 12 + 21) = 40$
 Questions to be answered correctly for 60% grade
 = $60\% \text{ of } 75 = 45$.
 $\therefore$ Required number of questions = $(45 - 40) = 5$.
3. (b) $(50/100) \times (a/100) \times (b) = (75/100) \times (b/100) \times (c) \rightarrow$
 $50a = 75c \rightarrow c = 0.667a$
4. (c) $0.18A + 0.15B + 0.19C = 0.17A + 0.17B + 0.17C$
 $0.01A - 0.02B + 0.02C = 0$... (1)
 $A - B = 500$... (2)
 $A - C = 3400$... (3)

Solving 1, 2 and 3 we get

$$A = 5800, B = 5300 \text{ and } C = 2400$$

$$A + B + C = 13500$$

5. (a) Income = Expenditure + Savings

$$\begin{pmatrix} 8x \\ 10x \end{pmatrix} = \begin{pmatrix} 5x + 3x \\ 8x + 2x \end{pmatrix} - x$$

$$\text{Now, the deficit} = (3x - 2x) = x = 3500$$

$$\therefore \text{the new salary} = 10x = 35,000$$

6. (d) The thought process would go like:

If we assume 100 students

Total : 60 boys and 40 girls.

Free waiver : 9 boys and 3 girls.

This means that a total of 12 people are getting a fee waiver. (But this figure is given as 90.)

Hence, 1 corresponds to 7.5.

Now, number of students not getting a fee waiver = 51 boys and 37 girls

50% concession $\rightarrow$ 25.5 boys and 18.5 girls (i.e. a total of 44.)

Hence, the required answer = $44 \times 7.5 = 330$

7. (b) Let A has x guavas and B has y guavas, the

$$x - \frac{1}{4}x = \left(y + \frac{1}{4}y\right) + 2 \quad \dots(1)$$

$$\text{and } y - \frac{7}{10}y = x + \frac{1}{10}y - 4 \quad \dots(2)$$

Solving 1 and 2, we get

$$x = 44 \quad y = 20$$

$$\text{Total price} = 44 + 20 = 64$$

8. (d)

Income $\rightarrow$	4	4.4	4.8	5.2]	18.4	lakh
Saving $\rightarrow$	2	1.76	1.44	1.04]	6.24	lakh
Exp. $\rightarrow$	2	2.64	3.36	4.16]	12.16	lakh

$$\text{So, } \frac{6.24}{12.16} \times 100 = 51\frac{6}{19}\%$$

9. (d) $A + B + C + D = 56$

$$B + C + D = 4.6A$$

$$\Rightarrow A + B + C + D = 5.6A \text{ (adding } A \text{ in both side)}$$

$$56 \text{ lakh} = 5.6A$$

$$\Rightarrow A = 10 \text{ lakh}$$

$$\text{Similarly, } A + C + D = \frac{11}{3}B$$

$$\Rightarrow A + B + C + D = \frac{14}{3}B$$

$$\Rightarrow B = 12 \text{ lakh}$$

$$\text{Similarly, } 4(A + B + D) = C$$

$$\Rightarrow A + B + D = 2.5C$$

$$\Rightarrow A + B + C + D = 3.5C$$

$$\Rightarrow C = 16 \text{ lakh}$$

$$\text{Therefore } D = (A + B + C + D) - (A + B + C) = 18 \text{ lakh}$$

10. (b) Let Ram replaces x litres of 12% sol. with 39% solution.

$$\text{Now, quality of 12\% sol. in 27 litre} = \frac{27 \times 12}{100}$$

$\therefore$ After replacing we have volume of 12% sol.

$$= \frac{27 \times 12}{100} - \frac{12x}{100} + \frac{39x}{100} = \frac{324 + 27x}{100}$$

This will be equal to 27 litre of 21% sol.

$$\therefore \frac{324 + 27x}{100} = \frac{21 \times 27}{100}$$

$$\therefore x = \frac{567 - 324}{27} = \frac{243}{27} = 9$$

11. (a)

Rice	Wheat
25	9
$\times x$	$\times 5x$
25x	45x

$$70x = 350$$

$$\Rightarrow x = 5$$

Hence the price of rice = ₹ 5 per kg

Price of wheat = ₹ 25 per kg

Now, the price of wheat = ₹ 30 per kg

Let the new amount of rice be M kg, then

$$M \times 5 + 9 \times 30 = 350$$

$$M = 16$$

Hence decrease (in%) of amount of rice

$$\frac{25 - 16}{25} \times 100 = 36\%$$

12. (a)

Year	Rate of Commission	Commission in values
1	20%	$0.2 \times 20,000 = 4000$
	25% (bonus)	$0.25 \times 4000 = 1000$
2	16%	$0.16 \times 20,000 = 3200$
3	12%	$0.12 \times 20,000 = 2400$
4	10%	$0.1 \times 20,000 = 2000$
5-10	4%	$6 \times 0.04 \times 20,000 = 4800$

$$\text{Total commission} = (4000 + 3200 + 2400 + 2000 + 4800) + (1000) = 17,400$$

13. (a) Solve using options.

Checking for option (a) will go as: According to this option 400 people have voted against the motion. Hence, originally 200 people must have favoured the motion. (Since, there is a 100% increase in the opponents)

This means that 200 people who were for the motion initially went against it.

This leaves us with 400 people who were for the motion initially (after the abduction.)

$\frac{1}{3}$ rd of the original having been abducted, they should amount to half what is left.

This means that 600 (for) and 200 (against) were the original distribution of 800.

This option fits perfectly (given all the constraints) and hence is the correct answer.

14. (b) Using statement ii only we can find the winner. In case of statement i, there may be two contestants with 49% valid votes each and remaining two getting 1% valid votes each.

15. (c) Let the initial times allotted be : 50, 40 and 20 hours. Then, the time used in each activity is: 20, 12 and 4 hours. Thus, 36 hours out of 110 are used in all.

$$\text{Hence, the answer is } 36/110 = 32.72\%$$

16. (a) Let the value of Index = 100

Value of Reliance share = 7

Value of HLL = 13

Value of Infosys = 15

Value of remaining = 65

New value of Reliance = 7.63

New value of HLL = 14.3

New value of Infosys = 15.60

New value of remaining = 68.47

$$\text{Hence, required percentage} = \frac{3.47}{65} \times 100 = 5.34\%$$

17. (b) Let the initial percentage of salt be 10% in 100 liters of sea water in the flask.

10 % of this is poured out (i.e. 10 liters are poured out) and the water heated so as to increase the percentage of salt in the beaker 5 times (we have assumed M as 5 here)

This means that there will be 30% salt in the beaker. Since, the salt concentration is increased by only evaporating water, the amount of salt remains the same.

Initially the salt was 10% of 100 liters (= worth 1 liter). Hence, the water must have been worth 9 liters.

Now, since this amount of salt becomes worth 50% of the total solution, the amount of water left after evaporation would have been 1 liter and the total would be 2 liters.

When the 2 liters are mixed back again: The new concentration of salt in sea water would go up. In this specific case by alligation we would get the following alligation situation:

Mix 90 liters of 10% salted sea water with 2 liters of 50% salted sea water.

The result using alligation will be : $[10 + 40/46]\%$ concentration of salted sea water. The value of the increase percentage will be $400/46$. (this will be the value of x)

Now, try to use the given options in order to match the fact that originally the flask contained 100 liters of sea water.

$$\text{Use } M = 5, x = 400/46,$$

Only option (b) matches the situation.

$$\frac{(9 \times 5 + 1)400/46}{(5 - 1)} = 100$$

$$18. (b) \frac{\frac{14.4}{80} - \frac{14.4}{104}}{\frac{14.4}{104}} = \frac{104 - 80}{80} = 30\%$$

19. (b) Hursh Sarma's savings:

Month	Salary	Savings
1	10800	1080
2	10800	900
3	10800	900
4	10800	0
5	12000	1500
6	12000	1800
7	12000	2400
8	12000	1200
9	12000	1000
10	12000	1500
11	12000	1800
12	12000	2400

Required Ratio = $4800/900 = 5.333$

20. (b) For 45% of the journey in city driving conditions, 54% of the fuel is consumed.

Hence, for the remaining 55% journey, 46% fuel is left.

Required increase in fuel efficiency

$$= \frac{\frac{55}{46} - \frac{45}{54}}{\frac{45}{54}} \times 100 = 43.5\%$$

21. (b) Let the capacity of the tank be 100 litres. Then,

Initially : A type petrol = 100 litres.

After first operation :

$$\text{A type petrol} = \left(\frac{100}{2}\right) = 50 \text{ litres;}$$

B type petrol = 50 litres.

After second operation :

$$\text{A type petrol} = \left(\frac{50}{2} + 50\right) = 75 \text{ litres;}$$

B type petrol = $(50/2) = 25$ litres

After third operation :

$$\text{A type petrol} = \left(\frac{75}{2}\right) = 37.5 \text{ litres;}$$

$$\text{B type petrol} = \left(\frac{25}{2} + 50\right) = 62.5 \text{ litres.}$$

$\therefore$ Required percentage = 37.5%.

22. (b) Required percentage

$$= \left(\frac{20\% \text{ of } 10 + 35\% \text{ of } 4}{10 + 4} \times 100\right)\%$$

$$= \left(\frac{3.4}{14} \times 100\right)\% = 24\frac{2}{7}\%$$

23. (b) Let the capacity of the tank be 100 litres. Then, Initially : A type petrol = 100 litres.

After first operation:

$$\text{A type petrol} = \left(\frac{100}{2}\right) = 50 \text{ litres;}$$

B type petrol = 50 litres.

After second operation:

$$\text{A type petrol} = \left(\frac{50}{2} + 50\right) = 75 \text{ litres;}$$

$$\text{B type petrol} = \left(\frac{50}{2}\right) = 25 \text{ litres.}$$

After third operation:

$$\text{A type petrol} = \left(\frac{75}{2}\right) = 37.5 \text{ litres; B type petrol}$$

$$= \left(\frac{25}{2} + 50\right) = 62.5 \text{ litres.}$$

$\therefore$ Required percentage = 37.5%

$$24. (c) \text{ Total money} = \left(600 \times \frac{25}{100} + 1200 \times \frac{50}{100}\right) = \text{₹} 750$$

$$25 \text{ paise coins removed} = \left(\frac{12}{100} \times 600\right) = 72$$

$$50 \text{ paise coins removed} = \left(\frac{24}{100} \times 1200\right) = 288$$

$$\text{Money removed} = \left(72 \times \frac{25}{100} + 288 \times \frac{50}{100}\right) = \text{₹} 162$$

$$\therefore \text{ Required percentage} = \left(\frac{162}{750} \times 100\right)\% = 21.6\%$$

25. (b) If $3x + 7 = x^2 + M = 7x + 5$

$$\text{ie, } 3x + 7 = 7x + 5$$

$$\text{or, } 4x = 2, \therefore x = \frac{1}{2}$$

$$\text{and } 3x + 7 = x^2 + M$$

$$\text{or, } \frac{1}{4} + M = \frac{3}{2} + 7 \Rightarrow M + \frac{1}{4} = 8 + \frac{1}{2}$$

$$\therefore M = 8\frac{1}{4}, 120\% \text{ of } M = 9.90$$

26. (c) The third gallery making the capacity half as large again means: an increase of 50% further it is given that: 4 (first + third) = 12 (second). In order to get to the correct answer, try to fit in the options into this situation.

Note here that the question is asking you to find the capacity of the second gallery as a percentage of the first.

If we assume option (a) as correct 70% the following solution follows:

If second is 70, then first is 100 and first + second is 170. Then third will be 85 (50% of first) + second.

Then the equation:

$4 \times (100 + 85)$ should be equal to 12×70 .

But this is not true.

27. (a) The net time lost over two weeks would be 0.25% of a week's time (since in the first week the clock loses $\frac{1}{2}\%$ and in the second week the clock gains $\frac{1}{4}\%$ on the true time.)

A week contains 168 hours. Hence, the clock loses 0.42 hours. i.e. 25.2 minutes or 25 minutes 12 seconds. Hence, the correct time would be 12 : 25 : 12

28. (b) Harish Sharma's savings:

Month	Salary	Savings
1	10800	1080
2	10800	900

3	10800	900
4	10800	0
5	12000	1500
6	12000	1800
7	12000	2400
8	12000	1200
9	12000	1000
10	12000	1500
11	12000	1800
12	12000	2400

Required Ratio = $4800/900 = 16 : 3$

29. (a) Let the value of index = 100

Value of Reliance share = 7

Value of HLL = 13

Value of Infosys = 15

Value of remaining = 65

New value of Reliance = 7.63

New value of HLL = 14.3

New value of Infosys = 15.60

New value of remaining = 68.47

Hence, Required percentage = $\frac{3.47}{65} \times 100 = 5.34\%$

Explanation of Test Yourself

1. (b) Let the numbers be x and y . Then, $\frac{x}{12} = \frac{y}{4} \Rightarrow x = 3y$.

$$\therefore \text{Required percentage} = \left(\frac{x-y}{y} \times 100 \right) \% \\ = \left(\frac{2y}{y} \times 100 \right) \% = 200\%$$

2. (a) Let the original price of the jewel be P and let the profit earned by the third seller be $x\%$.
Then, $(100+x)\%$ of 125% of 120% of $P = 165\%$ of P

$$\Rightarrow \left[\frac{(100+x)}{100} \times \frac{125}{100} \times \frac{120}{100} \times P \right] = \left(\frac{165}{100} \times P \right)$$

$$\Rightarrow (100+x) = \left(\frac{165 \times 100 \times 100}{125 \times 120} \right) = 110$$

$$\Rightarrow x = 10\%$$

OR

$$100 \xrightarrow{+65\%} 165$$

$$100 \xrightarrow{+20\%} 120 \xrightarrow{+25\%} 150$$

$$\text{So, third seller profit} = \frac{165-150}{150} \times 100 = 10\%$$

3. (b) The following structure would follow:
Passed all: 5%
Passed 4: 20% of 90% = 18%
Passed 1: 25% of 90% = 22.5%
Passed 2: 24.5%
Passed None: 5%
Passed 3: Rest $(100 - 5 - 18 - 22.5 - 24.5 - 5 = 25\%)$
But it is given that 300 people passed 3.
Hence, 25% = 300.
Hence, 1200 students must have appeared in the test.

4. (a) B's income is $\left(\frac{6\frac{1}{4}}{100 + 6\frac{1}{4}} \times 100 \right) \%$ less than A's

$$\text{i.e., } \left(\frac{6.25}{106.25} \times 100 \right) \% = 5.89\% \text{ less than A's income}$$

5. (b) Traveling for 2400 kms at 18 kmph, Vawal will use 133.33 litres of petrol every month. The increase in expenditure for Vawal will be $133.33 \times 0.07 \times 28 = \text{`} 262$ (approx).

6. (d) Let original rate be x per kg.

$$\text{Reduced rate} = \left[\left(100 - \frac{25}{4} \right) \times \frac{1}{100} x \right] = \frac{15x}{16} \text{ per kg}$$

$$\therefore \frac{120}{\frac{15x}{16}} - \frac{120}{x} = 1 \Leftrightarrow \frac{128}{x} - \frac{120}{x} = 1 \Leftrightarrow x = 8.$$

So, original rate = $\text{`} 8$ per kg
OR

$$\text{Original rate} = \frac{25}{4} \times \frac{100}{\left(100 - \frac{25}{4} \right) \times 1}$$

$$= \frac{25}{4} \times \frac{100 \times 4}{375} = \text{`} 8 \text{ per kg}$$

7. (c) Clearly, the numbers which have 1 or 9 in the unit's digit, have squares that end in the digit 1. Such numbers from 1 to 70 are 1, 9, 11, 19, 21, 29, 31, 39, 41, 49, 51, 59, 61, 69.
Number of such numbers = 14.

$$\therefore \text{Required percentage} = \left(\frac{14}{70} \times 100 \right) \% = 20\%$$

8. (c) I. Let the first number be x .
Then the second number = $1.25x$
Sum = $x + 1.25x = 2.25x$
Difference = $1.25x - x = 0.25x$
Also $9 \times 0.25x = 2.25x$.
Thus, [I] is true.

II. $\frac{56}{100} \times A = \frac{24}{100} \times B$

Multiplying by $\frac{49}{56}$ on both the sides.

$$\therefore \frac{49}{56} \times \frac{56}{100} \times A = \frac{49}{56} \times \frac{24}{100} \times B$$

$$\frac{49}{100} A = \frac{21}{100} B \Rightarrow 49\% \text{ of } A = 21\% \text{ of } B.$$

Thus, [II] is true.

III. $\frac{7}{11} x = \frac{2}{9} y \therefore 63x = 22y \Rightarrow \frac{63}{100} \times x = \frac{22}{100} y$

Multiplying by $\frac{81}{63}$ on both sides.

$$\frac{81}{63} \times \frac{63x}{100} = \frac{22}{100} \times \frac{81}{63} \times y$$

$$\therefore \frac{81}{100} \times x = \frac{198}{700} \times y = 28\frac{2}{7} \% \text{ of } y.$$

Thus, [III] is false.

9. (c) Let the population of females be x .
Then 110% of $x + 106\%$ of $(8000 - x) = 8600$

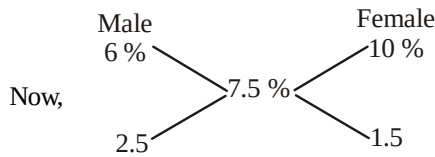
$$\text{or, } \frac{110x}{100} + \frac{106(8000 - x)}{100} = 8600$$

$$\text{or, } x(110 - 106) = 8600 \times 100 - 8000 \times 106$$

$$\therefore x = \frac{8600 \times 100 - 8000 \times 106}{110 - 106} = \frac{12,000}{4} = 3,000$$

By Method of Alligation

$$\text{Average \% of increase} = \frac{600}{8000} \times 100 = \frac{15}{2} = 7.5\%$$



$$\therefore \text{Male : Female} = 2.5 : 1.5 = 5 : 3$$

$$\therefore \text{The population of females} = \frac{8000}{5+3} \times 3 = 3000$$

10. (a) Ram's flat value after 2 years $= 1(1 + 0.1)^2 = 1.1^2$
Prem's flat value after 2 years $= 1.1(1 + 0.05)^2$
 $= 1.1(1.05)^2$
Difference $= 1.1(1.05)^2 - 1.1^2$
 $1.1(1.1025 - 1.1) = 1.1 \times 0.0025 \text{ lakh}$
 $1.1 \times 0.0025 \times 100000 = 25 \times 11 = \text{` 275}$
Ram paid ` 275 to Prem

11. (a) Voters to vote for $P = \frac{2}{5}V$
Voters to vote for $Q = \frac{3}{5}V$
On the last day
Voters voted for $P = \frac{85}{100} \times \frac{2}{5}V + \frac{25}{100} \times \frac{3}{5}V = \frac{245}{500}V$
Voters voted for $Q = \frac{75}{100} \times \frac{3}{5}V + \frac{15}{100} \times \frac{2}{5}V = \frac{255}{500}V$
As P lost by 2 votes,
 $\frac{255}{500}V - \frac{245}{500}V = \frac{V}{50} = 2 \text{ or } V = 100$

Alternatively

Let total voters be 100
Voters to vote for $P = 40$
Voters to vote for $Q = 60$

On the last day
Voters voted for

$$P = 40 - \frac{15}{100} \times 40 + \frac{25}{100} \times 60 = 40 - 6 + 15 = 49$$

Voters voted for

$$Q = 60 - \frac{25}{100} \times 60 + \frac{15}{100} \times 40 = 60 - 15 + 9 = 51$$

$\therefore P$ lost by $51 - 49 = 2$ votes
Hence, total voters = 100.

12. (c) If AC IInd costs 100, AC IInd would cost 120 and AC Ist would cost 190. 3 Tier ticket would cost : 47.5 and general ticket would cost 40.

$$\text{AC IInd} \rightarrow 780 = 120$$

Then the difference between 3 Tier and general ticket would be: $7.5 \times 780 = 48.75$

13. (c) Let maximum marks be 100
Hence his average = 60
Let his marks be $10x, 9x, 8x, 7x, 6x$ respectively in 5 subjects
i.e., $10x + 9x + 8x + 7x + 6x = 60\%$ of 5×100

$$= \frac{60}{100} \times 500 = 60 \times 5 \dots (i)$$

Solving, we get his marks were
75, 67.5, 60, 52.5, 45

Since passing marks are 50
He passed in 4 subjects

14. (b) 100 kg of fresh grapes have 90 kg water and 100 kg of dry grapes have 20 kg water

$$20 \text{ kg fresh grapes have } \frac{9}{10} \times 20 = 18 \text{ kg water}$$

i.e., 2 kg non-water

$$\text{For dry grapes non-water material} = \frac{80}{100}$$

$$\text{If any grapes are } x \text{ kg} \Rightarrow \frac{8}{10} \times x = 2 \Rightarrow x = 2.5 \text{ kg}$$

15. (c) Let total no. of employees be = 100
Employees who get 25000 plus salaries = 45
Male employees = 40
Female employees = 60
Male employees who get 25000 plus salaries = 30
Female employees who get 25000 plus salaries = 15
Female employees who get less than or equal to 25000
 $= (60 - 15) = 45$
i.e., $\frac{45}{60} = \frac{3}{4}$

6

PROFIT, LOSS AND DISCOUNT

- Introduction
- Total Cost Price (CP)
- Selling Price (SP)
- Profit (or Gain) and Loss
- Use of PCG (Percentage change graphic) in profit and loss
- Marked Price, List Price, Discount and Successive Discounts
- Contribution Margin (CM)
- Break-Even Point And Break-Even Sales

INTRODUCTION

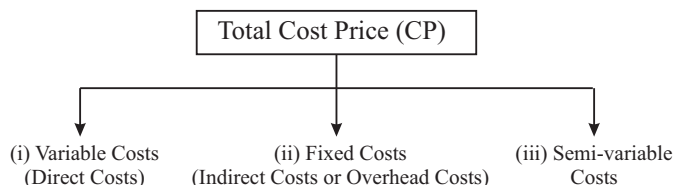
The concepts of this chapter Profit, Loss and Discount are used in day-to-day business. At least one question from this chapter is always asked in CAT and equivalent to CAT aptitude tests.

The concepts of profit, loss and discount are essential concepts for any aspiring CAT applicant. It is certainly a good idea to go through previous year papers of CAT and the other B-school entrance exams and try to figure out what common types of questions of this chapter are that come year after year and also kind of skills required to solve them.

TOTAL COST PRICE (CP)

The total amount paid or expended in either purchasing an object (or a service) or producing an object (or a service) is known as its Total Cost Price of that object (or the service) for purchaser or producer respectively.

Total cost price is subdivided into three parts as given below:



(I) Variable Costs (Direct Costs)

It is that part of the total cost that varies directly with the number of units of objects (or services) purchased or produced. For example price of raw material used in producing one unit of product. Wages to labour in producing one unit of the product when the wages are given on a piece rate basis, price per unit of an object at which a trader bought it, etc.

If price of raw material used for producing one unit of product is = ₹ 20. Then price of raw material used for producing five units of products = ₹ 20 × 5 = ₹ 100

If price of a note-book at which a trader bought it = ₹ 50

Then price of 10 note-books, which the trader paid to buy them = ₹ 50 × 10 = ₹ 500

Here, we clearly see that price of raw material varies directly with the number of product produced. Also price of note books that the trader paid to buy them varies directly with the number of note-books bought. Hence, they are the direct costs.

(II) Fixed Costs (Indirect Costs or Overhead Costs)

This is the part of the total cost which incurred irrespective of the number of items produced or purchased for sale. For example, irrespective of the number of units of a product produced or purchased for sale, the rent of the premises, salary of the employees, etc. are fixed and hence are fixed costs.

(III) Semi-variable Costs

Some costs like rent behave as fixed costs under normal circumstances but have to be increased after a certain level of production or purchase to sell, because the company needs additional space to accommodate the increased products and increased work load.

Here the rent is not fixed after a certain level therefore in the case mentioned here rent is neither completely fixed nor completely variable and hence rent is semi-variable costs.

Semi-variable costs are also a part of the total cost.

Thus

$$\text{Total Cost Price} = (\text{Variable Costs}) + (\text{Fixed Costs}) + (\text{Semi-variable costs})$$

In most of the problems; Fixed costs and Semi-variable costs are neither given nor are to be found out for these problems,

$$\text{Total Cost Price} = \text{Variable Costs Price}$$

Total Cost Price is simply called Cost Price.

SELLING PRICE (SP)

When a person (or an agency) P sells an object or a service to another person (or agency) Q at a price S , then P is called seller, Q is called purchaser and S is called Selling Price of the seller. S is also called variable cost of the purchaser. If there is no fixed cost and semi-variable cost of the purchaser, then S is called Total Cost Price or simply called cost price of the purchaser.

PROFIT (OR GAIN) AND LOSS

(1) When selling price is more than cost price (i.e. $SP > CP$) then profit has been incurred.

$$\begin{aligned} \text{(i)} \quad & \text{Profit} = SP - CP \\ \Rightarrow & \quad SP = CP + \text{Profit} \\ & \quad CP = SP - \text{Profit} \end{aligned}$$

$$\text{(ii)} \quad \text{Percentage Profit (or profit percent)} = \frac{\text{Profit}}{\text{Cost Price}} \times 100\%$$

Percentage profit means profit when cost price is ₹ 100. Percentage profit is always calculated on CP unless otherwise stated.

To understand the percentage profit clearly, suppose cost price (CP) and selling price (SP) of a book are ₹ 500 and ₹ 700

$$\text{Profit} = SP - CP = ₹ 700 - ₹ 500 = ₹ 200$$

Here, we see that, when CP is ₹ 500, then profit = ₹ 200

$$\Rightarrow \text{When CP will be ₹ 1, then profit} = ₹ \frac{200}{500}$$

$$\Rightarrow \text{When CP will be ₹ 100, then profit} = ₹ \frac{200}{500} \times 100$$

$$\text{or percentage profit} = \frac{200}{500} \times 100\% = 40\%$$

Hence, percentage profit (or profit per cent)

$$= \frac{\text{Profit}}{\text{Cost Price}} \times 100\%$$

Here profit and cost price means total profit and total cost price respectively.

$$\Rightarrow \text{Profit} = \frac{CP \times \text{Profit Per cent}}{100}, \text{ in terms of profit percent}$$

$$\text{(iii)} \quad SP = CP + \text{Profit}$$

If we substitute the value of profit in term of profit percent then

$$SP = CP + \frac{CP \times \text{Profit Percent}}{100}$$

$$\Rightarrow SP = \frac{CP (100 + \text{Profit Percent})}{100}$$

$$\Rightarrow CP = \frac{SP \times 100}{(100 + \text{Profit Percent})}$$

(2) When Selling Price is less than Cost Price (i.e. $SP < CP$), then loss has been incurred.

$$\begin{aligned} \text{(i)} \quad & \text{Loss} = CP - SP \\ & \quad SP = CP - \text{Loss} \\ & \quad CP = SP + \text{Loss} \end{aligned}$$

$$\text{(ii)} \quad \text{Percentage Loss (or Loss Percent)} = \frac{\text{Loss}}{CP} \times 100, \text{ here loss}$$

and CP means total loss and total CP respectively.

Percentage loss means loss when cost price is ₹ 100.

Percentage loss is always calculated on CP unless otherwise stated.

Loss in terms of loss percent,

$$\text{Loss} = \frac{CP \times \text{Loss Percent}}{100}$$

$$\text{(iii)} \quad \text{Since, } SP = CP - \text{Loss}$$

In terms of loss percent,

$$SP = CP - \frac{CP \times \text{Loss Percent}}{100}$$

$$\Rightarrow SP = \frac{CP (100 - \text{Loss Percent})}{100}$$

$$\Rightarrow CP = \frac{SP \times 100}{100 - \text{Loss Percent}}$$

USE OF PCG (PERCENTAGE CHANGE GRAPHIC) IN PROFIT AND LOSS

$$\text{(i)} \quad CP \xrightarrow{\text{Percentage Profit } \uparrow} SP$$

If $CP = ₹ 250$, Percentage profit = 10%,

$$250 \xrightarrow[+25]{10\% \uparrow} 275$$

(CP) (Profit) (SP)

$$\text{(ii)} \quad CP \xrightarrow{\text{Percentage Loss } \downarrow} SP$$

If $CP = ₹ 400$, percentage loss = 15%, then

$$400 \xrightarrow[-60]{15\% \downarrow} 340$$

(CP) (Loss) (SP)

$$\text{(iii)} \quad \text{To get the reverse relationship } A \rightarrow B \rightarrow A$$

Let profit percent = 25%, $SP = ₹ 1200$, $CP = ?$

If $CP = ₹ 100$, then

$$100 \xrightarrow[+25]{25\% \uparrow} 125 \xrightarrow[-25]{20\% \downarrow} 100$$

(CP) (Profit) (SP) (Reduction) (CP)

Hence required reduction in SP to get the $CP = 20\%$

$$\therefore 1200 \xrightarrow[-240]{20\% \downarrow} 960$$

(SP) (Reduction) (CP)

$$\therefore CP = ₹ 960$$

Illustration 1: By selling a table for ₹ 330, a trader gains 10%. Find the cost price of the table.

- (a) 300 (b) 363
(c) 297 (d) 270

Solution: (a) S.P. = ₹ 330, Gain = 10%

$$\therefore \text{C.P.} = \left(\frac{100}{100 + \text{Gain \%}} \right) \times \text{S.P.} = ₹ \frac{100}{100 + 10} \times 330$$

$$= \frac{100}{110} \times 330 = ₹ 300.$$

Illustration 2: If the cost price is 96% of the selling price, then what is the profit percent?

- (a) 4.5% (b) 4.2%
(c) 4% (d) 3.8%

Solution: (b) Let S.P. = ₹ 100. Then, C.P. = ₹ 96; Profit = ₹ 4.

$$\therefore \text{Profit \%} = \left(\frac{4}{96} \times 100 \right) \% = \frac{25}{6} \% = 4.17\% \approx 4.2\%.$$

Illustration 3: Arun got ₹ 0.70 as gain over ₹ 70. Find his gain percent.

- (a) 1% (b) 0.01%
(c) 0.1% (d) 7%

Solution: (a) Gain % = $\frac{0.70}{70} \times 100 = 1\%$.

Illustration 4: Vishal buys an old bike for ₹ 4700 and spends ₹ 800 on its repairs, then he sells it for ₹ 5800. Find his gain percent.

- (a) 5.2% (b) $4\frac{4}{7}\%$
(c) 5% (d) $5\frac{5}{11}\%$

Solution: (d) Total C.P. for Vishal = 4700 + 800 = ₹ 5500

$$\text{S.P.} = ₹ 5800$$

$$\therefore \text{Gain} = 5800 - 5500 = ₹ 300$$

$$\text{Gain \%} = \frac{300}{5500} \times 100 = 5\frac{5}{11}\%.$$

Illustration 5: P buys some toffees at 6 for a rupee and sells them at 4 for a rupee. Find his gain percent.

Solution: LCM of 6 and 4 is 12

$$\text{CP of 12 toffees} = \frac{1}{6} \times 12 = ₹ 2$$

$$\text{SP of 12 toffees} = \frac{1}{4} \times 12 = ₹ 3$$

$$\text{Gain} = 3 - 2 = ₹ 1$$

$$\therefore \text{Gain \%} = \frac{1}{2} \times 100 = 50\%.$$

MARKED PRICE, LIST PRICE, DISCOUNT AND SUCCESSIVE DISCOUNTS

(i) Marked Price (MP)

In big shops and departmental stores, every article is tagged with a card and its price is written on the card. This is called the marked price of the article. Mark price of an article is the retail price, which is decided by the retail shopkeeper. So the marked price of the same article can be different on different shops.

(ii) List Price

When a manufacturer decides the retail prices of its different products, then these retail prices are either printed on the products or a list of retail prices of different products is sent to all its retail shopkeepers. Since the list price is decided by the manufacturer and not by its retail shopkeeper, therefore it is the same at all retail shops.

(iii) Discount

In order to increase the sale or clear the old stock, sometimes the shopkeepers offer a certain percentage of rebate on the marked price or list price. This rebate is known as discount. Discount is always given on marked price or list price. Hence

$$\text{Selling price} = (\text{Marked price or List price}) - (\text{Discount})$$

Illustration 6: Marked price of a fan is ₹ 1200 and the shopkeeper allows a discount of 5% on it. Find the selling price of the fan.

$$\text{Solution: } 1200 \xrightarrow[-60]{5\% \downarrow} 1140$$

$$\text{SP} = \text{MP} - \text{Discount}$$

$$= 1200 - 60 = ₹ 1140.$$

Illustration 7: A trader marks his goods at 40% above the cost price and allows a discount of 25%. What is his gain per cent?

Solution:

$$100 \xrightarrow[-40]{40\% \uparrow} 140 \xrightarrow[-35]{25\% \downarrow} 105$$

(MP) (Mark up) (Discount) (SP)

$$\text{Gain\%} = (105 - 100)\% = 5\%.$$

Illustration 8: A trader purchased a washing machine for ₹ 10,000. He allows a discount of 12% on its marked price and still gains 10%. Find the marked price of the machine.

Solution:

$$\text{SP} = \text{CP} + \text{Gain}$$

$$= 10,000 \xrightarrow[-1000]{10\% \uparrow} 11000 \quad \dots (1)$$

(CP) (Gain) (SP)

$$\text{Let mark price} = ₹ x$$

$$\text{Then, } \text{SP} = \text{MP} - \text{Discount} = x - \frac{12x}{100} = \frac{88x}{100} \quad \dots (2)$$

From (1) and (2),

$$\frac{88x}{100} = 11000 \Rightarrow x = \frac{11000 \times 100}{88} = 12500$$

$$\text{Hence, MP} = ₹ 12500.$$

Illustration 9: How much per cent above the cost price should a shopkeeper mark his goods so that after allowing a discount of 25% on the marked price, he gain 20%?

Solution:

Let the shopkeeper mark $x\%$ above the cost price.

$$\therefore 100 \xrightarrow[-x]{x\% \uparrow} (100 + x) \xrightarrow[-\frac{1}{4}(100+x)]{25\% \downarrow} \frac{3}{4}(100 + x)$$

(CP) (Mark up) (MP) (Discount) (SP)

$$\begin{array}{ccccc} \text{Also} & 100 & \xrightarrow[+20]{20\% \uparrow} & 120 & \\ & (\text{MP}) & (\text{Gain}) & (\text{SP}) & \end{array}$$

$$\therefore \frac{3}{4} (100 + x) = 120 \Rightarrow 100 + x = 160 \Rightarrow x = 60$$

Hence, MP is 60% above the CP.

Illustration 10: After allowing a discount of $\frac{15}{2}\%$ on marked price, an article is sold for ₹ 555. Find its M.P.

Solution: Let M.P. = ₹ 100

$$\text{Discount} = \frac{15}{2}\%$$

$$\text{S.P.} = 100 - 7.50 = 92.50$$

If SP is ₹ 92.50, then M.P. = ₹ 100

$$\text{If SP is ₹ 555, then M.P.} = \frac{100}{92.50} \times 555 = ₹ 600.$$

Illustration 11: A garment dealer allows his customers 10% discount on marked price of the goods and still makes a profit of 25%. Find the cost price of a shirt if it is marked at ₹ 1250.

Solution:

Marked Price (MP) = ₹ 1250,

Discount = 10%, Profit = 25%

Let Cost Price (C.P.) = ₹ 'x'

Selling Price (S.P.) = M.P. - Discount

$$\text{SP} = 1250 - 10\% \text{ of } 1250$$

$$\text{SP} = 1250 - 125 = ₹ 1125$$

$$\text{Now, } \% \text{ Profit} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100$$

$$25 = \left(\frac{1125 - x}{x} \right) \times 100$$

$$\Rightarrow x = 1125 \times 4 - 4x$$

$$\Rightarrow 5x = 1125 \times 4$$

$$\Rightarrow x = \frac{1125 \times 4}{5} = 225 \times 4 = 900$$

$$\therefore x = ₹ 900$$

$$\text{C.P.} = ₹ 900.$$

Illustration 12: What price should Neha mark on a sari which cost her ₹ 3000, so as to gain 20% after allowing a discount of 10%?

Solution:

Let Marked Price (MP) of sari = ₹ x,

Discount = 10%

C.P. = ₹ 3000, % gain = 20%

SP = MP - Discount

$$\text{SP} = x - 10\% \text{ of } x = x - \frac{10}{100}x = \frac{9x}{10}$$

$$\% \text{ gain} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100$$

$$20 = \frac{\frac{9x}{10} - 3000}{3000} \times 100 \Rightarrow 20 \times 30 = \frac{9x}{10} - 3000$$

$$\Rightarrow \frac{9x}{10} = 3600$$

$$\Rightarrow x = \frac{3600 \times 10}{9} = 4000$$

$$\therefore \text{MP of Sari} = ₹ 4000.$$

Successive Discounts

If two or more discounts are allowed one after the other then such discounts are known as successive discounts or discounts in series.

Suppose a discount of 15% is given, then on the reduced price a discount of 10% is also given. In such a case, we say that the successive discounts of 15% and 10% are given.

Illustration 13: Find the single discount equivalent to two successive discounts of 20% and 10%.

Solution:

$$\begin{array}{ccccc} 100 & \xrightarrow[-20]{20\% \downarrow} & 80 & \xrightarrow[-8]{10\% \downarrow} & 72 \\ (\text{CP}) & (\text{First} & & (\text{Second} & (\text{SP}) \\ & \text{discount}) & & \text{discount}) & \end{array}$$

$\therefore$ Single discount equivalent to two given successive discounts

$$= (100 - 72)\% = 28\%.$$

Illustration 14: M.P. of a bed is ₹ 7500. The shopkeeper allows successive discounts of 8%, 5% and 2% on it. What is the net selling price?

Solution:

M.P. of bed = ₹ 7500

$$7500 \xrightarrow[-600]{8\% \downarrow} 6900 \xrightarrow[-345]{5\% \downarrow} 6555 \xrightarrow[-131.10]{2\% \downarrow} 6423.90$$

$$\therefore \text{Net selling price} = ₹ 6423.90$$

Illustration 15: Find a single discount equivalent to three successive discounts of 20%, 10% and 5%?

Solution:

$$\begin{array}{ccccc} 100 & \xrightarrow[-20]{20\% \downarrow} & 80 & \xrightarrow[-8]{10\% \downarrow} & 72 \xrightarrow[-3.60]{5\% \downarrow} 68.40 \\ (\text{CP}) & (\text{First} & & (\text{Second} & (\text{Third} & (\text{SP}) \\ & \text{discount}) & & \text{discount}) & \text{discount}) & \end{array}$$

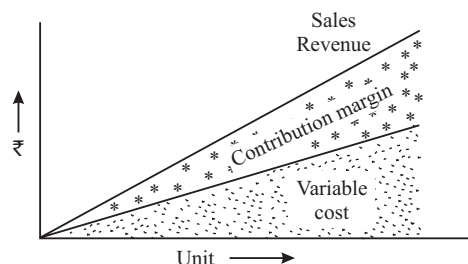
$\therefore$ Single discount equivalent to three given

successive discounts = $(100 - 68.40)$

$$= 31.60 = 31.6\%.$$

CONTRIBUTION MARGIN (CM)

Contribution margin is the amount by which sales revenue exceeds variable cost.



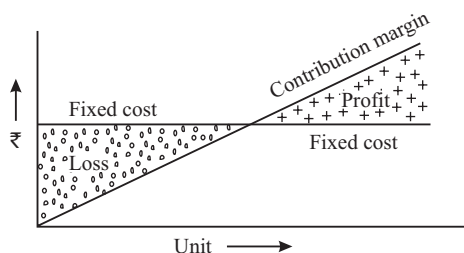
$$\text{Contribution margin} = (\text{Sales revenue}) - (\text{Variable cost})$$

Contribution margin per unit of sales

$$= (\text{Sales revenue per unit}) - (\text{Variable expenses per unit})$$

Actually, contribution margin is the portion of the sales revenue that is not consumed by variable cost and so contributes to the coverage of fixed cost. If fixed cost is less than the contribution margin, then there is a net profit in the business. If fixed cost is more than the contribution margin, then there is a net loss in the business.

If fixed cost is equal to the contribution margin, then there is no net profit or loss in the business.



If contribution margin is more than fixed cost, then

$$(\text{Contribution margin}) - (\text{Fixed cost}) = \text{Net profit}$$

If contribution margin is less than fixed cost, then

$$(\text{Fixed cost}) - (\text{Contribution margin}) = \text{Net Loss}$$

For a Company ABC in the Year 2012-2013

Given, Number of units sold = 250

Sales price per unit = ₹ 30,000

Variable cost per unit = ₹ 20,000

Fixed cost (or fixed expenses) = ₹ 15,00,000

Then, Total sales revenue = ₹ 30,000 × 250

$$= ₹ 75,00,000$$

Total variable cost of 250 units = ₹ 20,000 × 250

$$= ₹ 50,00,000$$

$$\text{Total contribution margin} = (\text{Total sales revenue}) - (\text{Total variable cost})$$

$$= ₹ 75,00,000 - ₹ 50,00,000$$

$$= ₹ 25,00,000$$

Since total contribution margin is more than the fixed cost, therefore

$$\text{Net Profit} = (\text{Total contribution margin}) - (\text{Fixed cost})$$

$$= ₹ 25,00,000 - ₹ 15,00,000$$

$$= ₹ 10,00,000$$

$$\text{Contribution margin per unit} = (\text{Sales price per unit}) -$$

$$(\text{Variable cost per unit})$$

$$= ₹ 30,000 - ₹ 20,000$$

$$= ₹ 10,000$$

$$\text{Profit per unit} = \frac{\text{Net profit}}{\text{Number of units sold}} = ₹ \frac{10,00,000}{250}$$

$$= ₹ 4000.$$

For a Company XYZ in the year 2012-2013

Given, Number of units sold = 200

Sales price per unit = ₹ 10,000

Variable cost per unit = ₹ 5,000

Fixed cost (or fixed expenses) = ₹ 12,00,000

Then, Total sales revenue = ₹ 10,000 × 200

$$= ₹ 20,00,000$$

Total variable cost = ₹ 5,000 × 200

$$= ₹ 10,00,000$$

$$\text{Total contribution margin} = (\text{Total sales revenue}) - (\text{Total variable cost})$$

$$= ₹ 20,00,000 - ₹ 10,00,000$$

$$= ₹ 10,00,000$$

Since total contribution margin is less than fixed cost, therefore

$$\text{Net loss} = (\text{Fixed cost}) - (\text{Total contribution margin})$$

$$= ₹ 12,00,000 - ₹ 10,00,000$$

$$= ₹ 2,00,000$$

BREAK-EVEN POINT AND BREAK-EVEN SALES

The break-even point is defined as the number of units of sale at which there is no profit or no loss.

At break-even point, total sales is called break-even sales.

If number of units sold is more than break-even point, then company starts earning a profit and if number of units sold is less than break-even point, then company makes losses.

At break even point,

$$\text{Total sales} = \text{Variable cost} + \text{Fixed cost} = \text{Total cost}$$

In case of profit,

$$\text{Profit} = \text{Actual sales} - \text{Break-even sales}$$

In case of loss,

$$\text{Loss} = \text{Break-even sales} - \text{Actual sales}$$

The break-even point is calculated on the basis of any time period. But this time period is normally annually or monthly.

All the formulas discussed before are used when number of goods sold equal to the number of goods purchased are the same. But when cost price of n goods is recovered by selling $(n - x)$ goods, then

$$\text{CP} = \text{SP of } (n - x) \text{ goods}$$

$$\text{Profit} = \text{SP of } x \text{ goods}$$

$$\therefore \text{Profit \%} = \frac{\text{SP of } x \text{ goods}}{\text{SP of } (n - x) \text{ goods}} \times 100$$

$$= \frac{x}{n - x} \times 100$$

$$\therefore \text{Gain \%} = \frac{3}{16} \times 100 = 18\frac{3}{4} \%$$

Practice Exercises



Foundation Level

- By selling 12 marbles for a rupee, a shopkeeper loses 20%. In order to gain 20% in the transaction, he should sell the marbles at the rate of how many marbles for a rupee?
 - 8
 - 4
 - 6
 - 3
- Three successive discounts of 10%, 12% and 15% amount to a single discount of:
 - 36.28%
 - 37%
 - 34.68%
 - None of these
- A reduction of 20% in the price of sugar enables a purchaser to obtain $2\frac{1}{2}$ kg more for ₹ 160. Find the original price per kg of sugar.
 - ₹ 12
 - ₹ 16
 - ₹ 20
 - ₹ 18
- Two motor cars were sold for ₹ 9,900 each, gaining 10% on one and losing 10% on the other. The gain or loss per cent in the whole transaction is:
 - Neither loss no gain
 - $\frac{1}{99}$ % gain
 - $\frac{100}{99}$ % profit
 - 1% loss
- A cycle agent buys 30 bicycles, of which 8 are first grade and the rest are second grade for ₹ 3150. Find at what price he must sell the first grade bicycles so that if he sells the second grade bicycles at third quarter of the price, he may make a profit of 40% on both the types of transactions?
 - ₹ 200
 - ₹ 180
 - ₹ 240
 - ₹ 210
- A dairyman pays ₹ 6.4 per litre of milk. He adds water and sells the mixture at ₹ 8 per litre, thereby making 37.5% profit. The proportion of water to milk received by the customers is:
 - 1 : 5
 - 1 : 20
 - 1 : 10
 - 1 : 12
- The cost price of 20 articles is the same as the selling price of x articles. If the profit is 25%, then the value of x is
 - 25
 - 16
 - 18
 - 15
- A departmental store receives a shipment of 1,000 shirts, for which it pays ₹ 9,000. The store sells the shirts at a price 80 per cent above the cost for one month, after which it reduces the price of the shirts to 20 per cent above the cost. The store sells 750 shirts for one month and 50 per cent of the remaining shirts afterwards. How much gross income did the sales of the shirts generate?
 - ₹ 10,000
 - ₹ 12,150
 - ₹ 10,800
 - ₹ 13,500
- A company blends two varieties of tea from two different tea gardens, one variety costing ₹ 20 per kg and other ₹ 25 per kg, in the ratio 5 : 4. He sells the blended tea at ₹ 23 per kg. Find his profit per cent:
 - 5% profit
 - 3.5% profit
 - 3.5% loss
 - No profit, no loss
- An article is listed at ₹ 65. A customer bought this article for ₹ 56.16 and got two successive discounts of which the first one is 10%. The other rate of discount of this scheme that was allowed by the shopkeeper was:
 - 3%
 - 6%
 - 4%
 - 2%
- Three partners altogether invested ₹ 1,14,000 in a business. At the end of the year, one got ₹ 337.50, the second ₹ 1,125.00 and the third, ₹ 675 as profit. What is the percentage of profit?
 - 5.8%
 - 1.8%
 - 4.8%
 - 3.8%
- A shopkeepers sells an article at $12\frac{1}{2}$ % loss. If he sells it for ₹ 92.50 more, then he gains 6%. What is the cost price of the article?
 - ₹ 510
 - ₹ 575
 - ₹ 500
 - ₹ 600
- Ramesh purchased a bicycle for ₹ 5,200 and spent ₹ 800 on its repairs. He had to sell it for ₹ 5,500. Find his profit or loss per cent.
 - ₹ 844.37
 - ₹ 588.47
 - ₹ 488.47
 - None of these
- Dhiraj purchased 150 kg of rice. He sold $\frac{1}{3}$ rd of it at 10% loss. At what per cent of profit must he sell the remaining rice so that he can make 10% profit on the whole?
 - 20%
 - 10%
 - 15%
 - None of these

15. A grocer purchased 20 kg of rice at the rate of ₹ 15 per kg and 30 kg of rice at the rate of ₹ 13 per kg. At what price per kg should he sell the mixture to earn $33\frac{1}{3}\%$ profit on the cost price?
(a) ₹ 28.00 (b) ₹ 20.00
(c) ₹ 18.40 (d) ₹ 17.40
16. A builder purchased a plot of land for ₹ 80 lakh and constructed a five-storey building inclusive of ground floor on it. How much should he charge for each flat to make 25% profit on his investment on land, if there are five flats on each storey?
(a) ₹ 50000 (b) ₹ 100000
(c) ₹ 500000 (d) None of these
17. The difference between a discount of 35% and two successive discounts of 20% and 20% on a certain bill was ₹ 22. Find the amount of the bill.
(a) ₹ 1,100 (b) ₹ 200
(c) ₹ 2,200 (d) None of these
18. A grocer purchased 80 kg of sugar at ₹ 13.50 per kg and mixed it with 120 kg sugar at ₹ 16 per kg. At what rate should he sell the mixture to gain 16%?
(a) ₹ 17 per kg (b) ₹ 17.40 per kg
(c) ₹ 16.5 per kg (d) ₹ 16 per kg
19. A sells a tube to B at a profit of 20% and B sells it to C at profit of 25%. If C pays ₹ 225 for it, what did A pay for it?
(a) ₹ 100 (b) ₹ 125
(c) ₹ 150 (d) ₹ 175
20. Prabhu purchased 30 kg of rice at the rate of ₹ 17.50 per kg and another 30 kg rice at a certain rate. He mixed the two and sold the entire quantity at the rate of ₹ 18.60 per kg and made 20 per cent overall profit. At what price per kg did he purchase the lot of another 30 kg rice?
(a) ₹ 14.50 (b) ₹ 12.50
(c) ₹ 15.50 (d) ₹ 13.50
21. A trader marks his goods at such a price that he can deduct 15% for cash and yet make 20% profit. Find the marked price of an item which costs him ₹ 90
(a) ₹ $135\frac{11}{13}$ (b) ₹ $105\frac{3}{21}$
(c) ₹ $127\frac{1}{17}$ (d) ₹ $95\frac{1}{21}$
22. A trader wants 10% profit on the selling price of a product whereas his expenses amount to 15% on sales. What should be his rate of mark up on an article costing ₹ 9?
(a) 20% (b) $66\frac{2}{3}\%$
(c) 30% (d) $\frac{100}{3}\%$
23. An article is listed at ₹ 65. A customer bought this article for ₹ 56.16 and got two successive discounts of which the first one is 10%. The other rate of discount of this scheme that was allowed by the shopkeeper was
(a) 3% (b) 4%
(c) 6% (d) 2%
24. The sale price of an article including the sales tax is ₹ 616. The rate of sales tax is 10%. If the shopkeeper has made a profit of 12%, then the cost price of the article is :
(a) ₹ 500 (b) ₹ 515
(c) ₹ 550 (d) ₹ 600
25. A man sold two watches for ₹ 1000 each. On one he gains 25% and on the other 20% loss. Find how much % does he gain or lose in the whole transaction?
(a) $\frac{100}{41}\%$ loss (b) $\frac{100}{41}\%$ gain
(c) No gain, no loss (d) Cannot be determined
26. The cost price of 20 articles is equal to the selling price of 25 articles. The loss percent in the transaction is
(a) 5 (b) 20
(c) 25 (d) 30
27. Rajni purchased a mobile phone and a refrigerator for ₹ 12000 and ₹ 10000 respectively. She sold the first at a loss of 12% and the second at a profit of 8%. What is her overall loss/profit?
(a) loss of ₹ 280 (b) profit of ₹ 2160
(c) loss of ₹ 240 (d) None of these
28. A property dealer sells a house for ₹ 6,30,000 and in the bargain makes a profit of 5%. Had he sold it for ₹ 5,00,000, then what percentage of loss or gain he would have made?
(a) $2\frac{1}{4}\%$ gain (b) 10% loss
(c) $12\frac{1}{2}\%$ loss (d) $16\frac{2}{3}\%$ loss
29. A manufacturer sells a car to a dealer at a profit of 50%, the dealer sells it to a customer at a profit of 20% and the customer sells it to a friend for ₹ 288000 at a loss of 20%. Find the cost of manufacturer.
(a) 200000 (b) 300000
(c) 400000 (d) 50000
30. A dishonest dealer professes to sell his goods at cost price, but he uses a weight of 960 gm for the kg weight. Find his gain percent.
(a) 2.8% (b) $4\frac{1}{6}\%$
(c) 4.16% (d) $3\frac{1}{3}\%$
31. A shopkeeper sold an article offering a discount of 5% and earned a profit of 23.5%. What would have been the percentage of profit earned if no discount was offered?
(a) 24.5 (b) 28.5
(c) 30 (d) None of these

32. A man sells an article at 5% profit. If he had bought it at 5% less and sold it for ₹ 1 less, he would have gained 10%. Find the cost price.
(a) 100 (b) 150
(c) 200 (d) 250
33. It is known that the shopkeeper takes a discount of 10% from his supplier and he disregards this discount while marking up (i.e., he marks up at the undiscounted price), find the percentage profit for the shopkeeper if there is no other change from the previous problem.
(a) 32% (b) 36.66%
(c) 40.33% (d) 46.66%
34. A shopkeeper marks up his goods by 40% and gives a discount of 10%. Apart from this, he uses a faulty balance also, which reads 1000 gm for 800 gm. What is his net profit percentage?
(a) 57.5% (b) 63.5%
(c) 42.5% (d) 36.5%
35. A supplier sells 20 pencils at the marked price of 16 pens to a retailer. The retailer, in turn, sells them at the marked price. What is the percentage profit or percentage loss of the retailer?
(a) Loss 25% (b) Profit 25%
(c) Loss 20% (d) Profit 20%
36. A milkman defrauds by means of a false measure to the tune of 20% in buying and also defrauds to the tune of 25% in selling. Find his overall % gain.
(a) 15% (b) 30%
(c) 50% (d) 45%
37. A businessman, while selling 20 articles, loses the cost price of 5 articles. Had he purchased the 20 articles for 25% less and sold them for $33\frac{1}{3}\%$ more than the original selling price, what is his gain?
(a) 5% (b) 75%
(c) $33\frac{1}{3}\%$ (d) 45%
38. $\frac{2}{3}$ of a consignment was sold at 6% profit and the rest at a loss of 3%. If there was an overall profit of ₹ 540, find the value of the consignment.
(a) ₹ 15,000 (b) ₹ 18,000
(c) ₹ 35,000 (d) ₹ 45,000
39. The ratio between the sale price and the cost price of an article is 7 : 5. What is the ratio between the profit and the cost price of that article?
(a) 2 : 7 (b) 5 : 2
(c) 7 : 9 (d) None of these
40. The percentage profit earned by selling an article for ₹ 1920 is equal to the percentage loss incurred by selling the same article for ₹ 1280. At what price should the article be sold to make 25% profit?
(a) ₹ 2000 (b) ₹ 2200
(c) ₹ 2400 (d) None of these
41. The profit by selling an item was 25%. If the item was marked 40% above the selling price then what is the ratio of the marked price to the cost price of the item?
(a) $\frac{5}{4}$ (b) $\frac{7}{4}$
(c) $\frac{3}{4}$ (d) $\frac{1}{4}$
42. Two dealers X and Y selling the same model of refrigerator mark them under the same selling prices. X gives successive discounts of 25% and 5% and Y gives successive discounts of 16% and 12%. From whom is it more profitable to purchase the refrigerator?
(a) From Y
(b) From X
(c) Indifferent between the two
(d) Cannot be determined
43. A shopkeeper marks up his goods by 20% and then gives a discount of 20%. Besides he cheats both his supplier and customer by 100 grams i.e., he takes 1100 gram from his supplier and sells only 900 grams to his customer. What is his net profit percentage?
(a) 24.5% (b) 17.33%
(c) 25% (d) 32.5%
44. Amit brought two cars. He then sold the first car at 10% profit and the second one at 25% profit. The selling price of the second car is 25% more than the selling price of the first car. What is the approximate profit per cent in both the cars together?
(a) 17.85% (b) 18.36%
(c) 16.19% (d) Cannot be determined
45. A pharmaceutical company made 3000 strips of tablets at a cost of ₹ 4800. The company gave away 1000 strips of tablets of doctors as free samples. A discount of 25% was allowed on the printed price. Find the ratio of profit if the price is raised from ₹ 3.25 to ₹ 4.25 per strip and if at the latter price, samples to doctors were done away with. (New profit/old profit)
(a) 55.5 (b) 63.5
(c) 75 (d) 99.25
46. A trader mixes three varieties of groundnuts costing ₹ 50, ₹ 20 and ₹ 30 per kg in the ratio 2 : 4 : 3 in terms of weight, and sells the mixture of ₹ 33 per kg. What percentage of profit does he make?
(a) 8% (b) 9%
(c) 10% (d) None of these
47. A manufacturer sells a pair of glasses to a wholesale dealer at a profit of 18%. The wholesaler sells the same to a retailer at a profit of 20%. The retailer in turn sells them to a customer for ₹ 30.09, thereby earning a profit of 25%. The cost price for the manufacturer is
(a) ₹ 16 (b) ₹ 20
(c) ₹ 17 (d) ₹ 24

48. The AMS magazine prints 5000 copies for ₹ 5,00,000 every month. In the July issue of the magazine, AMS distributed 500 copies free. Besides, it was able to sell $\frac{2}{3}$ of the remaining magazines were sold at the printed price of the magazine (which was ₹ 200). Find the percentage profit of AMS in the magazine venture in the month of July (assume a uniform 20% of the sale price as the vendor's discount and also assume that AMS earns no income from advertising for the issue).
- (a) 56% (b) 24%
(c) 28% (d) 22.6%
49. Samant bought a microwave oven and paid 10% less than the original price. He sold it with 30% profit on the price he had paid. What percentage of profit did Samant earn on the original price?
- (a) 17% (b) 20%
(c) 27% (d) 32%
50. If 5% more is gained by selling an article for ₹ 350 than by selling it for ₹ 340 the cost of the article is:
- (a) ₹ 50 (b) ₹ 160
(c) ₹ 200 (d) ₹ 225
51. A discount of 15% on one article is the same as a discount of 20% on another article. The costs of the two articles can be:
- (a) ₹ 40, ₹ 20 (b) ₹ 60, ₹ 40
(c) ₹ 80, ₹ 60 (d) ₹ 60, ₹ 40
52. A shopkeeper earns a profit of 12% on selling a book at 10% discount on the printed price. The ratio of the cost price to the printed price of the book is:
- (a) 45 : 56 (b) 50 : 61
(c) 55 : 69 (d) 99 : 125
53. By selling a watch at a profit of 10 per cent, a man got Rs 15 more than half its price. What is the price of the watch?
- (a) 10 (b) 15
(c) 25 (d) 5
54. A bookseller marks his books at an advance of 69% on the actual cost of production. He allows a discount of 15% and also given a copy free for every dozen sold at a time. What rate per cent profit does the bookseller make, if books are sold in lots of 12 ?
- (a) 32.6 (b) 47.5
(c) 24.9 (d) None of these

Standard Level

- A video magazine distributor made 3500 copies of the March issue of the magazine at a cost of ₹ 3,50,000. He gave 500 cassettes free to some key video libraries. He also allowed a 25% discount on the market price of the cassettes and gave one extra cassette free with every 29 cassettes bought at a time. In this manner, he was able to sell all the 3500 cassettes that were produced. If the market price of a cassette was ₹ 150, then what is his gain or loss per cent for the March issue of the video magazine?
(a) 3.4% loss (b) 15% gain
(c) 40% gain (d) 6.8% loss
- A cash payment that will settle a bill for 250 chairs at ₹ 50 per chair less 20% and 15% with a further discount of 5% on cash payment is
(a) ₹ 8075 (b) ₹ 7025
(c) ₹ 8500 (d) None of these
- An oil refinery takes 1000 L of crude oil as input and after refining for 1 h gives certain amount of output oil X L. This can be sold in the market at a profit of ₹ 30 per L. If this oil is further refined for $\frac{1}{2}$ h, it gives oil Y L. This can be sold at a profit of ₹ 50 per L. Output and input ratio at both the stages is 90%. The maximum amount that can be earned from 1000 L of crude input is
(a) ₹ 40000 (b) ₹ 30000
(c) ₹ 27000 (d) ₹ 40500
- A manufacturer sells a pair of glasses to a wholesale dealer at a profit of 18%. The wholesaler sells the same to a retailer at a profit of 20%. The retailer in turn sells them to a customer for ₹ 30.09, thereby earning a profit of 25%. The cost price for the manufacturer is
(a) ₹ 15 (b) ₹ 16
(c) ₹ 17 (d) ₹ 18
- A dealer offers a cash discount of 20% and still makes a profit of 20%, when he further allows 16 articles to a dozen to a particularly sticky bargainer. How much per cent above the cost price were his wares listed?
(a) 100% (b) 80%
(c) 75% (d) 66 $\frac{2}{3}$ %
- Instead of a metre scale cloth merchant uses a 120 cm scale while buying but uses an 80 cm scale while selling the same cloth. If he offers a discount of 20 per cent of cash payment, what is his overall per cent profit?
(a) 20% (b) 25%
(c) 40% (d) 15%
- A book is sold at profit of ₹ 20, which is 10% of its cost price. If its C.P. is increased by 50% and it is still sold at a profit of 10%, then find the new profit.
(a) ₹ 30 (b) ₹ 50
(c) ₹ 60 (d) ₹ 300
- A fruitseller sells mangoes at the rate of ₹ 9 per kg and thereby loses 20%. At what price per kg, he should have sold them to make a profit of 5%?
(a) ₹ 11.81 (b) ₹ 12
(c) ₹ 12.25 (d) ₹ 12.31
- A man would gain 20% by selling a chair for ₹ 47.5 and would gain 15% by selling a table for ₹ 57.5. He sells the chair for ₹ 36, what is the least price for which he must sell the table to avoid any loss on the two together
(a) ₹ 50.2 (b) ₹ 55.8
(c) ₹ 60 (d) ₹ 53.6
- By selling 5 dozen mangoes for ₹ 156 it was found that $\frac{3}{10}$ th of the outlay was gained. What should the retail price per mango be in order to gain 60%?
(a) ₹ 4 (b) ₹ 2
(c) ₹ 3.2 (d) ₹ 4.2
- An article is sold at 20 % profit. If its CP and SP are less by ₹ 10 and ₹ 5 respectively the percentage of profit increases by 10 %. Find the cost price.
(a) ₹ 40 (b) ₹ 80
(c) ₹ 60 (d) ₹ 50
- A man purchases two clocks A and B at a total cost of ₹ 650. He sells A with 20% profit and B at a loss of 25% and gets the same selling price for both the clocks. What are the purchasing prices of A and B respectively?
(a) ₹ 225; ₹ 425 (b) ₹ 250; ₹ 400
(c) ₹ 275; ₹ 375 (d) ₹ 300; ₹ 350
- A person purchases 100 pens at a discount of 10%. The net amount of money spent by the person to purchase the pens is ₹ 600. The selling expenses incurred by the person are 15% on the net cost price. What should be the selling price for 100 pens in order to earn a profit of 25%?
(a) ₹ 802.50 (b) ₹ 811.25
(c) ₹ 862.50 (d) ₹ 875
- A milkman buys milk contained in 10 vessels of equal size. If he sells his milk at ₹ 5 a litre, he loses ₹ 200; if he sells it at ₹ 6 a litre, he would gain ₹ 150 on the whole. Find the number of litres contained in each vessel.
(a) 20 litres (b) 30 litres
(c) 25 litres (d) 35 litres
- A shopkeeper purchased a table marked at ₹ 200 at successive discount of 10% and 15% respectively. He spent ₹ 7 on transportation and sold the table for ₹ 200. Find his gain %?
(a) No loss or gain (b) 25%
(c) 30% (d) 40%

16. A man buys 2 dozen bananas at ₹ 16 per dozen. After selling 18 bananas at the rate of ₹ 12 per dozen, the shopkeeper reduced the rate of ₹ 4 per dozen. The percent loss is:
 (a) 25.2% (b) 32.4%
 (c) 36.5% (d) 37.5%
17. A space research company wants to sell its two products A and B. If the product A is sold at 20% loss and the product B at 30% gain, the company will not lose anything. If the product A is sold at 15% loss and the product B at 15% gain, the company will lose ₹ 6 million in the deal. What is the cost of product B ?
 (a) ₹ 140 million (b) ₹ 120 million
 (c) ₹ 100 million (d) ₹ 80 million
18. Two-third of a consignment was sold at a profit of 5% and the remainder at a loss of 2%. If the total profit was ₹ 400, the value of the consignment (in rupees)
 (a) 20,000 (b) 15,000
 (c) 12,000 (d) 10,000
19. A dealer sold a radio at a loss of 2.5%. Had he sold it for ₹ 100 more, he would have gained $7\frac{1}{2}\%$. In order to gain $12\frac{1}{2}\%$, he should sell it for:
 (a) ₹ 850 (b) ₹ 925
 (c) ₹ 1,080 (d) ₹ 1,125
20. The raw material and manufacturing cost formed individually 70% and 30% of the total cost and the profit percentage is 14.28% of the raw material. If the cost of raw material increase by 20% and the cost of manufacturing is increased by 40% and the selling price is increased by 80%, then the new profit percentage is :
 (a) 57% (b) 65.8%
 (c) 60% (d) can't determined
21. A person purchased a cupboard and a cot for ₹ 18,000. He sold the cupboard at a profit of 20% and the cot at a profit of 30%. If his total profit was 25.833%, find the cost price of the cupboard.
 (a) ₹ 10,500 (b) ₹ 12,000
 (c) ₹ 7500 (d) ₹ 10,000
22. A sells a car priced at ₹ 36,000. He gives a discount of 8% on the first ₹ 20,000 and 5% on the remaining ₹ 16,000. His competitor B sells a car of the same make, priced at ₹ 36,000. If he wants to be competitive what percent discount should B offer on the marked price.
 (a) 5% (b) 5.5%
 (c) 6.66% (d) 8.33%
23. In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit?
 (a) 30% (b) 70%
 (c) 100% (d) 250%
24. A shopkeeper bought 150 calculators at the rate of ₹ 250 per calculator. He spent ₹ 2500 on transportation and packing. If the marked price of calculator is ₹ 320 per calculator and the shopkeeper gives a discount of 5% on the marked price then what will be the percentage profit gained by the shopkeeper?
 (a) 20% (b) 14%
 (c) 15% (d) 16%
25. A firm of readymade garments makes both men's and women's shirts. Its average profit is 6% of the sales. Its profit in men's shirts average 8% of the sales and women's shirts comprise 60% of the output. The average profit per sale rupee in women shirts is
 (a) 0.0466 (b) 0.0666
 (c) 0.0166 (d) 0.0366
26. A tradesman marks his goods at 25% above cost price and allows discount of 12.5 per cent for cash payment. What profit per cent does he make ?
 (a) $9\frac{3}{8}$ (b) $9\frac{1}{8}$
 (c) $9\frac{5}{8}$ (d) $9\frac{7}{8}$
27. A bookseller sells a book at a profit of 10%. If he had bought it at 4% less and sold it for ₹ 6 more, he would have gained $18\frac{3}{4}\%$ per cent. What did it cost him?
 (a) 120 (b) 130
 (c) 140 (d) 150
28. A watch passes through three hands and each gains 25%. If the third sells it for ₹ 250, what did the first pay for it?
 (a) 128 (b) 130
 (c) 145 (d) 150
29. I loss 9 per cent by selling pencils at the rate of 15 a rupee. How many for a rupee must I sell them to gain 5 per cent?
 (a) 10 (b) 13
 (c) 15 (d) 18
30. A tradesman marks an article at ₹ 205 more than the cost price. He allows a discount of 10% on the marked price. Find the profit per cent if the cost price is ₹ x.
 (a) $\frac{\left[\frac{x}{(18450)} - 10\right]}{x}$ (b) $\frac{[(18450)] - 10x}{x}$
 (c) $\frac{\left[\frac{x}{(18450)} - 100\right]}{x}$ (d) $\frac{\left[\frac{18450}{x} - 100\right]}{x}$

31. A manufacturer makes a profit of 15% by selling a colour TV for ₹ 5750. If the cost of manufacturing increases by 30% and the price paid by the retailer is increased by 20%, find the profit percent made by the manufacturer.
(a) $6\frac{2}{13}\%$ (b) $4\frac{8}{13}\%$
(c) $6\frac{1}{13}\%$ (d) $7\frac{4}{13}\%$
32. The profit earned when an article is sold for ₹ 800 is 20 times the loss incurred when it is sold for ₹ 275. At what price should the article be sold if it is desired to make a profit of 25%
(a) ₹ 300 (b) ₹ 350
(c) ₹ 375 (d) ₹ 400
33. Each of A and B sold their article at ₹ 1818 but A incurred a loss of 10% while B gained by 1%. What is the ratio of cost price of the articles of A to that of B?
(a) 101 : 90 (b) 85 : 89
(c) 81 : 75 (d) None of these
34. A manufacturer of a certain item can sell all he can produce at the selling price of ₹ 60 each. It costs him ₹ 40 in materials and labour to produce each item and he has overhead expenses of ₹ 3,000 per week in order to operate that plant. The number of units he should produce and sell in order to make a profit of at least ₹ 1,000 per week is
(a) 300 (b) 250
(c) 400 (d) 200
35. Dolly goes to a shop to purchase a doll priced at ₹ 400. She is offered 4 discount options by the shopkeeper. Which of these options should she opt for to gain maximum advantage of the discount offered?
(a) Single discount of 30%
(b) 2 successive discounts of 15% each
(c) 2 successive discounts of 20% and 10%
(d) 2 successive discounts of 20% and 12%
36. A trader sells goods to a customer at a profit of k% over the cost price, besides it he cheats his customer by giving 880 g only instead of 1 kg. Thus his overall profit percentage is 25%. Find the value of k?
(a) 8.33% (b) 8.25%
(c) 10% (d) 12.5%
37. A, B and C invest in the ratio of 3 : 4 : 5. The percentage of return on their investments are in the ratio of 6 : 5 : 4. Find the total earnings, if B earns ₹ 250 more than A :
(a) ₹ 6000 (b) ₹ 7250
(c) ₹ 5000 (d) None of these
38. A car mechanic purchased four old cars for ₹ 1 lakh. He spent total 2 lakh in the maintenance and repairing of these four cars. What is the average sale price of the rest three cars to get 50% total profit if he has already sold one of the four cars at ₹ 1.2 lakh?
(a) 1.5 lakh (b) 1.1 lakh
(c) 1.2 lakh (d) 1.65 lakh
39. The cost of setting up a magazine is ₹ 2800. The cost of paper and ink etc. is ₹ 80 per 100 copies and printing cost is ₹ 160 per 100 copies. In the last month 2000 copies were printed but only 1500 copies could be sold at ₹ 5 each. Total 25% profit on the sale price was realized. There is one more resource of income from the magazine which is advertising. What sum of money was obtained from the advertising in magazine?
(a) ₹ 1750 (b) ₹ 2350
(c) ₹ 1150 (d) ₹ 1975
40. A person purchases 90 clocks and sells 40 clocks at a gain of 10% and 50 clocks at gain of 20%. If he sold all of them at a uniform profit of 15%, then he would have got ₹ 40 less. The cost price of each clock is:
(a) ₹ 50 (b) ₹ 60
(c) ₹ 80 (d) ₹ 90
41. A tradesman fixed his selling price of goods at 30% above the cost price. He sells half the stock at this price, one quarter of his stock at a discount of 15% on the original selling price and rest at a discount of 30% on the original selling price. Find the gain percent altogether.
(a) 14.875% (b) 15.375%
(c) 15.575% (d) 16.375%
42. Cheap and Best, a kirana shop bought some apples at 4 per rupee and an equal number at 5 per rupee. He then sold the entire quantity at 9 for 2 rupees. What is his percentage profit or loss?
(a) 1.23% loss (b) 6.66%
(c) 8.888% (d) No profit no loss
43. Amar sold his moped to Bharat at 20% profit and Bharat sold it to Sridhar at 10% profit. Sridhar sold the same to a mechanic and received ₹ 2,316. If Amar had sold the same moped to the mechanic and receive the same amount the mechanic paid to Sridhar, what profit percentage would Amar have made?
(a) 52% (b) 48%
(c) 33.3% (d) Cannot be determined

Expert Level

- Rohit bought 20 soaps and 12 toothpastes. He marked-up the soaps by 15% on the cost price of each, and the toothpastes by ₹ 20 on the cost price each. He sold 75% of the soaps and 8 toothpastes and made a profit of ₹ 385. If the cost of a toothpaste is 60% the cost of a soap and he got no return on unsold items, what was his overall profit or loss?
 - Loss of ₹ 355
 - Loss of ₹ 210
 - Loss of ₹ 250
 - None of these
- A dealer buys dry fruit at the rate of ₹ 100, ₹ 80 and ₹ 60 per kg. He bought them in the ratio 12 : 15 : 20 by weight. He in total gets 20% profit by selling the first two and at last he finds he has no gain no loss in selling the whole quantity which he had. What was the percentage loss he suffered for the third quantity?
 - 40%
 - 20%
 - 30%
 - 50%
- The ratio of selling price of 3 articles A, B and C is 8 : 9 : 5 and the ratio of percentage profit is 8 : 7 : 14 respectively. If the profit percentage of A is 14.28% and the cost price of B is ₹ 400, what is the overall percentage gain?
 - 14.28%
 - 17.87%
 - 16.66%
 - None of these
- In an office the number of employees reduces in the ratio of 3 : 2 and the wages increases in the ratio of 20 : 27. What is the profit percentage of employees over the previous wages?
 - 10%
 - 9.09%
 - 11.11%
 - None of these
- The cost of servicing of a Maruti car at Maruti car Pvt. Ltd. is ₹ 400. Manager of service centre told me that for the second service within a year a customer can avail a 10% discount and further for third and fourth servicing he can avail 10% discount of the previous amount paid, within a year. Further if a customer gets more than 4 services within a year he has to pay just 60% of the servicing charges on these services. A customer availed 5 services from the same servicing station, what is the total percentage discount fetched by the customer?
 - 19.42%
 - 18.5%
 - 17.6%
 - 26%
- An article costing ₹ 20 was marked 25% above the cost price. After two successive discounts of the same percentage, the customer now pays ₹ 20.25. What would be the percentage change in profit had the price been increased by the same percentage twice successively instead of reducing it?
 - 3600%
 - 3200%
 - 2800%
 - 4000%
- By mixing two brands of tea and selling the mixture at the rate of ₹ 177 per kg, a shopkeeper makes a profit of 18%. If to every 2 kg of one brand costing ₹ 200 per kg, 3 kg of the other brand is added, then how much per kg does the other brand cost?
 - ₹ 110
 - ₹ 120
 - ₹ 140
 - None of these
- Jonny has two cycles and one rickshaw. The rickshaw is worth ₹ 96. If he sells the rickshaw along with the first cycle, he has an amount double that of the value of the second cycle. But if he decides to sell the rickshaw along with the second cycle, the amount received would be less than the value of first cycle by ₹ 306. What is the value of first cycle?
 - ₹ 900
 - ₹ 600
 - ₹ 498
 - None of these
- A dishonest dealer professes to sell his goods at cost price. But he uses a false weight and thus gains $6\frac{18}{47}\%$. For a kg, he uses a weight of:
 - 940 gms
 - 947 gms
 - 953 gms
 - 960 gms
- DSNL charges a fixed rental of ₹ 350 per month. It allows 200 calls free per month. Each call is charged at ₹ 1.4 when the number of calls exceeds 200 per month and it charges ₹ 1.6 when the number of calls exceeds 400 per month and so on. A customer made 150 calls in February and 250 calls in March. By how much per cent the each call is cheaper in March than each call in February?
 - 28%
 - 25%
 - 18.5%
 - None of these
- Tika Chand has a weighing balance in which there is a technical fault. The right pan of his balance measures always 200 g more than its left pan. Tika Chand as usual misutilise this balance in his business. While purchasing the articles he puts goods in the left pan and weight in the right pan while selling he reverse the order i.e., goods in the right pan and weight in the left pan. He uses only 2 kg weight for the measurement and to measure 2n kg weight he measures n times by 2-2 kg but he sells goods at cost price. What is his profit percentage?
 - 20%
 - $22\frac{2}{9}\%$
 - $18\frac{2}{11}\%$
 - None of these

12. Rotomac produces very fine quality of writing pens. Company knows that on an average 10% of the produced pens are always defective so are rejected before packing. Company promises to deliver 7200 pens to its wholesaler at ₹ 10 each. It estimates the overall profit on all the manufactured pens to be 25%. What is the manufacturing cost of each pen?
- (a) ₹ 6 (b) ₹ 7.2
(c) ₹ 5.6 (d) ₹ 8
13. A dishonest dealer purchases goods at 20% discount of the cost price of ₹ x and also cheats his wholesaler by getting his goods by 80% of x , but he gives a discount of 25% besides he cheats his customer by weighing 10% less than the required. What is his overall profit percentage?
- (a) 125% (b) 100%
(c) 98.66% (d) 120%
14. An egg seller sells his eggs only in the packs of 3 eggs, 6 eggs, 9 eggs, 12 eggs etc., but the rate is not necessarily uniform. One day Raju (which is not the same egg seller) purchased at the rate of 3 eggs for a rupee and the next hour he purchased equal number of eggs at the rate of 6 eggs for a rupee. Next day he sold all the eggs at the rate of 9 eggs for ₹ 2. What is his percentage profit or loss?
- (a) 10% loss (b) 11.11% loss
(c) 3% loss (d) 2.5% profit
15. A milkman purchases 10 litres of milk at ₹ 7 per litre and forms a mixture by adding freely available water which constitutes 16.66% of the mixture. Later on he replaced the mixture by some freely available water and thus the ratio of milk is to water is 2 : 1. He then sold the new mixture at cost price of milk and replaced amount of mixture at twice the cost of milk then what is the profit percentage?
- (a) 68% (b) 34%
(c) 40% (d) None of these
16. Raghav bought 25 washing machines and microwave ovens for ₹ 2,05,000. He sold 80% of the washing machines and 12 microwave ovens for a profit of ₹ 40,000. Each washing machine was marked up by 20% over cost and each microwave oven was sold at a profit of ₹ 2,000. The remaining washing machines and 3 microwave ovens could not be sold. What is Raghav's overall profit/loss?
- (a) ₹ 1000 profit (b) ₹ 2500 loss
(c) ₹ 1000 loss (d) Cannot be determined
17. A man buys apples at a certain price per dozen and sells them at eight times that price per hundred. His gain or loss per cent is _____.
- (a) 4% (b) - 4%
(c) 5% (d) - 5%
18. A person bought two clocks. The cost price of one of them exceeds the cost price of the other by $\frac{1}{4}$ th. He sold the dearer one at a gain of 10% and the other at a gain of 7.5% and thus got ₹ 98 in all as S.P. Find the cost price of the cheaper one.
- (a) ₹ 40 (b) ₹ 50
(c) ₹ 30 (d) ₹ 60
19. A trader buys a certain amount of goods worth ₹ 22520. He decides to make a profit of 5.36% on the sale of goods worth ₹ 5000 and increase the profit percent by 3.14% for sales upto ₹ 15000 and then increase the profit percent for the sale of remaining lot such that he is able to make a profit of 25% on the sale of the full lot. Find the profit that he makes on the third lot of goods.
- (a) ₹ 5620 (b) ₹ 4512
(c) ₹ 3212 (d) None of these
20. A person sells his table at a profit of $12\frac{1}{2}\%$ and the other hand if he sells the table at a loss of $8\frac{1}{3}\%$ but on the whole he gains ₹ 25. On the other hand if he sells the table at a loss of $8\frac{1}{3}\%$ and the chair at a profit of $12\frac{1}{2}\%$ then he neither gains nor loses. Find the cost price of the table.
- (a) ₹ 120 (b) ₹ 360
(c) ₹ 240 (d) ₹ 230
21. A man sells an article at 5% profit. If he had bought it at 5% less and sold it for ₹ 1 less, he would have gained 10%. The cost price of the article is :
- (a) ₹ 200 (b) ₹ 150
(c) ₹ 240 (d) ₹ 280
22. There are fifty successive percentage discounts given in a series of 2%, 4%, 6%, 8%...and so on. What is the net discount?
- (a) 98% (b) 2550%
(c) 100% (d) Infinite
23. A dishonest dealer marks up the price of his goods by 20% and gives a discount of 10% to the customer. He also uses a 900 gram weight instead of a 1 kilogram weight. Find his percentage profit due to these maneuvers.
- (a) 8% (b) 12%
(c) 20% (d) 16%
24. Three varieties of rice with Cost Price (in ₹/kg) 28, 36 and 45 are mixed in the ratio $a : b : c$ respectively. It is known that a , b and c are in Geometric Progression where $a < b < c$. The Cost Price (in ₹/kg) of the mixture becomes 40. What would have been the Cost Price (in ₹/kg) of the mixture had the three varieties been mixed in the ratio $c : a : b$?
- (a) 24 (b) 28
(c) 34 (d) 38
25. The cost price of four articles A, B, C and D are 'a', 'b', 'c' and 'd' respectively. A, B, C and D are sold at profits of 10%, 20%, 30% and 40% respectively. If the net profit on the sale of these four articles is 25%, 'a', 'b', 'c' and 'd' cannot be in the ratio
- (a) 4 : 1 : 4 : 3 (b) 1 : 2 : 2 : 1
(c) 2 : 3 : 6 : 1 (d) 5 : 2 : 7 : 3

26. Kadbury offers a packet of 5 chocolates at the list price of 4 chocolates and on purchasing 19 such packets gives one packet absolutely free. A trader receives 20 packets of the chocolates in the offer and sells each chocolate at its list price. What is his net percentage profit?
- (a) 24% (b) 31.58%
(c) 35% (d) 53.75%
27. A watch dealer sells watches at ₹ 600 per watch. However, he is forced to give two successive discounts of 10% and 5% respectively. However, he recovers the sales tax on the net sale price from the customer at 5% of the net price. What price does a customer have to pay him to buy the watch.
- (a) ₹ 539.75 (b) ₹ 539.65
(c) ₹ 538.75 (d) ₹ 538.65
28. A merchant makes a profit of 20% by selling an article. What would be the percentage change in the profit percent had he paid 10% less for it and the customer paid 10% more for it?
- (a) 120% (b) 125%
(c) 133.33% (d) 150%
29. Two thousand people lived in a Village of which 55% were male and the rest were female. The male population earned a profit of 5% and the female population earned 8% on an investment of ₹ 50 each. Find the change in the percentage profit of the village if the ratio of male to female gets reversed the next year, population remaining the same.
- (a) Drop of 0.3 (b) Increase of 0.3
(c) Increase of 0.45 (d) Drop of 0.45
30. Rupesh marks up an article by $p\%$, gives a discount of $\frac{p}{4}\%$ and gets a profit of $\frac{p}{4}\%$. Had he marked up by $\frac{p}{2}\%$ and given a discount $\frac{p}{6}\%$, what would be his profit percentage?
- (a) 25% (b) $33\frac{1}{3}\%$
(c) 50% (d) $66\frac{2}{3}\%$

Test Yourself

1. A dealer offers a cash discount of 20% and still makes a profit of 20%, when he further allows 16 articles to a dozen to a particularly sticky bargainer. How much per cent above the cost price were his wares listed?
(a) 100% (b) 80%
(c) 75% (d) 662/3%
2. Sunny marks up his goods by 40% and gives a discount of 10%. Apart from this, he uses a faulty balance which reads 800 gms for 1000 gms. What is his net profit/loss percentage?
(a) 8% (b) 57.2%
(c) 37.6% (d) None of these
3. A certain manufacturer sells a product to the distributor at 10% profit. Then the distributor sells it to the dealer and the dealer sells to the retailer at a mark up of 10% and 20% respectively. The retailer marks up his cost by 20% and then offers a 10% discount to the customer. If the customer had bought it from the distributor directly at the distributor's selling price, then how much reduction in price would he have got with respect to buying it from the retailer?
(a) 29.6% (b) 18%
(c) 32% (d) 22.8%
4. An article is listed at ₹ 65. A customer bought this article for ₹ 56.16 and got two successive discounts of which the first one is 10%. The other rate of discount of this scheme that was allowed by the shopkeeper was
(a) 3% (b) 4%
(c) 6% (d) 2%
5. A publisher printed 3,000 copies of 'Future Shock' at a cost of ₹ 2,400. He gave 500 copies free to different philanthropic institutions. He allowed a discount of 25% on the published price and gave one copy free for every 25 copies bought at a time. He was able to sell all the copies in this manner. If the published price is ₹ 3.25, then what is his overall gain or loss percentage in the whole transaction?
(a) 113% (b) 130%
(c) 162% (d) 144%
6. Mohan goes to furniture shop to buy a sofa set and a centre table. He bargains for a 10% discount on the centre table and 25% discount on sofa set. However, the shopkeeper, by mistake, interchanged the discount percentage figures while making the bill and Mohit paid accordingly. When compared to what he should pay for his purchases, what percentage did Mohit pay extra given that the centre table costs 40% as much as the sofa set.
(a) 12.3% (b) 7.2%
(c) 8.1% (d) 6.3%
7. A shopkeeper purchases a packet of 50 pens at ₹ 10 per pen. He sells a part of the packet at a profit of 30%. On the remaining part, he incurs a loss of 10%. If his overall profit on the whole packet is 10%, the number of pens he sold at profit is
(a) 25 (b) 30
(c) 20 (d) 15
8. A shopkeeper purchased 150 identical pieces of calculators at the rate of ₹ 250 each. He spent an amount of ₹ 2500 on transport and packing. He fixed the labelled price of each calculator at ₹ 320. However, he decided to give a discount of 5% on the labelled price. What is the percentage profit earned by him?
(a) 14% (b) 15%
(c) 16% (d) 20%
9. Mithilesh makes 750 articles at a cost of 60 paise per article. He fixes the selling price such that if only 600 articles are sold, he would have made a profit of 40% on the outlay. However, 120 articles got spoilt and he was able to sell 630 articles at this price. Find his actual profit percent as the percentage of total outlay assuming that the unsold articles are useless.
(a) 42% (b) 53%
(c) 47% (d) 46%
10. A trader wants 10% profit on the selling price of a product whereas his expenses amount to 15% on sales. What should be his rate of mark up on an article costing ₹ 9?
(a) 20% (b) 66 2/3 %
(c) 30% (d) 100/3 %
11. The cost of setting up the type of a magazine is ₹ 1000. The cost of running the printing machine is ₹ 120 per 100 copies. The cost of paper, ink and so on is 60 paise per copy. The magazines are sold at ₹ 2.75 each. 900 copies are printed, but only 784 copies are sold. What is the sum to be obtained from advertisements to give a profit of 10% on the cost?
(a) ₹ 730 (b) ₹ 720
(c) ₹ 726 (d) ₹ 736
12. A manufacturer sells goods to an agent at a profit of 20%. The agent's wholesale price to a shopkeeper is at a profit of 10% and the shopkeeper retails his goods at a profit of 12%. Find the retailer's price of an article which had cost the manufacturer ₹ 25.
(a) ₹ 37 (b) ₹ 40
(c) ₹ 44 (d) ₹ 46
13. A shopkeeper calculates percentage profit on the buying price and another on the selling price. What will be their difference in profits if both claim a profit of 20% on goods sold for ₹ 3000?
(a) ₹ 200 (b) ₹ 100
(c) ₹ 400 (d) ₹ 150
14. The accounts of a company show sales of ₹ 12,600. The primary cost is 35% of sales and trading cost accounts for 25% of the gross profit. Gross profit is arrived at by excluding the primary cost plus the cost of advertising expenses of ₹ 1400, director's salary of ₹ 650 per annum plus 2% of annual sales as miscellaneous costs. Find the percentage profit (approx) on a capital investment of ₹ 14,000?
(a) 35% (b) 31%
(c) 28% (d) Cannot be determined
15. A trader marked his goods at 20% above the cost price. He sold half the stock at the marked price, one quarter at a discount of 20% on the marked price and the rest at a discount of 40% on the marked price. His total gain is
(a) 2% (b) 4.5%
(c) 13.5% (d) 15%

Hints & Solutions

Foundation Level

1. (a) Given SP of 12 marbles = ₹ 1, loss = 20%

$$\therefore \text{CP of 12 marbles} = \frac{1}{0.8} = ₹ 1.25$$

$$\begin{aligned} \text{Now, SP of 12 marbles at a gain of 20\%} \\ = \text{CP} \times 1.2 = 1.25 \times 1.2 = ₹ 1.5 \end{aligned}$$

This implies that in order to gain 20%, he should sell 12 marbles for ₹ 1.5.

$$\Rightarrow \text{For ₹ 1, he should sell } \frac{12}{1.5} = 8 \text{ marbles.}$$

2. (d) Applying successive discounts of 10%, 12% and 15% on 100, we get $100 \times 0.9 \times 0.88 \times 0.85 = 67.32$

$$\Rightarrow \text{Single discount} = 100 - 67.32 = 32.68$$

Hence, none of the given options is correct.

3. (c) Total amount used for purchasing = ₹ 160. A reduction of 20% in the price means, now a person gets $\frac{5}{2}$ kg for ₹ 32 and this is the present price of the sugar.

$$\therefore \text{Present price per kg} = \frac{32}{5} \times 2 = ₹ 12.8$$

Let the original price be ₹ x . Then new price is arrived after reduction of 20% on it.

$$\Rightarrow x \times 0.8 = 12.8 \text{ or } x = ₹ 16.$$

4. (d) If any two transactions of SP is the same and also gain % and loss % are same then there is always a loss

$$\therefore \text{loss \%} = \left(\frac{\text{Common gain or loss \%}}{10} \right)^2 \left(\frac{10}{10} \right)^2 = 1\%$$

5. (c) Let he sells first grade cycle at a rate of ₹ z per bicycle.

$$\text{Then, } 8z + \frac{22 \times 3}{4} z = 3150 \times 1.4$$

$$\Rightarrow 32z + 66z = 17640$$

$$\Rightarrow 98z = 17640 \Rightarrow z = 180$$

He should sell the first grade bicycles at a rate of ₹ 180.

6. (b) Let the quantity of milk purchased be x litres and quantity of water added to it be y litres. Then ratio of water to milk will be $y : x$.

$$\text{Now, CP} = 6.4x \text{ and SP} = 8(x + y)$$

$$\text{and profit \%} = 37.5\%$$

$$\Rightarrow 8(x + y) = 6.4x \times 1.375$$

$$\Rightarrow 8x + 8y = 8.8x$$

$$\Rightarrow 8y = 0.8x \text{ or } \frac{x}{y} = \frac{80}{8} = \frac{10}{1}$$

$$\Rightarrow y : x = 1 : 10$$

7. (c) CP of 20 articles = SP of x articles = 1 (say)

$$\text{Therefore, CP of 1 article} = \frac{1}{20},$$

$$\text{And SP of 1 article} = \frac{1}{x}$$

$$\text{Now, gain\%} = \frac{\frac{1}{x} - \frac{1}{20}}{\frac{1}{20}} \times \frac{25}{100} \Rightarrow \left(\frac{20 - x}{20x} \right) \times 20 = \frac{25}{100}$$

$$\Rightarrow 80 - 4x = x \Rightarrow 5x = 80 \Rightarrow x = 16$$

8. (d) $750 \times (180\% \text{ of } ₹ 9) + 125 \times (120\% \text{ of } ₹ 9)$

$$= 750 \times 16.20 + 125 \times 10.80 = 12150 + 1350 = ₹ 13500$$

9. (c) Let the quantity of two varieties of tea be $5x$ kg and $4x$ kg, respectively.

$$\text{Now, SP} = 23 \times 9x = 207x$$

$$\text{and CP} = 20 \times 5x + 25 \times 4x = 200x$$

$$\text{Profit \%} = \frac{7x}{200x} \times 100 = 3.5\%$$

10. (b) Price of the article after first discount, $65 - \left(65 \times \frac{10}{100} \right)$
= ₹ 58.5

Therefore, the second discount

$$= \frac{58.5 - 56.16}{58.5} \times 100 = 4\%$$

11. (c) Total profit = $337.50 + 1125.00 + 675 = ₹ 2137.50$

$$\text{Percentage profit} = \frac{2137.50}{114000} \times 100 = 1.8\%$$

12. (b) S.P. = $100 - 12.5 = ₹ 87.5$

$$\text{S.P. after 6\% gain} = ₹ 106$$

$$\text{Difference} = ₹ 18.5$$

$$\therefore \text{C.P.} = \frac{92.5}{18.5} \times 100 = ₹ 500$$

13. (d) Cost price of bicycle = ₹ 5,200.

He spent ₹ 800 on its repairs

$$\therefore \text{C.P.} = 5200 + 800 = 6000.$$

$$\text{Selling price} = ₹ 5,500$$

$$\therefore \text{Loss} = 6000 - 5500 = 500$$

$$\text{Hence, loss \%} = \frac{500}{6000} \times 100 = 8.33\%$$

14. (a) Let the C.P. of 150 kg of rice be ₹ 150.

$$\therefore \text{S.P. of 50 kg of rice at 10\%}$$

$$\text{loss } \frac{90}{100} \times 50 = ₹ 45$$

For 10% of gain on the whole.

$$\text{S.P.} = 150 \times \frac{110}{100} = ₹ 165$$

$$\therefore 100 \text{ kg rice should be sold for ₹ 120.}$$

$$\therefore \text{Per cent gain} = 20$$

15. (c) $CP = 20 \times 15 + 30 \times 13 = ₹ 690$

$$\therefore SP = \frac{4}{3} \text{ of } 690 \times \frac{1}{50} = ₹ 18.40$$

16. (d) We do not know the total investment of builder, because in the question construction cost is not given. Hence, 'none of these' is the answer.

17. (c) Successive discount = $20\% + \frac{20 \times 80}{100}$

$$= 20 + 16 = 36\%$$

$$\text{Difference in discount} = 36 - 35 = 1\%$$

$$\therefore \text{Bill amount} = 22 \times 100 = ₹ 2200$$

18. (b) C.P. of 200 kg of mixture = $(80 \times 13.50 + 120 \times 16)$
= ₹ 3000.

$$S.P. = 116\% \text{ of } ₹ 3000 = ₹ \left(\frac{116}{100} \times 3000 \right) = ₹ 3480.$$

$$\therefore \text{Rate of S.P. of the mixture} = ₹ \left(\frac{3480}{200} \right) \text{ per kg}$$

$$= ₹ 17.40 \text{ per kg.}$$

19. (c) Let A paid = ₹ x
125 % of 120 % of x = 225

$$\Rightarrow \frac{125}{100} \times \frac{120}{100} \times x = 225$$

$$\Rightarrow x = \frac{225 \times 100 \times 100}{125 \times 120} = ₹ 150$$

20. (d) Let he purchase of ₹ x/kg.

$$\therefore (525 + 30x) \times \frac{120}{100} = 60 \times 18.60$$

$$\Rightarrow x = ₹ 13.5 / \text{kg.}$$

21. (c) $SP = 90 \times 1.2 = ₹ 108$

$$\text{Marked price} = \frac{108}{0.85} = ₹ 127.05$$

22. (d) Let the SP of the article be ₹ x

$$\text{Expenses} = 15\% \text{ of } x = ₹ 0.15x$$

$$\text{Profit} = 10\% \text{ of } x = ₹ 0.10x$$

$$CP = ₹ 9 \text{ (given)}$$

$$\text{Therefore, } 9 + 0.15x + 0.10x = x \Rightarrow x = 12$$

$$\therefore \% \text{ increase for marked price} = \frac{12 - 9}{9} \times 100$$

$$= \frac{100}{3} \%$$

23. (b) Price of the article after first discount

$$65 - 6.5 = ₹ 58.5$$

$$\text{Therefore, the second discount}$$

$$= \frac{58.5 - 56.16}{58.5} \times 100 = 4\%$$

24. (a) Let the CP of the article be ₹ x

$$\text{Then, } SP = x \times 1.12 \times 1.1$$

$$\text{Now, } x \times 1.12 \times 1.1 = 616$$

$$\Rightarrow x = \frac{616}{1.232} = ₹ 500$$

25. (b) When $S_1 = S_2$, then

$$\text{overall \% gain or \% loss}$$

$$= \left[100 - \frac{2(100 + x_1)(100 - x_2)}{(100 + x_1) + (100 - x_2)} \right] \%$$

$$\left(100 - \frac{2(125)(80)}{(125) + (80)} \right) \% = \left(100 - \frac{2 \times 125 \times 80}{205} \right) \%$$

$$\frac{100}{41} \% \text{ gain } (\because \text{it is +ve})$$

26. (c) Let C.P. of 1 article = ₹ 1

$$\text{then C.P. of 25 articles} = ₹ 25$$

$$\text{and S.P. of 25 articles} = ₹ 20$$

$$\therefore \text{loss \%} = \frac{25 - 20}{20} \times 100 = 25\%$$

27. (d) Total cost price of mobile phone and refrigerator

$$= ₹ (12000 + 10000) = ₹ 22000$$

$$SP \text{ of mobile phone} = (88\% \text{ of } 12000)$$

$$= ₹ \left(\frac{88}{100} \times 12000 \right) = ₹ 10560$$

$$SP \text{ of refrigerator} = 108\% \text{ of } 10000$$

$$= ₹ \left(\frac{108}{100} \times 10000 \right) = ₹ 10800$$

$$\text{Total SP of both the articles} = ₹ (10560 + 10800)$$

$$= ₹ 21360$$

$$\text{Loss} = ₹ (22000 - 21360) = ₹ 640.$$

28. (d) $C.P. = ₹ \left(\frac{100}{105} \times 630000 \right) = ₹ 600000.$

$$\therefore \text{Required loss \%} = \left(\frac{100000}{600000} \times 100 \right) \% = 16 \frac{2}{3} \%$$

29. (a) Let ₹ X be the C.P. of the manufacturer of the car

$$\Rightarrow X \times \frac{150}{100} \times \frac{120}{100} \times \frac{80}{100} = 288000$$

$$\Rightarrow X = 200000$$

30. (c) Supposing the goods cost the dealer ₹ 1 for the kg., he sells for ₹ 1 goods which cost him ₹ 0.96.

$$\text{Gain on } ₹ 0.96 = 0.04;$$

$$\% \text{ Gain} = \frac{0.04 \times 100}{0.96} = 4 \frac{1}{6} \% = 4.16\%$$

31. (c) Let C.P. be ₹ 100. Then, S.P. = ₹ 123.50.

Let marked price be ₹ x . Then, $\frac{95}{100}x = 123.50 \Rightarrow x$

$$= \left(\frac{12350}{95} \right) = ₹ 130.$$

Now, S.P. = ₹ 130, C.P. = ₹ 100

$\therefore$ Profit % = 30%

32. (c) Let cost price = x

$$\text{Then we have, } x \left(\frac{95}{100} \right) \left(\frac{110}{100} \right) = x \left(\frac{105}{100} \right) - 1$$

$$\text{or, } x = \frac{100 \times 100}{105 \times 100 - 95 \times 100} \times 200$$

$\therefore$ Cost price = ₹ 200

33. (d) The new situation is

Buying:

1100 grams for ₹ 900

Hence, 1320 grams for ₹ 1080

Selling: 900 grams for ₹ 1080

$$\text{Profit \%} = \frac{420}{900} \times 100 = 46.66\%$$

34. (a) Let us assume his CP/1000 gm = ₹ 100

So, his SP/kg (800 gm) = ₹ 126

So, his CP/800 gm = ₹ 80

So, profit = ₹ 46

So profit percentage = $\frac{46}{80} \times 100 = 57.5\%$

35. (b) MP of 1 Pencil = ₹ 1

For supplier, SP of 20 pencils = ₹ 16

For retailer, SP of 20 pencils = ₹ 20

$$\text{Profit percentage} = \frac{4}{16} \times 100 = 25\%$$

36. (c) The milkman defrauds 20% in buying and also defrauds 25% in selling, so his overall % gain will be

$$\left[\frac{(100 + 20\%)(100 + 25\%)}{100} - 100 \right] \% = 50\%$$

37. (c) Let the price of 1 article = ₹ 1

$$\Rightarrow \text{Loss} = 20 \text{ C.P.} - 20 \text{ S.P.}$$

$$\Rightarrow 5 \text{ C.P.} = 20 \text{ C.P.} - 20 \text{ S.P.} \Rightarrow 20 \text{ S.P.} = 15 \text{ C.P.}$$

$$\Rightarrow \text{CP}_1 \text{ of 20 articles} = ₹ 20$$

$$\Rightarrow \text{SP}_1 \text{ of 20 articles} = ₹ 15$$

Also given that, had he purchased the 20 articles for

25% less and sold them for $33\frac{1}{3}\%$ more, then

$$\Rightarrow \text{CP}_2 \text{ of 20 articles} = ₹ 15$$

$$\Rightarrow \text{CP}_2 \text{ of 20 articles} = ₹ 20$$

$$\therefore \text{Gain percentage} = \frac{20 - 15}{15} \times 100 = 33\frac{1}{3}\%$$

$$\begin{aligned} 38. (b) \text{ Value of consignment} &= \frac{540 \times 100}{\frac{2}{3} \times 6 + \frac{1}{3}(-3)} = \frac{540 \times 100}{4 - 1} \\ &= ₹ 18,000 \end{aligned}$$

39. (d) Let C.P. = ₹ $5x$ and S.P. = ₹ $7x$. Then, Gain = ₹ $2x$
 $\therefore$ Required ratio $2x : 5x = 2 : 5$

40. (a) Let C.P. be ₹ x .

$$\text{Then, } \frac{1920 - x}{x} \times 100 = \frac{x - 1280}{x} \times 100$$

$$\Rightarrow 1920 - x = x - 1280 \Rightarrow 2x = 3200 \Rightarrow x = 1600.$$

$\therefore$ Required S.P. = 125% of ₹ 1600

$$= \left(\frac{125}{100} \times 1600 \right) = ₹ 2000$$

41. (b) Let the cost price of an item = ₹ 100,

then, selling price = ₹ 125

($\because$ Profit by selling is 25%)

Now, marked price is 40% above the selling price

$$\therefore \text{M.P.} = 125 + 125 \times \frac{40}{100}$$

$$\Rightarrow \text{Marked price} = 125 \left(1 + \frac{40}{100} \right) = 175$$

$$\text{Hence, } \frac{\text{Marked price}}{\text{Cost price}} = \frac{175}{100} = \frac{7}{4}$$

42. (b) Assume marked price for both to be 100.

X's selling price = $100 \times 0.75 \times 0.95 = 71.25$

Y's selling price = $100 \times 0.84 \times 0.88 = 73.92$.

Buying from 'X' is more profitable.

43. (b) Assume his CP = ₹ 1000/1100 gm

MP = ₹ 1200 and SP = ₹ 960/900 gm

So, SP/1100 gm = ₹ 1173.33

So, profit = ₹ 173.33

Profit percentage = 17.33%

44. (a) $100 \xrightarrow{10\%} \text{First profit} \xrightarrow{25\%} \text{Net profit}$

45. (b) Find out the total revenue realization for both the cases:

Case 1: (Old) Total sales revenue

$$= 2000 \times 3.25 \times 0.75.$$

$$\text{Profit}_{\text{old}} = \text{Total sales revenue} - 4800$$

Case 2: (New) Total sales revenue

$$= 3000 \times 4.25 \times 0.75$$

$$\text{Profit}_{\text{new}} = \text{Total sales revenue} - 4800$$

The ratio of profit will be given by $\text{Profit}_{\text{new}} / \text{Profit}_{\text{old}}$

46. (c) Suppose he bought 2 kg, 4 kg and 3 kg of the three varieties. C.P. of 9 kg = ` (2 × 50 + 4 × 20 + 3 × 30) = ` 270.
S.P. of 9 kg = ` (9 × 33) = ` 297.

$$\therefore \text{Profit \%} = \left(\frac{27}{270} \times 100 \right) \% = 10\%.$$

47. (c) Let the cost price for the manufacturer be `x.
Then, 125% of 120% of 118% of x = 30.09.

$$\Rightarrow \frac{125}{100} \times \frac{120}{100} \times \frac{118}{100} x = \frac{3009}{100} \Rightarrow \frac{177}{100} x = \frac{3009}{100}$$

$$\Rightarrow x \left(\frac{3009}{177} \right) = 17.$$

48. (b) Total cost = 5 lacs
Total revenue = 3000 × 160 + 1500 × 200 – vendors discount of 20% of revenues
= 7.8 lacs – 1.56 lacs = 6.24 lacs
Profit per cent = (1.24 × 100)/5 = 24.8%

49. (a) Let original price = ` 100.

$$\text{Then C.P.} = ` 90, \text{ S.P.} = 130\% \text{ of } ` 90 = \left(\frac{130}{100} \times 90 \right)$$

$$= ` 117.$$

$$\therefore \text{Required percentage} = (117 - 100)\% = 17\%.$$

50. (c) Let C.P. be `x. Then, 5% of x = (350 – 340) = 10

$$\Rightarrow \frac{x}{20} = 10 \Rightarrow x = 200.$$

51. (c) Let the costs of the two articles be x and y. Then,

$$15\% \text{ of } x = 20\% \text{ of } y \Rightarrow \frac{x}{y} = \frac{20}{15} = \frac{4}{3}.$$

$$\text{So, } x \text{ and } y \text{ must be in the ratio of } 4 : 3. \\ ` 80, ` 60.$$

52. (a) Let cost price be ` 100. The, S.P. = ` 112.
Let printed price be `x.

$$90\% \text{ of } x = 112 \Rightarrow x = \left(\frac{112 \times 100}{90} \right) = \frac{1120}{9}$$

$$\therefore \text{Required ratio} = 100 : \frac{1120}{9} \quad 900 : 1120 = 45 : 56$$

53. (c) Price = `X
SP = ` 1.1x = 0.5x + 15, So, 0.6x = 15
So, x = 25

54. (a) CP = `X/dozen = 0.833/copy
So, MP = 1.69x/dozen
SP = ` 1.4365x/13 copies = 0.1105x/copy
So, profit = 32.6%

Standard Level

1. (d) CP of 3500 cassettes = ` 3,50,000
Given, discount 25% on marked price = 150

$$\therefore \text{Discount} = 150 \times \frac{25}{100} = 37.5$$

$$\therefore \text{SP of each set of 30} = (29 + 1) \text{ cassettes} \\ = 29 \times (150 - 37.5)$$

$$= ` 29 \times 112.50 = ` 3262.50$$

$$\therefore \text{SP of 3500 cassettes including 500 free cassettes} \\ = 3262.50 \times 100 = ` 3,26,250$$

$$(\because \text{cost of 1 cassette} = ` 100)$$

$$\therefore \text{Overall loss} = ` 3,50,000 - ` 3,26,250 \\ = ` 23,750$$

$$\therefore \% \text{ loss} = \frac{23750}{350000} \times 100 = 6.8$$

2. (a) Cost of one chair = ` 50

$$\therefore \text{Amount paid for 1 chair} = 50 \times \frac{100 - 20}{100} \times \frac{100 - 15}{100}$$

$$(\because \text{Given, 20\% and 15\% discount on chair})$$

$$= 50 \times \frac{80}{100} \times \frac{85}{100} = 34$$

$$\text{cost of 250 chair at the rate of 34 per chair is} \\ 250 \times 34 = ` 8500$$

$$\text{Further, 5\% discount on } ` 8500 \text{ is}$$

$$8500 \times \frac{5}{100} = ` 425$$

$$\therefore \text{Total payment after discount} \\ = 8500 - 425 = 8075$$

3. (d) First case,
(Refining for one hr)
Input = 1000 L

$$\text{Output} = 1000 \times \frac{90}{100} = 900L \Rightarrow x = 900L$$

$$\text{Profit} = 900 \times 30 = 27000$$

$$\text{Second case}$$

$$(\text{Refining for } \frac{1}{2} \text{ hr}).$$

$$\text{Input} = 9000 L$$

$$\text{Output } y = 900 \times \frac{90}{100} = 810 L$$

$$\text{Profit} = 810 \times 50 = ` 40500$$

4. (c) Let the cost price of manufactures is = P

$$\text{Selling price of manufacturer} = P + P \times \frac{18}{100} = \frac{59P}{50}$$

$$\text{Wholesaler selling price} = \frac{59P}{50} + \frac{59P}{50} \times \frac{20}{100}$$

$$= \frac{59P}{50} + \frac{59P}{250} = \frac{354P}{250}$$

$$\text{Retailer selling price} = \frac{354P}{250} + \frac{354P}{250} \times \frac{25}{100}$$

$$= \frac{354P}{250} + \frac{177P}{500} = \frac{805P}{500}$$

$$\text{Now, } \frac{805P}{500} = 30.09$$

$$\Rightarrow P = 17$$

Short

$$P \left(\frac{100}{118} \times \frac{100}{120} \times \frac{100}{125} \times 30.09 \right) = 17$$

5. (a) If listed price of article be ₹ 100 then discounted price be ₹ 80 (since discount = 20%)

After offering 16 articles to a dozen

$$\text{Price of 16 articles} = 80 \times 12$$

$$\text{Price of one article} = \frac{80 \times 12}{16} = 60$$

Profit = 20 %

$$\text{Cost price } x \text{ (say)} = 60 - x \times \frac{20}{100}$$

$$\Rightarrow x + \frac{x}{5} = 60 \Rightarrow x = 60 \times \frac{5}{6} = 50$$

$$\text{Per cent above the cost price} = \frac{100 - 50}{50} \times 100 = 100\%$$

6. (a) Let the cost of cloth per cm be ₹ x

As he uses 120 cm scale, so, he has 120 cm cloth cost incurred = $100x$. While selling he uses 80 cm scale, so

$$\text{actually he charges for } \frac{100}{80} \times 120 = 150 \text{ cm of cloth}$$

Amount obtained after 20% discount

$$= 0.8x \times 150 = 120x$$

$$\therefore \text{Profit} = \frac{20x}{100x} \times 100 = 20\%$$

7. (a) 10% of cost price = ₹ 20

$$\therefore \text{Original cost of book} = ₹ 200$$

$$\text{Now, revised cost} = 200 \times 1.5 = ₹ 300$$

$$\therefore \text{Profit} = 10\% \text{ of C.P.} = ₹ 30$$

8. (a) $80 : 9 = 105 : x$ or $x = \left(\frac{9 \times 105}{80} \right) = 11.81$

Hence, S.P. per k.g. = ₹ 11.81.

9. (d) C.P. of the chair = ₹ $47.5 \times \frac{100}{120} = ₹ 39.6$

$$\text{C.P. of the table} = ₹ 57.5 \times \frac{100}{115} = ₹ 50$$

$\Rightarrow$ The required S.P. for the table

$$= ₹ 39.6 + ₹ 50 - ₹ 36 = ₹ 53.6$$

10. (c) Let CP of 5 dozen mangoes be ₹ x . SP = ₹ 156 and Gain = 0.3 x

$$\therefore 156 - x = 0.3x \quad \therefore x = 120$$

$$\therefore \text{SP of 60 mangoes} = 120 \times 1.6 = ₹ 192$$

$$\therefore \text{SP per mango} = ₹ 3.2.$$

11. (b) $\frac{10(20+10) - 100(5-10)}{10} = \frac{800}{10} = ₹ 80$

12. (b) Let the cost price of clocks A and B be 'a' and (650-a) respectively.

Selling price for A = Selling price for B

$$a \left(1 + \frac{20}{100} \right) = (650 - a) \left(1 - \frac{25}{100} \right)$$

$$\left(\frac{120a}{100} \right) = (650 - a) \left(\frac{75}{100} \right)$$

$$a = 250$$

$$\text{Cost price for B} = 650 - 250 = 400$$

13. (c) Cost price = Money spent by the person to purchase + selling expenses

$$600 + \left(600 \times \frac{15}{100} \right) = 690$$

$$\text{Hence, selling price} = 690 \left(1 + \frac{25}{100} \right) = 862.50$$

14. (d) Difference in rupees by increasing the price by rupees 1 is ₹ 350. That means that the quantity of milk is 350 litre. Now, 10 equal containers will become 35 litre per container.

15. (b) C.P. $200 \left(1 - \frac{10}{100} \right) \left(1 - \frac{15}{100} \right) + 7$

$$\frac{200 \times 90 \times 85}{100 \times 100} + 7 = ₹ 153 + 7 = ₹ 160$$

$$\therefore \% \text{ gain} = \frac{200 - 160}{160} \times 100 = \frac{40}{160} \times 100 = 25\%$$

16. (d) C.P. = ₹ $(16 \times 2) = 32$. S.P. = ₹ $(12 \times 1.5 + 4 \times 0.5) = ₹ (18 + 2) = ₹ 20$.

$$\therefore \text{Loss\%} = \left(\frac{12}{32} \times 100 \right) \% = 37.5\%$$

17. (d) Since, selling price of both the products is same
 $\therefore$ % loss = % gain
 $\Rightarrow 20\% \text{ of } A = 30\% \text{ of } B \Rightarrow A/B = 3/2$
 Let cost of product $A = 3x$ and cost of product $B = 2x$.
 According to the question,

$$3x \times \frac{15}{100} - 2x \times \frac{15}{100} = 6$$

$$\Rightarrow 45x - 30x = 600 \Rightarrow x = \frac{600}{15} = 40$$

Hence, cost of product $B = 2 \times 40 = 80$ million

18. (b) Let the value of consignment be x .

when $\frac{2}{3}$ rd of consignment was sold at a profit of 5%,

$$\text{then Profit} = \frac{\frac{2}{3}x \times 5}{100}$$

When the remaining which is $\frac{x}{3}$ consignment was sold at a loss of 2%, then according to Ques, we have

$$\frac{\frac{2x}{3} \times 5}{100} - \frac{\frac{x}{3} \times 2}{100} = 400 \quad (\because \text{Total profit} = 400)$$

$$\Rightarrow 10x - 2x = 120000 \Rightarrow x = 15,000$$

19. (d) Let the cost price of radio be ₹ 1000.
 Dealer sold it at a loss of 2.5%
 i.e. selling price = 997.5
 When he sold it ₹ 100 more, then selling price = 1007.5

$$\left(\because \text{gain} = 7\frac{1}{2}\% \right)$$

Now, In order to gain $12\frac{1}{2}\%$,

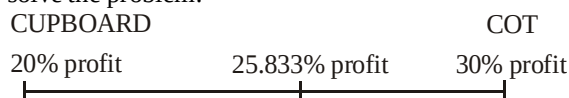
$$\text{Selling price will be } 1000 \times \frac{25}{200} + 1000 = 1125.$$

20. (a) $RM + MC = \text{Total cost}$
 Total cost + Profit = Sale price

$$\begin{array}{l} 70 + 30 = 100 \quad 100 + 10 = 110 \\ 84 + 42 = 126 \quad 126 + 72 = 198 \end{array} \quad \left. \begin{array}{l} \uparrow \\ \downarrow \end{array} \right\} + 80\%$$

$$\text{Therefore profit \%} = \frac{72}{126} \times 100 = 57.14\%$$

21. (c) The following alligation visualization would help us solve the problem:



$$\begin{aligned} \text{Ratio of cost of cupboard to cost of COT} \\ &= 4.1666 : 5.8333 \\ &= 25 : 35 \\ &= 5 : 7 \end{aligned}$$

$$\text{Cost of cupboard} = 5 \times 18000/12 = 7500.$$

22. (c) The total discount offered by A = 8% on 20000 + 5% on 16000 = 1600 + 800 = 2400.

If B wants to be as competitive, he should also offer a discount of ₹ 2400 on 36000. Discount percentage = $2400 \times 100/36000 = 6.66\%$ discount.

23. (b) Let the original cost price be ₹ 100.
 Then, profit = ₹ 320 and SP = ₹ 420
 New CP = ₹ 125
 New profit = SP - New CP = ₹ 295

$$\Rightarrow \text{Required profit} = \frac{295}{420} \times 100 = 70.23\% \approx 70\%$$

24. (b) CP of 150 calculators = $150 \times 250 = ₹ 37,500$.
 $\therefore$ total CP = 37,500 + 2500 = ₹ 40,000
 Marked price of 150 calculators = $150 \times 320 = ₹ 48,000$

$$\begin{aligned} \text{Selling price after discount} &= 48000 \times \frac{95}{100} \\ &= ₹ 45,600 \end{aligned}$$

$$\therefore \text{Percentage profit} = \frac{45,600 - 40,000}{40,000} \times 100 = 14\%$$

25. (a) Women's shirt comprise 60% of the output.
 $\therefore$ Men's shirts comprise $(100 - 60) = 40\%$ of the output.
 $\therefore$ Average profit from men's shirt = 8% of 40 = 3.2 out of 40

Overall average profit = 6 out of 100

$\therefore$ Average profit from women's shirts = 2.8 out of 60 i.e. 0.0466 out of each shirt.

26. (a) If the CP is 100, marked price = 125.
 But discount to the cash purchase

$$= 12.5\% \text{ of } ₹ 125 = ₹ 15\frac{5}{8}$$

Now the reduced price he gets from the cash purchaser

$$= 125 - 15\frac{5}{8} = 109\frac{3}{8}$$

$$\text{i.e., Gain percent} = 109\frac{3}{8} - 100 = 9\frac{3}{8}$$

27. (d) Let the actual cost price be ₹ 100
 Actual selling price at 10% profit = ₹ 110
 Supposed cost price at 4% less = ₹ 96

$$\text{Supposed selling price at } 18\frac{3}{4}\% \text{ profit}$$

$$= ₹ 96 \times \frac{118\frac{3}{4}}{100} = ₹ 114.$$

$\therefore$ The difference in the selling price

$$= ₹ 114 - ₹ 110 = ₹ 4$$

Now, use the unitary method.

If the difference is ₹ 4, the cost price = ₹ 100

If the difference is ₹ 6, the cost price

$$= ₹ \frac{100}{4} \times 6 = ₹ 150.$$

28. (a) By Rule fraction:

$$\text{First purchased for } 250 \left(\frac{100}{125} \right) \left(\frac{100}{125} \right) \left(\frac{100}{125} \right)$$

$$= 250 \left(\frac{4}{5} \right) \left(\frac{4}{5} \right) \left(\frac{4}{5} \right) = ₹ 128$$

29. (b) By the rule of fraction:

$$\text{He purchased } 15 \left(\frac{100-9}{100} \right) \text{ for a rupee.}$$

$$\text{Now to gain 5\%, he must sell } 15 \left(\frac{91}{100} \right) \left(\frac{100}{105} \right) = 13$$

for a rupee

30. (b) Cost price = x

$$\text{Marked price} = x + 205$$

$$\text{Selling price} = 0.9x + 184.5$$

$$\text{Percentage profit} = [(-0.1x + 184.5)x \times 100]$$

$$= \frac{18450 - 10x}{x}$$

31. (a) Original Cost Price = ₹ 5000

$$\text{New Cost Price} = 1.3 \times 5000 = ₹ 6500$$

$$\text{Price paid by retailer} = 1.2 \times 5750 = ₹ 6900$$

$$\text{Profit percentage} = (400/6500) \times 100 = 6(2/13)\%$$

32. (c) The interpretation of the first statement is that if the loss at 275 is 1L, the profit at 800 is 20L.

$$\text{Thus, } 21L = 800 - 275 = 525 \rightarrow L = 25.$$

$$\text{Thus, the cost price of the item is ₹ 300.}$$

$$\text{To get a profit of 25\%, the selling price should be } 1.25 \times 300 = 375.$$

$$33. (a) \text{ CP of A} = \frac{1818}{0.9} = 2020$$

$$\text{CP of B} = \frac{1818}{1.01} = 1800$$

$$\frac{\text{CP of A}}{\text{CP of B}} = \frac{2020}{1800} = \frac{101}{90} = 101 : 90$$

34. (d) Let x be no. of units.

$$\therefore \text{ Profit per unit } x = (60 - 40)x = 20x.$$

$$\text{Now, additional cost} = 3000$$

$$\therefore \text{ To make a profit of at least ₹ 1000}$$

we have

$$20x - 3000 = 1000 \Rightarrow 20x = 4000$$

$$\Rightarrow x = 200.$$

35. (a) She should opt for a straight discount of 30% as that gives her the maximum benefit.

$$36. (c) \text{ Profit \%} = \frac{25}{100} \frac{120 + k(\text{Profit})}{880 (\text{Sale})} \Rightarrow k = 100$$

$$\text{Therefore, net profit \%} = \frac{100}{1000} \times 100 = 10\%$$

37. (b)

	A	B	C
Investment	3x	4x	5x
Rate of return	6y%	5y%	4y%
Return	$\frac{18xy}{100}$	$\frac{20xy}{100}$	$\frac{20xy}{100}$

$$\text{Total} = (18 + 20 + 20) = \frac{58xy}{100}$$

$$\text{B's earnings} - \text{A's earnings} = \frac{2xy}{100} = 250$$

$$\text{Total earning} = \frac{58xy}{100} = 7250$$

38. (b) Total cost of 4 cars = 1 + 2 = 3 lakh

$$\text{Total SP of 4 cars} = 3 \times 1.5 = 4.5 \text{ lakh}$$

$$\text{SP of car} = 1.2 \text{ lakh}$$

$$\text{SP of rest 3 cars} = 4.5 - 1.2 = 3.3 \text{ lakh}$$

$$\text{Average SP of all the 3 cars} = 1.1 \text{ lakh}$$

39. (d) Setup cost = ₹ 2800

$$\text{Paper etc.} = ₹ 1600$$

$$\text{Printing cost} = ₹ 3200$$

$$\text{Total cost} = ₹ 7600$$

$$\text{Total sale price} = 1500 \times 5 = 7500$$

$$\text{Let the amount obtained from advertising is } x \text{ then}$$

$$(7500 + x) - (7600) = 25\% \text{ of } 7500$$

$$x = 1975$$

40. (c) Let C.P. of each clock be ₹ x. Then, C.P. of 90 clocks = ₹ 90x.

$$\therefore [(110\% \text{ of } 40x) + (120\% \text{ of } 50x)] - (115\% \text{ of } 90x) = 40$$

$$\Rightarrow 44x + 60x - 103.5x = 40 \Rightarrow 0.5x = 40 \Rightarrow x = 80$$

41. (b) Total cost (assume) = 100.

$$\text{Recovered amount} = 65 + 0.85 \times 32.5 + 0.7 \times 32.5$$

$$= 65 + 27.625 + 22.75 = 115.375$$

$$\text{Hence, profit percent} = 15.375\%$$

42. (a) Assume he bought 20 apples each. Net investment

$$\Rightarrow ₹ 5 + ₹ 4 = ₹ 9 \text{ for 40 apples. He would sell 40 apples}$$

$$\text{@ } (40 \times 2)/9 = ₹ 8.888 \rightarrow \text{Loss of ₹ 0.111 on ₹ 9 investment}$$

$$\text{Loss percentage} = 1.23\%$$

43. (d) Amar-100, Bharat-120, Sridhar-132

No profit or loss is mentioned about the deal between Sridhar and the mechanic. So the answer cannot be determined.

Expert Level

1. (a) Let C.P. of one soap = ₹ x
C.P. of one toothpaste = ₹ y
- ⇒ S.P. of one soap = ₹ $\frac{115x}{100}$
S.P. of one toothpaste = ₹ $(y + 20)$
and C.P. of one toothpaste = 60% C.P. of one soap
 $y = 0.6x$
- Profit on 75% of soap (i.e. 15 soap) and 8 toothpaste = ₹ 385
- ⇒ $\frac{115x}{100} \times 15 + 8 \times (y + 20) - (15x + 8y) = 385$
⇒ $x = ₹ 100$
 $y = ₹ 60 = 0.6x$
C.P. of 20 soaps and 12 toothpaste = $20 \times 100 + 12 \times 60$
 $= ₹ 2720$
- S.P. of 15 soaps and 8 toothpaste
 $= (15 \times 100 + 8 \times 60) + 385 = ₹ 2365$
and he get no return on unsold items.
Total S.P. = ₹ 2365
Loss = C.P. - S.P.
 $= ₹ (2720 - 2365) = ₹ 355$
2. (a) Let the quantity bought be $12x$, $15x$ and $20x$ kg respectively.
Total cost price = $100 \times 12x + 80 \times 15x + 60 \times 20x$
 $= 1200x + 1200x + 1200x = 3600x$
Selling price of first two quantities at profit of 20%
- $\frac{120}{100} \times 2400x = 2880x$
- Total selling price = $3600x$ (No profit no loss)
∴ Selling price of third quantity = $3600x - 2880x = 720x$
∴ Loss percentage = $\frac{1200x - 720x}{1200x} \times 100 = 40\%$
3. (d)
- | | | |
|--|--|--|
| A | B | C |
| $\frac{1}{7} \left(\frac{8}{7} \right) \frac{1}{8}$ | $\frac{1}{8} \left(\frac{9}{8} \right) \frac{1}{9}$ | $\frac{1}{4} \left(\frac{5}{4} \right) \frac{1}{5}$ |
- Since $14.28\% = \frac{1}{7}$
- So, the ratio of profit percentage of
- | | | | |
|---------------|---------------|---------------|---------|
| A | B | C | |
| 8 | 7 | 14 | (Given) |
| ↓ | ↓ | ↓ | |
| $\frac{1}{7}$ | $\frac{1}{8}$ | $\frac{1}{4}$ | |
- Thus the ratio of CP of A : B : C
 $7 : 8 : 4$

$$\text{Therefore \% profit} = \frac{(8+9+5)-(7+8+4)}{(7+8+4)} \times 100$$

$$\frac{3}{19} \times 100 = 15.78\%$$

4. (a)
- | | | | | |
|-------------|---|------------------|---|-------------------|
| Total wages | = | No. of employees | × | Wage per employee |
| $60xy$ | = | $3x$ | × | $20y$ |
| $65xy$ | = | $2x$ | × | $27y$ |
- Profit (%) = $\frac{60-54}{60} \times 100 = 10\%$
5. (a) Amount paid in Ist service = 100 (suppose)
Amount paid in IInd service = 90
Amount paid in IIIrd service = 81
Amount paid in IVth service = 72.9
Amount paid in Vth service = 60
- | | |
|-------------------|--------------|
| Total amount paid | <u>403.9</u> |
|-------------------|--------------|
- Discount = $500 - 403.9 = 96.1$
- Discount % = $\frac{96.1}{500} \times 100 = 19.42\%$
6. (d) The successive discounts must have been of 10% each. The required price will be got by reducing 25 by 10% twice consecutively. (use PCG application for successive change)
7. (d) Let the cost of the other brand be ₹ x per kg.
C.P. of 5 kg = ₹ $(2 \times 200 + 3 \times x) = ₹ (400 + 3x)$.
S.P. of 5 kg = ₹ $(5 \times 177) = ₹ 885$.
- ∴ $\frac{885 - (400 + 3x)}{400 + 3x} \times 100 = 18 \Leftrightarrow \frac{485 - 3x}{400 + 3x} = \frac{9}{50}$
 $\Leftrightarrow 24250 - 150x = 3600 + 27x \Leftrightarrow 177x = 20650$
 $\Leftrightarrow x = \left(\frac{350}{3} \right) 166 \frac{2}{3}$.
- So, cost of the other brand = ₹ 116.66.
8. (a) If we assume the value of the first cycle as ₹ 900. Then $900 + 96 = 996$ should be equal to twice the value of the second cycle. Hence, the value of the second cycle works out to be: 498.
Also $498 + 96 = 594$ which is ₹ 306 less than 900. Hence, option (a) fits the situation perfectly and will be the correct answer.
Note here that if you had tried to solve this through equations, you would have got stuck for a very long time.
9. (a) Let error = x gms. Then, $\frac{x}{1000-x} \times 100 = 6 \frac{18}{47}$
- ⇒ $\frac{100x}{1000-x} = \frac{300}{47} \Leftrightarrow 47x = 3(1000-x)$
⇒ $50x = 3000 \Leftrightarrow x = 60$
∴ Weight used = $(1000 - 60) = 940$ gms

$$10. (a) \text{ Charge of 1 call in February} = \frac{350}{150} \times \frac{7}{3}$$

$$\text{Charge of 1 call in March} = \frac{350 + 50 \times 1.4}{250}$$

$$= \frac{420}{250} = \frac{42}{25}$$

$$\% \text{ cheapness of a call in March} = \frac{\frac{7}{3} - \frac{42}{25}}{\frac{7}{3}} \times 100$$

$$= 28\%$$

11. (b) Let the CP and SP of 1 g = ₹ 1, then
He spends ₹ 2000 and purchase 2200 g
and he charges ₹ 2000 and sells 1800 g

$$\text{profit \%} = \frac{\text{goods left}}{\text{goods sold}} \times 100$$

$$\frac{400}{1800} \times 100 = 22\frac{2}{9}\%$$

12. (b) You must know that the company is able to deliver only 90% of the manufactured pens. So let K be the manufacturing price of a pen, then
total income (including 25% profit) = $(8000 \times K) \times 1.25$
also this same income is obtained by selling 90% manufactured pens at ₹ 10 which is equal to 7200×10
Thus $(8000 \times K) 1.25 = 7200 \times 10$
 $\Rightarrow K = ₹ 7.2$ (90% of $8000 = 7200$)
13. (a) Let the actual cost price of an article be ₹ 1 (in place of x). Now he purchases goods worth ₹ 120 and pays ₹ 80, since 20% discount is allowed.

$$\text{So the CP} = \frac{80}{120} \times \frac{2}{3}$$

Again MP = 180, SP = 135 (since 25% discount)
Thus the trader sells goods worth ₹ 90 instead of 100 g and charges ₹ 135. Therefore the effective

$$\text{SP} = \frac{135}{90} = \frac{3}{2}$$

$$\therefore \text{Profit (\%)} = \frac{\frac{3}{2} - \frac{2}{3}}{\frac{2}{3}} \times 100 = 125\%$$

14. (b) CP of one egg (in first case) = $\frac{1}{3} = 33.33$ paise

$$\text{CP of one egg (in second case)} = \frac{1}{6} = 16.66 \text{ paise}$$

$$\text{Average CP of one egg} = \frac{(33.33 + 16.66)}{2} = 25 \text{ paise}$$

$$\text{SP of one egg} = \frac{200}{9} \quad (\text{₹ 1} = 100 \text{ paise})$$

$$\text{Profit-loss} = \frac{25 - \frac{200}{9}}{25} \times 100 \quad \left(\text{loss\%} = \frac{\text{CP} - \text{SP}}{\text{CP}} \times 100 \right)$$

$$= 11.11\% \text{ loss}$$

15. (a) **Note :** First of all the price of milk does not matter. You can assume any convenient price. Besides it instead of 10 l of milk you can consider 100 l of milk to avoid calculations in decimal.
Now, since water is 16.66% in the mixture of milk, therefore with 100 l pure milk 20 l water is added. Again note that in replacement method the quantity of mixture does not increase except to the variation in ratio of contents.
Again by replacement formula

$$\frac{80}{120} \times \frac{100}{120} \left(1 - \frac{K}{120} \right) \Rightarrow K = 24l$$

Thus he replaces 24 l of mixture with water.
(Note the required ratio of milk is to water is 2 : 1. It means in 3 l of new mixture, there will be 2 l of water)

Thus if the price of new mixture be ₹ 1, then the price of replaced mixture be ₹ 2.

$$\text{Therefore, total SP} = 120 \times 1 + 24 \times 2 = 168$$

$$\text{and CP} = 100 \times 1 = 100$$

$$\therefore \text{Profit \%} = 68\%$$

16. (c) Total number of microwave ovens = 15
Hence, washing machines = 10
Thus, He sells 80% of both at a profit of ₹ 40,000 cost of 80% of the goods = $0.8 \times 2,05,000 = 1,64,000$
Total amount recovered = $1,64,000 + 40,000 = 2,04,000$
Hence, loss = ₹ 1000

17. (a) Let C.P. = ₹ x/dozen = $\frac{100x}{12}$ per hundred and S.P. 8x/hundred

$$\therefore \% \text{profit} = \frac{8x - \frac{100x}{12}}{\frac{100x}{12}} \times 100$$

$$= \frac{96x - 100x}{12 \times 100x} \times 100 \times 12 = -4\%$$

-ve sign shows that there is a loss of 4%.

Quicker Method (direct formula):

$$\% \text{ profit or loss} = 8 \times \text{dozen} - \text{Hundred} = 96 - 100 = -4\%$$

Since sign is -ve, there is a loss of 4%.

18. (b) Let CP's be X & $5X/4$. Now $5X/4 \times 1.1 + X \times 1.075 = 98$.

$$\Rightarrow X \left(\frac{55}{40} + \frac{1075}{1000} \right) = 98 \Rightarrow X \times \frac{1375 + 1075}{1000} = 98.$$

$$\Rightarrow X = \frac{98 \times 1000}{2450} \Rightarrow X = 40 \text{ and so } \frac{5X}{4} = 50.$$

19. (b) C.P. of goods for the trader = ₹ 22520
 S.P. = $125 \times 22520 = ₹ 28150$
 $\therefore$ Profit = ₹ 5630
 Now, $5630 = 0.0536 \times 5000 + 0.085 \times 10000$

$$+ \frac{x}{100} \times 7520$$

 $5630 = 268 + 850 + 75.2x$
 $75.2x = ₹ 4512$
 which is nothing but profit from sale of third lot.
20. (b) Suppose the cost price of table = ₹ T and cost price of a chair = ₹ C .
- Then; $12\frac{1}{2}\%$ of $T + \left(-8\frac{1}{3}\%\right)$ of $C = 25$
- and $\left(-8\frac{1}{3}\%\right)$ or $T + 12\frac{1}{2}\%$ of $C = 0$
- or, $\frac{25}{2}T - \frac{25}{3}C = 2500$ (1)
- $-\frac{25}{3}T + \frac{25}{2}C = 0$ (2)
- (1) $\div 2 +$ (2) $\div 3$ gives $\frac{25}{4}T - \frac{25}{9}T = 1250$
- or, $T \left[\frac{225 - 100}{36} \right] = 1250$
- $\therefore T = 360$ $\therefore$ Price of a table = ₹ 360
21. (a) Let the CP of the article be ₹ x .
- Then, SP = ₹ $\frac{105x}{100}$
- Now, new CP = ₹ $\frac{95x}{100}$ and new SP = ₹ $\frac{105x}{100} - 1$
- According to the question
- $$\frac{105x}{100} - 1 - \frac{95}{100} = \frac{10 \times 95x}{100 \times 100}$$
- $\therefore x = ₹ 200$
22. (c) Let MP = ₹ 100
 CP after 1st discount = ₹ 98
 CP after 2nd discount = ₹ 94.08
 But discount cannot be more than 100%
23. (c) If you assume that his cost price is ₹ 1 per gram, his cost for 1000 grams would be ₹ 1000. For supposed 1kg sale he would charge a price of 1080 (after an increase of 20% followed by a decrease of 10%). But, since he gives away only 900 grams the cost for him would be ₹ 900.
 Thus he is buying at 900 and selling at 1080 – a profit percentage of 20%

24. (c) $28a + 36b + 45c = 40(a + b + c)$
 $\Rightarrow 12a + 4b = 5c$
 Let the quantities (in kg) of the three varieties of rice be x , xr and xr^2 respectively, where r is the common ratio.
 $\therefore 12x + 4xr = 5xr^2$
 $\Rightarrow 5r^2 - 4r - 12 = 0$
 $\Rightarrow (5r + 6)(r - 2) = 0$
 $\Rightarrow r = 2$
 Therefore, $a : b : c = 1 : 2 : 4$.
- Required C.P. (in ₹/kg) = $\frac{28 \times 4 + 36 + 45 \times 2}{7} = 34$
25. (d) As per the information given in the question, we can conclude that
- $$\frac{0.1a + 0.2b + 0.3c + 0.4d}{a + b + c + d} = 0.25$$
- $\Rightarrow 0.05c + 0.15d = 0.05b + 0.15a$
 $\Rightarrow c + 3d = b + 3a$
 The ratio $5 : 2 : 7 : 3$ does not satisfy the given relation.
26. (b) Total Cost Price of all chocolates for the trader
 = List Price of 19×4 chocolates
 = List Price of 76 chocolates.
 Total Selling Price of all chocolates for the trader
 = List Price of 20×5 chocolates
 = List Price of 100 chocolates
 So Profit = List Price of $100 - 76 = 24$ chocolates
- Profit % = $\frac{24}{76} \times 100 = 31.58\%$
27. (d) $600 - 10\%$ of $600 = 540$. $540 - 5\%$ of $540 = 513$. $513 + 5\%$ of $513 = 538.65$
28. (c) Profit in original situation = 20%
 In new situation, the purchase price of 90 (buys at 10% less) would give a selling price of 132 (sells at 10% above 120).
 The new profit percent = $[(132 - 90) \times 100] / 90 = 46.66$
 Change in profit percent = $[(46.66 - 20) \times 100] / 20 = 133.33\%$
29. (b) In the first year, the profit percentage would be:
 Old Profit Percentage = $\frac{0.55 \times 5 + 0.45 \times 8}{1} = 6.35\%$
 New Profit Percentage = $\frac{0.55 \times 8 + 0.45 \times 5}{1} = 6.65$
30. (d) Alternatively, doing a bit of hit and trial gives the value of $p = 200$.
 Using this, we get the answer as option (d) = 66.66%

Explanation of Test Yourself

1. (a) If listed price of article be ₹ 100 then discounted price be ₹ 80 (since discount = 20%)
After offering 16 articles to a dozen
Price of 16 articles = 80×12
Price of one article $\frac{80 \times 12}{16} = 60$
Profit = 20 %
Let CP of one article
Cost price x (say) = $x + x \times \frac{20}{100}$
 $\Rightarrow x + \frac{x}{5} = 60 \Rightarrow x = 60 \times \frac{5}{6} = 50$
Per cent above the cost price $\frac{100 - 50}{50} \times 100 = 100\%$
2. (a) Assume CP/1000 gm = ₹ 1000
So, MP = ₹ 1400
SP = ₹ 1260
Now, when he sells 1000 gms, he actually obtains the money for only 800 grams (Go through the statement carefully).
So, when he sells articles worth ₹ 1000, money obtained after selling will be equal to = ₹ $1260 \times 0.8 = ₹ 1008$
So, profit percentage = 8%
3. (d) This whole sequence goes like this:
 $100 \rightarrow 10\% \text{ UP} \rightarrow 110 \rightarrow 10\% \text{ UP} \rightarrow 121 \rightarrow 20\% \text{ UP} \rightarrow 145.2 \rightarrow \text{UP by } 20\% \text{ and down by } 10\% \rightarrow 156.816$
Distributors' SP = 121
So, percentage reduction = 22.8%
4. (b) Price of the article after first discount
 $65 - 6.5 = ₹ 58.5$
Therefore, the second discount
 $= \frac{58.5 - 56.16}{58.5} \times 100 = 4\%$
5. (d) Cost ₹ 2400
Published Price ₹ 3.25
 $SP = 75/100 \times 3.25 = ₹ 2.4375$
No. of free copy = $(3000/25) = 120 + 500 = 620$
So, total SP = $2380 \times ₹ 2.4375 = ₹ 5801.25$
Hence percentage gain = $\frac{5801.25 - 2400}{2400} \times 100 = 144\%$
6. (c) Centre table – 40 sofa set – 100
According to Mohan, cost of centre table
 $90/100 \times 40 = 36$
Cost of sofa set – $75/100 \times 100$
According to shopkeeper,
CT – $75/100 \times 40 = 30$
7. (c) Through the method of alligation

+30	–10
+10	
20	20
1	1
8. (a) C.P. of 150 calculators
 $= 150 \times 250 + 2500 = 37500 + 2500 = ₹ 40000$
Labelled price of 150 calculators
 $= 150 \times 320 = ₹ 48000$
Discount allowed = 5%
 $\therefore$ S.P. of 150 calculators
 $= 48000 - 5\% \text{ of } 48000 = ₹ 45600$
 $\therefore$ Profit % = $\frac{5600}{40000} \times 100 = 14\%$
9. (c) Total outlay (initial investment) = $750 \times 0.6 = ₹ 450$.
By selling 600, he should make a 40% profit on the outlay. This means that the selling price for 600 should be $1.4 \times 450 \rightarrow ₹ 630$
Thus, selling price per article = $630/600 = 1.05$. Since, he sells only 630 articles at this price, his total recovery = $1.05 \times 630 = 661.5$
Profit percent (actual) = $(211.5/450) \times 100 = 47\%$
10. (d) Let the SP of the article be ₹ x
Expenses = 15% of $x = ₹ 0.15x$
Profit = 10% of $x = ₹ 0.10x$
CP = ₹ 9 (given)
Therefore, $9 + 0.15x + 0.1x = x \Rightarrow x = 12$
 $\therefore$ % increase for marked price = $\frac{12 - 9}{9} \times 100 = \frac{100}{3}\%$
11. (c) The total cost to print 900 copies would be given by:
Cost for setting up the type + cost of running the printing machine + cost of paper/ink etc
 $= 1000 + 120 \times 9 + 900 \times 0.6 = 1000 + 1080 + 540 = 2620$.
A 10% profit on this cost amounts to ₹ 262. Hence, the total amount to be recovered is ₹ 2882.
Out of this, 784 copies are sold for ₹ 2.75 each to recover ₹ 2156.
The remaining money has to be recovered through advertising.
Hence, The money to be recovered through advertising = $2882 - 2156 = ₹ 726$.

12. (a) Retailer's price = 112 % of 110% of (120 % of 25)

$$\frac{112}{100} \times \frac{110}{100} \times \frac{120}{100} \times 25 = \text{₹ } 36.96 \approx \text{₹ } 37$$

13. (b) The first one would get a profit of ₹ 500 (because his cost would be 2500 for him to get a 20% profit on cost price by selling at 3000).
The second one would earn a profit of 600 (20% of 3000).
Difference in profits = ₹ 100

14. (b) The following calculations will show the respective costs:

Primary cost: 35% of 12600 = 4410

Miscellaneous costs = 2% of 12600 = 252

$$\text{Gross profit} = 12600 - 4410 - 1400 - 650 - 252 = 5888$$

$$\text{Trading cost} = 0.25 \times 5888 = 1472$$

Hence, Net profit = 4416.

$$\text{Percentage profit} = 4416 \times 100 / 14000 = 31.54\%$$

15. (a) Let C.P. of whole stock = ₹ 100. Then, marked price of whole stock = ₹ 120.

$$\text{M.P. of } \frac{1}{2} \text{ stock} = \text{₹ } 60, \text{ M.P. of } \frac{1}{4} \text{ stock} = \text{₹ } 30.$$

$$\therefore \text{Total S.P.} = \text{₹ } [60 + (80\% \text{ of } 30) + (60\% \text{ of } 30)] \\ = \text{₹ } (60 + 24 + 18) = \text{₹ } 102$$

$$\text{Hence, gain\%} = (102 - 100)\% = 2\%.$$

7

INTEREST

- Introduction
- Interest

- Simple Interest (SI)
- Compound Interest (CI)

INTRODUCTION

Interest is an important chapter of commercial arithmetics. Interest is the fee paid by a borrower of assets to the owner as a compensation for the use of the assets. The concept of simple and compound interest is simple but it requires clear understanding of percentage calculation; clear understanding of percentage calculation will improve your speed in solving the questions of interests. Questions from this chapter seldom appear in CAT but it is necessary that a serious CAT aspirant should know the application of the concepts of this chapter because they are indirectly used in many questions.

INTEREST

If an agency (i.e. an individual, a firm or a bank etc.) borrow some money from any other agency, then the first agency is called the *borrower* and the second agency is called the *lender*. The borrowed money is called the *principal*.

If the borrower has to pay some additional money together with the borrowed money for the benefit of using borrowed money for a certain time period is called *loan period*, then this additional money is called the *interest* and the principal together with the interest is called the *amount* (i.e. Amount = Principal + Interest). When we deposite money in a bank, we earn interest, interest is calculated according to an agreement which specifies the rate of interest. Generally the rate of interest is taken as “percent per annum” which means “per ₹ 100 per year”. For example, a rate of 10% per annum means ₹ 10 on ₹ 100 for 1 year.

SIMPLE INTEREST (S.I.)

If the principal remains the same for whole loan period, then the interest is called the simple interest.

$$\text{S.I.} = \frac{PRT}{100}$$

where P = Principal, R = Rate of interest in percent per annum, T = Loan period (or whole time period in years)

In the formula of simple interest, by putting the value of any three unknowns out of the four unknowns $S.I.$, P , R , T ; you can find the remaining fourth unknown.

Simple rate of interest is generally written as rate of interest only i.e. if it is not mentioned whether the interest is simple or compound, then we should assume it as simple interest.

Illustration 1: At what rate percent by simple interest, will a sum of money double itself in 5 years 4 months ?

Solution: Let $P = ₹ x$

Then $A = ₹ 2x$

∴ $S.I. = A - P = ₹ 2x - ₹ x = ₹ x$

$T = 5 \text{ years } 4 \text{ months} = 5 \frac{4}{12} \text{ years}$

$= 5 \frac{1}{3} \text{ year} = \frac{16}{3} \text{ years}$

Let R be the rate percent per annum.

Using $R = \frac{S.I. \times 100}{P \times T}$, We get $R = \frac{x \times 100}{x \times \frac{16}{3}} = \frac{300}{16} = 18.75$.

Hence required rate = 18.75 % p.a.

Illustration 2: A sum of ₹ 450 amounts to ₹ 495 at simple interest in 2 years. In what time will the sum of ₹ 820 amount to ₹ 943 at the same rate ?

Solution: $P = ₹ 450$; $A = ₹ 495$; $T = 2 \text{ years}$;

$I = A - P = 495 - 450 = 45$

$I = \frac{PTR}{100}$

$R = \frac{100 \times I}{PT} = \frac{100 \times 45}{450 \times 2} = 5\% =$

Now take

$$P = ₹ 820; A = ₹ 943; R = 5\%; T = ?$$

$$I = A - P = 943 - 820 = ₹ 123.$$

$$I = \frac{PTR}{100}$$

$$T = \frac{100 \times I}{PR} = \frac{100 \times 123}{820 \times 5} \quad 3 \text{ years.} =$$

Illustration 3: Find the SI on ₹ 1800 from 21st Feb. 2003 to 12th April 2003 at 7.3% rate per annum.

Solution: $P = ₹ 1800; R = 7.3\%; I = ?$

No. of days = 7 + 31 + 12 = 50 days.

$$T = \frac{50}{365} \text{ years.}$$

$$I = \frac{PTR}{100} = \frac{1800 \times \frac{50}{365} \times 7.3}{100} = ₹ 18.$$

Illustration 4: Determine the rate per cent per annum if ₹ 25,000 amounts to 26,010 in 6 months, interest being compounded quarterly.

Solution: Here, $n = 2$ [$\because$ 6 months = 2 quarters]

Now, $A = P \left(1 + \frac{r}{100}\right)^n$, where r is the rate per cent per quarter.

$$\therefore 26010 = 25000 \left(1 + \frac{r}{100}\right)^2$$

$$\text{or } \left(1 + \frac{r}{100}\right)^2 = \frac{26010}{25000} = \frac{2601}{2500} \quad \left(\frac{51}{50}\right)^2$$

$$\text{or } \left(1 + \frac{r}{100}\right) = \frac{51}{50}$$

$$\text{or } \frac{r}{100} = \frac{51}{50} - 1 = \frac{51 - 50}{50} \quad \frac{1}{50} =$$

$$\text{or } r = \frac{1}{50} \times 100 = 2\%$$

Hence, the required rate is 2% p.a.

Illustration 5: A certain sum of money at C.I. amounts to ₹ 811.25 in 2 years and to ₹ 843.65 in 3 years. Find the sum of money.

Solution: Since $A = P \left[1 + \frac{R}{100}\right]^n$

$$\Rightarrow 811.25 = P \left[1 + \frac{R}{100}\right]^2 \quad \dots(1) \text{ and}$$

$$843.65 = P \left[1 + \frac{R}{100}\right]^3 \quad \dots(2)$$

$$\text{On dividing (2) by (1), we get: } \frac{843.65}{811.25} = 1 + \frac{R}{100}$$

$$\Rightarrow 1.04 = 1 + \frac{R}{100} \Rightarrow R = 4$$

Now, putting $R = 4$ into (1), we get

$$811.25 = P \left[1 + \frac{4}{100}\right]^2 \Rightarrow P = 750$$

$\Rightarrow$ The sum of money is ₹ 750.

COMPOUND INTEREST (C.I.)

If the borrower and the lender agree to fix up a certain interval of time (a year, a half year or a quarter of a year etc.) called conversion period, so that the amount (= principal + interest) at the end of an conversion period becomes the principal for the next conversion period, then the total interest over the whole loan period calculated in this way is called the compound interest.

Note: The main difference between the simple interest and the compound interest is that the principal in the case of simple interest remains constant throughout the loan period whereas in the case of compound interest, the principal changes periodically (i.e. after each conversion period) throughout the loan period.

Rate of interest is always given annually but it can be compounded annually, half yearly, quarterly or monthly.

Interest compounded annually means conversion period is one year and hence amount at the end of every one year becomes the principal for the next conversion period.

Interest compounded half yearly means conversion period is half year and hence amount at the end of every half year becomes the principal for the next conversion period.

Interest compounded quarterly means conversion period is a quarter of a year and hence amount at the end of every quarter of a year becomes the principal for the next conversion period.

Similarly, interest compounded monthly means conversion period is one month and hence amount at the end of every one month becomes the principal for the next conversion period.

1. Computation of Compound Interest When Interest is Compounded Annually

$$A = P \left(1 + \frac{r}{100}\right)^n$$

$$\therefore \text{C.I.} = A - P = P \left[\left(1 + \frac{r}{100}\right)^n - 1 \right]$$

Here A is the amount,

P is the principal, r is the rate of interest in percent per conversion period and n is the number of conversion periods in the whole loan period.

In the formula of compound interest by putting the value of any three unknowns out of the four unknowns A , P , r and n ; you can find the remaining fourth unknown.

Illustration 6: Roohi deposited ₹ 7,000 in a finance company for 3 years at an interest of 15% per annum compounded annually. What is the compound interest and the amount that Roohi gets after 3 years?

Solution: Principal, $P = ₹ 7000$, $n = 3$ years, $r = 15\%$ per annum
Amount of C.I.

$$A = P \left(1 + \frac{r}{100} \right)^n = 7000 \left(1 + \frac{15}{100} \right)^3 = 7000 \left(\frac{115}{100} \right)^3$$

$$= 7000 \times \frac{115}{100} \times \frac{115}{100} \times \frac{115}{100} = 10646.125 = ₹ 10646.13$$

$$= ₹ 10646 \text{ (approx.)}$$

$$\text{Compound interest} = A - P = 10646 - 7000 = ₹ 3646$$

2. Computation of Compound Interest When Interest is Compounded k Times Every Year

If r be the rate of interest in percent per year, then the rate of interest in percent per conversion period is $\frac{r}{k}$.

If n be the number of years in the whole loan period (or whole time period), then the number of conversion period is nk .

$$\therefore A = P \left(1 + \frac{r}{100k} \right)^{nk}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{100k} \right)^{nk} - 1 \right]$$

(a) In case of interest compounded half-yearly, $k = 2$

$$\therefore A = P \left(1 + \frac{r}{100 \times 2} \right)^{2n}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{100 \times 2} \right)^{2n} - 1 \right]$$

(b) In case of interest compounded quarterly, $k = 4$

$$\therefore A = P \left(1 + \frac{r}{100 \times 4} \right)^{4n}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{100 \times 4} \right)^{4n} - 1 \right]$$

(c) In case of interest compounded monthly, $k = 12$

$$\therefore A = P \left(1 + \frac{r}{100 \times 12} \right)^{12n}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{100 \times 12} \right)^{12n} - 1 \right]$$

Illustration 7: A sum of money is lent out at compound interest rate of 20 % per annum for 2 years. It would fetch ₹ 482 more if interest is compounded half-yearly. Find the sum.

Solution: Suppose the sum is ₹ P .

C.I. when interest is compounded yearly

$$= P \left[1 + \frac{20}{100} \right]^2 - P$$

C.I. when interest is compounded half-yearly

$$= P \left[1 + \frac{10}{100} \right]^4$$

Now, we have,

$$P \left[1 + \frac{10}{100} \right]^4 - P \left[1 + \frac{20}{100} \right]^2 = 482$$

$$\Rightarrow P \left[\{1.1\}^4 - \{1.2\}^2 \right] = 482$$

$$\Rightarrow P \left[\{(1.1)^2 - (1.2)\} \{(1.1)^2 + (1.2)\} \right] = 482$$

$$\Rightarrow P \left[\{1.21 - 1.2\} \{1.21 + 1.2\} \right] = 482$$

$$\Rightarrow P \left[(0.01) (2.41) \right] = 482$$

$$\therefore P = \frac{482}{2.41 \times 0.01} = ₹ 20,000$$

Illustration 8: In how many years will ₹ 800 amount to ₹ 882 at 5% per annum be compounded annually?

Solution: Here, $P = ₹ 800$

$A = ₹ 882$

$r = 5\% \text{ p.a.}$

Let number of years be n .

$$\text{Since, } A = P \left(1 + \frac{r}{100} \right)^n$$

$$\therefore 882 = 800 \left(1 + \frac{5}{100} \right)^n = 800 \left(1 + \frac{1}{20} \right)^n$$

$$\text{or } \frac{882}{800} = \left(1 + \frac{1}{20} \right)^n$$

$$\text{or } \frac{441}{400} = \left(\frac{21}{20} \right)^n$$

$$\text{or } \left(\frac{21}{20} \right)^2 = \left(\frac{21}{20} \right)^n \quad [\because 441 = 21^2 \text{ and } 400 = 20^2]$$

Since the bases are the same on both sides, hence $n = 2$

Since interest is compounded annually

$\therefore$ Time = 2 years

Illustration 9: If the simple interest on a certain sum of money for 3 yrs at 5% is ₹ 150, find the corresponding CI.

Solution: Whenever the relationship between CI and SI is asked for t yrs of time, we use the formula:

$$SI = \frac{rt}{100 \left[\left(1 + \frac{r}{100} \right)^t - 1 \right]} \times CI$$

$$\text{Hence, } 150 = \frac{5 \times 3}{100 \left[\left(1 + \frac{r}{100} \right)^3 - 1 \right]} \times CI$$

$$\therefore CI = \frac{150 \times 100 \left[\frac{9261 - 8000}{8000} \right]}{5 \times 3}$$

$$= \frac{150 \times 100 \times 1261}{5 \times 3 \times 8000} = \frac{1261}{8} = 157.62$$

Illustration 10: Find the compound interest on ₹ 1000 at the rate of 10% per annum for 18 months when interest is compounded half-yearly.

Solution: Here, $P = ₹ 1000$, $r = 10\%$ per annum and

$$n = \frac{18}{12} \text{ years} = \frac{3}{2} \text{ years}$$

$$\therefore \text{Amount after 18 months} = P \left(1 + \frac{r}{200} \right)^{2n}$$

$$= ₹ 1000 \times \left(1 + \frac{10}{200} \right)^{2 \times \frac{3}{2}} = ₹ 1000 \times \left(1 + \frac{1}{20} \right)^3$$

$$= ₹ 1000 \times \left(\frac{21}{20} \right)^3 = ₹ 1000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20}$$

$$= ₹ 1157.63 \text{ (approx.)}$$

Hence, Compound interest = Amount – Principal

$$= ₹ 1157.63 - ₹ 1000 = ₹ 157.63 \text{ (approx.)}$$

Illustration 11: Lussy deposited ₹ 7500 in a bank which pays him 12% interest per annum compounded quarterly. What is the amount which she will receive after 9 months?

Solution: Here, $P = ₹ 7500$, $r = 12\%$ per annum and

$$n = 9 \text{ months} = \frac{9}{12} \text{ years} = \frac{3}{4} \text{ years.}$$

$$\therefore \text{Amount after 9 months} = P \left(1 + \frac{r}{400} \right)^{4n}$$

$$= ₹ 7500 \times \left(1 + \frac{12}{400} \right)^{4 \times \frac{3}{4}} = ₹ 7500 \times \left(1 + \frac{3}{100} \right)^3$$

$$= ₹ 7500 \times \frac{103}{100} \times \frac{103}{100} \times \frac{103}{100} = ₹ 8195.45$$

Illustration 12: The difference between compound interest compounded annually and simple interest on a certain sum of money for 2 years at 5% per annum is ₹ 12.50. What is the compound interest on this sum for 2 years?

Solution: Let the sum be ₹ 100.

$$SI = \frac{100 \times 2 \times 5}{100} = ₹ 10$$

$$CI = 100 \left(1 + \frac{5}{100} \right)^2 - 100 = ₹ 10.20$$

$$CI - SI = 10.20 - 10 = 0.20$$

$$\text{Hence, sum} = \frac{100}{0.20} \times 10 = 5000$$

$$CI = 5000 \left(1 + \frac{5}{100} \right)^2 - 5000$$

$$= 5000 (1 + 0.05)^2 - 5000 = ₹ 512.5$$

Illustration 13: A sum of money placed at compound interest doubles itself in 4 years. In how many years will it amount to eight times itself?

Solution: We have

$$P \left(1 + \frac{r}{100} \right)^4 = 2P$$

$$\therefore \left(1 + \frac{r}{100} \right)^4 = 2$$

Cubing both sides, we get

$$\left(1 + \frac{r}{100} \right)^{12} = 2^3 = 8$$

$$\therefore P \left(1 + \frac{r}{100} \right)^{12} = 8P$$

Hence required time = 12 years.

Shortcut Approach:

x becomes $2x$ in 4 yrs.

$2x$ becomes $4x$ in 4 yrs.

$4x$ becomes $8x$ in 4 yrs.

Thus x becomes $8x$ in $4 + 4 + 4 = 12$ yrs.

Remember the follow result

If a sum becomes x times in y years at compound interest then it will be $(x)^n$ times in ny years

Thus if a sum becomes 3 times in y years at compound interest, it will be $(3)^2$ times in $2 \times 3 = 6$ years.

Illustration 14: If a sum deposited at compound interest becomes double in 4 years, when will it be 4 times at the same rate of interest?

Solution: Using the above conclusion, we say that the sum will be $(2)^2$ times in $2 \times 4 = 8$ years.

3. Computation of Compound Interest When Interest is Compounded Annually but Rate of Interest in Percent being Different for Different Years

$$A = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \dots \left(1 + \frac{R_n}{100} \right),$$

where $R_1, R_2, \dots, R_n$ are rate of interest in percent per year for different years.

Illustration 15: Ram Singh bought a refrigerator for ₹ 4000 on credit. The rate of interest for the first year is 5% and of the second years is 15%. How much will it cost him if he pays the amount after two years.

Solution: Here, $P = ₹ 4000$, $R_1 = 5\%$ per annum and $R_2 = 15\%$ per annum.

$$\therefore \text{Amount after 2 years} = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right)$$

$$= ₹ 4000 \times \left(1 + \frac{5}{100} \right) \left(1 + \frac{15}{100} \right)$$

$$= ₹ 4000 \times \left(1 + \frac{1}{20} \right) \left(1 + \frac{3}{20} \right)$$

$$= ₹ 4000 \times \frac{21}{20} \times \frac{23}{20} = ₹ 4830$$

Thus, the refrigerator will cost ₹ 4830 to Ram Singh.

4. If P be the value of an article (or population of a town or a country etc.) at a certain time and $R\%$ per annum is the rate of depreciation, then the value A at the end of n years is given by

$$A = P \left(1 - \frac{R}{100} \right)^n$$

Illustration 16: The population of a town 2 years ago was 62500. Due to migration to cities, it decreases every year at the rate of 4% per annum. Find its present population.

Solution: We have,

Population two years ago = 62500

Rate of decrease of population = 4% per annum.

$$\begin{aligned} \therefore \text{Present population} &= 62500 \times \left(1 - \frac{4}{100} \right)^2 \\ &= 62500 \times \left(1 - \frac{1}{25} \right)^2 \\ &= 62500 \times \left(\frac{24}{25} \right)^2 \\ &= 62500 \times \frac{24}{25} \times \frac{24}{25} = 57600 \end{aligned}$$

Hence, present population = 57600

5. If P be the population of a country (or value of an article etc.) at a certain time, which increases at the Rate $R_1\%$ per year for first n_1 years and decreases at the rate of $R_2\%$ per year for next n_2 years, then the population at the end of $(n_1 + n_2)$ years is given by

$$A = P \left(1 + \frac{R_1}{100} \right)^{n_1} \cdot \left(1 - \frac{R_2}{100} \right)^{n_2}$$

This formula can be extended for more than 2 different periods and rates.

Illustration 17: 10000 workers were employed to construct a river bridge in four years. At the end of first year, 10% workers were retrenched. At the end of the second year, 5% of the workers at the beginning of the second year were retrenched. However to complete the project in time, the number of workers was increased by 10% at the end of the third year. How many workers were working during the fourth year?

Solution: We have,

Initial number of workers = 10000

Reduction of workers at the end of first year = 10%

Reduction of workers at the end of second year = 5%

Increase of workers at the end of third year = 10%

$\therefore$ Number of workers working during the fourth year

$$= 10000 \left(1 - \frac{10}{100} \right) \left(1 - \frac{5}{100} \right) \left(1 + \frac{10}{100} \right)$$

$$= 10000 \times \frac{9}{10} \times \frac{19}{20} \times \frac{11}{10} = 9405$$

Hence, the number of workers working during the fourth year was 9405.

In CAT and CAT like competitions, number of conversion periods in the problems related to compound interest are generally 2, 3 or 4.

For more than 2 conversion periods, use of formula to find the amount, compound interest, depreciated value, depreciation, final population or change in population are time consuming but by using PCG (Percentage Change Graphic), we can find them in a faster way as illustrated in the following illustrations number 8 to 16.

Illustration 18: Find the amount of ₹ 1000 after 4 years at the rate of 10% per annum compounded annually.

Solution: Using Percentage Change Graphic (PCG), illustrated in the chapter of percentage.

$$\begin{aligned} 1000 &\xrightarrow{+10\%} 1100 \xrightarrow{+10\%} 1210 \xrightarrow{+10\%} 1331 \xrightarrow{+10\%} 1464.1 \end{aligned}$$

Hence required amount = ₹ 1464.10.

Illustration 19: Find the amount of ₹ 1500 after $1\frac{1}{2}$ years at the rate of 12% per annum compounded half-yearly.

Solution: Rate of interest half-yearly = $\frac{1}{2} \times 12\% = 6\%$

$$\text{Number of half years in } 1\frac{1}{2} \text{ years} = 2 \times 1\frac{1}{2} = 2 \times \frac{3}{2} = 3$$

$$\begin{aligned} 1500 &\xrightarrow{+6\%} 1590 \xrightarrow{+6\%} 1685.4 \xrightarrow{+6\%} 1786.8 \end{aligned}$$

(approximately one digit after decimal)

Hence required amount = ₹ 1786.8.

Illustration 20: Find the amount of ₹ 2400 after 3 years at the rate of 8% per annum compounded annually.

$$\begin{aligned} 2400 &\xrightarrow{+8\%} 2592 \xrightarrow{+8\%} 2799.2 \xrightarrow{+8\%} 3023.2 \end{aligned}$$

Hence required amount = ₹ 3023.2 (approximately one digit after decimal).

Illustration 21: Find the amount of ₹ 5000 after 9 months at the rate of 16% per annum compounded quarterly.

Solution: Rate of interest quarterly = $\frac{16}{4} = 4\%$

$$\text{Number of quarter years in 9 months} = \frac{9}{3} = 3$$

(Since in one quarter, there is 3 months)

$$5000 \xrightarrow[+(4 \times 50 = 200)]{4\% \uparrow} 5200 \xrightarrow[+(4 \times 52 = 208)]{4\% \uparrow} 5408 \xrightarrow[+(4 \times 54 = 216)]{4\% \uparrow} 5624$$

Hence required amount = ₹ 5624 (approximately)

Illustration 22: Find the amount of ₹ 8000 after 2 years at the rate of 12% per annum compounded annually.

Solution:

$$8000 \xrightarrow[+(10\% \text{ of } 8000 + 2 \times 1\% \text{ of } 8000 = 800 + 2 \times 80 = 960)]{12\% \uparrow} 8960 \xrightarrow[+(896 + 2 \times 89.6 = 896 + 179.2 = 1075.2)]{12\% \uparrow} 10035.2$$

Hence required amount = ₹ 10035.2.

Illustration 23: Find the amount of ₹ 12000 after 3 years at the rate of 12% per annum compounded annually.

$$\begin{aligned} \text{Solution: } 12000 &\xrightarrow[+(1200 + 2 \times 120 = 1440)]{12\% \uparrow} 13440 \\ &\xrightarrow[+(1344 + 268.8 = 1612.8)]{12\% \uparrow} 15052.8 \\ &\xrightarrow[+(1505.3 + 301.06 = 1806.36)]{12\% \uparrow} 16859.16 \end{aligned}$$

Hence required amount = ₹ 16859.16 (approx.)

Illustration 24: Calculate the amount and the compound interest on ₹ 8000 after 2 years when the rate of interest for successive years are 5% and 6% respectively.

$$\text{Solution: } 8000 \xrightarrow[+400]{5\% \uparrow} 8400 \xrightarrow[+504]{6\% \uparrow} 8904$$

Hence required amount = ₹ 8904

Illustration 25: A new car costs ₹ 360000. Its price depreciates at the rate of 10% a year during the first two years and at the rate of 20% a year thereafter. What will be the price of the car after 3 years ?

$$\begin{aligned} \text{Solution: } 360000 &\xrightarrow[-36000]{10\% \downarrow} 324000 \xrightarrow[-32400]{10\% \downarrow} 291600 \\ &\xrightarrow[-58320]{20\% \downarrow} 233280 \end{aligned}$$

Hence required price of the car after 3 years = ₹ 233280

Illustration 26: The population of a town was 250000 three years ago. It had increased by 3%, 4% and 6% in the last three years, find the present population of the town.

$$\begin{aligned} \text{Solution: } 250000 &\xrightarrow[+(3 \times 2500 = 7500)]{3\% \uparrow} 257500 \\ &\xrightarrow[+10300]{4\% \uparrow} 267800 \xrightarrow[+16068]{6\% \uparrow} 283868 \end{aligned}$$

Hence present population = 283868

Practice Exercises



Foundation Level

- To buy furniture for a new apartment, Sylvia Chang borrowed ₹ 5000 at 11% per annum simple interest for 11 months. How much interest will she pay?
(a) 500 (b) 504.17
(c) 6050 (d) 605
- Find the compound interest on ₹ 18,750 in 2 years the rate of interest being 4% for the first year and 8% for the second year.
(a) 2310 (b) 1130
(c) 3120 (d) None of these
- At a simple interest ₹ 800 becomes ₹ 956 in three years. If the interest rate, is increased by 3%, how much would ₹ 800 become in three years?
(a) ₹ 1020.80 (b) ₹ 1004
(c) ₹ 1028 (d) Data inadequate
- On retirement, a person gets 1.53 lakhs of his provident fund which he invests in a scheme at 20% p.a. His monthly income from this scheme will be
(a) ₹ 2,450 (b) ₹ 2,500
(c) ₹ 2,550 (d) ₹ 2,600
- In how many minimum number of complete years, the interest on ₹ 212.50 P at 3% per annum will be in exact number of rupees?
(a) 6 (b) 8
(c) 9 (d) 7
- A scooter costs ₹ 25,000 when it is brand new. At the end of each year, its value is only 80% of what it was at the beginning of the year. What is the value of the scooter at the end of 3 years?
(a) ₹ 10,000 (b) ₹ 12,500
(c) ₹ 12,800 (d) ₹ 12,000
- Village A has a population of 6800, which is decreasing at the rate of 120 per year. Village B has a population of 4200, which is increasing at the rate of 80 per year. In how many years will the population of the two villages will become equal ?
(a) 9 (b) 11
(c) 13 (d) 16
- A person invested some amount at the rate of 12% simple interest and a certain amount at the rate of 10% simple interest. He received yearly interest of ₹ 130. But if he had interchanged the amounts invested, he would have received ₹ 4 more as interest. How much did he invest at 12% simple interest ?
(a) ₹ 700 (b) ₹ 500
(c) ₹ 800 (d) ₹ 400
- A certain amount is lent at $x\%$ p.a. simple interest for two years. Instead, if the amount was lent at $2x\%$ p.a. simple interest for 'y' more years, then the interest would have been five times the earlier interest. What is the value of y?
(a) 2 years (b) 3 years
(c) 4 years (d) 5 years
- A certain sum of money triple itself in 8 years. In how many years it will be five times?
(a) 22 years (b) 16 years
(c) 20 years (d) 24 years
- The difference between CI and SI on a certain sum of money at 10% per annum for 3 years is ₹ 620. Find the principal if it is known that the interest is compounded annually.
(a) ₹ 200,000 (b) ₹ 20,000
(c) ₹ 10,000 (d) ₹ 100,000
- Michael Bolton has \$90,000 with him. He purchases a car, a laptop and a flat for \$15,000, \$13,000 and \$35,000 respectively and puts the remaining money in a bank deposit that pays compound interest @15% per annum. After 2 years, he sells off the three items at 80% of their original price and also withdraws his entire money from the bank by closing the account. What is the total change in his asset?
(a) -4.5% (b) +3.5%
(c) -4.32% (d) +5.5%
- An amount of ₹ 12820 due 3 years hence, is fully repaid in three annual instalments starting after 1 year. The first instalment is $\frac{1}{2}$ the second instalment and the second instalment is $\frac{2}{3}$ of the third instalment. If the rate of interest is 10% per annum, find the first instalment.
(a) ₹ 2400 (b) ₹ 1800
(c) ₹ 2000 (d) ₹ 2500
- What will be the ratio of simple interest earned by certain amount at the same rate of interest for 6 years and that for 9 years?
(a) 1 : 3 (b) 1 : 4
(c) 2 : 3 (d) None of these

15. A man borrows ₹ 6000 at 5% interest, on reducing balance, at the start of the year. If he repays ₹ 1200 at the end of each year, find the amount of loan outstanding, (in ₹), at the beginning of the third year.
(a) 3162.75 (b) 4155.00
(c) 4155.00 (d) 5100.00
16. Two equal sums were lent, one at the rate of 11% p.a. for five years and the other at the rate of 8% p.a. for six years, both under simple interest. If the difference in interest accrued in the two cases is ₹ 1008, find the sum.
(a) ₹ 11,200 (b) ₹ 5,600
(c) ₹ 12,600 (d) ₹ 14,400
17. A sum is invested at compound interest payable annually. The interest in two successive years was ₹ 225 and ₹ 236.25. Find the rate of interest
(a) 2% (b) 3%
(c) 5% (d) 9%
18. A person borrowed ₹ 500 at 3% per annum S.I. and ₹ 600 at $4\frac{1}{2}$ % per annum on the agreement that the whole sum, will be returned only when the total interest becomes ₹ 126. The number of years, after which the borrowed sum is to be returned, is :
(a) 2 (b) 3
(c) 4 (d) 5
19. A bank offers 5% compound interest calculated on half-yearly basis. A customer deposits ₹ 1600 each on 1st January and 1st July of a year. At the end of the year, the amount he would have gained by way of interest is
(a) ₹ 120 (b) ₹ 121
(c) ₹ 122 (d) ₹ 123
20. A sum of money invested at simple interest triples itself in 8 years. How many times will it become in 20 years time?
(a) 8 times (b) 7 times
(c) 6 times (d) 9 times
21. The population of a city is 200,000. If the annual birth rate and the annual death rate are 6% and 3% respectively, then calculate the population of the city after 2 years.
(a) 212,090 (b) 206,090
(c) 212,000 (d) 212,180
22. The population of Bangalore was 1283575 on 1 January 2001 and the growth rate of population was 10% in the last year and 5% in the years prior to it, the only exception being 1999 when because of a huge exodus there was a decline of 20% in population. What was the population of January 1, 1995 ?
(a) 1,000,000 (b) 1,200,000
(c) 1,250,000 (d) 1,500,000
23. A person bought a motorbike under the following scheme: Down payment of ₹ 15,000 and the rest amount at 8% per annum for 2 years. In this way, he paid ₹ 28,920 in total. Find the actual price of the motorbike. (Assume simple interest).
(a) ₹ 26,000 (b) ₹ 27,000
(c) ₹ 27,200 (d) ₹ 26,500
24. The ratio of the amount for two years under C.I. annually and for one year under S.I. is 6 : 5. When the rate of interest is same, then the value of rate of interest is
(a) 12.5% (b) 18%
(c) 20% (d) 16.66%
25. Mr. Bajaj invested $\frac{1}{7}$ of his total investment at 4% and $\frac{1}{2}$ at 5% and rest at 6% for the one year and received total interest of ₹ 730. What is the total sum invested?
(a) ₹ 70000 (b) ₹ 14000
(c) ₹ 24000 (d) ₹ 38000
26. Akram Ali left an amount of ₹ 340000 to be divided between his two sons aged 10 years and 12 years such that both of them would get an equal amount when each attain 18 years age. What is the share of elder brother if the whole amount was invested at 10% simple interest ?
(a) 120000 (b) 140000
(c) 160000 (d) 180000
27. A Sonata watch is sold for ₹ 440 cash or for ₹ 200 cash down payment together with ₹ 244 to be paid after one month. Find the rate of interest charged in the instalment scheme
(a) 10% (b) 15%
(c) 20% (d) 25%

Standard Level

- The compound interest on a certain sum for 2 years at 10% per annum is ₹ 1260. The simple interest on the same sum for double the time at half the rate per cent per annum is
 - ₹ 1200
 - ₹ 1160
 - ₹ 1208
 - ₹ 1175
- Manish borrowed a sum of ₹ 1150 from Anil at the simple rate of 6% per annum for 3 years. He then added some more money to the borrowed sum of lent it to Sunil for the same time at 9% per annum at simple interest. If Manish gains ₹ 274.95 by way of interest on the borrowed sum as well as his own amount from the whole transaction, then what is the sum lent by him to Sunil?
 - ₹ 1290
 - ₹ 1785
 - ₹ 1285
 - ₹ 1200
- The simple interest on a sum of money will be ₹ 300 after 5 years. In the next 5 years principal is trebled, what will be the total interest at the end of the 10th year?
 - 1200
 - 900
 - 600
 - 1500
- A person lent a certain sum of money at 4% simple interest; and in 8 years the interest amounted to ₹ 340 less than the sum lent. Find the sum lent.
 - 500
 - 600
 - 1000
 - 1500
- A sum was put at simple interest at a certain rate for 2 years. Had it been put at 1% higher rate, it would have fetched ₹ 24 more? The sum is
 - 1200
 - 1500
 - 1800
 - 2000
- A sum of money at compound interest amounts in two years to ₹ 2809, and in three years to ₹ 2977.54. Find the rate of interest and the original sum
 - 2000
 - 2100
 - 2200
 - 2500
- Consider the following statements
If a sum of money is lent at simple interest, then the
 - money gets doubled in 5 years if the rate of interest is $16\frac{2}{3}\%$.
 - money gets doubled in 5 years if the rate of interest is 20%.
 - money becomes four times in 10 years if it gets doubled in 5 years.
 Of these statements,
 - I and III are correct
 - II alone is correct
 - III alone is correct
 - II and III are correct
- Simple interest on a certain amount is $\frac{9}{16}$ of the principal. If the numbers representing the rate of interest in percent and time in years be equal, then time, for which the principal is lent out, is
 - $5\frac{1}{2}$ years
 - $6\frac{1}{2}$ years
 - 7 years
 - $7\frac{1}{2}$ years
- If the rate increases by 2%, the simple interest received on a sum of money increases by ₹ 108. If the time period is increased by 2 years, the simple interest on the same sum increases by ₹ 180. The sum is :
 - ₹ 1800
 - ₹ 3600
 - ₹ 5400
 - Data inadequate
- A man lends ₹ 10,000 in four parts. If he gets 8% on ₹ 2000; $7\frac{1}{2}\%$ on ₹ 4000 and $8\frac{1}{2}\%$ on ₹ 1400; what percent must he get for the remainder, if his average annual interest is 8.13% ?
 - 7%
 - 9%
 - $9\frac{1}{4}\%$
 - $10\frac{1}{2}\%$
- A man borrows ₹ 12,500 at 20% compound interest. At the end of every year he pays ₹ 2000 as part repayment. How much does he still owe after three such instalments?
 - ₹ 12,000
 - ₹ 12,864
 - ₹ 15,600
 - None of these
- A part of ₹ 38,800 is lent out at 6% per six months. The rest of the amount is lent out at 5% per annum after one year. The ratio of interest after 3 years from the time when first amount was lent out is 5 : 4. Find the second part that was lent out at 5%
 - ₹ 26,600
 - ₹ 28,800
 - ₹ 27,500
 - ₹ 28,000
- The difference between C.I. and S.I. on a certain sum of money at 10% per annum for 3 years is ₹ 620. Find the principal if it is known that the interest is compounded annually.
 - ₹ 200,000
 - ₹ 20,000
 - ₹ 10,000
 - ₹ 100,000
- We had 1000 goats at the beginning of year 2001 and the no. of goats each year increases by 10% by giving birth (compounded annually). At the end of each year, we double the no. of goats by purchasing the same no. of goats as there is the no. of goats with us at the time. What is the no. of goats at the beginning of 2004?
 - 10600
 - 10648
 - 8848
 - 8226
- The population of towns A and B is the ratio of 1 : 4. For the next 2 years, the population of A would increase and that of B would decrease by the same percentage every year. After 2 years, their population became equal. What is the percentage change in the population?
 - 33.33%
 - 66.66%
 - 25%
 - Not possible
- If the population of a town at the beginning of a year was 1530000, and the birth rate was 53.2, while the death rate was 31.2 per 1000 of the population, then the net increase in the population at the end of the year was
 - 336600
 - 363600
 - 366300
 - 330000
- Arun borrowed a sum of money from Jayant at the rate of 8% per annum simple interest for the first four years, 10%

- per annum for the next six years and 12% per annum for the period beyond ten years. If he pays a total of ₹ 12,160 as interest only at the end of 15 years, how much money did he borrow?
- (a) ₹ 8000 (b) ₹ 10,000
(c) ₹ 12,000 (d) ₹ 9,000
18. What will be the difference in simple and compound interest on ₹ 2000 after three years at the rate of 10 percent per annum?
- (a) ₹ 160 (b) ₹ 42
(c) ₹ 62 (d) ₹ 20
19. Aniket deposited two parts of a sum of ₹ 25000 in different banks at the rates of 15% per annum and 18% per annum respectively. In one year he got ₹ 4050 as the total interest. What was the amount deposited at the rate of 18% per annum?
- (a) ₹ 9000 (b) ₹ 18000
(c) ₹ 15000 (d) None of these
20. Mr. X invested an amount for 2 years at 15 percent per annum at simple interest. Had the interest been compounded, he would have earned ₹ 450/- more as interest. What was the amount invested?
- (a) ₹ 22000 (b) ₹ 24000
(c) ₹ 25000 (d) None of these
21. Mr Sridharan invested money in two schemes A and B, offering compound interest at 8 percent per annum and 9 percent per annum respectively. If the total amount of interest accrued through the two schemes together in two years was ₹ 4818.30 and the total amount invested was ₹ 27,000, what was the amount invested in Scheme A?
- (a) ₹ 15,000 (b) ₹ 13,500
(c) ₹ 12,000 (d) Cannot be determined
22. Parameshwaran invested an amount of ₹ 12,000 at the simple interest rate of 10 percent per annum and another amount at the simple interest rate of 20 percent per annum. The total interest earned at the end of one year on the total amount invested became 14 percent per annum. Find the total amount invested.
- (a) ₹ 22,000 (b) ₹ 25,000
(c) ₹ 20,000 (d) ₹ 24,000
23. A father left a will of ₹ 68,000 to be divided between his two sons aged 10 years and 12 years such that they may get equal amount when each attains the age of 18 years. If the money is reckoned at 10% p.a., find how much each gets at the time of the will.
- (a) ₹ 30,000, ₹ 38,000 (b) ₹ 28,000, ₹ 40,000
(c) ₹ 32,000, ₹ 36,000 (d) Cannot be determined
24. Two equal sums of money were invested, one at 4% and the other at 4.5%. At the end of 7 years, the simple interest received from the latter exceeded to that received from the former by ₹ 31.50. Each sum was
- (a) ₹ 1,200 (b) ₹ 600
(c) ₹ 750 (d) ₹ 900
25. A sum of ₹ 725 is lent in the beginning of a year at a certain rate of interest. After 8 months, a sum of ₹ 362.50 more is lent but at the rate twice the former. At the end of the year, ₹ 33.50 is earned as interest from both the loans. What was the original rate of interest?
- (a) 3.6% (b) 4.5%
(c) 5% (d) 3.46%
26. David invested certain amount in three different schemes A, B and C with the rate of interest 10% p.a., 12% p.a. and 15% p.a. respectively. If the the total interest accrued in one year was ₹ 3200 and the amount invested in Scheme C was 150 % of the amount invested in Scheme A and 240% of the amount invested in Scheme B, what was the amount invested in Scheme B?
- (a) ₹ 5000 (b) ₹ 6500
(c) ₹ 8000 (d) cannot be determined
27. Subash purchased a refrigerator on the terms that he is required to pay ₹ 1,500 cash down payment followed by ₹ 1,020 at the end of first year, ₹ 1,003 at the end of second year and ₹ 990 at the end of third year. Interest is charged at the rate of 10% per annum. Calculate the cash price
- (a) ₹ 3,000 (b) ₹ 2,000
(c) ₹ 4,000 (d) ₹ 5,000
28. A owes B ₹ 1,573, payable $1\frac{1}{2}$ years hence. Also B owes A ₹ 1,444.50, payable 6 months hence. If they want to settle the account forthwith, keeping 14% as the rate of interest, then who should pay whom and how much?
- (a) A to B, ₹ 28.50 (b) B to A, ₹ 37.50
(c) A to B, ₹ 50 (d) B to A, ₹ 50
29. Seema invested an amount of ₹ 16,000 for two years on compound interest and received an amount of ₹ 17,640 on maturity. What is the rate of interest?
- (a) 5% pa (b) 8% pa
(c) 4% pa (d) Data inadequate
30. A finance company declares that, at a certain compound interest rate, a sum of money deposited by anyone will become 8 times in three years. If the same amount is deposited at the same compound rate of interest, then in how many year will it become 16 times?
- (a) 5 years (b) 4 years
(c) 6 years (d) 7 years
31. Two friends A and B jointly lent out ₹ 81,600 at 4% per annum compound interest. After 2 years A gets the same amount as B gets after 3 years. The investment made by B was
- (a) ₹ 40,000 (b) ₹ 30,000
(c) ₹ 45,000 (d) ₹ 38,000
32. A money-lender, lends a part of his money at 10% per annum and the rest at 15% per annum. His annual income is ₹ 1900. However, if he had interchanged the rate of interest on the two sums, he would have earned ₹ 200 more. The amount lent will fetch what 15%?
- (a) ₹ 6000 (b) ₹ 4000
(c) ₹ 10000 (d) ₹ 4400
33. The simple interest on a sum of money is $\frac{1}{9}$ th of the principal, and the number of years is equal to the rate per cent per annum. Find the rate per cent.
- (a) $3\frac{1}{3}$ % (b) 3%
(c) 10% (d) None of these
34. Amin borrowed some money from Vishwas. The rate of interest for first two years is 8% p.a., for the next three years is 11 % p.a. and for the period beyond 5 years 14% p.a. Vishwas got an amount of ₹ 10920 as an interest at the end of eight years. Then what amount was borrowed by Amin?
- (a) ₹ 12000 (b) ₹ 15000
(c) ₹ 1400 (d) None of these

Expert Level

- Nikhilesh invested certain amount in three different schemes A, B and C with the rate of interest 10 percent per annum 12 percent per annum and 15 percent per annum respectively. If the total interest accrued in one year was ₹ 3200 and the amount invested in scheme C was 150% of the amount invested in scheme A and 240% of the amount invested in scheme B, what was the amount invested in scheme B?
(a) ₹ 8000 (b) ₹ 5000
(c) ₹ 6500 (d) Cannot be determined
- If there are three sum of money P, Q and R so that P is the simple interest of Q and Q is the simple interest of R, rate % and time are same in each case, then the relation of P, Q and R is given by
(a) $P^2 = QR$ (b) $Q^2 = PR$
(c) $R^2 = PQ$ (d) $PQR = 100$
- The difference between the simple interest received from two different sources on ₹ 1500 for 3 years is ₹ 13.50. The difference between their rates of interest is:
(a) 0.1% (b) 0.2%
(c) 0.3% (d) 0.4%
- A person invested in all ₹ 2600 at 4%, 6% and 8% per annum simple interest. At the end of the year, he got the same interest in all the three cases. The money invested at 4% is:
(a) ₹ 200 (b) ₹ 600
(c) ₹ 800 (d) ₹ 1200
- A sum of money is accumulating at compound interest at a certain rate of interest. If simple interest instead of compound were reckoned, the interest for the first two years would be diminished by ₹ 20 and that for the first three years, by ₹ 61. Find the sum.
(a) ₹ 7,000 (b) ₹ 8,000
(c) ₹ 7,500 (d) ₹ 6,500
- A man borrows ₹ 6000 at 10% compound rate of interest. He pays back ₹ 2000 at the end of each year to clear his debt. The amount that he should pay to clear all his dues at the end of third year is
(a) ₹ 6000 (b) ₹ 3366
(c) ₹ 3060 (d) ₹ 3066
- Arun invested a sum of money at a certain rate of simple interest for a period of 4 yrs, the total interest earned by him would have been 50% more than the earlier interest amount. What was the rate of interest per cent per annum?
(a) 4 (b) 8
(c) 5 (d) Cannot be determined
- Mr. Duggal invested ₹ 20000 with rate of interest @ 20% per annum. The interest was compounded half-yearly for the first 1 year and in the next year it was compounded yearly. What will be the total interest earned at the end of 2 years?
(a) ₹ 8800 (b) ₹ 9040
(c) ₹ 8040 (d) ₹ 9800
- The compound interest accrued on an amount of ₹ 25500 at the end of 3 year is ₹ 8440.50. What would be the simple interest accrued on the same amount at the same rate in the same period?
(a) ₹ 4650 (b) ₹ 5650
(c) ₹ 6650 (d) ₹ 7650
- A certain sum of money amounts to ₹ 756 in 2 years and to ₹ 873 in 3.5 years. Find the sum and the rate of interest.
(a) 11% (b) 13%
(c) 15% (d) 19%
- A sum of ₹ 1000 after 3 years at compound interest becomes a certain amount that is equal to the amount that is the result of a 3 year depreciation from ₹ 1728. Find the difference between the rates of C.I. and depreciation. (Given C.I. is 10% p.a.) (Approximately)
(a) 3.33% (b) 0.66%
(c) 3% (d) 2%
- A property dealer bought a rectangular plot (of land) in Noida 5 years ago at the rate of ₹ 1000 per m². The cost of plot is increases by 5% in every 6 years and the worth of a rupee falls down at a rate of 2% in every 5 years. What is the approximate value of the land per meter² 25 years hence?
(a) ₹ 995 (b) ₹ 1134
(c) ₹ 1500 (d) ₹ 1495
- Hari Lal and Hari Prasad have equal amounts. Hari Lal invested all his amount at 10% compounded annually for 2 years and Hari Prasad invested 1/4 at 10% compound interest (annually) and rest at r% per annum at simple interest for the same 2 years period. The amount received by both at the end of 2 years is same. What is the value of r?
(a) 14% (b) 12.5%
(c) 10.5% (d) 11%
- A person lent out some money for 1 year at 6% per annum simple interest and after 18 months, he again lent out the same money at a simple interest of 24% per annum. In both the cases, he got ₹ 4704. Which of these could be the amount that was lent out in each case if interest is paid half-yearly?
(a) ₹ 4000 (b) ₹ 4400
(c) ₹ 4200 (d) ₹ 3600
- Three persons Amar, Akbar and Anthony invested different amounts in a fixed deposit scheme for one year at the rate of 12% per annum and earned a total interest of ₹ 3,240 at the end of the year. If the amount invested by Akbar is ₹ 5000 more than the amount invested by Amar and the invested by Anthony is ₹ 2000 more than the amount invested by Akbar, what is the amount invested by Akbar?
(a) ₹ 12,000 (b) ₹ 10,000
(c) ₹ 7000 (d) ₹ 5000

16. According to the 2001 census, the population growth rate of Lucknow is going to be an increasing. AP with first year's rate as 5% and common difference as 5%, but simultaneously the migration, rate is an increasing GP with first term as 1% and common ratio as 2. If population on 31 December 2000 is 1 million, then find in which year will Lucknow witness its first fall in population?
- (a) 2005 (b) 2006
(c) 2007 (d) 2008
17. Extending this plan, ISBI further announced that widows of all the martyrs can get the loans in which the proportion of

soft loan will be double. This increase in the proportion of the soft loan component is only applicable for the first year. For all subsequent years, the soft loan component applicable on the loan, follows the values provided in the table. The widow of a soldier takes ` 40,000 under scheme 1 in one account for 1 year and ` 60,000 under scheme 2 for 2 years. Find the total interest paid by her over the 2 year period.

- (a) ` 11,600 (b) ` 10,000
(c) ` 8800 (d) None of these

Test Yourself

- A person invested in all ₹ 2600 at 4%, 6% and 8% per annum simple interest. At the end of the year, he got the same interest in all the three cases. The money invested at 4% is
 - ₹ 200
 - ₹ 600
 - ₹ 800
 - ₹ 1200
- A sum of money invested at simple interest triples itself in 8 years at simple interest. Find in how many years will it become 8 times itself at the same rate?
 - 24 years
 - 28 years
 - 30 years
 - 21 years
- If the simple interest is 10.5% annual and compound interest is 10% annual, find the difference between the interests after 3 years on a sum of ₹ 1000.
 - ₹ 15
 - ₹ 12
 - ₹ 16
 - ₹ 11
- Rajesh gave ₹ 1200 on loan. Some amount he gave at 4% per annum on simple interest and remaining at 5% per annum on simple interest. After two years, he got ₹ 110 as interest. Then the amounts given at 4% and 5% per annum on simple interest are, respectively
 - ₹ 500, ₹ 700
 - ₹ 400, ₹ 800
 - ₹ 900, ₹ 300
 - ₹ 1100, ₹ 1100
- Find the compound interest on ₹ 64,000 for 1 year at the rate of 10% per annum compounded quarterly (to the nearest integer).
 - ₹ 8215
 - ₹ 8205
 - ₹ 8185
 - None of these
- If a principal P becomes Q in 2 years when interest R% is compounded half-yearly. And if the same principal P becomes Q in 2 years when interest S% is compounded annually, then which of the following is true?
 - $R > S$
 - $R = S$
 - $R < S$
 - $R \leq D$
- A sum of ₹ 8000 is borrowed at 5% p.a. compound interest and paid back in 3 equal annual instalments. What is the amount of each instalment?
 - ₹ 2937.67
 - ₹ 3000
 - ₹ 2037.67
 - ₹ 2739.76
- The rate of interest on a sum of money for the first two years is 6% p.a., for the next two years it is 7% p.a. and 8% p.a. for the period exceeding four years; all at simple interest. If a person earns an interest of ₹ 7,536 by the end of the seven years, what is the amount at the end of the period of investment?
 - ₹ 15,072
 - ₹ 11,304
 - ₹ 22,608
 - ₹ 21,308
- Subash purchased a refrigerator on the terms that he is required to pay ₹ 1,500 cash down payment followed by ₹ 1,020 at the end of first year, ₹ 1,003 at the end of second year and ₹ 990 at the end of third year. Interest is charged at the rate of 10% per annum. Calculate the cash price:
 - ₹ 3,000
 - ₹ 2,000
 - ₹ 4,000
 - ₹ 5,000
- Three amounts x, y and z are such that y is the simple interest on x and z is the simple interest on y. If in all the three cases, rate of interest per annum and the time for which interest is calculated is the same, then find the relation between x, y and z.
 - $xyz = 1$
 - $x^2 = yz$
 - $z = x^2y$
 - $y^2 = xz$
- Hans Kumar borrows ₹ 7000 at simple interest from the village moneylender. At the end of 3 years, he again borrows ₹ 3000 and closes his account after paying ₹ 4615 as interest after 8 years from the time he made the first borrowing. Find the rate of interest.
 - 3.5%
 - 4.5%
 - 5.5%
 - 6.5%
- A sum is divided between A and B in the ratio of 1 : 2. A purchased a car from his part, which depreciates $14\frac{2}{7}\%$ per annum and B deposited his amount in a bank, which pays him 20% interest per annum compounded annually. By what percentage will the total sum of money increase after two years due to this investment pattern (approximately)?
 - 20%
 - 26.66%
 - 30%
 - 25%
- Adam borrowed some money at the rate of 6% p.a. for the first two years, at the rate of 9% p.a. for the next three years, and at the rate of 14% p.a. for the period beyond five years. If he pays a total interest of ₹ 11,400 at the end of nine years, how much money did he borrow?
 - ₹ 10,000
 - ₹ 12,000
 - ₹ 14,000
 - ₹ 16,000
- A sum of ₹ 725 is lent in the beginning of a year at a certain rate of interest. After 8 months, a sum of ₹ 362.50 more is lent but at the rate twice the former. At the end of the year, ₹ 33.50 is earned as interest from both the loans. What was the original rate of interest?
 - 3.6%
 - 4.5%
 - 5%
 - None of these
- If a sum of money at compound interest amounts to thrice itself in 3 years, then in how many years will it be 9 times itself?
 - 12 years
 - 6 years
 - 9 years
 - 15 years

Hints & Solutions

Foundation Level

1. (b) From the formula, $I = Prt$, with $P = 5000$, $r = .11$, and $t = 11/12$ (in years). The total interest she will pay is
 $I = 5000(.11)(11/12) = 504.17$
 or ₹ 504.17

2. (a) After first year the amount
 $= 18750 \left(1 + \frac{4}{100}\right) = 18750 \left(\frac{104}{100}\right)$
 After 2nd year the amount $= 18750 \left(\frac{104}{100}\right) \left(\frac{108}{100}\right)$
 $= 18750 \left(\frac{26}{25}\right) \left(\frac{27}{25}\right) = 21060$
 $\therefore \text{C.I.} = 21060 - 18750 = ₹ 2310.$

3. (c) Rate of interest $= \frac{956 - 800}{3 \times 800} \times 100 = 6.50\%$

$$\begin{aligned} \text{Amount} &= 800 + \frac{800 \times 9.5}{100} \times 3 \\ &= 800 + 228 = ₹ 1028 \end{aligned}$$

4. (c) Let S.I. = ₹ x

$$\frac{1.53 \times 10^5 \times 20}{100} = 30600$$

$$\text{Monthly income} = \frac{30600}{12} = ₹ 2550$$

5. (b) Interest for one year $= ₹ 212.50 \times \frac{3}{100} \times 1 = ₹ \frac{51}{8}$

Thus in 8 years, the interest is ₹ 51.

6. (c) After first year, the value of the scooter
 $= \frac{25000 \times 80}{100} = ₹ 20,000$

After second year, the value of scooter = ₹ 16,000

After third year, the value of scooter = ₹ 12,800

7. (c) Checking with options, we find that after 13 years, population of the village A = $6800 - 120 \times 13 = 5240$
 And that of village B = $4200 + 80 \times 13 = 5240$

8. (b) Let amount invested at 12% be x and amount invested at 10% be y .
 According to question

$$130 = \frac{x \times 12 \times 1}{100} + \frac{y \times 10 \times 1}{100}$$

$$\Rightarrow 13000 = 12x + 10y \quad \dots(1)$$

$$\text{And } 134 = \frac{x \times 10 \times 1}{100} + \frac{y \times 12 \times 1}{100}$$

$$\Rightarrow 13400 = 10x + 12y \quad (2)$$

From equations (1) and (2)

$$x = 500$$

9. (b) Let the sum be ₹ x .

$$ATQ = \frac{100 \times 2x \times (y+2)}{100} = \frac{5 \times 100 \times x \times 2}{100}$$

$$\Rightarrow y = 3 \text{ years}$$

10. (b) It triple itself in 8 years, which makes interest equal to 200% of principal.

So, 200% is added in 8 years

Hence, 400% which makes the whole amount equal to five times of the principal, which will be added in 16 years

11. (b) Go through trial and error of the options. You will get:
 $20000 \times (1.3) = 26000$ (@ simple interest)
 $20000 \times 1.1 \times 1.1 \times 1.1 = 26620$ @ compound interest.
 Thus 20000 is the correct answer.

12. (c) The final value would be:
 $0.8 \times 63000 + 27000 \times 1.15 \times 1.15 = 86107.5$
 $\rightarrow$ Drop in value = 4.32%

13. (c) Solve using options. Option (c) fits the situation as:
 $12820 = 2000 + 2 \text{ years interest on } 2000 + 4000 + 1 \text{ years interest on } 4000 + 6000$ (use 10% compound interest for calculation of interest) $\rightarrow$
 $12820 = 2000 + 420 + 4000 + 400 + 6000$.
 Thus, option (c) fits the situation perfectly.

14. (c) Let the principal be P and rate of interest be $R\%$.

$$\therefore \text{Required ratio} = \left[\frac{\left(\frac{P \times R \times 6}{100} \right)}{\left(\frac{P \times R \times 9}{100} \right)} \right] = \frac{6PR}{9PR} = \frac{6}{9} = 2:3.$$

15. (c) The amount man gets after one year

$$= 6000 + \frac{6000 \times 5 \times 1}{100} - 1200$$

$$= 6000 + 300 - 1200 = 5100$$

$\therefore$ Amount after two years i.e., at the beginning of the third year

$$= 5100 + \frac{5100 \times 5 \times 1}{100} - 1200$$

$$= 5100 + 255 - 1200 = 4155$$

Hence option (c)

16. (d) Let the sum be ₹ x .

$$\therefore \frac{x \times 11 \times 5}{100} - \frac{x \times 8 \times 6}{100} = 1008$$

$$\Rightarrow \frac{7x}{100} = 1008$$

$$\Rightarrow x = 14400$$

17. (c) Difference in interest = $236.25 - 225 = ₹ 11.25$
This difference is the simple interest over ₹ 225 for one year. Hence, rate of interest

$$= \frac{11.25 \times 100}{225 \times 1} \quad 5\%$$

18. (b) Let the time be x years. Then,

$$\left(\frac{500 \times 3 \times x}{100} \right) + \left(\frac{600 \times 9 \times x}{100 \times 2} \right) = 126$$

$$\Leftrightarrow 15x + 27x = 126 \Leftrightarrow 42x = 126 \Leftrightarrow x = 3.$$

$$\therefore \text{Required time} = 3 \text{ years}$$

19. (b) Amount = $\left[1600 \times \left(1 + \frac{5}{2 \times 100} \right)^2 \right]$
 $+ 1600 \times \left(1 + \frac{5}{2 \times 100} \right)$
 $= \left[1600 \times \frac{41}{40} \times \frac{41}{40} + 1600 \times \frac{41}{40} \right]$
 $= \left[1600 \times \frac{41}{40} \left(\frac{41}{40} + 1 \right) \right] = \left[\frac{1600 \times 41 \times 81}{40 \times 40} \right]$
 $= ₹ 3321.$

$$\text{C.I.} = ₹ (3321 - 3200) = ₹ 121$$

20. (c) Tripling in 8 years means that the interest earned in 8 years is equal to 200% of the capital value. Thus, interest per year (simple interest) is 25% of the capital. In 20 years, total interest earned = 500% of the capital and hence the capital would become 6 times its original value.
21. (d) The yearly increase in the population is 3%. Thus, the population would increase by 3% each year. 200000 would become 206000 while 206000 would become 212180.
22. (b) Solve through options to see that the value of 1200000 fits the given situation.
23. (b) Solve using options. If the price is 27000, the interest on 12000 (after subtracting the down payment) would be 16% of 12000 = 1920. Hence, the total amount paid would be 28920.
24. (c) On the second year (in terms of C.I.) is

$$\frac{P \left(1 + \frac{r}{100} \right)^2}{\left(P + \frac{Pr}{100} \right)} = \frac{6}{5} \Rightarrow \left(1 + \frac{r}{100} \right) = \frac{6}{5}$$

$$\Rightarrow r = 20\%$$

25. (b) Let the principal be x , then

$$\frac{\left(\frac{x}{7} \times 4 + \frac{x}{2} \times 5 + \frac{5x}{14} \times 6 \right)}{100} = 730$$

$$\Rightarrow x = 14000$$

Alternatively : Go through suitable options.

Choose any middlemost option so that if the chosen option is not correct, then you can determine that whether you have to increase or decrease the value of the choices given.

26. (d) Go through options

$$1.8 + \frac{1.8 \times 6 \times 10}{100} = 1.6 + \frac{1.6 \times 8 \times 10}{100}$$

Hence (d) is correct.

$$\text{Alternatively : } P_1 + \frac{P_1 \times 6 \times 10}{100} = P_2 + \frac{P_2 \times 8 \times 10}{100}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{9}{8}$$

$$\text{Share of elder brother} = \frac{340000 \times 9}{17} = ₹ 180000$$

27. (c) Principal for next month = $440 - 200 = 240$

$$\text{Amount paid after next month} = 244$$

$$\text{Therefore interest charged at ₹ 240} = 4$$

$$\therefore 4 = \frac{240 \times r \times 1}{12 \times 100}$$

$$r = 20\% \text{ per annum}$$

Standard Level

1. (a) Let the sum be ₹ P . Then,

$$\Rightarrow \left[P \left(1 + \frac{10}{100} \right)^2 - P \right] = 1260$$

$$\Rightarrow \left[P \left(\frac{11}{10} \right)^2 - 1 \right] = 1260$$

$$\therefore \text{Sum} = ₹ 6000$$

$$\text{So, S.I.} = \left(\frac{6000 \times 4 \times 5}{100} \right) = ₹ 1200$$

2. (b) Let Manish add ₹ x to the borrowed money.

$$\text{Then, } \frac{(1150 + x) \times 3 \times 9}{100} - \frac{1150 \times 3 \times 6}{100} = 274.95$$

$$\Rightarrow 27(1150 + x) - 18(1150) = 274.95 \times 100$$

$$\Rightarrow 27x = 17145$$

$$\Rightarrow x = 635$$

$$\text{So, money lent to Sunil} = ₹ (1150 + x)$$

$$= ₹ (1150 + 635) = ₹ 1785$$

3. (a) Simple interest for 5 years = ₹ 300

Now, when principal is trebled, the simple interest for 5 years will also treble the simple interest on original principal for the same period. Thus, S.I. for last 5 years when principal is trebled.

$$= 3 \times 300 = ₹ 900$$

$$\therefore \text{Total SI for 10 years} = 300 + 900 = ₹ 1200$$

4. (a) Let the sum be ₹ x .

$$\therefore \text{Interest} = \frac{x \times 8 \times 4}{100} = \frac{32x}{100}$$

$$x - \frac{32x}{100} = \frac{68x}{100}$$

When interest is $\frac{68x}{100}$ less, the sum is ₹ x .

$\therefore$ When interest is ₹ 340 less, the sum is

$$\frac{x}{68x} \times 100 \times 340 = ₹ 500$$

Direct Formula:

$$\text{Sum} = \frac{100}{100 - 8 \times 4} \times 340 = \frac{100 \times 340}{68} = ₹ 500$$

5. (a) $\text{Sum} = \frac{\text{Difference in interest} \times 100}{\text{Times} \times \text{Difference in rate}}$

$$= \frac{24 \times 100}{2 \times 1} = ₹ 1200$$

6. (d) Difference in amounts = $2977.54 - 2809 = ₹ 168.54$
Now, we see that ₹ 168.54 is the interest on ₹ 2809 in one year (it is either simple or compound interest because both are the same for a year).

$$\text{Hence, rate of interest} = \frac{168.54 \times 100}{2809} = 6\%$$

Now, for the original sum,

$$2809 = x \left(1 + \frac{6}{100} \right)^2$$

$$\text{Or, } 2809 = x \left(\frac{53}{50} \right)^2$$

$$\therefore x = \frac{2809 \times 50 \times 50}{53 \times 53} = ₹ 2500$$

7. (b) Let sum be x . Then, S.I. = x .

$$\text{I. Time} = \frac{100 \times x}{x \times \frac{50}{3}} = 6 \text{ years (False)}$$

$$\text{II. Time} = \frac{100 \times x}{x \times 20} = 5 \text{ years (True)}$$

III. Suppose sum = x . Then, S.I. = x and Time = 5 years.

$$\text{Rate} = \left(\frac{100 \times x}{x \times 5} \right) \% = 20\%$$

Now, sum = x , S.I. = $3x$ and Rate = 20%.

$$\therefore \text{Time} = \left(\frac{100 \times 3x}{x \times 20} \right) \text{ years} = 15 \text{ years (False)}$$

So, II alone is correct.

8. (d) Let sum = x . Then, S.I. = $\frac{9}{16}x$.

Let rate = $R\%$ and time = R years.

$$\therefore \left(\frac{x \times R \times R}{100} \right) = \frac{9x}{16} \Leftrightarrow R^2 = \frac{900}{16}$$

$$\Leftrightarrow R = \frac{30}{4} = 7\frac{1}{2}$$

Hence, time = $7\frac{1}{2}$ years.

9. (d) Let the sum be ₹ x , rate be $R\%$ p.a. and time be T years.

$$\text{Then, } \left[\frac{x \times (R+2) \times T}{100} \right] - \left(\frac{x \times R \times T}{100} \right) = 108$$

$$\Leftrightarrow 2xT = 10800 \quad \dots(i)$$

$$\left[\frac{x \times R \times (T+2)}{100} \right] - \left(\frac{x \times R \times T}{100} \right) = 180$$

$$\Leftrightarrow 2xR = 18000 \quad \dots(ii)$$

Clearly, from (i) and (ii), we cannot find the value of x .

So, the data is inadequate.

10. (b) Let the required rate be R . Then,

$$\left(\frac{20000 \times 8 \times 1}{100} \right) + \left(4000 \times \frac{15}{2} \times \frac{1}{100} \right)$$

$$+ \left(1400 \times \frac{17}{2} \times \frac{1}{100} \right) + \left(2600 \times R \times \frac{1}{100} \right)$$

$$= \left(\frac{813}{10000} \times 10000 \right)$$

$$\Leftrightarrow 160 + 300 + 119 + 26R = 813 \Leftrightarrow R = 9.$$

11. (d) $\text{Balance} = \left\{ \left[12500 \times \left(1 + \frac{20}{100} \right)^3 \right] - \left[2000 \times \left(1 + \frac{20}{100} \right)^2 + 2000 \times \left(1 + \frac{20}{100} \right) + 2000 \right] \right\}$

$$= \left\{ \left[12500 \times \frac{5}{6} \times \frac{6}{5} \times \frac{6}{5} \right] - \right.$$

$$\left. \left[2000 \times \frac{6}{5} \times \frac{6}{5} + 2000 \times \frac{6}{5} + 2000 \right] \right\}$$

$$= ₹ [21600 - (2880 + 2400 + 2000)] = ₹ 14320.$$

12. (b) $\frac{F \times (0.06) \times 6}{(38800 - F) \times 0.05 \times 2} = 5/4$

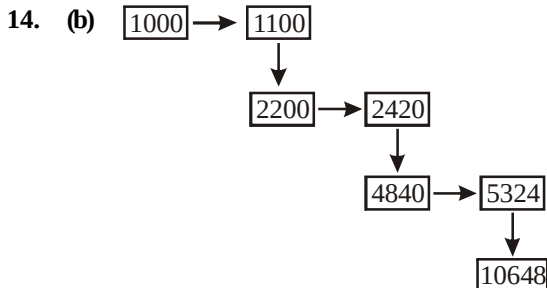
where F is the first part.

$$1.44F = 19400 - 0.5F$$

$$F = 19400/1.94 = 10000$$

Thus, the second part = $38800 - 10000 = 28800$

13. (b) Go through trial and error of the options. You will get:
 $20000 \times (1.3) = 26000$ @ simple interest
 $20000 \times 1.1 \times 1.1 \times 1.1 = 26620$ @ compound interest.
 Thus, 20000 is the correct answer.



15. (a) $x \left(1 + \frac{r}{100}\right)^2 = 4x \left(1 - \frac{r}{100}\right)^2$

$$\left(1 + \frac{r}{100}\right) = 2 \left(1 - \frac{r}{100}\right)$$

$$\frac{3r}{100} = 1$$

$$r = \frac{100}{3} = 33\frac{1}{3}\% = 33.33\%$$

16. (a) The original population was 1530 thousand
 $\Rightarrow$ No. of births was 53.2% of 1530 thousand
 $= 813960$
 $\Rightarrow$ No. of deaths was 31.2% of 1530 thousand $= 47360$
 $\Rightarrow$ Net increase in population $= 813960 - 47360$
 $= 336600$

Examination method : Net increase $= (53.2 - 31.2)\%$ of total

17. (a) Let the Principal $= P$

Then $\frac{P \times 8 \times 4}{100} + \frac{P \times 10 \times 6}{100} + \frac{P \times 12 \times 5}{100}$

$$= 12160$$

$$\Rightarrow 152P = 12160 \times 100$$

or $\frac{12160 \times 100}{152} = \text{` 8000}$

18. (c) For 3 years:

$$\text{Diff.} = \frac{\text{Sum} \times (\text{rate})^2 (300 + \text{rate})}{(100)^3}$$

$$= \frac{2000 \times 10 \times 10 \times 310}{100 \times 100 \times 100} = \text{` 62}$$

19. (d) Let the amount deposited at the rate of 15% per annum be x .
 $15\% \text{ of } x + 18\% \text{ of } (25000 - x) = 4050$
 or, $15\% \text{ of } x + 18\% \text{ of } 25000 - 18\% \text{ of } x = 4050$
 or, $3\% \text{ of } x = 4500 - 4050 = 450 \Rightarrow x = \text{` 15000}$
 $\therefore$ Amount deposited at 18%
 $= (25000 - 15000) = \text{` 10000}$

20. (d) $\frac{30p}{100} + 450 = \left[p \left(1 + \frac{15}{100}\right) - p \right]$

$$\Rightarrow p = \text{` 20,000.}$$

21. (c) Let, in scheme A, Sridharan invest x .
 Then, his investment in scheme B $= \text{` } (27000 - x)$.
 Now,

$$x \left(1 + \frac{8}{100}\right)^2 + (27000 - x) \left(1 + \frac{9}{100}\right)^2$$

$$- 27000 = 4818.30$$

$$\text{or, } x(1.08)^2 + (27000 - x)(1.09)^2 = 31818.30$$

$$\text{or, } 1.1664x + 32078.7 - 1.1881x = 31818.30$$

$$\text{or, } 0.0217x = 260.4$$

$$\text{or, } x = \frac{260.4}{0.0217} = \text{` 12000}$$

22. (c) Let the amount invested at 20% rate be x . According to the question,

$$12000 \times \frac{10}{100} + x \times \frac{20}{100} = (12000 + x) \times \frac{14}{100}$$

$$\text{or, } 1200 + \frac{x}{5} = 1680 + \frac{7}{50}x$$

$$\text{or, } \frac{x}{5} - \frac{7}{50}x = 480$$

$$\text{or, } \frac{3}{50}x = 480$$

$$\therefore x = \text{` 8000}$$

$$\therefore \text{Total amount invested} = (12000 + 8000) = \text{` 20000}$$

23. (c) Let one gets $= x$
 then, second gets $= \text{` } (68,000 - x)$
 Given : $A_1 = A_2$

$$x + \frac{x \times 10 \times 8}{100} = (68,000 - x) + \frac{(68000 - x) \times 10 \times 6}{100}$$

$$\Rightarrow x[100 + 80] = (68,000 - x)[100 + 60]$$

$$\Rightarrow \frac{180x}{160} = 68,000 - x$$

$$\Rightarrow 34x = 68000 \times 16 \Rightarrow x = \text{` 32,000}$$

$$\therefore \text{second gets} = \text{` 36,000}$$

24. (d) Difference of S.I. $= \text{` } \sqrt{31.50}$

Let each sum be x . Then

$$\frac{x \times 4 \times \frac{1}{2} \times 7}{100} - \frac{x \times 4 \times 7}{100} = 31.50$$

$$\text{or } \frac{7x}{100} \times \frac{1}{2} = \frac{63}{2}$$

$$\text{or } x = \text{` 900}$$

25. (d) Let the original rate be $R\%$. Then, new rate = $(2R)\%$.

$$\therefore \left(\frac{725 \times R \times 1}{100} \right) + \left(\frac{362.50 \times 2R \times 1}{100 \times 3} \right) = 33.50$$

$$\Rightarrow (2175 + 725)R = 33.50 \times 100 \times 3 = 10050$$

$$\Rightarrow R = \frac{10050}{2900} = 3.46\%$$

26. (a) Let x, y and z be the amounts invested in schemes A, B and C respectively. Then,

$$\left(\frac{x \times 10 \times 1}{100} \right) + \left(\frac{y \times 12 \times 1}{100} \right) + \left(\frac{z \times 15 \times 1}{100} \right) = 3200$$

$$\Rightarrow 10x + 12y + 15z = 320000 \quad \dots (1)$$

$$\text{Now, } z = 240\% \text{ of } y = \frac{12}{5}y \quad \dots (2)$$

$$\text{And, } z = 150\% \text{ of } x = \frac{3}{2}x$$

$$\Rightarrow x = \frac{2}{3}z = \left(\frac{2}{3} \times \frac{12}{5}y \right) = \frac{8}{5}y \quad \dots (3)$$

From (1), (2) and (3), we have :

$$16y + 12y + 36y = 320000 \Rightarrow 64y = 320000 \Rightarrow y = 5000.$$

$\therefore$ Sum invested in scheme B = ₹ 5000.

27. (c) Cash down payment = ₹ 1500

Let ₹ x becomes ₹ 1020 at the end of first year.

$$\text{Then, } 1020 = x \left(1 + \frac{10}{100} \right)$$

$$\text{or } x = \frac{1020 \times 100}{110} = ₹ 927.27$$

$$\text{Similarly, } 1003 = y \left(1 + \frac{10}{100} \right)^2$$

$$\text{or } y = \frac{1003 \times 20 \times 20}{22 \times 22} = ₹ 828.92$$

$$\text{and } z = \frac{990 \times 20 \times 20 \times 20}{22 \times 22 \times 22} = ₹ 743.80$$

$$\text{Hence, CP} = 1500 + 927.27 + 828.92 + 743.80 \\ = 3999.99 \text{ or } ₹ 4000.$$

28. (d) 14% in 1.5 yrs will be 21%
in 6 months will be 7%

$$\therefore \text{A's debt} = \frac{1573 \times 100}{121} = ₹ 1300$$

$$\text{B's debt} = \frac{1444.5 \times 100}{107} = ₹ 1350$$

Hence, B must pay ₹ 50 to A.

29. (a) Amount = ₹ 17640, Principal = ₹ 16000

Time = 2 yrs, Rate = R

$$17640 = 16000 \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow \frac{17640}{16000} = \left(1 + \frac{R}{100} \right)^2 \Rightarrow 1.1025 = \left(1 + \frac{R}{100} \right)^2$$

$$\Rightarrow 1 + \frac{R}{100} = 1.05 \Rightarrow \frac{R}{100} = 1.05 - 1 = 0.05$$

$$\Rightarrow R = 5\%$$

30. (b) Given $8P = P \left(1 + \frac{r}{100} \right)^3$

Where P = Principal amount,

r = Compound interest rate

$$\Rightarrow r = 100\%$$

$\therefore$ let the time in which the principal amount becomes 16 times be n

$$\text{Then } 16P = P \left(1 + \frac{100}{100} \right)^n$$

$$\Rightarrow 16 = 2^n \Rightarrow n = 4 \text{ yrs.}$$

31. (a) Let A lent ₹ x and B lent ₹ y

Since, A and B together lent out ₹ 81600

$$\therefore x + y = 81,600$$

Now, given (r) Rate = 4%

$$\therefore 1 + r = 1 + \frac{4}{100} = \frac{26}{25}$$

According to the question, we have

$$\frac{x}{y} = \left(\frac{26}{25} \right)^{3-2} = \frac{26}{25}$$

$$\therefore \text{Investment made by B} = 81600 \times \frac{25}{51} = 40,000$$

32. (a) Let the amount lent = P_1 at 15% and P_2 at 10%
According first condition.

$$\frac{P_1 \times 15 \times 1}{100} + \frac{P_2 \times 10 \times 1}{100} = 1900$$

$$15P_1 + 10P_2 = 1900 \times 100$$

....(1)

According to second condition.

$$\frac{P_1 \times 10 \times 1}{100} + \frac{P_2 \times 15 \times 1}{100} = 1900 + 200 = 2100$$

$$10P_1 + 15P_2 = 2100 \times 100 \quad (2) \times 10$$

$$15P_1 + 10P_2 = 1900 \times 100 \quad \dots(1) \times 15$$

$$100P_1 + 150P_2 = 2100000$$

$$225P_1 + 150P_2 = 2850000$$

$$\underline{\hspace{1cm}}$$

$$125P_1 = 750000$$

$$P_1 = 6000$$

33. (a) Let principal = P , time = t years, rate = t

$$\text{Then, } \frac{Ptt}{100} = \frac{P}{9}$$

$$\therefore t^2 = \frac{100}{9} \quad \therefore t = \frac{10}{3} = 3\frac{1}{3}$$

$$\therefore \text{rate} = 3\frac{1}{3}\%$$

Direct formula:

$$\text{Rate} = \text{time} = \sqrt{100 \times \frac{1}{9}} = \frac{10}{3} = 3\frac{1}{3}\%$$

34. (a) Let 'x' be the amount borrowed by Amin.

$$\therefore \frac{x \times 2 \times 8}{100} + \frac{x \times 3 \times 11}{100} + \frac{x \times 3 \times 14}{100} = 10920$$

$$\text{or, } \frac{91}{100}x = 10920 \text{ or } x = \frac{10920 \times 100}{91} = 12000$$

Expert Level

1. (b) Ratio of Nikhilesh's investments in different schemes

$$= 100 : \frac{150 \times 100}{240} : 150 = 8 : 5 : 12$$

Now, according to the question,

$$\frac{8k \times 10}{100} + \frac{5k \times 12}{100} + \frac{12k \times 15}{100} = 3200$$

$$\text{or, } 80k + 60k + 180k = 3200 \times 100$$

$$\text{or, } 320k = 3200 \times 100 \text{ or, } k = 1000$$

$$\therefore \text{amount invested in scheme B will be} \\ = 1000 \times 5 = ₹ 5000$$

2. (b) $P = \frac{Q \times r \times t}{100}$ and $Q = \frac{R \times r \times t}{100}$

$$\Rightarrow \frac{P}{Q} = \frac{R}{R} = \frac{r \times t}{100}$$

$$\therefore Q^2 = PR.$$

3. (c) $\left(\frac{1500 \times R_1 \times 3}{100} \right) - \left(\frac{1500 \times R_2 \times 3}{100} \right) = 13.50$

$$\Rightarrow 4500(R_1 - R_2) = 1350 \Rightarrow R_1 - R_2 = \frac{1350}{4500} = 0.3\%$$

4. (d) Let the parts be x , y and $[2600 - (x + y)]$. Then,

$$\frac{x \times 4 \times 1}{100} = \frac{y \times 6 \times 1}{100} = \frac{[2600 - (x + y)] \times 8 \times 1}{100}$$

$$\therefore \frac{y}{x} = \frac{4}{6} = \frac{2}{3} \text{ or } y = \frac{2}{3}x.$$

$$\text{So, } \frac{x \times 4 \times 1}{100} = \frac{\left(2600 - \frac{5}{3}x \right) \times 8}{100}$$

$$\Rightarrow 4x = \frac{(7800 - 5x) \times 8}{3} \Rightarrow 52x = (7800 \times 8)$$

$$\Rightarrow x = \left(\frac{7800 \times 8}{52} \right) = 1200.$$

$$\therefore \text{Money invested at } 4\% = ₹ 1200.$$

5. (b) Let $I_1 = \frac{P \times r \times t}{100}$

$$\text{and } I_2 = P(1+i)^t - P = P[(1+i)^t - 1]$$

According to the question,

$$20 = P[(1+i)^2 - 1 - 2i] \text{ and } 61 = P[(1+i)^3 - 1 - 3i]$$

On dividing, we get

$$\frac{20}{61} = \frac{P[(1+i)^2 - 1 - 2i]}{P[(1+i)^3 - 1 - 3i]} = \frac{P(i)^2}{P(i^3 + 3i^2)}$$

$$= \frac{i^2}{i^3 + 3i^2} = \frac{1}{3+i}$$

$$\Rightarrow 60 + 20i = 61 \Rightarrow 20i = 1 \Rightarrow i = \frac{1}{20}$$

$$\text{As we know, } \frac{r}{100} = i \Rightarrow \frac{1}{20} = \frac{r}{100} \Rightarrow r = 5$$

$$\text{Hence, } P = 20 \times \frac{1}{i} \times \frac{1}{i} = 20 \times 20 \times 20 = 8000.$$

6. (b) Amount = 6000

Rate = 10%

$$\text{First year Interest} = \frac{6000 \times 10 \times 1}{100} = ₹ 600$$

At the end of first year amount

$$= 6000 + 600 - 2000 = 4600$$

At the end of second year

$$\text{Interest} = \frac{4600 \times 10 \times 1}{100} = 460$$

At the second year amount

$$= 4600 + 460 - 2000 = 3060$$

At the end of third year

$$\text{Interest} = \frac{3060 \times 10 \times 1}{100} = 306$$

Amount at the end of third year

$$= 3060 + 306 = ₹ 3366$$

Amount refund in third year = ₹ 3366

7. (d) For $T = 4$ years, $P = P$ and $R = R\%$ per annum

$$S.I. = \frac{P \times T \times R}{100} = \left(\frac{P \times 4 \times R}{100} \right) = \left(\frac{PR}{25} \right)$$

For $T = 6$ years, $P = P$ and $R = R\%$ per annum

$$S.I. = \frac{P \times T \times R}{100} = \left(\frac{P \times 6 \times R}{100} \right) = \left(\frac{3PR}{50} \right)$$

$$\text{Now, } 150\% \text{ of } \frac{P \times R}{25} = \frac{3PR}{50}$$

$$\Rightarrow \frac{150}{100} \times \frac{P \times R}{25} = \frac{3PR}{50} \Rightarrow R = x$$

So, the given data is insufficient to find the rate of interest per cent per annum.

8. (b) $P = ₹20000$, $R\% = 20\%$ per annum.

$$A = P \left(1 + \frac{R}{100}\right)^{2n} \times \left(1 + \frac{R}{100}\right)^n$$

$$A = ₹ \left[20000 \times \left(1 + \frac{20}{100}\right)^{2 \times 1} \times \left(1 + \frac{20}{100}\right)^1 \right]$$

$$= ₹ \left[20000 \times \frac{110}{100} \times \frac{110}{100} \times \frac{120}{100} \right] = ₹29040$$

$$\text{C.I.} = A - P = ₹(29040 - 20000) = ₹9040$$

9. (d) $P = ₹25500$, $n = 3$ years, $\text{C.I.} = ₹8440.50$
 $A = P + \text{C.I.} = ₹(25500 + 8440.50) = ₹33940.50$

$$A = P \left(1 + \frac{R}{100}\right)^n$$

$$\Rightarrow 33940.50 = 25500 \left(1 + \frac{R}{100}\right)^3$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^3 = \frac{33940.50}{25500} = \left(\frac{11}{10}\right)^3$$

$$\Rightarrow \left(1 + \frac{R}{100}\right)^3 = \left(\frac{11}{10}\right)^3 \Rightarrow 1 + \frac{R}{100} = \frac{11}{10}$$

$$\Rightarrow \frac{R}{100} = \left(\frac{11}{10} - 1\right) = \frac{1}{10}$$

$$\Rightarrow R = 10\% \text{ per annum.}$$

Now, $P = ₹25500$, $T = 3$ years, $R\% = 10\%$ per annum

$$\text{SI} = \frac{P \times T \times R}{100} = ₹ \left(\frac{25500 \times 3 \times 10}{100} \right) = ₹7650$$

10. (b) $P + \text{S.I. for 3.5 years} = ₹873$
 $P + \text{S.I. for 2 years} = ₹756$
 On subtracting, $\text{S.I. for 1.5 years} = ₹117$

$$\text{Therefore, S.I. for 2 years} = ₹ \frac{117}{1.5} \times 2 = ₹156$$

$$\therefore P = 756 - 156 = ₹600 \text{ and rate} = \frac{100 \times 156}{600 \times 2}$$

$$= 13\% \text{ per annum}$$

11. (d) The amount @ 10% C.I. could become ₹1331. Also, ₹1728 depreciated at $R\%$ has to become ₹1331.
 Thus, $1728 \times \left[\frac{(100-R)}{100}\right]^3 = 1331$ (approximately).
 The closest value of $R = 8\%$
 Thus, the difference is 2%.

12. (b) Total time = 25 + 5 = 30 years
 Again no. of time periods for cost increment
 $= \frac{30}{6} = 5$
 and no. of time periods for rupee depreciation
 $= \frac{30}{5} = 6$

Now, the net value of the plot = $1000 \times (1.05)^5 \times (0.98)^6$
 $\approx ₹1130$

13. (c) Let the amount of investment with each one be ₹400, then

$$\begin{array}{cc} \text{Hari Lal} & \text{Hari Prasad} \\ [400(1.1)^2] & [100(1.1)^2] + \left[300 + \frac{300 \times r \times 2}{100}\right] \end{array}$$

$$\Rightarrow r = 10.5\%$$

14. (c) By using option (c) $4200 + (4\% \text{ of } 4200) 3 \text{ times} = 4200 + 0.04 \times 3 \times 4200 = 4704$
 15. (b) 12% Rate of interest on the amount invested gives an interest of ₹3240. This means that $0.12A = 3240 \rightarrow A = ₹27000$. The sum of the investments should be ₹27000. If Akbar invests x , Amar invests $x - 5000$ and Anthony invests $x + 2000$. Thus:
 $x + x - 5000 + x + 2000 = 27000 \rightarrow x = 10000$.
 16. (b) Population growth rate according to the problem:
 Year 1 = 5%, year 2 = 10%, year 3 = 15%
 Year 4 = 20%, year 5 = 25%, year 6 = 30%
 Population decrease due to migration:
 Year 1 = 1%, year 2 = 2%, year 3 = 4%
 Year 4 = 8%, year 5 = 16%, year 6 = 32%
 Thus, the first fall would happen in 2006
 17. (b) Interest she would pay under scheme 1:
 Year 1 the entire loan would be @ 4% – hence interest on 40000 = ₹1600.
 Total interest = 1600
 Interest on loan 2:

In year 1 : 80% of the loan (i.e., 48000) would be on 5%, 12000 would be @ 10% – hence total interest = 3600
 Year 2 : 40% of the loan (24000) would be on 5%, while the remaining loan would be on 10% – hence total interest = 4800.
 Thus, total interest on the two loans would be $1600 + 3600 + 4800 = 10000$.

Explanation of Test Yourself

1. (d) Let the parts be x, y and $[2600 - (x + y)]$. Then,

$$\frac{x \times 4 \times 1}{100} = \frac{y \times 6 \times 1}{100} = \frac{[2600 - (x + y)] \times 8 \times 1}{100}$$

$$\therefore \frac{y}{x} = \frac{4}{6} = \frac{2}{3} \text{ or } y = \frac{2}{3}x.$$

$$\text{So, } \frac{x \times 4 \times 1}{100} = \frac{\left(2600 - \frac{5}{3}x\right) \times 8}{100}$$

$$\Rightarrow 4x = \frac{(7800 - 5x) \times 8}{3} \Rightarrow 52x = (7800 \times 8)$$

$$\Rightarrow x = \left(\frac{7800 \times 8}{52}\right) = 1200.$$

$$\therefore \text{Money invested at } 4\% = \text{₹ } 1200.$$

2. (b) In 8 years, the interest earned = 200%

Thus, per year interest rate = $200/8 = 25\%$

To become 8 times we need a 700% increase

$$700/25 = 28 \text{ years.}$$

3. (c) At 10% compound interest the interest in 3 years would be $33.1\% = \text{₹ } 331$

At 10.5% simple interest the interest in 3 years would be $31.5\% = \text{₹ } 315$

Difference = ₹ 16

4. (a) Let the amount given 4% per annum be ₹ x .
then, amount given at 5% per annum = ₹ $(1200 - x)$

$$\text{Now, } \frac{x \times 4 \times 2}{100} + \frac{(1200 - x) \times 5 \times 2}{100} = 110$$

$$\Rightarrow x = \text{₹ } 500$$

And, the amount given at 5% per annum
= ₹ $(1200 - x) = \text{₹ } (1200 - 500) = \text{₹ } 700$

5. (d) $64000 \times (1.025)^4 = 70644.025$.

Interest 6644.025

6. (c) Since the interest is compounded half yearly at $R\%$ per annum, the value of R would be lesser than the value of S . (Remember, half yearly compounding is always profitable for the depositor).

7. (a) Let the repayment annually be X . Then:

$8000 + 3$ years interest on 8000 (on compound interest of 5%) = $X + 2$ years interest on $X + X + 1$ years interest on $X + X \rightarrow X = 2937.67$

8. (c) Let the principal be ₹ x .

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

$$\therefore \frac{x \times 2 \times 6}{100} + \frac{x \times 2 \times 7}{100} + \frac{x \times 8 \times 3}{100} = 7536$$

$$\Rightarrow \frac{50x}{100} = 7536$$

$$\Rightarrow x = 2 \times 7536 = \text{₹ } 15072$$

$$\therefore \text{Amount} = 15072 + 7536 = \text{₹ } 22608$$

9. (c) Cash down payment = ₹ 1500

Let ₹ x becomes ₹ 1020 at the end of first year.

$$\text{Then, } 1020 = x \left(1 + \frac{10}{100}\right)$$

$$\text{or } x = \frac{1020 \times 100}{110} = \text{₹ } 927.27$$

$$\text{Similarly, } 1003 = y \left(1 + \frac{10}{100}\right)^2$$

$$\text{or } y = \frac{1003 \times 20 \times 20}{22 \times 22} = \text{₹ } 828.92$$

$$\text{and } z = \frac{990 \times 20 \times 20 \times 20}{22 \times 22 \times 22} = \text{₹ } 743.80$$

$$\text{Hence, CP} = 1500 + 927.27 + 828.92 + 743.80 \\ = 3999.99 \text{ or } \text{₹ } 4000.$$

10. (d) You can think about this situation by taking some values. Let $x = 100, y = 10$ and $z = 1$ (at an interest rate of 10%). We can see that $10^2 = 100 \rightarrow y^2 = xz$

11. (d) The interest would be paid on

7000 for 3 years + 10000 for 5 years.

@ 6.5% the total interest for 8 years

$$= 1365 + 3250$$

$$= \text{₹ } 4615$$

12. (a) Let the amounts be ₹ 100 and ₹ 200 respectively. The value of the 100 would become $100 \times \frac{6}{7} \times \frac{6}{7} = 3600/49 = 73.46$

The other person's investment of 200 would become $200 \times 1.2 \times 1.2 = 288$

The total value would become $288 + 73.46 = 361.46$

This represents approximately a 20% increase in the value of the amount after 2 year.

13. (b) Let the sum borrowed be x . Then,

$$\left(\frac{x \times 6 \times 2}{100}\right) + \left(\frac{x \times 9 \times 3}{100}\right) + \left(\frac{x \times 14 \times 4}{100}\right) = 11400$$

$$\Rightarrow \left(\frac{3x}{25} + \frac{27x}{100} + \frac{14x}{25}\right) = 11400 \Rightarrow \frac{95x}{100} = 11400$$

$$\Rightarrow x = \left(\frac{11400 \times 100}{95}\right) = 12000.$$

Hence, sum borrowed = ₹ 12,000.

14. (d) Let the original rate be $R\%$. Then, new rate = $(2R)\%$.

$$\therefore \left(\frac{725 \times R \times 1}{100}\right) + \left(\frac{362.50 \times 2R \times 1}{100 \times 3}\right) = 33.50$$

$$\Rightarrow (2175 + 725)R = 33.50 \times 100 \times 3 = 10050$$

$$\Rightarrow R = \frac{10050}{2900} = 3.46\%$$

15. (b) Let sum be P

$$3P - P\left(1 + \frac{r}{100}\right)^3 \Rightarrow \left(1 + \frac{r}{100}\right)^3 = 3 \quad \dots(1)$$

$$\text{Now, let } P\left(1 + \frac{r}{100}\right)^n = 9P$$

$$\Rightarrow \left(1 + \frac{r}{100}\right)^n = 9 = 3^2 = \left\{\left(1 + \frac{r}{100}\right)^3\right\}^2 \quad [\text{By (1)}]$$

$$\Rightarrow \left(1 + \frac{r}{100}\right)^n = \left(1 + \frac{r}{100}\right)^6$$

$$\Rightarrow n = 6$$



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8

RATIO, PROPORTION AND VARIATION

- Introduction
- Ratio
- Decimal and Percentage Value of a Ratio
- Properties of Ratios
- Uses of Ratios
- Comparison of Ratios
- Calculation of Percentage Change in Ratio Using PCG (Percentage Change Graphic)
- Proportion
- Properties of Proportion
- Variations
- Types of Variations
- Compound Variations

INTRODUCTION

Concepts of this chapter are very useful in solving the problems of Data Interpretation, where ratio change and ratio comparison are very popular type questions. In CAT and its equivalent other aptitude tests, questions based on this chapter are regularly asked either directly or indirectly. The questions of this chapter are based on conceptual clarity and different applications of ratio, proportion and variation.

RATIO

Ratio is the comparison between two quantities in terms of their magnitudes. The ratio of two quantities is equivalent to a fraction that one quantity is of the other.

For example, let Swati has 5 note books and Priya has 7 note books. Then the ratio of the number of books that have with Swati to the number of books that have with Priya is 5 is to 7.

This ratio is expressed as 5 : 7 or $\frac{5}{7}$, which is a quotient of 5 and 7.

Ratio of any two numbers a and b is expressed as $a : b$ or $\frac{a}{b}$. The numbers that form the ratio is called the terms of the ratio. The numerator of the ratio is called the antecedent and the denominator is called the consequent of the ratio.

DECIMAL AND PERCENTAGE VALUE OF A RATIO

A ratio can be expressed in decimal and percentage.

Decimal value of $\frac{3}{5} = 0.6$

To express the value of a ratio as a percentage, we multiply the ratio by 100.

Hence $\frac{3}{5} = \frac{3}{5} \times 100\% = 60\%$

To find the decimal value of any ratio, you may calculate the percentage value using the percentage rule (discussed in the chapter Percentage) and then shift the decimal point 2 places towards left. Hence the decimal value of a ratio whose percentage value is 54.82% will be 0.5482.

PROPERTIES OF RATIOS

- (i) The value of a ratio does not change when the numerator and denominator both are multiplied by the same quantity

i.e. $\frac{a}{b} = \frac{ka}{kb} = \frac{la}{lb} = \frac{ma}{mb}$ etc.

For example, $\frac{2}{3} = \frac{4}{6} = \frac{6}{9}$ etc.

- (ii) The value of a ratio does not change when the numerator and denominator both are divided by the same quantity

i.e. $\frac{a}{b} = \frac{a/k}{b/k} = \frac{a/l}{b/l} = \frac{a/m}{b/m}$ etc.

- (iii) The ratio of two ratios (or fractions) can be expressed as a ratio of two numbers.

$$\frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

For example, $\frac{5/7}{3/4} = \frac{5}{7} \times \frac{4}{3} = \frac{20}{21}$

- (iv) If either or both the terms of a ratio are a surd quantity, then ratio will never evolve into integral numbers unless the surds are present in both numerator and denominator and these both surds are equal and present as a factor of numerator and denominator.

For example:

- (a) $\frac{\sqrt{7}}{9}, \frac{\sqrt{3}}{\sqrt{5}}, \frac{2\sqrt{6}}{\sqrt{3}}, \frac{5+\sqrt{3}}{\sqrt{3}}, \frac{2-\sqrt{5}}{3+\sqrt{5}}$ etc. will never evolve into integers.

- (b) $\frac{5\sqrt{3}}{2\sqrt{3}}$ is evolve into integral numbers as $\frac{5}{2}$.

- (c) The formula for the area of an equilateral triangle is $\frac{\sqrt{3}(\text{side})^2}{4}$. Here we can safely assume that the area of

any equilateral triangle will have $\sqrt{3}$ in its answer except the case when fourth root of three is $\sqrt[4]{3}$ [or $(3)^{1/4}$]

present as a factor of the length of a side.

Area of the equilateral triangle length of whose side is $(3)^{1/4}$

$$= \frac{\sqrt{3} \times [(3)^{1/4}]^2}{4} = \frac{\sqrt{3} \times (3)^{1/2}}{4} = \frac{\sqrt{3} \times \sqrt{3}}{4} = \frac{3}{4} \text{ square units.}$$

- (v) If a ratio $\frac{a}{b}$ is compounded with itself, the resulting ratios

$\frac{a^2}{b^2}, \frac{a^3}{b^3}$ etc. are called duplicate ratio, triplicate ratio etc respectively of the ratio $\frac{a}{b}$.

$\frac{(a)^{1/2}}{(b)^{1/2}}, \frac{(a)^{1/3}}{(b)^{1/3}}$ are called sub-duplicate and sub-triplicate of ratio respectively of the ratio $\frac{a}{b}$.

- (vi) $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \dots = \frac{a_1 + a_2 + a_3 + \dots}{b_1 + b_2 + b_3 + \dots}$

This means that if two or more ratios are equal, then the ratio whose numerator is the sum of the numerators of all the ratios and denominator is the sum of the denominators of all the ratios is equal to the original ratio.

Since $\frac{35}{50} = \frac{7}{10}$

$$\therefore \frac{35}{50} = \frac{7}{10} = \frac{35+7}{50+10} = \frac{42}{60}$$

- (vii) If $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \dots, \frac{a_n}{b_n}$ are unequal ratios (or fractions), then

$$\frac{a_1 + a_2 + a_3 + \dots + a_n}{b_1 + b_2 + b_3 + \dots + b_n} \text{ lies between the lowest and the}$$

highest of these ratios.

- (viii) If the ratio $\frac{a}{b} > 1$ and k is a positive number, then

$$\frac{a+k}{b+k} < \frac{a}{b} \text{ and } \frac{a-k}{b-k} > \frac{a}{b}$$

Similarly, if $\frac{a}{b} < 1$, then

$$\frac{a+k}{b+k} > \frac{a}{b} \text{ and } \frac{a-k}{b-k} < \frac{a}{b}$$

- (ix) If $\frac{c}{d} > \frac{a}{b}$, then $\frac{a+c}{b+d} > \frac{a}{b}$

and if $\frac{c}{d} < \frac{a}{b}$, then $\frac{a+c}{b+d} < \frac{a}{b}$

Illustration 1: Salaries of Rajesh and Sunil are in the ratio of 2 : 3. If the salary of each one is increased by ₹ 4000 the new ratio becomes 40 : 57. What is Sunil's present salary ?

- (a) ₹ 17000 (b) ₹ 20000
(c) ₹ 25500 (d) None of these

Solution: (d) Let the salaries of Rajesh and Sunil be ₹ $2x$ and ₹ $3x$ respectively.

Then, $\frac{2x+4000}{3x+4000} = \frac{40}{57}$

or $114x + 228000 = 120x + 160000$

or $6x = 68000$

or $3x = ₹ 34000$

Illustration 2: The ratio between the present ages of P and Q is 5 : 8. After four years, the ratio between their ages will be 2 : 3. What is Q 's age at present ?

- (a) 36 years (b) 20 years
(c) 24 years (d) None of these

Solution: (d) $\frac{P}{Q} = \frac{5}{8}$ or $P = \frac{5Q}{8}$... (1)

$$\frac{P+4}{Q+4} = \frac{2}{3}$$

or $3P + 12 = 2Q + 8$

or $2Q - 3P = 4$... (2)

Putting value of P from eq. (1),

$$2Q - 3 \times \frac{5}{8}Q = 4 \Rightarrow Q = 32.$$

USES OF RATIOS

(i) As a Bridge between three or more Quantities

If $a : b = N_1 : D_1$
 $b : c = N_2 : D_2$

$c : d = N_3 : D_3$
and $d : e = N_4 : D_4$
Then $a : b : c : d : e = N_1 N_2 N_3 N_4 : D_1 N_2 N_3 N_4 : D_1 D_2 N_3 N_4 : D_1 D_2 D_3 N_4 : D_1 D_2 D_3 D_4$

Here

a is correspond to the product of all four numerators

$(N_1 N_2 N_3 N_4)$

b is correspond to the first denominator and the last three numerators $(D_1 N_2 N_3 N_4)$

c is correspond to the first two denominators and the last two numerators $(D_1 D_2 N_3 N_4)$

d is correspond to the first three denominators and the last numerators $(D_1 D_2 D_3 N_4)$

e is correspond to the product of all four denominators

$(D_1 D_2 D_3 D_4)$

This method is applied for any three or more ratios.

This can be understood by following illustrations:

Illustration 3: Ratio of the age of A and B is $3 : 5$ and ratio of the age of B and C is $4 : 7$. Find the ratio of the age of A and C .

Solution: $A : B = 3 : 5 ; B : C = 4 : 7$
 $\Rightarrow A : B : C = 3 \times 4 : 5 \times 4 : 5 \times 7 = 12 : 20 : 35$

Here

A is correspond to the product of both numerators (3×4)

B is correspond to the product of first denominator and second numerator (5×4)

and C is correspond to the product of both denominators (5×7)

Hence ratio of the age of A and $C = 12 : 35$

Conventional Method

LCM of 5 and 4 (the two values corresponding B 's amount) is 20.

Now convert B 's value in both ratio to 20.

Hence $A : B = 3 \times 4 : 5 \times 4 = 12 : 20$

$B : C = 4 \times 5 : 7 \times 5 = 20 : 35$

$\Rightarrow A : B : C = 12 : 20 : 35$

$\Rightarrow A : C = 12 : 35$

This conventional method will be long for more than three ratios.

Illustration 4: If $A : B = 4 : 5 ; B : C = 3 : 7 ; C : D = 6 : 7$
 $D : E = 12 : 17$

then find the value of ratio $A : E$.

Solution: $A : B : C : D : E = (4 \times 3 \times 6 \times 12) : (5 \times 3 \times 6 \times 12) : (5 \times 7 \times 6 \times 12) : (5 \times 7 \times 7 \times 12) : (5 \times 7 \times 7 \times 17)$

$\therefore A : E = (4 \times 3 \times 6 \times 12) : (5 \times 7 \times 7 \times 17) = 864 : 4165$

Note that here we have found the ratio of $A : E$ directly without finding the consolidate ratio $(A : B : C : D : E)$ of A, B, C, D and E .

Illustration 5: If $A : B = 1 : 2, B : C = 3 : 4$ and $C : D = 5 : 6$, then find the value of $D : C : B$.

Solution: $A : B = 1 : 2, B : C = 3 : 4$

$\therefore A : B : C = 3 : 6 : 8$

Now $C : D = 5 : 6$

$\therefore A : B : C : D = 15 : 30 : 40 : 48$

$\therefore D : C : B = 48 : 40 : 30$

or $= 24 : 20 : 15$.

Illustration 6: If $A : B : C = 2 : 3 : 4$, then find $\frac{A}{B} : \frac{B}{C} : \frac{C}{A}$.

Solution: $A : B : C = 2 : 3 : 4$

$\therefore \frac{A}{B} : \frac{B}{C} : \frac{C}{A} = \frac{2}{3} : \frac{3}{4} : \frac{4}{2}$

$= \frac{2}{3} \times \frac{4}{4} : \frac{3}{4} \times \frac{3}{3} : \frac{4}{2} \times \frac{6}{6}$

$= \frac{8}{12} : \frac{9}{12} : \frac{24}{12}$

$= 8 : 9 : 24$

Hence required ratio $= 8 : 9 : 24$.

(ii) A Consolidate Relation between three unknowns (say x, y, z) when two equations in these three unknowns are given

Two equations in three unknowns cannot be solved without having a third equation in these unknowns but a consolidate ratio relation between these unknowns can be found out as follows:

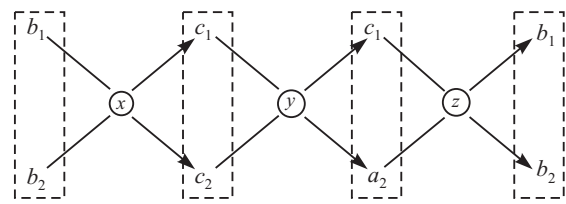
Let two equations containing three unknowns (x, y, z) are

$$a_1 x + b_1 y + c_1 z = 0$$

and

$$a_2 x + b_2 y + c_2 z = 0$$

Here $a_1, b_1, c_1, a_2, b_2, c_2$ are known coefficients.



Coefficients of middle terms Coefficients of last terms Coefficients of first terms Coefficients of middle terms

$$x : y : z = (b_1 c_2 - b_2 c_1) : (c_1 a_2 - c_2 a_1) : (a_1 b_2 - a_2 b_1)$$

or

$$\frac{x}{b_1 c_2 - b_2 c_1} = \frac{y}{c_1 a_2 - c_2 a_1} = \frac{z}{a_1 b_2 - a_2 b_1}$$

Denominator of the first ratio is obtained by subtracting the product of the coefficient b_2 and c_1 along the arrow pointing upward through x from the product of the coefficients b_1 and c_2 along the arrow pointing downward through x i.e. $(b_1 c_2 - b_2 c_1)$.

Similarly, we obtained the denominator of the second and third ratios as $c_1 a_2 - c_2 a_1$ and $a_1 b_2 - a_2 b_1$ respectively.

If $\frac{x}{b_1 c_2 - b_2 c_1} = \frac{y}{c_1 a_2 - c_2 a_1} = \frac{z}{a_1 b_2 - a_2 b_1} = k$, a constant.

Then $x = k(b_1 c_2 - b_2 c_1), y = k(c_1 a_2 - c_2 a_1)$ and

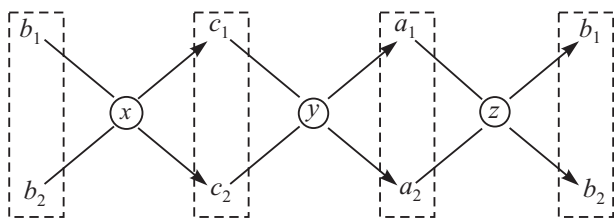
$z = k(a_1 b_2 - a_2 b_1)$

Illustration 7: Find a consolidate ratio relation between x, y and z , if

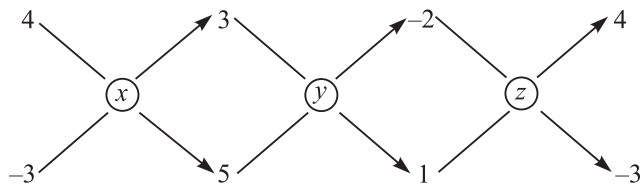
$$-2x + 4y + 3z = 0$$

$$x - 3y + 5z = 0$$

Solution:



Here $a_1 = -2, b_1 = 4, c_1 = 3, a_2 = 1, b_2 = -3, c_2 = 5$



$$x : y : z = (4 \times 5 - (-3) \times 3) : (3 \times 1 - 5 \times (-2)) : ((-2) \times (-3) - 1 \times 4)$$

or, $x : y : z = 29 : 13 : 2$

or, $\frac{x}{29} = \frac{y}{13} = \frac{z}{2} = k$ (let), a constant

$\therefore x = 29k, y = 13k$ and $z = 2k$

COMPARISON OF RATIOS

The value of a ratio is directly related to the value of numerator but inversely related to the value of denominator i.e. if (only numerator decrease)/(only denominator increases)/(numerator decreases and denominator increases) then the value of the ratio decreases and vice-versa.

There are eight cases in which we have to compare two ratios. In six out of these eight cases, we can easily compare the two ratios by keeping the above mentioned facts related to ratios in mind as shown in the following table.

S.No.	Cases	Comparison of Ratios	Comparison of Ratios (Example)
(i)	Numerator : Decreases Denominator : Fixed	(First Ratio) > (Second Ratio)	$\frac{5}{8} > \frac{3}{8}$
(ii)	Numerator : Increases Denominator : Fixed	(First Ratio) < (Second Ratio)	$\frac{4}{9} < \frac{7}{9}$
(iii)	Numerator : Fixed Denominator : Decreases	(First Ratio) < (Second Ratio)	$\frac{6}{7} < \frac{6}{5}$
(iv)	Numerator : Fixed Denominator : Increases	(First Ratio) > (Second Ratio)	$\frac{5}{8} > \frac{5}{9}$
(v)	Numerator : Decreases Denominator : Increases	(First Ratio) > (Second Ratio)	$\frac{6}{7} > \frac{5}{8}$
(vi)	Numerator : Increases Denominator : Decreases	(First Ratio) < (Second Ratio)	$\frac{3}{7} < \frac{5}{4}$

In the remaining two cases, we cannot compare the two ratios just by looking them.

The remaining two cases are

(vii) Numerator : Decreasing

Denominator : Decreasing

(viii) Numerator : Increasing

Denominator : Increasing

In both the remaining two cases (vii) and (viii), we can compare the two ratios by any one of the following four methods.

Method-I: Cross Multiplication Method

$$\frac{a}{b} > \frac{c}{d}, \text{ if } ad > bc$$

and $\frac{a}{b} < \frac{c}{d}, \text{ if } ad < bc$

For example $\frac{6}{7} > \frac{3}{5}$ because $6 \times 5 > 7 \times 3$

and $\frac{4}{5} < \frac{7}{8}$ because $4 \times 8 < 5 \times 7$

Method-II: Denominator Equating Method

By making the denominator of each ratio equal to the LCM of the denominators of both ratios, we can compare the two ratios by checking their numerators.

Illustration 8: Which of the two ratios $\frac{5}{6}$ and $\frac{8}{9}$ is greater.

Solution: LCM of 6 and 9 = 18

$$\frac{5}{6} = \frac{5 \times 3}{6 \times 3} = \frac{15}{18}$$

$$\frac{8}{9} = \frac{8 \times 2}{9 \times 2} = \frac{16}{18}$$

Since numerator of second ratio is greater than the numerator of first ratio,

$$\therefore \frac{16}{18} > \frac{15}{18} \Rightarrow \frac{8}{9} > \frac{5}{6}$$

Method-III: By Conversion of Ratios in Decimal Form

Consider two ratios $\frac{89}{76}$ and $\frac{68}{56}$

$$\begin{aligned} \frac{89}{76} &= \frac{76}{76} + \frac{130}{76 \times 10} = 1 + \frac{76}{76 \times 10} + \dots \\ &= 1 + 0.1 + \dots = 1.1 \text{ (approx.)} \end{aligned}$$

$$\begin{aligned} \frac{68}{56} &= \frac{56}{56} + \frac{120}{56 \times 10} = 1 + \frac{112}{56 \times 10} + \dots \\ &= 1 + 0.2 + \dots = 1.2 \text{ (approx.)} \end{aligned}$$

$$\therefore 1.2 > 1.1, \quad \therefore \frac{68}{56} > \frac{89}{76}$$

We find the value of the ratios upto that minimum place after decimal where, it will be easy to decide which one ratio is greater (or less).

Method IV: By Finding Percentage Change in Numerator and Percentage Change in Denominator

- (a) In case both numerator and denominator are decreases, if (Percentage decrease in numerator) <

(Percentage decrease in denominator)

then (First Ratio) < (Second Ratio)

And if (Percentage decrease in numerator) >

(Percentage decrease in denominator)

then (First Ratio) > (Second Ratio)

Consider the two ratios $\frac{175}{201}$ and $\frac{157}{180}$.

$$\text{Percentage decrease in numerator} = \frac{175 - 157}{175} \times 100 = \frac{1800}{175}$$

$$= \frac{1750}{175} + \frac{500}{175 \times 10} = 10 + \frac{350}{175 \times 10} + \frac{150}{175 \times 10}$$

$$= 10 + 0.2 + \frac{1500}{175 \times 100}$$

$$= 10.2 + \frac{1400}{175 \times 100} + \frac{1000}{175 \times 1000}$$

$$= 10.2 + 0.08 + \frac{1000}{175 \times 1000}$$

$$= 10.28\% \text{ (approx.)}$$

$$\text{Percentage decrease in denominator} = \frac{201 - 180}{201} \times 100$$

$$= \frac{2100}{201}$$

$$= \frac{2010}{201} + \frac{900}{201 \times 10} = 10 + \frac{804}{201 \times 10} + \frac{960}{201 \times 100}$$

$$= 10 + 0.4 + \frac{960}{201 \times 100} = 10.4\% \text{ (approx.)}$$

Since percentage decrease in numerator is less than the percentage decrease in denominator, hence first ratio $\left(\frac{175}{201}\right)$

is less than the second ratio $\left(\frac{157}{180}\right)$ i.e. $\frac{175}{201} < \frac{157}{180}$.

We find the percentage decrease in numerator and denominator upto which minimum place after decimal where it will be easy to decide percentage decrease in numerator is greater than or less than the percentage decrease in denominator.

- (b) In case both numerator and denominator increases, if (Percentage increase in numerator) >

(Percentage increase in denominator)

then (First Ratio) > (Second Ratio)

And if (Percentage increase in numerator) <

(Percentage increase in denominator)

then (First Ratio) < (Second Ratio)

Consider two ratios $\frac{79}{192}$ and $\frac{86}{208}$.

$$\text{Percentage increase in numerator} = \frac{86 - 79}{79} \times 100 = \frac{700}{79}$$

$$= \frac{632}{79} + \frac{680}{79 \times 10} = 8 + \frac{632}{79 \times 10} + \frac{480}{79 \times 100}$$

$$= 8 + 0.8 + \frac{480}{79 \times 100} = 8.8\% \text{ (approx.)}$$

$$\text{Percentage increase in denominator} = \frac{208 - 192}{192} \times 100$$

$$= \frac{1600}{192}$$

$$= \frac{1536}{192} + \frac{640}{192 \times 10} = 8 + \frac{576}{192 \times 10} + \frac{640}{192 \times 100}$$

$$= 8 + 0.3 + \frac{640}{192 \times 100} = 8.3\% \text{ (approx.)}$$

Since percentage increase in numerator is more than the percentage increase in denominator, therefore the first ratio is less than the second ratio

$$\text{i.e. } \frac{79}{192} < \frac{86}{208}$$

Out of the four methods discussed, this method-IV is less time consuming if you practise this method properly.

CALCULATION OF PERCENTAGE CHANGE IN RATIO USING PCG (PERCENTAGE CHANGE GRAPHIC)

We study the PCG (Percentage Change Graphic) in the chapter of Percentage. Using PCG, we can easily calculate the percentage change in a ratio.

Percentage change in between two ratios is found out in two stages as follow:

Original Ratio $\xrightarrow[\text{Effect of denominator}]{\text{Effect of numerator}}$ Intermediate Ratio $\rightarrow$ Final Ratio

For example, if ratio $\frac{25}{40}$ becomes $\frac{35}{50}$ then

Effect of numerator = $25 \rightarrow 35$ (40% increase)

Effect of denominator = $50 \rightarrow 40$ (20% decrease, reverse order)

Hence, overall effect on the ratio:

$$100 \xrightarrow[\substack{+40 \\ \text{(Numerator effect)}}]{40\% \uparrow} 140 \xrightarrow[\substack{-28 \\ \text{(Denominator effect)}}]{20\% \downarrow} 112$$

Therefore, overall effect = $(112 - 100 = 12)\%$ increase

PROPORTION

When two ratios are equal, the four quantities composing them are said to be proportionals. Hence, if $\frac{a}{b} = \frac{c}{d}$, then a, b, c, d are in proportional and is written as

$$a : b :: c : d$$

The terms a and d are called extremes while the terms b and c are called the means.

$$a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

Hence product of extremes = Product of means

Illustration 9: What must be added to each of the four numbers 10, 18, 22, 38 so that they become in proportion ?

Solution: Let the number to be added to each of the four numbers be x .

By the given condition, we get

$$(10 + x) : (18 + x) :: (22 + x) : (38 + x)$$

$$\Rightarrow (10 + x)(38 + x) = (18 + x)(22 + x)$$

$$\Rightarrow 380 + 48x + x^2 = 396 + 40x + x^2$$

Cancelling x^2 from both sides, we get

$$380 + 48x = 396 + 40x$$

$$\Rightarrow 48x - 40x = 396 - 380$$

$$\Rightarrow 8x = 16 \Rightarrow x = \frac{16}{8} = 2$$

Therefore, 2 should be added to each of the four given numbers.

Continue Proportion

(i) If $\frac{a}{b} = \frac{b}{c}$, then a, b, c , are said to be in continue proportion and vice-versa.

$$\text{Now } \frac{a}{b} = \frac{b}{c} \Rightarrow ac = b^2$$

Here b is called mean proportional and c is called third proportional of a and b .

(ii) If a, b, c and d are in continue proportion, then

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$$

$$\text{Also if } \frac{a}{b} = \frac{b}{c} = \frac{c}{d} = k \text{ (let), a constant}$$

Then $c = dk$

$$b = ck = dk \cdot k = dk^2$$

$$a = bk = dk^2 \cdot k = dk^3$$

PROPERTIES OF PROPORTION

(i) **Invertendo:** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$

(ii) **Alternando:** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$

(iii) **Componendo:** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a+b}{b} = \frac{c+d}{d}$

(iv) **Dividendo:** If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a-b}{b} = \frac{c-d}{d}$

(v) **Componendo and Dividendo:** If $\frac{a}{b} = \frac{c}{d}$, then

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Illustration 10: Find the value of $\frac{x+a}{x-a} + \frac{x+b}{x-b}$, if $x = \frac{2ab}{a+b}$

$$\text{Solution: } x = \frac{2ab}{a+b} \Rightarrow \frac{x}{a} = \frac{2b}{a+b}$$

By componendo – dividendo,

$$\frac{x+a}{x-a} = \frac{3b+a}{b-a}$$

$$\text{Similarly, } \frac{x}{b} = \frac{2a}{a+b}$$

$$\Rightarrow \frac{x+b}{x-b} = \frac{3a+b}{a-b}$$

$$\therefore \frac{x+a}{x-a} + \frac{x+b}{x-b} = \frac{3b+a}{b-a} + \frac{3a+b}{a-b} = \frac{-(3b+a)}{a-b} + \frac{3a+b}{a-b} = \frac{2a-2b}{a-b} = 2$$

VARIATIONS

We come across many situations in our day to day life where we see change in one quantity bringing change in the other quantity. For example:

- If the number of items purchased increases, its cost also increases.
- If the number of workers working to complete a job increases then days required to complete the job will decrease.

Here we observe that change in one quantity leads to change in other quantity. This is called variation.

TYPES OF VARIATIONS

There are three types of variations: Direct variation, Indirect variation and Compound variation.

(i) Direct Variations

There is a direct variation in two quantities if they are related in such a way that an increase in one causes an increase in the other in the same ratio or a decrease in one causes a decrease in the

other in the same ratio. This means that if one quantity becomes double then the other quantity also becomes double and if one quantity becomes half then the other quantity also becomes half etc. In other words if x and y are two variables then y varies

directly with x if the ratio $\frac{y}{x}$ is a constant.

' y varies directly with x ' is represented as $y \propto x$
 y varies directly as x is simply say that y varies as x .
 Here symbol ' $\propto$ ' means 'varies as'.

The representation $y \propto x$ can be converted to an equation $y = kx$, where k is a positive constant and called constant of proportionality.

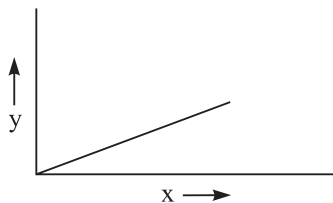
Hence $\frac{y}{x} = \text{constant}$

The equation $\frac{y}{x} = k$, means all ratios of a value of y with their corresponding value of x are equal.

If y_1, y_2 are two values of y corresponding to two values x_1 and x_2 of x , then $\frac{y_1}{x_1} = \frac{y_2}{x_2}$.

Graph

If y varies directly as x , then graph between x and y will be as shown below:



Some Examples of Direct Variations

- Number of persons $\propto$ Amount of work done
More number of persons, more work.
- Number of days $\propto$ Amount of work
More days, More work
- Working rate $\propto$ Amount of work
More working rate, more work
- Efficiency of worker $\propto$ Amount of work
More efficient worker, More work.

Illustration 11: A machine takes 5 hours to cut 120 tools. How many tools will it cut in 20 hours?

Solution: Here more time, more number of tools i.e. time and number of tools cut vary directly.

Let number of tools cut in 20 hours be ' x ', then

$$\frac{5}{120} = \frac{20}{x} \quad \left(\because \frac{y_1}{x_1} = \frac{y_2}{x_2} \right)$$

$$\Rightarrow x = \frac{20 \times 120}{5}$$

$$x = 480$$

Hence required number of tools = 480.

(ii) Inverse Variations

There is an inverse variation in two quantities if they are so related that an increase in one causes a decrease in the other in the same

ratio or vice-versa. This means that if one quantity becomes double then other quantity becomes half and if one quantity becomes one third then other quantity becomes thrice etc.

In other words if x and y are variables then y varies inversely with x , if xy is a constant.

' y varies inversely with x ' is represented as $y \propto \frac{1}{x}$.

Here symbol ' $\propto$ ' means 'varies as'. The representation $y \propto \frac{1}{x}$

can be converted to an equation $y = \frac{k}{x}$ or $xy = k$, where k is a positive constant, called constant of proportionality.

$\Rightarrow xy = \text{constant}$

The equation $xy = \text{constant}$, means all products of a value of y and their corresponding value of x are equal. That is if y_1, y_2 are two values of y corresponding to the values x_1, x_2 of x respectively, then $x_1 y_1 = x_2 y_2$

Graph

If y varies inversely as x , then graph between x and y will be as shown below:

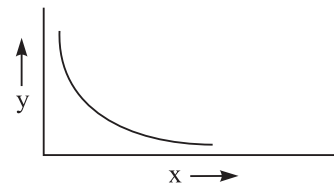


Illustration 12: If 900 persons can finish the construction of a building in 40 days, how many persons are needed to complete the construction of building in 25 days.

Solution: Let the required number of persons be ' x '. As the number of days required to complete the job is less, so more number of persons will be required. It is a case of inverse variation.

So $900 \times 40 = x \times 25$

$$\Rightarrow x = \frac{900 \times 40}{25} = 1440$$

Hence required number of persons = 1440.

COMPOUND VARIATIONS

In real life, there are many situations which involve more than one variation, i.e. change in one quantity depends on changes in two or more quantities either directly or inversely or by both.

Let x, y and z are variables, i.e. $y \propto x$

- (a) y varies directly as x when z is constant, i.e., $y \propto x$ and y varies directly as z when x is constant, i.e. $y \propto z$, then we say that y varies directly as the product of x and z .

Thus $y \propto xz$

or $y = k(xz)$, k is a positive constant

- (b) y varies directly as x when z is constant, i.e. $y \propto x$ and y varies inversely as z when x is constant i.e. $y \propto \frac{1}{z}$, then

$y \propto \frac{x}{z}$ or $y = k \left(\frac{x}{z} \right)$, where k is a positive constant.

(c) y varies inversely as x when z is constant i.e. $y \propto \frac{1}{x}$ and
 y varies inversely as z when x is constant then $y \propto \frac{1}{xz}$ or
 $y = \frac{k}{xz}$, where k is a positive constant.

Illustration 13: 25 horses eat 5 bags of corn in 12 days, how many bags of corn will 10 horses eat in 18 days ?

Solution: Here three quantities : number of horses (h), number of bags (b) and number of days (d) are involved.

Number of bags increases as number of hours increases. Also, number of bags increases as number of days increases.

$$\text{Hence } b \propto hd \Rightarrow \frac{b}{hd} = \text{constant}$$

$$\Rightarrow \frac{b_1}{h_1 d_1} = \frac{b_2}{h_2 d_2} \Rightarrow b_2 = \frac{b_1 h_2 d_2}{h_1 d_1}$$

$$\therefore b_2 = \frac{5 \times 10 \times 18}{25 \times 12} = 3$$

Hence number of bags required by 10 horses in 18 days = 3 bags.

Practice Exercises



Foundation Level

- Find the value of $\frac{x+a}{x-a} + \frac{x+b}{x-b}$, if $x = \frac{2ab}{a+b}$.
 (a) -2 (b) 2
 (c) 1 (d) -1
- A certain sum of money was divided among A, B and C in a certain way. C got half as much as A and B together got. A got one third of what B and C together got. What is the ratio of A's share to that of C's share?
 (a) 1:4 (b) 3:4
 (c) 4:1 (d) 3:5
- Two numbers are in the ratio of 3 : 4. If 5 is subtracted from each, the resulting numbers are in the ratio 2 : 3. Find the numbers
 (a) 12, 16 (b) 24, 32
 (c) 60, 80 (d) 15, 20
- The wages of labourers in a factory increased in the ratio 22 : 25 and there was a reduction in their number in the ratio 15 : 11. Find the original wage bill if the present bill is ₹ 5000.
 (a) ₹ 2500 (b) ₹ 3000
 (c) ₹ 5000 (d) ₹ 6000
- Which of the following numbers should be added to 11, 15, 17 and 23 so that they are in proportion?
 (a) 2 (b) 3
 (c) 5 (d) 1
- Find the fourth proportional to $12X^3$, $9aX^2$, $8a^3X$.
 (a) $4a^3$ (b) $6a^4$
 (c) $5a$ (d) $7a^5$
- Vijay decides to leave 100 acres of his land to his three daughters Vijaya, Sunanda and Anusuya in the proportion of one-third, one-fourth and one-fifth respectively. But Vijaya suddenly expires. Now how should Vijay divide the land between Sunanda and Anusuya?
 (a) $\frac{500}{9}$, $\frac{400}{9}$ (b) $\frac{450}{8}$, $\frac{350}{8}$
 (c) $\frac{420}{7}$, $\frac{280}{7}$ (d) $\frac{320}{7}$, $\frac{380}{7}$
- Find $a : b : c$, if $6a = 9b = 10c$.
 (a) 12 : 10 : 8 (b) 15 : 4 : 3
 (c) 15 : 18 : 9 (d) 15 : 10 : 9
- What is the least integer which when added to both terms of the ratio 5 : 9 will make a ratio greater than 7 : 10?
 (a) 6 (b) 8
 (c) 5 (d) 7
- If $a : b = 2 : 3$, $b : c = 3 : 4$, $c : d = 4 : 5$, find $a : b : c : d$.
 (a) 5 : 4 : 3 : 2 (b) 30 : 20 : 15 : 12
 (c) 2 : 3 : 4 : 6 (d) 2 : 3 : 4 : 5
- ₹ 1220 is divided, among A, B, C and D, such that B's share is $\frac{5}{9}$ th of A's; C's share is $\frac{7}{10}$ th of B's and D has $\frac{1}{3}$ as much as B and C together. Find A's share.
 (a) ₹ 540 (b) ₹ 802
 (c) ₹ 100 (d) ₹ 650
- In an examination, there are five subjects and each has the same maximum. A boy's marks are in the ratio 3 : 4 : 5 : 6 : 7 and his aggregate is $\frac{3}{5}$ th of the full marks. In how many subjects did he get more than 50% marks?
 (a) 1 (b) 2
 (c) 3 (d) 4
- Three friends started a business of renting out air conditioners by investing ₹ 20000, ₹ 24000 and ₹ 16000, respectively. C gets 20% of total profit for repair and maintenance of the air conditioner. If in a particular year, C gets ₹ 487.50 less than the total earnings of the other two, then the total profit for the year is :
 (a) ₹ 2812.50 (b) ₹ 3625.50
 (c) ₹ 4515.00 (d) None of these
- The ratio of the prices of two houses A and B was 4 : 5 last year. This year, the price of A is increased by 25% and that of B by ₹ 50000. If their prices are now in the ratio 9 : 10, the price of A last year was :
 (a) ₹ 3,60,000 (b) ₹ 4,50,000
 (c) ₹ 4,80,000 (d) ₹ 5,00,000
- The dimensions of a rectangular room when increased by 4 metres are in the ratio of 4 : 3 and when decreased by 4 metres, are in the ratio of 2 : 1. The dimensions of the room are
 (a) 6 m and 4 m (b) 12 m and 8 m
 (c) 16 m and 12 m (d) 24 m and 16 m

16. The sum of three numbers is 98. If the ratio of the first to the second is 2 : 3 and that of the second to the third is 5 : 8, then the second number is:
(a) 20 (b) 30
(c) 38 (d) 48
17. Two numbers are such as that square of one is 224 less than 8 times the square of the other. If the numbers are in the ratio of 3 : 4, they are
(a) 12, 16 (b) 6, 8
(c) 9, 12 (d) None of these
18. Tea worth ₹ 126 per kg and ₹ 135 per kg are mixed with a third variety in the ratio 1 : 1 : 2. If the mixture is worth ₹ 153 per kg, then the price of the third variety per kg is
(a) ₹ 169.50 (b) ₹ 170
(c) ₹ 175.50 (d) ₹ 180
19. In a mixture of 45 litres, the ratio of milk and water is 3 : 2. How much water must be added to make the ratio 9 : 11?
(a) 10 litres (b) 15 litres
(c) 17 litres (d) 20 litres
20. The ratio of the rate of flow of water in pipes varies inversely as the square of the radii of the pipes. What is the ratio of the rates of flow in two pipes of diameters 2 cm and 4 cm, respectively?
(a) 1 : 2 (b) 2 : 1
(c) 1 : 8 (d) 4 : 1
21. Given that 24 carat gold is pure gold. 18 carat gold is $\frac{3}{4}$ pure gold and 20 carat gold is $\frac{5}{6}$ pure gold. The ratio of the pure gold in 18 carat gold to the pure gold in 20 carat gold is :
(a) 3 : 8 (b) 9 : 10
(c) 15 : 24 (d) 8 : 5
22. If $\frac{y}{x-z} = \frac{y+x}{z} = \frac{x}{y}$, then find $x : y : z$.
(a) 1 : 2 : 3 (b) 3 : 2 : 1
(c) 4 : 2 : 3 (d) 2 : 4 : 7
23. Salaries of A, B and C were in the ratio 3 : 5 : 7, respectively. If their salaries were increased by 50%, 60% and 50% respectively, what will be the new ratio of their respective new salaries?
(a) 4 : 5 : 7 (b) 3 : 6 : 7
(c) 4 : 15 : 18 (d) 9 : 16 : 21
24. The average score of boys in an examination of a school is 71 and that of the girls is 73. The average score of the whole school in that examination is 71.8. Find the ratio of the number of boys to the number of girls that appeared in the examination.
(a) 4 : 5 (b) 3 : 2
(c) 3 : 5 (d) 5 : 2
25. Two casks of 48 L and 42 L are filled with mixtures of wine and water, the proportions in the two casks being respectively 13 : 7 and 18 : 17. If the contents of the two casks be mixed and 20 L of water is added to the whole, what will be the proportion of wine to water in the resultant solution?
(a) 21 : 31 (b) 12 : 13
(c) 13 : 12 (d) None of these
26. What amounts (in litres) of 90% and 97% pure acid solutions are mixed to obtain 21 L of 95% pure acid solution?
(a) 6 and 15 L (b) 14 and 15 L
(c) 12 and 15 L (d) 13 and 12 L
27. Arvind began a business with ₹ 550 and was joined afterwards by Brij with ₹ 330. When did Brij join, if the profits at the end of the year were divided in the ratio 10 : 3?
(a) After 4 months (b) After 6 months
(c) After 4.5 months (d) None of these
28. A, B and C are partners. A receives $\frac{9}{10}$ of the profit and B and C share the remaining profit equally. A's income is increased by ₹ 270 when the profit rises from 12 to 15%. Find the capital invested by B and C each
(a) ₹ 5000 (b) ₹ 1000
(c) ₹ 500 (d) ₹ 1500
29. A fort had provision of food for 150 men for 45 days. After 10 days, 25 men left the fort. The number of days for which the remaining food will last, is
(a) $29\frac{1}{5}$ (b) $37\frac{1}{4}$
(c) 42 (d) 54
30. In a mixture of 45 L, the ratio of milk and water is 2 : 1. If this ratio is to be 3 : 2, the quantity of water to be further added is
(a) 3 L (b) 5 L
(c) 8 L (d) None of these
31. If 40% of a number is equal to two-third of another number, what is the ratio of first number to the second number?
(a) 2 : 5 (b) 3 : 7
(c) 5 : 3 (d) 7 : 3
32. If the cost of printing a book of 320 leaves with 21 lines on each page and on an average 11 words in each line is ₹ 19, find the cost of printing a book with 297 leaves, 28 lines on each page and 10 words in each line.
(a) ₹ $22\frac{3}{8}$ (b) ₹ $20\frac{3}{8}$
(c) ₹ $21\frac{3}{8}$ (d) ₹ $21\frac{3}{4}$
33. A and B entered into a partnership with investments of ₹ 15000 and ₹ 40000 respectively. After 3 months A left from the business, at the same time C joins with ₹ 30000. At the end of 9 months, they got ₹ 7800 as profit. Find the share of B.
(a) ₹ 4800 (b) ₹ 600
(c) ₹ 2400 (d) ₹ 1200

34. The third proportional to $(x^2 - y^2)$ and $(x - y)$ is :
 (a) $(x + y)$ (b) $(x - y)$
 (c) $\frac{x+y}{x-y}$ (d) $\frac{x-y}{x+y}$
35. The sides of a triangle are in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$ and its perimeter is 104 cm. The length of the longest side is
 (a) 52 cm (b) 48 cm
 (c) 32 cm (d) 26 cm
36. Three friends A, B and C started a business by investing a sum of money in the ratio of 5 : 7 : 6. After 6 months C withdraws half of his capital. If the sum invested by 'A' is ₹ 40,000, out of a total annual profit of ₹ 33,000, C's share will be
 (a) ₹ 9,000 (b) ₹ 12,000
 (c) ₹ 11,000 (d) ₹ 10,000
37. The numbers of students speaking English and Hindi are in the ratio of 4:5. If the number of students speaking English increased by 35% and that speaking Hindi increased by 20%, what would be the new respective ratio?
 (a) 19:20 (b) 7:8
 (c) 8:9 (d) 9:10
38. The ratio of males and females in a city is 7 : 8 and the percentage of children among males and females is 25% and 20% respectively. If the number of adult females in the city is 156800 what is the total population?
 (a) 245000 (b) 367500
 (c) 196000 (d) 171500
39. A, B and C started a business with a total investment of ₹ 72000. A invests ₹ 6000 more than B and B invests ₹ 3000 less than C. If the total profit at the end of a year is ₹ 8640, find A's share.
 (a) ₹ 3240 (b) ₹ 2520
 (c) ₹ 2880 (d) ₹ 3360
40. A and B start a business with investments of ₹ 5000 and ₹ 4500 respectively. After 4 months, A takes out half of his capital. After two more months, B takes out one-third of his capital while C joins them with a capital of ₹ 7000. At the end of a year, they earn a profit of ₹ 5080. Find the share of each member in the profit.
 (a) A – ₹ 1400, B – ₹ 1900, C – ₹ 1780
 (b) A – ₹ 1600, B – ₹ 1800, C – ₹ 1680
 (c) A – ₹ 1800, B – ₹ 1500, C – ₹ 1780
 (d) A – ₹ 1680, B – ₹ 1600, C – ₹ 1800
41. A, B and C enter into a partnership. They invest ₹ 40,000, ₹ 80,000 and ₹ 1,20,000 respectively. At the end of the first year, B withdraws ₹ 40,000, while at the end of the second year, C withdraws ₹ 80,000. In what ratio will the profit be shared at the end of 3 years?
 (a) 2 : 3 : 5 (b) 3 : 4 : 7
 (c) 4 : 5 : 9 (d) None of these
42. Incomes of two companies A and B are in the ratio of 5 : 8. Had the income of company A been more by ₹ 25 lakh, the ratio of their incomes would have been 5 : 4. What is the income of company B?
 (a) ₹ 80 lakh (b) ₹ 50 lakh
 (c) ₹ 40 lakh (d) ₹ 60 lakh
43. Abhishek started a business investing ₹ 50,000. After one year he invested another ₹ 30,000 and Sudin also joined him with a capital of ₹ 70,000. If the profit earned in three years from the starting of business was ₹ 87,500, then find the share of Sudin in the profit.
 (a) ₹ 37,500 (b) ₹ 35,000
 (c) ₹ 38,281 (d) ₹ 52,500
44. In 1 kg mixture of sand and iron, 20% is iron. How much sand should be added so that the proportion of iron becomes 10%?
 (a) 1 kg (b) 200 gms
 (c) 800 gms (d) 1.8 kg
45. A started a business with ₹ 21,000 and is joined afterwards by B with ₹ 36,000. After how many months did B join if the profits at the end of the year are divided equally?
 (a) 3 (b) 4
 (c) 5 (d) 6
46. Mr AM, the magnanimous cashier at XYZ Ltd., while distributing salary, adds whatever money is needed to make the sum a multiple of 50. He adds ₹ 10 and ₹ 40 to A's and B's salary respectively and then he realises that the salaries of A, B and C are now in the ratio 4 : 5 : 7. The salary of C could be
 (a) ₹ 2300 (b) ₹ 2150
 (c) ₹ 1800 (d) ₹ 2100
47. When 30 percent of a number is added to another number the second number increases to its 140 per cent. What is the ratio between the first and the second number?
 (a) 3:4 (b) 4:3
 (c) 3:2 (d) None of these
48. The ratio of number of ladies to gents at a party was 1 : 2, but when 2 ladies and 2 gents left, the ratio became 1 : 3. How many people were originally present at the party?
 (a) 6 (b) 9
 (c) 12 (d) 10
49. A bag contains an equal number of one rupee, 50 paise and 25 paise coins respectively. If the total value is ₹ 35, how many coins of each type are there?
 (a) 20 coins (b) 30 coins
 (c) 28 coins (d) 25 coins

50. A and B invest ₹ 3,000 and ₹ 4,000 in a business. A receives ₹ 10 per month out of the profit as a remuneration for running the business and the rest of profit is divided in proportion to the investments. If in a year 'A' totally receives ₹ 390, what does B receive?
- (a) ₹ 375 (b) ₹ 360
(c) ₹ 350 (d) ₹ 260
51. If $f(x) = \frac{(x+1)}{(x-1)}$, then the ratio of x to $f(y)$ where $y = f(x)$ is
- (a) $x : y$ (b) $x^2 : y^2$
(c) $1 : 1$ (d) $y : x$
52. Three quantities A, B, C are such that $AB = KC$, where K is a constant. When A is kept constant, B varies directly as C; When B is kept constant, A varies directly as C and when C is kept constant, A varies inversely as B. Initially, A was at 5 and $A : B : C$ was $1 : 3 : 5$. Find the value of A when B equals 9 at constant C.
- (a) 8 (b) 8.33
(c) 9 (d) 9.5
53. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b}$, then each fraction is equal to
- (a) $(a+b+c)^2$ (b) $1/2$
(c) $1/4$ (d) 0
54. If $a : b = c : d$ then the value of $\frac{a^2 + b^2}{c^2 + d^2}$ is
- (a) $1/2$ (b) $\frac{a+b}{c+d}$
(c) $\frac{a-b}{c-d}$ (d) $\frac{ab}{cd}$
55. In Ramnagar Colony, the ratio of school going children to non-school going children is 5 : 4. If in the next year, the number of non-school going children is increased by 20%, making it 35,400, what is the new ratio of school going children to non-school going children?
- (a) 4 : 5 (b) 3 : 2
(c) 25 : 24 (d) None of these
56. In a journey of 45 km performed by tonga, rickshaw and cycle in that order, the distance covered by the three ways in that order are in the ratio of 8 : 1 : 3 and charges per kilometre in that order are in the ratio of 8 : 1 : 4. If the tonga charges being 24 paise per kilometre, the total cost of the journey is
- (a) ₹ 9.24 (b) ₹ 10
(c) ₹ 12 (d) None of these
57. If ₹ 1066 is divided among A, B, C and D such that $A : B = 3 : 4$, $B : C = 5 : 6$ and $C : D = 7 : 5$, who will get the maximum?
- (a) B (b) A
(c) C (d) D

Standard Level

- A man completes $\frac{5}{8}$ of a job in 10 days. At this rate, how many more days will it take him to finish the job?
(a) 5 (b) 6
(c) 7 (d) $7\frac{1}{2}$
- ₹ 1104 is divided between 3 men, 4 women and 6 boys, so that the share of a man, a woman and a boy are in the proportion of 3 : 2 : 1. How much does each boy get?
(a) ₹ 48 (b) ₹ 64
(c) ₹ 96 (d) Cannot be determined
- Seats of Physics, Chemistry and Mathematics in a school are in the ratio 4 : 5 : 6. There is a proposal to increase these seats by 75 in each department. What were the total number of seats in the school finally?
(a) 600 (b) 750
(c) 900 (d) None of these
- 60 kg of an alloy A is mixed with 100 kg of alloy B. If alloy A has lead and tin in the ratio 3 : 2 and alloy B has tin and copper in the ratio 1 : 4, then the amount of tin in the new alloy is
(a) 36 kg (b) 44 kg
(c) 53 kg (d) 80 kg
- A, B and C started a business. A invests $\frac{1}{2}$ capital for $\frac{1}{4}$ time, B invests $\frac{1}{8}$ capital for $\frac{1}{2}$ time and C invests the remaining capital for whole time. Find the share of B in the total profit of ₹ 9900.
(a) ₹ 2200 (b) ₹ 1100
(c) ₹ 6600 (d) ₹ 4400
- Two jars having a capacity of 3 and 5 litres respectively are filled with mixtures of milk and water. In the smaller jar 25% of the mixture is milk and in the larger 25% of the mixture is water. The jars are emptied into a 10 litre cask whose remaining capacity is filled up with water. Find the percentage of milk in the cask.
(a) 55% (b) 50%
(c) 45% (d) None of these
- The ratio of the number of students appearing for examination in the year 1998 in the states A, B and C was 3 : 5 : 6. Next year if the number of students in these states increases by 20%, 10% and 20% respectively, the ratio in states A and C would be 1 : 2. What was the number of students who appeared for the examination in the state A in 1998?
(a) 7200 (b) 6000
(c) 7500 (d) None of these
- A mixture of cement, sand and gravel in the ratio of 1 : 2 : 4 by volume is required. A person wishes to measure out quantities by weight. He finds that the weight of one cubic foot of cement is 94 kg, of sand 100 kg and gravel 110 kg. What should be the ratio of cement, sand and gravel by weight in order to give a proper mixture?
(a) 47 : 100 : 220 (b) 94 : 100 : 220
(c) 47 : 200 : 440 (d) None of these
- A, B, C subscribe ₹ 50,000 for a business. A subscribes ₹ 4000 more than B and ₹ 5000 more than C. Out of a total profit of ₹ 35,000, A receives :
(a) ₹ 8,400 (b) ₹ 11,900
(c) ₹ 13,600 (d) ₹ 14,700
- A, B and C jointly thought of engaging themselves in a business venture. It was agreed that A would invest ₹ 6500 for 6 months, B, ₹ 8400 for 5 months and C, ₹ 10,000 for 3 months. A wants to be the working member for which he was to receive 5% of the profits. The profit earned was ₹ 7400. Calculate the share of B in the profit.
(a) ₹ 1900 (b) ₹ 2660
(c) ₹ 2800 (d) ₹ 2840
- There is a ratio of 5 : 4 between two numbers. If 40 percent of the first number is 12 then what would be the 50 percent of the second number?
(a) 12 (b) 24
(c) 18 (d) None of the above
- In a partnership, A invests $\frac{1}{6}$ of the capital for $\frac{1}{6}$ of the time, B invests $\frac{1}{3}$ of the capital for $\frac{1}{3}$ of the time and C, the rest of the capital for whole time. Find A's share of the total profit of ₹ 2,300.
(a) ₹ 100 (b) ₹ 200
(c) ₹ 300 (d) ₹ 400
- A and B rent a pasture for 10 months; A puts in 80 cows for 7 months. How many can B put in for the remaining 3 months, if he pays half as much again as A?
(a) 120 (b) 180
(c) 200 (d) 280
- The resistance of a wire is proportional to its length and inversely proportional to the square of its radius. Two wires of the same material have the same resistance and their radii are in the ratio 9 : 8. If the length of the first wire is 162 cms., find the length of the other.
(a) 64 cm. (b) 120 cm.
(c) 128 cm. (d) 132 cm.

15. Two metals X and Y are to be used for making two different alloys. If the ratio by weight of $X : Y$ in the first alloy is $6 : 5$ and that in the second is $7 : 13$, how many kg of X metal must be melted along with 11 kg of the first alloy and 20 kg of the second so as to produce a new alloy containing 40% of metal Y ?
 (a) 11 (b) 12
 (c) 13 (d) 14
16. A diamond falls and breaks into three pieces whose weights are in the ratio $1 : 3 : 6$. The value of the diamond is proportional to the square of its weight. If the original value is ₹ 30,000, What is the loss in the value due to the breakage?
 (a) ₹ 13,800 (b) ₹ 16,200
 (c) ₹ 18,600 (d) ₹ 19,400
17. When a bus started from the first stop, the number of male passengers to the number of female passengers was $3 : 1$. At the stop 16 passengers get down and 6 more female passengers get into. Now the ratio of the male to female passengers becomes $2 : 1$. What was the total number of passengers in the bus when it started from the first stop?
 (a) 64 (b) 48
 (c) 54 (d) 72
18. In three vessels, the ratio of water and milk is $6 : 7$, $5 : 9$ and $8 : 7$, respectively. If the mixtures of the three vessels are mixed together, then what will be the ratio of water and milk?
 (a) 2431 : 3781 (b) 3691 : 4499
 (c) 4381 : 5469 (d) None of these
19. In two alloys, the ratio of iron and copper is $4 : 3$ and $6 : 1$, respectively. If 14 kg of the first alloy and 42 kg of the second alloy is mixed together to form a new alloy, then what will be the ratio of iron to copper in the new alloy?
 (a) 11 : 3 (b) 11 : 8
 (c) 8 : 1 (d) None of these
20. Mixture of milk and water has been kept in two separate containers. Ratio of milk to water in one of the containers is $5 : 1$ and that in the other container is $7 : 2$. In what ratio should the mixtures of these two containers be added together so that the quantity of milk in the new mixture may become 80%?
 (a) 3 : 2 (b) 2 : 3
 (c) 4 : 5 (d) None of these
21. Three containers of capacity 20 L, 5 L and 9 L contain mixture of milk and water with milk concentrations 90%, 80% and 70% respectively. The contents of three containers are emptied into a large vessel. What is the approximate ratio of milk to water in the resultant mixture?
 (a) 3 : 1 (b) 4 : 1
 (c) 5 : 1 (d) 2 : 1
22. Ratio of the earnings (in ₹) of A and B is $4 : 7$. If the earnings of A increase by 50% and those of B decrease by 25%, the new ratio of their earnings becomes $8 : 7$. How much is A earning?
 (a) ₹ 28000 (b) ₹ 21000
 (c) ₹ 26000 (d) Data inadequate
23. In the famous Bhojpur island, there are four men for every three women and five children for every three men. How many children are there in the island if it has 531 women?
 (a) 454 (b) 1180
 (c) 1070 (d) 389
24. If $a/b = 1/3$, $b/c = 2$, $c/d = 1/2$, $d/e = 3$ and $e/f = 1/4$, then what is the value of abc/def ?
 (a) $3/8$ (b) $27/8$
 (c) $3/4$ (d) $27/4$
25. If $a : b = c : d$, and $e : f = g : h$, then $(ae + bf) : (ae - bf) = ?$
 (a) $\frac{(e + f)}{(e - f)}$
 (b) $\frac{(cg + dh)}{(cg - dh)}$
 (c) $\frac{(cg + dh)}{(cg - dh)}$
 (d) $\frac{e - f}{e + f}$
26. The number of employees in a nationalised bank in a small town is 10, out of which 4 are female and the rest are males. A committee of 5 is to be formed. If m be the number of ways to form such a committee in which there is at least one female employee and n be the no. of ways to form such a committee which includes at least two male employees, then find the ratio $m : n$.
 (a) 3 : 2
 (b) 5 : 2
 (c) 1 : 1
 (d) 8 : 9

Expert Level

- Mr. Mehta and Mr. Yadav are neighbours in the 'Populated Colony'. The ratio of the number of sons and daughters Mr. Yadav has is equal to the duplicate of the sub triplicate ratio of the number of sons and daughters Mr. Mehta has. The daughters in any of the houses are more in number than the sons. If both the neighbours have an equal number of daughters, what is the minimum strength of the total children in both the houses?
(a) 18 (b) 19
(c) 14 (d) 12
- In a conference hall there are people in blue and yellow dresses. The ratio of the number of women in blue to the number of men in yellow is 3 : 2 and the ratio of the number of men in blue to the number of women in yellow is 3 : 5. If the ratio of the number of people in blue to the number of people in yellow is 21 : 23, then what is the ratio of the number of men to the number of women in the conference hall?
(a) 19 : 21 (b) 21 : 29
(c) 17 : 27 (d) Cannot be determined
- If the ratio of boys to girls in a class is B and the ratio of girls to boys is G , then $3(B + G)$ is :
(a) equal to 3 (b) less than 3
(c) more than 3 (d) less than $\frac{1}{3}$
- Two vessels contain mixtures of milk and water in the ratio of 8 : 1 and 1 : 5 respectively. The contents of both of these are mixed in a specific ratio into a third vessel. How much mixture must be drawn from the second vessel to fill the third vessel (capacity 26 gallons) completely in order that the resulting mixture may be half milk and half water?
(a) 12 gallons (b) 14 gallons
(c) 10 gallons (d) 13 gallons
- Two equal glasses are respectively $\frac{2}{3}$ and $\frac{1}{4}$ full of milk. They are then filled up with water and the contents are mixed in a tumbler. The ratio of milk and water in the tumbler is
(a) 5 : 6 (b) 11 : 13
(c) 13 : 11 (d) Cannot be determined
- The sum of the cubes of three numbers is 584 and the ratio of the first to second as also of second to the third is 1 : 2. What is the third number?
(a) 20 (b) 12
(c) 8 (d) None of these
- If $(a + b) : (b + c) : (c + a) = 6 : 7 : 8$ and $(a + b + c) = 14$, then the value of c is
(a) 6 (b) 7
(c) 8 (d) 14
- A sum of ₹1300 is divided amongst P , Q , R and S such that
$$\frac{P's\ share}{Q's\ share} = \frac{Q's\ share}{R's\ share} = \frac{R's\ share}{S's\ share} = \frac{2}{3}$$
. Then, P 's share is
(a) ₹140 (b) ₹160
(c) ₹240 (d) ₹320
- Two alloys of iron have different percentage of iron in them. The first one weighs 6 kg and second one weighs 12 kg. One piece each of equal weight was cut off from both the alloys and the first piece was alloyed with the second alloy and the second piece alloyed with the first one. As a result, the percentage of iron became the same in the resulting two new alloys. What was the weight of each cut-off piece?
(a) 4 kg (b) 2 kg
(c) 3 kg (d) 5 kg
- The ratio of the present ages of a son and his father is 1 : 5 and that of his mother and father is 4 : 5. After 2 years the ratio of the age of the son to that of his mother becomes 3 : 10. What is the present age of the father?
(a) 30 years (b) 28 years
(c) 37 years (d) 35 years
- The number of employees in Obelix Menhir Co. is a prime number and is less than 300. The ratio of the number of employees who are graduates and above, to that of employees who are not, can possibly be
(a) 101 : 88 (b) 87 : 100
(c) 110 : 111 (d) 97 : 84
- Three dogs are running in a park in such a way that when dog A takes 5 steps, dog B takes 6 steps and dog C takes 7 steps. But 6 steps of dog A are equal to 7 steps of dog B and 8 steps of dog C . What is the ratio of their speeds?
(a) 140 : 144 : 147 (b) 40 : 44 : 47
(c) 15 : 21 : 28 (d) 252 : 245 : 240

Test Yourself

- The salaries of A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be the new ratio of their salaries?
(a) 3 : 3 : 10 (b) 10 : 11 : 20
(c) 23 : 33 : 60 (d) Cannot be determined
- Given that 24 carat gold is pure gold, 18 carat gold is $\frac{3}{4}$ of pure gold and 20 carat gold is $\frac{5}{6}$ of pure gold, the ratio of the pure gold in 18 carat gold to the pure gold in 20 carat gold is:
(a) 3 : 8 (b) 9 : 10
(c) 15 : 24 (d) 8 : 5
- If 10 persons can clean 10 floors by 10 mops in 10 days, in how many days can 8 persons clean 8 floors by 8 mops?
(a) $12\frac{1}{2}$ days (b) 8 days
(c) 10 days (d) $8\frac{1}{3}$ days
- Three containers have their volumes in the ratio 3 : 4 : 5. They are full of mixtures of milk and water. The mixtures contain milk and water in the ratio of (4 : 1), (3 : 1) and (5 : 2) respectively. The contents of all these three containers are poured into a fourth container. The ratio of milk and water in the fourth container is:
(a) 4 : 1 (b) 151 : 48
(c) 157 : 53 (d) 5 : 2
- Three containers A, B and C are having mixtures of milk and water in the ratio of 1 : 5, 3 : 5 and 5 : 7 respectively. If the capacities of the containers are in the ratio 5 : 4 : 5, find the ratio of milk to water, if the mixtures of all the three containers are mixed together.
(a) 53 : 105 (b) 53 : 115
(c) 63 : 115 (d) 53 : 63
- A man ordered 4 pairs of black socks and some pairs of brown socks. The price of a black pair is double that of a brown pair. While preparing the bill, the clerk interchanged the number of black and brown pairs by mistake which increased the bill by 50%. The ratio of the number of black and brown pairs of sock in the original order was :
(a) 4 : 1 (b) 2 : 1
(c) 1 : 4 (d) 1 : 2
- Zinc and copper are melted together in the ratio 9 : 11. What is the weight of melted mixture, if 28.8 kg of zinc has been consumed in it?
(a) 58 kg (b) 60 kg
(c) 64 kg (d) 70 kg
- The Binary Ice-cream Shopper sells two flavours : Vanilla and Chocolate. On Friday, the ratio of Vanilla cones sold to Chocolate cones sold was 2 : 3. If the store had sold 4 more Vanilla cones, then, the ratio of Vanilla cones sold to the Chocolate cones sold would have been 3 : 4. How many Vanilla cones did the store sell on Friday ?
(a) 32 (b) 35
(c) 42 (d) 48
- ₹ 3650 is divided among 4 engineers, 3 MBAs and 5 CAs such that 3 CAs get as much as 2 MBAs and 3 Engineers as much as 2 CAs. Find the share of an MBA.
(a) 300 (b) 450
(c) 475 (d) None of these
- A bag contains 25 paise, 50 paise and 1 ₹ coins. There are 220 coins in all and the total amount in the bag is ₹ 160. If there are thrice as many 1 ₹ coins as there are 25 paise coins, then what is the number of 50 paise coins?
(a) 60 (b) 40
(c) 120 (d) 80
- In a co-educational school there are 15 more girls than boys. If the number of girls is increased by 10% and the number of boys is also increased by 16% there would be 9 more girls than boys. What is the number of students in the school?
(a) 140 (b) 125
(c) 265 (d) 255
- A milkman mixes 20 litres of water with 80 litres milk. After selling one-fourth of this mixture, he adds water to replenish the quantity that he has sold. What is the current ratio of water to milk?
(a) 2 : 3 (b) 1 : 2
(c) 1 : 3 (d) 3 : 4
- If $\frac{a}{b} = \frac{1}{3}$, $\frac{b}{c} = 2$, $\frac{c}{d} = \frac{1}{2}$, $\frac{d}{e} = 3$ and $\frac{e}{f} = \frac{1}{4}$, then what is the value of $\frac{abc}{def}$?
(a) $\frac{3}{8}$ (b) $\frac{27}{8}$
(c) $\frac{3}{4}$ (d) $\frac{27}{4}$
- A precious stone weighing 35 grams worth ₹ 12,250 is accidentally dropped and gets broken into two pieces having weights in the ratio of 2 : 5. If the price varies as the square of the weight then find the loss incurred.
(a) ₹ 5750 (b) ₹ 6000
(c) ₹ 5500 (d) ₹ 5000
- 40 men could have finished the whole project in 28 days but due to the inclusion of a few more men, work got done in $\frac{3}{4}$ of the time. Find the ratio of number of new men to number of old men.
(a) 12 : 19 (b) 20 : 27
(c) 27 : 20 (d) None of these

Hints & Solutions

Foundation Level

1. (b) $x \frac{2ab}{a+b} \Rightarrow \frac{x}{a} = \frac{2b}{a+b}$
 $\Rightarrow \frac{x+a}{x-a} = \frac{3b+a}{b-a}$ (componendo dividendo)
 Similarly, $\frac{x}{b} = \frac{2a}{a+b} \Rightarrow \frac{x+b}{x-b} = \frac{3a+b}{a-b}$
 $\therefore \frac{x+a}{x-a} + \frac{x+b}{x-a} = \frac{3b+a}{b-a} + \frac{3a+b}{a-b}$

$$-\frac{(3b+a)}{a-b} + \frac{3a+b}{a-b} = \frac{2a-2b}{a-b} = 2$$
2. (b) Let us represent their shares by the corresponding letter of their names.
 $A + B = 2C$ and $B + C = 3A$.
 $\Rightarrow A + 3A - C = 2C$ (since $B = 3A - C$)
 $4A = 3C \Rightarrow A : C = 3 : 4$
3. (d) Let $3X$ and $4X$ be the numbers $\Rightarrow \frac{3X-5}{4X-5} = \frac{2}{3}$
 $\Rightarrow 9X - 15 = 8X - 10 \Rightarrow X = 5$
 $\Rightarrow$ The required numbers are 15 and 20.
4. (d)

	Original	Present
Wages	22x	25x
Number	15y	11y

$$\text{Ratio of total wages} = \frac{22x \times 15y}{25x \times 11y} = \frac{6}{5}$$

If the present bill is ` 5000, the original was ` 6000.
5. (d) Let x is to be added
 $(11+x) : (15+x) = (17+x) : (23+x)$
 $\Rightarrow \frac{11+x}{15+x} = \frac{17+x}{23+x} \Rightarrow x = 1$
6. (b) Let r be the 4th proportional.
 Then $\frac{12X^3}{9aX^2} = \frac{8a^3X}{r} \Rightarrow r = 6a^4$
7. (a) The 100 acres should be divided between Sunanda and Ansuya in the ratio
 $\frac{1}{4} : \frac{1}{5}$ i.e. 5 : 4
 So, Sunanda gets $\frac{500}{9}$ acres and Ansuya gets $\frac{400}{9}$
8. (d) $\frac{a}{b} = \frac{9}{6} = 3 : 2 = 15 : 10$

$$\frac{b}{c} = \frac{10}{9} = 10 : 9$$

Hence, $a : b : c = 15 : 10 : 9$

9. (c) If x is the integer, $\frac{5+x}{9+x} > \frac{7}{10}$

$$\therefore 50 + 10x > 63 + 7x$$

$$\therefore 3x > 13$$

$$\therefore x > \frac{13}{3}$$

The least integer greater than $\frac{13}{3}$ is 5.

10. (d) Obviously the ratio is 2 : 3 : 4 : 5

11. (a) If A's share is 1, B's share = $\frac{5}{9} \times 1 = \frac{5}{9}$

$$C's \text{ share} = \frac{7}{10} \times \frac{5}{9} = \frac{7}{18};$$

$$D's \text{ share} = \frac{1}{3} \left(\frac{5}{9} + \frac{7}{18} \right) = \frac{17}{54}$$

$$\therefore A : B : C : D = 1 : \frac{5}{9} : \frac{7}{18} : \frac{17}{54} = 54 : 30 : 21 : 17.$$

$$\therefore A's \text{ share} = \frac{54}{122} \times 1220 = \text{Rs.} 540.$$

12. (c) If the maximum for each paper is 100, total marks = 500

$$\text{and his aggregate} = \frac{3}{5} \times 500 = 300. \text{ which when divided}$$

in the given ratio gives marks 36, 48, 60, 72 and 84 and so there are 3 subjects in which he gets more than 50.

13. (a) 80% of the total profit is divided in the ratio

$$20000 : 24000 : 16000 = 5 : 6 : 4$$

$$80\% \text{ of total profit} = 5x + 6x + 4x = 15x$$

$$\text{Total profit} = \frac{15x}{80\%} = 18.75x$$

$$\therefore \text{Share of C in profit} = 4x + 20\% \text{ of } 18.75x \\ = 4x + 3.75x = 7.75x$$

$$\text{Share of A in profit} = 5x$$

$$\text{Share of B in profit} = 6x$$

$$\therefore (6x + 5x) - 7.75x = 487.50$$

$$3.25x = 487.50 \Rightarrow x = 150$$

$$\therefore \text{Total profit} = 18.75 \times 150 = \text{` 2812.50.}$$

14. (a) Let the prices of two houses A and B be ₹ 4x and ₹ 5x, respectively for the last year.

Then, the prices of A this year = ₹ (1.25 × 4x) and that of B = ₹ (5x + 50,000)

This year, Ratio of their prices = 9 : 10

$$\therefore \frac{1.25 \times 4x}{5x + 50,000} = \frac{9}{10}$$

$$\Rightarrow 50x - 45x = 450000 \Rightarrow 5x = 4,50,000$$

$$\Rightarrow x = 90,000$$

Hence, the price of A last year was

$$4x = ₹ 3,60,000$$

15. (b) Let the length and breadth of the rectangular room be ℓ and b .

$$\text{We have, } \frac{\ell + 4}{b + 4} = \frac{4}{3}$$

$$\Rightarrow 3\ell + 12 = 4b + 16$$

$$\Rightarrow 3\ell - 4b = 4 \quad \dots(1)$$

$$\text{Again, we have } \frac{\ell - 4}{b - 4} = \frac{2}{1} \Rightarrow \ell - 4 = 2b - 8$$

$$\Rightarrow \ell - 2b = -4 \quad \dots(2)$$

Solving (1) and (2), we get $\ell = 12$ and $b = 8$.

16. (b) Let A, B and C be the first, second and third nos. respectively.

Then, $A : B = 2 : 3$ and $B : C = 5 : 8$

Consider, $A : B = 2 : 3 = 2 \times 5 : 3 \times 5 = 10 : 15$

and $B : C = 5 : 8 = 5 \times 3 : 8 \times 3 = 15 : 24$

$$\therefore A : B : C = 10 : 15 : 24$$

Let the required number be 10x, 15x and 24x.

Given, sum of three numbers = 98

Then,

$$\therefore 10x + 15x + 24x = 98$$

$$\Rightarrow 49x = 98 \Rightarrow x = 2$$

$$\Rightarrow \text{Second number} = 15x = 15 \times 2 = 30$$

17. (b) Given, ratio of numbers is 3 : 4

$$\therefore \text{The numbers are } 3x \text{ and } 4x.$$

Now, according to the question

$$16x^2 = 8(3x)^2 - 224$$

$$\Rightarrow 16x^2 = 72x^2 - 224 \Rightarrow 56x^2 = 224$$

$$\Rightarrow x = 2$$

$$\therefore \text{Required numbers} = 6, 8$$

18. (c) Let the third type of tea is priced at ₹ x per kg. Also suppose that the three types of tea mixed together are 1, 1 and 2 kg, respectively.

$$\text{Now, } \frac{126 \times 1 + 135 \times 1 + 2x}{1 + 1 + 2} = 153$$

$$\Rightarrow \frac{261 + 2x}{4} = 153 \Rightarrow 261 + 2x = 612$$

$$\Rightarrow x = \frac{351}{2} = ₹ 175.5 \text{ per kg.}$$

19. (b) Quantity of milk = $\frac{3}{5} \times 45 = 27$ litres

$$\text{Quantity of water} = \frac{2}{5} \times 45 = 18 \text{ litres}$$

Let x litres of water be added to make the ratio 9 : 11.

$$\therefore \frac{18}{27} = \frac{x + 11}{9} \Rightarrow 18 + x = 33 \Rightarrow x = 15$$

20. (d) Radius of the two pipes are 1 cm and 2 cm. Square of the radii of the pipes are 1 and 4.

$$\therefore \text{Required ratio of rates of flow in the two pipes} = 1 : \frac{1}{4} = 4 : 1$$

21. (b) 18 carat gold = $\frac{3}{4}$ pure gold = $\frac{3}{4} \times 24$
= 18 carat pure gold

$$20 \text{ carat gold} = \frac{5}{6} \text{ pure gold} = \frac{5}{6} \times 24$$

$$= 20 \text{ carat pure gold}$$

$$\therefore \text{Required ratio} = 18 : 20 = 9 : 10$$

22. (c) We have, $\frac{y}{x - z} = \frac{y + x}{z}$

$$\Rightarrow yz = xy + x^2 - yz - xz \quad \dots(1)$$

$$\text{Also, } \frac{x}{y} = \frac{y}{x - z} \Rightarrow x^2 - xz = y^2 \quad \dots(2)$$

From (1) and (2), we have

$$yz = xy - yz + y^2$$

$$\Rightarrow 2yz = xy + y^2$$

$$\therefore 2z = x + y \quad \dots(3)$$

Checking with the options, we find that the values given in option c satisfies the equation (3)

23. (d) Increased ratio of their respective salaries

$$= 3 \times \frac{150}{100} : 5 \times \frac{160}{100} : 7 \times \frac{150}{100}$$

$$= \frac{9}{2} : 8 : \frac{21}{2} \quad 9 : 16 : 21$$

24. (b) Let the no. of the boys and girls that appeared in the examination be x and y , respectively.

$$\text{Now, } 71.8 \frac{71x + 73y}{x + y}$$

$$\Rightarrow 71.8x + 71.8y = 71x + 73y$$

$$\Rightarrow (71.8 - 71)x = (73 - 71.8)y \Rightarrow 0.8x = 1.2y$$

$$\Rightarrow \frac{x}{y} = \frac{1.2}{0.8} = \frac{3}{2}$$

25. (b) In first cask,

$$\text{Quantity of water} = \frac{7}{20} \times 48 = 16.8\text{L}$$

$$\text{Quantity of wine} = \frac{13}{20} \times 48 = 31.2\text{L}$$

In second cask,

$$\text{Quantity of water} = \frac{17}{35} \times 42 = 20.6\text{L}$$

$$\text{Quantity of wine} = \frac{18}{35} \times 42 = 21.6\text{L}$$

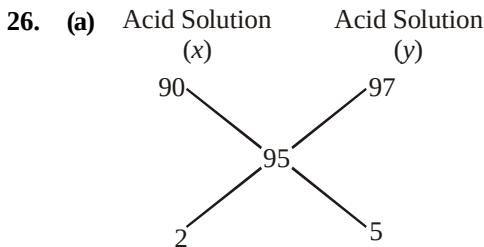
Now after mixing:

$$\text{Total quantity of wine} = 52.8\text{ L}$$

$$\text{Quantity of water} = 57.2\text{ L}$$

$$\text{Ratio after mixing} = \frac{52.8}{57.2} = \frac{528}{572} = \frac{12}{13}$$

$$= 12 : 13.$$



$\therefore$ Amount of the solution x and y in ratio $2 : 5$

$$\text{Amount of acid in solution } x = \frac{2}{7} \times 21 = 6\text{L}$$

$$\text{Amount of acid in solution } y = \frac{5}{7} \times 21 = 15\text{L}$$

6L acid in x and 15L in Y

27. (b) Profit ratio = $10 : 3$

$$\text{Time ratio} = t_1 : t_2$$

$$\text{Cost ratio} = 550 : 330$$

$$550 t_1 : 330 t_2 = 10 : 3$$

$$\frac{550 t_1}{330 t_2} = \frac{10}{3}$$

$$\frac{t_1}{t_2} = 2$$

$$\text{given } t_1 = 12 \text{ months}$$

$$\Rightarrow t_2 = 6 \text{ months}$$

28. (c) Let the profit = x

$$\text{Profit of } A = \frac{9x}{10}, \text{ Remaining profit} = \frac{x}{10}$$

$$\text{Profit of } B = \frac{x}{20}$$

$$\text{Profit of } C = \frac{x}{20}$$

$$\text{Ratio of profit} = \frac{9}{10} : \frac{1}{20} : \frac{1}{20} \\ = 18 : 1 : 1$$

A 's income is increased by ` 270 . When profit rises 3%

$$\text{Investment of } A = \frac{270}{3} \times 100 = \text{` } 9000.$$

If investment of A , B and $C = 18x$, x and x
 $18x = 9000$

$$x = 500$$

B investment = ` 500.

C investment = ` 500.

29. (c) After 10 days : 150 men had food for 35 days.

Suppose 125 men had food for x days. Now,

Less men, More days (Indirect Proportion)

Then,

men days

$$\begin{array}{cc} \uparrow 150 & 35 \downarrow \\ 125 & x \end{array}$$

$$\therefore 125 : 150 :: 35 : x \Rightarrow 125 \times x = 150 \times 35$$

$$\Rightarrow x = \frac{150 \times 35}{125} \Rightarrow x = 42.$$

Hence, the remaining food will last for 42 days.

30. (b) Quantity of milk = $\frac{2}{3} \times 45 = 30\text{ L}$

$$\text{Quantity of water} = \frac{1}{3} \times 45 = 15\text{ L}$$

Let the required quantity of water to be added be x litre.

$$\text{Then, } \frac{30}{15 + x} = \frac{3}{2}$$

$$\Rightarrow 3(15 + x) = 30 \times 2 = 60$$

$$\Rightarrow x = 5\text{L}$$

31. (c) Let 40% of $A = \frac{2}{3}B$. Then, $\frac{40A}{100} = \frac{2B}{3} \Leftrightarrow \frac{2A}{5} = \frac{2B}{3}$

$$\Leftrightarrow \frac{A}{B} = \left(\frac{2}{3} \times \frac{5}{2} \right) = \frac{5}{3}$$

$$\therefore A : B = 5 : 3$$

32. (c) Less leaves, less cost
(Direct Proportion)
More lines, more cost (Direct Proportion)
Less words, less cost (Direct Proportion)
- $$\left. \begin{array}{l} \text{leaves } 320 : 297 \\ \text{lines } 21 : 28 \\ \text{words } 11 : 10 \end{array} \right\} :: 19 : x$$
- $$\therefore 320 \times 21 \times 11 \times x = 297 \times 28 \times 10 \times 19$$
- $$\Rightarrow x = \frac{171}{8} = 21\frac{3}{8}$$
33. (a) Ratio of capitals of A, B and C
 $= (15000 \times 3) : (40000 \times 9) : (30000 \times 6)$
 $= 1 : 8 : 4$
- $$B's \text{ share} = \left(\frac{8}{13} \times 7800 \right) = ₹ 4800$$
34. (d) Let the third proportional to $(x^2 - y^2)$ and $(x - y)$ be z . Then,
 $(x^2 - y^2) : (x - y) :: (x - y) : z \Leftrightarrow (x^2 - y^2) \times z$
 $= (x - y)^2 \Leftrightarrow z = \frac{(x - y)^2}{(x^2 - y^2)} \cdot \frac{(x - y)}{(x + y)}$
35. (b) Ratio of sides $= \frac{1}{2} : \frac{1}{3} : \frac{1}{4} = 6 : 4 : 3$
- $$\text{Largest side} = \left(104 \times \frac{6}{13} \right) \text{ cm} = 48 \text{ cm}$$
36. (a) Sum invested by A, B and C is
- $$5 \times 12 : 7 \times 12 : 6 \times 6 + 3 \times 6$$
- $$\text{or, } 60 : 84 : 54 \quad \text{or, } 10 : 14 : 9$$
- $$\therefore \text{Share of C} = \frac{9}{33} \times 33,000 = ₹ 9,000$$
37. (d) Required ratio $= \frac{4 \times 135}{5 \times 120} = 9 : 10$
38. (b) Number of females $= 156800 \times \frac{100}{80} = 196000$
- $$\therefore \text{Number of males} = \frac{7}{8} \times 196000 = 171500$$
- $$\therefore \text{Total population} = 196000 + 171500 = 367500$$
39. (a) Let C's investment = ₹ x .
 B's investment = ₹ $(x - 3000)$
 A's investment = ₹ $(x - 3000 + 6000) = ₹ (x + 3000)$
 Now, $(A + B + C)$'s investment = ₹ 72000
 $\Rightarrow x + (x - 3000) + (x + 3000) = 72000$
 $\Rightarrow 3x = 72000 \Rightarrow x = 24000$
 Hence, A's investment = ₹ 27000
 B's investment = ₹ 21000
 C's investment = ₹ 24000
 Ratio of the capitals of A, B and C
 $= 27000 : 21000 : 24000$
 $= 9 : 7 : 8$
- $$A's \text{ share} = \left(\frac{9}{24} \times 8640 \right) = ₹ 3240$$
40. (b) $A : B : C = (5000 \times 4 + 2500 \times 8) : (4500 \times 6 + 3000 \times 6) : (7000 \times 6)$
 $= 40000 : 45000 : 42000 = 40 : 45 : 42$
- $$\therefore A's \text{ share} = \left(5080 \times \frac{40}{127} \right) = ₹ 1600;$$
- $$B's \text{ share} = \left(5080 \times \frac{45}{127} \right) = ₹ 1800;$$
- $$C's \text{ share} = \left(5080 \times \frac{42}{127} \right) = ₹ 1680.$$
41. (b) $A : B : C = (40000 \times 36) : (80000 \times 12 + 40000 \times 24) : (120000 \times 24 + 40000 \times 12)$
 $= 144 : 192 : 336 = 3 : 4 : 7$
42. (c) Let the incomes of two companies A and B be $5x$ and $8x$ respectively.
 From the question,
- $$\frac{5x + 25}{8x} = \frac{5}{4} \Rightarrow 20x + 100 = 40x \therefore x = 5$$
- $$\therefore \text{Income of company B} = 8x = ₹ 40 \text{ lakh}$$
43. (b) Ratio of Abhishek and Sudin for one month
 $= (50,000 \times 36) + (30,000 \times 24) : (70,000 \times 24)$
 $= (18,00,000 + 7,20,000) : 16,80,000 = 3 : 2$
 Hence share of Sudin in the profit earned from the business.
- $$= \frac{87,500}{(3+2)} \times 2 = ₹ 35,000.$$
44. (a) In 1 kg mixture quantity of iron = 200 gm
 Let x gm sand should be added, then
 10% of $(1000 + x) = 200$
 $\therefore x = 1000 \text{ gm} = 1 \text{ kg}$
45. (c) Suppose B joined after x months.
 Then, $21000 \times 12 = 36000 \times (12 - x) \Leftrightarrow 36x = 180$
 $\Leftrightarrow x = 5$.
 Hence, B joined after 5 months.
46. (d) From the given options, we just need to look for a multiple of 7. 2100 is the only option which is a multiple of 7 and is hence the correct answer.
47. (b) Let the first and the second numbers be x and y respect then
 $y + 30\% \text{ of } x = 140\% \text{ of } y$
 or, $y + 0.3x = 1.4y$
 or, $0.3x = 0.4y$
 $\therefore x : y = 0.4 : 0.3 = 4 : 3$
48. (c) Let number of ladies = x
 then, number of gents = $2x$
- $$\text{Now, } \frac{x-2}{2x-2} \cdot \frac{1}{3} \Rightarrow 3x - 6 = 2x - 2$$
- $$\Rightarrow x = 4$$
- $$\therefore \text{Total number of people originally present} = 4 + 8 = 12$$

49. (a) Let number of each type of coin = x . Then,
 $1 \times x + .50 \times x + .25x = 35$
 $\Rightarrow 1.75x = 35 \Rightarrow x = 20$ coins
50. (b) In a year, for A, total amount as a remuneration
 $= 10 \times 12 = ₹ 120$
 $\therefore$ Amount of A's profit = $390 - 120 = ₹ 270$
 Ratio of investment = $3 : 4$
 Let total profit = x
 Then, B's profit = $\frac{4}{7}x$
 $\therefore \frac{3}{3+4} \times x = 270 \Rightarrow x = 630$
 $\therefore$ B's profit = $630 - 270 = ₹ 360$
51. (c) Let $x = 5$
 Then $f(x) = 6/4 = 1.5 = y$
 And $f(y) = 2.5/0.5 = 5$.
 Thus, the ratio of $x : f(y) = 1 : 1$
Note: Even if you take some other value of y , you would still get the same answer.
52. (b) Initial values are 5, 15 and 25.
 Thus we have $5 \times 15 = K \times 25$.
 Hence, $K = 3$
 Thus, the equation is $AB = 3C$.
 For the problem, keep C constant at 25. Then, $A \times 9 = 3 \times 25$.
i.e. $A = 75/9 = 8.33$
53. (b) The given condition has a , b and c symmetrically placed. Thus, if we use $a = b = c = 2$ (say) we get each fraction as $1/2$.
54. (d) $1 : 2 = 3 : 6$, so $(a^2 + b^2)/(c^2 + d^2) = 5/45 = 1/9$
 From the given options, only ab/cd gives us this value.
55. (c) $5 : 4 \rightarrow 5 : 4.8 \rightarrow 25 : 24$
 Option (c) is correct.
56. (a) Total distances covered under each mode = 32, 4 and 12 km respectively.
 Total charges = $32 \times 24 + 4 \times 3 + 12 \times 12 = 924$ paise = ₹ 9.24.
57. (c) Since $A : B = 3 : 4$... (1)
 $B : C = 5 : 6$... (2)
 and $C : D = 7 : 5$... (3)
 Therefore, by, proportionating, (1) and (2)
 $A : B = 3 \times 5 : 4 \times 5 = 15 : 20$
 $B : C = 20 : 24$ and $C : D = 7 : 5$
 Hence, $A : B : C = 15 : 20 : 24$... (4)
 Now, $A : B : C = 15 \times 7 : 20 \times 7 : 24 \times 7$
 $= 105 : 140 : 168$
 $\therefore C : D = 24 \times 7 : 5 \times 7 = 168 : 120$
 [By proportionating (3) and (4)]
 Hence, $A : B : C : D = 105 : 140 : 168 : 120$
 Hence, C gets the maximum share.

Standard Level

1. (b) Work done = $\frac{5}{8}$. Balance work = $\left(1 - \frac{5}{8}\right) = \frac{3}{8}$.
 Less work, Less days (Direct Proportion)
 Let the required number of days be x . Then,

Work	days
$\uparrow \frac{5}{8}$	$\uparrow 10$
$\downarrow \frac{3}{8}$	$\downarrow x$

 Then, $\frac{5}{8} : \frac{3}{8} :: 10 : x \Rightarrow \frac{5}{8} \times x = \frac{3}{8} \times 10$
 $\Rightarrow x = \left(\frac{3}{8} \times 10 \times \frac{8}{5}\right) = 6$.
2. (a) Let each boy gets x , so the women gets $2x$ and a man gets $3x$.
 Now, $(3 \times 3x) + (4 \times 2x) + (6 \times x) = 1104$
 $\Rightarrow 23x = 1104 \Rightarrow x = 48$
 $\therefore$ Each boy gets ₹ 48.
3. (d) Let the number of seats in Physics, Chemistry and Mathematics be $4x$, $5x$ and $6x$.
 New ratio of seats = $(4x + 75) : (5x + 75) : (6x + 75)$
 $\therefore$ The given data is insufficient.
4. (b) Quantity of tin in 60 kg of A = $\left(60 \times \frac{2}{5}\right)$ kg = 24 kg
 Quantity of tin in 100 kg of B = $\left(100 \times \frac{1}{5}\right)$ kg = 20 kg
 Quantity of tin in the new alloy = $(24 + 20)$ kg = 44 kg.
5. (b) C's capital = $1 - \left(\frac{1}{2} + \frac{1}{8}\right) = 1 - \frac{5}{8} = \frac{3}{8}$
 Ratio of capitals of A, B and C
 $= \left(\frac{1}{2} \times \frac{1}{4}\right) : \left(\frac{1}{8} \times \frac{1}{2}\right) : \left(\frac{3}{8} \times 1\right)$
 $= \frac{1}{8} : \frac{1}{16} : \frac{3}{8} = 2 : 1 : 6$
 B's share = $\left(\frac{1}{9} \times 9900\right) = ₹ 1100$
6. (c) There will be a total of 4.5 litres of milk (25% of 3 + 75% of 5) giving a total of 4.5. Hence, 45%.
7. (d) Let the number of students appearing for examination in the year 1998 in the states A, B and C be $3x$, $5x$ and $6x$ respectively.
 According to the question,

$3x \times \frac{120}{100}$	$\frac{1}{2}$
$5x \times \frac{120}{100}$	$\frac{1}{2}$

 Hence data inadequate.
8. (a) $47 : 100 : 220$ would give: 0.5 cubic feet of cement, 1 cubic feet of sand and 2 cubic feet of gravel.
 required ratio $1 : 2 : 4$ is satisfied.

9. (d) Let $C = x$. Then, $B = x + 5000$ and $A = x + 5000 + 4000 = x + 9000$.
So, $x + x + 5000 + x + 9000 = 50000 \Leftrightarrow 3x = 36000$
 $\Leftrightarrow x = 12000$.
 $A : B : C = 21000 : 17000 : 12000 = 21 : 17 : 12$

$$\therefore A's \text{ share} = \left(35000 \times \frac{21}{50} \right) = \text{₹ } 14,700.$$

10. (b) For managing, A receives = 5% of ₹ 7400 = ₹ 370.
Balance = ₹ (74000 - 370) = ₹ 7030
Ratio of their investments = $(6500 \times 6) : (8400 \times 5) : (10000 \times 3)$
 $= 39000 : 42000 : 30000 = 13 : 14 : 10$.

$$\therefore B's \text{ share} = \left(7030 \times \frac{14}{37} \right) = \text{₹ } 2660.$$

11. (a) $\frac{a}{b} = \frac{5}{4}$, $b = \frac{4}{5}a$ Given $(40\% \text{ of } a) = \frac{2}{5}a = 12$

$$\therefore a = 5 \times 6 \text{ and } b = \frac{4}{5} \times 5 \times 6 = 24$$

$$\therefore 50\% \text{ of } b = \frac{24}{2} = 12$$

12. (a) Remaining capital $1 - \left(\frac{1}{6} + \frac{1}{3} \right) = \frac{1}{2}$

Ratio of their profit

$$\frac{1}{6} \times \left[\frac{1}{6} \times 12 \right] : \frac{1}{3} \times \left[\frac{1}{3} \times 12 \right] : \frac{1}{2} \times 12$$

$$= \frac{1}{3} : \frac{4}{3} : 6 = 1 : 4 : 18$$

$$\therefore A's \text{ share} = \frac{1}{1+4+18} \times 2300 = \text{₹ } 100$$

13. (d) Let B puts = x cows

$$\text{then amount paid by } B = \frac{3}{2} \times \text{amount paid by A.}$$

$$\therefore \frac{80 \times 7}{x \times 3} = \frac{\text{amount paid by A}}{3/2 \times \text{amount paid by A}}$$

$$\Rightarrow x = \frac{80 \times 7 \times 3}{3 \times 2} = 280 \text{ cows}$$

14. (c) If R is the resistance, l is the length and r is the radius.

$$R \propto \frac{l}{r^2}$$

$$\therefore R = \frac{k l}{r^2} \quad (\text{where } k \text{ is a constant})$$

$$\therefore \frac{R_1}{R_2} = \frac{\frac{k \times 162}{81}}{\frac{k \times \ell}{64}}; \text{ But } R_1 = R_2.$$

$$\therefore \frac{k \times 162}{81} = k \times \frac{\ell}{64} \quad \therefore \frac{162}{81} = \frac{\ell}{64}$$

$$\therefore \ell = 128 \text{ cms.}$$

15. (d) Let X kg of metal X be added.

$$\text{Wt. of metal } X \text{ in the 1st alloy} = \frac{6}{11} \times 11 \text{ kg} = 6 \text{ kg}$$

$$\text{Wt. of metal } X \text{ in the 2nd alloy} = \frac{7}{20} \times 20 \text{ kg} = 7 \text{ kg}$$

$$\text{Total wt. of the new alloy} = (11 + 20 + X) \text{ kg}$$

$$\text{Total wt. of metal } X \text{ in the new alloy} = (6 + 7 + X) \text{ kg}$$

$$\Rightarrow \frac{6+7+X}{11+20+X} = \frac{60}{100} \Rightarrow X = 14$$

16. (b) Let the weight of the three pieces be x , $3x$ and $6x$ and value (V) $\propto$ square of W

$$V \propto W^2 \Rightarrow \frac{V}{W^2} = \text{constant} = \frac{30000}{(10x)^2}$$

$$\therefore \text{loss due to breakage}$$

$$= \frac{30000(10x)^2 - (x)^2 + (6x)^2}{(10x)^2} = \frac{3000 \times 54x^2}{100x^2} = 16200$$

17. (a) Let initially, the number of males and females in the bus be $3x$ and x , respectively.

$$\therefore \text{Total no. of passengers} = 4x$$

At the first stop, the number of male passengers

$$(4x - 16) \times \frac{3}{4} = 3x - 12$$

$$\text{and female passengers} = (4x - 16) \times \frac{1}{4} + 6 = x + 2$$

$$\text{Now, } \frac{3x-12}{x+2} = \frac{2}{1} \Rightarrow 3x-12 = 2x+4 \Rightarrow x = 16$$

Hence, the total numbers of passengers, initially = $16 \times 4 = 64$

	Water	Milk	Total.
1st vessel	6	7	13
2nd vessel	5	9	14
3rd vessel	8	7	15

LCM of 13, 14 & 15 = 2730

Increase value of total to 2730 as follows.

1st vessel	1260	1470	2730
2nd vessel	975	1755	2730
3rd vessel	1456	1274	2730
Total	3691	4499	8190

$$\therefore \text{Required ratio} = \frac{3691}{4499}$$

19. (a)

	Iron	Copper	
Alloy I	8	6	14 kg.
Alloy II	36	6	42 kg.
	44	12	56 kg

20. (b) Let x and y be two containers.
Ratio of milk to water in container $x = 5 : 1$
and ratio of milk to water in container $y = 7 : 2$
It is given that quantity of milk should be 80% in new mixture. This means that quantity of water will be 20%
 $\therefore$ Ratio of milk to water in new mixture = $80 : 20 = 4 : 1$.

$$\therefore \frac{5}{6}x + \frac{7}{9}y : \frac{1}{6}x + \frac{2}{9}y = 4 : 1$$

$$\Rightarrow \frac{\frac{5x}{6} + \frac{7y}{9}}{\frac{x}{6} + \frac{2y}{9}} = \frac{4}{1}$$

$$\Rightarrow \frac{15x + 14y}{3x + 4y} = \frac{4}{1} \Rightarrow 15x + 14y = 12x + 16y$$

$$\Rightarrow 3x = 2y \Rightarrow x : y = 2 : 3$$

21. (c) In first container,

$$\text{Quantity of milk} = 20 \times \frac{90}{100} = 18\text{L}$$

$$\text{Quantity of water} = 2\text{L}$$

In second container,

$$\text{Quantity of milk} = 5 \times \frac{80}{100} = 4\text{L}$$

$$\text{Quantity of water} = 1\text{L}$$

In third container,

$$\text{Quantity of milk} = 9 \times \frac{70}{100} = 6.3\text{L}$$

$$\text{Quantity of water} = 2.7\text{L}$$

After mixing :

$$\text{total milk} = 18 + 4 + 6.3 = 28.3\text{L}$$

$$\text{total water} = 2 + 1 + 2.7 = 5.7\text{L}$$

Ratio milk and water after

$$\text{mixing} = 28.3\text{L} : 5.7\text{L} = 5 : 1 \text{ (approx)}$$

22. (d) Ratio of earning of A and B is $4 : 7$

$$A \text{ earning} = ₹ 4x$$

$$B \text{ earning} = ₹ 7x$$

A earning increase by 50%

$$\text{Now, A earning} = 4x + 4x \times \frac{50}{100} = 6x$$

B earning decreases by 25%

$$\text{Now B earning} = 7x - 7x \times \frac{25}{100} = \frac{21x}{4}$$

$$\text{Now, Ratio} = \frac{6x}{\frac{21x}{4}} = 8 : 7$$

Data inadequate

23. (b) Women : Men = $3 : 4$

$$\text{Men : Children} = 3 : 5$$

$$\rightarrow \text{Women : Men : Children} = 9 : 12 : 20$$

In the ratio, $9 \rightarrow 531$ Women

Thus, $20 \rightarrow 1180$ children.

24. (a) $a : b : c = 2 : 6 : 3$

$$a : b : c : d : e : f = 6 : 18 : 9 : 18 : 6 : 24$$

$$abc/def = 3/8$$

25. (b) Solve by taking values of a, b, c, d and e, f, g , and h independently of each other.

$$a = 1, b = 2, c = 3, d = 6$$

$$\text{and } e = 3, f = 9, g = 4$$

$$\text{and } h = 12$$

$$\text{gives } (ae + bf) : (ae - bf) = 21 : -15 = -7/5$$

$$\text{option (b) } (cg + dh)/(cg - dh) = 84/60 = -7/5.$$

26. (c) By using options, (c) is correct option where $m : n = 1 : 1$.

Expert Level

1. (b) As the number of sons, is less than the number of daughters in any of the families, let us assume the ratio of number of sons to number of daughters of Mr. Mehta to be $x^3 : y^3$ ($x < y$)

Ratio of number of sons to daughter of Mr. Yadav

$$= \left(\frac{\sqrt[3]{x^3}}{\sqrt[3]{y^3}} \right)^2 = \frac{x^2}{y^2}$$

The minimum strength can be considered only when

$$x = 1 \text{ and } y = 2. \therefore \frac{x^3}{y^3} = \frac{1}{8}; \frac{x^2}{y^2} = \frac{1}{4}$$

As number of daughters are equal in both the families, Mr. Yadav has $(1 : 4 = 2 : 8)$ Mr. Mehta has 1 son and eight daughters. Total strength = $2 + 8 + 1 + 8 = 19$.

2. (c)

	Blue	Yellow
Men	$3y$	$2x$
Women	$3x$	$5y$

Given that

$$\text{Blue : Yellow} = (3x + 3y) : (2x + 5y) = 21 : 23$$

$$\text{Let } 3x + 3y = 21n \quad \dots(1)$$

$$2x + 5y = 23n \quad \dots(2)$$

Solving (1) and (2)

$$\text{We get } x = 4n \text{ and } y = 3n$$

Men : Women

$$= (2x + 3y) : (3x + 5y)$$

$$= (8n + 9n) : (12n + 15n) = 17 : 27.$$

3. (c) Let the number of boys be x and the number of girls be y .

$$\text{Then, } 3(B + G) = 3\left(\frac{x}{y} + \frac{y}{x}\right) = \frac{3(x^2 + y^2)}{xy}$$

Clearly, which is greater than 3.

$$\therefore [(a - b)^2 \geq 0 \Rightarrow a^2 + b^2 - 2ab \geq 0]$$

$$\Rightarrow a^2 + b^2 \geq 2ab \Rightarrow \frac{a^2 + b^2}{ab} \geq 2$$

4. (b) Let x gallons of first mixture be mixed with y gallons of second mixture.

	Milk	Water
x gallons (1st)	$\frac{8}{9}x$	$\frac{1}{9}x$
y gallons (2nd)	$\frac{1}{6}y$	$\frac{5}{6}y$
Third vessel	$\frac{8}{9}x + \frac{1}{6}y$	$\frac{1}{9}x + \frac{5}{6}y$

Since the third vessel contains half milk and half water,

$$\frac{8}{9}x + \frac{1}{6}y = \frac{1}{9}x + \frac{5}{6}y \Rightarrow 16x + 3y = 2x + 15y$$

$$\Rightarrow 16x - 2x = 15y - 3y \Rightarrow \frac{x}{y} = \frac{12}{6} = \frac{2}{1}$$

$$\text{Hence } y = \frac{7}{13} \times 26 = 14 \text{ gallons}$$

5. (b) First glass contains milk = $\frac{2}{3}$ and water = $\frac{1}{3}$

$$\text{Second glass contains milk} = \frac{1}{4} \text{ and water} = \frac{3}{4}$$

$$\therefore \text{New tumbler contains milk} = \left(\frac{2}{3} + \frac{1}{4}\right) = \frac{11}{12}$$

$$\text{New tumbler contains water} = \left(\frac{1}{3} + \frac{3}{4}\right) = \frac{13}{12}$$

$$\therefore \text{Required ratio} = \frac{11}{12} : \frac{13}{12} = 11 : 13$$

6. (c) Let the three numbers be x , y and z .

$$\text{Then, } x^3 + y^3 + z^3 = 584 \quad \dots(1)$$

$$x : y = 1 : 2 \quad \dots(2)$$

$$y : z = 1 : 2 \quad \dots(3)$$

From Eq (2) and (3), we get $x : y : z = 1 : 2 : 4$

Let $x = k$, $y = 2k$ and $z = 4k$. Then,

$$k^3 + (2k)^3 + (4k)^3 = 584$$

$$\Rightarrow 73k^3 = 584$$

$$\Rightarrow k^3 = 8 \Rightarrow k = 2$$

$$\therefore \text{The third number} = z = 4k = (4 \times 2) = 8$$

7. (a) Let $(a + b) = 6k$, $(b + c) = 7k$ and $(c + a) = 8k$
Then, $2(a + b + c) = 21k \Leftrightarrow 2 \times 14 = 21k$

$$\Leftrightarrow k = \frac{28}{21} = \frac{4}{3}$$

$$\therefore (a + b) = \left(6 \times \frac{4}{3}\right) = 8 \Rightarrow c = (a + b + c) - (a + b) = (14 - 8) = 6$$

8. (b) Let $P = 2x$ and $Q = 3x$. Then, $\frac{Q}{R} = \frac{2}{3}$

$$\Rightarrow R = \frac{3}{2}Q = \left(\frac{3}{2} \times 3x\right) = \frac{9x}{2}$$

$$\text{Also, } \frac{R}{S} = \frac{2}{3} \Rightarrow S = \frac{3}{2}R = \left(\frac{3}{2} \times \frac{9x}{2}\right) = \frac{27x}{4}$$

$$\text{Thus, } P = 2x, Q = 3x, R = \frac{9x}{2} \text{ and } S = \frac{27x}{4}$$

$$\text{Now, } P + Q + R + S = 1300$$

$$\Leftrightarrow \left(2x + 3x + \frac{9x}{2} + \frac{27x}{4}\right) = 1300$$

$$\Leftrightarrow (8x + 12x + 18x + 27x) = 5200$$

$$\Leftrightarrow 65x = 5200 \Leftrightarrow x = \frac{5200}{65} = 80.$$

$$\therefore P's \text{ share} = (2 \times 80) = 160.$$

9. (a) The piece that is cut off should be such that the fraction of the first to the second alloy in each of the two new alloys formed should be equal.

If you cut off 4 kg, the respective ratios will be:

First alloy: 2 kg of first alloy and 4 kg of second alloy
second alloy : 4 kg of first alloy and 8 kg of the second alloy. It can easily be seen that the ratios are equal to 1 : 2 in each case.

10. (d) $\frac{S}{F} = \frac{1}{5} \Rightarrow F = 5S$

$$\frac{M}{F} = \frac{4}{5} \Rightarrow M = \frac{4}{5}F$$

$$\frac{S+2}{M+2} = \frac{3}{10}$$

$$\Rightarrow 10S + 20 = 30M + 6 = 3 \times \frac{4}{5} \times 5S + 6 = 12S + 6$$

$$\Rightarrow 2S = 14 \Rightarrow S = 7 \text{ years}$$

$$\Rightarrow F = 5S = 35 \text{ years}$$

11. (d) The sum of the numerator and denominator of the ratio should be a prime number.

Note that the number of employees is less than 300. Consider options.

- (a) 101 : 88

$$101 + 88 = 189$$

$189 \times 2 > 300 \Rightarrow$ Number of employees who are graduates and above = 101 and the others are 88 in number 189 is not a prime number. Option (a) is eliminated.

- (b) 87 : 100

$$87 + 100 = 187, 187 \times 2 > 300$$

187 is not a prime number.

- (c) 110 : 111

$$110 + 111 = 221, 221 \times 2 > 300$$

221 is not a prime number.

12. (a) Frequency of steps of $A : B : C = 5 : 6 : 7$

But in terms of size of steps; $6A = 7B = 8C$

$$\text{Ratio of the speeds of } A : B : C = (5/6) : (6/7) : (7/8) = 140 : 144 : 147$$

Explanation of Test Yourself

1. (c) Let $A = 2k$, $B = 3k$ and $C = 5k$.

$$A's \text{ new salary } \frac{115}{100} \text{ of } 2k \left(\frac{115}{100} \times 2k \right) = \frac{23}{10}k$$

$$B's \text{ new salary } \frac{110}{100} \text{ of } 3k \left(\frac{110}{100} \times 3k \right) = \frac{33}{10}k$$

$$C's \text{ new salary } \frac{120}{100} \text{ of } 5k \left(\frac{120}{100} \times 5k \right) = 6k$$

$$\therefore \text{ New ratio } = \frac{23k}{10} : \frac{33k}{10} : 6k = 23 : 33 : 60.$$

2. (b) 18 carat gold

$$= \frac{3}{4} \text{ pure gold} = \frac{3}{4} \times 24 = 18 \text{ carat gold}$$

20 carat gold

$$= \frac{5}{6} \text{ pure gold} = \frac{5}{6} \times 24 = 20 \text{ carat gold}$$

Required ratio = $18 : 20 = 9 : 10$

3. (c) Do not get confused by the distractions given in the problem. 10 men and 10 days means 100 man-days are required to clean 10 floors.

That is, 1 floor requires 10 man-days to get cleaned.

Hence, 8 floors will require 80 man-days to clean.

Therefore, 10 days are required to clean 8 floors.

4. (c) Let the three containers contain $3x$, $4x$ and $5x$ litres of mixtures, respectively.

$$\text{Milk in 1st mix.} = \left(3x \times \frac{4}{5} \right) \text{ litres} = \frac{12x}{5} \text{ litres.}$$

$$\text{Water in 1st mix.} = \left(3x - \frac{12x}{5} \right) \text{ litres} = \frac{3x}{5} \text{ litres.}$$

$$\text{Milk in 2nd mix.} = \left(4x \times \frac{3}{4} \right) \text{ litres} = 3x \text{ litres.}$$

$$\text{Water in 2nd mix.} = (4x - 3x) \text{ litres} = x \text{ litres.}$$

$$\text{Milk in 3rd mix.} = \left(5x \times \frac{5}{7} \right) \text{ litres} = \frac{25x}{7} \text{ litres.}$$

$$\text{Water in 3rd mix.} = \left(5x - \frac{25x}{7} \right) \text{ litres} = \frac{10x}{7} \text{ litres.}$$

Total milk in final mix.

$$= \left(\frac{12x}{5} + 3x + \frac{25x}{7} \right) \text{ litres} = \frac{314x}{35} \text{ litres.}$$

Total water in final mix.

$$= \left(\frac{3x}{5} + x + \frac{10x}{7} \right) \text{ litres} = \frac{106x}{35} \text{ litres.}$$

Required ratio of milk and water

$$= \frac{314x}{35} : \frac{106x}{35} \quad 157 : 53.$$

5. (b) Assume that there are 500, 400 and 500 litres respectively in the 3 containers.

Then we have, 83.33, 150 and 208.33 litres of milk in each of the three containers.

Thus, the total milk is 441.66 litres. Hence, the amount of water in the mixture is

$$1400 - 441.66 = 958.33 \text{ litres.}$$

Hence, the ratio of milk to water is

$$441.66 : 958.33 \times 53 : 115 \text{ (Using division by 0.33333)}$$

The calculation thought process should be:

$$(441 \times 3 + 2) : (958 \times 3 + 1) = 1325 : 2875.$$

Dividing by 25 $\rightarrow 53 : 115$.

6. (c) Let x pairs of brown socks were ordered.

Let P be the price of a brown pair.

Therefore, price of the black pair of sock = $2P$

$$\text{Now, } 4P + 2Px = 1.5(Px + 8P)$$

$$\Rightarrow 4P + 2Px = \frac{3}{2}(Px + 8P)$$

$$\Rightarrow 8P + 4Px = 3Px + 24P$$

$$\Rightarrow Px = 16P \Rightarrow x = 16$$

$$\therefore \text{ Required ratio } = \frac{4}{16} = 1 : 4$$

7. (c) For 9 kg zinc, mixture melted = $(9 + 11)$ kg.

For 28.8 kg zinc, mixture melted

$$= \left(\frac{20}{9} \times 28.8 \right) \text{ kg} = 64 \text{ kg.}$$

8. (a) $\frac{V}{C} \cdot \frac{2}{3}$ and $\frac{V+4}{C} \cdot \frac{3}{4} \dots(1)$

$$\therefore C = \frac{3V}{2} \Rightarrow \frac{V+4}{3V/2} \cdot \frac{3}{4}$$

[From (1)]

where V denoted for vanilla and C for chocolate.

$$\Rightarrow 4V + 16 = \frac{9V}{2} \Rightarrow 8V + 32 = 9V \Rightarrow V = 32$$

9. (b) $4E + 3M + 5C = 3650$

Also, $3C = 2M$, that is, $M = 1.5C$

and $3E = 2C$ that is, $E = 0.66C$

$$\text{Thus, } 4 \times 0.66C + 3 \times 1.5C + 5C = 3650$$

$$C = 3650/12.166$$

That is, $C = 300$

$$\text{Hence, } M = 1.5 \times C = 450$$

10. (a) The no. of coins of 1 ₹ = $3x$ and $25p = x$.
Conventionally, we can solve this using equations as follows.
 $A + B + C = 220$... (1)
 $A = 3C$... (2)
 $A + 0.5B + 0.25C = 160$... (3)
 We have a situation with 3 equations and 3 unknowns and we can solve for
 A (no. of 1 ₹ coins),
 B (no. of 50 paise coins)
 and C (no. of 25 paise coins)
 However, a much smarter approach would be to go through the options. If we check option (a) – no. of 50 paise coins = 60 we would get the number of 1 ₹ coins as 120 and the number of 25 paise coins as 40.
 $120 \times 1 + 60 \times 0.5 + 40 \times 0.25 = 160$
 This fits the conditions perfectly and is hence the correct answer.
11. (c) Solve using options. For option (c), we will get that initially there are 125 boys and 140 girls. After the given increases, the number of boys would be 145 and the number of girls would become 154 which gives a difference of 9 as required.
12. (a) Ratio of water and milk in mixture = $20 : 80 = 1 : 4$
 25% of mixture is sold
 $\therefore$ amount of water in mixture = 15l
 & amount of milk in mixture = 60l
 In new mixture ratio of water and milk
 = $(15 + 25) : 60 = 2 : 3$

13. (a) $\frac{a}{b} = \frac{1}{3} \Rightarrow$ for $a = 1, b = 3$
 $\frac{b}{c} = \frac{2}{1} \Rightarrow$ for $b = 3, c = \frac{3}{2}$
 $\frac{c}{d} = \frac{1}{2} \Rightarrow$ for $c = \frac{3}{2}, d = 3$
 $\frac{d}{e} = \frac{3}{1} \Rightarrow$ for $d = 3, e = 1$
 $\frac{e}{f} = \frac{1}{4} \Rightarrow$ for $e = 1, f = 4$
 $\Rightarrow \frac{abc}{def} = \frac{1 \times 3 \times 3/2}{3 \times 1 \times 4} = \frac{3}{8}$
14. (d) $P = K \times W^2 \rightarrow 12250 = K \times 35^2 \rightarrow K = 10$.
 Thus our price and weight relationship is: $P = 10W^2$.
 When the two pieces are in the ratio 2 : 5 (weight wise) then we know that their weights must be 10 grams and 25 grams respectively. Their values would be:
 10 gram piece: $10 \times 10^2 = ₹ 1000$;
 25 gram piece: $10 \times 25^2 = ₹ 6250$.
 Total Price = $1000 + 6250 = 7250$. From an initial value of 12250, this represents a loss of ₹ 5000.
15. (c) Since, the work gets done in 25% less time there must have been an addition of 33.33% men.
 This would mean 13.33 men extra $\rightarrow$ which would mean 14 extra men (in whole nos.)
 $\therefore$ Ratio = $54 : 40 = 27 : 20$

9

TIME AND WORK

- Introduction
- Concept of Efficiency
- Concept of Negative work
- Concept of Man-days
- Work Done
- Work Done Equation
- Work in Terms of Volume (special case as building a wall)
- Extension of the Concept of Time and Work

INTRODUCTION

Work and time is an important chapter for every aptitude test. This chapter plays an important role in CAT and other equivalent aptitude tests, on an average 2–3 questions from this chapter are regularly asked in CAT. The questions from this chapter are not directly based on formulae.

For solving the questions of this chapter, concepts of ratio and proportion are required. Students are advised to be highly attentive in solving the problems of this chapter.

CONCEPT OF EFFICIENCY

Efficiency means rate of doing work. This means that more the efficiency, less will be the number of days required to complete a certain work and less the efficiency, more will be the number of days required to complete a certain work.

Aliza is twice as efficient as Binny.

- ⇒ Aliza does twice as much work as Binny in the same time interval
- ⇒ Aliza will require half the time as required by Binny to do the same work.

CONCEPT OF NEGATIVE WORK

Suppose two persons A and B are working to build a wall while C is working to demolish the wall. If we consider the work as the building of the wall, then breaking the wall (by C) is negative work.

The concept of negative work generally appears in the problems based on pipes and cisterns, where there are inlet pipes and outlet pipes/leaks, which are working against each other.

If we consider the work of filling a tank, the inlet pipe does positive work while the outlet pipe/leaks does negative work.

Illustration 1: A can build a wall in 15 days and B can build it in 10 days, while C can completely demolish the wall in 12 days. If they start working at the same time, in how many days will the work be completed.

Solution: Work per day by $A = \frac{1}{15}$

Work per day by $B = \frac{1}{10}$

Work per day by $C = -\frac{1}{12}$

(negative sign is taken for negative work)

The net combined work per day by A, B and C

$$= \frac{1}{15} + \frac{1}{10} - \frac{1}{12} = \frac{1}{12}$$

Since, Total work done = (Work done per day) $\times$ (No. of days required to complete the work)

∴ No. of days required to complete the work

$$= \frac{\text{Total work done}}{\text{Work done per day}} = \frac{1}{\frac{1}{12}} = 12$$

CONCEPT OF MAN-DAYS

If ' M ' men working together can complete a work in ' D ' days, then the product of number of men (M) and number of days (D) i.e. $M \times D$ is known as the number of MAN-DAYS.

Number of man days to complete a specific task always remains constant.

Suppose 30 persons working together for 20 days to complete a job, then the total work done is equal to $(30 \times 20 = 600)$

man-days. If we change the number of days in which the work is to be completed, then the other factor i.e. the number of persons will change accordingly, so that the product of the factors becomes equal to 600 man-days.

WORK DONE

Consider a whole work as the unit work.

1. Work Done by Two Persons

Let A can do a whole work in x days and B can do the same one unit work in y days.

$$\text{Hence work done by } A \text{ in one day} = \frac{1}{x}$$

$$\text{and work done by } B \text{ in one day} = \frac{1}{y}$$

Then work done in one day when A and B work together

$$\begin{aligned} &= \frac{1}{x} + \frac{1}{y} \\ &= \frac{y+x}{xy} \quad \text{or} \quad \frac{x+y}{xy} \end{aligned}$$

Whole work = (Work done in one day) $\times$ (Number of days required to complete the whole work)

$$\begin{aligned} \text{Hence, number of days required to complete the whole work} \\ &= \frac{\text{Whole work}}{\text{Work done in one day}} \end{aligned}$$

$\Rightarrow$ Number of days required to complete the whole work when A and B are working together

$$= \frac{1}{\frac{1}{x} + \frac{1}{y}} = \frac{xy}{x+y}, \text{ because a whole work is considered as one}$$

unit of work.

Illustration 2: If A can do a work in 10 days and B can do the same work in 15 days, then how many days will they take to complete the work both while working together?

Solution: Work done by A in one day = $\frac{1}{10}$

$$\text{Work done by } B \text{ in one day} = \frac{1}{15}$$

Work done in one day when A and B work together

$$= \frac{1}{10} + \frac{1}{15} = \frac{3+2}{30} = \frac{5}{30} = \frac{1}{6}$$

$$\text{Hence required number of days} = \frac{1}{\frac{1}{6}} = 6$$

We can find the required number of days directly by using the formula,

$$\begin{aligned} \text{Number of days} &= \frac{xy}{x+y}, \text{ derived above} \\ &= \frac{10 \times 15}{10+15} = \frac{150}{25} = 6. \end{aligned}$$

Illustration 3: 'A' completes a work in 12 days. 'B' completes the same work in 15 days. 'A' started working alone and after 3 days B joined him. How many days will they now take together to complete the remaining work?

- (a) 5 (b) 8
(c) 6 (d) 4

Solution: (a) Work done by 'A' in 3 days = $\frac{1}{12} \times 3 = \frac{1}{4}$

$$\therefore \text{Remaining work} = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\text{Work done by } A \text{ and } B \text{ together} = \frac{12 \times 15}{27} \times \frac{20}{3} =$$

$$\therefore \text{Remaining work done by } A \text{ and } B \text{ together in} \\ = \frac{3}{4} \times \frac{20}{3} = 5 \text{ days}$$

2. Work Done by Three Persons

As we derived the formula for two persons, you can also derived the formula for three persons in the same way.

If A, B, C can do a work in x, y and z days respectively, then all of them working together can finish the work in $\frac{xyz}{xy + yz + zx}$ days.

Illustration 4: If A, B, C can do a work in 12, 15 and 20 days respectively, then how many days will they take to complete the work when all the three work together.

Solution:

$$\begin{aligned} \text{Required number of days} &= \frac{xyz}{xy + yz + zx} \\ &= \frac{12 \times 15 \times 20}{12 \times 15 + 15 \times 20 + 20 \times 12} \\ &= \frac{3600}{180 + 300 + 240} = \frac{3600}{720} = 5 \end{aligned}$$

Illustration 5: A and B can do a certain piece of work in 8 days, B and C can do it in 12 days and C and A can do it in 24 days. How long would each take separately to do it?

Solution: $(A + B)$'s one days's work = $1/8$,
 $(A + C)$'s one days's work = $1/24$,
 $(B + C)$'s one days's work = $1/12$,

Now add up all three equations :

$$2(A + B + C)\text{'s one days's work} = \frac{1}{8} + \frac{1}{24} + \frac{1}{12} = \frac{13}{72}$$

$$(A + B + C)\text{'s one days's work} = \frac{13}{144}$$

$$A\text{'s one days's work} = (A + B + C)\text{'s one days's work}$$

$$- (B + C)\text{'s one days's work} = \frac{13}{144} - \frac{1}{12} = \frac{1}{144}$$

Since A completes of the work in 1 day, he will complete 1 work in $\frac{144}{1} = 144$ days

By similar logic we can find that B needs days and C will require $\frac{144}{5}$ days.

If A and B together can do a piece of work in X days and A alone can do it in Y days, then B alone can do the work in $\frac{XY}{Y-X}$ days.

Illustration 6: Machine A can print one lakh books in 8 hours. Machine B can do the same job in 10 hours. Machine C can do the same job in 12 hours. All the three machines start job at 9.00 am. A breaks down at 11.00 am and the other two machines finish the job. Approximately at what time will the job be finished?

- (a) 12.00 noon (b) 1.30 pm
(c) 1.00 pm (d) 11.30 am

Solution: (c) Part of print done by A , B and C in 2

$$\text{hours} = 2 \left(\frac{1}{8} + \frac{1}{10} + \frac{1}{12} \right) = \frac{37}{60}$$

$$\text{Remaining} = 1 - \frac{37}{60} = \frac{23}{60}$$

If B and C print together, then they can print in $\frac{10 \times 12}{10 + 12}$ hrs.

Therefore, remaining part can be printed by

$$B \text{ and } C \text{ in } \frac{10 \times 12}{22} \times \frac{23}{60} \approx 2 \text{ hrs.}$$

Hence, the job will be finished at 9 am + 2 + 2 = 1.00 p.m.

3. If A and B Together Can do a Work in x Days and A Alone can do it in y Days, then B alone can do the Work in $\frac{xy}{y-x}$ Days

Illustration 7: A and B can do a work in 8 days and A alone can do it in 12 days. In how many days can B alone do it?

Solution:

$$\text{Work done by } A \text{ and } B \text{ working together in one day} = \frac{1}{8}$$

$$\text{Work done by } A \text{ in one day} = \frac{1}{12}$$

$$\therefore \text{Work done by } B \text{ in one day} = \frac{1}{8} - \frac{1}{12} = \frac{3-2}{24} = \frac{1}{24}$$

$$\begin{aligned} \text{Hence number of days in which } B \text{ alone can do the whole work} \\ = \frac{1}{\frac{1}{24}} = 1 \times \frac{24}{1} = 24 \end{aligned}$$

You can find required number of days directly by using the above formula as

$$\text{Required number of days} = \frac{xy}{y-x} = \frac{8 \times 12}{12-8} = \frac{8 \times 12}{4} = 24.$$

WORK DONE EQUATION

Work Done = Number of men $\times$ Number of Days

If M men can do W works in D days, then

$$\boxed{W = M \times D}$$

or $M \times D = W$

If M_1 men can do W_1 works in D_1 days and M_2 men can do W_2 works in D_2 days, then

$$M_1 D_1 = W_1 \quad \dots (1)$$

$$\text{and} \quad M_2 D_2 = W_2 \quad \dots (2)$$

On dividing equation (1) by (2), we get

$$\frac{M_1 D_1}{M_2 D_2} = \frac{W_1}{W_2}$$

$$\Rightarrow \boxed{M_1 D_1 W_2 = M_2 D_2 W_1}$$

Similarly if M_1 men can do W_1 works in D_1 days

Working T_1 hours per day and M_2 men can do W_2 works in D_2 days working T_2 hours per day, then

$$\boxed{M_1 D_1 T_1 W_2 = M_2 D_2 T_2 W_1}$$

Also if efficiency of each of M_1 men in first group is E_1 and efficiency of each of M_2 men in second group is E_2 , then

$$\boxed{M_1 D_1 T_1 E_1 W_2 = M_2 D_2 T_2 E_2 W_1}$$

This is the general equation in two work situations. Suffix 1 indicate first work situation while the suffix 2 indicate the second work situation. Thus in general,

$$M_1 D_1 T_1 E_1 W_2 = M_2 D_2 T_2 E_2 W_1$$

If one or more items in both work situations are the same, then no need to write them in the general equation in two work situations. For examples

- If E_1 and E_2 are the same, then $M_1 D_1 T_1 W_2 = M_2 D_2 T_2 W_1$
- If E_1 and E_2 and T_1 and T_2 are the same then $M_1 D_1 W_2 = M_2 D_2 W_1$
- If E_1 and E_2 , T_1 and T_2 and W_1 and W_2 are the same then $M_1 D_1 = M_2 D_2$

Illustration 8: 20 men can do a work in 35 days. How many men are needed to complete the same work in 25 days.

Solution: $M_1 D_1 = M_2 D_2$

$$\text{Here} \quad M_1 = 20, \quad D_1 = 35,$$

$$M_2 = ?, \quad D_2 = 25$$

$$\therefore 20 \times 35 = M_2 \times 25$$

$$\Rightarrow M_2 = \frac{20 \times 35}{25} = 28$$

Hence required number of men = 28.

Illustration 9: 12 men complete a work in 18 days. Six days after they had started working, 4 men joined them. How many days will all of them take to complete the remaining work?

- (a) 10 days (b) 12 days
(c) 15 days (d) 9 days

Solution: (d) In 1 day, work done by 12 men = $\frac{1}{18}$

$$\text{In 6 days, work done by 12 men} = \frac{6}{18} = \frac{1}{3}$$

$$\text{Remaining work} = \frac{2}{3}$$

$$\text{Now, } M_1 \times D_1 \times W_2 = M_2 \times D_2 \times W_1$$

$$\text{or } 12 \times 18 \times \frac{2}{3} = 16 \times D_2 \times 1$$

$$\text{or } D_2 = \frac{4 \times 18 \times 2}{16} \text{ days} = 9$$

Illustration 10: 25 men can make 120 chairs in 10 days. How many chairs can be made by 35 men in 15 days.

Solution:

$$M_1 D_1 W_2 = M_2 D_2 W_1$$

$$\text{Here } M_1 = 25, D_1 = 10, W_1 = 120$$

$$M_2 = 35, D_2 = 15, W_2 = ?$$

$$\therefore 25 \times 10 \times W_2 = 35 \times 15 \times 120$$

$$\Rightarrow W_2 = \frac{35 \times 15 \times 120}{25 \times 10} = 252$$

Hence required number of chairs = 252.

Illustration 11: 12 men can make 80 tables in 20 days working 8 hours a day. In how many days 30 men can make 120 tables working 6 hours a day?

Solution:

$$M_1 D_1 T_1 W_2 = M_2 D_2 T_2 W_1$$

$$\text{Here } M_1 = 12, D_1 = 20, T_1 = 8, W_1 = 80$$

$$M_2 = 30, D_2 = ?, T_2 = 6, W_2 = 120$$

$$\therefore 12 \times 20 \times 8 \times 120 = 30 \times D_2 \times 6 \times 80$$

$$\Rightarrow D_2 = \frac{12 \times 20 \times 8 \times 120}{30 \times 6 \times 80} = 16$$

Hence required number of days = 16 days.

Illustration 12: 5 men can prepare 150 toys in 5 days working 6 hours a day. In how many hours 10 boys can prepare 200 toys in 10 days, if a man works thrice as fast as a boy?

Solution: Since a man works thrice as fast as a boy.

$$\therefore (\text{Efficiency of a man}) : (\text{Efficiency of a boy}) = 3 : 1$$

$$\therefore E_1 : E_2 = 3 : 1 \text{ or } \frac{E_1}{E_2} = \frac{3}{1}$$

Since

$$M_1 D_1 T_1 W_2 = M_2 D_2 T_2 W_1$$

$$\text{Here } M_1 = 5, D_1 = 5, T_1 = 6, W_1 = 150$$

$$M_2 = 10, D_2 = 10, T_2 = ?, W_2 = 200$$

$$\therefore 5 \times 5 \times 6 \times 200 = 10 \times 10 \times T_2 \times 150$$

$$\Rightarrow T_2 = \frac{5 \times 5 \times 6 \times 200}{10 \times 10 \times 150} = 2$$

Hence required number of hours = 2 hours.

Illustration 13: 8 men can do a work in 12 days while 20 women can do it in 10 days. In how many days can 12 men and 15 women complete the same work.

Solution: 8 men can do a work in 12 days while 20 women can do it in 10 days.

$M_1 D_1 E_1 = M_2 D_2 E_2$, where E_1 and E_2 are efficiency of a man and a woman respectively

$$\text{Here } M_1 = 8, D_1 = 12, E_1 = ?$$

$$M_2 = 20, D_2 = 10, E_2 = ?$$

$$\therefore 8 \times 12 \times E_1 = 20 \times 10 \times E_2$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{25}{12} \text{ or } \frac{E_2}{E_1} = \frac{12}{25}$$

Since 12 men and 15 women have to complete the same work as can be done by 8 men in 12 days (or 20 women in 10 days)

$$\therefore (12 \times E_1 + 15 \times E_2) \times D_3 = 8 \times 12 \times E_1$$

$$\Rightarrow \frac{12 \times E_1 + 15 \times E_2}{E_1} \times D_3 = 8 \times 12$$

$$\Rightarrow \left(12 + 15 \times \frac{E_2}{E_1} \right) \times D_3 = 8 \times 12$$

$$\Rightarrow \left(12 + 15 \times \frac{12}{25} \right) \times D_3 = 8 \times 12$$

$$\Rightarrow D_3 = 5$$

Hence 12 men and 15 women can complete the work in 5 days.

Illustration 14: 12 men and 16 boys can do a piece of work in 5 days. 13 men and 24 boys can do the same work in 4 days. How long will 7 men and 10 boys take to do the same work?

Solution: 12 men and 16 boys can do a piece of work in 5 days 13 men and 24 boys can do it in 4 days.

If E_1 and E_2 are the efficiency of a man and a boy respectively, then

$$(12 \times E_1 + 16 \times E_2) 5 = (13 \times E_1 + 24 \times E_2) \times 4$$

$$60 E_1 + 80 E_2 = 52 E_1 + 96 E_2$$

$$\Rightarrow 8 E_1 = 16 E_2 \Rightarrow \frac{E_1}{E_2} = 2$$

Now 7 men and 10 boys have to complete the same work as can be done by 12 men and 16 boys in 5 days (or 13 men and 24 boys in 4 days)

$$\therefore (7 \times E_1 + 10 \times E_2) \times D_3 = (12 \times E_1 + 16 \times E_2) \times 5$$

$\Rightarrow$ Dividing both sides by E_2 , we get

$$\Rightarrow \left(7 \times \frac{E_1}{E_2} + 10 \right) \times D_3 = \left(12 \times \frac{E_1}{E_2} + 16 \right) \times 5$$

$$\Rightarrow (7 \times 2 + 10) \times D_3 = (12 \times 2 + 16) \times 5$$

$$\Rightarrow D_3 = \frac{25}{3} = 8 \frac{1}{3} \text{ days}$$

Illustration 15: A works twice as much as B in the same time period. Together, they finish the work in 14 days. In how many days can it be done by each separately?

Solution: Let E_1 and E_2 be the efficiency of A and B respectively.

$$\therefore E_1 : E_2 = 2 : 1, \Rightarrow \frac{E_1}{E_2} = \frac{2}{1} \text{ or } \frac{E_2}{E_1} = \frac{1}{2}$$

$$(1 \times E_1 + 1 \times E_2) \times 14 = 1 \times E_1 \times D_2,$$

where D_2 is the number of days in which A can complete the work separately.

$$\Rightarrow \left(1 + \frac{E_2}{E_1} \right) \times 14 = D_2 \Rightarrow D_2 = 21$$

Hence A can do the work separately in 21 days. Since A works twice as much as B in the same time period, hence B will take twice as much time as A. Therefore B can complete the work separately in 42 days.

Illustration 16: 10 men can finish a piece of work in 10 days, whereas 12 women can finish it in 10 days. If 15 men and 6 women undertake to complete the work, how many days they will take to complete it?

Solution: Let the efficiency of a man and a woman are E_1 and E_2 respectively.

Now 10 men can finish a piece of work in 10 days, whereas 12 women can finish it in 10 days.

$$\therefore 10 \times E_1 \times 10 = 12 \times E_2 \times 10 \quad [\because M_1 E_1 D_1 = M_2 E_2 D_2]$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{6}{5} \quad \text{or} \quad \frac{E_2}{E_1} = \frac{5}{6}$$

Now 15 men and 6 women have to complete the same work that 10 men can finish in 10 days (or 12 women can finish in 10 days).

$$\therefore (15 \times E_1 + 6 \times E_2) \times D_3 = 10 \times E_1 \times 10$$

$$\Rightarrow \left(15 + 6 \times \frac{E_2}{E_1} \right) \times D_3 = 10 \times 10$$

$$\Rightarrow \left(15 + 6 \times \frac{5}{6} \right) \times D_3 = 10 \times 10$$

$$\Rightarrow D_3 = 5$$

Hence required number of days = 5 days.

Illustration 17: A and B can do a work in 45 and 40 days respectively. They began the work together, but A left after some time and B finished the remaining work in 23 days. After how many days did A leave?

Solution: A and B can do a work in 45 and 40 days respectively

$$\therefore 1 \times E_1 \times 45 = 1 \times E_2 \times 40 \quad [\because M_1 D_1 E_1 = M_2 D_2 E_2]$$

$$\Rightarrow \frac{E_1}{E_2} = \frac{8}{9} \quad \text{or} \quad \frac{E_2}{E_1} = \frac{9}{8},$$

where E_1 and E_2 are efficiency of A and B respectively.

Since A and B began the work together to do the same work which A can do in 45 days (or B can do in 40 days), but A left after some time and B finished the remaining work in 23 days.

$\therefore E_1 \times D_3 + E_2 \times (D_3 + 23) = E_1 \times 45$, where D_3 is the number of days after which A left.

$$\Rightarrow D_3 + \frac{E_2}{E_1} \times (D_3 + 23) = 45$$

$$\Rightarrow D_3 + \frac{9}{8} \times (D_3 + 23) = 45$$

$$\Rightarrow D_3 + \frac{9}{8} D_3 + \frac{9}{8} \times 23 = 45$$

$$\Rightarrow 17 D_3 = 45 \times 8 - 9 \times 23 = 360 - 207 = 153$$

$$\therefore D_3 = 9$$

Hence required number of days = 9.

WORK IN TERMS OF VOLUME

(SPECIAL CASE AS BUILDING A WALL)

In some problems work is considered in terms of volume of work. For example volume of a wall of a certain length (L), breadth (B) and height (H) is LBH. In such type of problems the following equation is used:

If work = volume

$$\text{Then } M_1 T_1 D_1 W_1 = M_2 T_2 D_2 W_2$$

$$\Rightarrow \frac{W_1}{W_2} = \frac{M_1 T_1 D_1}{M_2 T_2 D_2}$$

$$\Rightarrow \frac{L_1 B_1 H_1}{L_2 B_2 H_2} = \frac{M_1 T_1 D_1}{M_2 T_2 D_2}$$

Where L_1, B_1, H_1 are the length, breadth and height of the wall to be built in first work situation and L_2, B_2, H_2 are the length, breadth and height of the wall to be built in second work situation.

M_1, T_1, D_1 are the number of men, number of hours and number of days in first work situation and M_2, T_2, D_2 are number of men, number of hours and number of days in second work condition.

Illustration 18: 5 men working 8 hours a day can completely build a wall of length 20 metres, breadth $\frac{1}{4}$ metre and height 6 metres in 3 days. How many days will 8 men working 6 hours a day require to build a wall of length 120 meters, breadth $\frac{1}{2}$ metre and height 4 metres.

$$\text{Solution: } \frac{L_1 B_1 H_1}{L_2 B_2 H_2} = \frac{M_1 T_1 D_1}{M_2 T_2 D_2}$$

$$\text{Here } L_1 = 20, B_1 = \frac{1}{4}, H_1 = 6, M_1 = 5, T_1 = 8, D_1 = 3,$$

$$L_2 = 120, B_2 = \frac{1}{2}, H_2 = 4, M_2 = 8, T_2 = 6, D_2 = ?$$

$$\therefore \frac{20 \times \frac{1}{4} \times 6}{120 \times \frac{1}{2} \times 4} = \frac{5 \times 8 \times 3}{8 \times 6 \times D_2} \Rightarrow D_2 = 20$$

Hence required number of days = 20 days.

EXTENSION OF THE CONCEPT OF TIME AND WORK

1. Pipes and Cisterns

Problems related to Pipes and Cisterns are almost the same as those of Time and Work. Statement 'pipes A and B can fill a tank in 2 hours and 3 hours working individually' is similar to the statement 'A and B can do a work in 2 hours and 3 hours respectively working individually'.

If a pipe fills a tank in 3 hours, then the pipe fills $\frac{1}{3}$ rd of the same tank in 1 hour.

The only difference with the pipes and cisterns problems is that there are inlets as well as outlets. Inlet is a pipe connected with a tank (or a cistern or a reservoir) that fills it. Outlet is a pipe connected with a tank (or a cistern or a reservoir) that empties it.

Hence, if we consider filling the tank by inlet as positive work, then emptying the tank by outlet will be considered as negative work.

(a) Let a pipe fill a tank in x hours and another pipe can empty the full tank in y hours. Then the net part of the tank filled in 1 hour, when both the pipes are opened, if x is less than y .

$$= \frac{1}{x} - \frac{1}{y} = \frac{y - x}{xy}$$

$\therefore$ time taken to fill the tank, when both the pipes are opened

$$= \frac{1}{\frac{y - x}{xy}} = \frac{xy}{y - x}$$

- (b) Let a pipe fill a tank in x hours while another fills the same tank in y hours but a third one empties the full tank in z hours. If all the three pipes are opened together, then the

$$\begin{aligned} \text{net part of the tank filled in 1 hour} &= \frac{1}{x} + \frac{1}{y} - \frac{1}{z} \\ &= \frac{yz + zx - xy}{xyz} \end{aligned}$$

$$\therefore \text{time taken to fill the tank} = \frac{xyz}{yz + zx - xy}$$

- (c) Let a pipe fill a tank in x hours but due to the leak in the bottom, the tank is filled in y hours and when the tank is filled, the time taken by the leak to empty the tank is z hours.

Net part of the tank filled in 1 hour by the pipe when there

$$\text{is the leak in the bottom} = \frac{1}{x} - \frac{1}{z} = \frac{z - x}{xz}$$

Since the tank will be filled completely in y hours by the pipe when there is the leak in the bottom, therefore

$$\left(\frac{z - x}{xz} \right) \times y = 1 \Rightarrow y = \frac{xz}{z - x} \Rightarrow yz - xy = xz$$

$$\Rightarrow z(y - x) = xy \Rightarrow z = \frac{xy}{y - x}$$

Hence, if a pipe can fill a tank in x hours but due to the leak in the bottom, the tank is filled in y hours, then the

fully filled tank will be emptied in $\frac{xy}{y - x}$ hours.

- (d) Let a pipe A fill a tank in x hrs while pipe B can fill the tank in y hrs alone. When both the pipes are opened together, then time required to fill the tank = $\frac{xy}{x + y}$ hrs.

- (e) Let pipes A , B and C fill a tank alone in x , y and z hrs respectively. When all the three pipes open together, then

$$\text{time required to fill the tank} = \frac{xyz}{xy + yz + zx} \text{ hrs.}$$

Illustration 19: If a pipe fills a tank in 4 hrs and another pipe can empty the full tank in 6 hrs. When both the pipes are opened together, then find the time required to completely fill the tank.

Solution: Required time = $\frac{xy}{y - x}$

Here $x = 4$, $y = 6$

$$\therefore \text{Required time} = \frac{4 \times 6}{6 - 4} = 12 \text{ hrs.}$$

Illustration 20: Pipe A can fill a tank in 6 hrs while pipe B alone can fill it in 5 hrs and pipe C can empty the full tank in 8 hrs. If all the pipes are opened together, how much time will be needed to completely fill the tank?

Solution: Required time

$$= \frac{xyz}{yz + zx - xy} = \frac{6 \times 5 \times 8}{5 \times 8 + 8 \times 6 - 6 \times 5}$$

$$= \frac{6 \times 5 \times 8}{58} = \frac{120}{29} = 4 \frac{4}{29} \text{ days.}$$

Illustration 21: A pipe can fill a tank in 10 hrs. Due to a leak in the bottom, it is filled in 15 hrs. If the tank is full, how much time will the leak take to empty it.

Solution: Required time = $\frac{xy}{y - x} = \frac{10 \times 15}{15 - 10} = 30 \text{ hrs.}$

Illustration 22: If three pipes A , B and C can fill the tank alone in 5, 6 and 8 hrs, then when all the three pipes are opened together, find the time to fill the tank completely.

Solution: Required time

$$\begin{aligned} &= \frac{xyz}{xy + yz + zx} = \frac{5 \times 6 \times 8}{5 \times 6 + 6 \times 8 + 8 \times 5} \\ &= \frac{240}{30 + 48 + 40} = \frac{240}{118} = 2 \frac{2}{59} \text{ hrs} \end{aligned}$$

2. Variable Work

When rate of doing work depends upon some external factors, the work is said to be variable work. For example, rate of flow a liquid through a pipe is directly proportional to the area of cross-section of the pipe. If pipe is cylindrical, its cross-section is a circle whose area depends upon square of its radius or diameter, because area of the circle

$$= \pi r^2 = \pi \left(\frac{d}{2} \right)^2 = \pi / 4 d^2$$

where r and d are the radius and diameter respectively of the circle.

Hence rate of flow of a liquid through a cylindrical pipe is directly proportional to the square of its radius or diameter.

This concept is used in solving various problems. You can understand the use of this concept by the following Illustration.

Illustration 23: There are three inlet taps in a water tank whose diameters are 2 cm, 3 cm and 5 cm respectively. The inlet tap of least diameter fill the empty water tank alone in 10 minutes. Find the time taken to fill the empty tank when all the three inlet taps are open together.

Solution: Since rate of flow is directly proportional to the square of the diameter.

i.e. Rate of flow = $K \times (\text{diameter})^2$, where K is a constant.

For the first tap, rate of flow = $K \times (2)^2 = 4K \text{ cm}^3/\text{minute}$

For the second tap, rate of flow = $K \times (3)^2 = 9K \text{ cm}^3/\text{minute}$

For the third tap, rate of flow = $K \times (5)^2 = 25K \text{ cm}^3/\text{minute}$

Now, capacity of the tank = (Rate of flow) $\times$ (time required to fill the tank)

$$\therefore \text{Capacity of the tank} = (4K) \text{ cm}^3/\text{minute} \times 10 \text{ minutes} = 40K \text{ cm}^3$$

Now volume of the water filled by all the three taps working together in 1 minute

$$= 4K + 9K + 25K = 38K \text{ cm}^3$$

Assume that all the taps working together take ' t ' minutes.

Then capacity of the tank = $38K \times t$

$$\therefore 38K \times t = 40K$$

$$\Rightarrow t = \frac{20}{19} = 1\frac{1}{19} \text{ minutes}$$

Hence, required time = $1\frac{1}{19}$ minutes.

3. Alternate Work

In some problems two or more people of different efficiencies work alternatively or in some particular pattern. You can understand the method to solve these types of problems through the following illustration.

Illustration 24: Sanjeev can build a wall in 20 days and Parveen can demolish the same wall in 30 days. If they work on alternate days with Sanjeev starting the job on the 1st day, then in how many days will the wall be built for the first time ?

Solution: Let us assume the total units of work
= 60 units (i.e. LCM of 20 and 30)

So, the wall built by Sanjeev in one day = 3 units

And wall demolished by Parveen in one day = 2 units

So, effectively in two days, total wall built = 1 unit

Now, they work on alternate days, so days taken to built 57 units = 57 days

On 58th day Sanjeev will add another 3 units and so completing the construction of wall in 58 days.

(This problem can be understood well with another very traditional problem—A frog climbs up a pole 3 inches in 1 minute and slips 2 inches in next minute. If height of the pole is 120 inches, then how much time is taken by the frog to reach the top of the pole ?)

Practice Exercises



Foundation Level

1. A and B together can do a job in 12 days. B alone can finish it in 28 days. In how many days can A alone finish the work?
(a) 21 days (b) 19 days
(c) 20 days (d) None of these
2. A can do $\frac{3}{4}$ of a work in 12 days. In how many days can he finish $\frac{1}{8}$ of the work?
(a) 6 days (b) 5 days
(c) 3 days (d) 2 days
3. A can finish a work in 18 days and B can do the same work in half the time taken by A. Then, working together, what part of the same work they can finish in a day?
(a) $\frac{1}{6}$ (b) $\frac{1}{9}$
(c) $\frac{2}{5}$ (d) $\frac{2}{7}$
4. A man is twice as fast as a woman. Together the man and the woman do the piece of work in 8 days. In how many days each will do the work if engaged alone?
(a) man-14 days, woman-28 days
(b) man-12 days, woman-24 days
(c) man-10 days, woman-20 days
(d) None of these
5. A is 30% more efficient than B. How much time will they, working together, take to complete a job which A alone could have done in 23 days?
(a) 11 days (b) 13 days
(c) $20\frac{3}{17}$ days (d) None of these
6. A contractor undertakes to build a walls in 50 days. He employs 50 peoples for the same. However after 25 days he finds that only 40% of the work is complete. How many more man need to be employed to complete the work in time?
(a) 25 (b) 30
(c) 35 (d) 20
7. 12 men complete a work in 18 days. Six days after they had started working, 4 men joined them. How many days will all of them take to complete the remaining work?
(a) 10 days (b) 12 days
(c) 15 days (d) 9 days
8. A man, a woman or a boy can do a job in 20 days, 30 days or 60 days respectively. How many boys must assist 2 men and 8 women to do the work in 2 days?
(a) 15 boys (b) 8 boys
(c) 10 boys (d) None of these
9. 10 men can complete a piece of work in 15 days and 15 women can complete the same work in 12 days. If all the 10 men and 15 women work together, in how many days will the work get completed?
(a) 6 (b) $6\frac{1}{3}$
(c) $6\frac{2}{3}$ (d) $7\frac{2}{3}$
10. After working for 8 days, Anil finds that only $\frac{1}{3}$ of the work has been done. He employs Rakesh who is 60% efficient as Anil. How many more days will Anil take to complete the job?
(a) 15 days (b) 12 days
(c) 10 days (d) 8 days
11. A can knit a pair of socks in 3 days. B can knit the same thing in 6 days. If they are knitting together, in how many days will they knit two pairs of socks?
(a) 4 days (b) 2 days
(c) $4\frac{1}{2}$ days (d) 3 days
12. A can build up a wall in 8 days while B can break it in 3 days. A has worked for 4 days and then B joined to work with A for another 2 days only. In how many days will A alone build up the remaining part of wall?
(a) $13\frac{1}{3}$ days (b) $7\frac{1}{3}$ days
(c) $6\frac{1}{3}$ days (d) 7 days
13. Sakshi can do a piece of work in 20 days. Tanya is 25% more efficient than Sakshi. The number of days taken by Tanya to do the same piece of work is
(a) 15 (b) 16
(c) 18 (d) 25
14. Three men, four women and six children can complete a work in seven days. A woman does double the work a man does and a child does half the work a man does. How many women alone can complete this work in 7 days?
(a) 7 (b) 8
(c) 12 (d) Cannot be determined

15. Sunil and Pradeep can complete a work in 5 days and 15 days respectively. They both work for one day and then Sunil leaves. In how many days in the remaining work completed by Pradeep ?
 (a) 11 days (b) 12 days
 (c) 15 days (d) 8 days
16. Suresh can finish a piece of work by himself in 42 days. Mahesh, who is $\frac{1}{5}$ times more efficient as Suresh, requires X days to finish the work by working all by himself. Then what is the value of X ?
 (a) 25 days (b) 30 days
 (c) 35 days (d) 20 days
17. If 6 BSF or 10 CRPF companies can demolish a terrorist outfit in Kashmir in 2 days, find how long will 4 BSF and 9 CRPF companies take to do the same ?
 (a) 1.27 days (b) 2.27 days
 (c) 3.27 days (d) 4.27 days
18. 2 men and 3 boys can do a piece of work in 10 days while 3 men and 2 boys can do the same work in 8 days. In how many days can 2 men and 1 boy do the work ?
 (a) $12\frac{1}{2}$ days (b) $11\frac{1}{2}$ days
 (c) $15\frac{1}{2}$ days (d) $13\frac{1}{2}$ days
19. Three pumps working 8 hours a day can empty a tank in 2 days. How many hours a day must 4 pumps work to empty the tank in 1 day.
 (a) 10 hours (b) 12 hours
 (c) 8 hours (d) None of these
20. If 18 binders bind 900 books in 10 days, how many binders will be required to bind 660 books in 12 days ?
 (a) 14 (b) 13
 (c) 22 (d) 11
21. If 27 men take 15 days to mow 225 hectares of grass, how long will 33 men take to mow 165 hectares ?
 (a) 9 days (b) 18 days
 (c) 6 days (d) 12 days
22. X and Y can do a piece of work in 72 days. Y and Z can do it in 120 days. X and Z can do it in 90 days. In how many days all the three together can do the work ?
 (a) 100 days (b) 150 days
 (c) 60 days (d) 80 days
23. If 6 men and 8 boys can do a piece of work in 10 days and 26 men and 48 boys can do the same work in 2 days, the time taken by 15 men and 20 boys to do the same type of work will be
 (a) 6 days (b) 4 days
 (c) 8 days (d) 7 days
24. The work done by man, a woman and a boy are in the ratio 3 : 2 : 1. There are 24 men, 20 women and 16 boys in a factory whose weekly wages amount to ₹ 224. What will be the yearly wages of 27 men, 40 women and 15 boys?
 (a) ₹ 16366 (b) ₹ 16466
 (c) ₹ 16066 (d) ₹ 16016
25. Two pipes can fill a cistern in 6 minutes and 7 minutes respectively. Both the pipes are opened alternatively for 1 minute each. In what time will they fill the cistern.
 (a) 6 minutes (b) $6\frac{2}{3}$ minutes
 (c) $6\frac{3}{7}$ minutes (d) $3\frac{1}{2}$ minutes
26. Three pipes A, B and C can fill a tank from empty to full in 30 minutes, 20 minutes and 10 minutes respectively. When the tank is empty, all the three pipes are opened. A, B and C discharge chemical solutions P, Q and R respectively. What is the proportion of solution R in the liquid in the tank after 3 minutes?
 (a) $\frac{3}{11}$ (b) $\frac{6}{11}$
 (c) $\frac{4}{11}$ (d) $\frac{7}{11}$
27. A and B can finish a work in 10 days while B and C can do it in 18 days. A started the work, worked for 5 days, then B worked for 10 days and the remaining work was finished by C in 15 days. In how many days could C alone have finished the whole work ?
 (a) 30 days (b) 15 days
 (c) 45 days (d) 24 days
28. A certain number of men can do a work in 60 days. If there were 8 men more it could be finished in 10 days less. How many men are there ?
 (a) 75 men (b) 40 men
 (c) 48 men (d) 45 men
29. A and B can do a job in 16 days and 12 days respectively. B has started the work alone 4 days before finishing the job, A joins B. How many days has B worked alone?
 (a) 6 days (b) 4 days
 (c) 5 days (d) 7 days
30. Two pipes A and B when working alone can fill a tank in 36 min. and 45 min. respectively. A waste pipe C can empty the tank in 30 min. First A and B are opened. After 7 min., C is also opened. In how much time will the tank be full ?
 (a) $\frac{1}{60}$ (b) $\frac{1}{30}$
 (c) $\frac{7}{20}$ (d) $\frac{13}{20}$
31. A can do a piece of work in 25 days and B in 20 days. They work together for 5 days and then A goes away. In how many days will B finish the remaining work ?
 (a) 17 days (b) 11 days
 (c) 10 days (d) 15 days

32. 12 men complete a work in 18 days. Six days after they had started working, 4 men joined them. How many days will all of them take to complete the remaining work ?
 (a) 10 days (b) 12 days
 (c) 15 days (d) 9 days
33. A can do a piece of work in 10 days, while B alone can do it in 15 days. They work together for 5 days and the rest of the work is done by C in 2 days. If they get ₹ 450 for the whole work, how should they divide the money ?
 (a) ₹ 225, ₹ 150, ₹ 75 (b) ₹ 250, ₹ 100, ₹ 100
 (c) ₹ 200, ₹ 150, ₹ 100 (d) ₹ 175, ₹ 175, ₹ 100
34. Raju can do a piece of work in 10 days, Vicky in 12 days and Tinku in 15 days. They all start the work together, but Raju leaves after 2 days and Vicky leaves 3 days before the work is completed. In how many days is the work completed?
 (a) 5 days (b) 6 days
 (c) 7 days (d) 8 days
35. A can do some work in 24 days, B can do it in 32 days and C can do it in 60 days. They start working together. A left after 6 days and B left after working for 8 days. How many more days are required to complete the whole work?
 (a) 30 (b) 25
 (c) 22 (d) 20
36. Mayank can do 50% more work than Shishu in the same time. Shishu alone can do a piece of work in 30 hours. Shishu starts working and he had already worked for 12 hours when Mayank joins him. How many hours should Shishu and Mayank work together to complete the remaining work?
 (a) 6 (b) 12
 (c) 4.8 (d) 9.6
37. Anand got an order from a boutique for 480 shirts. He brought 12 sewing machines and appointed some expert tailors to do the job. However, many did not report to duty. As a result, each of those who reported had to stitch 32 more shirts than was originally planned by Anand, with equal distribution of work. How many tailors had been appointed earlier and how many had not reported to work?
 (a) 12, 4 (b) 10, 3
 (c) 10, 4 (d) None of these
38. In a fort there was sufficient food for 200 soldiers for 31 days. After 27 days 120 soldiers left the fort. For how many extra days will the rest of the food last for the remaining soldiers?
 (a) 12 days (b) 10 days
 (c) 8 days (d) 6 days
39. Sambhu can do $\frac{1}{2}$ of the work in 8 days while Kalu can do $\frac{1}{3}$ of the work in 6 days. How long will it take for both of them to finish the work?
 (a) $\frac{88}{17}$ days (b) $\frac{144}{17}$ days
 (c) $\frac{72}{17}$ days (d) 8 days
40. A tank holds 100 gallons of water. Its inlet is 7 inches in diameter and fills the tank at 5 gallons/min. The outlet of the tank is twice the diameter of the inlet. How many minutes will it take to empty the tank if the inlet is shut off, when the tank is full and the outlet is opened? (Hint : Rate of filling or emptying is directly proportional to the diameter)
 (a) 7.14 min (b) 10.0 min
 (c) 0.7 min (d) 5.0 min
41. Three diggers dug a ditch of 324 m deep in six days working simultaneously. During one shift, the third digger digs as many metres more than the second as the second digs more than the first. The third digger's work in 10 days is equal to the first digger's work in 14 days. How many metres does the first digger dig per shift?
 (a) 15 m (b) 18 m
 (c) 21 m (d) 27 m
42. A can do a piece of work in 90 days, B in 40 days and C in 12 days. They work for a day each in turn, i.e., first day A does it alone, second day B does it alone and 3rd day C does it alone. After that the cycle is repeated till the work is finished. They get ₹ 240 for this job. If the wages are divided in proportion to the work each had done. Find the amount A will get?
 (a) 14 (b) 24
 (c) 34 (d) 36
43. Two forest officials in their respective divisions were involved in the harvesting of *tendu* leaves. One division had an average output of 21 tons from a hectare and the other division, which had 12 hectares of land less, dedicated to *tendu* leaves, got 25 tons of *tendu* from a hectare. As a result, the second division harvested 300 tons of *tendu* leaves more than the first. How many tons of *tendu* leaves did the first division harvest?
 (a) 3150 (b) 3450
 (c) 3500 (d) 3600
44. A and B can do a piece of work in 45 and 40 days respectively. They began the work together, but A leaves after some days and B finished the remaining work in 23 days. After how many days did A leave
 (a) 7 days (b) 8 days
 (c) 9 days (d) 11 days
45. There is sufficient food for 400 men for 31 days. After 28 days, 280 men leave the place. For how many days will the rest of the food last for the rest of the men?
 (a) 10 days (b) 12 days
 (c) 16 days (d) 20 days

46. A tyre has two punctures. The first puncture alone would have made the tyre flat in 9 minutes and the second alone would have done it in 6 minutes. If air leaks out at a constant rate, how long does it take both the punctures together to make it flat?
- (a) $1\frac{1}{2}$ minutes (b) $3\frac{1}{2}$ minutes
(c) $3\frac{3}{5}$ minutes (d) $4\frac{1}{4}$ minutes
47. 12 men and 16 boys can do a piece of work in 5 days, 13 men and 24 boys can do it in 4 days. Then the ratio of daily work done by a man to that of a boy is
- (a) 2 : 1 (b) 3 : 1
(c) 3 : 2 (d) 5 : 4
48. Two taps can fill a tank in 12 and 18 minutes respectively. Both are kept open for 2 minutes and the first is turned off. In how many minutes more will the tank be filled ?
- (a) 15 min. (b) 20 min.
(c) 11 min. (d) 13 min.
49. A cistern normally takes 6 hours to be filled by a tap but because of a leak, 2 hours more. In how many hours will the leak empty a full cistern ?
- (a) 20 hrs (b) 24 hrs
(c) 26 hrs (d) None of these
50. If 3 men or 4 women can reap a field in 43 days, how long will 7 men and 5 women take to reap it?
- (a) 7 days (b) 11 days
(c) 12 days (d) 16 days
51. If m man can do a work in r days, then the number of days taken by $(m + n)$ men to do it is:
- (a) $\frac{m+n}{mn}$ (b) $\frac{m+n}{mr}$
(c) $\frac{mr}{(m+n)}$ (d) $\frac{(m+n)r}{mn}$

Standard Level

- A pipe can fill a tank in 15 minutes and another one in 10 minutes. A third pipe can empty the tank in 5 minutes. The first two pipes are kept open for 4 minutes in the beginning and then the third pipe is also opened. In what time will the tank be emptied ?
(a) 35 min (b) 15 min
(c) 20 min (d) Cannot be emptied
- Filling pipe, if opened alone, takes 5 minutes to fill a cistern. Suddenly, during the course of filling, the waste pipe (which is of similar size and flow as of fill pipe) is opened for 2 minutes, then the cistern will be filled in
(a) $3\frac{1}{7}$ min (b) $3\frac{1}{3}$ min
(c) 5 min (d) 7 min
- Three taps A, B and C can fill a tank in 12, 15 and 20 hours respectively. If A is open all the time and B and C are open for one hour each alternately, then the tank will be full in :
(a) 6 hrs. (b) $6\frac{2}{3}$ hrs.
(c) 7 hrs. (d) $7\frac{1}{2}$ hrs.
- 1 man or 2 women or 3 boys can do a work in 44 days. Then, in how many days will 1 man, 1 woman and 1 boy do the work?
(a) 12 days (b) 24 days
(c) 18 days (d) 36 days
- A, B and C can do a work in 8, 16 and 24 days respectively. They all begin together. A continues to work till it is finished, C leaving off 2 days and B one day before its completion. In what time is the work finished?
(a) 3 days (b) 4 days
(c) 5 days (d) 8 days
- Two pipes A and B can fill a tank in 24 minutes and 32 minutes respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 18 minutes?
(a) 6 min. (b) 8 min.
(c) 12 min. (d) 14 min.
- A contractor undertook to do a piece of work in 9 days. He employed certain number of laboures but 6 of them were absent from the very first day and the rest could finish the work in only 15 days. Find the number of men originally employed .
(a) 15 (b) 6
(c) 13 (d) 9
- After working for 8 days, Anil finds that only $\frac{1}{3}$ of the work has been done. He employs Rakesh who is 60 % efficient as Anil. How many more days will Anil take to complete the job?
(a) 15 days (b) 12 days
(c) 10 days (d) 8 days
- A can build up a wall in 8 days while B can break it in 3 days. A has worked for 4 days and then B joined to work with A for another 2 days only. In how many days will A alone build up the remaining part of wall?
(a) $13\frac{1}{3}$ days (b) $7\frac{1}{3}$ days
(c) $6\frac{1}{3}$ days (d) 7 days
- A cistern has two taps which fill it in 12 minutes and 15 minutes respectively. There is also a waste pipe in the cistern. When all the three are opened, the empty cistern is full in 20 minutes. How long will the waste pipe take to empty the full cistern ?
(a) 10 min (b) 12 min
(c) 15 min (d) None of these
- A and B together can do a piece of work in 12 days which B and C together can do in 16 days. After A has been working at it for 5 days, and B for 7 days, C takes up and finishes it alone in 13 days. In how many days could each do the work by himself ?
(a) 8, 16, 24 (b) 16, 24, 48
(c) 16, 48, 24 (d) 8, 24, 48
- A pump can be operated both for filling a tank and for emptying it. The capacity of tank is 2400 m^3 . The emptying capacity of the pump is 10 m^3 per minute higher than its filling capacity. Consequently, the pump needs 8 minutes less to empty the tank to fill it. Find the filling capacity of pump.
(a) $50 \text{ m}^3/\text{min}$ (b) $60 \text{ m}^3/\text{min}$
(c) $58 \text{ m}^3/\text{min}$ (d) None of these
- A tank is filled in 5 hours by three pipes A, B and C. The pipe C is twice as fast as B and B is twice as fast as A. How much time will pipe A alone take to fill the tank ?
(a) 20 hrs (b) 25 hrs
(c) 35 hrs (d) Cannot be determind
- Two pipes A and B can fill a tank in 15 hours and 20 hours respectively while a third pipe C can empty the full tank in 25 hours. All the three pipes are opened in the begining. After 10 hours, C is closed. In how much time, will the tank be full?
(a) 12 hrs (b) 13 hrs
(c) 16 hrs (d) 18 hrs

15. 4 men and 6 women finish a job in 8 days, while 3 men and 7 women finish in 10 days. In how many days will 10 women finish it?
(a) 20 days (b) 30 days
(c) 40 days (d) 50 days
16. A can do a work in 25 days and B can do the same work in 20 days. They work together for 5 days and then A goes away. In how many days will B finish the work?
(a) 9 days (b) 11 days
(c) 15 days (d) 20 days
17. One man can do as much work in one day as a woman can do in 2 days. A child does one third the work in a day as a woman. If an estate-owner hires 39 pairs of hands, men, women and children in the ratio 6 : 5 : 2 and pays them in all ₹ 1113 at the end of the days work. What must be the daily wage of a child, if the wages are proportional to the amount of work done?
(a) ₹ 14 (b) ₹ 5
(c) ₹ 20 (d) ₹ 7
18. There is leak in the bottom of a tank. This leak can empty a full tank in 8 hours. When the tank is full, a tap is opened into the tank which admits 6 litres per hour and the tank is now emptied in 12 hours. What is the capacity of the tank?
(a) 28.8 litres (b) 36 litres
(c) 144 litres (d) Can't be determined
19. A company has a job to prepare certain no. of cans and there are three machines A, B & C for this job. A can complete the job in 3 days, B can complete the job in 4 days and C can complete the job in 6 days. How many days the company will take to complete job if all the machines are used simultaneously?
(a) 4 days (b) $4/3$ days
(c) 3 days (d) 12 days
20. 3 small pumps and a large pump are filling a tank. Each of the three small pumps works at $2/3$ rd the rate of the large pump. If all 4 pumps work at the same time, they should fill the tank in what fraction of the time that it would have taken the large pump alone?
(a) $4/7$ (b) $1/3$
(c) $2/3$ (d) $3/4$
21. A and B can do a job in 15 days and 10 days, respectively. They began the work together but A leaves after some days and B finished the remaining job in 5 days. After how many days did A leave?
(a) 2 days (b) 3 days
(c) 1 day (d) None of these
22. A contract is to be completed in 46 days and 117 men were set to work, each working 8 hours a day. After 33 days, $4/7$ of the work is completed. How many additional men may be employed so that the work may be completed in time, each man now working 9 hours a day ?
(a) 80 (b) 81
(c) 82 (d) 83
23. If 12 men or 15 women or 18 boys can do a piece of work in 15 days of 8 hours each, find how many men assisted by 5 women and 6 boys will finish the same work in 16 days of 9 hours each.
(a) 6 men (b) 2 men
(c) 8 men (d) 4 men
24. The work done by a man, a woman and a child is in the ratio of 3 : 2 : 1. There are 20 men, 30 women and 36 children in a factory. Their weekly wages amount to ₹ 780, which is divided in the ratio of work done by the men, women and children. What will be the wages of 15 men, 21 women and 30 children for 2 weeks?
(a) ₹ 585 (b) ₹ 292.5
(c) ₹ 1170 (d) ₹ 900
25. The diameter of three pipes are 1cm, $1\frac{1}{3}$ cm and 2 cm respectively. The quantity of water flowing through a pipe varies directly as the square of its diameter. If the pipe with 2 cm diameter can fill a tank in 61 minutes, in what time will all the three pipes together fill the tank?
(a) 36 min (b) 32 min
(c) 28 min (d) 40 min
26. x is 3 times as faster as y and is able to complete the work in 40 days less than y. Then the time in which they can complete the work together?
(a) 15 days (b) 10 days
(c) $7\frac{1}{2}$ days (d) 5 days
27. The Bubna dam has four inlets. Through the first three inlets, the dam can be filled in 12 minutes; through the second, the third and the fourth inlet, it can be filled in 15 minutes; and through the first and the fourth inlet, in 20 minutes. How much time will it take all the four inlets to fill up the dam?
(a) 8 min (b) 10 min
(c) 12 min (d) None of these
28. Two pipes A and B can fill up a half full tank in 1.2 hours. The tank was initially empty. Pipe B was kept open for half the time required by pipe A to fill the tank by itself. Then, pipe A was kept open for as much time as was required by pipe B to fill up $1/3$ of the tank by itself. It was then found that the tank was $5/6$ full. The least time in which any of the pipes can fill the tank fully is
(a) 4.8 hours (b) 4 hours
(c) 3.6 hours (d) 6 hours
29. Two pipes can fill a cistern in 14 and 16 hours respectively. The pipes are opened simultaneously and it is found that due to leakage in the bottom of the cistern, it takes 32 minutes extra for the cistern to be filled up. When the cistern is full, in what time will the leak empty it?
(a) 114 h (b) 112 h
(c) 100 h (d) 80 h

30. Each of A , B and C need a certain unique time to do a certain work. C needs 1 hour less than A to complete the work. Working together, they require 30 minutes to complete 50% of the job. The work also gets completed if A and B start working together and A leaves after 1 hour and B works for a further 3 hours. How much work does C do per hour?
- (a) 16.66% (b) 33.33%
(c) 50% (d) 66.66%
31. All the three taps were open and the emptying pipe is closed. At the time when the tank was supposed to be full, it was found that only $\frac{2}{5}$ th of the tank was full. It was discovered that all the residents had kept their water taps open during this period. At what rate were the residents of each house getting water? (Consider that each house has only one tap).
- (a) 1.1 litres/h (b) 2.22 litres/h
(c) 2.85 litres/h (d) 4.46 litres/h
32. A student studying the weather for d days observed that (i) it rained on 7 days. morning or afternoon; (ii) when it rained in the afternoon, it was clear in the morning; (iii) there were five clear afternoons and (iv) there were six clear morning. Then d equals
- (a) 3 (b) 7
(c) 11 (d) 9
33. The rate of flow of water (in litre per min) of three pipes are 2, N and 3, where $2 < N < 3$. The lowest and the highest flow rates are both decreased by a certain quantity x , while the intermediate rate is left unchanged. If the reciprocals of the three flow rates, in the order given above, are in arithmetic progression both before and after the change, then what is the quantity x (in litre per min)? (Negative flow rates indicate that the pipes act as emptying pipes instead of filling pipes).
- (a) 2.6 (b) -3.2
(c) 3.8 (d) -2.6
34. If 6 BSF or 10 CRPF companies can demolish a terrorist outfit in Kashmir in 2 days, find how long will 4 BSF and 9 CRPF companies take to do the same ?
- (a) 1.27 days (b) 2.27 days
(c) 3.27 days (d) 4.27 days
35. Three pumps working 8 hours a day can empty a tank in 2 day. How many hours a day must 4 pumps work to empty the tank in 1 day.
- (a) 10 hours (b) 12 hours
(c) 8 hours (d) None of these
36. A group of men decided to do a job in 4 days. But since 20 men dropped out every day, the job completed at the end of the 7th day. How many men were there at the beginning?
- (a) 240 (b) 140
(c) 280 (d) 150
37. The total number of men, women and children working in a factory is 18. They earn ₹ 4000 in a day. If the sum of the wages of all men, all women and all children is in the ratio of 18 : 10 : 12 and if the wages of an individual man, woman and child is in the ratio 6 : 5 : 3, then how much a woman earn in a day?
- (a) ₹ 400 (b) ₹ 250
(c) ₹ 150 (d) ₹ 120

Expert Level

- One man and six women working together can do a job in 10 days. The same job is done by two men in 'p' days and by eight women in $p + 5$ days. By what percentage is the efficiency of a man greater than that of a woman?
 - 300%
 - 500%
 - 600%
 - 700%
- The work done by 4 men in 12 days is equal to the work done by 6 women in 10 days and is also equal to the work done by 8 children in 9 days. A man, a woman and a child working together take 10 days to complete a particular job. In how many days will the same job be completed by 2 women and 5 children working together?
 - 5
 - 6
 - 4
 - 7
- C is twice efficient as A, B takes thrice as many days as C. A takes 12 days to finish the work alone. If they work in pairs (i.e., AB, BC, CA) starting with AB on the first day then BC on the second day and AC on the third day and so on, then how many days are required to finish the work?
 - $6\frac{1}{5}$ days
 - 4.5 days
 - $5\frac{1}{9}$ days
 - 8 days
- There was a leakage in the container of the refined oil. If 11 kg oil is leaked out per day then it would have lasted for 50 days, if the leakage was 15 kg per day, then it would have lasted for only 45 days. For how many days would the oil have lasted, if there was no leakage and it was completely used for eating purpose?
 - 80 days
 - 72 days
 - 100 days
 - 120 days
- According to a plan, a drilling team had to drill to a depth of 270 metres below the ground level. For the first three days the team drilled as per the plan. However, subsequently finding that their resources were getting underutilised according to the plan, it started to drill 8 metres more than the plan every day. Therefore, a day before the planned date they had drilled to a depth of 280 metres. How many metres of drilling was the plan for each day.
 - 38 metres
 - 30 metres
 - 27 metres
 - 28 metres
- Aman, Baman and Chaman can finish a job working alone in 15, 20 and 25 days respectively. However, while working with somebody the efficiency of Aman, Baman and Chaman reduces by 30%, 20% and 50% respectively. If none of them is allowed to work for three consecutive days, then what is the maximum possible fraction of the job that they can complete in four days?
 - $\frac{21}{50}$
 - $\frac{17}{50}$
 - $\frac{8}{25}$
 - $\frac{1}{3}$
- B and C are equally efficient, but the efficiency of A is half of each B and C. A and B started a work and 3 days later C joined them. If A alone can do the work in 14 days, then in how many more days the work will be completed?
 - 1
 - 2
 - 3
 - 4.5
- A finishes $\frac{6}{7}$ th of the work in $2z$ hours, B works twice as fast and finishes the remaining work. For how long did B work?
 - $\left(\frac{2}{3}\right)^z$
 - $\left(\frac{6}{7}\right)^z$
 - $\left(\frac{6}{49}\right)^z$
 - $\left(\frac{3}{18}\right)^z$
- 4 men and 2 boys can finish a piece of work in 5 days. 3 women and 4 boys can finish the same work in 5 days. Also 2 men and 3 women can finish the same work in 5 days. In how many days 1 man, 1 woman and 1 boy can finish the work, at their double efficiency?
 - $4\frac{8}{13}$
 - $4\frac{7}{13}$
 - $3\frac{7}{13}$
 - None of these
- The work done by 2 men in a day is equal to the work done by 3 children in a day. The work done by 3 men in a day is equal to the work done by 5 women in a day. It takes 10 days for a man, a woman and a child to complete a job working together. How many days will 2 children working together take to complete the same job?
 - 30
 - 15
 - 17
 - 34
- It takes 30 hours for an inlet pipe to fill an empty tank completely. When 5 identical inlet pipes and 4 identical outlet pipes operate together, the same empty tank get filled completely in 10 hours. How much time (in hours) will an outlet pipe take to empty the same tank when it's filled upto half its volume?
 - 15
 - 20
 - 24
 - 30
- A tank of capacity 25 litres has an inlet and an outlet tap. If both are opened simultaneously, the tank is filled in 5 minutes. But if the outlet flow rate is doubled and taps opened the tank never gets filled up. Which of the following can be outlet flow rate in liters/min?
 - 2
 - 6
 - 4
 - 3

13. Ashish, Binay and Joseph can do a job in 20, 30 and 40 days respectively. The three started the job together; Ashish left the job 4 days before it was completed and Binay left the job 3 days before it was completed. In how many days was the job completed?
(a) 14 (b) 12
(c) 16 (d) 15
14. Pawan and Qureshi working together can do a piece of work in 10 days whereas Qureshi and Rohit working together can do the same work in 12 days. All three work together to do a job for which they are paid ₹ 300. If Qureshi's share is ₹ 140, then what is Pawan's share?
(a) ₹ 100 (b) ₹ 60
(c) ₹ 80 (d) cannot be determined
15. Three cooks have to make 80 idlis. They are known to make 20 pieces every minute working together. The first cook began working alone and made 20 pieces having worked for sometime more than three minutes. The remaining part of the work was done by the second and the third cook working together. It took a total of 8 minutes to complete the 80 idlis. How many minutes would it take the first cook alone to cook 160 idlis for a marriage party the next day?
(a) 16 minutes (b) 24 minutes
(c) 32 minutes (d) 40 minutes
16. A cistern has a leak which would empty it in 6 hours. A tap is turned on which fills the cistern @ 10 liters per hour and then it is emptied in 15 hours. What is the capacity of the cistern?
(a) 100 litres (b) 166.66 litres
(c) 60.66 litres (d) None of these
17. Tap A can fill a tank in 20 hours, B in 25 hours but tap C can empty a full tank in 30 hours. Starting with A, followed by B and C each tap opens alternatively for one hour period till the tank gets filled up completely. In how many hour the tank will be filled up completely?
(a) $51\frac{11}{15}$ (b) $52\frac{2}{3}$
(c) $24\frac{4}{11}$ (d) None of these
18. Each of A, B and C need a certain unique time to do a certain work. C needs 1 hour less than A to complete the work. Working together, they require 30 minutes to complete 50% of the job. The work also gets completed if A and B start working together and A leaves after 1 hour and B works for a further 3 hours. How much work does C do per hour?
(a) 16.66% (b) 33.33%
(c) 50% (d) 66.66%
19. Two men and a woman are entrusted with a task. The second man needs three hours more to cope with the job than the first man and the woman would need working together. The first man, working alone, would need as much time as the second man and the woman working together. The first man, working alone, would spend eight hours less than the double period of time the second man would spend working alone. How much time would the two men and the woman need to complete the task if they all worked together?
(a) 2 hours (b) 3 hours
(c) 4 hours (d) 5 hours
20. Two pipes A and B can fill a cistern in 15 hours and 10 hours respectively. A tap C can empty the full cistern in 30 hours. All the three taps were open for 2 hours, when it was remembered that the emptying tap had been left open. It was then closed. How many hours more would it take for the cistern to be filled?
(a) 30 min. (b) 1.2 hours
(c) 24 min. (d) 35 min.
21. Working together B and C take 50% more number of days than A, B and C together take and A and B working together, take $\frac{8}{3}$ more number of days than A, B and C take together. If A, B and C all have worked together till the completion of the work and B has received ₹ 120 out of the total earning of ₹ 450, then in how many days did A, B and C together complete the whole work?
(a) 10 (b) 6
(c) 4 (d) 2
22. Eklavya can do the 6 times the actual work in 36 days while Faizal can do the one-fourth of the original work in 3 days. In how many days will both working together complete the 3 times of the original work?
(a) 6 (b) 10
(c) 12 (d) 15
23. Sixty-four men working 8 h a day plan to complete a piece of work in 9 days. However, 5 days later they found that they had completed only 40% of the work. They now wanted to finish the remaining portion of the work in 4 more days. How many hours per day should they need to work in order to achieve the target?
(a) 11 (b) 12
(c) 13 (d) 15
24. 4 pipes each of 3 cm diameter are to be replaced by a single pipe discharging the same quantity of water. What should be the diameter of the single pipe, if the speed of water is the same.
(a) 2 cm (b) 4 cm
(c) 6 cm (d) 8 cm
25. A can do a job in 3 days less time than B. A works at it alone for 4 days and then B takes over and completes it. If altogether 14 days were required to finish the job, then in how many days would each of them take alone to finish it?
(a) 17 days, 20 days (b) 12 days, 15 days
(c) 13 days, 16 days (d) None of these
26. Water flows at 3 metres per sec through a pipe of radius 4 cm. How many hours will it take to fill a tank 40 metres long, 30 metres broad and 8 metres deep, if the pipe remains full?
(a) 176.6 hours (b) 120 hour
(c) 135.5 hours (d) None of these
27. A ship 55 kms. from the shore springs a leak which admits 2 tones of water in 6 min ; 80 tones would suffer to sink her, but the pumps can throw out 12 tones an hour. Find the average rate of sailing that she may just reach the shore as she begins to sink.
(a) 5.5 km/h (b) 6.5 km/h
(c) 7.5 km/h (d) 8.5 km/h

Test Yourself

- One man can do as much work in one day as a woman can do in 2 days. A child does one third the work in a day as a woman. If an estate-owner hires 39 pairs of hands, men, women and children in the ratio 6 : 5 : 2 and pays them in all ₹ 1113 at the end of the days work. What must be the daily wage of a child, if the wages are proportional to the amount of work done?
(a) ₹ 14 (b) ₹ 5
(c) ₹ 20 (d) ₹ 7
- A water tank has three taps A, B and C. A fills four buckets in 24 minutes, B fills 8 buckets in 1 hour and C fills 2 buckets in 20 minutes. If all the taps are opened together a full tank is emptied in 2 hours. If a bucket can hold 5 litres of water, what is the capacity of the tank?
(a) 120 litres (b) 240 litres
(c) 180 litres (d) 60 litres
- There is leak in the bottom of a tank. This leak can empty a full tank in 8 hours. When the tank is full, a tap is opened into the tank which admits 6 litres per hour and the tank is now emptied in 12 hours. What is the capacity of the tank?
(a) 28.8 litres (b) 36 litres
(c) 144 litres (d) Can't be determined
- A company has a job to prepare certain no. of cans and there are three machines A, B & C for this job. A can complete the job in 3 days, B can complete the job in 4 days and C can complete the job in 6 days. How many days the company will take to complete job if all the machines are used simultaneously?
(a) 4 days (b) $4/3$ days
(c) 3 days (d) 12 days
- A can complete a piece of work in 4 days. B takes double the times taken by A, C takes double that of B, and D takes double that of C to complete the same task. They are paired in groups of two each. One pair takes two-third the time needed by the second pair to complete the work. Which is the first pair?
(a) A, B (b) A, C
(c) B, C (d) A, D
- 3 small pumps and a large pump are filling a tank. Each of the three small pumps works at $2/3$ rd the rate of the large pump. If all 4 pumps work at the same time, they should fill the tank in what fraction of the time that it would have taken the large pump alone?
(a) $4/7$ (b) $1/3$
(c) $2/3$ (d) $3/4$
- In nuts and bolts factory, one machine produces only nuts at the rate of 100 nuts per minute and needs to be cleaned for 5 minutes after production of every 1000 nuts. Another machine produces only bolts at the rate of 75 bolts per minute and needs to be cleaned for 10 minutes after production of every 1500 bolts. If both the machines start production at the same time, what is the minimum duration required for producing 9000 pairs of nuts and bolts?
(a) 130 minutes (b) 135 minutes
(c) 170 minutes (d) 180 minutes
- 8 men and 14 women are working together in a field. After working for 3 days, 5 men and 8 women leave the work. How many more days will be required to complete the work?
I. 19 men and 12 women together can complete the work in 18 days.
II. 16 men complete two-third of the work in 16 days
III. In a day, the work done by three men is equal to the work done by four women.
(a) I only (b) II only
(c) III only (d) I or II or III
- A can do 50% more work as B can do in the same time. B alone can do a piece of work in 20 hours. A, with the help of B, can finish the same work in how many hours?
(a) 12 (b) 8
(c) $13\frac{1}{3}$ (d) $5\frac{1}{2}$
- A can do a piece of work in 10 days, while B alone can do it in 15 days. They work together for 5 days and the rest of the work is done by C in 2 days. If they get ₹ 450 for the whole work, how should they divide the money?
(a) ₹ 225, ₹ 150, ₹ 75 (b) ₹ 250, ₹ 100, ₹ 100
(c) ₹ 200, ₹ 150, ₹ 100 (d) ₹ 175, ₹ 175, ₹ 100
- X can do a piece of work in 15 days. If he is joined by Y who is 50% more efficient, in what time will X and Y together finish the work?
(a) 10 days (b) 6 days
(c) 18 days (d) Data insufficient
- 12 men can complete a piece of work in 4 days, while 15 women can complete the same work in 4 days. 6 men start working on the job and after working for 2 days, all of them stopped working. How many women should be put on the job to complete the remaining work, If it is to be completed in 3 days?
(a) 15 (b) 18
(c) 22 (d) Data inadequate
- A contract is to be completed in 46 days and 117 men were set to work, each working 8 hours a day. After 33 days, $4/7$ of the work is completed. How many additional men may be employed so that the work may be completed in time, each man now working 9 hours a day?
(a) 80 (b) 81
(c) 82 (d) 83
- 10 horses and 15 cows eat grass of 5 acres in a certain time. How many acres will feed 15 horses and 10 cows for the same time, supposing a horse eats as much as 2 cows?
(a) $40/7$ acres (b) $39/8$ acres
(c) $40/11$ acres (d) $25/9$ acres
- The work done by a man, a woman and a child is in the ratio of 3 : 2 : 1. There are 20 men, 30 women and 36 children in a factory. Their weekly wages amount to ₹ 780, which is divided in the ratio of work done by the men, women and children. What will be the wages of 15 men, 21 women and 30 children for 2 weeks?
(a) ₹ 585 (b) ₹ 292.5
(c) ₹ 1170 (d) ₹ 900

Hints & Solutions

Foundation Level

1. (a) $(A + B)$'s 1 day's work = $\frac{1}{12}$ th part of whole work.

B 's 1 day's work = $\frac{1}{28}$ th part of whole work.

$\therefore A$'s 1 day's work = $\frac{1}{12} - \frac{1}{28} = \frac{1}{21}$ th part of whole work.

$\therefore A$ alone can finish the work in 21 days

2. (d) $\therefore A$ can do $\frac{3}{4}$ of the work in 12 days

$\therefore A$ can do $\frac{1}{8}$ of the work in $12 \times \frac{4}{3} \times \frac{1}{8}$ days = 2 days

3. (a) A 's 1 day's work = $\frac{1}{18}$ and B 's 1 day's work = $\frac{1}{9}$.

$\therefore (A + B)$'s 1 day's work = $\left(\frac{1}{18} + \frac{1}{9}\right) = \frac{1}{6}$.

4. (b) Let the man alone do the work in x days.

Then, the woman alone do the work in $2x$ days.

Their one day's work = $\frac{1}{8}$ th part of whole work

$$\text{i.e., } \frac{1}{x} + \frac{1}{2x} = \frac{1}{8}$$

$$\Rightarrow x = 12 \text{ days}$$

$\therefore$ man takes 12 days and woman $2x = 24$ days.

5. (b) Ratio of times taken by A and $B = 100 : 130 = 10 : 13$.

Suppose B takes x days to do the work.

$$\text{Then, } 10 : 13 :: 23 : x \Rightarrow x = \left(\frac{23 \times 13}{10}\right) \Rightarrow x = \frac{299}{10}.$$

$$A\text{'s 1 day's work} = \frac{1}{23}; B\text{'s 1 day's work} = \frac{10}{299}.$$

$$(A + B)\text{'s 1 day's work} = \left(\frac{1}{23} + \frac{10}{299}\right) = \frac{23}{299} = \frac{1}{13}.$$

$\therefore A$ and B together can complete the job in 13 days.

6. (a) 50 men complete 0.4 work in 25 days.

Applying the work rule, $m_1 \times d_1 \times w_2 = m_2 \times d_2 \times w_1$

we have,

$$50 \times 25 \times 0.6 = m_2 \times 25 \times 0.4$$

$$\text{or } m_2 = \frac{50 \times 25 \times 0.6}{25 \times 0.4} = 75 \text{ men}$$

Number of additional men required = $(75 - 50) = 25$

7. (d) In 1 day, work done by 12 men = $\frac{1}{18}$

$$\text{In 6 days, work done by 12 men} = \frac{6}{18} = \frac{1}{3}$$

$$\text{Remaining work} = \frac{2}{3}$$

$$\text{Now, } m_1 \times d_1 \times w_2 = m_2 \times d_2 \times w_1$$

$$\text{or } 12 \times 18 \times \frac{2}{3} = 16 \times d_2 \times 1$$

$$\text{or } d_2 = \frac{4 \times 18 \times 2}{16} = 9 \text{ days}$$

8. (b) Man's two day's work = $2 \times \frac{1}{20}$ th work $\frac{1}{10}$ th work

Woman's two days's work

$$2 \times \frac{1}{30} \text{ th work} = \frac{1}{15} \text{ th work}$$

$$\text{Boy's two day's work} = 2 \times \frac{1}{60} \text{ th work} = \frac{1}{30} \text{ th work}$$

Now, let 2 men, 8 women and x boys can complete work in 2 days. Then,

$$2 \text{ men's work} + 8 \text{ women's work} + x \text{ boy's work} = 1$$

$$2\left(\frac{1}{10}\right) + 8\left(\frac{1}{15}\right) + x\left(\frac{1}{30}\right) = 1$$

$$\Rightarrow x = \left(1 - \frac{1}{5} - \frac{8}{15}\right) \times 30 \Rightarrow x = 8 \text{ boys}$$

9. (c) 10 men's 1 day's work = $\frac{1}{15}$;

$$15 \text{ women's 1 day's work} = \frac{1}{12}.$$

(10 men + 15 women)'s 1 day's work

$$= \left(\frac{1}{15} + \frac{1}{12}\right) = \frac{9}{60} = \frac{3}{20}.$$

$\therefore$ 10 men and 15 women will complete the work in

$$\frac{20}{3} = 6\frac{2}{3} \text{ days.}$$

10. (c) In 8 days, Anil does $\frac{1}{3}$ rd work.
 $\therefore$ in 1 day, he does $\frac{1}{24}$ th work.
 $\therefore$ Rakesh's one day's work = 60% of $\frac{1}{24} = \frac{1}{40}$ th work.

$$\text{Remaining work } 1 - \frac{1}{3} = \frac{2}{3}$$

(Anil and Rakesh)'s one day's work

$$= \frac{1}{24} + \frac{1}{40} = \frac{1}{15} \text{th work}$$

Now, $\frac{1}{15}$ th work is done by them in one day

$$\therefore \frac{2}{3} \text{rd work is done by them in } 15 \times \frac{2}{3} = 10 \text{ days}$$

11. (a) A's one day's work $\frac{1}{3}$ rd work.

$$B's \text{ one day's work } \frac{1}{6} \text{th work.}$$

$$(A + B)'s \text{ one day's work } \frac{1}{3} + \frac{1}{6} = \frac{1}{2} \text{nd work}$$

$\therefore$ A and B together can complete the work (knit a pair of socks) in 2 days.

$\therefore$ They together knit two pair of socks in 4 days.

12. (b) A's one day's work $\frac{1}{8}$ th work

$$B's \text{ one day's work } \frac{1}{3} \text{rd work}$$

$$\therefore A's 4 \text{ day's work} = 4 \times \frac{1}{8} = \frac{1}{2} \text{nd work}$$

$$\therefore \text{In next two days, total wall } \frac{1}{2} + 2\left(\frac{1}{8}\right) - 2\left(\frac{1}{3}\right)$$

$$\frac{1}{12} \text{th wall}$$

$$\text{Remaining wall } 1 - \frac{1}{12} = \frac{11}{12} \text{th}$$

Now, $\frac{1}{8}$ th wall is built up by A in one day.

$$\therefore \frac{11}{12} \text{th wall is built up by A in } 8 \times \frac{11}{12} = 7\frac{1}{3} \text{ days.}$$

13. (b) Sakshi's one day's work $\frac{1}{20}$ th work

Tanya's one day's work

$$\frac{1}{20} + 25\% \text{ of } \frac{1}{20} = \frac{1}{16} \text{th work}$$

Hence, Tanya takes 16 days to complete the work.

14. (a) Let 1 woman's 1 day's work = x .

$$\text{Then, 1 man's 1 day's work} = \frac{x}{2}$$

$$\text{and 1 child's 1 day's work} = \frac{x}{4}.$$

$$\text{So, } \left(\frac{3x}{2} + 4x + \frac{6x}{4}\right) = \frac{1}{7} \Rightarrow x = \left(\frac{1}{7} \times \frac{4}{28}\right) = \frac{1}{49}.$$

$\therefore$ 1 woman alone can complete the work in 49 days.

So, to complete the work in 7 days, women required

$$= \left(\frac{49}{7}\right) = 7.$$

15. (a) Sunil takes 5 days and Pradeep takes 15 days to do the work.

In a day they would complete $\frac{1}{5} + \frac{1}{15}$ i.e., $\frac{4}{15}$ th work.

The remaining $11/15$ th work would be completed by

Pradeep in $\frac{11}{15} \times 15$ i.e. 11 days.

16. (c) Suresh, working alone 42 days = 1 unit of work. Mahesh is $1/5$ times more efficient than Suresh. So Mahesh is $6/5$ times as efficient as Suresh. Hence Mahesh should require $5/6$ th of the time, the time taken by Suresh.

Therefore time taken by Mahesh = $5/6 \times 42 = 35$ days.

17. (a) Given 6 BSF = 10 CRPF $\Rightarrow$ 4 BSF + 9 CRPF

$$= 4 + (9 \times 6/10) \text{ BSF} = \frac{94}{10} \text{ BSF}$$

$$\text{Now work} = 6 \times 2 \text{ BSF days} = \frac{94}{10} \times X \text{ BSF days}$$

$$\text{We have } 6 \times 2 = \frac{94}{10} \times X \Rightarrow X = 1.27 \text{ days}$$

18. (a) Let 1 man's 1 day's work = x and 1 boy's 1 day's work = y

$$\text{Then, } 2x + 3y = \frac{1}{10} \text{ and } 3x + 2y = \frac{1}{8}$$

$$\text{Solving, we get : } x = \frac{7}{200} \text{ and } y = \frac{1}{100}$$

$\therefore$ (2 men + 1 boy)'s 1 day's work

$$= \left(2 \times \frac{7}{200} + 1 \times \frac{1}{100}\right) = \frac{16}{200} = \frac{2}{25}$$

So, 2 men and 1 boy together can finish the work in

$$12\frac{1}{2} \text{ days.}$$

19. (b) Let the required number of working hours/day = x
More pumps, less working hrs per day (Indirect)
Less days, more working hrs per day (Indirect)

$$\left. \begin{array}{l} \text{Pumps } 4 : 3 \\ \text{Days } 1 : 2 \end{array} \right\} :: 8 : x$$

$$\therefore 4 \times 1 \times x = 3 \times 2 \times 8$$

$$\Rightarrow x = \frac{3 \times 2 \times 8}{4} = 12$$

20. (d) Let required number of binders be ' x '
Less books, less binders (direct)
More days, less binders (indirect)

$$\left. \begin{array}{l} \text{Books } 900 : 660 \\ \text{Days } 12 : 10 \end{array} \right\} :: 18 : x$$

$$900 \times 12 \times x = 660 \times 10 \times 18$$

$$x = \frac{660 \times 10 \times 18}{900 \times 12} = 11$$

21. (a) 27 men mow 225 hectares in 15 days
 $\therefore$ 1 man mow 225 hectares in (15×27) days (indirect)

$$\therefore 1 \text{ man mow 1 hectares in } \frac{15 \times 27}{225} \text{ days (direct)}$$

$$1 \text{ man mow 165 hectares in } \frac{15 \times 27}{225} \times 165 \text{ days (direct)}$$

$$\therefore 33 \text{ men mow 165 hectares in } \frac{15 \times 27 \times 165}{225 \times 33} = 9 \text{ days}$$

22. (c) $(X + Y)$'s one day work = $\frac{1}{72}$

$$(Y + Z)$$
's one day work = $\frac{1}{120}$

$$(Z + X)$$
's one day work = $\frac{1}{90}$

$$\therefore 2(X + Y + Z)\text{'s one day work} = \frac{1}{72} + \frac{1}{120} + \frac{1}{90} \\ = \frac{5+3+4}{360} = \frac{12}{360} = \frac{1}{30}$$

$$\therefore (X + Y + Z)\text{'s one day work} = \frac{1}{2} \times \frac{1}{30} = \frac{1}{60}$$

$\therefore$ They will complete the work in 60 days.

23. (b) Given $(6M + 8B) \times 10 = (26M + 48B) \times 2$
 $\Rightarrow 60M + 80B = 52M + 96B$
 $\Rightarrow 8M = 16B$
 $\Rightarrow 1M = 2B$
 $\therefore 15M + 20B = 30B + 20B = 50B$
 $6M + 8B = 12B + 8B = 20B$

Now	Boys	Days
	20	10
	50 ↓	x (Let)

$$\therefore x = \frac{20 \times 10}{50} = 4 \text{ days}$$

24. (d) 1 Man = 3 Boys and 1 Woman = 2 Boys
 $\therefore 24 \text{ Men} + 20 \text{ Women} + 16 \text{ Boys}$
 $= (24 \times 3) + (20 \times 2) + 16$
 $= 72 + 40 + 16$
 $= 128 \text{ Boys}$

$$27 \text{ Men} + 40 \text{ Women} + 15 \text{ Boys} = (27 \times 3) + (40 \times 2) + 15 \\ = 81 + 80 + 15 = 176 \text{ Boys.}$$

Now,

No. of Boys	Duration	Wages
128 ↑	1 ↑	224
176	52	x (Let)

$$\therefore x = \frac{176}{128} \times \frac{52}{1} \times 224$$

$$x = ₹ 16,016$$

25. (c) Part of the cistern filled by first pipe in 1 minute = $\frac{1}{6}$
Part of the cistern filled by second pipe in 2 minutes = $\frac{1}{7}$
Part of the cistern filled in first 2 minutes = $\frac{1}{6} + \frac{1}{7} = \frac{13}{42}$
Part of the cistern filled in 6 minutes = $\frac{3 \times 13}{42} = \frac{39}{42}$

$$\text{Remaining part} = 1 - \frac{39}{42} = \frac{3}{42} = \frac{1}{14}$$

$$\therefore \text{Time taken to fill } \frac{1}{14} \text{ parts} = \frac{6}{14} \times \frac{3}{7}$$

$$\therefore \text{Total time} = 6 + \frac{3}{7} = 6\frac{3}{7} \text{ minutes}$$

26. (b) Part filled by $(A + B + C)$ in 3 minutes

$$= 3 \left(\frac{1}{30} + \frac{1}{20} + \frac{1}{10} \right) = 3 \times \frac{11}{60} = \frac{11}{20}$$

$$\text{Part filled by } C \text{ in 3 minutes} = \frac{3}{10}$$

$$\therefore \text{Required ratio} = \frac{\frac{3}{10}}{\frac{11}{20}} = \frac{3}{10} \times \frac{20}{11} = \frac{6}{11}$$

27. (c) Let C completes the work in x days.

$$\text{Work done by } (A + B) \text{ in 1 day} = \frac{1}{10}$$

$$\text{Work done by } (B + C) \text{ in 1 day} = \frac{1}{18}$$

$$\begin{aligned} &A's 5 \text{ days' work} + B's 10 \text{ days' work} + C's 15 \text{ days' work} = 1 \\ \text{or } &(A + B)'s 5 \text{ days' work} + (B + C)'s 5 \text{ days' work} \\ &+ C's 10 \text{ days' work} = 1 \end{aligned}$$

$$\text{or } \frac{5}{10} + \frac{5}{18} + \frac{10}{x} = 1$$

$$\therefore x = 45 \text{ days}$$

28. (b) We have :
 x men to the work in 60 days and $(x + 8)$ men do the work in
 $(60 - 10 =) 50$ days.
 Then by "basic formula", $60x = 50(x + 8)$

$$\therefore x = \frac{50 \times 8}{10} = 40 \text{ men.}$$

29. (c) A's one day's work = $\frac{1}{16}$ th work

$$B's \text{ one day's work} = \frac{1}{12} \text{ th work}$$

Let the number of days B has worked alone = x days.
 Then,

$$A's \text{ amount of work} + B's \text{ amount of work} = 1$$

$$\Rightarrow 4\left(\frac{1}{16}\right) + (x+4)\left(\frac{1}{12}\right) = 1$$

$$\Rightarrow \frac{1}{4} + \frac{x+4}{12} = 1 \Rightarrow x = \frac{3}{4} \times 12 - 4 \Rightarrow x = 5 \text{ days}$$

30. (a) Part filled in 7 min. = $7 \times \left(\frac{1}{36} + \frac{1}{45}\right) = \frac{7}{20}$

$$\text{Remaining part} = \left(1 - \frac{7}{20}\right) = \frac{13}{20}$$

Part filled by $(A + B + C)$ in 1 min.

$$= \left(\frac{1}{36} + \frac{1}{45} - \frac{1}{30}\right) = \frac{1}{60}$$

31. (b) $(A + B)$'s 5 days' work

$$= 5 \left(\frac{1}{25} + \frac{1}{20}\right) = \frac{45}{100} = \frac{9}{20}$$

$$\text{Remaining work} = \left(1 - \frac{9}{20}\right) = \frac{11}{20}$$

$$\frac{11}{20} \text{ of the work would be finished by } B \text{ in}$$

$$\frac{11}{\frac{20}{1}} = 11 \text{ days.}$$

32. (d) In 1 day, work done by 12 men = $\frac{1}{18}$

$$\text{In 6 days, work done by 12 men} = \frac{6}{18} = \frac{1}{3}$$

$$\text{Remaining work} = \frac{2}{3}$$

$$\text{Now, } m_1 \times d_1 \times w_2 = m_2 \times d_2 \times w_1$$

$$\text{or } 12 \times 18 \times \frac{2}{3} = 16 \times d_2 \times 1$$

$$\text{or } d_2 = \frac{4 \times 18 \times 2}{16} = 9 \text{ days}$$

33. (a) Work done by A and B in 5 days = $\left(\frac{1}{10} + \frac{1}{15}\right) \times 5 = \frac{5}{6}$

$$\text{Work remaining} = 1 - \frac{5}{6} = \frac{1}{6}$$

$$\therefore C \text{ alone can do the work in } 6 \times 2 = 12 \text{ days}$$

$$\text{Ratio of their share work} = \frac{5}{10} : \frac{5}{15} : \frac{2}{12} = 3 : 2 : 1$$

$$\text{Share of wages} = `225, `150, `75.$$

34. (c) Raju = 10%, Vicky = 8.33% and Tinku = 6.66%. Hence, total work for a day if all three work = 25%. In 2 days they will complete, 50% work. On the third day onwards Raju doesn't work. The rate of work will become 15%. Also, since Vicky leaves 3 days before the actual completion of the work, Tinku works alone for the last 3 days (and must have done the last $6.66 \times 3 = 20\%$ work alone). This would mean that Vicky leaves after 80% work is done. Thus, Vicky and Tinku must be doing 30% work together over two days. Hence, total time required = 2 days (all three) + 2 days (Vicky and Tinku) + 3 days (Tinku alone).

35. (c) In 6 days A would do 25% of the work and in 8 days B would do 25% of the work himself. So C has to complete 50% of the work by himself. In all C would require 30 days to do 50% of the work. So, he would require 22 more days.

36. (d) Ratio of efficiency of Mayank and Shishu = $3/2$
 So ratio of time taken by Mayank and Shishu = $2/3$
 So if Shishu takes 30 hours, then Mayank will take 20 hours

$$\text{Shishu in 6 hours} = 1/5 \text{ the work.}$$

$$\text{Remaining work} = 1 - 1/5 = 4/5 \text{ the work,}$$

$$\text{Shishu and Mayank together} = \frac{1}{20} + \frac{1}{30} = \frac{1}{12}$$

$$\text{So required time} = \frac{4/5}{1/12} = 9.6 \text{ hours}$$

37. (c) Solve using option
38. (d) After 27 days, food left = $4 \times 200 = 800$ soldier days worth of food. Since, now there are only 80 soldiers, this food would last for $800/80 = 10$ days. Number of extra days for which the food lasts = $10 - 4 = 6$ days.
39. (b) Sambhu requires 16 days to do the work while Kalu requires 18 days to do the work.
 $(1/16 + 1/18) \times n = 1$
 $\rightarrow n = 288/34 = 144/17$
40. (b) The outlet pipe will empty the tank at a rate which is double the rate of filling. If the inlet is shut off, the tank will get emptied of 100 gallons of water in ten minutes.
41. (a) The per day digging of all three combined is 54 meters. Hence, their average should be 18. This means that the first should be $18 - x$, the second, 18 and the third $18 + x$.
 The required conditions are met if we take the values as 15, 18, 21 metres for the first, second and third diggers respectively.
42. (b) Let the total amount of work = 180 unit
 A does = 2 units/day
 B does = 4.5 units/day
 C does = 15 units/day
 To finish 180 units of work, every body has to work for 9 days.
 In 9 days A will do = $9 \times 2 = 18$ units work.
 So A's contribution = $\dots = \frac{18}{180} \dots \frac{1}{10}$
 So out of 240, A's share = $240 \times 1/10 = 24$
43. (a) $25(n - 12) = 21n + 300$. Solving this equation, $n = 150$. Hence, the first division harvest 3150 tons.
44. (c) $n(1/45 + 1/40) + 23/40 = 1 \rightarrow n = 9$
45. (a) The rest of the food will last for $(31 - 28 = 3)$ days if nobody leaves the place.

Thus, the rest of the food will last for $3\left(\frac{400}{120}\right)$ days for the 120 men left.

$$\therefore = 3\left(\frac{400}{120}\right) = 10 \text{ days}$$

46. (c) 1 minute's work of both the punctures = $\left(\frac{1}{9} + \frac{1}{6}\right) = \frac{5}{18}$.
 So, both the punctures will make the tyre flat in $\frac{18}{5} 3\frac{3}{5}$ min.
47. (a) Let 1 man's 1 days work = x
 1 boy's 1 day's work = y
 $12x + 16y = \frac{1}{5}$
 $13x + 24y = \frac{1}{4}$

Solving these two equation we get,

$$x = \frac{1}{100}, y = \frac{1}{200}$$

Required ratio = $2 : 1$

48. (d) Part filled by first tap in one min $\frac{1}{12}$ th

Part filled by second tap in one min $\frac{1}{18}$ th

$$\text{Now, } 2\left[\frac{1}{12} + \frac{1}{18}\right] + \text{unfilled part} = 1$$

$$\Rightarrow \text{unfilled part} = \frac{13}{18} \text{ th}$$

$\therefore \frac{1}{18}$ th part of tank is filled by second tap in 1 min.

$\therefore \frac{13}{18}$ th part of tank is filled by second tap in 1 min.

$$= 18 \times \frac{13}{18} \text{ min} = 13 \text{ min.}$$

49. (b) $\therefore$ Cistern fill in 6 hours.

$\therefore$ in 1 hour, filled part $\frac{1}{6}$ th

Now, due to leakage, filled part in 1 hour $\frac{1}{8}$ th

Part of the cistern emptied, due to leakage in 1 hour

$$= \frac{1}{6} - \frac{1}{8} = \frac{1}{24} \text{ th}$$

$\therefore$ The leakage will empty the full cistern in 24 hrs.

50. (c) 3 men reap $\frac{1}{43}$ rd of the field in 1 day.

$\therefore$ 1 man reaps $\frac{1}{43 \times 3}$ rd of the field in 1 day.

4 women reap $\frac{1}{43}$ rd of the field in 1 day.

$\therefore$ 1 woman reaps $\frac{1}{43 \times 3}$ th of the field in 1 day.

$$\therefore 7 \text{ men and } 5 \text{ women reap } \left(\frac{7}{43 \times 3} + \frac{5}{43 \times 4}\right) =$$

$$\frac{1}{12} \text{ th of the field in 1 day}$$

$\therefore$ 7 men and 5 women will reap the whole field in 12 days.

51. (c) $M_1 \times D_1 = M_2 \times D_2$
 $m \times r = (m + n) \times \bar{D}_2$

$$D_2 = \frac{mr}{(m + n)}$$

Standard Level

1. (c) Proportion of the volume of the tank filled by both the

$$\text{pipes in 4 min} = 4 \left(\frac{1}{15} + \frac{1}{10} \right) = \frac{2}{3} \text{ rd of the tank.}$$

Volume of the tank filled by all the pipes working

$$\text{together} = \frac{1}{15} + \frac{1}{10} - \frac{1}{5} = \frac{-1}{30}$$

$$\text{i.e., } \frac{1}{30} \text{ tank is emptied in 1 min.}$$

$$\therefore \frac{2}{3} \text{ rd of the tank can be emptied in } \frac{2 \times 30}{3} = 20 \text{ min}$$

2. (d) Since, flow of waste pipe = flow of filling pipe.

$\Rightarrow$ Filled part in one min = emptied part in one min.

$\therefore$ After opening the waste pipe for 2 min, cistern will be full in $(5 + 2) = 7$ min.

3. (c) $(A + B)$'s 1 hour's work = $\left(\frac{1}{12} + \frac{1}{15} \right) = \frac{9}{60} = \frac{3}{20}$

$$(A + C)\text{'s 1 hour's work} = \left(\frac{1}{12} + \frac{1}{20} \right) = \frac{8}{60} = \frac{2}{15}$$

$$\text{Part filled in 2 hrs} = \left(\frac{3}{20} + \frac{2}{15} \right) = \frac{17}{60}$$

$$\text{Part filled in 6 hrs} = \left(3 \times \frac{17}{60} \right) = \frac{17}{20}$$

$$\text{Remaining part} = \left(1 - \frac{17}{20} \right) = \frac{3}{20}$$

Now, it is the turn of A and B and $\frac{3}{20}$ part is filled by

A and B in 1 hour.

$\therefore$ Total time taken to fill the tank = $(6 + 1)$ hrs = 7 hrs.

4. (b) Thus, by our extended formula, number of required days

$$= \frac{1}{\frac{1}{44 \times 1} + \frac{1}{44 \times 2} + \frac{1}{44 \times 3}} = \frac{44 \times 1 \times 2 \times 3}{6 + 3 + 2} = 24 \text{ days}$$

5. (c) Let the work be finished in x days.

Then, A's x day's work + B's $(x - 1)$ day's work + C's $(x - 2)$ day's work = 1

$$\text{or, } \frac{x}{8} + \frac{x-1}{16} + \frac{x-2}{24} = 1$$

$$\text{or, } \frac{6x + 3x - 3 + 2x - 4}{48} = 1$$

$$\text{or, } 11x = 55$$

$$\therefore x = 5 \text{ days}$$

6. (b) Let B be closed after x minutes. Then, part filled by $(A + B)$ in x min. + part filled by A in $(18 - x)$ min = 1.

$$\therefore x \left(\frac{1}{24} + \frac{1}{32} \right) + (18 - x) \times \frac{1}{24} = 1$$

$$\text{or, } \frac{7x}{96} + \frac{18 - x}{24} = 1 \text{ or, } 7x + 4(18 - x) = 96$$

$$\text{or, } 3x = 24 \quad \therefore x = 8.$$

So, B should be closed after 8 min.

Direct Formula:

$$\text{Pipe B should be closed after } \left(1 - \frac{18}{24} \right) \times 32 = 8 \text{ min.}$$

7. (a) Let the number of men originally employed be x .

$$9x = 15(x - 6)$$

$$\text{or } x = 15$$

8. (c) In 8 days, Anil does $\frac{1}{3}$ rd work.

$$\therefore \text{in 1 day, he does } \frac{1}{24} \text{ th work.}$$

$$\therefore \text{Rakesh's one day's work} = 60\% \text{ of } \frac{1}{24} = \frac{1}{40} \text{ th work.}$$

$$\text{Remaining work } 1 - \frac{1}{3} = \frac{2}{3}$$

(Anil and Rakesh)'s one day's work

$$= \frac{1}{24} + \frac{1}{40} = \frac{1}{15} \text{ th work}$$

Now, $\frac{1}{15}$ th work is done by them in one day.

$$\therefore \frac{2}{3} \text{ rd work is done by them in } 15 \times \frac{2}{3} = 10 \text{ days}$$

9. (b) A's one day's work $\frac{1}{8}$ th work

$$B\text{'s one day's work } \frac{1}{3} \text{ rd work}$$

$$\therefore A\text{'s 4 day's work} = 4 \times \frac{1}{8} = \frac{1}{2} \text{ nd work}$$

$$\therefore \text{In next two days, total wall } \frac{1}{2} + 2 \left(\frac{1}{8} \right) - 2 \left(\frac{1}{3} \right) = \frac{1}{12} \text{ th wall}$$

$$\text{Remaining wall } 1 - \frac{1}{12} = \frac{11}{12} \text{ th}$$

Now, $\frac{1}{8}$ th wall is built up by A in one day.

$$\therefore \frac{11}{12} \text{ th wall is built up by A in } 8 \times \frac{11}{12} = 7 \frac{1}{3} \text{ days.}$$

10. (a) Work done by the waste pipe in 1 minutes

$$= \frac{1}{20} - \left(\frac{1}{12} + \frac{1}{15} \right) = -\frac{1}{10} \text{ [-ve sign means emptying]}$$

∴ Waste pipe will empty the full cistern in 10 minutes.

11. (c) A and B in 1 day do $\frac{1}{12}$ th work.

B and C in 1 day do $\frac{1}{16}$ th work.

Now, from the question,

A's 5 days' + B's 7 days' C's 13 days's work = 1

or, A's 5 days' + B's 5 days' + B's 2 days' + C's

2 days' + C's 11 days' work = 1

(A + B)'s 5 days' + (B + C)'s 2 days' + C's 11 days'

work = 1

$$\therefore \frac{5}{12} + \frac{2}{16} + \text{C's 11 days' work} = 1$$

$$\therefore \text{C's 11 day's work} = 1 - \left(\frac{5}{12} + \frac{2}{16} \right) = \frac{11}{24}$$

$$\therefore \text{C's 1 day's work} = \frac{11}{24 \times 11} = \frac{1}{24}$$

$$\therefore \text{B's 1 day's work} = \frac{1}{16} - \frac{1}{24} = \frac{1}{48}$$

$$\therefore \text{A's 1 day's work} = \frac{1}{12} - \frac{1}{48} = \frac{1}{16}$$

∴ A, B and C can do the work in 16, 48 and 24 days respectively.

12. (a) Let the filling capacity of pump be $x \text{ m}^3/\text{min}$.

Then, emptying capacity of pump = $(x + 10) \text{ m}^3/\text{min}$.

$$\therefore \frac{2400}{x} - \frac{2400}{x+10} = 8$$

$$\Rightarrow x^2 + 10x - 3000 = 0$$

$$\Rightarrow (x - 50)(x + 60) = 0 \Rightarrow x = 50 \text{ m}^3/\text{min}.$$

13. (c) Suppose pipe A alone takes x hours to fill the tank.

Then, pipes B and C will take $\frac{x}{2}$ and $\frac{x}{4}$ hours respectively to fill the tank.

$$\therefore \frac{1}{x} + \frac{2}{x} + \frac{4}{x} = \frac{1}{5} \Rightarrow \frac{7}{x} = \frac{1}{5} \Rightarrow x = 35 \text{ hrs.}$$

14. (a) Part filled in 10 hours = $10 \left(\frac{1}{15} + \frac{1}{20} - \frac{1}{25} \right) = \frac{23}{30}$.

$$\text{Remaining part} = \left(1 - \frac{23}{30} \right) = \frac{7}{30}.$$

$$(A + B)\text{'s 1 hour's work} = \left(\frac{1}{15} + \frac{1}{20} \right) = \frac{7}{60}.$$

$$\frac{7}{60} : \frac{7}{30} :: 1 : x \text{ or } x = \left(\frac{7}{30} \times 1 \times \frac{60}{7} \right) = 2 \text{ hours.}$$

∴ The tank will be full in $(10 + 2)$ hrs = 12 hrs.

15. (c) **Method I.** Considering one day's work:

$$4m + 6w = \frac{1}{8} \quad \dots(1)$$

$$3m + 7w = \frac{1}{10} \quad \dots(2)$$

$(1) \times 3 - (2) \times 4$ gives

$$18w - 28w = \frac{3}{8} - \frac{4}{10} \quad \text{or, } 10w = \frac{1}{40}$$

∴ 10 women can do the work in 40 days.

Method II. We find that

$$8(4m + 6w) = 10(3m + 7w)$$

$$\text{or, } 2m = 22w$$

$$\therefore 4m = 44w$$

∴ 4 men + 6 women = 50 women do in 8 days

$$\therefore 10 \text{ women do in } \frac{8 \times 50}{10} = 40 \text{ days}$$

16. (b) A + B can do the work in 5 days = $5 \left[\frac{1}{25} + \frac{1}{20} \right]$

$$= \frac{5 \times 45}{25 \times 20} = \frac{9}{20}$$

$$\text{Rest of the work} = 1 - \frac{9}{20} = \frac{11}{20}$$

$$B \text{ will do the rest of the work in } 20 \times \frac{11}{20} = 11 \text{ days.}$$

17. (d) One day work of man = 1

$$\text{One day work of woman} = \frac{1}{2}$$

$$\text{One day work of child} = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

$$\text{One day work ratio} = 1 : \frac{1}{2} : \frac{1}{6} \quad \text{or } 6 : 3 : 1$$

Ratio of men, women and children = 6 : 5 : 2

Their wages ratio = 6 : 3 : 1

Let wage per child = x

Then, wage per woman = $3x$

And wage per man = $6x$

Let y = number of children

There are 39 pairs of hands

$$\text{Therefore, } 6y + 5y + 2y = 39$$

$$\text{or } 13y = 39 \Rightarrow y = 3$$

Hence, man = 18, woman = 15 and children = 6

Amount paid, $(6x)18 + (3x)15 + (x)6 = 1113$

$$108x + 45x + 6x = 1113$$

$$\therefore x = 7$$

18. (c) Let the capacity of tank be x litres

In one hour tank empties = $\frac{1}{8}$ of x litre

In one hour, tap admits 6 litres
after opening tap tank is emptied in 12 hours.

So in one hour tank empties by $\frac{1}{12}$ of $x = \frac{x}{12}$ litres.

Therefore equation becomes $6 - \frac{x}{8} = -\frac{x}{12}$

$$6 - \frac{x}{8} - \frac{x}{12} = \frac{4x}{96} = \frac{x}{24}$$

$\therefore x = 144$ litres

19. (b) Let work done by A in one day be a , similarly, for B, b and for C, c

So, $3a = 1$, $4b = 1$, $6c = 1$ [Total work be 1 unit]

So, Total work done by the 3 Machines in one day

$$= \frac{1}{3} + \frac{1}{4} + \frac{1}{6} = \frac{3}{4}$$

Therefore, time taken to complete the work is

$$\frac{1}{3/4} = \frac{4}{3} \text{ days.}$$

20. (b) Suppose large pump takes t hours to fill a tank

$\therefore$ 1 hour work of large pump fills $\frac{1}{t}$ part

1 hour work of each small pump fills $= \frac{1}{t} \times \frac{2}{3}$

1 hour work of all 4 pumps fill $\frac{1}{t} + 3 \times \frac{2}{3t} = \frac{3}{t}$

Therefore, $\frac{3}{t}$ part is filled by all 4 pumps in 1 hour

$\therefore$ Whole tank would be filled in $1 \times \frac{t}{3} = \frac{t}{3} h$ this is

1/3 of the time taken by large pump i.e., t hour

21. (b) A's one day's work = $\frac{1}{15}$ th work.

B's one day's work $\frac{1}{10}$ th work.

(A + B)'s one day's work $\frac{1}{15} + \frac{1}{10} = \frac{1}{6}$ th work.

Let A left after x days.

$\therefore$ (A + B)'s x days' work $\frac{x}{6}$ th work.

Remaining work $1 - \frac{x}{6} = \frac{6-x}{6}$ th work.

Now, in 5 days, work done by B $\frac{6-x}{6}$ th work.

$\therefore$ In 1 day work done by B $\frac{6-x}{30}$ th work

$$\text{and } \frac{6-x}{30} = \frac{1}{10}$$

$\therefore x = 3$ days

22. (b) Let x additional men employed.
117 men were supposed to finish the whole work in
 $46 \times 8 = 368$ hours.

But 117 men completed $\frac{4}{7}$ of the work in 33×8

$= 264$ hours

$\therefore$ 117 men could complete the work in 462 hours.

Now $(117 + x)$ men are supposed to do $\frac{3}{7}$ of the work,

working 9 hours a day, in $13 \times 9 = 117$ hours, so as to finish the work in time.

i.e., $(117 + x)$ men are supposed to complete the whole

work in $117 \times \frac{7}{3} = 273$ hours.

$$\therefore (117 + x) \times 273 = 117 \times 462$$

$$\Rightarrow (117 + x) \times 7 = 3 \times 462$$

$$\Rightarrow x + 117 = 3 \times 66 = 198 \Rightarrow x = 81$$

$\therefore$ Required number of additional men to finish the work in time = 81.

23. (b) Given 12 men $\equiv$ 15 women $\equiv$ 18 boys
 $\therefore$ 1 Man $\equiv$ 1.5 boys, 1 woman = 6/5 boys.

Now, $5W + 6B = 12B$.

Required answer is calculated as follows :

Total no. of boys reqd. = $18 \times [(15/16) \times (8/9)]$
 $= 15$ boys

The number of boys already present = 12.

Hence, 3 boys more required.

But 3 boys = 2 men.

So, 2 men are required.

24. (c)

	Men	Women	Children
Work	3	2	1
Numbers	20	30	36
Ratio of wages	$= (3 \times 20) : (2 \times 30) : (1 \times 36) = 5 : 5 : 3$		

Total wages of men = $\frac{5}{13} \times 780 = ₹ 300$

$\therefore$ Wages of a man = ₹ 15

Similarly, wages of woman = ₹ 10

and wages of child = ₹ 5

Total wages of 15 men, 21 women and 30 children
 $= 15 \times 15 + 21 \times 10 + 30 \times 5 = ₹ 585$

Total wages for 2 weeks = ₹ 1170

25. (a) Diameter of three pipes say A, B, C are in the ratio
 $1 : \frac{4}{3} : 2$

The ratio of flow can in the ratio $1^2 : \left(\frac{4}{3}\right)^2 : 2^2$

$$= 1 : \frac{16}{9} : 4$$

Time taken by each pipe separately to fill the tank

$$= 1 : \frac{9}{16} : 4$$

If the pipe with diameter 2 cm takes 61 min. to fill the tank, then pipe A will take 61×4 minutes and pipe B will take

$$61 \times 4 \times \frac{9}{16} = \frac{61 \times 9}{4} \text{ min}$$

$\therefore$ In 1 min all the 3 pipes will fill

$$\frac{1}{61} + \frac{1}{61 \times 4} + \frac{4}{61 \times 9} = \frac{4 \times 9 + 9 + 4 \times 4}{61 \times 4 \times 9}$$

$$= \frac{1}{36} \text{ of the tank}$$

$\therefore$ Time taken by all the three pipes to fill the tank = 36 mins.

26. (a) If x complete a work in x days. y will do the same task in $3x$ days.

$$3x - x = 40$$

$$\Rightarrow x = 20$$

y will finish the task in 60 days

$(x + y)$'s 1 days work

$$= \frac{1}{20} + \frac{1}{60} = \frac{1}{15}$$

Both of them will complete the work in 15 days.

27. (b) Let the inlets be A, B, C and D .

$$A + B + C = 8.33\%$$

$$B + C + D = 6.66\%$$

$$A + D = 5\%$$

$$\text{Thus } 2A + 2B + 2C + 2D = 20\%$$

$$\text{and } A + B + C + D = 10\%$$

$\rightarrow$ 10 minutes would be required to fill the tank completely.

28. (b) $A + B = \frac{50\%}{1.2} = 41.66\%$

(where A and B represent the work per hour of pipes A and B respectively).

Solve using options to see which one fits the remaining conditions for the problem.

For example, if we check option b (4 hrs), then we get that the work of the faster pipe (say A) = 25%.

Then $B = 16.66\%$.

Then B was open for $4/2 = 2$ hours and A was open for $6/3 = 2$ hours.

$$\text{Work done} = 25\% \times 2 + 16.66\% \times 2 = 83.33\% = \frac{5}{6} \text{ of work.}$$

$\therefore$ Total time = 272 = 4h

29. (b) The 32 minutes extra represents the extra time taken by the pipes due to the leak.

Normal time for the pipes $\rightarrow n \times (1/14 + 1/16) = 1 \rightarrow n = 112/15 = 7 \text{ hrs } 28 \text{ minutes.}$

Thus, with 32 minutes extra, the pipes would take 8 hours to fill the tank.

$$\begin{aligned} \text{Thus, } 8(1/14 + 1/16) - 8 \times (1/L) &= 1 \rightarrow 8/L \\ &= 8(15/112) - 1 \\ 1/L &= 15/112 - 1/8 \\ &= 1/112. \end{aligned}$$

Thus, $L = 112$ hours.

30. (c) $0.5(A + B + C) = 50\%$ of the work.

Means $\rightarrow A, B$ and C can do the full work in 1 hour.

Thus, $(A + B + C) = 100\%$

From this point it is better to solve through options, Option (c) gives the correct answer based on the following thought process.

If $c = 50\%$ work per hour, it means C takes 2 hours to complete the work.

Consequently, A would take 3 hours and hence do 33.33% work per hour.

Since, $A + B + C = 100\%$, this gives us B 's hourly work rate 16.66%.

For this option to be correct these nos. should match the second instance and the information given there.

According to the second condition:

$A + 4B$ should be equal to 100%. Putting $A = 33.33\%$ and $B = 16.66\%$ we see that the condition of the problem.

31. (c) If emptying pipe is closed, then it takes 100/19 hours to fill the tank.

But, only 2/5th of 50,000 = 20,000 litres is filled.

So in 100/19 hours, 30,000 litres water drained

Hence in 1 hour, 5700 litres water drained

So rate at which each house gets water.

$$= \frac{5700}{2000} = 2.85 \text{ litres/hours}$$

32. (d) Let x = Number of days it rained in the morning and had clear afternoons.

y = Number of days it rained in the afternoon and had clear mornings.

z = Number of days it rained in the morning or afternoon

So according to question, $x + y = 7$

$$x + z = 5$$

$$y + z = 6$$

Adding all three equations, $x + y + z = 9$

So, $d = 9$ days

33. (a) According to the question, $1/2, 1/a, 1/3$ are in A.P. So 'a' can be calculated.

$$\text{Now, } \frac{1}{2-x}, \frac{1}{a}, \frac{1}{3-x}$$

So x can be calculated.

$$x = 2.6.$$

34. (a) Given $6 \text{ BSF} \equiv 10 \text{ CRPF} \Rightarrow 4 \text{ BSF} + 9 \text{ CRPF}$

$$= 4 + (9 \times 6/10) \text{ BSF} = \frac{94}{10} \text{ BSF}$$

$$\text{Now work} = 6 \times 2 \text{ BSF days} = \frac{94}{10} \times X \text{ BSF days}$$

$$\text{We have } 6 \times 2 \equiv \frac{94}{10} \times X \Rightarrow X = 1.27 \text{ days}$$

35. (b) Let the required number of working hours/day = x
More pumps, less working hrs per day (Indirect)
Less days, more working hrs per day (Indirect)

$$\left. \begin{array}{l} \text{Pumps } 4 : 3 \\ \text{Days } 1 : 2 \end{array} \right\} :: 8 : x$$

$$\therefore 4 \times 1 \times x = 3 \times 2 \times 8$$

$$\Rightarrow x = \frac{3 \times 2 \times 8}{4} = 12$$

36. (b) Go through option
 $140 \times 4 = (140 + 120 + 100 + \dots + 20)$
 $560 = 560$

Alternatively: Let n be the initial number of worker then

$$n \times 4 = n + (n - 20) + (n - 40) + \dots + (n - 120)$$

$$4n = 7n - 420$$

$$\Rightarrow 3n = 420$$

$$\Rightarrow n = 140 \text{ workers}$$

37. (b) Ratio of number of men, women and children

$$= \frac{18}{6} : \frac{10}{5} : \frac{12}{3} = 3x : 2x : 4x$$

$$\therefore (3x + 2x + 4x) = 18$$

$$\therefore x = 2$$

Therefore, number of women = 4

$$\text{Share of all women} = \frac{10}{40} \times 4000 = 1000$$

$$(\because 18 + 10 + 12 = 40)$$

$$\therefore \text{Share of each woman} = \frac{1000}{4} = 250$$

Expert Level

1. (b) Let the work (in units) done by a man and a woman in one day be M and W respectively. Total work (in units) = $10(M + 6W) = 10M + 60W$

$$\Rightarrow \frac{10M + 60W}{8W} - \frac{10M + 60W}{2M} = 5$$

$$\frac{5M}{4W} - \frac{30W}{M} = \frac{5}{2}$$

On putting $\frac{M}{W} = x$, we get

$$\frac{5x}{4} - \frac{30}{x} = \frac{5}{2} \Rightarrow x = 6 \text{ or } \frac{M}{W} = 6$$

$\therefore$ The efficiency of a man is greater than that of a woman by 500%.

2. (a) Let the amount of work (in units) completed by a man, a woman and a child in a day be M , W and C respectively. The amount of work (in units) completed by 4 men in

$$12 \text{ days} = 4 \times 12 \times M = 48M.$$

The amount of work (in units) completed by 6 women in 10 days = $6 \times 10 \times W = 60W$.

The amount of work (in units) completed by 8 children in 9 days = $8 \times 9 \times C = 72C$.

$$\text{So } 48M = 60W = 72C$$

$$\text{or } 4M = 5W = 6C = 60K \text{ (say)}$$

$$\text{Hence, } M = 15K, W = 12K \text{ and } C = 10K.$$

The amount of work (in units) completed by a man, a woman and a child together in 10 days = $(15 + 12 + 10)K \times 10 = 370K$.

The amount of work (in units) completed by 2 women and 5 children together in a day = $(2 \times 12 + 5 \times 10)K = 74K$.

$$\text{Hence, the answer} = \frac{374}{74} = 5 \text{ days.}$$

3. (c)

	A		B		C
Efficiency	3	:	2	:	6
Number of days	2	:	3	:	1

$\therefore$ Number of days taken by A = 12

Number of days taken by B = 18

Number of days taken by C = 6

$$1 \text{ day's work of } (A + B) = \frac{5}{36}$$

$$1 \text{ day's work of } (B + C) = \frac{8}{36}$$

$$1 \text{ day's work of } (C + A) = \frac{9}{36}$$

Days $\rightarrow$	1	2	3	4	5	6
Work $\rightarrow$	$5/36$	$8/36$	$9/36$	$5/36$	$8/36$	$1/36$
			$35/36$			

$$\text{In 5 days total work done} = \frac{35}{36}$$

Now, the rest work ($1/36$), which is done by AC.

$\therefore$ Number of days taken by AC for the rest work

$$= \frac{1/36}{9/36} = \frac{1}{9}$$

$$\text{Therefore, total time} = 5 + \frac{1}{9} = 5\frac{1}{9} \text{ days}$$

4. (b) Let x kg of oil is used for eating purpose, daily, then

$$(x + 11) \times 50 = (x + 15) \times 45$$

$$x = 25$$

$$\therefore \text{Total quantity of oil} = (25 + 11) \times 50 = 1800$$

$$\therefore \text{Required number of days} = \frac{1800}{25} = 72 \text{ days}$$

5. (b) Let n be the number of metres planned per day. Start from the options to find the number of planned days. In the options the 2 feasible values are 30 metres and 27 metres (as these divide 270). Suppose, we check for 30 metres per day, the work would have got completed in 9 days as per the original plan. In the new scenario:
 $3n + 5(n + 8) = 280 \rightarrow n = 30$ too. Hence, this option is correct.

Note that if we tried with 27 metres per day the final equation would not match as we would get:

$3n + 6(n + 8) = 280 \rightarrow$ which does not give us the value of n as 27 and hence this option is rejected.

6. (b) Let the job be of L.C.M.(15, 20, 25) = 300 units. The number of units completed by Aman, Baman and Chaman in a day while working alone on the job are 20, 15 and 12 respectively.

The number of units completed by Aman, Baman and Chaman in a day while working on the job with somebody else are 14, 12 and 6 respectively. Aman, Baman and Chaman together complete $14 + 12 + 6 = 32$ units of work in a day.

The work must be distributed in either of the following ways to ensure the maximum output in four consecutive days:

	DAY 1	DAY 2	DAY 3	DAY 4
WAY 1	(A, B, C)	A/(B, C)	(B, C)/A	(A, B, C)
WAY 2	(A, B, C)	C/(A, B)	(A, B)/C	(A, B, C)

The number of units completed will be $32 + 20 + 18 + 32 = 102$.

Hence, the answer = $\frac{102}{300} = \frac{17}{50}$.

7. (a) Number of days taken by A to complete work alone = 14 days
 Number of days taken by B to complete work alone = 7 days
 Number of days taken by C to complete work alone = 7 days

One day's work of A and B = $\frac{1}{14} + \frac{1}{7} = \frac{3}{14}$

and one day's work of A, B and C = $\frac{1}{14} + \frac{1}{7} + \frac{1}{7} = \frac{5}{14}$

3 day's work of A and B = $3 \times \frac{3}{14} = \frac{9}{14}$

remaining work = $\frac{5}{14} \left(1 - \frac{9}{14}\right)$

This remaining work will be done by A, B and C

$$= \frac{5/14}{5/14} = 1 \text{ day}$$

8. (d) Since A finishes $6/7^{\text{th}}$ of the work in $2z$ hours. B would finish $12/7^{\text{th}}$ of the work in $2z$ hours. Thus, to do $1/7^{\text{th}}$ of the work (which represents the remaining work), B would require $2z/12 = z/6$ hours. Option (d) is correct.

9. (d) Efficiency of 4 men and 2 boys = 20%
 Efficiency of 3 women and 4 boys = 20%
 Efficiency of 2 men and 3 women = 20%
 $\therefore$ Efficiency of 6 men, 6 women and 6 boys = 60%
 $\therefore$ Efficiency of 1 man, 1 woman and 1 boy = 10%
 Now, since they will work at double their efficiency
 $\therefore$ Efficiency of 1 man, 1 woman and 1 boy = 20%
 $\therefore$ Required number of days = 5

10. (c) Let the number of units of work completed by a man, a woman and a child in one day be M , W and C respectively.

Hence, $2M = 3C$ and $3M = 5W$.

Let $M = 15U$. Hence, $C = 10U$ and $W = 9U$.

The amount of work completed by a man, a woman and a child together in 10 days = $10(15U + 10U + 9U) = 340U$.

The amount of work completed by 2 children in a day = $20U$.

Hence, the answer = $\frac{340}{20} = 17$ days.

11. (d) Let the volume filled (in units) by an inlet pipe in an hour = V .

The total volume (in units) of the tank = $30V$.

Let the volume emptied (in units) by an outlet pipe in an hour = U .

Hence, $10(5V - 4U) = 30V$ or $V = 2U$.

The time taken by an outlet pipe to empty $15V$ volume

$$= \frac{15V}{U} = 30 \text{ hours}$$

12. (b) The net inflow when both pipes are opened is 5 litres a minute.

The outlet flow should be such that if its rate is doubled the net inflow rate should be negative or 0. Only an option greater than or equal to '5' would satisfy this condition.

Option (b) is the only possible value.

13. (b) Let's assume that the job consists of 120 (L.C.M. of 20, 30 and 40) units of work.

Therefore, Ashish, Binay and Joseph can do 6, 4 and 3 units of work respectively in a day.

If the job took x days to complete, Ashish, Binay and Joseph worked for $x - 4$, $x - 3$, and x days respectively.

Hence, $6(x - 4) + 4(x - 3) + 3x = 120$.

$\Rightarrow x = 12$.

14. (a) Let the work (in units) done in a day by Pawan, Qureshi and Rohit be P , Q and R respectively. Let the total work done be (L.C.M. of 10 and 12) 60 units.
 $\therefore P + Q = 6$ units and $Q + R = 5$ units.
 Let Pawan's share be x . So Rohit's share will be $(160 - x)$.
 Ratio of the amount received by Pawan and Qureshi together to the amount received by Qureshi and Rohit together
- $$\frac{x + 140}{140 + 160 - x} = \frac{6}{5} \text{ or } x = 100$$
15. (c) From the condition of the problem and a little bit of trial and error we can see that the first cook worked for 4 minutes and the 2nd and 3rd cooks also worked for 4 minutes.
 As $4(A + B + C) = 4(A + B + C)$ and we know that $A + B + C = 20$ idlis per minute.
 Thus, the first cook make 20 idlis in 4 minutes.
 To make 160 idlis he would take 32 minutes.
16. (a) Efficiency of only leakage = 16.66%
 Effective efficiency of leakage = 6.66%
 It means the capacity of filling pipe = 10%
 Therefore, the inlet pipe can fill the tank in 10 hours hence the capacity of tank = 100 L
17. (a) Efficiency of $A = 5\%$
 Efficiency of $B = 4\%$
 Efficiency of $C = -3.33\%$
 It means in every 3 consecutive hours taps A , B and C can fill 5.66% ($= 5 + 4 - 3.33$).
 Therefore in 51 hours ($= 3 \times 17$) taps A , B and C can fill 96.33% ($= 5.66 \times 17$)
 $\therefore$ the remaining part i.e., 3.66% ($= 100 - 96.33$) can be filled up by A in $\frac{11}{15}$ hours $\left(\frac{3.66}{5} \right)$, since it is now A 's turn.
 Hence, the total time required $= 51 + \frac{11}{15} = 51\frac{11}{15}$
18. (c) $0.5(A + B + C) = 50\%$ of the work.
 Means $\rightarrow A$, B and C can do the full work in 1 hour.
 Thus, $(A + B + C) = 100\%$
 From this point it is better to solve through options.
 Option (c) gives the correct answer based on the following thought process.
 If $c = 50\%$ work per hour, it means C takes 2 hours to complete the work.
 Consequently, A would take 3 hours and hence do 33.33% work per hour.
 Since, $A + B + C = 100\%$, this gives us B 's hourly work rate = 16.66%
 For this option to be correct these nos, should match the second instance and the information given there.
 According to the second condition:
 $A + 4B$ should be equal to 100%. Putting $A = 33.33\%$ and $B = 16.66\%$, we see that the condition is satisfied.
 Hence, this option is correct.
19. (a) In order to solve this question, if we look at the first statement, we could think of the following scenarios:
 If the time taken by the first man and the woman is 1 hour (100% work per hour), the time taken by the second man would be 4 hours (25% work per hour). In such a case, the total time taken by all three to complete the task would be $100/125 = 0.8$ hours. But this value is not there in the options. Hence, we reject this set of values.
 If the time taken by the first man and the woman is 2 hours (50% work per hour), the time taken by the second man would be 5 hours (20% work per hour). In such a case, the total time taken by all three to complete the task would be $100/70 = 10/7$ hours. But this value is not there in the options. Hence, we reject this set of values.
 If the time taken by the first man and the woman is 3 hours (33.33% work per hour), the time taken by the second man would be 6 hour (16.66% work per hour). In such a case, the total time taken by all three to complete the task would be $100/50 = 2$ hours. Since this value is there in the options we should try to see whether this set of values meets the other conditions in the question.
 In this case, it is given that the first man working alone takes as much time as the second man and the woman. Since, the work of all three is 50%, this means that the work of the first man is 25%. Consequently the work of the woman is 8.33%.
 Looking at the third condition given in the problem – the time taken by the first man to do the work alone (@ 25% per hour he would take 4 hour) should be 8 hours less than double the time taken by the second man. This condition can be seen to be fulfilled here because the second man would take 6 hours to complete his work (@ 16.66% per hour) and hence, double his time would be 12 hours-which satisfies the difference of 8 hours.
 Thus, the total time taken is 2 hours.
20. (c) Time taken by pipes A and B to fill the whole tank
 $= \frac{100}{16.66} = 6$ hours
 Capacity filled in 2 hours by pipes A , B and C
 $= 2 \times 13.33 = 26.66\%$
 Remaining capacity = 73.33%
 This remaining capacity can be filled be A and B
 $= \frac{73.33}{16.66} = 4\frac{2}{5}$ hours
 So, the total time required $= 2 + 4\frac{2}{5} = 6$ hours 24 minutes
 Thus, in this case 24 minutes extra are required.
21. (c) From the first statement:
- | | | |
|------------|-----------|---------------|
| | $(B + C)$ | $(A + B + C)$ |
| Efficiency | $2x$ | $3x$ |
| Days | $3y$ | $2y$ |

Thus, we can say that efficiency of A is $\frac{1}{3}$ the efficiency of (A + B + C).

From the last statement:

$$\text{Share of B out of total amount} = \frac{120}{450} = \frac{4}{15}$$

From these two results, we can conclude that:

$$\begin{array}{ccc} A & : & B & : & C \\ \text{Ratio of efficiency} & \frac{5}{15} \left(\frac{1}{3} \right) & : & \frac{4}{15} & : & \frac{6}{15} \\ \Rightarrow & 5 & : & 4 & : & 6 \end{array}$$

$$\begin{array}{ccc} \text{Ratio of number of days} \Rightarrow & \frac{1}{5} & : & \frac{1}{4} & : & \frac{1}{6} \\ \Rightarrow & 12x & : & 15x & : & 10x \end{array}$$

$$\text{One day's work of A and B} = \frac{1}{12x} + \frac{1}{15x} = \frac{9}{60x}$$

$\therefore$ A and B will take $\frac{60x}{9}$ days to complete the whole work.

Again one day's work of A, B and C

$$= \frac{1}{12x} + \frac{1}{15x} + \frac{1}{10x} = \frac{15}{60x}$$

$\therefore$ A, B and C working together complete the work in

$$\frac{60x}{15} \text{ days}$$

$$\therefore \frac{60x}{9} - \frac{60x}{15} = \frac{8}{3}$$

(Since, A and B take $\frac{8}{3}$ days more than A, B and C)

$$\Rightarrow x = 1$$

$\therefore$ Number of days required to complete the whole work by

$$A, B \text{ and } C = 4x = 4 \times 1 = 4 \text{ days}$$

22. (c) Efficiency of Eklavya = 16.66%

Efficiency of Faizal = 8.33%

Total efficiency of Eklavya and Faizal = 25%

So, they can do actual work in 4 days

$\therefore$ 3 times work requires 12 days.

23. (d)

Men	Working Hours	Day	Work done
64	8	5	40%
64	n(say)	4	60%

40% work in 40 hours $\rightarrow$ 60% work in 60 hours.

Hence, working hours = $60/4 = 15$ hours.

24. (c) Let h be the length of water column discharged in 1 hour or 1 minute.

Volume discharged by the 4 pipes

= Volume discharged by the single pipe.

$$4 \times \pi \times (1.5)^2 \times h = \pi \times (r)^2 \times h$$

$$\therefore r^2 = 9 \quad \therefore r = 3, \text{ Diameter} = 6 \text{ cm.}$$

25. (b) Let B can finish the work in x days.

Then A can finish the work in $(x-3)$ days.

$$B's \text{ one day's work} = \frac{1}{x} \text{ th work}$$

$$A's \text{ one day's work} = \frac{1}{x-3} \text{ th work}$$

$$A's 4 \text{ days' work} = \frac{4}{x-3} \text{ th work}$$

$$\text{Remaining work} = 1 - \frac{4}{x-3} = \frac{x-7}{x-3} \text{ th work}$$

The remaining work done by B in $14 - 4 = 10$ days.

$$\text{Now, in 10 days, work done by B} = \frac{x-7}{x-3} \text{ th work}$$

$$\therefore \text{In 1 day, work done by B} = \frac{1}{10} \left(\frac{x-7}{x-3} \right) \text{ th work}$$

$$\text{and } \frac{1}{10} \left(\frac{x-7}{x-3} \right) = \frac{1}{x}$$

$$\Rightarrow x = 15 \text{ days}$$

$\therefore$ B will finish in 15 days and A will finish in 12 days

26. (a) Radius of the pipe (r) = 4 cm = 0.04 metre

Volume of water flowing out per sec

$$= \pi r^2 \times \text{rate of flow}$$

$$= \frac{22}{7} \times 0.04^2 \times 3 \text{ cu metres} = 0.0151 \text{ cubic m}$$

$$\text{Time taken to fill the tank} = 40 \times 30 \times \frac{8}{0.0151} \text{ sec}$$

$$= \frac{40 \times 30 \times 8}{0.01} \times \frac{1}{3600} \text{ hours} = 176.6 \text{ hours}$$

27. (a) Rate of admission of water

$$= \frac{2}{6} \text{ tonnes/min} = \frac{1}{3} \text{ tonnes/min}$$

Rate of pumping out of water

$$= \frac{12}{60} \text{ tonnes/min} = \frac{1}{5} \text{ tonnes/min.}$$

$$\text{Rate of accumulation} = \left(\frac{2}{6} - \frac{12}{60} \right) \text{ tonnes/min.}$$

Time to accumulate 80 tonnes of water

$$= \frac{\text{Amount of water}}{\text{Accumulation rate}} = \frac{80}{\left(\frac{1}{3} - \frac{1}{5} \right)} = 600 \text{ min.} = 10 \text{ hours}$$

$\therefore$ Average sailing rate so as avoid sinking

$$= \frac{\text{Distance}}{\text{Time}} = \frac{55}{10} \text{ km/h} = 5.5 \text{ km/h}$$

Explanation of Test Yourself

1. (d) One day work of man = 1

$$\text{One day work of woman} = \frac{1}{2}$$

$$\text{One day work of child} = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

$$\text{One day work ratio} = 1 : \frac{1}{2} : \frac{1}{6} \text{ or } 6 : 3 : 1$$

$$\text{Ratio of men, women and children} = 6 : 5 : 2$$

$$\text{Their wages ratio} = 6 : 3 : 1$$

$$\text{Let wage per child} = x$$

$$\text{Then, wage per woman} = 3x$$

$$\text{And wage per man} = 6x$$

$$\text{Let } y = \text{number of children}$$

$$\text{There are 39 pairs of hands}$$

$$\text{Therefore, } 6y + 5y + 2y = 39$$

$$\text{or } 13y = 39 \Rightarrow y = 3$$

$$\text{Hence, man} = 18, \text{ woman} = 15 \text{ and children} = 6$$

$$\text{Amount paid, } (6x)18 + (3x)15 + (x)6 = 1113$$

$$108x + 45x + 6x = 1113$$

$$\therefore x = 7$$

2. (b) Tap A fills 4 buckets ($4 \times 5 = 20$ litres) in 24 min.

$$\text{In 1 hour tap A fills } \frac{20}{24} \times 60 = 50 \text{ litres}$$

$$\text{In 1 hour tap B fills} = 8 \times 5 = 40 \text{ litres}$$

$$\text{In 1 hour tap C fills } \frac{2 \times 5}{20} \times 60 = 30 \text{ litres}$$

$$\text{If they open together they would fill}$$

$$50 + 40 + 30 = 120 \text{ litres in one hour}$$

$$\text{but full tank is emptied in 2 hours}$$

$$\text{So, tank capacity would be } 120 \times 2 = 240 \text{ litres.}$$

3. (c) Let the capacity of tank be x litres

$$\text{In one hour tank empties} = \frac{1}{8} \text{ of } x = \frac{x}{8} \text{ litre}$$

$$\text{In one hour, tap admits 6 litres}$$

$$\text{after opening tap tank is emptied in 12 hours.}$$

$$\text{So in one hour tank empties by } \frac{1}{12} \text{ of } x = \frac{x}{12} \text{ litres.}$$

$$\text{Therefore equation becomes } 6 - \frac{x}{8} = -\frac{x}{12}$$

$$6 - \frac{x}{8} = -\frac{x}{12} \Rightarrow \frac{x}{8} - \frac{x}{12} = \frac{x}{24}$$

$$\therefore x = 144 \text{ litres}$$

4. (b) Let work done by A in one day be a , similarly, for B, b and for C, c

$$\text{So, } 3a = 1, 4b = 1, 6c = 1 \text{ [Total work be 1 unit]}$$

$$\text{So, Total work done by the 3 machines in one day}$$

$$= \frac{1}{3} + \frac{1}{4} + \frac{1}{6} = \frac{3}{4}$$

$$\text{Therefore, time taken to complete the work is}$$

$$\frac{1}{3/4} = \frac{4}{3} \text{ days.}$$

5. (d) A's 1 day work $\frac{1}{4}$; B's $\frac{1}{8}$; C's $\frac{1}{16}$; D's $\frac{1}{32}$

$$\text{Consider pairs A, B and C, D}$$

$$\text{For A, B combined, 1 day work } \frac{3}{8} \text{ per day}$$

$$\text{For C, D combined, 1 day work } \frac{3}{32} \text{ per day}$$

$$\text{For A, C combined, 1 day work } \frac{5}{16} \text{ per day}$$

$$\text{For B, D combined, 1 day work } \frac{5}{32} \text{ per day}$$

$$\text{For B, C combined, 1 day work } \frac{3}{16} \text{ per day}$$

$$\text{For A, D combined, 1 day work } \frac{9}{32} \text{ per day}$$

$$\text{Hence, we see that A, D is first pair B, C is second}$$

6. (b) Suppose large pump takes t hours to fill a tank

$$\therefore 1 \text{ hour work of large pump fills } \frac{1}{t} \text{ part}$$

$$1 \text{ hour work of each small pump fills} = \frac{1}{t} \times \frac{2}{3}$$

$$1 \text{ hour work of all 4 pumps fill } \frac{1}{t} + 3 \times \frac{2}{3t} = \frac{3}{t}$$

$$\text{Therefore, } \frac{3}{t} \text{ part is filled by all 4 pumps in 1 hour}$$

$$\therefore \text{ Whole tank would be filled in } 1 \times \frac{t}{3} = \frac{t}{3} \text{ h this is}$$

$$1/3 \text{ of the time taken by large pump i.e., } t \text{ hour}$$

7. (c) **Machine I :**

Time to produce 9000 nuts

$$\frac{9000}{100} + 8 \times 5 = 130 \text{ minutes}$$

Machine II :

Time to produce 9000 bolts

$$\frac{9000}{75} + 5 \times 10 = 120 + 50 = 170 \text{ minutes}$$

So minimum time required for the production of 9000 nuts and bolts = 170 minutes.

8. (d) Clearly, I only gives the answer
Similarly, II only gives the answer
And, III only gives the answer

9. (b) B alone can do a work in 20 hours.

$\therefore$ A alone can do $\frac{3}{2}$ of the work in 20 hours.

i.e., A alone can do the same work in $\frac{40}{3}$ hours

$\therefore$ (A + B)'s one hour's work $\frac{3}{40} + \frac{1}{20} = \frac{5}{40} = \frac{1}{8}$
 $\Rightarrow$ A and B together can finish the whole work in 8 hours.

10. (a) Work done by A and B in 5 days = $\left(\frac{1}{10} + \frac{1}{15}\right) \times 5 = \frac{5}{6}$

Work remaining = $1 - \frac{5}{6} = \frac{1}{6}$

$\therefore$ C alone can do the work in $6 \times 2 = 12$ days

Ratio of their share work = $\frac{5}{10} : \frac{5}{15} : \frac{2}{12} = 3 : 2 : 1$

Share of wages = ` 225, ` 150, ` 75.

11. (b) X's one day's work $\frac{1}{15}$ th work.

Y's one day's work $\frac{1}{15} + 50\%$ of $\frac{1}{15} = \frac{1}{10}$ th work

$\therefore$ (X + Y)'s one day's work $\frac{1}{15} + \frac{1}{10} = \frac{1}{6}$ th work

Hence, they together finish the work in 6 days.

12. (a) 1 man's 1 day's work = $\frac{1}{48}$;

1 woman's 1 day's work = $\frac{1}{60}$.

6 men's 2 day's work = $\left(\frac{6}{48} \times 2\right) = \frac{1}{4}$.

Remaining work = $\left(1 - \frac{1}{4}\right) = \frac{3}{4}$.

Now, $\frac{1}{60}$ work is done in 1 day by 1 woman.

So, $\frac{3}{4}$ work will be done in 3 days by

$$\left(60 \times \frac{3}{4} \times \frac{1}{3}\right) = 15 \text{ women.}$$

13. (b) Let x additional men employed.

117 men were supposed to finish the whole work in $46 \times 8 = 368$ hours.

But 117 men completed $\frac{4}{7}$ of the work in

$33 \times 8 = 264$ hours

$\therefore$ 117 men could complete the work in 462 hours.

Now (117 + x) men are supposed to do $\frac{3}{7}$ of the work,

working 9 hours a day, in $13 \times 9 = 117$ hours, so as to finish the work in time.

i.e., (117 + x) men are supposed to complete the whole

work in $117 \times \frac{7}{3} = 273$ hours.

$\therefore$ (117 + x) $\times$ 273 = 117 $\times$ 462

$\Rightarrow$ (117 + x) $\times$ 7 = 3 $\times$ 462

$\Rightarrow$ x + 117 = 3 $\times$ 66 = 198 $\Rightarrow$ x = 81

$\therefore$ Required number of additional men to finish the work in time = 81.

14. (a) 1 horse = 2 cows, 10 horses = 20 cows.

$\Rightarrow$ 10 horses + 15 cows = 20 + 15 = 35 cows.

15 horses + 10 cows = 40 cows. Now 35 cows eat 5 acres.

$\Rightarrow$ 40 cows eat $5 \times \frac{40}{35} = 5 \frac{5}{7}$ acres.

Here we have converted everything in terms of cows, you can work in terms of horses also.

15. (c)

	Men	Women	Children
Work	3	2	1
Numbers	20	30	36

Ratio of wages = (3 $\times$ 20) : (2 $\times$ 30) : (1 $\times$ 36) = 5 : 5 : 3

Total wages of men = $\frac{5}{13} \times 780 = \text{` } 300$

$\therefore$ Wages of a man = ` 15

Similarly, wages of woman = ` 10

and wages of child = ` 5

Total wages of 15 men, 21 women and 30 children

= 15 $\times$ 15 + 21 $\times$ 10 + 30 $\times$ 5 = 585

Total wages for 2 weeks = ` 1170

10

TIME, SPEED AND DISTANCE

- Introduction
- Motion or movement
- Conversion kmph (kilometer per hour) to m/s (meter per second) and vice-versa
- Direct and Inverse Proportionality Between any two of the speed(S), Time(T) and Distance(D) When the Third One is Constant
- Average Speed
- Relative Speed
- To and Fro Motion in a Straight Line Between Two Points A and B
- Uniform Acceleration and Uniform Deceleration
- Application of Alligation in the Problems Related to Time, Speed and Distance
- Concept Related to Motion of Trains
- Boats and Streams
- Basic Terminology Related to Races
- Circular Motion
- Clocks

INTRODUCTION

For a CAT aspirant, a problem on time, speed and distance means solving complex situation with the help of many equations. Ability to solve the problems of this chapter depends only on the depth of your understanding of this chapter. Concepts of this chapters are used in solving questions based on motion in a straight line, relative motion, circular motion, train and boat etc. In CAT and other equivalent aptitude tests, each year 2 to 4 questions are generally asked. So this chapter is very important from the point of view of CAT and other equivalent aptitude tests.

MOTION OR MOVEMENT

When a body changes its position with respect to any external stationary body then it is said that the body is in motion or the body is moving with respect to the stationary body. Thus when a body travels from one place to another place, we say that the body is in motion or the body is moving. To move from a point A to another point B situated at a distance (D) from the point A with some speed (S) by a body takes some time (T).

Speed is defined as the rate at which distance is covered during the motion. It is measured in terms of distance per unit time. Unit of speed may have any combination of unit of distance and unit of time in the numerator and denominator respectively. For example unit of speed can be metre/sec, km/hour, metre/min., km/min., km/day, km/sec, feet/sec, miles/hr etc.

The relation between speed (S), distance (D) and time (T) is given below :

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$\text{or, Speed} \times \text{Time} = \text{Distance} \quad \text{i.e. } S \times T = D$$

In the above relation, the unit used for measuring the distance (D) covered during the motion and the unit of time (T) i.e. duration to cover the distance (D) will be the same as in numerator and denominator respectively of the unit used for the speed.

CONVERSION OF KMPH (KILOMETER PER HOUR) TO M/S (METRE PER SECOND) AND VICE-VERSA

$$1 \text{ kmph or } 1 \text{ km/h} = \frac{1 \text{ km}}{1 \text{ hr}} = \frac{1000 \text{ m}}{60 \times 60 \text{ sec}} = \frac{5 \text{ m}}{18 \text{ sec}} = \frac{5}{18} \text{ m/s}$$

$$\Rightarrow x \text{ kmph} = \frac{5x}{18} \text{ m/s} \quad \text{and vice-versa } x$$

$$\text{m/s} = \frac{18x}{5} \text{ kmph} \quad \text{or} \quad \frac{18x}{5} \text{ km/h}$$

i.e. to convert km/hr to m/sec, multiply by $\frac{5}{18}$ and to convert m/sec to km/hr multiply by $\frac{18}{5}$.

Illustration 1: Convert 90 km/h into m/s.

$$\text{Solution: } 90 \text{ km/h} = 90 \times \frac{5}{18} = 25 \text{ m/s.}$$

Illustration 2: The driver of a Maruti car driving at the speed of 68 km/h locates a bus 40 metres ahead of him. After 10 seconds, the bus is 60 metres behind. The speed of the bus is

- (a) 30 km/h (b) 32 km/h
(c) 25 km/h (d) 38 km/h

Solution: (b) Let speed of Bus = S_B km/h.

Now, in 10 sec., car covers the relative distance
= (60 + 40) m = 100 m

$$\therefore \text{Relative speed of car} = \frac{100}{10} = 10 \text{ m/s}$$

$$= 10 \times \frac{18}{5} = 36 \text{ km/h}$$

$$\therefore 68 - S_B = 36$$

$$\Rightarrow S = 32 \text{ km/h}$$

DIRECT AND INVERSE PROPORTIONALITY BETWEEN ANY TWO OF THE SPEED(S), TIME(T) AND DISTANCE(D) WHEN THE THIRD ONE IS CONSTANT

We know that $S \times T = D$

- (i) Time (T) is directly proportional to distance (D) when speed (S) is constant

$T \propto D$, when speed (S) is constant

$$\Rightarrow T = kD, \text{ where } k \text{ is the constant, called constant of proportionality}$$

Thus when a body covers d_1 and d_2 distances in time t_1 and t_2 respectively with constant (i.e. uniform) speed, then

$$t_1 = kd_1 \text{ and } t_2 = kd_2$$

$$\therefore \frac{t_1}{t_2} = \frac{kd_1}{kd_2} \Rightarrow \boxed{\frac{t_1}{t_2} = \frac{d_1}{d_2}}$$

- (ii) Speed (S) is directly proportional to distance (D) when time (T) is constant i.e.

$S \propto D$, when time T is constant

$S = mD$, where m is the constant, called constant, of proportionality

Thus, when a body covers d_1 and d_2 distances with speeds s_1 and s_2 respectively in constant time (i.e. duration), then

$$s_1 = md_1 \text{ and } s_2 = md_2$$

$$\therefore \frac{s_1}{s_2} = \frac{md_1}{md_2} \Rightarrow \boxed{\frac{s_1}{s_2} = \frac{d_1}{d_2}}$$

- (iii) Speed (S) is inversely proportional to time (T) when distance (D) is constant i.e.

$S \propto \frac{1}{T}$, when distance (D) is constant

$S = n \frac{1}{T}$, where n is the constant, called constant of proportionality.

$$\Rightarrow ST = n$$

Thus when a body covers a fixed distance with speeds s_1 and s_2 in time t_1 and t_2 respectively, then

$$s_1 t_1 = n \text{ and } s_2 t_2 = n$$

$$\therefore \boxed{s_1 t_1 = s_2 t_2}$$

Illustration 3: Two cyclists travel with a constant speed. One cyclist travels 30 kms in 2 hrs, then in how many hours the other cyclist travels 40 kms.

Solution: Since speed of the two cyclists is the same, hence

$$\frac{t_1}{t_2} = \frac{d_1}{d_2}$$

Here $t_1 = 2$ hrs, $d_1 = 30$ kms, $t_2 = ?$, $d_2 = 40$ kms

$$\therefore \frac{2}{t_2} = \frac{30}{40} \Rightarrow 3t_2 = 8 \Rightarrow t_2 = \frac{8}{3} = 2\frac{2}{3} \text{ hrs}$$

Hence required time = $2\frac{2}{3}$ hrs.

Illustration 4: A car travels from Delhi to Jaipur at a speed of 50 km/hr and another car travels from Delhi to Ludhiana at a speed 60 km/hr. If the time taken by both the cars is the same and the distance of Jaipur from Delhi is 270 kms, then find the distance of Ludhiana from Delhi.

Solution: Since the time taken by both the cars is the same, therefore

$$\frac{s_1}{s_2} = \frac{d_1}{d_2}$$

Here $s_1 = 50$, $d_1 = 270$, $s_2 = 60$, $d_2 = ?$

$$\therefore \frac{50}{60} = \frac{270}{d_2} \Rightarrow 5d_2 = 270 \times 6$$

$$\Rightarrow d_2 = \frac{270 \times 6}{5} = 324 \text{ kms.}$$

Hence distance of Ludhiana from Delhi = 324 kms.

Illustration 5: A cyclist goes to the post-office from his village at 12 km/h and reaches the post-office 15 minutes before 10 a.m. When he goes to the post-office from his village at 10 km/h, reaches the post-office 30 minutes after 10 a.m. Find the distance of the post-office from the village of the cyclist.

Solution: Since the distance of the post-office from the village of the cyclist is constant. Therefore

$$s_1 t_1 = s_2 t_2$$

Here $s_1 = 12$ km/h, $t_2 = t_1 + \frac{15}{60} + \frac{30}{60} = t_1 + \frac{3}{4}$

$s_2 = 10$ km/h, $t_1 = ?$

$$\therefore 12 \times t_1 = 10 \times \left(t_1 + \frac{3}{4} \right)$$

$$\Rightarrow 6t_1 = 5 \times \left(\frac{4t_1 + 3}{4} \right)$$

$$\Rightarrow 4t_1 = 15 \Rightarrow t_1 = \frac{15}{4} \text{ hrs}$$

Distance = Speed $\times$ Time

$$= 12 \times \frac{15}{4} = 45 \text{ kms}$$

Hence distance of the district centre = 45 kms.

Illustration 6: A man travels 120 km by ship, 450 km by rail and 60 km by horse taking altogether 13 hrs 30 min. The speed of the train is 3 times that of the horse and $1\frac{1}{2}$ times that of the ship. Find the speed of the train.

Solution: If the speed of the horse is x km/hr, that of the train is $3x$ and that of the ship is $\frac{3x}{1\frac{1}{2}} = 2x$ km/hr

$$\therefore \frac{120}{2x} + \frac{450}{3x} + \frac{60}{x} = \frac{27}{2}$$

$$\therefore \frac{60}{x} + \frac{150}{x} + \frac{60}{x} = \frac{27}{2}, \therefore \frac{270}{x} = \frac{27}{2}$$

$$\therefore x = 20 \therefore \text{Speed of the train} = 60 \text{ km/hr.}$$

AVERAGE SPEED

Average speed is defined as the ratio of total distance covered to the total time taken by an object i.e.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

If an object travels $d_1, d_2, d_3, \dots, d_n$ distances with different speeds $s_1, s_2, s_3, \dots, s_n$ in time $t_1, t_2, t_3, \dots, t_n$ respectively; then average speed (S_a) is given by

$$S_a = \frac{d_1 + d_2 + d_3 + \dots + d_n}{t_1 + t_2 + t_3 + \dots + t_n} \quad \dots (1)$$

Since, Distance = Speed $\times$ Time

$$\therefore d_1 = s_1 t_1, d_2 = s_2 t_2, d_3 = s_3 t_3, \dots, d_n = s_n t_n$$

Hence from (1),

$$S_a = \frac{s_1 t_1 + s_2 t_2 + s_3 t_3 + \dots + s_n t_n}{t_1 + t_2 + t_3 + \dots + t_n}$$

$$\text{Since Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\therefore t_1 = \frac{d_1}{s_1}, t_2 = \frac{d_2}{s_2}, t_3 = \frac{d_3}{s_3}, \dots, t_n = \frac{d_n}{s_n}$$

Hence from (1),

$$S_a = \frac{d_1 + d_2 + d_3 + \dots + d_n}{\frac{d_1}{s_1} + \frac{d_2}{s_2} + \frac{d_3}{s_3} + \dots + \frac{d_n}{s_n}}$$

Special Cases

In chapter of Averages, we studied that

(i) If with two different speeds s_1 and s_2 the same distance d is covered, then

$$\text{Average Speed} = \frac{2s_1 \cdot s_2}{s_1 + s_2}$$

(ii) If with three different speeds s_1, s_2 and s_3 the same distance d is covered, then

$$\text{Average Speed} = \frac{3s_1 \cdot s_2 \cdot s_3}{s_1 \cdot s_2 + s_2 \cdot s_3 + s_3 \cdot s_1}$$

Illustration 7: A car moves 300 km at a speed of 45 km/h and then it increases its speed to 60 km/h to travel another 500 km. Find average speed of car.

Solution:

$$\text{Average speed} = \frac{d_1 + d_2}{\frac{d_1}{s_1} + \frac{d_2}{s_2}} = \frac{300 + 500}{\frac{300}{45} + \frac{500}{60}} = \frac{800}{\frac{45}{3}} = \frac{160}{3}$$

$$= 53\frac{1}{3} \text{ km/h}$$

Illustration 8: A person goes to his office at $\frac{1}{3}$ rd of the speed at which he returns from his office. Average speed during the whole trip (i.e. one round) is 12 km/h. What is the speed of the person while he was going to his office?

Solution: Let $s_2 = 3k$, then $s_1 = k$

$$\text{Average speed} = \frac{2s_1 \cdot s_2}{s_1 + s_2}$$

$$\Rightarrow 12 = \frac{2 \times k \times 3k}{k + 3k} = \frac{3k}{2}$$

$$\therefore k = 8$$

Hence required speed = 8 km/h.

Illustration 9: A covers $\frac{1}{3}$ rd of the journey at the speed of 10 km/h and half of the rest at the speed of 20 km/h and rest at the speed of 30 km/h. What is the average speed of A?

Solution:

Distance covered at 10 km/h = $\frac{1}{3}$ rd of the whole journey

Distance covered at 20 km/h = $\left(\left(1 - \frac{1}{3}\right) \times \frac{1}{2} = \frac{1}{3}\right)$ rd of the whole journey

Distance covered at 30 km/h = $\left(1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}\right)$ rd of the whole journey

Since the distances covered with each of the three given speeds are the same, therefore

$$\begin{aligned} \text{Average speed} &= \frac{3s_1 \cdot s_2 \cdot s_3}{s_1 \cdot s_2 + s_2 \cdot s_3 + s_3 \cdot s_1} \\ &= \frac{3 \times 10 \times 20 \times 30}{10 \times 20 + 20 \times 30 + 30 \times 10} \\ &= 16\frac{4}{11} \text{ km/h.} \end{aligned}$$

Illustration 10: A man makes his upward journey at 16 km/h and downward journey at 28 km/h. What is his average speed?

(a) 32 km/h

(b) 56 km/h

(c) 20.36 km/h

(d) 22 km/h

Solution: (c) Let the distance travelled during both upward and down-ward journey be x km.

$$\begin{aligned} \text{Average speed} &= \frac{\text{Total distance covered}}{\text{Total time taken}} \\ &= \frac{x + x}{\frac{x}{16} + \frac{x}{28}} = \frac{2}{\frac{1}{16} + \frac{1}{28}} = \frac{2 \times 28 \times 16}{44} = 20.36 \text{ km/h} \end{aligned}$$

Illustration 11: On a journey across Bombay, a tourist bus averages 10 km/h for 20% of the distance, 30 km/h for 60% of it and 20 km/h for the remainder. The average speed for the whole journey was

(a) 10 km/h

(b) 30 km/h

(c) 5 km/h

(d) 20 km/h

Solution: (d) Let the average speed be x km/h. and total distance = y km. then,

$$\frac{0.2}{10}y + \frac{0.6}{30}y + \frac{0.2}{20}y = \frac{y}{x}$$

$$\Rightarrow x = \frac{1}{0.05} = 20 \text{ km/h}$$

RELATIVE SPEED

Generally, when we talk about the speed of a body, we mean the speed of the body with respect to a stationary point (or object), which we have already discussed. In many cases, we need to determine the speed of a body with respect to an independent moving point (or body). In such cases, we have to take into account the speed of the independent body with respect to which we want to find the speed of another body.

The speed of a body 'A' with respect to an independent moving body 'B' is called relative speed of the body A with respect to the body 'B'.

Formulae of Relative Speed

- If two bodies are moving in opposite directions at speeds s_1 and s_2 respectively, then relative speed of any one body with respect to other body is $(s_1 + s_2)$.
- If two bodies are moving in the same direction at speeds s_1 and s_2 respectively, then relative speed of any one body with respect to other body is given by
 $s_1 - s_2$, when s_1 is greater than s_2
 and $s_2 - s_1$ when s_2 is greater than s_1 .

Illustration 12: A car X starts from Delhi and another car Y starts from Moradabad at the same time to meet each other. Speed of car X is 40 km/h while speed of car Y is 50 km/h. If the distance between Delhi and Moradabad is 210 kms, when will they meet?

Solution: Effective speed = Relative speed = $40 + 50$
 $= 90 \text{ km/h}$

$$\text{Time taken} = \frac{210}{90} = 2\frac{1}{3} \text{ hrs.}$$

Illustration 13: A car X starts running from a place at a uniform speed of 40 km/h in a particular direction. After one and half hour, another car Y starts running in the same direction from the same place at a uniform speed and overtakes car X after 1 hour 36 minutes. Find the speed of car Y.

Solution: Distance covered by X in one and half hours

$$= 40 \times \frac{3}{2} \text{ km} = 60 \text{ kms.}$$

To overtake the car X by car Y, the distance of 60 kms will be covered by car Y with relative speed of $(S_y - 40)$ km/h in 1 hour 36 minutes, where S_y is the speed of car Y.

$$\text{Now } 1 \text{ hour } 36 \text{ minutes} = \left(1 + \frac{36}{60}\right) \text{ hrs} = \frac{3}{5} \text{ hrs}$$

$$\text{Now } \text{Speed} \times \text{Time} = \text{Distance}$$

$$\Rightarrow (S_y - 40) \times \frac{8}{5} = 60$$

$$\Rightarrow S_y = 77.5$$

Hence, required speed = 77.5 km/h.

Illustration 14: Two men A and B start from a place P walking at 3 km and $3\frac{1}{2}$ kmph respectively. How many km apart will they be at the end of $2\frac{1}{2}$ hours?

- If they walk in opposite directions?
- If they walk in the same direction?
- What time will they take to be 16 km apart if.
 (a) they walk in opposite directions?
 (b) in the same direction?

Solution:

- When they walk in opposite directions, they will be

$$\left(3 + 3\frac{1}{2}\right) = 6\frac{1}{2} \text{ km apart in 1 hour.}$$

$$\therefore \text{In } 2\frac{1}{2} \text{ hours they will be } 6\frac{1}{2} \times \frac{5}{2} = 16\frac{1}{4} \text{ km apart.}$$

- If they walk in the same direction, they will be

$$3\frac{1}{2} - 3 = \frac{1}{2} \text{ km apart in 1 hour.}$$

$$\Rightarrow \text{In } 2\frac{1}{2} \text{ hours they will be } \frac{1}{2} \times \frac{5}{2} = 1\frac{1}{4} \text{ km apart.}$$

- Time to be 16 km apart while walking in opposite directions = $\frac{16}{3 + 3\frac{1}{2}} = 2\frac{6}{13}$ hours.

But if they walk in the same direction, time

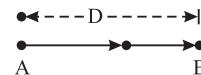
$$= \frac{16}{3\frac{1}{2} - 3} = 32 \text{ hours}$$

TO AND FRO MOTION IN A STRAIGHT LINE BETWEEN TWO POINTS A AND B

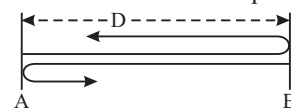
Two and fro motion in a straight line between two points A and B means motion of one or more bodies between two fixed points A and B such that when any body reached at any end point A or B, they start moving towards the opposite end point.

1. When two bodies start moving towards each other from two points A and B

- If distance between A and B is D , then the two bodies together have to cover D unit of distance for the first meeting.



- For the next number of meeting (i.e. second, third, fourth meeting and so on) both A and B together have to cover $2D$ distance more from the previous meeting.



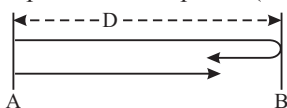
Hence to meet the fifth time they have to cover together $D + (4 \times 2D) = 9D$ unit of distances. Similarly for the

ninth meeting they have to cover together $D + (8 \times 2D) = 17D$ units of distance. Thus, for the n th meeting they have to cover together $D + (n - 1) \times 2D$ i.e. $(2n - 1)D$ units of distance.

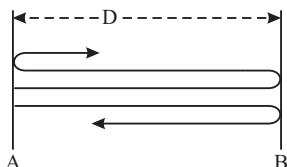
(c) At any point of time ratio of the distances covered by the two bodies will be equal to the ratio of their speeds.

2. When two bodies start moving towards the same direction from the point A

(a) Since the faster body reaches the next end (or opposite end) first than the slower body and the faster body starts returning before the slower body reaches the same opposite end and hence the two bodies meet somewhere between the two ends. For the first meeting after they start to move they have to cover $2D$ distance, where D is the distance between two particular end points (i.e. A and B)



(b) For every subsequent meeting they have to cover together $2D$ unit distance more from the previous meeting.



Thus, for the n th meeting they have to cover together $(n \times 2D)$ units of distance.

(c) At any point of time ratio of the distances covered by the two bodies will be equal to the ratio of their speeds.

Illustration 15: Two runners Shiva and Abhishek start running to and fro between opposite ends A and B of a straight road towards each other from A and B respectively. They meet first time at a point $0.75D$ from A , where D is the distance between A and B . Find the point of their 6th meeting.

Solution: At the time when Shiva and Abhishek meet first time,

$$\begin{aligned} \text{Ratio of their speeds} &= \text{Ratio of distance covered by them} \\ &= 0.75 : 0.25 \\ &= 3 : 1 \end{aligned}$$

Total distance covered by Shiva and Abhishek together till they meet at 6th time $= D + 5 \times 2D = 11D$

Total distance covered by Shiva till he meets Abhishek 6th time $= \frac{3}{3+1} \times 11D = 8.25D$

After covering a distance of $8.25D$, Shiva will be at a point at a distance of $0.25D$ from A or $0.75D$ from B .

UNIFORM ACCELERATION AND UNIFORM DECELERATION

Acceleration is the rate of increase of speed. If acceleration is constant (i.e. uniform), then acceleration is called uniform acceleration. Deceleration is the rate of decrease of speed. Deceleration is also called negative acceleration. If deceleration

is constant (i.e. uniform), then deceleration is called uniform deceleration.

For example, if speed of a car increases 2 km/h in each successive hour then the car has an uniform acceleration of 2 km/h^2 . If speed of the car decreases 5 km/h in each successive hour then the car has a uniform deceleration of 5 km/h^2 .

Unit of acceleration and deceleration are km/h^2 , m/sec^2 , etc.

i.e. the unit of acceleration is the ratio of unit of distance and square of the unit of time.

If a car is moving with the speed of 50 km/h having an acceleration 3 km/h^2 , then its speed after 4 hours will be

$$50 + 3 \times 4 = 62 \text{ km/h}$$

Here the speed 50 km/h is called initial speed and the speed 62 km/h is called final speed of the car. Thus

$$\text{Final Speed} = \text{Initial Speed} + \text{Acceleration} \times \text{Time}$$

$$\text{Also Final Speed} = \text{Initial Speed} - \text{Deceleration} \times \text{Time}$$

If on applying the brakes, a car running with a speed of 40 km/h stopped in 5 minutes, then the deceleration produced due to the brakes will be found out as

$$\text{Final Speed} = \text{Initial Speed} - \text{Deceleration} \times \text{Time}$$

$$\therefore 0 = 40 - \text{Deceleration} \times \frac{5}{60}$$

$$\Rightarrow \text{Deceleration} = 480 \text{ km/h}^2$$

Note that after applying the brakes, the car is finally stopped, hence we have taken the found speed of the car $= 0$

AN APPLICATION OF ALLIGATION IN THE PROBLEMS RELATED TO TIME, SPEED AND DISTANCE

Suppose a cyclist goes from P to Q at an average speed of S_1 and then comes back from Q to P at an average speed of S_2 . You have already studied the shortcut formula for the average speed of the whole journey when the distances covered by two different speeds are the same as given below :

$$\text{Average speed for the whole journey} = \frac{2S_1S_2}{S_1 + S_2}$$

However, we can find the average speed of the whole journey even if the distance travelled by two different speeds is unequal very easily by using the process of alligation which you have studied in the chapter of Alligation.

(i) To understand the process of alligation to find the average speed of the whole journey when distance travelled by two different speeds is equal. Suppose a bus travels from Delhi to Agra at a speed of 50 km/h and from Agra to Delhi at 75 km/h .

Here the distance travel with different speeds is equal, therefore average speed for the whole journey will be the weighted average of the two speeds (weighted on the basis of the time taken to travelled at each speed).

The speed ratio is $2 : 3$. The value of the time ratio used for calculating the weighted average is the inverse ratio of the ratio of speeds. Hence ratio of times is $3 : 2$.

$$\begin{array}{ccc} 50 & & 75 \\ & \searrow & \nearrow \\ & A_w & \\ & \nearrow & \searrow \\ 75 - A_w & : & A_w - 50 \\ 3 & : & 2 \\ \hline \frac{75 - A_w}{A_w - 50} = \frac{3}{2} & \Rightarrow & A_w = 60 \end{array}$$

Hence average speed of the whole journey is 60 km/h.

(ii) To understand the process of alligation to find the average speed of the whole journey when distances travelled by two different speeds are unequal. Suppose a bus travels from a city A to city B at a speed 30 km/h and city B to city C at a speed of 40 km/h. If distance between city A and B be 60 km and between B and C be 120 km.

Here the ratio of the speeds is 3 : 4.

Inverse of the ratio of speeds is 4 : 3.

Since distances covered with different speeds are not equal but they are in the ratio 1 : 2. Therefore we cannot take 4 : 3 as ratio of times but the product of 4 : 3 and 1 : 2 will be the ratio of times *i.e.*

$$\begin{aligned} \text{Ratio of times} &= (\text{Inverse of the ratio of speeds}) \times \\ &\quad (\text{Ratio of distances}) \\ &= (4 : 3) \times (1 : 2) \\ &= 4 \times 1 : 3 \times 2 = 4 : 6 = 2 : 3 \end{aligned}$$

CONCEPT RELATED TO MOTION OF TRAINS

The following things need to be kept in mind before solving questions on trains.

(i) When the train is crossing a moving object, the speed of the train has to be taken as the relative speed with respect to the object.

$$\begin{array}{c} \text{A} \quad \text{object} \quad \text{B} \\ \hline \text{P} \quad \quad \quad \text{Q} \quad \quad \quad \text{R} \quad \text{Train} \quad \text{S} \\ \text{The train just} \quad \quad \quad \text{The train has} \\ \text{start crossing} \quad \quad \quad \text{just crossed} \\ \text{the object} \quad \quad \quad \text{the object} \end{array}$$

$$\left(\text{Relative speed of the train with respect to the object} \right) \times \left(\text{Time taken by the train to cross the object} \right) = \left(\text{Distance travelled by the train} \right)$$

For object moving in opposite direction of the train,

$$\left(\text{Relative speed of the train with respect to the object} \right) = \left(\text{Speed of the train} \right) + \left(\text{Speed of the object} \right)$$

And for object moving in the same direction of the train,

$$\left(\text{Relative speed of the train with respect to the object} \right) = \left(\text{Speed of the train} \right) - \left(\text{Speed of the object} \right)$$

$$\begin{aligned} (\text{Distance travelled by the train when crossing the object}) &= \text{Distance travelled by the engine from } Q \text{ to } S \\ &= QR + RS \\ &= AB + RS \\ &= \text{Length of the object} + \text{Length of the train} \end{aligned}$$

In the case of a train crossing a man, tree or a pole, the length of the man, tree or pole is actually its diameter (or width) which is generally considered as negligible *i.e.* a man, a tree, a pole or a point etc. has no length.

The various situations of motion of the train in which the questions are asked in CAT and all other aptitude examinations and formulae used in various situations are given in the following table:

S. No.	Situations	Basic Formulae	Expanded Form of Basic Formulae	Expanded Formulae in Symbolic Form
1.	When a train crossing a moving object with length in opposite direction	Relative Speed $\times$ Time = Distance	$\left[\left(\text{Speed of the train} \right) + \left(\text{Speed of the object} \right) \right] \times \left(\text{Time taken by the train to cross the moving object} \right) = \left(\text{Length of the train} \right) + \left(\text{Length of the object} \right)$	$(S_T + S_0) \times t = (L_T + L_0)$
2.	When a train crossing a moving object with length in the same direction	Relative Speed $\times$ Time = Distance	$\left[\left(\text{Speed of the train} \right) - \left(\text{Speed of the object} \right) \right] \times \left(\text{Time taken by the train to cross the moving object} \right) = \left(\text{Length of the train} \right) + \left(\text{Length of the object} \right)$	$(S_T - S_0) \times t = (L_T + L_0)$
3.	When a train crossing a moving object without length like a man, a tree, a pole, a point etc. in opposite direction	Relative Speed $\times$ Time = Distance	$\left[\left(\text{Speed of the train} \right) + \left(\text{Speed of the object} \right) \right] \times \left(\text{Time taken by the train to cross the moving object} \right) = \left(\text{Length of the train} \right)$	$(S_T + S_0) \times t = L_T$
4.	When a train crossing a moving object without length in the same direction	Relative Speed $\times$ Time = Distance	$\left[\left(\text{Speed of the train} \right) - \left(\text{Speed of the object} \right) \right] \times \left(\text{Time taken by the train to cross the moving object} \right) = \left(\text{Length of the train} \right)$	$(S_T - S_0) \times t = L_T$
5.	When a train crossing a stationary object with length	Speed $\times$ Time = Distance	$\left(\text{Speed of the train} \right) \times \left(\text{Time taken to cross the stationary object} \right) = \left[\left(\text{Length of the train} \right) + \left(\text{Length of the object} \right) \right]$	$S_T \times t = L_T + L_0$

S. No.	Situations	Basic Formulae	Expend Form of Basic Formulae	Expend Formulae in Symbolic Form
6.	When a train crossing a stationary object without length	Speed $\times$ Time = Distance	$\left(\text{Speed of the train} \right) \times \left(\text{Time taken to cross the stationary object} \right) = \left(\text{Length of the train} \right)$	$S_T \times t = L_T$

S_T = Speed of the train, S_0 = Speed of the object, L_T = Length of the train, L_0 = Length of the object, t = time taken by the train to cross the object

Illustration 16: A train passes an electric pole in 10 seconds and a platform 120 m long in 18 seconds. Find the length of the train.

Solution: Let the length of the train be = x m

$$\text{Speed} = \frac{x}{10} = \frac{120 + x}{18} \Rightarrow x = 150$$

Hence length of the train = 150 m.

Illustration 17: A train of length 100 m takes 1/6 hour to pass over another train 150 m long coming from the opposite direction. If the speed of first train is 60 km/h, then find speed of the second train.

Solution: Let speed of the second train be x km/h.

Relative Speed = Sum of speed of two trains

$$= (60 + x) \text{ km/h} = (60 + x) \frac{5}{18} \text{ m/s}$$

$$\text{Time} = \frac{\text{Sum of length of two trains}}{\text{Relative Speed}}$$

$$\Rightarrow 10 = \frac{250 \times 18}{(60 + x) \times 5} \Rightarrow x = 30 \text{ km/h.}$$

Illustration 18: Two trains 137 metres and 163 metres in length are running towards each other on parallel lines, one at the rate of 42 kmph and another at 48 kmph. In what time will they be clear of each other from the moment they meet?

- (a) 10 sec (b) 12 sec
(c) 14 sec (d) cannot be determined

Solution: (b) Relative speed of the trains

$$= (42 + 48) \text{ kmph} = 90 \text{ kmph}$$

$$= \left(90 \times \frac{5}{18} \right) \text{ m/sec} = 25 \text{ m/sec.}$$

Time taken by the trains to pass each other

$$= \text{Time taken to cover } (137 + 163) \text{ m at } 25 \text{ m/sec}$$

$$= \left(\frac{300}{25} \right) \text{ sec} = 12 \text{ seconds.}$$

Illustration 19: A train 110 m in length travels at 60 km/h. How much time does the train take in passing a man walking at 6 km/h against the train?

- (a) 6 s (b) 12 s
(c) 10 s (d) 18 s

Solution: (a) Relative speeds of the train and the man

$$= (60 + 6) = 66 \text{ km/h} = \frac{66 \times 5}{18} \text{ m/s}$$

Distance = 110 m

Therefore, time taken in passing the men

$$= \frac{110 \times 18}{66 \times 5} = 6 \text{ s}$$

BOATS AND STREAMS

In still water, a boat moves with its own speed which is called speed of the boat in still water (S_B).

When the boat is moving against the flow of the water or with the flow of the water, the speed of movement of the boat depends on the speed of flow of water [*i.e.* speed of stream (S_S)].

Speed of the boat moving against the flow of water (*i.e.* moving in upstream)

$$\begin{aligned} &= \text{Speed of boat in still water} - \text{Speed of stream} \\ &= S_B - S_S \end{aligned}$$

Speed of the boat moving with flow of water (*i.e.* moving in downstream)

$$\begin{aligned} &= \text{Speed of boat in still water} + \text{Speed of stream} \\ &= S_B + S_S \end{aligned}$$

The basic formula used for solving the problems of boats and streams is

$$\text{Speed} \times \text{Time} = \text{Distance}$$

Illustration 20: A man can row a boat in downstream at 12 km/h and in upstream at 8 km/h. Find the speed of the boat that the man can row in still water.

Solution:

$$S_B + S_S = 12 \quad \dots (1)$$

$$S_B - S_S = 8 \quad \dots (2)$$

On adding (1) and (2), we get

$$2 S_B = 20 \Rightarrow S_B = 10$$

Hence speed of boat in still water = 10 km/h.

Illustration 21: A boat covers 48 km in upstream and 72 km in downstream in 12 hours, while it covers 72 km in upstream and 48 km in downstream in 13 hours. Find the speed of the stream.

$$\text{Solution: } \frac{48}{S_B - S_S} + \frac{72}{S_B + S_S} = 12$$

$$\frac{72}{S_B - S_S} + \frac{48}{S_B + S_S} = 13$$

$$\text{Let } \frac{1}{S_B - S_S} = x \text{ and } \frac{1}{S_B + S_S} = y$$

$$\text{Then } 48x + 72y = 12 \quad \dots (1)$$

$$\text{and } 72x + 48y = 13 \quad \dots (2)$$

$$\text{On adding (1) and (2),} \\ 120x + 120y = 25 \Rightarrow x + y = \frac{5}{24} \quad \dots (3)$$

$$\text{On subtracting (2) from (1),} \\ 24y - 24x = -1 \Rightarrow x - y = \frac{1}{24} \quad \dots (4)$$

$$\text{On adding (3) and (4),} \\ 2x = \frac{1}{4} \Rightarrow x = \frac{1}{8} \Rightarrow S_B - S_S = 8 \quad \dots (5)$$

$$\text{On subtracting (4) from (3),} \\ 2y = \frac{1}{6} \Rightarrow y = \frac{1}{12} \Rightarrow S_B + S_S = 12 \quad \dots (6)$$

$$\text{Subtracting (5) from (6),} \\ 2S_S = 4 \Rightarrow S_S = 2$$

Hence speed of stream = 2 km/h.

Illustration 22: A motor boat takes 12 hours to go downstream and it takes 24 hours to return the same distance. Find the ratio of the speed of boat in still water to the speed of stream.

Solution: Distance = Speed $\times$ Time

$$\text{Distance travelled in downstream} \\ = \text{Distance travelled in upstream}$$

$$(S_B + S_S) \times 12 = (S_B - S_S) \times 24 \\ \Rightarrow S_B + S_S = 2S_B - 2S_S \\ \Rightarrow 3S_S = S_B \Rightarrow \frac{S_B}{S_S} = \frac{3}{1} \Rightarrow S_B : S_S = 3 : 1$$

Hence required ratio = 3 : 1.

Illustration 23: P, Q, and R are the three towns on the bank of a river which flows uniformly. Q is equidistant from P and R. I row from P to Q and back in 10 hours and I can row from P to R in 4 hours. Compare the speed of my boat in still water with that of the river.

- (a) 4 : 3 (b) 5 : 3
(c) 6 : 5 (d) 7 : 3

Solution: (c) Let the speed of the boat be v_1 and the speed of the current be v_2 and d be the distance between the cities.

$$\text{Now, } \frac{d}{v_1 + v_2} = 4 \text{ and } \frac{d}{v_1 - v_2} = 6$$

$$\Rightarrow \frac{v_1 + v_2}{v_1 - v_2} = \frac{6}{4}$$

$$\text{or } \frac{2v_1}{2v_2} = \frac{10}{2} \text{ or } \frac{v_1}{v_2} = 5 : 1$$

$$\text{Required ratio} = (5 + 1) : 5 = 6 : 5$$

Illustration 24: Vikas can row a certain distance downstream in 6 hours and return the same distance in 9 hours. If the stream flows at the rate of 3 km/h, find the speed of Vikas in still water.

Solution:

By the formula,

$$\text{Vikas's speed in still water} = \frac{3(9 + 6)}{9 - 6} = 15 \text{ km/h}$$

BASIC TERMINOLOGY RELATED TO RACES

1. Startup or Head Start

When a runner allows to another runner to stay ahead in the same race, then it is said that there is a startup in the race.

For example if A allows B to go ahead before starting the race, then it is said that A gives startup to B and B has the startup. If before starting the race B goes ahead of x metre, then we can say A gives x metre startup to B or B has startup (or headstart) of x metre.

2. Dead Heat

When the runners reach the finishing line (or the final post) then it is said that these runners finish (or end) the race in dead heat.

Some Useful Concepts

- When it is said that A can give B a startup x metre in y metre race, then it means in y metre race B runs x metre less than A in the same time.
- When A beats B by t second in a race of y metre then it means B is the loser and A is the winner and when A reaches the finishing line, B is still some distance back to A , from which B takes t seconds to cover the remaining distance. Hence we can calculate the speed of loser B .
- The ratio of speed of the runners is always maintained throughout the race.

Illustration 25: In 2 km race A gives a startup of 300 m to B . Despite this, A wins the race by 400 m. Find the ratio of speed of A and B .

Solution: A and B covers 2000 m and 1300 m respectively in same time intervals.

Since time period for both runners A and B are the same, hence ratio of speeds of A and B = 2000 : 1300 = 20 : 13

Illustration 26: In a 2 km race A wins over B by 200 m or 20 seconds. B can give a startup 500 m to C in 2 km race. Find out by how much time A will win over C ?

Solution:

$$\text{Ratio of speeds of } A \text{ and } B = 2000 : 1800 = 10 : 9$$

$$\text{Ratio of speeds of } B \text{ and } C = 2000 : 1500 = 4 : 3$$

$$\text{Ratio of speeds of } A, B \text{ and } C = 2000 : 1800 : 1350$$

$$\text{Speed of } B = \frac{200}{20} = 10 \text{ m/s}$$

$$\therefore \text{Speed of } C = 7.5 \text{ m/s} [\because \text{Ratio of speed of } B \text{ to } C = 4 : 3]$$

Now C has to cover 650 m in extra time. Therefore, the time taken by C to cover the remaining distance = $\frac{650}{7.5} = 86.7$ seconds.

Hence, required time = 1 minute 27 second (approx)

CIRCULAR MOTION

When two bodies start moving from a place on a circular track simultaneously in the same direction, the faster body keeps increasing the distance by which the slower body is behind the faster body. When the distance by which the faster body is in front of the slower body becomes equal to the circumference of the track, the faster body meets the slower body first time i.e. faster body comes in line with the slower body.

- (i) When two bodies are moving in the opposite directions, their relative speed is equal to the sum of their individual speeds.
- (ii) When two bodies are moving in the same direction, their relative speed is equal to the difference of the speeds of the two bodies.

First Meeting

- (i) Let A and B are two runners.
Time taken by A and B to meet for the first time

$$= \frac{\text{Circumference of the circular Track}}{\text{Relative speed}}$$
- (ii) When there are more than two runners, suppose A is the fastest runner and A meets B first time in time t_{AB} , A meets C first time in time t_{AC} , A meets D first time in time t_{AD} and so on. Then time taken by all of them to meet for the first time is the LCM of t_{AB} , t_{AC} , t_{AD} , etc.

First Meeting at the Starting Point

Let A take, t_A time, B takes t_B time, C takes t_C times and so on, to complete one round, then the time taken to meet all the runners for the first time at the starting point

$$= \text{LCM of } t_A, t_B, t_C \text{ etc.}$$

Illustration 27: The jogging track in a sports complex is 726 metres in circumference. Pradeep and his wife start from the same point and walk in opposite directions at 4.5 km/h and 3.75 km/h, respectively. They will meet for the first time in

- (a) 5.5 min (b) 6.0 min
- (c) 5.28 min (d) 4.9 min

Solution: (c) Let the husband and the wife meet after x minutes 4500 metres are covered by Pradeep in 60 minutes.

In x minutes, he will cover $\frac{4500}{60}x$ metres.

Similarly,

In x minutes, his wife will cover $\frac{3750}{60}x$ m.

Now, $\frac{4500}{60}x + \frac{3750}{60}x = 726$

$\Rightarrow x = \frac{726 \times 60}{8250} = 5.28 \text{ min}$

Illustration 28: A , B and C start running on a circular track simultaneously from the same place of the circular track at the speed of 30 m/s, 60 m/s and 40 m/s respectively in the same direction. The circumference of the track is 1200 m.

- (i) When will they be together again for the first time?
- (ii) When will they be together again for the first time at the starting point?

Solution:

- (i) Speed of B is more than speed of A and C .
Relative speed of B with respect to $A = 60 - 30 = 30$ m/s
Relative speed of B with respect to $C = 60 - 40 = 20$ m/s

B meets A after every $\frac{1200}{30} = 40$ seconds

B meets C after every $\frac{1200}{20} = 60$ seconds

LCM of 40 and 60 = 120 seconds

Hence A , B and C will meet first time after 120 seconds i.e. 2 minutes.

- (ii) Time taken by A to complete one round on the track

$$= \frac{1200}{30} = 40 \text{ seconds}$$

Time taken by B to complete one round on the track

$$= \frac{1200}{60} = 20 \text{ seconds}$$

Time taken by C to complete one round on the track

$$= \frac{1200}{40} = 30 \text{ seconds}$$

LCM of 40, 20 and 30 = 120

Hence A , B and C will meet first time at the starting point after 120 seconds i.e. 2 minutes.

Illustration 29: A and B run on a circular track of circumference 800 m in the opposite direction. Speeds of A and B are 50 m/s and 30 m/s respectively. Initially A and B are diametrically opposite to each other.

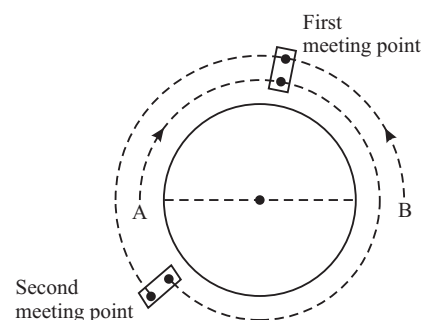
- (i) When will they meet for the first time?
- (ii) What is the ratio of distances covered by each one to meet for the first time?

Solution:

- (i) Relative speed of A with respect to $B = 50 + 30 = 80$ m/s
Initially A and B are diametrically opposite to each other means B is 400 m ahead of A in the race.

Time taken by A to meet B first time $= \frac{400}{80} = 5$ s

- (ii) To meet second time A and B have to cover 800 m



Hence time taken to meet second time $= \frac{800}{80} = 10$ seconds

CLOCKS

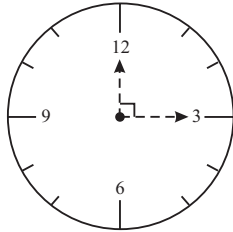
Problems on clocks are based on the movement of the minute hand and hour hand. We consider the dial of a clock as a circular track having a circumference of 60 km. minute hand and hour hand are two runners running with the speed of 60 km/h and 5 km/hr respectively in the same direction. Hence relative speed of minute hand with respect to hour hand is 55 km/h. This means that for every hour elapsed, the minute hand goes 55 km more than the hour hand.

Degree Concept of a Clock

Total angle subtended at the centre of a clock = 360°

Angle made by hour hand at the centre = 30° per hour
= 0.5° per minute

Angle made by minute hand at the centre = 360° per hour
= 6° per minute



Number of Right Angles and Straight Angles Formed by Minute Hand and Hour Hand

A right angle is formed by hour hand and minute hand when distance between tip of hour hand and tip of minute hand is 15 km. A straight line is formed by hour hand and minute hand when distance between their tips is 30 km.

A clock makes two right angles in every hour. Thus there are 2 right angles between marked 1 to 2, 2 to 3, 3 to 4 and so on the dial.

Two straight lines are formed by hour hand and minute hand in every hour.

Thus two straight lines are formed by hour hand and minute hand between marked 1 to 2, 2 to 3, 3 to 4 and so on.

(iii) Hour hand and minute hand of a clock are together after every $65\frac{5}{11}$ minutes. So, if hour hand and minute hand of a clock are meeting in less than $65\frac{5}{11}$ minutes, then the clock is running fast and if hour hand and minute hand are meeting in more than $65\frac{5}{11}$ minutes, then clock is running slow.

Illustration 30: Between 5 O' clock and 6 O' clock, when hour hand and minute hand of a clock overlap each other ?

Solution: At 5 O' clock, distance between tips of two hands = 25 km

Relative speed = 55 km/h

Required time to overlap the two hands

$$= \frac{25 \text{ km}}{55 \text{ km/h}} = \frac{5}{11} \text{ h}$$

$$= \frac{5 \times 60}{11} \text{ min}$$

$$= 27 \text{ min} + \frac{3 \times 60}{11} \text{ sec}$$

$$= 27 \text{ min} + 16 \text{ sec.}$$

$$= 27 \text{ minutes } 16 \text{ seconds.}$$

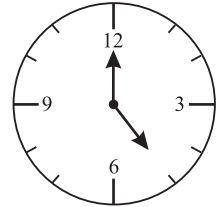


Illustration 31: Mrs. Veena Gupta goes for marketing between 5 P.M. and 6 P.M. When she comes back, she finds that the hour hand and the minute hand have interchanged their positions. For how much time was she out of her house ?

Solution: Since two hands interchange their positions, so sum of the angles subtended at the centre by hour hand and minute hand = 360°

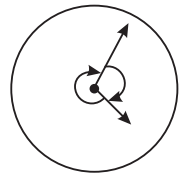
Let us suppose that she was out of house for 't' minutes.

So, the sum of the angles subtended at the centre by the hour hand and minute hand = $(0.5 \times t)^\circ + (6t)^\circ$

$$\therefore 0.5t + 6t = 360$$

$$\Rightarrow 6.5t = 360 \Rightarrow t = 55.4 \text{ (app.)}$$

Hence required time = 55.4 minutes.



Practice Exercises



Foundation Level

- A certain distance is covered by a train with a certain speed. If half the distance is covered in double time, then the ratio of this speed to that of the original one is
 (a) 1 : 4 (b) 4 : 1
 (c) 1 : 2 (d) 2 : 1
- A man makes his upward journey at 16 km/h and downward journey at 28 km/h. What is his average speed?
 (a) 32 km/h (b) 56 km/h
 (c) 20.36 km/h (d) 22 km/h
- Sound is said to travel in air at about 1100 feet per second. A man hears the axe striking the tree, $\frac{11}{5}$ seconds after he sees it strike the tree. How far is the man from the wood chopper?
 (a) 2197 ft (b) 2420 ft
 (c) 2500 ft (d) 2629 ft
- A salesman travels a distance of 50 km in 2 hours and 30 minutes. How much faster, in kilometres per hour, on an average, must he travel to make such a trip in $\frac{5}{6}$ hour less time?
 (a) 10 (b) 20
 (c) 30 (d) None of these
- Two persons A and B started from two different places towards each other. If the ratio of their speed be 3 : 5, then what is the ratio of distance covered by A and B respectively till the point of meeting?
 (a) 1 : 2 (b) 3 : 4
 (c) 3 : 5 (d) 5 : 3
- If a man travels at 30 km/h, he reaches his destination late by 10 minutes but if he travels at 42 km/h then he reaches 10 minutes earlier. The distance travelled by him is
 (a) 30 km (b) 35 km
 (c) 45 km (d) 36 km
- Two trains each of 120 m in length, run in opposite directions with a velocity of 40 m/s and 20 m/s respectively. How long will it take for the tail ends of the two trains to meet each other during the course of their journey?
 (a) 20 s (b) 3 s
 (c) 4 s (d) 5 s
- Two trains starting at the same time from two stations, 200 km apart and going in opposite directions, cross each other at a distance of 110 km from one of them. What is the ratio of their speeds?
 (a) 11 : 20 (b) 9 : 20
 (c) 11 : 9 (d) 19 : 20
- Two runner start running together for a certain distance, one at 8 km/h and another at 5 km/h. The former arrives one and half an hour, before the latter. The distance (in km) is:
 (a) 12 (b) 20
 (c) 25 (d) 36
- A can complete a journey in 10 hours. He travels first half of the journey at the rate of 21 km/hr and second half at the rate of 24 km/hr. Find the total journey in km.
 (a) 220 km (b) 224 km
 (c) 230 km (d) 234 km
- A train is moving at a speed of 132 km/h. If the length of the train is 110 metres, how long will it take to cross a railway platform, 165 metres long?
 (a) 5 s (b) 7.5 s
 (c) 10 s (d) 15 s
- A person travels equal distances with speeds of 3km/hr, 4 km/hr and 5km/hr and takes a total time of 47 minutes. The total distance (in km) is:
 (a) 2 (b) 3
 (c) 4 (d) 5
- A and B travel the same distance at 9 km/h and 10 km/h respectively. If A takes 20 minutes longer than B, the distance travelled by each is:
 (a) 16 (b) 20
 (c) 30 (d) None of these
- A passenger train takes two hours less for a journey of 300 km if its speed is increased by 5 km/h from its normal speed. The normal speed of the train is
 (a) 35 km/h (b) 50 km/h
 (c) 25 km/h (d) 30 km/h
- A gun is fired at a distance of 3.32 km from Chauhan. He hears its sound 10 seconds later. Find the speed of the sound.
 (a) 301 m/s (b) 302 m/s
 (c) 332 m/s (d) 340 m/s
- A walks around a circular field at the rate of one round per hour while B runs around it at the rate of six rounds per hour. They start in the same direction from the same point at 7.30 a.m. They shall first cross each other at:
 (a) 7.42 a.m. (b) 7.48 a.m.
 (c) 8.10 a.m. (d) 8.30 a.m.

17. A car driver travels from the plains to a hill station, which are 200 km apart at an average speed of 40 km/h. In the return trip he covers the same distance at an average speed of 20 km/h. The average speed of the car over the entire distance of 400 km is
(a) 16.56 km/h (b) 17.89 km/h
(c) 26.67 km/h (d) 35 km/h
18. Two trains of equal lengths are running on parallel tracks in the same direction at 46 km/h and 36 km/h, respectively. The faster train passes the slower train in 36 sec. The length of each train is
(a) 50 m (b) 80 m
(c) 72 m (d) 82 m
19. In a 800 m race around a stadium having the circumference of 200 m, the top runner meets the last runner on the 5th minute of the race. If the top runner runs at twice the speed of the last runner, what is the time taken by the top runner to finish the race?
(a) 20 min (b) 15 min
(c) 10 min (d) 5 min
20. Excluding stoppages, the speed of a train is 45 km/h and including stoppages, it is 36 km/h. For how many minutes does the train stop per hour?
(a) 10 min. (b) 12 min.
(c) 15 min. (d) 18 min.
21. The driving wheel of a locomotive engine, 2.1 m in radius, makes 75 revolutions in one minute. Find the speed of the train in km/h.
(a) 60 km/h (b) 59.4 km/h
(c) 61.5 km/h (d) None of these
22. A train covers 180 km distance in 4 hours. Another train covers the same distance in 1 hour less. What is the difference in the distances covered by these trains in one hour?
(a) 45 km (b) 9 km
(c) 40 km (d) None of these
23. Speed of a speed-boat when moving in the direction parallel to the direction of the current is 16 km/hr. Speed of the current is 3 km/hr. So the speed of the boat against the current will be (in km/hr)
(a) 22 (b) 9.5
(c) 10 (d) None of these
24. A plane left 30 minutes later than the scheduled time and in order to reach the destination 1500 km away in time, it had to increase the speed by 250 km/h from the usual speed. Find its usual speed.
(a) 720 km/h (b) 740 km/h
(c) 730 km/h (d) 750 km/h
25. Two trains are 2 km apart and their lengths are 200 m and 300 m. They are approaching towards each other with a speed of 20 m/s and 30 m/s, respectively. After how much time will they cross each other?
(a) 50 s (b) 100 s
(c) 25/3 s (d) 150 s
26. A train 300 m long is running at a speed of 90 km/hr. How many seconds will it take to cross a 200 m long train running in the opposite direction at a speed of 60 km/hr?
(a) $7\frac{1}{5}$ (b) 60
(c) 12 (d) 20
27. A boat travels upstream from B to A and downstream from A to B in 3 hours. If the speed of the boat in still water is 9 km/hr and the speed of the current is 3 km/hr, the distance between A and B is
(a) 4 km (b) 8 km
(c) 6 km (d) 12 km
28. A motor boat can travel at 10 km/h in still water. It traveled 91 km downstream in a river and then returned, taking altogether 20 hours. Find the rate of flow of the river.
(a) 6 km/hr (b) 5 km/hr
(c) 8 km/hr (d) 3 km/hr
29. Two men starting from the same place walk at the rate of 5 km/h and 5.5 km/h respectively. What time will they take to be 8.5 km apart, if they walk in the same direction?
(a) 16 h (b) 8 h 30 min
(c) 4h / 5min (d) 17 h
30. Speed of a boat in standing water is 9 km/h and the speed of the stream is 1.5 km/h. A man rows to a place at a distance of 105 km and comes back to the starting point. The total time taken by him is
(a) 20 h (b) 18 h
(c) 16 h (d) 24 h
31. An aeroplane travels distances 2500 km, 1200km and 500km at the rate of 500 km/hr, 400 km/hr, and 250 km/hr, respectively. The average speed is
(a) 420 km/hr (b) 405 km/hr
(c) 410 km/hr (d) 575 km/hr
32. There are 20 poles with a constant distance between each pole. A car takes 24 seconds to reach the 12th pole. How much time will it take to reach the last pole?
(a) 25.25 s (b) 17.45 s
(c) 35.75 s (d) 41.45 s
33. A man walks half of the journey at 4 km/h by cycle does one third of journey at 12 km/h and rides the remainder journey in a horse cart at 9 km/h, thus completing the whole journey in 6 hours and 12 minutes. The length of the journey is
(a) 36 km (b) $\frac{1332}{67}$ km
(c) 40 km (d) 28 km
34. A train covers 180 km distance in 4 hours. Another train covers the same distance in 1 hour less. What is the difference in the distances covered by these trains in one hour?
(a) 45 km (b) 9 km
(c) 40 km (d) None of these

35. The jogging track in a sports complex is 726 metres in circumference. Pradeep and his wife start from the same point and walk in opposite directions at 4.5 km/h and 3.75 km/h, respectively. They will meet for the first time in
(a) 5.5 min (b) 6.0 min
(c) 5.28 min (d) 4.9 min
36. A boat goes 24 km upstream and 28 km downstream in 6 hours. It goes 30 km upstream and 21 km downstream in 6 hours and 30 minutes. The speed of the boat in still water is :
(a) 10 km/h (b) 4 km/h
(c) 14 km/h (d) 6 km/h
37. Two trains for Mumbai leave Delhi at 6 a.m. and 6 : 45 am and travel at 100 kmph and 136 kmph respectively. How many kilometres from Delhi will the two trains be together
(a) 262.4 km (b) 260 km
(c) 283.33 km (d) 275 km
38. Two points A and B are located 48 km apart on the riverfront. A motorboat must go from A to B and return to A as soon as possible. The river flows at 6 km/h. What must be the least speed of the motorboat in still water for the trip from A to B and back again to be completed in not more than six hours (assume that the motorboat does not stop at B)?
(a) 18 km/h (b) 16 km/h
(c) 25 km/h (d) 46 km/h
39. A 200 m-long train passes a 350 m long platform in 5 s. If a man is walking at a speed of 4 m/s along the track and the train is 100 m away from him, how much time will it take to reach the man?
(a) Less than 1 s (b) 1.04 s
(c) More than 2s (d) Data insufficient
40. A clock gains 15 minutes per day. It is set right at 12 noon. What time will it show at 4.00 am, the next day?
(a) 4 : 10 am (b) 4 : 45 am
(c) 4 : 20 am (d) 5 : 00 am
41. During a journey of 80 km a train covers first 60 km with a speed of 40 km/h and completes the remaining distance with a speed of 20 km/h. What is the average speed of the train during the whole journey?
(a) 30 km/h (b) 32 km/h
(c) 36 km/h (d) 40 km/h
42. A travels from B to C, a distance of 250 miles, in 5.5 hours. He returns to B in 4 hours 40 minutes. His average speed is
(a) 44 (b) 46
(c) 48 (d) 50
43. A race course is 400 metres long. A and B run a race and A wins by 5 metres. B and C run over the same course and B wins by 4 metres. C and D run over it and D wins by 16 metres. If A and D run over it, then who would win and by how much ?
(a) A by 8.4 metres (b) D by 8.4 metres
(c) D by 7.3 metres (d) A by 7.3 metres
44. A circular running path is 726 metres in circumference. Two men start from the same point and walk in opposite directions at 3.75 km/h and 4.5 km/h, respectively. When will they meet for the first time ?
(a) After 5.5 min (b) After 6.0 min
(c) After 5.28 min (d) After 4.9 min
45. R and S start walking each other at 10 AM at the speeds of 3 km/hr and 4 km/hr respectively. They were initially 17.5 km apart. At what time do they meet?
(a) 2 : 30 PM (b) 11 : 30 AM
(c) 1 : 30 PM (d) 12 : 30 PM
46. A person travels from P to Q at a speed of 40 kmph and returns by increasing his speed by 50%. What is his average speed for both the trips?
(a) 36 kmph (b) 45 kmph
(c) 48 kmph (d) 50 kmph
47. A car travels first half distance between two places with a speed of 40 km/h and the rest of the half distance with a speed of 60 km/h. The average speed of the car is
(a) 48 km/h (b) 37 km/h
(c) 44 km/h (d) None of these
48. Two cyclists start on a circular track from a given point but in opposite directions with speeds of 7 m/sec and 8 m/sec respectively. If the circumference of the circle is 300 metres, after what time will they meet at the starting point ?
(a) 100 sec (b) 20 sec
(c) 300 sec (d) 200 sec
49. If a train runs at 40 kmph, it reaches its destination late by 11 minutes but if it runs at 50 kmph, it is late by 5 minutes only. The correct time for the train to complete its journey is:
(a) 13 min. (b) 15 min.
(c) 19 min. (d) 21 min.
50. A man while returning from his factory, travels $\frac{2}{3}$ of the distance by bus and $\frac{3}{4}$ of the rest by car, and the remaining by foot. If he travels 2 km on foot, find the distance covered by him.
(a) 24 km (b) 22 km
(c) 28 km (d) 26 km
51. A car driver, driving in a fog, passes a pedestrian who was walking at the rate of 2 km/hr in the same direction. The pedestrian could see the car for 6 minutes and it was visible to him up to a distance of 0.6 km. What was the speed of the car?
(a) 15 km/hr (b) 30 km/hr
(c) 20 km/hr (d) 8 km/hr
52. A plane left 30 min later than its scheduled time to reach its destination 1500 km away. In order to reach in time it increases its speed by 250 km/h. What is its original speed?
(a) 1000 km/h (b) 750 km/h
(c) 600 km/h (d) 800 km/h

53. Bombay Express left Delhi for Bombay at 14.30 hrs, travelling at a speed of 60 kmph and Rajdhani Express left Delhi for Bombay on the same day at 16.30 hrs, travelling at a speed of 80 kmph. How far away from Delhi will the two trains meet?
 (a) 120 km (b) 360 km
 (c) 480 km (d) 500 km
54. A person can swim at a speed of 9 km per hour in still water. If the speed of the stream is 6 km per hour, then how long does he take to swim up to a distance of 9 km and return at the starting point?
 (a) 2 hours (b) $2\frac{1}{2}$ hours
 (c) $3\frac{3}{5}$ hours (d) $3\frac{3}{4}$ hours
55. A thief goes away with a Maruti car at a speed of 40 km/h. The theft has been discovered after half an hour and the owner sets off in another car at 50 km/h. When will the owner overtake the thief from the start.
 (a) $2\frac{1}{2}$ hours (b) 2 hr 20 min
 (c) 1 hr 45 min (d) cannot be determined
56. A man is walking at a speed of 10 km per hour. After every kilometre, he takes rest for 5 minutes. How much time will he take to cover a distance of 5 kilometres?
 (a) 48 min. (b) 50 min.
 (c) 45 min. (d) 55 min.
57. One-fourth of a certain journey is covered at the rate of 25 km/h, one-third at the rate of 30 km/h and the rest at 50 km/h. Find the average speed for the whole journey.
 (a) 600/53 km/h (b) 1200/53 km/h
 (c) 1800/53 km/h (d) 1600/53 km/h
58. A railway passenger counts the telegraph poles on the rail road as he passes them. The telegraph poles are at a distance of 50 meters. What will be his count in 4 hours if the speed of the train is 45 km per hour?
 (a) 2500 (b) 600
 (c) 3600 (d) 5000
59. A long distance runner runs 9 laps of a 400 metres track everyday. His timings (in minutes) for four consecutive days are 88, 96, 89 and 87 respectively. On an average, how many metres/minute does the runner cover ?
 (a) 40 m/min (b) 45 m/min
 (c) 38 m/min (d) 49 m/min
60. A dog starts chasing a cat 2 hours later. It takes 2 hours to dog to catch the cat. If the speed of the dog is 30 km/h, what is the speed of cat?
 (a) 10 km/h (b) 15 km/h
 (c) 20 km/h (d) Can't be determined

Standard Level

1. A and B can run 200 m in 22 seconds and 25 seconds, respectively. How far is B from the finishing line when A reaches in ?
(a) 8 m (b) 12 m
(c) 16 m (d) 24 m
2. If a man walks at the rate of 5 kmph, he misses a train by 7 minutes. However, if he walks at the rate of 6 kmph, he reaches the station 5 minutes before the arrival of the train. Find the distance covered by him to reach the station.
(a) 4 km (b) 6 km
(c) 5 km (d) 7 km
3. The speed of a car increases by 2 kms after every one hour. If the distance travelled in the first one hour was 35 kms, what was the total distance travelled in 12 hours?
(a) 456 kms (b) 482 kms
(c) 552 kms (d) None of these
4. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the speed of the car is
(a) 4 : 3 (b) 3 : 4
(c) 3 : 2 (d) 2 : 3
5. Wheels of diameters 7 cm and 14 cm start rolling simultaneously from X and Y which are 1980 cm apart towards each other in opposite directions. Both of them make the same number of revolutions per second. If both of them meet after 10 seconds, the speed of the smaller wheel is
(a) 22 cm/s (b) 44 cm/s
(c) 66 cm/s (d) 132 cm/s
6. A person has to cover a distance of 6 km in 45 minutes, If he covers one-half of the distance in two-thirds of the total time; to cover the remaining distance in the remaining time, his speed (in km/hr) must be:
(a) 6 (b) 8
(c) 12 (d) 15
7. An aeroplane first flew with a speed of 440 km/h and covered a certain distance. It still had to cover 770 km less than what it had already covered, but it flew with a speed of 660 km/h. The average speed for the entire flight was 500 km/h. Find the total distance covered.
(a) 3250 km (b) 2750 km
(c) 4400 km (d) 1375 km
8. A car travels the first one-third of a certain distance with a speed of 10 km/hr, the next one-third distance with a speed of 20 km/hr, and the last one-third distance with a speed of 60 km/hr. The average speed of the car for the whole journey is
(a) 18 km/hr (b) 24 km/hr
(c) 30 km/hr (d) 36 km/hr
9. A train starts from Delhi at 6 : 00 AM and reaches Ambala Cantt at 10 AM. The other train starts from Ambala Cantt at 8 AM and reaches Delhi at 11:30 PM. If the distance between Delhi and Ambala Cantt. is 200 km, then at what time did the two trains meet each other ?
(a) 8 : 56 AM (b) 8 : 46 AM
(c) 7 : 56 AM (d) 8 : 30 AM
10. Rahul can row a certain distance downstream in 6 hours and return the same distance in 9 hours. If the speed of Rahul in still water is 12 km/hr, find the speed of the stream.
(a) 2 km/hr (b) 2.4 km/hr
(c) 3 km/hr (d) Data inadequate
11. A man can row 4.5 km/hr in still water and he finds that it takes him twice as long to row up as to row down the river. Find the rate of the stream.
(a) 1.5 km/hr (b) 2 km/hr
(c) 2.5 km/hr (d) 1.75 km/hr
12. A man sitting in a train travelling at the rate of 50 km/hr observes that it takes 9 sec for a goods train travelling in the opposite direction to pass him. If the goods train is 187.5 m long, find its speed.
(a) 40 km/hr (b) 25 km/hr
(c) 35 km/hr (d) 36 km/hr
13. Two trains are moving in opposite directions at speeds of 60 km/hour and 90 km/hour. Their lengths are 1.10 km and 0.9 km respectively. The time taken by the slower train to cross the faster train in seconds is
(a) 36 (b) 49
(c) 45 (d) 48
14. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the car is
(a) 2 : 3 (b) 3 : 2
(c) 3 : 4 (d) 4 : 3
15. The distance between two cities A and B is 330 km. A train starts from A at 8 a.m. and travels towards B at 60 km/hr. Another train starts from B at 9 a.m. and travels towards A at 75 km/hr. At what time do they meet?
(a) 10 a.m. (b) 10.30 a.m.
(c) 11 a.m. (d) 11.30 a.m.

16. A and B run a 5 km race on a round course of 400 m. If their speeds be in the ratio 5 : 4, how often does the winner pass the other?
- (a) $4\frac{1}{2}$ times (b) $2\frac{3}{4}$ times
- (c) $3\frac{1}{2}$ times (d) $2\frac{1}{2}$ times
17. A motorcyclist covered two thirds of a total journey at his usual speed. He covered the remaining distance at three fourth of his usual speed. As a result, he arrived 30 minutes later than the time he would have taken at usual speed. If the total journey was 180 km, the what is his usual speed?
- (a) 40 kmph (b) 36 kmph
- (c) 30 kmph (d) 32 kmph
18. A man can row a certain distance against the stream in six hours. However, he would take two hours less to cover the same distance with the current. If the speed of the current is 2 kmph, then what is the rowing speed in still water?
- (a) 10 kmph (b) 12 kmph
- (c) 14 kmph (d) 8 kmph
19. If I walk at 4 km/h, I miss the bus by 10 minutes. If I walk at 5 km/h, I reach 5 minutes before the arrival of the bus. How far I walk to reach the bus stand ?
- (a) 5 km (b) 4.5 km
- (c) $5\frac{1}{4}$ km / h (d) Cannot be determined
20. A man covers a certain distance on a toy train. If the train moved 4 km/h faster, it would take 30 minutes less. If it moved 2 km/h slower, it would have taken 20 minutes more. Find the distance.
- (a) 60 km (b) 58 km
- (c) 55 km (d) 50 km
21. An aeroplane flies along the four sides of a square at the speeds of 200, 400, 600 and 800 km/h. Find the average speed of the plane around the field.
- (a) 384 km/h (b) 370 km/h
- (c) 368 km/h (d) None of these
22. A thief steals a car at 2 : 30 p.m. and drives it at 60 kmph. The theft is discovered at 3 p.m. and the owner sets off in another car at 75 kmph. When will he overtake the thief ?
- (a) 4 : 30 p.m. (b) 4 : 45 p.m.
- (c) 5 p.m. (d) 5 : 15 p.m.
23. Points A and B are 70 km apart on a highway. One car starts from A and the another one from B at the same time. If they travel in the same direction, they meet in 7 hours. But if they travel towards each other, they meet in one hour. The speeds of the two cars are, respectively.
- (a) 45 and 25 km/h (b) 70 and 10 km/h
- (c) 40 and 30 km/h (d) 60 and 40 km/h
24. A river 3 m deep and 40 m wide is flowing at the rate of 2 km per hour. How much water (in litres) will fall into the sea in a minute?
- (a) 4,00,000 (b) 40,00,000
- (c) 40,000 (d) 4,000
25. Vinay fires two bullets from the same place at an interval of 12 minutes but Raju sitting in a train approaching the place hears the second report 11 minutes 30 seconds after the first. What is the approximate speed of train (if sound travels at the speed of 330 metre per second)?
- (a) 660/23 m/s (b) 220/7 m/s
- (c) 330/23 m/s (d) 110/23 m/s
26. A dog sees a cat. It estimates that the cat is 25 leaps away. The cat sees the dog and starts running with the dog in hot pursuit. If in every minute, the dog makes 5 leaps and the cat makes 6 leaps and one leap of the dog is equal to 2 leaps of the cat. Find the time in which the cat is caught by the dog (assume an open field with no trees)
- (a) 12 minutes (b) 15 minutes
- (c) 12.5 minutes (d) None of these
27. A train of 300 m is travelling with the speed of 45 km/h when it passes point A completely. At the same time, a motorbike starts from point A with the speed of 70 km/h. When it exactly reaches the middle point of the train, the train increases its speed to 60 km/h and motorbike reduces its speed to 65 km/h. How much distance will the motorbike travel while passing the train completely?
- (a) 2.52 km (b) 2.37 km
- (c) 2 km (d) None of these
28. A group of soldiers are marching with a speed of 5 m/s. The distance between the first and the last row of soldiers is 100 m. A dog starts running from the last row and moves towards the first row, turns and comes back to the last row. If the dog has travelled 400 m, the speed of the dog is
- (a) $5\sqrt{2}$ m/s (b) $3\sqrt{5}$ m/s
- (c) $6\sqrt{5}$ m/s (d) $6\sqrt{2}$ m/s
29. Ram runs $\frac{7}{4}$ times as fast as Sham, If Ram gives Sham a start of 300 m, how far must the winning post be if both Ram and Sham have to end the race at the same time?
- (a) 1400 m (b) 700 m
- (c) 350 m (d) 210 m

30. A watch, which gains time uniformly, was 5 minutes behind the correct time when it showed 11:55 AM on Monday. It was 10 minutes ahead of the correct time when it showed 06:10 PM on the next day. When did the watch show the correct time?
- (a) 6 AM, Tuesday (b) 6 PM, Monday
(c) 2 PM, Tuesday (d) 10 PM, Monday
31. Pankaj went to the post-office at the speed of 60 km/hr while returning for his home he covered the half of the distance at the speed of 10 km/hr, but suddenly he realized that he was getting late so he increased the speed and reached the home by covering rest half of the distance at the speed of 30 km/hr. The average speed of the Pankaj in the whole length of journey is:
- (a) 5.67 km/hr (b) 24 km/hr
(c) 22.88 km/hr (d) 5.45 km/hr
32. With an average speed of 40 km/h, a train reaches its destination in time. If it goes with an average speed of 35 km/h, it is late by 15 minutes. The length of the total journey is:
- (a) 40 km (b) 70 km
(c) 30 km (d) 80 km
33. A student rides on a bicycle at 8 km/h and reaches his school 2.5 minutes late. The next day he increases his speed to 10 km/h and reaches the school 5 minutes early. How far is the school from his house?
- (a) 1.25 km (b) 8 km
(c) 5 km (d) 10 km
34. Two rockets approach each other, one at 42000 mph and the other at 18000 mph. They start 3256 miles apart. How far are they apart (in miles) 1 minute before impact ?
- (a) 1628 (b) 1000
(c) 826 (d) 1200
35. Two guns were fired from the same place at an interval of 10 minutes and 30 seconds, but a person in the train approaching the place hears the second shot 10 minutes after the first. The speed of the train (in km/hr), supposing that speed travels at 330 metres per second, is
- (a) 19.8 (b) 58.6
(c) 59.4 (d) 111.80
36. Train A running at 60 km/h leaves Mumbai for Delhi at 6 p.m. Train B running at 90 km/h also leaves for Delhi at 9 p.m. Train C leaves Delhi for Mumbai at 9 p.m. If all the three trains meet at the same time between Mumbai and Delhi, then what is the speed of train C, if distance between Delhi and Mumbai is 1260 km ?
- (a) 60 km/h (b) 90 km/h
(c) 120 km/h (d) 135 km/h
37. A boat, while going downstream in a river covered a distance of 50 mile at an average speed of 60 miles per hour. While returning, because of the water resistance, it took one hour fifteen minutes to cover the same distance . What was the average speed of the boat during the whole journey?
- (a) 40 mph (b) 48 mph
(c) 50 mph (d) 55 mph
38. A man takes 5 hour 45 min. in walking to a certian place and riding back. He would have gained 2 hours by riding both ways. The time he would take to walk both ways, is
- (a) 3 hrs 45 min (b) 7 hrs 30 min
(c) 7 hrs 45 min (d) 11 hrs 45 min
39. A boatman rows to a place 45 km distant and back in 20 hours. He finds that he can row 12 km with the stream in same time as 4 km against the stream . Find the speed of the stream.
- (a) 3 km/hr (b) 2.5 km/hr
(c) 4 km/hr (d) Cannot be determined
40. A man goes 15 metres due west and then 8 metres due north. How far is he from the starting point?
- (a) 19 metres (b) 16 metres
(c) 17 metres (d) 15 metres
41. Two trains, 130 m and 110 m long, are going in the same direction. The faster train takes one minute to pass the other completely. If they are moving in opposite directions, they pass each other completely in 3 seconds. Find the speed of each train.
- (a) 38 m/sec, 36 m/sec (b) 42 m/sec, 38 m/sec
(c) 36 m/sec, 42 m/sec (d) None of these
42. A man who can swim 48 m/min in still water swims 200 m against the current and 200 m with the current. If the difference between those two times is 10 minutes, find the speed of the current.
- (a) 30 m/min (b) 29 m/min
(c) 31 m/min (d) 32 m/min
43. A train after travelling 150 km meets with an accident and then proceeds with $\frac{3}{5}$ of its former speed and arrives at its destination 8 h late. Had the accident occurred 360 km further, it would have reached the destination 4 h late. What is the total distance travelled by the train?
- (a) 840 km (b) 960 km
(c) 870 km (d) 1100 km
44. A man who can swim 48 m/min in still water swims 200 m against the current and 200 m with the current. If the difference between those two times is 10 min, what is the speed of the current?
- (a) 30 m/min (b) 31 m/min
(c) 29 m/min (d) 32 m/min

45. A man walks a certain distance and rides back in $6\frac{1}{4}$ h. He can walk both ways in $7\frac{3}{4}$ h. How long it would take to ride both ways ?
 (a) 5 hours (b) $4\frac{1}{2}$ hours
 (c) $4\frac{3}{4}$ hours (d) 6 hours
46. An accurate clock shows 8 o'clock in the morning. Through how many degrees will the hour hand rotate when the clock shows 2 o'clock in the afternoon?
 (a) 144° (b) 150°
 (c) 168° (d) 180°
47. Shyam's house, his office and his gym are all equidistant from each other. The distance between any 2 of them is 4 km. Shyam starts walking from his gym in a direction parallel to the road connecting his office and his house and stops when he reaches a point directly east of his office. He then reverses direction and walks till he reaches a point directly south of his office. The total distance walked by Shyam is
 (a) 6 km (b) 9 km
 (c) 16 km (d) 12 km
48. A dog after travelling 50 km meets a swami who counsels him to go slower. He then proceeds at $\frac{3}{4}$ of his former speed and arrives at his destination 35 minutes late. Had the meeting occurred 24 km further the dog would have reached its destination 25 minutes late. The speed of the dog is
 (a) 48 km/h (b) 36 km/h
 (c) 54 km/h (d) 58 km/h
49. Ramesh and Somesh are competing in a 100 m race. Initially, Ramesh runs at twice the speed of Somesh for the first fifty m. After the 50 m mark, Ramesh runs at $\frac{1}{4}$ th his initial speed while Somesh continues to run at his original speed. If Somesh catches up with Ramesh at a distance of 'N' m from the finish line, then N is equal to
 (a) 35 (b) 10
 (c) 45 (d) None of these
50. A, B, and C are three participants in a kilometer race. If A can give B a start of 40 metres and B can give C a start of 25 metres, how many metres of a start can A give to C?
 (a) 60 m (b) 64 m
 (c) 62 m (d) 66 m
51. A monkey ascends a greased pole 12 metres high. He ascends 2 metres in first minute and slips down 1 metre in the alternate minute. In which minute, he reaches the top ?
 (a) 21st (b) 22nd
 (c) 23rd (d) 24th
52. Mallah can row 40 km upstream and 55 km downstream in 13 h and 30 km upstream and 44 km downstream in 10 hours. What is the speed of Mallah in still water?
 (a) 6 km/h (b) 12 km/h
 (c) 3 km/h (d) 8 km/h
53. A passenger sitting in a train of length 100 m, which is running with speed of 60 km/h passing through two bridges, notices that he crosses the first bridge and the second bridge in time intervals which are in the ratio of 7 : 4 respectively. If the length of first bridge be 280 m, then the length of second bridge is:
 (a) 490 m (b) 220 m
 (c) 160 m (d) Can't be determined
54. A man can cross a downstream river by steamer in 40 minutes and same by boat in 1 hour. If the time of crossing the river in upstream direction by steamer is 50% more than downstream time by the steamer and the time required by boat to cross the same river by boat in upstream is 50% more than the time required in downstream by boat. What is the time taken for the man to cross the river downstream by steamer and then return to same place by boat half the way and by steamer the rest of the way?
 (a) 85 min (b) 115 min
 (c) 120 min (d) 125 min
55. A tiger is 50 of its own leaps behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 m and 5 m per leap respectively, what distance will the tiger have to run before it catches the deer?
 (a) 600 m (b) 700 m
 (c) 800 m (d) 1000 m
56. A candle of 6 cm long burns at the rate of 5 cm in 5 h and another candle of 8 cm long burns at the rate of 6 cm in 4h. What is the time required by each candle to remain of equal lengths after burning for some hours, when they start to burn simultaneously with uniform rate of burning?
 (a) 1 h (b) 1.5 h
 (c) 2 h (d) None of these
57. Two persons start from the opposite ends of a 90 km straight track and run to and fro between the two ends. The speed of first person is 30 m/s and the speed of other is $125/6$ m/s. They continue their motion for 10 hours. How many times they pass each other?
 (a) 10 (b) 9
 (c) 12 (d) None of these
58. At what time after 3:10 am, the acute angle made by the minute and hour-hand is double to that of a 3:10 am, for the first time?
 (a) 4 h 43 min (b) 3 h 48 min
 (c) $3\text{h}\frac{320}{11}\text{min}$ (d) None of these

59. A swiss watch is being shown in a museum which has a very peculiar property. It gains as much in the day as it loses during night between 8 pm to 8 am. In a week how many times will the clock show the correct time?
 (a) 6 times (b) 14 times
 (c) 7 times (d) 8 times
60. The metro service has a train going from Mumbai to Pune and Pune to Mumbai every hour, the first one at 6 a.m. The trip from one city to other takes $4\frac{1}{2}$ hours, and all trains travel at the same speed. How many trains will you pass while going from Mumbai to Pune if you start at 12 noon?
 (a) 8 (b) 10
 (c) 9 (d) 13
61. A wall clock gains 2 minutes in 12 hours, while a table clock loses 2 minutes in 36 hours; both are set right at noon on Tuesday. The correct time when they both show the same time next would be
 (a) 12 : 30 night (b) 12 noon
 (c) 1 : 30 night (d) 12 night
62. Two ants start simultaneously from two ant holes towards each other. The first ant covers 8% of the distance between the two ant holes in 3 hours, the second ant covered $\frac{7}{120}$ of the distance in 2 hours 30 minutes. Find the speed (feet/h) of the second ant if the first ant travelled 800 feet to the meeting point.
 (a) 15 feet/h (b) 25 feet/h
 (c) 45 feet/h (d) 35 feet/h
63. A watch loses $\frac{2}{3}\%$ time during the 1st week and gains $\frac{1}{3}\%$ time during the next week. If on a Sunday noon, it showed the right time, what time will it show at noon on the Saturday after the next.
 (a) 11 : 26 : 24 a.m. (b) 10 : 52 : 18 a.m.
 (c) 10 : 52 : 48 a.m. (d) 11 : 36 : 24 a.m.

Expert Level

1. My Scooty gives an average of 40 kmpl of petrol. But after recent filling at the new petrol pump, its average dropped to 38 kmpl. I investigated and found out that it was due to adulterated petrol. Petrol pumps add kerosene, which is $\frac{2}{3}$ cheaper than petrol, to increase their profits. Kerosene generates excessive smoke and knocking and gives an average of 18 km per 900 ml. If I paid Rs. 30 for a litre of petrol, what was the additional amount the pump-owner was making ?
 (a) ` 1.75 (b) ` 1.80
 (c) ` 2.30 (d) ` 2
2. I have to reach a certain place at a certain time and I find that I shall be 15 min too late, if I walk at 4 km an hour, and 10 min too soon, if I walk at 6 km an hour. How far have I to walk?
 (a) 25 km (b) 5 km
 (c) 10 km (d) None of these
3. On a journey across Bombay, a tourist bus averages 10 km/h for 20% of the distance, 30 km/h for 60% of it and 20 km/h for the remainder. The average speed for the whole journey was
 (a) 10 km/h (b) 30 km/h
 (c) 5 km/h (d) 20 km/h
4. The average speed of a train in the onward journey is 25% more than that in the return journey. The train halts for one hour on reaching the destination. The total time taken for the complete to and fro journey is 17 hours, covering a distance of 800 km. The speed of the train in the onward journey is:
 (a) 45 km/hr (b) 47.5 km/hr
 (c) 52 km/hr (d) 56.25 km/hr
5. Pankaj walked at 5 km/h for certain part of the journey and then he took an auto for the remaining part of the journey travelling at 25 km/h. If he took 10 hours for the entire journey. What part of journey did he travelled by auto if the average speed of the entire journey be 17 km/h:
 (a) 750 km (b) 100 km
 (c) 150 km (d) 200 km
6. Train X starts from point A for point B at the same time that train Y starts from B to A. Point A and B are 300 km apart. The trains are moving at a constant speed atleast at 25 km/h. The trains meet each other 3 hours after they start. If the faster train takes atleast 2 more hours to reach the destination. By which time will the slower train have definitely reached its destination? (Ignoring the length of trains in crossing).
 (a) 4 hours after the start
 (b) 7.5 hours after the start
 (c) 6 hours after the start
 (d) None of the above
7. A boat takes 7 hours to go from P to R, through a midpoint Q, but it takes 8 hours to go from P to Q, and then return from Q to P. How long it would take to go from R to P?
 (a) 7 h (b) 8 h
 (c) 9 h (d) None of these
8. A beats B by 100 m in a race of 1200 m and B beats C by 200 m in a race of 1600 m. Approximately by how many metres can A beat C in a race of 9600 m?
 (a) 1600 m (b) 1800 m
 (c) 1900 m (d) 2400 m
9. A gives both B and C a start of 60 m in a 1500 m race. However, while B finishes with him, C is 15 m behind them when A and B cross the finishing line. How much start can B give C for the 1500 m race course?
 (a) $7\frac{6}{23}$ m (b) $15\frac{5}{8}$ m
 (c) $7\frac{11}{16}$ m (d) $5\frac{5}{24}$ m
10. Due to the technical snag in the signal system two trains start approaching each other on the same rail track from two different stations, 240 km away from each other. When the two trains at 60 km/h touching each time each train. The bird is initially sitting on the top of the engine of one of the trains and it moves so till these trains collide. If these trains collide one and a half hour after the start, then how many kilometers bird travels till the time of collision of trains?
 (a) 90 km (b) 130 km
 (c) 120 km (d) None of these
11. A surveillance plane is moving between two fixed places Pukhwara and Kargil at 120 km/hr. The distance between two places is 600 km. After 18 hour what will be the distance between the Kargil and its position if it is starts moving from Pukhwara?
 (a) 360 km (b) 300 km
 (c) 240 km (d) None of these
12. There are three runners Tom, Dick and Harry with their respective speeds of 10 km/h, 20 km/h and 30 km/h. They are initially at P and they have to run between the two points P and Q which are 10 km apart from each other. They start their race at 6 am and end at 6 pm on the same day. If they run between P and Q without any break, then how many times they will be together either at P and Q during the given time period?
 (a) 5 (b) 7
 (c) 4 (d) 12

13. A soldier fired two bullets at an interval of 335 seconds moving at a uniform speed v_1 . A terrorist who was running ahead of the soldier in the same direction, hears the two shots at an interval of 330 seconds? If the speed of sound is 1188 km/h, then who is the faster and by how much?
- (a) Soldier, 22 km/h (b) Terrorist, 25 km/h
(c) Soldier, 18 km/h (d) Terrorist, 20 km/h
14. A man goes to the fair in Funcity with his son and faithful dog. Unfortunately man misses his son which he realises 20 minutes later. The son comes back towards his home at the speed of 20 m/min and man follows him at 40 m/min. The dog runs to the son(child) and comes back to the man (father) to show him the direction of his son. It keeps moving to and fro at 60 m/min between son and father, till the man meets the son. What is the distance travelled by the dog in direction of the son?
- (a) 800 m (b) 1675 m
(c) 848 m (d) 1000 m
15. A thief sees a jeep at a distance of 250 m, coming towards him at 36 km/h. Thief takes 5 seconds to realise that there is nothing but the police is approaching him by the jeep and start running away from police at 54 km/hr. But police realise after 10 seconds, when the thief starts running away, that he is actually a thief and gives chase at 72 km/h. How long after thief saw police and catchup with him and what is the distance police had to travel to do so?
- (a) 50 s, 1000 m (b) 65s, 1150 m
(c) 65 s, 1300 m (d) 45 s, 1050 m
16. In a circus there was a leopard and a tiger walking in the two different rings of same radii. There I observed that when leopard moved 3 steps, tiger moved 5 steps in the same time, but the distance traversed by leopard in 5 steps is equal to the distance traversed by tiger in 4 steps. What is the number of rounds that a leopard made when tiger completed 100 rounds
- (a) 120 (b) 48
(c) 75 (d) None of these
17. Arti and Barkha start swimming towards each other from the deep end and shallow end respectively of a swimming pool in Funcity. They start their swimming simultaneously in the length of 300 m pool. The ratio of their speeds is 1 : 2 respectively. Each swimmer rests for 6 seconds once she reaches the other end and starts swimming back. Where will they meet for the second time in the still water of swimming pool?
- (a) 30 m from the shallow end
(b) at the shallow end
(c) at the depend
(d) can't be determined
18. If the two incorrect watches are set at 12:00 noon at correct time, when will both the watches show the correct time for the first time given that the first watch gains 1 min in 1 hour and second watch loses 4 min in 2 hours:
- (a) 6 pm, 25 days later
(b) 12:00 noon, 30 days later
(c) 12 noon, 15 days later
(d) 6 am 45 days later
19. Ramu purchased a second hand swiss watch which is very costly. In this watch the minute-hand and hour hand coincide after every $65\frac{3}{11}$ minutes. How much time does the watch lose or gain per day?
- (a) 4 min (b) 5 min
(c) 4 min, 20 sec (b) None of these
20. Kumbhakarna starts sleeping between 1 am and 2 am and he wakes up when his watch shows such a time that the two hands (i.e., hour-hand and minute-hand) interchanging the respective places. He wakes up between 2 am and 3 am on the same night. How long does he sleep?
- (a) $55\frac{5}{13}$ min (b) $110\frac{10}{13}$ min
(c) $54\frac{6}{13}$ min (d) None of these
21. A faulty clock gains 10 minutes every hour. If the time is set correctly at 12 Noon on 1st Jan 2010, then how many times will its minute-hand and hour-hand meet in the next 24 hours ?
- (a) 22 (b) 26
(c) 24 (d) 25
22. Progressive express left for New Delhi, increasing its speed in each hour. It started its journey from Lucknow, but after four hours of its journey it met with accident. Its speed in the fourth hour was $\frac{7}{5}$ times that of the third hour and the speed in the third hour was $\frac{10}{7}$ times that of the second hour and in the second hour it was $\frac{7}{5}$ times that of the first hour. If it would have been travelled with the half of the speed that of the third hour, then it would have gone 160 km less in the same time (i.e., in four hours). The average speed of the train during the journey of 4 hours was:
- (a) 50 km/hr (b) 90 km/hr
(c) 80 km/hr (d) can't be determined
23. Two rifles are fired from the same place at a difference of 11 minutes 45 seconds. But a man who is coming towards the place in a train hears the second sound after 11 minutes. Find the speed of train.
- (a) 72 km/h (b) 36 km/h
(c) 81 km/h (d) 108 km/h
24. Two people A and B start from P and Q (distance = D) at the same time towards each other. They meet at a point R, which is at a distance 0.4 D from P. They continue to move to and fro between the two points. Find the distance from point P at which the fourth meeting takes place.
- (a) 0.8 D (b) 0.6 D
(c) 0.3 D (d) 0.4 D

25. Two riders on the horseback with a gun and a bullet proof shield were moving towards each other at a constant speed of 20 km/h and 5 km/h respectively. When they were 100 km apart, they started firing bullets at each other at the speed of 10 km/h. When a bullet of rider 1 hits the shield of rider 2, rider 2 fires a bullet and the process continues vice versa. Neglecting the time lag at the instant when the bullet hits the shield and the rider fires the shot, find the total distance covered by all the bullets shot by both the riders.
(a) 50 km (b) 40 km
(c) 25 km (d) None of these
26. A passenger train departs from Ahmedabad at 6 pm for Bombay. At 9 p.m. an express train, whose average speed exceeds that of the passenger train by 15 km/h, leaves Bombay for Ahmedabad. Two trains meet each other mid-route. At what time do they meet, given that the distance between the cities is 1080 km?
(a) 4 pm (b) 2 pm
(c) 12 midnight (d) 6 am
27. A car covers a distance of 715 km at a constant speed. If the speed of the car had been 10 km/h more, then it would have taken 2 h less to cover the same distance. What is the original speed of the car?
(a) 55 km/h (b) 50 km/h
(c) 45 km/h (d) 65 km/h
28. A train leaves station X at 5 a.m. and reaches station Y at 9 a.m. Another train leaves station Y at 7 a.m. and reaches station X at 10:30 a.m. At what time do the two trains cross each other?
(a) 7:36 am (b) 7:56 am
(c) 8:36 am (d) 8:56 am
29. A train covered a certain distance at a uniform speed. If the train had been 6 km/h faster, then it would have taken 4 hours less than the scheduled time. And, if the train were slower by 6 km/h, then the train would have taken 6 hours more than the scheduled time. The length of the journey is
(a) 700 km (b) 740 km
(c) 720 km (d) 760 km
30. A man swimming in a stream which flows $1\frac{1}{2}$ km/hr., finds that in a given time he can swim twice as far with the stream as he can against it. At what rate does he swim?
(a) $5\frac{1}{2}$ km/hr (b) $4\frac{1}{2}$ km/hr
(c) $7\frac{1}{2}$ km/hr (d) None of these
31. In a 400 metres race, A gives B a start of 5 seconds and beats him by 15 metres. In another race of 400 metres, A beats B by $7\frac{1}{7}$ seconds. Find their speeds.
(a) 8 m/sec, 7 m/sec (b) 7 m/sec, 6 m/sec
(c) 6 m/sec, 5 m/sec (d) 5 m/sec, 4 m/sec
32. The speeds of three cars are in the ratio 2 : 3 : 4. The ratio between the times taken by these cars to travel the same distance is
(a) 4 : 3 : 2 (b) 2 : 3 : 4
(c) 4 : 3 : 6 (d) 6 : 4 : 3
33. Anand travelled 300 km by train and 200 km by taxi. It took him 5 h and 30 min. However, if he travels 260 km by train and 240 km by taxi, he takes 6 min more. The speed of the train is
(a) 100 km/h (b) 120 km/h
(c) 80 km/h (d) 110 km/h
34. A boat takes 19 h for travelling downstream from point A to point B and coming back to a point C midway between A and B. If the velocity of the stream is 4 km/h and the speed of the boat in still water is 14 km/h, what is the distance between A and B?
(a) 200 km (b) 180 km
(c) 160 km (d) 220 km
35. A car travels 25 km/h faster than a bus for a journey of 500 km. If the bus takes 10 h more than the car, then the speeds of the car and the bus are
(a) 25 km/h and 40 km/h (b) 50 km/h and 25 km/h
(c) 25 km/h and 60 km/h (d) None of these
36. Speed of a speed-boat when moving in the direction perpendicular to the direction of the current is 16 km/h. Speed of the current is 3 km/h. So the speed of the boat against the current will be (in km/h)
(a) 22 (b) 9.5
(c) 10 (d) None of these
37. Two ants start simultaneously from two ant holes towards each other. The first ant covers 8% of the distance between the two ant holes in 3 hours, the second ant covered $\frac{7}{120}$ of the distance in 2 hours 30 minutes. Find the speed (feet/h) of the second ant if the first ant travelled 800 feet to the meeting point.
(a) 15 feet/h (b) 25 feet/h
(c) 45 feet/h (d) 35 feet/h
38. Two Indian tourists in the US cycled towards each other, one from point A and the other from point B. The first tourist left point A 6 hrs later than the second left point B, and it turned out on their meeting that he had travelled 12 km less than the second tourist. After their meeting, they kept cycling with the same speed, and the first tourist arrived at B 8 hours later and the second arrived at A 9 hours later. Find the speed of the faster tourist.
(a) 4 km/h (b) 6 km/h
(c) 9 km/h (d) 2 km/h
39. A motorcyclist left point A for point B. Two hours later, another motorcyclist left A for B and arrived at B at the same time as the first motorcyclist. Had both motorcyclists started simultaneously from A and B travelling towards each other, they would have met in 80 minutes. How much time did it take the faster motorcyclist to travel from A to B?
(a) 6 hours (b) 3 hours
(c) 2 hours (d) 4 hours

40. Shaurya and Arjit take a straight route to the same terminal point and travel with constant speeds. At the initial moment, the positions of the two and the terminal point form an equilateral triangle. When Arjit covered a distance of 80 km, the triangle become right-angled. When Arjit was at a distance of 120 km from the terminal point, the Shaurya arrived at the point. Find the distance between them at the initial moment assuming that there are integral distances throughout the movements described.
 (a) 300 km (b) 240 km
 (c) 200 km (d) 225 km
41. Three cars started simultaneously from Ajmer to Benaras along the same highway. The second car travelled with a speed that was 10 km/h higher than the first car's speed and arrived at Benaras 1 hour earlier than the first car. The third car arrived at Benaras 33.33 minutes earlier than the first car, travelling half the time at the speed of the first car and the other half at the speed of the second car. Find the total distance covered by these three cars during their journey between Ajmer and Benaras.
 (a) 360 km (b) 600 km
 (c) 540 km (d) 840 km
42. Two towns are at a distance of 240 km from each other. A motorist takes 8 hours to cover the distance if he travels at a speed of V_0 km/h from town A to an intermediate town C, and then continues on his way with an acceleration of x km/hr². He needs the same time to cover the whole distance if he travels from A to C at V_0 km/h and from C to B at V_1 km/h or from A to C at V_1 km/h from C to B at V_0 km/h. Find V_0 if the acceleration 'x' is double V_0 in magnitude and $V_0 \neq V_1$.
 (a) 15 km/h (b) 10 km/h
 (c) 20 km/h (d) 8 km/h
43. A pedestrian and a cyclist left Nagpur for Buti Bori at the same time. Having reached Buti Bori, the cyclist turned back and met the pedestrian an hour after the start. After their meeting, the pedestrian continued his trip to Buti Bori and cyclist turned back and also headed for Buti Bori. Having reached Buti Bori, the cyclist turned back again and met the pedestrian 30 mins after their first meeting. Determine what time it takes the pedestrian 30 mins after their first meeting. Determine what time it takes the pedestrian to cover the distance between Nagpur and Buti Bori.
 (a) 1 hour (b) 2 hours
 (c) 2.5 hours (d) 3 hours
44. Two people started simultaneously from points A and B towards each other. At the moment the person who started from A had covered two-thirds of the way, the other person had covered 2 km less than half the total distance. If it is known that when the person who started from B had covered 1/4 of the way, the other person was 3 km short of the mid point. Find the distance between A and B. The speeds of the two people were constant.
 (a) $(15 - 3\sqrt{17})$ km (b) $(15 + 3\sqrt{17})$ km
 (c) Both (a and b) (d) $3\sqrt{17} - 5$ km

Test Yourself

- A racetrack is in the form of a right triangle. The longer of the legs of the track is 2 km more than the shorter of the legs (both these legs being on a highway). The start and end points are also connected to each other through a side road. The escort vehicle for the race took the side road and rode with a speed of 30 km/h and then covered the two intervals along the highway during the same time with a speed of 42 km/h. Find the length of the racetrack.
(a) 14 km (b) 10 km
(c) 24 km (d) 36 km
- Two trains 137 metres and 163 metres in length are running towards each other on parallel lines, one at the rate of 42 kmph and another at 48 kmph. In what time will they be clear of each other from the moment they meet?
(a) 10 sec (b) 12 sec
(c) 14 sec (d) cannot be determined
- Two planes move along a circle of circumference 1.2 km with constant speeds. When they move in different directions, they meet every 15 seconds and when they move in the same direction, one plane overtakes the other every 60 seconds. Find the speed of the slower plane.
(a) 0.04 km/s (b) 0.03 km/s
(c) 0.05 km/s (d) 0.02 km/s
- An ant moved for several seconds and covered 3 mm in the first second and 4 mm more in each successive second than in its predecessor. If the ant had covered 1 mm in the first second and 8 mm more in each successive second, then the difference between the path it would cover during the same time and the actual path would be more than 6 mm but less than 30 mm. Find the time for which the ant moved (in seconds)
(a) 5 s (b) 4 s
(c) 6 s (d) 2 s
- A train leaves station X at 5 a.m. and reaches station Y at 9 a.m. Another train leaves station Y at 7 a.m. and reaches station X at 10:30 a.m. At what time do the two trains cross each other?
(a) 7:36 am (b) 7:56 am
(c) 8:36 am (d) 8:56 am
- Rahim sets out to cross a forest. On the first day, he completes $\frac{1}{10}$ th of the journey. On the second day, he covers $\frac{2}{3}$ rd of the distance travelled the first day. He continues in this manner, alternating the days in which he travels $\frac{1}{10}$ th of the distance still to be covered, with days on which he travels $\frac{2}{3}$ of the total distance already covered. At the end of seventh day, he finds that $22\frac{1}{2}$ km more will see the end of his journey. How wide is the forest?
(a) $66\frac{2}{3}$ km (b) 100 km
(c) 120 km (d) 150 km
- Two ducks move along the circumference of a circular pond in the same direction and come alongside each other every 54 minutes. If they moved with the same speeds in the opposite directions, they would meet every 9 minutes. It is known that when the ducks moved along the circumference in opposite directions, the distance between them decreased from 54 to 14 feet every 48 seconds. What is the speed of the slower duck?
(a) 20 feet/min (b) 15 feet/min
(c) 30 feet/min (d) 20.83 feet/min
- An athlete runs to and fro between points A and B at a speed of 10 km/h. A second athlete simultaneously runs from point B to A and back at a speed of 15 km/h. If they cross each other 12 min after the start, after how much time will they cross each other?
(a) 18 min (b) 24 min
(c) 36 min (d) 48 min
- A train's journey is disrupted due to an accident on its track after it has travelled 30 km. Its speed then comes down to $\frac{4}{5}$ th of its original and consequently it runs 45 min late. Had the accident taken place 18 km farther away, it would have been 36 min late. Find the original speed of the train.
(a) 25 km/h (b) 36 km/h
(c) 30 km/h (d) 20 km/h
- A tank of 4800 m^3 capacity is full of water. The discharging capacity of the pump is $10 \text{ m}^3/\text{min}$ higher than its filling capacity. As a result the pump needs 16 min less to discharge the fuel than to fill up the tank. Find the filling capacity of the pump.
(a) $50 \text{ m}^3/\text{min}$ (b) $25 \text{ m}^3/\text{min}$
(c) $55 \text{ m}^3/\text{min}$ (d) $24 \text{ m}^3/\text{min}$
- Karan and Arjun run a 100-metre race, where Karan beats Arjun by 10 metres. To do a favour to Arjun, Karan starts 10 metres behind the starting line in a second 100-metre race. They both run at their earlier speeds. Which of the following is true in connection with the second race?
(a) Karan and Arjun reach the following line simultaneously
(b) Arjun beats Karan by 1 metre
(c) Arjun beats Karan by 11 metre
(d) Karan beats Arjun by 1 metre
- A train X departs from station A at 11.00 am for station B, which is 180 km away. Another train Y departs from station B at 11.00 am for station A. Train X travels at an average speed of 70 kms/hr and does not stop anywhere until it

arrives at station *B*. Train *Y* travels at an average speed of 50 kms/hr, but has to stop for 15 minutes at station *C*, which is 60 kms away from station *B* enroute to station *A*. Ignoring the lengths the train, what is the distance, to the nearest km, from station *A* to the point where the trains cross each other?

- (a) 112 (b) 118
(c) 120 (d) None of these

13. The vehicle of Mr. Ghosh needs 30% more fuel at the speed of 75 kmph than it needs at the speed of 50 kmph. At a speed of 50 kmph, Mr. Ghosh can go to a distance of 195 kms. At the speed of 75 kmph, he will be able to travel a distance of
(a) 120 kms (b) 150 kms
(c) 160 kms (d) 140 kms
14. I started climbing up the hill at 6 am and reached the temple at the top at 6 pm. Next day I started coming down at 6 am and reached the foothill at 6 pm. I walked on the same road.

The road is so short that only one person can walk on it. Although I varied my pace on my way, I never stopped on my way. Then which of the following must be true

- (a) My average speed downhill was greater than that uphill
(b) At noon, I was at the same spot on both the days.
(c) There must be a point where I reached at the same time on both the days.
(d) There cannot be a spot where I reached at the same time on both the days.

15. In a watch, the minute hand crosses the hour hand for the third time exactly after every 3 hrs., 18 min., 15 seconds of watch time. What is the time gained or lost by this watch in one day?
(a) 14 min. 10 seconds lost
(b) 13 min. 44 seconds lost
(c) 13min. 20 seconds gained
(d) 14 min. 40 seconds gained

Hints & Solutions

Foundation Level

1. (a) Let a distance x be covered in time t .

$$\text{Required ratio} = \frac{\frac{x/2}{t}}{\frac{x}{t}} = \frac{1}{4} = 1:4$$

2. (c) Let the distance travelled during both upward and downward journey be x km.

$$\begin{aligned} \text{Average speed} &= \frac{\text{Total distance covered}}{\text{Total time taken}} \\ &= \frac{x+x}{\frac{x}{16} + \frac{x}{28}} = \frac{2}{\frac{28+16}{28 \times 16}} \\ &= \frac{2 \times 28 \times 16}{44} = 20.36 \text{ km/h} \end{aligned}$$

3. (b) Distance = $\left(1100 \times \frac{11}{5}\right)$ feet = 2420 feet.

4. (a) Time required = (2 hrs 30 min – 50 min) = 1 hr 40 min
 $= 1\frac{2}{3}$ hrs

$$\therefore \text{Required speed} = \left(50 \times \frac{3}{5}\right) \text{ km/hr} = 30 \text{ km/hr.}$$

$$\text{Original speed} = \left(50 \times \frac{2}{5}\right) \text{ km/hr} = 20 \text{ km/hr.}$$

$$\therefore \text{Difference in speed} = (30 - 20) \text{ km/hr} = 10 \text{ km/hr.}$$

5. (c) When time is constant the distance covered by A and B will be in the ratio of their speeds, respectively.

6. (b) Let the distance travelled be x km.

Then, the correct time at a speed of 30 km/h

$$= \frac{x}{30} - \frac{10}{60} \text{ and}$$

$$\text{the correct time at a speed of 42 km/h} = \frac{x}{42} + \frac{10}{60}$$

$$\text{Now, } \frac{x}{30} - \frac{10}{60} = \frac{x}{42} + \frac{10}{60}$$

$$\text{or } \frac{x}{30} - \frac{x}{42} = \frac{2}{6} \text{ or } \frac{12x}{1260} = \frac{2}{6} \text{ or } x = 35 \text{ km}$$

7. (c) Relative speed of the trains = $(40 + 20) = 60$ m/s
 Distance = $(120 + 120) = 240$ m

Time taken by trains to cross each other completely

$$= \frac{240}{60} = 4 \text{ s}$$

$\therefore$ Larger the no. of cogs (tooth of wheel) of wheel, lesser will be that no. of revolution made by it.

8. (c) Let the speed of trains be x km/h and y km/h, respectively.

When the trains cross each other, time taken by both the trains will be equal.

$$\text{i.e. } \frac{110}{x} = \frac{90}{y} \Rightarrow \frac{x}{y} = \frac{110}{90} \Rightarrow x:y = 11:9$$

9. (b) Required distance = $\frac{S_1 S_2}{(S_1 + S_2)} \times \text{Time difference}$

$$= \frac{8 \times 5}{3} \times \frac{3}{2} = 20 \text{ km}$$

10. (b) Let the total distance be x km. Then,

$$\frac{\frac{1}{21}x}{\frac{1}{21}} + \frac{\frac{1}{24}x}{\frac{1}{24}} = 10 \Rightarrow \frac{x}{21} + \frac{x}{24} = 20$$

$$\Rightarrow 15x = 168 \times 20 \Rightarrow x = \left(\frac{168 \times 20}{15}\right) = 224 \text{ km.}$$

11. (b) Speed of the train = $132 \text{ km/h} = \frac{132 \times 5}{18} \text{ m/s}$

$$\text{Distance} = (110 + 165) = 275 \text{ m}$$

Time required to cross the railway platform

$$= \frac{275 \times 18}{132 \times 5} = 7.5 \text{ s}$$

12. (b) Let the total distance be $3x$ km.

$$\text{Then, } \frac{x}{3} + \frac{x}{4} + \frac{x}{5} = \frac{47}{60} \Leftrightarrow \frac{47x}{60} = \frac{47}{60} \Leftrightarrow x = 1.$$

$$\therefore \text{Total distance} = (3 \times 1) \text{ km} = 3 \text{ km.}$$

13. (c) $\frac{x}{9} - \frac{x}{10} = \frac{20}{60}$

$$\text{or, } \frac{10x - 9x}{90} = \frac{20}{60}$$

$$\therefore x = 30 \text{ km}$$

14. (c) Let the normal speed = x km/h

Then, the new speed = $(x + 5)$ km/h.

$$\text{Now, } \frac{300}{x} - 2 = \frac{300}{(x+5)} \text{ or } \frac{300}{x} - \frac{300}{(x+5)} = 2$$

Checking with options, we see that $x = 25$ km/h.

15. (c) Distance between Chauhan and the gun
 $= 3.32 \times 1000 = 3320 \text{ m}$
 Time taken = 10 s
 $\Rightarrow \text{Speed} = \frac{3320}{10} = 332 \text{ m/s}$
16. (a) Since A and B move in the same direction along the circle, so they will first meet each other when there is a difference of one round between the two.
 Relative speed of A and B = $(6 - 1) = 5$ rounds per hour.
 Time taken to complete one round at this speed = $\frac{1}{5}$ hr = 12 min.
 $\therefore$ They meet at 7:42 a.m.
17. (c) Average speed = $\frac{\text{Total distance covered}}{\text{Total time taken}}$
 $= \frac{2 \times 200}{\frac{200}{40} + \frac{200}{20}} = \frac{2 \times 40 \times 20}{40 + 20} = \frac{2 \times 40 \times 20}{60} = \frac{80}{3} = 26.67 \text{ km/h.}$
18. (a) Let the length of each train be x metres.
 Then, the total distance covered = $(x + x) = 2x \text{ m}$
 Relative speed = $(46 - 36) = 10 \text{ km/h} = \frac{10 \times 5}{18} \text{ m/s}$
 Now, $36 \frac{2x \times 18}{50}$
 or $x = 50 \text{ m}$
19. (c) After 5 minutes (before meeting), the top runner covers 2 rounds i.e., 400 m and the last runner covers 1 round i.e., 200 m.
 $\therefore$ Top runner covers 800 m race in 10 minutes.
20. (b) Due to stoppages the train travels $(45 - 36) = 9 \text{ km}$ less in an hour than it could have travelled without stoppages.
 Thus train stops per hour for $\frac{9}{45} \times 60 = 12 \text{ min.}$
21. (b) Distance travelled by the train in 1 hour
 $= 2 \times \frac{22}{7} \times 2.1 \times 75 \times 60 \text{ m.} = \frac{132 \times 450}{1000} = 59.4 \text{ km}$
 i.e. speed of the train = 59.4 km/h.
22. (d) First train's speed is 45 km/hr.
 $\left(\text{Using speed} = \frac{\text{Distance}}{\text{Time}} \right)$
 Second train's speed is 60 km/hr.
 Difference in the distance covered by these trains in 1 hr. is 15 km.
23. (c) Speed of speed-boat = $16 - 3 = 13 \text{ km/hr.}$
 $\therefore$ Speed of boat against the current = $13 - 3 = 10 \text{ km/hr.}$
24. (d) Let the usual speed be $x \text{ km/hr}$, then
 $\frac{1500}{x} - \frac{1500}{x + 250} = \frac{1}{2}$
 $\Rightarrow x = 750 \text{ km/hr}$
25. (a) Relative velocity = $20 + 30 = 50 \text{ m/s.}$
 Distance = 2.5 kms. = 2500 m.
 $t = 2500/50 = 50 \text{ s.}$
26. (c) Relative speed = $90 + 60 = 150 \text{ km/hr.}$
 Total distance to be covered = $300 + 200 = 500 \text{ m}$
 Time required = $\frac{500}{150 \times 1000} \times 3600 = 12 \text{ sec.}$
27. (d) Required distance between A and B
 $= \frac{3((9)^2 - (3)^2)}{2(9)} = \frac{3(81 - 9)}{18} = \frac{72}{6} = 12 \text{ km.}$
28. (d) Total distance covered = $2 \times 91 \text{ km} = 182 \text{ km}$
 Time taken = 20 hours
 $\therefore$ Average speed = $\frac{182}{20} = 9.1 \text{ km/h}$
 Let the speed of flow of the river = $x \text{ km/hr}$
 then, $\frac{10^2 - x^2}{10} = 9.1 \Rightarrow 100 - 91 = x^2 \Rightarrow x = \pm 3$
 Hence, rate of flow of the river = 3 km/h
29. (d) Relative speed = $5.5 - 5 = 0.5 \text{ km/h.}$
 Required time = $\frac{8.5}{0.5} = 17 \text{ h}$
30. (d) x (speed of boat in standing water) = 9 km/hr
 speed of stream = 1.5 km/h
 Total time taken by him = $\frac{105}{10.5} + \frac{105}{7.5}$
 $= 10 + 14 = 24 \text{ h}$

31. (a) Given, distances are 2500 km, 1200 km and 500 km.
Given, speeds are 500 km/h, 400 km/h and 250 km/h

$$\begin{aligned}\therefore \text{Total time} &= \frac{2500}{500} + \frac{1200}{400} + \frac{500}{250} \\ &= 5 + 3 + 2 = 10 \text{ hr.} \\ \therefore \text{Average speed} &= \frac{\text{Total distance}}{\text{Total time}} \\ &= \frac{2500 + 1200 + 500}{10} = \frac{4200}{10} \\ &= 420 \text{ km/hr}\end{aligned}$$

32. (d) Let the distance between each pole be x m.
Then, the distance up to 12th pole = $11x$ m

$$\begin{aligned}\text{Speed} &= \frac{11x}{24} \text{ m/s} \\ \text{Time taken to covers the total distance of } 19x \\ &= \frac{19x \times 24}{11x} = 41.45 \text{ s}\end{aligned}$$

33. (a) Let the length of the journey = x km.

$$\therefore \text{Journey rides by horse cart } x \left(1 - \frac{1}{2} - \frac{1}{3}\right)$$

$$\frac{1}{6} x \text{ km.}$$

Then, total time taken to complete journey $\frac{31}{5}$ hr

$$\begin{aligned}\Rightarrow t_1 + t_2 + t_3 &= \frac{31}{5} \\ \Rightarrow \frac{x}{2} \times \frac{1}{4} + \frac{x}{3} \times \frac{1}{12} + \frac{x}{6 \times 9} &= \frac{31}{5} \\ \Rightarrow x &= \frac{31}{5} \times \frac{216}{37} = 36.2 \text{ km} \approx 36 \text{ km}\end{aligned}$$

34. (d) Required difference = $\frac{180}{3} - \frac{180}{4} = 15$ km

35. (c) Let the husband and the wife meet after x minutes.
4500 metres are covered by Pradeep in 60 minutes.

$$\text{In } x \text{ minutes, he will cover } \frac{4500}{60} x \text{ metres.}$$

Similarly,

$$\text{In } x \text{ minutes, his wife will cover } \frac{3750}{60} x \text{ m.}$$

$$\text{Now, } \frac{4500}{60} x + \frac{3750}{60} x = 726$$

$$\Rightarrow x = \frac{726 \times 60}{8250} = 5.28 \text{ min}$$

36. (a) Let speed of the boat in still water be x km/h and speed of the current be y km/h.

Then, upstream speed = $(x - y)$ km/h
and downstream speed = $(x + y)$ km/h

$$\text{Now, } \frac{24}{(x - y)} + \frac{28}{(x + y)} = 6 \quad \dots(1)$$

$$\text{and } \frac{30}{(x - y)} + \frac{21}{(x + y)} = \frac{13}{2} \quad \dots(2)$$

Solving (1) and (2), we have
 $x = 10$ km/h and $y = 4$ km/h

37. (c) The train that leaves at 6 am would be 75 km ahead of the other train when it starts. Also, the relative speed being 36 kmph, the distance from Mumbai would be $(75/36) \times 136 = 283.33$ km

38. (a) Solve through options. At 18 kmph the motorboat would take exactly 6 hours.

39. (a) The train can cover $(200 + 350)$ m distane in five seconds which means the speed of the train is 110 m/s. Relative speed of man and trian is 114 m/s. To cover the distance of 100 metre, it will take less than one second.

40. (a) The clock gains 15 min in 24 hours.
Therefore, in 16 hours, it will gain 10 minutes.
Hence, the time shown by the clock will be 4.10 am.

41. (b) Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$$= \frac{80}{\frac{60}{40} + \frac{20}{20}} = \frac{80}{2.5} = 32 \text{ km/h}$$

42. (d) Total distance = $250 \times 2 = 500$ km

$$\text{Total time} = 5\frac{1}{2} \text{ hrs} + 4\frac{2}{3} \text{ hrs} = 10\frac{1}{6} \text{ hrs}$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}} = \frac{500}{10\frac{1}{6}} = \frac{3000}{61} \text{ hrs}$$

$$= 49.18 \text{ hours} \approx 50 \text{ hours (approx.)}$$

43. (c) The statements in the question can be reformulated as follows:

If A covers 400 m, B covers 395 m.

If B covers 400 m, C covers 396 m.

If D covers 400 m, C covers 384 m.

Therefore, if B covers 395 m, then C will cover,

$$\frac{396}{400} \times 395 = 391.05 \text{ m}$$

Again, If C covers 391.05 m, then D will cover

$$\frac{400}{384} \times 391.05 = 407.34 \text{ m.}$$

Thus, if A and D run over 400 m, then D wins by 7.3 m.

44. (c) Their relative speeds = $(4.5 + 3.75) = 8.25$ km/h

$$\text{Distance} = 726 \text{ metres} = \frac{726}{1000} = 0.726 \text{ km}$$

$$\text{Required time} = \frac{0.726}{8.25} \times 60 = 5.28 \text{ min}$$

45. (d) Since they are moving in opposite direction, therefore their relative speed will be $4 + 3 = 7$ km/hr.

$$\text{Time } \frac{d}{s} = \frac{17.5}{7} = 2.5 \text{ hrs.}$$

(where d is distance and s is speed).

$\therefore$ They should meet at 12.30 PM.

46. (c) Speed on return trip = 150% of 40 = 60 kmph.

$$\begin{aligned} \therefore \text{Average speed} &= \left(\frac{2 \times 40 \times 60}{40 + 60} \right) \text{ km/hr} \\ &= \left(\frac{4800}{100} \right) \text{ km/hr} = 48 \text{ km/hr.} \end{aligned}$$

47. (a) Average speed = $\frac{2 \times V_1 \times V_2}{V_1 + V_2} = \frac{2 \times 40 \times 60}{40 + 60}$

$$= 48 \text{ km/h}$$

48. (c) The speeds of the two cyclists are different. Hence, when one of the cyclist has covered one round more than the other cyclist, only then they will meet at the starting point.

$$\begin{aligned} \therefore \text{Time when the two cyclists will meet} \\ &= 300 \text{ m} \times (\text{difference in speeds}) \\ &= 300 \times (8 - 7) \text{ sec} = 300 \text{ seconds.} \end{aligned}$$

49. (c) Let the correct time to complete the journey be x min.
Distance covered in $(x + 11)$ min. at 40 kmph
= Distance covered in $(x + 5)$ min. at 50 kmph

$$\therefore \frac{(x + 11)}{60} \times 40 = \frac{(x + 5)}{60} \times 50 \Leftrightarrow x = 19 \text{ min.}$$

50. (a) Let x be the total distance.
 $\therefore$ According to the question,

$$\text{Distance covered by him on foot} = \frac{1}{3}x \times \frac{1}{4} = \frac{x}{12}$$

But given he travels on foot = 2 km

$$\therefore \frac{x}{12} = 2 \Rightarrow x = 24 \text{ km.}$$

51. (d) Let speed of car = x km / hr
Let speed of pedestrian = $y = 2$ km / hr
 $\therefore$ Relative speed = $(x - 2)$ km / hr
 $\therefore$ According to the question,

$$(x - 2) \times \frac{6}{60} = 0.6 \Rightarrow x - 2 = 6 \Rightarrow x = 8 \text{ km / h}$$

52. (b) Let the original time be T hours and original speed be x km/h

$$\frac{1500}{x} = T \quad \dots(1)$$

$$\frac{1500}{x + 250} = T - \frac{30}{60} \quad \dots(2)$$

Solving equations (1) and (2), we get

Speed of plane = $x = 750$ or -1000 (not possible)

$$\therefore x = 750 \text{ km/h}$$

53. (c) Suppose they meet x hours after 14.30 hrs.

Then, $60x = 80(x - 2)$ or $x = 8$.

$$\therefore \text{Required distance} = (60 \times 8) = 480 \text{ km.}$$

54. (c) Total time taken

$$= \left(\frac{9}{9 + 6} + \frac{9}{9 - 6} \right) \text{ hour}$$

$$= \left(\frac{3}{5} + 3 \right) \text{ hours} = 3\frac{3}{5} \text{ hours}$$

55. (a) Distance to be covered by the thief and by the owner is same.

Let after time ' t ', owner catches the thief.

$$\therefore 40 \times t = 50 \left(t - \frac{1}{2} \right)$$

$$\Rightarrow 10t = 25 \Rightarrow t = \frac{5}{2} \text{ hr} = 2\frac{1}{2} \text{ hr}$$

56. (b) Rest time = Number of rest $\times$ Time for each rest
 $= 4 \times 5 = 20$ minutes

Total time to cover 5 km

$$= \left(\frac{5}{10} \times 60 \right) \text{ minutes} + 20 \text{ minutes} = 50 \text{ minutes.}$$

57. (c) Assume that the distance is 120 km. Hence, 30 km is covered @ 25 kmph, 40 @ 30 kmph and so on. Then average speed is 120/total time

58. (c) Time taken to cross a pole = $\frac{50}{1000} \times \frac{1}{45} \text{ hr}$

$$\therefore \text{No. of counts} = \frac{4 \times 1000 \times 45}{50} = 80 \times 45 = 3600.$$

59. (a) Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

$$\begin{aligned} &= \frac{400 \times 4 \times 9}{88 + 96 + 89 + 87} = \frac{400 \times 4 \times 9}{360} \\ &= 40 \text{ metres / minutes} \end{aligned}$$

60. (b) Time = $\frac{\text{Distance advanced}}{\text{Relative speed}}$

$$\begin{aligned} &= \frac{2 \times x}{(30 - x)} \\ &\Rightarrow x = 15 \text{ km/h} \end{aligned}$$

Standard Level

1. (d) When A covers 200 metres, B covers

$$200 \times \frac{22}{25} = 176 \text{ m}$$

So, B is $(200 - 176) = 24$ m far away from the end point when A reaches in.

2. (b) Let the required distance be x km.

Difference in the times taken at two speeds

$$= 12 \text{ min} = \frac{1}{5} \text{ hr.}$$

$$\therefore \frac{x}{5} - \frac{x}{6} = \frac{1}{5} \Leftrightarrow 6x - 5x = 6 \Leftrightarrow x = 6$$

Hence, the required distance is 6 km.

3. (c) Total distance travelled in 12 hours = $(35 + 37 + 39 + \dots \text{ upto 12 terms})$.

This is an A.P. with first term, $a = 35$, number of terms, $n = 12$, common difference, $d = 2$.

$$\therefore \text{Required distance} = \frac{12}{2} [2 \times 35 + (12 - 1) \times 2]$$

$$= 6(70 + 22) = 552 \text{ km.}$$

4. (b) Let the speed of the train and the car be x km/h and y km/h, respectively.

$$\text{Now, } \frac{120}{x} + \frac{480}{y} = 8 \quad \dots(1)$$

$$\text{and } \frac{200}{x} + \frac{400}{y} = \frac{25}{3} \quad \dots(2)$$

$$\text{From (1), } 120y + 480x = 8xy \text{ and } \dots(3)$$

$$\text{From (2), } 200y + 400x = \frac{25}{3}xy \quad \dots(4)$$

From (3) and (4),

$$\frac{120y + 480x}{8} = \frac{3(200y + 400x)}{25}$$

$$\text{or } 15y + 60x = 24y + 48x$$

$$\text{or } 12x = 9y \text{ or } \frac{x}{y} = \frac{3}{4} \text{ or } x : y = 3 : 4$$

5. (c) Circumference of the wheel starting from

$$X = 2 \times \frac{22}{7} \times 3.5 = 22 \text{ cm}$$

Circumference of the wheel starting from

$$Y = 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$$

Let both the wheels make n revolutions in one second.

$$\text{Distance covered by both the wheels in 1 sec}$$

$$= 22n + 44n = 66n \text{ cm}$$

$$\Rightarrow \text{Distance covered by both the wheels in}$$

$$10 \text{ sec} = 660n \text{ cm}$$

$$\text{Now, } 660n = 1980 \Rightarrow n = 3$$

$$\text{Speed of the smaller wheel} = 22n \text{ cm/s} = 66 \text{ cm/s}$$

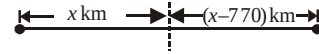
6. (c) Remaining distance = 3 km and Remaining time

$$= \left(\frac{1}{3} \times 45 \right) \text{ min} = 15 \text{ min} = \frac{1}{4} \text{ hour.}$$

$$\therefore \text{Required speed} = (3 \times 4) \text{ km/hr} = 12 \text{ km/hr.}$$

7. (b) Let the aeroplane covers x km at a speed of 440 km/h and $(x - 770)$ km at a speed of 660 km/h.

Hence, it covers a total distance of $(2x - 770)$ km at a speed of 500 km/h.



$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

$$\Rightarrow 500 = \frac{2x - 770}{\frac{x}{440} + \frac{x - 770}{660}}$$

$$\text{or } \frac{2x - 770}{500} = \frac{x}{440} + \frac{x - 770}{660}$$

$$\text{or } x = 1760$$

Therefore, the total distance covered = $2x - 770$

$$= 2 \times 1760 - 770 = 2750 \text{ km}$$

8. (a) Let the whole distance travelled be x km and the average speed of the car for the whole journey be y km/hr.

$$\text{Then, } \frac{(x/3)}{40} + \frac{(x/3)}{20} + \frac{(x/3)}{60} = \frac{x}{y}$$

$$\Leftrightarrow \frac{x}{30} + \frac{x}{60} + \frac{x}{180} = \frac{x}{y}$$

$$\Leftrightarrow \frac{1}{18}y = 1$$

$$\Leftrightarrow y = 18 \text{ km/hr.}$$

9. (a) Speed of first train = 50 km/hr.

$$\text{Speed of second train} = \frac{400}{7} \text{ km/hr.}$$

At 8:00 AM distance between two trains is 100 kms.

Relative velocity

$$= 50 + \frac{400}{7} = \frac{350 + 400}{7} = \frac{750}{7} \text{ km/h}$$

$$\text{Time taken } \frac{100 \times 7}{750} \times 60 = 56 \text{ min. Hence, the two}$$

trains meet each other at 8:56 AM.

10. (b) Let the speed of the stream be x km/hr and distance travelled be S km. Then,

$$\frac{S}{12 + x} = 6 \text{ and } \frac{S}{12 - x} = 9$$

$$\Rightarrow \frac{12 - x}{12 + x} = \frac{6}{9} \Rightarrow 108 - 9x = 72 + 6x$$

$$\Rightarrow 15x = 36 \Rightarrow x = \frac{36}{15} = 2.4 \text{ km/hr.}$$

11. (a) If the rate of the stream is x , then $2(4.5 - x) = 4.5 + x$
 $\Rightarrow 9 - 2x = 4.5 + x \Rightarrow 3x = 4.5 \Rightarrow x = 1.5 \text{ km/hr}$

12. (b) Distance covered = 187.5m, Time = 9 secs

$$\text{Relative speed} = \frac{187.5}{9} \times \frac{3600}{1000} = 75 \text{ km/hr}$$

As the trains are travelling in opposite directions, speed of goods train = $75 - 50 = 25 \text{ km/hr}$.

13. (d) Relative speed of both trains = $60 + 90 = 150 \text{ km/h}$
 Total distance = $1.10 + 0.9 = 2 \text{ km}$

$$\therefore \text{Required time} = \frac{2 \times 60 \times 60}{150} = 48 \text{ seconds.}$$

14. (c) Let the speed of the train be $x \text{ km/hr}$ and that of the car be $y \text{ km/hr}$.

$$\text{Then, } \frac{120}{x} + \frac{480}{y} = 8 \text{ or } \frac{1}{x} + \frac{4}{y} = \frac{1}{15} \quad \dots(1)$$

$$\text{And, } \frac{200}{x} + \frac{400}{y} = \frac{25}{3} \text{ or } \frac{1}{x} + \frac{2}{y} = \frac{1}{24} \quad \dots(2)$$

Solving (1) and (2), we get $x = 60$ and $y = 80$.

$\therefore$ Ratio of speeds = $60 : 80 = 3 : 4$.

15. (c) Suppose they meet x hrs after 8 a.m. Then,
 (Distance moved by first in x hrs) + [Distance moved by second in $(x - 1)$ hrs] = 330

$$\therefore 60x + 75(x - 1) = 330$$

$$\Rightarrow x = 3$$

So, they meet at $(8 + 3)$, i.e. 11 a.m.

16. (d) Given, ratio of speeds of A and B is $5 : 4$.

$\therefore$ B makes 4 rounds when A makes 5 rounds.

Now, distance covered by A in 5 rounds

$$= \left(5 \times \frac{400}{1000} \right) 2 \text{ km}$$

and distance covered by B in 4 rounds

$$\left(4 \times \frac{400}{1000} \right) \text{ km} = 1.6 \text{ km}$$

It is clear that in 5 hours, A passes

B only once. (i.e., 1 time).

In other words, in covering 2 km, A passes B 1 time.

$\therefore$ In covering 5 km, A passes B in $\left(\frac{1}{2} \times 5 \right)$ times

i.e., $2\frac{1}{2}$ times.

17. (a) Total journey = 180 km

$$\frac{1}{3} \text{ rd of journey } \frac{180}{3} = 60 \text{ km.}$$

If usual speed be $x \text{ kmph}$, then

$$\frac{60}{3x} - \frac{60}{x} = \frac{1}{2}$$

$$\Rightarrow \frac{80}{x} - \frac{60}{x} = \frac{1}{2}$$

$$\Rightarrow \frac{80}{x} - \frac{60}{x} = \frac{1}{2}$$

$$\Rightarrow x = 40 \text{ kmph}$$

18. (a) If the rowing speed in still water be $x \text{ kmph}$, and the distance by $y \text{ km}$, then

$$\frac{y}{x-2} = 6$$

$$\Rightarrow y = 6(x-2) \quad \dots(1)$$

$$\text{and, } \frac{y}{x+2} = 4$$

$$\Rightarrow y = 4(x+2) \quad \dots(2)$$

$$\Rightarrow 6(x-2) = 4(x+2)$$

$$\Rightarrow x = 10 \text{ kmph}$$

19. (a) d product of speed $\left[\frac{\text{difference of time}}{\text{difference of speed}} \right]$

$$d = \frac{4 \times 5 \left[10 - (-5) \right]}{60 \left[5 - 4 \right]} \quad \text{[Here, -ve sign indicates before the schedule time]}$$

$$\Rightarrow d = 5 \text{ km}$$

20. (a) Let the distance be $x \text{ km}$. Let speed of train be $y \text{ km/h}$. Then by question, we have

$$\frac{x}{y+4} = \frac{x}{y} - \frac{30}{60} \quad \dots(1)$$

$$\text{and } \frac{x}{y-2} = \frac{x}{y} + \frac{20}{60} \quad \dots(2)$$

On solving (1) and (2), we get $x = 3y$

Put $x = 3y$ in (1) we get

$$\frac{3y}{y+4} = 3 - \frac{1}{2} \Rightarrow y = 20$$

Hence, distance = $20 \times 3 = 60 \text{ km}$.

21. (a) Let each side of the square be $x \text{ km}$ and let the average speed of the plane around the field be $y \text{ km/h}$. Then,

$$\frac{x}{200} + \frac{x}{400} + \frac{x}{600} + \frac{x}{800} = \frac{4x}{y}$$

$$\Rightarrow \frac{25x}{2400} = \frac{4x}{y} \Rightarrow y = \left(\frac{2400 \times 4}{25} \right) = 384.$$

$\therefore$ Average speed = 384 km/h .

22. (c) Here, distance to be covered by the thief and by the owner is same.

Let after 2 : 30 p. m., owner catches the thief in t hrs.

$$\text{Then, } 60 \times t = 75 \left(t - \frac{1}{2} \right) \Rightarrow t = \frac{5}{2} \text{ hrs}$$

So, the thief is overtaken at 5 p.m.

23. (c) Let the speed of the cars be x km/h and y km/h, respectively.
Their relative speeds when they are moving in same direction = $(x - y)$ km/h.
Their relative speeds when they are in opposite directions = $(x + y)$ km/h.

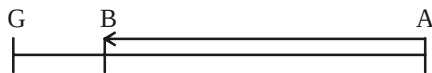
$$\text{Now, } \frac{70}{x + y} = 1 \text{ or } x + y = 70 \quad \dots(1)$$

$$\text{and } \frac{70}{(x - y)} = 7 \text{ or } x - y = 10 \quad \dots(2)$$

Solving (1) and (2), we have
 $x = 40$ km/h and $y = 30$ km/h.

24. (b) Volume of water flowed in an hour
= $2000 \times 40 \times 3$ cubic metre = 240000 cubic metre
 $\therefore$ volume of water flowed in 1 minute
= $\frac{240000}{60} = 4000$ cubic metre = 40,00,000 litre

25. (c)



In the above figure, the train travels from A to B in 11 : 30 minutes.

Suppose, you denote the time at which the first gunshot is heard as $t = 0$. Also, if you consider the travel of the sound of the second the gunshot is heard at point B at $t = 11 : 30$ minutes. Also, the second gunshot should reach point B at $t = 12$ minutes. Hence, the sound of the 2nd gunshot would take 30 seconds to travel from B to A.

$$\text{Thus, } \frac{S_{\text{train}}}{S_{\text{sound}}} = \frac{t_{\text{sound}}}{t_{\text{train}}}$$

$$S_{\text{train}} = 330 \times \frac{30}{690} = \frac{330}{23} \text{ m/s.}$$

26. (c) Initial distance = 25 dog leaps.
Per minute $\rightarrow$ dog makes 5 dog leaps
Per minute $\rightarrow$ Cat makes 6 cat leaps = 3 dog leaps.
Relative speed = 2 dog leaps/minutes.
An initial distance of 25 dog leaps would get covered in 12.5 minutes.
27. (b) Speed of train while passing point
 $A = 70 \times (5/18) \text{ m/s} = V_1$
Speed of bike initially = $70 \times (5/18) \text{ m/s} = V_2$
Time taken by the bike to reach at the mid-point of the train = $150/(V_2 - V_1)$
Again find out the new speeds of train and bike, and calculate the time taken by the bike to cover the rest 150 m distance relative to the train.
28. (a) Form the equations first and then use the options.

29. (b)

	Ram	:	Sham
Speed	7	:	4
Time	4	:	7
Distance	4	:	7

$$\text{Now, } 7x - 4x = 300$$

$$\text{Means } x = 100$$

Therefore, the winning post is $7 \times 100 = 700$ m away from the starting point

30. (d) The watch gains $(5 + 10) = 15$ min in 30 hours (12 Noon to 6 PM next day). This means that it will show the correct time when it gains 5 min in 10 hours or at 10 PM on Monday.
31. (b) Average speed when Pankaj was returning

$$= \frac{2 \times 10 \times 30}{40} = 15 \text{ km/hr}$$

Now the average speed of the whole journey

$$= \frac{2 \times 15 \times 60}{75} = 24 \text{ km/hr}$$

32. (b) The train needs to travel 15 minutes extra @35 kmph. Hence, it is behind by 8.75 kms. The rate of losing distance is 5 kmph. Hence, the train must have travelled for $8.75/5 = 1$ hour 45 minutes. @40 kmph $\rightarrow 70$ km.
Alternatively, you can also see that 12.5% drop in speed results in 14.28% increase in time. Hence, total time required is 105 minutes @ 40 kmph $\rightarrow 70$ kilometers.
Alternatively, solve through options.
33. (c) Let the distance between the school and the home be x km.

$$\text{Then, } \frac{x}{8} - \frac{2.5}{60} = \frac{x}{10} + \frac{5}{60} \text{ or } \frac{x}{8} - \frac{x}{10} = \frac{5}{60} + \frac{2.5}{60}$$

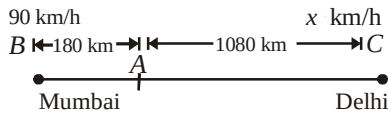
$$\text{or } \frac{2x}{80} = \frac{7.5}{60} \text{ or } x = \frac{7.5 \times 80}{2 \times 60} = 5 \text{ km}$$

34. (b) Relative speed of rockets
= $(42000 + 18000) = 60000$ mile/h
It means both of them together cover a distance of 60000 miles between themselves in 60 minutes or 1000 miles in 1 minute.
Hence, they should be 1000 miles apart, 1 minute before impact.
35. (c) Let the speed of the train be x m/sec. Then,
Distance travelled by the train in 10 min. = Distance travelled by sound in 30 sec.
 $\Leftrightarrow x \times 10 \times 60 = 330 \times 30$
 $\Leftrightarrow x = 16.5$.

$$\therefore \text{Speed of the train} = 16.5 \text{ m/sec} = \left(16.5 \times \frac{18}{5}\right) \text{ km/hr} = 59.4 \text{ km/hr}$$

36. (c) Let the speed of train C be x km/h.
At 9 p.m. the train A will have covered a distance of 180 km.
For trains A and B relative speed = $(90 - 60) = 30$ km/h
Distance between them = 180 km

$$\text{Time after which they meet} = \frac{180}{30} = 6 \text{ hrs}$$



For trains A and C relative speeds = $(60 + x)$ km/h
Distance between them = 1080 km.

$$\text{Time after which they meet} = \frac{1080}{(60 + x)} \text{ hrs}$$

As the time of meeting of all the three trains is the same, we have $\frac{1080}{(60 + x)} = 6$

$$\text{or } x = 120 \text{ km/h}$$

37. (b) Time taken by the boat during downstream

$$\text{journey} = \frac{50}{60} = \frac{5}{6} \text{ h}$$

$$\text{Time taken by the boat in upstream journey} = \frac{5}{4} \text{ h}$$

$$\text{Average speed} = \frac{2 \times 50}{\frac{5}{6} + \frac{5}{4}} = \frac{100 \times 24}{50} = 48 \text{ mph}$$

38. (c) Let the distance be x km. Then,
(Time taken to walk x km) + (Time taken to ride x km)
 $= \frac{23}{4}$ hrs.
 $\Rightarrow$ (Time taken to walk $2x$ km) + (Time taken to ride $2x$ km) = $\frac{23}{2}$ hrs.

$$\text{But, time taken to ride } 2x \text{ km} = \frac{15}{4} \text{ hrs.}$$

$$\therefore \text{Time taken to walk } 2x \text{ km} = \left(\frac{23}{2} - \frac{15}{4} \right) \text{ hrs} = \frac{31}{4} \text{ hrs}$$

$$= 7 \text{ hrs } 45 \text{ min.}$$

39. (a) Let the speed of the boatman be x km/hr and that of stream be y km/hr. Then

$$\frac{12}{x+y} - \frac{4}{x-y}$$

$$\Rightarrow 12x - 12y = 4x + 4y$$

$$\Rightarrow 8x = 16y \Rightarrow x = 2y$$

$$\text{Now } \frac{45}{x+y} + \frac{45}{x-y} = 20$$

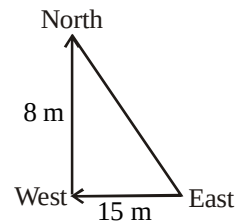
$$\Rightarrow 45 + 135 = 60y \Rightarrow 180 = 60y \Rightarrow y = 3 \text{ km/hr.}$$

40. (c) Required distance

$$= \sqrt{8^2 + 15^2}$$

$$= \sqrt{64 + 225}$$

$$\sqrt{289} = 17 \text{ m}$$



41. (b) Let the Speed of faster train be x and speed of slower train be y .

Now, when both the train move in same direction their relative speed = $x - y$

$$\text{Now, total distance covered} = 130 + 110 = 240$$

$$\text{Now, distance} = \text{speed} \times \text{time}$$

$$\therefore 240 = (x - y) \times 60 (\because 1 \text{ min } = 60 \text{ sec})$$

$$\Rightarrow x - y = 4 \quad \dots(1)$$

When the trains move in opposite direction

then their relative speed = $x + y$

$$\therefore 240 = (x + y) \times 3$$

$$\Rightarrow 80 = x + y \quad \dots(2)$$

on solving eqs (1) and (2), we get $x = 42$ m/sec

and $y = 38$ m/sec

42. (d) Let v_m = velocity of man = 48 m/min

Let v_c = velocity of current

then t_1 = time taken to travel 200 m against the current.

$$\text{i.e., } t_1 = \frac{200}{v_m - v_c} \quad \dots(1)$$

and t_2 time taken to travel 200 m with the current

$$\text{i.e., } t_2 = \frac{200}{v_m + v_c} \quad \dots(2)$$

$$\text{Given : } t_1 - t_2 = 10 \text{ min}$$

$$\therefore \frac{200}{v_m - v_c} - \frac{200}{v_m + v_c} = 10$$

$$\Rightarrow v_m^2 - v_c^2 = 40v_c \Rightarrow v_c^2 + 40v_c - (48)^2 = 0$$

$$\Rightarrow v_c = 32, -72$$

Hence, speed of the current = 32 ($\because v_c \neq -72$).

43. (c) Let the total distance to be travelled = x km

Speed of train = v km/h

and time taken = t hr.

$$\frac{150}{v} + \frac{x-150}{\left(\frac{3v}{5}\right)} = (t+8) \quad \dots(1)$$

$$\frac{510}{v} + \frac{x-510}{\frac{3}{5}v} = (t+4) \quad \dots(2)$$

Eq (2) – Eq (1)

$$\frac{510}{v} - \frac{150}{v} + \frac{x-510}{\frac{3}{5}v} - \frac{x-150}{\frac{3v}{5}} = -4$$

$$\frac{360}{v} - \frac{360 \times 5}{3v} = -4 \Rightarrow v = 60 \text{ km/hr.}$$

$$t = \frac{x}{60}$$

Put in eqn (1)

$$\frac{150}{60} + \frac{x-150}{\frac{3 \times 60}{5}} = \left(\frac{x}{60} + 8 \right)$$

$$\frac{5}{2} + \frac{x-150}{36} = \frac{x}{60} + 8$$

$$\frac{x-150}{36} - \frac{x}{60} = 8 - \frac{5}{2} = \frac{11}{2}$$

$$\frac{10x - 1500 - 6x}{360} = \frac{11}{2}$$

$$\Rightarrow 4x - 1500 = \frac{360 \times 11}{2} = 1980 \Rightarrow 4x = 3480$$

$$x = \frac{3480}{4} \text{ km} = 870 \text{ km}$$

44. (d) Let speed of current = v m/min

$$\frac{200}{48-v} - \frac{200}{48+v} = 10$$

$$20(48+v) - 20(48-v) = 48^2 - v^2$$

$$40v = 48^2 - v^2$$

$$v^2 + 40v - 2304 = 0$$

$$v = 32 \text{ m/min.}$$

45. (c) We know that, the relation in time taken with two different modes of transport is

$$t_{\text{walk both}} + t_{\text{ride both}} = 2(t_{\text{walk}} + t_{\text{ride}})$$

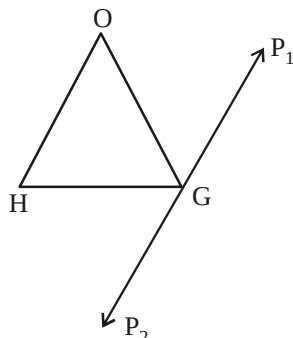
$$\frac{31}{4} + t_{\text{ride both}} = 2 \times \frac{25}{4}$$

$$\Rightarrow t_{\text{ride both}} = \frac{25}{2} - \frac{31}{4} = \frac{19}{4} = 4\frac{3}{4} \text{ hrs}$$

46. (d) Time difference between 8 am and 2 pm = 6 hrs.
Angle traced by the hour hand in 6 hours

$$\left(\frac{360}{12} \times 6 \right)^\circ = 180^\circ$$

47. (d)



From the figure above we see that Shyam would have walked a distance of $4 + 4 + 4 = 12$ km. (G to P_1 , P_1 to G and G to P_2).

48. (a) The dog loses $1/3$ rd of his normal time from the meeting point. (Thus normal time = $35 \times 3 = 105$ minutes)

If the meeting occurred 24 km further, the dog loses 25 minutes.

This means that the normal time for the new distance would be 75 minutes. Thus, normally the dog would cover this distance of 24 km in 30 minutes.

Thus, normal speed = 48 km/hr.

49. (d) This question gives us the freedom to assume any value of speeds of Ramesh and Somesh. Let us assume the initial speed of Somesh = 20 m/s, then the initial speed of Ramesh = 40 m/s.

Till 50 m they are running with this speed only. Time taken by Ramesh in covering 50m = 1.25sec.

In the same time Somesh is covering 25m. After this stages, speed of Somesh is 20m/s, whereas speed of Ramesh = 10 m/s. Now relative speed = 10m/s and distance = 25m. At 75m from the starting, both of them will be meeting.

50. (b) When A covers 1000 m, B covers 960 m.

Similarly, when B covers 1000 m, C covers 975 m.

$$\therefore \text{When B covers 960 m, C covers } \frac{975}{1000} \times 960 = 936 \text{ m.}$$

Thus, A can give a start to C by a distance = $(1000 - 936) \text{ m} = 64 \text{ m}$.

51. (a) In 2 minutes, he ascends = 1 metre
 $\therefore$ 10 metres, he ascends in 20 minutes.
 $\therefore$ He reaches the top in 21st minute.

$$52. (d) \frac{40}{(B-S)} + \frac{55}{(B+S)} = 13$$

$$\frac{30}{(B-S)} + \frac{44}{(B+S)} = 10$$

On solving these, we get $B = 8 \text{ km/h}$, $S = 5 \text{ km/h}$

$\therefore$ speed of Mallah in still water = 8 km/h

53. (c) Note here the length of the train in which passenger is travelling is not considered since we are concerned with the passenger instead of train. So, the length of the bridge will be directly proportional to the time taken by the passenger respectively.

$t \rightarrow$ Time

$l \rightarrow$ Length of bridge

$$\text{Therefore, } \frac{t_1}{t_2} = \frac{l_1}{l_2}$$

$$\frac{7}{4} = \frac{280}{2}$$

$$\Rightarrow x = 160 \text{ m}$$

54. (b) Downstream (Steamer) = 40 min
Downstream (Boat) = 60 min
Upstream (Steamer) = 60 min
Upstream (Boat) = 90 min
Required time = 40 + 30 + 45 = 115 min.
55. (c) Speed of tiger = 40 m/min
Speed of deer = 20 m/min
Relative speed = 40 - 20 = 20 m/min
Difference in distances = 50 × 8 = 400 m
 $\therefore$ Time taken in overtaking (or catching) = $\frac{400}{20} = 20$ min
 $\therefore$ Distance travelled in 20 min = 20 × 40 = 800 m
56. (d) $(6 - x) = (8 - 1.5x)$
 $\Rightarrow x = 4$ cm
So, it will take 4 hours to burn in such a way that they remain equal in length.
57. (c) The speeds of two persons is 108 km/h and 75 km/h.
The first person covers 1080 km in 10 hours and thus he makes 12 rounds. Thus, he will pass over another person 12 times in any one of the direction.
58. (c) Angle between two hands at 3 : 10 am
= $(90 + 5) - 60 = 35^\circ$
So, the required angle = 70° , after 3:10 am
Total time required to make 70° angle when minute-hand is ahead of hour-hand.
 $= \frac{90 + 70}{11/2} = \frac{320}{11}$ min
So at 3h $\frac{320}{11}$ min the required angle will be formed.
59. (d) $(n + 1)$ times in n days.
60. (c) If you start at 12 noon, you would reach at 4 : 30 PM.
You would be able to meet the train which left Mumbai at 8 AM, 9 AM, 10 AM, 11 AM, 12 Noon, 1 PM, 2 PM, 3 PM and 4 PM – a total of 9 trains.
61. (b) In 36 hours, there would be a gap of 8 minutes. The two watches would show the same time when the gap would be exactly 12 hours or 720 minutes.
The no. of 36 hour time frames required to create this gap = $720/8 = 90$.
Total time = $90 \times 36 = 3240$ hours. Since this is divisible by 24, the watches would show 12 noon.
62. (d) Assume the distance between the two ant holes is 600 feet. Then, the first ant's speed is 16 feet/hr while the second ant's speed is 14 feet/hr.
If the first ant covers 800 feet, the second will cover 700 feet (since, distance is proportional to speed).
Hence total distance is 1500 feet and required speed is $14 \times 2.5 = 35$ feet/hr.
63. (c) The net time loss is $1/3\%$ of 168 hours.

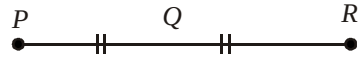
Expert Level

1. (d) Average of Kerosene $\frac{18}{900} \times 1000 = 20$ km / ltr.
Cost of petrol = ₹ 30 / ltr;
 $\therefore$ Cost of Kerosene $\frac{2}{3}$ of petrol = ₹ 10/ltr.
Let the quantity of Kerosene be x in 1 ltr. of mixture.
 $\therefore 20(x) + 40(1 - x) = 38$
 $\Rightarrow x = 0.1$ ltr.
 $\therefore$ Cost of mixture = $10(0.1) + 30(0.9) = 28/-$
Hence the additional amount that pump owner was charging = $30 - 28 = ₹ 2$.
2. (b) Distance (D) = Speed (S) × Time (T)
 $D = 4 \times \left(T + \frac{15}{60}\right)$
 $D = 4T + 1$... (1)
 $D = 6 \left(T - \frac{10}{60}\right)$
 $D = 6T - 1$... (2)
Solving equations (1) and (2), we get
 $T = 1$ h
 $D = 4 \times 1 + 1 = 5$ km
3. (d) Let the average speed be x km/h.
and Total distance = y km. Then,
 $\frac{0.2}{10}y + \frac{0.6}{30}y + \frac{0.2}{20}y = \frac{y}{x}$
 $\Rightarrow x = \frac{1}{0.05} = 20$ km / h
4. (d) Let the speed in return journey be x km/hr.
Then, speed in onward journey = $\frac{125}{100}x = \left(\frac{5}{4}x\right)$ km/hr.
Average speed = $\left(\frac{2 \times \frac{5}{4}x \times x}{\frac{5}{4}x + x}\right)$ km/hr = $\frac{10x}{9}$ km/hr.
 $\therefore \left(800 \times \frac{9}{10x}\right) = 16 \Leftrightarrow x = \left(\frac{800 \times 9}{16 \times 10}\right) = 45$.
So, speed in onward journey = $\left(\frac{5}{4} \times 45\right)$ km/hr
= 56.25 km/hr
5. (c) Let he walked for x hours, then
 $5x + 25(10 - x) = 17 \times 10$
 $\Rightarrow x = 4$
 $\therefore 10 - x = 6$ h
Hence, distance travelled by auto = $25 \times 6 = 150$ km.

6. (b) Let the speed of X and Y be the x km/h and y km/h respectively. Since they meet after 3 hours, so $x + y = 100$.

Since, the faster train takes at least $3 + 2 = 5$ hours to complete the 300 km journey. Hence, minimum possible speed for the slower train = 40 km/h at which speed it will take 7.5 h to complete the journey

$$\left(\begin{matrix} 7.5 & 300 \\ & 40 \end{matrix} \right)$$

7. (c) 

$$PQ = QR$$

$$P \rightarrow Q \rightarrow R (7h)$$

It means $P \rightarrow Q$ (3.5 h)

Again $\{P \rightarrow Q \text{ and } Q \rightarrow P\}$ (8 h)

It means $Q \rightarrow P$ (4.5 h)

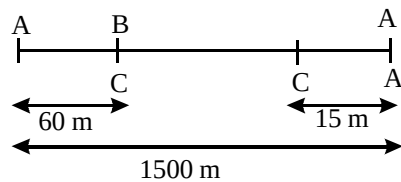
Therefore $R \rightarrow Q$ (4.5 h)

Thus, from R to P boat will take 9 hours

Hint: $P \rightarrow R$ (Downstream)

$R \rightarrow P$ (Upstream)

8. (c) Ratio of speed of A : B = 12 : 11
and ratio of speeds of B : C = 8 : 7
Therefore ratio of speeds of A : B : C = 96 : 88 : 77
So in 9600 m race A will beat C by 1900 m

9. (b) 

In the same time, when A covers 1500 m, B covers 1440 m and C covers 1425 m.

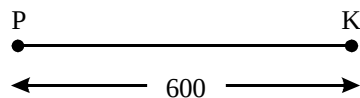
So, in 1440 m race B can give a start of 15 m.

$\therefore$ In 1500 m race B will give a start of

$$\frac{15}{1440} \times 1500 = 15 \frac{5}{8} \text{ m}$$

10. (a) Time taken to collide the two trains = $\frac{3}{2}$ h

So, in $\frac{3}{2}$ h bird travels $\frac{3}{2} \times 60 = 90$ km

11. (c) 

In 18 h plane will cover $18 \times 120 = 2160$ km

Now, $2160 = (600 \times 2) + 600 + 360$

So, the plane will be 360 km away from Kargil it means it will be $(600 - 360) = 240$ km away from Pukhwara.

12. (b) 
They will be together at every two hours. Therefore in 12h they will be $(6 + 1) = 7$ times together at P and they will never meet together at Q.

13. (c)

Speed of wind (Sound)
Relative speed of soldier and terrorist

$\frac{\text{Time utilised}}{\text{Difference in time}}$

$$\frac{1188}{x} = \frac{330}{5}$$

$$\Rightarrow x = 18 \text{ km/h}$$

14. (d) In 20 minutes the difference between man and his son = $20 \times 20 = 400$ m
Distance travelled by dog when he goes towards son

$$= \frac{400}{40} \times 60$$

= 600 m and time required is 10 minutes

In 10 minutes the remaining difference between man and son.

$$400 - (20 \times 10) = 200 \text{ m}$$

Total distance travelled by dog = $600 + 400 = 1000$ m

15. (b) Initial speed of police = 10 m/s
Increased speed of police = 20 m/s
Speed of thief = 15 m/s
Initial difference between thief and police = 250 m
After 5 seconds difference between thief and police = $250 - (5 \times 10) = 200$ m
After 10 seconds more the difference between thief and police = $200 + (5 \times 10) = 250$ m.
Now, the time required by police to catch the thief

$$= \frac{250}{5} = 50 \text{ s}$$

Distance travelled = $50 \times 20 = 1000$ m

Total time = $50 + 15 = 65$ s

Total distance = $1000 + (15 \times 10) = 1150$ m

16. (b) The ratio of speeds
= The ratio of distances, when time is constant.
 $\therefore$ The ratio of distances covered by leopard to the tiger = 12 : 25
Again, ratio of rounds made by leopard to the tiger = 12 : 25
Hence, leopard makes 48 rounds, when tiger makes 100 rounds.

17. (b) Since both rest for 6 seconds so when B is just about to start the journey A reaches there at the shallow end so they meet at they shallow end.

18. (b) **For the first watch:** When a watch creates the difference of 12 hours, it shows correct time.
So to create the difference of 12 h required time

$$= \frac{60 \times 12}{24} = 30 \text{ days}$$

For the second watch: To create the difference of 12 h required time.

$$= \frac{30 \times 12}{24} = 15 \text{ days}$$

So, after 30 days at the same time both watches show the correct time.

19. (a) You must know that a correct watch coincide just after

$$65\frac{5}{11} \text{ min.}$$

Therefore in every $65\frac{5}{11}$ hours the watch gains $\frac{2}{11}$

Hence, in 24 hours it will gain $\frac{2}{11} \times \frac{11}{720} \times 24 \times 60 = 4 \text{ min}$

20. (a) To exchange the position both hands to cover 360°

together. In one minute, hour-hand moves $\frac{1^\circ}{2}$ and in one minute, minute-hand moves 6° . Let the required time be t min, then

$$6t + \frac{1}{2}t = 360$$

$$\Rightarrow t = \frac{360}{13} \times 2 = \frac{720}{13} = 55\frac{5}{13} \text{ min}$$

21. (d) The minute-hand of a normal clock covers $\frac{360}{60} = 6^\circ$ per minute. The hour-hand of a normal clock covers

$\frac{30}{60} \frac{1^\circ}{2}$ per minute. So once they are together, in

every minute the minute hand gains $6 - \frac{1}{2} = \frac{11}{2}$ over the hour hand.

So, time between two meetings = $\frac{360}{\frac{11}{2}} = \frac{720}{11}$ minutes.

So, in any clock the hour-hand and the minute-hand

meet after every $\frac{720}{11}$ minutes.

If 60 minutes have passed in a normal clock then time passed in the faulty clock is 70 minutes.

If 24 hrs (or 24×60 minutes) have passed in a normal clock then time passed in the faulty clock must be $24 \times 70 = 1680$ minutes.

Number of times the hands meet –

$$\frac{1680}{\frac{720}{11}} = 25.67 = 25$$

22. (b) Let the speed for the first hour be x km/hr

then the speed for the second hour be $\frac{7}{5}x$ km/hr

then the speed for the third hour be

$$\frac{10}{7} \times \frac{7}{5}x = 2x \text{ km/hr}$$

then the speed for the fourth hour be

$$2x \times \frac{7}{5} = \frac{14x}{5} \text{ km/hr}$$

Therefore total distance in four hours

$$= x + \frac{7}{5}x + 2x + \frac{14x}{5} = \frac{36x}{5} \text{ km}$$

$$\therefore \text{Average speed} = \frac{\text{Total Distance}}{\text{Total time}} = \frac{\left(\frac{36x}{5}\right)}{4}$$

$$= \frac{9x}{5} \text{ km/hr}$$

Again the distance in 4 hours @ speed of x km/hr which is half of the third hour's speed is $4x$ km

$$\text{Hence } \frac{36x}{5} - 4x = 160 \text{ km}$$

$$\Rightarrow x = 50$$

Hence, the average speed = $\frac{9 \times 50}{5} = 90 \text{ km/hr}$

23. (c) If we assume the speed of the sound as 330 m/s, we can see that the distance traveled by the sound in 45 seconds is the distance traveled by the train in 11 minutes.

$$330 \times 45 = 660 \times s \rightarrow s = 22.5 \text{ m/s} = 81 \text{ kmph}$$

24. (a) The ratio of speeds of A to B would be 2 : 3.



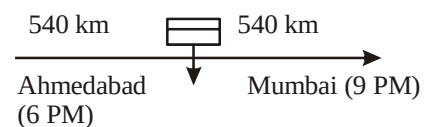
0.4 D means ratio of speeds = 2:3

The 4th meeting would occur after a combined movement of $D + 6D = 7D$. $2/5$ th of this distance would be covered by A and $3/5$ th of this distance would be the distance covered by B. Thus, distance covered by A would be $2/5$ th of $7D$ —: distance covered by A = $2.8D$ – which means that the 4th meeting occurs at a distance of $0.8 D$ from P.

25. (b) We can see that it takes them 4 hours to reach each other. And this is the same time for which bullets will cover some distance.

So, the total distance covered by the bullet = $4 \times 10 = 40 \text{ km}$

26. (d)



Now using options can get us the result.

Take the option 6 A.M. which means the train from Ahmedabad takes 12 hours to cover 540 km. In this way, the speed will be 45 km/h and train from Mumbai takes 9 hours to cover 540 km which means the speed is 60km/h: It is written in the question that the difference between the speed of the train from Ahmedabad and that from Mumbai is 15 km/h. Hence, this is the answer.

27. (a) Let the original speed of car = v km/hr.

$$\frac{715}{v} - \frac{715}{v+10} = 2$$

$$\frac{v+10-v}{v(v+10)} = \frac{2}{715}$$

$$v(v+10) = 715 \times 5$$

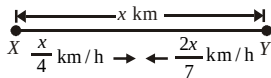
$$v^2 + 10v - 3575 = 0$$

$$(v+65)(v-55) = 0$$

$$\therefore v = 55 \text{ km/hr.}$$

28. (b) Let the distance between X and Y be x km. Then, the

speed of A is $\frac{x}{4}$ km/h and that of B is $\frac{2x}{7}$ km/h.



Relative speeds of the trains

$$= \left(\frac{x}{4} + \frac{2x}{7} \right) = \frac{15x}{28} \text{ km/h}$$

Therefore the distance between the trains at 7 a.m.

$$= x - \frac{x}{2} = \frac{x}{2} \text{ km}$$

Hence, time taken to cross each other

$$= \frac{\frac{x}{2}}{\frac{15x}{28}} \text{ hr} = \frac{x}{2} \times \frac{28}{15x} \text{ hr} = \frac{14}{15} \times 60 \text{ min} = 56 \text{ min}$$

Thus, both of them meet at 7.56 a.m.

29. (c) Let the speed of train be x km/h and actual time taken is t hrs.

$$\text{In first case, distance} = (x+6)(t-4) \text{ km} \quad \dots(1)$$

$$\text{In second case, distance} = (x-6)(t+6) \text{ km} \quad \dots(2)$$

Also distance = xt from (1) and (2)

$$(x+6)(t-4) = (x-6)(t+6) \quad \dots(3)$$

$$\Rightarrow \frac{x+6}{x-6} \cdot \frac{t-4}{t+6} = \frac{x}{x-6} \cdot \frac{2t+2}{10}$$

$$\Rightarrow \frac{x}{6} \cdot \frac{t+1}{5}$$

$$\Rightarrow 5x = 6t + 6 \Rightarrow 5x - 6t = 6$$

$$\Rightarrow t = \frac{5x-6}{6}$$

Putting the value of ' t ' in eqn. (3), we get

$$x = 30 \text{ km/hr}$$

$$\therefore t = 24 \text{ hr}$$

$$\text{Thus, distance} = 30 \times 24 = 720$$

Alternatively :

The speed difference between slow-speed and fast-speed train is 12 km/hr. and the time difference is 10

hrs. Speed difference of 12km/hr. hints that the distance should be divisible by 12. Only option (c) is divisible by 12.

By conventional method following equation will help solve the problem.

$$\frac{d}{s} = \frac{d}{s+12} + 10$$

Easier method is as follows. Speed difference of 12 km/hr hints that the distance should be divisible by 12. Only option (c) is divisible by 12.

$$\frac{720}{12} = 60 \text{ hrs.}, \frac{720}{24} = 30 \text{ hrs.}, \frac{720}{36} = 20 \text{ hrs.}$$

So, fastest speed is 36km/hr. slowest speed is 24km/hr.

30. (b) Let the speed of swimmer be x km/hr

When he swim with the flow

then speed = $(x + 3/2)$ km/h.

$$\therefore S_1 = \left(x + \frac{3}{2} \right) \times t$$

When he swim against the flow of stream

$$\text{then speed} = \left(x - \frac{3}{2} \right) t$$

$$\therefore S_2 = \left(x - \frac{3}{2} \right) t$$

According to the ques

$$S_1 = 2S_2$$

$$\left(x + \frac{3}{2} \right) t = 2 \left(x - \frac{3}{2} \right) t$$

$$\left(x + \frac{3}{2} \right) t = 2t \left(\frac{2x-3}{2} \right)$$

$$\Rightarrow \left(\frac{2x+3}{2} \right) = 2x-3$$

$$\Rightarrow 2x+3 = 4x-6 \Rightarrow 9 = 2x \Rightarrow x = \frac{9}{2} = 4\frac{1}{2} \text{ km/hr}$$

31. (a) Let V_A and t_A be the speed and time of A respectively. and V_B , t_B be the speed and time of B respectively.

Now, total (length) distance = 400 m

Now, B beats A by 15 metres.

$\therefore$ Distance covered by B = 400 - 15 and $t_B = t_A + 5$

$$\therefore V_B = \frac{400-15}{t_A+5} \quad \dots(1)$$

$$\text{Similarly, } V_A = \frac{400}{t_A}$$

$$\text{In another race of 400 m, } V_B = \frac{400}{t_A + \frac{50}{7}} \quad \dots(2)$$

Equations (1) and (2), we get

$$\frac{400-15}{t_A+5} = \frac{400}{t_A + \frac{50}{7}}$$

$$\Rightarrow 385 \left(t_A + \frac{50}{7} \right) = 400(t_A + 5)$$

$$\Rightarrow 77 \left(t_A + \frac{50}{7} \right) = 80(t_A + 5)$$

$$\Rightarrow 150 = 3t_A \Rightarrow t_A = 50 \text{ sec.}$$

$$\therefore V_A = \frac{400}{50} = 8 \text{ m/sec and } V_B = \frac{385}{55} = 7 \text{ m/sec.}$$

32. (d) Let the distance be x .

Ratio of speeds of 3 cars = 2 : 3 : 4

$$\therefore S_1 = \frac{2}{9}, S_2 = \frac{3}{9}, S_3 = \frac{4}{9}$$

Now, as we know, distance = speed $\times$ time

$$\therefore x = \frac{2}{9}T_1, x = \frac{3}{9}T_2, x = \frac{4}{9}T_3$$

$$\Rightarrow \frac{T_1}{x} = \frac{9}{2}, \frac{T_2}{x} = \frac{9}{3}, \frac{T_3}{x} = \frac{9}{4}$$

$$\therefore \frac{T_1}{x} : \frac{T_2}{x} : \frac{T_3}{x} = \frac{9}{2} : \frac{9}{3} : \frac{9}{4} = 108 : 72 : 54 = 6 : 4 : 3$$

$\therefore$ Required ratio = 6 : 4 : 3.

33. (a) Let the speed of train = v_1 km/h
and speed of taxi = v_2 km/h

$$\frac{300}{v_1} + \frac{200}{v_2} = \frac{11}{2} \Rightarrow \frac{3}{v_1} + \frac{2}{v_2} = \frac{11}{200} \quad \dots(1)$$

$$\frac{260}{v_1} + \frac{240}{v_2} = \frac{336}{60} \Rightarrow \frac{26}{v_1} + \frac{24}{v_2} = \frac{336}{600} \quad \dots(2)$$

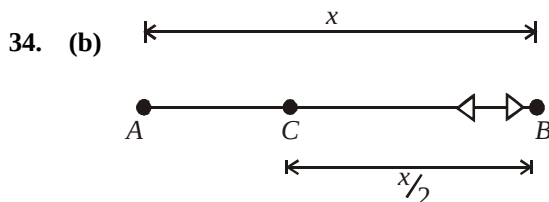
From eqs (1) and (2)

$$\frac{36}{v_1} + \frac{24}{v_2} = \frac{11 \times 12}{200}$$

$$\frac{36}{v_1} + \frac{24}{v_2} = \frac{336}{600}$$

$$\Rightarrow \frac{10}{v_1} - \frac{132}{200} - \frac{336}{600} = \frac{396 - 336}{600} = \frac{1}{10}$$

$$v_1 = 100 \text{ km/h}$$



Speed of boat in still water is 14 km/h.

Velocity of stream = 4 km/h.

Let the distance between point A and B = x km.

$$\frac{x}{18} + \frac{x}{2(10)} = 19$$

$$\frac{x}{18} + \frac{x}{20} = 19$$

$$\frac{20x + 18x}{360} = 19$$

$$38x = 19 \times 360$$

$$x = \frac{19 \times 360}{38}$$

$$\therefore x = 180 \text{ km/h}$$

35. (b) Let the speed of car = V km/h

then speed of bus = $V - 25$ km/h

Journey distance = 500 km

$$\text{Now, } \frac{500}{V - 25} - \frac{500}{V} = 10$$

$$\Rightarrow 500V - 500(V - 25) = V(V - 25) \times 10$$

$$\Rightarrow 500V - 500V + 12500 = (V^2 - 25V) 10$$

$$\Rightarrow V^2 - 25V - 1250 = 0$$

$$\Rightarrow V(V - 25) = 1250$$

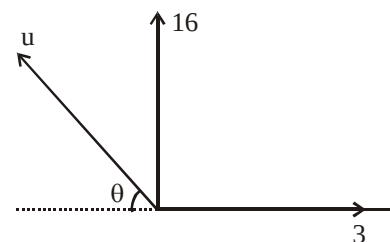
$$\Rightarrow V(V - 25) = 50 \times 25$$

$$\Rightarrow V = 50 \text{ km/h}$$

Speed of car = 50 km/h

Speed of bus = 25 km/h

36. (d)



Let the speed of the boat be u km per hour.

$$\therefore u \cos \theta = 3, u \sin \theta = 16$$

$$\Rightarrow \tan \theta = \frac{16}{3} \Rightarrow \sin \theta = \frac{16}{\sqrt{265}}$$

Since, $u \sin \theta = 16$

$$\Rightarrow u \cdot \frac{16}{\sqrt{265}} = 16$$

$$\Rightarrow u = \sqrt{265} = 16.28 \text{ km per hour}$$

$\therefore$ Speed of the boat against the current

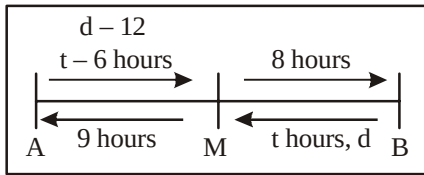
$$= u - 3 = 16.28 - 3 = 13.28 \text{ km per hour.}$$

37. (d) Since the second ant covers $7/120$ of the distance in 2 hours 30 minutes, we can infer that it covers $8.4/120 = 7\%$ of the distance in 3 hours. Thus, in 3 hours both ants together cover 15% of the distance $\rightarrow 5\%$ per hour $\rightarrow$ they will meet in 20 hours.

Also, ratio of speeds = 8 : 7.

So, the second ant would cover 700 ft to the meeting point in 20 hours and its speed would be 35 feet/hr.

38. (b) This is a complex trial and error based question and the way you would have to think in this is:

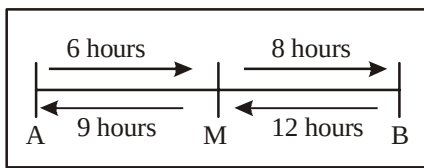


From the figure above, it is clear that A is faster as he takes only $t + 2$ hours while B has taken $t + 9$ hours to complete the journey.

Then, we get: $(t - 6)/9 = 8/t$

Solving for t , we get $t = -6$ (not possible)

Or $t = 12$. Putting this value of t in the figure it change to:



We also get ratio of speeds = 3 : 2 (inverse of ratio of times)

The next part of the puzzle is to think of the 12km less traveled by the first person till the meeting point.

If the speed of the faster person is $3s$, that of the slower person = $2s$.

Further

$$12 \times 2s - 6 \times 3s = 12 \text{ km}$$

$$s = 2 \text{ kmph.}$$

39. (c) Give that they meet in 80 minutes, when moving towards each other, the sum of their speeds should be such that they cover 1.25% of the distance per minute (i.e., 75% of the distance per hour).

40. (b) If the side of the initial equilateral triangle is S , then when Arjit covers $(S - 120)$ kms, Shaurya covers S kilometres. Also, when Arjit covers a distance of 80 kilometers, Shaurya covers a distance such that the resultant triangle is right angled.

Check these conditions through options.

41. (b) If S_1 is the speed of the first car, then $(S_1 + 10)$ will be the second car's speed. If t_1 hours is the time required for the first car, then $(t_1 - 1)$ hours is the time required for the second car in covering the same distance, while

$$\text{that of the third car is } \left(t_1 - \frac{33.33}{60} \right) \text{ hours.}$$

Check these conditions through options.

42. (c) Let the distance AC = d

$$\text{Then, } \frac{d}{V_0} + \frac{240-d}{V_1} = \frac{d}{V_1} + \frac{240-d}{V_0}$$

If $V_0 \neq V_1$, then the above condition will be satisfied only if $d = 120$ km.

43. (b) Suppose A and B are the points where the first and the second meeting took place.

The total distance covered by the pedestrian and the cyclist before the first meeting = Twice the distance between Nagpur and Buti Bori.

Total time taken is 1 hour.

Total distance cover by pedestrian and the cyclist between the two meetings = Twice the distance between A and Buti Bori.

and time taken is half an hour.

Hence, A is the mid-point. This will result in a GP.

44. (c) If $2d$ is the distance between A and B, then

$$\frac{\frac{2}{3} \times 2d}{d-2} = \frac{d-3}{2d \times \frac{1}{4}}$$

Explanation of Test Yourself

1. (a) The requisite conditions are met on a Pythagoras triplet 6, 8, 10. Since the racetrack only consists of the legs of the right triangle the length must be $6 + 8 = 14$ km.
 2. (b) Relative speed of the trains

$$= (42 + 48) \text{ kmph} = 90 \text{ kmph}$$

$$= \left(90 \times \frac{5}{18}\right) \text{ m/sec} = 25 \text{ m/sec.}$$

Time taken by the trains to pass each other

$$= \text{Time taken to cover } (137 + 163) \text{ m at } 25 \text{ m/sec}$$

$$= \left(\frac{300}{25}\right) \text{ sec} = 12 \text{ seconds.}$$
 3. (b) The sum of speeds would be 0.08 m/s (relative speed in opposite direction). Also if we go by option (b), the speeds will be 0.03 and 0.05 m/s respectively. At this speed the overlapping would occur every 60 seconds.
 4. (b) The movement of the ant in the two cases would be 3, 7, 11, 15, 19, 23 and 1, 9, 17, 25, 33, 41. It can be seen that after 3 seconds the difference is 6 mm after 4 seconds, the difference is 16 mm and after 5 seconds the difference is 30 mm. Thus, it is clearly seen that the ant moved for 4 seconds.
 5. (b) Let the distance between X and Y be x km. Then, the speed of A is $\frac{x}{4}$ km/h and that of B is $\frac{2x}{7}$ km/h.
-
- Relative speeds of the trains
- $$= \left(\frac{x}{4} + \frac{2x}{7}\right) = \frac{15x}{28} \text{ km/h}$$
- Therefore the distance between the trains at 7 a.m.
- $$= x - \frac{x}{2} = \frac{x}{2} \text{ km}$$
- Hence, time taken to cross each other
- $$= \frac{\frac{x}{2}}{\frac{15x}{28}} = \frac{x}{2} \times \frac{28}{15x} = \frac{14}{15} \times 60 = 56 \text{ min}$$
- Thus, both of them meet at 7 : 56 a.m.
6. (c) The distances covered in percentage would be,
 $10\% + 6.66\% + 8.33\% + 16.66\% + 5.833\% + 31.666\% + 2.0833 = 81.25\%$
 $(22.5/18.75) \times 100 = 120 \text{ km}$
 7. (d) The sum of the speeds of the ducks is 50 feet/min. Hence circumference $= 9 \times 50 = 450$ feet and difference of speeds $= \frac{450}{54} = 8.33$.
 $\therefore$ Speed of slower duck $= \frac{50 - 8.33}{2} = 20.83 \text{ feet/min.}$
 8. (c)

Both the athlete are crossing each other after 12 minutes which means the distance between them is 5 km. It will be easy to go through the ratio of the speed which is 2 : 3. The answer is 36 minutes.
 9. (c) Let the original speed be X km/h
According to the question, $18/(4/5x) - 18/x = 9/60$ hr
 $x = 30 \text{ km/h}$
 10. (a) Solve this through options as: For option (a)
 $4800/60 - 4800/50 = 16$ minutes
 11. (d) When Karan runs 100m, Arjun runs only 90m
So, in the new situation,
Karan has to run 110 m
Hence, distance covered by Arjun when Karan covers
- $$110 \text{ m} \times \frac{90}{100} = 99 \text{ m}$$
- Therefore, Karan beats Arjun by 1m
12. (a)

Time taken by Y for distance cover from B to C with stoppages

$$\left(\frac{6}{5} + \frac{1}{4}\right) \text{ hrs} = \frac{24+5}{20} = \frac{29}{20} \text{ hrs.}$$

Say they cross each other at x distance from A

$$\therefore \frac{x}{70} = \frac{29}{20} + \frac{120-x}{50}$$

$$\therefore \frac{x}{50} + \frac{x}{70} = \frac{29}{20} + \frac{12}{5}$$

$$\Rightarrow \frac{12x}{350} = \frac{29+48}{20} \Rightarrow \frac{12x}{35} = \frac{77}{2}$$

$$\therefore x = \frac{77}{2} \times \frac{35}{12} = 112.29 \approx 112 \text{ km}$$

13. (b) The only thing which matters in this problem is mileage or kms per litre of the fuel. At 50 kmph 195 kms can be covered. According to condition 1.3 times the fuel will be required at 75kmph.
Therefore, distance travelled will be $195/1.3 = 150$ kms.
14. (c) 1st day he climbing up at 6.00 a.m. and reached at 6.00 p.m.
2nd day he coming down at 6.00 a.m. and reached the foothill 6.00 p.m.
Hence, average speed of both path is same.
At noon it is not necessary that he was at same spot.
There must be a point where he reached at the same time on both the days.
15. (b) When watch, runs correct the minute hand should cross the hour hand once in every $65 + \frac{5}{11}$ minutes.

So, they should ideally cross 3 times once in

$$3 \times \left(\frac{720}{11} \right) = \frac{2160}{11} \text{ minutes} = 196.36 \text{ minutes.}$$

But in the watch under consideration, they meet after every 3 hour, 18 minutes and 15 seconds, i.e.,

$$\left(3 \times 60 + 18 + \frac{15}{60} \right) \times \frac{793}{4} \text{ minutes}$$

In 24 hours a watch has 1440 minutes.

Thus, our watch is actually losing time (as it is slower than the normal watch). Hence, when our watch

$$\text{elapsed} \left(1440 \times \frac{196.36}{198.25} \right) = 1426.27 \text{ minutes.}$$

Hence, the amount of time lost by our watch in one day = $(1440 - 1426.27) = 13.73$ i.e., 13 minutes and 44 seconds (approx).



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UNIT-III

Algebra

Chapter 11	Progressions
Chapter 12	Linear Equations
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11

PROGRESSIONS

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INTRODUCTION

For the CAT and CAT like aptitude tests this chapter is very important. The problems related to this chapter are solved using the logic of arithmetic progressions (more commonly) and geometric progressions. So many problems of this chapter are solved using the options. This chapter is actually the extension of the chapter number system.

ARITHMETIC PROGRESSIONS (A.P.)

A sequence of numbers which are either continuously increased or continuously decreased by a common difference found by subtracting any term of the sequence from the next term.

The following sequences of numbers are arithmetic progressions:

- (i) 5, 8, 11, 14, ...
- (ii) -6, -1, 4, 9, 14, ...
- (iii) 10, 7, 4, 1, -2, -5, ...
- (iv) $p, p + q, p + 2q, p + 3q, \dots$

In the arithmetic progression (i); 5, 8, 11 and 14 are first term, second term, third term and fourth term respectively. Common difference of this A.P. is found out either by subtracting 5 from 8, 8 from 11 or 11 from 14. Thus common difference = 3. Similarly, common difference of arithmetic progression (ii), (iii) and (iv) are 5, -3 and q respectively. First term and common difference of an A.P. are denoted by a and d respectively. Hence

- d of (i) A.P. = 3, d of (ii) A.P. = 5,
- d of (iii) A.P. = -3 and d of (iv) A.P. = q

n^{th} TERM OF AN A.P.

To find an A.P. if first term and common difference are given, we add the common difference to first term to get the second term and add the common difference to second term to get the third term and so on.

The standard for m of an A.P. is

$$a, a + d, a + 2d, a + 3d, \dots$$

Here ' a ' is the first term and ' d ' is the common difference. Also we see that coefficient of d is always less by one than the position of that term in the A.P. Thus n^{th} term of the A.P. is given by

$$T_n = a + (n - 1)d \quad \dots(1)$$

This equation (1) is used as a formula to find any term of the A.P.

If l be the last term of a sequence containing n terms, then

$$l = T_n = a + (n - 1)d$$

To find any particular term of any A.P., generally we put the value of a , n and d in the formula (i) and then calculate the required term.

For example to find the 25th term of the A.P. 6, 10, 14, 18, ... ; using the formula (i), we put the value of $a = 6$, $n = 25$ and $d = 4$ in formula and calculate as

$$T_{25} = 6 + (25 - 1) \times 4 = 6 + 24 \times 4 = 6 + 96 = 102$$

However if you consider the formula (i) as "To find the n^{th} term add the $(n - 1)$ times the common difference to the first term", you will get the answer much faster. See some examples:

To find 25th term of the A.P. 6, 10, 14, 18, ... ; we add 24 (one less than 25) times the common difference to the first term 6. Thus

$$T_{25} = 96 + 6 = 102$$

Similarly, 16th term of the A.P. 3, 8, 13, 18, ... ; $T_{16} = 75 + 3 = 78$
22nd term of the A.P. -16, -10, -4, 2, 8, ... ; $T_{22} = -16 + 126 = 110$

Illustration 1: In an A.P. if $a = -7.2$, $d = 3.6$, $a_n = 7.2$, then find the value of n .

Solution: $a_n = a + (n-1)d$
 $\Rightarrow 7.2 = -7.2 + (n-1)(3.6)$
 $\Rightarrow 14.4 = (n-1)(3.6)$
 $\Rightarrow n-1 = 4 \Rightarrow n = 5.$

Illustration 2: Which term of the A.P. 21, 42, 63, ... is 420 ?

Solution: $420 = a_n = a + (n-1)d$
[Here $a = 21$, $d = 42 - 21 = 21$]
 $= 21 + (n-1)21$
 $= 21n$
 $\therefore n = \frac{420}{21} = 20$
 $\therefore$ required term is 20th term.

Illustration 3: Is -150 a term of the A.P. 11, 8, 5, 2, ... ?

Solution: Here $a = 11$, $d = -3$
 $-150 = a_n = a + (n-1)d$
 $= 11 + (n-1)(-3)$
 $= 11 - 3n + 3$
 $= 14 - 3n$
 $3n = 14 + 150$
 $n = \frac{164}{3} = 54\frac{2}{3},$

which is not possible because n is +ve integer.
 $\therefore$ -150 is not a term of the given A.P.

SUM OF FIRST n TERMS OF AN A.P.

Sum of first n terms means sum of from first term to n^{th} term.

Consider an A.P. whose first term and common difference are 'a' and 'd' respectively. Sum of first n terms S_n of this A.P. is given by

$$S_n = \frac{n}{2} [2a + (n-1)d] \quad \dots(1)$$

If last term of an A.P. containing n terms be l , then n^{th} term $= l = a + (n-1)d$.

$$\therefore S_n = \frac{n}{2} [2a + (n-1)d] = \frac{n}{2} [a + \{a + (n-1)d\}]$$

$$\Rightarrow S_n = \frac{n}{2} (a + l) \quad \dots(2)$$

In any A.P., the terms equidistant from beginning and from end form a pair of corresponding terms.

Note that if we say only ' r^{th} term', then it means r^{th} term from beginning not from end. Consider an A.P. having six terms: 1, 4, 7, 10, 13, 16. There are three pair of corresponding terms.

First pair $\rightarrow$ First term from the beginning and first term from the end i.e., first term and last term, i.e., 1 and 16.

$$\text{Average of first pair of corresponding terms} = \frac{1+16}{2} = \frac{17}{2}$$

Sum of term numbers of the corresponding terms $= 1 + 6 = 7$

Second pair $\rightarrow$ Second term from the beginning and second term from the end i.e., second term and fifth term i.e., 4 and 13.

$$\text{Average of second pair of corresponding terms} = \frac{4+13}{2} = \frac{17}{2}$$

Sum of the term numbers of the corresponding terms $= 2 + 5 = 7$

Third pair $\rightarrow$ Third term from the beginning and third term from the end i.e., third term and fourth term i.e., 7 and 10.

$$\text{Average of third pair of corresponding terms} = \frac{7+10}{2} = \frac{17}{2}$$

Sum of term numbers of the corresponding terms $= 3 + 4 = 7$

Thus we see that average of all pairs of corresponding terms are the same. This is true for all A.P.s.

Also we see that sum of term numbers for the terms in a corresponding pair is one more than the number of terms of the A.P. This rule holds true for all A.P.s. Using this result, you can easily find the term number of the corresponding term of any term of an A.P. if its total number of terms is known.

For example, if an A.P. has 27 terms, then to find the term number of the corresponding term of 5th term, we subtract the 5 from 28 (one more than 27), which gives the 23 as the term number of the corresponding term of 5th term.

From equation (ii), sum of all terms of an A.P.,

$$S_n = \frac{n}{2} (a + l) = n \left(\frac{a + l}{2} \right)$$

$$\Rightarrow S_n = n \times (\text{Average of first and last term})$$

Since first and last terms are corresponding terms of all A.P. and average of all pairs of corresponding terms of an A.P. are the same, we can replace the 'average of first and last term' by 'average of any pair of corresponding terms'. Then sum of all terms of an A.P.,

$$S_n = n \times (\text{Average of any pair of corresponding terms})$$

$$S_n = (\text{No. of terms}) \times (\text{Average of any pair of corresponding terms})$$

If you consider the sum of all terms of an A.P. as product of average of any pair of corresponding terms and number of terms, then you will get the answer much faster. See some examples:

To find the sum of all terms of an A.P. of 31 terms whose 6th term is 26 and 11th term is 46. First we will find the common difference.

The difference between 11th term and 6th term is five times the common difference. Therefore, common difference $= \frac{46-26}{5} = 4$

Now, we find the term number of the corresponding term of 11th term, which $31 + 1 - 11 = 21$. Thus 21st term is the corresponding term of 11th term.

Now, we find the 21st term. To find it, we add 10 times the common difference to 11th term.

$$\text{Hence } 21^{\text{st}} \text{ term} = 46 + 10 \times 4 = 86.$$

Now sum of all terms $= 31$ times the average of 11th and 21st term

$$= 31 \times \left(\frac{46+86}{2} \right) = 31 \times 66$$

$$= 31 \times 60 + 31 \times 6 = 1860 + 186 = 2046$$

The whole process can be shown in a single line as

$$(46-26) \div 5 = 4 \rightarrow (31+1)-11 = 21 \rightarrow 46 + 10 \times 4 = 86 \rightarrow$$

$$31 \times \left(\frac{46+86}{2} \right) = 31 \times 66 = 31 \times 60 + 31 \times 6 = 1860 + 186 = 2046.$$

Illustration 4: Find the sum of an A.P. of 17 terms, whose 3rd term is 8 and 8th term is 28.

Solution: $(28 - 8) \div 5 = 4 \rightarrow (17 + 1) - 8 = 10 \rightarrow 28 + 2 \times 4 = 36 \rightarrow 17 \times \left(\frac{28 + 36}{2} \right) \rightarrow 17 \times 32 = 17 \times 30 + 17 \times 2 = 510 + 34 = 544$

Illustration 5: Find the sum of first 25 terms of an A.P. whose 4th term is 13 and 22nd term is 67.

Solution: 4th and 22nd terms are corresponding terms of an A.P. of 25 terms. Hence required sum $= 25 \times \left(\frac{13 + 67}{2} \right) = 25 \times 40 = 1000$.

CONSIDERING THE TERMS IN AN A.P.

If sum of three consecutive terms of an A.P. is given, then if required consider the three consecutive terms as $(a - d)$, a and $(a + d)$. This reduces one unknown d thereby making the solution easier.

Similarly, we consider the four consecutive terms as $(a - 3d)$, $(a - d)$, $(a + d)$, $(a + 3d)$ and five consecutive terms as $(a - 2d)$, $(a - d)$, a , $(a + d)$ and $(a + 2d)$; if their sums are given otherwise consider three terms as a , $a + d$, $a + 2d$; four terms as a , $a + d$, $a + 2d$, $a + 3d$ and five terms as a , $a + d$, $a + 2d$, $a + 3d$, $a + 4d$.

SPECIAL CASES OF A.P.s IN WHICH SUM UPTO DIFFERENT TERMS ARE THE SAME

Case I: When first term is -ve and common difference is +ve, then there will be a possibility of the sum of first n_1 terms being the same as the sum of first n_2 terms. However, this will not necessarily occur. This will be clear through the following examples:

- (a) -9, -6, -3, 0, 3, 6, 9, 12, ...
- (b) -14, -7, 0, 7, 14, 21, 28, ...
- (c) -8, -2, 4, 10, 16, ...
- (d) -16, -11, -6, -1, 4, 9, 14, 19, ...
- (e) -6, -4, -2, 0, 2, 4, 6, 8, 10, ...
- (f) -15, -9, -3, 3, 9, 15, ...

Check the above all A.P.s. Although first term of each A.P. is -ve but common difference of each A.P. is +ve. You can see in each of the A.P. (a), (b), (e) and (f); sum up to different number of terms can be the same but in each of the A.P. (c) and (d), sum up to different number of terms can not be the same.

In (a), $S_3 = S_4$, $S_2 = S_5$, $S_1 = S_6$

In (b), $S_2 = S_3$, $S_1 = S_4$

In (e), $S_3 = S_4$, $S_2 = S_5$, $S_1 = S_6$

In (f), $S_2 = S_4$, $S_1 = S_5$

You can observe that

- (i) Sum up to two different number of terms is the same if there are balances in the A.P. about the number zero.
- (ii) Sum of the term numbers upto which sums are equal is constant for a given A.P. as in the A.P. (a) $3 + 4 = 2 + 5 = 1 + 6 = 7$.
- (iii) In the case where 0 is a term of the A.P., the sum up to two different number of terms when equal then one term number is odd and other term number is even as in the A.P.s, (a), (b) and (e).

- (iv) In the case where 0 is not a term of the A.P., the sum up to two different number of terms when equal then either both term numbers are odd or both term numbers are even as in the A.P. (f).

See the use of one the observations in a problem in which sum of first 8 terms of an A.P. equals the sum of first 17 terms and we have to find the sum of first 25 terms of the A.P.

As you observe in (ii) sum of the term numbers up to which sums are equal is constant for a given A.P.

Therefore, $S_8 = S_{17} \Rightarrow S_{(8-8=0)} = S_{(17+8=25)}$

Since, $S_0 = S_{25}$, and we know that $S_0 = 0$

$\therefore S_{25} = 0$

Thus we find the solution in few seconds without any using formula and long calculation. We use the different observations in different problem situations as the one discussed.

Case II: When first term is +ve and common difference is -ve. The observations of this case are the same as of case-I. You can also see in the A.P.s:

- (a) 6, 3, 0, -3, -6, -9, ...
- (b) 4, 0, -4, -8, ...
- (c) 5, 1, -3, -7, ...
- (d) 10, 6, 2, -2, -6, -10, ...

ARITHMETIC MEAN OF n NUMBERS

Arithmetic mean of n numbers $a_1, a_2, a_3, a_4, \dots, a_n$

$$= \frac{a_1 + a_2 + a_3 + a_4 + \dots + a_n}{n}$$

To Find a given Number of Arithmetic Mean(s) between Two given Numbers

Between any two given numbers, it is always possible to insert any number of terms such that whole series thus formed shall be an A.P. The terms thus inserted are called Arithmetic Means.

(a) A Single Arithmetic Mean Between any Two Given Numbers

If A be the arithmetic mean between any two given numbers a and b ; then a, A, b will be in A.P.

$$\therefore b - A = A - a \Rightarrow A = \frac{a + b}{2}$$

(b) More Than One Arithmetic Mean Between Any Two Given Numbers

If a and b are two given numbers and n be the number of arithmetic means between them, then a will be the first term and b will be the $(n + 2)^{\text{th}}$ term of the A.P. formed when n arithmetic means inserted between a and b .

$\therefore b = a + (n + 1)d$, where d is the common difference

$$\Rightarrow d = \frac{b - a}{n + 1}$$

Since required arithmetic means are the second, third, fourth, ..., $(n + 1)^{\text{th}}$ term of the A.P. whose first term is a and common difference is $\frac{b - a}{n + 1}$, hence the required arithmetic means are

$$a + \frac{b - a}{n + 1}, a + \frac{2(b - a)}{n + 1}, a + \frac{3(b - a)}{n + 1}, \dots, a + \frac{n(b - a)}{n + 1}.$$

Illustration 6: Find three arithmetic means between 3 and 19.

Solution: $d = \frac{b-a}{n+1} = \frac{19-3}{3+1} = 4$

∴ First arithmetic mean = $3 + 4 = 7$
Second arithmetic mean = $7 + 4 = 11$
and Third arithmetic mean = $11 + 4 = 15$.

Illustration 7: Find the A.M. between $(x-y)$ and $(x+y)$.

Solution: Let A be A.M. between $(x-y)$ and $(x+y)$

∴ $A = \frac{x-y + x+y}{2} = x$

Illustration 8: For what value of n , A.M. between a and b is

$\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$?

Solution: A.M. between a and b is $\frac{a+b}{2}$

∴ $\frac{a^{n+1} + b^{n+1}}{a^n + b^n} = \frac{a+b}{2}$

⇒ $2(a^{n+1} + b^{n+1}) = (a^n + b^n)(a+b)$

⇒ $a^n = b^n$

∴ $n = 0$

GEOMETRIC PROGRESSION (G.P.)

A sequence of numbers whose each term (except first term) is found out by multiplying the just previous term by the same number. The number by which we multiply to any term to get its next term is called common ratio of the G.P.

For example, 5, 10, 20, 40, ... is a G.P. whose first term is 5, second term is 10, third term is 20 and so on. Its common ratio is 2, because to get any term (except first term) we multiply its just previous term by 2.

Common ratio is also found out by dividing any term (except first term) by its just previous term, thus

common ratio = $\frac{10}{5} = \frac{20}{10} = \frac{40}{20} = \dots = 2$

First term of a G.P. is denoted by ' a ' and its common ratio is denoted by r .

∴ $a = 5, r = 2$

Standard form of a G.P. is

$a, ar, ar^2, ar^3, ar^4, \dots$

(i) n^{th} term of a G.P., $a_n = ar^{n-1}$

(ii) Sum of first n terms of a G.P.,

$S_n = \frac{a(r^n - 1)}{r - 1}$, if $|r| > 1$

and $S_n = \frac{a(1 - r^n)}{1 - r}$, if $|r| < 1$

(iii) If $|r| < 1$, then sum of infinite terms of the G.P.,

$S_\infty = \frac{a}{1 - r}$

If $|r| \geq 1$, then sum of infinite terms cannot exist.

Illustration 9: Which term of the G.P. 2, 1, $\frac{1}{2}$, $\frac{1}{4}$, ... is $\frac{1}{128}$?

Solution: Let the n^{th} term be $\frac{1}{128}$. Then,

$a_n = \frac{1}{128}$

⇒ $ar^{n-1} = \frac{1}{128}$

⇒ $2\left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^7 \Rightarrow \left(\frac{1}{2}\right)^{n-2} = \left(\frac{1}{2}\right)^7$

∴ $n - 2 = 7$

⇒ $n = 9$.

Illustration 10: The third term of a G.P. is 4. Find the product of its first five terms.

Solution: Let a be the first term and r the common ratio. Then,

$a_3 = 4 \Rightarrow ar^2 = 4$

Product of first five terms = $a_1 a_2 a_3 a_4 a_5 = a(ar)(ar^2)(ar^3)(ar^4)$
= $a^5 r^{10} = (ar^2)^5 = (4)^5 = 1024$.

Illustration 11: Find the sum of the series

$2 + 6 + 18 + \dots + 4374$.

Solution: Required sum = $\frac{a(r^n - 1)}{r - 1} = \frac{(ar^{n-1})r - a}{r - 1}$

= $\frac{4374 \times 3 - 2}{3 - 1} = 6560$.

[Here $a = 2, r = 3, ar^{n-1} = 4374$]

Illustration 12: Find the sum of the G.P. $5 + 55 + 555 + \dots$ to n terms.

Solution:

$5 + 55 + 555 + \dots$ to n terms

= $\frac{5}{9} [9 + 99 + 999 + \dots \text{to } n \text{ terms}]$

= $\frac{5}{9} [(10 - 1) + (10^2 - 1) + (10^3 - 1) + \dots + (10^n - 1)]$

= $\frac{5}{9} [(10 + 10^2 + 10^3 + \dots + 10^n) - (1 + 1 + 1 \dots \text{to } n \text{ terms})]$

= $\frac{5}{9} \left[10 \times \frac{(10^n - 1)}{10 - 1} - n \right] = \frac{5}{81} [10^{n+1} - 10n - 10]$.

Illustration 13: Find the sum of all terms of the G.P.

$4, 2, 1, \frac{1}{2}, \frac{1}{2^2}, \frac{1}{2^3}, \dots$

Solution: $a = 4, |r| = \frac{1}{2} < 1$

Sum of all terms, $S_\infty = \frac{4}{1 - \frac{1}{2}} = \frac{4}{\frac{1}{2}} = 8$

Illustration 14: Find the sum of infinite terms of the G.P.

$\frac{4}{5}, -\frac{8}{5}, \frac{16}{5}, -\frac{32}{5}, \dots$

$$\text{Solution: } a = \frac{4}{5}, r = \frac{-8}{\frac{4}{5}} = -2$$

$$\therefore |r| = |-2| = 2 > 1$$

Hence sum of infinite terms of the given G.P. will not exist.

CONSIDERING THE TERMS IN A G.P.

If product of three consecutive terms of a G.P. is given, then if required consider the three consecutive terms of the G.P. as $\frac{a}{r}, a$ and ar . This reduces one unknown, r , which makes the solution easier.

Similarly, we consider the four consecutive terms as $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$ and five consecutive terms as $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$; if their products are given. Otherwise we consider three terms as a, ar, ar^2 ; four terms as a, ar, ar^2, ar^3 and five terms as a, ar, ar^2, ar^3, ar^4 .

Illustration 15: If the sum of three consecutive terms of a G.P. is 38 and their product is 1728, then find these three consecutive terms.

Solution: Let the three consecutive terms be $\frac{a}{r}, a, ar$. Then,

$$\frac{a}{r} \cdot a \cdot ar = 1728 \Rightarrow a^3 = 1728$$

$$\Rightarrow a = 12$$

$$\text{And } \frac{a}{r} + a + ar = 38$$

$$a \left(\frac{1}{r} + 1 + r \right) = 38$$

$$6r^2 - 13r + 6 = 0$$

$$r = 3/2 \text{ or } r = 2/3$$

$$\therefore \text{Numbers are } 8, 12, 18 \text{ or } 18, 12, 8.$$

GEOMETRIC MEAN OF n NUMBERS

Geometric mean of n positive numbers $a_1, a_2, a_3, a_4, \dots, a_n$ is $(a_1 \cdot a_2 \cdot a_3 \cdot a_4 \dots a_n)^{1/n}$.

To Find a given Number of Geometric Mean(s) Between two Given Numbers

Between any two given numbers, it is always possible to insert any number of terms such that whole series thus formed will be a G.P. The terms thus inserted are called Geometric Means.

(a) A Single Geometric Mean Between any Two Given Numbers

Let G be the geometric mean (G.M.) between any two given numbers a and b ; then a, G, b are in G.P.

$$\therefore \frac{b}{G} = \frac{G}{a}$$

$$\Rightarrow G = \sqrt{ab}$$

(b) More Than one Geometric Means Between Any Two Given Numbers

If a and b are two given numbers and n be the number of geometric means between them, then a will be the first term and b will be

the $(n+2)^{\text{th}}$ term of the G.P. formed when n geometric means inserted between a and b .

$$\therefore b = a \cdot r^{n+1} \Rightarrow r^{n+1} = \frac{b}{a} \Rightarrow r = \left(\frac{b}{a} \right)^{\frac{1}{n+1}};$$

where r is the common ratio.

Since required geometric means are the second, third, fourth, ..., $(n+1)^{\text{th}}$ term of the G.P. whose first term is a and common

ratio is $\left(\frac{b}{a} \right)^{\frac{1}{n+1}}$, hence required geometric means are

$$a \left(\frac{b}{a} \right)^{\frac{1}{n+1}}, a \left(\frac{b}{a} \right)^{\frac{2}{n+1}}, a \left(\frac{b}{a} \right)^{\frac{3}{n+1}}, \dots, a \left(\frac{b}{a} \right)^{\frac{n}{n+1}}$$

Illustration 16: Find 5 G.M. between 576 and 9.

Solution: Let G_1, G_2, G_3, G_4, G_5 be 5 G.M. between 576 and 9.

$$\therefore r = \left(\frac{9}{576} \right)^{\frac{1}{5+1}} = \left(\frac{1}{64} \right)^{\frac{1}{6}} = \frac{1}{2}$$

$$G_1 = ar = 576 \times \frac{1}{2} = 288; G_2 = ar^2 = 576 \times \frac{1}{4} = 144$$

$$G_3 = ar^3 = 576 \times \frac{1}{8} = 72, G_4 = ar^4 = 576 \times \frac{1}{16} = 36;$$

$$G_5 = ar^5 = 576 \times \frac{1}{32} = 18$$

HARMONIC PROGRESSION (H.P.)

Harmonic progression is defined as a sequence, reciprocal of whose terms in order are in A.P.

Thus, if $a, b, c, d, \dots$ are in H.P., then $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d}, \dots$ are in A.P.

The standard form of a H.P. is

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \dots$$

Remember that a, b, c are in H.P. $\Leftrightarrow b = \frac{2ac}{a+c}$

General Term of a H.P.

General term (n^{th} term) of a H.P. is given by $T_n = \frac{1}{a + (n-1)d}$

There is no formula and procedure for finding the sum of any number of terms in H.P.

Questions based on H.P. are generally solved by inverting the terms (i.e., converting H.P. into A.P.) and use of formula and properties of the A.P.

Harmonic Mean of n Numbers

Harmonic mean of n numbers (or quantities) $a_1, a_2, a_3, a_4, \dots, a_n$

$$= \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4} + \dots + \frac{1}{a_n}}$$

To Find a Harmonic Mean Between Two Given Numbers

Let H be the harmonic mean between two given numbers a and b ;

then a, H, b are in H.P. or $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ are in A.P.

$$\therefore \frac{1}{H} - \frac{1}{a} = \frac{1}{b} - \frac{1}{H} \Rightarrow \frac{2}{H} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow H = \frac{2ab}{a+b}$$

RELATIONS BETWEEN ARITHMETIC MEAN (A.M.), GEOMETRIC MEAN (G.M.) AND HARMONIC MEAN (H.M.)

If A , G , H are the arithmetic, geometric, and harmonic means between a and b , then we have

$$A = \frac{a+b}{2} \quad \dots(1)$$

$$G = \sqrt{ab} \quad \dots(2)$$

$$H = \frac{2ab}{a+b} \quad \dots(3)$$

Therefore, $A \times H = \frac{(a+b)}{2} \times \frac{2ab}{(a+b)} = ab = G^2$. Hence G is the geometric mean between A and H . From these results we see that

$$A - G = \frac{a+b}{2} - \sqrt{ab} = \frac{a+b-2\sqrt{ab}}{2}$$

$$= \left[\frac{\sqrt{a}-\sqrt{b}}{\sqrt{2}} \right]^2, \text{ which is positive if } a \text{ and } b \text{ are positive.}$$

Therefore, the arithmetic mean of any two positive numbers is greater than their geometric mean.

$$\text{Also } G^2 = AH$$

Hence G is the intermediate in value between A and H , therefore $A > G > H$.

Illustration 17: Find two numbers whose A.M. is 34 and G.M. is 16.

Solution: Let two numbers be a and b .

$$\text{A.M.} = 34 = \frac{a+b}{2} \Rightarrow a+b = 68 \quad \dots(1)$$

$$\text{G.M.} = 16 = \sqrt{ab} \Rightarrow ab = 256$$

$$a-b = \sqrt{(a+b)^2 - 4ab} = \sqrt{4624 - 4 \times 256} = \sqrt{3600}$$

$$\therefore a-b = 60 \quad \dots(2)$$

By (1) and (2)

$$a = 64, b = 4$$

$\therefore$ Required numbers are 64 and 4.

Convergent Series

Consider a series,

$$\frac{1}{5}, \frac{3}{5^2}, \frac{5}{5^3}, \frac{7}{5^4}, \dots$$

you can observe that subsequent terms of this series keep getting smaller. If taken to infinite terms, the sum of this series will reach a value which it will never cross i.e. the sum of this series reaches to a limit. Such type of series are called convergent series.

Some other examples of convergent series are

$$(a) \frac{1}{10}, \frac{2}{10^2}, \frac{3}{10^3}, \frac{4}{10^4}, \dots$$

$$(b) \frac{1}{1^2}, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{4^2}, \dots$$

$$(c) \frac{1}{3^2}, \frac{2}{3^3}, \frac{3}{3^4}, \frac{4}{5^5}, \dots$$

This type of series cannot be strictly said to be under the domain of progressions. But since questions on finding sum of infinite terms of convergent series are very commonly asked in CAT and CAT like competitive exams, we are studying this series in this chapter.

Let's see an example based on multiple choice.

Illustration 18: Sum of infinite terms of the series

$$1 + \frac{4}{7} + \frac{9}{7^2} + \frac{16}{7^2} + \frac{25}{7^4} + \dots, \text{ is}$$

$$(a) 27/14$$

$$(b) 21/13$$

$$(c) 49/27$$

$$(d) 256/147$$

Solution: There are two methods to solve the problem. One method requires lengthy mathematical process which we do not advise you.

The other process is one where we try to predict the approximate value of the sum by taking into account the first few significant terms. (This approach is possible to use because of the fact that in such series we invariably reach the point where the value of the next term becomes insignificant and does not add substantially to the sum). After adding the significant terms we are in a position to guess the approximate value of the sum of the series.

Let us look at the above question in order to understand the process.

In the given series the values of the terms are:

$$\text{First term} = 1$$

$$\text{Second term} = 4/7 = 0.57$$

$$\text{Third term} = 9/63 = 0.14$$

$$\text{Fourth term} = 16/343 = 0.04$$

$$\text{Fifth term} = 25/2401 = 0.01$$

Addition upto the fifth term is approximately 1.76

Options (b) and (d) are smaller than 1.76 in value and hence cannot be correct.

That leaves us with options (a) and (c)

Option (a) has a value of 1.92 approximately while option (c) has a value of 1.81 approximately.

At this point you need to make a decision about how much value the remaining terms of the series would add to 1.76 (sum of the first 5 terms)

Looking at the pattern we can predict that the sixth term will be $36/7^5 = 36/16807 = 0.002$ (approx.)

Value of the remaining terms are insignificant. So the answer will not reach 1.92 and will be restricted to 1.81.

Hence the correct option is (c).

Using the above process, solve some other questions of divergent series.

USEFUL RESULTS

- (i) If the same number be added or subtracted from each term of an A.P., the resulting terms are also in A.P. with the same common difference as earlier.

(ii) If each term of an A.P. be multiplied or divided by the same quantity x , then resulting terms are also in A.P., whose common difference is obtained on multiplying or dividing by x in the earlier common difference respectively.

(iii) (A) If we count in step of x from term number n_1 to n_2 including both term numbers n_1 and n_2 , we get $\frac{n_2 - n_1}{x} + 1$ terms.

For examples, (a) If we count in step of 2 from 24th to 64th term, including both 24th and 64th term, we get $\frac{64 - 24}{2} + 1 = 21$ terms.

(b) If we count in step of 1 from 20th to 46th term, including both 20th and 46th term, we get $\frac{46 - 20}{1} + 1 = 27$ terms

(B) If we count in steps x from term number n_1 to n_2 , including one of the term n_1 and n_2 , we get $\frac{n_2 - n_1}{x}$ terms.

For example, if we count in step of 3 from 13th term to 40th term, including 13th term but not 40th term, we get $= \frac{40 - 13}{3} = 9$ terms.

(C) If we count in steps x from term n_1 to n_2 , excluding the terms n_1 and n_2 , we get $\left(\frac{n_2 - n_1}{x} - 1\right)$ terms.

Illustration 19: Find the number of terms in the sequence 95, 99, 103, 107, ... 335.

Solution: Here numbers are counted in steps 4 ($= 99 - 95$).

$$\text{Hence required numbers} = \frac{335 - 95}{4} + 1 = \frac{240}{4} + 1 = 61$$

Illustration 20: Find how many terms of the sequence 114, 121, 128, ... below 245.

Solution: $\frac{245 - 114}{7} + 1 = \frac{131}{7} + 1 = 19.7$, which is not a natural number. This mean 245 is not a term of the given sequence. In such type of cases, we take greatest integer less than 19.7, which is 19.

(iv) (a) Sum of first n natural numbers

$$= 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

(b) Sum of first n odd natural numbers

$$= 1 + 3 + 5 + \dots + (2n-1) = n^2$$

(c) Sum of first n even natural numbers
 $= 2 + 4 + 6 + \dots + 2n = n(n+1)$

(d) Sum of odd numbers $\leq n$

$$= \begin{cases} \left(\frac{n+1}{2}\right)^2, & \text{if } n \text{ is odd} \\ \left(\frac{n}{2}\right)^2, & \text{if } n \text{ is even} \end{cases}$$

(e) Sum of even numbers $\leq n$

$$= \begin{cases} \frac{n}{2} \left(\frac{n}{2} + 1\right), & \text{if } n \text{ is odd} \\ \left(\frac{n-1}{2}\right) \left(\frac{n+1}{2}\right), & \text{if } n \text{ is even} \end{cases}$$

(v) (a) Sum of squares of first n natural numbers
 $= 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

(b) Sum of cubes of first n natural numbers

$$= 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

= Square of the sum of first n natural numbers.

(vi) (a) $T_n = S_n - S_{n-1}$

(b) For A.P., $d = S_2 - 2S_1$

(vii) (a) In an A.P., the sum of terms equidistant from the beginning and end is constant and equal to the sum of first term and last term.

(b) If in an A.P. sum of p terms is equal to sum of q terms, then sum of $(p+q)$ terms is zero.

(c) If in an A.P., p^{th} term is q and q^{th} term is p then n^{th} term is $(p+q-n)$.

(d) If in an A.P., sum of p terms is q and sum of q terms is p , then sum of $(p+q)$ terms is $-(p+q)$.

(viii) If each term of a G.P. be multiplied or divided by the same quantity, the resulting sequence will be also in G.P. with the same common ratio as before.

(ix) If $a, b, c, d, \dots$ are in G.P., then they are also in continue proportion.

$$\text{Since, } \frac{b}{a} = \frac{c}{b} = \frac{d}{c} = \dots = r \text{ (common ratio)}$$

$\Rightarrow \frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \dots = \frac{1}{r}$; which shows $a, b, c, d \dots$; are in continued proportion.

Hence quantities in continued proportion may be represented as $x, xr, xr^2, xr^3, \dots$

Practice Exercises



Foundation Level

- If the 4th term of an arithmetic progression is 14 and 12th term is 70, then the first term is
(a) -10 (b) -7
(c) +7 (d) +10
- The fourth, seventh and tenth terms of a G.P. are p, q, r respectively, then :
(a) $p^2 = q^2 + r^2$ (b) $q^2 = pr$
(c) $p^2 = qr$ (d) $pqr + pq + 1 = 0$
- Find the sum of all numbers in between 10–50 excluding all those numbers which are divisible by 8. (include 10 and 50 for counting.)
(a) 1070 (b) 1220
(c) 1320 (d) 1160
- Find the general term of the GP with the third term 1 and the seventh term 8.
(a) $(2^{3/4})^{n-3}$ (b) $(2^{3/2})^{n-3}$
(c) $(2^{3/4})^{3-n}$ (d) $(2^{3/4})^{2-n}$
- In an infinite geometric progression, each term is equal to 3 times the sum of the terms that follow. If the first term of the series is 8, find the sum of the series?
(a) 12 (b) 32/3
(c) 34/3 (d) Data inadequate
- How many 3-digit numbers are completely divisible by 6?
(a) 149 (b) 150
(c) 151 (d) 166
- $(11^2 + 12^2 + 13^2 + \dots + 20^2) = ?$
(a) 385 (b) 2485
(c) 2870 (d) 3255
- A sequence is generated by the rule that the n^{th} term is $n^2 + 1$ for each positive integer n . In this sequence, for any value $n > 1$, the value of $(n + 1)^{\text{th}}$ term less the value of n^{th} term is
(a) $2n^2 + 1$ (b) $n^2 + 1$
(c) $2n + 1$ (d) $n + 2$
- On March 1st 2016, Sherry saved ₹ 1. Everyday starting from March 2nd 2016, he save ₹ 1 more than the previous day. Find the first date after March 1st 2016 at the end of which his total savings will be a perfect square.
(a) 17th March 2016 (b) 18th April 2016
(c) 26th March 2016 (d) None of these
- A man arranges to pay off a debt of ₹ 3,600 in 40 annual instalments which form an AP. When 30 of the instalments are paid, he dies leaving one-third of the debt unpaid. Find the value of the first instalment.
(a) 55 (b) 53
(c) 51 (d) 49
- A number 15 is divided into three parts which are in AP and the sum of their squares is 83. Find the smallest number.
(a) 5 (b) 3
(c) 6 (d) 8
- A boy agrees to work at the rate of one rupee on the first day, two rupees on the second day, four rupees on the third day and so on. How much will the boy get if he starts working on the 1st of February and finishes on the 20th of February ?
(a) 2^{20} (b) $2^{20} - 1$
(c) $2^{19} - 1$ (d) 2^{19}
- What is the sum of all the two-digit numbers which when divided by 7 gives a remainder of 3?
(a) 94 (b) 676
(c) 696 (d) None of these
- If $1 + 10 + 10^2 + \dots$ upto n terms $= \frac{10^n - 1}{9}$, then the sum of the series $4 + 44 + 444 + \dots$ upto n term is
(a) $\frac{4}{9}(10^n - 1) - \frac{4n}{9}$ (b) $\frac{4}{81}(10^n - 1) - \frac{4n}{9}$
(c) $\frac{40}{81}(10^n - 1) - \frac{4n}{9}$ (d) $\frac{40}{9}(10^n - 1) - \frac{4n}{9}$
- A man starts going for morning walk every day. The distance walked by him on the first day was 2 kms. Everyday he walks half of the distance walked on the previous day. What can be the maximum total distance walked by him in his life time?
(a) 4 kms. (b) 120 kms.
(c) 18 kms. (d) Data inadequate
- If sixth term of a H. P. is $\frac{1}{61}$ and its tenth term is $\frac{1}{105}$, then the first term of that H.P. is
(a) $\frac{1}{28}$ (b) $\frac{1}{39}$
(c) $\frac{1}{6}$ (d) $\frac{1}{17}$
- The sum of the 6th and 15th terms of an arithmetic progression is equal to the sum of 7th, 10th and 12th terms of the same progression. Which term of the series should necessarily be equal to zero ?
(a) 10th (b) 8th
(c) 1st (d) None of these

18. In a geometric progression the sum of the first and the last term is 66 and the product of the second and the last but one term is 128. Determine the first term of the series.
 (a) 64 (b) 64 or 2
 (c) 2 or 32 (d) 32
19. Four geometric means are inserted between $1/8$ and 128. Find the third geometric mean.
 (a) 4 (b) 16
 (c) 32 (d) 8
20. How many terms of the series $1 + 3 + 5 + 7 + \dots$ amount to 123454321?
 (a) 11101 (b) 11011
 (c) 10111 (d) 11111
21. An equilateral triangle is drawn by joining the midpoints of the sides of another equilateral triangle. A third equilateral triangle is drawn inside the second one joining the midpoints of the sides of the second equilateral triangle, and the process continues infinitely. Find the sum of the perimeters of all the equilateral triangles, if the side of the largest equilateral triangle is 24 units.
 (a) 288 units (b) 72 units
 (c) 36 units (d) 144 units
22. The sum to infinity of the progression $9 - 3 + 1 - \frac{1}{3} + \dots$ is
 (a) 9 (b) $9/2$
 (c) $27/4$ (d) $15/2$
23. The sequence $[x_n]$ is a GP with $x_2/x_4 = 1/4$ and $x_1 + x_4 = 108$. What will be the value of x_3 ?
 (a) 42 (b) 48
 (c) 44 (d) 56
24. The 1st, 8th and 22nd terms of an AP are three consecutive terms of a GP. Find the common ratio of the GP, given that the sum of the first twenty-two terms of the AP is 385.
 (a) Either 1 or $1/2$ (b) 2
 (c) 1 (d) Either 1 or 2
25. If the m th term of an AP is $1/n$ and n th term is $1/m$, then find the sum to mn terms.
 (a) $(mn - 1)/4$ (b) $(mn + 1)/4$
 (c) $(mn + 1)/2$ (d) $(mn - 1)/2$
26. Find the value of $1 - 2 - 3 + 2 - 3 - 4 + \dots$ upto 100 terms.
 (a) -694 (b) -626
 (c) -624 (d) -549
27. If $\log\left(\frac{5c}{a}\right)$, $\log\left(\frac{3b}{5c}\right)$ and $\log\left(\frac{a}{3b}\right)$ are in an A.P., where a , b and c are in a GP, then a , b and c , are the lengths of sides of
 (a) An isosceles triangle
 (b) An equilateral triangle
 (c) A scalene triangle
 (d) None of these
28. If $\frac{1}{x} + \frac{1}{z} + \frac{1}{x-y} + \frac{1}{z-y} = 0$, Which of the following statements is true?
 (a) x, y, z are in H.P. or $x, \frac{y}{2}, z$ are in A.P.
 (b) x, y, z are in A.P. or x, y, z are in H.P.
 (c) $x, \frac{y}{2}, z$ are in H.P. or x, y, z are in G.P.
 (d) x, y, z are in G.P. or x, y, z are in A.P.
29. Let $n > 1$, be a positive integer. Then the largest integer m , such that $(n^m + 1)$ divides $(1 + n + n^2 + n^3 + \dots + n^{127})$ is :
 (a) 127 (b) 63
 (c) 64 (d) 32
30. The sum of an infinite GP whose common ratio is numerically less than 1 is 32 and the sum of the first two terms is 24. What will be the third term ?
 (a) 2 (b) 16
 (c) 8 (d) 4
31. The sum of the series $1/(\sqrt{2} + \sqrt{1}) + 1/(\sqrt{2} + \sqrt{3}) + \dots + 1/(\sqrt{120} + \sqrt{121})$ is:
 (a) 10 (b) 11
 (c) 12 (d) None of these
32. What will be the value of $x^{1/2} \cdot x^{1/4} \cdot x^{1/8} \dots$ to infinity.
 (a) x^2 (b) x
 (c) $x^{3/2}$ (d) x^3
33. Find the sum of n terms of the series $11 + 103 + 1005 + \dots$
 (a) $10/9(10^n - 1) - 1$
 (b) $100/99(10^n - 1) + n^2$
 (c) $10/9(10^n - 1) + n^2$
 (d) None of these
34. Three distinct numbers x, y, z , form a GP in that order and the numbers $x + y, y + z, z + x$ form an AP in that order. Find the common ratio of the GP.
 (a) 1 (b) -2
 (c) 2 (d) Either (a) or (b)
35. Two AMs. A_1 and A_2 , two GMs. G_1 and G_2 and two HMs. H_1 and H_2 are inserted between any two numbers. Then find the arithmetic mean between H_1 and H_2 in terms of A_1, A_2, G_1, G_2 .
 (a) $\frac{A_1 + A_2}{2G_1G_2}$ (b) $\frac{A_1 - A_2}{2G_1G_2}$
 (c) $\frac{A_1 + A_2}{2 + G_1G_2}$ (d) $\frac{G_1 + G_2}{2A_1A_2}$
36. The sum of all terms of the arithmetic progression having ten terms except for the first term, is 99, and except for the sixth term, 89. Find the third term of the progression if the sum of the first and the fifth term is equal to 10.
 (a) 15 (b) 5
 (c) 8 (d) 10

Standard Level

- Three numbers a, b, c , non-zero, form an arithmetic progression. Increasing a by 1 or increasing c by 2 results in a geometric progression. Then b equals :
 (a) 16 (b) 14
 (c) 12 (d) 10
 - If a, b, c are three unequal numbers such that a, b, c are in A.P. and $b - a, c - b, a$ are in G.P. then $a : b : c$ is
 (a) 1 : 2 : 3 (b) 3 : 4 : 5
 (c) 2 : 3 : 4 (d) 5 : 7 : 9
 - Each of the series $13 + 15 + 17 + \dots$ and $14 + 17 + 20 + \dots$ is continued to 100 terms. Find how many terms are identical between the two series?
 (a) 35 (b) 34
 (c) 32 (d) 33
 - If $\log_x a, a^{x/2}$, and $\log_b x$ are in GP, then x is
 (a) $\log_a (\log_b a)$
 (b) $\log_a (\log_e a) + \log_a (\log_e b)$
 (c) $-\log_a (\log_b a)$
 (d) $\log_a (\log_e b) - \log_a (\log_e a)$
 - Determine the value of :

$$\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots + \frac{1}{\sqrt{120} + \sqrt{121}}$$
 (a) $\sqrt{120}$ (b) 10
 (c) $12\sqrt{12}$ (d) 8
 - $(666 \dots n \text{ digits})^2 + (888 \dots n \text{ digits})$ is equal to
 (a) $(10^n - 1) \times \frac{4}{9}$
 (b) $(10^{2n} - 1) \times \frac{4}{9}$
 (c) $\frac{4(10^n - 10^{n-1} - 1)}{9}$
 (d) $\frac{4(10^n + 1)}{9}$
 - If $x + y + z = 1$ and x, y, z are positive numbers such that $(1 - x)(1 - y)(1 - z) \geq kxyz$, then $k =$
 (a) 2 (b) 4
 (c) 8 (d) 16
 - The sum of thirty-two consecutive natural numbers is a perfect square. What is the least possible sum of the smallest and the largest of the thirty-two numbers?
 (a) 81 (b) 36
 (c) 49 (d) 64
 - If a, b and c are in HP, then $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in.
 (a) AP
 (b) GP
 (c) HP
 (d) Cannot be determined uniquely
 - The middle term of arithmetic series 3, 7, 11...147, is
 (a) 71 (b) 75
 (c) 79 (d) 83
 - If a man saves ` 4 more each year than he did the year before and if he saves ` 20 in the first year, after how many years will his savings be more than ` 1000 altogether?
 (a) 19 years (b) 20 years
 (c) 21 years (d) 18 years
 - What is the maximum sum of the terms in the arithmetic progression 25, $24\frac{1}{2}$, 24,?
 (a) $637\frac{1}{2}$ (b) 625
 (c) $662\frac{1}{2}$ (d) 650
 - $\left(1 - \frac{1}{n}\right) + \left(1 - \frac{2}{n}\right) + \left(1 - \frac{3}{n}\right) + \dots$ upto n terms = ?
 (a) $\frac{1}{2}n$ (b) $\frac{1}{2}(n-1)$
 (c) $\frac{1}{2}n(n-1)$ (d) None of these
 - If $1^3 + 2^3 + \dots + 9^3 = 2025$, then the value of $(0.11)^3 + (0.22)^3 + \dots + (0.99)^3$ is close to:
 (a) 0.2695 (b) 0.3695
 (c) 2.695 (d) 3.695
 - How many terms are identical in the two APs 1, 3, 5, ... up to 120 terms and 3, 6, 9, ... up to 80 terms?
 (a) 38 (b) 39
 (c) 40 (d) 41
 - If the sum of the first $2n$ terms of the AP 2, 5, 8 ... is equal to the sum of first n terms of the AP 57, 59, 6 ..., then what is the value of n ?
 (a) 7 (b) 9
 (c) 11 (d) 13
- Directions (Questions. 17 – 18) :** It is possible to arrange eight of the nine numbers 2, 3, 4, 5, 7, 10, 11, 12, 13 in the vacant squares of the 3 by 4 array shown below so that the arithmetic average of the numbers in each row and column is the same integer.
- | | | | |
|---|---|----|----|
| 1 | | | 15 |
| | 9 | | |
| | | 14 | |
- The arithmetic average is
 (a) 6 (b) 7
 (c) 8 (d) 9

18. Which one of the nine numbers must be left out when completing the array ?

- (a) 4 (b) 5
(c) 7 (d) 10

Directions (Qs. 19-22) : Seven integers A, B, C, D, E, F and G are to be arranged in an increasing order such that

- I. First four numbers are in arithmetic progression.
II. Last four numbers are in geometric progression.
III. There exists one number between E and G .
IV. There exist no numbers between A and B .
V. D is the smallest number and E is the greatest.

VI. $\frac{A}{D} \cdot \frac{G}{C} = \frac{F}{A} > 1$

VII. $E = 960$

19. $\frac{E}{A} = ?$

- (a) 2 (b) 8
(c) 4 (d) 5

20. $D = ?$

- (a) 30 (b) 25
(c) 22 (d) 20

21. The common difference in the A.P. is

- (a) 20 (b) 22
(c) 25 (d) 30

22. The position and value of A is

- (a) 5th highest and 100
(b) 4th highest and 100
(c) 4th highest and 110
(d) None of these

23. If the positive real numbers a, b and c are in Arithmetic Progression, such that $abc = 4$, then minimum possible value of b is :

(a) $2^{\frac{3}{2}}$

(b) $2^{\frac{2}{3}}$

(c) $2^{\frac{1}{3}}$

(d) None of these

24. After striking a floor a rubber ball rebounds $(7/8)^{\text{th}}$ of the height from which it has fallen. Find the total distance that it travels before coming to rest, if it is gently dropped from a height of 420 meters?

- (a) 2940 (b) 6300
(c) 1080 (d) 3360

25. The sum of $\frac{1}{1^3} + \frac{2}{1^3 + 2^3} + \frac{2}{1^3 + 2^3} + \frac{3}{1^3 + 2^3 + 3^3} + \dots$ upto n terms is equal to

(a) $\frac{n-1}{n}$

(b) $\frac{n}{n+1}$

(c) $\frac{n+1}{n+2}$

(d) $\frac{n+1}{n}$

26. The sum of the first three terms of the arithmetic progression is 30 and the sum of the squares of the first term and the second term of the same progression is 116. Find the seventh term of the progression if its fifth term is known to be exactly divisible by 14.

- (a) 36 (b) 40
(c) 43 (d) 22

27. The sum of an infinite GP is 162 and the sum of its first n terms is 160. If the inverse of its common ratio is an integer, then how many values of common ratio is/are possible, common ratio is greater than 0?

- (a) 0 (b) 1
(c) 2 (d) 3

Expert Level

1. In a list of 7 integers, one integer, denoted as x is unknown. The other six integers are 20, 4, 10, 4, 8 and 4. If the mean, median, and mode of these seven integers are arranged in increasing order, they form an arithmetic progression. The sum of all possible ways of x is
- (a) 26 (b) 32
(c) 40 (d) 38

Directions (Questions. 2 and 3) :

Let $A_1, A_2, \dots, A_n$ be the n points on the straight-line $y = px + q$. The coordinates of A_k is (x_k, y_k) , where $k = 1, 2, \dots, n$ such that $x_1, x_2, \dots, x_n$ are in arithmetic progression. The coordinates of A_2 is $(2, -2)$ and A_{24} is $(68, 31)$.

2. The y-ordinates of A_8 is
- (a) 13 (b) 10
(c) 7 (d) 5.5
3. The number of point(s) satisfying the above mentioned characteristics and not in the first quadrant is/are
- (a) 1 (b) 2
(c) 3 (d) 7
4. Let $\{A_n\}$ be a unique sequence of positive integers satisfying the following properties:
 $A_1 = 1, A_2 = 2, A_4 = 12$, and $A_{n+1} \cdot A_{n-1} = A_n^2 \pm 1$ for $n = 2, 3, 4, \dots$. Then, A_7 is
- (a) 60 (b) 120
(c) 149 (d) 169
5. If $X = \frac{a}{(1+r)} + \frac{a}{(1+r)^2} + \dots + \frac{a}{(1+r)^n}$, then what is the value of $a + a(1+r) + \dots + a(1+r)^{n-1}$?
- (a) $X[(1+r) + (1+r)^2 + \dots + (1+r)^n]$
(b) $X(1+r)^n$
(c) $X[(1+r)^n - 1/r]$
(d) $X(1+r)^{n-1}$
6. Suppose a, b and c are in Arithmetic Progression and a^2, b^2 , and c^2 are in Geometric Progression. If $a < b < c$ and $a + b + c = \frac{3}{2}$ then the value of $a =$
- (a) $\frac{1}{2\sqrt{2}}$ (b) $\frac{1}{2\sqrt{3}}$
(c) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (d) $\frac{1}{2} - \frac{1}{\sqrt{2}}$
7. In a nuclear power plant a technician is allowed an interval of maximum 100 minutes. A timer with a bell rings at specific intervals of time such that the minutes when the timer rings are not divisible by 2, 3, 5 and 7. The last

alarm rings with a buzzer to give time for decontamination of the technician. How many times will the bell ring within these 100 minutes and what is the value of the last minute when the bell rings for the last time in a 100 minute shift?

(a) 25 times, 89 (b) 21 times, 97
(c) 22 times, 97 (d) 19 times, 97

8. If $\sum_{r=1}^{\infty} \frac{1}{(2r-1)^2} = \frac{\pi^2}{8}$ then the value of $\sum_{r=1}^{\infty} \frac{1}{r^2}$ is
- (a) $\frac{\pi^2}{8} + \frac{1}{4}x$ (b) $\frac{\pi^2}{4} + \frac{1}{3}x$
(c) $\frac{\pi^2}{8} - \frac{1}{4}x$ (d) $\frac{\pi^2}{4} + \frac{2}{3}x$
9. The internal angles of a plane polygon are in AP. The smallest angle is 100° and the common difference is 10° . Find the number of sides of the polygon?
- (a) 8 (b) 9
(c) either 8 or 9 (d) None of these
10. Ramesh starts a simple calculation. He multiplies the integers $1, 2, 3, \dots, n$ two at a time and adds up the products. If n is 10, the final sum will be
- (a) 1320 (b) 2640
(c) 660 (d) 782
11. If A is the sum of the squares of the first n natural numbers (where $n < 100$), then for how many values of n will A be divisible by 5?
- (a) 40 (b) 60
(c) 59 (d) 39
12. If a, b and c are distinct positive real numbers and $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ is
- (a) less than 1 (b) equal to 1
(c) greater than 1 (d) any real number
13. If the 10th term of the sequence, $a, a-b, a-2b, a-3b, \dots$ is 20 and the 20th term is 10, then the x th term of the series is
- (a) $10-x$ (b) $20-x$
(c) $29-x$ (d) $30-x$
14. Two numbers A and B are such that their GM is 20% lower than their AM. Find the ratio between the numbers.
- (a) 3 : 2 (b) 4 : 1
(c) 2 : 1 (d) 3 : 1
15. If a, b, c, d, e, f are in A.P., then $e - c$ is equal to
- (a) $2(c-a)$ (b) $2(d-c)$
(c) $2(f-d)$ (d) $(d-c)$
16. If $\frac{a_2 a_3}{a_1 a_4} = \frac{a_2 + a_3}{a_1 + a_4} = 3 \left(\frac{a_2 - a_3}{a_1 - a_4} \right)$ then a_1, a_2, a_3, a_4 are in
- (a) A.P. (b) G.P.
(c) H.P. (d) None of these

17. A number of saplings are lying at a place by the side of a straight road. These are to be planted in a straight line at a distance interval of 10 meters between two consecutive saplings. Mithilesh, the country's greatest forester, can carry only one sapling at a time and has to move back to the original point to get the next sapling. In this manner he covers a total distance of 1.32 kms. How many saplings does he plant in the process if he ends at the starting point?

(a) 15 (b) 14
(c) 13 (d) 12

18. Consider the expression

$$\frac{(a^2 + a + 1)(b^2 + b + 1)(c^2 + c + 1)(d^2 + d + 1)(e^2 + e + 1)}{abcde}$$

where a, b, c, d and e are positive numbers. The minimum value of the expression is

(a) 3 (b) 243
(c) 10 (d) 100

19. a, b, c, d and e are integers such that $1 \leq a < b < c < d < e$. If a, b, c, d and e are geometric progression and $\text{lcm}(m, n)$ is the least common multiple of m and n , then the maximum value of

$$\frac{1}{\text{lcm}(a, b)} + \frac{1}{\text{lcm}(b, c)} + \frac{1}{\text{lcm}(c, d)} + \frac{1}{\text{lcm}(d, e)}$$
 is

(a) 1 (b) $\frac{15}{16}$
(c) $\frac{79}{81}$ (d) $\frac{7}{8}$

20. Suppose a, x, y, z and b are in A.P. where $x + y + z = 15$, and $a, \alpha, \beta, \gamma, b$ are in H.P. where $1/\alpha + 1/\beta + 1/\gamma = 5/3$. Find a and b .

(a) 1 and 9 (b) 3 and 7
(c) 2 and 8 (d) 5 and 6

21. If the arithmetic mean between a and b equals n times their geometric mean, then find the ratio $a : b$.

(a) $(2n^2 + 1) \pm 2n\sqrt{n^2 - 1}$
(b) $(2n^2 - 1) \pm 2n\sqrt{n^2 - 1}$

(c) $(2n^2 - 1) \pm 2n\sqrt{n^2 + 1}$

(d) None of these

22. An arithmetic progression P consists of n terms. From the progression three different progressions P_1, P_2 and P_3 are created such that P_1 is obtained by the 1st, 4th, 7th terms of P , P_2 has the 2nd, 5th, 8th, terms of P and P_3 has the 3rd, 6th, 9th, terms of P . It is found that of P_1, P_2 and P_3 two progressions have the property that their average is the following can be a possible value of n ?

(a) 20 (b) 26
(c) 36 (d) Both (a) and (b)

23. Rohit drew a rectangular grid of 529 cells, arranged in 23 rows and 23 columns, and filled each cell with a number. The numbers with which he filled each cell were such that the numbers of each row taken from left to right formed an arithmetic series and the numbers of each column taken from top to bottom also formed an arithmetic series. The seventh and the seventeenth numbers of the fifth row were 47 and 63 respectively, while the seventh and the seventeenth numbers of the fifteenth row were 53 and 77 respectively. What is the sum of all the numbers in the grid?

(a) 32798 (b) 65596
(c) 52900 (d) None of these

24. An arithmetic series consists of $2n$ terms, and the first term equals the value of the common difference. If a new series is formed taking the 1st, 3rd, 5th, ... $(2n - 1)$ th term of the old series, find the ratio of the sum of the new series to that of the sum of the terms of the old series.

(a) $\frac{n+1}{2(2n+1)}$ (b) $\frac{n}{2n+1}$
(c) $\frac{1}{2}$ (d) Cannot be determined

25. Let $a = 111 \dots 1$ (55 digits),
 $b = 1 + 10 + 10^2 + 10^3 + 10^4$,
 $c = 1 + 10^5 + 10^{10} + 10^{15} + \dots + 10^{50}$, then

(a) $a = b + c$ (b) $a = bc$
(c) $b = ac$ (d) $c = ab$

Test Yourself

- If $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P. then which of the following is in A.P. ?
 (a) a, b, c (b) a^2, b^2, c^2
 (c) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ (d) None of these
- A person is entitled to receive an annual payment which for each year is less by one tenth of what it was for the year before. If the first payment is 100, then find the maximum possible payment which he can receive, however long he may live:
 (a) 900 (b) 9999
 (c) 1000 (d) None of these
- In an infinite geometric progression, each term is equal to 3 times the sum of the terms that follow. If the first term of the series is 8, find the sum of the series?
 (a) 12 (b) $32/3$
 (c) $34/3$ (d) Data inadequate
- Find the value of the expression $1 - 6 + 2 - 7 + 3 - 8 + \dots$ to 100 terms
 (a) -250 (b) -500
 (c) -450 (d) -300
- How many terms of the series $-12, -9, -6, \dots$ must be taken that the sum may be 54?
 (a) 6 (b) 9
 (c) 12 (d) 24
- The sum of all odd numbers between 1 and 1000 which are divisible by 3 is
 (a) 83667 (b) 90000
 (c) 83660 (d) None of these
- After striking the floor, a rubber ball rebounds to $4/5$ th of the height from which it has fallen. Find the total distance that it travels before coming to rest if it has been gently dropped from a height of 120 metres.
 (a) 540 metres (b) 960 metres
 (c) 1080 metres (d) 1020 metres
- A and B set out to meet each other from two places 165 km apart. A travels 15 km the first day, 14 km the second day, 13 km the third day and so on. B travels 10 km the first day, 12 km the second day, 14 km the third day and so on. After how many days will they meet?
 (a) 8 days (b) 5 days
 (c) 6 days (d) 7 days
- The interior angles of a polygon are in AP. The smallest angle is 120° and the common difference is 5° . Find the number of sides of the polygon.
 (a) 7 (b) 8
 (c) 9 (d) 10
- The fourth, seventh and tenth terms of a G.P. are p, q, r respectively, then :
 (a) $p^2 = q^2 + r^2$ (b) $q^2 = pr$
 (c) $p^2 = qr$ (d) $pqr + pq + 1 = 0$
- The first term of an infinite G.P is 1 and any term is equal to the sum of all the succeeding terms. find the series.
 (a) $1, \frac{1}{2}, \frac{1}{8}, \frac{1}{16}, \dots, \infty$ (b) $1, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots, \infty$
 (c) $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots, \infty$ (d) None of these
- Sum of n terms of the series $8 + 88 + 888 + \dots$ equals
 (a) $\frac{8}{81} [10^{n+1} - 9n - 10]$ (b) $\frac{8}{81} [10^n - 9n - 10]$
 (c) $\frac{8}{81} [10^{n+1} - 9n + 10]$ (d) None of these
- A geometric progression consists of 500 terms. Sum of the terms occupying the odd places is P_1 and the sum of the terms occupying the even places is P_2 . Find the common ratio.
 (a) P_2/P_1 (b) P_1/P_2
 (c) $P_2 + P_1/P_1$ (d) $P_2 + P_1/P_2$
- The middle points of the sides of a triangle are joined forming a second triangle. Again a third triangle is formed by joining the middle points of this second triangle and this process is repeated infinitely. If the perimetres of the outer triangle are P and A respectively, what will be the sum of perimetres of triangles thus formed?
 (a) 2^P (b) P^2
 (c) 3^P (d) $P^2/2$
- If a be the arithmetic mean and b, c be the two geometric means between any two positive numbers, then $(b^3 + c^3) / abc$ equals
 (a) $(ab)^{1/2}/C$ (b) 1
 (c) a^2c/b (d) None of these

Hints & Solutions

Foundation Level

1. (a) $a_4 = a + (4-1) \times d$
 $14 = a + 3d \Rightarrow a = 14 - 3d$... (1)
 Also, $70 = a + 11d$... (2)
 After putting the value of a from equation (1) in equation (2)
 $14 - 3d + 11d = 70$
 $8d = 70 - 14$
 $\therefore d = 8$
 $\therefore a = 14 - 24 = -10$
2. (b) Let a be the first term and r be common ratio.
 i.e., given conditions
 Fourth term of G.P. : $p = T_4 = ar^3$... (1)
 Seventh term of G.P. : $q = T_7 = ar^6$... (2)
 Tenth term of G.P. : $r = T_{10} = ar^9$... (3)
 Equ. (1) $\times$ Equ. (3) :
 $pr = ar^3 \times ar^9 \Rightarrow pr = a^2 r^{12} \Rightarrow pr = (ar^6)^2 \Rightarrow pr = q^2$
3. (a) The answer will be given by:
 $[10 + 11 + 12 + \dots + 50] - [16 + 24 + \dots + 48]$
 $= 41 \times 30 - 32 \times 5$
 $= 1230 - 160 = 1070$.
4. (a) Go through the options. The correct option should give value as 1, when $n = 3$ and as 8 when $n = 8$.
 Only option (a) satisfies both conditions.
5. (a) The series would be 8, $8/3$, $8/9$ and so on. The sum of the infinite series would be $8(1 - 1/3) = 8 \times 3/2 = 12$.
6. (b) 3-digit numbers divisible by 6 are 102, 108, 114, ..., 996
 This is an A.P. in which $a = 102$, $d = 6$ and $\ell = 996$
 Let the number of terms be n . Then $t_n = 996$.
 $\therefore a + (n-1)d = 996 \Rightarrow 102 + (n-1) \times 6 = 996$
 $\Rightarrow 6 \times (n-1) = 894 \Rightarrow (n-1) = 149 \Rightarrow n = 150$
 $\therefore$ Number of terms = 150.
7. (b) $(11^2 + 12^2 + 13^2 + \dots + 20^2) = (1^2 + 2^2 + \dots + 30^2) - (1^2 + 2^2 + \dots + 10^2)$
 $= \frac{20 \times 21 \times 41}{6} - \frac{10 \times 11 \times 21}{6}$
 $\left[\because (1^2 + 2^2 + \dots + n^2) = \frac{1}{6} n(n+1)(2n+1) \right]$
 $= 2870 - 385 = 2485$
8. (c) $(n+1)^{\text{th}}$ term $- n^{\text{th}}$ term $= (n+1)^2 + 1 - (n^2 + 1)$
 $= n^2 + 2n + 1 + 1 - n^2 - 1 = 2n + 1$
9. (d) $n(n+1)/2$ should be a perfect square. The first value of n when this occurs would be for $n = 8$. Thus, on the 8th of March the required condition would come true.
10. (c) Sum of 40 instalments $= S_{40} = 3600 = 20(2a + 39d)$
 or $2a + 39d = 180$... (1)
 Sum of 30 instalments $= S_{30} = 2400 = 15(2a + 29d)$
 or $2a + 29d = 160$... (2)
 From (1) and (2), we get $a = 51$ and $d = 2$
 The value of first instalment = 51
11. (b) The three parts are 3, 5 and 7 since $3^2 + 5^2 + 7^2 = 83$. Since, we want the smallest number, the answer would be 3.
12. (b) Sum of a G.P. with first term 1 and common ratio 2 and number of terms 20.
 $\frac{1 \times (2^{20} - 1)}{(2 - 1)} = 2^{20} - 1$
13. (b) This series is like $\rightarrow 10, 17, 21, \dots, 94$.
 Here $n = 13$, $d = 7$ and $a = 10$
 Using the formula for the sum
 $S_n = \frac{n}{2} [2a + (n-1)d]$, sum = 676
 Alternatively, using the average method,
 average = (1st number + last number)/2
 Average = $\frac{10 + 94}{2} = 52$
 So, the sum = average $\times$ number of numbers
 $= 52 \times 13 = 676$
14. (c) Expression $= 4 + 44 + 444 + \dots$ to n terms
 $= 4(1 + 11 + 111 + \dots$ to n terms)
 $= \frac{4}{9}(9 + 99 + 999 + \dots$ to n terms)
 $= \frac{4}{9}[(10-1) + (100-1) + (1000-1) + \dots$ to n terms]
 $= \frac{4}{9}[(10 + 10^2 + 10^3 + \dots$ to n terms) $- n]$
 $= \frac{4}{9}[10(1 + 10 + 10^2 + \dots$ to n terms) $- n]$
 $= \frac{40}{9} \cdot \frac{(10^n - 1)}{9} - \frac{4}{9}n$
 $[1 + 10 + 10^2 + \dots$ to n terms $= \frac{10^n - 1}{9}]$
 $= \frac{40}{81}(10^n - 1) - \frac{4}{9}n$

15. (a) The distance walked on the first day = 2 kms.
The distance walked on subsequent days is half the distance walked on the previous day.

$$\therefore \text{Total distance walked} = 2 + 1 + \frac{1}{2} + \frac{1}{4} + \dots$$

This is a geometric series whose first term, $a = 2$ and common ratio, $r = \frac{1}{2}$

Maximum total distance walked by the person in his life-time means the number of terms in the series would be infinite.

Hence, the series would be an infinite geometric series. Sum of an infinite geometric series is given by

$$s = \frac{a}{1-r} \text{ or } s = \frac{2}{1-\frac{1}{2}}$$

$$s = \frac{2}{\frac{1}{2}} \text{ or } s = 4 \text{ kms.}$$

16. (c) Let six term of H.P. = $\frac{1}{61}$
 $\Rightarrow$ six term of A.P. = 61
Similarly, tenth term of A.P. = 105
Let first term of AP is a and common diff. = d
 $\therefore a + 5d = 61$
and $a + 9d = 105$
solving these equation, we get
 $a = 6, d = 11$
Hence, first term of H.P. = $\frac{1}{6}$
17. (b) Let the first term and common difference of the AP be a and d , respectively.
Now, $(a + 5d) + (a + 14d)$
 $= (a + 6d) + (a + 9d) + (a + 11d)$
or $2a + 19d = 3a + 26d$
or $a + 7d = 0$
i.e., 8th term is 0.
18. (b) Let a be the first term and r be the common ratio of the GP.
Also assume that n^{th} term is the last term of the GP.

$$\text{Then, } a + ar^{n-1} = 66 \quad \dots\dots\dots (1)$$

$$\text{and } ar \cdot ar^{n-2} = 128$$

$$\text{or } a^2 r^{n-1} = 128 \quad \dots\dots\dots (2)$$

$$\text{From (1) and (2), } a + \frac{128}{a} = 66$$

$$\text{or } a^2 - 66a + 128 = 0 \quad \therefore a = 64, 2.$$

19. (d) $1/8 \times r^5 = 128 \rightarrow r^5 = 128 \times 8 = 1024 \rightarrow r = 4$.
Thus, the series would be $1/8, 1/2, 2, 8, 32, 128$.
The third geometric mean would be 8.
20. (d) It can be seen that for the series the average of two terms is 2, for 3 terms the average is 3 and so on. Thus, the sum to 2 terms is 2^2 , for 3 terms it is 3^2 and so on. For 11111 terms it would be $11111^2 = 123454321$.
21. (d) The side of the first equilateral triangle being 24 units, the first perimeter is 72 units. The second perimeter would be half of that and so on. 72, 36, 18 ...
22. (c) Given series is,
 $9 - 3 + 1 - \frac{1}{3} + \dots$ an infinite G.P.
Here $a = 9, r = -\frac{1}{3}$
where a = first term and r = common ratio
We know, $S_{\infty} = \frac{a}{1-r} = \frac{9}{1-\left(-\frac{1}{3}\right)} = \frac{9}{\frac{4}{3}} = \frac{27}{4}$
23. (b) $r = 2$ and $a + ar^3 = 108$.
24. (b) Since the sum of 22 terms of the AP is 385, the average of the numbers in the AP would be $385/22 = 17.5$. This means that the sum of the first and last terms of the AP would be $2 \times 17.5 = 35$. Trial and error gives us the terms of the required GP can be 2.
25. (c) The A.P. will become:
 $1/6, 1/3, 1/2, 2/3, 5/6, 1$
or in decimal terms, 0.166, 0.333, 0.5, 0.666, 0.833, 1
Sum to 6 terms = 3.5
Check the option with $m = 2$ and $n = 3$. Only option (c) gives 3.5.
26. (b) The first 100 terms of this series can be viewed as:
 $(1 - 2 - 3) + (2 - 3 - 4) + \dots + (33 - 34 - 35) + 34$
The first 33 terms of the above series (indicated inside the brackets) will give an A.P.: $-4, -5, -6 \dots -36$
Sum of this A.P. = $33 \times -20 = -660$
Answer = $-660 + 34 = -626$
27. (d) For a, b, c to be the length of the sides of the triangle, it's AM should not be 0.

$$28. (a) \frac{1}{x} + \frac{1}{z-y} + \frac{1}{z} + \frac{1}{x-y} = 0$$

$$\Rightarrow \frac{x+z-y}{x(z-y)} + \frac{x+z-y}{z(x-y)} = 0$$

$$\Rightarrow xz - xy + zx - zy = 0$$

$$2xz = y(x+z)$$

$$\frac{2}{y} = \frac{1}{x} + \frac{1}{z}$$

Hence, x, y, z are in H.P. and $x, \frac{y}{2}, z$ are in A.P.

29. (c) Let $S = 1 + n + n^2 + \dots + n^{127}$

$$= \frac{1(n^{128} - 1)}{(n - 1)} \left[\because S_n = \frac{a(r^n - 1)}{r - 1}; r > 1 \right]$$

$$= \frac{n^{128} - 1}{(n - 1)} \frac{(n^{64} - 1)(n^{64} + 1)}{(n - 1)}$$

Thus at $m = 64$ the given expression is divisible by $(n^m + 1)$

30. (d) Trying to plug in values we can see that the infinite sum of the G.P. 16, 8, 4, 2 ... is 32 and hence the third term is 4.

31. (a) The first term sum of the series is $\sqrt{2} - 1$, for 2 terms we have the sum as $\sqrt{3} - 1$ and so on. For the given series of 120 terms the sum would be $\sqrt{121} - 1 = 10$.

32. (b) The expression can be written as

$$x^{(1/2 + 1/4 + 1/8 + 1/16 + \dots)} = x^{\text{INFINITESUM OF THE GP}} = x^1$$

33. (c) Checking option (a), Put $n = 1$.

$10/9(10^n - 1) - 1 = 9$, so it is not correct.

Checking option (b), Put $n = 1$

$100/99(10^n - 1) + n^2$ is not equal to 11, so this is also not correct.

Checking option (c). Put $n = 1$.

$10/9(10^n - 1) + n^2 = 11$. But just because this option satisfies $n = 1$, it should not be assumed to be correct.

Let us check it for $n = 2$.

Option (c) gives us 104. So, this is the answer.

Normally, in these cases, checking the options till $n = 2$ guarantees the answer, but sometimes we need to check it till $n = 3$.

34. (d) Since x, y, z are in GP so $y/x = z/y$ and also $x + y, y + z, z + x$ are in AP so $2(y + z) = 2x + y + z$ i.e., $z = 2x - y$.

So, $y/x(2x - y)/y \Rightarrow y/x = (2x/y) - 1$

Since y/x is the ratio, assume $y/x = r$

and $r = -2, 1$ $r^2 + r - 2 = 0$

35. (a) a, A_1, A_2, b are in A.P.

$$\therefore A_1 - a = A_2 - A_1 = b - A_2$$

$$A_1 + A_2 = a + b$$

a, G_1, G_2, b are in G.P.

$$\Rightarrow \frac{G_1}{a} \cdot \frac{G_2}{G_1} \cdot \frac{b}{G_2} \Rightarrow G_1 G_2 = ab$$

a, H_1, H_2, b are in H.P.

$$\Rightarrow \frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{b} \text{ are in A.P.}$$

$$\frac{1}{H_1} - \frac{1}{a} = \frac{1}{H_2} - \frac{1}{H_1} = \frac{1}{b} - \frac{1}{H_2}$$

$$\Rightarrow \frac{1}{H_1} + \frac{1}{H_2} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{A_1 + A_2}{G_1 G_2}$$

$$\text{So arithmetic mean} = \frac{A_1 + A_2}{2G_1 G_2}$$

36. (b) Sum of the first term and the fifth term = 10

$$\text{or } a + a + 4d = 10$$

$$\text{or } a + 2d = 5 \quad (1)$$

and, the sum of all terms of the A.P. except for the 1st term = 99

$$\text{or } 9a + 45d = 99$$

$$a + 5d = 11 \quad \dots(2)$$

Solve (1) and (2) we $a = 1, b = 2$ to get the answer.

Standard Level

1. (c) a, b, c form an AP.

$$\therefore 2b = a + c$$

Increasing a by 1 or c by 2 results in a GP

$$\therefore b^2 = (a + 1)c \quad \dots(1)$$

$$\text{and } b^2 = a(c + 2) \quad \dots(2)$$

$$\therefore (a + 1)c = a(c + 2)$$

$$\therefore ac + c = ac + 2a$$

$$\therefore c = 2a$$

$$\text{Now, } 2b = a + c$$

$$\therefore 2b = a + 2a$$

$$\therefore b = \frac{3a}{2}$$

Putting this in (1), we get

$$\frac{9a^2}{4} = (a + 1)2a$$

$$\therefore \frac{9a}{4} = 2a + 2$$

$$\therefore 9a = 8a + 8$$

$$\therefore a = 8$$

$$\therefore b = \frac{3a}{2} = \frac{3 \times 8}{2} = 12$$

2. (a) Given $b - a = c - b$ and $(c - b)^2 = a(b - a)$

$$\Rightarrow (b - a)^2 = a(b - a)$$

$$\Rightarrow b - a = a (\because b \neq a) \quad b = 2a \text{ and } c = 3a$$

$$\therefore a : b : c = 1 : 2 : 3.$$

3. (d) The two series till their hundredth terms are 13, 15, 17, 211 and 14, 17, 20 ... 311. The common terms of the series would be given by the series 17, 23, 29 ... 209. The number of terms in this series of common terms would be $192/6 + 1 = 33$.

4. (a) Since, $\log_x a, a^{x/2}$ and $\log_b x$ are in GP,

$$\therefore (a^{x/2})^2 = (\log_x a) \times (\log_b x)$$

$$\text{or } a^x = \log_b a$$

Taking log of both sides with base a , we get,

$$x \log_a a = \log_a (\log_b a)$$

$$\text{or } x = \log_a (\log_b a)$$

$$\begin{aligned} 5. (b) & \frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \dots + \frac{1}{\sqrt{120}+\sqrt{121}} \\ &= \frac{(\sqrt{2}-\sqrt{1})}{(\sqrt{2})^2-(\sqrt{1})^2} + \frac{(\sqrt{3}-\sqrt{2})}{(\sqrt{3})^2-(\sqrt{2})^2} + \frac{(\sqrt{4}-\sqrt{3})}{(\sqrt{4})^2-(\sqrt{3})^2} + \\ & \dots + \frac{(\sqrt{121}-\sqrt{120})}{(\sqrt{121})^2-(\sqrt{120})^2} \\ &= (\sqrt{2}-\sqrt{1}) + (\sqrt{3}-\sqrt{2}) + (\sqrt{4}-\sqrt{3}) + \\ & \dots + (\sqrt{120}-\sqrt{119}) + (\sqrt{121}-\sqrt{120}) \\ &= -\sqrt{1} + \sqrt{121} = 11 - 1 = 10 \end{aligned}$$

6. (b) For 1 term, the value should be:

$$6^2 + 8 = 44$$

Only option (b) gives 44 for $n = 1$

7. (c) Since A.M. $\geq$ G.M.

$$\therefore \frac{y+z}{2} \geq \sqrt{yz} \quad \dots(1)$$

$$\frac{z+x}{2} \geq \sqrt{zx} \quad \dots(2)$$

$$\frac{y+x}{2} \geq \sqrt{yx} \quad \dots(3)$$

Multiplying (1), (2) and (3), we get

$$\frac{(y+z)(z+x)(x+y)}{8} \geq xyz$$

$$\text{or } (1-x)(1-y)(1-z) \geq 8xyz \quad (\because x+y+z=1)$$

8. (c) Let the numbers be $a, a+1, a+2, \dots, a+31$.

$$\text{Sum of these numbers } 32a + \frac{31 \times 32}{2} = 16(2a+31)$$

As 16 is a perfect square, the least possible value of $2a+31=49$.

Therefore, $a=9$ and $a+31=40$.

The least possible sum = 49.

9. (c) a, b, c are in H.P, so, $\frac{1}{a}, \frac{1}{b}$ and $\frac{1}{c}$ will be in A.P.

$$\text{or, } \frac{a+b+c}{a}, \frac{a+b+c}{b} \text{ and } \frac{a+b+c}{c} \text{ will be in A.P}$$

$$\text{or, } 1 + \frac{b+c}{c}, 1 + \frac{a+c}{b} \text{ and } 1 + \frac{a+b}{c} \text{ will be in A.P.}$$

$$\text{Hence } \frac{b+c}{a}, \frac{a+c}{b} \text{ and } \frac{a+b}{c} \text{ are in A.P.}$$

$$\text{So, } \frac{a}{b+c}, \frac{b}{c+a} \text{ and } \frac{c}{a+b} \text{ will be in H.P.}$$

Let us take 1, 1/2, 1/3 (which are in H.P)

$$\frac{a}{b+c} = \frac{6}{5}, \frac{b}{c+a} = \frac{3}{8}, \frac{c}{a+b} = \frac{2}{9}$$

Now, when we check these values of A.P, G.P and H.P,

$$\text{we find that } \frac{8}{3} \text{ is the AM of } \frac{5}{6} \text{ and } \frac{9}{2}.$$

10. (b) 3, 7, 11 147

It is an arithmetic series whose

first term, $a=3$

last term, $x_n=147$

common difference, $d=4$

$$x_n = a + (n-1)d$$

$$147 = 3 + (n-1) \times 4$$

$$n-1 = \frac{147-3}{4}$$

$$n-1 = 36, n = 37$$

The given series consists of 37 terms. Therefore, its middle term will be

$$\frac{37+1}{2} = 19\text{th term}$$

$$x_{19} = 3 + (19-1)4$$

$$= 3 + 18 \times 4 = 75$$

$\therefore$ The middle term of the given arithmetic series is 75.

11. (a) We need the sum of the series $20 + 24 + 28$ to cross 1000. Trying out the options, we can see that in 20 years the sum of his savings would be: $20 + 24 + 28 + \dots + 96$. The sum of this series would be $20 \times 58 = 1160$. If we remove the 20th year we will get the series for savings for 19 years. The series would be $20 + 24 + 28 + \dots + 92$. Sum of the series would be $1160 - 96 = 1064$. If we remove the 19th year's savings the savings would be $1064 - 92$ which would go below 1000. Thus, after 19 years his savings would cross 1000.

12. (a) The maximum sum would occur when we take the sum of all the positive terms of the series.

The series 25, 24.5, 24, 23.5, 23, 1, 0.5, 0 has 51 terms. The sum of the series would be given by:

$$n \times \text{average} = 51 \times 12.5 = 637.5$$

13. (b) Given sum = $(1 + 1 + 1 + \dots \text{ to } n \text{ terms})$

$$= \left(\frac{1}{n} + \frac{2}{n} + \frac{3}{n} + \dots \text{ to } n \text{ terms} \right)$$

$$= n - \frac{n}{2} \left(\frac{1}{n} + 1 \right) \quad [\because \ell = \text{nth terms} = \frac{n}{n} = 1]$$

$$= n - \frac{n+1}{2} = \frac{1}{2}(n-1)$$

14. (c) $(0.11)^3 + (0.22)^3 + \dots + (0.99)^3 = (0.11)^3 (1^3 + 2^3 + \dots + 9^3)$

$$= 0.001331 \times 2025 = 2.695275 = 2.695.$$

15. (c) The first series is 3, 5, 7 239

While the second series is 3, 6, 9 ... 240

Hence, the last common term is 237.

$$\text{Thus our answer becomes } \frac{237-3}{6} + 1 = 40$$

16. (c) The equation can be written as:

$$\frac{2n}{2} [4 + (2n-1)3] = \frac{n}{2} [114 + (n-1)2]$$

$$\therefore n = 11$$

17. (c) Let us add all the 13 numbers

$$\Rightarrow 1 + 9 + 14 + 15 + [2 + 3 + 4 + 5 + 7 + 10 + 11 + 12 + 13] = 106$$

As there are 4 columns and 3 rows so the sum of the 12 numbers has to be divisible by 12, i.e. the sum should be $96 (12 \times 8)$.

$$\text{So the sum of all the numbers in a row} = \frac{96}{3} = 32$$

$$\text{and in a column} = \frac{96}{4} = 24$$

Further the arithmetic mean of the numbers in a row or

$$\text{column} = \frac{32}{4} \text{ or } \frac{24}{3} = 8$$

18. (d) Clearly 10 has to be left out.

- 19-22. $D \dots GFE$

Given $E = 960$, which is of the form ax^3 .

$$960 = 2^3 \times 2^3 \times 3 \times 5$$

So the common ratio (r) of the G.P. of last 4 numbers is either 2 or 4.

	$r = 4$	$r = 2$
E	960	960
F	240	480
G	60	240
$-$	15	120
$-$		
D		

$$\text{For } r = 4; \frac{G}{C} = \frac{F}{A} \Rightarrow \frac{60}{C} = \frac{240}{A} \text{ or } A = 4C \text{ not possible}$$

$$\text{For } r = 2; \frac{G}{C} = \frac{F}{A} \Rightarrow \frac{240}{C} = \frac{480}{A} \text{ or } A = 2C \text{ possible}$$

$$\Rightarrow r = 2$$

As $A = 2C$, so C takes the 2nd lowest position as B and A have to be together.

Again $A > B$, as in the other case the A.P. will be $D = 0, C = C, A = 2C, B = 3C$, which is not possible as D is an integer. The possible values are :

D	C	B	A	G	F	E
30	60	90	120	240	480	960

$$19. (b) \frac{E}{A} = \frac{960}{120} = 8.$$

$$20. (a) D = 30.$$

$$21. (d) \text{ Common difference in A.P.} = 60 - 30 = 30.$$

$$22. (d) A \text{ is 4}^{\text{th}} \text{ highest and the value is 120.}$$

$$23. (b) \text{ Product of three numbers } a, b \text{ and } c \text{ in A.P. is 4} \\ \text{i.e., } abc = 4, \text{ a constant.}$$

$$\text{Hence, the minimum possible value of } b = (4)^{1/3} = (2)^{2/3}$$

$$24. (b) \text{ The sum of the total distance it travels would be given by the infinite sum of the series:}$$

$$420 \times 8/1 + 367.5 \times 8/1 = 3360 + 2940 = 6300.$$

$$25. (b) \text{ The general term is}$$

$$T_n = \frac{\frac{n}{2} \cdot \frac{n+1}{2}}{1^3 + 2^3 + 3^3 + \dots + n^3} = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$$\therefore S_n = 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

26. (b) Since the sum of the first three terms of the A.P. is 30, the average of the A.P. till 3 terms would be $30/3 = 10$. The value of the second terms would be equal to this average and hence the second term is 10. Using the information about the sum of squares of the first and second terms being 116., we have that the first term must be 4. Thus, the A.P. has a first term of 4 and a common difference of 6. The seventh term would be 40.

$$27. (c) a/(1-r) = 162 \text{ and } a(1-r^n)/(1-r) = 160$$

$$\Rightarrow 1-r^n = 160/162 \Rightarrow r^n = 1/81$$

Hence, there will be only two values of r , i.e., 2 and 4.

Expert Level

1. (c) The given integers are 4, 4, 4, 8, 10, 20 and x
Let $x < 4$

$$\text{Mean} = \frac{50+x}{7}, \text{Median} = 4, \text{Mode} = 4$$

If mean, median and mode are in AP then
Here mean = 4

$$\frac{50+x}{7} = 4 \Rightarrow x = -22$$

Consider (b) $4 < x < 8$

$$\text{Mean} = \frac{50+x}{7}, \frac{54}{7} < \text{Mean} < \frac{58}{7}$$

$$\text{Mean} = x$$

$$\text{Mode} = 4$$

As these are in AP Mean = 8

$$\therefore x = 6$$

Consider (c) $8 < x$

$$\text{Mean} = \frac{50+x}{7} > \frac{58}{7}$$

$$\text{Median} = 8$$

$$\text{Mode} = 4$$

As these are in AP, Mean = 12 i.e., $x = 34$

$\therefore x$ can be $-22, 6$ or 34

The sum of these is 18, which is not given in the options.

If negative value of -22 is ignored then

mean = median = mode = 4.

$\therefore$ We have to select 40.

answer in the given choices.

2. (c) $y = px + q$

Given the second term of AP = $A_2 = 2$

and the 24th term of AP = $A_{24} = 68$

Let the common difference of AP = d

$$\Rightarrow A_2 = A_1 + d \Rightarrow A_1 = 2 - d \quad \dots(1)$$

$$A_{24} = A_1 + (24-1)d$$

$$68 = 2 - d + (24-1)d = 2 + 22d$$

$$\Rightarrow d = \frac{66}{22} = 3 \Rightarrow A_1 = 2 - 3 = -1$$

$$\Rightarrow A_8 = A_1 + (8-1)d = -1 + 7 \times 3 = 20$$

Further the two points A_2 and A_{24} will lie on $y = px + q$

$$\Rightarrow -2 = 2p + q \quad \dots(2)$$

$$\& 31 = 68p + q \quad \dots(3)$$

Subtracting (2) from (3),

$$33 = 66p \Rightarrow p = \frac{1}{2}$$

$$\text{Putting } p \text{ in (1)} \Rightarrow -2 = 2 \times \frac{1}{2} + q \Rightarrow q = -3$$

The equation becomes, $y = \frac{1}{2}x - 3$

$\therefore$ y ordinate of A_8 , where $x = 20$, is

$$y = \frac{1}{2} \times 20 - 3 = 7$$

3. (c) From the given equation $y = \frac{x}{2} - 3$, the points satisfying it and lying in the first quadrant will be those for which $x > 0$ & $y > 0$

$$\Rightarrow \frac{x}{2} - 3 > 0 \text{ or } x > 6$$

So for $x \leq 6$, the points will not lie in 1st quadrants.

The x -coordinate can be found from the AP whose terms can be written as

$-1, 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32, 35, 38, 41, 44, 47, 50, 53, 56, 59, 62, 65, 68$.

So in all there are 3 points which do not lie in first quadrant.

4. (d) $A_{n+1} \cdot A_{n-1} = A_n^2 \pm 1$

$$\text{For } n = 2, A_3 \cdot A_1 = A_2^2 \pm 1$$

$$\Rightarrow A_3 \cdot 1 = 2^2 \pm 1 = 4 \pm 1$$

$$\Rightarrow A_3 = 3 \text{ or } 5$$

$$\text{For } n = 3, A_4 \cdot A_2 = A_3^2 \pm 1$$

$$\Rightarrow 12 \times 2 = A_3^2 \pm 1$$

$$\Rightarrow A_3^2 = 24 + 1 = 25 \quad (\because \{A_n\} \in I^t)$$

$$\Rightarrow A_3 = 5$$

$$\text{For } n = 4, A_5 \cdot A_3 = A_4^2 \pm 1$$

$$\Rightarrow A_5 \times 5 = 12^2 \pm 1 = 144 \pm 1$$

$$\Rightarrow A_5 = \frac{144+1}{5} = 29$$

$$\text{For } n = 5, A_6 \cdot A_4 = A_5^2 \pm 1$$

$$\Rightarrow A_6 \cdot 12 = 29^2 \pm 1 = 841 \pm 1$$

$$\Rightarrow A_6 = \frac{841-1}{12} = 70$$

$$\text{For } n = 6, A_7 \cdot A_5 = A_6^2 \pm 1$$

$$\Rightarrow A_7 \cdot 29 = 70^2 \pm 1$$

$$\text{or } A_7 = \frac{4900+1}{29} = 169.$$

5. (b) Let $X = \frac{a}{(1+r)} + \frac{a}{(1+r)^2} + \dots + \frac{a}{(1+r)^n}$

By taking L.C.M., we get

$$X = \frac{a(1+r)^{n-1} + a(1+r)^{n-2} + \dots + a}{(1+r)^n}$$

$$X(1+r)^n = a + a(1+r) + \dots + a(1+r)^{n-1}.$$

Aliter : We can multiply by $(1+r)^n$ on both side, we get

$$X(1+r)^n = a(1+r)^{n-1} + a(1+r)^{n-2} + \dots + a.$$

6. (d) Since a, b, c are in A.P.

$$\therefore a = b - d \text{ and } c = b + d$$

Also $(b-d)^2, b^2, (b+d)^2$ are in G.P.

$$\therefore b^4 = (b^2 - d^2)^2$$

$$\Rightarrow (b + d^2 - b^2)(b^2 - d^2 + b^2) = 0$$

As $d \neq 0$ ($a < b < c$),

$$\therefore 2b^2 = d^2 \text{ or } d = \sqrt{2}b.$$

Hence, the 3 numbers are $(1 - \sqrt{2})b, b, (1 + \sqrt{2})b$.

$$\therefore 3b = \frac{3}{2}, \text{ or } b = \frac{1}{2}$$

$$\therefore \frac{a}{2} - (1 - \sqrt{2})\frac{1}{2} = \frac{1}{2} - \frac{\sqrt{2}}{2}$$

7. (c) In order to find how many times the alarm rings we need to find the number of numbers below 100 which are not divisible by 2, 3, 5 or 7. This can be found by:

100 – (numbers divisible by 2) – (numbers divisible by 3 but not by 2) – (numbers divisible by 7 but not by 2 or 3 or 5).

Numbers divisible by 2 up to 100 would be represented by the series 2, 4, 6, 8, 10 .. 100 → A total of 50 numbers.

Number divisible by 3 but not by 2 up to 100 would be represented by the series 3, 9, 15, 21 ... 99 finding the number of number in this series:

$$[(\text{last term} - \text{first term}) / \text{common difference}] + 1 = [99 - 3] / 6 + 1 = 16 + 1 = 17.$$

Numbers divisible by 5 but not by 2 or 3: Numbers divisible by 5 but not by 2 up to 100 would be represented by the series 5, 15, 25, 35 ... 95 → A total of 10 numbers. But from these numbers, the numbers 15, 45 and 75 are also divisible by 3. Thus,

we are left with $10 - 3 = 7$ new numbers which are divisible by 5 but not by 2 and 3.

Numbers divisible by 7, but not by 2, 3 or 5:

Numbers divisible by 7 but not by 2 upto 100 would be represented by the series 7, 21, 35, 49, 63, 77, 91 → A total of 7 numbers. But from these numbers we should not count 21, 35 and 63 as they are divisible by either 3 or 5. Thus a total of $7 - 3 = 4$ numbers are divisible by 7 but not by 2, 3 or 5

8. (a) Here $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

$$\text{Let } \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = x$$

$$\text{Then } x = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$= \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) + \left(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \dots \right)$$

$$= \frac{\pi^2}{8} + \frac{1}{4} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots \right) = \frac{\pi^2}{8} + \frac{1}{4}x$$

9. (c) The sum of the interior angles of a polygon are multiples of 180 and are given by $(n - 1) \times 180$ where n is the number of sides of the polygon. Thus, the sum of interior angles of a polygon would be a member of the series: 180, 360, 540, 720, 900, 1080, 1260. The sum of the series with first term 100 and common difference 10 would keep first term 100 and common difference 10 would keep increasing when we take more and more terms of the series. In order to see the number of sides of the polygon, we should get a situation where the sum of the series represented by $100 + 110 + 120 \dots$ should become a multiple of 180. The number of sides in the polygon would then be the number of terms in the series 100, 110, 120 at that point.

If we explore the sums of the series represented by $100 + 110 + 120 \dots$

We realize that the sum of the series becomes a multiple of 180 for 8 terms as well as for 9 terms.

It can be seen in: $100 + 110 + 120 + 130 + 140 + 150 + 160 + 170 = 1080$.

or $100 + 110 + 120 + 130 + 140 + 150 + 160 + 170 + 180 = 1260$

10. (a) We know that $2[ab + ac + ad + \dots + bc + bd + \dots] = [a + b + c + \dots]^2 - [a^2 + b^2 + c^2 + \dots + \sum N]^2$

Also by rules of A.P.,

$\Rightarrow a^3 + b^3 + c^3 + \dots = (a+b+c+\dots)^2$ Combining the two,

we get :

$$(ab + ac + ad + \dots + bc + bd + \dots)$$

$$\frac{1}{2}[(a^3 + b^3 + c^3 + \dots) - (a^2 + b^2 + c^2 + \dots)]$$

$$\text{Now } \sum n^3 = \left[\frac{n(n+1)}{2} \right]^2 = 55^2 = 3025 \text{ and}$$

$$\sum n^2 = \frac{n(n+1)(2n+1)}{6} = 10 \times 11 \times \frac{21}{6} = 385$$

$$\Rightarrow \text{Answer} = \frac{1}{2}[3025 - 385] = 1320$$

11. (c) For $n = 2, n = 4$ and $n = 5$ the values that A assumes are $12 + 22, 12 + 22 + 32 + 42, 12 + 22 + 32 + 42 + 52$ respectively. Each of these is divisible by 5. For $n = 1$ or 3 , A takes values 12 and $12 + 22 + 32$ respectively both of which are not divisible by 5. So in the set of the 1st 5 natural numbers, 3 numbers are divisible by 5. For $n = 6, 7, 8, 9, 10$ A behaves in exactly the same manner as for $n = 1, 2, 3, 4, 5$ respectively. This pattern repeats for the next set of 5 natural numbers and so on. So for $n = 1$ to $n = 100$, A is divisible by 5, in three-fifths of cases. So for 60 values of n A would be divisible by 5. Since $n < 100$ and for $n = 100$, A is divisible by 5, the total number of values that satisfy the condition would be 59.

Alternate solution:

Sum of the squares of first n natural numbers is

$$\frac{n(n+1)(2n+1)}{6} \quad A$$

Now n can take 5 types of values i.e., $5k, 5k + 1, 5k + 2, 5k - 2$ and $5k - 1$.

Let's put all the values in A:

If $n = 5k$, A will be divisible by 5.

$$\text{If } n = 5k + 1, A = \frac{(5k+1)(5k+2)(10k+3)}{6}.$$

So A is not divisible by 5.

$$\text{If } n = 5k + 2, A = \frac{(5k+2)(5k+3)(10k+5)}{6}.$$

So A is divisible by 5.

$$\text{If } n = 5k - 2, A = \frac{(5k-2)(5k-1)(10k-3)}{6}.$$

So A is not divisible by 5.

$$\text{If } n = 5k - 1, A = \frac{(5k-1)(5k)(10k-1)}{6}.$$

So A is divisible by 5.

So all the numbers of the type $5k, 5k + 2$ and $5k - 1$ i.e., 3 numbers out of every 5 consecutive numbers will satisfy the given condition. So 57 out of the first 95 natural numbers will satisfy the condition. 97 and 99 also satisfy the given condition. So total numbers are $57 + 2 = 59$.

$$12. (a) \text{ Since } a \text{ and } b \text{ are unequal, } \frac{a^2 + b^2}{2} > \sqrt{a^2 b^2}$$

(A.M. > G.M. for unequal numbers)

$$\Rightarrow a^2 + b^2 > 2ab$$

$$\text{Similarly } b^2 + c^2 > 2bc \text{ and } c^2 + a^2 > 2ca$$

$$\text{Hence } 2(a^2 + b^2 + c^2) > 2(ab + bc + ca)$$

$$\Rightarrow ab + bc + ca < 1$$

13. (d) $a, a - b, a - 2b, \dots$ is an AP with first term = a and common difference = $-b$

Now,

$$t_{10} = a + (10 - 1)(-b) \Rightarrow 20 = a - 9b \quad \dots(1)$$

$$t_{20} = a + (20 - 1)(-b) \Rightarrow 10 = a - 19b \quad \dots(2)$$

From equations (1) and (2),

$$20 - 10 = a - 9b - a + 19b$$

$$\Rightarrow 10b = 10 \Rightarrow b = 1$$

From equation (1),

$$20 = a - 9 \Rightarrow a = 29$$

$$\therefore t_x = 29 + (x - 1) \times -1$$

$$= 29 - x + 1 = 30 - x$$

14. (b) Trial and error gives us that for option (b): With the ratio 4 : 1, the numbers can be taken as $4x$ and $1x$. Their AM would be $2.5x$ and their GM would be $2x$. The GM can be seen to be 20% lower than the A.M. Option (b) is thus the correct answer.

15. (b) Let x be the common difference of the A.P. a, b, c, d, e, f .

$$\therefore e = a + (5 - 1)x \quad [\because a_n = a + (n - 1)d]$$

$$\Rightarrow e = a + 4x \quad \dots(1)$$

$$\text{and } c = a + 2x \quad \dots(2)$$

$\therefore$ using equations (1) and (2), we get

$$e - c = a + 4x - a - 2x$$

$$\Rightarrow e - c = 2x = 2(d - c).$$

$$16. (a) \frac{a_1 + a_4}{a_1 a_4} = \frac{a_2 + a_3}{a_2 a_3},$$

$$\text{So } \frac{1}{a_4} + \frac{1}{a_1} = \frac{1}{a_3} + \frac{1}{a_2} \text{ or } \frac{1}{a_4} - \frac{1}{a_3} = \frac{1}{a_2} - \frac{1}{a_1} \quad \dots(1)$$

$$\text{Also } \frac{3(a_2 - a_3)}{a_2 a_3} = \frac{a_1 - a_4}{a_1 a_4};$$

$$\text{So } 3\left(\frac{1}{a_3} - \frac{1}{a_2}\right) = \frac{1}{a_4} - \frac{1}{a_1} \quad \dots(2)$$

Clearly, (1) and (2)

$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2} = \frac{1}{a_4} - \frac{1}{a_3};$$

So $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}$ are in A.P.

17. (d) To plant the 1st sapling, Mithilesh will cover 20 m; to plant the 2nd sapling he will cover 40 m and so on. But for the last sapling, he will cover only the distance from the starting point to the place where the sapling has to be planted.

18. (b) $\frac{a^2 + a + 1}{3} \geq (a^2 \times a \times 1)^{1/3}$ By AM – GM relation

$$\text{Hence, } \frac{a^2 + a + 1}{3} \geq a \text{ or } \frac{a^2 + a + 1}{a} \geq 3$$

Similarly, a similar relation for b, c, d and e and then multiplying, we get

$$\frac{(a^2 + a + 1)(b^2 + b + 1)(c^2 + c + 1)(d^2 + d + 1)(e^2 + e + 1)}{abcde}$$

$$\geq 3^5 = 243$$

19. (b) As a, b, c, d & e are in GP. Thus they can be expressed as a, ax, ax^2, ax^3, ax^4 where x is the common ratio of GP.

$$\ell\text{cm}(a, b) \quad \ell\text{cm}(a, ax) = ax = b$$

$$\text{Similarly, } \ell\text{cm}(b, c) = c$$

$$\ell\text{cm}(c, d) = d \text{ and } \ell\text{cm}(d, e) = e$$

Thus the expression

$$\frac{1}{\ell\text{cm } a, b} + \frac{1}{\ell\text{cm } b, c} + \frac{1}{\ell\text{cm } c, d} + \frac{1}{\ell\text{cm } d, e}$$

$$\frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{1}{e}$$

For the expression to be maximum, b, c, d and e should have minimum value. It is possible only when a is minimum i.e., $= 1$.

Thus the GP with integers having minimum value with first term $= 1$ will be 1, 2, 4, 8, 16.

Thus

$$\frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{1}{e} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{8+4+2+1}{16} = \frac{15}{16}$$

20. (a) Since a, x, y, z and b are in AP, we have $a + b = x + z = 2y$

$$\text{Also } x + y + z = 15 \Rightarrow 2y + y = 15$$

$$\Rightarrow 3y = 15 \text{ or } y = 5 \therefore a + b = 10$$

Since a, β, γ and b are in HP, we have

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{\alpha} + \frac{1}{\gamma} = \frac{2}{\beta}$$

$$\text{Also, } \frac{1}{\alpha} + \frac{1}{\gamma} + \frac{1}{\beta} = \frac{5}{3} \Rightarrow \frac{3}{\beta} = \frac{5}{3} \text{ or } \frac{1}{\beta} = \frac{5}{9}$$

$$\text{Thus } \frac{1}{a} + \frac{1}{b} = \frac{10}{9} \Rightarrow \frac{b+a}{ab} = \frac{10}{9} \Rightarrow ab = 9$$

Hence, either $a = 1, b = 9$ or $a = 9, b = 1$.

21. (b) We are given $\frac{\text{A.M}}{\text{G.M}} = n \Rightarrow \frac{(a+b/2)}{\sqrt{ab}} = n$.

$$\Rightarrow \frac{a+b}{\sqrt{ab}} = 2n \Rightarrow \sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}} = 2n$$

$$\text{Multiplying by } \sqrt{\frac{a}{b}}, \text{ we get } \frac{a}{b} + 1 = 2n \left(\sqrt{\frac{a}{b}} \right)$$

$$\Rightarrow \frac{a}{b} - 2n \left(\sqrt{\frac{a}{b}} \right) + 1 = 0.$$

This is a quadratic equation in $\sqrt{\frac{a}{b}}$. Solving for $\sqrt{\frac{a}{b}}$, we get

$$\sqrt{\frac{a}{b}} = \frac{2n \pm 2\sqrt{n^2 - 1}}{2} = n \pm \sqrt{n^2 - 1}$$

$$\Rightarrow \frac{a}{b} = \left(n \pm \sqrt{n^2 - 1} \right)^2 = n^2 + n^2 - 1 \pm 2n\sqrt{n^2 - 1}$$

$$= (2n^2 - 1) \pm 2n\sqrt{n^2 - 1}.$$

22. (d) The key to this question is what you understand from the statement- 'for two progressions out of P_1, P_2 , and P_3 , the average is itself a term of the original the Progression P .' For option (a) which tells us that the Progression P has 20 terms, we can see that P_1 would have 7 terms, P_2 would have 7 terms and P_3 would have 6 terms. Since, both P_1 and P_2 have an odd number of terms (being the middle terms for an AP with 7 terms) would be equal to their average. Since, all terms of P_1, P_2 and P_3 have been taken out of the original AP P , we can see that for P_1 and P_2 their average itself would be a term of the original

progression P . This would not occur for P_3 as P_3 has an even number of terms. Thus, 20 is a correct value for n .

Similarly, if we go for $n = 26$ from the second option we get:

P_1 , P_2 and P_3 would have 9, 9 and 8 terms respectively and the same condition would be met here too. For $n = 36$ from the third option, the three progressions would have 12 terms each and none of them would have an odd number of terms.

Thus, option (d) is correct as both option (a) and (b) satisfy the conditions given in the problem.

23. (a) 1st row – average 51 – total = 23×51

2nd row – average 52 – total 23×52

23rd row – average 73 – total 23×73

The overall total can be got by using averages as:

$23 \times 23 \times 62 = 32798$

24. (b) The series consist of $2n$ terms,

first term = a , common diff = a , no of terms = $2n$

$$\text{Sum of all terms} = \frac{2n}{2[2a + (2n-1)a]} \quad \dots(1)$$

for the new series taking 1st, 3rd, 5th, $(2n-1)$ th term of old series.

First term = a , common difference = $2a$, number of terms = n

$$\text{Sum of all terms} = \frac{n}{2}[2a + (2n-1)2a] \quad \dots(2)$$

Dividing (1) by (2) get the required ratio.

25. (b) Since, $a = 1111 \dots 1$ (55 digits)

$$b = 1 + 10 + 10^2 + 10^3 + 10^4 = \frac{1(10^5 - 1)}{10 - 1} = \frac{10^5 - 1}{9} \text{ and}$$

$$c = 1 + 10^5 + 10^{10} + 10^{15} + \dots 10^{50} = 1 \left[\frac{(10^5)^{11} - 1}{10^5 - 1} \right]$$

$$bc = (10^5 - 1)/9 \times (10^{55} - 1)/(10^5 - 1)$$

$$= (9999 \dots 55 \text{ digits})/9 = a$$

Explanation of Test Yourself

1. (c) $\frac{b+c-a}{a}, \frac{c+a-b}{b}, \frac{a+b-c}{c}$ are in A.P.
 $\therefore \frac{b+c-a}{a} + 2, \frac{c+a-b}{b} + 2, \frac{a+b-c}{c} + 2$
 are in A.P. (adding 2 in each term)
 or $\frac{a+b+c}{a}, \frac{c+a+b}{b}, \frac{a+b+c}{c}$ are in A.P.
 or $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.
2. (c) His first payment = ₹ 100
 His second payment = ₹ 90 = $\left(100 \times \frac{9}{10}\right)$
 His third payment = ₹ 81 etc. = $\left(90 \times \frac{9}{10}\right)$
 $\therefore$ The annual payments are 100, 90, 81, ... which are in
 G.P. with common ratio $\frac{9}{10} (< 1)$
 Therefore the sum to infinity of this G.P.
 $= 100 + 90 + 81 + \dots \infty$
 $= \frac{100}{1 - \frac{9}{10}} = \frac{100}{\frac{1}{10}} = 1000$
 Hence the person can receive maximum amount of ₹ 1000.
3. (a) The series would be 8, 8/3, 8/9 and so on. The sum of the infinite series would be $8/(1 - 1/3) = 8 \times 3/2 = 12$.
4. (a) The series $(1 - 6 + 2 - 7 + 3 - 8 + \dots)$ to 100 terms can be rewritten as:
 $\Rightarrow (1 + 2 + 3 + \dots \text{to } 50 \text{ terms}) - (6 + 7 + 8 + \dots \text{to } 50 \text{ terms})$
 Both these are AP's with values of a and d as $\rightarrow a = 1, n = 50$ and $d = 1$ and $a = 6, n = 50$ and $d = 1$ respectively.
 Using the formula for sum of an AP we get:
 $\rightarrow 25(2 + 49) - 25(12 + 49)$
 $\rightarrow 25(51 - 61) = -250$
 Alternatively, we can do this faster by considering $(1 - 6), (2 - 7)$, and so on as one unit or one term.
 $1 - 6 = 2 - 7 = \dots = -5$. Thus the above series is equivalent to a series of fifty -5 's added to each other.
 So, $(1 - 6) + (2 - 7) + (3 - 8) + \dots 50 \text{ terms} = -5 \times 50 = -250$
5. (c) Here $S = 54, a = -12, d = 3, n$ is unknown and has to be calculated. To do so we use the formula for the sum of an A.P. and get

$$54 = \frac{n}{2} [2(-12) + (n-1)3]$$

$$\text{or } 108 = -24n - 3n + 3n^2 \text{ or } 3n^2 - 27n - 108 = 0$$

$$\text{or } n^2 - 9n - 36 = 0, \Rightarrow (n+3)(n-12) = 0$$

The value of n (the number of terms) cannot be negative, Hence: -3 is rejected

So we have $n = 12$

Alternatively, we can directly add up individual terms and keep adding manually till we get a sum of 54. We will observe that this will occur after adding 12 terms. In this case, as also in all cases where the number of terms is mentally manageable, mentally adding the terms till we get the required sum will turn out to be much faster than the equation based process.

6. (a) Sum of odd numbers between 1 and 1000, which is divisible by 3 = $3 + 9 + 15 + 21 + 27 + \dots + 999 = S$ (let)
 $\therefore$ Let n be the number of terms in series and a is first term

$$\therefore l = a + (n-1)d,$$

where l is the last term and d is the common difference.

$$999 = 3 + (n-1) \times 6$$

$$n-1 = \frac{999-3}{6} = \frac{996}{6}$$

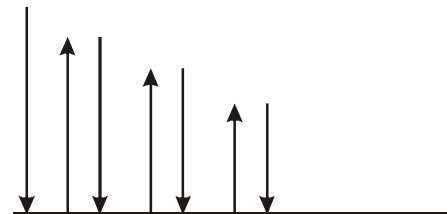
$$n-1 = 166 \Rightarrow n = 167$$

$$\therefore S = \frac{n}{2} [2a + (n-1)d]$$

$$\frac{167}{2} [2 \times 3 + (167-1) \times 6]$$

$$\frac{167}{2} [1002] = 167 \times 501 = 83667$$

7. (c)



In the figure above, every bounce is $4/5$ th of the previous drop.

In the above movement, there are two infinite G.P.s (The G.P. representing the falling distances and the G.P. representing the rising distances.)

The required answer: (Using $a/(1-r)$ formula)

$$\frac{120}{1/5} + \frac{96}{1/5} = 1080$$

8. (c) The combined travel would be 25 on the first day, 26 on the second day, 27 on the third day, 28 on the fourth day, 29 on the fifth day and 30 on the sixth day. They meet after 6 days.

9. (c) Sum of the A.P. for n sides = Sum of interior angles of a polygon of n sides.

$$\frac{n}{2} \times [2a + (n-1)d] = (2n-4) \times 90$$

where $a = 120^\circ$ and $d = 5^\circ$, then $n = 9$

10. (b) Let a be the first term and r be common ratio.

i.e., given conditions

Fourth term of G.P. : $p = T_4 = \dots(1)$

Sixth term of G.P. : $q = T_6 = ar^5 \dots(2)$

Tenth term of G.P. : $r = T_{10} = ar^9 \dots(3)$

Eqn. (1) $\times$ Eqn. (3)

$$pr = ar^3 \times ar^9 \Rightarrow pr = a^2 r^{12} \Rightarrow pr = (ar^6)^2 \Rightarrow pr = q^2$$

11. (c) Given that $T_p = (T_{p+1} + T_{p+2} + \dots \infty)$

or, $ar^{p-1} = ar^p + ar^{p+1} + ar^{p+2} + \dots$

$$\therefore r^{p-1} = \frac{r^p}{1-r} \text{ [sum of an infinite G.P.]}$$

$$\therefore 1 - r = r \Rightarrow r = \frac{1}{2}.$$

Hence the series is $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots \infty$.

12. (a) $\text{Sum} = \frac{8}{9} [9 + 99 + 999 + \dots n \text{ terms}]$

$$= \frac{8}{9} [(10-1) + (100-1) + (1000-1) + \dots n \text{ terms}]$$

$$= \frac{8}{9} [(10 + 10^2 + 10^3 + \dots + 10^n) - n]$$

$$= \frac{8}{9} \left[\frac{10(10^n - 1)}{10 - 1} - n \right]$$

$$= \frac{8}{81} [10^{n+1} - 9n - 10]$$

13. (a) Assume a series having a few number of terms e.g., 1, 2, 4, 8, 16, 32

Now sum of all the terms at the even places = 42 (P_2)

and sum of all the terms at the odd places = 21 (P_1)

$$\text{common ratio of this series} = \frac{42}{21} = 2 = P_2 / P_1.$$

14. (a) The length of sides of successive triangles form a GP with common ratio $1/3$.

15. (d) Take any values for the numbers.

Say, the two positive numbers are 1 and 27.

Then, $a = 14$, $b = 3$ and $c = 9$.

12

LINEAR EQUATIONS

- Linear Equations
- Steps to be followed to solve a word problem using linear equation(s)

LINEAR EQUATIONS

Many times in mathematics, we have to find the value of an unknown. In this case we represent the unknown by using some letters like p, q, r, x, y etc. These letters are then called as the variable representations of the unknown quantity.

Let's see a problem: A man says, "I am thinking of a number, when I divide it by 3 and then add 5, my answer is twice the number thought of". Find the number.

Although you do not have the actual number in your mind, you can still move ahead to solve the problem by assuming a variable to represent the number.

The information given in the problem related to the number ultimately will give the value of the unknown i.e., the number in this particular problem. See the process involved in solving the above problem:

Let the number be x .

On dividing the number x by 3, we get $\frac{x}{3}$.

On adding 5 to $\frac{x}{3}$, we get $\frac{x}{3} + 5$.

According to the information given in the problem, $\frac{x}{3} + 5$ is twice the number i.e., $2x$.

$$\therefore \frac{x}{3} + 5 = 2x$$

$$\Rightarrow 5 = 2x - \frac{x}{3}$$

$$\Rightarrow 5 = \frac{6x - x}{3}$$

$$\Rightarrow 5 = \frac{5x}{3}$$

$$\Rightarrow 5 \times 3 = 5x$$

$$\Rightarrow \frac{5 \times 3}{5} = x \Rightarrow x = 3$$

Hence, required number = 3

Here, ' $\frac{x}{3} + 5 = 2x$ ' is the mathematical statement of equality involving the variable x .

Each mathematical statement of equality involving any number of variables is called an equation. Note that in the above equation there is a single variable x , but according to the given and required information, you may have to suppose more than one variable to move ahead to solve the problem and hence, an equation may have one or more than one variable. If all the variables in the equation are in numerator, no product or quotient (of the expressions including variable(s)) is available in the equation and the power of each variable is unity, then the equation is called linear equation.

Linear equations are commonly used in CAT and Cat like Aptitude tests.

See the following illustration, whose solutions will be found out by converting the statements of the problems into linear equation(s).

Illustration 1: Find the two consecutive even numbers whose sum is 76.

Solution: Let one of the two consecutive even numbers be x .

As we know that the difference between any two consecutive even number is always 2. Therefore the next consecutive even number will be $(x + 2)$.

According to the question, sum of the two consecutive even numbers is 76.

$$\therefore x + (x + 2) = 76$$

$$\Rightarrow 2x + 2 = 76, \Rightarrow 2x = 76 - 2 = 74$$

$$\Rightarrow x = \frac{74}{2} = 37$$

Hence the two consecutive numbers are 37 and 39.

Note that 'the difference between any two consecutive even numbers is always 2' is an information related to the variable x is an extra information because it is not given in the problem, but without this information, we would not form the equation required for solving the problem. Thus you must use the extra information, which helps in formation of equation, if needed.

Illustration 2: Sanjay starts his job with a certain monthly salary and earns a fixed increment every year. If his salary was ₹ 31,000 after four years of service and ₹ 40,000 after 10 years, find his initial salary and annual increment.

Solution: Let the initial salary be ₹ x and fixed increment every year be ₹ y .

$$\therefore x + 4y = 31000 \quad \dots(1)$$

$$\text{and } x + 10y = 40000 \quad \dots(2)$$

On subtracting equation (1) from (2), we get

$$6y = 9000 \Rightarrow y = 1500$$

Now putting the value of y in equation (1), we get

$$x + 6000 = 31000 \Rightarrow x = 25000$$

Hence initial salary = ₹ 25000

and fixed annual increment = ₹ 1500.

Illustration 3: If a number is decreased by 4 and divided by 6, the result is 8. What would be the result if 2 was subtracted from the number and then it was divided by 5?

$$(a) 9\frac{2}{3} \quad (b) 10$$

$$(c) 10\frac{1}{5} \quad (d) 11\frac{1}{5}$$

Solution: (b) Let the number be x . Then,

$$\frac{x-4}{6} = 8 \Rightarrow x-4 = 48 \Rightarrow x = 52$$

$$\therefore \frac{x-2}{5} = \frac{52-2}{5} = \frac{50}{5} = 10.$$

Illustration 4: If three numbers are added in pairs, the sums equal 10, 19 and 21. The numbers are

$$(a) 4, 6, 10 \quad (b) 6, 4, 15$$

$$(c) 3, 5, 10 \quad (d) 2, 5, 15$$

Solution: (b) Let the numbers be x, y and z . Then,

$$x + y = 10 \quad \dots(1) \quad y + z = 19 \quad \dots(2)$$

$$\dots(3) \quad x + z = 21 \quad \dots(4)$$

Adding (1), (2) and (3), we get : $2(x + y + z) = 50$

$$\text{or } x + y + z = 25.$$

Thus,

$$x = 25 - 19 = 6; y = 25 - 21 = 4;$$

$$z = 25 - 10 = 15.$$

Hence, the required numbers are 6, 4 and 15.

Illustration 5: If the sum of two numbers is 42 and their product is 437, then find the absolute difference between the numbers.

$$(a) 4$$

$$(b) 7$$

$$(c) 9$$

$$(d) \text{ Cannot be determined}$$

Solution: (a) Let the numbers be x and y . Then, $x + y = 42$ and $xy = 437$.

$$x - y = \sqrt{(x + y)^2 - 4xy} = \sqrt{(42)^2 - 4 \times 437}$$

$$= \sqrt{1764 - 1748} = \sqrt{16} = 4.$$

$\therefore$ Required difference = 4.

Note that depending upon the number of variables in a problem, a linear equation may have one, two or even more variables. But to get the value of the variables the number of equations should be always equal to the number of variables.

STEPS TO BE FOLLOWED TO SOLVE A WORD PROBLEM USING LINEAR EQUATION(S)

Step (i): Read the problem carefully and note what is/are given and what is/are required.

Step (ii): Denote the unknown quantity by some letters, say p, q, r, x, y etc.

Step (iii): Translate the statements of the problem into mathematical statements i.e., equations using the condition(s) given in the problem and extra information(s) related to the variable(s) derived from the statement(s) in the problem.

Step (iv): Solve the equation(s) for the unknown(s).

Step (v): Check whether the solution satisfies the equation(s).

Most of the time in solving the word problem you get stuck. It could be due to one or more of the following four reasons:

Reason (i): You are not able to interpret one or more statements in the problem. In this case you concentrate on developing your ability to decode the mathematical meaning of the statement(s) in the problems.

Reason (ii): You have either not used all the information given in the problem or have used them in the incorrect order.

In such a case, go back to the problem and try to identify each statement and see whether you have utilized it or not. If you have already used all the information, then check whether you have used the information given in the problem in the correct order.

Reason (iii): Even though you might have used all the information given in the problem, you have not utilized some of the information completely.

In such a case, you need to review each part of each information given in the problem and look at whether any additional details can be derived out from the same informations. If derived any additional details, use them in forming or solving the equation(s). Sometimes a statement can be used for more than one perspective. In this case, if you have used that statement for one perspective, then using it in the other perspective will solve the problem.

Reason (iv): You are struck because the problem does not have a solution. In such a case, check the solution once and if it is correct go back to reason (i), (ii) and (iii).

Illustration 6: Find the two odd numbers whose sum is 12.

Solution: Let the two odd numbers are x and y .

$$\text{Then } x + y = 12$$

There is no other information about the two variable x and y .

Hence, there will be no other equation between the variable x and y . So, we can not find the exact solution of the problem. The equation formed above yields a set of possibilities for the value of x and y as (1, 11), (3, 9), (5, 7), (7, 5), (9, 3), (11, 1). One of these possibilities has to be the correct answer.

Illustration 7: A piece of wire is 80 metres long. It is cut into three pieces. The longest piece is 3 times as long as the middle-sized and the shortest piece is 46 metres shorter than the longest piece. Find the length of the shortest piece (in metres).

Solution: Let the length of the longest piece = a metres

Length of middle-sized piece = b metres

Since sum of the length of three pieces of wire = 80 metres

$\therefore$ length of shortest piece = $80 - (a + b)$ metres

$$\text{Now } a = 3b \quad \dots(1)$$

$$\text{and } 80 - (a + b) = a - 46 \quad \dots(2)$$

From (1) and (2),

$$80 - \left(a + \frac{a}{3}\right) = a - 46$$

$$\Rightarrow 80 - \frac{3a + a}{3} = a - 46$$

$$\Rightarrow 80 + 46 = a + \frac{4a}{3}$$

$$\Rightarrow \frac{7a}{3} = 126 \Rightarrow a = 126 \times \frac{3}{7} = 54$$

$$\therefore b = \frac{a}{3} = \frac{54}{3} = 18,$$

$$\text{and } 80 - (a + b) = 80 - (54 + 18) = 8$$

Hence length of shortest piece = 8 metres.

Illustration 8: Mohan took five papers in an examination, where each paper was of 250 marks. His marks in these papers were in the ratio 6 : 8 : 10 : 12 : 15. In all papers together, Mohan obtained 70% of the total marks. Then find the number of papers in which he got more than 80% marks.

Solution: Ratio of marks obtained in five papers are

$$6 : 8 : 10 : 12 : 15.$$

Let marks obtained in five papers are $6x$, $8x$, $10x$, $12x$ and $15x$.

$$\therefore 6x + 8x + 10x + 12x + 15x = 5 \times 250 \times \frac{70}{100}$$

$$\Rightarrow 51x = 125 \times 7 \Rightarrow x = \frac{125 \times 7}{51} = 17 \text{ (approx.)}$$

$$\text{Now } 80\% \text{ of } 250 = 250 \times \frac{80}{100} = 200$$

$$\text{Now } 10x = 170, 12x = 12 \times 17 = 204$$

$$\text{Hence } 6x, 8x, 10x < 200$$

$$\text{and } 12x, 15x > 200$$

Therefore Mohan got more than 80% in only two subjects.

Illustration 9: The sum of the digits of a two digit number is 16. If the number formed by reversing the digits is less than the original number by 18. Find the original number.

Solution: Let unit digit be x . Then tens digit = $16 - x$

$$\therefore \text{Original number} = 10 \times (16 - x) + x \\ = 160 - 9x.$$

On reversing the digits, we have x at the tens place and $(16 - x)$ at the unit place.

$$\therefore \text{New number} = 10x + (16 - x) = 9x + 16$$

$$\text{Original number} - \text{New number} = 18$$

$$(160 - 9x) - (9x + 16) = 18$$

$$160 - 18x - 16 = 18$$

$$-18x + 144 = 18$$

$$-18x = 18 - 144 \Rightarrow 18x = 126$$

$$\Rightarrow x = 7$$

$\therefore$ In the original number, we have unit digit = 7

$$\text{Tens digit} = 16 - 7 = 9$$

Thus, original number = 97

Illustration 10: The denominator of a rational number is greater than its numerator by 4. If 4 is subtracted from the numerator and 2 is added to its denominator, the new number becomes $\frac{1}{6}$. Find the original number.

Solution: Let the numerator be x .

Then, denominator = $x + 4$

$$\therefore \frac{x - 4}{x + 4 + 2} = \frac{1}{6}$$

$$\Rightarrow \frac{x - 4}{x + 6} = \frac{1}{6}$$

$$\Rightarrow 6(x - 4) = x + 6$$

$$\Rightarrow 6x - 24 = x + 6 \Rightarrow 5x = 30$$

$$\therefore x = 6$$

Thus, Numerator = 6, Denominator = $6 + 4 = 10$.

$$\text{Hence the original number} = \frac{6}{10}.$$

Illustration 11: A man covers a distance of 33 km in $3\frac{1}{2}$ hours; partly on foot at the rate of 4 km/hr and partly on bicycle at the rate of 10 km/hr. Find the distance covered on foot.

Solution: Let the distance covered on foot be x km.

$$\therefore \text{Distance covered on bicycle} = (33 - x) \text{ km}$$

$$\therefore \text{Time taken on foot} = \frac{\text{Distance}}{\text{Speed}} = \frac{x}{4} \text{ hr.}$$

$$\therefore \text{Time taken on bicycle} = \frac{33-x}{10} \text{ hr.}$$

$$\text{The total time taken} = \frac{7}{2} \text{ hr.}$$

$$\frac{x}{4} + \frac{33-x}{10} = \frac{7}{2}$$

$$\frac{5x+66-2x}{20} = \frac{7}{2}$$

$$6x+132 = 140$$

$$6x = 140 - 132$$

$$6x = 8$$

$$x = \frac{8}{6} = 1.33 \text{ km.}$$

$\therefore$ The distance covered on foot is 1.33 km.

Illustration 12: The total age of A and B is 12 years more than the total age of B and C . C is how many years younger than A ?

(a) 12

(b) 24

(c) C is elder than A

(d) Data inadequate

Solution: (a) $(A+B) - (B+C) = 12 \Rightarrow A - C = 12$.

C is 12 year younger than A .

Illustration 13: The sum of four numbers is 64. If you add 3 to the first number, 3 is subtracted from the second number, the third is multiplied by 3 and the fourth is divided by 3, then all the results are equal. What is the difference between the largest and the smallest of the original numbers?

(a) 21

(b) 27

(c) 32

(d) Cannot be determined

Solution: (c) Let the four numbers be A , B , C and D .

Let $A+3 = B-3 = 3C = D/3 = x$ (let)

Then, $A = x-3$, $B = x+3$, $C = x/3$ and $D = 3x$.

$$A+B+C+D = 64 \Rightarrow (x-3) + (x+3) + x/3 + 3x = 64$$

$$\Rightarrow 5x + x/3 = 64 \Rightarrow 16x = 192 \Rightarrow x = 12$$

Thus, the numbers are 9, 15, 4 and 36.

$$\therefore \text{Required difference} = (36-4) = 32.$$

Practice Exercises



Foundation Level

- If $(x - 3)(2x + 1) = 0$, then the possible values of $2x + 1$ are:
(a) 0 only (b) 0 and 3
(c) $-\frac{1}{2}$ and 3 (d) 0 and 7
- Father is 5 years older than the mother and mother's age now is thrice the age of the daughter. The daughter is now 10 years old. What was father's age when the daughter was born?
(a) 20 years (b) 15 years
(c) 25 years (d) 30 years
- A father told his son, "I was as old as you are at present, at the time of your birth," If the father is 38 years old now, what was the son's age five years back?
(a) 19 years (b) 14 years
(c) 38 years (d) 33 years
- When 24 is subtracted from a number, it reduces to its four-seventh. What is the sum of the digits of that number?
(a) 1 (b) 9
(c) 11 (d) Data inadequate
- If the sum of one-half and one-fifth of a number exceeds one-third of that number by $7\frac{1}{3}$, the number is
(a) 15 (b) 18
(c) 20 (d) 30
- A driver's income consists of his salary and tips. During one week his tips were $\frac{5}{4}$ of his salary. What fraction of his income came from tips?
(a) $\frac{4}{9}$ (b) $\frac{5}{9}$
(c) $\frac{5}{8}$ (d) $\frac{5}{4}$
- In a certain party, there was a bowl of rice for every two guests, a bowl of broth for every three of them and a bowl of meat for every four of them. If in all there were 65 bowls of food, then how many guests were there in the party?
(a) 65 (b) 24
(c) 60 (d) 48
- Two numbers are such that the square of one is 224 less than 8 times the square of the other. If the numbers be in the ratio of 3 : 4, the numbers are
(a) 36 (b) 48
(c) 56 (d) 64
- Ram and Mohan are friends. Each has some money. If Ram gives ₹ 30 to Mohan, then Mohan will have twice the money left with Ram. But if Mohan gives ₹ 10 to Ram, then Ram will have thrice as much as is left with Mohan. How much money does each have?
(a) ₹ 62, ₹ 34 (b) ₹ 6, ₹ 2
(c) ₹ 170, ₹ 124 (d) ₹ 43, ₹ 26
- The sum of two numbers is 25 and their difference is 13. Find their product.
(a) 104 (b) 114
(c) 315 (d) 325
- There are two examination rooms A and B. If 10 candidates are sent from room A to room B, the number of candidates in each room is the same, while if 20 are sent from room B to room A, the number of candidates in room A becomes double the number in room B. The number of candidates in each room are, respectively:
(a) 80 and 100 (b) 100 and 80
(c) 80 and 120 (d) 100 and 60
- A person on tour has ₹ 360 for his daily expenses. He decides to extend his tour programme by 4 days which leads to cutting down daily expenses by ₹ 3 a day. The number of days of his tour programme is
(a) 15 (b) 20
(c) 18 (d) 16
- The difference between the squares of two numbers is 256000 and the sum of the numbers is 1000. The numbers are:
(a) 600, 400 (b) 628, 372
(c) 640, 360 (d) None of these
- The sum of three consecutive odd numbers is 20 more than the first of these numbers. What is the middle number?
(a) 7 (b) 9
(c) 11 (d) Data inadequate
- The autorickshaw fare consists of a fixed charge together with the charge for the distance covered. For a journey of 10 km, the charge paid is ₹ 85 and for a journey of 15 km, the charge paid is ₹ 120. The fare for a journey of 25 km will be
(a) ₹ 175 (b) ₹ 190
(c) ₹ 180 (d) ₹ 225
- The denominator of a rational number is greater than its numerator by 4. If 4 is subtracted from the numerator and 2 is added to its denominator, the new number becomes $\frac{1}{6}$. Find the original number.
(a) $\frac{1}{6}$ (b) $\frac{6}{10}$
(c) $\frac{10}{6}$ (d) 6

17. The present ages of Vikas and Vishal are in the ratio 15 : 8. After ten years, their ages will be in the ratio 5 : 3. Find their present ages.
(a) 60 years, 32 years (b) 32 years, 60 years
(c) 15 years, 8 years (d) 8 years, 15 years
18. The sum of three consecutive multiples of 3 is 72. What is the largest number ?
(a) 21 (b) 24
(c) 27 (d) 36
19. Two-fifths of one-fourth of three-sevenths of a number is 15. What is half of that number?
(a) 96 (b) 196
(c) 94 (d) None of these
20. The sum of the ages of a father and his son is 4 times the age of the son. If the average age of the father and the son is 28 years, what is the son's age?
(a) 14 years (b) 16 years
(c) 12 years (d) Data inadequate
21. The product of two numbers is 192 and the sum of these two numbers is 28. What is the smaller of these two numbers?
(a) 16 (b) 14
(c) 12 (d) 18
22. The sum of three consecutive even numbers is 14 less than one-fourth of 176. What is the middle number?
(a) 8 (b) 10
(c) 6 (d) Data inadequate
23. The difference between the numerator and the denominator of a fraction is 5. If 5 is added to its denominator, the fraction is decreased by $1\frac{1}{4}$. Find the value of the fraction.
(a) $\frac{1}{6}$ (b) $2\frac{1}{4}$
(c) $3\frac{1}{4}$ (d) 6
24. The sum of three numbers is 300. If the ratio between first and second be 5 : 9 and that between second and third be 9 : 11, then second number is
(a) 12 (b) 60
(c) 108 (d) 132
25. When 20 is subtracted from a number, it reduces to seven-twelve of the number. What is the sum of the digit of the number?
(a) 40 (b) 44
(c) 46 (d) 48
26. If the number obtained on the interchanging the digits of two-digit number is 18 more than the original number and the sum of the digits is 8, then what is the original number?
(a) 50 (b) 51
(c) 52 (d) 53
27. There are two numbers such that sum of twice the first number and thrice the second number is 100 and the sum of thrice the first number and twice the second number is 120. Which is the larger number?
(a) 32 (b) 12
(c) 14 (d) 35
28. There are two number such that the sum of twice the first number and thrice the second number is 300 and the sum of thrice the first number and twice the second number is 265. What is the larger number?
(a) 24 (b) 39
(c) 85 (d) 74
29. If the digits of a two-digit number are interchanged, the number formed is greater than the original number by 45. If the difference between the digits is 5, then what is the original number?
(a) 16 (b) 27
(c) 38 (d) Cannot be determined
30. A railway half ticket cost half the full fare. But the reservation charge on the half ticket is the same as that on full ticket. One reserved first class ticket for a journey between two stations is ` 525 and the cost of one full and one half reserved first class tickets is ` 850. What is the reservation charges?
(a) ` 125 (b) ` 2003
(c) ` 145 (d) Cannot be determined
31. Krishna has some hens and some goats. If the total number animal heads are 81 and the total number of animal legs are 234, how many goats does Krishna have?
(a) 45 (b) 24
(c) 36 (d) Cannot be determined
32. The average age of father and his son is 22 years. The ratio of their ages is 10 : 1 respectively. What is the age of the son?
(a) 24 (b) 4
(c) 40 (d) 14
33. The sum of third, fourth and fifth part of a number exceeds half of the number by 34. Find the number.
(a) 60 (b) 120
(c) 30 (d) None of these
34. A series of books was published at seven years interval. When the seventh book was issued, the sum of the publication year was 13,524. When was the first book published?
(a) 1932 (b) 1942
(c) 1911 (d) 1917
35. In a two-digit number the digit in the unit's place is three times the digit in the tenth's place. The sum of the digits is equal to 8. Then, what is the number ?
(a) 20 (b) 26
(c) 39 (d) 13
36. The number obtained by interchanging the two digits of a two-digit number is lesser than the original number by 54. If the sum of the two-digit number is 10, then what is the original number ?
(a) 28 (b) 39
(c) 82 (d) Cannot be determined
37. The age of the father 5 years ago was 5 times the age of his son. At present the father's age is 3 times that of his son. What is the present age of the father?
(a) 33 years (b) 30 years
(c) 45 years (d) None of these

38. If the numerator of a fraction is increased by 150% and denominator of the fraction is increased by 350%. The resultant fraction is $\frac{25}{31}$ what is the original fraction?
- (a) $\frac{11}{7}$ (b) $\frac{11}{15}$
(c) $\frac{15}{17}$ (d) $\frac{13}{15}$
39. The denominator of a fraction is 2 more than thrice its numerator. If the numerator as well as denominator is increased by one, the fraction becomes $\frac{1}{3}$. What was the original fraction?
- (a) $\frac{4}{13}$ (b) $\frac{3}{11}$
(c) $\frac{5}{13}$ (d) $\frac{5}{11}$
40. Smita was asked to multiply a certain number by 36. She multiplied it by 63 instead and got an answer of 3834 more than the correct one. What was the number to be multiplied?
- (a) 152 (b) 126
(c) 142 (d) 148
41. Ravi has spent a quarter $\left(\frac{1}{4}\right)$ of his life as a boy, one-fifth $\left(\frac{1}{5}\right)$ as a youth, one-third $\left(\frac{1}{3}\right)$ as man and thirteen (13) years in old age. What is his present age?
- (a) 70 years (b) 80 years
(c) 60 years (d) 65 years
42. In a group of equal number of cows and herdsmen the number of legs was 28 less than four times the number of heads. The number of herdsmen was
- (a) 7 (b) 28
(c) 21 (d) 14
43. The ratio of the present ages of a mother and daughter is 7 : 1. Four years ago the ratio of their ages was 19 : 1. What will be the mother's age four years from now?
- (a) 42 years (b) 38 years
(c) 46 years (d) 36 years
44. A number when subtracted by $\frac{1}{7}$ of itself gives the same value as the sum of all the angles of a triangle. What is the number?
- (a) 224 (b) 210
(c) 140 (d) 350
45. Farah got married 8 years ago. Today her age is $1\frac{2}{7}$ times her age at the time of her marriage. At present her daughter's age is one-sixth of her age. What was her daughter's age 3 years ago?
- (a) 6 years (b) 4 years
(c) 3 years (d) None of these
46. There are some parrots and some tigers in a forest. If the total number of animal heads in the forest is 858 and the total number of animal legs is 1,846, what is the number of parrots in the forest?
- (a) 845 (b) 833
(c) 800 (d) 793
47. The ratio between a two-digit number and the sum of the digits of that number is 4 : 1. If the digit in the unit's place is 3 more than the digit in the ten's place, what is the number?
- (a) 36 (b) 63
(c) 39 (d) 93
48. The ratio of two numbers is 4 : 7. If each of these numbers increases by 30, their ratio will become 5 : 8. What is the average of these two numbers?
- (a) 135 (b) 145
(c) 155 (d) 165
49. A number of two digits has 3 for its unit's digit, and the sum of digits is $\frac{1}{7}$ of the number itself, The number is
- (a) 43 (b) 53
(c) 63 (d) 73
50. A number is doubled and 9 is added. If the resultant is trebled, it becomes 75. What is that number?
- (a) 3.5 (b) 6
(c) 8 (d) None of these
51. The difference between a two-digit number and the number obtained by interchanging the position of its digits is 36. What is the difference between the two digits of that number?
- (a) 3 (b) 4
(c) 9 (d) Cannot be determined
52. 54 is to be divided into two parts such that the sum of 10 times the first and 22 times the second is 780. The bigger part is:
- (a) 24 (b) 34
(c) 30 (d) 32
53. The sum of five whole numbers is 146. If m is the largest of the five numbers, then which is the smallest value that m can have
- (a) 30 (b) 35
(c) 28 (d) 27
54. A man has ₹ 480 in the denominations of one-rupee notes, five-rupee notes and ten-rupee notes. The number of notes of each denomination is equal. What is the total number of notes that he has ?
- (a) 45 (b) 60
(c) 75 (d) 90
55. If the numerator of a fraction is increased by 200% and the denominator is increased by 200%, then resultant fraction is $2\frac{4}{5}$. What is the original fraction?
- (a) $\frac{4}{7}$ (b) $\frac{13}{12}$
(c) $\frac{11}{12}$ (d) None of these

Standard Level

- The sum of the digits of a three-digit number is 16. If the tens digit of the number is 3 times the units digit and the units digit is one-fourth of the hundredth digit, then what is the number ?
(a) 446 (b) 561
(c) 682 (d) 862
- A two digit number is such that the product of its digits is 14. When 45 is added to the number, then the digits interchange their places. Find the number.
(a) 72 (b) 27
(c) 37 (d) 14
- When Ranjeev was born, his father was 32 years older than his brother and his mother was 25 years older than his sister. If Ranjeev's brother is 6 years older than Ranjeev and his mother is 3 years younger than his father, how old was Ranjeev's sister when he was born?
(a) 15 years (b) 14 years
(c) 7 years (d) 10 years
- In an exercise room some discs of denominations 2 kg and 5 kg are kept for weightlifting. If the total number of discs is 21 and the weight of all the discs of 5 kg is equal to the weight of all the discs of 2 kg, find the weight of all the discs together.
(a) 80 kg (b) 90 kg
(c) 56 kg (d) None of these
- One-third of Ramesh's marks in Arithmetic is equal to half his marks in English. If he gets 150 marks in the two subjects together, how many marks has he got in English?
(a) 60 (b) 120
(c) 30 (d) 50
- The sum of four numbers is 64. If you add 3 to the first number, 3 is subtracted from the second number, the third is multiplied by 3 and the fourth is divided by 3, then all the results are equal. What is the difference between the largest and the smallest of the original numbers?
(a) 21 (b) 27
(c) 32 (d) Cannot be determined
- In a family, a couple has a son and a daughter. The age of the father is three times of his daughter and the age of the son is half of his mother. The wife is nine years younger to her husband and the brother is seven years older than his sister. What is the age of the mother?
(a) 40 years (b) 50 years
(c) 45 years (d) 60 years
- The sum of the numerator and denominator of a fraction is 11. If 1 is added to the numerator and 2 is subtracted from the denominator it becomes $\frac{3}{2}$. The fraction is
(a) $\frac{3}{8}$ (b) $\frac{5}{6}$
(c) $\frac{7}{4}$ (d) $\frac{9}{2}$
- A uniform one metre long rod AB of weight 17 kg is suspended horizontally from fixed supports by two vertical strings attached to points C and D on the rod at distances of 12 cm and 16 cm from A and B respectively. The strings at C and D can support weights of 10 kg and 9 kg respectively without breaking. Then the position where a weight of 2 kg can be attached to the rod without breaking either of the strings is
(a) 10 cm from A (b) 12 cm from A
(c) 13 cm from A (d) None of these
- In an objective examination of 90 questions, 5 marks are allotted for every correct answer and 2 marks are deducted for every wrong answer. After attempting all the 90 questions a student got a total of 387 marks. Find the number of questions that he attempted wrong.
(a) 36 (b) 18
(c) 9 (d) 27
- Two different natural numbers are such that, their product is less than their sum. Then one of the number must be
(a) 3 (b) 1
(c) 2 (d) 0
- Out of total number of students in a college 12% are interested in sports. $\frac{3}{4}$ of the total number of students are interested in dancing. 10% of the total number of students are interested in singing and the remaining 15 students are not interested in any of the activities. What is the total number of students in the college?
(a) 450 (b) 500
(c) 600 (d) Cannot be determined
- A number consists of two digits such that the digit in the ten's place is less by 2 than the digit in the unit's place. Three times the number added to $\frac{6}{7}$ times the number obtained by reversing the digits equals 108. The sum of digits in the number is :
(a) 8 (b) 9
(c) 6 (d) 7

14. When the numerator and the denominator of a fraction are increased by 1 and 2 respectively, the fraction becomes $\frac{2}{3}$, and when the numerator and the denominator of the same fraction are increased by 2 and 3 respectively, the fraction becomes $\frac{5}{7}$. What is the original fraction?
- (a) $\frac{5}{6}$ (b) $\frac{3}{4}$
(c) $\frac{3}{5}$ (d) $\frac{6}{7}$
15. If three numbers are added in pairs, the sums equal 10, 19 and 21. Find the numbers.
- (a) 6, 4, 15 (b) 1, 9, 12
(c) 9, 10, 2 (d) 5, 6, 10
16. Find the number of positive integer solutions of the equation $\frac{2}{x} + \frac{15}{y} = 5$.
- (a) 0 (b) 1
(c) 2 (d) 3
17. Of the three numbers, the sum of the first two is 45; the sum of the second and the third is 55 and the sum of the third and thrice the first is 90. The third number is
- (a) 20 (b) 25
(c) 30 (d) 3
18. In a certain factory, each day the expected number of accidents is related to the number of overtime hours by a linear equation. Suppose that on one day there were 1000 overtime hours logged and 8 accidents reported and on another day, there were 400 overtime hours logged and 5 accidents. What are the expected number of accidents when no overtime hours are logged?
- (a) 2 (b) 3
(c) 4 (d) 5
19. One of the angles of a triangle is two-third angle of sum of adjacent angles of parallelogram. Remaining angles of the triangle are in ratio 5 : 7 respectively. What is the value of second largest angle of the triangle?
- (a) 25° (b) 40°
(c) 35° (d) Cannot be determined

Expert Level

- There are three baskets of fruits. First basket has twice the number of fruits in the 2nd basket. Third basket has $\frac{3}{4}$ th of the fruits in the first. The average of the fruits in all the baskets is 30. What is the number of fruits in the first basket?
(a) 20 (b) 30
(c) 35 (d) 40
- Large, medium and small ships are used to bring water. 4 large ships carry as much water as 7 small ships. 3 medium ships carry the same amount of water as 2 large ships, and 1 small ship. 15 large, 7 medium and 14 small ships each made 36 journeys and brought a certain quantity of water. In how many journeys would 12 large, 14 medium and 21 small ships bring the same quantity?
(a) 32 (b) 25
(c) 29 (d) 49
- If the sum of two numbers is 52 and their product is 672, then find the absolute difference between the numbers.
(a) 73 (b) 4
(c) 32 (d) 6
- A man ordered 4 pairs of black socks and some pairs of brown socks. The price of a black pair is double that of a brown pair. While preparing the bill, the clerk did a mistake and interchanged the number of black and brown pairs. This increased the bill by 50%. The ratio of the number of black and brown pairs of socks in the original order was
(a) 4 : 1 (b) 2 : 1
(c) 1 : 4 (d) 1 : 2
- A certain number of tennis balls were purchased for ₹ 450. Five more balls could have been purchased for the same amount if each ball was cheaper by ₹ 15. Find the number of balls purchased.
(a) 15 (b) 20
(c) 10 (d) 25
- In a certain office, $\frac{1}{3}$ of the workers are women, $\frac{1}{2}$ of the women are married and $\frac{1}{3}$ of the married women have children. If $\frac{3}{4}$ of the men are married and $\frac{2}{3}$ of the married men have children, then what part of workers are without children?
(a) $\frac{5}{18}$ (b) $\frac{4}{9}$
(c) $\frac{11}{18}$ (d) $\frac{17}{36}$
- The sum of three consecutive odd numbers and three consecutive even numbers together is 231. also, the smallest odd number is 11 less than the smallest even number. What is the sum of the largest odd number and the largest even number?
(a) 81 (b) 83
(c) 74 (d) 87
- The sum of four consecutive even numbers is 44. What is the sum of the squares of these numbers?
(a) 288 (b) 504
(c) 696 (d) 920
- Two different two-digit natural numbers are written beside each other such that the larger number is written on the left. When the absolute difference of the two numbers is subtracted from the four-digit number so formed, the number obtained is 5481. What is the sum of the two two-digit numbers?
(a) 70 (b) 71
(c) 72 (d) 73
- A number consists of 3 digits whose sum is 10. The middle digit is equal to the sum of the other two and the number will be increased by 99 if its digits are reversed. The number is :
(a) 145 (b) 253
(c) 370 (d) 352
- A number consists of two-digits such that the digit in the ten's place is less by 2 than the digit in the unit's place. Three times the number added to $\frac{6}{7}$ times the number obtained by reversing the digits equals 108. The sum of the digits in the number is:
(a) 6 (b) 7
(c) 8 (d) 9
- Albela, Bob and Chulbul have to read a document of seventy eight pages and make a presentation next day. They realize that the article is difficult to understand and they would require team work to finish the assignment. Albela can read a page in 2 minutes, Bob in 3 minutes, and Chulbul in 4 minutes. If they divide the article into 3 parts so that all three of them spend the equal amount of time on the article, the number of pages that Bob should read is
(a) 24 (b) 25
(c) 26 (d) 27
- Nikhil's mother asks him to buy 100 pieces of sweets worth ₹ 100/-. The sweet shop has 3 kinds of sweets, *kajubarfi*, *gulabjamun* and *sandesh*. *Kajubarfi* costs ₹ 10/- per piece, *gulabjamun* costs ₹ 3/- per piece and *sandesh* costs 50 paise per piece. If Nikhil decides to buy at least one sweet of each type. How many *gulabjamuns* should he buy?
(a) 1 (b) 2
(c) 3 (d) 4

Test Yourself

- If $y \neq 0$ then the number of values of the pair (x, y) such that $x + y + \frac{x}{y} = \frac{1}{2}$ and $(x + y)\frac{x}{y} = -\frac{1}{2}$, is:
 - 1
 - 2
 - 0
 - None of these
- On Children's Day, sweets were to be equally distributed among 175 children in a school. Actually on the Children's Day 35 children were absent and therefore, each child got 4 sweets extra. How many sweets were available in all for distribution?
 - 2480
 - 2680
 - 2750
 - None of these
- A two-digit number is seven times the sum of its digits. If each digit is increased by 2, the number thus obtained is 4 more than six times the sum of its digits. Find the number.
 - 42
 - 24
 - 48
 - None of these
- A woman sells to the first customer half her stock of apples and half an apple extra; to the second customer, she gives half her remaining stock and half an apple extra, and so also to a third and then to a fourth customer. She find that she has now 15 apples left. How many she had at first?
 - 250
 - 254
 - 255
 - 375
- If the numerator of a fraction is increased by 2 and the denominator is increased by 1, the fraction becomes $\frac{5}{8}$ and if the numerator of the same fraction is increased by 3 and the denominator is increased by 1 the fraction becomes $\frac{3}{4}$. What is the original fraction?
 - $\frac{2}{7}$
 - $\frac{4}{7}$
 - $\frac{3}{7}$
 - None of these
- Out of total number of students in a college 12% are interested in sports. $\frac{3}{4}$ of the total number of students are interested in dancing. 10% of the total number of students are interested in singing and the remaining 15 students are not interested in any of the activities. What is the total number of students in the college?
 - 450
 - 500
 - 600
 - None of these
- The sum of four numbers is 64. If you add 3 to the first number, 3 is subtracted from the second number, the third is multiplied by 3 and the fourth is divided by 3, then all the results are equal. What is the difference between the largest and the smallest of the original numbers?
 - 32
 - 27
 - 21
 - None of these
- A piece of string is 40 centimeters long. It is cut into three pieces. The longest piece is 3 times as long as the middle-sized piece and the shortest piece is 23 centimeters shorter than the longest piece. Find the length of the shortest piece.
 - 27
 - 5
 - 4
 - 9
- If the ages of P and R are added to twice the age of Q , the total becomes 59. If the ages of Q and R are added to thrice the age of P , the total becomes 68. And if the age of P is added to thrice the age of Q and thrice the age of R , the total becomes 108. What is the age of P ?
 - 15 years
 - 19 years
 - 17 years
 - 12 years
- When you reverse the digits of the number 13, the number increases by 18. How many other two-digit numbers increase by 18 when their digits are reversed?
 - 5
 - 6
 - 7
 - 8
- If the numerator of a fraction is increased by 25% and the denominator is doubled, the fraction thus obtained is $\frac{5}{9}$. What is the original fraction?
 - $\frac{2}{3}$
 - $\frac{4}{9}$
 - $\frac{8}{9}$
 - None of these
- Out of three numbers the sum of the first and the second number is 73 and the sum of the second and the third number is 77. The sum of third and thrice the first number is 104. What is the third number?
 - 25
 - 39
 - 48
 - None of these
- Two-third of the first number is equal to cube of the second number. If the second number is equal to 12% of 50. What is the sum of the first and second numbers?
 - 330
 - 360
 - 390
 - 372
- A railway half ticket costs half the full fare. But the reservation charge on the half ticket is the same as that on full ticket. One reserved first class ticket for a journey between two stations is ₹ 525 and the cost of one full and one half reserved first class tickets is ₹ 850. What is the reservation charge?
 - ₹ 125
 - ₹ 2003
 - ₹ 145
 - Cannot be determined
- Out of three numbers the sum of the first and the second number is 73 and the sum of the second and the third number is 77. The sum of third and thrice the first number is 104. What is the third number?
 - 25
 - 39
 - 48
 - None of these

Hints & Solutions

Foundation Level

1. (d) $(x - 3)(2x + 1) = 0 \Rightarrow (x - 3) = 0$ or $(2x + 1) = 0$
when $x - 3 = 0$, $x = 3$

when $2x + 1 = 0 \Rightarrow x = -\frac{1}{2}$

When $x = 3$, then $(2x + 1) = 7$ and when $x = -\frac{1}{2}$,
then $2x + 1 = 0$,
Possible values of $(2x + 1)$ are 0 and 7.
2. (c) Let father's, mother's and daughter's present age be F, M, D respectively.
We have, $F = M + 5$, $M = 3D$ and $D = 10$
 $\Rightarrow M = 3 \times 10 = 30$ years and $F = 30 + 5 = 35$ years
The father's age at the time of birth of the daughter
 $= 35 - 10 = 25$ years
3. (b) Let the present age of the son be x years, then
 $x = 38 - x$ or $x = 19$ years
Five years back, son's age $= x - 5 = 19 - 5 = 14$ years
4. (c) Let the number be x , Then,
$$x - 24 = \frac{4}{7}x \Leftrightarrow x - \frac{4}{7}x = 24 \Leftrightarrow \frac{3}{7}x = 24$$

$$\Leftrightarrow x = \left(\frac{24 \times 7}{3}\right) = 56.$$
5. (c) Let the number be x .
Then, $\left(\frac{1}{2}x + \frac{1}{5}x\right) - \frac{1}{3}x = \frac{22}{3} \Leftrightarrow \frac{11x}{3}$
$$= \frac{22}{3} \Leftrightarrow x = \left(\frac{22 \times 30}{3 \times 11}\right) = 20$$
6. (b) Let the salary of the driver be x .
Then, his income during one week $= x + \frac{5}{4}x = \frac{9x}{4}$
$$\text{Required fraction} = \frac{\frac{5}{4}x}{\frac{9x}{4}} = \frac{5}{9}$$
7. (c) Let the number of rice bowls be x ,
the number of broth bowls be y
and the number of meat bowls be z .
Now, $x + y + z = 65$... (1)
and $2x = 3y = 4z$... (2)
From (1) and (2), we have $x = 30$, $y = 20$, $z = 15$
Thus, the total number of guests $= 2x = 3y = 4z = 60$
8. (c) Let the numbers be $4x$ and $7x$. Then, $\frac{4x+4}{7x+4} = \frac{3}{5}$
$$\Leftrightarrow 5(4x + 4) = 3(7x + 4) \Leftrightarrow x = 8$$

 $\therefore$ Larger number $= 7x = 56$
9. (a) We have, $(M + 30) = 2(R - 30)$
or $M + 30 = 2R - 60$
or $M - 2R = -90$... (1)
Again $(R + 10) = 3(M - 10)$
or $R + 10 = 3M - 30$
or $R - 3M = -30 - 10$
or $R - 3M = -40$... (2)
Solving (1) and (2), we have $M = 34$ and $R = 62$.
10. (b) Let the numbers be x and y . Then, $x + y = 25$ and
 $x - y = 13$.
 $\Rightarrow xy = 114$
11. (b) According to question, we have
 $A - 10 = B + 10 \Rightarrow A - B = 20$... (1)
and $2(B - 20) = (A + 20)$
 $\Rightarrow 2B - 40 = A + 20$ or $A - 2B = -60$... (2)
From (1) and (2), we get $A = 100$ and $B = 80$
12. (b) Person's daily expenses $= x$
Number of days tour last $= y$ days
So, $x \times y = 360$... (1)
 $(x - 3)(y + 4) = 360$... (2)
Solving equations (1) and (2), we get
 $y = 20$ or -24 (not possible)
 $\therefore y = 20$ days
13. (b) Let the numbers be x and y . Then, $x^2 - y^2 = 256000$
and $x + y = 1000$.
On dividing, we get $x + y = 256$.
Solving $x + y = 1000$ and $x - y = 256$.
We get $x = 628$ and $y = 372$.
14. (b) Let the numbers be x , $x + 2$ and $x + 4$
Then, $x + (x + 2) + (x + 4) = x + 20 \Leftrightarrow 2x = 14$
 $\Leftrightarrow x = 7$.
 $\therefore$ Middle number $= x + 2 = 9$.
15. (b) Let fixed charge $= x$
and charge for 1 km is y
$$\begin{array}{r} x + 10y = 85 \\ x + 15y = 120 \\ \hline -5y = -35 \\ y = 7 \text{ per km} \\ x = 15 \end{array}$$

Charges for 25 km $= 15 + 25 \times 7 = 190$

16. (b) Let the numerator be x .
Then, denominator = $x + 4$
$$\therefore \frac{x-4}{x+4+2} = \frac{1}{6}$$
$$\Rightarrow \frac{x-4}{x+6} = \frac{1}{6}$$
$$\Rightarrow 6(x-4) = x+6$$
$$\Rightarrow 6x-24 = x+6 \Rightarrow 5x = 30$$
$$\therefore x = 6$$

Thus, Numerator = 6, Denominator = $6 + 4 = 10$.
Hence, the original number = $\frac{6}{10}$.

17. (a) Let the present ages of Vikas and Vishal be $15x$ years and $8x$ years.
After 10 years,
Vikas's age = $15x + 10$ and
Vishal's age = $8x + 10$
$$\therefore \frac{15x+10}{8x+10} = \frac{5}{3}$$
$$\Rightarrow 3(15x+10) = 5(8x+10)$$
$$\Rightarrow 45x+30 = 40x+50$$
$$\Rightarrow 5x = 20 \Rightarrow x = \frac{20}{5} = 4$$
$$\therefore \text{Present age of Vikas} = 15x = 15 \times 4 = 60 \text{ years}$$
$$\text{Present age of Vishal} = 8x = 8 \times 4 = 32 \text{ years.}$$

18. (c) Let the numbers be $3x$, $3x + 3$ and $3x + 6$
Then, $3x + (3x + 3) + (3x + 6) = 72$
$$\Leftrightarrow 9x = 63 \Leftrightarrow x = 7.$$
$$\therefore \text{Largest number} = 3x + 6 = 27$$

19. (d)
$$\frac{2}{5} \times \frac{1}{4} \times \frac{3}{7} \times x = 15$$
$$\therefore \frac{x}{2} \times \frac{5 \times 7 \times 2 \times 5}{2} = 25 \times 7 = 175$$

20. (a) $F + S = 4S$
or, $F = 3S \Rightarrow F : S = 3 : 1$
The ages of father and son = 56 years
$$\therefore \text{Son's age} = \frac{1}{4} \times 56 = 14 \text{ years}$$

21. (c) Let the two numbers be x and y .
$$\therefore xy = 192, x + y = 28 \quad \dots(1)$$
$$\therefore (x-y)^2 = (x+y)^2 - 4xy = 784 - 768 = 16$$
$$\therefore x-y = 4 \quad \dots(2)$$

Combining (1) and (2), $x = 16$, and $y = 12$.

22. (b) Let the middle no. = x
$$(x-2) + x + (x+2) = \frac{176}{4} - 14$$

or $3x = \frac{120}{4}$ or, $x = 10$

23. (b) Let the denominator be x . Then, numerator = $x + 5$.
Now,
$$\frac{x+5}{x} - \frac{x+5}{x+5} = \frac{5}{4} \Leftrightarrow \frac{x}{x} - \frac{5}{4} = 1 - \frac{9}{4} = -\frac{5}{4}$$

So, the fraction is $2\frac{1}{4}$

24. (c) Let the three numbers be $5x$, $9x$ and $11x$ respectively.
Then, $25x = 300 \Rightarrow x = 12$
So, the second number is $9x = 9 \times 12 = 108$

25. (d) Let the number be x . Then,
$$x - 20 = \frac{7x}{12} \Rightarrow x - \frac{7x}{12} = 20 \Rightarrow \frac{5x}{12} = 20$$
$$\Rightarrow x = 48$$

26. (d) Let the unit's digit be y and the ten's digit be x .
Then, the number is $10x + y$.
Interchanging the numbers, the new number is $10y + x$.
Then, $10x + y = 10y + x + 18 \Rightarrow 9x - 9y = 18$
$$\Rightarrow x - y = 2 \text{ and given } x + y = 8$$

Solving $x = 5$, $y = 3$
Then, the original number is 53.

27. (a) Let the two numbers be x and y . Then,
$$2x + 3y = 100 \quad \dots(1)$$
$$\text{and } 3x + 2y = 120 \quad \dots(2)$$

Solving eqs (1) and (2), we get $y = 12$ and $x = 32$
So, the larger of the numbers is 32.

28. (d) Let the two numbers be x and y respectively. Then,
$$2x + 3y = 300 \quad \dots(1)$$
$$\text{and } 3x + 2y = 265 \quad \dots(2)$$

Solving eqs (1) and (2), we get $x = 39$ and $y = 74$
So, the larger number is 74.

29. (a) From the options, the required two-digit number is 16.

30. (a) $T + R = 525 \quad \dots(1)$
and
$$\frac{3}{2}T + 2R = 850 \quad \dots(2)$$

Solving eqs (1) and (2), we get $R = 125$

31. (c) $h + g = 81 \quad \dots(1)$
$$\text{and } 2h + 4g = 234 \quad \dots(2)$$

Solving eqs (1) and (2), we get $h = 45$ and $g = 36$

32. (c) Suppose the age of son is x years.
Therefore, age of father = $10x$ years
According to question
$$\frac{10x+x}{2} = 22$$
$$\Rightarrow 11x = 44$$
$$\therefore x = \frac{44}{11} = 4 \text{ years}$$

Age of father = $10 \times 4 = 40$ years

33. (b) Let X be the given number. Then
$$X/3 + X/4 + X/5 - X/2 = 34.$$

Solving this, we get $X = 120$.

34. (c) Let first book published in year x
According to question
 $x + x + 7 + x + 14 + x + 21 + x + 28 + x + 35 + x + 42$
 $= 13524$
 $147 + 7x = 13524$
 $7x = 13524 - 147 = 13377$

$$\therefore x = \frac{13377}{7} = 1911$$

35. (b) Let the ten's digit be x . Then, the unit's digit is $3x$.
Then, $x + 3x = 8 \Rightarrow x = 2$. So, ten's digit is 2 and unit's digit is 6. So, number is 26.

36. (c) From the option, 82 is the right choice
as $82 - 28 = 54$

37. (b) Let the present age of father = x year and Son's present age = y years.

5 years ago, father's age = $x - 5$ and

Son's age = $y - 5$

According to the question,

$$x - 5 = 5(y - 5) \quad \dots (1)$$

$$\text{and } x = 3y \quad \dots (2)$$

$\therefore$ From eqs (1) and (2), we have

$$y = 10 \quad \text{and} \quad x = 30 \text{ years.}$$

Hence, father's present age = 30 years.

38. (c) Let the original fraction be $\frac{x}{y}$.

$$\text{Then, } \frac{\frac{250}{100} \times x}{\frac{450}{100} \times y} = \frac{25}{51} \Rightarrow \frac{250x}{450y} = \frac{25}{51}$$

$$\Rightarrow \frac{x}{y} = \frac{450 \times 25}{250 \times 51} = \frac{15}{17}$$

39. (b) By trial and error method.

40. (c) Let the number be x .

$$63x - 36x = 3834 \Rightarrow 27x = 3834 \Rightarrow x = 142$$

41. (c) Suppose his present age is x years.

According to question

$$\frac{x}{4} + \frac{x}{5} + \frac{x}{3} = x - 13$$

$$\Rightarrow \frac{15x + 12x + 20x}{60} = x - 13$$

$$\Rightarrow 47x = 60x - 780$$

$$\Rightarrow 60x - 47x = 780$$

$$\Rightarrow 13x = 780$$

$$\therefore x = \frac{780}{13} = 60 \text{ years}$$

42. (d) Suppose the number of cows = x

Therefore, the number of herdsmen = x

According to question,

$$4 \times 2x - 28 = x \times 2 + x \times 4$$

$$\Rightarrow 8x - 28 = 2x + 4x$$

$$\Rightarrow 8x - 6x = 27$$

$$\therefore x = \frac{27}{2} = 14$$

43. (c) Let the present age of mother and daughter be $7x$ and x .

Then,

$$\frac{7x - 4}{x - 4} = \frac{19}{1} \Rightarrow 7x - 4$$

$$= 19x - 76$$

$$= 12x = 72$$

$$= x = 6$$

$$\therefore \text{mother's age 4 yrs hence} = 7 \times 6 + 4 = 46 \text{ yrs.}$$

44. (b) $\frac{6x}{7} = 180 \Rightarrow x = 210$

45. (c) Age of Farah = $x = (x - 8) \times \frac{9}{7}$

$$\Rightarrow x = \frac{9(x - 8)}{7}$$

$$\Rightarrow 7x = 9x - 72$$

$$\Rightarrow 2x = -72$$

$$\Rightarrow \frac{x}{2} = 36$$

years

$$\frac{36}{6} = 6 \text{ years}$$

$$\therefore \text{Age of daughter 3 years ago} = 6 - 3 = 3 \text{ years}$$

46. (d) Let the number of parrots be p and the number of tigers be t . Then

$$p + t = 858 \quad \dots (1)$$

$$2p + 4t = 1846 \quad \dots (2)$$

After rearranging equation (2), we get

$$p + 2t = 923 \quad \dots (3)$$

Solving (1) & (2) we get

$$t = 65 \text{ \& } p = 793$$

47. (a) Let the ten's digit be x . Then, units digit = $(x + 3)$. Sum of the digits = $x + (x + 3) = 2x + 3$.

$$\text{Number} = 10x + (x + 3) = 11x + 3.$$

$$\therefore \frac{11x + 3}{2x + 3} = \frac{4}{1} \Leftrightarrow 11x + 3 = 4(2x + 3)$$

$$\Leftrightarrow 3x = 9 \Leftrightarrow x = 3$$

$$\text{Hence, required number} = 11x + 3 = 36$$

48. (d) Let the numbers be x and y .

$$\therefore \frac{x}{y} = \frac{4}{7}$$

$$\therefore 7x = 4y \quad \dots (1)$$

$$\frac{x + 30}{y + 30} = \frac{5}{8}$$

$$\therefore 8x - 5y = -90 \quad \dots (2)$$

From eqn (2), $32x - 20y = -360$

From eqn (1), $35x = 20y$

$$\therefore 32x - 35x = -360$$

$$\therefore x = \frac{360}{3} = 120$$

$$y = 210$$

$$\therefore \text{Average} = \frac{330}{2} = 165$$

49. (c) Let the ten's digit be x . Then, number = $10x + 3$ and sum of digits = $(x + 3)$.

$$\text{So, } (x + 3) = \frac{1}{7} (10x + 3) \Leftrightarrow 7x + 21 = 10x + 3$$

$$\Leftrightarrow 3x = 18 \Leftrightarrow x = 6.$$

Hence, the number is 63.

50. (c) Let the number be x . Then, $3(2x + 9) = 75$

$$\Rightarrow 2x + 9 = 25 \Rightarrow 2x = 16 \Rightarrow x = 8.$$

51. (b) Let the ten's digits be x and unit's digit be y .

$$\text{Then, } (10x + y) - (10y + x) = 36 \Leftrightarrow 9(x - y) = 36$$

$$\Leftrightarrow x - y = 4.$$

52. (b) Let the two parts be $(54 - x)$ and x .

$$\text{Then, } 10(54 - x) + 22x = 780 \Leftrightarrow 12x = 240$$

$$\Leftrightarrow x = 20.$$

$$\therefore \text{Bigger part} = (54 - x) = 34.$$

53. (a) Greatest of the five numbers will be least if remaining four numbers are less than m and as large as possible

$\Rightarrow$ The remaining four numbers are same.

$$4(m - 1) + m = 146$$

$$\Rightarrow m = 30$$

54. (d) Let number of notes of each denomination be x .

$$\text{Then, } x + 5x + 10x = 480 \Leftrightarrow 16x = 480 \Leftrightarrow x = 30.$$

Hence, total number of notes = $3x = 90$.

55. (d) Let the original fraction be $\frac{x}{y}$. Then,

$$\frac{\frac{300x}{100}}{\frac{300y}{100}} = \frac{14}{5} \Rightarrow \frac{x}{y} = \frac{14}{5}$$

Standard Level

1. (d) Let a , b and c be the digits at the hundredth, tens and units places, respectively.

$$\text{Now, } a + b + c = 16, \quad \dots(1)$$

$$b = 3c \quad \dots(2)$$

$$\text{and } c = \frac{1}{4}a \quad \dots(3)$$

From (2) and (3),

$$b = \frac{3}{4}a \quad \dots(4)$$

From (1), (3) and (4), we have

$$a + \frac{3}{4}a + \frac{1}{4}a = 16$$

$$\text{or } a = 8, b = 6 \text{ and } c = 2$$

Hence, the three digit number is 862.

2. (b) Let the digit at units place be y and that at the tens place be x .

$$\text{Number} = 10x + y$$

$$\text{We have, } xy = 14 \text{ and } 10x + y + 45 = 10y + x$$

$$\Rightarrow 9x - 9y = -45 \text{ or } x - y = -5 \quad \dots(1)$$

$$\text{Now, } (x + y)^2 = (x - y)^2 + 4xy = (-5)^2 + 4 \times 14 = 25 + 56 = 81$$

$$\Rightarrow x + y = 9 \quad \dots(2)$$

$$\text{From (1) + (2)}$$

$$2x = 4 \text{ or } x = 2$$

$$\Rightarrow \text{Number} = 27.$$

3. (d) Let present age of Ranjeev = x years

$$\text{Present age of Ranjeev's brother} = (x + 6) \text{ years}$$

$$\text{Present age of Ranjeev's father} = (x + 6 + 32) \text{ years}$$

$$= (x + 38) \text{ years}$$

$$\therefore \text{Present age of Ranjeev's mother} = (x + 38 - 3) \text{ years}$$

$$= (x + 35) \text{ years}$$

$$\therefore \text{Present age of Ranjeev's sister} = (x + 35 - 25) \text{ years}$$

$$= (x + 10) \text{ years}$$

$$\therefore \text{Age of Ranjeev's sister when he was born}$$

$$= (x + 10 - x) = 10 \text{ years.}$$

4. (d) Let the total number of discs of 2 kg and 5 kg be ' a ' and ' b ' respectively.

$$\text{Then, } a + b = 21 \text{ and } 5b = 2a$$

$$\text{Solving the above two equations, we get } a = 15, b = 6$$

$$\therefore \text{Weight of all discs together}$$

$$= 15 \times 2 + 6 \times 5 = 60 \text{ kg}$$

5. (a) $\frac{1}{3}A = \frac{E}{2}$

$$\Rightarrow \frac{A}{3} - \frac{E}{2} = 0$$

$$\Rightarrow 2A - 3E = 0 \quad \dots(1)$$

$$A + E = 150 \quad \dots(2)$$

From equations (1) and (2)

$$E = 60$$

6. (c) Let the four numbers be A , B , C and D .

$$\text{Let } A + 3 = B - 3 = 3C = D/3 = x.$$

$$\text{Then, } A = x - 3, B = x + 3, C = x/3 \text{ and } D = 3x.$$

$$A + B + C + D = 64$$

$$\Rightarrow (x - 3) + (x + 3) + x/3 + 3x = 64$$

$$\Rightarrow 5x + x/3 = 64 \Rightarrow 16x = 192 \Rightarrow x = 12$$

Thus, the numbers are 9, 15, 4 and 36.

$$\therefore \text{Required difference} = (36 - 4) = 32.$$

7. (d) Suppose husband's age be H years.
Then wife's age $W = H - 9$

$$\text{Son's age } S = \frac{H-9}{2}$$

$$\text{Daughter's age } D = \frac{H}{3}$$

According to question,

$$\frac{H}{3} + 7 = \frac{H-9}{2} \Rightarrow 2H + 42 = 3H - 27$$

$$\Rightarrow H = 42 + 27 = 69$$

$$\therefore W = 60.$$

Solving through option (c).

$$45 + 9 = 54 H.$$

$D = 18$
 $S = 22.5$ → Difference is 4.5 years, so this is incorrect

Solving through option (d) matches all conditions.

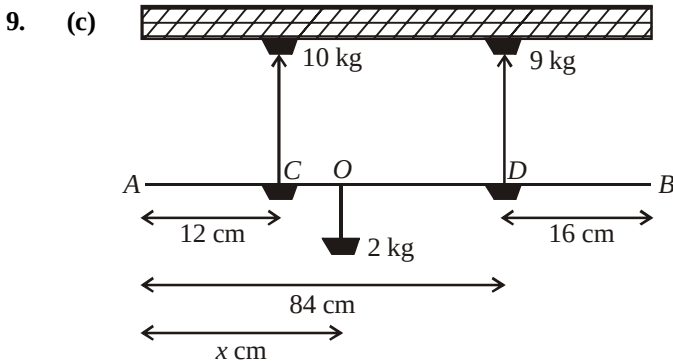
8. (b) Let the fraction is $\frac{a}{b}$, then $a + b = 11$

$$\text{and } \frac{a+1}{b-2} = \frac{3}{2} \Rightarrow 2a - 2 = 3b - 6$$

$$\Rightarrow 2a - 3b = -8$$

Solving both $a + b = 11$ and $2a - 3b = -8$

$$\Rightarrow a = 5, b = 6 \therefore \text{fraction} = \frac{5}{6}$$



Situation is depicted in the figure above.

Let we place 2 kg weight at point O, at a distance x cm from A. Taking moment of all forces about A, we have

$$10 \times 12 + 9 \times 84 = 2x + 17 \times 50$$

$$876 = 2x + 850$$

$$\therefore 2x = 26 \Rightarrow x = 13 \text{ cm}$$

10. (c) Let the number of questions that he attempted wrongly be n , so the questions attempted correctly will be $(90 - n)$.

According to the question,

$$5 \times (90 - n) + (-2) \times n = 387 \Rightarrow 450 - 7n = 387$$

$$\Rightarrow n = 9$$

11. (b) Since, $1 \times x < 1 + x$, So, one of the number is 1.

12. (b) Let 'x' be the total number of students in college

$$x - \left[\frac{12x}{100} + \frac{3x}{4} + \frac{10x}{100} \right] = 15$$

$$x - \left[\frac{48x + 300x + 40x}{400} \right] = 15 \therefore x = 500$$

13. (c) Let the unit's digit be x .

$$\therefore \text{Ten's digit} = x - 2$$

$$\therefore \text{Number} = 10(x - 2) + x = 10x - 20 + x = 11x - 20$$

New number obtained after reversing the digits

$$= 10x + x - 2 = 11x - 2$$

According to the question,

$$3(11x - 20) + \frac{6}{7}(11x - 2) = 108$$

$$\Rightarrow (11x - 20) + \frac{2}{7}(11x - 2) = 36$$

$$\Rightarrow 77x - 140 + 22x - 4 = 252$$

$$\Rightarrow 99x = 252 + 144 \Rightarrow x = \frac{396}{99} = 4$$

$$\therefore \text{Number} = 11x - 20 = 11 \times 4 - 20 = 24$$

$$\therefore \text{Sum of digits} = 2 + 4 = 6$$

14. (b) $\frac{x+1}{y+2} = \frac{2}{3} \Rightarrow 3x - 2y = 1$

$$\frac{x+2}{y+3} = \frac{5}{7} \Rightarrow 7x - 5y = 1$$

$$\text{or, } 3x - 2y = 7x - 5y \Rightarrow 3y = 4x \Rightarrow \frac{x}{y} = \frac{3}{4}$$

15. (a) Let the numbers be x, y and z . Then,

$$x + y = 10 \quad \dots(1)$$

$$y + z = 19 \quad \dots(2)$$

$$x + z = 21 \quad \dots(3)$$

Adding (1), (2) and (3), we get : $2(x + y + z) = 50$

$$\text{or } (x + y + z) = 25.$$

$$\text{Thus, } x = (25 - 19) = 6; y = (25 - 21) = 4;$$

$$z = (25 - 10) = 15.$$

Hence, the required numbers are 6, 4 and 15.

16. (b) We have $\frac{2}{x} + \frac{15}{y} = 5$

$$\Rightarrow 2y + 15x = 5xy$$

$$\Rightarrow 5xy - 2y - 15x = 0$$

$$\Rightarrow (y - 3)(5x - 2) = 6$$

Now, 6 can be written as $2 \times 3, -2 \times -3, 1 \times 6$

or -1×-6 .

The only possible case is $5x \cdot 2 = 3$ and $y \cdot 3 = 2$.

Therefore, $x = 1$ and $y = 3$.

17. (c) Let the numbers be x , y and z . Then, $x + y = 45$,
 $y + z = 55$ and $3x + z = 90$
 $\Rightarrow y = 45 - x$, $z = 55 - y = 55 - (45 - x) = 10 + x$.
 $\therefore 3x + 10 + x = 90$ or $x = 20$.
 $y = (45 - 20) = 25$ and $z = (10 + 20) = 30$.
 $\therefore$ Third number = 30.

18. (b) Let y = number of accidents reported and
 x = number of overtime hours logged.

then $y = mx + c$

(where, m and c are constants)

$$8 = 1000m + c \quad \dots(1)$$

$$5 = 400m + c \quad \dots(2)$$

On solving (1) and (2), we get $m = \frac{1}{200}$ and $c = 3$

So, number of overtime hour = 0 $\Rightarrow x = 0 \Rightarrow y = c$

So expected number of accidents when no overtime hours are logged = 3.

19. (c) An angle is a triangle

$$= \frac{2}{3} \times 180^\circ = 120^\circ$$

Remaining $180^\circ - 120^\circ = 60^\circ$ is the ratio of 5 : 7.

$$\text{So, } 5x + 7x = 60$$

$$12x = 60$$

$$x = 5$$

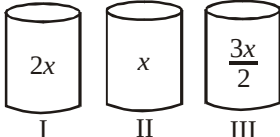
So, angles are $5 \times 5 = 25^\circ$

and $7 \times 5 = 35^\circ$

and 120°

So, value of second largest angle of triangle is 35° .

Expert Level

1. (d) 

Suppose the number of fruits in the second basket = x
 Number of fruits in the first basket = $2x$

$$\text{Number of fruits in the third basket} = 2x \times \frac{3}{4} = \frac{3x}{2}$$

$$\text{Now, } 2x \times x + \frac{3}{4} = 30 \times 3$$

$$= \frac{4x + 2x + 3x}{2} = 30 \times 3$$

$$= 9x = 30 \times 3 \times 2$$

$$\therefore x = \frac{30 \times 3 \times 2}{9} = 20$$

$$2x = 2 \times 20 = 40$$

2. (c) Let the quantity carried by large, medium and small ships be L , M and S respectively.

$$4L = 7S, 3M = 2L + S$$

$$\Rightarrow L = \frac{7}{4}S \text{ and } 3M = 2 \times \frac{7}{4}S + S = \frac{7}{2}S + S = \frac{9}{2}S$$

$$\Rightarrow M = \frac{3}{2}S$$

$$\text{Now } 15L + 7M + 14S = \frac{15}{4} \times 15S + \frac{3}{2} \times 7S + 14S$$

$$= \frac{105S}{4} + \frac{21}{2}S + 14S$$

$$= \frac{105S + 42S + 56S}{4} = \frac{203}{4}S$$

$$\frac{203}{4}S \times 36 = 1827S$$

$$\text{Again, } 12L + 14M + 21S = \frac{7}{4} \times 12S + \frac{3}{2} \times 14S + 21S$$

$$= 21S + 21S + 21S = 63S$$

$$\therefore \text{Number of journeys} = \frac{1827S}{63S} = 29$$

3. (b) Let the numbers be x and y .

Then, $x + y = 52$ and $xy = 672$

$$x - y = \sqrt{(x + y)^2 - 4xy} = \sqrt{(52)^2 - 4 \times 672}$$

$$= \sqrt{2704 - 2688} = \sqrt{16} = 4$$

$\therefore$ Required difference = 4

4. (c) Let he purchase x pairs of brown socks.

Price of black socks and brown socks be $2a$ and a per pair respectively.

$$4 \times 2a + x \times a = x \times 2a + 4 \times a$$

$$12a + \frac{3}{2}xa = 2xa + 4a$$

$$12 + \frac{3}{2}x = 2x + 4$$

$$\frac{x}{2} = 8$$

$$\therefore \text{Required ratio} = \frac{4}{16} = \frac{1}{4}$$

5. (c) Let the no. of balls = b

$$\text{Rate} = \frac{450}{b}$$

$$(b + 5) \left(\frac{450}{b} - 15 \right) = 450$$

$$\text{or, } 450 - 15b + \frac{2250}{b} - 75 = 450$$

$$\text{or, } b^2 + 5b - 150 = 0$$

$$\text{or, } (b + 15)(b - 10) = 0$$

or, $b = 10$ (Neglecting negative value)

6. (c) Suppose total number of workers in the office = x

$$\text{Number of woman workers} = \frac{x}{3}$$

$$\therefore \text{Number of man workers} = x - \frac{x}{3} = \frac{3x - x}{3} = \frac{2x}{3}$$

$$\text{Number of married woman workers} = \frac{x}{3} \times \frac{1}{2} = \frac{x}{6}$$

Number of married woman workers who have children

$$= \frac{x}{6} \times \frac{1}{3} = \frac{x}{18}$$

$$\text{Number of married man workers} = \frac{2x}{3} \times \frac{3}{4} = \frac{x}{2}$$

Number of married man workers who have children

$$= \frac{x}{2} \times \frac{2}{3} = \frac{x}{3}$$

$$\text{Number of workers who have children} = \frac{x}{3} + \frac{x}{18}$$

$$= \frac{6x + x}{18} = \frac{7x}{18}$$

Number of workers without children

$$= x - \frac{7x}{18} = \frac{18x - 7x}{18} = \frac{11x}{18}$$

7. (a) Let the smallest even number be $(x + 11)$

Then, the three consecutive even numbers are $(x + 11)$, $(x + 13)$ and $(x + 15)$ respectively.

Hence, the smallest odd number will be x .

Then, the three consecutive odd numbers are x , $(x + 2)$ and $(x + 4)$ respectively.

Now,

$$(x + 11) + (x + 13) + (x + 15) + x + (x + 2) + (x + 4) = 231$$

$$\Rightarrow 6x + 45 = 231 \Rightarrow x = 31$$

Sum of the largest odd number and largest even number is

$$(x + 15) + (x + 4) = 2x + 19 = 81$$

8. (b) Let the four consecutive even numbers be $(x - 2)$, x , $(x + 2)$ and $(x + 4)$ respectively.

$$\text{Then, } (x - 2) + x + (x + 2) + (x + 4) = 44$$

$$\Rightarrow 4x + 4 = 44$$

$$\Rightarrow x = 10$$

$\therefore$ The numbers are 8, 10, 12 and 14 respectively.

The sum of their squares

$$= (8^2 + 10^2 + 12^2 + 14^2)$$

$$= (64 + 100 + 144 + 196) = 504$$

9. (d) As the larger number is written on the left, the larger number is either 54 or 55.

Let the smaller number be x .

Case I: The larger number is 54.

$$5400 + x = 5481 + 54 - x$$

$$2x = 5535 - 5400 = 135$$

(In this case x will not be a natural number.)

Case II: The larger number is 55.

$$5500 + x = 5481 + 55 - x$$

$$2x = 5536 - 5500 = 36$$

$$\Rightarrow x = 18$$

Hence, the required sum = 73.

10. (b) Let the middle digit be x . Then, $2x = 10$ or $x = 5$. So, the number is either 253 or 352. Since the number increases on reversing the digits, so the hundred's digit is smaller than the unit's digit. Hence, required number = 253.

11. (a) Let the unit's digit be x . Then, ten's digit = $(x - 2)$.

$$\therefore 3[10(x - 2) + x] + \frac{6}{7}[10x + (x - 2)] = 108$$

$$\Leftrightarrow 231x - 420 + 66x - 12 = 756 \Leftrightarrow 297x = 1188$$

$$\Leftrightarrow x = 4.$$

Hence, sum of the digits = $x + (x - 2) = 2x - 2 = 6$

12. (a) Let 't' be the total time taken by them to read 78 pages.

Let the speeds with which they read be $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ pages per minutes respectively.

$$\text{Then, } t \times \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) = 78$$

$$t \times \left(\frac{6 + 4 + 3}{12} \right) = 78$$

$$t = \frac{12 \times 78}{13} \Rightarrow 72 \text{ minutes}$$

Hence, the number of pages that Bob should read

$$= 72 \times \frac{1}{3} = 24 \text{ pages.}$$

13. (a) Let Nikhil buy x , y , and z pieces of kajubarfi, gulabjamun and sandesh respectively.

$$\therefore x + y + z = 100 \quad \dots (1)$$

$$\text{Also: } 10x + 3y + \frac{1}{2}z = 100 \quad \dots (2)$$

Eliminating z from (1) and (2), we get

$$\therefore 19x + 5y = 100$$

$$\therefore y = \frac{100 - 19x}{5}$$

for $x = 1$, we are not getting any natural value of y . The only value of x for natural value of y is $x = 5$

$$\therefore y = \frac{5}{5} = 1$$

$\therefore$ Nikhil must buy 1 gulabjamun.

Explanation of Test Yourself

1. (b) $(x + y) + \left(\frac{x}{y}\right) = \frac{1}{2}$ and $(x + y) = \frac{-1}{2}$

Solving these two equations the values of

$(x + y)$ and $\left(\frac{x}{y}\right)$ will be $(1, -1/2)$

When $x + y = 1$ and $\frac{x}{y} = -1/2$

$(x, y) = (2, -1)$

When $x + y = -1/2$ and $\frac{x}{y} = 1$

$(x, y) = \left(-\frac{1}{4}, -\frac{1}{4}\right)$.

$\therefore$ No. of possible pairs = 2.

2. (d) Let the original number of sweets be x .
According to the question,

$$\frac{x}{140} - \frac{x}{175} = 4$$

$$\text{or, } 175x - 140x = 4 \times 140 \times 175$$

$$\text{or, } x = \frac{4 \times 140 \times 175}{35} = 2800$$

3. (a) Let the two-digit number be $10x + y$.

$$10x + y = 7(x + y) \quad \text{P} \quad x = 2y \quad \dots(1)$$

$$10(x + 2) + y + 2 = 6(x + y + 4) + 4$$

$$\text{or } 10x + y + 22 = 6x + 6y + 28 \quad \text{P} \quad 4x - 5y = 6 \quad \dots(2)$$

Solving equations (1) and (2), we get $x = 4$ and $y = 2$

$\therefore$ Number = 42

4. (c) Begin with the fourth customer :

Her stock before the 4th customer

$$\left(15 + \frac{1}{2}\right) \times 2 \text{ or } 31.$$

Her stock before the 3rd customer $\left(31 + \frac{1}{2}\right) \times 2$ or 63.

Her stock before the 2nd customer $\left(63 + \frac{1}{2}\right) \times 2$ or 127.

Her stock before the 1st customer

$$\left(127 + \frac{1}{2}\right) \times 2 = 255.$$

5. (c) Let the original fraction be $\frac{x}{y}$.

$$\text{Then } \frac{x+2}{y+1} = \frac{5}{8} \text{ or, } 8x - 5y = -11 \quad \dots(1)$$

$$\text{Again, } \frac{x+3}{y+1} = \frac{3}{4} \text{ or, } 4x - 3y = -9 \quad \dots(2)$$

Solving, (1) and (2) we get $x = 3$ and $y = 7$

$$\therefore \text{ fraction} = \frac{3}{7}$$

6. (b) Let 'x' be the total number of students in college

$$x - \left[\frac{12x}{100} + \frac{3x}{4} + \frac{10x}{100}\right] = 15$$

$$x - \left[\frac{48x + 300x + 40x}{400}\right] = 15 \quad \therefore x = 500$$

7. (a) Let the first, second, third and fourth numbers be a, b, c and d respectively.

According to the question,

$$a + b + c + d = 64 \quad \dots(1)$$

$$\text{and } a + 3 = b - 3 = 3c = \frac{d}{3}$$

$$\text{i.e., } a + 3 = b - 3 \Rightarrow b = a + 6 \quad \dots(2)$$

$$\text{Also, } c = \frac{a+3}{3} \text{ and } d = 3(a+3)$$

Solving the above eqns, we get

$$a = 9, b = 15, c = 4 \text{ and } d = 36$$

$\therefore$ Difference between the largest and the smallest numbers = $36 - 4 = 32$

8. (c) Let the length of shortest piece be x cm.

Then length of longest piece be $23 + x$

and length of middle piece be $\frac{23+x}{3}$

$$\text{According to question, } 23 + x + \frac{23+x}{3} + x = 40$$

$$69 + 3x + 23 + x + 3x = 120$$

$$\text{or } 92 + 7x = 120$$

$$\text{or } x = 4 \text{ cm.}$$

9. (d) $P + R + 2Q = 59$;

$$Q + R + 3P = 68$$

$$\text{and } P + 3(Q + R) = 108$$

Solving the above two equations, we get $P = 12$ years.

10. (b) Let one of the number be $10x + y$
Reversing the digits it become $10y + x$.
As per question.
 $10y + x - (10x + y) = 18$
 $\Rightarrow 9(y - x) = 18$ or $y - x = 2$
So, it will be possible in all the cases where the difference between the two digits = 2.
So, the numbers are 13, 24, 35, 46, 57, 68, 79.
Hence the number of such two-digit numbers apart from 13 is 6.

11. (c) Let the original fraction be $\frac{x}{y}$, Then,

$$\frac{\frac{125x}{100}}{\frac{2y}{9}} \cdot \frac{5}{9} \Rightarrow \frac{125x}{100 \times 2y} \cdot \frac{5}{9} \Rightarrow \frac{x}{y} = \frac{5 \times 100 \times 2}{125 \times 9} = \frac{8}{9}$$

12. (d) Let the three numbers be x, y and z respectively
Then, $x + y = 73$... (1)
 $y + z = 77$... (2)
and $3x + z = 104$... (3)
 $\Rightarrow 3(73 - y) + (77 - y) = 104$
 $\Rightarrow 219 - 3y + 77 - y = 104$
 $\Rightarrow 296 - 4y = 104$
 $\Rightarrow 4y = 192 \Rightarrow y = 48$
Hence, $z = 77 - 48 = 29$

13. (a) The second number = 12% of 50 = $\frac{12}{100} \times 50 = 6$
Let the first number be x . Then,

$$\frac{2}{3} \times x = 6^3 \Rightarrow x = \frac{6 \times 6 \times 6 \times 3}{2} = 324$$

Required sum = $324 + 6 = 330$

14. (a) $T + R = 525$... (1)

and $\frac{3}{2}T + 2R = 850$... (2)

Solving eqs (1) and (2), we get $R = 125$

15. (d) Let the three numbers by x, y and z respectively

Then, $x + y = 73$... (1)

$y + z = 77$... (2)

and $3x + z = 104$... (3)

$\Rightarrow 3(73 - y) + (77 - y) = 104$

$\Rightarrow 219 - 3y + 77 - y = 104$

$\Rightarrow 296 - 4y = 104$

$\Rightarrow 4y = 192 \Rightarrow y = 48$

Hence, $z = 77 - 48 = 29$

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FUNCTIONS

- Introduction
- Function
- Rules for Finding the Domain of a Function
- Methods of Representation of Functions

- Some Special Functions
- Shifting of Graphs
- Combination of Shifting of a Graph

INTRODUCTION

Function in mathematics is an equation or rule that defines a relationship between the two variables; one of them is dependent variable and other is independent variable. This chapter is very important from the point of view of CAT and other equivalent aptitude tests. The number of questions being asked from this topic is almost constant. Basically on an average 3–4 problems are asked from this chapter. A deep understanding of the concepts of this chapter is required to solve the problems.

FUNCTION

A function is a rule which relates two or more than two variables. Out of these variables one is dependent variable and others are independent variables. If y is dependent variable and x is independent variable, then the function is symbolically expressed as

$$y = f(x)$$

$y = f(x)$ is read as y is the function of x . But f denotes the rule by which y varies with x .

In the function $y = f(x)$, there is a unique real value of y for each real value of x . A set D of all real values of x for which the value of y is a unique real value is called domain of the function $y = f(x)$. A set R of all unique real values of y corresponding to each value of x from set D is called Range of the function $y = f(x)$.

The concept of the function can be easily understood by the following examples:

- (i) The function between diameter d of a circle and radius r is
- $$d = 2r$$

Here d is a dependent variable and r is an independent variable, because d and r both are variable but value of d is dependent upon the value of r .

Here domain is a set of all positive real values, because value of r cannot be non-positive and for each positive real value of r , the value of d is a unique positive real number.

Range is also a set of all positive real values, because the diameter, which is twice the length of the radius will be all the positive real numbers for all positive real value of r .

- (ii) The function between the volume V of a cuboid with its side length x is

$$V = x^3$$

Here V is dependent variable and x is independent variable
Domain = Set of all positive real numbers.

Range = Set of all positive real numbers.

- (iii) The function between the area A of the circle with its radius r is

$$A = \pi r^2$$

Here A and r are dependent and independent variables respectively.

Since value of r can be any positive real number and for all positive real values of r , values of A will be all positive real numbers, hence

Domain = Set of all positive real numbers.

Range = Set of all positive real numbers.

- (iv) For the function $y = x^2$, y is a dependent variable and x is an independent variable,

Domain = Set of all real numbers

But Range = Set of all non-negative real numbers, because value of y cannot be negative for any value of x for the given function.

Illustration 1: If $f(x) = -2x + 7$ and $g(x) = x^2 - 5x + 6$, find $f(3)$, $f(-4)$, $g(2)$, and $g(-1)$.

Solution:

$$\begin{aligned} f(x) &= -2x + 7, & g(x) &= x^2 - 5x + 6 \\ f(3) &= -2(3) + 7 = 1 & g(2) &= 2^2 - 5(2) + 6 = 0 \\ f(-4) &= -2(-4) + 7 = 15 & g(-1) &= (-1)^2 - 5(-1) + 6 = 12 \end{aligned}$$

RULES FOR FINDING THE DOMAIN OF A FUNCTION

1. Domain of Algebraic Functions

- (i) Denominator should be non-zero

For the function $y = \frac{2x}{x-3}$, the value of x can be any real number but can not be 3, because for $x = 3$, denominator of the function will be zero.

Hence domain of the function is the set of all real numbers except 3 i.e. domain = $R - \{3\}$.

- (ii) Expression under the even root (i.e. square root, fourth root, etc.) should be non-negative.

For the function $y = \sqrt{5-x}$,

$$5-x \geq 0 \Rightarrow x \leq 5$$

Hence domain = Set of all real numbers which are equal or less than 5.

2. Domain of Logarithmic Functions

$\log_b a$ is defined when $a > 0$, $b > 0$ but $b \neq 1$.

For the function $y = \log_2(x-4)$

$$x-4 > 0 \Rightarrow x > 4$$

Hence domain = Set of all real numbers greater than 4.

3. Domain of Exponential Functions

a^x is defined for all real values of x , where $a > 0$.

For the function $y = (3x-2)^x$,

$$3x-2 > 0 \Rightarrow x > \frac{2}{3}$$

Hence domain = Set of all real numbers greater than $\frac{2}{3}$.

Note: If a and b are two real numbers such that $a > b$, then

- Interval $[a, b]$ means all real numbers equal or greater than a but equal or less than b .
- Interval $[a, b)$ means all real numbers equal or greater than a but less than b .
- Interval $(a, b]$ means all real numbers greater than a but equal or less than b .
- Interval (a, b) means all real numbers greater than a but less than b .
- $(a, b) \cup (c, d)$ means all real numbers greater than a but less than b or greater than c but less than d .

Illustration 2: The domain of the function $f(x) = \frac{1}{\sqrt{x^2 - 3x + 2}}$ is

- (a) $(-\infty, 1)$ (b) $(-\infty, 1) \cup (2, \infty)$
(c) $(-\infty, 1] \cup [2, \infty)$ (d) $(2, \infty)$

Solution: (b) For $f(x)$ to be defined, we must have $x^2 - 3x + 2 = (x-1)(x-2) > 0 \Rightarrow x < 1$ or $x > 2$
Domain of $f = (-\infty, 1) \cup (2, \infty)$.

METHODS OF REPRESENTATION OF FUNCTIONS

A function is represented mainly in three ways as given below.

1. Analytical Representation

When a function is represented by a uniform equation for the entire domain or by several equations which are different for different parts of the domain.

For example

(a) $y = 5x^2 + 2x$

This is the uniform function for entire domain

(b) $y = \begin{cases} x^2 + 4, & \text{if } x \leq 2 \\ x - 3, & \text{if } x > 2 \end{cases}$

This is the function which is represented by two equations which are different for different parts of the domain as given above.

2. Tabular Representation

When a function is represented by a sequence of values of the independent variable with the corresponding values of the dependent variable, then this representation is called Tabular representation of the function.

For example,

(a)

x	1	2	3	4	5	6
y	1	4	9	16	25	36

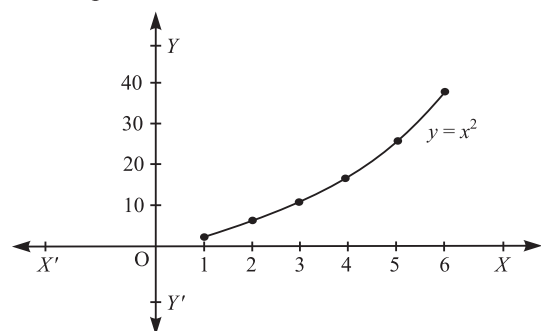
(b)

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1

3. Graphical Representation

When a function is represented by a graph taking different values of dependent variable along x -axis and corresponding values of independent variable along y -axis in a cartesian plane, then this representation of function is called graphical representation of function.

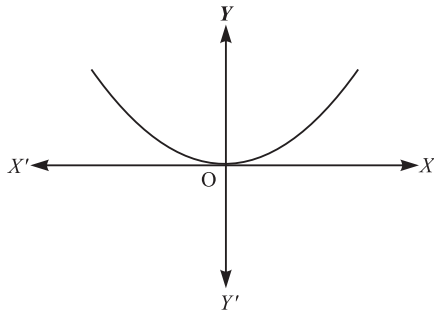
For example



SOME SPECIAL FUNCTIONS

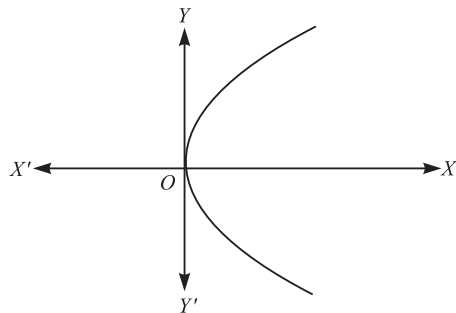
1. Even and Odd Functions

- (i) **Even functions:** If a function $y = f(x)$ be such that $f(-x) = f(x)$, then the function is called an even function. Graph of the even function $y = f(x)$ is symmetrical about the y -axis. For example the graph of even function $y = x^2$ is symmetrical about y -axis.



However, if y is independent variable and x is dependent variable, then the even function $x = f(y)$ is symmetrical about the x -axis.

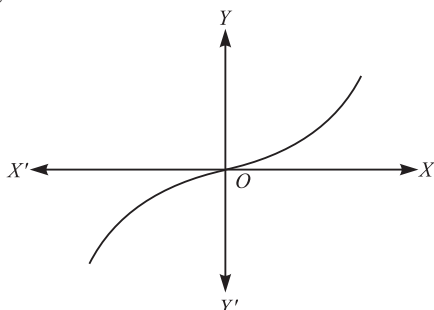
For example the graph of even function $x = y^2$ is symmetrical about x -axis.



Some examples of even functions are $y = x^2 - 5$, $y = x^6 + 2$, etc.

Sum, difference, product and quotient of even functions are also even.

- (ii) **Odd functions:** If a function $y = f(x)$ is such that $f(-x) = -f(x)$, then the function is called an odd function. For example graph of the odd function $y = x^3$ is shown in the figure.



Graph of odd functions are two-fold graphs i.e., on folding the graph paper twice, once along x -axis and then along y -axis, one part of the graph overlaps the other part of the graph.

Some examples of odd functions are $y = x^3 - 2x$, $y = x^5$, $y = x^3 + \frac{1}{x}$, etc.

- Sum and difference of two odd functions is odd function.
- Product of two odd functions is an even function.
- Sum of even and odd function is neither even nor odd function.
- Product of an even and an odd function is odd function.
- Every function can be expressed as the sum of an even function and an odd function.
- A function may be even, odd or neither even nor odd.

For example $4x^3 + 3x^2 + 5$ is neither an even function nor an odd function.

Illustration 3: The function $f(x) = x \frac{a^x - 1}{a^x + 1}$ is odd or even ?

Solution:

$$\text{Since } f(-x) = -x \cdot \frac{a^{-x} - 1}{a^{-x} + 1} = -x \cdot \frac{1 - a^x}{1 + a^x} = x \frac{a^x - 1}{a^x + 1} = f(x)$$

$\therefore f(x)$ is an even function.

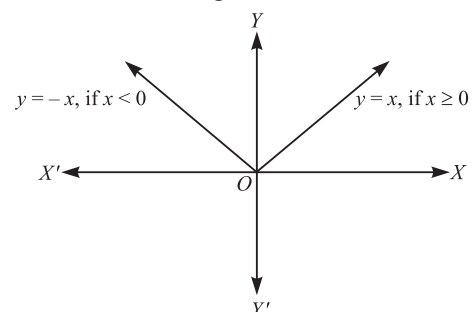
2. Modulus Function

$$f(x) = |x|$$

$$\text{or } f(x) = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Domain = Set of all real numbers

Range = Set of all non-negative real numbers



Note that $|x|$ is always equal or greater than zero i.e. $|x| \geq 0$

For example,

$$|0| = 0$$

$$|5| = 5, \text{ since } 5 > 0$$

$$|-5| = -(-5) = 5, \text{ since } -5 < 0$$

Illustration 4: If $|6x - 4| = 5$, find the value of x .

Solution: **Case-I:** $6x - 4 = 5$, if $6x - 4 \geq 0$

$$\Rightarrow x = \frac{3}{2}, \text{ if } x \geq \frac{2}{3}$$

Case-II: $-(6x - 4) = 5$, if $6x - 4 < 0$

$$\Rightarrow 6x = -1, \text{ if } x < \frac{2}{3}$$

$$\Rightarrow x = -\frac{1}{6}, \text{ if } x < \frac{2}{3}$$

Illustration 5: Find the value of x if $2x^2 + 6|x| + 3 = 0$.

Solution: Since $2x^2$ and $6|x|$ is non-negative and 3 is positive, therefore their sum cannot be equal to zero.

Hence, there is no value of x for which $2x^2 + 6|x| + 3 = 0$

3. Composite Function

If two or more functions are composed into one function, then the resulting function is called composite function.

For example, if

$y = f(x)$ and $y = g(x)$ are two functions then

$f(g(x))$ and $g(f(x))$ are composite functions

Let $f(x) = 2x - 3$ and $g(x) = -3x^2$

Then $f(g(x)) = 2(-3x^2) - 3 = -6x^2 - 3$

and $g(f(x)) = -3(2x - 3)^2$

Illustration 6: Given $f(x) = 2x + 1$ and $g(x) = x^2 + 2x - 1$, find $(f - g)(x)$. Then evaluate the difference when $x = 2$.

Solution: The difference of the functions f and g is given by

$$\begin{aligned}(f - g)(x) &= f(x) - g(x) \\ &= (2x + 1) - (x^2 + 2x - 1) = -x^2 + 2.\end{aligned}$$

When $x = 2$, the value of this difference is

$$(f - g)(2) = -(2)^2 + 2 = -2.$$

4. Inverse Function

Let a function $y = f(x)$ be defined for the domain D and has a range R is such that for each value of y from the range R of the function there is a unique value of x in the domain D , then inverse function of

$y = f(x)$ exists and is given by $x = g(y)$.

$x = g(y)$ is read as x is the function of y .

Here x is the dependent variable and y is independent variable.

The domain of $y = f(x)$ is the range of $x = g(y)$.

The range of $y = f(x)$ is the domain of $x = g(y)$.

Let's see how the inverse of the function $y = f(x)$, where

$$f(x) = \frac{2x - 1}{1 - x} \text{ is found out.}$$

$$\text{Let } \frac{2x - 1}{1 - x} = y$$

$$\Rightarrow y - xy = 2x - 1$$

$$\Rightarrow y + 1 = x(2 + y)$$

$$\Rightarrow x = \frac{y + 1}{y + 2}$$

$$\Rightarrow x = g(y), \text{ where } g(y) = \frac{y + 1}{y + 2}$$

$$\text{Hence inverse of the function } y = f(x), \text{ where } f(x) = \frac{2x - 1}{1 - x}$$

$$\text{is } x = g(y), \text{ where } g(y) = \frac{y + 1}{y + 2}.$$

The inverse of the function is also written as following

replacing $g(y)$ by $f^{-1}(x)$ and y in $\frac{y + 1}{y + 2}$ by x .

$$\therefore f^{-1}(x) = \frac{x + 1}{x + 2}$$

Illustration 7: The inverse of the function $f(x) = \frac{a^x - a^{-x}}{a^x + a^{-x}}$ is

(a) $\log_a \left(\frac{1 - x}{1 + x} \right)$

(b) $\frac{1}{2} \log_a \left(\frac{1 + x}{1 - x} \right)$

(c) $\log_a \left(\frac{1 + x}{1 - x} \right)$

(d) None of these

Solution: (b). Let $y = \frac{a^x - a^{-x}}{a^x + a^{-x}} = \frac{a^{2x} - 1}{a^{2x} + 1}$

$$\Rightarrow \frac{y - 1}{y + 1} = \frac{(a^{2x} - 1) - (a^{2x} + 1)}{(a^{2x} - 1) + (a^{2x} + 1)}$$

(Using componendo and dividendo)

$$\Rightarrow \frac{y - 1}{y + 1} = \frac{-2}{2a^{2x}} \Rightarrow a^{2x} = \frac{1 + y}{1 - y}$$

$$\Rightarrow 2x \log_a a = \log_a \left(\frac{1 + y}{1 - y} \right) \Rightarrow x = \frac{1}{2} \log_a \left(\frac{1 + y}{1 - y} \right)$$

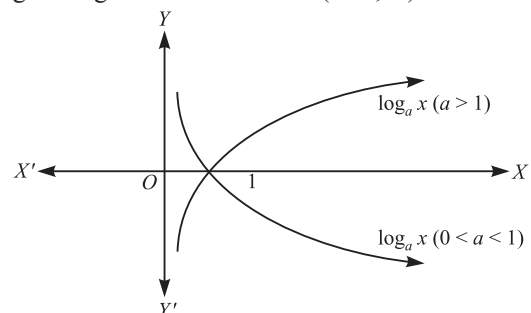
5. Logarithmic Function

If $a > 0$ and $a \neq 1$, then the function $y = \log_a x$ is called logarithmic function, where x is any positive real number.

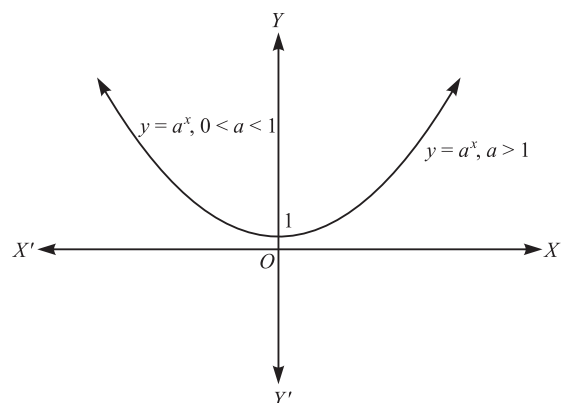
If $a = e$ (a number called exponential number which is approximately equal to 2.71), then the logarithmic function is denoted by $\ln x$.

Domain of logarithmic function = $(0, \infty)$

Range of logarithmic function = $(-\infty, \infty)$



6. Exponential Function



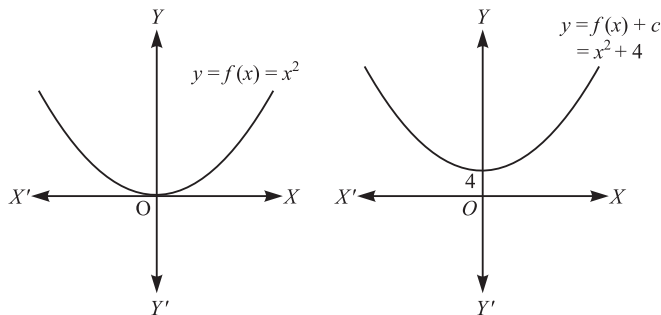
If $a > 0, a \neq 1$, then the function defined by $y = a^x$ is called exponential function with base a , where x is any real number.
Domain of the exponential function = Set of all real numbers
Range of the exponential function = $(0, \infty)$.

SHIFTING OF GRAPHS

Visualising the graph of a function and how graph will shift when expression of the function changed is very important to solve the questions based on functions.

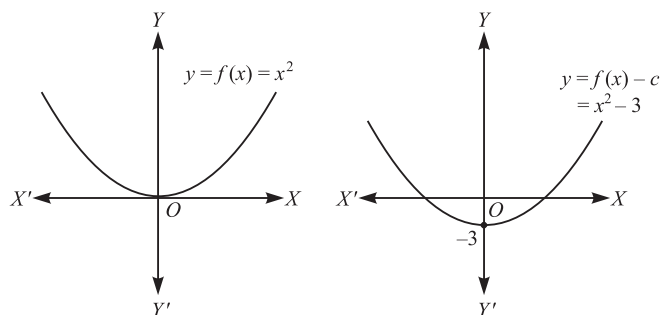
Let's see some tips about how the graphs will change when expression of the function is changed.

1. Graph of $y = f(x)$ and $y = f(x) + c$, where c is a Positive Constant



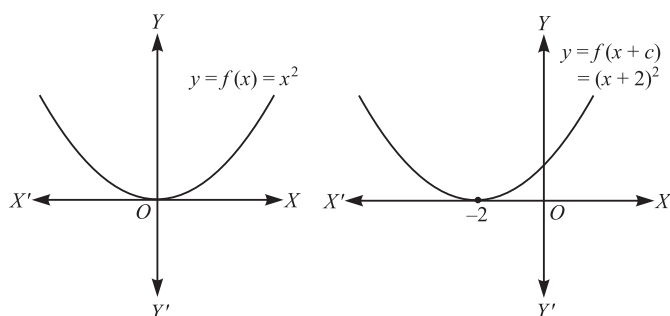
Shape of the graph of $y = f(x) + c$ will be the same as that of the graph of $y = f(x)$ but the graph of $y = f(x) + c$ will be c units above the graph of $y = f(x)$.

2. Graph of $y = f(x)$ and $y = f(x) - c$, where c is a Positive Constant



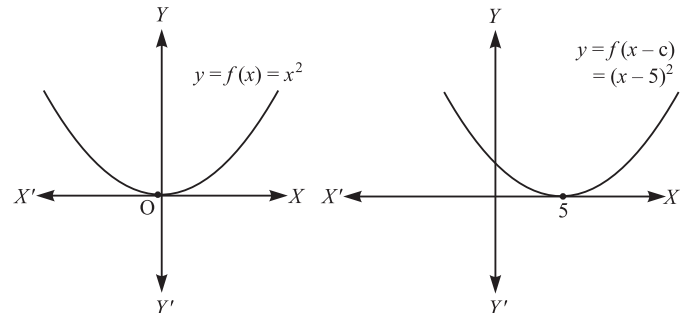
Shape of the graph of $y = f(x) - c$ will be the same as that of the graph of $y = f(x)$ but the graph of $y = f(x) - c$ will be c units below the graph of $y = f(x)$.

3. Graph of $y = f(x)$ and $y = f(x + c)$, where c is a Positive Constant



Shape of the graph of $y = f(x + c)$ will be the same as that of the graph $y = f(x)$ but the graph of $y = f(x + c)$ will be c units left of the graph of $y = f(x)$.

4. Graph of $y = f(x)$ and $y = f(x - c)$, where c is a Positive Constant



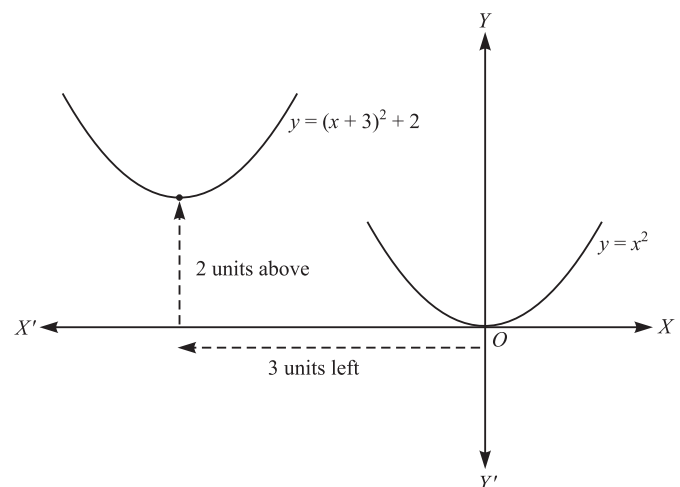
Shape of the graph of $y = f(x - c)$ will be the same as that of $y = f(x)$ but the graph of $y = f(x - c)$ will be c units right of the graph of $y = f(x)$.

COMBINATION OF SHIFTING OF A GRAPH

To visualize the graph of the function $y = x^2 + 6x + 11$, convert $x^2 + 6x + 11$ into $(x + 3)^2 + 2$ i.e. $y = (x + 3)^2 + 2$.

$$[\because x^2 + 6x + 11 = (x^2 + 2 \times x \times 3 + 3^2) + 2 = (x + 3)^2 + 2]$$

Shape of the graph of $y = (x + 3)^2 + 2$ is the same as that of $y = x^2$ but the graph of $y = (x + 3)^2 + 2$ will be 3 units left and 2 units above the graph of $y = x^2$.



Practice Exercises



Foundation Level

- A function f is defined by $f(x) = x + \frac{1}{x}$. Consider the following.
 - $(f(x))^2 = f(x^2) + 2$
 - $(f(x))^3 = f(x^3) + 3f(x)$
 Which of the above is/are correct?
 - 1 only
 - 2 only
 - Both 1 and 2
 - Neither 1 nor 2
 - What is the range of the function $f(x) = \frac{|x|}{x}, x \neq 0$?
 - Set of all real numbers
 - Set of all integers
 - $\{-1, 1\}$
 - $\{-1, 0, 1\}$
 - The domain of the function $f(x) = \frac{\sqrt{(x+1)(x-3)}}{x-2}$ is
 - $[-1, 2) \cup [3, \infty)$
 - $(-1, 2) \cup [3, \infty)$
 - $[-1, 2] \cup [3, \infty)$
 - None of these
 - If $f(x) = \sqrt{x^3}$, then $f(3x)$ will be equal to
 - $\sqrt{3x^3}$
 - $3\sqrt{x^3}$
 - $3\sqrt{3x^3}$
 - $3\sqrt{x^5}$
 - If $f(x) = e^x$, then the value of $7f(x)$ will be equal to
 - e^{7x}
 - $7e^x$
 - $7e^{7x}$
 - e^x
 - If $f(x) = \frac{x+1}{x-1}, x \neq 1$, find $f(f(f(f(f(2))))))$
 - 2
 - 3
 - 4
 - 6
 - Let $f: (-\infty, 1] \rightarrow (-\infty, 1]$ such that $f(x) = x(2-x)$, then $f^{-1}(x)$ is
 - $1 + \sqrt{1-x}$
 - $1 - \sqrt{1-x}$
 - $\sqrt{1-x}$
 - None of these
 - If $f(x) = x^2 - x + 1$, then find the inverse of the $f(x)$
 - $\left(x - \frac{1}{2}\right)$
 - $\sqrt{x - \frac{3}{4}} + \frac{1}{2}$
 - $\sqrt{x - \frac{3}{4}} - \frac{1}{2}$
 - None of these
 - $f(x) = y + \frac{1}{y}$, where $y > 0$. If y increases in value, then $f(x)$
 - Fluctuates up and down in value
 - Decreases in values
 - Increases in value but at a much higher rate than y
 - Increases in value and tends towards y
 - For what value of x , $x^2 + 10x + 11$ will give the minimum value?
 - 5
 - +10
 - 5
 - 10
- Directions for questions 11 and 12:** Define the following functions:
- $a @ b = \frac{a+b}{2}$
 - $a \# b = a^2 - b^2$
 - $a ! b = \frac{a-b}{2}$
- Find the value of $\{[(3@4)!(3\#2)]@[(4!3)@(2\#3)]\}$.
 - 0.75
 - 1
 - 1.5
 - 2.25
 - Which of the following expressions has a value of 4 for $a = 5$ and $b = 3$?
 - $\frac{(a!b)}{(a\#b)}$
 - $(a!b)(a@b)$
 - $\frac{(a\#b)}{(a!b)(a@b)}$
 - Both (b) and (c)
 - Find $f \circ f$ if $f(t) = t/(1+t^2)^{1/2}$.
 - $1/(1+2t^2)^{1/2}$
 - $t/(1+2t^2)^{1/2}$
 - $(1+2t^2)$
 - None of these
 - If $0 < x < 1000$ and $\left[\frac{x}{2}\right] + \left[\frac{x}{3}\right] + \left[\frac{x}{5}\right] = \frac{31}{30}x$, where $[x]$ is the greatest integer less than or equal to x , the number of possible values of x is
 - 34
 - 32
 - 33
 - None of these
 - $f(x) = 3x^2, g(x) = h(x) = 3x^3 + 3$. The value of $f(x)g(x)$ differ from the corresponding values of $h(x)$ approximately by what value
 - 9
 - 5
 - 3
 - Cannot be determined

16. If $f(x) = |x|$ and $g(x) = [x]$, then value of $f \circ g \left(-\frac{1}{4} \right) + g \circ f \left(-\frac{1}{4} \right)$ is
- (a) 0 (b) 1
(c) -1 (d) $1/4$
17. If $f(x)$ is an even function, then the graph $y = f(x)$ will be symmetrical about
- (a) x-axis (b) y-axis
(c) Both the axes (d) None of these
18. Find the maximum value of the function $1/(x^2 - 3x + 2)$.
- (a) $11/4$ (b) $1/4$
(c) 0 (d) None of these
19. The domain of definition of $y = \left[\log_{10} \left(\frac{5x - x^2}{4} \right) \right]^{1/2}$ is
- (a) $[1, 4]$ (b) $[-4, -1]$
(c) $[0, 5]$ (d) $[-1, 5]$
20. If $f(t) = \sqrt{t}$, $g(t) = t/4$ and $h(t) = 4t - 8$, then the formula for $g(f(h(t)))$ will be
- (a) $\frac{\sqrt{t-2}}{4}$ (b) $2\sqrt{t} - 8$
(c) $\frac{\sqrt{(4t-8)}}{4}$ (d) $\frac{\sqrt{(t-8)}}{4}$
21. If $f(x) = 5x^3$ and $g(x) = 3x^5$, then $f(x) \cdot g(x)$ will be
- (a) Even function (b) Odd function
(c) Both (d) None of these
22. If $f(x) = \begin{cases} 1-x, & 0 \leq x \leq 2 \\ x-1, & 2 \leq x \leq 4 \\ 1, & 4 \leq x \leq 6 \end{cases}$ then find,
- $f(0) + f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{45}{18}\right)$
- (a) 1 (b) 2
(c) 3 (d) None of these
23. Given $f(x) = \log\left(\frac{1+x}{1-x}\right)$ and $g(x) = \frac{3x+x^3}{1+3x^2}$, then $f \circ g(x)$ is
- (a) $-f(x)$ (b) $3f(x)$
(c) $[f(x)]^3$ (d) None of these
24. If $3f(x) + 5f\left(\frac{1}{x}\right) = \frac{1}{x} - 3, \forall x \neq 0 \in R$, then $f(x) =$
- (a) $\frac{1}{16}\left(\frac{3}{x} + 5x - 6\right)$ (b) $\frac{1}{16}\left(-\frac{3}{x} + 5x - 6\right)$
(c) $\frac{1}{14}\left(-\frac{3}{x} + 5x + 6\right)$ (d) None of these
25. The graph of $y = (x+3)^3 + 1$ is the graph of $y = x^3$ shifted
- (a) 3 units to the right and 1 unit down
(b) 3 units to the left and 1 unit down
(c) 3 units to the left and 1 unit up
(d) 3 units to the right and 1 unit up
26. Which of the following is not an even function?
- (a) $f(x) = e^x + e^{-x}$ (b) $f(x) = e^x - e^{-x}$
(c) $f(x) = e^{2x} + e^{-2x}$ (d) None of these
27. Let $f(x) = |x-2| + |x-3| + |x-4|$ and $g(x) = f(x+1)$. Then
- (a) $g(x)$ is an even function
(b) $g(x)$ is an odd function
(c) $g(x)$ is neither even nor odd
(d) None of these
28. Which of the following functions is inverse of itself?
- (a) $f(x) = \frac{1-x}{1+x}$ (b) $g(x) = 5^{\log x}$
(c) $h(x) = 2^{x(x-1)}$ (d) None of these
29. Find the value of $f(f(-2))$, if $f(x) = \frac{x}{x+1}$
- (a) $3/2$ (b) $4/3$
(c) $2/3$ (d) None of these
30. Find the value of $f(f(f(3))) + f(f(1))$, if
- $$f(x) = \begin{cases} \frac{x}{x+1}; & \text{if } x \text{ is an integer} \\ \frac{1}{x-(x)}; & \text{if } x \text{ is not an integer} \end{cases}$$
- (a) 4 (b) 5
(c) 6 (d) 7
31. If $f(x) = \log\left\{\frac{1+x}{1-x}\right\}$, then $f(x) + f(y)$ is
- (a) $f(x+y)$ (b) $f\left\{\frac{x+y}{1+xy}\right\}$
(c) $(x+y)f\left\{\frac{1}{1+xy}\right\}$ (d) $\frac{f(x)+f(y)}{1+xy}$
32. Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ is
- (a) maximized whenever $a > 0, b > 0$
(b) maximized whenever $a > 0, b < 0$
(c) minimized whenever $a > 0, b > 0$
(d) minimized whenever $a > 0, b < 0$
33. Let $f(x)$ be a function satisfying $f(x)f(y) = f(xy)$ for all real x, y . If $f(2) = 4$, then what is the value of $f\left(\frac{1}{2}\right)$?
- (a) 0 (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) cannot be determined

34. Which of the following functions is an odd function?
 (a) $2^{-x} \cdot x$ (b) $2^{x-x \cdot x \cdot x \cdot x}$
 (c) Both (a) and (b) (d) Neither (a) nor (b)
35. If $f(t) = t^2 + 2$ and $g(t) = (1/t) + 2$, then for $t = 2$, $f[g(t)] - g[f(t)] = ?$
 (a) 1.2 (b) 2.6
 (c) 4.34 (d) None of these
36. Given $f(t) = kt + 1$ and $g(t) = 3t + 2$. If $fog = gof$, find k .
 (a) 2 (b) 3
 (c) 5 (d) 4
37. Find the domain of the definition of the function
 $y = \log_{10} \left[(x-5)/(x^2 - 10x + 24) \right] - (x+4)^{1/2}$
- (a) $x > 6$ (b) $4 < x < 5$
 (c) Both (a) and (b) (d) None of these
38. If $f(x) = e^x$ and $g(x) = \log_e x$, then value of fog will be
 (a) x (b) 0
 (c) 1 (d) e
39. If $f(x) = \frac{x-1}{x+1}$, then $f(ax)$ in terms of $f(x)$ is equal to
 (a) $\frac{f(x)+a}{1+af(x)}$ (b) $\frac{(a-1)f(x)+a+1}{(a+1)f(x)+a-1}$
 (c) $\frac{(a+1)f(x)+a-1}{(a-1)f(x)+a+1}$ (d) None of these

Standard Level

- Read the instructions below.
 $A * B = A^3 - B^3$
 $A + B = A - B$
 $A - B = A/B$ and
 find the value of $(3 * 4) - (8 + 12)$.
 (a) 9 (b) 9.25
 (c) -9.25 (d) None of these
 - Which of the following two functions are identical?
 (i) $f(x) = x^2/x$ (ii) $g(x) = (\sqrt{x})^2$
 (iii) $h(x) = x$
 (a) (i) and (ii) (b) (ii) and (iii)
 (c) (i) and (iii) (d) None of these
 - If $f(x) = \log x^4$ and $g(x) = 4 \log x$, then the domain for which $f(x)$ and $g(x)$ are identical?
 (a) $(-\infty, \infty)$ (b) $[0, \infty)$
 (c) $(0, \infty)$ (d) None of these
 - Let $f(n) = \left[\frac{1}{2} + \frac{n}{100} \right]$ where $[x]$ denotes the integral part of x . Then the value of $\sum_{n=1}^{100} f(n)$ is
 (a) 50 (b) 51
 (c) 1 (d) None of these
- Directions for questions 5 – 8 :** Read the information given below and answer the questions that follow :
- If $f(x) = 2x + 3$ and $g(x) = \frac{x-3}{2}$, then
 $fog(x) =$
 (a) 1 (b) $gof(x)$
 (c) $\frac{15x+9}{16x-5}$ (d) $\frac{1}{x}$
 - For what value of x ; $f(x) = g(x-3)$?
 (a) -3 (b) $1/4$
 (c) -4 (d) None of these
 - What is value of $(gofogogof)(x)(fogofog)(x)$?
 (a) x (b) x^2
 (c) $\frac{5x+3}{4x-1}$ (d) $\frac{(x+3)(5x+3)}{(4x-5)(4x-1)}$
 - What is the value of $fo(fog)ogof(x)$?
 (a) x (b) x^2
 (c) $2x+3$ (d) $\frac{x+3}{4x-5}$
 - Let $g(x) = \max(5-x, x+2)$. The smallest possible value of $g(x)$ is
 (a) 4.0 (b) 4.5
 (c) 1.5 (d) None of these
 - $f(x)$ is any function and $f^{-1}(x)$ is known as inverse of $f(x)$, then $f^{-1}(x)$ of $f(x) = \frac{1}{x} + 1$ is
 (a) $\frac{1}{x} - 1$ (b) $x - 1$
 (c) $\frac{1}{(x-1)}$ (d) $\frac{1}{x+1}$
 - A function is defined as $f(x) = \frac{25}{(x+5)^2}$, then the value of $f(x)$
 (a) decreases as the value of x increases, only if x is negative
 (b) increases as the value of x increases, only if x is negative
 (c) decreases as the value of x increases, only if x is positive
 (d) both (a) and (c) are true
 - If $f(x) = \cos(\ln x)$, then $f(x)f(y) - \frac{1}{2} \left[f\left(\frac{x}{y}\right) + f(xy) \right]$ has the value
 (a) -1 (b) $1/2$
 (c) -2 (d) None of these
 - The function $f(x) = \max\{(1-x), (1+x), 2\}$ is equivalent to
 (a) $f(x) = \begin{cases} 1-x & x \leq -1 \\ 2, & -1 < x < 1 \\ 1+x, & x \geq 1 \end{cases}$
 (b) $f(x) = \begin{cases} 1+x, & x \leq -1 \\ 2, & -1 < x < 1 \\ 1-x, & x \geq 1 \end{cases}$
 (c) $f(x) = \begin{cases} 1-x, & x \leq -1 \\ 1, & -1 < x < 1 \\ 1+x, & x \geq 1 \end{cases}$
 (d) None of these
 - If $f(x+y) = f(x) + f(y) - xy - 1$ for all $x, y \in R$ and $f(1) = 1$ then $f(n) = n, n \in N$ is true if
 (a) $n = 1$ (b) $n = 1$ and $n = 2$
 (c) n is odd (d) any value of n

15. If $f(s) = (b^s + b^{-s})/2$, where $b > 0$. Find $f(s+t) + f(s-t)$.
- (a) $f(s) - f(t)$ (b) $2f(s) \cdot f(t)$
 (c) $4f(s) \cdot f(t)$ (d) $f(s) + f(t)$

16. The inverse of $f(x) = \frac{2}{3} \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$ is

- (a) $\frac{1}{3} \log_{10} \frac{1+x}{1-x}$
 (b) $\frac{1}{2} \log_{10} \frac{2+3x}{2-3x}$
 (c) $\frac{1}{3} \log_{10} \frac{2+3x}{2-3x}$
 (d) $\frac{1}{6} \log_{10} \frac{2-3x}{2+3x}$

Directions for questions 17 – 20 : Read the information given below and answer the questions that follow :

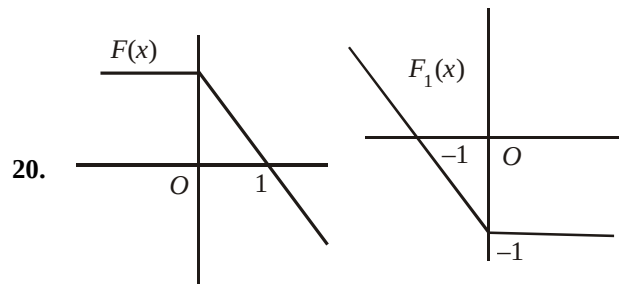
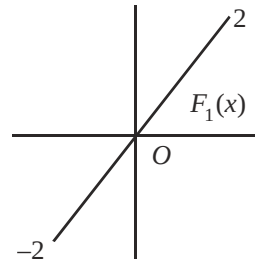
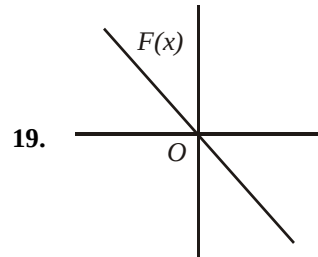
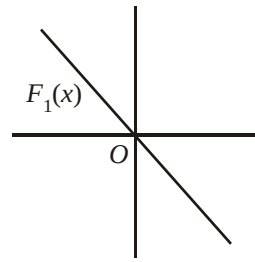
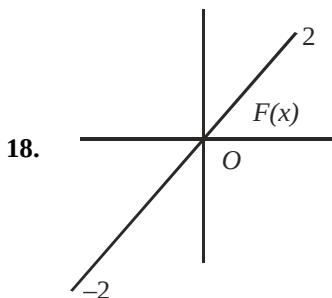
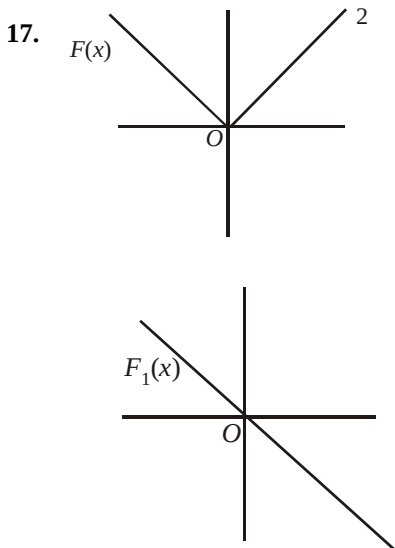
Any function has been defined for a variable x , where range of $x \in (-2, 2)$.

Mark (a) if $F_1(x) = -F(x)$

Mark (b) if $F_1(x) = F(-x)$

Mark (c) if $F_1(x) = -F(-x)$

Otherwise mark (d).



21. If $f(x) = x^3 - 4x + p$, and $f(0)$ and $f(1)$ are of opposite signs, then which of the following is necessarily true?
- (a) $-1 < p < 2$ (b) $0 < p < 3$
 (c) $-2 < p < 1$ (d) $-3 < p < 0$
22. Let $g(x)$ be a function such that $g(x+1) + g(x-1) = g(x)$ for every real x . Then for what value of p is the relation $g(x+p) = g(x)$ necessarily true for every real x ?
- (a) 5 (b) 3
 (c) 2 (d) 6
23. $f(x)$ is any function and $f^{-1}(x)$ is known as inverse of $f(x)$, then $f^{-1}(x)$ of $f(x) = x/(x-1)$, $x \neq 1$ is
- (a) $x/(1+x)$ (b) $\frac{x}{x^2-1}$
 (c) $x/(x-1)$ (d) $-x/(x+1)$

24. If $f(x)$ is a function satisfying $f(x) \cdot f(1/x) = f(x) + f(1/x)$ and $f(4) = 65$, what will be the value of $f(6)$?
- (a) 37 (b) 217
(c) 64 (d) None of these

Directions for questions number 25 - 28: Following questions are based on the given information for the following functions $f(x)$

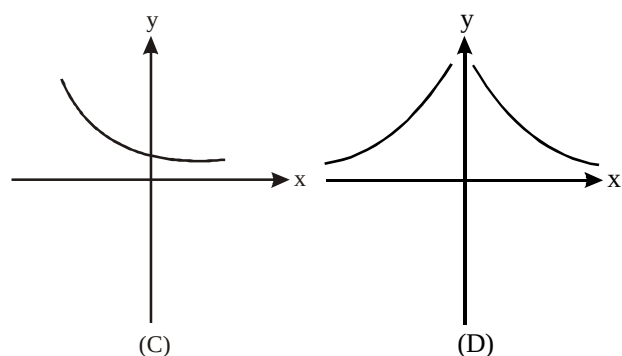
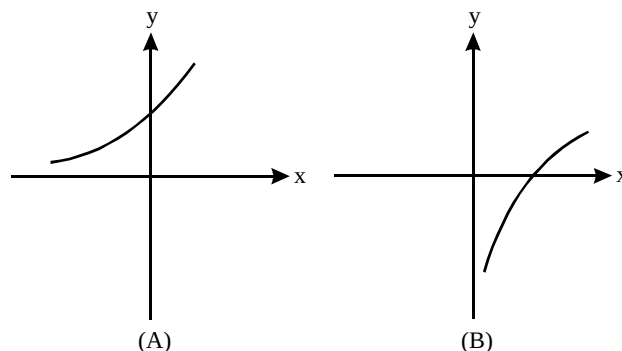
$$f(x) = 2bx + f(-x); \text{ if } x < 0$$

$$f(x) = a; \text{ if } x = 0$$

$$f(x) = b + c - 2cx + f(x-1); \text{ if } x > 0$$

25. $f(8)$ equals:
- (a) $a + 8b - 32c$ (b) $a + 8(b - 8c)$
(c) $8(a + b - c)$ (d) None of these
26. $f(-19)$ equals:
- (a) $a - 19b + 361c$ (b) $a + 19(b - 19c)$
(c) $a - 19(b + 19c)$ (d) None of these
27. If $a = 15$, $b = 11$, $c = -3$, then $f(7)$ equals:
- (a) 239 (b) 115
(c) -147 (d) None of these
28. If $a = 4$, $b = -17$ and $c = -18$, then for what value of x , $f(x) = 0$?
- (a) 4 or 9 (b) $\frac{1}{2}$ or $\frac{4}{9}$
(c) -1 or 18 (d) None of these
29. If $a = 12$, $b = 10$ and $c = 8$, then for what value of x , $f(x) < 0$?
- (a) $\frac{3}{4}$ (b) 1
(c) -2 (d) None of these

30. If $x = 10^y$, then the graph of inverse of this function is



- (a) A (b) B
(c) C (d) D

Expert Level

1. Which of the following is an even functions?

- (a) $|x^2| - 5x$ (b) $x^4 + x^5$
(c) $e^{2x} + e^{-2x}$ (d) $|x|^2/x$

2. Which of the following pairs are identical?

- (a) $f(x) = \sqrt{x^2}, g(x) = (\sqrt{x})^2$
(b) $f(x) = \frac{1}{\sqrt{x^2}}, g(x) = \frac{x}{x^2}$
(c) $f(x) = \log(x-1) + \log(x-2), g(x) = \log(x-1)(x-2)$
(d) None of these

3. If $f(x) = 1 - f(1-x)$, then the value of

$$f\left(\frac{1}{999}\right) + f\left(\frac{2}{999}\right) + \dots + f\left(\frac{998}{999}\right) \text{ is}$$

- (a) 998 (b) 1
(c) 499 (d) None of these

Directions for questions 4 – 6 : Read the information given below and answer the questions that follow :

The following functions have been defined :

$$la(x, y, z) = \min(x + y, y + z)$$

$$le(x, y, z) = \max(x - y, y - z)$$

$$ma(x, y, z) = (\frac{1}{2}) [le(x, y, z) + la(x, y, z)]$$

4. Given that $x > y > z > 0$, which of the following is necessarily true?

- (a) $la(x, y, z) < le(x, y, z)$
(b) $ma(x, y, z) < la(x, y, z)$
(c) $ma(x, y, z) < le(x, y, z)$
(d) Cannot be determined

5. What is the value of $ma(10, 4, le(la(10, 5, 3), 5, 3))$?

- (a) 7.0 (b) 6.5
(c) 8.0 (d) 7.5

6. For $x = 15, y = 10$ and $z = 9$, find the value of :

$$le(x, \min(y, x - z), le(9, 8, ma(x, y, z)))$$

- (a) 5 (b) 12

- (c) 9 (d) 4

7. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$. Then for all $x, f(g(x))$ is equal to

- (a) x (b) 1
(c) $f(x)$ (d) $g(x)$

8. If $a < b < c < d < e$ and $f(x) = (x-a)^2(x-b)(x-c)(x-d)(x-e)$ which of the following is true?

- (a) $f(x) > 0$, for $(x < a)$ as well as for $(d < x < e)$
(b) $f(x) < 0$, for $(a < x < b)$ as well as for $(d < x < e)$
(c) $f(x) < 0$, for $(b < x < c)$ as well as for $(d < x < e)$
(d) None of these

9. If $F(x) = \begin{cases} \left(x - 10 \left[\frac{x}{10}\right]\right) \cdot 10^{\lceil \log_{10} x \rceil} + F\left(\left[\frac{x}{10}\right]\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$

where $[x]$ stands for the greatest integer not exceeding 'x', then $F(7752) =$

- (a) 2222 (b) 7777
(c) 7752 (d) 2577

10. Let f be a function with domain $[-3, 5]$ and let $g(x) = |3x + 4|$, Then the domain of $(f \circ g)(x)$ is

- (a) $\left(-3, \frac{1}{3}\right)$ (b) $\left[-3, \frac{1}{3}\right]$
(c) $\left[-3, \frac{1}{3}\right)$ (d) None of these

11. The function $y = 1/x$ shifted 1 unit down and 1 unit right is given by

- (a) $y - 1 = 1/(x + 1)$ (b) $y - 1 = 1/(x - 1)$
(c) $y + 1 = 1/(x - 1)$ (d) $y + 1 = 1/(x + 1)$

Test Yourself

- Which of the following functions satisfies the condition $f\left(\frac{x+y}{x-y}\right) = \frac{f(x)+f(y)}{f(x)-f(y)}$?
 (a) $f(x) = x$ (b) $f(x) = ax + b$
 (c) $f(x) = 2^x$ (d) $f(x) = x^2$
- Let $f(x) = 1 + |x|$, $x < -1$
 $[x]$, $x \geq -1$, where $[.]$ denotes the greatest integer function.
 Then $f\{f(-2.3)\}$ is equal to
 (a) 4 (b) 2
 (c) -3 (d) 3
- What is the maximum value of the function $y = \min(12 - x, 8 + x)$?
 (a) 12 (b) 10
 (c) 11 (d) 8
- If $f(x) = \frac{2^x + 2^{-x}}{2}$, then $f(x+y) \cdot f(x-y)$ is equal to
 (a) $\frac{1}{2} [f(x+y) + f(x-y)]$
 (b) $\frac{1}{2} [f(2x) + f(2y)]$
 (c) $\frac{1}{2} [f(x+y) \cdot f(x-y)]$
 (d) None of these
- A polynomial function $f(x)$ satisfies $f(x) = f\left(\frac{1}{x}\right)$
 $= f(x) + f\left(\frac{1}{x}\right)$. If $f(10) = 1001$, then what is the value of $f(20)$?
 (a) 2002 (b) 8004
 (c) 8001 (d) None of these
- Let a, b and c be fixed positive real numbers. Let $f(x) = \frac{ax}{b+cx}$ for $x \leq 1$. Then as x increases,
 (a) $f(x)$ increases
 (b) $f(x)$ decreases
 (c) $f(x)$ increases first and then decreases
 (d) None of these
- Let $f(x) = |x-2| + |x-3| + |x-4|$ and $g(x) = f(x+1)$. Then
 (a) $g(x)$ is an even function
 (b) $g(x)$ is an odd function
 (c) $g(x)$ is neither even nor odd
 (d) None of these
- If $f(x) = 2x^2 + 6x - 1$, then the value of $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1}$ is
 (a) $11/13$ (b) $35/3$
 (c) $45/29$ (d) None of these
- The graph of $y = (x+3)^3 + 1$ is the graph of $y = x^3$ shifted
 (a) 3 units to the right and 1 unit down
 (b) 3 units to the left and 1 unit down
 (c) 3 units to the left and 1 unit up
 (d) 3 units to the right and 1 unit up
- If $f(x) = x^3$ and $g(x) = x^2/5$, then $f(x) - g(x)$ will be
 (a) Odd function (b) Even function
 (c) Neither (a) nor (b) (d) Both (a) and (b)
- $f(x)$ is any function and $f^{-1}(x)$ is known as inverse of $f(x)$, then $f^{-1}(x)$ of $f(x) = x/(x-1)$, $x \neq 1$ is
 (a) $x/(1+x)$ (b) $\frac{x}{x^2-1}$
 (c) $x/(x-1)$ (d) $-x/(x+1)$
- Which of the following equations will best fit for the given data ?

x	1	2	3	4	5	6
y	4	8	14	22	32	44

 (a) $y = ax + b$ (b) $y = a + bx + cx^2$
 (c) $y = e^{ax+b}$ (d) None of these
- If $f(s) = (b^s + b^{-s})/2$, where $b > 0$. Find $f(s+t) + f(s-t)$.
 (a) $f(s) - f(t)$ (b) $2f(s) \cdot f(t)$
 (c) $4f(s) \cdot f(t)$ (d) $f(s) + f(t)$
- Let $f(x) = |x-2| + |2.5-x| + |3.6-x|$, where x is a real number, attains a minimum at?
 (a) $x = 2.3$ (b) $x = 2.5$
 (c) $x = 2.7$ (d) None of these
- The function $y = 1/x$ shifted 1 unit down and 1 unit right is given by
 (a) $y - 1 = 1/(x+1)$ (b) $y - 1 = 1/(x-1)$
 (c) $y + 1 = 1/(x-1)$ (d) $y + 1 = 1/(x+1)$

Hints & Solutions

Foundation Level

1. (c) $f(x^2) + 2 = x^2 + \frac{1}{x^2} + 2$

$$\left(x + \frac{1}{x}\right)^2 = \{f(x)\}^2$$

and $f(x^3) + 3f(x)$

$$= x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)^3 = \{f(x)\}^3$$

Thus, both 1 and 2 are correct.

2. (c) As we know

$$|x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\therefore f(x) = \frac{|x|}{x} = \begin{cases} \frac{x}{x} & \text{if } x > 0 \\ \frac{-x}{x} & \text{if } x < 0 \end{cases}$$

$$\begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Hence, range = $\{-1, 1\}$.

3. (a)

4. (c) $f(x) = \sqrt{x^3} \Rightarrow f(3x) = \sqrt{(3x)^3} = 3\sqrt{3x^3}$

5. (b) $7f(x) = 7e^x$.

6. (b) $f(2) = \frac{2+1}{2-1} = 3$

$$f(f(2)) = f(3) = \frac{3+1}{3-1} = 2$$

$$f(f(f(2))) = f(f(3)) = f(2) = \frac{2+1}{2-1} = 3$$

$$f(f(f(f(2)))) = f(3) = \frac{3+1}{3-1} = 2$$

$$f(f(f(f(f(2)))) = f(2) = \frac{2+1}{2-1} = 3$$

7. (b) Let $f(x) = y = x(2-x)$

$$\Rightarrow x^2 - 2x - y = 0$$

$$\Rightarrow x = \frac{2 \pm 2\sqrt{1-y}}{2} = 1 \pm \sqrt{1-y}$$

But $x < 1$

$$\text{So } x = 1 - \sqrt{1-y}$$

$$\therefore f^{-1}(x) = 1 - \sqrt{1-x}$$

8. (b) $y = x^2 - x + 1$

$$\Rightarrow y = x^2 - x + \frac{1}{4} + \frac{3}{4}$$

$$\Rightarrow y = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4}$$

$$\Rightarrow y - \frac{3}{4} = \left(x - \frac{1}{2}\right)^2$$

$$\Rightarrow \left(x - \frac{1}{2}\right) = \sqrt{y - \frac{3}{4}}$$

$$\Rightarrow x = \sqrt{y - \frac{3}{4}} + \frac{1}{2}$$

$$\Rightarrow y = \sqrt{x - \frac{1}{2}} + \frac{1}{2}$$

9. (d) Draw the graph of y or see it by assuming different values of y .

10. (c) $dy/dx = 2x + 10 = 0 \Rightarrow x = -5$

11. (c) $\{(3)@4!(3\#2)@[(4!3)@]2\#4\}$
 $\{(3, 5)!(5) @ [(0.5) @ (-5)]\}$
 $\{[-0.75] @ [-2.25]\} = -1.5$.

12. (d) $b = (1)(4) = 4$.

13. (b) $f(f(t)) = f[t/(1+t^2)^{1/2}] = t/(1+2t^2)^{1/2}$

14. (c) $x/2 + x/3 + x/5 = 31x/30$

It means $[x/2] = x/2$, $[x/3] = x/3$ and $[x/5] = x/5$

Now $[N] = N$ is possible only if N is an integer. Hence, $x/2$, $x/3$ and $x/5$ are integers. So, x is divisible by 2, 3 and 5. Or, x is divisible by 30.

Total 33 values are possible.

15. (c) $h(x) = 3x^3 + 3 = (3x^2)(x) + 3 = f(x)g(x) + 3$

Thus, for every x , the corresponding values of $f(x)$, $g(x)$ and $h(x)$ differ by 3.

16. (b) $\therefore fog\left(-\frac{1}{4}\right) = f\left[g\left(-\frac{1}{4}\right)\right] = f(-1) = 1$

$$\text{and } gof\left(-\frac{1}{4}\right) = g\left[f\left(-\frac{1}{4}\right)\right] = g\left(\frac{1}{4}\right) = [1/4] = 0$$

$\therefore$ Required value = $1 + 0 = 1$

17. (b) y -axis by definition.

18. (d) Since the denominator $x^2 - 3x + 2$ has real roots, the maximum value would be infinity.

19. (a) $\frac{5x-x^2}{4} \geq 1 \Rightarrow 1 \leq x \leq 4$

20. (c) $g(f(h(t))) = g(f(4t-8)) = g(\sqrt{4t-8})$
 $= \frac{\sqrt{4t-8}}{4}$

21. (a) $f(x) \cdot g(x) = 15x^8$, which is an even function. Thus, option (a) is correct.

22. (c) $f(0) = 1 - 0 = 1$, $f\left(\frac{1}{2}\right) = 1 - \frac{1}{2} = \frac{1}{2} = 0.5$

$f(1) = 1 - 1 = 0$, $f\left(\frac{45}{18}\right) = 2.5 - 1 = 1.5$

$\therefore f(0) + f\left(\frac{1}{2}\right) + f(1) + f\left(\frac{45}{18}\right) = 1 + 0.5 + 0 + 1.5 = 3$

23. (b) We have $(fog)(x) = f(g(x))$

$= \log \left\{ \frac{1+g(x)}{1-g(x)} \right\} \left[\text{since } f(x) = \log \left(\frac{1+x}{1-x} \right) \right]$

$\log \left\{ \frac{1+\left(\frac{3x+x^3}{1+3x^2}\right)}{1-\left(\frac{3x+x^3}{1+3x^2}\right)} \right\} \left[\text{Substituting for } g(x) \right]$

$\log \left\{ \frac{1+3x^2+3x+x^3}{1+3x^2-3x-x^3} \right\}$

$\log \left\{ \frac{(1+x)^3}{(1-x)^3} \right\} = 3 \log \left(\frac{1+x}{1-x} \right) = 3f(x)$

24. (b) We have, $3f(x) + 5f\left(\frac{1}{x}\right) = \frac{1}{x} - 3, \forall x (\neq 0) \in \mathbb{R} \dots (1)$

$\Rightarrow 3f\left(\frac{1}{x}\right) + 5f(x) = x - 3 \dots (2)$

$\left[\text{Replacing } x \text{ by } \frac{1}{x} \right]$

Multiplying (1) by 3 and (2) by 5 and subtracting, we get

$\Rightarrow f(x) = \frac{1}{16} \left(-\frac{3}{x} + 5x - 6 \right), \forall x (\neq 0) \in \mathbb{R}$

25. (c) $(x+3)^3$ would be shifted 3 units to the left and hence $(x+3)^3 + 1$ would shift 3 units to the left and 1 unit up.

26. (b) Is not even since $e^x - e^{-x} \neq e^{-x} - e^x$.

27. (c) $g(x) = f(x+1) = |x-2+1| + |x-3+1| + |x-4+1|$
 $= |x-1| + |x-2| + |x-3|$

28. (a) Since $f \circ f(x) = f(f(x))$

$= f\left(\frac{1-x}{1+x}\right) = \frac{1-\left(\frac{1-x}{1+x}\right)}{1+\left(\frac{1-x}{1+x}\right)} = x$, for all x

So, inverse of f is f itself. It can be easily seen that $g \circ g(x) \neq x$ and $h \circ h(x) \neq x$

29. (c) $f(-2) = \frac{-2}{-2+1} = 2$

$f(f(-2)) = f(2) = \frac{2}{2+1} = \frac{2}{3}$

30. (b) $f(3) = \frac{3}{3+1} = \frac{3}{4}$

$f(f(3)) = f\left(\frac{3}{4}\right) = \frac{1}{\frac{3}{4} - \left[\frac{3}{4}\right]} = \frac{1}{\frac{3}{4} - 0} = \frac{4}{3}$

$f(f(f(3))) = f\left(\frac{4}{3}\right) = \frac{1}{\frac{4}{3} - \left[\frac{4}{3}\right]} = \frac{1}{\frac{1}{3}} = 3$

$f(1) = \frac{1}{1+1} = \frac{1}{2}$

$f(f(1)) = \frac{1}{\frac{1}{2} - \left[\frac{1}{2}\right]} = 2 \Rightarrow 3+2=5$

31. (b) $f(x) = \log \left(\frac{1+x}{1-x} \right)$ and $f(y) = \log \left(\frac{1+y}{1-y} \right)$

$\therefore f(x) + f(y) = \log \left(\frac{1+x}{1-x} \right) + \log \left(\frac{1+y}{1-y} \right)$

$\log \left\{ \left(\frac{1+x}{1-x} \right) \left(\frac{1+y}{1-y} \right) \right\} = \log \left(\frac{1+x+y+xy}{1-x-y+xy} \right)$

$\log \frac{(1+xy) \left(1 + \frac{x+y}{1+xy} \right)}{(1+xy) \left(1 - \frac{x+y}{1+xy} \right)}$

[Divide and multiply the numerator & denominator by $(1+xy)$]

$\log \frac{1 + \frac{x+y}{1+xy}}{1 - \frac{x+y}{1+xy}} = f\left(\frac{x+y}{1+xy}\right)$

32. (d) $f(x) = ax^2 - bx$. In this function, x^2 and x are always positive.

The value thus depends on a and b . $f(0) = a - b$. Using different options, we find that $a - b$ will be positive if $a > 0$ and $b < 0$.

The minimum value of a positive function is 0. Hence (d) is correct option.

33. (b) $f(x) \cdot f(y) = f(x \cdot y)$
 $\Rightarrow p(0) \cdot p(1) = p(0)$
 $\therefore p(1) = 1$

$$\text{Now, } p(2) \cdot p\left(\frac{1}{2}\right) = p(1)$$

$$\Rightarrow 4 \times p\left(\frac{1}{2}\right) = 1$$

$$\therefore p\left(\frac{1}{2}\right) = \frac{1}{4}$$

34. (d) Neither $2^{-x \cdot x}$ nor $2^{x - x \cdot x \cdot x}$ is an odd function as for neither of them is $f(x) = -f(-x)$

35. (d) $f(g(t)) - g(f(t)) = f(2.5) - g(6) = 8.25 - 2.166 = 6.0833$.

36. (a) $fog = f(3t + 2) = K(3t + 2) + 1$
 $gof = g(kt + 1) = 3(kt + 1) + 2$
 $K(3t + 2) + 1 = 3(kt + 1) + 2$
 $\Rightarrow 2k + 1 = 5$
 $\Rightarrow k = 2$

37. (c) For the function to exist, the argument of the logarithmic function should be positive. Also, $(x + 4) \geq 0$ should be obeyed simultaneously.

For $\frac{(x-5)}{(x^2 - 10x + 24)}$ to be positive both numerator and denominator should have the same sign. Considering all this, we get:

$4 < x < 5$ and $x > 6$.

38. (a) $fog = f(\log_e x) = e^{\log_e x} = x$.

39. (c)

Standard Level

- 1 (b) Such problems should be solved by the BODMAS rule for sequencing of operations.
 Solving, thus, we get $(3 \cdot 4) - (8 + 12)$
 $= -37 - (-4)$. [Note here that the '-' sign between -37 and -4 is the operation defined above.]
 $= 37/4 = 9.25$
2. (d) For two functions to be identical, their domains should be equal.
 Checking the domains of $f(x)$, $g(x)$ and $h(x)$,
 $f(x) = x^2/x$, x should not be equal to zero.

So, domain will be all real numbers except at $x = 0$.

$$g(x) = (\sqrt{x})^2, x \text{ should be non-negative.}$$

So, domain will be all positive real numbers.

$h(x) = x$, x is defined every where,

So, we can see that none of them have the same domain.

3. (c) Domain $f(x)$ is $R - \{0\}$ i.e., $(-\infty, 0) \cup (0, \infty)$

and Domain $g(x)$ is R^+ i.e., $(0, \infty)$

$\therefore$ Common domain of $f(x)$ and $g(x)$ is $(0, \infty)$

Hence, if $x \in (0, \infty)$, then $f(x) = g(x)$

4. (b) For all the values of $n < 50$, $f(n) = 0$
 And for all the $n \geq 50$, $f(n) = 1$. Hence, 51 such values are there.

5. (b) $fog(x) = f\{g(x)\} = f\left(\frac{x-3}{2}\right) = 2\left(\frac{x-3}{2}\right) + 3 = x$

$$\text{And } gof(x) = g\{f(x)\} = g(2x+3) = \frac{2x+3-3}{2} = x$$

Clearly $fog(x) = gof(x)$.

6. (c) $f(x) = g(x-3)$

$$2x+3 = \frac{x-3-3}{2} \Rightarrow 2x+3 = \frac{x-6}{2}$$

$$4x+6 = x-6 \text{ or } 3x = -12 \text{ or } x = -4.$$

7. (b) $\{go \text{ fo } go \text{ go } go \text{ fo } f(x)\} \{fo \text{ go } g(x)\}$

From Q. 3, we have $fog(x) = gof(x) = x$

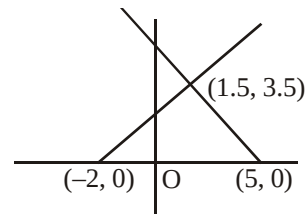
Therefore above expression becomes $(x) \cdot (x) = x^2$.

8. (c) $fo(fog)o(gof)(x)$

we have, $fo \text{ } g(x) = go \text{ } f(x) = x$

So given expression reduces to $f(x)$ that is $2x + 3$.

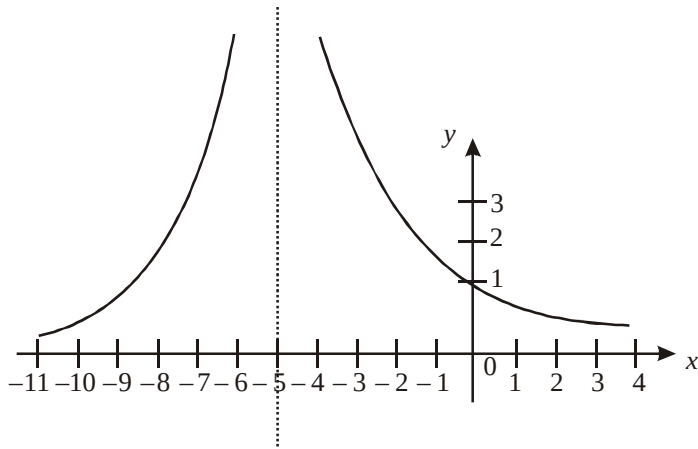
9. (d) $g(x) = \max(5-x, x+2)$. Drawing the graph,



The lines representing the function $g(x)$ intersect one another at a unique point. It clearly shows that the smallest value of $g(x) = 3.5$.

10. (c) $y = 1/x + 1$
 Hence, $y - 1 = 1/x$
 $\Rightarrow x = 1/(y - 1)$
 Thus $f^{-1}(x) = 1/(x - 1)$

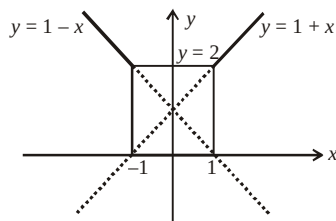
11. (c)



12. (d) $f(x) = \cos(\log x)$

$$\begin{aligned} \therefore f(x)f(y) &= \frac{1}{2} \left[f\left(\frac{x}{y}\right) + f(xy) \right] \\ &= \cos(\log x) \cos(\log y) - \frac{1}{2} [\cos(\log x - \log y) + \cos(\log x + \log y)] \\ &= \cos(\log x) \cos(\log y) - \frac{1}{2} [2 \cos(\log x) \cos(\log y)] \\ &= 0 \end{aligned}$$

13. (a) We draw the graphs of $y = 1 - x$, $y = 1 + x$ and $y = 2$



From the graph, we get

$$f(x) = \begin{cases} 1 - x & \text{if } x \leq -1 \\ 2 & \text{if } -1 < x < 1 \\ 1 + x & \text{if } x \geq 1 \end{cases}$$

14. (a) Since $f(x+y) = f(x) + f(y) \quad \forall x, y \in \mathbb{R}$

$$\therefore f(x+1) = f(x) + f(1) - x - 1 \quad [\text{Putting } y = 1]$$

$$\Rightarrow f(x+1) = f(x) - x \quad [\because f(1) = 1]$$

$$\therefore f(n+1) = f(n) - n < f(n) \Rightarrow f(n+1) < f(n)$$

So,

$$f(n) < f(n-1) < f(n-2) \dots < f(3) < f(2) < f(1) = 1$$

$$\therefore f(n) = n \text{ holds only for } n = 1$$

15. (b) Let $s = 1$, $t = 2$ and $b = 3$

$$\begin{aligned} \text{Then, } f(s+t) + f(s-t) &= f(3) + f(-1) (3^3 + 3^{-3})/2 + (3^{-1} + 3^1)/2 \\ &= 27 + 1/27)/2 + [3 + (1/3)]/2 \\ &= 730/54 + 10/6 \\ &= 820/54 = 410/27 \\ 2f(s) \times f(t) &\text{ gives the same value.} \end{aligned}$$

16. (b) If $y = \frac{2}{3} \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$, $10^{2x} = \frac{3y+2}{2-3y}$

$$\text{or } x = \frac{1}{2} \log_{10} \frac{2+3y}{2-3y} \therefore f^{-1}(x) = \frac{1}{2} \log_{10} \frac{2+3x}{2-3x}.$$

17. (d) From the graph $F_1(x) = F(x)$ for $x \in (-2, 0)$

$$\text{but, } F_1(x) = -F(x) \text{ for } x \in (0, 2).$$

No option of (a, b, c) satisfy this condition.

18. (d) From the graphs, $F_1(x) = -F(x)$ and also $F_1(x) = F(-x)$. So, both (a) and (b) are satisfied which is not given in any of the option.

19. (d) By observation $F_1(x) = -F(x)$ and also $F_1(x) = F(-x)$. So, both (a) and (b) are satisfied. Since no option is given, mark (d) as the answer.

20. (c) By observation $F_1(x) = -F(-x)$. This can be checked by taking any value of x say 1, 2. So, answer is (c).

21. (b) $f(x) = x^3 - 4x + p$

$$f(0) = p$$

$$\text{Let } p > 0$$

.....(1)

$$f(1) = p - 3 \text{ (which will be negative)}$$

$$\Rightarrow p - 3 < 0 \text{ or } p < 3$$

.....(2)

$$\text{From (1) and (2)}$$

$$0 < p < 3.$$

$$\text{Again let } p < 0 \text{ (3), then } p - 3 > 0 \text{ (iv)}$$

$$\text{From (3) and (4) :}$$

$$3 < p < 0$$

which is not possible

22. (d) $g(x+1) + g(x-1) = g(x)$

$$\Rightarrow g(x+1) = g(x) - g(x-1)$$

$$\text{Using } x = x + 5$$

$$\Rightarrow g(x+6) = g(x+5) - g(x+4)$$

$$= g(x+4) - g(x+3) - g(x+4) = -g(x+3)$$

$$= -[g(x+2) - g(x+1)]$$

$$= -g(x+2) + g(x+1)$$

$$= -g(x+1) + g(x) + g(x+1) = g(x)$$

$$\text{Hence, } p = 6.$$

23. (c) $y = x/(x-1)$
 $\Rightarrow (x-1)/x = 1/y$
 $\Rightarrow 1 - (1/x) = 1/y$
 $\Rightarrow 1/x = 1 - 1/y$
 $\Rightarrow 1/x = (y-1)/y$
 $\Rightarrow x = y/(y-1)$
Hence, $f^{-1}(x) = x/(x-1)$
24. (b) We have $f(x) \cdot f(1/x) = f(x) + f(1/x)$
 $\Rightarrow f(1/x) [f(x) - 1] = f(x)$
For $x = 4$, we have $f(1/4) [f(4) - 1] = f(4)$
 $\Rightarrow f(1/4) [64] = 54$
 $\Rightarrow f(1/4) = 54/64 = 1/64 + 1$
This mean $f(x) = x^3 + 1$
For $f(6)$ we have $f(6) = 216 + 1 = 217$

Solutions for questions number 26-28: In case of $x > 0$, we get the following pattern.

- $f(1) = b + c - 2c + a = a + b - c$
 $f(2) = b + c - 4c + a + b - c = a + 2b - 4c$
 $f(3) = b + c - 6c + a + 2b - 4c = a + 3b - 9c$
 $f(4) = b + c - 8c + a + 3b - 9c = a + 4b - 16c$
(i.e., $f(x) = a + bx - cx^2$)
25. (b) Hence, $f(8) = a + 8b - 64c = a + 8(b - 8c)$
Hence (b) is correct.
26. (c) $f(-19) = 2b \times (-19) + f(-(-19))$
 $= -38b + f(19)$
 $= -38b + a + 19b - 361c$
 $= a - 19b - 361c = a - 19(b + 19c)$
Hint: For $x < 0$; i.e., $f(-x) = a + b(-x) - c(-x)^2$
27. (a) $f(7) = a + 7b - 49c$
 $\therefore$ when $a = 15$, $b = 11$ and $c = -3$
 $f(7) = 15 + 7 \times 11 - 49(-3)$
 $= 15 + 77 + 147 = 239$
28. (b) $f(x) = a + b(x) - c(x)^2$ for every x
 $\therefore 0 = 4 - 17x + 18x^2$
Now, for convenience go through options.
29. (c) $f(x) < 0$
 $\Rightarrow a + b(x) - c(x)^2 < 0$
 $\Rightarrow 12 + 10(x) - 8(x)^2 < 0$..(1)
Now, for convenience go through options.
Alternatively: Solve the quadratic inequality (1) and then get the required set of values.
30. (b) Inverse of an exponential function is a logarithmic function
 $\therefore x = 10^y \Rightarrow \log_{10} x = y$
graph (A) is an exponential function
graph (B) is a logarithmic function
graph (C) is also an exponential function
graph (D) represent inverse function like $\left| \frac{1}{x} \right|$
logarithmic function is defined only for positive values
have choice (a), (c) and (d) are wrong. Also the increase function (i.e., $f^{-1}(x)$ of a function $f(x)$) is symmetric about $x = y$.

Expert Level

1. (c) Use options for solving.
If a function is even it should satisfy the equation $f(x) = f(-x)$.
We now check the four options to see which of them represents an even function.
Check option (a) $f(x) = |x^2| - 5x$.
Putting $-x$ in the place of x .
 $f(x) = |(-x)^2| - 5(-x)$
 $= |x^2| + 5(x) \neq f(x)$
Checking option (b) $f(x) = x^4 + x^5$.
Putting $-x$ at the place of x ,
 $f(-x) = (-x)^4 + (-x)^5 = x^4 - x^5 \neq f(x)$
Checking option (c) $f(x) = e^{2x} + e^{-2x}$
Putting $-x$ at the place of x .
 $f(-x) = e^{-2x} + e^{-(-2x)} = e^{-2x} + e^{2x} = f(x)$
So (c) is the answer.
You do not need to go further to check for d . However, if you had checked, you would have been able to disprove it as follows:
Checking option (d), $f(x) = |x|^2/x$
Putting $f(x)$ at the place of x .
 $f(-x) = |-x|^2 / -x = |x|^2 / -x \neq f(-x)$
2. (d) Domain of $f(x) = \sqrt{x^2}$ is R i.e., $x \in (-\infty, \infty)$
Domain of $g(x) = (\sqrt{x})^2$ is $R^+ \cup \{0\}$ i.e., $x \in [0, \infty)$
 $\therefore \sqrt{x^2} \neq (\sqrt{x})^2$
Again co-domain of $f(x) = \frac{1}{\sqrt{x^2}}$ is $R^+ \sim \{0\}$
and co-domain of $g(x) = \frac{x}{x^2}$ is R^+
i.e., $g(x) \in (0, \infty)$
i.e., $g(x) \in (-\infty, 0) \cup (0, \infty)$
 $\therefore \frac{1}{\sqrt{x^2}} \neq \frac{x}{x^2}$
Again domain of $f(x) = \log(x-1) + \log(x-2)$ is $(2, \infty)$ and Domain of $g(x) = \log(x-1)(x-2)$ is $(-\infty, 1) \cup (2, \infty)$
 $\therefore \log(x-1) + \log(x-2) \neq \log(x-1)(x-2)$
Since in each point domain of definition of $f(x)$ is unequal, therefore the given pairs of functions are not identical.

3. (c) $f(x) = 1 - f(1-x)$
 $\Rightarrow f(x) + f(1-x) = 1$
 Now, $f\left(\frac{1}{999}\right) + f\left(\frac{2}{999}\right) + \dots + f\left(\frac{997}{999}\right) + f\left(\frac{998}{999}\right)$
 $= \left[f\left(\frac{1}{999}\right) + f\left(\frac{998}{999}\right) \right] + \left[f\left(\frac{2}{999}\right) + f\left(\frac{997}{999}\right) \right] + \dots$
 $= \left[f\left(\frac{1}{999}\right) + f\left(1 - \frac{1}{999}\right) \right] + \left[f\left(\frac{2}{999}\right) + f\left(1 - \frac{2}{999}\right) \right] + \dots$
 $= 1 + 1 + 1 + \dots \text{499 times} = 499$
4. (d) Since $x > y > z > 0$
 $\therefore la(x, y, z) = y + z$
 and $le = \max(x - y, y - z)$
 we cannot find the value of le . Therefore we can't say whether $la > le$ or $le > la$
 Hence, we can't comment, as data is insufficient.
5. (b) $la(10, 5, 3) = 8$; $le(8, 5, 3) = 3$
 $ma(10, 4, 3) = \frac{1}{2}[6 + 7] = \frac{13}{2} = 6.5$
6. (c) $ma(15, 10, 9) = \frac{1}{2}[5 + 19] = 12$
 $\min(10, 6) = 6$; $le(9, 8, 12) = 1$; $le(15, 6, 1) = 9$.
7. (b) $g(x) = 1 + x - [x]$;

$$f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

 For integral values of x ; $g(x) = 1$
 For $x < 0$; (but not integral value) $x - [x] > 0 \Rightarrow g(x) > 1$
 For $x > 0$; (but not integral value) $x - [x] > 0 \Rightarrow g(x) > 1$
 $\therefore g(x) \geq 1, \forall x$ $\therefore f(g(x)) = 1, \forall x$
8. (d) Use the following:
 If $a < x$, then $(x - a) > 0$, $b < x$, then $(x - b) > 0$, $c < x$, then $(x - c) > 0$ and $d < x$, then $(x - d) > 0$, $e < x$, then $(x - e) > 0$. And $a > x$, then $(x - a) < 0$, similarly other also can be calculated. Now use the options. Hence, the answer is option (d) None of these.
9. (d) $F(7752) = (7752 - 10 \times 775) \cdot 10^3 + F(775)$
 $= 2000 + F(775)$
 $2000 + (775 - 10 \times 77) \cdot 10^2 + F(77)$
 $2000 + 500 + F(77)$
 $2000 + 500 + (77 - 10 \times 7) \cdot 10^1 + F(7)$
 $2000 + 500 + 70 + F(7)$
 $2000 + 500 + 70 + (7 - 10 \times 0) \cdot 10^0 + F(0)$
 $2000 + 500 + 70 + 7 + 0 = 2577$
10. (b) $(fog)(x) = f[g(x)] = f(|3x + 4|)$.
 since the domain of f is $[-3, 5]$,
 $\therefore -3 \leq |3x + 4| \leq 5$
 $\Rightarrow |3x + 4| \leq 5 \Rightarrow -5 \leq 3x + 4 \leq 5$
 $\Rightarrow -9 \leq 3x \leq 1 \Rightarrow -3 \leq x \leq 1/3$.
 $\therefore$ Domain of fog is $\left[-3, \frac{1}{3}\right]$
11. (c) Looking at the options, one unit right means x is replaced by $(x - 1)$. Also, 1 unit down means -1 on the RHS.
 Thus, $(y + 1) = 1/(x - 1)$

Explanation of Test Yourself

1. (a) Using the options, if $f(x) = x$, then $f[(x+y)/(x-y)] = (x+y)/(x-y)$ and similarly it can be seen that RHS will be same as LHS.

2. (d) $f(x) = 1 + |x|$, $x < -1$ and $[x]$, $x \geq -1$,
So, $f(-2.3) = 1 + |-2.3| = 1 + 2.3 = 3.3$
 $f\{f(-2.3)\} = f(3.3) = [3.3] = 3$

3. (a) Equate $12 - x = 8 + x$ to give you the intersection point between the two lines $12 - x$ and $8 + x$. The intersection occurs at a value of x as 2. It can be visualized by plotting both these lines that the maximum value of the given function both these lines that the maximum value of the given function would occur at $x = 2$. Hence, the correct answer would be 10.

4. (b) $f(x+y) \cdot f(x-y)$

$$= \frac{2^{x+y} + 2^{-x-y}}{2} \cdot \frac{2^{x-y} + 2^{-x-y}}{2}$$

$$= \frac{2^{2x} + 2^{2y} + 2^{-2x} + 2^{-2y}}{2 \times 2}$$

$$= \frac{1}{2} \left[\frac{2^{2x} + 2^{-2x}}{2} + \frac{2^{2y} + 2^{-2y}}{2} \right]$$

$$= \frac{1}{2} [f(2x) + f(2y)]$$

5. (c) Let $f(x) = x^n + 1$, so $f(1/x) = (1/x^n) + 1$

Checking it for $f(x) f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ satisfies the value of $f(x)$ assumed.

$$f(x) = x^n + 1 = 1001, \text{ so } x^n = 1000$$

$$\text{Hence, } n = 3$$

$$f(20) = 20^3 + 1 = 8001$$

6. (a) For any increase in the value of x , increase in the numerator will be more than the increase in the denominator. This can be verified through taking a few values of x . Alternatively, this can be verified through plotting the graphs. The slope of the graph of the numerator will be more than the slope of the graph in the denominator.

Hence, $f(x)$ will increase.

7. (c) $g(x) = f(x-1) = |x-2+1| + |x-3+1| + |x-4+1| = |x-1| + |x-2| + |x-3|$

Obviously, this is neither odd nor even.

Alternatively, we know the graph of this function will neither be symmetrical to axis or origin. [See the topic Graphs and Maxima Minima]

8. (d) Applying componendo and dividendo in $\frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1}$
 $[2.f(3/4)]/2 = f(3/4)$

$$f(3/4) = 2(3/4)^2 + 6(3/4) - 1$$

$$f(x) = 2x^2 + 6x - 1, \text{ then the value of } \frac{f\left(\frac{3}{4}\right)+1}{f\left(\frac{3}{4}\right)-1} \text{ is}$$

So, the answer is option (d).

9. (c) $(x+3)^3$ would be shifted 3 units to the left and hence $(x+3)^3 + 1$ would shift 3 units to the left and 1 unit up.

10. (c) $(x^3 - x^2/5) = f(x) - g(x)$ is neither even nor odd.

11. (c) $y = x/(x-1)$
 $\Rightarrow (x-1)/x = 1/y$
 $\Rightarrow 1 - (1/x) = 1/y$
 $\Rightarrow 1/x = 1 - 1/y \Rightarrow 1/x = (y-1)/y$
 $\Rightarrow x = y/(y-1)$

$$\text{Hence, } f^{-1}(x) = x/(x-1).$$

12. (b) At $x = 1$, $y = 4$; and $x = 1$, $y = 8$

$$4 = a + b \text{ and } 8 = 2a + b$$

$$\Rightarrow a = 4, b = 0$$

$$\text{So, } y = ax + b \Rightarrow y = 4x$$

The other values do not satisfy this last equation. so option (a) is not fit.

Similarly, we may find that option (c) is also not fit. But option (b) is absolutely fit.

13. (b) Let $s = 1$, $t = 2$ and $b = 3$

$$\begin{aligned} \text{Then, } f(s+t) + f(s-t) \\ &= f(3) + f(-1) = (3^3 + 3^{-3})/2 + (3^{-1} + 3^1)/2 \\ &= [27 + 1/27]/2 + [3 + (1/3)]/2 \\ &= 730/54 + 10/6 \\ &= 820/54 = 410/27 \end{aligned}$$

Option (b) $2f(s) \times f(t)$ gives the same value.

14. (b) $f(x) = |x-2| + |2.5-x| + |3.6-x|$ can attain minimum value when either of the three terms = 0.

$$\text{Case I : When } |x-2| = 0 \Rightarrow x = 2,$$

$$\text{Value of } f(x) = 0.5 + 1.6 = 2.1.$$

$$\text{Case II : When } |2.5-x| = 0 \Rightarrow x = 2.5$$

$$\text{Value of } f(x) = 0.5 + 0 + 1.1 = 1.6.$$

$$\text{Case III : When } |3.6-x| = 0 \Rightarrow x = 3.6$$

$$\text{Value of } f(x) = 1.6 + 1.1 + 0 = 2.7.$$

Hence the minimum value of $f(x)$ is 1.6 at $x = 2.5$.

15. (c) Looking at the options one unit right means x is replaced by $(x-1)$. Also, 1 unit down means -1 on the RHS.

$$\text{Thus, } (y+1) = 1/(x-1)$$

14

QUADRATIC AND CUBIC EQUATIONS

- Introduction
- Quadratic Polynomials
- Quadratic Equations
- Graph of a Quadratic Expression
- Geometrical Meaning of Roots or Solutions of a Quadratic Equation
- Sign of a Quadratic Expression
- Sum and Product of Roots
- Formation of an Equation with Given Roots
- Greatest and Least Value of a Quadratic Expression
- Cubic Equations
- Bi-Quadratic Equation

INTRODUCTION

A quadratic equation is a polynomial equation of degree two and a cubic equation is a polynomial equation of degree three. This chapter is very important for CAT and other equivalent aptitude tests. Every year 2–3 questions are asked from this chapter. The questions based on this chapter have been asked directly or indirectly. Usually, the questions from quadratic equation are asked. But a good CAT aspirant must have knowledge of cubic and other polynomial questions because questions may be asked from cubic and other types of polynomial equation as well.

QUADRATIC POLYNOMIALS

An expression in the form of $ax^2 + bx + c$, where a, b, c are real numbers but $a \neq 0$, is called a quadratic polynomial. For examples $2x^2 - 5x + 3$, $-x^2 + 2x$, $3x^2 - 7$, $\sqrt{2}x^2 + 7x + 2$, etc.

QUADRATIC EQUATIONS

A quadratic expression when equated to zero is called a quadratic equation. Hence an equation in the form of $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$, is called a quadratic equation. For examples,

$$2x^2 - 5x + 3 = 0, -x^2 + 2x = 0,$$

$$3x^2 - 7 = 0 \text{ and } \sqrt{2}x^2 + 7x + 2 = 0, \text{ etc.}$$

Illustration 1: Which of the following is not a quadratic equation?

- (a) $x^2 - 2x + 2(3 - x) = 0$
 (b) $x(x + 1) + 1 = (x - 2)(x - 5)$

(c) $(2x - 1)(x - 3) = (x + 5)(x - 1)$

(d) $x^3 - 4x^2 - x + 1 = (x - 2)^3$

Solution: (b) **Hint:** $x(x + 1) + 1 = (x - 2)(x - 5)$

$$\Rightarrow x^2 + x + 1 = x^2 - 7x + 10$$

$$\Rightarrow 8x - 9 = 0, \text{ which is not a quadratic equation.}$$

Discriminant (D)

For the quadratic equation $ax^2 + bx + c = 0$,
 $D = b^2 - 4ac$

Here, D is the symbol of discriminant.

Roots or Solution of a Quadratic Equation

- (i) If $D > 0$, then the quadratic equation $ax^2 + bx + c = 0$ has two distinct roots given by

$$\alpha = \frac{-b + \sqrt{D}}{2a} \text{ and } \beta = \frac{-b - \sqrt{D}}{2a}$$

Here α and β are symbols of roots of the quadratic equation.

- (ii) If $D = 0$, then the quadratic equation $ax^2 + bx + c = 0$ has two equal roots given by

$$\alpha = \beta = -\frac{b}{2a}$$

- (iii) If $D < 0$, then the quadratic equation $ax^2 + bx + c = 0$ has two roots in the form of $a + ib$ and $a - ib$, where a and b are real numbers and $i = \sqrt{-1}$. The roots in the form of $a + ib$ and $a - ib$ are known as imaginary roots. Imaginary roots $a + ib$ and $a - ib$ are also known as complex conjugate roots.

Note that imaginary roots means roots are not real numbers.

Note that if

$$ax^2 + bx + c = a(x - \alpha)(x - \beta),$$

Then α and β satisfy the equation $ax^2 + bx + c = 0$ and hence α and β are the roots of

$$ax^2 + bx + c = 0.$$

Illustration 2: If $ax^2 + bx + c = 0$ has equal roots, then $c =$

- (a) $-\frac{b}{2a}$ (b) $\frac{b}{2a}$
(c) $-\frac{b^2}{4a}$ (d) $\frac{b^2}{4a}$

Solution: (d) $ax^2 + bx + c = 0$ has equal roots if disc. $b^2 - 4ac = 0$

$$\Rightarrow b^2 = 4ac$$

$$\Rightarrow c = \frac{b^2}{4a}$$

Illustration 3: Find the condition that the quadratic equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ may have a common root.

Solution: Let α be a common root of the given equations.

$$\text{Then } \alpha^2 + a\alpha + b = 0 \text{ and } \alpha^2 + b\alpha + a = 0$$

By the method of cross-multiplication, we get

$$\frac{\alpha^2}{a^2 - b^2} = \frac{\alpha}{b - a} = \frac{1}{b - a}$$

$$\text{This gives } \alpha^2 = \frac{a^2 - b^2}{b - a} = -(a + b) \text{ and } \alpha = 1$$

$$\Rightarrow (1)^2 = -(a + b) \Rightarrow 1 = -a - b$$

$$\Rightarrow a + b + 1 = 0 \text{ is the required condition.}$$

Illustration 4: Solve $\sqrt{4x^2 + 4x + 1} < 3 - x$

$$\text{Solution: } \sqrt{4x^2 + 4x + 1} < 3 - x \Rightarrow \sqrt{(2x + 1)^2} < 3 - x$$

$$\Rightarrow \pm(2x + 1) < 3 - x$$

$$\Rightarrow 2x + 1 < 3 - x \text{ or } -(2x + 1) < 3 - x$$

$$\Rightarrow 3x < 2 \text{ or } 2x + 1 > x - 3$$

$$\Rightarrow x < \frac{2}{3} \text{ or } x > -4$$

$$\text{Hence, } -4 < x < \frac{2}{3}$$

Illustration 5: Find the solution set of the equation

$$x + 5 - \frac{8}{x + 5} = 7.$$

$$\text{Solution: } x + 5 - \frac{8}{x + 5} = 7$$

Multiply both sides by $(x + 5)$, we get,

$$(x + 5)^2 - 8 = 7(x + 5)$$

$$\text{i.e., } (x + 5)^2 - 7(x + 5) - 8 = 0$$

$$\text{Put } u = x + 5$$

$$\text{The equation reduces to } u^2 - 7u - 8 = 0$$

$$\text{i.e., } (u - 8)(u + 1) = 0$$

$$\therefore u = 8 \text{ or } u = -1$$

$$u = x + 5$$

$$\text{i.e., } x + 5 = 8 \Rightarrow x = 8 - 5 = 3$$

$$x + 5 = -1 \Rightarrow x = -1 - 5 = -6$$

$$\therefore \text{ roots are } x = 3 \text{ and } x = -6.$$

$$\text{The solution set} = \{-6, 3\}.$$

Illustration 6: The real roots of the equation $x^{2/3} + x^{1/3} - 2 = 0$ are

- (a) 1, 8 (b) -1, -8
(c) -1, 8 (d) 1, -8

Solution: (d) The given equation is $x^{2/3} + x^{1/3} - 2 = 0$

$$\text{Put } x^{1/3} = y, \text{ then } y^2 + y - 2 = 0$$

$$\Rightarrow (y - 1)(y + 2) = 0$$

$$\Rightarrow y = 1 \text{ or } y = -2$$

$$\Rightarrow x^{1/3} = 1 \text{ or } x^{1/3} = -2$$

$$\therefore x = (1)^3 \text{ or } x = (-2)^3 = -8$$

Hence, the real roots of the given equations are 1, -8.

Illustration 7: If $x^2 + 4x + k = 0$ has real roots, then

- (a) $k \geq 4$ (b) $k \leq 4$
(c) $k \leq 0$ (d) $k \geq 0$

Solution: (b) Since $x^2 + 4x + k = 0$ has real roots.

$$\therefore \text{ Disc. } (4)^2 - 4k \geq 0$$

$$\Rightarrow 16 - 4k \geq 0$$

$$\Rightarrow 4k \leq 16$$

$$\Rightarrow k \leq 4$$

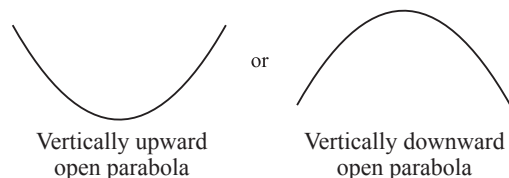
Properties of Quadratic Equations and Their Roots

- If D is a perfect square then roots are rational otherwise irrational.
- If $p + \sqrt{q}$ is one root of a quadratic equation, then their conjugate $p - \sqrt{q}$ must be the other root and vice-versa, where p is rational and $\sqrt{q}$ is a surd.
- If a quadratic equation in x has more than two roots, then it is an identity in x .

GRAPH OF A QUADRATIC EXPRESSION

Graph of $y = ax^2 + bx + c$; where a, b, c are real numbers but $a \neq 0$ is always a parabola.

The shape of the parabola is like

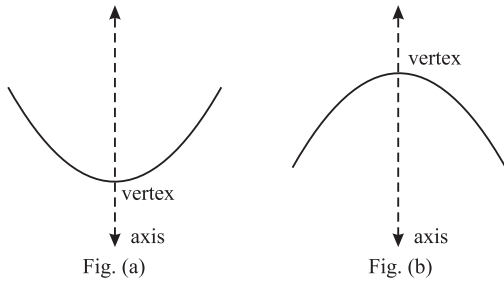


If $a > 0$, then parabola is vertically upward open and if $a < 0$, then parabola is vertically downward open.

Axis of a parabola is a vertical line which divides it in two halves.

A point on the parabola where the graph turn down to up or up to down is called vertex of the parabola. Coordinate of the vertex

$$\text{of the parabola is always } \left(-\frac{b}{2a}, \frac{4ac - b^2}{4a} \right).$$



Vertex is either the lowest point on the parabola as in the fig (a) or the highest point on the parabola as in the fig (b).

GEOMETRICAL MEANING OF ROOTS OR SOLUTIONS OF A QUADRATIC EQUATION

x -coordinate of the points where the graph of the quadratic expression $y = ax^2 + bx + c$ intersects or touches the x -axis are called roots of the quadratic equation $ax^2 + bx + c = 0$. If the parabola intersects the x -axis at two distinct points, then there are two different real roots of the quadratic equation.

If the parabola only touches the x -axis at a point, then the quadratic equation has two real equal roots.

If the parabola does not touch or intersect the x -axis then there are two different imaginary roots. The imaginary roots means roots are not real numbers.

SIGN OF A QUADRATIC EXPRESSION

Let $y = ax^2 + bx + c$, where $ax^2 + bx + c$ is a quadratic expression.

Case-I: If $a > 0$ and

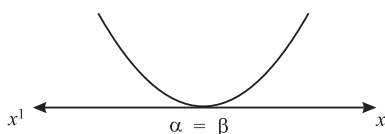
- (i) $D > 0$, the graph of $y = ax^2 + bx + c$ is a vertically upward open parabola which intersects the x -axis at two different points.



In the figure, α and β are the values of the x -coordinates of the points where the parabola intersects the x -axis. Hence α and β are two roots of the quadratic equation $ax^2 + bx + c = 0$, such that $\alpha < \beta$.

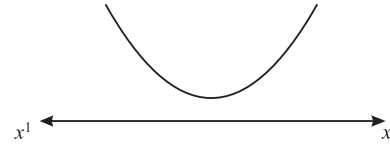
It is clear from the graph,

- (a) $ax^2 + bx + c$ will be positive for all real values of x which are less than α or greater than β i.e., $ax^2 + bx + c > 0$ for $x \in (-\infty, \alpha) \cup (\beta, \infty)$
 (b) $ax^2 + bx + c$ will be negative for all real values of x which lie between α and β i.e., $ax^2 + bx + c < 0$ for $x \in (\alpha, \beta)$
 (ii) $D = 0$, the graph of $y = ax^2 + bx + c$ is vertically upward open parabola which touches the x -axis at only one point. Hence both roots α and β of $ax^2 + bx + c = 0$ are the same i.e. $\alpha = \beta$.



It is clear from the graph, the value of $ax^2 + bx + c$ will be positive for all real values of x except $x = \alpha$ or β .

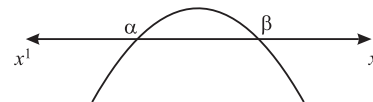
- (iii) $D < 0$, then graph of $y = ax^2 + bx + c$ is vertically upward open parabola, which does not intersect or touch the x -axis at any point.



It is clear from graph, the value of $ax^2 + bx + c$ will be positive for all values of x i.e., $ax^2 + bx + c > 0$ for $x \in (-\infty, \infty)$.

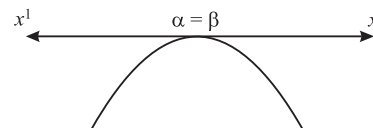
Case-II: If $a < 0$ and

- (i) $D > 0$, then graph of $y = ax^2 + bx + c$ is the vertically downward open parabola, which intersects the x -axis at two different points.



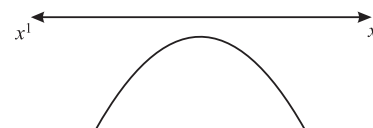
In the figure α and β are the value of the x -coordinates of the points where the graph intersects the x -axis. Hence, α and β are two roots of the quadratic equation $ax^2 + bx + c = 0$, such that $\alpha < \beta$.

- (a) $ax^2 + bx + c$ will be positive for all real values of x which are greater than α but less than β i.e., $ax^2 + bx + c > 0$ for $x \in (\alpha, \beta)$
 (b) $ax^2 + bx + c$ will be negative for all real values of x which i.e. $ax^2 + bx + c < 0$ for $x \in (-\infty, \alpha) \cup (\beta, \infty)$
 (ii) $D = 0$, the graph of $y = ax^2 + bx + c$ is vertically downward open parabola which touches the x -axis at only one point. Hence both roots α and β are the same i.e. $\alpha = \beta$.



It is clear from the graph, the value of $ax^2 + bx + c$ will be negative for all real values of x except $x = \alpha$ or β .

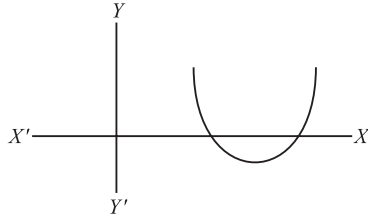
- (iii) $D < 0$, then graph of $y = ax^2 + bx + c$ is vertically downward open parabola which does not intersect or touch the x -axis at any point.



It is clear from the graph, the value of $ax^2 + bx + c$ will be negative for all real values of x i.e., $ax^2 + bx + c < 0$ for $x \in (-\infty, \infty)$.

Illustration 8: For the below figure of $ax^2 + bx + c = 0$

- (a) $a < 0$ (b) $b > 0$
 (c) $D > 0$ (d) None of these



Solution: (c)

SUM AND PRODUCT OF ROOTS

If α and β are the roots of a quadratic equation $ax^2 + bx + c = 0$,
Then,

$$\text{Sum of roots, } \alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$$

$$\text{Product of roots, } \alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

Illustration 9: Find the sum and product of roots of $-2x^2 + 3x - 5 = 0$.

Solution: Sum of roots $= -\frac{b}{a} = -\frac{3}{-2} = \frac{3}{2}$

Product of roots $= \frac{c}{a} = \frac{-5}{-2} = \frac{5}{2}$

FORMATION OF AN EQUATION WITH GIVEN ROOTS

If α and β are the roots of a quadratic equation, then the quadratic equation will be

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\text{i.e., } x^2 - (\text{Sum of the roots})x + \text{Product of the roots} = 0$$

Illustration 10: If α and β are the roots of the equation $3x^2 - x + 4 = 0$, then find the quadratic equation whose

roots are $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.

Solution: $\alpha + \beta = -\frac{-1}{3} = \frac{1}{3}, \alpha\beta = \frac{4}{3}$

Now, $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

$$= \frac{\frac{1}{3}}{\frac{4}{3}} = \frac{1}{4}$$

$$\frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

Hence required quadratic equation,

$$x^2 - \frac{1}{4}x + \frac{3}{4} = 0$$

$$\Rightarrow 4x^2 - x + 3 = 0$$

Illustration 11: If α, β are the roots of $x^2 + ax + b = 0$, find the equation for which $\alpha^2 + \beta^2$ and $\alpha^{-2} + \beta^{-2}$ are the roots.

Solution: $\alpha + \beta = -a, \alpha\beta = b$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = a^2 - 2b$$

$$\alpha^{-2} + \beta^{-2} = \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} = \frac{a^2 - 2b}{b^2}$$

Required equation,

$$x^2 - x \left\{ (a^2 - 2b) + \frac{a^2 - 2b}{b^2} \right\} + (a^2 - 2b) \frac{(a^2 - 2b)}{b^2} = 0$$

$$\Rightarrow x^2 - x \left\{ \frac{b^2(a^2 - 2b) + a^2 - 2b}{b^2} \right\} + \frac{(a^2 - 2b)(a^2 - 2b)}{b^2} = 0$$

$$\Rightarrow b^2x^2 - x\{b^2(a^2 - 2b) + a^2 - 2b\} + (a^2 - 2b)^2 = 0$$

Illustration 12: Form the quadratic equations for the given roots.

$$\frac{3 + \sqrt{5}}{4}, \frac{3 - \sqrt{5}}{4}$$

Solution: Here, $S = \frac{3 + \sqrt{5}}{4} + \frac{3 - \sqrt{5}}{4} = \frac{6}{4} = \frac{3}{2}$

and $P = \left(\frac{3 + \sqrt{5}}{4} \right) \left(\frac{3 - \sqrt{5}}{4} \right) = \frac{9 - 5}{16} = \frac{1}{4}$

$\therefore$ The required equation is $x^2 - Sx + P = 0$

i.e., $x^2 - \frac{3}{2}x + \frac{1}{4} = 0 \Rightarrow 4x^2 - 6x + 1 = 0$

GREATEST AND LEAST VALUE OF A QUADRATIC EXPRESSION

(i) If $a > 0$, then least value of the quadratic expression

$$ax^2 + bx + c \text{ is } -\frac{D}{4a} = \frac{4ac - b^2}{4a} \text{ at } x = -\frac{b}{2a}$$

Note that there is no greatest value of the quadratic expression $ax^2 + bx + c$ if $a > 0$.

(ii) If $a < 0$, then the greatest value of the quadratic expression

$$ax^2 + bx + c \text{ is } -\frac{D}{4a} = \frac{4ac - b^2}{4a} \text{ at } x = -\frac{b}{2a}$$

Note that there is no least value of the quadratic expression $ax^2 + bx + c$.

CUBIC EQUATIONS

An equation in the form of $ax^3 + bx^2 + cx + d = 0$, where a, b, c, d are real numbers but $a \neq 0$, is called a cubic equation. For example,

$$2x^3 - 4x^2 + 3x + 5 = 0, -x^3 - 4x + 7 = 0, 5x^3 = 0,$$

$$x^3 - 5 = 0, x^3 + 3x^2 = 0, \text{ etc.}$$

Any cubic equation has three roots. If α , β and γ are three roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$, then

$$(i) \alpha + \beta + \gamma = -\frac{b}{a} = -\frac{\text{coefficient of } x^2}{\text{coefficient of } x^3}$$

$$(ii) \alpha \cdot \beta + \beta \cdot \gamma + \gamma \cdot \alpha = \frac{c}{a} = \frac{\text{coefficient of } x}{\text{coefficient of } x^3}$$

$$(iii) \alpha \cdot \beta \cdot \gamma = -\frac{d}{a} = -\frac{\text{constant term}}{\text{coefficient of } x^3}$$

Illustration 13: If α , β , γ are the roots of the equation $2x^3 - 3x^2 + 6x + 1 = 0$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to

- (a) $-15/4$ (b) $15/4$
(c) $9/4$ (d) 4

Solution: (a) Given equation $2x^3 - 3x^2 + 6x + 1 = 0$,

$$\alpha + \beta + \gamma = \frac{3}{2}, \quad \alpha\beta\gamma = \frac{-1}{2}, \quad \sum \alpha\beta = 3$$

$$(\alpha^2 + \beta^2 + \gamma^2) = (\alpha + \beta + \gamma)^2 - 2(\sum \alpha\beta)$$

$$= \left(\frac{3}{2}\right)^2 - 2 \cdot 3 = \frac{9}{4} - 6 = \frac{-15}{4}$$

BI-QUADRATIC EQUATION

An equation in the form of $ax^4 + bx^3 + cx^2 + dx + e = 0$, where a , b , c , d and e are real numbers but $a \neq 0$, is called bi-quadratic equation. For example,

$$5x^4 + 2x^3 - x^2 + 3x + 8 = 0,$$

$$-x^4 - 3x^2 + 2 = 0,$$

$$4x^4 - x = 0,$$

$$2x^4 + 3x^3 + 8 = 0, \text{ etc.}$$

Any bi-quadratic equation has four roots.

If α , β , γ , and δ are the four roots of a bi-quadratic equation, then

$$(i) \alpha + \beta + \gamma + \delta = -\frac{b}{a} = -\frac{\text{coefficient of } x^3}{\text{coefficient of } x^4}$$

$$(ii) \alpha\beta + \beta\gamma + \gamma\delta + \delta\alpha = \frac{e}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$$

$$(iii) \alpha\beta\gamma + \beta\gamma\delta + \gamma\delta\alpha + \delta\alpha\beta = -\frac{d}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^4}$$

$$(iv) \alpha \beta \gamma \delta = \frac{e}{a} = \frac{\text{constant term}}{\text{coefficient of } x^4}$$

Practice Exercises



Foundation Level

1. Which of the following is a quadratic equation ?
 (a) $x^{\frac{1}{2}} + 2x + 3 = 0$
 (b) $(x - 1)(x + 4) = x^2 + 1$
 (c) $x^4 - 3x + 5 = 0$
 (d) $(2x + 1)(3x - 4) = 2x^2 + 3$

 2. Solve $x - \frac{1}{x} = 1\frac{1}{2}$
 (a) $-\frac{1}{2}, 2$
 (b) $\frac{1}{2}, 2$
 (c) $\frac{1}{2}, \frac{2}{3}$
 (d) None of these

 3. If $2x^2 - 7xy + 3y^2 = 0$, then the value of $x : y$ is
 (a) 3 : 2
 (b) 2 : 3
 (c) 3 : 1 or 1 : 2
 (d) 5 : 6

 4. Father's age is 4 less than five times the age of his son and the product of their ages is 288. Find the father's age.
 (a) 40 years
 (b) 36 years
 (c) 26 years
 (d) 42 years

 5. The sum of a rational number and its reciprocal is $\frac{13}{6}$, find the number.
 (a) $\frac{2}{3}$ or $\frac{3}{2}$
 (b) $\frac{3}{4}$ or $\frac{4}{3}$
 (c) $\frac{2}{5}$ or $\frac{5}{2}$
 (d) None of these

 6. $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}} = ?$
 (a) 2.3
 (b) 3
 (c) 6
 (d) 6.3

 7. If $x^2 + 2 = 2x$, then the value of $x^4 - x^3 + x^2 + 2$ is
 (a) 1
 (b) 0
 (c) -1
 (d) $\sqrt{2}$

 8. Minimum value of $x^2 + \frac{1}{x^2 + 1} - 3$ is
 (a) 0
 (b) -1
 (c) -3
 (d) -2

 9. One root of $x^2 + kx - 8 = 0$ is square of the other. Then the value of k is
 (a) 2
 (b) 8
 (c) -8
 (d) -2

 10. If the roots, x_1 and x_2 , of the quadratic equation $x^2 - 2x + c = 0$ also satisfy the equation $7x_2 - 4x_1 = 47$, then which of the following is true?
 (a) $c = -15$
 (b) $x_1 = -5, x_2 = 3$
 (c) $x_1 = 4.5, x_2 = -2.5$
 (d) None of these

 11. For what value of k , are the roots of the quadratic equation $(k + 1)x^2 - 2(k - 1)x + 1 = 0$ real and equal?
 (a) $k = 0$ only
 (b) $k = -3$ only
 (c) $k = 0$ or $k = 3$
 (d) $k = 0$ or $k = -3$

 12. If α, β are the roots of the equation $2x^2 - 3x - 6 = 0$, find the equation whose roots are $\alpha^2 + 2$ and $\beta^2 + 2$.
 (a) $4x^2 + 49x + 118 = 0$
 (b) $4x^2 - 49x + 118 = 0$
 (c) $4x^2 - 49x - 118 = 0$
 (d) $4x^2 + 49x - 118 = 0$

 13. If the roots of the equation $(a^2 + b^2)x^2 - 2ab(a + c)x + (b^2 + c^2) = 0$ are equal, then which one of the following is correct?
 (a) $2b = a + c$
 (b) $b^2 = ac$
 (c) $b + c = 2a$
 (d) $b = ac$

 14. If α and β are the roots of the equation $x^2 - 2x + 4 = 0$, then what is the value of $\alpha^3 + \beta^3$?
 (a) 16
 (b) -16
 (c) 8
 (d) -8

 15. If p and q are the roots of the equation $x^2 - px + q = 0$, then what are the values of p and q respectively?
 (a) 1, 0
 (b) 0, 1
 (c) -2, 0
 (d) -2, 1

 16. What is the value of $\sqrt{5\sqrt{5\sqrt{5\sqrt{5\dots\infty}}}}$?
 (a) 5
 (b) $\sqrt{5}$
 (c) 1
 (d) $(5)^{1/4}$

 17. If r and s are roots of $x^2 + px + q = 0$, then what is the value of $\frac{1}{r^2} + \frac{1}{s^2}$?
 (a) $p^2 - 4q$
 (b) $\frac{p^2 - 4q}{2}$
 (c) $\frac{p^2 - 4q}{q^2}$
 (d) $\frac{p^2 - 2q}{q^2}$

18. Find the solution of $\frac{2x}{x-3} + \frac{1}{2x+3} + \frac{3x+9}{(x-3)(2x+3)} = 0$
- (a) 0 (b) -1
(c) 3 (d) -3
19. Sum of the areas of two squares is 468 m^2 . If the difference of their perimeters is 24 m , find the sides of the two squares.
- (a) 9m, 6m (b) 18m, 12m
(c) 18m, 6m (d) 9m, 12m
20. If the roots of $x^2 - kx + 1 = 0$ are non-real, then
- (a) $-3 < k < 3$ (b) $-2 < k < 2$
(c) $k > 2$ (d) $k < -2$
21. If $ax^2 + bx + c = 0$ has real and different roots, then
- (a) $b^2 - 4ac = 0$ (b) $b^2 - 4ac > 0$
(c) $b^2 - 4ac < 0$ (d) $b^2 - 4ac \leq 0$
22. If $\sqrt{3x^2 + x + 5} = x - 3$, then the given equation has solution/solutions.
- (a) $x = -4$ (b) $x = \frac{1}{2}$
(c) $x = -4, \frac{1}{2}$ (both) (d) No solution
23. The sum of two numbers p and q is 18 and the sum of their reciprocals is $\frac{1}{4}$. Then the numbers are
- (a) 10, 8 (b) 12, 6
(c) 9, 9 (d) 14, 4
24. If the roots of the equation $x^2 - bx + c = 0$ differ by 2, then which of the following is true?
- (a) $c^2 = 4(c+1)$ (b) $b^2 = 4c+4$
(c) $c^2 = b+4$ (d) $b^2 = 4(c+2)$
25. The sum of a number and its reciprocal is one-fifth of 26. What is the sum of that number and its square?
- (a) 3 (b) 4
(c) 5 (d) 6
26. Two numbers are such that the square of greater number is 504 less than 8 times the square of the other. If the numbers are in the ratio 3 : 4. Find the number.
- (a) 15 and 20 (b) 6 and 8
(c) 12 and 16 (d) 9 and 12
27. The equation $x + \sqrt{x-2} = 4$ has
- (a) two real roots and one imaginary root
(b) one real and one imaginary root
(c) two imaginary roots
(d) one real root
28. The roots of the equation $x^2 - 2x = 0$ can be obtained graphically by finding the abscissas of the points of intersection of each of the following pairs of equations except
- (a) $y = x, y = x-2$ (b) $y = x^2, y = 2x$
(c) $y = x^2 - 2x + 1, y = 1$ (d) $y = x^2 - 2x, y = 0$
29. If in applying the quadratic formula to a quadratic equation $f(x) = ax^2 + bx + c = 0$ it happens that $c = \frac{b^2}{4a}$. Then the graph of $y = f(x)$ will certainly
- (a) have a maximum (b) have a minimum
(c) be a tangent to x -axis (d) be a tangent to the y -axis
30. The equation $\sqrt{x+10} - \frac{6}{\sqrt{x+10}} = 5$ has
- (a) an extraneous root between -5 and -1
(b) an extraneous root between -10 and -6
(c) two extraneous roots
(d) a real root between 20 and 25
[An extraneous root means a root which does not satisfy the equation.]
31. If $\log_{10}(x^2 - 3x + 6) = 1$, then the value of x is
- (a) 10 or 2 (b) 4 or -2
(c) 4 only (d) 4 or -1
32. The roots of the equation $2\sqrt{x} + 2x - \frac{1}{2} = 5$ can be found by solving
- (a) $4x^2 - 25x + 4 = 0$ (b) $4x^2 + 25x - 4 = 0$
(c) $4x^2 - 17x + 4 = 0$ (d) None of these
33. Two numbers whose sum is 6 and the absolute value of whose difference is 8 are the roots of the equation
- (a) $x^2 - 6x + 7 = 0$ (b) $x^2 - 6x - 7 = 0$
(c) $x^2 + 6x - 8 = 0$ (d) $x^2 - 6x + 8 = 0$
34. The roots of the equation $x^2 + 2\sqrt{3}x + 3 = 0$ are
- (a) real and equal (b) rational and equal
(c) rational and unequal (d) imaginary
35. The roots of the equation $ax^2 + bx + c = 0$ will be reciprocal if
- (a) $a = b$ (b) $a = bc$
(c) $c = a$ (d) $b = c$
36. If $\frac{b}{x-a} - \frac{x+a}{b}$ then the value of x in terms of a and b is
- (a) $\pm\sqrt{a^2 + b^2}$ (b) $+\sqrt{a^2 + b^2}$
(c) $-\sqrt{a^2 + b^2}$ (d) None of these
37. For what value of b and c would the equation $x^2 + bx + c = 0$ have roots equal to b and c .
- (a) (0, 0) (b) (1, -2)
(c) (1, 2) (d) Both (a) and (b)

Standard Level

- The least value of $ax^2 + bx + c$ ($a > 0$) is
 (a) $\frac{-b}{2a}$ (b) $\frac{4ac - b^2}{4a}$
 (c) $\frac{c}{a}$ (d) cannot be determined
- The discriminant of $ax^2 - 2\sqrt{2}x + c = 0$ with a, c are real constants is zero. The roots must be
 (a) equal and integral (b) rational and equal
 (c) real and equal (d) imaginary
- If one root of the equation $ax^2 + bx + c = 0$ is three times the other, then _____
 (a) $b^2 = 16ac$ (b) $b^2 = ac$
 (c) $3b^2 = 16ac$ (d) None of these
- If the product of roots of the equation $x^2 - 3(2a + 4)x + a^2 + 18a + 81 = 0$ is unity, then a can take the values as
 (a) 3, -6 (b) 10, -8
 (c) -10, -8 (d) -10, -6
- If the roots of the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ are equal, then which of the following is true?
 (a) $ab = cd$ (b) $ad = bc$
 (c) $ad = \sqrt{bc}$ (d) $ab = \sqrt{cd}$
- For what values of c in the equation $2x^2 - (c^3 + 8c - 1)x + c^2 - 4c = 0$ the roots of the equation would be opposite to signs?
 (a) $c \in (0, 4)$ (b) $c \in (-4, 0)$
 (c) $c \in (0, 3)$ (d) $c \in (-4, 4)$
- If $x^2 - 3x + 2$ is a factor of $x^4 - ax^2 + b = 0$, then the values of a and b are
 (a) -5, -4 (b) 5, 4
 (c) -5, 4 (d) 5, -4
- If $f(x) = x^2 + 2x - 5$ and $g(x) = 5x + 30$, then the roots of the quadratic equation $g[f(x)]$ will be
 (a) -1, -1 (b) 2, -1
 (c) $-1 + \sqrt{2}, -1 - \sqrt{2}$ (d) 1, 2
- The quadratic equation $g(x) = (px^2 + qx + r)$, $p \neq 0$, attains its maximum value at $x = 7/2$. Product of the roots of the equation $g(x) = 0$ is equal to 10. What is the value of $p \times q \times r$?
 (a) 70 (b) -70
 (c) 0 (d) Cannot be determined
- If $a + b + c = 0$ and a, b, c , are rational numbers then the roots of the equation $(b + c - a)x^2 + (c + a - b)x + (a + b - c) = 0$ are
 (a) rational (b) irrational
 (c) non-real (d) None of these
- If α and β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then the value of $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$ is
 (a) $\frac{3bc - a^3}{b^2c}$ (b) $\frac{3abc - b^3}{a^2c}$
 (c) $\frac{3abc - b^2}{a^3c}$ (d) $\frac{ab - b^2c}{2b^2c}$
- If a, b are the two roots of a quadratic equation such that $a + b = 24$ and $a - b = 8$, then the quadratic equation having a and b as its roots is
 (a) $x^2 + 2x + 8 = 0$ (b) $x^2 - 4x + 8 = 0$
 (c) $x^2 - 24x + 128 = 0$ (d) $2x^2 + 8x + 9 = 0$
- If $m + \frac{1}{m-2} = 4$ then, what is value of $(m-2)^2 + \frac{1}{(m-2)^2}$?
 (a) -2 (b) 0
 (c) 2 (d) 4
- If $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} = 4$, then the value of $x^2 + y^2$ is
 (a) 2 (b) 4
 (c) 8 (d) 16
- If $3x^3 - 9x^2 + kx - 12$ is divisible by $x - 3$, then it is also divisible by :
 (a) $3x^2 - 4$ (b) $3x^2 + 4$
 (c) $3x - 4$ (d) $3x + 4$
- If $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = 2\frac{1}{6}$ then the value of x is
 (a) $\frac{6}{13}$ or $\frac{4}{143}$ (b) $\frac{3}{2}$ or $\frac{2}{3}$
 (c) $\frac{5}{2}$ or $\frac{2}{3}$ (d) $\frac{9}{13}$ or $\frac{4}{13}$

17. If the equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$, have one root in common, then find the value of $(a + b)$
- (a) 0 (b) 1
(c) -1 (d) 2
18. Let x, y be two positive numbers such that $x + y = 1$. Then, the minimum value of $\left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2$ is
- (a) 12 (b) 20
(c) 12.5 (d) 13.3
19. Solve the simultaneous equations
- $$\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{5}{2}; x + y = 10$$
- (a) 8, 6 (b) 8, 2
(c) 4, 6 (d) 5, 5
20. If roots of an equation $ax^2 + bx + c = 0$ are positive, then which one of the following is correct?
- (a) Signs of a and c should be like
(b) Signs of b and c should be like
(c) Signs of a and b should be like
(d) None of the above
21. If the sum of the squares of the roots of $x^2 - (p - 2)x - (p + 1) = 0$ ($p \in R$) is 5, then what is the value of p ?
- (a) 0 (b) -1
(c) 1 (d) $\frac{3}{2}$
22. If α and β are the roots of the equation $x^2 + 6x + 1 = 0$, then what is $|\alpha - \beta|$ equal to?
- (a) 6 (b) $3\sqrt{2}$
(c) $4\sqrt{2}$ (d) 12
23. If $\frac{1}{2 - \sqrt{-2}}$ is one of the roots of $ax^2 + bx + c = 0$, where a, b, c are real, then what are the values of a, b, c respectively?
- (a) 6, -4, 1 (b) 4, 6, -1
(c) 3, -2, 1 (d) 6, 4, 1
24. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\frac{1}{\alpha + \beta}, \frac{1}{\alpha} + \frac{1}{\beta}$ is equal to
- (a) $acx^2 + (a^2 + bc)x + bc = 0$
(b) $b cx^2 + (b^2 + ac)x + ab = 0$
(c) $abx^2 + (c^2 + ab)x + ca = 0$
(d) None of these
25. Find the roots of the equation $a^3x^2 + abcx + c^3 = 0$
- (a) $\alpha^2\beta, \beta^2\alpha$ (b) α, β^2
(c) $\alpha^2\beta, \beta\alpha$ (d) $\alpha^3\beta, \beta^3\alpha$
26. A natural number when increased by 12, equals 160 times its reciprocal. Find the number.
- (a) 3 (b) 5
(c) 8 (d) 16
27. Solve: $\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}; a \neq 0, x \neq 0$
- (a) a, b (b) $-a, b$
(c) $0, a$ (d) $-a, -b$
28. Which is not true?
- (a) Every quadratic polynomial can have at most two zeros.
(b) Some quadratic polynomials do not have any zero. [i.e. real zero]
(c) Some quadratic polynomials may have only one zero. [i.e. one real zero]
(d) Every quadratic polynomial which has two zeros.
29. The expression $a^2 + ab + b^2$ is _____ for $a < 0, b < 0$
- (a) $\neq 0$ (b) < 0
(c) > 0 (d) $= 0$
30. For what value of c the quadratic equation $x^2 - (c + 6)x + 2(2c - 1) = 0$ has sum of the roots as half of their product?
- (a) 5 (b) -4
(c) 7 (d) 3
31. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$ is
- (a) $abx^2 + b(c + a)x + (c + a)^2 = 0$
(b) $(c + a)x^2 + b(c + a)x + ac = 0$
(c) $cax^2 + b(c + a)x + (c + a)^2 = 0$
(d) $cax^2 + b(c + a)x + c(c + a)^2 = 0$
32. If $x^2 + ax + b$ leaves the same remainder 5 when divided by $x - 1$ or $x + 1$, then the values of a and b are respectively
- (a) 0 and 4 (b) 3 and 0
(c) 0 and 5 (d) 4 and 0
33. The condition that both the roots of quadratic equation $ax^2 + bx + c = 0$ are positive is
- (a) a and c have an opposite sign that of b
(b) b and c have an opposite sign that of a
(c) a and b have an opposite sign that of c
(d) None of these

34. If the equation $x^2 - bx + 1 = 0$ does not possess real roots, then which one of the following is correct?
- (a) $-3 < b < 3$ (b) $-2 < b < 2$
 (c) $b > 2$ (d) $b < -2$
35. If $\sin \theta = x + \frac{a}{x}$ for all $x \in \mathbb{R} - \{0\}$, then which one of the following is correct?
- (a) $a \geq 4$ (b) $a \geq \frac{1}{2}$
 (c) $a \leq \frac{1}{4}$ (d) $a \leq \frac{1}{2}$
36. If the roots of the quadratic equation $3x^2 - 5x + p = 0$ are real and unequal, then which one of the following is correct?
- (a) $p = 25/12$ (b) $p < 25/12$
 (c) $p > 25/12$ (d) $p \leq 25/12$
37. If the roots of the equation $x^3 - ax^2 + bx - c = 0$ are three consecutive integers, then what is the smallest possible value of b ?
- (a) $-\frac{1}{\sqrt{3}}$ (b) -1
 (c) 0 (d) 1

Expert Level

- If $x = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1}}$, value of $x^2 + x - 1$ is
 - $\sqrt{5}$
 - 2
 - $-\sqrt{5}$
 - $\sqrt{5} + 1$
- Find the minimum value of $\sqrt{x^2 + y^2}$ if $5x + 12y = 60$.
 - $\frac{60}{13}$
 - $\frac{13}{5}$
 - $\frac{13}{12}$
 - 1
- If one root of $x^2 + px + 12 = 0$ is 4, while the equation $x^2 + px + q = 0$ has equal roots, then the value of q is
 - 49/4
 - 4/49
 - 4
 - 1/4
- If the equation $x^3 - ax^2 + bx - a = 0$ has three real roots then which of the following is true?
 - $a = 11$
 - $a \neq 1$
 - $b = 1$
 - $b \neq 1$
- If the equation $x^2 + kx + 1 = 0$ has the roots α and β , then what is the value of $(\alpha + \beta) \times (\alpha^{-1} + \beta^{-1})$?
 - k^2
 - $\frac{1}{k^2}$
 - $2k^2$
 - $\frac{1}{(2k^2)}$
- If α , β and γ are the roots of the equation $9x^3 - 7x + 6 = 0$, then the equation whose roots are $3\alpha + 2$, $3\beta + 2$ and $3\gamma + 2$ is
 - $x^3 - 6x^2 + 5x + 24 = 0$
 - $9x^3 - x + 16 = 0$
 - $2x^3 - 27x - 8 = 0$
 - None of these
- Quadratic equations $(2p-1)z^2 + (2p+1)z + c = 0$ and $(q+1)y^2 + (4q+1)y + 3c = 0$ have the same pair of roots. Given that $c \neq 0$, what is the value of $(p+q)$?
 - 3
 - 4
 - 2
 - Cannot be determined
- If the roots of the equation $x^2 + px + q = 0$ differ from the roots of the equation $x^2 + qx + p = 0$ by the same quantity, then what is the value of $(p+q)$?
 - 4
 - 4
 - 0
 - 8
- If $a = b = c$, then the roots of the equation $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are
 - real and unequal
 - imaginary
 - real and equal
 - None of these
- If the roots of the equation $a_1x^2 + a_2x + a_3 = 0$ are in the ratio $r_1 : r_2$ then
 - $r_1 \cdot r_2 \cdot a_1^2 = (r_1 + r_2) a_3^2$
 - $r_1 \cdot r_2 \cdot a_2^2 = (r_1 + r_2)^2 a_1 \cdot a_3$
 - $r_1 \cdot r_2 \cdot a_2 = (r_1 + a_2)^2 a_1 \cdot a_2$
 - $r_1 \cdot r_2 \cdot a_1^2 = (r_1 + r_2)^2 a_3 \cdot a_2$
- For what value of a do the roots of the equation $2x^2 + 6x + a = 0$, satisfy the conditions $\left(\frac{\alpha}{\beta}\right) + \left(\frac{\beta}{\alpha}\right) < 2$.
 - $a < 0$ or $a > \frac{9}{2}$
 - $a > 0$
 - $-1 < a < 0$
 - $-1 < a < 1$
- If both the roots of the quadratic equation $ax^2 + bx + c = 0$ lie in the interval $(0, 3)$ then a lies in
 - $(1, 3)$
 - $(-1, -3)$
 - $(-\sqrt{121}/91, -\sqrt{8})$
 - None of these
- The value of p for which the sum of the square of the roots of $2x^2 - 2(p-2)x - p - 1 = 0$ is least is
 - 1
 - $\frac{3}{2}$
 - 2
 - 1
- The set of values of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is
 - $(-\infty, 0)$
 - $(0, 1)$
 - $(1, \infty)$
 - $(0, \infty)$
- Quadratic equation $x^2 + bx + c = 0$ has roots a and b , such that LCM of $(a, b) = 24$ and HCF of $(a, b) = 2$. What is the total number of such quadratic equations if a and b are natural numbers?
 - 1
 - 2
 - 3
 - 4
- There are two quadratic expressions $a_1x^2 + b_1x + c_1$ and $a_2x^2 + b_2x + c_2$. Both of them have the same roots. If the ratio of a_1 to a_2 is 1 : 2, what is the ratio of the maximum values of the two quadratic expressions? ($a_1, a_2 < 0$)
 - 1 : 3
 - 3 : 1
 - 1 : 5
 - None of these
- If the equation $ax^2 + bx + c = 0$ has a root less than -2 and a root greater than 2, and $a > 0$, then which of the following is true?
 - $4a + 2|b| + c < 0$
 - $4a + 2|b| + c > 0$
 - $4a + 2|b| + c = 0$
 - None of these

Test Yourself

- If $\alpha \neq \beta$ but $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$ then the equation having α/β and β/α as its roots is
 (a) $3x^2 - 19x + 3 = 0$ (b) $3x^2 + 19x - 3 = 0$
 (c) $3x^2 - 19x - 3 = 0$ (d) $x^2 - 5x + 3 = 0$
- Sum of the real roots of the equation $x^2 + 5|x| + 6 = 0$
 (a) Equals to 5 (b) Equals to 10
 (c) Equals to -5 (d) None of these
- If the expression $ax^2 + bx + c$ is equal to 4 when $x = 0$ leaves a remainder 4 when divided by $x + 1$ and a remainder 6 when divided by $x + 2$, then the values of a, b and c are respectively
 (a) 1, 1, 4 (b) 2, 2, 4
 (c) 3, 3, 4 (d) 4, 4, 4
- If two quadratic equations $ax^2 + ax + 3 = 0$ and $x^2 + x + b = 0$ have a common root $x = 1$ then which of the following statements hold true?
 (A) $a + b = -3.5$ (B) $ab = 3$
 (C) $\frac{a}{b} = \frac{3}{4}$ (D) $a - b = -0.5$
 (a) A, B, C (b) B, C, D
 (c) A, C, D (d) A, B, D
- The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root is common. The other roots of the first and second equations are integers in the ratio 4 : 3. Then the common root is
 (a) 1 (b) 4
 (c) 3 (d) 2
- If the roots of the equation $(x - a)(x - b) + (x - b)(x - c) + (x - c)(x - a) = 0$, (where a, b, c are real numbers) are equal, then:
 (a) $b^2 - 4ac = 0$ (b) $a = b = c$
 (c) $a + b + c = 0$ (d) None of these
- If the sum of the roots of the quadratic equations $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2} =$
 (a) 0 (b) -1
 (c) 1 (d) 2
- Find all the values of p for which one root of the equation $x^2 - (p + 1)x + p^2 + p - 8 = 0$, is greater than 2 and of the other root is smaller than 2.
 (a) $-2 < p < 3$ (b) $-3 < p < 2$
 (c) $-3 < p$ (d) $p < 2$
- Roots of the equation $x^2 + bx - c = 0 (b, c > 0)$ are
 (a) both positive (b) both negative
 (c) of opposite sign (d) None of these
- If α, β, γ be the roots of the equation $x(1 + x^2) + x^2(6 + x) + 2 = 0$ then the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ is
 (a) -3 (b) $\frac{1}{2}$
 (c) $-\frac{1}{2}$ (d) None of these
- If the roots of the equation $x^2 - px + q = 0$ differ by unity then
 (a) $q^2 = 4p$ (b) $p^2 = 4q + 1$
 (c) $q^2 = 4p + 1$ (d) $p^2 = 4q - 1$
- The roots of the equation $x^4 - 2x^3 + x = 380$ are
 (a) $5, -4, \frac{1 \pm 5\sqrt{-3}}{2}$ (b) $-5, 4, \frac{-1 \pm 5\sqrt{-3}}{2}$
 (c) $5, 4, \frac{-1 \pm 5\sqrt{-3}}{2}$ (d) $-5, -4, \frac{1 \pm 5\sqrt{-3}}{2}$
- If the roots of the equations $px^2 + 2qx + r = 0$ and $qx^2 - 2\sqrt{pr}x + q = 0$ be real, then
 (a) $p = q$ (b) $q^2 = pr$
 (c) $p^2 = qr$ (d) $r^2 = pr$
- In writing a quadratic equation of the form $x^2 + bx + c = 0$, a student writes are coefficient of x incorrectly and finds the roots as 7 and 8. Another student makes a mistake in writing the constant term and gets the roots as 8 and -3. Find the correct quadratic equation.
 (a) $x^2 - 5x + 56 = 0$ (b) $x^2 - 5x + 21 = 0$
 (c) $x^2 - 21x + 56 = 0$ (d) $x^2 - 7x + 21 = 0$
- If α and β are the roots of $2x^2 + 7x + c = 0$ and $|\alpha^2 - \beta^2| = \frac{7}{4}$ then c is equal to
 (a) 2 (b) 3
 (c) 6 (d) 5

Hints & Solutions

Foundation Level

1. (d) Equations in options (a) and (c) are not quadratic equations as in (a) max. power of x is fractional and in (c), it is not 2 in any of the terms.

For option (b), $(x-1)(x+4) = x^2 + 1$

$$\text{or } x^2 + 4x - x - 4 = x^2 + 1$$

$$\text{or } 3x - 5 = 0$$

which is not a quadratic equations but a linear.

For option (d), $(2x+1)(3x-4) = 2x^2 + 3$

$$\text{or } 6x^2 - 8x + 3x - 4 = 2x^2 + 3$$

$$\text{or } 4x^2 - 5x - 7 = 0$$

which is clearly a quadratic equation.

2. (a) $x - \frac{1}{x} = 1 \Rightarrow \frac{x^2 - 1}{x} = \frac{3}{2}$

$$\Rightarrow 2(x^2 - 1) = 3x \Rightarrow 2x^2 - 2 = 3x$$

$$\Rightarrow 2x^2 - 3x - 2 = 0$$

$$\Rightarrow 2x^2 - 4x + x - 2 = 0$$

$$\Rightarrow 2x(x-2) + 1(x-2) = 0$$

$$\Rightarrow (2x+1)(x-2) = 0$$

Either $2x+1=0$ or $x-2=0$

$$\Rightarrow 2x = -1 \text{ or } x = 2$$

$$\Rightarrow x = \frac{-1}{2} \text{ or } x = 2$$

$$\therefore x = \frac{-1}{2}, 2 \text{ are solutions.}$$

3. (c) $2x^2 - 7xy + 3y^2 = 0$

$$2\left(\frac{x}{y}\right)^2 - 7\left(\frac{x}{y}\right) + 3 = 0$$

$$\frac{x}{y} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{7 \pm \sqrt{49 - 24}}{2 \times 2} = \frac{7 \pm 5}{4} = 3, \frac{1}{2}$$

$$\Rightarrow \frac{x}{y} = \frac{3}{1} \text{ or } \frac{x}{y} = \frac{1}{2}$$

4. (b) Let the son's age be x years.
So, father's age = $5x - 4$ years.

$$\therefore x(5x-4) = 288$$

$$\Rightarrow 5x^2 - 4x - 288 = 0 \Rightarrow 5x^2 - 40x + 36x - 288 = 0$$

$$\Rightarrow 5x(x-8) + 36(x-8) = 0$$

$$\Rightarrow (5x+36)(x-8) = 0$$

$$\text{Either } x-8=0 \text{ or } 5x+36=0 \Rightarrow x=8 \text{ or } x=\frac{-36}{5}$$

x cannot be negative; therefore, $x=8$ is the solution.
 $\therefore$ Son's age = 8 years and Father's age = $5x - 4 = 36$ years.

5. (a) Let the number be x .

$$\text{Then, } x + \frac{1}{x} = \frac{13}{6} \Rightarrow \frac{x^2 + 1}{x} = \frac{13}{6} \Rightarrow 6x^2 - 13x + 6 = 0$$

$$\Rightarrow 6x^2 - 9x - 4x + 6 = 0 \Rightarrow (3x-2)(2x-3) = 0$$

$$\Rightarrow x = \frac{2}{3} \text{ or } x = \frac{3}{2}$$

Hence, the required number is $\frac{2}{3}$ or $\frac{3}{2}$.

6. (b) $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$

$$\sqrt{6+x} = x$$

$$6+x = x^2$$

$$x^2 - x - 6 = 0$$

$$x^2 - 3x + 2x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$x = 3$$

7. (b) $x^2 + 2 = 2x \Rightarrow x^2 - 2x + 2 = 0$

$$x^2 - 2x + 2 \Rightarrow x^4 - x^3 + x^2 + 2(x^2 + x + 1)$$

$$x^4 - 2x^3 + 2x^2$$

$$\begin{array}{r} - \quad + \quad - \\ x^3 - x^2 + 2 \end{array}$$

$$x^3 - 2x^2 + 2x$$

$$\begin{array}{r} - \quad + \quad - \\ x^2 - 2x + 2 \end{array}$$

$$x^2 - 2x + 2$$

$$\begin{array}{r} - \quad + \quad - \\ x^2 - 2x + 2 \end{array}$$

$$0$$

$$\therefore x^4 - x^3 + x^2 + 2$$

$$= (x^2 - 2x + 2)(x^2 + x + 1) = 0$$

8. (d) $x^2 \geq 0$

$\therefore$ Minimum value

$$= 0 + \frac{1}{1} - 3 = -2$$

9. (d) Given $x^2 + kx - 8 = 0$

Let a and b be the roots of given equation and $b = a^2$ (given)

$$\text{Sum of roots} = a + b = -k = a + a^2 \quad \dots(1)$$

$$\text{Product of roots} = ab = -8 = a^3 \Rightarrow a = -2$$

$$\text{Using } a = -2 \text{ in (1), } -k = -2 + 4 = 2 \text{ or } k = -2$$

10. (a) $7x_2 - 4x_1 = 47$
 $x_1 + x_2 = 2$
 Solving $11x_2 = 55$
 $x_2 = 5$ & $x_1 = -3$
 $\therefore c = -15$
11. (c) Since, the roots of the equation $(k+1)x^2 - 2(k-1)x + 1 = 0$ are real and equal.
 $\therefore \{-2(k-1)\}^2 - 4(k+1) = 0$ ($\because b^2 - 4ac = 0$)
 $\Rightarrow 4(k^2 - 2k + 1) - 4(k+1) = 0$
 $\Rightarrow k^2 - 2k + 1 - 1 = 0$
 $\Rightarrow k^2 - 3k = 0$
 $\Rightarrow k = 0, k = 3$
12. (b) Since, α, β are root of the equation $2x^2 - 3x - 6 = 0$
 $\therefore \alpha + \beta = \frac{3}{2}$ and $\alpha\beta = -3$
 $\Rightarrow \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= \frac{9}{4} + 6 = \frac{33}{4}$
 Now, $(\alpha^2 + 2) + (\beta^2 + 2) = (\alpha^2 + \beta^2) + 4$
 $= \frac{33}{4} + 4 = \frac{49}{4}$
 and $(\alpha^2 + 2)(\beta^2 + 2) = \alpha^2\beta^2 + 2(\alpha^2 + \beta^2) + 4$
 $= (-3)^2 + 2\left(\frac{33}{4}\right) + 4 = \frac{59}{2}$
 So, the equation whose roots are $\alpha^2 + 2$ and $\beta^2 + 2$ is
 $x^2 - x\{(\alpha^2 + 2) + (\beta^2 + 2)\} + (\alpha^2 + 2)(\beta^2 + 2) = 0$
 $\Rightarrow x^2 - \frac{49}{4}x + \frac{59}{2} = 0$
 $\Rightarrow 4x^2 - 49x + 118 = 0$
13. (b) Since roots of the given equation are equal.
 $\therefore D = 0$
 On solving we get $b^2 = ac$
14. (b) Use $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
15. (a)
16. (a) Let $x = \sqrt{5\sqrt{5\sqrt{5}\dots\infty}}$
 $\Rightarrow x^2 = 5x \Rightarrow x = 0, 5$
17. (d) $\frac{1}{r^2} + \frac{1}{s^2} = \frac{s^2 + r^2}{(rs)^2}$
 $= \frac{(s+r)^2 - 2sr}{(rs)^2} = \frac{p^2 - 2q}{q^2}$
18. (b) Clearly, the given equation is valid if $x - 3 \neq 0$ and
 $2x + 3 \neq 0$ i.e., when $x \neq \frac{-3}{2}, 3$
 Now, $\frac{2x}{x-3} + \frac{1}{2x+3} + \frac{3x+9}{(x-3)(2x+3)} = 0$

$$\Rightarrow 2x(2x+3) + (x-3) + 3x+9 = 0$$

$$[\text{Multiplying throughout by } (x-3)(2x+3)]$$

$$\Rightarrow 4x^2 + 6x + x - 3 + 3x + 9 = 0$$

$$\Rightarrow 4x^2 + 10x + 6 = 0$$

$$\Rightarrow 2x^2 + 5x + 3 = 0$$

$$\Rightarrow 2x^2 + 2x + 3x + 3 = 0$$

$$\Rightarrow 2x(x+1) + 3(x+1) = 0$$

$$\Rightarrow (2x+3)(x+1) = 0$$

$$\Rightarrow x+1 = 0 \Rightarrow x = -1$$
 [$\because 2x+3 \neq 0$]

Hence, $x = -1$ is the only solution of the given equation.

19. (b) Let first square has side x , $\therefore$ Area = x^2 , Perimeter = $4x$ and let second square has side y ,
 $\therefore$ Area = y^2 , Perimeter = $4y$
 Let $x > y$ so that $4x > 4y$
 Given, $x^2 + y^2 = 468$... (1)
 and $4x - 4y = 24 \Rightarrow x - y = 6 \Rightarrow y = x - 6$... (2)
 Using (2) in (1), we get $x^2 + (x-6)^2 = 468$
 $\Rightarrow x^2 + x^2 - 12x + 36 = 468 \Rightarrow 2x^2 - 12x - 432 = 0$
 $\Rightarrow x^2 - 6x - 216 = 0 \Rightarrow x = \frac{6 \pm \sqrt{36 + 864}}{2} = \frac{6 \pm \sqrt{900}}{2}$

$$= \frac{6 \pm 30}{2} = \frac{36}{2}, \frac{-24}{2} = 18, -12$$

But x being length cannot be negative $\therefore x = 18$
 put $x = 18$ in (2), we get $y = x - 6 = 18 - 6 = 12$
 $\therefore$ sides of the two squares = $x, y = 18 \text{ m}, 12 \text{ m}$

20. (b) Since the roots of $x^2 - kx + 4 = 0$ are non-real.
 $\therefore$ Disc., $(-k)^2 - 4 < 0 \Rightarrow k^2 - 4 < 0$
 $\Rightarrow k^2 < 4 \Rightarrow |k| < 2 \Rightarrow -2 < k < 2$
21. (b)
22. (d) Square both sides, we shall get $x = -4, \frac{1}{2}$. But both of them do not satisfy the given equation.
23. (b) $p + q = 18$... (1)
 and $\frac{1}{p} + \frac{1}{q} = \frac{1}{4}$... (2) (Given)
 i.e. $\frac{p+q}{pq} = \frac{1}{4} \Rightarrow \frac{18}{pq} = \frac{1}{4}$
 $\Rightarrow pq = 72$... (3)
 From (1) and (3), $p(18-p) = 72$
 $\Rightarrow p^2 - 18p + 72 = 0 \Rightarrow (p-6)(p-12) = 0$
 $\Rightarrow p = 6, 12$ when $p = 6, q = 12$; when $p = 12, q = 6$
 Hence the numbers are 12, 6.
24. (b) Let the roots be α and $\alpha + 2$.
 Then $\alpha + \alpha + 2 = b \Rightarrow \alpha = \frac{(b-2)}{2}$ (1)
 and $\alpha(\alpha + 2) = c \Rightarrow \alpha^2 + 2\alpha = c$ (2)
 Putting the value of α from (1) in (2).
 $\left(\frac{(b-2)}{2}\right)^2 + 2\left(\frac{(b-2)}{2}\right) = c$
 $\Rightarrow \frac{(b^2 + 4 - 4b)}{4} + b - 2 = c$
 $\Rightarrow b^2 + 4 - 8 = 4c$
 $\Rightarrow b^2 = 4c + 4$

25. (c) Let the number be x . Then,

$$x + \frac{1}{x} = \frac{26}{5}$$

$$\Rightarrow \frac{x^2 + 1}{x} = \frac{26}{5}$$

$$\Rightarrow 5x^2 - 26x + 5 = -0$$

$$\Rightarrow (x - 5)(5x - 1) = 0$$

$$\Rightarrow x = 5 \text{ or } \frac{1}{5}$$

26. (d)

27. (d) $x + \sqrt{x-2} = 4$

$$\sqrt{x-2} = 4 - x$$

Squaring on the both sides

$$x - 2 = 16 + x^2 - 8x$$

$$x^2 - 9x + 18 = 0$$

$$(x - 6)(x - 3) = 0$$

$$x = 6 \text{ or } 3$$

But by checking, only $x = 3$ satisfies the equation.

28. (a) Except (a) the remaining equations show the equation

$$x^2 - 2x = 0$$

For (b) $y = x^2 = 2x$

$$\Rightarrow x^2 - 2x = 0$$

For (c) $y = x^2 - 2x + 1 = 1$

$$\Rightarrow x^2 - 2x = 0$$

For (d) $y = x^2 - 2x = 0$

$$\Rightarrow x^2 - 2x = 0$$

29. (c) Since $c = \frac{b^2}{4a}$

$$\Rightarrow b^2 = 4ac$$

$$\Rightarrow b^2 - 4ac = 0$$

This implies that $f(x) = 0$ has equal roots, i.e., the curve touches the x -axis at exactly one point.

30. (b) $\sqrt{x+10} - \frac{6}{\sqrt{x+10}} = 5$

$$x + 10 - 6 = 5\sqrt{x+10}$$

$$x + 4 = 5\sqrt{x+10}$$

Squaring on both sides,

$$x^2 + 8x + 16 = 25x + 250$$

$$x^2 - 17x - 234 = 0$$

$$x^2 - 26x + 9x - 234 = 0$$

$$x(x - 26) + 9(x - 26) = 0$$

$$(x - 26)(x + 9) = 0$$

$$x = 26 \text{ (or) } -9$$

Here $x = -9$ is not satisfying. So it is extraneous.

31. (d) $\log_{10}(x^2 - 3x + 6) = 1$

$$x^2 - 3x + 6 = 10^1$$

$$x^2 - 3x - 4 = 0$$

$$(x - 4)(x + 1) = 0$$

$$x = 4 \text{ or } -1$$

32. (c) $2\sqrt{x} + \frac{2}{\sqrt{x}} = 5$

$$2x + 2 = 5\sqrt{x}$$

$$\Rightarrow 4x^2 + 8x + 4 = 25x$$

$$\Rightarrow 4x^2 - 17x + 4 = 0$$

33. (b) Let α and β are the roots

$$\alpha + \beta = 6$$

$$\alpha - \beta = 8$$

$$2\alpha = 14$$

$$\alpha = 7$$

$$\beta = -1$$

$$\alpha + \beta = 6, \alpha\beta = -7$$

The quadratic equation is $x^2 - 6x - 7 = 0$

34. (a) $b^2 - 4ac = (2\sqrt{3})^2 - 4(1)(3) = 0$. So the roots are real and equal.

35. (c) Since roots are reciprocal,

$$\text{product of the roots} = 1 \Rightarrow \frac{c}{a} = 1$$

$$\Rightarrow c = a.$$

36. (a) $\frac{b}{x-a} = \frac{x+a}{b}$

$$x^2 - a^2 = b^2$$

$$x^2 = b^2 + a^2$$

$$x = \pm\sqrt{a^2 + b^2}$$

37. (d) Solve using options. It can be seen that $b = 0$ and $c = 0$ the condition is satisfied. It is also satisfied at $b = 1$ and $c = -2$.

Standard Level

1. (b) If $a > 0$ then, $ax^2 + bx + c$ has a minimum

$$\text{value at } x = \frac{-b}{2a} \text{ and is equal to } \frac{4ac - b^2}{4a}.$$

If $a < 0$, then $ax^2 + bx + c$ has a maximum

$$\text{value at } x = \frac{-b}{2a} \text{ and is equal to } \frac{4ac - b^2}{4a}.$$

2. (c) $ax^2 - 2\sqrt{2}x + c = 0$

$$(2\sqrt{2})^2 - 4ac = 0$$

$$4ac = 8$$

$$ac = 2$$

$$c = \frac{2}{a}$$

Let α, β be the roots.

$$\alpha + \beta = \frac{2\sqrt{2}}{a}, \alpha\beta = \frac{c}{a} \cdot \frac{2}{a^2}$$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= \frac{8}{a^2} - \frac{8}{a^2} = 0$$

$$\alpha = \beta$$

$$\text{So, } \alpha = \beta = \frac{\sqrt{2}}{a}$$

Hence the roots are real and equal.

3. (c) Let $\alpha, 3\alpha$ are the roots.

$$\alpha + 3\alpha = \frac{-b}{a} \Rightarrow 4\alpha = \frac{-b}{a}$$

$$\Rightarrow \alpha = \frac{-b}{4a} \quad \dots(1)$$

$$\alpha \times 3\alpha = \frac{c}{a} \Rightarrow 3\alpha^2 = \frac{c}{a}$$

$$\frac{3b^2}{16a^2} = \frac{c}{a} \quad [\text{by (1)}]$$

$$3b^2 = 16ac.$$

4. (c) The product of the roots is given by: $(a^2 + 18a + 81)/1$. Since product is unity we get: $a^2 + 18a + 81 = 1$

$$\text{Thus, } a^2 + 18a + 80 = 0$$

Solving, we get $a = -10$ and $a = -8$.

5. (b) Solve this by assuming each option to be true and then check whether the given expression has equal roots for the option under check.

Thus, if we check for option (b).

$$ad = bc.$$

We assume $a = 6, d = 4, b = 12, c = 2$ (any set of values that satisfies $ad = bc$)

$$\text{Then } (a^2 + b^2)x^2 - 2(ac + bc)x + (c^2 + d^2) = 0$$

$$180x^2 - 120x + 20 = 0$$

We can see that this has equal roots. Thus, option (b) is a possible answer. The same way if we check for a, c and d we see that none of them gives us equal roots and can be rejected.

6. (a) For the roots to be opposite in sign, the product of roots should be negative.

$$(c^2 - 4c)/2 < 0 \Rightarrow 0 < c < 4$$

7. (b) $x^2 - 3x + 2 = 0$ gives its roots as $x = 1, 2$. Put these values in the equation and then use the options.

8. (a) $g(f(x)) = 5x^2 + 10x + 5$

Roots are -1 and -1

9. (d) Value of p cannot be calculated, so $p \times q \times r$ cannot be determined.

10. (a) The sum of coefficients

$$= (b + c - a) + (c + a - b) + (a + b - c) = a + b + c = 0$$

(given)

$\therefore x = 1$ is a root of the equation

$\therefore$ The other root is $\frac{a+b-c}{b+c-a}$, which is rational as

a, b, c , are rational

Hence, both the roots are rational.

ALTERNATIVE:

$$D = (c + a - b)^2 - 4(b + c - a)(a + b - c)$$

$$(-2b)^2 - 4(-2a)(-2c) = 4b^2 - 16ac$$

$$4(a + c)^2 - 16ac = 4[(a + c)^2 - 4ac] = [2(a - c)]^2$$

D is a perfect square. Hence, the roots of the equation are rational.

11. (b) Here, $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$

$$\text{Thus, } \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3 + \beta^3}{\alpha\beta}$$

$$\frac{(\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2)}{\alpha\beta} \quad \dots(1)$$

$$\text{Now, } (\alpha^2 + \beta^2 - \alpha\beta) = [(\alpha + \beta)^2 - 2\alpha\beta - \alpha\beta]$$

$$= [(\alpha + \beta)^2 - 3\alpha\beta]$$

Hence (1) becomes

$$\Rightarrow \frac{(\alpha + \beta)[(\alpha + \beta)^2 - 3\alpha\beta]}{\alpha\beta} = \frac{-\frac{b}{a} \left[\frac{b^2}{a^2} - \frac{3c}{a} \right]}{\frac{c}{a}}$$

$$= \frac{-b}{c} \left[\frac{b^2 - 3ac}{a^2} \right] = \frac{3abc - b^3}{a^2c}$$

12. (c) $a + b = 24$ and $a - b = 8$

$$\Rightarrow a = 16 \text{ and } b = 8 \Rightarrow ab = 16 \times 8 = 128$$

A quadratic equation with roots a and b is

$$x^2 - (a + b)x + ab = 0 \text{ or } x^2 - 24x + 128 = 0$$

13. (c) $m + \frac{1}{m-2} = 4$

$$m^2 - 2m - 3 = 0$$

$$(m - 3)(m + 1) = 0$$

$$m = 3$$

$$m - 2 = 1$$

$$\text{Now } (m - 2)^2 + \frac{1}{(m - 2)^2}$$

$$= 1^2 + \frac{1}{1^2} = 2$$

14. (a) $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} - 4 = 0$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 + y^2 + \frac{1}{y^2} - 2 = 0$$

$$\Rightarrow \left(x - \frac{1}{x}\right)^2 + \left(y - \frac{1}{y}\right)^2 = 0$$

$$\Rightarrow x - \frac{1}{x} = 0$$

$$\Rightarrow x^2 - 1 = 0 \Rightarrow x = 1$$

Similarly,

$$y = 1$$

$$\therefore x^2 + y^2 = 1 + 1 = 2$$

15. (b) Given, $3x^3 - 9x^2 + kx - 12$ is divisible by $x - 3$

$$\therefore f(3) = 0$$

$$\therefore 3 \times 3^3 - 9 \times 3^2 + 3k - 12 = 0$$

$$\therefore 81 - 81 + 3k - 12 = 0$$

$$\therefore 3k = 12$$

$$\therefore k = 4$$

So, the equation will be

$$3x^3 - 9x^2 + 4x - 12 = 0$$

$$\therefore (x - 3)(3x^2 + 4) = 0$$

Thus, $3x^2 + 4$ is a factor of the given equation.

16. (d) Given $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$

$$\Rightarrow \sqrt{\frac{x}{x\left(\frac{1}{x}-1\right)}} + \sqrt{\frac{x\left(\frac{1}{x}-1\right)}{x}} = \frac{13}{6}$$

$$\Rightarrow \sqrt{\frac{1}{\frac{1}{x}-1}} + \sqrt{\frac{1}{x}-1} = \frac{13}{6}$$

$$\Rightarrow \frac{1}{\sqrt{\frac{1}{x}-1}} + \sqrt{\frac{1}{x}-1} = \frac{13}{6}$$

$$\Rightarrow 1 + \frac{1}{x} - 1 = \frac{13}{6} \left(\sqrt{\frac{1}{x}-1} \right)$$

$$\Rightarrow \frac{1}{x} = \frac{13}{6} \left(\sqrt{\frac{1}{x}-1} \right)$$

By squaring on both side, we get

$$169x^2 - 169x + 36 = 0$$

$$\Rightarrow x = \frac{169 \pm 65}{338} \Rightarrow x = \frac{9}{13} \text{ or } \frac{4}{13}$$

17. (c) Let root of $x^2 + ax + b = 0$ be (α, β) and root of $x^2 + bx + a = 0$ be (α, γ)

Now, by subtracting, we get

$$(x^2 + ax + b) - (x^2 + bx + a) = 0$$

$$\Rightarrow (a - b)x = (a - b)$$

$$\Rightarrow x = \frac{a-b}{a-b} = 1 \quad \dots(1)$$

Thus from the first equation, we have

$$1 + a + b = 0$$

$$\Rightarrow a + b = -1.$$

18. (c) Given, $x + y = 1$

$$\text{Then, } \left(x + \frac{1}{x}\right)^2 + \left(y + \frac{1}{y}\right)^2 = x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} + 4$$

Minimum value of $x^2 + y^2$ occur when $x = y$

$$[\because x + y = 1]$$

$$\text{Put } x = y = \frac{1}{2}$$

$$\text{Minimum value} = \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2 = \frac{25}{2} = 12.5$$

19. (b) We have $\sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{5}{2}$...(1)

$$\text{and } x + y = 10 \quad \dots(2)$$

$$\text{Now, } \sqrt{\frac{x}{y}} + \sqrt{\frac{y}{x}} = \frac{5}{2} \Rightarrow \frac{x+y}{\sqrt{xy}} = \frac{5}{2}$$

$$\Rightarrow \frac{10}{\sqrt{xy}} = \frac{5}{2}$$

[using eq. (2)]

$$\Rightarrow \sqrt{xy} = 4 \Rightarrow xy = 16$$

Thus, the given system of simultaneous equations reduces to

$$x + y = 10 \text{ and } xy = 16$$

$$\Rightarrow y = 10 - x$$

$$\text{and } xy = 16$$

$$\Rightarrow x(10 - x) = 16$$

$$\Rightarrow x^2 - 10x + 16 = 0$$

$$\Rightarrow (x - 2)(x - 8) = 0 \Rightarrow x = 2 \text{ or } x = 8$$

$$\text{Now, } x = 2 \text{ and } x + y = 10 \Rightarrow y = 8$$

$$\text{and } x = 8 \text{ and } x + y = 10 \Rightarrow y = 2$$

Hence, the required solution are $x = 2, y = 8$

and $x = 8, y = 2$

20. (a) If roots of an equation $ax^2 + bx + c = 0$ are positive, then signs of a and c should be like.

21. (c) Let α and β be the roots of $x^2 - (p - 2)x - (p + 1) = 0$

$$\text{Then, } \alpha + \beta = p - 2$$

$$\text{and } \alpha\beta = -(p + 1)$$

$$\therefore \alpha^2 + \beta^2 = 5$$

$$\Rightarrow (\alpha + \beta)^2 - 2\alpha\beta = 5$$

$$\Rightarrow (p - 2)^2 + 2(p + 1) = 5$$

$$\Rightarrow p^2 - 4p + 4 + 2p + 2 = 5$$

$$\Rightarrow p^2 - 2p + 1 = 0 \Rightarrow (p - 1)^2 = 0$$

$$\Rightarrow p = 1$$

22. (c) $\therefore \alpha$ and β are the roots of the equation $x^2 + 6x + 1 = 0$

$$\therefore \alpha + \beta = -6 \text{ and } \alpha\beta = 1$$

$$\text{Now, } (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= (-6)^2 - 4$$

$$= 36 - 4 = 32$$

$$\Rightarrow |\alpha - \beta| = \sqrt{32} = 4\sqrt{2}$$

23. (a) The given root is

$$= \frac{1}{2 - \sqrt{-2}} \cdot \frac{2 + \sqrt{2}i}{6}$$

$$\therefore \text{Another root} = \frac{2 - \sqrt{2}i}{6}$$

Now, find sum and product of the roots and put in

$$x^2 - (\text{sum of the roots})x + (\text{multiplication of the roots}) = 0$$

$$24. (b) S = \frac{1}{\alpha + \beta} + \frac{\alpha + \beta}{\alpha\beta} = -\frac{a}{b} - \frac{b}{c} = -\frac{(ac + b^2)}{bc}$$

$$P = \frac{1}{\alpha + \beta} \cdot \frac{\alpha + \beta}{\alpha\beta}$$

$$= \frac{1}{\alpha\beta} \cdot \frac{a}{c}$$

Put the values of P and S in $x^2 - Sx + P = 0$, we get the required result.

25. (a) Dividing the equation $a^3x^2 + abcx + c^3 = 0$ by c^2 , we get

$$a\left(\frac{ax}{c}\right)^2 + b\left(\frac{ax}{c}\right) + c = 0$$

$$\Rightarrow \frac{ax}{c} = \alpha, \beta$$

$$\Rightarrow x = \frac{c}{a}\alpha, \frac{c}{a}\beta$$

$$\Rightarrow x = \alpha^2\beta, \alpha\beta^2$$

$$[\because \frac{c}{a} \alpha\beta = \text{product of roots}]$$

Hence, $\alpha^2\beta$ and $\alpha\beta^2$ are the roots of the equation $a^3x^2 + abcx + c^3 = 0$.

26. (b) Let the natural number be x .

$$\text{By the given condition: } x + 12 = \frac{160}{x} (x \neq 0)$$

$$\Rightarrow x^2 + 12x - 160 = 0 \Rightarrow x = \frac{-12 \pm \sqrt{144 + 640}}{2}$$

$$= \frac{-12 \pm \sqrt{784}}{2} = \frac{-12 \pm 28}{2} = \frac{40}{2} \text{ or } \frac{16}{2}$$

$$= -10 \text{ or } 5. \text{ But } x \text{ is a natural number } \therefore x = 5.$$

$$27. (d) \frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x} \Rightarrow \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow \frac{x-a-b-x}{(a+b+x)x} = \frac{b+a}{ab}$$

$$\Rightarrow \frac{-(a+b)}{(x^2+ax+bx)} = \frac{(a+b)}{ab}$$

$$\Rightarrow \frac{-1}{x^2+ax+bx} = \frac{1}{ab}$$

$$\Rightarrow x^2+ax+bx = -ab \Rightarrow x(x+a) + b(x+a) = 0$$

$$\Rightarrow (x+a)(x+b) = 0 \Rightarrow x = -a \text{ or } x = -b$$

28. (d) (a) is clearly true.

(b) $x^2 + 1$ is a quadratic polynomial which has no real value of x for which $x^2 + 1$ is zero.

$[\because x^2 \geq 0 \Rightarrow x^2 + 1 > 0 \text{ for all real } x] \therefore$ (b) is true.

(c) The quadratic polynomial

$$x^2 - 2x + 1 = (x-1)^2 \text{ has only one zero i.e. } 1$$

$\therefore$ (c) is true.

$[\because (x-1)^2 > 0 \text{ at } x \neq 1 \text{ and for } x = 1, (x-1)^2 = 0]$

(d) is false

$[\because \text{of (b), (c)}]$

Hence (d) holds.

29. (c) For a, b negative the given expression will always be positive since, a^2, b^2 and ab are all positive.

30. (c) $(c+6) = 1/2 \times 2(2c-1) \Rightarrow c+6 = 2c-1 \Rightarrow c = 7$

31. (c) Assume any equation:

$$\text{Say } x^2 - 5x + 6 = 0$$

The roots are 2, 3.

We are now looking for the equation, whose roots are:

$$(2 + 1/3) = 2.33 \text{ and } (3 + 1/2) = 3.5.$$

Also $a = 1, b = -5$ and $c = 6$.

Put these values in each option to see which gives 2.33 and 3.5 as its roots.

32. (c) $f(x) = x^2 + ax + b$

$$f(1) = f(-1) = 5$$

$$\Rightarrow a + b = -a + b = 5$$

$$\Rightarrow a = 0, b = 5$$

33. (a) For both the roots: (α, β) to be positive

$$\alpha + \beta > 0 \text{ and } \alpha\beta > 0$$

$$\Rightarrow \frac{-b}{a} > 0 \text{ and } \frac{c}{a} > 0$$

i.e., b and a are of opposite sign and c and a are of same sign.

34. (b) Given quadratic equation is $x^2 - bx + 1 = 0$

It has no real roots. It means, equation has imaginary roots.

Which is possible when $B^2 - 4AC < 0$

Here, $B = -b, A = 1, C = 1$

$$\Rightarrow b^2 - 4 < 0 \Rightarrow b^2 < 4 \Rightarrow -2 < b < 2$$

35. (c) Given equation is

$$\sin \theta = x + \frac{a}{x}, x \in \mathbb{R} - \{0\}$$

$$\Rightarrow x^2 + a = x \sin \theta$$

$$\Rightarrow x^2 - x \sin \theta + a = 0$$

$$\text{Now, discriminant} = \sqrt{\sin^2 \theta - 4a}$$

For x to be real root,
discriminant ≥ 0

$$\Rightarrow \sqrt{\sin^2 \theta - 4a} \geq 0$$

$$\Rightarrow \sin^2 \theta - 4a \geq 0 \Rightarrow \sin^2 \theta \geq 4a$$

$$\Rightarrow \frac{1}{\sin^2 \theta} \leq \frac{1}{4a} \Rightarrow a \leq \frac{\sin^2 \theta}{4}$$

$$\Rightarrow a \leq \frac{1}{4} (\because \sin^2 \theta \text{ lies between } 0 \text{ and } 1)$$

36. (b) The given equation is,

$$3x^2 - 5x + p = 0$$

We have, $a = 3, b = -5, c = p$

$$D = b^2 - 4ac = 25 - 12p$$

For Real and unequal, $D > 0$

$$\therefore 25 - 12p > 0$$

$$\Rightarrow 25 > 12p \Rightarrow p < \frac{25}{12}$$

37. (b) Let roots are $(n-1), n$ and $(n+1)$

Sum of the roots = b

$$(n-1)n + n(n+1) + (n+1)(n-1) = b$$

$$\Rightarrow n^2 - n + n^2 + n + n^2 - 1 = b$$

$$\Rightarrow 3n^2 - 1 = b$$

The value of b will be minimum when the value of n^2 is minimum i.e., $n^2 = 0$

Hence, minimum value of $b = -1$.

Expert Level

1. (d) $x = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1}} = \sqrt{\frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}}$

$$= \sqrt{\frac{(\sqrt{5}+1)^2}{5-1}} = \frac{\sqrt{5}+1}{2}$$

$$\therefore x^2 + x - 1$$

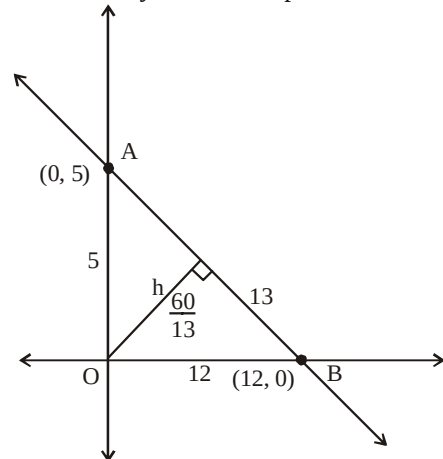
$$= \left(\frac{\sqrt{5}+1}{2}\right)^2 + \frac{\sqrt{5}+1}{2} - 1 = \frac{5+1+2\sqrt{5}}{4} + \frac{\sqrt{5}+1}{2} - 1$$

$$= \frac{3+\sqrt{5}}{2} + \frac{\sqrt{5}+1}{2} - 1 = \frac{3+\sqrt{5}+\sqrt{5}+1-2}{2}$$

$$= \sqrt{5} + 1$$

2. (a) $\sqrt{x^2 + y^2}$ is the radius of a circle with x and y
such that $5x + 12y = 60$

The line $5x + 12y = 60$ can be plotted as follows:



All the points on the line AB satisfy $5x + 12y = 60$

$\sqrt{x^2 + y^2}$ will be minimum when the distance between O and line AB is minimum.

$$A(\triangle ABC) = \frac{1}{2} \times OB \times AO = \frac{1}{2} \times AB \times h$$

$$\therefore \frac{1}{2} \times 12 \times 5 = \frac{1}{2} \times 13 \times h$$

$$\therefore h = \frac{60}{13}$$

3. (a) Given $x^2 + px + 12 = 0$

Since, $x = 4$ is the one root of the equation, therefore $x = 4$ will satisfy this equation

$$\therefore 16 + 4p + 12 = 0 \Rightarrow p = -7$$

Other quadratic equation becomes $x^2 - 7x + q = 0$

(By putting value of p)

Its roots are equal, so, $b^2 = 4ac$

$$\Rightarrow 49 = 4q \text{ or } q = \frac{49}{4}$$

4. (d) Let $f(x) = x^3 - ax^2 + bx - a = 0$

In the given equation, there are 3 sign changes, therefore, there are at most 3 positive real roots.

In $f(-x)$, there is no sign change.

Thus, there is no negative real root. i.e. if α, β and γ are the roots then they are all positive and we have

$$f(x) = (x - \alpha)(x - \beta)(x - \gamma) = 0$$

$$x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \gamma\alpha)x - \alpha\beta\gamma$$

$$\Rightarrow b = \alpha\beta + \beta\gamma + \gamma\alpha \text{ and } \Rightarrow a = \alpha + \beta + \gamma = \alpha\beta\gamma$$

$$\Rightarrow \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = 1$$

$$\Rightarrow \frac{1}{\alpha\beta} + \frac{1}{\alpha\gamma} + \frac{1}{\beta\gamma} = 1$$

$$\Rightarrow \alpha\beta, \alpha\gamma, \beta\gamma > 1$$

$$\Rightarrow b = \alpha\beta + \beta\gamma + \gamma\alpha > 1+1+1 = 3.$$

Thus, $b \neq 1$.

5. (a) $\therefore$ The roots of the equations $x^2 + kx + 1 = 0$ are α and β .

$$\therefore \alpha + \beta = -k \text{ and } \alpha\beta = 1$$

$$\text{Now, } (a+b)(a-1+b-1) = (\alpha+\beta)\left(\frac{1}{\alpha} + \frac{1}{\beta}\right)$$

$$= (\alpha+\beta)\left(\frac{\alpha+\beta}{\alpha\beta}\right) = \frac{(\alpha+\beta)^2}{\alpha\beta} = \frac{(-k)^2}{1} = k^2$$

6. (a) Since each of the roots are changing symmetrically from $\alpha \rightarrow 3\alpha + 2$

Thus, to get the required equation, just replace x with

$$\frac{x-2}{3} \text{ in the given equation.}$$

Thus, we get

$$9\left(\frac{x-2}{3}\right)^3 - 7\left(\frac{x-2}{3}\right) + 6 = 0$$

$$\Rightarrow x^3 - 6x^2 + 5x + 24 = 0$$

7. (a) For the equations to have same pair of roots

$$\frac{2p-1}{q+1} = \frac{2p+1}{4q+1} = \frac{c}{3c}$$

$$\therefore 3(2p-1) = q+1 \Rightarrow 6p-q=4$$

$$\text{and, } 3(2p+1) = 4q+1 \Rightarrow 6p-4q=-1$$

Solving two equation $q = 2$ and $p = 1$

$$\therefore (p+q) = 3$$

8. (b) Let x_1, x_2 be the roots of the equation $x^2 + px + q = 0$ and x_3, x_4 be the roots of the equation $x^2 + qx + p = 0$.

$$\text{Hence, } x_1 + x_2 = -p, x_1 \times x_2 = q, x_3 + x_4 = -q,$$

$$x_3 \times x_4 = p \quad \dots(1)$$

According to the question, $x_1 - x_2 = x_3 - x_4$

$$\text{or, } (x_1 - x_2)^2 = (x_3 - x_4)^2 \text{ or } (x_1 + x_2)^2 - 4x_1 \times x_2$$

$$= (x_3 + x_4)^2 - 4x_3 \times x_4 \quad \dots(2)$$

Putting the values from (1) in (2), we obtain $(p-q)$

$$(p+q+4) = 0$$

Hence, either $p = q$ (not possible otherwise both the equations will become same) or $p + q = -4$.

9. (c) $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$

$$3x^2 - 2x(a+b+c) + (ab+bc+ca) = 0$$

$$D = b^2 - 4ac$$

$$= [2(a+b+c)]^2 - 4(3)(ab+bc+ca)$$

$$= 4(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= 2(2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca)$$

$$= 2[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

$$= 0 \text{ (since } a = b = c)$$

So, the roots are real and equal.

10. (b) Assume the equation to be $(x-1)(x-2) = 0$ which gives $a_1 = 1, a_2 = -3$ and $a_3 = 2$ and $r_1 = 1, r_2 = 2$. With his information check the options.

11. (d) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} < 2 \rightarrow \frac{\alpha^2 + \beta^2}{\alpha\beta} < 2$

$$\rightarrow \frac{(\alpha+\beta)^2 - 2\alpha^2\beta^2}{\alpha\beta} < 2$$

Use the formulae for sum of the roots and product of the roots.

12. (d) For each of the given options it can be seen that the roots do not lie in the given interval. Thus, option (d) is correct.

13. (b) We have to minimize : $R_1^2 + R_2^2$ or $(R_1 + R_2)^2 - 2R_1R_2$
 $\Rightarrow (p-2)^2 - 2 \times (-p+1)/2 = p^2 - 4p + 4 + p + 1$
 $= p^2 - 3p + 5.$

14. (b) $p(p-1)/3 < 0$ (Product of roots should be negative).
 $\Rightarrow p(p-1) < 0$
 $p^2 - p < 0.$

This happens for $0 < p < 1$.

15. (b) HCF of $\alpha, \beta = 2$, so we can assume $\alpha = 2x$ and $\beta = 2y$.
 So, $2x \times 2y = 24 \times 2$, or $xy = 12$
 Different values of x and y possible are (12, 1), (4, 3) and hence different values of roots of the equation will be (24, 2), (8, 6).

16. (d) $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$

$$\text{For same pair of roots } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = \frac{1}{2} \text{ (given)}$$

$$\text{Expression (2)} = a_2x^2 + b_2x + c_2 = 2(a_1x^2 + b_1x + c_1) = 2 \text{ (expression (1))}$$

Then the ratio of the maximum value is 1 : 2

17. (a) Let the equation be $x^2 - x - 12 = 0$ ($x = -3, 4$)
 Here $a = 1, b = -1, c = -12$

Explanation of Test Yourself

1. (a) We have $\alpha^2 = 5\alpha - 3$ and $\beta^2 = 5\beta - 3$;
 $\Rightarrow \alpha$ & β are roots of equation, $x^2 = 5x - 3$
or $x^2 - 5x + 3 = 0$
 $\therefore \alpha + \beta = 5$ and $\alpha\beta = 3$
Thus, the equation having $\frac{\alpha}{\beta}$ & $\frac{\beta}{\alpha}$ as its roots is

$$x^2 - x\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right) + \frac{\alpha\beta}{\alpha\beta} = 0$$

$$\Rightarrow x^2 - x\left(\frac{\alpha^2 + \beta^2}{\alpha\beta}\right) + 1 = 0 \text{ or } 3x^2 - 19x + 3 = 0$$
2. (d) The equation is:
 $x^2 + 5x + 6 = 0$ and $x^2 - 5x + 6 = 0$
– Sum of roots = $-5 + 5 = 0$.
3. (a) We get $c = 4$ (by putting $x = 0$)
Then, at $x = -1$, $a - b + 4 = 4$, So $a - b = 0$.
At $x = -2$, $4a - 2b + 4 = 6 \Rightarrow 4a - 2b = 2 \Rightarrow 2a = 2$
 $\Rightarrow a = 1$, Thus, option (a) is correct.
4. (a) Use the value of $x = 1$ in each of the two quadratic equations to get the value of a and b respectively. With these values check the options for their validity.
5. (d) Let the roots of equation $x^2 - 6x + a = 0$ be α and 4β and that of the equation $x^2 - cx + 6 = 0$ be α and 3β . Then
 $\alpha + 4\beta = 6$; $4\alpha\beta = a$
and $\alpha + 3\beta = c$; $3\alpha\beta = 6$
 $\Rightarrow a = 8$
 $\therefore$ The equation becomes $x^2 - 6x + 8 = 0$
 $\Rightarrow (x-2)(x-4) = 0 \Rightarrow$ roots are 2 and 4
 $\Rightarrow \alpha = 2, \beta = 1 \therefore$ Common root is 2.
6. (b) Solving equation, we get
 $3x^2 - 2(a+b+c)x + ab + bc + ca = 0$
For roots to be equal, $a = b = c$.
7. (d) Let the roots be α and β .
 $\therefore \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$
 $\Rightarrow \frac{-b}{a} \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2} \Rightarrow \frac{-b}{a} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2\beta^2}$
 $\Rightarrow \frac{-b}{a} = \frac{(b/a)^2 - \frac{2c}{a}}{c^2/a^2}$
 $\Rightarrow \frac{-b}{a} \frac{b^2 - 2ca}{c^2} \Rightarrow b^2a + bc^2 = 2ca^2$
 $\Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$
8. (a) Let the roots of the given equation be α, β
Now for roots $(\alpha - \beta), (\beta - 2)$ the equation can be deduced by replacing x with $(x + 2)$
 $\therefore$ The deduced equation would be
 $\Rightarrow (x+2)^2 - (p+1)(x+2) + p^2 + p - 8 = 0$
 $\Rightarrow x^2 + (3-p)x + p^2 - p - 6 = 0$
 $\Rightarrow x^2 + (p-3)x + (p+2)(p-3) = 0$
Now, $\alpha > 2$ and $\beta < 2$
 $\therefore (\alpha - 2) > 0$ and $(\beta - 2) < 0$
 $\therefore (\alpha - 2)(\beta - 2) < 0 \Rightarrow (p+2)(p-3) < 0$
 $\therefore (\alpha - 2)(\beta - 2) < 0 \Rightarrow (p+2)(p-3) < 0$
 $\therefore -2 < p < 3$.
9. (c) Since $b, c > 0$
Therefore $\alpha + \beta = -b < 0$ and $\alpha\beta = -c < 0$
Since product of the roots is $-ve$ therefore roots must be of opposite sign.
10. (c) $x(1+x^2) + x^2(6+x) + 2 = 0$
 $\Rightarrow 2x^3 + 6x^2 + x + 2 = 0$
For roots : α, β, γ

$$\alpha + \beta + \gamma = -\left(\frac{6}{2}\right) = -3$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{1}{2}$$

$$\alpha\beta\gamma = -\frac{2}{2} = -1$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{1/2}{-1} = -\frac{1}{2}$$
11. (b) Given equation : $x^2 - px + q = 0$
The roots of given equation differ by unity
i.e. let a and $(a + 1)$ be the roots
 $\therefore$ Sum of the roots = $a + a + 1 = -\frac{-p}{1} = p$
 $\Rightarrow 2a + 1 = p \Rightarrow a = \frac{p-1}{2}$
And product of the roots = $a(a + 1) = q$
 $\Rightarrow a^2 + a = q$
Put the value of a in $a^2 + a = q$.
 $\Rightarrow \left(\frac{p-1}{2}\right)^2 + \frac{p-1}{2} = q$
 $\Rightarrow \frac{p^2 + 1 - 2p}{4} + \frac{p-1}{2} = q$
 $\Rightarrow p^2 + 1 - 2p + 2p - 2 = 4q$
 $\Rightarrow p^2 - 1 = 4q \Rightarrow p^2 = 4q + 1$.

12. (a) Given equation is $x^4 - 2x^3 + x - 380 = 0$

$$\Rightarrow (x^2 - x - 20)(x^2 - x + 19) = 0$$

$$\Rightarrow (x - 5)(x + 4)(x^2 - x + 19) = 0$$

Hence, the required roots of the equation are

$$5, -4, \frac{1 \pm 5\sqrt{-3}}{2}.$$

13. (b) Consider both equations

$$px^2 + 2qx + r = 0 \quad \dots(1)$$

$$\text{and } qx^2 - 2\sqrt{pr} \cdot x + q = 0 \quad \dots(2)$$

Since, both the equations are quadratic and have real roots, therefore from equation (1), we have

$$\therefore 4q^2 - 4pr \geq 0 \quad (\text{using discriminant})$$

$$\Rightarrow q^2 \geq pr \quad \dots(3)$$

and from second equation $4pr - 4q^2 \geq 0$

$$\Rightarrow pr \geq q^2 \quad \dots(4)$$

From eqs. (3) and (4) we get $q^2 = pr$.

14. (a) Coefficient of $x^2 = 2$

In first case the product of the root $= 8 \times 7 = 56$ which is the constant term. So $c = 56$

In the second case the sum of the roots $= -b = 8 - 3 = 5$

$$\Rightarrow b = -5$$

$\therefore$ The correct quadratic equation is $x^2 - 5x + 56 = 0$

15. (c) $|\alpha^2 - \beta^2| = |(\alpha + \beta)| |\alpha - \beta|$

$$\Rightarrow \frac{7}{4} = |\alpha - \beta| \cdot \frac{7}{2}$$

$$\Rightarrow |\alpha - \beta|^2 = \frac{1}{4} \Rightarrow |(\alpha + \beta)^2 - 4\alpha\beta| = \frac{1}{4}$$

$$\Rightarrow \left| \frac{49}{4} - \frac{4c}{2} \right| = \frac{1}{4}$$

$$\Rightarrow 49 - 8c = 1 \Rightarrow c = \frac{48}{8} = 6$$

15

INEQUALITIES

● Introduction ● Inequality ● Types of Inequalities ● Some Properties of Inequality ● Important Results ● Solution of an Inequality ● Equivalent Inequalities	● Notation and Ranges ● Solutions of Linear Inequalities in one Unknown ● Solutions of Quadratic Inequalities ● Solution of System of Inequalities in one Variable ● Inequalities Containing a Modulus
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INTRODUCTION

Inequality, in mathematics, states that a mathematical expression is less than or greater than some other expression. An inequality is not as specific as an equation but it does contain information about the expression involved.

Inequality is considered to be one of the most important topics from the point of view of CAT and other equivalent aptitude tests. This fact seems to be true simply because inequalities can be clubbed with many other Algebra topics and hence finds favour with the CAT examination Paper Setters.

INEQUALITY

Two real numbers, two algebraic expressions or an algebraic expression and a real number related by the symbol $>$, $<$, $\geq$ or $\leq$ form an inequality.

' $>$ ' means 'greater than'. Hence $a > b$ read as a is greater than b .

' $<$ ' means 'less than'. Hence $a < b$ read as a is less than b .

' $\geq$ ' means 'greater than or equal to'. Hence $a \geq b$ is read as a is greater than or equal to b .

' $\leq$ ' means 'less than or equal to'. Hence $a \leq b$ is read as a is less than or equal to b .

TYPES OF INEQUALITIES

1. Numerical Inequalities

Inequalities which does not contain any variable are called numerical, inequalities.

2. Literal Inequalities

Inequalities which does not contain any variable are called literal inequalities. For examples, $8 > 6$, $-7 < 0$, etc.

3. An inequality may contain more than one variable. For examples $2xy < 8$, $x + 3y \geq 20$, etc.

An inequality in one variable may be linear, quadratic or cubic etc. For examples $2x + 5 < 10$, $x^2 + 4x + 3 \geq 0$, $-x^3 + 2x^2 - 4 \leq 8$, etc.

4. Strict Inequalities

Inequalities involving the symbol ' $>$ ' or ' $<$ ' are called strict inequalities.

5. Slack Inequalities

Inequalities involving the symbol ' $\geq$ ' or ' $\leq$ ' are called slack inequalities.

$$a \geq b \text{ means } a > b \text{ or } a = b$$

$$a \leq b \text{ means } a < b \text{ or } a = b$$

Note that simultaneous relation between any three different quantities a , b and c will be either $a < b < c$, $a < b \leq c$, $a \leq b < c$ or $a \leq b \leq c$

SOME PROPERTIES OF INEQUALITY

- If $a > b$, then evidently $b < a$ i.e. if the sides of an equality be transposed, the sign of equality must be reversed.
- Sign of inequality does not change when equal numbers added to (or subtracted from) both sides of an inequality.

- i.e. $a > b$
 $\Rightarrow a + 5 > b + 5$
 and also $a - 4 > b - 4$
- (iii) Sign of inequality does not change when both sides of an inequality can be multiplied (or divided) by the same positive number. But when both sides are multiplied or divided by a negative number, then the sign of inequality is reversed.
- i.e. $a \leq b$
 $\Rightarrow 3a \leq 3b$
 and also $\frac{a}{5} \geq \frac{b}{5}$
 But $-3a \geq -3b$
 and also $\frac{a}{-5} \leq \frac{b}{-5}$
- (iv) If $a > b$ and $b > c$, then $a > c$. Since $5 > 4$ and $4 > 2$, therefore $5 > 2$.
- (v) If $a > b > 0$ then $\frac{1}{a} < \frac{1}{b}$
 Since $6 > 2 > 0$, therefore $\frac{1}{6} < \frac{1}{2}$.
- (vi) If $a > b > 0$ and $n > 0$ then $a^n > b^n$ and $(a)^{1/n} > (b)^{1/n}$
 Since $3 > 2 > 0$ and $4 > 0$, therefore $(3)^4 > (2)^4$ and also $(3)^{1/4} > (2)^{1/4}$
- (vii) If $x > y > 0$ and $a > 1$, then $a^x > a^y$
 Since $5 > 3 > 0$ and $6 > 1$, therefore $(6)^5 > (6)^3$
- (viii) If $x > y > 0$ and $0 < a < 1$ then $a^x < a^y$
 Since $6 > 4 > 0$ and $0 < \frac{2}{3} < 1$, therefore $\left(\frac{2}{3}\right)^6 < \left(\frac{2}{3}\right)^4$.

IMPORTANT RESULTS

- (i) Square of any real number is always equal or greater than 0. i.e. if a is a real number, then $a^2 \geq 0$.
- (ii) For any real number a ,
 $|a| \geq 0$
- (iii) If a is a positive real number and $|x| \leq a$, then
 $-a \leq x \leq a$
- (iv) If a is a positive real number and $|x| \geq a$, then
 $x \leq -a$ or $x \geq a$
- (v) $|a + b| \leq |a| + |b|$
 In general
 $|a_1 + a_2 + a_3 + \dots + a_n| \leq |a_1| + |a_2| + |a_3| + \dots + |a_n|$
- (vi) $|a - b| \geq |a| - |b|$
- (vii) $a^2 + b^2 \geq 2ab$
- (viii) $ax^2 + bx + c \geq 0$, if $a > 0$ and $b^2 - 4ac \leq 0$
 $ax^2 + bx + c = 0$, if $b^2 - 4ac = 0$ and $x = -\frac{b}{2a}$
- (ix) $a^2 + b^2 + c^2 \geq ab + bc + ca$
- (x) $(a + b)(b + c)(c + a) \geq 8abc$, if $a, b, c \geq 0$
- (xi) $a^3 + b^3 \geq ab(a + b)$

(xii) $\frac{a}{b} + \frac{b}{a} \geq 2$ if a and b are of same sign.

- (xiii) Arithmetic mean of any number of positive quantities is more than their geometric mean i.e.
 Arithmetic mean $\geq$ Geometric mean

$$\therefore \frac{a+b}{2} \geq \sqrt{ab}$$

Also $\frac{a_1 + a_2 + a_3 + \dots + a_n}{n} > (a_1, a_2, a_3 \dots a_n)^{1/n}$

or $\left(\frac{a_1 + a_2 + a_3 + \dots + a_n}{n}\right)^n > a_1 \cdot a_2 \cdot a_3 \dots a_n$

- (xiv) If the sum of two positive quantities is given (i.e. constant), then their product is greatest when they are equal but if the product of two positive quantities is given, their sum is least when they are equal.

For example if sum of two numbers is 100, then 50×50 is greatest out of 50×50 , 51×49 , 52×48 , ..., 98×2 and 99×1 .

Also you can see that

$$50 \times 50 > 51 \times 49 > 52 \times 48 > \dots > 98 \times 2 > 99 \times 1$$

Thus if the sum of two numbers is given, then their product is maximum when they are equal and their product decreases as the difference between the two numbers increases.

SOLUTION OF AN INEQUALITY

The solution of an inequality is the all values of the variable present in the given inequality for which the given inequality is true. An inequality has no solution if there is no such value of the variable present in the inequality for which the inequality is true.

EQUIVALENT INEQUALITIES

Two inequalities are said to be equivalent if any solution of one is also a solution of the other and vice-versa. If both inequalities have no solution, then they are also regarded to be equivalent.

In other words two inequalities are said to be equivalent if the correctness of one of them implies the correctness of the other and vice-versa.

For example, if $x > a$ then $x + 3 > a + 3$, because sign of inequality does not change when equal number added to both sides of inequality.

Since $x > a \Rightarrow x + 3 > a + 3$, therefore $x > a$ and $x + 3 > a + 3$ are equivalent inequalities.

NOTATION AND RANGES

If a, b, c, d are four numbers such that $a < b < c < d$, then

- (i) $x \in (a, b)$ means $a < x < b$
 (ii) $x \in [a, b]$ means $a \leq x \leq b$
 (iii) $x \in [a, b)$ means $a \leq x < b$
 (iv) $x \in (a, b]$ means $a < x \leq b$
 (v) $x \in (a, b) \cup (c, d)$ means $a < x < b$ or $c < x < d$

SOLUTIONS OF LINEAR INEQUALITIES IN ONE UNKNOWN

Inequalities of the form $ax + b > 0$, $ax + b \geq 0$, $ax + b < 0$ and $ax + b \leq 0$ are called linear inequalities.

(i) $ax + b > 0$

$\Rightarrow x > -\frac{b}{a}$, if $a > 0$

and $x < -\frac{b}{a}$, if $a < 0$

(ii) $ax + b \geq 0$

$\Rightarrow x \geq -\frac{b}{a}$, if $a > 0$

and $x \leq -\frac{b}{a}$, if $a < 0$

(iii) $ax + b < 0$

$\Rightarrow x < -\frac{b}{a}$, if $a > 0$

and $x > -\frac{b}{a}$, if $a < 0$

(iv) $ax + b \leq 0$

$\Rightarrow x \leq -\frac{b}{a}$, if $a > 0$

and $x \geq -\frac{b}{a}$, if $a < 0$

Illustration 1: Solve $2(x - 3) + 4 \geq 4 - x$

Solution: $2(x - 3) + 4 \geq 4 - x$
 $\Rightarrow 2x - 6 + 4 \geq 4 - x \Rightarrow 2x + x - 2 \geq 4$
 $\Rightarrow 3x \geq 4 + 2 \Rightarrow 3x \geq 6$
 $\Rightarrow x \geq \frac{6}{3} \Rightarrow x \geq 2$

This solution can also be written as $x \in [2, \infty)$.

Illustration 2: Solve $3(x + 4) + 1 < 2(3x + 1) + 15$

Solution: $3(x + 4) + 1 < 2(3x + 1) + 15$
 $\Rightarrow 3x + 12 + 1 < 6x + 2 + 15$
 $\Rightarrow 3x - 6x < 17 - 13$
 $\Rightarrow -3x < 4$
 $\Rightarrow x > -\frac{4}{3} \Rightarrow x > -\frac{4}{3}$

This solution can also be written as $x \in \left(-\frac{4}{3}, \infty\right)$.

Illustration 3: Solve the following inequations:

$\frac{2x+4}{x-1} \geq 5$

Solution: We have,

$\frac{2x+4}{x-1} \geq 5$

$\Rightarrow \frac{2x+4}{x-1} - 5 \geq 0$

$\Rightarrow \frac{x-3}{x-1} \leq 0$ [Dividing both sides by 3]

$\Rightarrow 1 < x \leq 3$

$\Rightarrow x \in (1, 3]$

Hence, the solution set of the given inequation is $(1, 3]$.

Illustration 4: Solve: $-5 \leq \frac{2-3x}{4} \leq 9$.

Solution: We have,

$-5 \leq \frac{2-3x}{4} \leq 9$

$\Rightarrow \frac{22}{3} \geq x \geq \frac{-34}{3}$

$\Rightarrow \frac{-34}{3} \leq x \leq \frac{22}{3}$

$\Rightarrow x \in [-34/3, 22/3]$

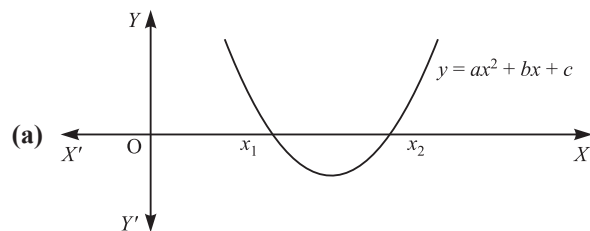
Hence, the interval $[-34/3, 22/3]$ is the solution set of the given system of inequations.

SOLUTIONS OF QUADRATIC INEQUALITIES

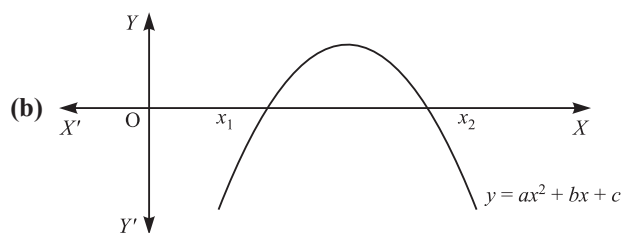
Inequalities of the form $ax^2 + bx + c > 0$, $ax^2 + bx + c \geq 0$, $ax^2 + bx + c < 0$ and $ax^2 + bx + c \leq 0$, where a, b, c are real numbers but $a \neq 0$, are called quadratic inequalities.

Case-(i): If D (i.e. $b^2 - 4ac$) > 0

and x_1, x_2 are the roots of the quadratic equation $ax^2 + bx + c = 0$, such that $x_1 < x_2$, then

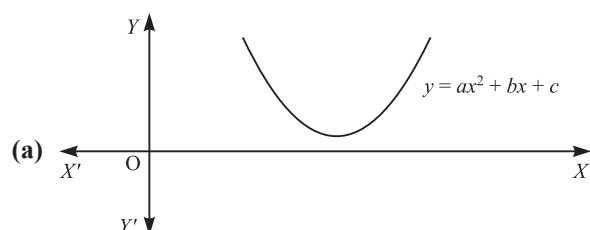


- For $a > 0$; solutions of $ax^2 + bx + c > 0$ are $x < x_1$ or $x > x_2$ i.e. the interval $(-\infty, x_1) \cup (x_2, \infty)$.
- For $a > 0$; solutions of $ax^2 + bx + c < 0$ are $x_1 < x < x_2$ i.e. (x_1, x_2) .

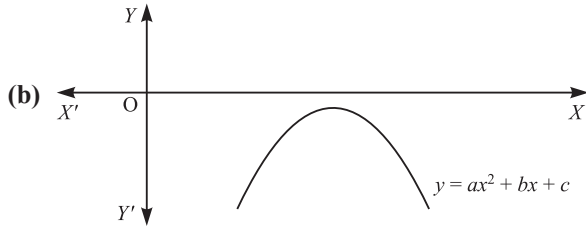


- For $a < 0$; solutions of $ax^2 + bx + c < 0$ are $x < x_1$ or $x > x_2$ i.e. the interval $(-\infty, x_1) \cup (x_2, \infty)$.
- For $a < 0$; solutions of $ax^2 + bx + c > 0$ are $x_1 < x < x_2$.

Case-(ii): If D (i.e. $b^2 - 4ac$) < 0

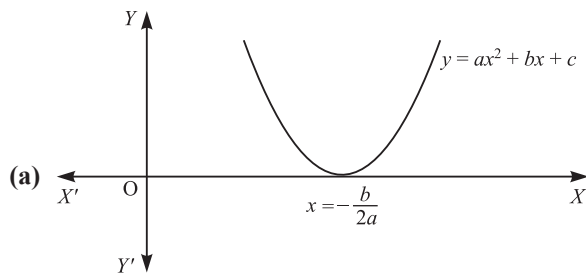


- For $a > 0$; solutions of $ax^2 + bx + c > 0$ or $ax^2 + bx + c \geq 0$ are all numbers i.e. $-\infty < x < \infty$ or the interval $(-\infty, \infty)$.
- For $a > 0$; $ax^2 + bx + c < 0$ or $ax^2 + bx + c \leq 0$ has no solution.

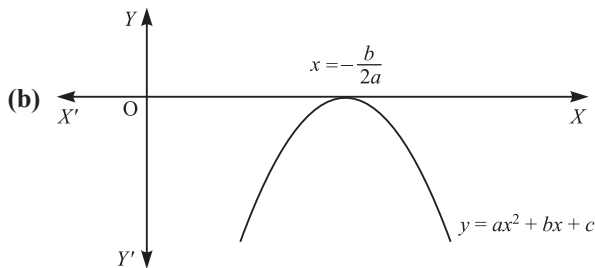


- For $a < 0$; solution of $ax^2 + bx + c < 0$ or $ax^2 + bx + c \leq 0$ are all numbers i.e. $-\infty < x < \infty$ or the interval $(-\infty, \infty)$.
- For $a < 0$; $ax^2 + bx + c > 0$ or $ax^2 + bx + c \geq 0$ has no solution.

Case-(iii): If D (i.e. $b^2 - 4ac$) = 0



- For $a > 0$; solution of $ax^2 + bx + c > 0$ are all real numbers except $-\frac{b}{2a}$.
- For $a > 0$; solution of $ax^2 + bx + c \geq 0$ are all real numbers.
- For $a > 0$; $ax^2 + bx + c < 0$ has no solution.
- For $a > 0$; $ax^2 + bx + c \leq 0$ has unique solution $x = -\frac{b}{2a}$.



- For $a < 0$; solution of $ax^2 + bx + c < 0$ are all real numbers except $-\frac{b}{2a}$.
- For $a < 0$; solution of $ax^2 + bx + c \leq 0$ are all real numbers.
- For $a < 0$; $ax^2 + bx + c > 0$ has no solution.
- For $a < 0$; $ax^2 + bx + c \geq 0$ has an unique solution $x = -\frac{b}{2a}$.

Illustration 5: Solve $2x^2 + 5x - 3 > 0$

Solution: $D = 25 + 4 \times 2 \times 3 > 0$, $a = 2 > 0$

Also $2x^2 + 5x - 3 = 0$
 $\Rightarrow 2x^2 + 6x - x - 3 = 0$

$$\Rightarrow 2x(x+3) - 1(x+3) = 0$$

$$\Rightarrow (x+3)(2x-1) = 0$$

Hence roots of $2x^2 + 5x - 3 = 0$ are -3 and $\frac{1}{2}$.

Hence solutions of $2x^2 + 5x - 3 > 0$ are

$$x < -3 \text{ or } x > \frac{1}{2}$$

$$\text{i.e. } x \in (-\infty, -3) \cup \left(\frac{1}{2}, \infty\right).$$

Illustration 6: Solve $-3x^2 + 2x + 8 \leq 0$

Solution: $D = 4 + 4 \times 3 \times 8 > 0$, $a = -3 < 0$

Also $-3x^2 + 2x + 8 = 0$

$$\Rightarrow -3x^2 + 6x - 4x + 8 = 0$$

$$\Rightarrow -3x(x-2) - 4(x-2) = 0$$

$$\Rightarrow (x-2)(-3x-4) = 0$$

$$\Rightarrow (x-2)(3x+4) = 0$$

Hence root of $-3x^2 + 2x + 8 = 0$ are $x = 2$, $-\frac{4}{3}$.

Hence solutions of $-3x^2 + 2x + 8 < 0$ are

$$x < -\frac{4}{3} \text{ and } x > 2$$

Now solution of $-3x^2 + 2x + 8 \leq 0$ means all the solutions of $-3x^2 + 2x + 8 < 0$ and $-3x^2 + 2x + 8 = 0$ separately.

Hence solution of $-3x^2 + 2x + 8 \leq 0$ are

$$x \leq -\frac{4}{3} \text{ and } x \geq 2 \text{ i.e. } x \in \left(-\infty, -\frac{4}{3}\right] \cup [2, \infty)$$

Illustration 7: Solve $-5x^2 + 3x - 2 > 0$

Solution: $D = 9 - 4 \times (-5) \times (-2) = 9 - 40 = -31 < 0$
 $a = -5 < 0$

Hence no solution of $-5x^2 + 3x - 2 > 0$

Illustration 8: Solve $9x^2 + 12x + 4 \leq 0$

Solution: $D = 144 - 4 \times 9 \times 4 = 0$
 $a = 9 > 0$

Hence solution of $9x^2 + 12x + 4 \leq 0$ is

$$x = -\frac{12}{2 \times 9} = -\frac{2}{3}$$

Illustration 9: Solve the inequality, $\frac{3x^2 - 7x + 8}{x^2 + 1} \leq 2$

Solution: Domain: $x \in \mathbb{R}$

given inequality is equivalent to

$$\frac{3x^2 - 7x + 8}{x^2 + 1} - 2 \leq 0$$

$$\Rightarrow \frac{3x^2 - 7x + 8 - 2x^2 - 2}{x^2 + 1} \leq 0$$

$$\Rightarrow \frac{3x^2 - 7x + 6}{x^2 + 1} \leq 0 \Rightarrow \frac{(x-1)(x-6)}{x^2 + 1} \leq 0$$

$$\Rightarrow x \in [1, 6]$$

Illustration 10: $\frac{8x^2 + 16x - 51}{(2x-3)(x+4)} > 3$, if x satisfies

- (a) $x < -4$ (b) $-3 < x < 3/2$
(c) $x > 5/2$ (d) All of these

Solution: (d) Consider $\frac{8x^2 + 16x - 51}{(2x-3)(x+4)} - 3 > 0$

$$\Rightarrow \frac{2x^2 + x - 15}{2x^2 + 5x - 12} > 0 \Rightarrow \frac{(2x-5)(x+3)}{(2x-3)(x+4)} > 0$$

Hence both Nr and Dr are positive if $x < -4$ or $x > 5/2$ and both negative if $-3 < x < 3/2$

Hence all the statements are true.

SOLUTION OF SYSTEM OF INEQUALITIES IN ONE VARIABLE

If two or more inequalities in one variable are given and we have to find all those values of the variable that satisfied all the inequalities, then all the given inequalities together called system of inequalities and all the values of the variable that satisfied all the inequalities are called solution of the system of inequalities.

Illustration 11: Solve the system of inequalities

$$\begin{aligned} 2x + 4 &> 4x - 3 \\ 3x - 2 &\leq 12 - 2x \\ -5x + 1 &\geq x + 3 \end{aligned}$$

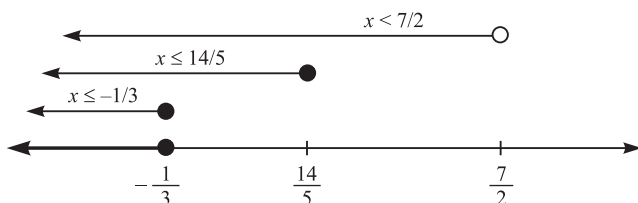
Solution: $2x + 4 > 4x - 3$
 $\Rightarrow 2x - 4x > -3 - 4$
 $\Rightarrow -2x > -7$
 $\Rightarrow 2x < 7$

$\therefore x < \frac{7}{2}$... (1)

$3x - 2 \leq 12 - 2x$
 $\Rightarrow 3x + 2x \leq 12 + 2$
 $\Rightarrow 5x \leq 14$
 $\therefore x \leq \frac{14}{5}$... (2)

And $-5x + 1 \geq x + 3$
 $\Rightarrow -5x - x \geq 3 - 1$
 $\Rightarrow -6x \geq 2$
 $\Rightarrow 6x \leq -2$
 $\Rightarrow x \leq -\frac{1}{3}$... (3)

We represent the values of x in (1), (2) and (3); on the number line as shown below:



Note that the dark circle represents that the number corresponding to it included and blank circle represents that the number corresponding to it is not included.

Now all values of x which satisfied each of the three inequalities of the given system of inequations is $x \leq -\frac{1}{3}$, which is

represented by thick dark ray with a dark circle on the number line.

Hence required solution is $x \leq -\frac{1}{3}$ i.e. $x \in \left(-\infty, -\frac{1}{3}\right]$.

Illustration 12: Solve $\frac{4x-3}{2-x} > -5$

$$\begin{aligned} 3 - 2x &< 4 \\ 5x - 3 &\geq 7 \end{aligned}$$

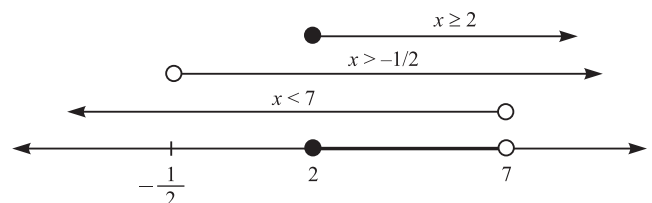
Solution: $\frac{4x-3}{2-x} > -5$

$$\begin{aligned} \Rightarrow 4x - 3 &> -5(2-x) \\ \Rightarrow 4x - 3 &> -10 + 5x \\ \Rightarrow 4x - 5x &> -10 + 3 \\ \Rightarrow -x &> -7 \\ \Rightarrow x &< 7 \end{aligned} \quad \dots (1)$$

$$\begin{aligned} \Rightarrow 3 - 2x &< 4 \\ \Rightarrow -2x &< 4 - 3 \\ \Rightarrow -2x &< 1 \\ \Rightarrow 2x &> -1 \\ \Rightarrow x &> -\frac{1}{2} \end{aligned} \quad \dots (2)$$

And $5x - 3 \geq 7$
 $\Rightarrow 5x \geq 10 \Rightarrow x \geq 2$... (3)

We represent the values of x in (1), (2) and (3) on the number line as shown below:



Now all values of x which satisfied each of the three inequalities of the given system of inequalities is $2 \leq x < 7$, which is represented by thick dark line segment with one dark circle (on the left side) and a blank circle (on the right side) on the number line.

Hence required solution is $2 \leq x < 7$
i.e. $x \in [2, 7)$

INEQUALITIES CONTAINING A MODULUS

- (i) • If $a > 0$, then $|x| \leq a \Rightarrow -a \leq x \leq a$
• If $a > 0$, then $|x| < a \Rightarrow -a < x < a$
(ii) • If $a > 0$, then $|x| \geq a \Rightarrow x \leq -a$ and $x \geq a$
• If $a > 0$, then $|x| > a \Rightarrow x < -a$ and $x > a$
• If $a < 0$, then $|x| \geq a \Rightarrow x \leq a$ and $x \geq -a$
• If $a < 0$, then $|x| > a \Rightarrow x < a$ and $x > -a$

Illustration 13: Solve $|x-3| \geq 4$

Solution: $|x-3| \geq 4$
 $\Rightarrow (x-3) \leq -4$ and $(x-3) \geq 4$
 $\Rightarrow x \leq -4 + 3$ and $x \geq 4 + 3$
 $\Rightarrow x \leq -1$ and $x \geq 7$
i.e. $x \in (-\infty, -1] \cup [7, \infty)$

Illustration 14: Solve $|5 - 4x| < -2$

Solution: $|5 - 4x| < -2$

$$\Rightarrow (5 - 4x) < -2 \text{ and } (5 - 4x) > 2$$

$$\Rightarrow -4x < -2 - 5 \text{ and } -4x > 2 - 5$$

$$\Rightarrow -4x < -7 \text{ and } -4x > -3$$

$$\Rightarrow 4x > 7 \text{ and } 4x < 3$$

$$\Rightarrow x > \frac{7}{4} \text{ and } x < \frac{3}{4}$$

$$\text{i.e. } x \in \left(-\infty, \frac{3}{4}\right) \cup \left(\frac{7}{4}, \infty\right)$$

Illustration 15: Solve the inequation:

$$\left|\frac{2}{x-4}\right| > 1, \quad x \neq 4$$

Solution: We have,

$$\Rightarrow \left|\frac{2}{x-4}\right| > 1, \quad x \neq 4$$

$$\Rightarrow \frac{2}{|x-4|} > 1 \quad \left[\because \left|\frac{a}{b}\right| = \frac{|a|}{|b|}\right]$$

$$\Rightarrow 2 > |x-4| \quad [\because |x-4| > 0 \text{ for all } x \neq 4]$$

$$\Rightarrow 4 - 2 < x < 4 + 2 \quad [\because |x-a| < r \Leftrightarrow a-r < x < a+r]$$

$$\Rightarrow 2 < x < 6$$

$$\Rightarrow x \in (2, 6)$$

But, $x \neq 4$.

Hence, the solution set of the given inequation is $(2, 4) \cup (4, 6)$

Practice Exercises



Foundation Level

1. The solution set of $x - 2y \geq 0; 2x - y \leq -2; x \geq 0; y \geq 0$ is
 - (a) Empty
 - (b) Bounded
 - (c) Neither empty nor bounded
 - (d) None of these
2. If $x^3 \geq 0, y^3 \geq 0$ and $(x + y) \leq 1$, then the maximum value of $(2x + 3y)$ is
 - (a) 2
 - (b) 3
 - (c) 4
 - (d) 5
3. Let $y^2 = \frac{(x+5)(x-3)}{x-1}$, then all the real values of x , for which y has non-zero real values, are :
 - (a) $-5 < x < 1$
 - (b) $x > 15$
 - (c) $-5 < x < 1$ or $x > 3$
 - (d) None of these
4. If $0 < x < 5$ and $1 < y < 2$, then which of the following is true?
 - (a) $x + y < 0$
 - (b) $-3 < 2x - 3y < 4$
 - (c) $-6 < 2x - 3y < 7$
 - (d) $-3 < 3x - y < 2$
5. Which of the following is the solution set of $|2x - 3| < 7$
 - (a) $\{x : -5 < x < 2\}$
 - (b) $\{x : -5 < x < 5\}$
 - (c) $\{x : -2 < x < 5\}$
 - (d) $\{x : x < -5 \text{ or } x > 2\}$
6. $|x^2 - 4x| < 5$
 - (a) $-1 \leq x \leq 5$
 - (b) $1 \leq x \leq 5$
 - (c) $-1 \leq x \leq 1$
 - (d) $-1 < x < 5$
7. If $1 \leq x \leq 3$ and $2 \leq y \leq 4$, what is the maximum value of $\left(\frac{x}{y}\right)$?
 - (a) $\frac{2}{3}$
 - (b) 4
 - (c) $\frac{3}{2}$
 - (d) 2
8. If both the inequalities given below are true, then what are the values of x and y which satisfy the inequalities?

$$\frac{x}{4} + \frac{2}{3} < \frac{2x}{3} - \frac{1}{6}, \frac{1}{6} + \frac{3}{4y} > \frac{7}{8}$$
 - (a) $x > 2, y < 2$
 - (b) $x < 2 < y$
 - (c) $x > 2, y > 2$
 - (d) $x < 2, y < 2$
9. The solution set of $\frac{1}{4} + \frac{1}{x} < \frac{1}{2} + \frac{1}{3}$, when $x \in R$, is:
 - (a) $\frac{1}{x} < \frac{7}{12}$
 - (b) $x \geq \frac{12}{7}$
 - (c) $x < \frac{14}{7}$
 - (d) $\frac{1}{x} > \frac{3}{2}$
10. $x > 2, y > -1$, then which of the following holds good?
 - (a) $xy > -2$
 - (b) $xy < -2$
 - (c) $x > -\frac{2}{y}$
 - (d) None of these
11. If $x > 5$ and $y < -1$, then which of the following statements is true?
 - (a) $(x + 4y) > 1$
 - (b) $x > -4y$
 - (c) $-4x < 5y$
 - (d) None of these
12. If $13x + 1 < 2z$ and $z + 3 = 5y^2$, then
 - (a) x is necessarily less than y
 - (b) x is necessarily greater than y
 - (c) x is necessarily equal to y
 - (d) None of the above is necessarily true
13. If a and b are negative, and c is positive, which of the following statements is/are true?
 - I. $a - b < a - c$
 - II. if $a < b$, then $\frac{a}{c} < \frac{b}{c}$
 - III. $\frac{1}{b} < \frac{1}{c}$
 - (a) I only
 - (b) II only
 - (c) III only
 - (d) II and III only
14. If $x + y > 5$ and $x - y > 3$, then which of the following gives all possible values of x ?
 - (a) $x > 3$
 - (b) $x > 4$
 - (c) $x > 5$
 - (d) $x < 5$

Standard Level

1. If $x^2 + x + \frac{1}{x^2} + \frac{1}{x} < 0$, then which of the following is true?

- (a) $x + \frac{1}{x} > -2$ (b) $x + \frac{1}{x} < -2$
(c) $x + \frac{1}{x} < 1$ (d) Both (a) and (c)

2. Find the area of the quadrilateral formed by the solution set of the inequations $2x + 3y \leq 12$, $x \geq 0$, $y \geq 0$ and $x \leq 3$.

- (a) 6 (b) 8
(c) 9 (d) 12

3. The solution set of inequality $|x-1|+|x+1| < 4$ is

- (a) $(-\infty, -2)$ (b) $(-2, 2)$
(c) $(2, \infty)$ (d) $(-\infty, \infty)$

4. If $x \in \mathbb{R}$, and $\alpha = \frac{x^2}{(1+x^4)}$, then

- (a) $0 \leq \alpha \leq 2$ (b) $0 \leq \alpha \leq 1$
(c) $0 \leq \alpha \leq \frac{1}{4}$ (d) $0 \leq \alpha \leq \frac{1}{2}$

5. If x is real find the limits between which x must lie; when

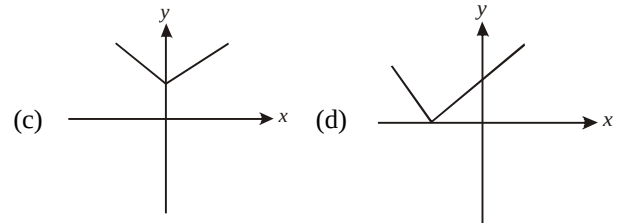
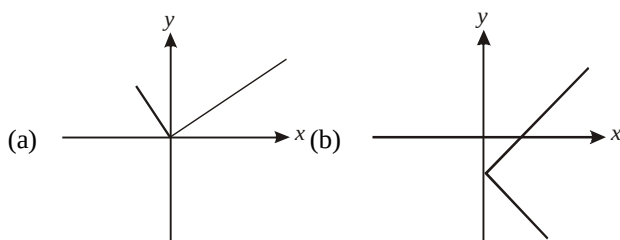
$$\left| \frac{x^2 - 3x - 1}{x^2 + x + 1} \right| < 3$$

- (a) All values of x (b) $-2 < x < -1$
(c) $(-\infty, 2) \cup (4, \infty)$ (d) $(-\infty, -2) \cup (-1, \infty)$

6. If $6 \geq x \geq -2$ and $4 \geq y \geq -4$, find the limits for $\frac{y}{x}$, where x and y are non-zero integers.

- (a) $\frac{y}{x} \geq 2, \frac{y}{x} \leq \frac{2}{3}$ (b) $\frac{y}{x} \geq \frac{-2}{3}, \frac{y}{x} \leq 2$
(c) $\frac{y}{x} \geq \frac{-2}{3}, \frac{y}{x} \leq \frac{1}{4}$ (d) $\frac{y}{x} \geq -4, \frac{y}{x} \leq 4$

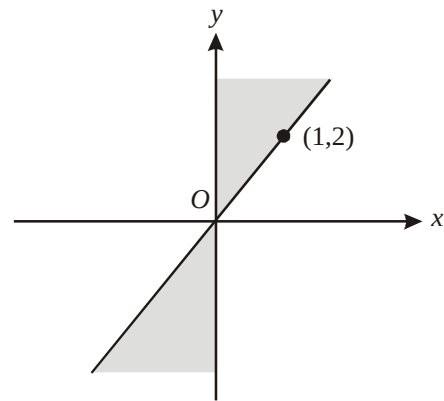
7. Which of the following shaded regions shows the graph of the inequality $y \leq |x+2|$?



8. If x satisfies $|x-1| + |x-2| + |x-3| \geq 6$, then

- (a) $0 \leq x \leq 4$ (b) $x \leq -2$ or $x \geq 4$
(c) $x \leq 0$ or $x \geq 4$ (d) $x \leq -1$ or $x \geq 5$

9. The shaded portion of figure shows the graph of which of the following?



- (a) $x(y-2x) \geq 0$ (b) $x(y-2x) \leq 0$
(c) $x\left(y + \frac{1}{2}x\right) \geq 0$ (d) $x\left(y - \frac{1}{2}x\right) \leq 0$

10. The set of all real numbers x for which $x^2 - [x+2] + x > 0$, is

- (a) $(-\infty, -2) \cup (2, \infty)$ (b) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
(c) $(-\infty, -1) \cup (1, \infty)$ (d) $(\sqrt{2}, \infty)$

11. For the real numbers p, q, r, x, y , let $p < x < q$ and $p < y < r$. Which one of the following is correct?

- (a) $p < x < y < r$ (b) $p < x < q < r$
(c) $p < y < x < q$ (d) None of these

12. What is the value of m which satisfies $3m^2 - 21m + 30 < 0$?

- (a) $m < 2$ or $m > 5$ (b) $m > 2$
(c) $2 < m < 5$ (d) $m < 5$

13. Which of the following values of x do not satisfy the inequality $(x^2 - 3x + 2 > 0)$ at all?

- (a) $1 \leq x \leq 2$ (b) $2 < x < 4$
(c) $-5 \leq x \leq -1$ (d) $5 < x \leq 10$

14. Given that $-1 \leq v \leq 1$, $-2 \leq u \leq -0.5$ and $-2 \leq z \leq -0.5$ and $w = \frac{vz}{u}$, then which of the following is necessarily true?
- (a) $-0.5 \leq w \leq 2$ (b) $-4 \leq w \leq 4$
 (c) $-4 \leq w \leq 2$ (d) $-2 \leq w \leq -0.5$
15. A real number x satisfying $1 - \frac{1}{n} < x \leq 3 + \frac{1}{n}$, for every positive integer n , is best described by
- (a) $1 < x < 4$ (b) $1 < x \leq 3$
 (c) $0 < x \leq 4$ (d) $1 \leq x \leq 3$
16. If $|b| \geq 1$ and $x = -|a|b$, then which one of the following is necessarily true?
- (a) $a - xb < 0$ (b) $a - xb \geq 0$
 (c) $a - xb > 0$ (d) $a - xb \leq 0$
17. The number of solutions of the equation $2x + y = 40$ where both x and y are positive integers and $x \leq y$
- (a) 7 (b) 13
 (c) 14 (d) 18
18. If a , b and c are three real numbers, then which of the following is NOT true?
- (a) $|a+b| \leq |a| + |b|$
 (b) $|a-b| \leq |a| + |b|$
 (c) $|a-b| \leq |a| - |b|$
 (d) $|a-c| \leq |a-b| + |b-c|$
19. x and y are real numbers satisfying the conditions $2 < x < 3$ and $-8 < y < -7$. Which of the following expressions will have the least value?
- (a) x^2y (b) xy^2
 (c) $5xy$ (d) None of these

Expert Level

- If $ab > 0$, then the minimum value of $(a+b)\left(\frac{1}{a} + \frac{1}{b}\right)$ is
 - < 1
 - > 4
 - < 7
 - can't say
- What are the limits of $\frac{2^n(n+1)^n}{n^n}$; where n is a positive integer?
 - $2 \leq \frac{2^n(n+1)^n}{n^n} \leq 3$
 - $2^{n+1} \leq \frac{2^n(n+1)^n}{n^n} \leq 3.2^n$
 - $2^n \leq \frac{2^n(n+1)^n}{n^n} \leq 3^n$
 - None of these
- If $\frac{13x^2 + 30x + 17}{3x^2 + 14x + 8} > 3$, then find the value of x , if $x \in R$ and $x \neq \frac{-2}{3}, -4$.
 - $(-\infty, -4) \cup \left(\frac{-2}{3}, \frac{7}{2}\right) \cup \left(\frac{9}{2}, \infty\right)$
 - $(-\infty, -4) \cup \left(\frac{-2}{3}, \frac{-1}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$
 - $\left(-4, \frac{-2}{3}\right) \cup \left(\frac{-1}{2}, \frac{7}{2}\right)$
 - None of these
- The solution of the equation $2^{|x+2|} - |2^{x+1} - 1| = 2^{x+1} + 1$ is
 - $\{-3, -1\}$
 - $\{-3\} \cup [1, \infty)$
 - $\{-3\} \cup [-1, \infty)$
 - $\{-\infty, -3\} \cup \{-1\}$
- If $(y^2 + 5y + 3)(x^2 + x + 1) < 2x$ for all $x \in R$, then y lies in the interval
 - $(-\infty, -2]$
 - $\left[\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right]$
 - $\left[\frac{5+\sqrt{5}}{2}, \infty\right)$
 - $\left[\frac{5+\sqrt{5}}{2}, -\frac{2}{3}\right]$
- If $b^2 > 4ac$, then $ax^2 + 4x + 4)^2 + b(x^2 + 4x + 4) + c = 0$ has distinct real roots if
 - $b < a < 0 < c$
 - $a < b < 0 < c$
 - $b < 0 < a < c$
 - $c < 0 < b < a$
- The largest interval for which $x^{12} - x^9 + x^4 - x + 1 > 0$ is
 - $-4 < x \leq 0$
 - $0 < x < 1$
 - $-100 < x < 100$
 - $-\infty < x < \infty$
- If x is real and $x^2 - 3x + 2 > 0$, $x^2 - 3x - 4 \leq 0$, then which one of the following is correct?
 - $-1 \leq x \leq 4$
 - $2 \leq x \leq 4$
 - $-1 < x \leq 1$
 - $-1 \leq x < 1$ or $2 < x \leq 4$
- m is the smallest positive integer such that for any integer $n \leq m$, the quantity $n^3 - 7n^2 + 11n - 5$ is positive. What is the value of m ?
 - 4
 - 5
 - 8
 - None of these
- If n is such that $36 \leq n \leq 72$, then $x = \frac{n^2 + 2\sqrt{n}(n+4) + 16}{n + 4\sqrt{n} + 4}$ satisfies
 - $20 < x < 54$
 - $23 < x < 58$
 - $25 < x < 64$
 - $28 < x < 60$
- What values of x satisfy $x^{2/3} + x^{1/3} - 2 \leq 0$?
 - $-8 \leq x \leq 1$
 - $-1 \leq x \leq 8$
 - $1 < x < 8$
 - $1 \leq x \leq 8$

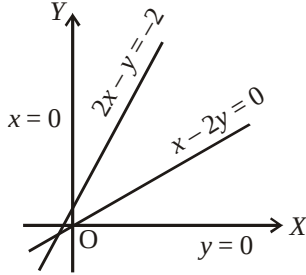
Test Yourself

- $|2x - 3| < |x + 5|$, then x belongs to
 (a) $(-3, 5)$ (b) $(5, 9)$
 (c) $\left(-\frac{2}{3}, 8\right)$ (d) $\left(-8, \frac{2}{3}\right)$
- The set of all x satisfying the inequality $\frac{4x-1}{3x+1} \geq 1$
 (a) $\left(-\infty, -\frac{1}{3}\right) \cup \left[\frac{1}{4}, \infty\right)$
 (b) $\left(-\infty, -\frac{2}{3}\right) \cup \left[\frac{5}{4}, \infty\right)$
 (c) $\left(-\infty, -\frac{1}{3}\right) \cup [2, \infty)$
 (d) $\left(-\infty, -\frac{2}{3}\right] \cup [4, \infty)$
- If $x^2 + x + \frac{1}{x^2} + \frac{1}{x} < 0$, then which of the following is true?
 (a) $x + \frac{1}{x} > -2$ (b) $x + \frac{1}{x} < -2$
 (c) $x + \frac{1}{x} < 1$ (d) Both (a) and (c)
- If a, b, c are real numbers such that
 $a - b + c < 1$, $a + b + c > -1$ and $9a + 3b + c < -4$,
 then what is the sign of ' a '?
 (a) +ve only (b) -ve only
 (c) +ve or -ve (d) Cannot be said
- If $x + y > 5$ and $x - y > 3$, then which of the following inequalities gives all the possible values of x ?
 (a) $x > 3$ (b) $x > 4$
 (c) $x > 5$ (d) $x < 5$
- How many numbers in the set $\{-4, -3, 0, 2\}$ satisfy the conditions $|y - 4| \leq 6$ and $|y + 4| < 5$?
 (a) 3 (b) 1
 (c) 2 (d) None of these
- What is the solution set of the equation
 $x^3 - x^2 + x - 1 > 0$?
 (a) $x < -1$ (b) $-1 < x < 1$
 (c) $x > 1$ (d) $x > 0$
- For all ' x ', $x^2 + 2ax + 10 - 3a > 0$, then the interval in which ' a ' lies is
 (a) $a < -5$ (b) $-5 < a < 2$
 (c) $a > 5$ (d) $2 < a < 5$
- For positive numbers a, b, c the least value of
 $(a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$ is
 (a) 3 (b) 9
 (c) $\frac{27}{4}$ (d) None of these
- If for $x \in R$, $\frac{1}{3} < \frac{x^2 - 2x + 4}{x^2 + 2x + 4} < 3$, then $\frac{9 \cdot 3^{2x} - 6 \cdot 3^x + 4}{9 \cdot 3^{2x} + 6 \cdot 3^x + 4}$ lies between
 (a) $\frac{1}{2}$ and 2 (b) $\frac{1}{3}$ and 3
 (c) 0 and 2 (d) None of these
- If p, q, r are positive and are in AP, the roots of quadratic equation $px^2 + qx + r = 0$ are real for:
 (a) $\left|\frac{r}{p} - 7\right| \geq 4\sqrt{3}$ (b) $\left|\frac{p}{r} - 7\right| \geq 4\sqrt{3}$
 (c) all p and r (d) no p and r
- The solution of the inequation $4^{-x+0.5} - 7 \cdot 2^{-x} < 4$, $x \in R$ is:
 (a) $(-2, \infty)$ (b) $(2, \infty)$
 (c) $\left(2, \frac{7}{2}\right)$ (d) None of these
- The solution set of $\left|\frac{x+1}{x}\right| + |x+1| = \frac{(x+1)^2}{|x|}$ is:
 (a) $\{x \mid x \geq 0\}$ (b) $\{x \mid x > 0\} \cup \{-1\}$
 (c) $\{-1, 1\}$ (d) $\{x \mid x \geq 1 \text{ or } x \leq -1\}$
- If $b > a$, then the equation $(x - a)(x - b) - 1 = 0$ has:
 (a) both roots in $[a, b]$
 (b) both roots in $(-\infty, 0)$
 (c) both roots in $(b, +\infty)$
 (d) one root in $(-\infty, a)$ and other root in $(b, +\infty)$
- The integer k for which the inequality $x^2 - 2(4k - 1)x + 15k^2 - 2k - 7 > 0$ is valid for any x is:
 (a) 2 (b) 3
 (c) 4 (d) 6

Hints & Solutions

Foundation Level

1. (a) Plotting the given inequations, we get the following graph :



There is no common region. Hence, the solution set is empty.

2. (b) It is given that $x^3 > 0$, $y^3 > 0$ and $x + y \leq 1$.
 $x + y \leq 1 \Rightarrow 2(x + y) \leq 2 \Rightarrow 2x + 2y \leq 2$
 $\Rightarrow 2x + 2y + y \leq 2 + y$
 $\Rightarrow 2x + 3y \leq 2 + 1 = 3$. (since $y \leq 1$).
3. (c) If y is real and non-zero $y^2 > 0$, i.e., $\frac{(x+5)(x-3)}{(x-1)} > 0$
Case (i) : If numerator and denominator are > 0
 $\Rightarrow x > 3$ or $x < -5$ and $x > 1$
Case (ii) : If numerator and denominator are < 0
 $\Rightarrow -5 < x < 3$ and $x < 1$.
4. (c) $0 < x < 5$... (1)
 $0 < 2x < 10$... (2) (multiply (1) by 2)
 $1 < y < 2$... (3)
 $-6 < -3y < -3$... (4) (multiply (3) by -3)
 Subtracting (4) from (2)
 $0 - 6 < 2x - 3y < 10 - 3$ i.e., $-6 < 2x - 3y < 7$.
5. (c) If the expression between the absolute value bars is positive. It's less than +7 or, if the expression between the bars is negative, it's greater than -7. In other words, $2x - 3$ is between -7 and +7
 $-7 < 2x - 3 < 7$
 $-4 < 2x < 10$
 $-2 < x < 5$
6. (d) At $x = 0$, inequality is satisfied, option (b) is rejected.
 At $x = 2$, inequality is satisfied, option (c) is rejected.
 At $x = 5$, LHS = RHS.
 At $x = -1$, LHS = RHS.
 Thus, option (d) is correct.
7. (c) $\text{Max} \left(\frac{x}{y} \right) = \frac{\text{Max}(x)}{\text{Min}(y)} = \frac{3}{2}$
8. (a) $\frac{x}{4} + \frac{2}{3} < \frac{2x}{3} - \frac{1}{6} \Rightarrow \frac{5}{6} < \frac{8x-3x}{12}$

$$\Rightarrow \frac{5}{6} < \frac{5x}{12} \Rightarrow x > 2 \text{ and } \frac{1}{y} + \frac{1}{2} > \frac{7}{8} \Rightarrow \frac{4+3}{4y} > \frac{7}{8}$$

$$\Rightarrow \frac{1}{y} > \frac{1}{2} \Rightarrow y < 2.$$

9. (a) $\frac{1}{4} + \frac{1}{x} < \frac{5}{6} \quad \therefore \frac{1}{x} < \frac{5}{6} - \frac{1}{4}$
 $\therefore \frac{1}{x} < \frac{7}{12}.$
10. (d) Given, $y > -1$ i.e., if y is positive no. then product of x and y also positive.
 But any option does not give xy is +ve.
 By putting different values of x and y , we see that none of these three hold good.
11. (d) $x > 5$ and $y < -1 \Rightarrow 4y < -4$
 (i) $x > 5$ and $4y < -4$ so $x + 4y < 1$
 (ii) Let $x > -4y$ be true $\Rightarrow 4y < -4$ or $-4y > 4$
 So, $x > 4$, which is not true as given $x > 5$.
 So, $x > -4y$ is not necessarily true.
 (iii) $x > 5 \Rightarrow -4x < -20$ and $5y < -5$
 It is not necessary that $-4x < 5y$ as $-4x$ can be greater than $5y$, since $5y < -5$.
 Hence, none of the options is true.
12. (d) $13x + 1 < 2z$ and $z + 3 = 5y^2$
 $\Rightarrow 13x + 1 < 2(5y^2 - 3)$
 $\Rightarrow 13x + 7 < 10y^2 \Rightarrow 10y^2 > 13x + 7$
 In the above equation, all the options a , b & c are possible but not necessarily true.
13. (d) Let $a = x$, $b = -y$ and $c = z$
 Statement I : $-x + y < -x - z$
 So I is not true.
 Statement II : $\frac{-x}{z} < \frac{-y}{z}$
 Since $x < y$ so II is true
 Statement III : $\frac{1}{-y} < \frac{1}{z}$
 Since y is negative and z is positive.
 So III is true
 Hence, statements II and III are true.
14. (b) $x + y > 5$... (1)
 $x - y > 3$... (2)
 Adding inequations (1) and (2), we get
 $2x > 8$ i.e. $x > 4$

Standard Level

1. (d) Given that $\left(x^2 + \frac{1}{x^2}\right) + \left(x + \frac{1}{x}\right) < 0$

$$\left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 2 < 0$$

Substituting $x + \frac{1}{x} = y$, we get

$$y^2 + y - 2 < 0$$

$$\Rightarrow (y-1)(y+2) < 0$$

$$\therefore \text{either } y-1 < 0; y+2 > 0$$

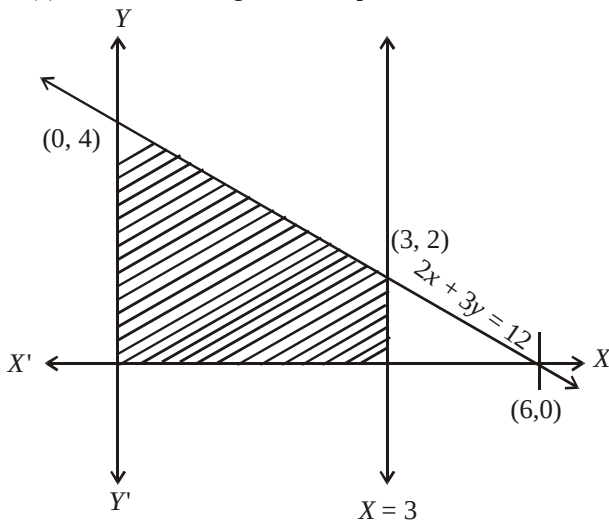
$$\text{or } y+2 < 0; y-1 > 0.$$

$$\begin{aligned} \text{i.e., } y < 1, y > -2 \text{ or } y < -2; \quad y > 1 \\ \Downarrow \\ \text{not possible} \end{aligned}$$

Therefore, $-2 < y < 1$

$$\text{i.e., } -2 < \left(x + \frac{1}{x}\right) < 1.$$

2. (c) The shaded region is a trapezium.



The lengths of the parallel sides are 4 and 2 and the distance between them is equal to 3.

$$\begin{aligned} \therefore \text{Required area} &= \frac{1}{2} \times (4 + 2) \times 3 = \frac{1}{2} \times (6) \times 3 \\ &= 9 \text{ sq. units} \end{aligned}$$

3. (b) As solved in examples (1) and (2), the solution will consist of three intervals

$$x < -1, -1 \leq x < 1 \text{ and } x \geq 1$$

$$\text{If } x < -1, \text{ then } |x-1| = -(x-1) \text{ and } |x+1| = -(x+1).$$

$$\text{We get, } -(x-1) - (x+1) < 4 \Rightarrow x > -2$$

$$\therefore \text{We have } x < -1 \text{ and } x > -2 \Rightarrow -2 < x < -1$$

$$\Rightarrow x \in (-2, -1)$$

$$\text{If } -1 \leq x < 1, \text{ then } |x-1| = -(x-1) \text{ and } |x+1| = x+1.$$

$$\text{We get, } -(x-1) + (x+1) < 4 \Rightarrow 2 < 4, \text{ which is valid}$$

$$\therefore \text{Inequality holds true for all } x \text{ satisfying}$$

$$-1 \leq x < 1 \Rightarrow x \in [-1, 1)$$

$$\text{If } x \geq 1, \text{ then } |x-1| = (x-1) \text{ and } |x+1| = x+1,$$

$$\text{we get } x-1+x+1 < 4 \Rightarrow x < 2.$$

$$\therefore \text{We have } x \geq 1 \text{ and}$$

$$x < 2 \Rightarrow 1 \leq x < 2 \Rightarrow x \in [1, 2)$$

Combining the solutions of the above three cases, we get

$$x \in (-2, -1) \cup [-1, 1) \cup [1, 2) \Rightarrow (-2, 2)$$

4. (d) For real x , we have $x^2/(1+x^4) \geq 0$ also

$$1+x^4-2x^2 = (1-x^2)^2 \geq 0$$

$$\Rightarrow 1+x^4 \geq 2x^2$$

$$\therefore \left(\frac{x^2}{1+x^4}\right) \leq \frac{1}{2}$$

$$\text{So } 0 \leq \frac{x^2}{1+x^4} \leq \frac{1}{2} \Rightarrow 0 \leq x \leq \frac{1}{2}$$

5. (d) Given inequality can be written as,

$$-3 < \left(\frac{x^2-3x-1}{x^2+x+1}\right) < +3$$

Since (x^2+x+1) is positive for all x , therefore multiplying by (x^2+x+1) , we get

$$-3(x^2+x+1) < (x^2-3x-1) < 3(x^2+x+1) \quad \dots(1)$$

From first part of (1), we get

$$-3(x^2+x+1) < x^2-3x-1$$

$$\Rightarrow 4x^2+2 > 0 \quad \dots(2)$$

This will be true for all real x .

Now, solving second part of inequality (1)

$$(x^2-3x-1) < 3(x^2+x+1)$$

$$\Rightarrow (x+1)(x+2) > 0$$

$$x \in (-\infty, -2) \cup (-1, \infty) \quad \dots(3)$$

Combining (2) and (3), we find the solution as

$$x \in (-\infty, -2) \cup (-1, \infty).$$

6. (d)
$$\begin{array}{cccccccc} y & 4 & 3 & 2 & 1 & -1 & -3 & -4 \\ x & 6 & 5 & 4 & 3 & 1 & -1 & -2 \end{array}$$

$$\text{Hence, minimum value of } \frac{y}{x} = \frac{4}{-1} = -4$$

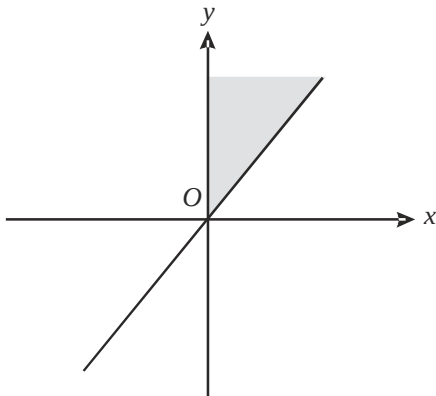
$$\text{and maximum value of } \frac{y}{x} = \frac{4}{1} = 4$$

$$\therefore \frac{y}{x} \geq -4 \text{ and } \frac{y}{x} \leq 4.$$

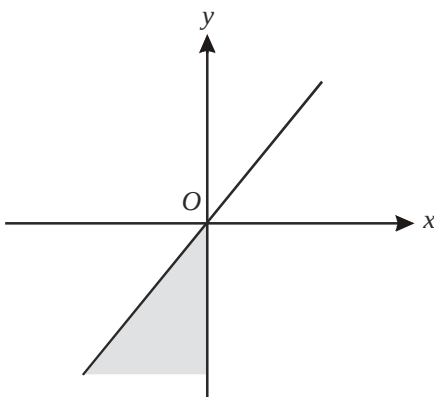
7. (d)

8. (c) For $x \leq 1$, the given in equation becomes
 $1 - x + 2 - x + 3 - x \geq 6 \Rightarrow -3x \geq 0 \Rightarrow x \leq 0$
 and for $x \geq 3$, the given equation becomes
 $x - 1 + x - 3 + x - 3 \geq 6 \Rightarrow 3x \geq 12 \Rightarrow x \geq 4$
 For $1 < x \leq 2$ we get
 $x - 1 + 2 - x + 3 - x \geq 6$
 $\Rightarrow -x + 4 \geq 6$
 i.e., $-x \geq 2 \Rightarrow x \leq -2$ not possible.
 For $2 < x < 3$, we get $x - 1 + x - 2 + 3 - x \geq 6$
 $\Rightarrow x \geq 6$, not possible
 Hence solution set is
 $(-\infty, 0] \cup [4, \infty)$, i.e. $x \leq 0$ or $x \geq 4$.

9. (a) Each of the answer choices in the form of the product of two factors on the left and a " ≥ 0 " or " ≤ 0 " on the right.
 The product will be negative when the two factors have opposite signs, and it will be positive when the factors have the same sign. Choice (1), for example, has a " ≥ 0 ", so you'll be looking other factors to have the same sign.
 Either : $x \geq 0$ and $y - 2x \geq 0 \Rightarrow x \geq 0$ and $y \geq 2x$
 or $x \leq 0$ and $y - 2x \leq 0 \Rightarrow x \leq 0$ and $y \leq 2x$
 The graph of $x \geq 0$ and $y \geq 2x$ looks like this :



The graph of $x \leq 0$ and $y \leq 2x$ looks like this



Together they make the graph in the figure.

10. (b) For $x \geq -2$, $x^2 - x - 2 + x > 0$
 $\Rightarrow x^2 > 2 \Rightarrow x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
 $\Rightarrow x \in [-2, -\sqrt{2}) \cup (\sqrt{2}, \infty)$ For $x < -2$

$$x^2 + x + 2 + x > 0 \text{ or } x^2 + 2x + 2 > 0$$

which is true for all x . Hence $x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

11. (b) For the real number p, q, r, x and y
 $p < x < q$ and $p < y < r$
 $p < x < q < r$

12. (c) $3m^2 - 21m + 30 < 0$

$$\text{or } m^2 - 7m + 10 < 0$$

$$\text{or } m^2 - 5m - 2m + 10 < 0 \quad (\text{factorize})$$

$$\text{or } m(m-5) - 2(m-5) < 0$$

$$\text{or } (m-2)(m-5) < 0$$

Case I : $m-2 > 0$ and $m-5 < 0$

$$\Rightarrow m > 2 \text{ and } m < 5 \Rightarrow 2 < m < 5$$

Case II : $m-2 < 0$ and $m-5 > 0 \Rightarrow m < 2$ and $m > 5$

nothing common

Hence, $2 < m < 5$

13. (a) Given inequality is $x^2 - 3x + 2 > 0$

$$\Rightarrow x^2 - 2x - x + 2 > 0$$

$$\Rightarrow x(x-2) - 1(x-2) > 0$$

$$\Rightarrow (x-2)(x-1) > 0 \Rightarrow x < 1, x > 2$$

14. (b) Substitute the extreme values in the given equation :
 $v = 1, u = -0.5, z = -2$.

$$\text{Then } w = \frac{vz}{u} = 4.$$

Only option (b) gives this.

15. (c) Given inequality is

$$1 - \frac{1}{n} < x \leq 3 + \frac{1}{n} \quad \dots(1)$$

For any positive n ,

$$0 < \frac{1}{n} \leq 1 \quad \dots(2)$$

Add 3 to each part in inequality (2), we get

$$3 < 3 + \frac{1}{n} \leq 4 \quad \dots(3)$$

Again from inequality (2), we get

$$0 > -\frac{1}{n} \geq -1 \quad (\text{multiply } (-1) \text{ in each part})$$

$$\Rightarrow 1 > 1 - \frac{1}{n} \geq 0 \quad (\text{Add 1 in each part})$$

$$\Rightarrow 0 \leq 1 - \frac{1}{n} < 1 \quad \dots(4)$$

From inequalities (1), (3) and (4), we get $0 < x \leq 4$

16. (b) $|b| \geq 1 \Rightarrow b \geq 1$ or $b \leq -1$

$x - |a|b$ (given)

Consider $a - xb = a - (-|a|b)b$

$a + |a|b^2 \geq 0$ since, $b^2 \geq 1$

17. (b) $2x + y = 40$; $x \leq y$; $x, y \in I^+$

This problem can be solved by putting various values for x and y . Starting from $x = 1$. The above equation can be solved till $x = 13$.

At $x = 13$, $y = 14$ which is $> x$. But above this value of x , it becomes greater than y so the condition $x \leq y$ is violated.

18. (c) This can be checked by taking arbitrary values of a and b in the given terms. Taking $a = 2$ and $b = 3$, we conclude that (c) is not true.

19. (c) $2 < x < 3$ and $-8 < y < -7$

$4 < x^2 < 9$

and $-8 < y < -7$

$-32 < x^2y < -63$

While $-80 < 5xy < -105$

Hence, $5xy$ is the least because xy^2 is positive

Expert Level

1. (b) $(a+b)\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{(a+b)^2}{ab}$

$= \frac{a^2}{ab} + \frac{b^2}{ab} + \frac{2ab}{ab} = \frac{a}{b} + \frac{b}{a} + 2$

When $ab > 0 \Rightarrow a$ and b have the same sign

$\therefore \frac{a}{b} + \frac{b}{a}$ will have a minimum value of 2

$\therefore \frac{a}{b} + \frac{b}{a} + 2$ will have minimum of 4.

2. (b) $\frac{2^n(n+1)^n}{n^n} = 2^n\left(\frac{n+1}{n}\right)^n = 2^n\left(1 + \frac{1}{n}\right)^n$

Now, we have $2 \leq \left(1 + \frac{1}{n}\right)^n \leq 3$

$\therefore 2 \cdot 2^n \leq 2^n\left(1 + \frac{1}{n}\right)^n \leq 3 \cdot 2^n$

i.e., $2^{n+1} \leq 2^n\left(1 + \frac{1}{n}\right)^n \leq 3 \cdot 2^n$.

3. (b) $\frac{13x^2 + 30x + 17}{3x^2 + 14x + 8} - 3 > 0$

$\frac{13x^2 + 30x + 17 - 9x^2 - 42x - 24}{3x^2 + 14x + 8} > 0$

$\frac{4x^2 - 12x - 7}{3x^2 + 14x + 8} > 0$

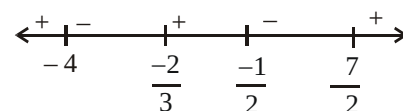
$\frac{(2x+1)(2x-7)}{(3x+2)(x+4)} > 0$

Multiplying both the numerator and the denominator by $(3x+2)(x+4)$, we get

$(2x+1)(2x-7)(3x+2)(x+4) > 0$

(The sign does not change as $(3x+2)^2(x+4)^2$ is always positive).

Representing the above inequality on the number line we get,



The region where the inequality is satisfied is

$(-\infty, -4) \cup \left(-\frac{2}{3}, -\frac{1}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$

4. (c) $|x+2|$ vanishes at $x = -2$ and $|2^{x+1} - 1|$ vanishes at $x = -1$, hence we divide the problem into three intervals :

(i) If $x < -2$, then $|x+2| = -(x+2)$ also $x+1 \leq -1$

$\Rightarrow 2^{x+1} < 2^{-1} = \frac{1}{2} \Rightarrow 2^{x+1} < 1$

$\Rightarrow |2^{x+1} - 1| = -(2^{x+1} - 1)$

$\therefore$ Equation is $2^{-x-2} + 2^{x+1} - 1 = 2^{x+1} + 1$

$\Rightarrow 2^{-x-2} = 2 \Rightarrow x = -3$

(ii) If $-2 \leq x < -1$, then $|x+2| = x+2$ also,

$x+1 < 0 \Rightarrow 2^{x+1} < 1 \Rightarrow |2^{x+1} - 1| = -(2^{x+1} - 1)$

$\therefore$ Equation is $2^{x+2} + 2^{x+1} - 1 = 2^{x+1} + 1$

$\Rightarrow 2^{x+2} = 2 \Rightarrow x = -1 \notin [-2, -1)$

(iii) If $x \geq -1$, then $|x+2| = x+2$ and

$$|2^{x+1} - 1| = 2^{x+1} - 1$$

$\therefore$ Equation is $2^{x+2} - 2^{x+1} + 1 = 2^{x+1} + 1$,

$$\Rightarrow 2^{x+2} = 2^{x+1} \text{ which is identity}$$

$\therefore$ all x such that $x \geq -1$ satisfy the equation

Hence, the solution set is $x \in \{-3\} \cup [-1, \infty)$

5. (b) Given $(y^2 + 5y + 3)(x^2 + x + 1) < 2x$

$$\Rightarrow y^2 - 5y + 3 < \frac{2x}{x^2 + x + 1} \quad \dots(1)$$

$$[\because x^2 + x + 1 > 0 \quad \forall x \in \mathbb{R}]$$

$$\text{Let } \frac{2x}{x^2 + x + 1} = z \Rightarrow zx^2 + (z-2)x + z = 0$$

$$\because x \in \mathbb{R} \Rightarrow (z-2)^2 - 4z \geq 0$$

$$\Rightarrow 3z^2 + 4z - 4 \leq 0 \Rightarrow -2 \leq z \leq \frac{2}{3}$$

$$\Rightarrow -2 \leq \frac{2x}{x^2 + x + 1} \leq \frac{2}{3} \text{ clearly the inequality (1) holds}$$

$$\text{if } y^2 - 5y + 3 < -2 \Rightarrow y^2 - 5y + 5 < 0 \Rightarrow \frac{5-\sqrt{5}}{2} < y < \frac{5+\sqrt{5}}{2}$$

6. (c) Let $x^2 + 4x + 4 = y$, then the equation becomes

$$ay^2 + by + c = 0 \quad \dots(1)$$

$\because b^2 - 4ac > 0 \Rightarrow$ equation (1) has two distinct real roots, say α and β

$$\therefore y = \alpha \text{ or } \beta \Rightarrow x^2 + 4x + 4 = \alpha \text{ or } \beta$$

$$\Rightarrow (x+2)^2 = \alpha \text{ or } \beta \quad \dots(2)$$

Clearly equation (2) will give four distinct real values of x if α and β are positive. That is if equation (1) has positive roots. For this a and c should have the same sign and the sign of b should be negative. Only the option (c) satisfies this condition.

7. (d) Given expression $x^{12} - x^9 + x^4 - x + 1 = f(x)$ (say)

For $x < 0$ put $x = -y$ where $y > 0$

then we get $f(x) = y^{12} + y^9 + y^4 + y + 1 > 0$ for $y > 0$

$$\text{For } 0 < x < 1, \quad x^9 < x^4 \Rightarrow -x^9 + x^4 > 0$$

Also $1 - x > 0$ and $x^{12} > 0$

$$\Rightarrow x^{12} - x^9 + x^4 + 1 - x > 0 \Rightarrow f(x) > 0$$

For $x > 1$

$$f(x) = x(x^3 - 1)(x^8 + 1) + 1 > 0$$

So $f(x) > 0$ for $-\infty < x < \infty$.

8. (d) Consider first: $x^2 - 3x + 2 > 0$

$$\Rightarrow (x-1)(x-2) > 0$$

$$\Rightarrow x < 1 \text{ or } x > 2 \quad \dots(1)$$

$$\text{and } x^2 - 3x - 4 \leq 0$$

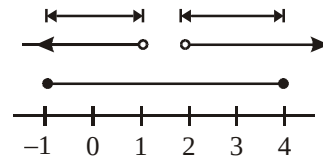
$$\Rightarrow (x-4)(x+1) \leq 0$$

$$\Rightarrow -1 \leq x \leq 4 \quad \dots(2)$$

Combining (1) and (2)

$$-1 \leq x < 1 \text{ or } 2 < x \leq 4$$

Drawing on number line :



9. (d) Let $y = n^3 - 7n^2 + 11n - 5$

$$= (n-1)(n^2 - 6n + 5) = (n-1)^2(n-5)$$

Now, $(n-1)^2$ is always positive. And for $n < 5$, the expression gives a negative quantity. Therefore, the least value of n will be 6. Hence $m = 6$.

10. (d) $x = \frac{n^2 + 2\sqrt{n}(n+4) + 16}{n + 4\sqrt{n} + 4}$

$$\text{Let } \sqrt{n} = t$$

$$\Rightarrow x = \frac{t^4 + 2t(t^2 + 4) + 16}{t^2 + 4t + 4}$$

$$\frac{(t+2)(t^3+8)}{(t+2)^2} = \frac{t^3+8}{t+2}$$

$$\Rightarrow x = t^2 - 2t + 4 \quad \dots(1)$$

For $t = 6$ to $t = 6\sqrt{2}$

$$(40-12) < x < (72+4-12\sqrt{2})$$

[putting in equation (1)]

$$\Rightarrow 28 < x < 76 - 12\sqrt{2} \text{ or } 28 < x < 60$$

11. (a) $x^{2/3} + x^{1/3} - 2 \leq 0$

$$\text{Put } x^{1/3} = y \text{ or } x = y^3 \Rightarrow y^2 + y - 2 \leq 0$$

$$\text{or } y^2 + 2y - y - 2 \leq 0$$

$$\text{or } (y-1)(y+2) \leq 0 \Rightarrow -2 \leq y \leq 1$$

$$\text{or } (-2)^3 \leq x \leq 1^3 \text{ or } -8 \leq x \leq 1$$

Explanation of Test Yourself

1. (c) We have, $|2x - 3| < |x + 5|$

$$\Rightarrow |2x - 3| - |x + 5| < 0$$

$$\Rightarrow \begin{cases} 3 - 2x + x + 5 < 0, x \leq -5 \\ 3 - 2x - x - 5 < 0, -5 < x \leq \frac{3}{2} \\ 2x - 3 - x - 5 < 0, x > \frac{3}{2} \end{cases}$$

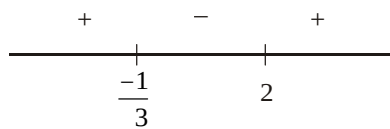
$$\Rightarrow \begin{cases} x > 8, x \leq -5 \\ x > -\frac{2}{3}, -5 < x \leq \frac{3}{2} \\ x < 8, x > \frac{3}{2} \end{cases}$$

$$\Rightarrow x \in \left(-\frac{2}{3}, \frac{3}{2}\right) \cup \left(\frac{3}{2}, 8\right)$$

$$\Rightarrow x \in \left(-\frac{2}{3}, 8\right)$$

2. (c) $\frac{4x-1}{3x+1} \geq 1 \Rightarrow \frac{4x-1}{3x+1} - 1 \geq 0 \Rightarrow \frac{x-2}{3x+1} \geq 0$

By using wavy-curve method



$$\therefore x \leq -\frac{1}{3}, x \geq 2 \text{ but } x \neq -\frac{1}{3}$$

$$\therefore x < -\frac{1}{3}, x \geq 2, \therefore \left(-\infty, -\frac{1}{3}\right) \cup [2, \infty)$$

3. (d) Given that $\left(x^2 + \frac{1}{x^2}\right) + \left(x + \frac{1}{x}\right) < 0$

$$\left(x + \frac{1}{x}\right)^2 + \left(x + \frac{1}{x}\right) - 2 < 0$$

Substituting $x + \frac{1}{x} = y$, we get

$$y^2 + y - 2 < 0$$

$$\Rightarrow (y-1)(y+2) < 0$$

$$\therefore \text{either } y-1 < 0; y+2 > 0$$

$$\text{or } y+2 < 0; y-1 > 0.$$

$$\text{i.e., } y < 1, y > -2 \text{ or } y < -2; \quad y > 1$$

$\Downarrow$

not possible

Therefore, $-2 < y < 1$

$$\text{i.e., } -2 < \left(x + \frac{1}{x}\right) < 1.$$

4. (b) We have, $a - b + c < 1$ and $-a - b - c < 1$

By subtracting, we get

$$\Rightarrow 2a + 2c < 0$$

$$\text{or } a + c < 0 \quad \dots(1)$$

$$\text{Also, } -3a - 3b - 3c < 3$$

$$\text{And, } 9a + 3b + c < -4$$

By adding above two inequalities,

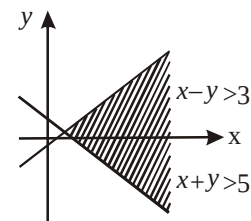
$$\Rightarrow 6a - 2c < -1 \quad \dots(2)$$

$$\text{From (1)} \times 2 + (2),$$

$$\text{We have, } 8a < -1$$

$$\text{or } a < -\frac{1}{8}.$$

5. (b) Plotting the given inequations on the same graph paper, we get the following solution region (shaded).



Solving, $x + y = 5$ and $x - y = 3$,

We get, $x = 4, y = 1$

$$\Rightarrow x > 4$$

6. (d) $|y - 4| \leq 6 \Rightarrow -2 \leq 10$

$$|y + 4| < 5 \Rightarrow -9 < y < 1$$

$\therefore$ Combining the two region we get.

$$-2 \leq y < 1$$

$\therefore$ Only one element satisfies the condition i.e., $= 0$

Alternative solution:

Go through the options.

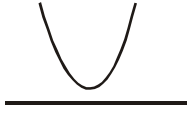
7. (c) $x^3 - x^2 + x - 1 > 0$

$$x^2(x-1) + (x-1) > 0$$

$$(x^2 + 1)(x-1) > 0$$

As, $x^2 + 1 > 0$ for all x , so, $x > 1$.

8. (b) $x^2 + 2ax + 10 - 3a > 0 \quad \forall x$



$$\begin{aligned} \Rightarrow D &< 0 \\ \Rightarrow 4a^2 - 4(10 - 3a) &< 0 \\ \Rightarrow a^2 + 3a - 10 &< 0 \\ \Rightarrow (a + 5)(a - 2) &< 0 \\ \Rightarrow a &\in (-5, 2) \end{aligned}$$

9. (b) Since a, b, c are three positive numbers. As we know that $AM \geq GM$

$$\Rightarrow \frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

Their reciprocals are $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$

Again applying

$AM \geq GM$

$$\Rightarrow \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} \geq \sqrt[3]{\frac{1}{abc}}$$

Multiplying eqs. (1) and (2), we get

$$(a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$$

10. (b) $\frac{9 \cdot 3^{2x} - 6 \cdot 3^x + 4}{9 \cdot 3^{2x} + 6 \cdot 3^x + 4} \cdot \frac{(3^{(x+1)})^2 - 2(3^{(x+1)}) + 4}{(3^{(x+1)})^2 + 2(3^{(x+1)}) + 4}$

$$\frac{t^2 - 2t + 4}{t^2 + 2t + 4} \quad (\text{where } t = 3^{x+1})$$

$$\text{Since, } \frac{1}{3} < \frac{x^2 - 2x + 4}{x^2 + 2x + 4} < 3$$

$\therefore$ From (1), the given expression lies between $\frac{1}{3}$ and 3.

11. (b) $\therefore p, q, r$ are in AP.

$$\therefore q = \frac{p+r}{2} \quad [\because p+r=2q]$$

For the real roots $q^2 - 4pr \geq 0$

$$\Rightarrow \left(\frac{p+r}{2} \right)^2 - 4pr \geq 0$$

$$\Rightarrow p^2 + r^2 - 14pr \geq 0$$

$$\Rightarrow \left(\frac{p}{r} \right)^2 - 14 \left(\frac{p}{r} \right) + 1 \geq 0$$

$$\Rightarrow \left(\frac{p}{r} - 7 \right)^2 \geq 48$$

$$\Rightarrow \left| \frac{p}{r} - 7 \right| \geq 4\sqrt{3}$$

12. (a) The given inequation is

$$4^{-x+0.5} - 7 \cdot 2^{-x} < 4, \quad x \in R$$

Let $2^{-x} = t$

$$\therefore 2t^2 - 7t < 4$$

$$\Rightarrow 2t^2 - 7t - 4 < 0$$

$$\Rightarrow (2t+1)(t-4) < 0$$

$$\Rightarrow -\frac{1}{2} < t < 4$$

$$\Rightarrow 0 < t < 4 \quad [\because t = 2^{-x} > 0]$$

$$\Rightarrow 0 < 2^{-x} < 4$$

$$\Rightarrow -2 < x < \infty$$

$$\therefore x \in (-2, \infty)$$

13. (b) We have, $\frac{|x+1|}{|x|} + |x+1| = \frac{|x+1|^2}{|x|}$

Since, $|a| + |b| = |a+b|$, iff $ab \geq 0$

$$\therefore \frac{(x+1)}{x} \cdot (x+1) \geq 0 \Rightarrow \frac{(x+1)^2}{x} \geq 0$$

$$\Rightarrow x > 0 \text{ or } \{-1\}$$

14. (d) Let us assume $a = 3, b = 4$, given that $a < b$ then the given equation becomes

$$(x-3)(x-4) - 1 = 0$$

$$x^2 - 7x + 11 = 0$$

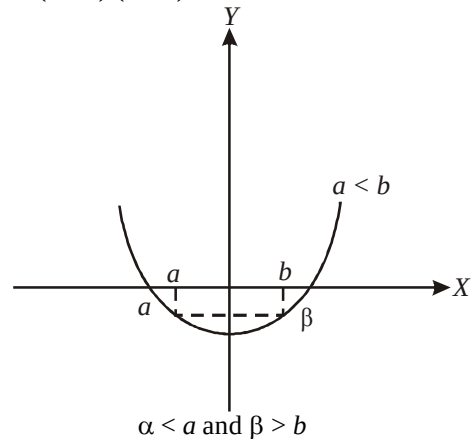
$$\therefore x = \frac{7 \pm \sqrt{49 - 44}}{2} \Rightarrow x = \frac{7 \pm \sqrt{5}}{2}$$

$$\Rightarrow x = \frac{7 + \sqrt{5}}{2} > 4 \text{ and } \frac{7 - \sqrt{5}}{2} < 3$$

Hence only option (d) is satisfied, hence correct.

Alternatively: Since $f(x) = (x-a)(x-b) - 1 = 0$

$$\Rightarrow (x-a)(x-b) = 1$$



$\alpha < a$ and $\beta > b$

It means x -axis is shifted to -1 unit below the original position. So it is clear from the graph that $\alpha < a$ and $\beta > b$

15. (b) Let $f(x) = x^2 - 2(4k-1)x + 15k^2 - 2k - 7$. Then $f(x) > 0 \Rightarrow D < 0$ [$\because$ Coefficient of $x^2 > 0$]

$$\Rightarrow 4(4k-1)^2 - 4(15k^2 - 2k - 7) < 0$$

$$\Rightarrow k^2 - 6k + 8 < 0 \Rightarrow 2 < k < 4$$



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LOGARITHMS

- Introduction
- Definition
- Laws of Logarithm

- Some Important Properties
- Characteristics and Mantissa
- Very Useful Results

INTRODUCTION

Logarithms were introduced by John Napier in the early 17th century as a means to simplify calculations. They were rapidly adopted by navigators, engineers and others to perform computations more easily.

Logarithm is a very important topic in every entrance exam. From CAT point of view, this topic is also important. Usually 1–2 problems are asked in CAT exams, but logarithm concept may be applied in other questions asked in CAT exams, too.

Students are advised not to be scared with the term logarithm since concept of logarithms is simple and this simple formula is sufficient to solve the questions easily.

DEFINITION

If $x = a^m$, then $\log_a x = m$, where 'a' and 'x' both are positive real numbers but 'a' not equal to 1 i.e., $a, x > 0$, but $a \neq 1$.

Here log is the short form of logarithm.

$\log_a x$ is read as log of x to the base a.

For example,

- (i) Since, $10 = 10^1$, $100 = 10^2$, $1000 = 10^3$, etc.
Hence, $\log_{10} 10 = 1$, $\log_{10} 100 = 2$, $\log_{10} 1000 = 3$, etc.
- (ii) Since, $8 = 2^3$, $16 = 2^4$, $32 = 2^5$, etc.
Hence, $\log_2 8 = 3$, $\log_2 16 = 4$, $\log_2 32 = 5$, etc.
- (iii) Since, $\frac{1}{8} = (2)^{-3}$, $\frac{1}{16} = (2)^{-4}$, etc.
Hence, $\log_2 \left(\frac{1}{8}\right) = -3$, $\log_2 \left(\frac{1}{16}\right) = -4$, etc.
- (iv) Since, $0.01 = (10)^{-2}$, $0.001 = (10)^{-3}$, etc.
Hence $\log_{10}(0.01) = -2$, $\log_{10}(0.001) = -3$, etc.

LAWS OF LOGARITHM

(i) $\log_a (m \times n) = \log_a m + \log_a n$

In general,

$$\log_a (m \times n \times p \times \dots) = \log_a m + \log_a n + \log_a p + \dots$$

For example:

$$\log_2 (4 \times 5 \times 6) = \log_2 4 + \log_2 5 + \log_2 6$$

Note that

$$\log_a m + \log_a n \neq \log_a (m + n)$$

(ii) $\log_a \left(\frac{m}{n}\right) = \log_a m - \log_a n$

For example:

$$\log_4 \left(\frac{8}{15}\right) = \log_4 8 - \log_4 15$$

Note that

$$\log_a m - \log_a n \neq \log_a (m - n)$$

(iii) $\log_a (m)^n = n \log_a m$

For example:

$$\log_3 (5)^4 = 4 \log_3 5$$

(iv) $\log_b a = \frac{\log_c a}{\log_c b}$ [Change of base rule]

For example,

$$\log_5 20 = \frac{\log_2 20}{\log_2 5} = \frac{\log_4 20}{\log_4 5} = \frac{\log_7 20}{\log_7 5} = \dots \text{ etc.}$$

(v) $\log_b a = \frac{1}{\log_a b}$

For example,

$$\log_{10} 100 = \frac{1}{\log_{100} 10}$$

(vi) $\log_b a \cdot \log_c b = \log_c a$ [Chain Rule]

In general,

$$\log_b a \cdot \log_c b \cdot \log_d c \dots \log_n m = \log_n a$$

For example,

$$\log_{24} 256 \cdot \log_{10} 24 \cdot \log_2 10 = \log_2 256$$

Illustration 1: $\log_a 4 + \log_a 16 + \log_a 64 + \log_a 256 = 10$. Then $a = ?$

- (a) 4 (b) 2
(c) 8 (d) 5

Solution: (a) The given expression is:

$$\log_a (4 \times 16 \times 64 \times 256) = 10$$

$$\text{i.e. } \log_a 4^{10} = 10$$

Thus, $a = 4$.

Illustration 2: Find x if $\log x = \log 1.5 + \log 12$

- (a) 12 (b) 8
(c) 18 (d) 15

Solution: (c) $\log x = \log 18 \Rightarrow x = 18$

Illustration 3: Find x , if $\log (2x - 2) - \log (11.66 - x) = 1 + \log 3$

- (a) 452/32 (b) 350/32
(c) 11 (d) 11.33

Solution: (c) $\log (2x - 2)/(11.66 - x) = \log 30$

$$\Rightarrow (2x - 2)/(11.66 - x) = 30$$

$$2x - 2 = 350 - 30x$$

$$\text{Hence, } 32x = 352 \Rightarrow x = 11.$$

Illustration 4: Solve for x :

$$\log \frac{75}{35} + 2 \log \frac{7}{5} - \log \frac{105}{x} - \log \frac{13}{25} = 0$$

- (a) 90 (b) 65
(c) 13 (d) 45

Solution: (c) $(75/35) \times (49/25) \times (x/105) \times (25/13) = 1$

$$\Rightarrow x = 13$$

SOME IMPORTANT PROPERTIES

- (i) $x = a^m \Rightarrow \log_a x = m$
and $\log_a x = m \Rightarrow x = a^m$
Here equation $x = a^m$ is in exponential form and equation $\log_a x = m$ is in logarithmic form.
- (ii) If base of log is not mentioned, then we assume the base as 10.
 $\therefore \log m = \log_{10} m$
log to the base 10 is called common log.
- (iii) Since, $10000 = (100)^2 = (10)^4$
 $\therefore \log_{100} 10000 = 2, \log_{10} 10000 = 4$
Thus value of log of a number on different bases is different i.e., value of log of a number depends on its base.
- (iv) (a) Since, $a = a^1$, hence $\log_a a = 1$
For example, $\log_5 5 = 1, \log_{10} 10 = 1$
Thus log of any number to the same base is always 1.
(b) Since, $1 = a^0$, hence, $\log_a 1 = 0$
For example, $\log_8 1 = 0$
Thus log of 1 to any base always equal to 0.

(v) $a^{(\log_a x)} = x$

For example,

$$20^{(\log_{20} 50)} = 50$$

(vi) (a) log of zero and negative numbers is not defined.

(b) Base of log is always positive but not equal to 1.

Illustration 5: Find x , if $0.01^x = 2$

- (a) $\log 2/2$ (b) $2/\log 2$
(c) $-2/\log 2$ (d) $-\log 2/2$

Solution: (d) $x = \log_{0.01} 2 = -\log 2/2$.

Illustration 6: If $\log 3 = .4771$, find $\log (.81)^2 \times \log \left(\frac{27}{10}\right)^{\frac{2}{3}} \div \log 9$.

- (a) 2.689 (b) -0.0552
(c) 2.2402 (d) 2.702

Solution: (b) $2 \log (81/100) \times 2/3 \log (27/10) \div \log 9$
 $= 2 [\log 3^4 - \log 100] \times 2/3 [(\log 3^3 - \log 10)] \div 2 \log 3$
 $= 2 [\log 3^4 - \log 100] \times 2/3 [(3 \log 3 - 1)] \div 2 \log 3$
Substitute $\log 3 = 0.4771 \Rightarrow -0.0552$.

CHARACTERISTICS AND MANTISSA

The integral part of the value logarithm is called characteristic and its decimal part is called mantissa. Logarithms to the base 10 are called common logarithms. The characteristic of common logarithm of a number greater than 1 is a number less by one than the number of digits in the integral part of the number.

For example,

$$\log_{10} 235.78 = 2.3725$$

Here $235.78 > 1$ and characteristic is equal to 2, which is clearly one less than the number of digits (3) of integral part (235) of 235.78.

$$\text{Mantissa} = .3725$$

Characteristic of a decimal number less than 1 is one greater than the number of consecutive zeros immediately after the decimal point and is a negative.

$$\log_{10} 0.00578 = \bar{3}.7620 \text{ i.e., } (-3 + .7620)$$

Here $0.00578 < 1$ and characteristic is equal to -3 , which is clearly one greater than the number of consecutive zeros (2) immediately after the decimal point and is negative.

$$\text{Mantissa} = 0.7620$$

Note that mantissa is always written as positive.

In case the value of the logarithm of a number is negative

like $\log_{10} 0.00578 = -2.2380$, then to make the mantissa positive, we subtract 1 from the integral part and add 1 to the decimal part.

$$\begin{aligned} \text{Thus } \log_{10} 0.00578 &= -2 - .2380 + 1 - 1 \\ &= (-2 - 1) + (1 - .2380) \\ &= -3 + .7620 \end{aligned}$$

The characteristic may be positive or negative but mantissa is always positive. When the characteristic is negative, it is represented by putting a bar on the number. Thus instead of writing the characteristic -3 , we write the characteristic as $\bar{3}$.

Hence, we write

$$\log_{10} 0.00578 = -3 + .7620$$

As $\log_{10} 0.00578 = \bar{3}.7620$ i.e., $(\bar{3} + .7620)$

Hence if we write

$$\log_{10} 0.00578 = -2.2380 \text{ (i.e., } -2 - .2380\text{)}$$

We cannot take -2 as characteristic and $-.2380$ as mantissa.

To find the characteristic and mantissa we write

$$\log_{10} 0.00578 = -2.2380$$

As $\log_{10} 0.00578 = \bar{3}.7620$ (i.e., $-3 + .7620$)

Then we have got -3 as characteristic and $.7620$ as mantissa.

VERY USEFUL RESULTS

- (i) Characteristic of common logarithm of any positive number less than 1 is negative.
- (ii) Characteristic of common logarithm of any number greater than 1 is positive.
- (iii) If the logarithm of x to any base b gives the characteristic n , then we can say that the possible number of integral values of x will be $a^{n+1} - a^n$.

Thus, if $\log_{10} x = 3.x_1 x_2 x_3 \dots$,

then the number of integral values that x can take is given by $10^{3+1} - 10^3 = 10000 - 1000 = 9000$.

This can be verified as follows:

log of base 10 gives a characteristic 3 for all 4 digits numbers with the lowest being 1000 and the highest being 9000. Hence, there are 9000 integral values possible for x .

- (iv) If $-n$ is the characteristic of $\log_{10} y$, then the number of zeroes between the decimal and the first significant number after the decimal will be $n - 1$.

Thus, if the log of a number y has a characteristic -4 , then the first three decimal places after the decimal point will be zeroes.

Hence, y will be in the form,

$$y = 0.000 y_1 y_2 y_3 \dots$$

where y_1 is the positive integer and $y_2, y_3, \dots$ are non-negative integers. Hence one of the value of y may be

$$y = 0.00025013$$

Practice Exercises



Foundation Level

- Find the value of $\log_5 10 \times \log_{10} 15 \times \log_{15} 20 \times \log_{20} 25$.
(a) $5/2$ (b) 5
(c) 2 (d) $\log\left(\frac{5}{2}\right)$
- If $\log_3 a = 4$, find value of a .
(a) 27 (b) 3
(c) 9 (d) 81
- Find the value of $\log \frac{9}{8} - \log \frac{27}{32} + \log \frac{3}{4}$
(a) 0 (b) 1
(c) 3 (d) $\log(3/4)$
- Evaluate : $3^{2-\log_3 5}$
(a) $\frac{9}{5}$ (b) 45
(c) $5/9$ (d) $9 \log_{35}$
- The value of $\left[\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)} \right]$ is equal to
(a) 1 (b) 2
(c) 3 (d) 4
- If $\log_2 [\log_3 (\log_2 x)] = 1$, then x is equal to
(a) 512 (b) 128
(c) 12 (d) 0
- Find the value of $\log_{27} \frac{1}{81}$
(a) $-4/3$ (b) -3
(c) -1 (d) $-1/3$
- Find the value of $\frac{8 \log_8 8}{2 \log_{\sqrt{8}} 8}$
(a) 1 (b) 2
(c) 3 (d) 4
- If $\log 2 = 0.30103$, find the number of digits in 2^{56} .
(a) 17 (b) 31
(c) 100 (d) 200
- $\log_3 (5+x) + \log_8 8 = 2^2$
(a) 22 (b) 33
(c) 11 (d) 44
- $\log 216\sqrt{6}$ to the base 6 is equal to
(a) 3 (b) $3/2$
(c) $7/2$ (d) None of these
- If $\log_k x \times \log_5 k = 3$, then find the value of x .
(a) k^5 (b) $5k^3$
(c) 243 (d) 125
- $\log_a \left(\frac{m}{n} \right)$ is equal to
(a) $\log_a (m-n)$ (b) $\log_a m - \log_a n$
(c) $\frac{(\log_a m)}{n}$ (d) $\log_a m \div \log_a n$
- If $\log_5 [\log_3 (\log_2 x)] = 1$ then x is
(a) 2^{234} (b) 243
(c) 2^{243} (d) None of these
- The value of $\left[3 \log \left(\frac{81}{80} \right) + 5 \log \left(\frac{25}{24} \right) + 7 \log \left(\frac{16}{15} \right) \right]$ is
(a) $\log 3$ (b) $\log 5$
(c) $\log 7$ (d) $\log 2$
- If $\log_{10} a + \log_{10} b = c$, then the value of a is
(a) bc (b) $\frac{c}{b}$
(c) $\frac{(10)^c}{b}$ (d) $\frac{10b}{c}$
- If $\log_y x = 8$ and $\log_{10y} 16x = 4$, then find the value of y .
(a) 1 (b) 2
(c) 3 (d) 5

18. $\log 0.0867 = ?$
 (a) $\log 8.67 + 2$ (b) $\log 8.67 - 2$
 (c) $\frac{\log 867}{1000}$ (d) $-2 \log 8.67$
19. Find x , if $0.01^x = 2$
 (a) $\log 2/2$ (b) $2/\log 2$
 (c) $-2/\log 2$ (d) $-\log 2/2$
20. If $2^x \cdot 3^{2x} = 100$, then the value of x is
 ($\log 2 = 0.3010$, $\log 3 = 0.4771$)
 (a) 2.3 (b) 1.59
 (c) 1.8 (d) 1.41
21. The mantissa of $\log 3274$ is 0.5150. The value of $\log (0.3274)$ is
 (a) $\bar{1}.5150$ (b) 1.5150
 (c) $\bar{2}.5150$ (d) None of these
22. If $\log_{10} a = b$, then find the value of 10^{3b} in terms of a .
 (a) a^3 (b) $3a$
 (a) $a \times 1000$ (d) $a \times 100$
23. If $\log 3 = 0.4771$, find $\log (0.81)^2 \times \log \left(\frac{27}{10} \right)^{\frac{2}{3}} \div \log 9$.
 (a) 2.689 (b) -0.0552
 (c) 2.2402 (d) 2.702
24. $\log_{10} 10 + \log_{10} 10^2 + \dots + \log_{10} 10^n$
 (a) $n^2 + 1$ (b) $n^2 - 1$
 (c) $\left(\frac{n^2 + n}{3} \right)$ (d) $\frac{n^2 + n}{2}$
25. If a, b and c are distinct positive number ($\neq 1$) such that
 $(\log_b a \log_c a - \log_a a) + (\log_a b \log_c b - \log_b b) + (\log_a c \log_b c - \log_c c) = 0$. What is the value of abc ?
 (a) 1 (b) 0
 (c) -1 (d) None of these
26. What is the value of x in the following expression
 $\log_{3/4} \log_2 (x^2 + 7) \log_{1/4} (x^2 + 7)^{-1} = -2$?
 (a) $+3$ (b) -3
 (c) ± 3 (d) None of these

Standard Level

- If $\log_{10} 2 = 0.3010$, then the value of $\log_{10} 80$ is :
 (a) 1.9030 (b) 1.6020
 (c) 3.9030 (d) 2.9030
- Which of the following is true ?
 (a) $\log_{17} 275 = \log_{19} 375$ (b) $\log_{17} 275 < \log_{19} 375$
 (c) $\log_{17} 275 > \log_{19} 375$ (d) Cannot be determined
- The value of $\log_{2\sqrt{3}} (1728)$ is
 (a) 3 (b) 5
 (c) 6 (d) 9
- If $\log 2 = 0.30103$, then the number of digits in 4^{50} is
 (a) 30 (b) 31
 (c) 100 (d) 200
- Number of digits in 60^{12}
 (a) 25 (b) 22
 (c) 23 (d) 24
- Find the value of $\log_{3^2} 5^4 \times \log_{5^2} 3^4$.
 (a) 5 (b) 3
 (c) 4 (d) 2
- If $a = b^x$, $b = c^y$ and $c = a^z$, then the value of xyz is equal to
 (a) -1 (b) 0
 (c) 1 (d) abc
- If $\log_7 \log_5 (\sqrt{x} + 5 + \sqrt{x}) = 0$, find the value of x .
 (a) 1 (b) 0
 (c) 2 (d) None of these
- If $\log_3 [\log_3 [\log_3 x]] = \log_3 3$, then what is the value of x ?
 (a) 3 (b) 27
 (c) 3^9 (d) 3^{27}
- What is $\log \left(a + \sqrt{a^2 + 1} \right) + \log \left(\frac{1}{a + \sqrt{a^2 + 1}} \right)$ is equal to?
 (a) 1 (b) 0
 (c) 2 (d) $\frac{1}{2}$
 $\frac{1}{\log_a bc + 1} + \frac{1}{\log_b ac + 1} + \frac{1}{\log_c ab + 1}$ is equal to
 (a) 1 (b) 2
 (c) 0 (d) abc
- If $p = \log_3 5$ and $q = \log_{17} 25$, which one of the following is correct?
 (a) $p < q$ (b) $p = q$
 (c) $p > q$ (d) can't say
- If $\log_{10} x = a$, $\log_{10} y = b$ and $\log_{10} z = c$, then $\text{antilog}(pa + qb - rc) = ?$
 (a) $\frac{pxqy}{rz}$ (b) $px + qy - rz$
 (c) $\frac{x^p y^q}{z^r}$ (d) $x^p y^q z^r$
- If a, b, c are three consecutive integers, then $\log(ac + 1)$ has the value
 (a) $\log b$ (b) $(\log b)^2$
 (c) $2 \log b$ (d) $\log 2b$
- Find the value of $(7^3)^{-2 \log_7 8}$
 (a) 8^{-7} (b) 6^{-8}
 (c) 8^{-6} (d) None of these
- Find the values of x satisfying $\log_{x^2+6x+8} \log_{2x^2+2x+3} (x^2 - 2x) = 0$ is
 (a) 0 (b) -1
 (c) 2 (d) -3
- If $\log_{0.3} (x - 1) < \log_{0.09} (x - 1)$, then x lies in the interval
 (a) $(2, \infty)$ (b) $(1, 2)$
 (c) $(-2, -1)$ (d) None of these
- If $(\log_3 x)^2 + \log_3 x < 2$, then which one of the following is correct ?
 (a) $0 < x < \frac{1}{9}$ (b) $\frac{1}{9} < x < 3$
 (c) $3 < x < \infty$ (d) $\frac{1}{9} \leq x \leq 3$
- What is the value of x in the following expression? $x + \log_{10} (1 + 2^x) = x \log_{10} 5 + \log_{10} 6$.
 (a) 1 (b) 0
 (c) -1 (d) 3
- How many values of $x (x > 1)$ satisfy the following equation: $\log_2 x \times \log_4 x \times \log_6 x = \log_2 x \cdot \log_4 x + \log_2 x \cdot \log_6 x + \log_4 x \cdot \log_6 x$?
 (a) 0 (b) 1
 (c) 2 (d) More than 2
- The equation $\log_5 x + \left(\log_{(x^2+3)} 25 \right)^{-1} = \log_{25} 10$ has
 (a) no solution (b) one solution
 (c) two solutions (d) four solutions
- If $\log_{10} x - \log_{10} \sqrt{x} = 2 \log_x 10$, then a possible value of x is given by
 (a) 10 (b) 1/100
 (c) 1/1000 (d) None of these

23. What is the value of $\frac{\log_{27} 9 \times \log_{16} 64}{\log_4 \sqrt{2}}$?
 (a) $\frac{1}{6}$ (b) $\frac{1}{4}$
 (c) 8 (d) 4
24. If $(\log_x x)(\log_3 2x)(\log_{2x} y) = \log_x x^2$, then what is the value of y ?
 (a) $9/2$ (b) 9
 (c) 18 (d) 27
25. What is the value of $\log_{10} \left(\frac{9}{8} \right) - \log_{10} \left(\frac{27}{32} \right) + \log_{10} \left(\frac{3}{4} \right)$?
 (a) 3 (b) 2
 (c) 1 (d) 0
26. The value of $25^{(-1/4 \log_5 25)}$ is equal to
 (a) $\frac{1}{5}$ (b) $-\frac{1}{25}$
 (c) -25 (d) None of these
27. If $\log_{10} x, \log_{10} y, \log_{10} z$ are in AP then x, y, z are in
 (a) AP (b) GP
 (c) HP (d) None of these
28. Find the value of $\frac{\log \sqrt{27} + \log \sqrt{8} - \log \sqrt{125}}{\log 6 - \log 5}$
 (a) $\frac{2}{3}$ (b) $\frac{1}{3}$
 (c) $\frac{3}{2}$ (d) None of these
29. Find the value of x and y respectively for $\log_{10} (x^2 y^3) = 7$ and $\log_{10} (x/y) = 1$
 (a) $x = 10, y = 100$ (b) $x = 100, y = 10$
 (c) $x = 10, y = 20$ (d) None of these
30. Arrange the following in an ascending order
 A $\log_7 2401$, B $\log_7 \sqrt{7} \times 343$, C $= \log_{\sqrt{6}} 216$, D $= \log_2 32$
 (a) ABCD (b) BDCA
 (c) BDAD (d) BADC
31. If $3 \log_{(3x^2)} 27 - 2 \log_{(3x)} 9 = 0$, then what is the value of x ?
 (a) $1/243$ (b) $1/7$
 (c) $1/49$ (d) None of these
32. If $\log_k N = 6$, and $\log_{25k} (8N) = 3$, then k is
 (a) 12.5 (b) $(12.5)^2$
 (c) $\left(\frac{5}{2} \right)^{2/3}$ (d) $(12.5)^{2/3}$
33. What is the value of $\log_3 2, \log_4 3, \log_5 4, \dots, \log_{16} 15$?
 (a) $1/2$ (b) $1/3$
 (c) $2/3$ (d) $1/4$
34. Find the value of x , if $\log (2x - 2) - \log (11.66 - x) = 1 + \log 3$
 (a) $452/32$ (b) $350/32$
 (c) 11 (d) 11.33
35. If $\log_4 5 = a$ and $\log_5 6 = b$ then what is the value of $\log_3 2$?
 (a) $\frac{1}{2a+1}$ (b) $\frac{1}{2b+1}$
 (c) $2ab + 1$ (d) $\frac{1}{2ab-1}$
36. What is the value of x if
 $\log_3 x + \log_9 x + \log_{27} x + \log_{81} x = \frac{25}{4}$?
 (a) 9 (b) 27
 (c) 81 (d) None of these
37. What is the value of $\log_{32} 27 \times \log_{243} 8$?
 (a) $\frac{\log 9}{\log 4}$ (b) $\frac{\log 3}{\log 2}$
 (c) $\log 27$ (d) None of these
38. What is the value of x in the following expression?
 $\log_7 \log_5 [\sqrt{(x+5)} + \sqrt{x}] = 0$
 (a) 1 (b) 2
 (c) 3 (d) 4
39. If $x = \log_a (bc), y = \log_b (ca)$ and $z = \log_c (ab)$, then which of the following is equal to 1?
 (a) $x + y + z$
 (b) $(1+x)^{-1} + (1+y)^{-1} + (1+z)^{-1}$
 (c) xyz
 (d) None of these
40. Express $\log \frac{\sqrt[3]{a^2}}{b^5 \sqrt{c}}$ or $\frac{a^{2/3}}{b^5 \sqrt{c}}$ in terms of $\log a, \log b$ and $\log c$.
 (a) $\frac{3}{2} \log a + 5 \log b - 2 \log c$
 (b) $\frac{2}{3} \log a - 5 \log b - \frac{1}{2} \log c$
 (c) $\frac{2}{3} \log a - 5 \log b + \frac{1}{2} \log c$
 (d) $\frac{3}{2} \log a + 5 \log b - \frac{1}{2} \log c$
41. If $\log 2 = 0.301, \log 3 = 0.477$, find the number of digits in $(108)^{10}$.
 (a) 21 (b) 27
 (c) 20 (d) 18
42. $\log a^n / b^n + \log b^n / c^n + \log c^n / a^n$
 (a) 1 (b) n
 (c) 0 (d) 2
43. $\log_2 (9 - 2^x) = 10^{\log (3-x)}$, solve for x .
 (a) 0 (b) 3
 (c) Both (a) and (b) (d) 0 and 6

Expert Level

- What is the number of zeroes coming immediately after the decimal point in the value of $(0.2)^{25}$? ($\log_{10} 2 = 0.30103$)
(a) 15 (b) 16 (c) 17 (d) None of these
- If $\log_{10} \left[1 - \left\{ 1 - (1 - x^2)^{-1} \right\}^{-1} \right]^{-1/2} = 1$, then which of the following is the value of x .
(a) 1 (b) 2 (c) 10 (d) None of these
- $N = n!$, where $n > 2$. Find the value of $(\log_2 N)^{-1} + (\log_3 N)^{-1} + (\log_4 N)^{-1} + \dots + (\log_n N)^{-1}$.
(a) 0 (b) 1 (c) 10 (d) N
- What is the value of $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \dots + \frac{1}{\log_{40} n}$?
(a) $\frac{1}{\log_{(40!)} n}$ (b) $\log_{(40!)} n$ (c) 1 (d) None of these
- Which of the following options represents the value of $\log \sqrt{128}$ to the base 0.625?
(a) $\frac{2 + \log_8 2}{\log_8 5 - 1}$ (b) $\frac{\log_8 128}{2 \log_8 0.625}$ (c) $\frac{2 + \log_8 2}{2(\log_8 5 - 1)}$ (d) Both (b) and (c)
- Let $u = (\log_2 x)^2 - 6 \log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$, has
(a) no solution for x (b) exactly one solution for x (c) exactly two distinct solutions for x (d) exactly three distinct solutions for x
- If $5^x = (0.5)^y = 1000$, then the value of $\left(\frac{1}{x} - \frac{1}{y} \right)$ is
(a) $\frac{1}{4}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) 1
- The least value of expression $2 \log_{10} x - \log_x (1/100)$ for $x > 1$ is?
(a) 2 (b) 3 (c) 4 (d) 5
- What is the value of P if $\log_e 2 \cdot \log_p 625 = \log_{10} 16 \cdot \log_e 10$?
(a) 2 (b) 4 (c) 5 (d) 7
- If $\log_{12} 27 = a$, then $\log_6 16$ is
(a) $(3 - a)/4(3 + a)$ (b) $(3 + a)/4(3 - a)$ (c) $4(3 + a)/(3 - a)$ (d) $4(3 - a)/(3 + a)$
- If $2[\log(x + y) - \log 5] = \log x + \log y$, then what is the value of $x^2 + y^2$?
(a) $20 - xy$ (b) $23xy$ (c) $25 - xy$ (d) $28xy$
- The number of solutions of the equation $\log_{x-3} (x^3 - 3x^2 - 4x + 8) = 3$ is
(a) 1 (b) 2 (c) 3 (d) 4
- What is the value of 'a' if it is known that the fourth term of the expansion of $(a + a^{\log a})^5$ is equal to 10,00,000?
(a) 10 (b) -5/2 (c) 2 (d) 100
- Solve : $3^{2x-1} = 4^{x+2}$.
(a) 2.774 (b) 3.774 (c) 1.774 (d) 4.774
- Solve for x : $\log_5 (5^{1/x} + 125) = \log_5 6 + 1 + 1/2x$
(a) 1/4, 1/6 (b) 1/2, 1/4 (c) 1/3, 1/4 (d) 1, 1/2
- If $\log(0.57) = \bar{1}.756$, then the value of $\log 57 + \log (0.57)^3 + \log \sqrt{0.57}$ is
(a) 0.902 (b) $\bar{2}.146$ (c) 1.902 (d) $\bar{1}.146$
- If $\log_2 [\log_7 (x^2 - x + 37)] = 1$, then what could be the value of x ?
(a) 3 (b) 5 (c) 4 (d) None of these
- If $\frac{1}{3} \log_3 M + 3 \log_3 N = 1 + \log_{0.008} 5$, then
(a) $M^9 = \frac{9}{N}$ (b) $N^9 = \frac{9}{M}$ (c) $M^3 = \frac{3}{N}$ (d) $N^9 = \frac{3}{M}$
- If $\log_3 2, \log_3 (2^x - 5), \log_3 (2^x - 7/2)$ are in arithmetic progression, then the value of x is equal to
(a) 5 (b) 4 (c) 2 (d) 3

20. If $x \geq y$ and $y > 1$, then the value of the expression $\log_x \left(\frac{x}{y} \right) + \log_y \left(\frac{y}{x} \right)$ can never be
- (a) -1 (b) -0.5
(c) 0 (d) 1
21. If $\log_y x = (a \cdot \log_z y) = (b \cdot \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?
- (a) $(-2, 1/2)$ (b) $(1, 1)$
(c) $(0.4, 2.5)$ (d) $(2, 2)$
22. What is the value of $\frac{\log_{\sqrt{\alpha\beta}}(H)}{\log_{\sqrt{\alpha\beta\gamma}}(H)}$?
- (a) $\log_{\alpha\beta}(\alpha)$ (b) $\log_{\alpha\beta\gamma}(\alpha\beta)$
(c) $\log_{\alpha\beta}(\alpha\beta\gamma)$ (d) $\log_{\alpha\beta}(\beta)$
23. $\frac{1}{\log_2 a} + \frac{1}{\log_4 a} + \frac{1}{\log_8 a} + \dots$ to n terms $= \frac{n(n+1)}{k}$, then k is equal to
- (a) $\log_2 a^2$ (b) $\log_2 \left(\frac{a}{2} \right)$
(c) $\log_a 2$ (d) None of these
24. If $(x^4 - 2x^2y^2 + y^4)^{a-1} = (x-y)^{2a}(x+y)^{-2}$, then the value of a is
- (a) $(x^2 - y^2)$ (b) $\frac{x-y}{x+y}$
(c) $\frac{\log(x+y)}{\log(x-y)}$ (d) $\frac{\log(x-y)}{\log(x+y)}$
25. If $u = v^2 = w^3 = z^4$, then $\log_u(uvwz)$ is equal to
- (a) $1 + \frac{1}{2} - \frac{1}{3} - \frac{1}{4}$ (b) 24
(c) $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}$ (d) $\frac{1}{24}$
26. The first term and the last term of a GP are a and k respectively. If the number of terms be n , then n is equal to ($r \rightarrow$ common ratio)
- (a) $1 - \frac{\log k - \log a}{\log r}$ (b) $1 + \frac{\log a + \log k}{\log r}$
(c) $1 + \frac{\log k - \log a}{\log r}$ (d) $\log r = \log k - \log a$
27. If $y = a^{\left(\frac{1}{1-\log_a x} \right)}$ and $z = a^{\left(\frac{1}{1-\log_a y} \right)}$, then x is equal to:
- (a) $1 + \log yz$ (b) $\frac{1}{1-\log_a z}$
(c) $a^{\left(\frac{1}{1-\log_a z} \right)}$ (d) $\frac{y}{z}$
28. Solve the following equations for x and y
- $\log_{100} |x+y| = \frac{1}{2}$, $\log_{10} y - \log_{10} |x| = \log_{100} 4$
- (a) $\left(\frac{8}{3}, \frac{16}{3} \right)$ (b) $\left(\frac{10}{3}, \frac{20}{3} \right)$
(c) $\left(-\frac{10}{3}, \frac{20}{3} \right)$ (d) None of these
29. The greatest possible value of n could be if $9^n < 10^8$, given that $\log 3 = 0.4771$ and $n \in \mathbb{N}$:
- (a) 7 (b) 8
(c) 9 (d) 10
30. Find the solution set of x , for the given inequality $\log_m n > 1$, where $m = \frac{(5)^2 - (x)^2}{(4)^2}$ and $n = \frac{2(12-x) - x^2}{14}$
- (a) $(-5, 4)$ (b) $\{-3, 3\}$
(c) $(-3, 1) \cup (3, 4)$ (d) None of these
31. The equation $x^4 \frac{3(\log_2 x)^2 + \log_2 x - 5}{4} \sqrt{2}$ has
- (a) at least two integral roots
(b) exactly three real solutions
(c) exactly two irrational solutions
(d) complex roots
32. If $\log_{1/2} \frac{x^2 + 6x + 9}{2(x+1)} < -\log_2(x+1)$, then x lies in the interval
- (a) $(-1, -1 + 2\sqrt{2})$ (b) $(1 - 2\sqrt{2}, 2)$
(c) $(-1, \infty)$ (d) None of these
33. What is the value of x in the following expansion?
- $1 - \log_{10} 5 = \frac{1}{3} \left(\log_{10} \frac{1}{2} + \log_{10} x + \frac{1}{3} \log_{10} 5 \right)$
- (a) 1 (b) $16 \times 5^{-1/3}$
(c) $16 \times 5^{1/3}$ (d) None of these
34. If $\log_{0.04}(x-1) \geq \log_{0.2}(x-1)$ then x belongs to the interval
- (a) $(1, 2]$ (b) $(-\infty, 2]$
(c) $[2, +\infty)$ (d) None of these
35. If $\log_{10} 3 = x$ and $\log_{30} 5 = y$, then $\log_8 30$ is equal to
- (a) $3(1-x-y)$ (b) $\frac{1}{3(1-x-y)}$
(c) $\frac{3}{(1-x-y)}$ (d) $\frac{1-x-y}{3}$

Test Yourself

- If $\log_7 \log_5 (\sqrt{x} + 5 + \sqrt{x}) = 0$, find the value of x .
(a) 1 (b) 0
(c) 2 (d) None of these
- $\left(\frac{21}{10}\right)^x = 2$. Then $x = ?$
(a) $\frac{\log 2}{\log 3 + \log 7 - 1}$ (b) $\frac{\log 2}{\log 3 + \log 7 + 1}$
(c) $\frac{\log 3}{\log 2 + \log 7 - 1}$ (d) $\frac{\log 2}{\log 3 - \log 7 + 1}$
- $\frac{\log_8 17}{\log_9 23} - \frac{\log_{2\sqrt{2}} 17}{\log_3 23} =$
(a) $\frac{17}{8}$ (b) $\frac{2}{3}$
(c) $\frac{8}{9}$ (d) 0
- The smallest positive value of x satisfying $\log_{\cos x} \sin x + \log_{\sin x} \cos x = 2$ is
(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$
- Simplify :
 $\left[\frac{1}{\log_{xy}(xyz)} + \frac{1}{\log_{yz}(xyz)} + \frac{1}{\log_{zx}(xyz)} \right]$
(a) 4 (b) 5
(c) 3 (d) 2
- If $\log_a b = \frac{1}{2}$, $\log_b c = \frac{1}{3}$ and $\log_c a = \frac{K}{5}$, then the value of K is
(a) 25 (b) 35
(c) 30 (d) 20
- If $a = \log_{24} 12$, $b = \log_{36} 24$, $C = \log_{48} 36$. Then $1 + abc$ is equal to
(a) $2ac$ (b) $2bc$
(c) $2ab$ (d) None of these
- Which of the following is true?
(a) $\log_{17} 275 = \log_{19} 375$
(b) $\log_{17} 275 < \log_{19} 375$
(c) $\log_{17} 275 > \log_{19} 375$
(d) None of these
- If $A = \log_2 \log_2 \log_4 256 + 2 \log_{\sqrt{2}} 2$, then A is equal to
(a) 2 (b) 3
(c) 5 (d) 7
- Calculate: $\log_2 (2/3) + \log_4 (9/4)$
(a) 1 (b) 2
(c) 0 (d) $3/2$
- If $\log_3 2$, $\log_3 (2^x - 5)$, $\log_3 (2^x - 7/2)$ are in arithmetic progression, then the value of x is equal to
(a) 5 (b) 4
(c) 2 (d) 3
- Which of the following options represents the value of $\log_{\sqrt{128}}$ to the base. 625?
(a) $\frac{2 + \log_8 2}{\log_8 5 - 1}$ (b) $\frac{\log_8 128}{2 \log_8 0.625}$
(c) $\frac{2 + \log_8 2}{2(\log_8 5 - 1)}$ (d) Both (b) and (c)
- If 1, $\log_{81} (3^x + 48)$ and $\log_9 \left(3^x - \frac{8}{3} \right)$ are in A.P., then find x
(a) 1 (b) 2
(c) 9 (d) 3
- If $\log 2 = 0.3010$ then the Arithmetic mean of $\log 40$ and $\log 5$ is
(a) 2.3010 (b) 30.10
(c) 3.0103 (d) 1.1505
- Which of the following is true?
(a) $\log_{11} 1650 > \log_{13} 1950$
(b) $\log_{11} 1650 < \log_{13} 1950$
(c) $\log_{11} 1650 = \log_{13} 1950$
(d) None of these

Hints & Solutions

Foundation Level

1. (c) $\log_5 10 \times \log_{10} 15 \times \log_{15} 20 \times \log_{20} 25$
 $= (\log 10 / \log 5) \times (\log 15 / \log 10) \times (\log 20 / \log 15)$
 $\times (\log 25 / \log 20)$
 $= \log 25 / \log 5 = 2 \log 5 / \log 5 = 2$.
2. (d) $\because \log_3 a = 4 \therefore 3^4 = a \Rightarrow a = 81$
3. (a) Given $\log \frac{9}{8} - \log \frac{27}{32} + \log \frac{3}{4} = \log \left(\frac{9}{8} \div \frac{27}{32} \right) + \log \frac{3}{4}$
 $= \log \left(\frac{9}{8} \times \frac{32}{27} \times \frac{3}{4} \right) = \log 1 = 0 \quad \therefore \log_a 1 = 0$
4. (a) Given : $3^{2-\log_3 5} = 3^2 \cdot 3^{-\log_3 5} \quad (\because a^{m+n} = a^m \cdot a^n)$
 $= 9 \cdot 3^{\log_3 5^{-1}} \quad 9 \times 5^{-1} = \frac{9}{5}$
5. (b) Given expression
 $= \log_{xyz} (xy) + \log_{xyz} (yz) + \log_{xyz} (zx)$
 $= \log_{xyz} (xy \times yz \times zx) = \log_{xyz} (xyz)^2$
 $= 2 \log_{xyz} (xyz) = 2 \times 1 = 2$
6. (a) $\log_2 [\log_3 (\log_2 x)] = 1 = \log_2 2$
 $\Rightarrow \log_3 (\log_2 x) = 2$
 $\Rightarrow \log_2 x = 3^2 = 9$
 $\Rightarrow x = 2^9 = 512$
7. (a) Let $\log_{27} \left(\frac{1}{81} \right) = x$
 $\therefore (27)^x = \frac{1}{81}$
 $\therefore 3^{3x} = 3^{-4} \Rightarrow 3x = -4 \Rightarrow x = -\frac{4}{3}$
8. (b) $\frac{8 \log_8 8}{2 \log_{\sqrt{8}} 8} = \frac{8 \times 1}{2 \log_{\sqrt{8}} (\sqrt{8})^2} = \frac{8}{4 \log_{\sqrt{8}} \sqrt{8}} = \frac{8}{4} = 2$
9. (a) $\log (2^{56}) = 56 \log 2 = (56 \times 0.30103) = 16.85768$.
 Its characteristic is 16. Hence, the number of digits in 2^{56} is 17.
10. (a) $\log_3 (5+x) + \log_8 8 = 2^2$
 $\log_3 (5+x) + 1 = 4$
 $\log_3 (5+x) = 3$
 $3^3 = 5+x$
 $5+x = 27$
 $x = 27-5 = 22$.
11. (c) $\log_6 216\sqrt{6} = \log_6 (6)^3 (6)^{1/2} = \log_6 (6)^{7/2}$
 $= \frac{7}{2} \log_6 6 = \frac{7}{2} \quad (\because \log_a a = 1)$
12. (d) Given, $\log_5 k \log_k x = 3$
 $\frac{\log k}{\log 5} \cdot \frac{\log x}{\log k} = 3 \Rightarrow \frac{\log x}{\log 5} = 3$
 $\Rightarrow \log x = 3 \log 5 \Rightarrow \log x = \log 5^3$
 $\Rightarrow x = 5^3 \Rightarrow x = 125$
13. (b) $\log_a \left(\frac{m}{n} \right) = \log_a m - \log_a n$
14. (c) $\log_5 [\log_3 (\log_2 x)] = 1 = \log_5 5$
 $\Rightarrow \log_3 (\log_2 x) = 5 = \log_3 3^5$
 $\Rightarrow \log_2 x = 3^5 = 243$
 $\Rightarrow 2^{243} = x$
15. (d) $3 \log \frac{81}{80} + 5 \log \frac{25}{24} + 7 \log \frac{16}{15}$
 $= \log \left[\left(\frac{81}{80} \right)^3 \times \left(\frac{25}{24} \right)^5 \times \left(\frac{16}{15} \right)^7 \right]$
 $= \log \left(\frac{3^{12} \times 5^{10} \times 2^{28}}{2^{12} \times 5^3 \times 2^{15} \times 3^5 \times 3^7 \times 5^7} \right)$
 $= \log 2$
16. (c) $\log_{10} a + \log_{10} b = c$
 $\Rightarrow \log_{10} (ab) = c$
 $\Rightarrow 10^c = ab$
 $\Rightarrow a = \frac{(10)^c}{b}$
17. (d) $\log_y x = 8 \Rightarrow y^8 = x \quad \dots(1)$
 $\log_{10} y = 4 \Rightarrow 10^4 y^4 = 16x \quad \dots(2)$
 Dividing (2) by (1) $10^4 y^{-4} = 16 \Rightarrow y = 5$
18. (b) $\log 0.0867 = \log (8.67/100) = \log 8.67 - \log 100$
 $\log 8.67 - 2$

19. (d) $x = \log_{0.01} 2 = -\log 2/2$.
20. (b) $2^x \cdot 3^{2x} = 100$
 $\Rightarrow x \log 2 + 2x \log 3 = \log 100$
 $\Rightarrow x(0.3010 + 2 \times 0.4771) = 2$
 $\Rightarrow x = \frac{1}{1.2552} = 1.59$
21. (a) Since, 0.3274 gives characteristic $\bar{1}$. Therefore value of $\log(0.3274) = \bar{1}.5150$
22. (a) $\log_{10} a = b \Rightarrow 10^b = a \Rightarrow$ By definition of logs.
 Thus $10^{3b} = (10^b)^3 = a^3$.
23. (b) $2 \log(81/100) \times 2/3 \log(27/10) \div \log 9$
 $= 2 [\log 3^4 - \log 100] \times 2/3 [(\log 3^3 - \log 10)] \div 2 \log 3$
 $= 2 [\log 3^4 - \log 100] \times 2/3 [(3 \log 3 - 1)] \div 2 \log 3$
 Substitute $\log 3 = 0.4771 \Rightarrow -0.0552$
24. (d) $\log_{10} 10 + \log_{10} 10^2 + \dots + \log_{10} 10^n$
 $= 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$
25. (a) $\log_b a \log_e a + \log_a b \log_a b + \log_a c \log_b c = 3$
 $\Rightarrow \frac{(\log a)^2}{\log b \cdot \log c} + \frac{(\log b)^2}{\log a \log c} + \frac{(\log c)^2}{\log a \log b} = 3$
 $\Rightarrow \frac{(\log a)^3 + (\log b)^3 + (\log c)^3}{\log a \cdot \log b \cdot \log c} = 3$
 $\Rightarrow (\log a)^3 + (\log b)^3 + (\log c)^3 = 3 \log a \cdot \log b \cdot \log c$
 We know, $x^3 + y^3 + z^3 = 3xyz$ when $x + y + z = 0$
 $\log a + \log b + \log c = 0 \Rightarrow \log abc = 0 \Rightarrow abc = 1$.
26. (c) Go through the options.

Standard Level

1. (a) $\log_{10} 80 = \log_{10} (8 \times 10) = \log_{10} 8 + \log_{10} 10$
 $= \log_{10} 2^3 + 1$
 $= (3 \log_{10} 2) + 1 = (3 \times 0.3010) + 1 = 1.9030$
2. (b) $19^2 = 361$; $375 > 361$
 and $17^2 = 289 = 289 > 275$
 $\therefore \log_{17} 275 : \log_{17} (289 - 14)$ – Which will be less than 2.
 $\log_{19} 375 = \log_{19} (361 + 14)$ – Which will be greater than 2.
 $\therefore \log_{17} 275 < \log_{19} 375$.
3. (c) Let $\log_{2\sqrt{3}} (1728) = x$.
 Then, $(2\sqrt{3})^x = 1728 = (12)^3$
 $= \left[(2\sqrt{3})^x \right]^3 = (2\sqrt{3})^6$
 $\therefore x = 6$, i.e., $\log_{2\sqrt{3}} (1728) = 6$.

4. (b) $\log 4^{50} = 50 \log 4 = 50 \log 2^2$
 $= (50 \times 2) \log 2 = 100 \times \log 2$
 $= (100 \times 0.30103) = 30.103$
 So the number of digits = 31.
5. (b) Let $x = 60^{12}$
 Applying log on both sides.
 $\log x = 12 \log 60 = 12 [\log (6 \times 10)] = 12 [\log 6 + \log 10]$
 $= 12 [\log (2 \times 3) + 1] = 12 [\log 2 + \log 3 + 1]$
 $= 12 [0.303 + 0.4771 + 1] = 12 [1.7801] = 21.3612$
 In 10^2 we have 3 digits.
 10^3 we have 4 digits.

 10^n we have $(n + 1)$ digit
 So $\log_{10} x = 21.3612$ or $x = 10^{21.3612}$
 $\Rightarrow$ Number of digits = 22.
6. (c) $\log_{3^2} 5^4 \times \log_{5^2} 3^4$
 $= \frac{4}{2} \log_3 5 \times \frac{4}{2} \log_5 3$
 $= 4 (\log_3 5 \times \log_5 3) = 4 \times 1 = 4$.
7. (c) $a = b^x, b = c^y, c = a^z$
 $\Rightarrow x = \log_b a, y = \log_c b, z = \log_a c$
 $\Rightarrow xyz = (\log_b a) \times (\log_c b) \times (\log_a c)$
 $\Rightarrow xyz = \left(\frac{\log a}{\log b} \times \frac{\log b}{\log c} \times \frac{\log c}{\log a} \right) = 1$.
8. (b) $\log_7 \log_5 (\sqrt{x} + 5 + \sqrt{x}) = 0$
 use $\log_a x = b$
 $\Rightarrow a^b = x$
 $\therefore \log_5 (\sqrt{x} + 5 + \sqrt{x}) = 7^0 = 1$
 $\sqrt{x} + 5 + \sqrt{x} = 5^1 = 5 \Rightarrow 2\sqrt{x} = 0 \therefore x = 0$
9. (d) Consider $\log_3 [\log_3 [\log_3 x]] = \log_3 3$
 $\Rightarrow \log_3 [\log_3 x] = 3$
 $\Rightarrow \log_3 x = 3^3$
 $\Rightarrow \log_3 x = 27 \Rightarrow x = 3^{27}$
10. (b) Let $\log(a + \sqrt{a^2 + 1}) + \log\left(\frac{1}{a + \sqrt{a^2 + 1}}\right)$
 $= \log(a + \sqrt{a^2 + 1}) + \log 1 - \log(a + \sqrt{a^2 + 1})$
 $= \log(a + \sqrt{a^2 + 1}) - \log(a + \sqrt{a^2 + 1})$
 $= 0$

$$\begin{aligned} & \frac{1}{\log_a bc + 1} + \frac{1}{\log_b ac + 1} + \frac{1}{\log_c ab + 1} \\ &= \frac{1}{\log_a bc + \log_a a} + \frac{1}{\log_b ac + \log_b b} + \frac{1}{\log_c ab + \log_c c} \\ &= \frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc} \\ &= \log_{abc} a + \log_{abc} b + \log_{abc} c \\ &= \log_{abc} abc = 1 \end{aligned}$$

12. (c) $q = \log_{17}(5)^2 = 2 \log_{17} 5$

$$\Rightarrow \frac{1}{q} = \frac{1}{2} \log_5 17$$

And $\frac{1}{p} = \log_5 3 = \frac{1}{2} \times (2 \log_5 3)$

$$= \frac{1}{2} (\log_5 9)$$

$$\therefore \frac{1}{p} < \frac{1}{q} \Rightarrow p > q$$

13. (c) $(pa + qb - rc) = p \log_{10} x + q \log_{10} y - r \log_{10} z$

$$= \log_{10} (x^p) + \log_{10} (y^q) - \log_{10} (z^r)$$

$$= \log_{10} \left(\frac{x^p y^q}{z^r} \right)$$

$$\Rightarrow \text{antilog } (pa + qb - rc) = \frac{x^p y^q}{z^r}$$

14. (c) a, b, c are consecutive integers

$$\therefore b = a + 1 \text{ and } c = a + 2$$

$$\therefore \log (ac + 1) = \log [a(a + 2) + 1]$$

$$= \log [(b - 1)(b - 1 + 2) + 1]$$

$$= \log b^2 = 2 \log b$$

$$[\because a = b - 1]$$

15. (c) $(7^3)^{-2 \log_7 8} = 7^{-6 \log_7 8} = 7^{(\log_7 8^{-6})}$

$$= 8^{-6} = \frac{1}{8^6}$$

16. (b) $x^2 + 6x + 8 > 0$ and $2x^2 + 2x + 3 > 0$

$$\Rightarrow (x + 4)(x + 2) > 0 \text{ and } \left(x + \frac{1}{2}\right)^2 + \frac{5}{4} > 0$$

$$\Rightarrow x \in (-\infty, -4) \cup (-2, \infty)$$

$$..(1)$$

The given equation can be written as

$$\log_{(2x^2+2x+3)} (x^2 - 2x) = 1$$

$$\Rightarrow x^2 - 2x = 2x^2 + 2x + 3$$

$$\Rightarrow x^2 + 4x + 3 = 0$$

$$\Rightarrow x = -1, -3$$

But at $x = -3$, $\log_{(x^2+6x+8)}$ is not defined

Hence, $x = -1$

17. (a) First of all for $\log (x - 1)$ to be defined, $x - 1 > 0$

$$\Rightarrow x > 0$$

$$....(1)$$

$$\text{Now, } \log_{0.3} (x - 1) < \log_{0.09} (x - 1)$$

$$\Rightarrow \log_{0.3} (x - 1) < \log_{(0.3)^2} (x - 1)$$

$$\Rightarrow \log_{0.3} (x - 1) < \frac{1}{2} \log_{0.3} (x - 1)$$

$$\Rightarrow 2 \log_{0.3} (x - 1) < \log_{0.3} (x - 1)$$

$$\Rightarrow \log_{0.3} (x - 1)^2 < \log_{0.3} (x - 1)$$

$$\Rightarrow (x - 1)^2 > (x - 1)$$

[The inequality is reversed since base lies between 0 and 1]

$$\Rightarrow (x - 1)^2 - (x - 1) > 0 \Rightarrow (x - 1)(x - 2) > 0(2)$$

Combining (1) and (2), we get $x > 2$

$$\therefore x \in (2, \infty)$$

18. (b) Given equation is $(\log_3 x)^2 + \log_3 x < 2$

$$\Rightarrow (\log_3 x)^2 + (\log_3 x) - 2 < 0$$

$$\Rightarrow (\log_3 x + 2)(\log_3 x - 1) < 0$$

$$\Rightarrow -2 < \log_3 x < 1$$

$$\Rightarrow \log_3 3^{-2} < \log_3 x < \log_3 3$$

$$\Rightarrow \frac{1}{9} < x < 3$$

19. (a) $x + \log_{10} (1 + 2^x) = x \log_{10} 5 + \log_{10} 6$

$$\Rightarrow \log_{10} 10^x + \log_{10} (1 + 2^x) = \log_{10} 5^x \cdot 6$$

$$\Rightarrow 10^x (1 + 2^x) = 5^x \cdot 6 \Rightarrow 2^x (1 + 2^x) = 6$$

20. (b) $\log_2 x \cdot \log_4 x \cdot \log_6 x = \log_2 x \cdot \log_4 x + \log_2 x \cdot \log_6 x + \log_4 x \cdot \log_6 x$

Dividing LHS and RHS by $\log_2 x \cdot \log_4 x \cdot \log_6 x$

$$\Rightarrow 1 \cdot \frac{1}{\log_6 x} + \frac{1}{\log_4 x} + \frac{1}{\log_2 x}$$

$$\Rightarrow 1 \cdot \log_x 6 + \log_x 4 + \log_x 2$$

$$\Rightarrow \log_x 6 \times 4 \times 2 = 1$$

$$\Rightarrow x = 48. \therefore \text{only 1 value of } x.$$

21. (b) The equation can be written as

$$\log x + \frac{1}{2} \log_5 (x^2 + 3) = \frac{1}{2} \log_5 10$$

$$\text{leading to } x \sqrt{x^2 + 3} = \sqrt{10} \text{ i.e., } (x^2 + 5)(x^2 - 2) = 0$$

Of the two values $x = \pm \sqrt{2}$ $\log x$ exists only when

$$x = \sqrt{2}.$$

22. (b) Let $\log_{10} x - \log_{10} \sqrt{x} = 2 \log_x 10$
 $\Rightarrow \frac{1}{2} \log_{10} x = 2 \log_x 10 \Rightarrow \log_{10} x = \log_x 10^4$
 $\Rightarrow \frac{\log_{10} x}{\log_x 10} = 4 \Rightarrow (\log_{10} x)^2 = 4$
 $\Rightarrow \log_{10} x = \pm 2$
 $\Rightarrow x = 10^2 \text{ or } 10^{-2}$

23. (d) The given logarithm expression

$$\frac{\log_{27} 9 \log_{16} 64}{\log_4 \sqrt{2}}$$

is simplified as :

$$\frac{\log 9}{\log 27} \times \frac{\log 64}{\log 16} \times \frac{\log 4}{\log \sqrt{2}}$$

$$= \frac{2 \log 3}{3 \log 3} \times \frac{6 \log 2}{4 \log 2} \times \frac{2 \log 2}{\frac{1}{2} \log 2}$$

$$\frac{2}{3} \times \frac{6}{4} \times 4 = 4$$

24. (b) $(\log_x x)(\log_3 2x)(\log_{2x} y) = \log_x x^2$
 $\Rightarrow 1(\log_3 2x)(\log_{2x} y) = 2 \quad (\because \log_x x^2 = 2 \log_x x)$
 $\Rightarrow \left(\frac{\log 2x}{\log 3} \right) \left(\frac{\log y}{\log 2x} \right) = 2$
 $\Rightarrow \frac{\log y}{\log 3} = 2 \Rightarrow \log y = 2 \log 3$
 $\Rightarrow \log y = \log 3^2 \Rightarrow y = 3^2 \Rightarrow y = 9$

25. (d) Consider, $\log \frac{9}{8} - \log \frac{27}{32} + \log \frac{3}{4}$
 $= \log \left(\frac{9}{8} \times \frac{32}{27} \right) + \log \frac{3}{4}$
 $= \log \left(\frac{4}{3} \right) + \log \frac{3}{4} = \log \left(\frac{4}{3} \times \frac{3}{4} \right) = \log 1 = 0$

26. (a) $25^{[(-1/4) \log_5 25]} = 5^{[2(-1/4) \log_5 25]}$
 $= 5^{(-1/2) \log_5 25} = 5^{\log_5 (25)^{-1/2}}$
 $= 25^{-1/2} = \frac{1}{5}$

27. (b) $\log_{10} x, \log_{10} y, \log_{10} z$ are in AP
 $\therefore 2 \log_{10} y = \log_{10} x + \log_{10} z$
 $\Rightarrow \log_{10} y^2 = \log_{10} (xz)$
 $\Rightarrow y^2 = xz$
 $\Rightarrow x, y, z$ are in GP

28. (c) $\frac{\log \sqrt{27} + \log \sqrt{8} - \log \sqrt{125}}{\log 6 - \log 5}$
 $= \frac{\log (3^{3/2} \times 2^{3/2}) - \log (5)^{3/2}}{\log 6 - \log 5}$
 $= \frac{\log 6^{3/2} - \log 5^{3/2}}{\log 6 - \log 5}$
 $= \frac{\frac{3}{2} \log \left(\frac{6}{5} \right)}{\log \left(\frac{6}{5} \right)} = \frac{3}{2}$

29. (b) Best way is to go through options

Alternatively: $\log_{10} x^2 y^3 = 7$

$$\Rightarrow x^2 y^3 = 10^7 \quad \dots(1)$$

$$\text{and } \log_{10} \left(\frac{x}{y} \right) = 1$$

$$\Rightarrow \frac{x}{y} = 10 \quad \dots(2)$$

$$\frac{x^2 y^3}{x/y} = \frac{10^7}{10^2} \Rightarrow y^5 = 10^5$$

$$\Rightarrow y = 10 \therefore x = 100$$

30. (d) $A = \log_7 2401 = 4$

$$B = \log_7 (\sqrt{7} \cdot 343)$$

$$\Rightarrow B = \log_7 (7^{1/2} \times 7^3) = 3 \frac{1}{2}$$

$$C = \log_{\sqrt{6}} 216 = 2 \cdot \log_6 6^3 = 3$$

$$D = \log_2 32 = 5$$

$\therefore$ Arranging in ascending order: BADC

31. (a) $3 \log_{(3x^2)} 27 - 2 \log_{(3x)} 9 = 0$

$$\Rightarrow 3 \log_{3x^2} 3^3 - 2 \log_{3x} 3^2 = 0$$

$$\Rightarrow 9 \log_{3x^2} 3 = 4 \log_{3x} 3$$

$$\Rightarrow \frac{9}{\log_{3x} 3} = \frac{4}{\log_{3x^2} 3}$$

$$\Rightarrow 9 \log_3 3x = 4 \cdot \log_3 3x^2$$

$$\Rightarrow \log_3 (3x)^9 = \log_3 (3x^2)^4$$

$$\Rightarrow 3^9 \times x^9 = 3^4 \times x^8 \Rightarrow x = 3^{-5}$$

$$\therefore x = \frac{1}{243}$$

32. (a) $\log_k N = 6 \Rightarrow K^6 = N$
 $\log_{25k} (8N) = 3 \Rightarrow (25K)^3 = 8N$

$$\therefore \frac{25^3 K^3}{K^6} \cdot 8 \Rightarrow 25^3 \cdot K^3 = 2^3$$

$$\therefore K = \frac{25}{2} = 12.5$$

33. (d) $\log_3 2, \log_4 3, \log_5 4, \dots, \log_{16} 15$

$$= \frac{\log 2}{\log 3} \cdot \frac{\log 3}{\log 4} \cdot \frac{\log 4}{\log 5} \cdot \frac{\log 15}{\log 16} = \frac{\log 2}{\log 16} = \frac{\log 2}{\log 2^4}$$

$$= \frac{\log 2}{4 \log 2} = \frac{1}{4}$$

34. (c) $\log (2x - 2)/(11.66 - x) = \log 30$

$$\Rightarrow (2x - 2)/(11.66 - x) = 30$$

$$2x - 2 = 350 - 30x$$

$$\text{Hence, } 32x = 352 \Rightarrow x = 11$$

35. (d) ${}_4 a$ and $\log_5 6 = b$

$$\Rightarrow \log_4 5 \times \log_5 6 = ab$$

$$\Rightarrow \log_4 6 = ab \Rightarrow \frac{1}{2} \log_2 6 = ab$$

$$\Rightarrow (1 + \log_2 3) = 2ab$$

36. (b) $\log_3 x + \log_9 x + \log_{27} x + \log_{81} x = \frac{25}{4}$

$$\Rightarrow \log_3 x + \frac{1}{2} \log_3 x + \frac{1}{3} \log_3 x + \frac{1}{4} \log_3 x = \frac{25}{4}$$

$$\Rightarrow \log_3 x [1 + 1/2 + 1/3 + 1/4] = \frac{25}{4}$$

$$\Rightarrow \frac{25}{2} \times \log_3 x = \frac{25}{4}$$

$$\Rightarrow \log_3 x = 3 \Rightarrow x = 27$$

37. (d) $\log_{32} 27 \times \log_{243} 8 = \log_{2^5} 3^3 \times \log_{3^5} 2^3$

$$= \frac{3}{5} \log_2 3 \times \frac{3}{5} \log_3 2$$

$$= \left(\frac{3}{5}\right)^2 \log_2 3 \times \log_3 2 = \frac{9}{25}$$

38. (d) Go through options. (Use $\log 1 = 0$)

39. (b) $x = \log_a bc \therefore 1 + x = \log_a abc$

$$y = \log_b ca \therefore 1 + y = \log_b abc$$

$$z = \log_c bc \therefore 1 + z = \log_c abc$$

$$\therefore \frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z} = \log_{abc} a + \log_{abc} b + \log_{abc} c$$

$$= \log_{abc} abc = 1$$

40. (b) $2/3 \log a - 5 \log b - 1/2 \log c$

41. (a) Let the number be y

$$y = 108^{10}$$

$$\Rightarrow \log y = 10 \log 108$$

$$\log y = 10 \log (27 \times 4)$$

$$\log y = 10 [3 \log 3 + 2 \log 2]$$

$$\log y = 10 [1.43 + 0.602]$$

$$\text{Hence, } \log y = 10 [2.03] = 20.3$$

$$\text{Thus, } y \text{ has 21 digits}$$

42. (c) $\log(a^n b^n c^n / a^n b^n c^n) = \log 1 = 0$

43. (a) For $x = 0$, we have LHS

$$\log_2 8 = 3.$$

$$\text{RHS: } 10^{\log 3} = 3$$

$$\text{We do not get LHS = RHS for either } x = 3 \text{ or } x = 6.$$

$$\text{Thus, option (a) is correct.}$$

Expert Level

1. (c) $\log(0.2)^{25} = 25 \log\left(\frac{2}{10}\right)$

$$= 25 (\log 2 - \log 10)$$

$$= -25 (1 - 0.30103)$$

$$= -25 \times 0.69897 = -17.47425$$

$$\therefore \text{Number of zeroes after the decimal point is 17.}$$

2. (c) $\log_{10} \left[1 - \left\{ 1 - (1 - x^2)^{-1} \right\}^{-1} \right]^{-1/2} = 1$

$$\Rightarrow \log_{10} \left[1 - \left\{ 1 - \frac{1}{1 - x^2} \right\}^{-1} \right]^{-1/2} = 1$$

$$\Rightarrow \log_{10} \left[1 - \left\{ \frac{-x^2}{1 - x^2} \right\}^{-1} \right]^{-1/2} = 1$$

$$\Rightarrow \left[\frac{1}{x^2} \right]^{-1/2} = 10 \Rightarrow x = 10$$

3. (b) $N = n!$ (given)

$$\frac{1}{\log_2 N} + \frac{1}{\log_3 N} + \frac{1}{\log_4 N} + \dots + \frac{1}{\log_n N}$$

$$= \log_N 2 + \log_N 3 + \log_N 4 + \dots + \log_N n$$

$$= \log_N n! = \log_N N = 1$$

4. (a) $\frac{1}{\log_2 n} + \frac{1}{\log_3 n} + \frac{1}{\log_4 n} + \dots + \frac{1}{\log_{40} n}$

$$= \log_n 2 + \log_n 3 + \log_n 4 + \dots + \log_n 40$$

$$= \log_n (40!)$$

$$\frac{1}{\log_{40!} n}$$

5. (d) $0.5 \log_{0.625} 128$

$$= 0.5 [\log_8 128 / \log_8 0.625]$$

$$= 1/2 [\log_8 128 / \log_8 0.625]$$

$$\frac{\log_8 128}{2(\log_8 5 - \log_8 8)} = \frac{\log_8 128}{2[\log_8 5 - 1]}$$

6. (b) $x^u = 256$

Taking log to the base 2 on both sides

$$u \log_2 x = \log_2 256 = \log_2 2^8 = 8 \log_2 2 = 8$$

$$(\because \log_a a = 1)$$

$$\Rightarrow u = \frac{8}{\log_2 x}$$

$$\text{Let } \log_2 x = a$$

$$\Rightarrow \text{Equation becomes, } \frac{8}{a} a^2 - 6a + 12$$

(from given value of 'a')

$$\Rightarrow a^3 - 6a^2 + 12a - 8 = 0 \Rightarrow (a - 2)^3 = 0$$

$$\Rightarrow a = 2 \Rightarrow \log_2 x = 2 \Rightarrow u = \frac{8}{2} = 4$$

$\therefore$ There is only one solution.

7. (b) $(5)^x = 1000$

$$\Rightarrow 5 \cdot 10^{3/x}$$

$$\text{And } (0.5)^y = 1000$$

$$\Rightarrow \left(\frac{5}{10}\right)^y = 10^3$$

$$\Rightarrow 5^y = 10^3 \times 10^y$$

$$\Rightarrow 5 \cdot 10^{\frac{(3+y)}{y}}$$

From eqs (1) and (2)

$$5 = 10^{3/x} = 10^{\frac{(3+y)}{y}}$$

$$\Rightarrow \frac{3}{x} = \frac{3+y}{y}$$

$$\Rightarrow \frac{1}{x} - \frac{1}{y} = \frac{1}{3}$$

8. (c) $2 \log_{10} x - \log_x \frac{1}{100} = 2 \log_{10} x - \frac{\log_{10} 10^{-2}}{\log_{10} x}$

$$= 2 \log_{10} x + \frac{2}{\log_{10} x}$$

$$= 2 \left(\log_{10} x + \frac{1}{\log_{10} x} \right)$$

$$\text{Since, } x > 1 \Rightarrow \log_{10} x > 0$$

But since $AM \geq GM$

$$\therefore \frac{\left(\log_{10} x + \frac{1}{\log_{10} x} \right)}{2} \geq \sqrt{\log_{10} x \times \frac{1}{\log_{10} x}}$$

$$\Rightarrow \log_{10} x + \frac{1}{\log_{10} x} \geq 2$$

$$\Rightarrow 2 \left(\log_{10} x + \frac{1}{\log_{10} x} \right) \geq 4$$

$$\text{For } x = 10, 2 \left(\log_{10} x + \frac{1}{\log_{10} x} \right) \geq 4$$

Hence the least value of $\left(\log_{10} x - \log_x \frac{1}{100} \right)$ is 4.

9. (c) $\log_e 2 \cdot \log_p 625 = \log_{10} 16 \cdot \log_e 10$

$$\Rightarrow \log_p 625 = \log_{10} 2^4 \cdot \frac{\log_e 10}{\log_e 2}$$

$$\Rightarrow \log_p 5^4 = 4 \log_{10} 2 \cdot \log_2 10$$

$$\text{or } 4 \log_p 5 = 4$$

$$\therefore \log_p 5 = 1 \Rightarrow P = 5$$

... (1) 10. (d) $\log_{12} 27 = a$

$$\Rightarrow \log_{12} 3^3 = a \Rightarrow \log_{12} 3 = \frac{a}{3}$$

$$\Rightarrow \log_3 12 = \frac{3}{a} \Rightarrow \log_3 4 + \log_3 3 = \frac{3}{a}$$

$$\Rightarrow 2 \log_3 2 = \frac{3-a}{a}$$

$$\Rightarrow \log_3 2 = \frac{3-a}{2a}$$

... (1)

$$\text{Now, } \log_6 16 = 4 \log_6 2 = A \text{ (let)}$$

$$\Rightarrow \log_6 2 = \frac{A}{4} \Rightarrow \log_2 6 = \frac{4}{A} \Rightarrow \log_2 3 = \frac{4-A}{A}$$

$$\Rightarrow \log_3 2 = \frac{A}{4-A}$$

... (2)

From (1) and (2)

$$\frac{3-a}{2a} = \frac{A}{4-A} \text{ or } \frac{2a}{3-a} = \frac{4-A}{A}$$

$$\text{or } \frac{2a}{3-a} + 1 = \frac{4-A}{A} + 1 \text{ or } \frac{2a+3-a}{3-a} = \frac{4-a+A}{A}$$

$$\text{or } \frac{a+3}{3-a} = \frac{4}{A} \text{ or } A = \frac{4(3-a)}{3+a}$$

11. (b) $2[\log(x+y) - \log 5] = \log x + \log y$

$$\Rightarrow \log \frac{(x+y)^2}{5^2} = \log xy$$

$$\Rightarrow \frac{(x+y)^2}{25} = xy \Rightarrow x^2 + y^2 + 2xy = 25xy$$

$$\Rightarrow x^2 + y^2 = 23xy$$

12. (a) The given equation can be written as
 $x^3 - 3x^2 - 4x + 8 = (x-3)^3 = x^3 - 9x^2 + 27x - 27$

$$\text{i.e., } 6x^2 - 31x + 35 = (3x-5)(2x-7) = 0$$

$$x = \frac{7}{2} \text{ is the only solution as the base } x-3 \text{ is } -ve$$

$$\text{for } x = \frac{5}{3}.$$

13. (a) By binomial expansion,

$$(a + a^{\log a})^5 = a^5 + {}^5C_1 a^4 (a^{\log a})^1 + {}^5C_2 a^3 (a^{\log a})^2 + {}^5C_3 a^2 (a^{\log a})^3 + {}^5C_4 a^1 (a^{\log a})^4 + (a^{\log a})^5$$

$$\text{Now, the fourth term is } {}^5C_3 a^2 (a^{\log a})^3 = 10,00,000$$

$$\therefore 10a^2 (a^{\log a})^3 = 10^6$$

$$a^2 (a^{\log a})^3 = 10^5$$

Now substitute from the options for 'a', we get $a = 10$.

14. (d) Taking logarithms can be both side

$$(2x-1)(\ln 3) = (x+2)(\ln 4)$$

$$2x(\ln 3) - 1(\ln 3) = x(\ln 4) + 2(\ln 4)$$

$$2x(\ln 3) - x(\ln 4) = 2(\ln 4) + 1(\ln 3)$$

Factor out x on the left side to get

$$[2\ln(3) - \ln 4]x = 2(\ln 4) + \ln 3$$

Divide both sides by the coefficient of x

$$x = \frac{2(\ln 4) + \ln 3}{2(\ln 3) - \ln 4} = 4.774$$

15. (b) $\log_5(5^{1/x} + 125) = \log_5 6 + 1 + (1/2x)$

$$\therefore \log_5(5^{1/x} + 5^3) = \log_5 6 + \log_5 5 + \log_5 5^{1/2x}$$

$$\therefore \log_5(5^{1/x} + 5^3) = \log_5(30 \times 5^{1/2x})$$

$$\therefore 5^{1/x} + 125 = 30 \times 5^{1/2x}$$

$$\text{Put } 5^{1/2x} = a$$

$$\therefore a^2 + 125 = 30a$$

$$\Rightarrow a^2 - 30a + 125 = 0$$

$$\Rightarrow (a-5)(a-25) = 0$$

$$\Rightarrow a = 5 \text{ or } 25$$

$$\therefore 5^{1/2x} = 5^1 \text{ or } 5^2$$

$$\Rightarrow 1/(2x) = 1 \text{ or } 2$$

$$\Rightarrow x = 1/2 \text{ or } 1/4$$

16. (a) $\log(0.57) = \bar{1}.756 \Rightarrow \log 57 = 1.756$

[$\therefore$ mantissa will remain the same]

$$\therefore \log 57 + \log(0.57)^3 + \log \sqrt{0.57}$$

$$= \log 57 + 3 \log \left(\frac{57}{100} \right) + \log \left(\frac{57}{100} \right)^{1/2}$$

$$= \log 57 + 3 \log 57 - 3 \log 100 + \frac{1}{2} \log 57 - \frac{1}{2} \log 100$$

$$= \frac{9}{2} \log 57 - \frac{7}{2} \log 100$$

$$= \frac{9}{2} \times 1.756 - \frac{7}{2} \times 2 = 7.902 - 7$$

$$= 0.902.$$

17. (c) Given $\log_2[\log_7(x^2 - x + 37)] = 1$

Use $\log_p x = y$

$$\Rightarrow p^y = x$$

$$\therefore 2 = \log_7(x^2 - x + 37)$$

$$7^2 = x^2 - x + 37$$

$$\Rightarrow 49 = x^2 - x + 37 \Rightarrow x^2 - x - 12 = 0$$

$$\Rightarrow (x-4)(x+3) = 0$$

$$\Rightarrow x = 4, -3$$

$$\therefore x = 4$$

18. (b) $\log_3 M^{\frac{1}{3}} N^3 = \log_3 3 + \log_{0.008} 5$ ($\therefore \log_a a = 1$)

$$M^{\frac{1}{3}} N^3 = 3^1 \times 3^{\log_{0.008} 5}$$

$$\Rightarrow N^9 = \frac{27}{M} \left(3^{3 \log_{0.008} 5} \right)$$

$$\frac{1}{M} (27) (3^{\log(0.2)^3 5^3}) = \frac{1}{M} (27) 3^{(\log_{0.2} 5)}$$

$$= \frac{1}{M} (27) 3^{(\log_{1/5} 5)} = \frac{1}{M} (27) (3)^{-1}$$

$$= \frac{1}{M} \frac{27}{3} = \frac{9}{M}$$

19. (d) In an AP, the three terms a, b and c are related as $2b = a + c$

$$\text{Hence, } 2[\log_3(2^x - 5)] = \log_3 2 + \log_3 \left(2^x - \frac{7}{2} \right)$$

$$\log_3(2^x - 5)^2 = \log_3(2^{x+1} - 7)$$

$$\Rightarrow (2^x - 5)^2 = (2^{x+1} - 7)$$

Substitute the choices, only $x = 3$ satisfies the above condition.

20. (d) Let $A = \log_x \left(\frac{x}{y} \right) + \log_y \left(\frac{y}{x} \right)$

$$\log_x x - \log_x y + \log_y y - \log_y x$$

$$\Rightarrow A = 2 - (\log_y x + \log_x y) = 2 - \left(\log_y x + \frac{1}{\log_y x} \right)$$

We know, $a + \frac{1}{a} > 1$

$$\Rightarrow A = 2 - (\text{a real no more than one}) \text{ or } A < 1$$

So, A can now be 1.

21. (d) $\log_y x = a, \log_z y = b, \log_x z = ab$

$$\Rightarrow a \log_z y = ab \text{ or } \log_z y = b$$

$$\text{or } y = z^b \quad \dots(1)$$

$$\text{and } b \log_x z = ab \text{ or } \log_x z = a$$

$$\text{or } z = x^a \quad \dots(2)$$

$$\text{and } \log_y x = ab \Rightarrow x = y^{ab} \quad \dots(3)$$

$$x = y^{ab} = (z^b)^{ab} = (x^a)^{bab} = x^{a^2 b^2}$$

$$\text{or } a^2 b^2 = 1 \quad \dots(4)$$

Putting the options in condition 4, we see that it is not satisfied only when $a = 2$ and $b = 2$.

22. (c) Consider $\frac{\log_{\sqrt{\alpha\beta}} H}{\log_{\sqrt{\alpha\beta\gamma}} H} = \frac{\log_H \sqrt{\alpha\beta\gamma}}{\log_H \sqrt{\alpha\beta}}$

$$\log_{\sqrt{\alpha\beta}} \sqrt{\alpha\beta\gamma} = \log_{\alpha\beta} (\alpha\beta\gamma)$$

23. (a) $\frac{1}{\log_2 a} + \frac{1}{\log_4 a} + \frac{1}{\log_8 a} + \dots + \frac{1}{\log_{2^n} a} = \frac{n(n+1)}{k}$

$$= \log_a 2 + \log_a 2^2 + \log_a 2^3 + \dots + \log_a 2^n = \frac{n(n+1)}{k}$$

$$= (\log_a 2) (1 + 2 + 3 + \dots + n) = \frac{n(n+1)}{k}$$

$$\Rightarrow (\log_a 2) \left[\frac{n(n+1)}{2} \right] = \frac{n(n+1)}{k}$$

$$\Rightarrow \frac{1}{2} \log_a 2 = \frac{1}{k}$$

$$\Rightarrow \log_a 2 = \frac{2}{k}$$

$$\Rightarrow \log_2 a = \frac{k}{2}$$

$$\Rightarrow 2 \log_2 a = k$$

$$\Rightarrow \log_2 a^2 = k$$

24. (d) $\left[(x^2 - y^2)^2 \right]^{a-1} = (x-y)^{2a} (x+y)^{-2}$

$$\Rightarrow (x^2 - y^2)^{2(a-1)} = (x-y)^{2a} (x+y)^{-2}$$

$$\Rightarrow (x-y)^{2(a-1)} (x+y)^{2(a-1)} = (x-y)^{2a} (x+y)^{-2}$$

$$\Rightarrow \frac{(x-y)^{2a-2} (x+y)^{2a-2}}{(x-y)^{2a} (x+y)^{-2}} = 1$$

$$\Rightarrow (x-y)^{-2} (x+y)^{2a} = 1$$

$$\Rightarrow -2 \log (x-y) + 2a \log (x+y) = \log 1$$

$$\Rightarrow 2a \log (x+y) = 2 \log (x-y)$$

$$a = \frac{\log (x-y)}{\log (x+y)}$$

25. (c) $v = u^{1/2}, w = u^{1/3}, z = u^{1/4}$

$$\therefore uvwz = u^{\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right)}$$

$$\therefore \log_u uvwz = \log_u u^{\left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right)}$$

$$= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right) \log_u u = \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \right)$$

26. (c) Last term $k = a \cdot r^{n-1}$, $r \rightarrow$ common ratio

$$\therefore \log k = \log (a \cdot r^{n-1})$$

$$\Rightarrow \log k = \log a + (n-1) \log r$$

$$\Rightarrow \frac{\log k - \log a}{\log r} = n-1$$

$$\Rightarrow n = 1 + \left(\frac{\log k - \log a}{\log r} \right)$$

27. (c) $y = a^{\left(\frac{1}{1-\log_a x} \right)}$

$$\Rightarrow \log_a y = \frac{1}{1-\log_a x} \quad (\text{taking log of both sides})$$

$$\Rightarrow \log_a x = 1 - \frac{1}{\log_a y} \quad \dots(1)$$

$$\text{Again } z = a^{\left(\frac{1}{1-\log_a y} \right)}$$

$$\Rightarrow \log_a z = \frac{1}{1-\log_a y}$$

$$\Rightarrow \log_a y = 1 - \frac{1}{\log_a z} \quad \dots(2)$$

∴ From eqs (1) and (2), we get

$$\log_a x = 1 - \frac{1}{\left(1 - \frac{1}{\log_a z}\right)}$$

$$\log_a x = \frac{1}{1 - \log_a z}$$

$$\Rightarrow x = a^{\frac{1}{1 - \log_a z}}$$

28. (b) $\log_{100} |x + y| = \frac{1}{2}$

$$\Rightarrow |x + y| = 10 \quad (-10 \text{ is inadmissible})$$

and $\log_{10} y - \log_{10} |x| = \log_{10} 4$

$$\Rightarrow \log_{10} \left(\frac{y}{|x|} \right) = \log_{10} 4$$

$$\Rightarrow \frac{y}{|x|} = 4$$

$$y = 4|x|$$

$$|x + 2|x|| = 10$$

Now, if $x > 0$, then we have

$$3x = 10 \Rightarrow x = \frac{10}{3}$$

and if $x < 0$, then

$$|-x| = 10$$

$$\Rightarrow x = -10$$

$$x = \frac{10}{3}, y = \frac{20}{3}$$

and $x = -10, y = 20$

29. (b) Given that $9^n < 10^8$

Taking log to both sides

$$\log 9^n < \log 10^8$$

$$\Rightarrow 2n \log 3 < 8 \log 10$$

$$\Rightarrow 2n \times 0.4771 < 8$$

$$\Rightarrow n \times 0.9542 < 8$$

$$\Rightarrow n < \frac{8}{0.9542}$$

$$n < 8.3839$$

$$\therefore n = 8$$

$$(\because n \in \mathbb{N})$$

30. (c) $m > 0$ and $n > 0$ and $m \neq 1$

i.e., $25 - x^2 > 0$ and $x \neq \pm 3$

and $24 - 2x - x^2 > 0$

$$\Rightarrow -5 < x < 5, x \neq \pm 3$$

and $x^2 + 2x - 24 < 0$

$$-5 < x < 5, x \neq \pm 3$$

$$\text{and } -6 < x < 4$$

$$x \in (-5, 4) - \{-3, 3\} \quad \dots(1)$$

Case 1: $0 < m < 1 \Rightarrow 9 < x^2 < 25$

$$\Rightarrow x \in (-5, -3) \cup (3, 5) \quad \dots(2)$$

∴ The given inequality be written as

$$\frac{24 - 2x - x^2}{14} < \frac{25 - x^2}{16}$$

$$\Rightarrow x^2 + 16x - 17 > 0$$

$$\Rightarrow (x + 17)(x - 1) > 0$$

$$\Rightarrow x < -17 \text{ or } x > 1$$

From (1) and (b), we have

$$x \in (3, 4)$$

Case 2: If $m > 1$, i.e., $\frac{25 - x^2}{16} > 1$

$$\Rightarrow x \in (-3, 3) \quad \dots(3)$$

The given inequality reduces to

$$\frac{24 - 2x - x^2}{14} > \frac{25 - x^2}{16}$$

$$\Rightarrow x^2 + 16x - 17 < 0$$

$$\Rightarrow -17 < x < 1$$

Thus combining with (3), we get

$$x \in (-3, 1)$$

but $x \in (-5, 4) - \{-3, 3\}$ by (1)

thus $x \in (-3, 1)$

Hence the required value of x should lie in

$$(-3, 1) \cup (3, 4)$$

31. (b) The L.H.S of the equation

$$\frac{3}{x^4} (\log_2 x)^2 + \log_2 x - \frac{5}{4} \sqrt{2} \text{ is defined}$$

If $x > 0$ and $x \neq 1$. Taking log of both the sides at the base 2,

$$\left[\frac{3}{4} (\log_2 x)^2 + \log_2 x - \frac{5}{4} \right] \log_2 x = \log_2 2^{\frac{1}{2}}$$

$$\Rightarrow \frac{3}{4} t^3 + t^2 - \frac{5}{4} t = \frac{1}{2}, \text{ where } t = \log_2 x$$

$$\Rightarrow 3t^3 - 4t^2 - 5t + 2 = 0$$

$$\Rightarrow (t - 1)(t + 2)(3t + 1) = 0 \Rightarrow t = 1, -2 \text{ or } -\frac{1}{3}$$

$$\therefore \log_2 x = 1, -2 \text{ or } -\frac{1}{3} \Rightarrow x = 2, 2^{-2} \text{ or } 2^{-\frac{1}{3}}$$

$$\Rightarrow x = 2, \frac{1}{4} \text{ or } \frac{1}{\sqrt[3]{2}}$$

$\therefore$ The equation has exactly three real solutions, hence at least one real solution. Also there is exactly one irrational solution.

32. (a) The log functions are defined if $\frac{x^2+6x+9}{2(x+1)} > 0$ and

$$x+1 > 0 \Rightarrow \frac{(x+3)^2}{2(x+1)} > 0 \text{ and } x+1 > 0 \Rightarrow x > -1$$

Now the inequality is

$$\log_{2^{-1}} \frac{x^2+6x+9}{2(x+1)} < -\log_2(x+1)$$

$$\Rightarrow -\log_2 \frac{x^2+6x+9}{2(x+1)} < -\log_2(x+1)$$

$$\Rightarrow \log_2 \frac{x^2+6x+9}{2(x+1)} > \log_2(x+1)$$

$$\Rightarrow \frac{x^2+6x+9}{2(x+1)} > (x+1)$$

$$\Rightarrow \frac{-x^2+2x+7}{2(x+1)} > 0 \Rightarrow (x+1)(x^2-2x-7) < 0$$

$$\Rightarrow x^2-2x-7 < 0 \quad [\because x+1 > 0]$$

$$\Rightarrow -1-2\sqrt{2} < x < -1+2\sqrt{2},$$

$$\text{but } x > -1 \Rightarrow -1 < x < -1+2\sqrt{2}$$

$$33. (b) 1 - \log_{10} 5 = \frac{1}{3} \left[\log_{10} \left(\frac{x \cdot 5^{1/3}}{2} \right) \right]$$

$$= 3[\log_{10} 10 - \log_{10} 5] = \log_{10} \left(\frac{x \cdot 5^{1/3}}{2} \right)$$

$$\Rightarrow \log_{10} 2^3 = \log_{10} \left(\frac{x \cdot 5^{1/3}}{2} \right)$$

$$\Rightarrow 2^3 = \frac{x \cdot 5^{1/3}}{2}$$

$$\therefore x = 16 \times 5^{-1/3}$$

$$34. (c) \log_{0.04}(x-1) \geq \log_{0.2}(x-1)$$

$$\Rightarrow \frac{\log_e(x-1)}{\log_e 0.04} \geq \frac{\log_e(x-1)}{\log_e 0.2}$$

$$\Rightarrow \frac{\log_e(x-1)}{\log_e(0.2)^2} \geq \frac{\log_e(x-1)}{\log_e 0.2}$$

$$\therefore \log_e 0.2 < 0$$

$$\therefore \log_3(x-1) \leq 2 \log_3(x-1)$$

$$\therefore x-1 \geq 1$$

$$\therefore x \geq 2 \Rightarrow x \in [2, \infty)$$

$$35. (b) \log_{30} 3 = x, \log_{30} 5 = y$$

$$\therefore x + y = \log_{30} 15$$

$$\Rightarrow x + y = \log_{30} (30/2)$$

$$\Rightarrow (x + y) = 1 - \log_{30} 2 \Rightarrow \log_{30} 2 = 1 - x - y$$

$$\Rightarrow 3 \log_{30} 2 \Rightarrow 3(1 - x - y) \Rightarrow \log_{30} 8 = 3(1 - x - y)$$

$$\log_8 30 = \frac{1}{3(1-x-y)}$$

Explanation of Test Yourself

1. (b) $\log_7 \log_5(\sqrt{x} + 5 + \sqrt{x}) = 0$

use $\log_a x = b$

$$\Rightarrow a^b = x$$

$$\therefore \log_5(\sqrt{x} + 5 + \sqrt{x}) = 7^0 = 1$$

$$\sqrt{x} + 5 + \sqrt{x} = 5^1 = 5 \Rightarrow 2\sqrt{x} = 0 \therefore x = 0$$

2. (a) $\log x = \log_{(21/10)} 2$

$$\therefore \log x \text{ as } \log_a b = \frac{\log b}{\log a}$$

$$= \frac{\log 2}{\log 21 - \log 10} \cdot \frac{\log 2}{[\log 3 + \log 7 - 1]} = [\because \log 10 = 1]$$

3. (d) Using, $\log_{a^k} m = \frac{1}{k} \log_a m$

$$\frac{\log_8 17}{\log_9 23} = \frac{\log_{(2\sqrt{2})^2} 17}{\log_{3^2} 23} = \frac{\frac{1}{2} \log_{2\sqrt{2}} 17}{\frac{1}{2} \log_3 23} = \frac{\log_{2\sqrt{2}} 17}{\log_3 23}$$

Hence,

$$\frac{\log_8 17}{\log_9 23} - \frac{\log_{2\sqrt{2}} 17}{\log_3 23} = \frac{\log_{2\sqrt{2}} 17}{\log_3 23} - \frac{\log_{2\sqrt{2}} 17}{\log_3 23} = 0$$

4. (c) Given equation implies

$$\frac{(\log \sin x)^2 + (\log \cos x)^2}{\log \cos x \cdot \log \sin x} = 2 \quad \therefore \log_a b = \frac{\log b}{\log a}$$

$$\Rightarrow (\log \sin x)^2 + (\log \cos x)^2 - 2 \log \cos x \cdot \log \sin x = 0$$

$$\Rightarrow (\log \sin x - \log \cos x)^2 = 0$$

$$\Rightarrow \log \sin x = \log \cos x$$

$$\Rightarrow \sin x = \cos x \Rightarrow x = \frac{\pi}{4}$$

5. (d) Given expression

$$= \frac{1}{\frac{\log xyz}{\log xy}} + \frac{1}{\frac{\log xyz}{\log yz}} + \frac{1}{\frac{\log xyz}{\log zx}}$$

$$= \frac{\log xy + \log yz + \log zx}{\log xyz} = \frac{\log x^2 y^2 z^2}{\log xyz}$$

$$= \frac{2 \log xyz}{\log xyz} = 2$$

6. (c) $\log_a b = \frac{1}{2}, \log_b c = \frac{1}{3}, \log_c a = \frac{K}{5}$

$$\Rightarrow \frac{\log b}{\log a} = \frac{1}{2}, \frac{\log c}{\log b} = \frac{1}{3}, \frac{\log a}{\log c} = \frac{K}{5}$$

$$\Rightarrow \frac{1}{2} \times \frac{1}{3} \times \frac{K}{5} = 1 \Rightarrow K = 30$$

7. (b) $abc = \frac{\log 12}{\log 24} \cdot \frac{\log 24}{\log 36} \cdot \frac{\log 36}{\log 48} = \frac{\log 12}{\log 48}$

$$\therefore 1 + abc = \frac{\log 48 + \log 12}{\log 48} = \frac{\log(48 \cdot 12)}{\log 48}$$

$$\frac{\log 24^2}{\log 48} = 2 \cdot \frac{\log 24}{\log 48} = 2bc$$

8. (b) Let L.H.S. = $\log_{17} 275$

Changing to base 10

$$\Rightarrow \frac{\log 275}{\log 17} \approx 16.18$$

R.H.S. = $\log_{19} 375$

Changing to base 10

$$\Rightarrow \frac{\log 375}{\log 19} \approx 19.74$$

Hence, $\log_{17} 275 < \log_{19} 375$

9. (c) $A = \log_2 \log_2 \log_4 (4)^4 + 2 \log_{\sqrt{2}} (\sqrt{2})^2$

$$= \log_2 \log_2 4 + 2 \times 2 \quad \because \log_a a = 1$$

$$= \log_2 \log_2 (2)^2 + 4 = \log_2 2 + 4 = 1 + 4 = 5$$

10. (c) $\log_2 (2/3) + \log_4 (9/4)$

$$= \log_2 (2/3) + \log_2 (9/4) \log_2 4$$

$$= \log_2 (2/3) + 1/2 \log_2 (9/4)$$

$$= \log_2 (2/3) + 1/2 (2 \log_2 3/2)$$

$$= \log_2 (2/3) + \log_2 3/2 = \log_2 1 = 0$$

11. (d) In an AP, the three terms a, b, c are related as
 $2b = a + c$

$$\text{Hence, } 2 [\log_3(2^x - 5)] = \log_3 2 + \log_3 \left(2^x - \frac{7}{2} \right)$$

$$\log(2^x - 5)^2 = (2^{x+1} - 7)$$

Substitute the options, only $x = 3$ satisfies the conditions.

12. (d) $0.5 \log_{0.625} 128$

$$= 0.5 [\log_8 128 / \log_8 0.625]$$

$$= 1/2 [\log_8 128 / \log_8 0.625]$$

$$\frac{\log_8 128}{2(\log_8 5 - \log_8 8)} \quad \frac{\log_8 128}{2[\log_8 5 - 1]}$$

13. (b) The three numbers are 1, $\log_{9^2} (3^x + 48)$ and

$$\log_9 \left(3^x - \frac{8}{3} \right),$$

i.e., $\log_9 9, \frac{1}{2} \log_9 (3^x + 48), \log_9 \left(3^x - \frac{8}{3} \right)$ are in A.P.

$$\Rightarrow \left\{ \left(3^x + 48 \right)^{\frac{1}{2}} \right\}^2 = 9 \left(3^x - \frac{8}{3} \right)$$

$$\Rightarrow 8 \cdot 3^x = 72 \Rightarrow 3^x = 9 \Rightarrow x = 2.$$

14. (d) $AM = \frac{\log 40 + \log 5}{2} = \frac{1}{2} \log 200$

$$= \frac{1}{2} (\log 100 + \log 2) = \frac{1}{2} (2 + 0.3070) = 1.1505$$

15. (a) $\log_{11} 1650 > 3$

$$\log_{13} 1950 < 3$$

Hence, $\log_{11} 1650 > \log_{13} 1950$

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SET THEORY

- Introduction
- Sets
- Representations of Sets
- Standard Symbols of Some Special Sets
- Types of Sets
- Subsets
- Intervals as Subsets of a Set of Real Numbers (R)
- Power Set of a Set
- Universal Set
- Venn Diagrams
- Operation on Sets
- Disjoint Sets
- Cardinal Number
- Situation Based Venn Diagrams

INTRODUCTION

Sets are used to define the concepts of relation and functions. Set theory is an important chapter for CAT and other MBA entrance exams. Usually 2–3 questions from this chapter are asked in CAT exams.

It is advised to students to give proper attention to understand the concepts of set theory. Having understood the concept soundly, you can clear the concept of Function and Relation easily.

SETS

A set is a well-defined collection of different objects.

In everyday life, we often speak about the collection of objects of particular kind such as a cricket team, the rivers of India, the vowels in the English alphabet etc. Each of these collection is well-defined collection of objects in the sense that we can definitely decide whether a given particular object belongs to a given collection or not. For example, we say that 10 does not belong to the given collection of all odd natural numbers. On the other hand, 15 belongs to this given collection.

Note that

- (i) Objects, elements and members of a set are synonymous terms.
- (ii) Sets are usually denoted by capital letters A, B, C, D, E, F , etc.
- (iii) The elements of a set are represented by small letters a, b, c, d, e, f , etc.

- (iv) Each element in a set comes only once i.e. repetition of any element is not allowed.

If a is an element of a set A , we say that "a belongs to A ". The Greek symbol $\in$ (epsilon) is used to denote the phrase 'belongs to'.

Thus, we write $a \in A$. If ' b ' is not an element of a set A , we write $b \notin A$ and read " b does not belong to A ".

If V be the set of vowels of English alphabet, then $a \in V$ but $b \notin V$. In the set P of prime factors of 30, $3 \in P$ but $15 \notin P$.

REPRESENTATIONS OF SETS

There are two methods of representing a set:

- (i) Roster or tabular form (ii) Set-builder form.

Roster or Tabular Form

- (i) In roster form, all the elements of a set are listed within a bracket $\{ \}$ and separated by commas. For example, the set of all even positive integers less than 7 is described in roster form as $\{2, 4, 6\}$.
- (ii) In roster form, the order in which the elements are listed is immaterial.

Set-builder Form

The set $\{a, e, i, o, u\}$ in roster form can be written as set in builder form as $\{x : x \text{ is a vowel of English alphabet}\}$. Here the set written in set builder form is read as ' x is an element of the set such that x is a vowel of English alphabet'. Here the colon ($:$) read as 'such that'. In set-builder, a common property which possesses all the elements of the set is written after colon ($:$).

Statement	Roster form	Set-builder form
(1) The set of currencies used in USA, England, Japan, Germany and Russia.	{Dollar, Pound, Yen, Euro, Rouble}	$\{x : x \text{ is the currencies used in USA, England, Japan, Germany and Russia}\}$
(2) The set of Capital of Kerala, Karnataka, Tamilnadu, Andhra Pradesh and Gujarat	{Tiruvananthapuram, Bangalore, Chennai, Hyderabad and Gandhi Nagar}	$\{x : x \text{ is the capitals of Kerala, Karnataka, Tamilnadu, Andhra Pradesh and Gujarat}\}$
(3) The set of all distinct letters used in the word student.	$\{s, t, u, d, e, n\}$	$\{x : x \text{ is the distinct letters used in the word student.}\}$
(4) The set of all the states of India beginning with the letter A.	{Andhra Pradesh, Arunachal Pradesh, Assam}	$\{x : x \text{ is the state of India beginning with the letter A}\}$
(5) The set of six presidents of India since 1980.	{Neelam Sanjeeva Reddy, Gyani Zail Singh, Radha Swami Venkat Raman, Dr. Shankar Dayal Sharma, K.R. Narayan, A.P.J. Abdul Kalam}	$\{x : x \text{ is the presidents of India since 1980}\}$
(6) The set of all natural numbers between 11 and 15.	{12, 13, 14}	$\{x : x \in N, 11 < x < 15\}$

Illustration 1: Write the set $X = \left\{1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \frac{1}{25}, \dots\right\}$ in the set-builder form.

Solution: We observe that the elements of set X are the reciprocals of the squares of all natural numbers. So, the set X in set builder form is

$$X = \left\{\frac{1}{n^2} : n \in N\right\}.$$

Illustration 2: Write the following intervals in set builder form

- (i) $(-3, 0)$ (ii) $[6, 12]$ (iii) $(6, 12]$ (iv) $[-23, 5)$

Solution: The following intervals are written in set builder form as :

- (i) $(-3, 0)$ is an open interval which does not include both -3 and 0 . So, it can be shown in the set builder form as : $\{x : x \in R, -3 < x < 0\}$.
 (ii) $[6, 12]$ is a closed interval which includes both 6 and 12 . So it can be shown in the set builder form as $\{x : x \in R, 6 \leq x \leq 12\}$

- (iii) $(6, 12]$ is an interval open at the first end and closed at the second end i.e. it excludes 6 but includes 12 . So it is shown in the set builder form as : $\{x : x \in R, 6 < x \leq 12\}$
 (iv) $[-23, 5)$ is an interval closed at the first end point but open at the second end point. It means that the interval includes -23 but excludes 5 . It is written in the set builder form as : $\{x : x \in R, -23 \leq x < 5\}$.

STANDARD SYMBOLS OF SOME SPECIAL SETS

- N : Set of all natural numbers
 Z : Set of all integers
 Q : Set of all rational numbers
 R : Set of all real numbers
 Z^+ : Set of all positive integers
 Q^+ : Set of all positive rational numbers, and
 R^+ : Set of all positive real numbers.

The symbols for the special sets given above will be referred throughout the chapter.

TYPES OF SETS

Empty Set

A set which does not contain any element is called an empty set, null set or void set.

The empty set is denoted by the symbol ϕ or $\{\}$.

Given below are few examples of empty sets.

- (i) If $A = \{x : 1 < x < 2, x \text{ is a natural number}\}$, then A is the empty set, because there is no natural number between 1 and 2 .
 (ii) If $B = \{x : x^2 - 2 = 0 \text{ and } x \text{ is rational number}\}$, then B is the empty set, because the equation $x^2 - 2 = 0$ is not satisfied by any rational value of x .
 (iii) If $C = \{x : x \text{ is an even prime number greater than } 2\}$, then C is the empty set, because 2 is the only even prime number.
 (iv) If $D = \{x : x^2 = 4, x \text{ is odd}\}$, then D is the empty set, because the equation $x^2 = 4$ is not satisfied by any odd value of x .

Finite and Infinite Sets

A set which is empty or consists of a definite number of elements is called finite otherwise, the set is called infinite.

Consider some examples:

- (i) Let W be the set of the days of the week, then W is the finite set.
 (ii) Let S be the set of solutions of the equation $x^2 - 16 = 0$, then S is the finite set.
 (iii) Let G be the set of points on a line, then G is infinite set.

When we represent a set in the roster form, we write all the elements of the set within bracket $\{\}$. But it is not possible to write all the elements of an infinite set within bracket $\{\}$, because the number of elements of such a set is not finite. So, we represent the infinite set in the roster form by writing a few elements which clearly indicate the structure of the set followed (or preceded)

by some dots. For example, $\{1, 2, 3, \dots\}$ is the set of natural numbers, $\{1, 3, 5, 7, \dots\}$ is the set of odd natural numbers, $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is the set of integers. All these sets are infinite sets.

All infinite sets cannot be described in the roster form. For example, set of all real numbers cannot be described in roster form.

Equal Sets

Two sets A and B are said to be equal if they have exactly the same elements and we write $A = B$. Otherwise, the sets are said to be unequal and we write $A \neq B$.

- (i) Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 1, 4, 2\}$. then $A = B$, because elements of both sets are the same. Only order of the elements in the two sets is different but it is not considered in a set.
- (ii) Let A be the set of prime numbers less than 6 and P the set of prime factors of 30. Then A and P are equal, since 2, 3 and 5 are the only prime factors of 30 and also these are less the only prime numbers than 6.

Singleton Set

Set A set, consisting of a single element is called a singleton set. The sets $\{0\}$, $\{5\}$, $\{-7\}$ are singleton sets.

$\{x : x + 6 = 0, x \in \mathbb{Z}\}$ is a singleton set, because this set contains only one integer namely, -6 .

Illustration 3: State which of the following sets are finite and which are infinite:

- (i) $A = \{x : x \in \mathbb{Z} \text{ and } x^2 - 5x + 6 = 0\}$
- (ii) $B = \{x : x \in \mathbb{Z} \text{ and } x^2 \text{ is even}\}$
- (iii) $C = \{x : x \in \mathbb{Z} \text{ and } x^2 = 36\}$
- (iv) $D = \{x : x \in \mathbb{Z} \text{ and } x > -10\}$

Solution: We have,

$$(i) A = \{x : x \in \mathbb{Z} \text{ and } x^2 - 5x + 6 = 0\} = \{2, 3\}$$

So, A is a finite set

$$(ii) B = \{x : x \in \mathbb{Z} \text{ and } x^2 \text{ is even}\} \\ = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}$$

Clearly, B is an infinite set.

$$(iii) C = \{x : x \in \mathbb{Z} \text{ and } x^2 = 36\} \\ = \{6, -6\}$$

Clearly, C is a finite set.

$$(iv) D = \{x : x \in \mathbb{Z} \text{ and } x > -10\} \\ = \{-9, -8, -7, \dots\}$$

Clearly, D is an infinite set.

Illustration 4: Find the pairs of equal sets, from the following sets, if any, giving reasons:

$$A = \{0\}, B = \{x : x > 15 \text{ and } x < 5\}, C = \{x : x - 5 = 0\}, \\ D = \{x : x^2 = 25\} \\ E = \{x : x \text{ is an integral positive root of the equation } x^2 - 2x - 15 = 0\}.$$

Solution: We have,

$$A = \{0\},$$

$$B = \{x : x > 15 \text{ and } x < 5\} = \phi,$$

$$C = \{x : x - 5 = 0\} = \{5\},$$

$$D = \{x : x^2 = 25\} = \{-5, 5\},$$

$$\text{and } E = \{5\}.$$

$$\text{Clearly, } C = E.$$

SUBSETS

Set A is said to be a subset of a set B if every element of set A is also an element of set B . Here set B is called superset of set A . A is a subset of B , is represented $A \subset B$. Thus $A \subset B$ if whenever $a \in A$, then $a \in B$. It is often convenient to use the symbol " $\Rightarrow$ " which means implies. Using this symbol, we can write the definition of subset as follows: $A \subset B$ if $a \in A \Rightarrow a \in B$.

We read the above statement as " A is a subset of B if a is an element of A implies a is also an element of B ". If A is not a subset of B , we write $A \not\subset B$. For example:

- (i) The set Q of rational numbers is a subset of the set R of real numbers, so we write $Q \subset R$.
- (ii) If A is the set of all divisors of 56 and B the set of all prime divisors of 56, then B is a subset of A so we write $B \subset A$.
- (iii) Let $A = \{1, 3, 5\}$ and $B = \{x : x \text{ is an odd natural number less than } 6\}$. Then $A \subset B$ and $B \subset A$ and hence $A = B$.
- (iv) Let $A = \{a, e, i, o, u\}$ and $B = \{a, b, c, d\}$. Then A is not a subset of B . Also B is not a subset of A .

Important Points about Subsets

- (i) Every set is a subset of itself.
- (ii) Empty set is a subset of every set.
- (iii) Total number of subsets of a finite set containing n element is 2^n .

INTERVALS AS SUBSETS OF A SET OF REAL NUMBERS (R)

Let $a, b \in \mathbb{R}$ and $a < b$. Then the set of real numbers $\{y : a < y < b\}$ is called an open interval and is denoted by (a, b) . All the real numbers between a and b belong to the open interval (a, b) but a, b themselves do not belong to this interval.

The interval which contains the end points also is called closed interval and is denoted by $[a, b]$. Thus

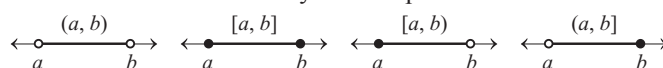
$$[a, b] = \{x : a \leq x \leq b\}$$

We can also have intervals closed at one end and open at the other end, i.e.,

$[a, b) = \{x : a \leq x < b\}$ is a semi open interval between a and b , including a but excluding b .

$(a, b] = \{x : a < x \leq b\}$ is a semi open interval between a and b including b but excluding a .

These sets can be shown by the dark portion of the number line.



Dark small circle on the number line means the point is included and small blank circle on the number line means the point is not included.

Illustration 5: Write the following intervals in the set-builder form

- (i) $(-7, 0)$
- (ii) $[6, 12]$
- (iii) $(6, 12]$
- (iv) $[-20, 3)$

Solution: We have,

- (i) $(-7, 0) = \{x : x \in R \text{ and } -7 < x < 0\}$
- (ii) $[6, 12] = \{x : x \in R \text{ and } 6 \leq x \leq 12\}$
- (iii) $(6, 12] = \{x : x \in R \text{ and } 6 < x \leq 12\}$
- (iv) $[-20, 3) = \{x : x \in R \text{ and } -20 \leq x < 3\}$

POWER SET OF A SET

The set of all subsets of a set A is called the power set of A . It is denoted by $P(A)$.

Let $A = \{a, b\}$, then $P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

In $P(A)$, every element is a set.

If A has n elements then its power set has 2^n elements.

Illustration 6: If $A = \{a, \{b\}\}$, find $P(A)$.

Solution: Let $B = \{b\}$. Then, $A = \{a, B\}$.

$\therefore P(A) = \{\phi, \{a\}, \{B\}, \{a, B\}\} = \{\phi, \{a\}, \{\{b\}\}, \{a, \{b\}\}\}.$

UNIVERSAL SET

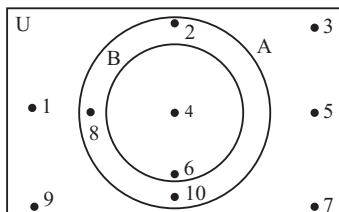
If there are some sets under consideration, and out of these sets, there is a set which is the superset of all other given sets i.e., all other sets under consideration are subsets of this set. Such a set is known as the universal set, denoted by U .

For example,

- (i) In the context of human population studies, the universal set consists of all the people in the world.
- (ii) If $\{1, 2, 3, 4\}$, $\{2, 5, 6\}$, $\{1, 3, 7, 8, 9\}$ and $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ are the sets under consideration, then set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ can be considered as universal set because all other three sets are the subsets of this set.

VENN DIAGRAMS

In order to illustrate universal sets, subsets and certain operations on sets in a clear and simple way, we use geometric figures. These figures are called Venn-Diagrams. In Venn Diagrams, a universal set is represented by a rectangle and any other set is represented by a circle.



In the Venn-diagrams, the elements of the sets are written in their respective circles.

In the Venn-diagrams, $U = \{1, 2, 3, \dots, 10\}$ is the universal set of which $A = \{2, 4, 6, 8, 10\}$ and $B = \{4, 6\}$ are subsets, and also $B \subset A$.

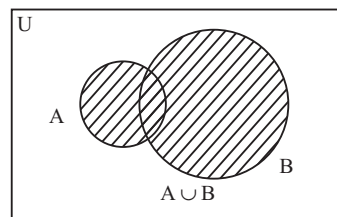
OPERATION ON SETS

Union of Sets

Union of two sets A and B is the set which consists of all those elements which are either in A or in B (including those which are in both sets A and B). In symbols, we write

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

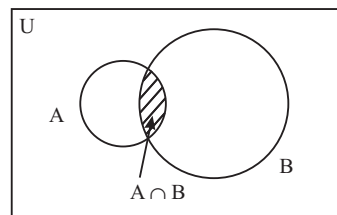
The union of two sets A and B can be represented by a Venn diagram as shown in figure by shaded portion



Intersection of Sets

The intersection of two sets A and B is the set of all those elements which belong to both sets A and B . Symbolically, we write

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$



The shaded portion in figure indicates the intersection of sets A and B .

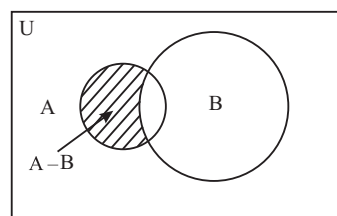
Difference of Sets

The difference of the sets A and B (in the order A minus B) is the set of elements which belong to A but not to B . Symbolically, we write $A - B$ and read as " A minus B ".

In the set builder notation, we can write

$$A - B = \{x : x \in A \text{ but } x \notin B\}$$

The difference of two sets A and B is represented in Venn diagram by shaded portion.

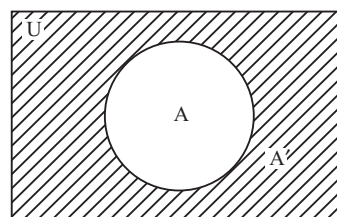


Complements of a Set

Let U be the universal set and A be a subset of U . Then the complement of A is the set of all elements of U which are not the elements of set A . Symbolically, we write A' or A^c to denote the complement of set A .

Thus, $A' = \{x : x \in U \text{ but } x \notin A\}.$

Obviously $A' = U - A$



Complement of set A i.e. A' is represented in Venn diagram by shaded region.

Some Properties of Complement of a Set

1. Complement laws:
 - (i) $A \cup A' = U$
 - (ii) $A \cap A' = \phi$
 2. De Morgan's law:
 - (i) $(A \cup B)' = A' \cap B'$
 - (ii) $(A \cap B)' = A' \cup B'$
 3. Law of double complementation: $(A')' = A$
 4. Laws of empty set and universal set: $\phi' = U$ and $U' = \phi$.
- These laws can be verified by using Venn diagrams.

Illustration 7: If $A = \{x : x = 3n, n \in \mathbb{Z}\}$ and $B = \{x : x = 4n, n \in \mathbb{Z}\}$, then find $A \cap B$.

Solution: We have,

$$\begin{aligned} x \in A \cap B &\Leftrightarrow x = 3n, n \in \mathbb{Z} \text{ and } x = 4n, n \in \mathbb{Z} \\ &\Leftrightarrow x \text{ is a multiple of 3 and } x \text{ is a multiple of 4} \\ &\Leftrightarrow x \text{ is a multiple of 3 and 4 both} \\ &\Leftrightarrow x \text{ is a multiple of 12.} \\ &\Leftrightarrow x = 12n, n \in \mathbb{Z} \end{aligned}$$

Hence, $A \cap B = \{x : x = 12n, n \in \mathbb{Z}\}$.

If A and B are two sets, then $A \cap B = A$, if $A \subset B$ and $A \cap B = B$, if $B \subset A$.

Illustration 8: Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 2, 3, 4\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 4, 5, 6\}$. Find

- (i) A'
- (ii) $(A \cup B)'$
- (iii) $(A')'$
- (iv) $(B - C)'$

Solution:

- (i) $\{5, 6, 7, 8, 9\}$
- (ii) $\{5, 7, 9\}$
- (iii) A
- (iv) $\{1, 3, 4, 5, 6, 7, 9\}$

Illustration 9: Find the union of each of the following Pairs of sets:

- (i) $A = \{x : x \text{ is a natural number and } 1 < x \leq 6\}$
 $B = \{x : x \text{ is a natural number and } 6 < x \leq 10\}$
- (ii) $A = \{1, 2, 3\}$, $B = \phi$.

Solution:

- (i) $A = \{x : x \text{ is a natural number and } 1 < x \leq 6\}$
 $\Rightarrow A = \{2, 3, 4, 5, 6\}$
 $B = \{x : x \text{ is a natural number and } 6 < x \leq 10\}$
 $\Rightarrow B = \{7, 8, 9, 10\}$
 $\therefore A \cup B = \{2, 3, 4, 5, 6\} \cup \{7, 8, 9, 10\}$
 $\Rightarrow A \cup B = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- (ii) We have, $A = \{1, 2, 3\}$, $B = \phi$
 $\Rightarrow A \cup B = \{1, 2, 3\} \cup \phi$
 $\Rightarrow A \cup B = \{1, 2, 3\}$

Illustration 10: If $A = \{x : x = 3n, n \in \mathbb{Z}\}$ and $B = \{x : x = 4n, n \in \mathbb{Z}\}$, then find $(A \cap B)$.

Solution: Let $x \in (A \cap B) \Leftrightarrow x \in A$ and $x \in B$

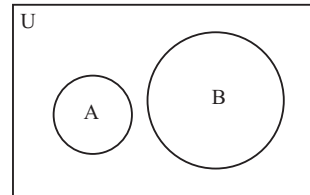
$$\begin{aligned} &\Leftrightarrow x \text{ is a multiple of 3 and } x \text{ is a multiple of 4.} \\ &\Leftrightarrow x \text{ is a multiple of 3 and 4 both} \\ &\Leftrightarrow x \text{ is a multiple of 12} \\ &\Leftrightarrow x = 12n, n \in \mathbb{Z} \end{aligned}$$

Hence $A \cap B = \{x : x = 12n, n \in \mathbb{Z}\}$

DISJOINT SETS

If A and B are two sets such that $A \cap B = \phi$, then A and B are called disjoint sets.

For example, let $A = \{2, 4, 6, 8\}$ and $B = \{1, 3, 5, 7\}$. Here A and B are disjoint sets, because there is no element common to both sets A and B .



In the Venn diagram, A and B are disjoint sets.

CARDINAL NUMBER

Number of element in a set A is called cardinal number of set A . It is represented by $n(A)$. If $A = \{a, b, c, d, e, f\}$, then $n(A) = 6$

1. If A and B are finite sets then

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

2. If A , B and C are three finite sets, then

$$\begin{aligned} n(A \cup B \cup C) &= n(A) + n(B) + n(C) - n(A \cap B) \\ &\quad - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C) \end{aligned}$$

Illustration 11: In a political survey, 78% of the politicians favour at least one proposal, 50% of them are in favour of proposal A , 30% are in favour of proposal B and 20% are in favour of proposal C . 5% are in favour of all three proposals. what is the percentage of people favouring more than one proposal?

- (a) 16
- (b) 17
- (c) 18
- (d) 19

Solution: (b)

$$\begin{aligned} n(A \cup B \cup C) &= n(A) + n(B) + n(C) - n(A \cap B) \\ &\quad - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C) \\ \text{or } 78 &= 50 + 30 + 20 - \Sigma n(A \cap B) + 5 \\ \text{or } \Sigma n(A \cap B) &= 27 \\ \text{This includes } n(A \cap B \cap C) &\text{ three times.} \\ \therefore \text{Percentage of people favouring more than one proposal} &= 27 - 5 \times 2 = 17 \end{aligned}$$

Illustration 12: If X and Y are two sets such that $X \cup Y$ has 50 elements, X has 28 elements and Y has 32 elements, how many elements does $X \cap Y$ have?

Solution:

$$\begin{aligned} \text{Given that } n(X \cup Y) &= 50, \quad n(X) = 28, \\ n(Y) &= 32, \quad n(X \cap Y) = ? \end{aligned}$$

By using the formula,

$$n(X \cup Y) = n(X) + n(Y) - n(X \cap Y),$$

$$\begin{aligned} \text{We find that } n(X \cap Y) &= n(X) + n(Y) - n(X \cup Y) \\ &= 28 + 32 - 50 = 10 \end{aligned}$$

Illustration 13: In a class of 35 students, 24 like to play cricket and 16 like to play football. Also, each student likes to play at least one of the two games. How many students like to play both cricket and football?

Solution: Let X be the set of students who like to play cricket and Y be the set of students who like to play football. Then $X \cup Y$ is the set of students who like to play at least one of the two games, and $X \cap Y$ is the set of students who like to play both games.

Given $n(X) = 24$, $n(Y) = 16$, $n(X \cup Y) = 35$, $n(X \cap Y) = ?$

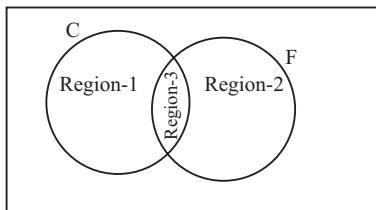
Using the formula $n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$,

We get $35 = 24 + 16 - n(X \cap Y)$

Thus, $n(X \cap Y) = 5$ i.e., 5 students like to play both games.

SITUATION BASED VENN DIAGRAMS

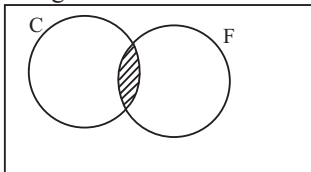
1. Suppose set C represents the people who like cricket and F represents the people who like football.



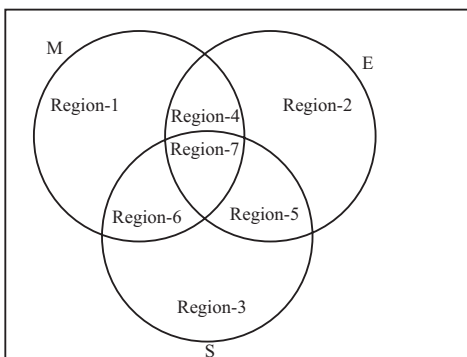
In the above Venn-diagram,

- Region- 1:** Represents the people who like cricket only (means people who like cricket but not football.)
Region- 2: Represents the people who like football only (means people who like football but not cricket.)
Region- 3: Represents the people who like both cricket and football.

The people who like both cricket and football is represented by the common shaded region of set A and set B in the Venn diagram.



2. Let M represent the students who passed in mathematics, E represents the students who passed in English and S represents the students who passed in Science. Then students who passed in both Mathematics and English are represented by common region of the sets M and E .



Students who passed in both English and Science are represented by the common region of set E and S . Students who passed in both Science and Mathematics represented by the common region of set S and M . Students who passed in both Mathematics

and English are represented by the common region of sets M and E . Students who passed in all the three subjects, Mathematics, English and Science are represented by common region of all the three sets M , E and S .

- Region- 1:** Represents the students who passed in Mathematics only (means the students who passed in Mathematics but not passed in English and Science).
Region- 2: Represents the students who passed in English only (means the students who passed in English but not passed in Science and mathematics).
Region- 3: Represents the students who passed in Science only (means the students who passed in science but not passed in Mathematics and English).
Region- 4: Represents the students who passed in both Mathematics and English only (means the students who passed in both Mathematics and English but not in Science).
Region- 5: Represents the students who passed in both English and Science only (means the students who passed in both English and Science but not passed in Mathematics).
Region- 6: Represents the students who passed in both Science and Mathematics only (means the students who passed in both Science and Mathematics, but not passed in English).
Region- 7: Represents the students who passed in all the three subjects Mathematics, English and Science.

Note that

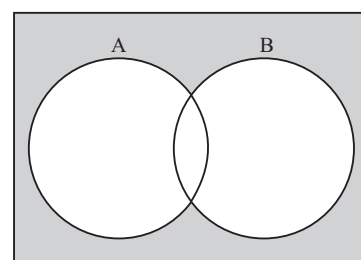
- Students who passed in Mathematics are represented by the sum of the regions 1, 4, 6 and 7.
- Students who passed in English are represented by the sum of the regions 2, 4, 5 and 7.
- Students who passed in Science are represented by the sum of the regions 3, 5, 6 and 7.
- Students who passed in both Mathematics and English are represented by the sum of the regions 4 and 7.
- Students who passed in both English and Science are represented by the sum of the regions 5 and 7.
- Students who passed in both Science and Mathematics are represented by the sum of the regions 6 and 7.

Illustration 14: Draw the appropriate Venn diagram for each of the following:

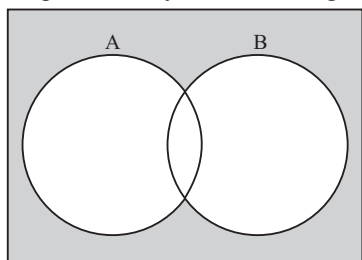
- (i) $(A \cup B)'$ (ii) $A' \cap B'$ (iii) $(A \cap B)'$ (iv) $A' \cup B'$

Solution:

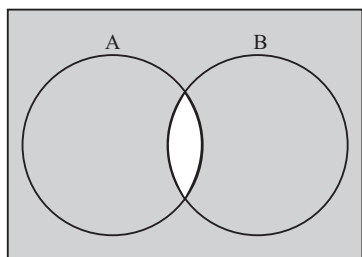
- (i) $(A \cup B)'$ is represented by the shaded region.



(ii) $A' \cap B'$ is represented by the shaded region.



(iii) $(A \cap B)'$ is represented by the shaded region.



(iv) $A' \cup B'$ is represented by the shaded region.

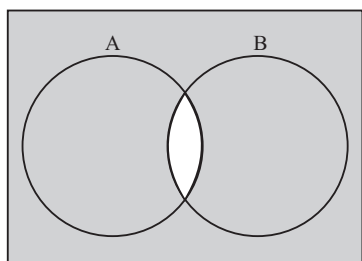
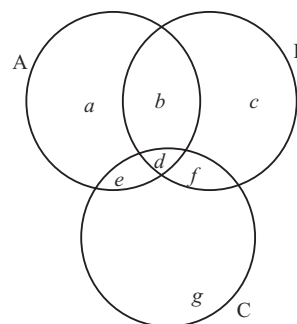


Illustration 15: Out of 10000 people surveyed, 3700 liked city A , 4000 liked city B and 5000 liked city C . 700 people liked A and B 1200 liked A and C and 1000, liked B and C . Each person liked at least one city. Then find

- The number of people liking all the three cities.
- The number of persons liking at least two cities as a % of number of people liking exactly one city.
- The number of persons liking exactly two cities as a percentage of the number of people liking at least one city.
- The number of persons liking A and B but not C .

Solution: Refer the figure given

$$\begin{aligned} n(A \cup B \cup C) &= n(A) + n(B) + n(C) - n(A \cap B \cap C) \\ &\Rightarrow 10000 = 3700 + 4000 + 5000 - 700 - 1000 - 1200 + d \\ &\Rightarrow d = 200 \end{aligned}$$



Once the value of d is known, all other values will be determined fastly.

e.g. $b + d = 700$ (given) $\Rightarrow b = 500$

Similarly $e = 1000, f = 800, a = 2000, c = 2500, g = 3000$

A. $d = 200$.

B. At least two cities $b + d + e + f = 2500$

Exactly one city $a + c + g = 7500$

$\Rightarrow \% = 2500/7500 \times 100 \% = 33.33\%$

C. Exactly two cities $= b + e + f = 2300$

At least one city = 10000

$\Rightarrow$ Required $\% = 23 \%$.

D. $b = 500$.

Illustration 16: In a survey of 100 students, the number of students studying the various languages were found to be: English only 18, English but not Hindi 23, English and Sanskrit 8, English 26, Sanskrit 48, Sanskrit and Hindi 8, no language 24. Find:

(i) How many students were studying Hindi?

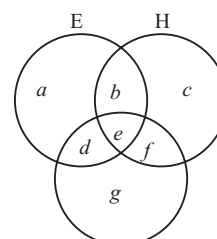
(ii) How many students were studying English and Hindi?

Solution:

We have, $a = 18, a + b = 23, d + e = 8, a + b + d + e = 26, d + e + f + g = 48, e + f = 8, a + b + c + d + e + f + g = 100 - 24 = 76$
 $\therefore a = 18, b = 0, c = 10, d = 5, e = 3, f = 5$ and $g = 35$

(i) $n(H) = b + c + e + f = 18$

(ii) $n(H \cap E) = b + e = 3$



Practice Exercises



Foundation Level

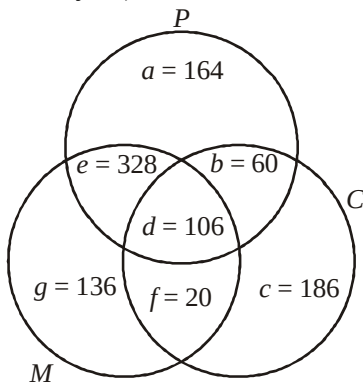
- In a group of students, 100 students know Hindi, 50 know English and 25 know both. Each of the students knows either Hindi or English. How many students are there in the group?
(a) 100 (b) 115
(c) 110 (d) 125
- If X and Y are two sets such that $n(X) = 17$, $n(Y) = 23$ and $n(X \cup Y) = 38$, then $n(X \cap Y)$
(a) 2 (b) 3
(c) 3 (d) 4
- If X and Y are two sets such that X has 40 elements, $X \cup Y$ has 60 elements and $X \cap Y$ has 10 elements, how many elements does Y have?
(a) 25 (b) 35
(c) 30 (d) 40
- Let $S = \{0, 1, 5, 4, 7\}$. Then the total number of subsets of S is
(a) 64 (b) 32
(c) 40 (d) 20
- If two sets A and B are having 99 elements in common, then the number of elements common to each of the sets $A \times B$ and $B \times A$ are
(a) 2^{99} (b) 99^2
(c) 100 (d) 19
- In an examination out of 100 students, 75 passed in English, 60 passed in Mathematics and 45 passed in both English and Mathematics. What is the number of students passed in exactly one of the two subjects?
(a) 45 (b) 60
(c) 75 (d) 90
- If A and B are two disjoint sets, then which one of the following is correct?
(a) $A - B = A - (A \cap B)$ (b) $B - A' = A \cap B$
(c) $A \cap B = (A - B) \cap B$ (d) All of these
- If $X = \{4^n - 3n - 1 : n \in N\}$ and $Y = \{9(n-1) : n \in N\}$, then precisely:
(a) $X \subset Y$ (b) $X \subseteq Y$
(c) $X = Y$ (d) $X \supseteq Y$
- $A \cup B = A$, then
(a) $A \subset B$ (b) $B \subset A$
(c) $A \not\subset B$ and $B \not\subset A$ (d) None of these
- If A, B, C , are any three sets, then
(a) $A - (B \cup C) = (A - B) \cup (A - C)$
(b) $A - (B \cup C) = (A - B) \cap (A - C)$
(c) $A - (B \cap C) = (A - B) \cup (A - C)$
(d) $A - (B \cap C) = (A \cup B) - (A \cup C)$
- Consider the following statements:
For any two sets A and B
(1) $(A - B) \cup B = A$ (2) $(A - B) \cup A = A$
(3) $(A - B) \cap B = \phi$ (4) $A \subseteq B \Rightarrow A \cup B = B$
Of these statements:
(a) 1, 2, 3 are correct (b) 2, 3, 4 are correct
(c) 1, 3, 4 are correct (d) 1, 2, 4 are correct
- $A - (B \cup C)$ equals
(a) $(A - B) \cup (A - C)$ (b) $(A - B) \cap (A \cap C)$
(c) $(A - B) \cup C$ (d) $(A - C) \cup B$
- If A and B are disjoint sets, then $A \cap (A' \cup B)$ is equal to which one of the following?
(a) ϕ (b) A
(c) $A \cup B$ (d) $A - B'$
- If A, B and C are three finite sets, then what is $[(A \cup B) \cap C]$ equal to?
(a) $A' \cup B' \cap C'$ (b) $A' \cap B' \cap C'$
(c) $A' \cap B' \cup C'$ (d) $A \cap B \cap C$

Standard Level

Directions for Qs. 1–3 : Read the information given below and answer the questions that follow.

The result of an exam is given below. Out of 1000 students who appeared

- (i) 658 failed in Physics
 - (ii) 166 failed in Physics and Chemistry
 - (iii) 372 failed in Chemistry, 434 failed in Physics and Maths
 - (iv) 590 failed in Maths, 126 failed in Maths & Chemistry.
- Find the number of people who failed in (assuming that none is passed in all subjects).



We have the following equations :

$$a + b + c + d + e + f + g = 1000$$

$$a + b + d + e = 658, b + d = 166$$

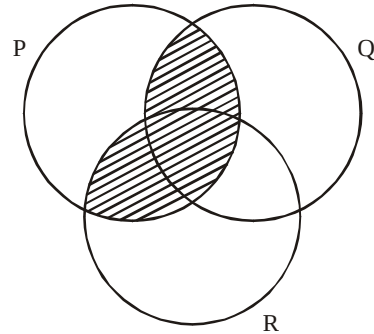
$$b + d + c + f = 372.$$

$$d + e = 434 \text{ as in the figure.}$$

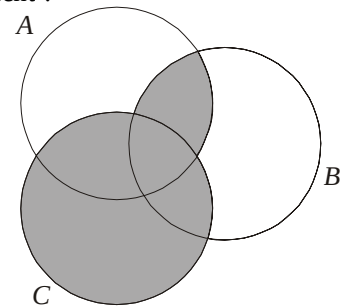
$$d + e + f + g = 590, d + f = 126. \text{ Find the values.}$$

1. Chemistry but not in Physics.
 - (a) 318
 - (b) 198
 - (c) 213
 - (d) 206
2. Physics or Maths but not in Chemistry.
 - (a) 558
 - (b) 718
 - (c) 628
 - (d) None of these
3. Physics but neither Chemistry nor Maths.
 - (a) 164
 - (b) 228
 - (c) 196
 - (d) None of these
4. Let T be the set of integers $\{3, 11, 19, 27, \dots, 451, 459, 467\}$ and S be a subset of T such that the sum of no two elements of S is 470. The maximum possible number of elements in S is
 - (a) 32
 - (b) 28
 - (c) 29
 - (d) 30
5. 70 per cent of the employees in a multinational corporation have VCD players, 75 per cent have microwave ovens, 80 per cent have ACs and 85 per cent have washing machines. At least what percentage of employees has all four gadgets?
 - (a) 15
 - (b) 5
 - (c) 10
 - (d) Cannot be determined

6. What does the shaded region represent in the figure given below ?



- (a) $(P \cup Q) - (P \cap Q)$
 - (b) $P \cap (Q \cap R)$
 - (c) $(P \cap Q) \cap (P \cap R)$
 - (d) $(P \cap Q) \cup (P \cap R)$
7. What does the shaded region in the Venn diagram given below represent ?



- (a) $C \cap (A' \cap B')$
 - (b) $C \cup (C' \cap A \cap B)$
 - (c) $C \cup (C \cap A) \cup (C \cap B)$
 - (d) $C \cup (A/B)$
8. If A, B, C are three sets, then what is $A - (B - C)$ equal to?
 - (a) $A - (B \cap C)$
 - (b) $(A - B) \cup C$
 - (c) $(A - B) \cup (A \cap C)$
 - (d) $(A - B) \cup (A - C)$
9. If $A = \{1, 2, 5, 6\}$ and $B = \{1, 2, 3\}$, then what is $(A \times B) \cap (B \times A)$ equal to?
 - (a) $\{(1, 1), (2, 1), (6, 1), (3, 2)\}$
 - (b) $\{(1, 1), (1, 2), (2, 1), (2, 2)\}$
 - (c) $\{(1, 1), (2, 2)\}$
 - (d) $\{(1, 1), (1, 2), (2, 5), (2, 6)\}$
10. Which one of the following is a null set ?
 - (a) $\{0\}$
 - (b) $\{\{\}\}$
 - (c) $\{\{\}\}$
 - (d) $\{x | x^2 + 1 = 0, x \in R\}$
11. In a car agency one day 120 cars were decorated with three different accessories viz., power window, AC and music system. 80 cars were decorated with power windows, 65 cars were decorated with AC and 80 cars were decorated with music systems. What is the minimum and maximum number of cars which were decorated with all of three accessories?
 - (a) 10, 61
 - (b) 10, 45
 - (c) 25, 35
 - (d) None of these

12. In a certain office, 72% of the workers prefer tea and 44% prefer coffee. If each of them prefers tea or coffee and 40 like both, the total number of workers in the office is :
 (a) 200 (b) 240
 (c) 250 (d) 320
13. A survey show that 63% of the Indians like cheese whereas 76% like apples. If $x\%$ of the Indians like both cheese and apples, then find the range of x .
 (a) $0 \leq x \leq 23\%$ (b) $0 \leq x \leq 39\%$
 (c) $4 \leq x \leq 35\%$ (d) $6 \leq x \leq 33\%$
14. $A = \{x | x \text{ is a prime number} \leq 100\}$
 $B = \{x | x \text{ is an odd number} \leq 100\}$
 What is the ratio of the number of subsets of set A to set B ?
 (a) 2^{25} (b) 2^{-25}
 (c) 2 (d) $\frac{50^2}{25^2}$
15. If A and B are two sets such that A has 12 elements, B has 17 elements, and $A \cup B$ has 21 elements, how many elements does $A \cap B$ have ?
 (a) 7 (b) 8
 (c) 9 (d) 10
16. In an examination 70% of the candidates passed in English, 65% in Mathematics, 27 % failed in both the subjects. Find the total number of candidates.
 (a) 200 (b) 400
 (c) 300 (d) 100
17. If the set A has p elements, B has q elements, then the number of elements in $A \times B$ is
 (a) $p + q + 1$ (b) pq
 (c) p^2 (d) $p + q$
18. Let $A = \{(n, 2n) : n \in N\}$ and $B = \{(2n, 3n) : n \in N\}$. What is $A \cap B$ equal to ?
 (a) $\{(n, 6n) : n \in N\}$ (b) $\{(2n, 6n) : n \in N\}$
 (c) $\{(n, 3n) : n \in N\}$ (d) ϕ
19. If $X = \{x : x > 0, x^2 < 0\}$, and $Y = \{\text{flower, Churchill, moon, Kargil}\}$, then which one of the following is a correct statement?
 (a) X is well defined but Y is not a well defined set
 (b) Y is well defined but X is not a well defined set
 (c) Both X and Y are well defined sets
 (d) Neither X nor Y is a well defined set
20. If $n(A) = 115$, $n(B) = 326$, $n(A - B) = 47$, then what is $n(A \cup B)$ equal to?
 (a) 373 (b) 165
 (c) 370 (d) 394
21. What is the percentage of persons who read all the three papers?
 (a) 20% (b) 25%
 (c) 30% (d) 40%
22. What is the percentage of persons who read only two papers?
 (a) 19% (b) 31%
 (c) 44% (d) None of these
23. What is the percentage of persons who read only one paper?
 (a) 38% (b) 48%
 (c) 51% (d) None of these
24. What is the percentage of persons who read only A but neither B nor C ?
 (a) 4% (b) 3%
 (c) 1% (d) None of these
25. If A and B are any two sets, then what is $A \cap (A \cup B)$ equal to?
 (a) Complement of A (b) Complement of B
 (c) B (d) A
26. Let $A = \{3, \pi, \sqrt{2}, -5, 3 + \sqrt{7}, 2/7\}$. The subset of A containing all the elements from it which are irrational numbers is
 (a) $\{\pi, \sqrt{2}, 3 + \sqrt{7}\}$ (b) $\{3, \pi, 2/7, -5, 3 + \sqrt{7}\}$
 (c) $\{3, 2/7, -5\}$ (d) $\{3, -5\}$
27. 40% of the people read newspaper X , 50% read newspaper Y and 10% read both the papers. What percentage of the people read neither newspaper?
 (a) 10% (b) 15%
 (c) 20% (d) 25%

Directions for questions 28–31: Read the passage below and solve the questions based on it.

5% of the passengers do not like coffee, tea and lassi and 10% like all the three, 20% like coffee and tea, 25% like lassi and coffee and 25% like lassi and tea. 55% like coffee, 50% like tea, and 50% like lassi.

28. The passengers who like only coffee is greater than the passengers who like only lassi by
 (a) 25% (b) 100%
 (c) 75% (d) 0%
29. The percentage of passengers who like both tea and lassi but not coffee, is
 (a) 15 (b) 25
 (c) 40 (d) 75
30. The percentage passengers who like at least 2 of the coffee, tea and lassi, is
 (a) 30 (b) 45
 (c) 50 (d) 60
31. If the number of passengers is 180, then the number of passengers who like lassi only, is
 (a) 10 (b) 18
 (c) 27 (d) 36

For the next Four (21–24) questions that follow:

In a city, three daily newspapers A , B , C are published, 42% read A ; 51% read B ; 68% read C ; 30% read A and B ; 28% read B and C ; 36% read A and C ; 8% do not read any of the three newspapers.

21. What is the percentage of persons who read all the three papers?

Expert Level

- Set A has 4 elements and set B has 7 elements. What can be the minimum number of elements in $A \cup B$?
(a) 6 (b) 7
(c) 8 (d) 9
 - In a political survey, 78% of the politicians favour at least one proposal. 50% of them are in favour of proposal A, 30% are in favour of proposal B and 20% are in favour of proposal C. 5% are in favour of all three proposals. What is the percentage of people favouring more than one proposal?
(a) 16 (b) 17
(c) 18 (d) 19
 - Which one of the following is correct?
(a) $A \cup (B - C) = A \cap (B \cap C)$
(b) $A - (B \cup C) = (A \cap B') \cap C'$
(c) $A - (B \cap C) = (A \cap B') \cap C$
(d) $A \cap (B - C) = (A \cap B) \cap C$
 - In a referendum about three proposals, 78% of the people were against at least one of the proposals, 50% of the people were against proposal 1st, 30% against proposal 2nd and 20% against proposal 3rd. If 5% of the people were against all the three proposals, what percentage of people were against more than one of the three proposals?
(a) 10 (b) 12
(c) 17 (d) 22
 - A survey among 151 persons is conducted regarding their favourite channel of radio – Radio city, Radio Mirchi and Radio life. It was found that every listener of Radio Mirchi also listen either Radio city or Radio life. The number of persons listening all the radio channels is the same as the number of persons who listen none of the channels. 55 persons listen exactly two channels and 70 persons listen only one channel. The number of people who listen all the three channels?
(a) 16 (b) 13
(c) 9 (d) data insufficient
 - $A = \{x \mid 3x^2 - 7x - 6 = 0\}$, $B = \{x \mid 6x^2 - 5x - 6 = 0\}$. Find $A \cap B$
(a) $\left\{-\frac{2}{3}\right\}$ (b) $\left\{-\frac{1}{3}\right\}$
(c) $\left\{-\frac{4}{3}\right\}$ (d) $\left\{-\frac{5}{3}\right\}$
 - If $X = \{8^n - 7n - 1 : n \in N\}$ and $Y = \{49(n - 1) : n \in N\}$, then
(a) $X \subseteq Y$ (b) $Y \subseteq X$
(c) $X = Y$ (d) None of these
 - Of the members of three athletic teams in a certain school, 21 are on the basket ball team, 26 on the hockey team, and 29 on the football team, 14 play hockey and basketball. 15 play hockey and football, and 12 play football and basketball, 8 are on all the three teams. How many members are there altogether?
(a) 38 (b) 47
(c) 51 (d) 43
 - A group of 50 students appeared for the two examinations one in Physics and the other in Mathematics. 38 students passed in Physics and 37 in Mathematics. If 30 students passed in both subjects, determine how many students failed in both the subjects.
(a) 2 (b) 3
(c) 4 (d) 5
- Directions for Questions 10 & 11 :** Read the information given below and answer the questions that follow :
- There are three different cable channels namely Ahead, Luck and Bang. In a survey it was found that 85% of viewers respond to Bang, 20% to Luck, and 30% to Ahead. 20% of viewers respond to exactly two channels and 5% to none.
- What percentage of the viewers responded to all three?
(a) 10 (b) 12
(c) 14 (d) None of these
 - Assuming 20% respond to Ahead and Bang and 16% respond to Bang and Luck, what is the percentage of viewers who watch only Luck?
(a) 20 (b) 10
(c) 16 (d) None of these
 - There are two disjoint sets S_1 and S_2 where
 $S_1 = \{f(1), f(2), f(3), \dots\}$
 $S_2 = \{g(1), g(2), g(3), \dots\}$ such that $S_1 \cup S_2$ forms the set of natural number.
Also $f(1) < f(2) < f(3), \dots$ & $g(1) < g(2) < g(3), \dots$ and $f(n) = g(g(n)) + 1$ then what is $g(a)$?
(a) 0 (b) 1
(c) 2 (d) can't be determined
 - Consider the sets $T_n = \{n, n + 1, n + 2, n + 3, n + 4\}$, where $n = 1, 2, 3, \dots, 96$. How many of these sets contains 6 or any integral multiple thereof (i.e., any one of the numbers 6, 12, 18,.....)?
(a) 80 (b) 81
(c) 82 (d) 83

14. A survey was conducted of 100 people to find out whether they had read recent issues of Golmal, a monthly magazine. The summarized information regarding readership in 3 months is given below:
Only September : 18; September but not August: 23;
July : 48; July and August : 10;
September and July : 8;
September : 28;
None of the three months : 24.
What is the number of surveyed people who have read exactly two consecutive issues (out of the three)?
(a) 7 (b) 9
(c) 12 (d) 14
15. Which one of the following operations on sets is not correct where B' denotes the complement of B ?
(a) $(B' - A') \cup (A' - B') = (A \cup B) - (A \cap B)$
(b) $(A - B) \cup (B - A) = (A' \cup B') - (A' \cap B')$
(c) $(B' - A') \cap (A' - B') = (B - A) \cap (A - B)$
(d) $(B' - A') \cap (A' - B') = (B - A') \cup (A' - B)$
16. For non-empty sets A , B and C , the following two statements are given:
Statement P : $A \cap (B \cup C) = (A \cap B) \cup C$
Statement Q : C is a subset of A
Which one of the following is correct ?
(a) $P \Leftarrow Q$
(b) $P \Leftrightarrow Q$
(c) $P \Rightarrow Q$
(d) Nothing can be said about the correctness of the above three with certainty
17. If A , B and C are three sets and U is the universal set such that $n(U) = 700$, $n(A) = 200$, $n(B) = 300$ and $n(A \cap B) = 100$, then what is the value of $(A' \cap B')$?
(a) 100 (b) 200
(c) 300 (d) 400
18. If $A = \{4n + 2 \mid n \text{ is a natural number}\}$ and $B = \{3n \mid n \text{ is a natural number}\}$, then what is $(A \cap B)$ equal to?
(a) $\{12n^2 + 6n \mid n \text{ is a natural number}\}$
(b) $\{24n - 12 \mid n \text{ is a natural number}\}$
(c) $\{60n + 30 \mid n \text{ is a natural number}\}$
(d) $\{12n - 6 \mid n \text{ is a natural number}\}$
19. In a survey among B-school students, 68% of those surveyed were in favour of at least one of the three magazines A , B and C . 38% of those surveyed favoured magazine A , 26% favoured magazine B and 36% favoured magazine C . If 11% of those surveyed favoured all three magazines. What percent of those surveyed favoured more than one of the three magazines?
(a) 25% (b) 33%
(c) 21% (d) 26%
20. In a hotel, 60% had vegetarian lunch while 30% had non-vegetarian lunch and 15% had both types of lunch. If 96 people were present, how many did not eat either type of lunch ?
(a) 20 (b) 24
(c) 26 (d) 28
21. In a group of 80 employees, the number of employees who are engineers is twice that of the employees who are MBAs. The number of employees who are not engineers is 32 and employees who are both engineers and MBAs is twice that of the employees who are only MBAs. How many employees are neither engineer (B. Tech) nor MBAs?
(a) 24 (b) 38
(c) 36 (d) can't determined
- Directions for Questions 22–24:** There are 60 workers out of which 25 are women. Also:
(i) 28 workers are married:
(ii) 26 workers are graduate
(iii) 20 married workers are graduate of which 9 are men
(iv) 15 men are graduate
(v) 15 men are married
22. How many unmarried women are graduate?
(a) 20 (b) 8
(c) 0 (d) can't be determined
23. How many unmarried women work in the company?
(a) 11 (b) 12
(c) 9 (d) None of these
24. How many graduate men are married?
(a) 9 (b) 15
(c) 13 (d) None of these

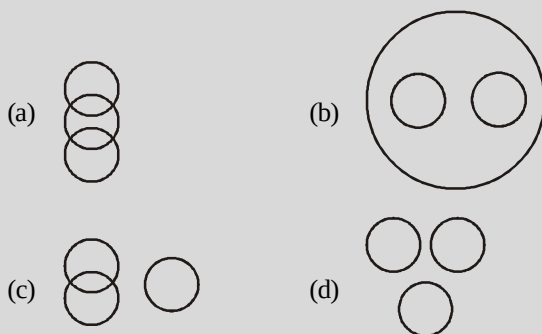
Test Yourself

Directions (Qs. 1–3): Refer to the data below and answer the questions that follow:

In the survey among students at all the IIMs, it was found that 48% preferred coffee, 54% liked tea and 64% smoked. Of the total, 28% liked coffee and tea, 32% smoked and drank tea and 30% smoked and drank coffee. Only 6% did none of these. If the total number of students is 2000 then find.

- The ratio of the number of students who like only coffee to the number who like only tea is
(a) 5 : 3 (b) 8 : 3
(c) 2 : 3 (d) 3 : 2
- Number of students who like coffee and smoking but not tea is
(a) 600 (b) 240
(c) 280 (d) 360
- The percentage of those who like coffee or tea but not smoking among those who like at least one of these is
(a) more than 30 (b) less than 30
(c) less than 25 (d) None of these
- If $A = \{1, 2, 3, 4, 5\}$, then the number of proper subsets of A is
(a) 31 (b) 38
(c) 48 (d) 54
- Let $A = \{x : x \in \mathbb{R}, |x| < 1\}$; $B = \{x : x \in \mathbb{R}, |x-1| \geq 1\}$ and $A \cup B = \mathbb{R} - D$, then the set D is
(a) $\{x : 1 < x \leq 2\}$ (b) $\{x : 1 \leq x < 2\}$
(c) $\{x : 1 \leq x \leq 2\}$ (d) None of these
- Let $X = \{1, 2, 3, 4, 5\}$. The number of different ordered pairs (Y, Z) that can be formed such that $Y \subseteq X$, $Z \subseteq X$ and $Y \cap Z$ is empty is
(a) 5^2 (b) 3^5
(c) 2^5 (d) 5^3

Directions (Qs. 7 & 8): Given below are four diagrams one of which describes the relationship among the three classes given in each of the two questions that follow. You have to decide which of the diagrams is the most suitable for a particular set of classes.



- Elephants, tigers, animals
- Gold, platinum, ornaments
- Sets A and B have 5 and 7 elements respectively. What can be the minimum number of elements in $A \cup B$?
(a) 5 (b) 7
(c) 12 (d) 35

- In a town of 10,000 families it was found that 40% family buy newspaper A , 20% buy newspaper B and 10% families buy newspaper C , 5% families buy A and B , 3% buy B and C and 4% buy A and C . If 2% families buy all the three newspapers, then number of families which buy A only is
(a) 3100 (b) 3300
(c) 2900 (d) 1400
- The number of elements in the set $\{(a, b) : 2a^2 + 3b^2 = 35, a, b \in \mathbb{Z}\}$, where $\mathbb{Z}$ is the set of all integers, is
(a) 2 (b) 4
(c) 8 (d) 12
- In a survey among B-school students, 68% of those surveyed were in favour of at least one of the three magazines- A , B and C . 38% of those surveyed favoured magazine A , 26% favoured magazine B and 36% favoured magazine C . If 11% of those surveyed favoured all three magazines. What per cent of those surveyed favoured more than one of the three magazines?
(a) 25% (b) 33%
(c) 21% (d) 26%
- In our coaching there were 200 students enrolled for DI, 150 for English and 150 for Maths of these 80 students enrolled for both DI and English. 60 students enrolled for Maths and English, while 70 students enrolled for DI and Maths. Some of these students enrolled for all the three subjects. Diwakar teaches those students who are enrolled for DI classes only. Priyanka teaches those students who are enrolled for English only and Varun teaches those students who are enrolled for Maths only. Sarvesh is a senior most faculty therefore, he can teach all the three subjects. Students always prefer a specialist for their respective subjects. If Diwakar teaches 80 students then the other three faculty can be arranged in terms of the number of students, taught as:
(a) Sarvesh > Varun > Priyanka
(b) Sarvesh > Priyanka > Varun
(c) Varun > Sarvesh > Priyanka
(d) None of these
- In a certain zoo, there are 42 animals in one sector, 34 in the second sector and 20 in the third sector. Out of this, 24 graze in sector one and also in sector two. 10 graze in sector two and sector three. 12 graze in sector one and sector three. These figures also include four animals grazing in all the three sectors. If all these animals are now transported to another zoo, then find the total number of animals?
(a) 38 (b) 56
(c) 54 (d) None of these
- A survey was conducted at a coaching institution and it was found that there were 34 students who appeared in MAT. There were 37 students who appeared in CAT of which 17 students appeared in MAT. 30 students appeared in XAT of which 13 students appeared in MAT. Of the XAT applicants (i.e., appeared students) 14 appeared in CAT and these 6 appeared in MAT. How many students appeared in CAT but not in MAT or XAT?
(a) 9 (b) 10
(c) 12 (d) None of these

Hints & Solutions

Foundation Level

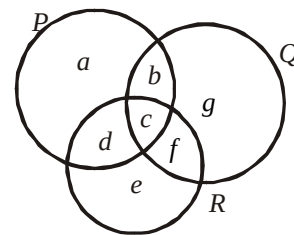
- (d) Total number of students = $100 + 50 - 25 = 125$
- (a)
- (c)
- (b)
- (b)
- (a) Total number of students = 100
Let E denote the students who have passed in English.
Let M denote the students who have passed in Maths.
 $\therefore n(E) = 75, n(M) = 60$ and $n(E \cap M) = 45$
we know $n(E \cup M) = n(E) + n(M) - n(E \cap M)$
 $= 75 + 60 - 45 = 90$
Required number of students = $90 - 45 = 45$
- (a) Since, A and B are two disjoint therefore $A \cap B = \phi$
 $\therefore A - B = A - (A \cap B)$
- (a)
- (b)
- (b)
- (b)
- (b)
- (a) $A \cap (A' \cup B) = A \cap A' = \phi$
- (c) We know that
 $[(A \cup B) \cap C]' = A' \cap B' \cup C'$

Standard Level

- (d) Chemistry but not Physics = $c + f = 206$.
- (c) Physics and Maths but not Chemistry = $a + e + g = 628$.
- (a) Physics but neither Maths nor Chemistry = $a = 164$.
- (d) $T = \{3, 11, 19, 27, \dots, 467\}$ is an AP with $a = 3$ and $d = 8$.
To find number of terms, we use the formula for n th term:
 $a + (n - 1)d = 3 + (n - 1)8 = 467$.
Hence, $n = 59$. S = subset in which not sum of two elements = 470.
So, S can be a set in which either the first half or the second half of the terms are present. So number of maximum possible elements in $S = \frac{59}{2} = 29.5 \approx 30$.
- (c) Employees who doesn't have VCD = $100 - 70 = 30\%$
Employees who doesn't have MWO = $100 - 75 = 25\%$
Employees who doesn't have AC = $100 - 80 = 20\%$

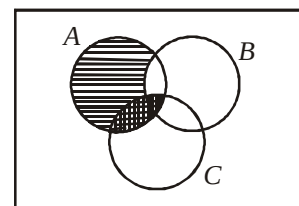
Employees who doesn't have WM = $100 - 85 = 15\%$
 $\therefore$ Total employees who doesn't have atleast one of the four equipments = $30 + 25 + 20 + 15 = 90\%$
 $\therefore$ Percentage of employees having all four gadgets = $100 - 90 = 10\%$.

- (d) The shaded region represents $(P \cap Q) \cup (P \cap R)$.
Let the intersecting sets P, Q, R divide it into 7 regions marked, a to g as shown below.



The shaded part contains regions b, c , and d .

- $(P \cup Q) - (P \cap Q) \equiv$ regions $a, b, c, d, f, g, -b, c \equiv a, d, f, g, \therefore$ not correct.
 - $(P \cap (Q \cap R)) \equiv a, b, c, d, \cap c, f \equiv c, b, c, d$ not correct.
 - $(P \cap Q) \cap (P \cap R) \equiv$ regions $b, c, \cap$ region, $c, d \equiv c$, so not correct
 - $(P \cap Q) \cup (P \cap R) \equiv$ regions $b, c, \cup c, d \equiv b, c, d$ so correct.
- (c) In the given Venn diagram shaded region is
 $C \cup (C \cap A) \cup (C \cap B)$.
 - (c) Following venn diagram shows the relation
 $A - (B - C)$



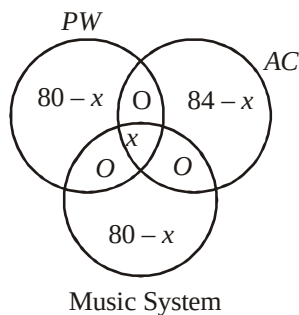
In the above venn diagram, horizontal lines shows $(A - B)$ and vertical lines show $(A \cap C)$

$$\therefore (A - B) \cup (A \cap C) = A - (B - C)$$

- (b) Let $A = \{1, 2, 5, 6\}$ and $B = \{1, 2, 3\}$
 $\therefore A \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (5, 1), (5, 2), (5, 3), (6, 1), (6, 2), (6, 3)\}$
and $B \times A = \{(1, 1), (1, 2), (1, 5), (1, 6), (2, 1), (2, 2), (2, 5), (2, 6), (3, 1), (3, 2), (3, 5), (3, 6)\}$
 $\Rightarrow (A \times B) \cap (B \times A)$
 $= \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

10. (d) Consider the set given in option 'd'.
 $\{x | x^2 + 1 = 0, x \in R\}$
 Let $x^2 + 1 = 0 \Rightarrow x^2 = -1 \Rightarrow x = \pm i$ which is complex.
 But $x \in R$. Hence for, any $x \in R$, $x^2 + 1$ can not be zero.
11. (d) 80 cars were decorated with power windows it means at least 40 cars were decorated with AC or music system or both.
 84 cars were decorated with ACs, it means at least 36 cars were decorated with power windows or music systems.
 80 cars were decorated with music system it means at least 40 cars were decorated with power windows or ACs.
 It means if there is no intersection in these three, then at most $40 + 36 + 40 = 116$ cars had been decorated with one or two accessories.
 Hence at least 4 cars would have been decorated with all three accessories.

For maximum value of x :

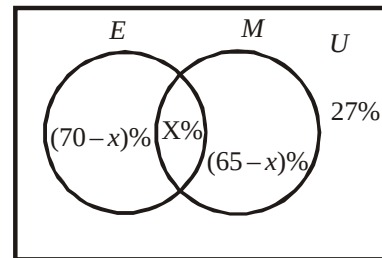


Total number of cars = $(80 - x) + (84 - x) + (80 - x) + x$
 $120 = 244 - 2x$
 $\Rightarrow 2x = 124 \Rightarrow x = 62$
 minimum $\rightarrow 4$ cars and maximum $\rightarrow 62$ cars

12. (c) Let total number be x . Then
 $n(A) = \frac{72}{100}x = \frac{18x}{25}$, $n(B) = \frac{44}{100}x = \frac{11x}{25}$ and
 $n(A \cap B) = 40$ $n(A \cap B) = n(A) + n(B) - n(A \cup B)$
 $\Rightarrow x \cdot \frac{18x}{25} + \frac{11x}{26} - 40 \Rightarrow \frac{29x}{25} - x = 40$
 $\Rightarrow \frac{4x}{25} = 40 \Rightarrow x = 250$
13. (b) $n(C) = 63\%$
 $n(A) = 76\%$
 $n(C \cup A) = n(C) + n(A) - n(C \cap A)$
 $100\% = 63\% + 76\% - X\%$
 $X\% = 39\%$
14. (b) We know that there are 25 prime number below 100.
 $n(A) = 25$
 The total number of subsets of sets $A = 2^{25}$

There are 50 odd numbers below 100.
 The total number of subsets of $B = 2^{50}$
 Required ratio = $\frac{2^{25}}{2^{50}} = 2^{-25}$.

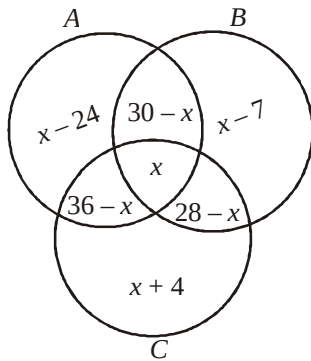
15. (b) $n(A) = 12, n(B) = 17, n(A \cup B) = 21$
 $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 $21 = 12 + 17 - n(A \cap B)$ or $n(A \cap B)$
 $= 12 + 17 - 21 = 8$
 $\Rightarrow A \cap B$ has 8 elements.
16. (b) Let the set E and M represent students who passed in English and Mathematics respectively.
 $n(E \cup M) = (100 - 27)\% = 73\%$
 $n(E \cup M) = n(E) + n(M) - n(E \cap M)$
 $73\% = 70\% + 65\% - x\%$
 $x\% = 62\%$
 Now, $62\% \equiv 248$



$$\therefore \text{Total number of candidates} = \frac{248 \times 100}{62} = 400$$

17. (b) As A has p elements and B has q elements so, $A \times B$ has pq elements.
18. (d) $A = \{(n, 2n) : n \in N\}$ and $B = \{(2n, 3n) : n \in N\}$
 Listing few members of each set
 $A = \{(1, 2), (2, 4), (3, 6), \dots\}$
 $B = \{(2, 3), (4, 6), (6, 9), \dots\}$
 There is no member common to both these sets, hence,
 $A \cap B = \phi$
19. (c) $X = \{x : x > 0, x^2 < 0\}$
 We know that the square of each number greater than zero is always greater than zero. So, X contains no member and so, X is null set but a well defined set.
 Also, $Y = \{\text{flower, Churchill, Moon, Kargil}\}$ is well defined. So, Y is also a well defined set.
20. (a) We know, for two sets A and B
 $A - B = A - (A \cap B)$
 $\therefore n(A - B) = n(A) - n(A \cap B)$
 Given, $n(A) = 115, n(B) = 326$ and $n(A - B) = 47$
 $\Rightarrow 47 = 115 - n(A \cap B)$
 $\Rightarrow n(A \cap B) = 68$
 Consider $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
 $= 115 + 326 - 68 = 373$

(21-24).



Let $x\%$ people read all the three newspapers.

Since 8% people do not read any newspapers.

$$\therefore (x-24) + (x-7) + (x+4) + (30-x) + (36-x) + (28-x) + x = 92$$

$$\Rightarrow x + 98 - 31 = 92$$

$$\Rightarrow x = 92 - 67 = 25$$

21. (b) Hence people who read all the three newspapers = 25%

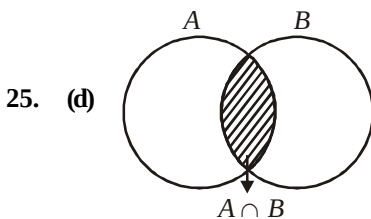
22. (d) $(30-x) + (36-x) + (28-x) = 94 - 3x$
 $= 98 - 3 \times 25 = 23$

Hence percentage of people who read only two newspapers = 23%

23. (b) $(x-24) + (x-7) + (x+4) = 3x-27$
 $= 3 \times 25 - 27 = 48$

Hence percentage of people who read only one newspaper = 48%

24. (c) $x-24 = 25-24 = 1$
 Hence percentage of people who read only Newspaper A but neither B nor C = 1%



25. (d)

$$A \cap (A \cup B) = (A \cap A) \cup (A \cap B) = A \cup (A \cap B)$$

$$= A \text{ (By diagram)}$$

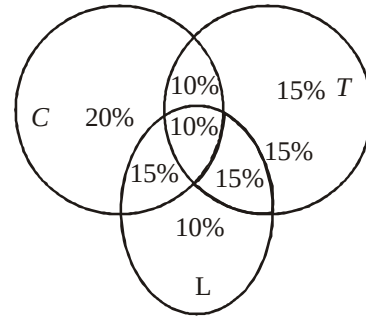
$$\text{Thus, } A \cap (A \cup B) = A$$

26. (a)

27. (c) $n(A) = 40, n(B) = 50, n(A \cap B) = 10$.
 $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 40 + 50 - 10 = 80$.
 $\therefore$ Percentage reading either or both newspapers = 80%.

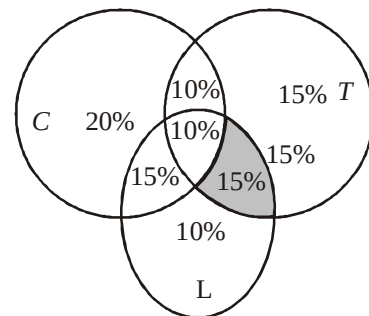
Hence, percentage reading neither newspaper
 $= (100 - 80)\% = 20\%$

(28-31)

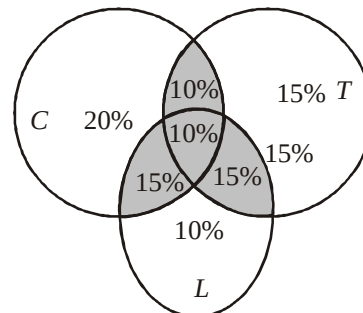


Where C = Coffee, T = Tea and L = Lassi

28. (b) The passengers who like only coffee = 20% and the passengers who like only lassi = 10%
 Required passengers = 100%
29. (a) It can be seen that the percentage of passengers who like both tea and lassi but not coffee = 15%. This is the figure representing this area



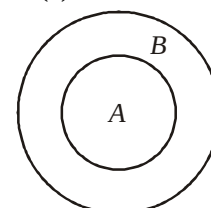
30. (c) The percentage of passengers who like at least 2 of the coffee, tea and lassi can be seen in the below figure:



31. (b) 10% of the people like only lassi. So, the number of persons = 18

Expert Level

1. (b) $A \cup B$ will have minimum number of elements, if set A is a subset of set B.
 $\therefore n(A \cup B) = n(B) = 7$



$$2. \quad (b) \quad n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

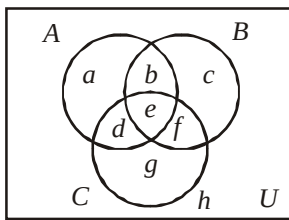
$$\text{or } 78 = 50 + 30 + 20 - \sum n(A \cap B) + 5$$

$$\text{or } \sum n(A \cap B) = 27$$

This includes $n(A \cap B \cap C)$ three times.

$\therefore$ Percentage of people favouring more than one proposal $= 27 - 5 \times 2 = 17$

3. (b) Let a Venn-diagram be drawn taking three intersecting sets A , B and C under a universal set U . This makes 8 regions a to h as shown.



A has regions a, b, d, e

B has regions b, c, e, f

C has regions d, e, f, g

C' has regions a, b, c, h

B' has regions a, d, g, h

$$(a) : A \cup (B - C) = A \cap (B \cap C)$$

$$\text{LHS} \equiv (a, b, e, d) \cup b, c \equiv a, b, c, d, e.$$

$$\text{RHS} \equiv a, b, d, e \cap e, f \equiv e,$$

So, statement (a) is not correct.

$$(b) : A - (B \cup C) = (A \cap B') \cap C'$$

$$\text{LHS} \equiv (a, b, d, e) - (b, c, d, e, f, g) \equiv a.$$

$$\text{RHS} \equiv (a, b, d, e \cap a, d, g, h) \cap (a, b, c, h) \equiv a,$$

So, statement (b) is correct.

Correct statement is :

$$A - (B \cup C) = (A \cap B') \cap C'$$

$$4. \quad (c) \quad n(1 \cup 2 \cup 3) = n(1) + n(2) + n(3) - n(1 \cap 2) - n(2 \cap 3) - n(1 \cap 3) + n(1 \cap 2 \cap 3)$$

$$\text{or, } 78\% = 50\% + 30\% + 20\% - n(1 \cap 2) - n(2 \cap 3)$$

$$- n(1 \cap 3) + 5\%$$

$$n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3) = 27\%$$

$$\text{Only } n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3) = 27\% - 15\%$$

$$= 12\%, \text{ since } n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3)$$

$$= 27\% - 15\%$$

Percentage of people against more than one of the three proposals $= \text{Only } [n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3)]$

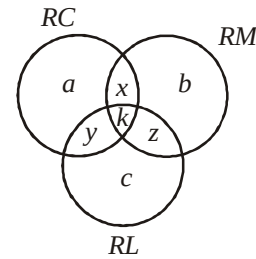
$$+ n(1 \cap 2 \cap 3)$$

$$= 12\% + 5\% = 17\%$$

$$5. \quad (b) \quad \beta = (x + y + z) = 55$$

$$\alpha = (a + b + c) = 70$$

$$\gamma = k$$



Let m people listen none of the three channels, then $m = \gamma = k$

$$(\alpha + \beta + \gamma) + m = 151$$

$$\Rightarrow \alpha + \beta + \gamma + \gamma = 151$$

$$\Rightarrow (55 + 70) + 2\gamma = 151$$

$$\Rightarrow \gamma = 13$$

Hence, there are 13 people listen all three channels.

$$6. \quad (a) \quad 3x^2 - 7x - 6 = 0$$

$$\therefore 3x^2 - 9x + 2x - 6 = 0$$

$$3x(x - 3) + 2(x - 3) = 0$$

$$\therefore (3x + 2)(x - 3) = 0$$

$$\therefore x = -\frac{2}{3} \text{ or } x = 3 \quad \therefore A \equiv \left\{-\frac{2}{3}, 3\right\}$$

$$6x^2 - 5x - 6 = 0$$

$$6x^2 - 9x + 4x - 6 = 0$$

$$3x(2x - 3) + 2(2x - 3) = 0$$

$$(3x + 2)(2x - 3) = 0$$

$$x = -\frac{2}{3} \text{ or } x = \frac{2}{3}$$

$$\therefore B = \left\{-\frac{2}{3}, \frac{2}{3}\right\} \quad \therefore A \cap B = \left\{-\frac{2}{3}\right\}$$

$$7. \quad (a) \quad \text{Since } 8^n - 7n - 1 = (7 + 1)^n - 7n - 1 = 7^n + {}^nC_1 7^{n-1} + {}^nC_2 7^{n-2} + \dots + {}^nC_{n-1} 7 + {}^nC_n - 7n - 1$$

$$= {}^nC_2 7^2 + {}^nC_3 7^3 + \dots + {}^nC_n 7^n$$

$$({}^nC_0 = {}^nC_n, {}^nC_1 = {}^nC_{n-1} \text{ etc.})$$

$$= 49 [{}^nC_2 + {}^nC_3 (7) + \dots + {}^nC_n 7^{n-2}]$$

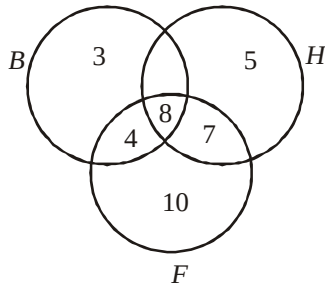
$$\therefore 8^n - 7n - 1 \text{ is a multiple of } 49 \text{ for } n \geq 2.$$

$$\text{For } n = 1, 8^n - 7n - 1 = 8 - 7 - 1 = 0$$

$$\text{For } n = 2, 8^n - 7n - 1 = 64 - 14 - 1 = 49$$

$$\therefore 8^n - 7n - 1 \text{ is a multiple of } 49 \text{ for all } n \in N.$$

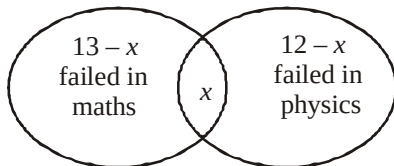
8. (d) Let B, H, F denote the sets of members who are on the basketball team, hockey team and football team respectively. Then $n(B) = 21, n(H) = 26, n(F) = 29$
 $n(H \cap B) = 14, n(H \cap F) = 15,$
 $n(F \cap B) = 12$ and $n(B \cap H \cap F) = 8$
 We have to find $n(B \cup H \cup F)$.
 Using the result
 $n(B \cup H \cup F) = n(B) + n(H) + n(F) - n(B \cap H) - n(H \cap F) - n(F \cap B) + n(B \cap H \cap F)$
 We get $n(B \cap H \cap F) = 21 + 26 + 29 - 14 - 15 - 12 + 8$
 $= 84 - 41 = 43$



Total members $= 3 + 6 + 5 + 4 + 8 + 7 = 43$

9. (d) $n(P) = 38, n(M) = 37, n(P \cap M) = 30$
 $n(P \cup M) = n(P) + n(M) - n(P \cap M)$
 $= 38 + 37 - 30 = 75 - 30 = 45$
 Number of students who failed, i.e.,
 $n(P \cap M) = n(\cup) - n(P \cup M) = 50 - 45 = 5$.
 Hence, number of students failed are 5.

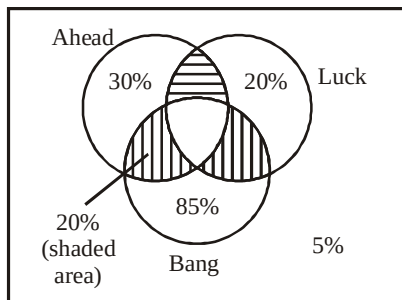
Examination method : Let x failed in both



Now, total fail $= 50 - 30 = 20$

$$13 - x + x + 12 - x = 30 \Rightarrow x = 5$$

10. (a)



Here, $n(A \cup L \cup B) = 100 - 5 = 95$

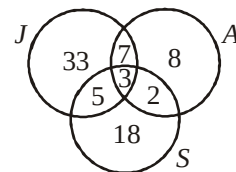
$$n(A \cap L) + n(L \cap B) + n(B \cap A)$$

$$- 3n(A \cap L \cap B) = 20$$

$$\begin{aligned} & n(A \cup L \cup B) \\ &= n(A) + n(L) + n(B) - n(A \cap L) - n(L \cap B) \\ & \quad - n(B \cap A) + n(A \cap L \cap B) \\ & 95 = 85 + 30 + 20 - [n(A \cap L) + n(L \cap B) \\ & \quad + n(B \cap A) - 3n(A \cap L \cap B)] \\ & \quad - 3n(A \cap L \cap B) + n(A \cap L \cap B) \\ & \Rightarrow 95 = 135 - 20 - 2n(A \cap L \cap B) \end{aligned}$$

$$\text{or } n(A \cap L \cap B) = \frac{20}{2} = 10$$

11. (d) Percentage of viewers who watch only Luck
 $n(\text{only } L) = n(L) - n(L \cap B) - n(L \cap A) + n(L \cap B \cap A)$
 $= 20 - (16) - [20 - \{16 - 10\} - \{20 - 10\} + 10] + 10$
 $= 20 - 16 - [20 - 6 - 10 + 10] + 10 = 20 - 16 - 14 + 10 = 0$
12. (b) It is given that $f(n) = g(g(n)) + 1$
 Therefore, $f(n) > g(g(n))$
 Also, $g(1) < g(2) < g(3)$ shows that the function $g(x)$ is an increasing function. So for a natural number n ,
 $g(n) \geq n$
 $\Rightarrow g(g(n)) \geq g(n)$
 Thus, $f(n) > g(n)$ for every n
 or $f(1) > g(1) \Rightarrow g(1)$ is the least number in $S_1 \cup S_2$.
 Now, $S_1 \cup S_2 =$ set of natural numbers.
 Therefore, 1 in $S_1 \cup S_2$ is the smallest number.
 Thus, $g(a) = 1$.
13. (a) Sets starting from 1, 7, 13..... does not contain multiple of 6.
 Now 1, 7, 13, 19. ... forms an A.P.
 $\Rightarrow T_n = 1 + (n-1)6 \leq 96 \Rightarrow 6n \leq 101 \Rightarrow n = 16$
 $\therefore$ No. of sets which doesn't contain the multiple of 6
 $= 96 - 16 = 80$.
14. (b) Putting the given information in the form of a Venn diagram, we get



$$n(J \cup A \cup S) = 100 - 24 = 76$$

$$n(S \cap J) = 8; n(\text{only } S) = 18$$

$$n(S \text{ but not } A) = 23 = \overbrace{n(\text{only } S)}^{18} + \underbrace{n(S) - n(S \cap A)}_5$$

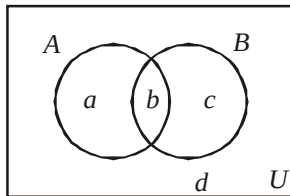
$$n(S \cap A \cap J) = n(S \cap J) - 5 = 8 - 5 = 3$$

To find the people who have read exactly 2 consecutive issues (out of 3) we shall find the people reading J & A and A & S.

Hence required no. = $7 + 2 = 9$.

15. (d) Let there be two sets A and B and universal set of A and B , be U .

Then drawing these sets on a Venn-diagram, four regions are created as shown in the figure :

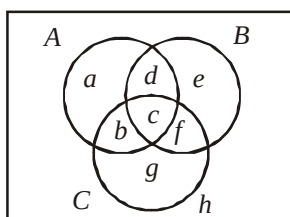


$B \equiv$ regions b, c
 $B' \equiv$ regions a, d
 $A \equiv$ regions a, b
 $A' \equiv$ regions c, d
 $A - B \equiv$ region a
 $B - A \equiv$ region c
 $B' - A' \equiv$ region a
 $A' - B' \equiv$ region c
 $B - A' \equiv$ region b
 $A' - B \equiv$ region d

From these we check the operations given in the choice. choices (a), (b) and (c) are correct.

- (d) LHS = region $a \cap c = \phi$
 RHS' = region $b \cup$ region $d = b, d$.
 So, for (d) LHS $\neq$ RHS

16. (b) A Venn diagram is drawn for 3 intersecting set A, B, C under a universal set U ; creating 8 regions in total named, a to h as shown



$$\text{Statement } P: A \cap (B \cup C) = (A \cap B) \cup C$$

$$\text{LHS} \equiv a, b, c, d, \cap (b, c, d, e, f, g, h) \equiv b, c, d$$

$$\text{RHS} \equiv (c, d) \cup (b, c, f, g) = b, c, d, f, g$$

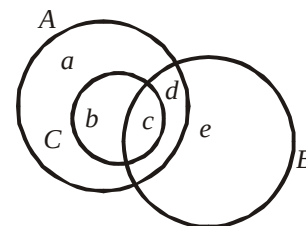
If P is correct then region f, g do not lie in set C and set C has regions b, c only.

This follows that C is subset of A . Since, Set A has regions a, b, c, d and C has regions b, c .

Thus, $P \Rightarrow Q$.

Also, if C is a subset of A , Q is true, then the Venn diagram appears as below:

$$\{A \cap (B \cup C)\}$$



LHS of P statement gives

$$\text{region } a, b, c, d \cap \text{region } b, c, d, e \equiv b, c, d.$$

$$\text{RHS: } \{(A \cap B) \cup C\} \text{ gives : region } c, d, \cup \text{ region } b, c. \\ \equiv b, c, d$$

and LHS = RHS shows $Q \Rightarrow P$,

Comparing both gives $P \Leftrightarrow Q$.

17. (c) From the given data
 $n(U) = 700, n(A) = 200, n(B) = 300$ and
 $n(A \cap B) = 100$.

We know that,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) \\ = 200 + 300 - 100 = 400$$

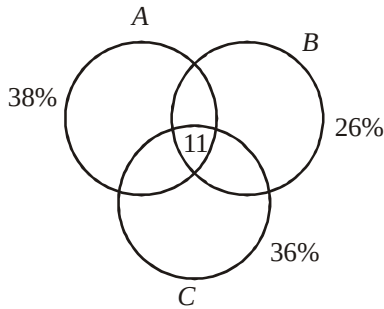
$$\text{Now, } n(A' \cap B') = n(U) - n(A \cup B) \\ = 700 - 400 = 300$$

18. (d) Let $A = \{4n + 2 : n \in \mathbb{N}\}$
 and $B = \{3n : n \in \mathbb{N}\}$
 $\Rightarrow A = \{6, 10, 14, 18, 22, 26, 30, 34, 38, 42, \dots\}$
 and $B = \{3, 6, 9, 12, 15, 18, 21, 24, 27, 30, \dots\}$
 $\therefore A \cap B = \{6, 18, 30, 42, \dots\}$
 $= 6 + 12n - 12 = 12n - 6$.

Hence, $A \cap B = \{12n - 6 : n \text{ is a natural number}\}$.

19. (c) $\alpha + \beta + \gamma = 68$
 $\alpha + 2\beta + 3\gamma = (38 + 26 + 36) = 100$
 and $\gamma = 11$
 $\therefore (\alpha + 2\beta + 3\gamma) - [(\alpha + \beta + \gamma) + \gamma] = \beta + \gamma$
 $= 100 - [68 + 11]$
 $= 21$

Hence 21% favoured more than one magazine.



($\alpha \rightarrow$ favoured only one magazine, $\beta \rightarrow$ favoured only two magazines, $\gamma \rightarrow$ favoured all three magazines)

20. (b) $n(A) = \left(\frac{60}{100} \times 96\right) = \frac{288}{5}$, $n(B) = \left(\frac{30}{100} \times 96\right) = \frac{144}{5}$,

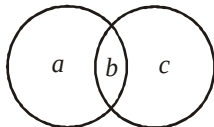
$$n(A \cap B) = \left(\frac{15}{100} \times 96\right) = \frac{72}{5}$$

$$\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\frac{288}{5} + \frac{144}{5} - \frac{72}{5} = \frac{360}{5} = 72$$

So, people who had either or both types of lunch = 72.
Hence, people who had neither type of lunch = $(96 - 72 = 24)$.

21. (a) Let a be the number of engineers only



c be the number of MBAs only

b be the number of employees who are both engineers and MBAs and

d be the number of employees who are neither engineer nor MBA

$$\therefore a + b + c + d = 80 \quad \dots(1)$$

$$(a + b) = 2(b + c) \Rightarrow (a - b) = 2c \quad \dots(2)$$

$$\text{and } c + d = 32 \quad \dots(3)$$

$$\text{and } a + d = 56 \quad \dots(4)$$

$$\text{and } b = 2c \quad \dots(5)$$

From eqs (2) and (5), we get

$$a = 2b$$

From eqs (1) and (3), we get

$$a + b = 48$$

From eq. (6) we get

$$b = 16$$

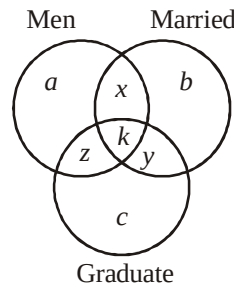
$$\therefore a = 32 \quad \text{(from eq. 6)}$$

$$\text{and } c = 8 \quad \text{(from eq. 5)}$$

$$\text{and } d = 24$$

Hence 24 employees are neither engineer nor MBAs.

Solution for questions number 22-24:



Total number of employees = 60

Women = 25

Men = 35

Married workers = 28

Graduate workers = 26

$a \rightarrow$ unmarried men who are not graduate

$b \rightarrow$ married women who are not graduate

$c \rightarrow$ unmarried women who are graduate

$x \rightarrow$ married men who are not graduate

$y \rightarrow$ married women who are graduate

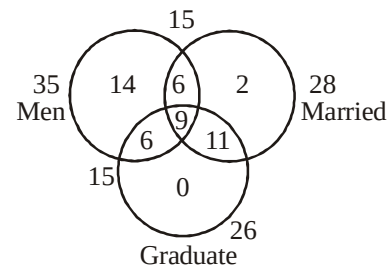
$z \rightarrow$ unmarried men who are graduate

$k \rightarrow$ married men who are graduate

$p \rightarrow$ unmarried women who are not graduate

According to the given information the Venn diagram can be completed as given below.

22. (c) No one unmarried woman is graduate. Hence (c)

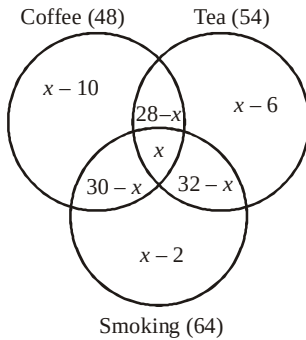


23. (b) Number of unmarried women
= $60 - [14 + 2 + 6 + 6 + 11 + 9] = 12$.

24. (a) There are 9 graduate men who are married.

Explanation of Test Yourself

Solutions for Questions 1 to 3: If you try to draw a figure for this question, the figure would be something like:



We can then solve this as:

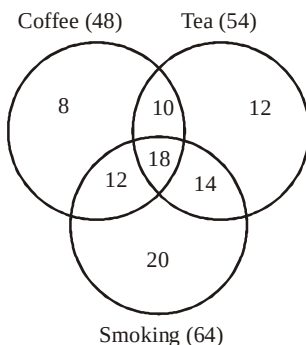
$$x - 10 + 28 - x + x + 30 - x + x + 2 + 32 - x + x - 6 = 94 \rightarrow x + 76 = 94 \rightarrow x = 18.$$

Note: In this question, since all the values for the use of the set theory formula are given, we can find the missing value of students who liked all three as follows:

$$94 = 48 + 54 + 64 - 28 - 32 - 30 + \text{All three} \rightarrow \text{All three} = 18$$

As you can see this is a much more convenient way of solving this question, and the learning you take away for the 3 circle situation is that whenever you have all the values known and the only unknown value is the centre value—it is wiser and more efficient to solve for the unknown using the formula rather than trying to solve through a venn diagram.

Based on this value of x , we get the diagram completed as:



1. (c) $8 : 12 = 2 : 3$

2. (b) $12\% \text{ of } 2000 = 240.$

3. (a) $30/94 \rightarrow \text{more than } 30\%.$

4. (a) **Note :** Number of proper subsets of $A = 2^n - 1$

Given : $A = \{1, 2, 3, 4, 5\}$

Here $n = 5$

$\therefore$ no. of proper subsets $= 2^5 - 1$

5. (b) $A = \{x : x \in R, -1 < x < 1\}$

$B = \{x : x \in R : x - 1 \leq -1 \text{ or } x - 1 \geq 1\}$

$= \{x : x \in R : x \leq 0 \text{ or } x \geq 2\}$

$\therefore A \cup B = R - D$, where $D = \{x : x \in R, 1 \leq x < 2\}$

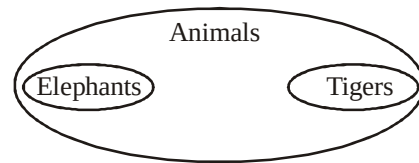
6. (b) Let $X = \{1, 2, 3, 4, 5\}$

Total no. of elements $= 5$

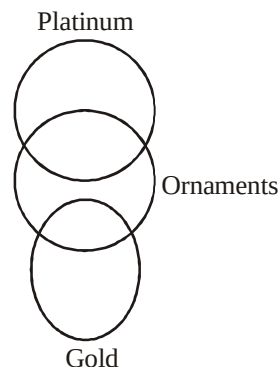
Each element has 3 options. Either set Y or set Z or none. ($\because Y \cap Z = \phi$)

So, number of ordered pairs $= 3^5$

7. (b)



8. (a)



9. (b) Here $n(A) = 5, n(B) = 7$

$\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 5 + 7 - n(A \cap B)$

Minimum number of elements in $A \cap B$ is 5.

i.e., $n(A \cap B) = 5$

$\therefore$ Minimum number in $n(A \cup B) = 5 + 7 - 5 = 7$

10. (b) $n(A) = 40\% \text{ of } 10,000 = 4,000$

$n(B) = 20\% \text{ of } 10,000 = 2,000$

$n(C) = 10\% \text{ of } 10,000 = 1,000$

$n(A \cap B) = 5\% \text{ of } 10,000 = 500$

$n(B \cap C) = 3\% \text{ of } 10,000 = 300$

$n(C \cap A) = 4\% \text{ of } 10,000 = 400$

$$n(A \cap B \cap C) = 2\% \text{ of } 10,000 = 200$$

$$\text{We want to find } n(A \cap B^c \cap C^c) = n[A \cap (B \cup C)^c]$$

$$= n(A) - n[A \cap (B \cup C)]$$

$$= n(A) - n[(A \cap B) \cup (A \cap C)]$$

$$= n(A) - [n(A \cap B) + n(A \cap C) - n(A \cap B \cap C)]$$

$$= 4000 - [500 + 400 - 200] = 4000 - 700 = 3300.$$

11. (c) Given set is $\{(a, b) : 2a^2 + 3b^2 = 35, a, b \in \mathbb{Z}\}$

$$\text{We can see that, } 2(\pm 2)^2 + 3(\pm 3)^2 = 35$$

$$\text{and } 2(\pm 4)^2 + 3(\pm 1)^2 = 35$$

$$\therefore (2, 3), (2, -3), (-2, -3), (-2, 3), (4, 1), (4, -1), (-4, -1), (-4, 1) \text{ are 8 elements of the set. } \therefore n = 8.$$

12. (c) $\alpha + \beta + \gamma = 68$

$$\alpha + 2\beta + 3\gamma = (38 + 26 + 36) = 100$$

$$\text{and } \gamma = 11$$

$$\therefore (\alpha + 2\beta + 3\gamma) - [(\alpha + \beta + \gamma) + \gamma] = \beta + \gamma$$

$$= 100 - [68 + 11]$$

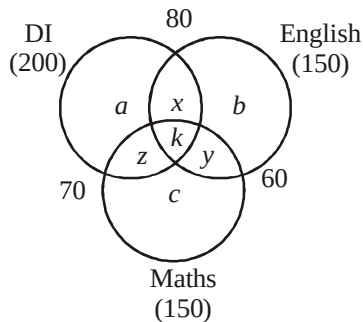
$$= 21$$

Hence 21% favoured more than one magazine.

13. (a) $a + x + k + z = 200$... (1)
 $b + x + k + y = 150$... (2)
 $c + y + k + z = 150$... (3)

But since Diwakar teaches only 80 students of DI.

Therefore, $a = 80$

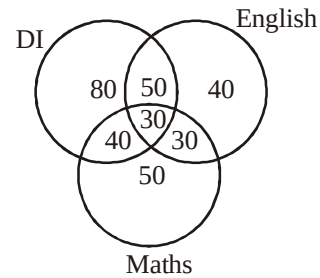


$$\text{Hence, } x + k + z = 120$$

$$\text{but } (x + k) + (k + z) = 150$$

$$\therefore k = 30$$

$$\text{Hence, } x = 50, z = 40, y = 30, b = 40$$

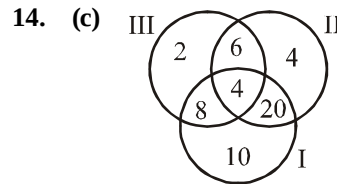


$$\therefore \text{No. of students taught by Diwakar} = a = 80$$

$$\text{No. of students taught by Priyanka} = b = 40$$

$$\text{No. of students taught by Varun} = c = 50$$

$$\text{No. of students taught by Sarvesh} = x + y + z + k = 150$$



From the Venn diagram it follows:

$$n(\text{sector I}) = 42, n(\text{sector II}) = 34,$$

$$n(\text{sector III}) = 20$$

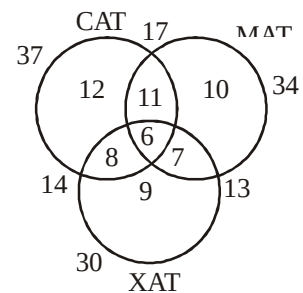
$$n(I \cap II) = 24, n(II \cap III) = 10,$$

$$n(I \cap III) = 12, n(I \cap II \cap III) = 4$$

Now using the formula,

$$\text{we get } n(I \cup II \cup III) = 42 + 34 + 20 - 24 - 10 - 12 + 4 = 54.$$

15. (c)



Hence there are 12 students who appeared in CAT but not in MAT or XAT.

UNIT-IV

Geometry

Chapter 18	Geometry
Chapter 19	Mensuration
Chapter 20	Coordinate Geometry

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18

GEOMETRY

- Introduction
- Points, Lines, Line Segment, Ray and Plane
- Lines and Angles
- Polygons
- Triangles
- Basic Properties and Some Important Theorems of Triangles
- Important Terms Related to a Triangle
- Congruency of Two Triangles
- Similarity of Two Triangles
- Quadrilaterals
- Circles
- Basic Pythagorean Triplets
- Determination of Nature of Triangle
- Important Points
- Locus
- Sine and Cosine Rule

INTRODUCTION

Geometry is a very important chapter for every aptitude test. The level of questions is moderate to tough, which requires clear understanding of definitions, theorems and its related properties. In CAT and other aptitude tests 5–8 questions are generally asked from geometry alone.

In order to crack CAT, a very good command over geometry is essential.

POINTS, LINES, LINE SEGMENT, RAY AND PLANE

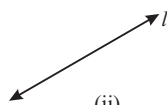
Point: A point is like a dot marked by a very sharp pencil on a plane paper. A point is named by a capital letter like P . In the figure P is a point. Length, breadth and height of a point are negligible and hence cannot be measured.

P

Line: A line is defined as a group of points. Which are straight one after another. Each line is extended infinitely in two directions. Examples:



(i)



(ii)

A line is named by either any two points on it or by a single small letter. In figure (i), AB is a line. In figure (ii), l is a line.

Arrows on both sides of a line indicate that the line is extended both sides infinitely. A line has only length. It does not have any width or height.

Line Segment: If a part of the line is cut out, then this cut out piece of the line is called a line segment. A line segment has no arrow at its any end.

This means that no line segment is extended infinitely in any direction.

Ray: A ray is a part of a line extended infinitely in any one direction only. Example:



A ray is named by two points, one of which is the end point on the ray called initial point and other point is any point on the ray.

In the figure, AB is a ray. The point A is called the initial point. Arrow of the ray indicates that the ray is extended infinitely towards arrow head.

Plane: It is a flat surface extended infinitely. It has only length and breadth but no thickness. Surface of a black board, surface of a wall, surface of a table are some examples of parts of planes because they are flat surfaces but not extended infinitely.

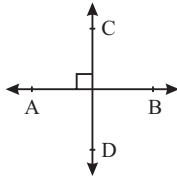
LINES AND ANGLES

Intersecting Lines: If two or more lines intersect each other, then they are called intersecting lines. In the figure AB and CD are intersecting lines.



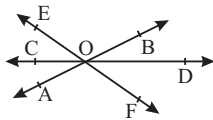
If two lines intersect at right angles, then two lines are called perpendicular lines

In the following figure AB and CD are perpendicular lines.



Symbolically it is represented as $AB \perp CD$ or $CD \perp AB$.

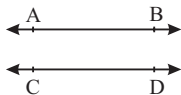
Concurrent Lines: If three or more lines pass through a point, then they are called concurrent lines and the point through which these all lines pass is called point of concurrent.



In the figure, AB , CD and EF are concurrent lines and point O is the point of concurrent.

Parallel Lines: Two straight lines are parallel if they lie in the same plane and do not intersect even if they produced.

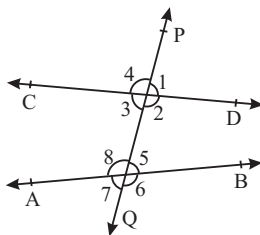
Perpendicular distances between two parallel lines are the same at all places.



In the figure AB and CD are parallel lines.

Symbol for parallel lines is $\parallel$. Hence parallel lines AB and CD represented symbolically as $AB \parallel CD$.

Transversal Line: A line which intersects two or more given lines at distinct points is called a transversal of the given lines.

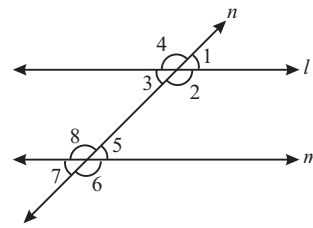


In figure straight lines AB and CD are intersected by a transversal PQ .

- (i) **Corresponding angles:** In the figure $\angle 1$ and $\angle 5$, $\angle 4$ and $\angle 8$, $\angle 2$ and $\angle 6$, $\angle 3$ and $\angle 7$ are four pairs of corresponding angles.
- (ii) **Alternate interior angles:** $\angle 3$ and $\angle 5$, $\angle 2$ and $\angle 8$, are two pairs of alternate interior angles.
- (iii) **Alternate exterior angles:** $\angle 1$ and $\angle 7$, $\angle 4$ and $\angle 6$ are two pairs of alternate exterior angles.
- (iv) **Consecutive interior angles:** In the figure, $\angle 2$ and $\angle 5$, $\angle 5$ and $\angle 8$, $\angle 8$ and $\angle 3$, $\angle 3$ and $\angle 2$ are four pairs of consecutive interior angles.

Interior angles on the same side of a transversal are called co-interior angles. In the fig. $\angle 2$ and $\angle 5$, $\angle 3$ and $\angle 8$ are two pairs of co-interior angles.

When a transversal intersects two parallel lines:



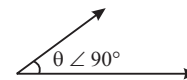
In the figure two parallel lines l and m are intersected by a transversal line n , then

- (a) Two angles of each pair of corresponding angles are equal i.e. $\angle 1 = \angle 5$; $\angle 2 = \angle 6$; $\angle 4 = \angle 8$; $\angle 3 = \angle 7$
- (b) Two angles of each pair of alternate interior angles are equal i.e. $\angle 2 = \angle 8$; $\angle 3 = \angle 5$
- (c) Two angles of each pair of alternate exterior angles are equal i.e. $\angle 1 = \angle 7$; $\angle 4 = \angle 6$
- (d) Any two consecutive interior angles are supplementary. i.e. their sum is 180° . Hence $\angle 2 + \angle 5 = 180^\circ$; $\angle 5 + \angle 8 = 180^\circ$; $\angle 8 + \angle 3 = 180^\circ$; $\angle 3 + \angle 2 = 180^\circ$

Note that

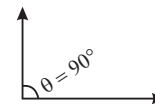
- (i) If two angles of any pair of corresponding angles are equal, then the two lines are parallel.
- (ii) If two angles of any pair of alternate interior angles are equal, then the two lines are parallel.
- (iii) If two angles of any pair of alternate exterior angles are equal, then the two lines are parallel.
- (iv) If any two consecutive interior angles are supplementary (i.e. their sum is 180°), then the two lines are parallel.

Acute angle: An angle is said to be acute angle if it is less than 90° .



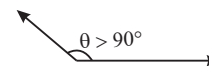
Here $0^\circ < \theta < 90^\circ$, hence θ is acute angle.

Right angle: An angle is said to be right angle if it is of 90° .



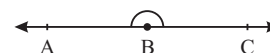
Here θ is right angle.

Obtuse angle: An angle is said to be obtuse angle if it is of more than 90° .



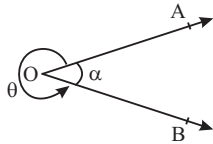
Here θ is obtuse angle.

Straight angle: An angle is said to be straight angle if it is of 180° .



Here θ is a straight angle.

Reflex angle: An angle is said to be reflex angle if it is of greater than 180° .



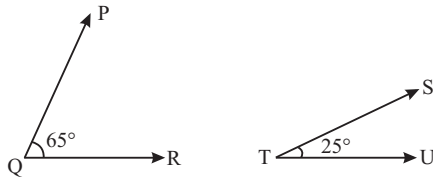
Here θ is the reflex angle.

Reflex angle θ is written as

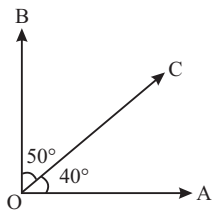
$$\theta = 360^\circ - \angle AOB \text{ (or } 360^\circ - \alpha)$$

Here $\angle AOB$ or α is less than 180°

Complementary angles: Two angles, the sum of whose measures is 90° , are called the complementary angles.

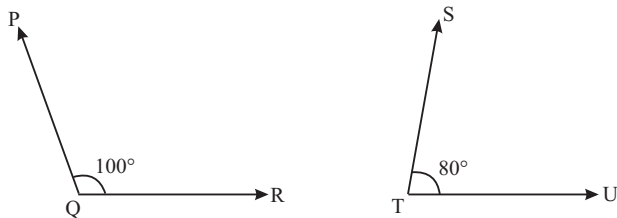


$\angle PQR$ and $\angle STU$ are complementary angles.

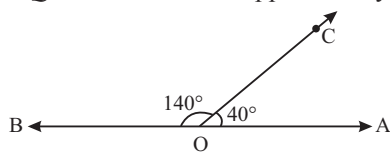


In figure $\angle AOC$ and $\angle BOC$ are also complementary angles.

Supplementary angles: Two angles, the sum of whose measures is 180° , are called the supplementary angles.



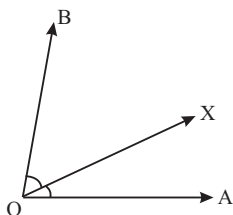
In figure, $\angle PQR$ and $\angle STU$ are supplementary angles.



In figure, $\angle AOC$ and $\angle BOC$ are also supplementary angles.

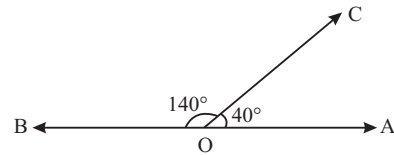
Adjacent angles: Two angles are called adjacent angles, if

- they have the same vertex
- they have a common arm and
- non-common arms are on either side of the common arm.



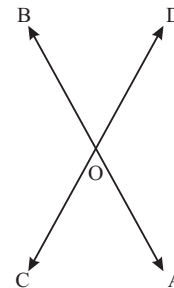
In figure, $\angle AOX$ and $\angle BOX$ are adjacent angles because O is the common vertex, OX is common arm, non-common arm OA and OB are on either side of OX .

Linear pair of angles: Two adjacent angles are said to form a linear pair of angles, if their non common arms are two opposite rays. In other words if the sum of two adjacent angles is 180° , then they are said to form a linear pair of angles.



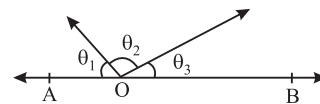
In figure, $\angle AOC$ and $\angle BOC$ are linear pair angles.

Vertically opposite angles: Two angles are called a pair of vertically opposite angles, if their arms form two intersecting lines.



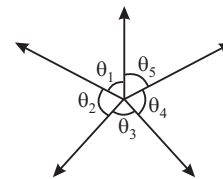
In figure, $\angle AOC$ and $\angle BOD$ form a pair of vertically opposite angles. Also $\angle AOD$ and $\angle BOC$ form a pair of vertically opposite angles.

Angles on one side of a line at a point on the line: Sum of all the angles on any one side of a line at a point on the line is always 180° .



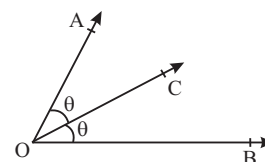
Here AOB is a straight line, hence in figure, $\theta_1 + \theta_2 + \theta_3 = 180^\circ$.

Angle around a point: Sum of all the angles around a point is always 360° .



Here $\theta_1, \theta_2, \theta_3, \theta_4$ and θ_5 are the angles around a point. Hence $\theta_1 + \theta_2 + \theta_3 + \theta_4 + \theta_5 = 360^\circ$

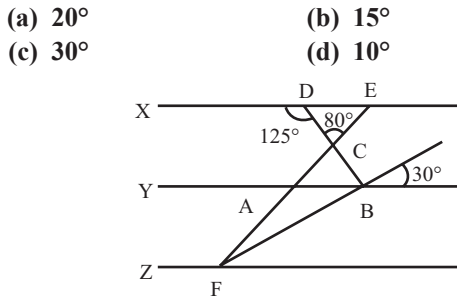
Angle bisector: An angle bisector is a ray which bisects the angle whose initial point be the vertex of the angle.



Since $\angle AOC = \angle BOC = \theta$

Hence ray OC is the bisector of $\angle AOB$.

Illustration 1: Three straight lines, X, Y and Z are parallel and the angles are as shown in the figure above. What is $\angle AFB$ equal to?



Solution: (b) $\angle CDE = 180^\circ - 125^\circ = 55^\circ$

In $\triangle DCE$,

$$\angle CED = 180^\circ - 55^\circ - 80^\circ = 45^\circ$$

and $\angle ABF = 30^\circ$ (vertically opposite)

Also, $\angle ABF = \angle BFM = 30^\circ$ (alternate angle)

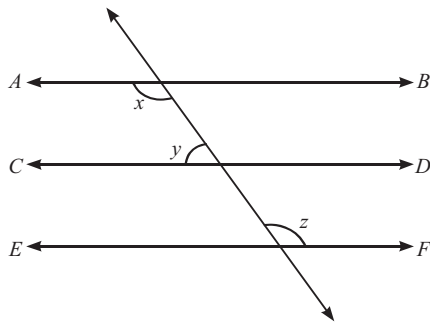
and, $\angle DEF = \angle EFM$ (alternate angle)

$$\angle EFM = 45^\circ$$

$$\Rightarrow \angle EFB + \angle BFM = 45^\circ \Rightarrow \angle EFB = 45^\circ - 30^\circ$$

$$\Rightarrow \angle AFB = 15^\circ$$

Illustration 2: In figure, if $AB \parallel CD$, $CD \parallel EF$ and $y : z = 3 : 7$, $x = ?$



(a) 112°

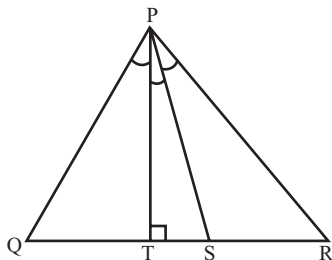
(b) 116°

(c) 96°

(d) 126°

Solution: (d) As $y + z = 180^\circ$, $\therefore y = 54^\circ$
 $x + y = 180^\circ$
 $x = 180 - 54 = 126^\circ$

Illustration 3: In the $\triangle PQR$, PS is the bisector of $\angle P$ and $PT \perp QR$, then $\angle TPS$ is equal to



(a) $\angle Q + \angle R$ (b) $90^\circ + \frac{1}{2} \angle Q$

(c) $90^\circ - \frac{1}{2} \angle R$ (d) $\frac{1}{2} (\angle Q - \angle R)$

Solution: (d) PS is the bisector of $\angle QPR$

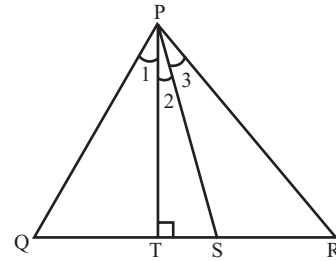
$$\therefore \angle 1 + \angle 2 = \angle 3$$

$$\Rightarrow \angle Q = 90^\circ - \angle 1$$

$$\angle R = 90^\circ - \angle 2 - \angle 3$$

...(1)

$$\begin{aligned} \text{So, } \angle Q - \angle R &= [90^\circ - \angle 1] - [90^\circ - \angle 2 - \angle 3] \\ \Rightarrow \angle Q - \angle R &= \angle 2 + \angle 3 - \angle 1 \\ &= \angle 2 + (\angle 1 + \angle 2) - \angle 1 \quad [\text{From Eq. (1)}] \end{aligned}$$



$$\Rightarrow \angle Q - \angle R = 2\angle 2 \Rightarrow \frac{1}{2} (\angle Q - \angle R) = \angle TPS$$

POLYGONS

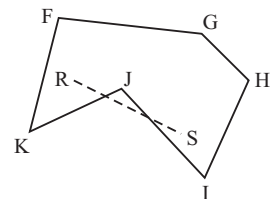
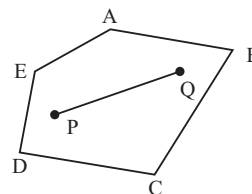
Polygons are closed plane figures formed by series of line segments, e.g. triangles, rectangles, etc.

Polygons can also be classified into convex and concave polygons.

A convex polygon is a polygon in which any line segment joining any two points of the polygon always lies completely inside the polygon, otherwise the polygon is concave.

$ABCDE$ is a convex polygon because any line segment joining any two points of the polygon completely lies inside the polygon.

$FGHIJK$ is a concave polygon because line segment joining two points R and S of the polygon does not lie completely inside the polygon.



Convex polygons can be classified into regular and irregular polygons.

(a) **Regular polygon:** A convex polygon whose all the sides are equal and also all the angles equal is called a regular polygon.

A regular polygon is simply called polygon.

(b) **Irregular Polygon:** A convex polygon in which all the sides are not equal or all the angles are not of the same measure is called an irregular polygon.

Polygons can also be divided on the basis of number of sides they have

No. of sides of the polygon	Name of the polygon
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon

7	Heptagon
8	Octagon
9	Nonagon
10	Decagon
⋮	⋮
etc.	etc.

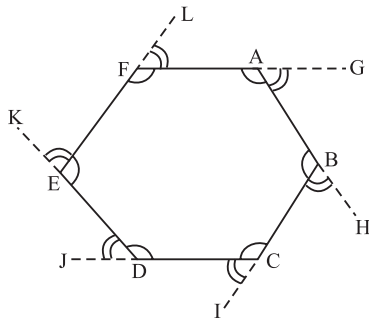
Interior and Exterior Angles of a Polygon

An angle inside a polygon between any two adjacent sides at a vertex of the polygon is called an interior angle of the polygon. An angle outside a polygon made by a side of the polygon with its adjacent side produced is called an exterior angle of the polygon.

In the figure $ABCDEF$ is a polygon.

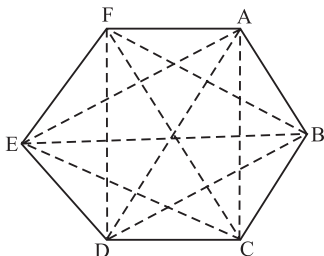
$\angle FAB, \angle ABC, \angle BCD, \angle CDE, \angle DEF$ and

$\angle EFA$ are interior angles of the polygon $ABCDEF$.



$\angle BAG, \angle CBH, \angle DCI, \angle EDJ, \angle FEK$ and $\angle AFL$ are exterior angles of the polygon $ABCDEF$.

Diagonals of a Polygon

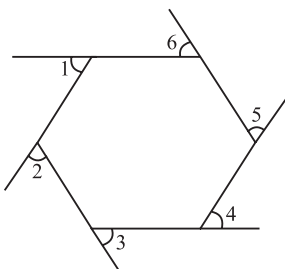


A diagonal of a polygon is a line segment connecting two non-consecutive vertices of the Polygon.

In the figure, diagonals are drawn by dotted line segments.

Properties of Polygons

- Sum of all the interior angles of a polygon with ' n ' sides $= (n - 2) 180^\circ$
- Sum of all the exterior angles of a polygon $= 360^\circ$
 $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 = 360^\circ$



- Perimeter of a regular polygon with a side length of $a = n \times a$

- No. of sides of a regular polygon = an exterior angle $= \frac{360^\circ}{\text{An exterior angle}}$

- Number of diagonals of a polygon with n sides $= \frac{n(n-3)}{2}$

Illustration 4: An interior angle of a regular polygon is 135° . Find the number of sides of the polygon.

Solution: Since interior angle of the regular polygon $= 135^\circ$, hence exterior angle $= 180^\circ - 135^\circ = 45^\circ$

$$\therefore \text{No. of sides} = \frac{360^\circ}{\text{An exterior angle}} = \frac{360^\circ}{45^\circ} = 8$$

$$\therefore \text{No. of sides} = 8$$

Illustration 5: An interior angle of a regular polygon is 100° more than its an exterior angle. Find the number of sides the polygon.

Solution: Let measure of each exterior angle be x° .

Then measure of each interior angle $= (x + 100)$

$$\text{Now } x + (x + 100) = 180$$

$$\Rightarrow 2x = 80 \Rightarrow x = 40$$

$$\text{Now number of sides} = \frac{360}{\text{An exterior angle}} = \frac{360}{40} = 9$$

TRIANGLES

A triangle is a convex polygon having three sides.

A triangle is represented by the symbol Δ .

Triangles can be classified on the basis of their sides or angles.

On the basis of sides, triangles are of the following types

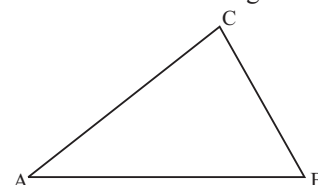
- Equilateral triangle:** All the three sides are equal
- Isosceles triangle:** Two sides are equal
- Scalene triangle:** All the three sides are unequal.

On the basis of angles, triangles are of the following types

- Acute angled triangle:** Each interior angle is less than 90° .
- Right angled triangle:** One of the interior angle is equal 90° .
- Obtuse angled triangle:** One of the interior angle is more than 90° .

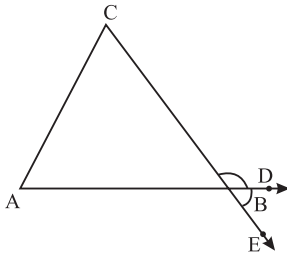
BASIC PROPERTIES AND SOME IMPORTANT THEOREMS OF TRIANGLES

- Sum of measures of the interior angles of a triangle is 180° .



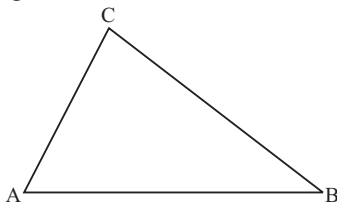
In $\triangle ABC$, $\angle CAB + \angle ABC + \angle ACB = 180^\circ$
or $\angle A + \angle B + \angle C = 180^\circ$

2. The exterior angle of a triangle is equal to the sum of the opposite (not adjacent) interior angles



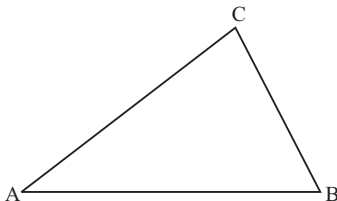
In $\triangle ABC$, $\angle CBD = \angle A + \angle C = \angle ABE$

3. Sum of the lengths of any two sides of a triangle is greater than the length of the third side.



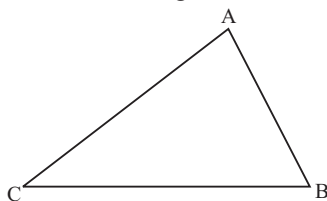
- (i) $AB + AC > BC$ (ii) $AC + BC > AB$
(iii) $AB + BC > AC$

4. Difference between the lengths of any two sides of a triangle is smaller than the length of the third side.



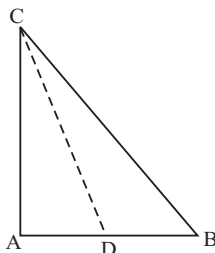
- (i) $|AB - BC| < AC$ (ii) $|AC - AB| < BC$
(iii) $|AC - BC| < AB$

5. In any triangle, side opposite to greatest angle is largest and side opposite to smallest angle is smallest.



In $\triangle ABC$, if $\angle A > \angle B > \angle C$, then BC is the largest side and AB is the smallest side.

6. In any triangle line joining any vertex to the mid point of its opposite side is called a median of the opposite side of the triangle.

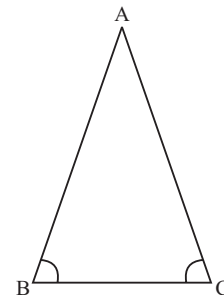


In $\triangle ABC$ D is the mid point of AB
Hence CD is a median of $\triangle ABC$.

A triangle can have 3 medians.

Any median of a triangle divides the triangle into two triangles of equal areas.

7. Sides opposite to equal angles in a triangle are equal.

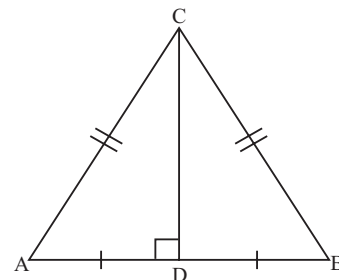


In $\triangle ABC$, $\angle B = \angle C$
 $\therefore AB = AC$

Converse of this property is also true.

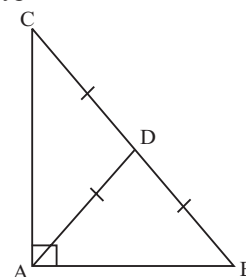
8. In an isosceles triangle, if a perpendicular is drawn to unequal side from its opposite vertex, then

- (a) The perpendicular is the median
(b) The perpendicular bisects the vertex angle.



$\triangle ABC$ is an isosceles triangle in which $AC = BC$.
 CD is perpendicular to AB , hence CD is a median and $\angle ACD = \angle BCD$

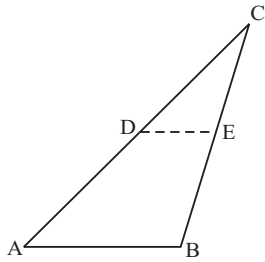
9. In a right angled triangle, the line joining the vertex of the right angle to the mid point of the hypotenuse is half the length of the hypotenuse.



In $\triangle ABC$, $\angle BAC = 90^\circ$ and D is the mid point of BC , then

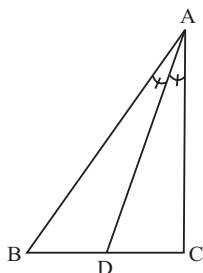
$$AD = \frac{1}{2} BC = BD = CD$$

10. **Mid-point theorem:** In any triangle, line segment joining the mid points of any two sides is parallel to the third side and equal to half of the length of third side.



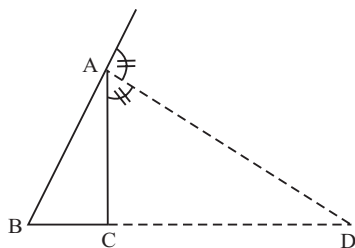
In $\triangle ABC$, D and E are mid points of sides AC and BC , then DE is parallel to AB i.e. $DE \parallel AB$ and $DE = \frac{1}{2} AB$

- 11. Angle Bisector Theorem:** Bisector of an angle (internal or external) of a triangle divides the opposite side (internally or externally) in the ratio of the sides containing the angle.
For example:



In figure AD is the bisector of exterior $\angle BAC$

$$\therefore \frac{AB}{AC} = \frac{BD}{DC}$$

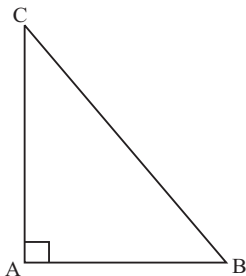


In figure AD is the bisector of exterior $\angle BAC$.

$$\therefore \frac{AB}{AC} = \frac{BD}{DC}$$

Converse of the angle bisector theorem is also true.

- 12. Pythagoras Theorem:** In a right angled triangle. Square of longest or hypotenuse = Sum of square of other two sides.



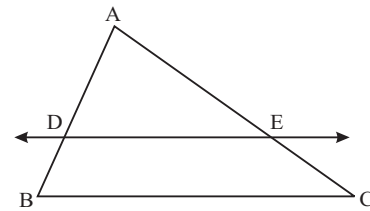
In figure $\triangle ABC$ is a triangle right angled at A .

$$\therefore (BC)^2 = (AB)^2 + (AC)^2$$

Converse of this theorem is also true.

- 13. Basic Proportionality Theorem (BPT):** If a line is drawn parallel to one side of a triangle which intersects the other

two sides in distinct points, the other two sides are divided in the same ratio.



In $\triangle ABC$, $DE \parallel BC$,

$$\text{Then, } \frac{AD}{DB} = \frac{AE}{EC}$$

This theorem is also known as Thales theorem.

Converse of this theorem is also true.

Illustration 6: In a triangle ABC , $\angle A = x$, $\angle B = y$, and $\angle C = y + 20$.

If $4x - y = 10$, then the triangle is :

- (a) Right-angled (b) Obtuse-angled
(c) Equilateral (d) None of these

Solution: (a) We have, $x + y + (y + 20) = 180$

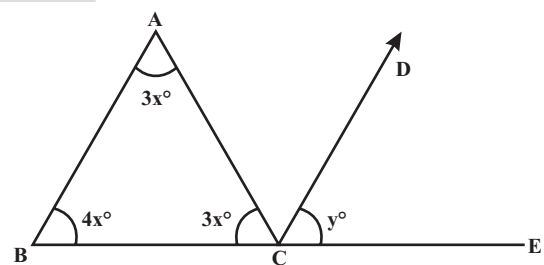
$$\text{or } x + 2y = 160 \quad \dots(1)$$

$$\text{and } 4x - y = 10 \quad \dots(2)$$

$$\text{From (1) and (2), } y = 70, x = 20$$

Angles of the triangles are $20^\circ, 70^\circ, 90^\circ$. Hence the triangle is a right angled.

Illustration 7: In the given figure, $CD \parallel AB$. Find y .



- (a) 79° (b) 72°
(c) 74° (d) 77°

Solution: (b) In $\triangle ABC$,

$$\angle ABC + \angle BCA + \angle CAB = 180^\circ$$

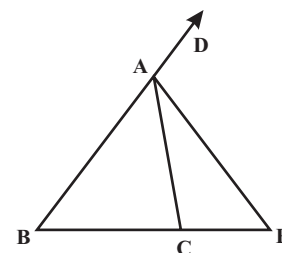
$$\Rightarrow 4x + 3x + 3x = 180^\circ \Rightarrow 10^\circ x = 180^\circ \Rightarrow x = 18^\circ$$

$$\text{Now, } \angle ABC = \angle DCE$$

(corresponding angles are equal)

$$\Rightarrow \angle DCE = 4x^\circ \Rightarrow y = 4 \times 18^\circ = 72^\circ$$

Illustration 8: In the adjoining figure, AE is the bisector of exterior $\angle CAD$ meeting BC produced in E . If $AB = 10$ cm, $AC = 6$ cm and $BC = 12$ cm, then CE is equal to



- (a) 6 cm (b) 12 cm
(c) 18 cm (d) 20 cm

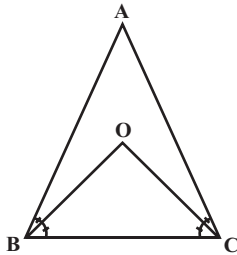
Solution: (c) $\frac{BE}{CE} = \frac{AB}{AC}$ as AE is an exterior angle bisector.

Let $CE = x$, $BE = BC + EC = 12 + x$

$$\Rightarrow \frac{12+x}{x} = \frac{10}{6} \Rightarrow (12+x)6 = 10x$$

$$\Rightarrow 72 + 6x = 10x \Rightarrow 4x = 72 \Rightarrow x = 18 \text{ cm}$$

Illustration 9: OB and OC are respectively the bisectors of $\angle ABC$ and $\angle ACB$. Then, $\angle BOC$ is equal to

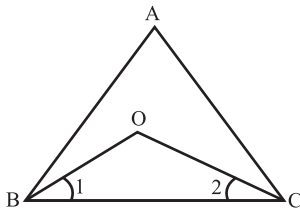


(a) $90^\circ - \frac{1}{2} \angle A$ (b) $90^\circ + \angle A$

(c) $90^\circ + \frac{1}{2} \angle A$ (d) $180^\circ - \frac{1}{2} \angle A$

Solution: (c) In $\triangle BOC$,
 $\angle 1 + \angle 2 + \angle BOC = 180^\circ$
 $\angle A + \angle B + \angle C = 180^\circ$.

.... (1)



$$\Rightarrow \frac{1}{2} \angle A + \frac{1}{2} \angle B + \frac{1}{2} \angle C = 90^\circ$$

$$\Rightarrow \frac{1}{2} (\angle A) + \angle 1 + \angle 2 = 90^\circ \Rightarrow \angle 1 + \angle 2 = 90^\circ - \frac{1}{2} \angle A$$

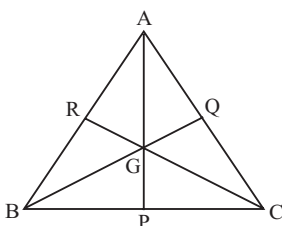
Put $\angle 1 + \angle 2$ in Eq. (1), we get

$$\angle BOC = 180^\circ - 90^\circ - \left(90^\circ - \frac{1}{2} \angle A\right)$$

$$= 90^\circ + \frac{1}{2} \angle A$$

IMPORTANT TERMS RELATED TO A TRIANGLE

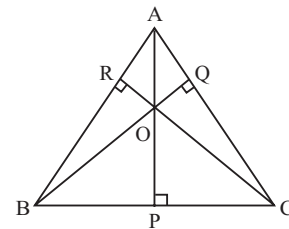
- Medians and Centroid:** We know that a line segment joining the mid point of a side of a triangle to its opposite vertex is called a median.



AP , BQ and CR are medians of $\triangle ABC$ where P , Q and R are mid points of sides BC , CA and AB respectively.

- Three medians of a triangle are concurrent. The point of concurrency of three medians is called Centroid of the triangle denoted by G .
- Centroid of the triangle divides each median in the ratio $2 : 1$
 i.e. $AG : GP = BG : GQ = CG : GR = 2 : 1$, where G is the centroid of $\triangle ABC$.

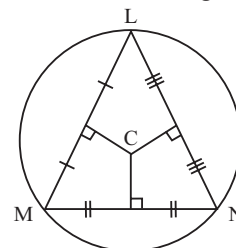
- Altitudes and Orthocentre:** A perpendicular drawn from any vertex of a triangle to its opposite side is called altitude of the triangle. There are three altitudes of a triangle. In the figure, AP , BQ and CR are altitudes of $\triangle ABC$. The altitudes of a triangle are concurrent (meet at a point) and the point of concurrency of altitudes is called Orthocentre of the triangle, denoted by O .



In figure, AP , BQ and CR meet at O , hence O is the orthocentre of the triangle ABC .

Note: The angle made by any side at the orthocentre and at the vertex opposite to the side are supplementary angle.
 Hence, $\angle BAC + \angle BOC = \angle ABC + \angle AOC = \angle ACB + \angle AOB = 180^\circ$.

- Perpendicular Bisectors and Circumcentre:** A line which is perpendicular to a side of a triangle and also bisects the side is called a perpendicular bisector of the side.
 - Perpendicular bisectors of sides of a triangle are concurrent and the point of concurrency is called circumcentre of the triangle, denoted by ' C '.
 - The circumcentre of a triangle is centre of the circle that circumscribes the triangle.
 - Angle formed by any side of the triangle at the circumcentre is twice the vertical angle opposite to the side.

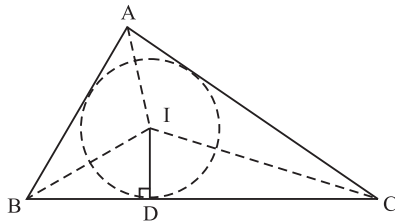


In figure, perpendicular bisectors of sides LM , MN and NL of $\triangle LMN$ meet at C . Hence C is the circumcentre of the triangle LMN .

$$\angle MCN = 2 \angle MLN$$

- Angle Bisectors and Incentre:** Lines bisecting the interior angles of a triangle are called angle bisectors of triangle.

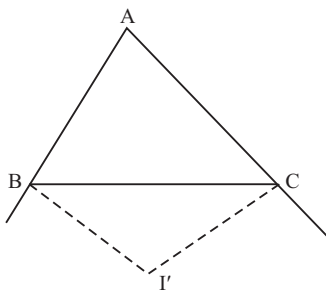
- (i) Angle bisectors of a triangle are concurrent and the point of concurrency is called Incentre of the triangle, denoted by I .
- (ii) With I as centre and radius equal to length of the perpendicular drawn from I to any side, a circle can be drawn touching the three sides of the triangle. So this is called incircle of the triangle. Incentre is equidistant from all the sides of the triangle.
- (iii) Angle formed by any side at the incentre is always 90° more than half the vertex angle opposite to the side.



In figure AI, BI, CI are angle bisectors of $\triangle ABC$. Hence I is the incentre of the $\triangle ABC$ and

$$\angle BIC = 90^\circ + \frac{1}{2} \angle A, \angle AIC = 90^\circ + \frac{1}{2} \angle B$$

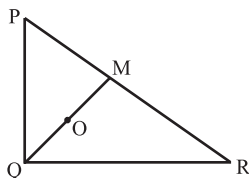
and $\angle AIB = 90^\circ + \frac{1}{2} \angle C$



If BI' and CI' be the angle bisectors of exterior angles at B and C , then

$$\angle BI'C = 90^\circ - \frac{1}{2} \angle A.$$

Illustration 10: If in the given figure $\angle PQR = 90^\circ$, O is the centroid of $\triangle PQR$, $PQ = 5$ cm and $QR = 12$ cm, then OQ is equal to



- (a) $3\frac{1}{2}$ cm
- (b) $4\frac{1}{3}$ cm
- (c) $4\frac{1}{2}$ cm
- (d) $5\frac{1}{3}$ cm

Solution: (b) By Pythagoras theorem,

$$PR = \sqrt{PQ^2 + QR^2} = \sqrt{5^2 + 12^2} = 13 \text{ cm}$$

$\therefore O$ is centroid $\Rightarrow QM$ is median and M is mid-point of PR .

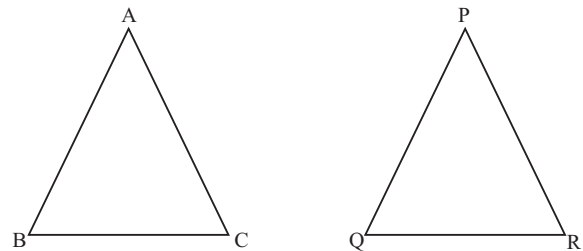
$$QM = PM = \frac{13}{2}$$

$\therefore$ Centroid divides median in ratio $2 : 1$.

$$\therefore OQ = \frac{2}{3} QM = \frac{2}{3} \times \frac{13}{2} = \frac{13}{3} \therefore OQ = 4\frac{1}{3} \text{ cm}$$

CONGRUENCY OF TWO TRIANGLES

Two triangles are congruent if they are of the same shape and size i.e. if any one of them can be made to superpose on the other it will cover exactly.



If two triangles ABC and PQR are congruent then 6 elements (i.e. three sides and three angles) of one triangle are equal to corresponding 6 elements of other triangle.

$$(i) \angle A = \angle P, \angle B = \angle Q, \angle C = \angle R$$

$$(ii) AB = PQ, BC = QR, AC = PR$$

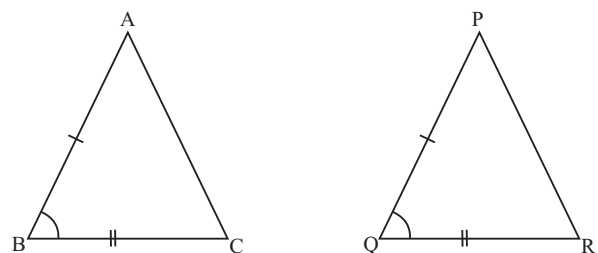
This is symbolically written as $\triangle ABC \cong \triangle PQR$

Note: In two congruent triangles, sides opposite to equal angles are corresponding sides and angles opposite to equal sides are corresponding angles.

Conditions of Congruency

There are 4 conditions of congruency of two triangles.

1. **SAS (Side-Angle-Side) Congruency:** If two sides and the included angle between these two sides of one triangle is equal to corresponding two sides and included angle between these two sides of another triangle, then the two triangles are congruent.



In $\triangle ABC$ and $\triangle PQR$

$$AB = PQ,$$

$$BC = QR$$

$$\text{and } \angle ABC = \angle PQR$$

$$\therefore \triangle ABC \cong \triangle PQR \quad [\text{by SAS congruency}]$$

Here $\cong$ is the sign of congruency.

2. **ASA (Angle-Side-Angle) Congruency:** If two angles and included side between these two angles of one triangle are equal to corresponding angles and included side between these two angles of another triangle, then two triangles are congruent.

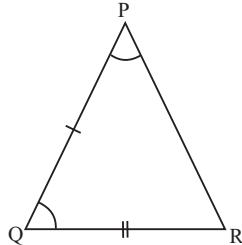
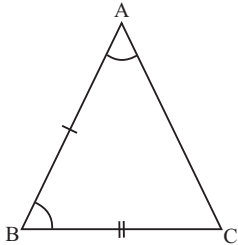
In $\triangle ABC$ and $\triangle PQR$

$$\angle A = \angle P$$

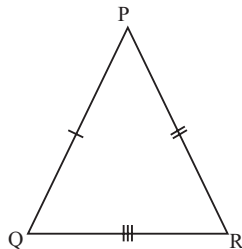
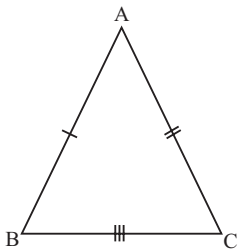
$$\angle B = \angle Q$$

$$AB = PQ$$

$\therefore \triangle ABC \cong \triangle PQR$ [by ASA congruency]



3. **SSS (Side-Side-Side) Congruency:** If three sides of one triangle are equal to corresponding three sides of another triangle, the two triangles are congruent.



In $\triangle ABC$ and $\triangle PQR$

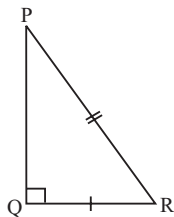
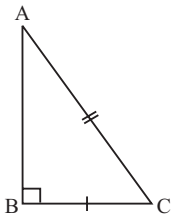
$$AB = PQ$$

$$BC = QR$$

$$CA = RP$$

$\therefore \triangle ABC \cong \triangle PQR$ [by SSS congruency]

4. **RHS (Rightangle-Hypotenuse-Side) Congruency:** Two right angled triangles are congruent to each other if hypotenuse and one side of one triangle are equal to hypotenuse and corresponding side of another triangle.



In $\triangle ABC$ and $\triangle PQR$

$$\angle ABC = \angle PQR = 90^\circ$$

$$AC = PR$$

$$BC = QR$$

$\therefore \triangle ABC \cong \triangle PQR$ [by RHS congruency]

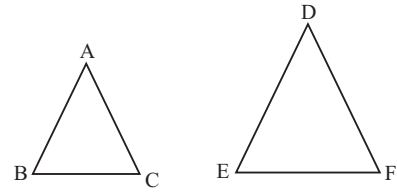
SIMILARITY OF TWO TRIANGLES

Two triangles are said to be similar, if their shapes are the same but their size may or may not be equal.

When two triangles are similar, then

- all the corresponding angles are equal and
- all the corresponding sides are in the same ratio (or proportion)

Note: In two similar triangles, sides opposite to equal angles are called corresponding sides. And angles opposite to side proportional to each other are called corresponding angles.



If $\triangle ABC$ and $\triangle DEF$ are similar, then

$$\angle A = \angle D$$

$$\angle B = \angle E$$

$$\angle C = \angle F$$

$$\text{and } \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

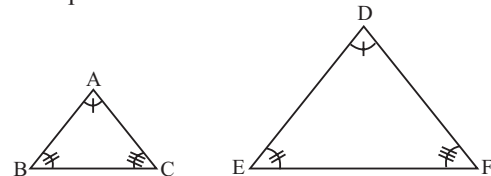
$\triangle ABC \sim \triangle DEF$, read as triangle ABC is similar to triangle DEF .

Here $\sim$ is the sign of similarity.

Conditions of Similarity

There are 4 conditions of similarity.

- AAA (Angle-Angle-Angle) Similarity:** Two triangles are said to be similar, if their all corresponding angles are equal. For example:



In $\triangle ABC$ and $\triangle DEF$, if

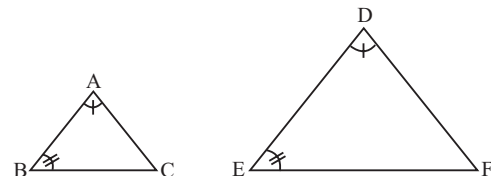
$$\angle A = \angle D$$

$$\angle B = \angle E$$

$$\angle C = \angle F$$

Then $\triangle ABC \sim \triangle DEF$ [By AAA Similarity]

Corollary AA (Angle-Angle) Similarity: If two angles of one triangle are respectively equal to two angles of another triangles, then two triangles are similar.



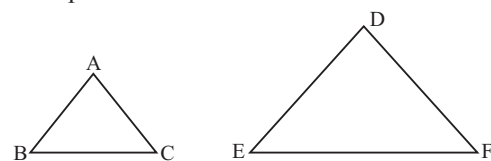
In $\triangle ABC$ and $\triangle DEF$, if

$$\angle A = \angle D$$

$$\angle B = \angle E$$

then $\triangle ABC \sim \triangle DEF$ [By AA Similarity]

- SSS (Side-Side-Side) Similarity:** Two triangles are said to be similar, if sides of one triangle are proportional (or in the same ratio of) to the sides of the other triangle: For example:



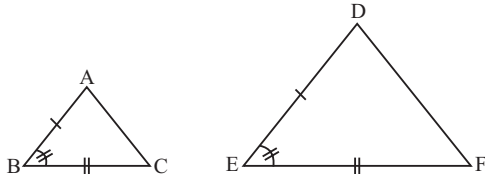
In $\triangle ABC$ and $\triangle DEF$, if

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$$

Then $\triangle ABC \sim \triangle DEF$ [By SSS Similarity]

3. **SAS (Side-Angle-Side) Similarity:** Two triangles are said to be similar if two sides of a triangle are proportional to the two sides of the other triangle and the angles included between these sides of two triangles are equal.

For example:



In $\triangle ABC$ and $\triangle DEF$, if

$$\frac{AB}{DE} = \frac{BC}{EF}$$

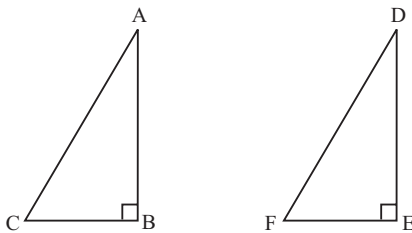
and

$$\angle B = \angle E$$

Then, $\triangle ABC \sim \triangle DEF$ [By SAS Similarity]

4. **RHS (Rightangle-Hypotenuse-Side) Similarity:** Two triangles are said to be similar if one angle of both triangle is right angle and hypotenuse of both triangles are proportional to any one other side of both triangles respectively.

For example:



In $\triangle ABC$ and $\triangle DEF$, if

$$\angle B = \angle E [= 90^\circ]$$

$$\frac{AC}{DF} = \frac{AB}{DE}$$

Then $\triangle ABC \sim \triangle DEF$ [By RHS similarity]

Note: In similar triangles,

Ratio of medians = Ratio of corresponding heights
= Ratio of circumradii
= Ratio of inradii

Theorem

If two triangles are similar, then ratio of areas of two similar triangle is equal to the ratio of square of corresponding sides.

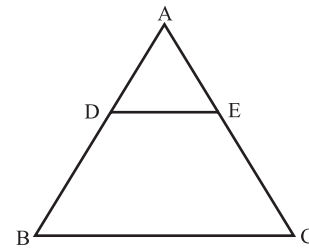
Illustration 11: D and E are the points on the sides AB and AC respectively of a $\triangle ABC$ and $AD = 8$ cm, $DB = 12$ cm, $AE = 6$ cm and $EC = 9$ cm, then BC is equal to

- (a) $\frac{2}{5} DE$ (b) $\frac{5}{2} DE$
(c) $\frac{3}{2} DE$ (d) $\frac{2}{3} DE$

Solution: (b) As in $\triangle ADE$ and $\triangle ABC$

$$\frac{AD}{AB} = \frac{8}{20} = \frac{2}{5}, \frac{AE}{AC} = \frac{6}{15} = \frac{2}{5} \quad \Rightarrow$$

So, $\frac{AD}{AB} = \frac{AE}{AC}$



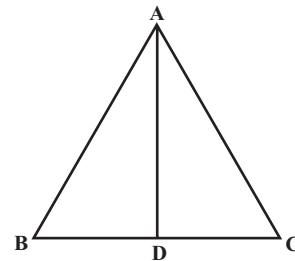
and $\angle A = \angle A$ (common)

$\triangle ADE \sim \triangle ABC$

$$\therefore \frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{DE}{BC} = \frac{2}{5}$$

$$\Rightarrow BC = \frac{5}{2} DE$$

Illustration 12: In a right angled $\triangle ABC$ in which $\angle A = 90^\circ$. If $AD \perp BC$, then the correct statement is



- (a) $AB^2 = BD \times DC$ (b) $AB^2 = BD \times AD$
(c) $AB^2 = BC \times DC$ (d) $AB^2 = BC \times BD$

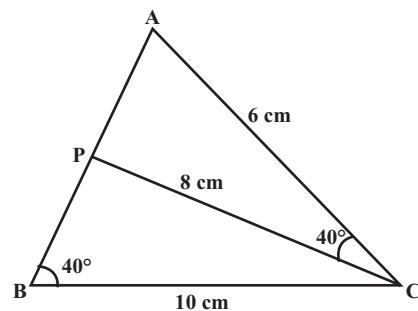
Solution: (d) Clearly, $\triangle ABD \sim \triangle CBA$

$$\Rightarrow \frac{AB}{BD} = \frac{CB}{BA}$$

$$\Rightarrow AB^2 = BC \times BD$$

Illustration 13: From the adjoining diagram, calculate

- (i) AB (ii) AP



Solution: In $\triangle APC$ and $\triangle ABC$,

$$\angle ACP = \angle ABC$$

$$\angle A = \angle A$$

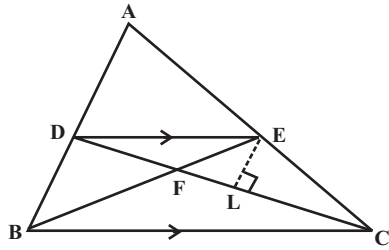
$$\Rightarrow \triangle APC \sim \triangle ABC \Rightarrow \frac{AP}{AC} = \frac{PC}{BC} = \frac{AC}{AB}$$

$$\therefore \frac{AP}{6} = \frac{8}{10} = \frac{6}{AB}$$

$$\Rightarrow AP = 6 \times \frac{8}{10} = 4.8 \text{ and } AB = \frac{60}{8} = 7.5$$

$$\Rightarrow AP = 4.8 \text{ cm and } AB = 7.5 \text{ cm}$$

Illustration 14: In the adjoining figure, $DE \parallel BC$ and $AD : DB = 4 : 3$



Find $\frac{AD}{AB}$ and then $\frac{DE}{BC}$

Solution: Since the sides of similar triangles are proportional, we have

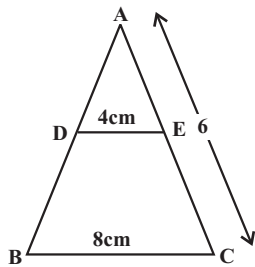
$$\frac{AD}{AB} = \frac{DE}{BC}$$

$$\text{But, } \frac{AD}{DB} = \frac{4}{3} \Rightarrow \frac{AD}{AD+DB} = \frac{4}{4+3} = \frac{AD}{AB} = \frac{4}{7}$$

$$\therefore \frac{DE}{BC} = \frac{AD}{AB} = \frac{4}{7} =$$

Illustration 15: In the given figure, DE parallel to BC . If $AD = 2$ cm, $DB = 3$ cm and $AC = 6$ cm, then AE is

- (a) 2.4 cm (b) 1.2 cm
(c) 3.4 cm (d) 4.8 cm



Solution: (a) The triangles ADE and ABC are similar.

$$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC}$$

$$\text{or } \frac{2}{5} = \frac{AE}{6}$$

$$\therefore AE = \frac{12}{5} = 2.4 \text{ cm}$$

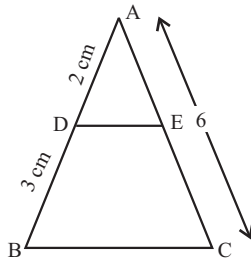
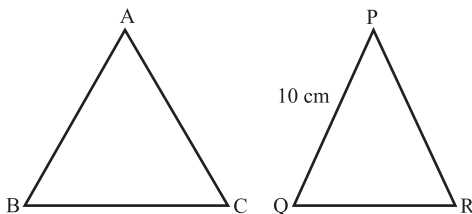


Illustration 16: The perimeters of two similar triangles ABC and PQR are 36 cm, and 24 cm, respectively. If $PQ = 10$ cm, then the length of AB is :

- (a) 16 cm (b) 12 cm
(c) 14 cm (d) 15 cm

Solution: (d)



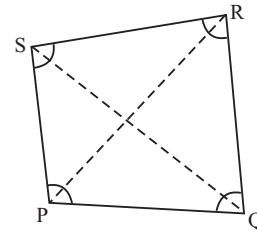
$\triangle ABC$ and $\triangle PQR$ are similar.

$$\frac{AB}{PQ} = \frac{\text{Perimeter of } \triangle ABC}{\text{Perimeter of } \triangle PQR} \Rightarrow \frac{AB}{PQ} = \frac{36}{24}$$

$$\text{or } AB = \frac{36}{24} \times 10 = 15$$

QUADRILATERALS

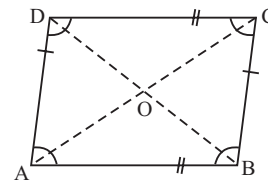
Quadrilateral is a plane figure bounded by four straight lines. The line segment which joins the opposite vertices of a quadrilateral is called diagonal of the quadrilateral. In figure, $PQRS$ is a quadrilateral and PR , QS are its two diagonals.



Sum of angles of a quadrilateral = 360°
i.e. $\angle P + \angle Q + \angle R + \angle S = 360^\circ$

Types of Quadrilaterals

- 1. Parallelogram:** A parallelogram is a quadrilateral with opposite sides parallel and equal.

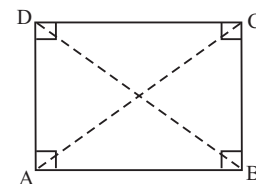


In figure, $ABCD$ is a parallelogram in which AC and BD as diagonals which intersect each other at O .

Properties:

- Opposite sides are equal i.e.
 $AB = DC$, $AD = BC$
- Opposite sides are parallel i.e.
 $AB \parallel DC$ and $AD \parallel BC$
- Opposite angles are equal i.e.
 $\angle BAD = \angle BCD$ and $\angle ABC = \angle ADC$
- Diagonals bisect each other, i.e.
 $OA = OC$, $OB = OD$
- Sum of pair of consecutive angles is 180° i.e.,
 $\angle A + \angle B = 180^\circ$, $\angle B + \angle C = 180^\circ$,
 $\angle C + \angle D = 180^\circ$, $\angle D + \angle A = 180^\circ$.

- 2. Rectangle:** A rectangle is a parallelogram with all angles equal to 90° .



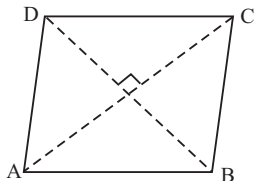
In figure, $\angle A = \angle B = \angle C = \angle D = 90^\circ$

Properties:

- (i) In a rectangle
Length of diagonal, are equal i.e.

$$AC = \sqrt{AB^2 + BC^2} = BD$$

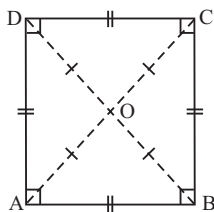
- (ii) In a rectangle diagonals bisect each other.
(iii) All rectangles are parallelogram but all parallelograms are not rectangles.
3. **Rhombus:** A parallelogram is a rhombus if its all sides are equal.



In rhombus $ABCD$, $AB = BC = CD = DA$

Properties:

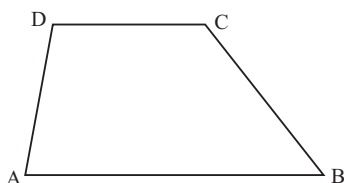
- (i) In a rhombus diagonals bisect each other at right angles i.e. angle between AC and DB is 90° .
(ii) All rhombus are parallelogram but all parallelograms are not rhombus.
4. **Square:** A parallelogram is a square if all the four sides are equal and also all the four angles are equal (i.e. 90°).



In figure, $ABCD$ is a square in which $AB = BC = CD = DA$ and $\angle A = \angle B = \angle C = \angle D = 90^\circ$

Properties:

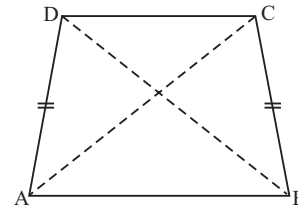
- (i) In a square diagonals are equal i.e. $AC = BD$
(ii) In a square diagonals bisect each other at right angle, i.e.
 $OA = OC$, $OB = OD$
and $\angle AOB = \angle BOC = \angle COD = \angle DOA = 90^\circ$.
(iii) All square are rhombus but rhombus may or may not be a square.
5. **Trapezium:** A quadrilateral is a trapezium if one pair of opposite sides are parallel.
In trapezium $ABCD$, $AB \parallel DC$.



If lateral sides (i.e. non-parallel sides) of a trapezium are equal, then it is called isosceles trapezium.

Properties of isosceles trapezium

In the figure $ABCD$ is an isosceles trapezium, then



- (i) $AB \parallel DC$
(ii) $AD = BC$
(iii) Diagonals are equal i.e. $AC = BD$

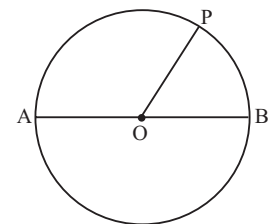
Diagonal Properties of all Parallelograms

Sr. No.	Diagonal Properties	Type of Parallelogram			
		Parallelogram	Rectangle	Rhombus	Square
1	Diagonals bisect each other	✓	✓	✓	✓
2	Diagonals are equal	×	✓	×	✓
3	Diagonals are at 90° to each other	×	×	✓	✓

CIRCLES

A circle is a locus i.e. path of a point in a plane which moves in such a way that its distance from a fixed point always remains constant.

In figure, 'O' is the fixed point and P is a moving point in the same plane. The path traced by P is called a circle. Fixed point O is the centre of the circle and the constant distance OP is called radius of the circle.

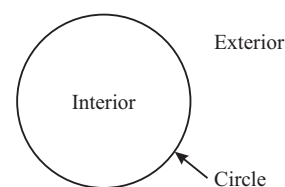


A diameter is a line segment passing through the centre and joins the two points on the circle in the figure.

AB is the diameter as it passes through the centre and joins the two points on the circle. Diameter = $2 \times$ radius.

A circle divides the plane in which it lies into three parts.

- (i) Inside the circle, called interior of the circle
(ii) The circle
(iii) Outside the circle, called the exterior of the circle.



The circle and its interior make up the circular region.

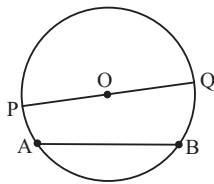
Circumference

Length of a complete circle is called its circumference.

In figure, AB is tangent to circle of radius 'r', which touches the circle at point P.

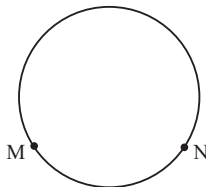
P is called the point of contact of tangent to the circle. Radius through the point of contact is always perpendicular to the tangent at the point of contact i.e. $OP \perp AB$.

Chord: A line segment joining any two points on the circle is called chord of the circle. A chord which passes through the centre is the diameter of the circle.

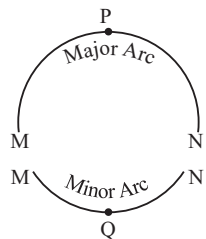


In the figure, O is the centre of the circle.
 AB and PQ both are chords.
 But PQ is the diameter (longest chord) also.

Arcs: A piece of a circle between two points is called an arc. Consider two points M and N on the circle. We find that there are two pieces of circle between M and N . One is longer and other is smaller.



The longer piece is called major arc and smaller piece is called minor arc

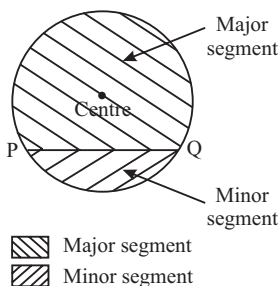


Major arc is denoted by $\widehat{MPN}$ and minor arc is denoted by $\widehat{MQN}$.

When M and N are ends of a diameter then both the arcs are equal and both are called semicircle.

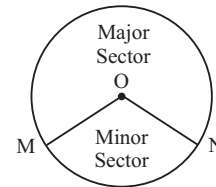
Segment: The region between a chord and an arc of a circle is called a segment.

There are two segments corresponding to two arcs, major segment and minor segment. Major segment is the segment enclosed by major arc. Centre of the circle lies in the major segment. Minor segment is the segment enclosed by minor arc. Centre of the circle does not lie in the minor segment.



If two arcs are equal, then both segments are semi-circles.

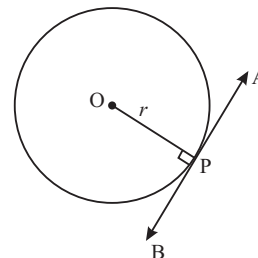
Sector: The region between an arc and the two radii joining the centre to the end point of the arc is called a sector. There are two sectors Minor and Major Sectors.



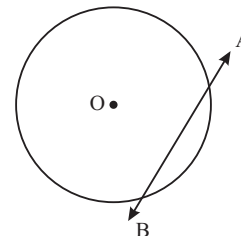
The sector which is larger than semicircular region is called major sector and the region less than the semicircular region is called minor sector.

If both sectors are equal, then each sector is a semi-circle.

Tangent: A tangent is a straight line which touches the circumference of a circle at only one point. A tangent does not intersect the circumference, if produced infinitely on either sides.

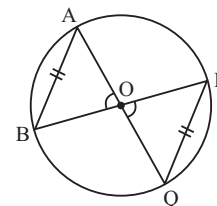


Secant: A secant is a straight line of infinite length which intersects the circumference of a circle at two different points. In figure, AB is a secant.



Basic Properties of a Circle

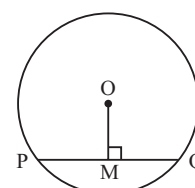
1. Equal chord of a circle subtend equal angles at the centre.



If $AB = PQ$, then $\angle AOB = \angle POQ$

The converse is also true.

2. The perpendicular from the centre of a circle to a chord of the circle bisects the chord.



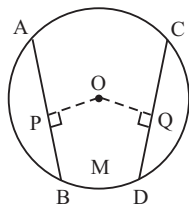
In figure, PQ is chord of a circle with centre ' O ', OM is perpendicular to PQ therefore $PM = MQ$. The converse is also true.

3. One and only one circle can pass through given three non-collinear points.

If three or more points lie on a line, then they are called collinear points otherwise called non-collinear points.

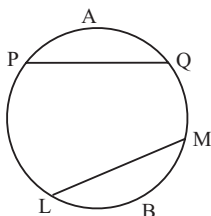
4. Equal chords of a circle are equidistant from the centre of the circle.

In the figure, if $AB = CD$, then $OP = OQ$



The converse is also true.

5. Two equal chords have equal corresponding arcs.

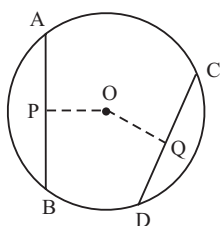


If $PQ = LM$ then

(a) $\widehat{PAQ} = \widehat{LBM}$ (Minor Arc)

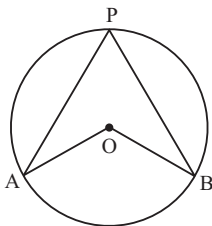
(b) $\widehat{PBQ} = \widehat{LAM}$ (Major Arc)

6. The greater of the two chords is nearer to the centre.



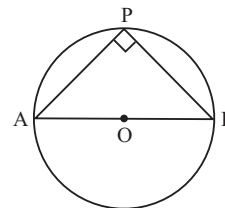
If $AB > CD$, then $OP < OQ$

7. The angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.



minor arc $\widehat{AB}$ subtend $\angle AOB$ at the centre O and also subtend $\angle APB$ at point P (situated on remaining part of circle). So $\angle AOB = 2 \angle APB$

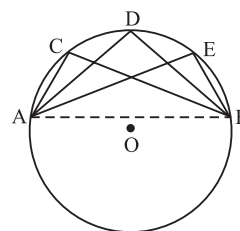
8. Angle in a semicircle is a right angle.



In figure, AOB is a diameter, hence $AOBPA$ is a semicircle, therefore $\angle APB = 90^\circ$.

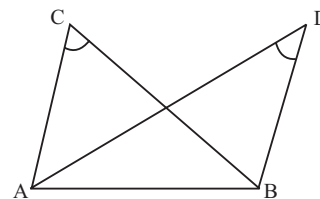
9. Angles in the same segment of a circle are equal.

$\angle ACB$, $\angle ADB$, $\angle AEB$ are in the same segment $ACDEBA$ of the circle.



$\therefore \angle ACB = \angle ADB = \angle AEB$.

10. If in a plane a line segment joining two points subtends equal angles at two other points lying on the same side of a line containing the line segment, the four points lie on a circle i.e. they are concyclic.

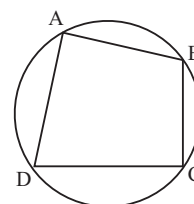


In figure, if $\angle ACB = \angle ADB$, then points A, B, D, C lie on a circle.

11. The sum of either pair of opposite angles of a cyclic quadrilateral is 180° .

A cyclic quadrilateral is the quadrilateral whose four vertices are concyclic i.e. the four vertices lie on a circle.

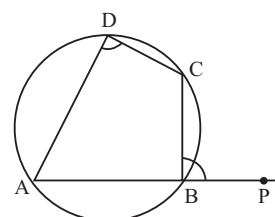
In figure, $ABCD$ is a cyclic quadrilateral,



$\therefore \angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$

The converse is also true.

12. If a side of a cyclic quadrilateral is produced the exterior angle so formed is equal to the interior opposite angle.

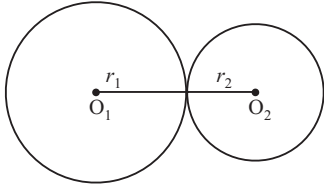


In figure, $ABCD$ is a cyclic quadrilateral,

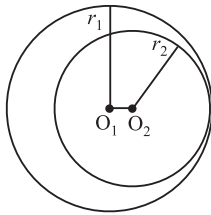
$$\therefore \angle CBP = \angle CDA$$

13. Two circles C_1 with centre O_1 , radius r_1 and C_2 with centre O_2 , radius r_2 will touch

(a) Externally, if and only if $O_1O_2 = r_1 + r_2$



(b) Internally, if and only if $O_1O_2 = |r_1 - r_2|$

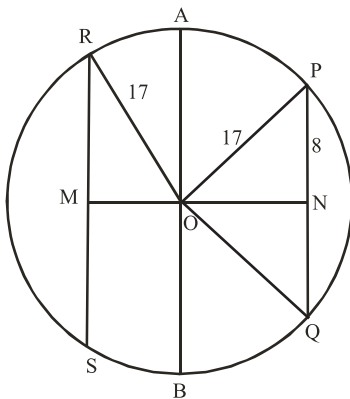


14. Two circles are congruent if their radii are equal.

Illustration 17: In a circle of radius 17 cm, two parallel chords are drawn on opposite sides of a diameter. The distance between the chords is 23 cm. If length of one chord is 16 cm, then the length of the other one is :

- (a) 15 cm (b) 23 cm
(c) 30 cm (d) 34 cm

Solution: (c) Let PQ and RS be two parallel chords of the circle on the opposite sides of the diameter $AB = 16$ cm



Now, $PN = 8$ (Since ON is the perpendicular bisector)

In $\triangle PON$

$$ON^2 = OP^2 - PN^2 \\ = (17)^2 - (8)^2 = 289 - 64 = 225$$

$$\text{or } ON = 15 \Rightarrow \therefore OM = 23 - 15 = 8$$

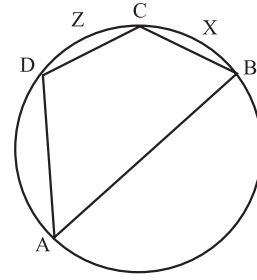
In $\triangle ORM$,

$$RM^2 = OR^2 - OM^2 \\ 17^2 - 8^2 = 289 - 64 = 225$$

$$\text{or } RM = 15 \Rightarrow RS = 15 \times 2 = 30 \text{ cm}$$

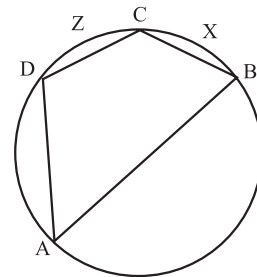
Illustration 18: In the cyclic quadrilateral $ABCD$, $\angle BCD = 120^\circ$, $m(\text{arc } DZC) = 7^\circ$, find $\angle DAB$ and $m(\text{arc } CXB)$.

- (a) $60^\circ, 70^\circ$ (b) $60^\circ, 40^\circ$
(c) $60^\circ, 50^\circ$ (d) $60^\circ, 60^\circ$



Solution: (c) $m\angle DAB + 180^\circ - 120^\circ = 60^\circ$ (Opposite angles of a cyclic quadrilateral)

$$m(\text{arc } BCD) = 2m\angle DAB = 120^\circ.$$



$$\therefore m(\text{arc } CXB) = m(\text{arc } BCD) - m(\text{arc } DZC) \\ = 120^\circ - 70^\circ = 50^\circ.$$

BASIC PYTHAGOREAN TRIPLETS

A Pythagorean triplet is a set of three natural numbers a , b and c , which are length of the sides of a right angled triangle.

Hence, if $a^2 + b^2 = c^2$, $b^2 + c^2 = a^2$ or $c^2 + a^2 = b^2$, then the set of natural numbers a , b and c is a Pythagorean triplet.

Since $3^2 + 4^2 = 5^2$, hence 3, 4, 5 form a Pythagorean triplet.

General Rule To Find Pythagorean Triplet: If r and s are two natural numbers such that $r > s$, $r - s$ is odd and GCD of r and s is 1, then the Pythagorean triplet a , b , c are defined by

$$a = r^2 - s^2, \quad b = 2rs \quad \text{and} \quad c = r^2 + s^2.$$

Note: If each term of any Pythagorean triplet is multiplied or divided by such a positive number that the products or quotients obtained respectively are natural numbers then the new products or quotients are also form Pythagorean triplets.

Since 3, 4, 5 form a Pythagorean triplet, therefore 9, 12 and 15 also form a Pythagorean triplet.

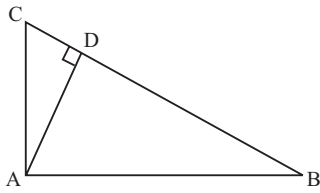
DETERMINATION OF NATURE OF TRIANGLE

Let length of three sides of a triangle are a , b and c .

- If c be the length of longest side and $c^2 = a^2 + b^2$, then the triangle is right-angled triangle.
- If c be the length of longest side and $c^2 > a^2 + b^2$, then the triangle is an obtuse-angled triangle.
- If c be the length of longest side and $c^2 < a^2 + b^2$, then the triangle is an acute-angled triangle.

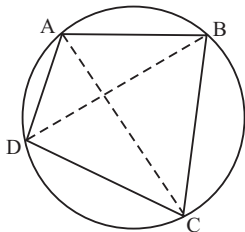
IMPORTANT POINTS

1. In $\triangle ABC$ right angled at A , if AD is perpendicular to BC .

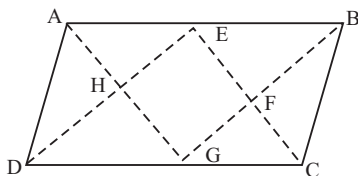


$\triangle ABC \sim \triangle DBA \sim \triangle DAC$
 Now $\triangle ABC \sim \triangle DBA$
 $\Rightarrow \frac{AB}{BC} = \frac{DB}{BA} \Rightarrow AB^2 = DB \times BC$
 And $\triangle ABC \sim \triangle DAC$
 $\Rightarrow \frac{AC}{BC} = \frac{DC}{AC} \Rightarrow AC^2 = DC \times BC$
 And $\triangle DBA \sim \triangle DAC$
 $\Rightarrow \frac{DA}{DB} = \frac{DC}{DA} \Rightarrow DA^2 = DB \times DC$

2. In a cyclic quadrilateral, product of the diagonals is equal to the sum of the products to the opposite sides,
 $AC \times BD = (AD \times BC) + (AB \times CD)$

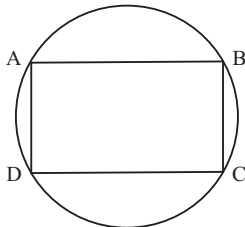


3. Bisectors of the angles of a parallelogram or a rectangle form a rectangle.

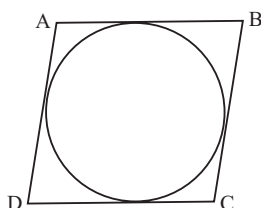


In parallelogram $ABCD$, AG , BG , CE and DE are the bisectors of $\angle A$, $\angle B$, $\angle C$ and $\angle D$ respectively. Hence in the figures $EFGH$ is a rectangle.

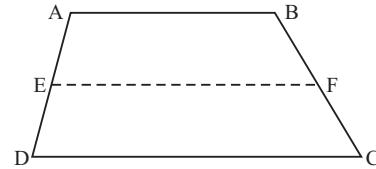
4. A parallelogram inscribed in a circle is a rectangle.
 In figure, $ABCD$ is a rectangle.



5. A parallelogram circumscribed a circle is a rhombus. In figure, $ABCD$ is a rhombus.



6. Median of a trapezium is the line segment joining mid-points of non-parallel sides of the trapezium.
 In the figure E and F are the mid points of non-parallel sides AB and CD respectively. Hence EF is the median of trapezium $ABCD$.

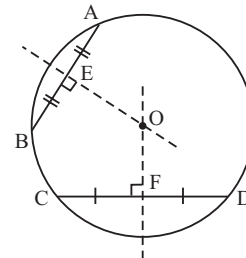


$$EF = \frac{1}{2} (AB + CD)$$

$$\text{Also } EF = \frac{a \times (AB) + b \times (DC)}{AD},$$

where $AE = a$ and $ED = b$

7. Perpendicular bisectors of two chords of a circle intersect at its centre of the circle.

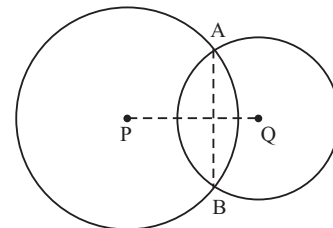


In figure, OE and OF are perpendicular bisectors of chords AB and CD , OE and OF meet at point O . Hence O is the centre of the circle.

8. If two circles intersect each other at two points then the line through the centres is the perpendicular bisector of the common chord.

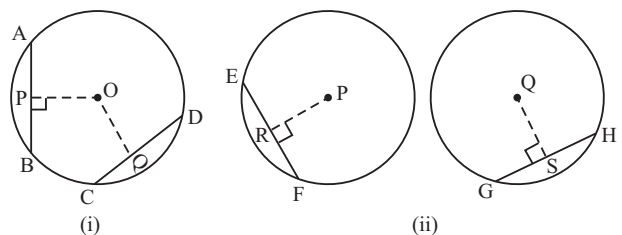
In figure, two circles with centre P and Q intersect each other at two points A and B .

Hence AB is the common chord of the two circles.



Therefore, PQ is the perpendicular bisector of common chord AB .

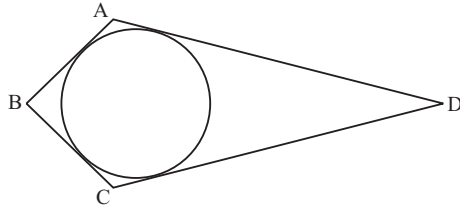
9. Equal chords of a circle or congruent circles are equidistant from the centre.



In figure (i), AB and CD are two equal chords of a circle, therefore their perpendicular distances OP and OQ respectively from the centre O are equal.

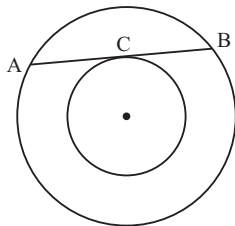
In figure (ii), two circles are congruent i.e. their radii are equal. EF and GH are two equal chords. Hence their perpendicular distances from centre P and Q respectively are equal.

10. If a circle touches all the four sides of a quadrilateral then the sum of the two opposite sides is equal to the sum of other two.



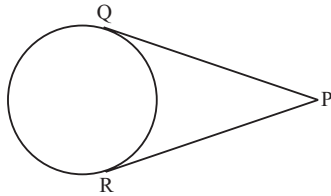
$$AB + DC = AD + BC$$

11. In two concentric circles, if a chord of the larger circle is also tangent to the smaller circle, then the chord is bisected at the point of contact.



Hence in the figure, $AC = CB$

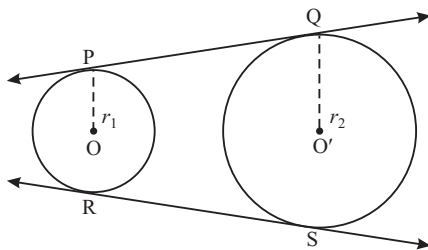
12. Length of two tangents from an exterior point to a circle are equal.



In figure PQ and PR are two tangents drawn from an exterior point to a circle.

$$\therefore PQ = PR$$

13. **Direct common tangent:** A tangent to two circles are such that the two circles lie on the same side of the tangent, then the tangent is called direct tangent to the two circles.



In the figure, PQ and RS are two direct common tangent to the same two circles. Length of these two common tangents to the same two circles are equal.

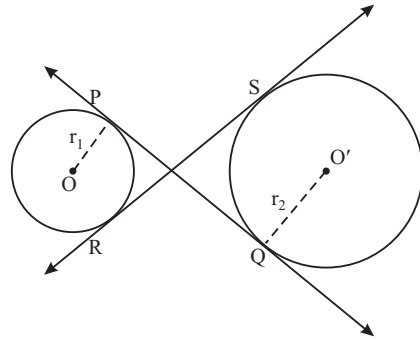
$$\text{i.e. } PQ = RS$$

$$\text{Also } PQ = RS = \sqrt{(OO')^2 + (r_2 - r_1)^2}$$

Here O, O' are the centres and r_1, r_2 are the radii of the two circles respectively. Also $r_2 > r_1$.

14. **Indirect or Transverse Common Tangent:** If a tangent to two circles is such that the two circles lie on opposite sides of the tangent, then the tangent is called indirect tangent.

Length of two indirect tangents to two circles is equal.



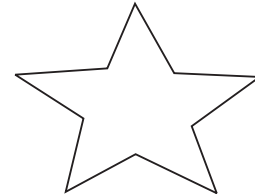
In the figure, PQ and RS are two indirect common tangents to the same two circles.

$$\therefore PQ = RS$$

$$\text{Also } PQ = RS = \sqrt{(OO')^2 - (r_1 + r_2)^2}$$

Here O, O' are centres r_1, r_2 are radii of the two circles respectively.

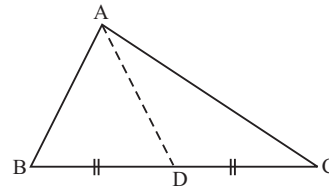
15. **Star:** A star has a shape like given in the figure.



If a star has n sides, then

$$\text{Sum of its all angles} = (n - 4) \times 180^\circ$$

16. In a triangle, the sum of the square of any two sides of a triangle is equal to twice the sum of the square of the median to the third side and square of half the third side.

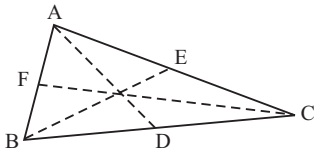


In the figure, AD is the median.

$$\therefore AB^2 + AC^2 = 2 \left[AD^2 + \left(\frac{BC}{2} \right)^2 \right]$$

17. In a triangle,

$$3 \times \left(\text{Sum of square of three sides of a triangle} \right) = 4 \times \left(\text{Sum of the square of three medians of the triangle} \right)$$

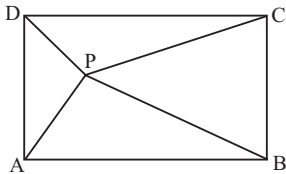


In figure AD , BE and CF are medians of $\triangle ABC$.

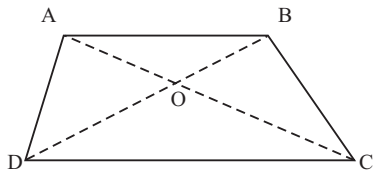
$$\therefore 3 \times (AB^2 + BC^2 + CA^2) = 4 \times (AD^2 + BE^2 + CF^2)$$

18. In the figure given below, if P is any point inside the rectangle $ABCD$, then

$$PA^2 + PC^2 = PB^2 + PD^2$$

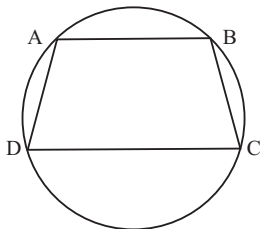


19. Diagonals of a trapezium divide each other in the ratio of the parallel sides of the trapezium. In trapezium $ABCD$, $AB \parallel DC$



$$\therefore \frac{AO}{OC} = \frac{BO}{OD} = \frac{AB}{CD}$$

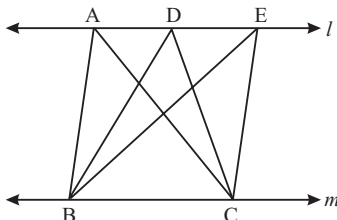
20. If a trapezium is inscribed inside a circle, then the trapezium is an isosceles trapezium i.e. its non-parallel sides are equal.



In the figure, $ABCD$ is a trapezium in which $AB \parallel CD$

$$\therefore AD = BC$$

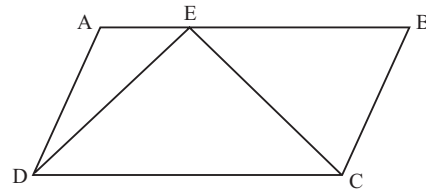
21. Area of triangles on the same base and lie between the same pair of parallel lines are equal.



In the figure, $\triangle ABC$, $\triangle DBC$ and $\triangle EBC$ are on the same base BC and lie between the same pair of parallel lines l and m .

$$\therefore \text{area of } \triangle ABC = \text{area of } \triangle DBC = \text{area of } \triangle EBC$$

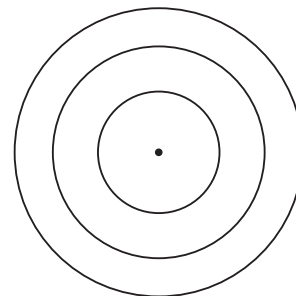
22. If a parallelogram and a triangle are on the same base and lie between the same pair of parallel lines, then area of the parallelogram is twice the area of the triangle.



In the figure, $ABCD$ a parallelogram and EDC a triangle are on the same base and lie between the same pair of parallel lines AB and CD .

$$\therefore \text{area of parallelogram } ABCD = 2 \times (\text{area of } \triangle EDC)$$

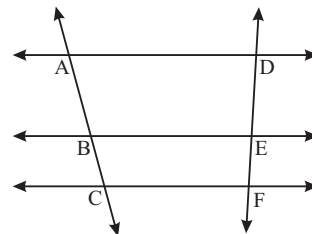
23. **Concentric circles:** Two or more circles in a plane are said to be concentric, if they have the same centre.



Concentric circles

24. Intercepts made by three or more parallel lines on two or more lines are in the same ratios.

In the figure three parallel lines AD , BE and CF made intercepts AB , BC and DE , EF on two lines AC and DF respectively.



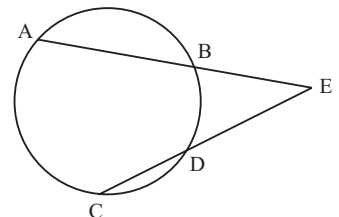
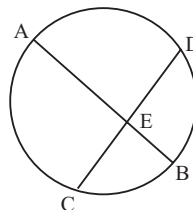
$$\therefore \frac{AB}{BC} = \frac{DE}{EF}$$

25. (a) In an equilateral triangle centroid, incentre, circum-centre, orthocentre coincide at the same point.

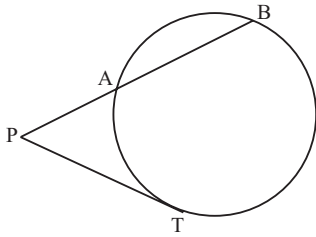
$$(b) \text{Circumradius} = 2 \times \text{in radius}$$

26. A parallelogram is a rectangle if its diagonals are equal.

27. If two chords AB and CD of a circle intersect inside a circle (or outside a circle when produced) at point E , then $AE \times EB = CE \times ED$.

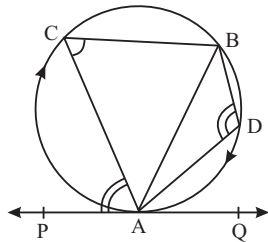


28. If PB is a secant which intersects the circle at A and B and PT is a tangent at T to the circle, then



$$PA \times PB = PT^2$$

29. Angles in the alternate segment:



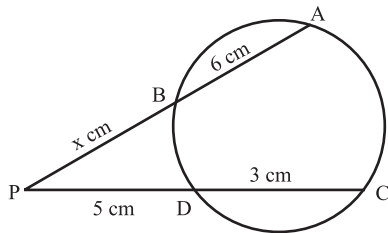
In the figure, AB is a chord of a circle. PQ is a tangent at an end point A of the chord to the circle. C is any point on arc AB and D is any point on arc BA .

$\angle BAQ$ and $\angle ACB$ are angles in the alternate segments
 $\angle BAP$ and $\angle ADB$ are angles in the alternate segments.

Angles in the alternate segments of a circle are equal i.e.

$$\angle BAQ = \angle ACB \text{ and } \angle BAP = \angle ADB$$

Illustration 19: In the given figure, chords AB and CD of a circle intersect externally at P . If $AB = 6$ cm, $CD = 3$ cm and $PD = 5$ cm, then $PB = ?$

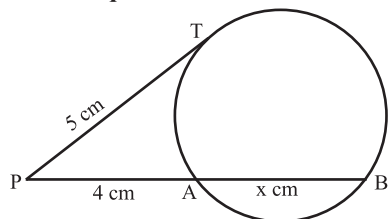


- (a) 5 cm (b) 6.25 cm
 (c) 6 cm (d) 4 cm

Solution: (d) $PA \times PB = PC \times PD$ (According to property of circle)

$$\begin{aligned} \Rightarrow (x+6) \times x &= 8 \times 5 \\ \Rightarrow x^2 + 6x - 40 &= 0 \\ \Rightarrow (x+10)(x-4) &= 0 \Rightarrow x = 4 \\ \therefore PB &= 4 \text{ cm} \end{aligned}$$

Illustration 20: In the given figure, PAB is a secant and PT is a tangent to the circle from P . If $PT = 5$ cm, $PA = 4$ cm and $AB = x$ cm, then x is equal to



- (a) 2.5 cm (b) 2.6 cm
 (c) 2.25 cm (d) 2.75 cm

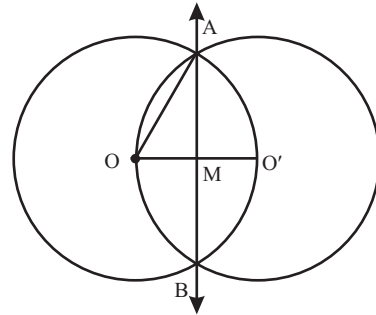
Solution: (c) $PA \times PB = PT^2 \Rightarrow 4 \times (4+x) = 25$

$$\Rightarrow 4+x = \frac{25}{4} = 6.25 \Rightarrow x = 2.25 \text{ cm}$$

Illustration 21: Two equal circles pass through each other's centre. If the radius of each circle is 5 cm, what is the length of the common chord?

- (a) $5\sqrt{3}$ (b) $10\sqrt{3}$
 (c) $\frac{5\sqrt{3}}{2}$ (d) 5

Solution: (a)



Given, distance between the centres of two circle = 5 cm

$$OO' = 5 \text{ cm}$$

$$\therefore OM = \frac{5}{2} \text{ cm}$$

In $\triangle OAM$,

$$OA^2 = OM^2 + AM^2$$

$$(5)^2 = \left(\frac{5}{2}\right)^2 + AM^2$$

$$AM = \sqrt{25 - \frac{25}{4}} = \frac{5\sqrt{3}}{2} \text{ cm}$$

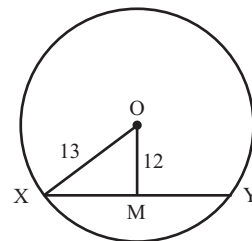
$\therefore$ The length of common chord, $AB = 2 \times AM$

$$= 2 \times \frac{5\sqrt{3}}{2} = 5\sqrt{3} \text{ cm}$$

Illustration 22: The radius of a circle is 13 cm and xy is a chord which is at a distance of 12 cm from the centre. The length of the chord is

- (a) 12 cm (b) 10 cm
 (c) 20 cm (d) 15 cm

Solution: (b) From figure,



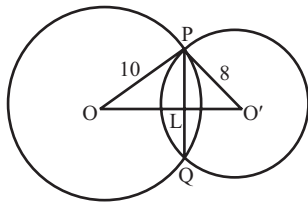
$$\begin{aligned} XM &= \sqrt{13^2 - 12^2} \\ &= \sqrt{169 - 144} = 5 \end{aligned}$$

$\therefore$ Length of the chord = $2 \times XM$
 $= 2 \times 5 = 10 \text{ cm}$

Illustration 23: Two circles of radii 10 cm and 8 cm. intersect and length of the common chord is 12 cm. Find the distance between their centres.

- (a) 13.8 cm (b) 13.29 cm
(c) 13.2 cm (d) 12.19 cm

Solution: (b) Here, $OP = 10$ cm; $O'P = 8$ cm



$$PQ = 12 \text{ cm}$$

$$\therefore PL = \frac{1}{2} PQ \Rightarrow PL = \frac{1}{2} \times 12 \Rightarrow PL = 6 \text{ cm}$$

In rt. $\triangle OLP$, $OP^2 = OL^2 + LP^2$

(using Pythagoras theorem)

$$\Rightarrow (10)^2 = OL^2 + (6)^2 \Rightarrow OL^2 = 64; OL = 8$$

$$\text{In } \triangle O'LP, (O'L)^2 = O'P^2 - LP^2 = 64 - 36 = 28$$

$$O'L^2 = 28 \Rightarrow O'L = \sqrt{28}$$

$$O'L = 5.29 \text{ cm}$$

$$\therefore OO' = OL + O'L = 8 + 5.29$$

$$OO' = 13.29 \text{ cm}$$

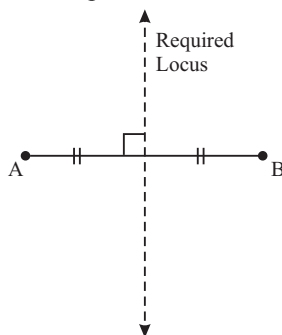
LOCUS

The locus of a point is the path traced out by a moving point under given geometrical conditions. Alternatively, the locus is the set of all those points which satisfy the given geometrical conditions.

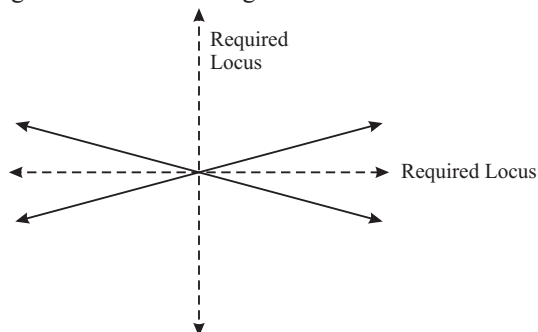
The plural of locus is loci and is read as 'Losai'.

The Locus of a Point in Different Conditions

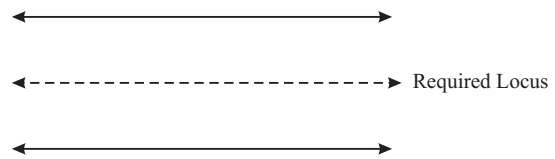
- (i) The locus of a point which is equidistant from two fixed points is the perpendicular bisector of the line segment joining the two fixed points.



- (ii) The locus of a point which is equidistant from two intersecting straight lines is a pair of straight lines which bisect the angles between the two given lines.

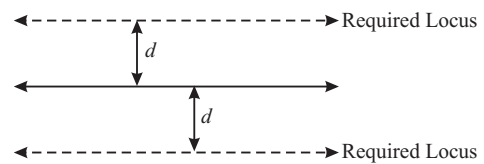


- (iii) The locus of a point equidistant from two given parallel straight lines is a straight line parallel to the given straight lines and midway between them.



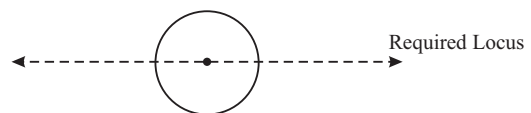
- (iv) The locus of a point which is equidistant from a fixed point in a plane is a circle.

- (v) The locus of a point, which is at a given distance from a given straight line, is a pair of parallel straight lines either side to the given line at a given distance from it.

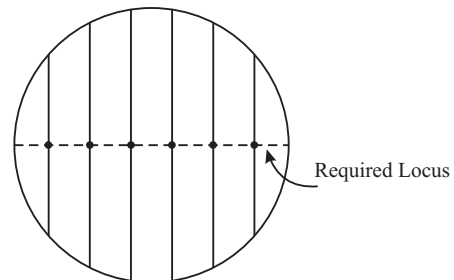


Here d is the given distance.

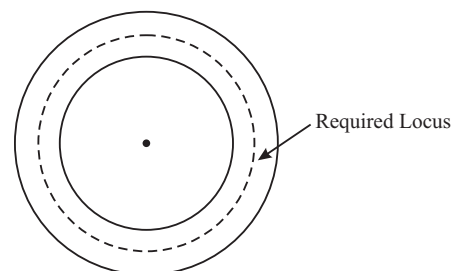
- (vi) The locus of the centre of a wheel moving on a straight horizontal road, is a straight line parallel to the road and at a height equal to the radius of the wheel.



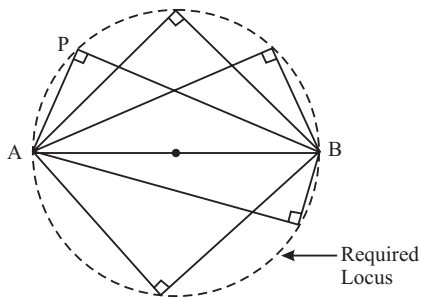
- (vii) The locus of mid-points of all parallel chords of a circle, is the diameter of the circle which is perpendicular to the given parallel chords.



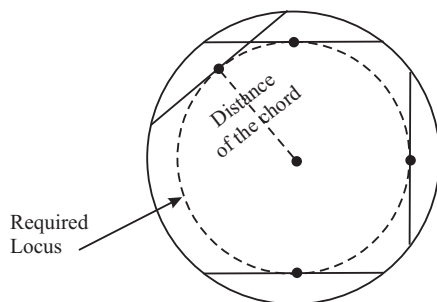
- (viii) The locus of a point which is equidistant from two concentric circles is the circumference of the circle concentric with the given circles and midway between them.



- (ix) If A and B are two fixed points, then the locus of a point P such that $\angle APB = 90^\circ$, is the circle with AB as diameter.



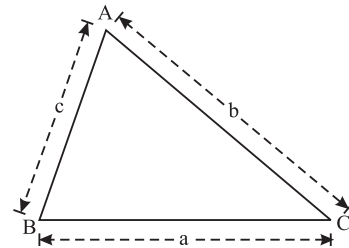
- (x) The locus of midpoints of all equal chords of a circle is the circumference of the circle concentric with the given circle and radius equal to the distance of equal chords from the centre of the given circle.



SINE AND COSINE RULE

If in a ΔABC ; a , b and c are the length of the sides opposite to vertices A , B and C respectively, then

$$(i) \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \text{ (sine rule)}$$



$$(ii) \left. \begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos A \\ b^2 &= c^2 + a^2 - 2ca \cos B \\ c^2 &= a^2 + b^2 - 2ab \cos C \end{aligned} \right\} \text{ (Cosine rule)}$$

Note that $\sin 0^\circ = 0$, $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{1}{\sqrt{2}}$,

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \sin 90^\circ = 1$$

$$\cos 0^\circ = 1, \cos 30^\circ = \frac{\sqrt{3}}{2}, \cos 45^\circ = \frac{1}{\sqrt{2}},$$

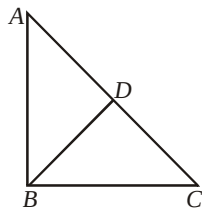
$$\cos 60^\circ = \frac{1}{2}, \cos 90^\circ = 0$$

Practice Exercises

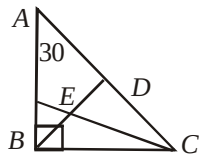


Foundation Level

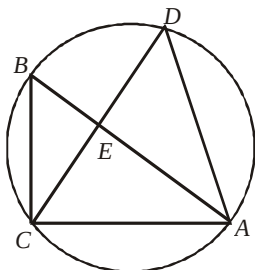
1. In triangle ABC , angle B is a right angle. If (AC) is 6 cm, and D is the mid-point of side AC . The length of BD is



- (a) 4 cm (b) $\sqrt{6}$ cm
(c) 3 cm (d) 3.5 cm
2. $AB \perp BC$ and $BD \perp AC$. And CE bisects the angle C . $\angle A = 30^\circ$. The, what is $\angle CED$.



- (a) 30° (b) 60°
(c) 45° (d) 65°
3. If two parallel lines are cut by two distinct transversals, then the quadrilateral formed by these four lines will always be a :
- (a) parallelogram (b) rhombus
(c) square (d) trapezium
4. In the adjoining figure, points A, B, C and D lie on the circle. $AD = 24$ and $BC = 12$. What is the ratio of the area of the triangle CBE to that of the triangle ADE



- (a) 1 : 4 (b) 1 : 2
(c) 1 : 3 (d) Insufficient data
5. In $\triangle ABC$, AD is the bisector of $\angle A$ if $AC = 4.2$ cm., $DC = 6$ cm., $BC = 10$ cm., find AB .

- (a) 2.8 cm (b) 2.7 cm
(c) 3.4 cm (d) 2.6 cm

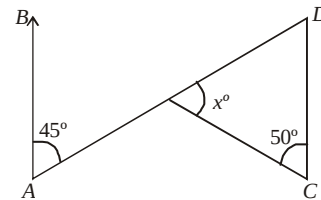
6. How many sides a regular polygon has with its sum of interior angles eight times its sum of exterior angles?

- (a) 16 (b) 24
(c) 18 (d) 30

7. A point P is 26 cm away from the centre O of a circle and the length PT of the tangent draw from P to the circle is 10cm. Find radius of the circle

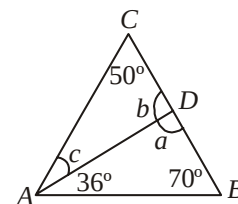
- (a) 2.4 cm (b) 3.2 cm
(c) 2.2 cm (d) 4.2 cm

8. In the given figure, $AB \parallel CD$, $\angle BAE = 45^\circ$, $\angle DCE = 50^\circ$ and $\angle CED = x$, then find the value of x .



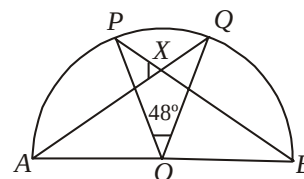
- (a) 85° (b) 95°
(c) 60° (d) 20°

9. Given the adjoining figure. Find a, b, c



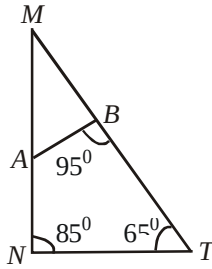
- (a) $74^\circ, 106^\circ, 20^\circ$ (b) $90^\circ, 20^\circ, 24^\circ$
(c) $60^\circ, 30^\circ, 24^\circ$ (d) $106^\circ, 24^\circ, 74^\circ$

10. In the figure given below, AB is a diameter of the semicircle $APQB$, centre O , $\angle POQ = 48^\circ$ cuts BP at X , calculate $\angle AXP$.

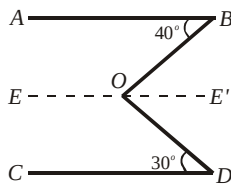


- (a) 50° (b) 55°
(c) 66° (d) 40°

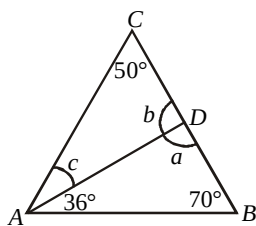
11. In the figure, if $\frac{NT}{AB} = \frac{9}{5}$ and if $MB = 10$, find MN .



- (a) 5 (b) 4
(c) 28 (d) 18
12. In the given figure, $AB \parallel CD$, $\angle ABO = 40^\circ$ and $\angle CDO = 30^\circ$. If $\angle DOB = x$, then find the value of x .

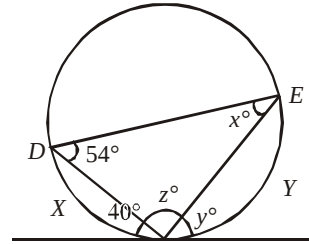


- (a) 10° (b) 70°
(c) 110° (d) 20°
13. M and N are points on the sides PQ and PR respectively of a $\triangle PQR$. For each of the following cases state whether MN is parallel to QR
- A. $PM = 4$, $QM = 4.5$, $PN = 4$, $NR = 4.5$
B. $PQ = 1.28$, $PR = 2.56$, $PM = 0.16$, $PN = 0.32$
- (a) only in case A
(b) only in case B
(c) both in the case A & B
(d) None of these
14. Given the adjoining figure. Find a , b , c .

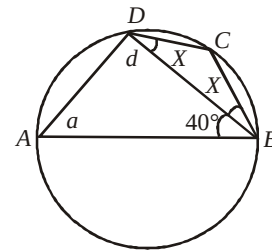


- (a) $74^\circ, 106^\circ, 24^\circ$ (b) $90^\circ, 20^\circ, 24^\circ$
(c) $60^\circ, 30^\circ, 24^\circ$ (d) $106^\circ, 24^\circ, 74^\circ$
15. The perimeters of two similar $\triangle s ABC$ and PQR are respectively 36 cm and 24 cm. If $PQ = 10$ cm, then AB is equal to
- (a) 5 cm (b) 10 cm
(c) 15 cm (d) 9 cm
16. In the triangle ABC , AD bisects $\angle BAC$, $BC = 6.4$, $AB = 5$ and $AC = 3$, then the length of BD is equal to
- (a) 3.5 (b) 5.5
(c) 3.2 (d) 4

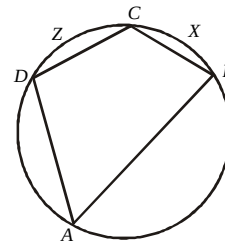
17. In the given figure, $m\angle EDC = 54^\circ$, $m\angle DCA = 40^\circ$. Find x , y and z respectively.



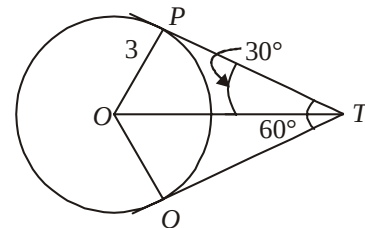
- (a) $20^\circ, 27^\circ, 86^\circ$ (b) $40^\circ, 54^\circ, 86^\circ$
(c) $20^\circ, 27^\circ, 43^\circ$ (d) $40^\circ, 54^\circ, 43^\circ$
18. In the adjoining figure, $ABCD$ is a cyclic quadrilateral. If AB is a diameter, $BC = CD$ and $\angle ABD = 40^\circ$, find the measure of $\angle DBC$.



- (a) 65 (b) 25
(c) 45 (d) 60
19. In the cyclic quadrilateral $ABCD$, $\angle BCD = 120^\circ$, $m(\text{arc } DZC) = 70^\circ$, find $\angle DAB$ and $m(\text{arc } CXB)$.

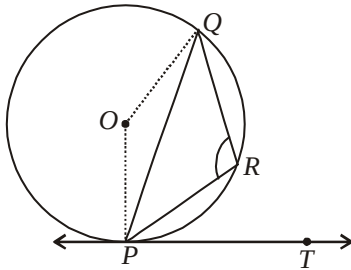


- (a) $60^\circ, 70^\circ$ (b) $60^\circ, 40^\circ$
(c) $60^\circ, 50^\circ$ (d) $60^\circ, 60^\circ$
20. In the above figure (ii), angle c is –
- (a) 270° (b) 70°
(c) 105° (d) 45°
21. If two tangents inclined at an angle 60° are drawn to a circle of radius 3 cm, then length of each tangent is equal to

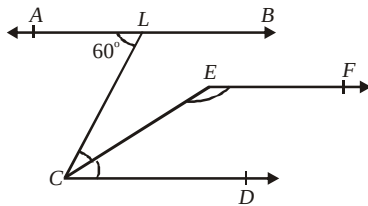


- (a) $\frac{3}{2}\sqrt{2}$ cm (b) 6 cm
(c) 3 cm (d) $3\sqrt{3}$ cm

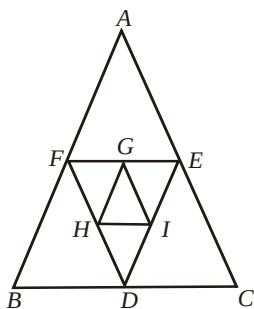
22. In the given fig. PQ is a chord of a circle and PT is the tangent at P such that $\angle QPT = 60^\circ$. Then $\angle PRQ$ is equal to



- (a) 135° (b) 150°
(c) 120° (d) 110°
23. If four sides of a quadrilateral $ABCD$ are tangential to a circle, then.
(a) $AC + AD = BD + CD$ (b) $AD + BC = AB + CD$
(c) $AB + CD = AC + BC$ (d) $AC + AD = BC + DB$
24. In the given figure, $AB \parallel CD$, $\angle ALC = 60^\circ$, EC is the bisector of $\angle LCD$ and $EF \parallel AB$. Then, find the measure of $\angle CEF$.



- (a) 80° (b) 130°
(c) 120° (d) 150°
25. D, E, F are midpoints of BC, CA and AB respectively. G, H, I are midpoints of FE, FD, DE respectively. Areas of $\triangle DHI$ and $\triangle AFE$ are in the ratio

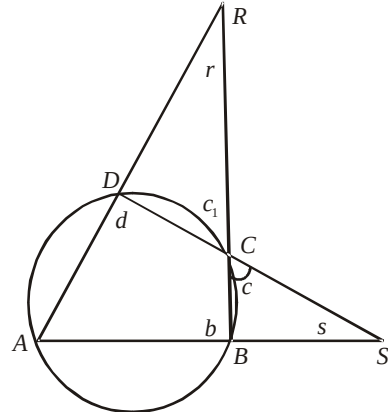


- (a) $1 : 3$ (b) $1 : 4$
(c) $1 : 9$ (d) $1 : 16$
26. John wishes to determine the distance between two objects A and B , but there is an obstacle between the two objects which prevents him from making a direct measurement. He designed an ingenious way to overcome this difficulty. First, he fixes a pole at convenient point O so that from O , both ends are visible. Then he fixes another pole at a point D on the line AO (produced) such that $AO = DO$. In a similar way, he fixes a third pole at a point C on the line BO

(produced) such that $BO = CO$. Then he measures CD and finds that $CD = 170$ cm. Find the distance between the objects A and B .

- (a) 90 cm (b) 170 cm
(c) 140 cm (d) 150 cm

27. In the adjoining figure, $ABCD$ is a cyclic quadrilateral. Then $r + s$ is equal to



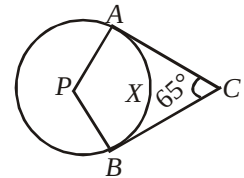
- (a) 180° (b) $2c$
(c) $180^\circ + 2c$ (d) $180^\circ - 2c$

28. Angle between the hour and minute hand of a clock at 11-15 is –

- (a) 90° (b) 112.5°
(c) 120° (d) 150°

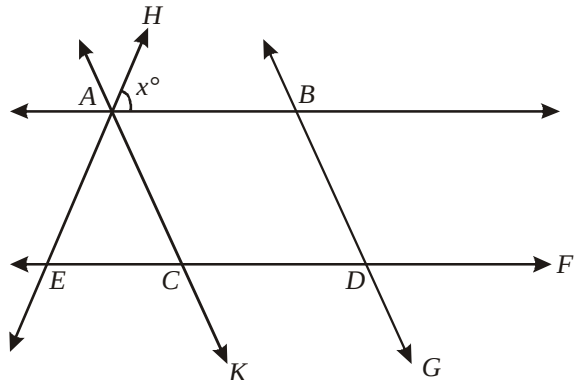
29. P is the centre of the circle $m \angle ACB = 65^\circ$. Find $m(\text{arc } AXB)$

- (a) 105°
(b) 115°
(c) 65°
(d) 245°

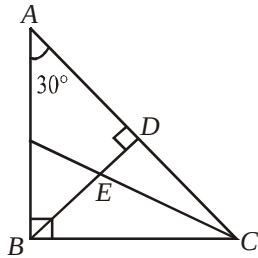


30. The centroid, circumcenter, orthocenter in a triangle–
(a) are always coincident.
(b) are always collinear.
(c) are always the inside the triangular area.
(d) always coincide in an equilateral triangle and otherwise collinear.

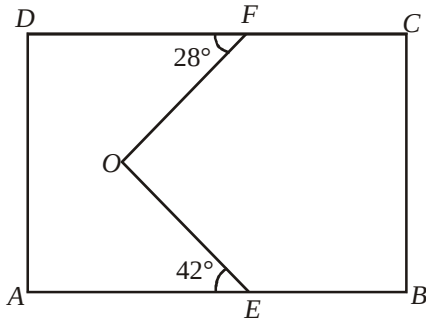
31. In the given figure $AB \parallel CD$ and $AC \parallel BD$. If $\angle EAC = 40^\circ$, $\angle FDG = 55^\circ$, $\angle HAB = x^\circ$, then find the value of x .



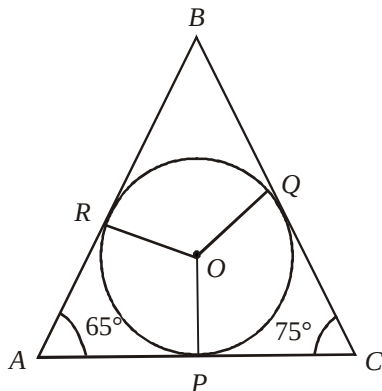
- (a) 85° (b) 75°
(c) 65° (d) 55°
32. Which one of the following cannot be the ratio of angles in a right angled triangle?
(a) $1 : 2 : 3$ (b) $1 : 1 : 2$
(c) $1 : 3 : 6$ (d) None of these
33. In $\triangle ABC$, $AB \perp BC$ and $BD \perp AC$. And CE bisects the angle C . $\angle A = 30^\circ$. What is $\angle CED$?



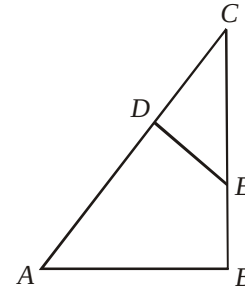
- (a) 30° (b) 60°
(c) 45° (d) 65°
34. In the adjoining figure $ABCD$ is a rectangle and $DF = CF$ also, $AE = 3BE$. What is the value of $\angle EOF$, if $\angle DFO = 28^\circ$ and $\angle AEO = 42^\circ$?



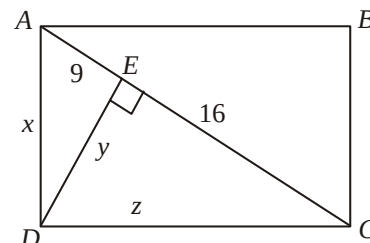
- (a) 14° (b) 42°
(c) 70° (d) 90°
35. Each interior angle of a regular polygon exceeds its exterior angle by 132° . How many sides does the polygon have?
(a) 9 (b) 15
(c) 12 (d) None of these
36. In a triangle ABC , O is the centre of incircle PQR , $\angle BAC = 65^\circ$, $\angle BCA = 75^\circ$, find $\angle ROQ$:



- (a) 80° (b) 120°
(c) 140° (d) can't be determined
37. ABC and CDE are right angled triangle. $\angle ABC = \angle CDE = 90^\circ$. D lies on AC and E lies on BC . $AB = 24$ cm, $BC = 60$ cm. If $DE = 10$ cm, then CD is:

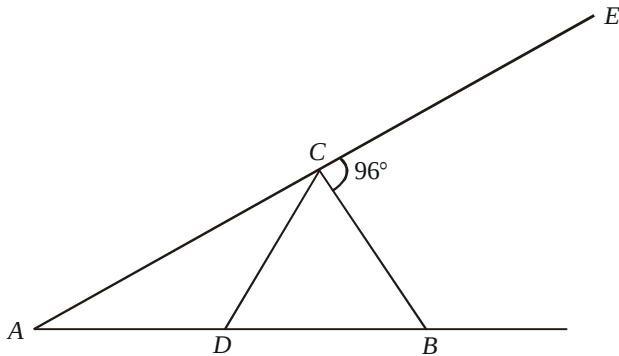


- (a) 28 cm
(b) 35 cm
(c) 25 cm
(d) can't be determined
38. The largest angle of a triangle of sides 7 cm, 5 cm and 3 cm is
(a) 45° (b) 60°
(c) 90° (d) 1200
39. $ABCD$ is a parallelogram in which $\angle B = 70^\circ$. Find the number of points X in the plane of the parallelogram such that it is equidistant from its vertices.
(a) zero (b) one
(c) two (d) three
40. $PQRS$ is trapezium, in which PQ is parallel to RS , and $PQ = 3(RS)$. The diagonal of the trapezium intersect each other at X , then the ratio of $\triangle PXQ$ and $\triangle RXS$ is
(a) $6 : 1$ (b) $3 : 1$
(c) $9 : 1$ (d) $7 : 1$
41. C is the midpoint of DE . Area of parallelogram $ABCD = 16$ sq. cm. Find the area of $\triangle BCDE$.
(a) 8 sq.cm (b) 16 sq. cm
(c) 32 sq. cm (d) 24 sq. cm
42. What are the respective value of x , y and z in the given rectangle $ABCD$?



- (a) 15, 12, 20
(b) 12, 15, 20
(c) 8, 10, 12
(d) None of these

43. In the figure (not drawn to scale) given below, if $AD = CD = BC$, and $\angle BCE = 96^\circ$, how much is $\angle DBC$?

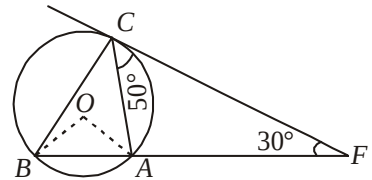


- (a) 32° (b) 84°
(c) 64° (d) Cannot be determined

44. In a trapezium $ABCD$, $AB \parallel CD$ and $AD = BC$. If P is point of intersection of diagonals AC and BD , then all of the following is wrong except.

- (a) $PA \cdot PB = PC \cdot PD$
(b) $PA \cdot PC = PB \cdot PD$
(c) $PA \cdot AB = PD \cdot DC$
(d) $PA \cdot PD = AB \cdot DC$

45. Find $\angle BOA$.



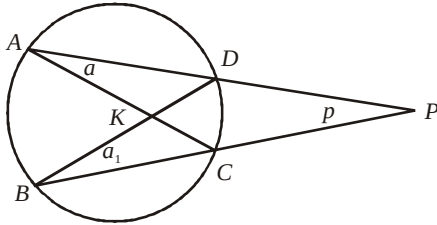
- (a) 100° (b) 150°
(c) 80° (d) Indeterminate

Standard Level

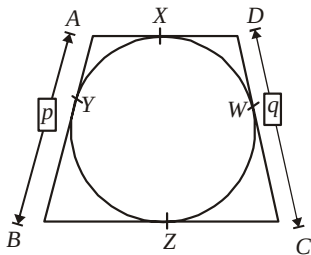
1. Here XY has been divided into 5 congruent segments and semicircles have been drawn. But suppose XY were divided into millions of congruent segments and semicircles were drawn, what would the sum of the lengths of the arcs be?



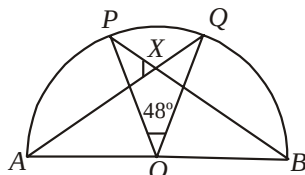
- (a) $2YX$ (b) $5XY$
(c) XY (d) None of these
2. In the adjoining figure, chord AD and BC of a circle are produced to meet at P , $PA = 10$ cm, $PB = 8$ cm, $PC = 5$ cm, $AC = 6$ cm. Find BD , PD .



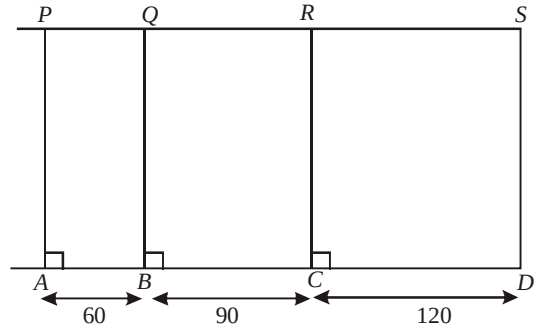
- (a) 5.8, 3 (b) 3.8, 5
(c) 2.8, 6 (d) 4.8, 4
3. In the adjoining figure the circle touches the side of the quadrilateral $ABCD$. If $AB = p$, express $(AD + BC)$ in terms of p and



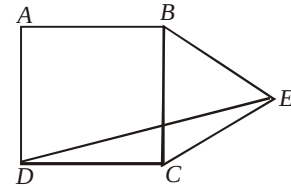
- (a) $p + q$ (b) $\frac{1}{2} p + q$
(c) $2(p - q)$ (d) $3(p - q)$
4. In the figure given below, AB is a diameter of the semi-circle $APQB$, centre O , $\angle POQ = 48^\circ$ cuts BP at X , calculate $\angle AXP$.



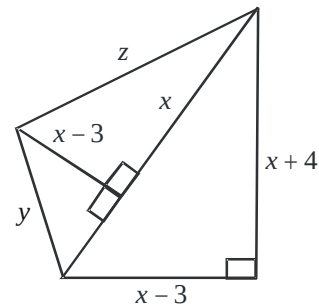
5. In the figure, if $PS = 360$, find PQ , QR and RS .



- (a) 150° (b) 160°
(c) 180° (d) 190°
6. If $ABCD$ is a square and BCE is an equilateral triangle, what is the measure of the angle DEC ?



- (a) 15° (b) 30°
(c) 20° (d) 45°
7. Based on the figure below, what is the value of x , if $y = 10$

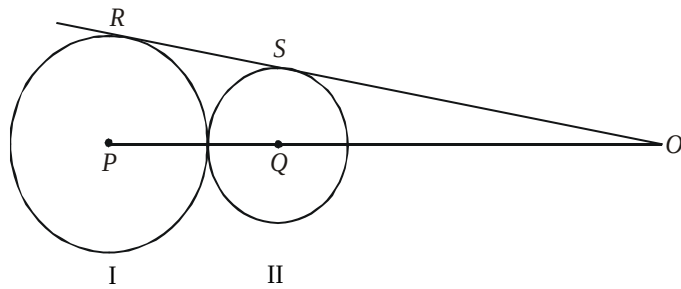


- (a) 10 (b) 11
(c) 12 (d) None of these
8. In a triangle ABC , the internal bisector of the angle A meets BC at D . If $AB = 4$, $AC = 3$ and $\angle A = 60^\circ$, then the length of AD is

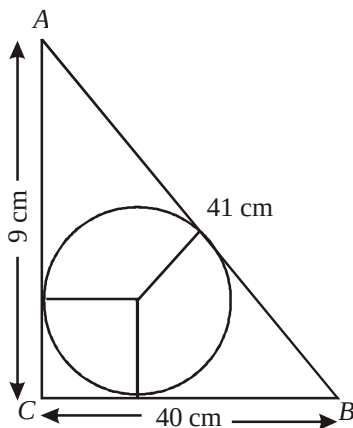
- (a) $2\sqrt{3}$ (b) $\frac{12\sqrt{3}}{7}$
(c) $15\sqrt{\frac{3}{8}}$ (d) $6\sqrt{\frac{3}{7}}$

Directions for Questions 9–11: Answer the questions on the basis of the information given below.

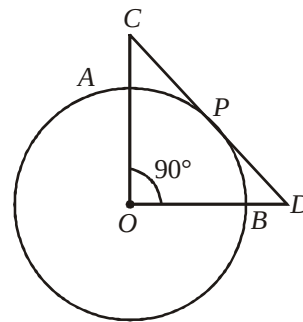
In the adjoining figure, I and II are circles with centers P and Q respectively. The two circles touch each other and have a common tangent that touches them at points R and S respectively. This common tangent meets the line joining P and Q at O . The diameters of I and II are in the ratio 4 : 3. It is also known that the length of PO is 28 cm.



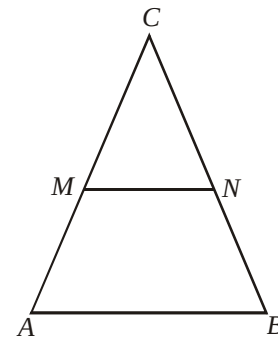
9. What is the ratio of the length of PQ to that of QO ?
 - (a) 1 : 4
 - (b) 1 : 3
 - (c) 3 : 8
 - (d) 3 : 4
10. What is the radius of the circle II?
 - (a) 2 cm
 - (b) 3 cm
 - (c) 4 cm
 - (d) 5 cm
11. The length of SO is
 - (a) $8\sqrt{3}$ cm
 - (b) $10\sqrt{3}$ cm
 - (c) $12\sqrt{3}$ cm
 - (d) $14\sqrt{3}$ cm
12. What is the inradius of the incircle shown in the figure?



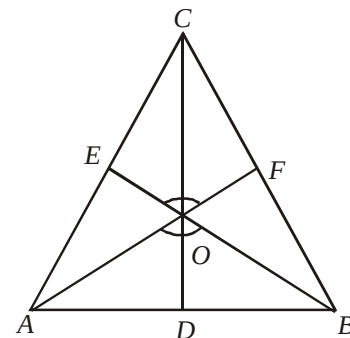
- (a) 9 cm
 - (b) 4
 - (c) can't be determined
 - (d) None of these
13. In a circle O is the centre and $\angle COD$ is right angle. $AC = BD$ and CD is the tangent at P . What is the value of $AC + CP$, if the radius of the circle is 1 metre?



- (a) 105 cm
 - (b) 141.4 cm
 - (c) 138.6 cm
 - (d) can't be determined
14. In the triangle ABC , MN is parallel to AB . Area of trapezium $ABNM$ is twice the area of triangle CMN . What is ratio of $CM : AM$?

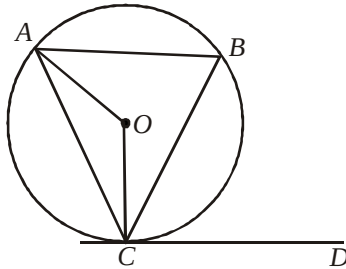


- (a) $\frac{1}{\sqrt{3}+1}$
 - (b) $\frac{\sqrt{3}-1}{2}$
 - (c) $\frac{\sqrt{3}+1}{2}$
 - (d) None of these
15. ABC is a triangle in which $\angle CAB = 80^\circ$ and $\angle ABC = 50^\circ$, AE , BF and CD are the altitudes and O is the orthocentre. What is the value of $\angle AOB$?



- (a) 65°
- (b) 70°
- (c) 50°
- (d) 130°

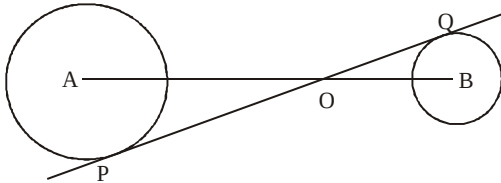
16. In the given diagram O is the centre of the circle and CD is a tangent. $\angle CAB$ and $\angle ACD$ are supplementary to each other $\angle OAC 30^\circ$. Find the value of $\angle OCB$:



- (a) 30° (b) 20°
(c) 60° (d) None of these
17. The sides of a triangle are in the ratio of $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$. If the perimeter is 52 cm, then the length of the smallest side is

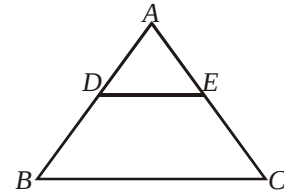
- (a) 9 cm (b) 10 cm
(c) 11 cm (d) 12 cm
18. The ratio of the area of a square to that of the square drawn on its diagonal is
- (a) 1 : 4 (b) 2 : 1
(c) 1 : 2 (d) 1 : 3

19. PQ is a tangential to circles with centers A and B at P and Q respectively. If $AB = 10$ cm. and $PQ = 8$ cm, find the radius of the bigger circle. Given that area of triangle APO is four times the area of triangle OQB –

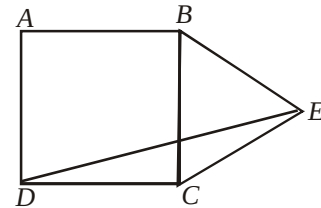


- (a) 2 cm (b) 4 cm
(c) 6 cm (d) 8 cm
20. Instead of walking along two adjacent sides of a rectangular field, a boy took a short cut along the diagonal and saved a distance equal to half the longer side. Then the ratio of the shorter side to the longer side is
- (a) $1/2$ (b) $2/3$
(c) $1/4$ (d) $3/4$
21. Two circles touch each other internally. Their radii are 2 cm and 3 cm. The biggest chord of the outer circle which is outside the inner circle is of length
- (a) $2\sqrt{2}$ cm (b) $3\sqrt{2}$ cm
(c) $2\sqrt{3}$ cm (d) $4\sqrt{2}$ cm
22. The sum of the interior angles of a polygon is 1620° . The number of sides of the polygon are :
- (a) 9 (b) 11
(c) 15 (d) 12

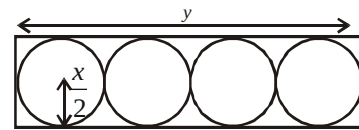
23. In $\triangle ABC$, $DE \parallel BC$ and $\frac{AD}{DB} = \frac{3}{5}$. If $AC = 5.6$ cm, find AE .



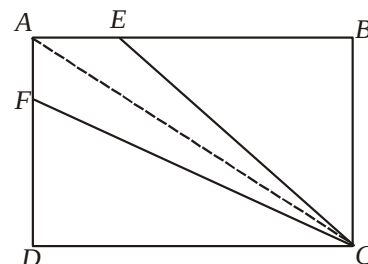
- (a) 2.1 cm (b) 3.1 cm
(c) 1.2 cm (d) 2.3 cm
24. If one of the diagonals of a rhombus is equal to its side, then the diagonals of the rhombus are in the ratio:
- (a) $\sqrt{3} : 1$ (b) $\sqrt{2} : 1$
(c) 3 : 1 (d) 2 : 1
25. If $ABCD$ is a square and BCE is an equilateral triangle, what is the measure of the angle DEC ?



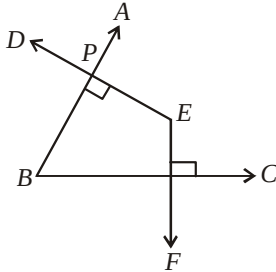
- (a) 15° (b) 30°
(c) 20° (d) 45°
26. $ABCD$ is a square, F is the mid-point of AB and E is a point on BC such that BE is one-third of BC . If area of $\triangle FBE = 108 \text{ m}^2$, then the length of AC is :
- (a) 63 m (b) $36\sqrt{2}$ m
(c) $63\sqrt{2}$ m (d) $72\sqrt{2}$ m
27. Arc ADC is a semicircle and $DB \perp AC$. If $AB = 9$ and $BC = 4$, find DB .
- (a) 6 (b) 8
(c) 10 (d) 12
28. In the figure below, which of the following is the relationship between 'x' and 'y' if the equal circles shown are tangents to each other and to the sides of the rectangle



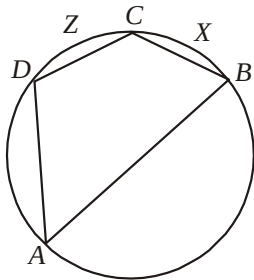
- (a) $x = \frac{1}{4}y$ (b) $x = \frac{y}{\pi}$
(c) $x = \pi y^2$ (d) $x = 2\pi y$
29. In the given figure given below, E is the mid-point of AB and F is the midpoint of AD . if the area of $FAEC$ is 13, what is the area of $ABCD$?



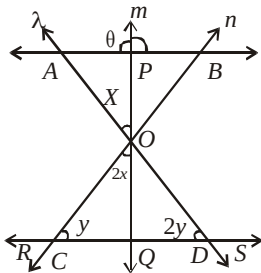
- (a) 19.5 (b) 26
(c) 39 (d) None of these
30. In the given figure, $\angle ABC$ and $\angle DEF$ are two angles such that $BA \perp ED$ and $EF \perp BC$, then find value of $\angle ABC + \angle DEF$.



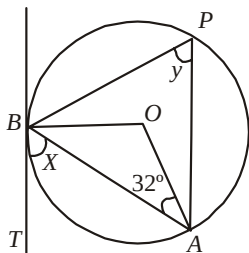
- (a) 120° (b) 180°
(c) 150° (d) 210°
31. In the cyclic quadrilateral $ABCD$, $\angle BCD = 120^\circ$, $m(\text{arc } DZC) = 7^\circ$, find $\angle DAB$ and $m(\text{arc } CXB)$.



- (a) $60^\circ, 70^\circ$ (b) $60^\circ, 40^\circ$
(c) $60^\circ, 50^\circ$ (d) $60^\circ, 60^\circ$
32. Give that segment AB and CD are parallel, if lines ℓ , m and n intersect at point O . Find the ratio of θ to $\angle ODS$



- (a) 2 : 3 (b) 3 : 2
(c) 3 : 4 (d) Data insufficient
33. In the given figure, AB is chord of the circle with centre O , BT is tangent to the circle. The values of x and y are



- (a) $52^\circ, 52^\circ$ (b) $58^\circ, 52^\circ$
(c) $58^\circ, 58^\circ$ (d) $60^\circ, 64^\circ$
34. The distance between two parallel chords of length 8 cm each in a circle of diameter 10 cm is
- (a) 6 cm (b) 7 cm
(c) 8 cm (d) 5.5 cm
35. The internal bisectors of the angles B and C of a triangle ABC meet at O . Then find the measure of $\angle BOC$.

- (a) $90^\circ - \frac{\angle A}{2}$ (b) $180^\circ - \frac{\angle A}{2}$
(c) $90^\circ + \frac{\angle A}{2}$ (d) $180^\circ + \frac{\angle A}{2}$

36. In a $\triangle ABC$, angle C is 68° , the perpendicular bisector of AB at R meets BC at P . If $\angle PAC = 42^\circ$ then $\angle ABC$ is equal to
- (a) 45° (b) 42°
(c) 35° (d) 34°

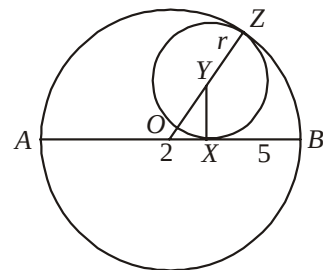
37. If in a $\triangle ABC$, $\angle B = 120^\circ$, then which of the following is true? [$\cos 120^\circ = -\frac{1}{2}$]

- (a) $a^2 + c^2 = b^2 + ac$ (b) $a^2 + c^2 = b^2 - ac$
(c) $a^2 + c^2 = b^2 + 2ac$ (d) $a^2 + c^2 = b^2 - 2ac$

38. A chord of length 14 cm is at a distance of 6 cm from the centre of a circle. Find the length of another chord at a distance of 2 cm from the centre of the circle.

- (a) 18 cm (b) 16 cm
(c) 10 cm (d) 12 cm

39. In the adjoining figure x is a point on diameter AB of the circle with centre O , such that $AX = 9$ cm, $XB = 5$ cm. Find the radius of the circle (centre Y) which touches the diameter at X and touches the circle, centre O , internally at Z .

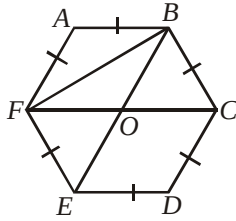


- (a) $3\frac{3}{14}$ cm. (b) $3\frac{1}{14}$ cm.
(c) $1\frac{1}{14}$ cm. (d) $2\frac{3}{14}$ cm.

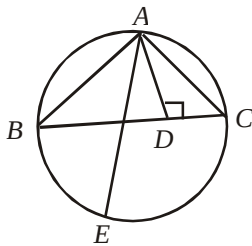
40. In $\triangle ABC$, $AB = AC = 8$, PR and PQ are parallel to lines AC and AB respectively. P is the midpoint of BC . Find the perimeter of $\square PRAQ$.

- (a) 16 (b) 18
(c) 20 (d) 12

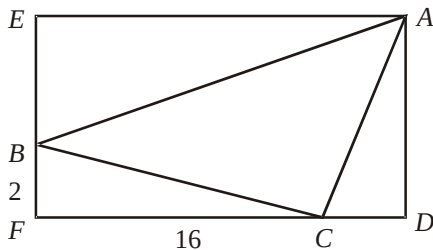
41. The height of the hexagon whose side is a



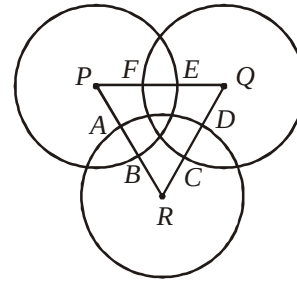
- (a) $\frac{3\sqrt{3}}{2}a$ (b) $\frac{3\sqrt{3}}{4}$
 (c) $\sqrt{3}a$ (d) None of these
42. In $\triangle ABC$, $AB = 8$, $AC = 6$, Altitude $AD = 4.8$ AE is the diameter of the circumcircle. Find the circumradius.



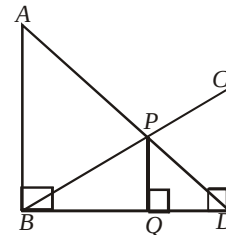
- (a) 5 (b) 10
 (c) 15 (d) Cannot be determined
43. The length of a ladder is exactly equal to the height of the wall it is resting against. If lower end of the ladder is kept on a stool of height 3 m and the stool is kept 9 m away from the wall the upper end of the ladder coincides with the tip of the wall. Then, the height of the wall is
- (a) 12 m. (b) 15 m.
 (c) 18 m. (d) 11 m.
44. In the given figure, $EADF$ is a rectangle and ABC is a triangle whose vertices lie on the sides of $EADF$. $AE = 22$, $BE = 6$, $CF = 16$ and $BF = 2$. Find the length of the line joining the mid-points of the sides AB and BC .



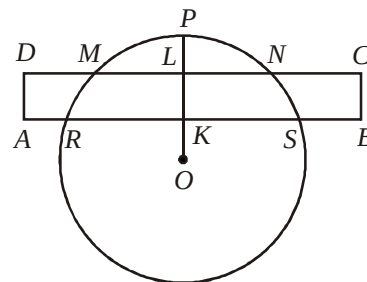
- (a) $4\sqrt{2}$ (b) 5
 (c) 3.5 (d) None of these
45. Three circles, each of radius 20 and centres at P , Q , R . further, $AB = 5$, $CD = 10$ and $EF = 12$. What is the perimeter of the triangle PQR ?



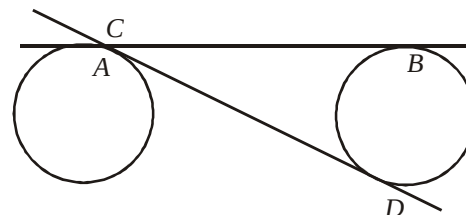
- (a) 120 (b) 66
 (c) 93 (d) 87
46. In the diagram given below, $\angle ABD = \angle CDB$ $\angle PQD = 90^\circ$. If $AB : CD = 3 : 1$, the ratio of $CD : PQ$ is



- (a) 1 : 0.69 (b) 1 : 0.75
 (c) 1 : 0.72 (d) None of these
47. What is the distance in cm between two parallel chords of lengths 32 cm and 24 cm in a circle of radius 20 cm?
- (a) 1 or 7 (b) 2 or 14
 (c) 3 or 21 (d) 4 or 28
48. In the adjoining figure O is the centre of the circle. The radius OP bisects a rectangle $ABCD$, at right angle. $DM = NC = 2$ cm and $AR = SB = 1$ cm and $KS = 4$ cm and $OP = 5$ cm. What is the area of the rectangle?

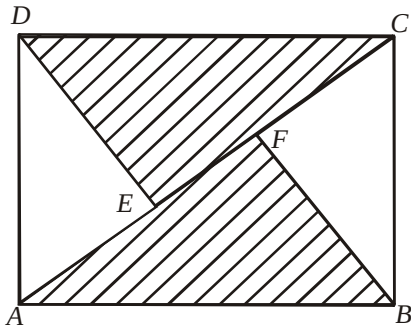


- (a) 8 cm^2 (b) 10 cm^2
 (c) 12 cm^2 (d) None of these
49. There are two circles each with radius 5 cm. Tangent AB is 26 cm. The length of tangent CD is:

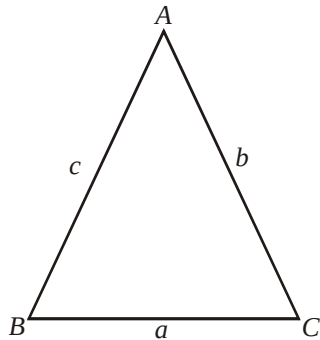


- (a) 15 cm (b) 21 cm
 (c) 24 cm (d) can't be determined

50. $ABCD$ is a rectangle of dimensions $6 \text{ cm} \times 8 \text{ cm}$. DE and BF are the perpendiculars drawn on the diagonal of the rectangle. What is the ratio of the shaded to that of unshaded region?

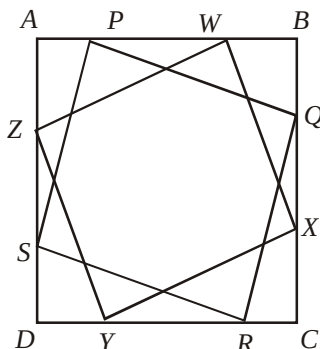


- (a) 7 : 3 (b) 16 : 9
(c) $4 : 3\sqrt{2}$ (d) Data insufficient
51. In the given triangle ABC , the length of sides AB and AC is same (i.e., $b = c$) and $60^\circ < A < 90^\circ$, then the possible length of BC is

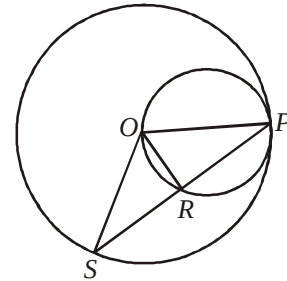


- (a) $b < a < 2b$ (b) $\frac{c}{3} < a < 3a$
(c) $b < a < b\sqrt{3}$ (d) $c < a < c\sqrt{2}$
52. The angles of a triangle are in the ratio of 4 : 1 : 1. Then the ratio of sine of the largest angle to the smallest angle is the largest side to the perimeter is $[\sin 120^\circ = \sin 60^\circ]$
- (a) $\frac{2}{3}$ (b) $\frac{1}{2+\sqrt{3}}$
(c) $\frac{\sqrt{3}}{1}$ (d) $\frac{2}{1+\sqrt{3}}$

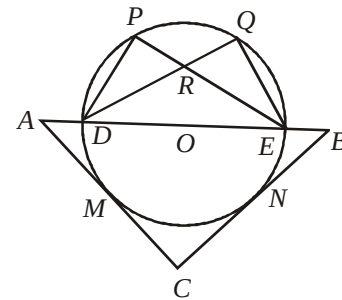
53. In the adjoining figure $ABCD$, $PQRS$ and $WXYZ$ are three squares. Find number of triangles and quadrilaterals in the figure:



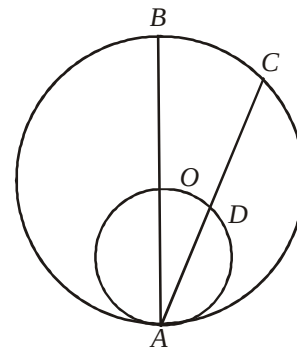
- (a) 24 and 16 (b) 28 and 15
(c) 27 and 16 (d) None of these
54. In the adjoining figure the diameter of the larger circle is 10 cm and the smaller circle touches internally the larger circle at P and passes through O , the centre of the larger circle. Chord SP cuts the smaller circle at R and OR is equal to 4 cm. What is the length of the chord SP ?



- (a) 9 cm (b) 12 cm
(c) 6 cm (d) $8\sqrt{2}$ cm
55. ABC is an isosceles triangle and AC , BC are the tangents at M and N respectively. DE is the diameter of the circle. $\angle ADP = \angle BEQ = 100^\circ$. What is the value of $\angle PRD$?

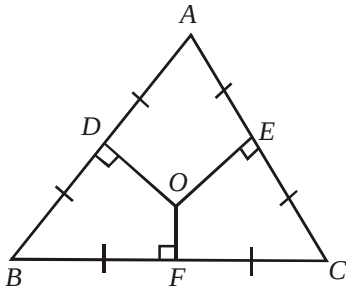


- (a) 60° (b) 50°
(c) 20° (d) can't be determined
56. What is the sum of all the angles of a 9 pointed star (i.e., $\angle 1 + \angle 2 + \angle 3 + \dots + \angle 8 + \angle 9$):
- (a) 909° (b) 900°
(c) 720° (d) 540°
57. A smaller circle touches internally to a larger circle at A and passes through the centre of the larger circle. O is the centre of the larger circle and BA , OA are of the diameters of the larger and smaller circles respectively. Chord AC intersects the smaller circle at a point D . If $AC = 12 \text{ cm}$, then AD is:

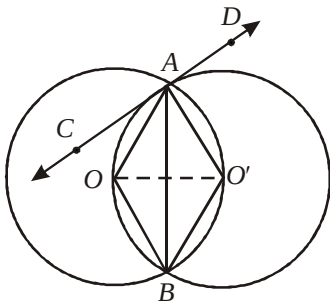


- (a) 4 cm (b) 6 cm
(c) 5.6 cm (d) Data insufficient

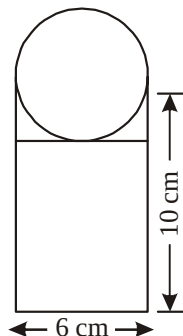
58. OD , OE and OF are perpendicular bisectors to the three sides of the triangle. What is the relationship between $m \angle BAC$ and $m \angle BOC$?



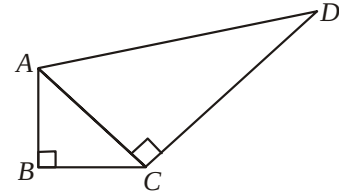
- (a) $m \angle BAC = 180^\circ - m \angle BOC$
 (b) $m \angle BOC = 90^\circ + \frac{1}{2} m \angle BAC$
 (c) $m \angle BAC = 90^\circ + \frac{1}{2} m \angle BOC$
 (d) $m \angle BOC = 2 m \angle BAC$
59. Two circles $C(O, r)$ and $C(O', r')$ intersect at two points A and B and O lies on $C(O', r')$. A tangent CD is drawn to the circle $C(O', r')$ at A . Then



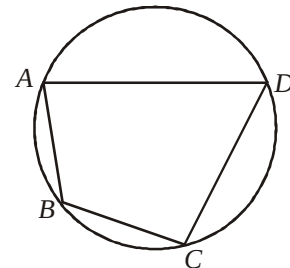
- (a) $\angle OAC = \angle OAB$ (b) $\angle OAB = \angle AO'O$
 (c) $\angle AO'B = \angle AOB$ (d) $\angle OAC = \angle AOB$
60. Find the perimeter of the given figure.



- (a) $(32 + 3\pi)$ cm (b) $(36 + 6\pi)$ cm
 (c) $(46 + 3\pi)$ cm (d) $(26 + 3\pi)$ cm
61. In the given figure, $\triangle ABC$ and $\triangle ACD$ are right angle triangles and $AB = x$ cm, $BC = y$ cm, $CD = z$ cm and $x \cdot y = z$ and x, y and z has minimum integral value. Find the area of $ABCD$
- (a) 36 cm^2 (b) 64 cm^2
 (c) 24 cm^2 (d) 25 cm^2



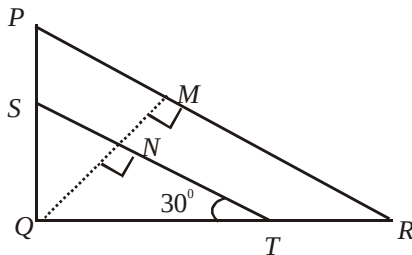
62. $\triangle ABC$ has sides AB, AC measuring 2001 and 1002 units respectively. How many such triangles are possible with all integral sides?
- (a) 2001 (b) 1002
 (c) 2003 (d) 1004
63. In the figure given, $ABCD$ is a cyclic quadrilateral and $AB = 25$ cm, $BC = 39$ cm, $CD = 52$ cm and $AD = 60$ cm. What is the diameter of the circle?



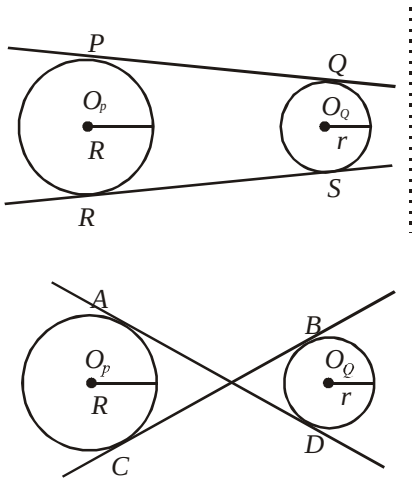
- (a) 60 cm (b) 65 cm
 (c) 72 cm (d) 78 cm
64. A polygon is said to be concave if at least one of the angles of the polygon is greater than 180° . A concave polygon is such that each of its internal angles measures either 60° or 300° . If this polygon contains twenty 300° angles, find the number of 60° angles in it.
- (a) 26 (b) 18
 (c) 23 (d) 24

Expert Level

1. In the adjoining figure $SQ = TR = a$, $QT = b$
 $OM \perp SR$, $ST \parallel PR$.
 $m \angle STQ = 30^\circ$.
 $m \angle SQT = 90^\circ$.
 Find QM



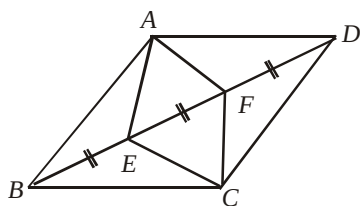
- (a) $\frac{a+b}{2}$ (b) $2(a+b)$
 (c) $\frac{2a+b}{2}$ (d) $\frac{a+2b}{2}$
2. For the figures shown below



$$O_p O_q = O_A O_B = 15.$$

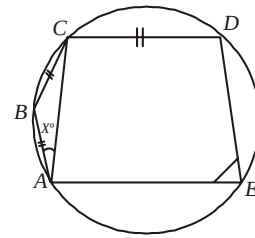
If $R = 9$ and $r = 2$, which one of the following relations is true?

- (a) $PQ = 13.3$ (b) $AD = 10.2$
 (c) $PQ > AD$ (d) All of these
3. E and F are the points of trisection of the diagonal BD of parallelogram ABCD of area 90 sq. cm. Find $A(\square AECF)$

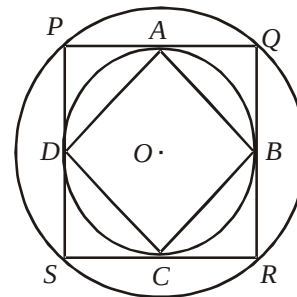


- (a) 45 sq. cm
 (b) 20 sq. cm
 (c) 15 sq. cm
 (d) 30 sq. cm

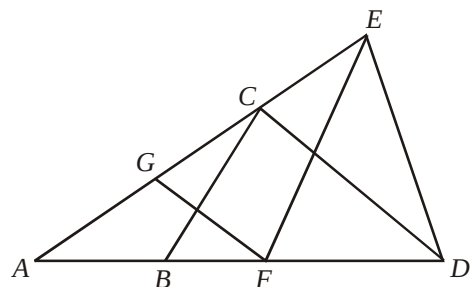
4. In the adjoining figure AB, BC, CD are equal chords of a circle. If $\angle BAC = x^\circ$, then the measure of $\angle AED$ is



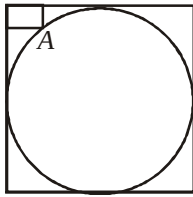
- (a) $2x^\circ$
 (b) $3x^\circ$
 (c) $180^\circ - 2x^\circ$
 (d) $180^\circ - 3x^\circ$
5. The figure below shows two concentric circles with centre O. PQRS is a square inscribed in the outer circle. It also circumscribes the inner circle, touching it at point B, C, D and A. What is the ratio of the perimeter of the outer circle to that of polygon ABCD?



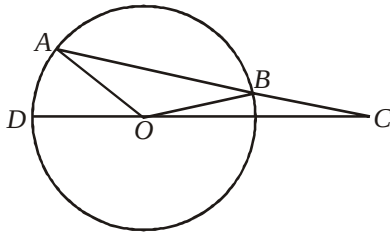
- (a) $\frac{\pi}{3}$ (b) π
 (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$
6. In the figure below, $AB = BC = CD = DE = EF = FG = GA$. Then, $\angle DAE$ is approximately



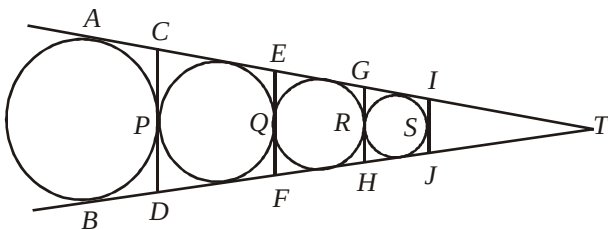
- (a) 15° (b) 30°
(c) 20° (d) 25°
7. If the perimeter of a triangle is 14 and the sides are integers, then how many different triangles are possible?
(a) 6 (b) 5
(c) 4 (d) 3
8. a , b and c are sides of a triangle. If $a^2 + b^2 + c^2 = ab + bc + ac$ then the triangle will be
(a) equilateral (b) isosceles
(c) right angled (d) obtuse angle
9. In the figure below, the rectangle at the corner measures $10 \text{ cm} \times 20 \text{ cm}$. The corner A of the rectangle is also a point on the circumference of the circle. What is the radius of the circle in cm?



- (a) 10 cm (b) 40 cm
(c) 50 cm (d) None of these
10. In the figure given below, AB is the chord of a circle with centre O . AB is extended to C such that $BC = OB$. The straight line CO is produced to meet the circle at D . If $\angle ACD = y$ degrees and $\angle AOD = x$ degrees such that $x = ky$, then the value of k is

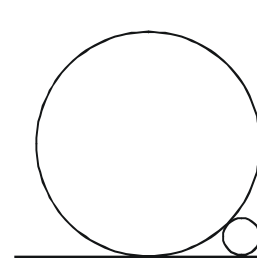


- (a) 3 (b) 2
(c) 1 (d) None of these
11. In the adjoining figure AT and BT are the two tangents at A and B respectively CD is also a tangent at P . There are some more circles touching each other and the tangents AT and BT also. Which one of the following is true?



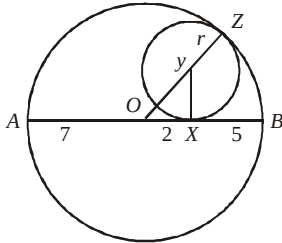
- (a) $PC + CT = PD + DT$
(b) $RG + GT = RH + HT$
(c) $PC + QE = CE$

- (d) All of these
12. In an isosceles right angled triangle ABC , $\angle B$ is right angle. Angle bisector of $\angle BAC$ is AN cut at M to the median BO . Point ' O ' lies on the hypotenuse, OM is 20 cm, then the value of AB is:
(a) 38.96 cm (b) 24.18 cm
(c) 34.134 cm (d) None of these
13. The biggest possible regular hexagon H is cut out of an equilateral triangle X . The biggest possible equilateral triangle Y is cut out from the hexagon H . What is the ratio of the areas of the equilateral triangles X and Y ?
(a) 5 : 1 (b) 6 : 1
(c) 8 : 1 (d) 3 : 1
14. In any quadrilateral $ABCD$, the diagonal AC and BD intersect at a point X . If E , F , G and H are the mid-points of AX , BX , CX and DX respectively, then what is the ratio of $(EF + FG + GH + GE)$ to $(AD + DC + CB + BA)$?
(a) $\frac{1}{2}$ (b) $\frac{3}{2}$
(c) $\frac{3}{4}$ (d) Data insufficient
15. A square, whose side is 2 meters, has its corners cut away so as to form an octagon with all sides equal. Then, the length of the each side of the octagon, in meters is:
(a) $\frac{\sqrt{2}}{\sqrt{2}+1}$ (b) $\frac{2}{\sqrt{2}-1}$
(c) $\frac{2}{\sqrt{2}-1}$ (d) $\frac{2}{\sqrt{2}+1}$
16. Let C_1 and C_2 be the inscribed and circumscribed circles of a triangle with sides 3 cm, 4 cm and 5 cm. The $\frac{\text{area of } C_1}{\text{area of } C_2}$ equals.
(a) $\frac{16}{25}$ (b) $\frac{4}{25}$
(c) $\frac{9}{25}$ (d) $\frac{9}{16}$
17. A circle with radius 2 is placed against a right angle. Another smaller circle is also placed as shown in the adjoining figure. What is the radius of the smaller circle?

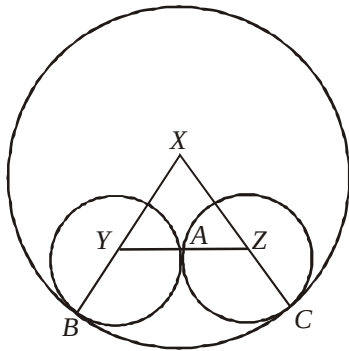


- (a) $3 - 2\sqrt{2}$ (b) $4 - 2\sqrt{2}$
(c) $7 - 4\sqrt{2}$ (d) $6 - 4\sqrt{2}$

18. In the figure below, X is a point on diameter AB of the circle with centre O , such that $AX = 9$ cm, $XB = 5$ cm. Find the radius of the circle (centre O) which touches the diameter at X and touches the circle, centre O , internally at Z .

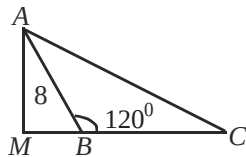


- (a) $3\frac{3}{14}$ cm (b) $3\frac{1}{14}$ cm
(c) $2\frac{1}{14}$ cm (d) $2\frac{3}{14}$ cm
19. In the adjoining figure, 2 circles with centres Y and Z touch each other externally at point A .



Another circle, with centre X , touches the other 2 circles internally at B and C . If $XY = 6$ cm, $YZ = 9$ cm and $ZX = 7$ cm, then find the radius of the circle with centre y .

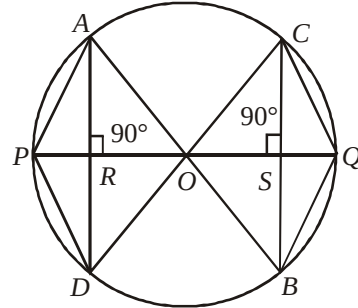
- (a) 13 (b) 5
(c) 17 (d) 8
20. In the figure, $AB = 8$, $BC = 7$, $m\angle ABC = 120^\circ$. Find AC .



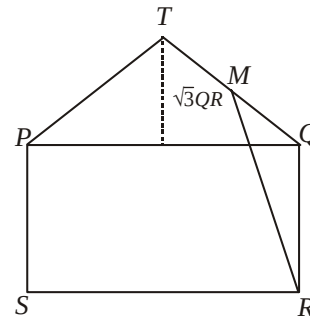
- (a) 11 (b) 12
(c) 13 (d) 14
21. An equilateral triangle BPC is drawn inside a square $ABCD$. What is the value of the angle APD in degrees?
- (a) 75 (b) 90
(c) 120 (d) 150
22. Consider obtuse-angled triangles with sides 8 cm, 15 cm and x cm. If x is an integer, then how many such triangles exist?

- (a) 5 (b) 21
(c) 10 (d) 15

23. In the adjoining figure O is the centre of the circle. $\angle AOD = 120^\circ$. If the radius of the circle be ' r ', then find the sum of the areas of quadrilaterals $AODP$ and $OBQC$:



- (a) $\frac{\sqrt{3}}{2}r^2$ (b) $3\sqrt{3}r^2$
(c) $\sqrt{3}r^2$ (d) None of these
24. In the above question (number 48) what is the ratio of $CE : BE$:
- (a) $\sqrt{29} : (12 - \sqrt{29})$ (b) $12 : \sqrt{29}$
(c) $7 : \sqrt{21}$ (d) None of these
25. $PQRS$ is a rectangle with $PQ = 2QR$. T is a point such that the shortest distance from PQ is $\sqrt{3}$ time QR . Let M be the mid point of QT . The measure of the angle QMR is

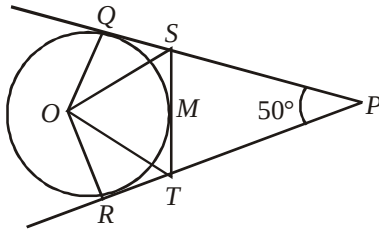


- (a) 90° (b) 60°
(c) 30° (d) 15°
26. The radius of a circle with centre O is $\sqrt{50}$ cm. A and C two points on the circle, and B is a point inside the circle. The length of AB is 6 cm, and the length of BC is 2 cm. The angle ABC is a right angle. Find the square of the distance OB .
- (a) 26 (b) 25
(c) 24 (d) 23
27. In a right angled triangle $\angle B$ and $\angle A$ are acute angles. If $\angle A - \angle B = k$, where A and B are integers, then how many integer values can ' k ' take?
- (a) 80 (b) 88
(c) 45 (d) 89

28. In a right angled triangle ABC , $\angle B$ is right angle, side AB is half of the hypotenuse. AE is parallel to median BD and CE is parallel to BA . What is the ratio of length of BC to that of EC ?

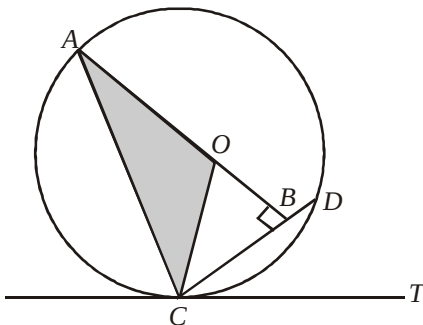
(a) $\sqrt{2}:1$ (b) $\sqrt{3}:2$
(c) $\sqrt{5}:\sqrt{3}$ (d) can't be determined

29. In the adjoining figure 'O' is the centre of the circle and PQ , PR and ST are the three tangents. $\angle QPR = 50^\circ$, then the value of $\angle SOT$ is:



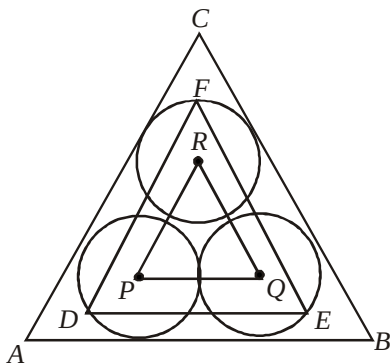
(a) 30° (b) 75°
(c) 65° (d) can't be determined

30. In the given diagram CT is tangent at C , making an angle of $\frac{\pi}{4}$ with CD . O is the centre of the circle. $CD = 10$ cm. What is the perimeter of the shaded region (ΔAOC) approximately?



(a) 27 cm (b) 30 cm
(c) 25 cm (d) 31 cm

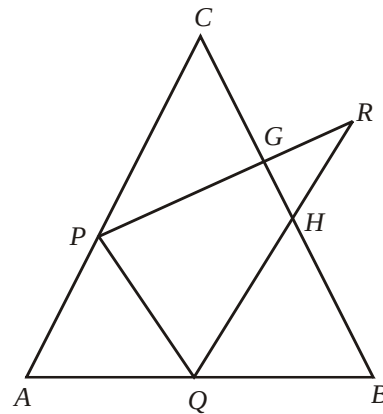
31. In the adjoining figure three congruent circles are touching each other. Triangle ABC circumscribes all the three circles. Triangle PQR is formed by joining the centres of the circle. There is a third triangle DEF . Points A, D, P and B, E, Q and C, F, R lie in the same straight line respectively.



What is the ratio of perimeters of $\Delta ABC : \Delta DEF : \Delta PQR$

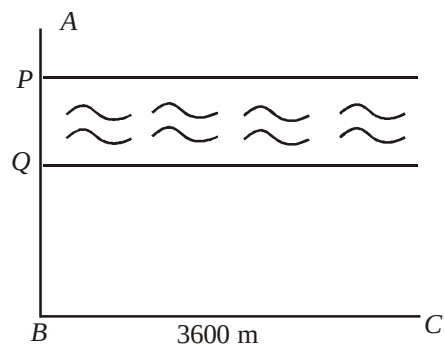
(a) $3\sqrt{2}:2\sqrt{2}:1$
(b) $2(4+\sqrt{3}):(2+\sqrt{3}):3$
(c) $2(1+\sqrt{3}):(2+\sqrt{3}):2$
(d) $2(1+\sqrt{3}):2\sqrt{3}:\sqrt{3}$

32. In the adjoining figure, P and Q are the mid-points of AC and AB . Also, $PG = GR$ and $HQ = HR$. What is the ratio of area of ΔPQR : area of ΔABC ?



(a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) $\frac{3}{5}$ (d) None of these

33. In the given diagram, river PQ is just perpendicular to the national highway AB . At a point B highway just turns to right angle and reaches to C . $PA = 500$ m and $BQ = 700$ m and width of the uniformly wide river (i.e., PQ) is 300 m. Also $BC = 3600$ m. A bridge has to be constructed across the river perpendicular to its stream in such a way that a person can reach from A to C via bridge covering least possible distance. PQ is the widthness of the river, then what is the minimum possible required distance from A to C including the length of bridge?



(a) 4100 m (b) 3900 m
(c) $3000\sqrt{2}$ m (d) None of these

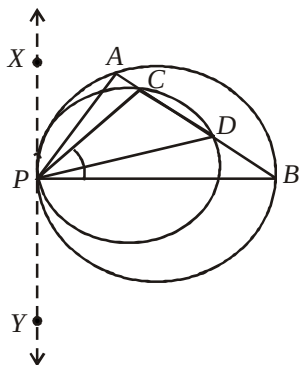
34. Let S be an arbitrary point on the side PQ of an acute angle ΔPQR . Let T be the point of intersection of QR extended with the straight line PT drawn parallel to SR through P . Let U be the point of intersection of PR extended with the straight line QU drawn parallel to SR through Q . If $PT = a$ and $QU = b$, then the length of SR is

- (a) $\frac{a+b}{ab}$ (b) $\frac{a-b}{ab}$
(c) $\frac{ab}{a+b}$ (d) $\frac{ab}{a-b}$

35. If a sphere of the maximum volume is placed inside a hollow right circular cone with radius ' r ' and slant height ' l ' such that the base of the cone touches the sphere, then the volume of the sphere is

- (a) $\frac{4}{3}\pi\left(\frac{l+r}{l-r}\right)^3$ (b) $\frac{4}{3}\pi r^3\left(\frac{l-r}{l+r}\right)^{\frac{3}{2}}$
(c) $\frac{4}{3}\pi\left(\frac{l-r}{l+r}\right)^3$ (d) $\frac{4}{3}\pi r^3\left(\frac{l+r}{l-r}\right)^{\frac{3}{2}}$

36. Two circles touch internally at point P and a chord AB of the circle of larger radius intersects the other circle in C and D . Which of the following holds good?



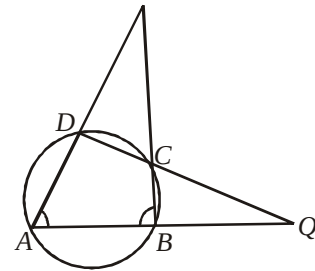
- (a) $\angle CPA = \angle DPB$
(b) $2\angle CPA = \angle CPD$
(c) $\angle APX = \angle ADP$
(d) $\angle BPY = \angle CPD + \angle CPA$

37. All of the following is true except:

- (a) The points of intersection of direct common tangents and indirect common tangents of two circles divide the line segment joining the two centres respectively externally and internally in the ratio of their radii
(b) In a cyclic quadrilateral $ABCD$, if the diagonal CA bisects the angle C , then diagonal BD is parallel to the tangent at A to the circle through A, B, C, D
(c) If TA, TB are tangent segments to a circle $C(O, r)$ from an external point T and OT intersects the circle in P , then AP bisects the angle TAB .
(d) If in a right triangle ABC , BD is the perpendicular on the hypotenuse AC , then

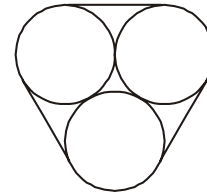
- (i) $AC \cdot AD = AB^2$ and
(ii) $AC \cdot AD = BC^2$

38. In the adjoining figure $\angle A = 60^\circ$ and $\angle ABC = 80^\circ$, $\angle BQC$



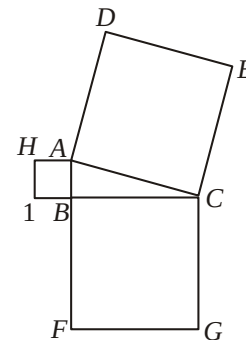
- (a) 40° (b) 80°
(c) 20° (d) 30°

39. The diagram below represents three circular garbage cans, each of diameter 2 m. The three cans are touching as shown. Find, in metres, the perimeter of the rope encompassing the three cans.



- (a) $2\pi + 6$ (b) $3\pi + 4$
(c) $4\pi + 6$ (d) $6\pi + 6$

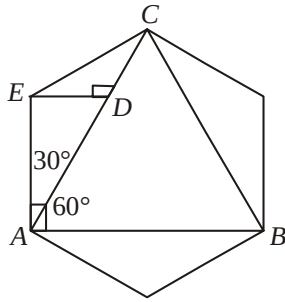
40. All the three quadrilaterals $ADEC$, $ABIH$ and $BCGF$ are squares and $\angle ABC = 90^\circ$. If the area of $ADEC = x^2$ and the area of $AHIB = y^2$ ($x^2 > y^2$), then the area of $BCGF$ is:



- (a) $(x+y)(x-y)$ (b) $(x+y)^2$
(c) $(x-y)^2$ (d) None of these

41. In the figure below, if the perimeter of ΔABC is p , then the perimeter of the regular hexagon is

- (a) $\frac{3p}{\sqrt{2}}$ (b) $\frac{\sqrt{2}p}{3}$
(c) $\frac{\sqrt{3}p}{2}$ (d) $\frac{2p}{\sqrt{3}}$



42. Euclid has a triangle in mind. Its longest side has length 20 and another of its side has length 10. Its area is 80. What is the exact length of its third side?

(a) $\sqrt{260}$ (b) $\sqrt{250}$
(c) $\sqrt{240}$ (d) $\sqrt{270}$

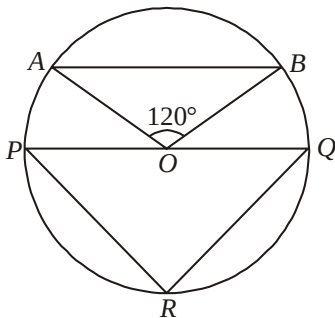
43. Rajat cut out two identical triangular pieces of cardboard, each of area 300 sq. cm, and then placed them upon a table, one on top of the other such that the triangles completely coincide with each other. Now, if he rotated one of the two triangles by 180° about a vertical axis passing through its centroid, find the area that is common to both the triangles.

(a) 100 cm^2 (b) 150 cm^2
(c) 200 cm^2 (d) $133 \frac{1}{3} \text{ cm}^2$

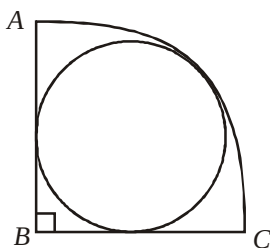
44. PQ is the diameter of a circle with centre O . Point R on the circumference of the circle is equidistant from P and Q . A and B are two points in the opposite segment such that (arc AB) is $\frac{1}{3}$ times the circumference; also $AP = BQ$. The ratio of area of ΔPRQ to ΔAOB is:

(a) $4/\sqrt{3}$ (b) $\sqrt{3}/4$
(c) $\sqrt{3}/2$ (d) $2/\sqrt{3}$

The following diagram could be drawn: $\angle AOB = 120^\circ$. $\angle PRQ = 90^\circ$, $PQ = 2r$, where 'r' is the radius of the circle.

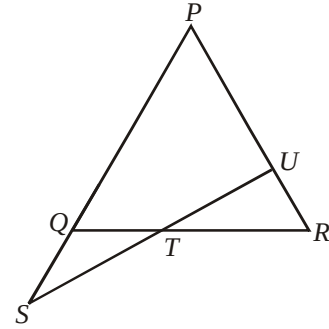


45. If ABC is a quarter circle and a circle is inscribed in it and if $AB = 1 \text{ cm}$, find radius of smaller circle.



(a) $\sqrt{2} - 1$ (b) $(\sqrt{2} + 1)/2$
(c) $\sqrt{2} - 1/2$ (d) $1 - 2\sqrt{2}$

46. Through T , the mid-point of the side QR of a ΔPQR , a straight line is drawn to meet PQ produced to S and PR at U , so that $PU = PS$. If length of $UR = 2$ units then the length of QS is:



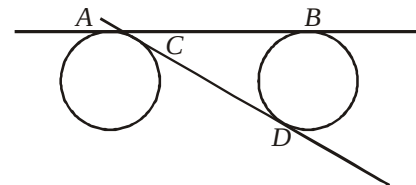
(a) $2\sqrt{2}$ Units (b) $\sqrt{2}$ Units
(c) 2 Units (d) Cannot be determined

47. Two chords of lengths a and b of a circle subtend 60° and 90° angles at the centre respectively. Which of the following is correct?

(a) $b = \sqrt{2}a$ (b) $b = \sqrt{2}b$
(c) $a = 2b$ (d) $b = 2a$

48. If two equal circles of radius 5 cm have two common tangent AB and CD which touch the circle on A, C and B, D respectively and if $CD = 24 \text{ cm}$, find the length of AB .

(a) 27 cm (b) 25 cm
(c) 26 cm (d) 30 cm



Direction for Questions 49 and 50: Read the passage below and solve the question based on it.

There are three equal circles of unit radii touching each other.

49. Find the area of the triangle circumscribing the three circles.

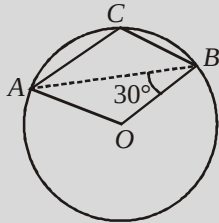
(a) $\sqrt{3}(\sqrt{3} + 1)^2$ (b) $(\sqrt{3} + 1)^2$
(c) $\sqrt{3}(\sqrt{3} + 1)$ (d) None of these

50. What would be the area of the remaining portion if the same three circles are circumscribed by another circle?

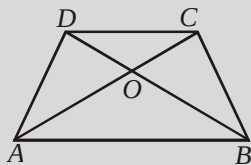
(a) $\pi \left(\frac{2}{\sqrt{3}} + 1 \right)^2$ (b) $\pi \left(\frac{1}{\sqrt{3}} + 1 \right)^2 - 3\pi$
(c) $\left(\frac{3}{\sqrt{2}} + 1 \right)^2 \pi - 3\pi$ (d) $\left(\frac{2}{\sqrt{3}} + 1 \right)^2 \pi - 3\pi$

Test Yourself

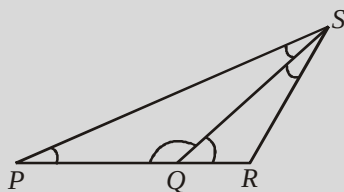
1. In the following figure, O is the centre of the circle and $\angle ABO = 30^\circ$, find $\angle ACB$.



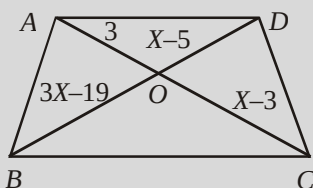
- (a) 60° (b) 120°
(c) 75° (d) 90°
2. In the adjoining figure, $ABCD$ is a trapezium in which $AB \parallel DC$ and $AB = 3 DC$. Determine the ratio of the areas of $(\triangle AOB$ and $\triangle COD)$.



- (a) $9 : 1$ (b) $1 : 9$
(c) $3 : 1$ (d) $1 : 3$
3. In the figure below, $PQ = QS$, $QR = RS$ and angle $SRQ = 100^\circ$. How many degrees is angle QPS ?

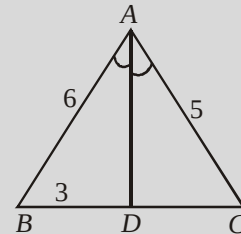


- (a) 20° (b) 40°
(c) 15° (d) 30°
4. $ABCD$ is a cyclic quadrilateral in which $BC \parallel AD$, $\angle ADC = 110^\circ$ and $\angle BAC = 50^\circ$ find $\angle DAC$
- (a) 60° (b) 45°
(c) 90° (d) 120°
5. In the given figure, $AD \parallel BC$. Find the value of x .



- (a) $x = 8, 9$ (b) $x = 7, 8$
(c) $x = 8, 10$ (d) $x = 7, 10$

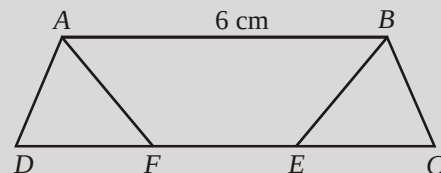
6. In the given figure, AD is the bisector of $\angle BAC$, $AB = 6$ cm, $AC = 5$ cm and $BD = 3$ cm. Find DC .



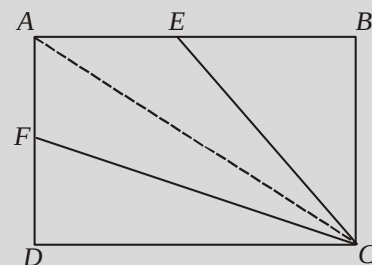
- (a) 11.3 cm (b) 2.5 cm
(c) 3.5 cm (d) 4 cm
7. $ABCD$ is a square, F is the mid-point of AB and E is a point on BC such that BE is one-third of BC . If area of $\triangle FBE = 108 \text{ m}^2$, then the length of AC is :

- (a) 63 m (b) $36\sqrt{2}$ m
(c) $63\sqrt{2}$ m (d) $72\sqrt{2}$ m

8. $ABCD$ is a trapezium in which AB is parallel to DC , $AD = BC$, $AB = 6$ cm, $AB = EF$ and $DF = EC$. If two lines AF and BE are drawn so that area of $ABEF$ is half of $ABCD$. Find DF/CD .



- (a) $1/4$ (b) $1/3$
(c) $2/5$ (d) $1/6$
9. In the given figure given below, E is the mid-point of AB and F is the midpoint of AD . if the area of $FAEC$ is 13, what is the area of $ABCD$?



- (a) 19.5 (b) 26
(c) 39 (d) None of these

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- A circle with points A , B , C , and D on its circumference. Chords AC and BD intersect at point E . Angle CBE is labeled as 35° . Angle ADE is labeled with a question mark.

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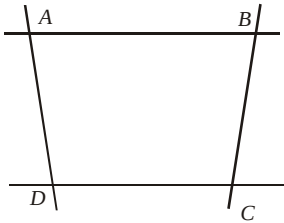
Hints & Solutions

Foundation Level

1. (c) In a right angled Δ , the length of the median is $\frac{1}{2}$ the length of the hypotenuse. Hence $BD = \frac{1}{2} AC = 3$ cm.

2. (b) In ΔABC , $\angle C = 180 - 90 - 30 = 60^\circ$
 $\therefore \angle DCE = \frac{60}{2} = 30^\circ$

3. (d) Again in ΔDEC , $\angle CED = 180 - 90 - 30 = 60^\circ$
 The quadrilateral obtained will always be a trapezium as it has two lines which are always parallel to each other.



4. (a) $AD = 24$, $BC = 12$

In ΔBCE & ΔADE

since $\angle CBA = \angle CDA$ (Angles by same arc)

$\angle BCE = \angle DAE$ (Angles by same arc)

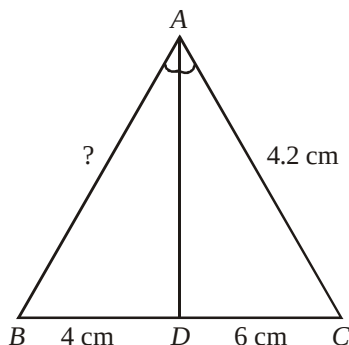
$\angle BEC = \angle DEA$ (Opp. angles)

$\therefore \angle BCE$ & $\angle DAE$ are similar Δ s

with sides in the ratio 1 : 2

Ratio of area = 1:4 (i.e. square of sides)

5. (a)



$\Delta ABD \sim \Delta ACD$

$$\frac{AC}{DC} = \frac{AB}{BD} \Rightarrow \frac{4.2}{6} = \frac{AB}{4}$$

$$\therefore AB = 2.8 \text{ cm}$$

6. (c) Let n be the number of sides of the polygon
 Now, sum of interior angles = $8 \times$ sum of exterior angles

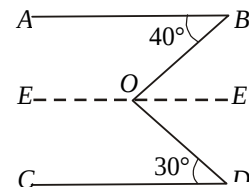
$$\text{i.e. } (2n - 4) \times \frac{\pi}{2} = 8 \times 2\pi$$

$$\text{or } (2n - 4) = 32$$

$$\text{or } n = 18$$

7. (a) 2.4 cm
 8. (a) $\angle EDC = \angle BAD = 45^\circ$ (alternate angles)
 $\therefore x = \angle DEC = 180^\circ - (50^\circ + 45^\circ) = 85^\circ$
 9. (a) $a + 36^\circ + 70^\circ = 180^\circ$ (sum of angles of triangle)
 $\Rightarrow a = 180^\circ - 36^\circ - 70^\circ = 74^\circ$
 $b = 36^\circ + 70^\circ$ (Ext. angle of triangle) = 106°
 $c = a - 50^\circ$ (Ext. angle of triangle) = $74^\circ - 50^\circ = 24^\circ$
 10. (c) $b = \frac{1}{2} (48^\circ)$
 ($\angle$ at centre = 2 at circumference on same PQ) 24°
 $\angle AQB = 90^\circ$ ($\angle$ in semi-circle)
 $\angle QXB = 180^\circ - 90^\circ - 24^\circ$ ($\angle$ sum of Δ) = 66°
 11. (d) $\angle MBA = 180^\circ - 95^\circ = 85^\circ$
 $\angle AMB = \angle TMN$... (Same angles with different names)
 $\therefore \Delta MBA \sim \Delta MNT$ (AA test for similarity)
 $\frac{MB}{MN} = \frac{AB}{NT}$ (proportional sides)
 $\frac{10}{MN} = \frac{5}{9} \therefore MN = \frac{90}{5} = 18$.

12. (b) Through O draw EOE' parallel to AB & so to CD .

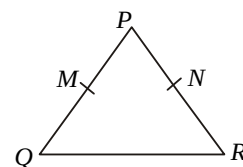


$$\therefore \angle BOE' = \angle ABO = 40^\circ \text{ (alternate angles)}$$

$$\angle E'OD = \angle CDO = 30^\circ \text{ (alternate angles)}$$

$$\therefore \angle BOD = (40^\circ + 30^\circ) = 70^\circ. \text{ So, } x = 70.$$

13. (c) The triangle PQR is isosceles
 $\Rightarrow MN \parallel QR$ by converse of Proportionality Theorem.



- (b) Again by Converse of Proportionality theorem,
 $MN \parallel QR$.

14. (a) $a + 36^\circ + 70^\circ = 180^\circ$ (sum of angles of triangle)
 $\Rightarrow a = 180^\circ - 36^\circ - 70^\circ = 74^\circ$
 $b = 36^\circ + 70^\circ$ (Ext. angle of triangle) = 106°
 $c = a - 50^\circ$ (Ext. angle of triangle) = $74^\circ - 50^\circ = 24^\circ$.

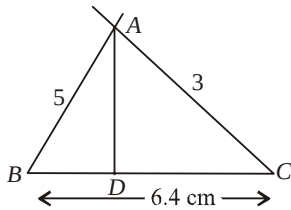
15. (c) Perimeter of $\triangle ABC = 36$ cm.
Perimeter of $\triangle PQR = 24$ cm and $PQ = 10$ cm.
We have to find AB . Perimeter of $\triangle ABC = AB + BC + AC$.
Perimeter of $\triangle PQR = PQ + QR + PR$. Since $\triangle ABC \sim \triangle PQR$.

$$\therefore \frac{AB}{PQ} = \frac{BC}{QR} = \frac{AC}{PR} = \frac{AB+BC+AC}{PQ+QR+PR} = \frac{36}{24}$$

$$\Rightarrow \frac{AB}{PQ} = \frac{36}{24} \Rightarrow AB = \frac{36}{24} \times PQ = \frac{36}{24} \times 10 = 15 \text{ cm.}$$

16. (d) AD is the bisector of $\angle A$.

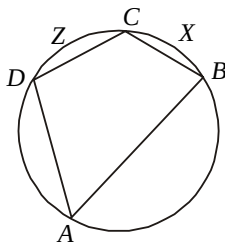
$$\therefore \frac{AB}{AC} = \frac{BD}{DC} = \frac{5}{3}$$



$$\Rightarrow \frac{DC}{BD} = \frac{3}{5} \Rightarrow \frac{DC+BD}{BD} = \frac{3+5}{5}$$

$$\Rightarrow \frac{BC}{BD} = \frac{8}{5} \Rightarrow BD = BC \times \frac{5}{8} = 6.4 \times \frac{5}{8} = 4$$

17. (b) $m\angle ACD = m\angle DEC$
 $\therefore m\angle DEC = x = 40^\circ$
 $\therefore m\angle ECB = m\angle EDC$
 $\therefore m\angle ECB = y = 54^\circ$
 $54^\circ + x + z = 180^\circ \dots (\text{sum of all the angles of a triangle})$
 $54^\circ + 40^\circ + z = 180^\circ$
 $\therefore z = 86^\circ$
18. (b) In $\triangle BCD$, $BC = CD$, $\angle BDC = \angle CBD = x$
In cyclic quadrilateral $ABCD$, $\angle ABC + \angle ADC = 180^\circ$
 $40^\circ + x + 90^\circ + x = 180^\circ \Rightarrow x = 25^\circ$.
19. (c) $m\angle DAB = 180^\circ - 120^\circ = 60^\circ \dots (\text{opposite angles of a cyclic quadrilateral})$
 $m(\text{arc } BCD) = 2m\angle DAB = 120^\circ$.



$$\therefore m(\text{arc } CXB) = m(\text{arc } BCD) - m(\text{arc } DZC)$$

$$= 120^\circ - 70^\circ = 50^\circ.$$

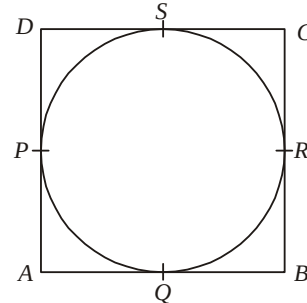
20. (d)

21. (d) $\frac{OP}{PT} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow PT = \sqrt{3} \quad OP = 3\sqrt{3} \text{ cm.}$

22. (c) $\angle OPQ = \angle OQP = 30^\circ$, i.e., $\angle POQ = 120^\circ$.
Also,

$$\angle PRQ = \frac{1}{2} \text{ reflex } \angle POQ$$

23. (b) Since $ABCD$ is a quadrilateral
Again AP, AQ are tangents to the circle from the point A .



$$\therefore AP = AQ$$

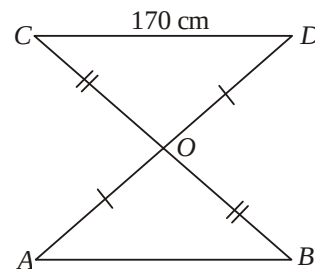
Similarly $BR = BQ$
 $CR = CS$
 $DP = DS$
 $\therefore (AP + DP) + (BR + CR) = AQ + DS + BQ + CS$
 $= (AQ + BQ) + (CS + DS)$
 $\Rightarrow AD + BC = AB + CD$

24. (d) $\angle LCD = \angle ALC = 60^\circ$ (alternate angles)

$$\angle DCE = \frac{1}{2} \angle LCD = 30^\circ. (EC \text{ is the angle bisector})$$

$$\therefore \angle FEC = (180^\circ - 30^\circ) = 150^\circ.$$

25. (b) We have area of triangle $AFE = A/4$. (If A = Area of triangle ABC) and area of triangle $DHI = (A/4)/4 = A/16$. Hence, ratio = 1 : 4.
26. (b) In $\triangle AOB$ and $\triangle COD$



$$AO = OD, BO = OC$$

$$\angle AOB = \angle COD \text{ (vertically opposite angles)}$$

$$\therefore \triangle AOB \cong \triangle COD$$

$$\therefore AB = CD = 170 \text{ cm.}$$

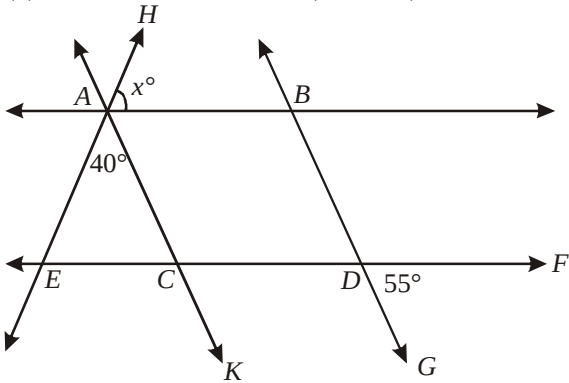
27. (d) $c = c_1$ (Vert. opp. $\angle$ s). $b = c + s$ (Ext. $\angle$).
 $d = c_1 + r$ (Ext. $\angle$)
But $b + d = 180^\circ$ (Opp. $\angle$ s, cyclic quad.)
 $\Rightarrow c + s + c_1 + r = 180^\circ$
 $\Rightarrow r + s + 2c = 180^\circ \Rightarrow r + s = 180^\circ - 2c$.

28. (b)

29. (b) $m\angle PAC = m\angle PBC = 90^\circ$
.....(Tangent perpendicularity theorem)
 $m\angle PAC + m\angle PBC + m\angle ACB = 360^\circ$
 $\therefore m\angle APB = 360 - (90 + 90 + 65) = 115^\circ$
 $\therefore m(\angle AXB) = 115^\circ$.

30. (d) Basic concept

31. (a) $\angle DCK = \angle FDG = 55^\circ$ (corr. $\angle$ s)



$$\therefore \angle ACE = 180^\circ - (\angle EAC + \angle ACE)$$

$$\therefore \angle HAB = \angle AEC = 85^\circ \text{ (corr. } \angle \text{s)}$$

Hence, $x = 85^\circ$

32. (c) Clearly option (a) shows the angles would be 30, 60 and 90. It can be the ratio of angle in a right angled triangle.

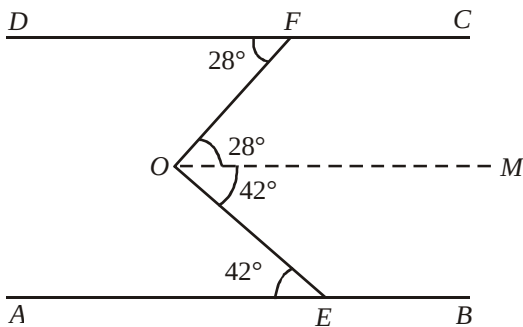
Option (b) shows the angles would be 45, 45 and 90, then it can be the ratio of angle in a right angled triangle. But option (c) cannot form the ratio of angles of right angled triangle.

33. (b) In $\triangle ABC$, $\angle C = 180 - 90 - 30 = 60^\circ$

$$\therefore \angle DCE = \frac{60}{2} = 30^\circ$$

Again in $\triangle DEC$, $\angle CED = 180 - 90 - 30 = 60^\circ$

34. (c) $\angle DFO = \angle FOM$
and $\angle AEO = \angle EOM$ (since $CD \parallel AB$)



$$\therefore \angle FOE = (28^\circ + 42^\circ) = 70^\circ$$

35. (b) Go through option for quicker answer

$$\text{Exterior angle} = \frac{360}{15} = 24^\circ \text{ (for } n = 15)$$

$$\therefore \text{Interior angle} = 180^\circ - 24^\circ = 156^\circ$$

$$\therefore \text{Interior} - \text{Exterior} = 156 - 24 = 132^\circ$$

Hence, option (b) is correct.

36. (c) $\angle ABC = 180 - (65 + 75) = 40^\circ$

$$\angle ORB = \angle OQB = 90^\circ$$

$$\therefore \angle ROQ = 360 - (90 + 90 - 40)$$

$$\therefore \angle ROQ = 140^\circ$$

37. (c) $\triangle ABC$ is similar to $\triangle EDC$

$$\therefore \frac{AB}{ED} = \frac{BC}{DC} = \frac{AC}{EC}$$

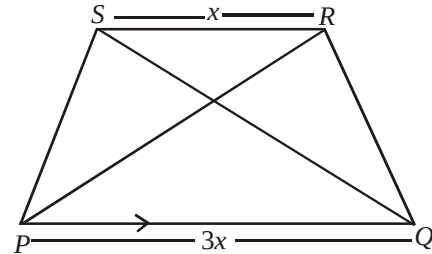
$$\therefore \frac{AB}{DE} = \frac{BC}{DC} \Rightarrow \frac{24}{10} = \frac{60}{DC}$$

$$\Rightarrow DC = 25 \text{ cm}$$

38. (d) Clearly, triangle is obtuse, So (d) is the correct option.

39. (a) No such point is possible

40. (c)



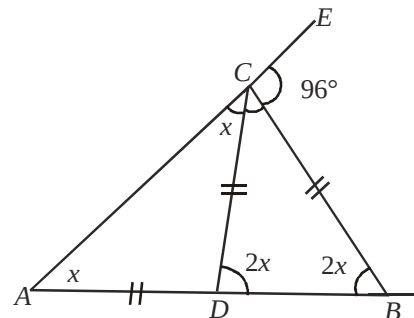
$$\frac{ar(\triangle P \times Q)}{(\triangle R \times S)} = \frac{PQ^2}{RS^2} = \frac{(3x)^2}{x^2} = 9:1$$

41. (a) The parallelogram $ABCD$ and $\triangle BCE$ lies between the same parallel lines AB and DE and has base of equal

$$\text{length. } \therefore A(\triangle BCE) = \frac{1}{2} A(\square ABCD) = \frac{1}{2} \times 16 = 8 \text{ sq. cm.}$$

42. (a) Form the figure given in the question, we get $x^2 - y^2 = 81$, $x^2 + y^2 = 625$ and $y^2 + 256 = z^2$
Form the option the only triplet satisfying the three equations is 15, 12, 20

43. (c)



$$\text{Let } \angle CAD = \angle ACD = x$$

At point C,

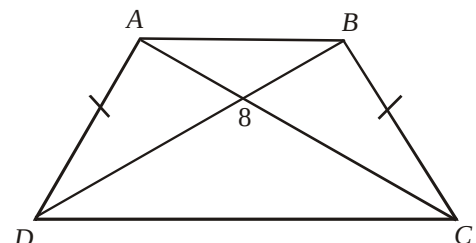
$$x + (180^\circ - 4x) + 96^\circ = 180^\circ$$

$$\Rightarrow 180^\circ - 3x + 96^\circ = 180^\circ$$

$$\therefore x = 32^\circ$$

$$\text{Hence, } \angle DBC = 2 \times 32 = 64^\circ$$

44. (b)



$$\triangle APD \sim \triangle BPC$$

$$\therefore \frac{PA}{PB} = \frac{PD}{PC}$$

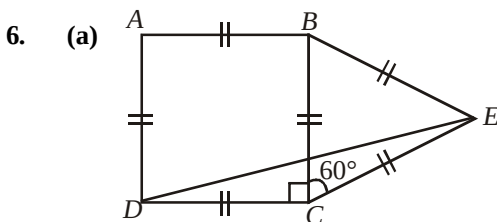
$$\text{i.e., } PA \cdot PC = PB \cdot PD.$$

$$\therefore \text{option (b)}$$

45. (a) $\angle CAF = 100^\circ$. Hence $\angle BAC = 80^\circ$
 Also, $\angle OCA = (90 - \angle ACF) = 90 - 50 = 40^\circ = \angle OAC$
 (Since the triangle OCA is isosceles)
 Hence $\angle OAB = 40^\circ$
 In isosceles $\triangle OAB$, $\angle OBA$ will also be 40°
 Hence, $\angle BOA = 180 - 40 - 40 = 100^\circ$

Standard Level

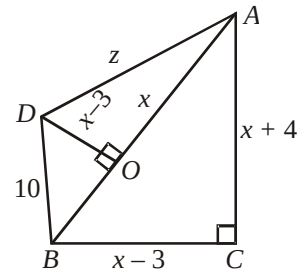
1. (c) Should be XY since you divide XY into millions of congruent portions, each portion which is the diameter of the semicircle is very small. So the sum of all the arcs should be XY.
2. (d) In Triangles ACP and BDP; $a = a_1$ ($\angle$ in same seg.); $p = p$ (common)
 $\angle ACP = \angle BDP$ (3rd $\angle$ of triangle)
 $\Rightarrow$ Triangle ACP $\sim$ Triangle BDP (A.A.A.)
 $BD/BP = AC/AP$ (corr. sides of $\sim$ triangles)
 $\Rightarrow BD/8 = 6/10 \Rightarrow BD = 4.8$ cm
 $PD/BP = PC/AP$ (Corr. sides) $PD/8 = 5/10$
 $\Rightarrow PD = 40/10 = 4$ cm.
3. (a) Let $AY = AY = a$
 $BY = BZ = b$
 $CZ = CW = c$ (tangents from ext. pt.)
 $DW = DX = d$
 $AD + BC = a + d + b + c = a + b + d + c = p + q$.
4. (c) $b = \frac{1}{2} (48^\circ)$
 ($\angle$ at centre = 2 at circumference on same PQ) 24°
 $\angle AQB = 90^\circ$ ($\angle$ in semi-circle)
 $\angle QXB = 180^\circ - 90^\circ - 24^\circ$ ($\angle$ sum of Δ) = 66°
5. (b) PA, AB, RC and SD are perpendicular to AD. Hence they are parallel. So, the intercepts are proportional.
 $\therefore \frac{AB}{BD} = \frac{PQ}{QS} \therefore \frac{60}{210} = \frac{x}{360 - x}$
 $\therefore \frac{2}{7} = \frac{x}{360 - x} \therefore x = \frac{720}{9} = 80$
 $\therefore PQ = 80 \therefore QS = 360 - 80 = 280$
 Again, $\frac{BC}{CD} = \frac{QR}{RS} \therefore \frac{90}{120} = \frac{y}{280 - y}$
 $\therefore \frac{3}{4} = \frac{y}{280 - y} \therefore 7y = 280 \times 3 \therefore y = 120$
 $\therefore QR = 120$
 $\therefore SR = 280 - 120 = 160$
Another method: $60 : 90 : 120 = 2 : 3 : 4$
 $\therefore$ Divide 360 in the ratio 2 : 3 : 4
 $\Rightarrow PQ = 80, QR = 120$ and $RS = 160$



In $\triangle DEC$, $\angle DCE = 90 + 60 = 150^\circ$

$$\angle CDE = \angle DEC = \frac{180 - 150}{2} = 15^\circ$$

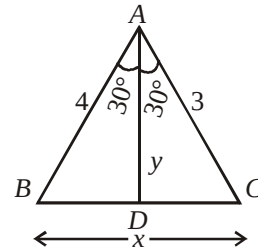
7. (b)



$$AB^2 = (x + 4)^2 + (x - 3)^2 = 2x^2 + 25 + 2x$$

Since solving this equation is very difficult. So, it is a better approach (Time saving) to put the values given in the options and try to find out a solution.
 Hence, trying out we get 11 as the value of x.

8. (b)



Let $BC = x$ and $AD = y$

Using the theorem of angle bisector,

$$\frac{BD}{DC} = \frac{AB}{AC} = \frac{4}{3} \Rightarrow BD = \frac{4}{7}x \text{ and } DC = \frac{3}{7}x$$

In $\triangle ABD$, by sine rule, $\frac{\sin 30}{4/7x} = \frac{\sin B}{y} \dots(1)$

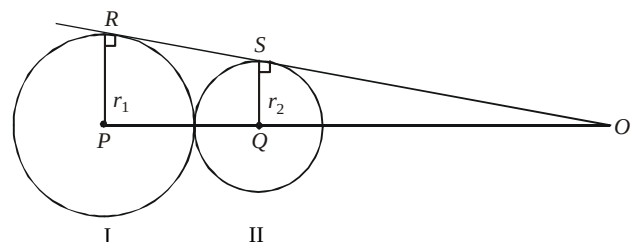
In $\triangle ABC$, by sine rule; $\frac{\sin 60}{x} = \frac{\sin B}{3}$

$$\Rightarrow \frac{\sqrt{3}}{2x} = \frac{\sin 30 \cdot y}{4/7x \times 3}$$

[Putting value of $\sin B$ from (1)]

$$\Rightarrow y = \frac{\sqrt{3}}{2x} \times \frac{4}{7}x \times 3 \times \frac{2}{1} = \frac{12\sqrt{3}}{7}$$

9-11.



9. (b) In $\triangle SOQ$ and $\triangle ROP$

$\angle O$ is common

$\angle S = \angle R = 90^\circ$ (tangent at circle)

$$\therefore \Delta SOQ \sim \Delta ROP$$

$$\Rightarrow \frac{RP}{SQ} = \frac{OP}{OQ} = \frac{PQ+OQ}{OQ} = \frac{PQ}{OQ} + 1$$

$$\Rightarrow \frac{4}{3} = \frac{PQ}{OQ} + 1 \text{ or } \frac{PQ}{OQ} = \frac{4}{3} - 1 = \frac{1}{3}$$

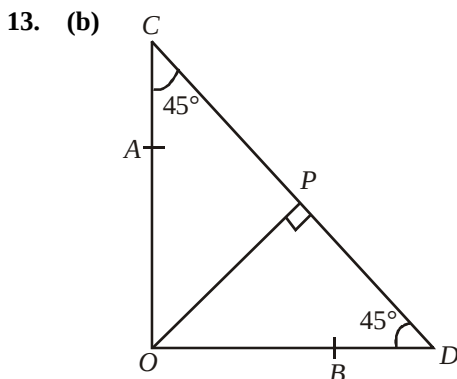
$$\Rightarrow PQ = 7 \text{ and } OQ = 21$$

$$\therefore \text{Required ratio} = \frac{7}{21} = \frac{1}{3}$$

10. (b) $PQ = r_1 + r_2 = 7$
As the ratio of radii is 4 : 3.
So, the only value which satisfies the radii of circle II = 3

11. (c) In ΔSOQ ,
 $\Rightarrow SO^2 + SQ^2 = OQ^2$
 $\Rightarrow SO^2 = 21^2 - 3^2 = (21-3)(21+3) = 18 \times 24 = 432$
 $\Rightarrow SO = 12\sqrt{3}$

12. (b) Inradius of right angled triangle
 $= \frac{AB + BC - AC}{2}$
 $= \frac{9 + 40 - 41}{2} = 4 \text{ cm}$



$$OC = OD \text{ and } OA = OP = OB$$

$$OP = 1 \text{ m}$$

$$\therefore PC = 1 \text{ m}$$

$$OC = \sqrt{2} \text{ m}$$

$$\therefore AC = OC - OA$$

$$= (\sqrt{2} - 1) \text{ m}$$

$$\text{and } AC + CP = (\sqrt{2} - 1) + 1$$

$$= \sqrt{2} \text{ m}$$

$$= 1.414 \text{ m} = 141.4 \text{ cm}$$

14. (c) $\frac{\text{ar}(\Delta CMN)}{\text{ar}(\Delta ABNM)} = \frac{1}{2}$

$$\therefore \frac{\text{ar}(\Delta CMN)}{\text{ar}(\Delta CAB)} = \frac{1}{3}$$

$$\Rightarrow \frac{MN}{AB} = \frac{CM}{CA} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{CM}{MA} = \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2} \quad MA = (CA - CM)$$

15. (d) $\angle ACB = 50^\circ$
 $\angle CFO = \angle CEO = 90^\circ$
 $\therefore \angle FOE = 360^\circ - (90^\circ + 90^\circ + 50^\circ) = 130^\circ$
but $\angle AOB = \angle FOE = 130^\circ$

16. (a) $\angle OCD = 90^\circ$
 $\angle OAC = \angle OCA = 30^\circ$
 $\angle ACD = \angle ACO + \angle OCD = 30^\circ + 90^\circ = 120^\circ$
 $\therefore \angle BAC = 180^\circ - 120^\circ = 60^\circ$
 $\Rightarrow \angle BCD = 60^\circ$ ($\angle BCD = \angle BAC$)
 $\Rightarrow \angle OCB = \angle OCB - \angle BCD$
 $= 90^\circ - 60^\circ = 30^\circ$

17. (d) $\frac{1}{2} : \frac{1}{3} : \frac{1}{4} = 6 : 4 : 3$
 $6x + 4x + 3x = 52$, or $13x = 52$, or $x = 4$
Required length = 12 cm.

18. (c) Required ratio = $\frac{a^2}{(\sqrt{2}a)^2} = 1 : 2$

19. (b) Δ 's APO and BQO are similar
($\angle APO = \angle BQO = 90^\circ$, tangent is $\perp r$ to radius
 $\angle AOP = \angle QOB$, vertically opposite angles).
 $\therefore AO : OB :: 2 : 1$ and $OP : OQ :: 2 : 1$, $AB = 10$

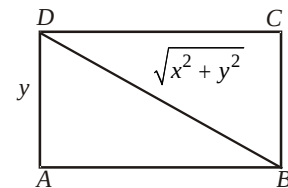
$$\Rightarrow AO = \frac{2}{3} \times AB = \frac{20}{3} \text{ and } OP = \frac{2}{3} \times PQ = \frac{16}{3}$$

$$AP^2 = OA^2 - OP^2 \text{ (In } \Delta OAP, \angle APO = 90^\circ)$$

$$= \frac{1}{3^2} (20^2 - 16^2) = \frac{1}{3^2} \times 144$$

$$\therefore AP = \frac{12}{3} = 4$$

20. (d)



According to question,

$$(x+y) - \sqrt{x^2 + y^2} = \frac{x}{2}$$

$$(x+y) - \frac{x}{2} = \sqrt{x^2 + y^2}$$

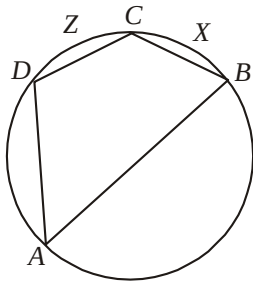
$$\left(\frac{x}{2} + y\right)^2 = x^2 + y^2$$

$$\frac{x^2}{4} + y^2 + xy = x^2 + y^2$$

$$x^2 + 4xy = 4x^2$$

$$4xy - 3x^2 = 4y = 3x \Rightarrow \frac{y}{x} = \frac{3}{4}$$

31. (c) $m \angle DAB + 180^\circ - 120^\circ = 60^\circ$
(Opposite angles of a cyclic quadrilateral)
 $m(\text{arc } BCD) = 2m \angle DAB = 120^\circ$.



$$\therefore m(\text{arc } CXB) = m(\text{arc } BCD) - m(\text{arc } DZC) \\ = 120^\circ - 70^\circ = 50^\circ$$

32. (c) Let the line m cut AB and CD at point P and Q respectively

$$\angle DOQ = x \text{ (exterior angle)}$$

Hence, $Y + 2x$ (corresponding angle)

$$\therefore y = x$$

...(1)

Also, $\angle DOQ = x$ (vertically opposite angles)

In $\triangle OCD$, sum of the angles = 180°

$$\therefore y + 2y + 2x + x = 180^\circ$$

$$3x + 3y = 180^\circ$$

$$x + y = 60$$

...(2)

From (1) and (2) $x = y = 30 = 2y = 60$

$$\therefore \angle ODS = 180 - 60 = 120^\circ$$

$$\therefore \theta = 180 - 3x = 180 - 3(30) = 180 - 90 = 90^\circ$$

$\therefore$ The required ratio = $90 : 120 = 3 : 4$.

33. (c) Given AB is a circle and BT is a tangent, $\angle BAO = 32^\circ$
Here, $\angle OBT = 90^\circ$

[$\because$ Tangent is $\perp$ to the radius at the point of contact]

$$OA = OB$$

[Radii of the same circle]

$$\therefore \angle OBA = \angle OAB = 32^\circ$$

[Angles opposite to equal side are equal]

$$\therefore \angle OBT = \angle OBA + \angle ABT = 90^\circ \text{ or } 32^\circ + x = 90^\circ$$

$$\angle x = 90^\circ - 32^\circ = 58^\circ$$

$$\text{Also, } \angle AOB = 180^\circ - \angle OAB - \angle OBA$$

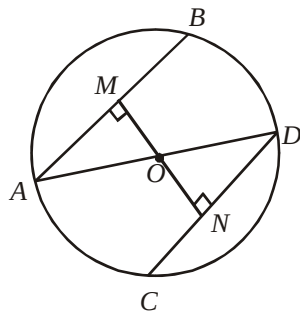
$$= 180^\circ - 32^\circ - 32^\circ = 116^\circ$$

$$\text{Now } Y = \frac{1}{2} \angle AOB$$

[Angle formed at the center of a circle is double the angle formed in the remaining part of the circle]

$$= \frac{1}{2} \times 116^\circ = 58^\circ$$

34. (a)



Two parallel chords AB & CD & $AB = CD = 8$ cm
Diameter of circle = $AD = 10$ cm.

$$\therefore \text{radius} = AO = OD = \frac{10}{2} = 5 \text{ cm}$$

$$AM = MB = \frac{AB}{2} = 4 \text{ cm.}$$

$\triangle AOM$ is right angle $\triangle$,

$$AO^2 = AM^2 + OM^2$$

$$5^2 = 4^2 + OM^2$$

$$OM^2 = 25 - 16 = 9$$

$$\Rightarrow OM = 3 \text{ cm.}$$

Similarly,

$$OM = ON = 3 \text{ cm}$$

$\therefore$ Distance between parallel chords = MN

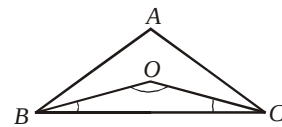
$$= OM + ON$$

$$= 3 + 3 = 6 \text{ cm}$$

35. (c) $\angle A + \angle B + \angle C = \angle 180^\circ$

$$\Rightarrow \frac{1}{2} \angle B + \frac{1}{2} \angle C = 90 - \frac{1}{2} \angle A$$

$$\Rightarrow \frac{1}{2} \angle B + \frac{1}{2} \angle C + \angle BOC = 180^\circ$$

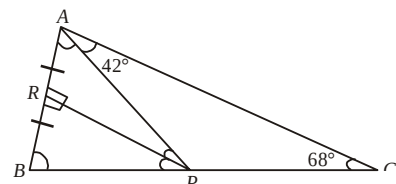


$$\Rightarrow 90^\circ - \frac{1}{2} \angle A + \angle BOC = 180^\circ$$

$$\Rightarrow \angle BOC = 180^\circ - \left(90^\circ - \frac{\angle A}{2} \right) = \left(90^\circ + \frac{\angle A}{2} \right)$$

36. (c) $\angle APB = 42^\circ + 68^\circ = 110^\circ$

(Exterior angle of a triangle is equal to sum of opposite interior $\angle$ s).



$$\triangle APR \cong \triangle BPR \text{ [SAS condition]}$$

$$\therefore \angle RPB = \angle RPA = \frac{110^\circ}{2} = 55^\circ$$

$$\therefore \text{In } \triangle BRP, \angle ABC = 90^\circ - 55^\circ = 35^\circ$$

37. (b) $\cos B = (a^2 + c^2 - b^2)/2ac$

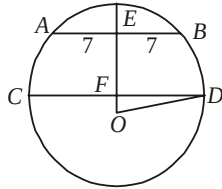
$$\angle B = 120^\circ \text{ and } \cos 120^\circ = -\frac{1}{2},$$

$$-\frac{1}{2} = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\Rightarrow -ac = a^2 + c^2 - b^2$$

$$\Rightarrow a^2 + c^2 = b^2 - ac$$

38. (a) Let AB be the chord of length 14 cm.
at a distance of 6 cm from the centre O .
Draw $OE \perp AB$. Then, $BE = 7$ cm and $OE = 6$ cm.
 $\therefore OB^2 = OE^2 + BE^2 = (6^2 + 7^2) = 85$.
Let CD be the chord at a distance of 2 cm from O .
Now, $OF = 2$ cm, $OD^2 = OB^2 = 85$.



$$\therefore FD = 9 \text{ cm.}$$

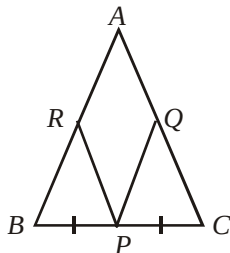
$$\therefore CD = 2 \times FD = 18 \text{ cm.}$$

39. (a) Let $YX = YZ = r$ (same radii); OYZ is a straight line (contact of circles)
 $YX \perp AB$ (Tangent $\perp$ to radius);
 $AX = 9$, $XB = 5$ (given)
 $\Rightarrow AB = 14$, $OB = OZ = 7$ (Same radii)
 $OX + 7 - 5 = 2$
In triangle OXY , $OY = 7 - r$; $YX + r$, $OX = 2$
 $\Rightarrow OY^2 = YX^2 + OX^2$ (Pythagoras' Theorem)
 $(7 - r)^2 = r^2 + 2^2$ $49 - 14r + r^2 = r^2 + 4$
 $14r = 45$ $r = 3\frac{3}{14}$ cm.

40. (a) R is the midpoint of side AB and Q is the midpoint of side AC . (midpoint theorem)

$$\therefore RP = \frac{1}{2} AC = \frac{1}{2} AB$$

$$PQ = \frac{1}{2} AB$$



Perimeter of $\square PRQ = 2(RP + PQ)$

$$= 2 \left(\frac{1}{2} AB + \frac{1}{2} AB \right)$$

$$= 2AB = 2 \times 8 = 16$$

41. (c) The given hexagon is regular hexagon.
 $\therefore AF = AO = FO = OE = FE$
Height of the hexagon = distance AE .

$$= \frac{\sqrt{3}}{2} a + \frac{\sqrt{3}}{2} a = \sqrt{3} a.$$

42. (a) In $\triangle ABD$ and $\triangle AEC$
 $m \angle ADB = m \angle ACE$ (both 90°)

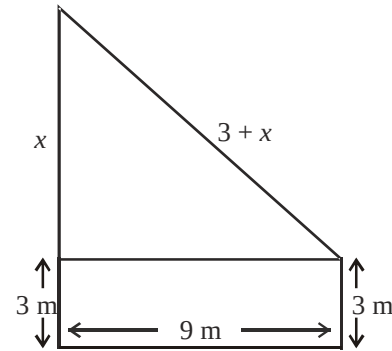
$m \angle ABC = m \angle ABC$ (intercept the same area AC)
 $\therefore \triangle ABD \sim \triangle AEC$ (AA similarity)

$$\frac{AB}{AE} = \frac{AD}{AC} \quad (\text{Proportional sides})$$

$$\therefore AE = \frac{AB}{AD} \times AC = \frac{8}{4.8} \times 6 = 10$$

$$\therefore \text{Circumradius} = 5,$$

43. (b)

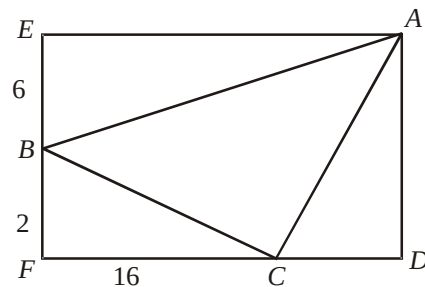


Using Pythagoras theorem, $x^2 + 81 = (3 + x)^2$

$$\Rightarrow x^2 + 81 = 9 + x^2 + 6x \Rightarrow 6x = 72 \Rightarrow x = 12 \text{ m}$$

Height of wall = $12 + 3 = 15$ m

44. (b) Since $EADF$ is a rectangle



$$\therefore DF = AE = 22$$

$$CD = 22 - 16 = 6$$

$$AD = EF = 8$$

$$\therefore AC = 10 \quad (\text{By Pythagoras theorem})$$

Also since line joining mid-points of two sides is half the length of the third side.

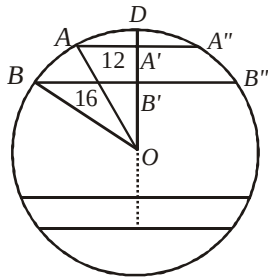
$$\text{Hence, required length} = \frac{1}{2} \times AC = 5$$

45. (c) $PR = PB + AR = 5 + 20 = 25$ [$\because AB = 5$ cm]
So, perimeter = $PR + PQ + QR$
 $= 25 + (20 - 5) + 20 + (20 - 10) + 20 + (20 - 12)$
 $= 35 + 30 + 28 = 93$

46. (b) Using the quality of similar triangles, $\frac{CP}{PB} = \frac{CD}{AB} = \frac{1}{3}$;

$$\text{In } \triangle BPQ \text{ and } \triangle BCD, \frac{CD}{PQ} = \frac{BC}{BP} = \frac{4}{3} = 1:0.75$$

47. (d) Remember that a perpendicular from the centre to a chord divides it into two equal parts.



$$\begin{aligned} \text{In } \triangle OBB', \\ OB^2 &= BB'^2 + OB'^2 \\ \Rightarrow 20^2 &= 16^2 + OB'^2 \end{aligned}$$

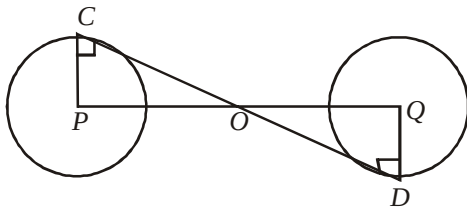
$$\Rightarrow OB' = \sqrt{20^2 - 16^2} = 12$$

Similarly in $\triangle OAA'$, $OA' = \sqrt{20^2 - 12^2} = 16$
 $\therefore$ Distance between the two parallel chords
 $= 16 - 12 = 4$ cm or $16 + 12 = 28$ cm

48. (b) $(OS)^2 = (OK)^2 + (KS)^2$
 $25 = OK^2 + 16 \Rightarrow OK = 3$
 and $(OS)^2 = (OL)^2 + (LN)^2$
 $25 = (OL)^2 + 9$
 $\Rightarrow OL = 4$ cm
 $\therefore KL = OL - OK = 1$ cm
 $\therefore$ Area of rectangle $= 1 \times 10 = 10$ cm²

49. (c) $AB = PQ = 26$ cm
 and $PO = OQ = 13$ CM

$$CO = \sqrt{(PO)^2 - (PC)^2}$$



$$CO = \sqrt{(13)^2 - (5)^2}$$

$$CO = 12$$
 cm

$$CD = 2CO = 24$$
 cm

Alternatively: Solve by using the formula of tangents.

$$\begin{aligned} 50. (b) \frac{\text{Area of } \triangle DAE}{\text{Area of } \triangle DEC} &= \frac{\frac{1}{2} \times DE \times AE}{\frac{1}{2} \times DE \times CE} \\ &= \frac{AE}{CE} \cdot \left(\frac{AD}{DC} \right)^2 = \left(\frac{6}{8} \right)^2 = \frac{9}{16} \end{aligned}$$

$$\text{Similarly, in } \triangle ABC, \frac{\text{Area of } \triangle BCF}{\text{Area of } \triangle BFA} = \frac{9}{16}$$

$$\therefore \text{The area of shaded to unshaded region} = \frac{16}{9}$$

51. (d) At $\angle A = 60^\circ$, $BC = b = c$
 and at $\angle A = 90^\circ$, $BC = \sqrt{2}b = \sqrt{2}c$
 $\therefore 60^\circ < A < 90^\circ$, $BC = c < a < c\sqrt{2}$

52. (c) $\angle A : \angle B : \angle C = 4 : 1 : 1$
 Hence we can suppose
 $\angle A = 4x$, $\angle B = x$, $\angle C = x$
 $\therefore 4x + x + x = 180$
 $x = 30$
 $\therefore A = 120$, $B = 30$, $C = 30$

$$\text{Now, } \frac{\sin A}{\sin B} = \frac{\sin 120^\circ}{\sin 30^\circ} = \frac{\sin 60^\circ}{\sin 30^\circ} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \frac{\sqrt{3}}{1}$$

53. (b) Calculate them physically (or manually)
 54. (c) Notice $\angle ORP = 90^\circ$ ($\because OP$ is a diameter of smaller circle)
 $OS = 5$ cm and $OR = 4$ cm

$$\therefore SR = \sqrt{(5)^2 - (4)^2} = 3$$
 cm

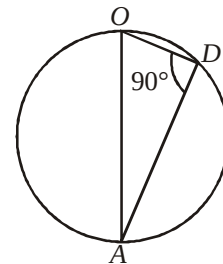
$$\therefore SP = 2(SR) = 6$$
 cm

55. (c) $\angle PDB = \angle QEA = 80^\circ$
 $\therefore \angle PED = \angle QDE = 10^\circ$ ($\because \angle DPE = \angle DQE = 90^\circ$)
 $\therefore \angle DRE = 180 - (10 + 10) = 160^\circ$
 $\therefore \angle PRD = 180 - \angle DRE = 20^\circ$

56. (b) $9 \times 180 - 2 \times 360$
 $= 180 \times 5 = 900^\circ$

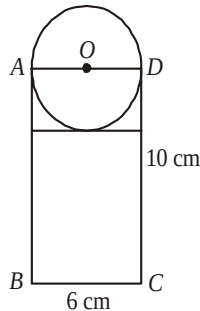
$$\therefore \begin{cases} n \times 180 - 2 \times 360 \\ = 180(n - 4) \end{cases}$$

57. (b) $\angle ADO$ is a right angle (angle of semicircle)
 Again when OD is perpendicular on the chord AC and OD passes through the centre of circle ABC , then it must bisect the chord AC at D .
 $\therefore AD = CD = 6$ cm



58. (d) As the point 'O' is formed by the $\perp$ bisects to the three sides of the Δ , so point 'O' is the circumcenter. This means that virtually, points A, B and C are on the circumference of the circle.
 Thus $m \angle BOC = 2 m \angle BAC$ ($\because$ angle subtended by an arc at the centre of the circle is twice the angle subtended at the circumference).
 59. (a) $OB = OA$ – radius of circle
 $\Rightarrow \angle CAO = \angle OBA$
 (angles in alternate segments are equal)
 Now, if $\angle CAO = \angle OBA$
 $\therefore \angle OAC = \angle OAB$
 $\therefore$ option (a) is correct

60. (d) Perimeter of the figure = $10 + 10 + 6 + 3\pi$
 $= 26 + 3\pi$ cm



61. (a) Find the indivisible area of $\triangle ABC$ and $\triangle ACD$.
 Add them then;

$$\frac{1}{2} AB \times BC + \frac{1}{2} AC \times CD \Rightarrow \frac{1}{2} xy + \frac{1}{2} z \sqrt{x^2 + y^2}$$

$\Rightarrow$ replace x, y, z by the lowest integral values.

Here as we are talking about right angles, so we have to take the smallest Pythagorean triplet, which in this case would be = 3, 4, 5

Answer $\Rightarrow 36$ cm³

62. (c) Value of BC will lie in between 999 and 3003. Hence $999 < BC < 3003$.
 So, the total values possible for $BC = 2003$.
63. (b) BC, CD and BD constitute a right = angle triangle ($13 \times 3, 13 \times 4$) and 13×5). If $BD = 65$, then it satisfies the other set of values of AB and AD also.
64. (c) Let the no. of 60° angles be N . So, total number of angles in the triangle = $20 + N$. Now, sum of interior angles = $(2n - 4)90^\circ$. Sum of angles in this case = $(20 + N - 2) \times 180^\circ$. And $(20 + N - 2) \times 180^\circ = (60^\circ) + 20 \times 300$
 $\therefore N = 23$

Expert Level

1. (a) $\triangle PQR$ is a right-angled triangle with QM as the perpendicular dropped on the hypotenuse PR .
 $\therefore PQ \times QR = QM \times PR$
 $m \angle STQ$
 $QR = a + b$
 $PQ = QR \tan 30 = \frac{QR}{\sqrt{3}}$
 $PR = \frac{QR}{\cos 30} = 2 \frac{QR}{\sqrt{3}}$
 $\therefore QM = \frac{PQ \times QR}{PR} = \frac{QR}{\sqrt{3}} \times QR \times \frac{\sqrt{3}}{2QR} = \frac{a+b}{2}$
2. (c) Length of direct common tangent.
 $PQ = \sqrt{15^2 - (9-2)^2} = \sqrt{225 - 49} = \sqrt{176} = 13.3$
 Thus, A is true.
 Length of transverse common tangent
 $AD = \sqrt{15^2 (9+2)^2} = \sqrt{225 - 121} = \sqrt{104} = 10.2$
 Thus, B is true, $PQ > AD$. Thus (c) is true

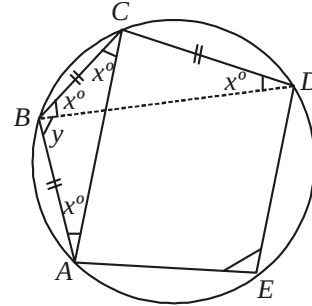
3. (d) $ar(\triangle ABD) = \frac{1}{2} \times ar(\square ABCD) = 45$ sq. cm.
(Diagonal bisects the area)
 $ar(\triangle ABE) = ar(\triangle AEF) = ar(\triangle AFD)$
 (Triangles of equal base & height)

$$\therefore ar(\triangle AEF) = \frac{45}{3} = 15 \text{ sq. cm}$$

Similarly, $ar(\triangle EFC) = 15$ sq. cm

$$\therefore ar(\square AECF) = 30 \text{ sq. cm.}$$

4. (b) Join BD , $AB = BC = CD$ (eq. chords cut off eq. arcs)



$\therefore \angle CDB = \angle BCA = \angle BAC = \angle CBD$
 $= x^\circ$ (eq. arcs subtend eq. $\angle$ S at circumference)
 $y = 180^\circ - x^\circ - x^\circ - x^\circ$ ($\angle$ sum of $\triangle$) = $180^\circ - 3x^\circ$
 $AED = 180^\circ - y^\circ$ (opp. $\angle$ s, cyclic quad.)
 $= 3x^\circ$

5. (c) Let the diagonal of $PQRS$ be $2r$.

Therefore, side = $r\sqrt{2}$.

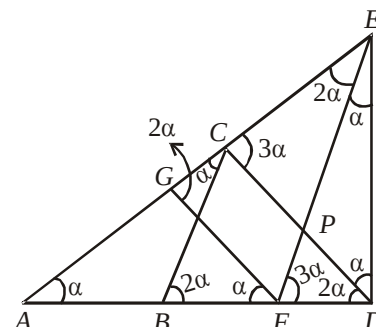
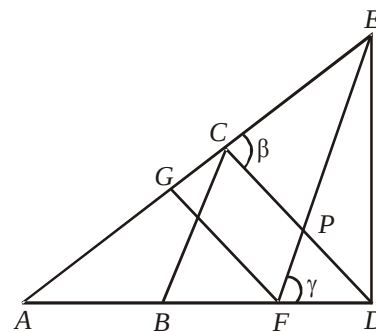
Now, $ABCD$ is a square. And side $\frac{r\sqrt{2}}{2} \times \sqrt{2} = r$.

Perimeter of $ABCD = 4r$.

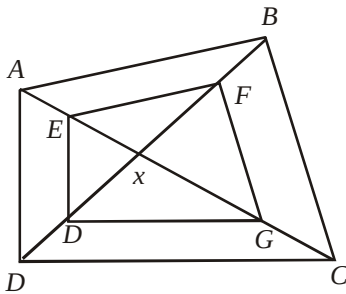
Circumference of bigger circle = $2\pi r$.

Therefore, required ratio = $\frac{\pi}{2}$

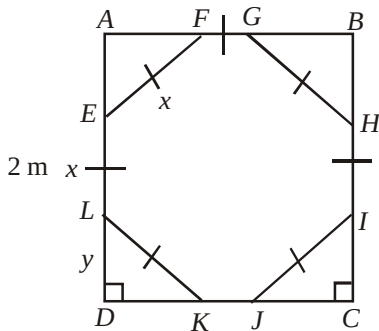
6. (d) Let $\angle EAD = \alpha$, then, $\angle AFG = \alpha$ and also $\angle ACB = \alpha$.
 Hence $\angle CBD = 2\alpha$ (exterior angle to $\triangle ABC$).
 Since $CB = CD$, hence $\angle CDB = 2\alpha$



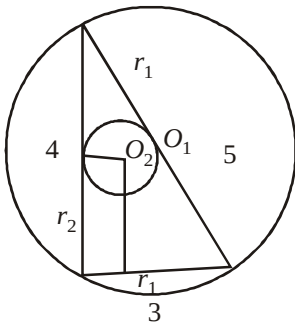
14. (a) This problem is based on mid-point theorem.



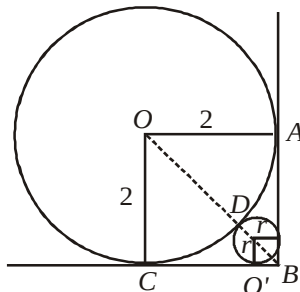
15. (a) $x + 2y = 2$
In $\triangle AEF$, $x^2 = 2y^2$
Use (1) and (2) to get the answer.



16. (b) $r_1 = 2.5$ cm
 $r = 1$ cm
Get the answer.



17. (d)



OABC is square with side = 2

$$\therefore OB = \sqrt{2^2 + 2^2} = 2\sqrt{2}$$

$$OB = 2\sqrt{2} = OD + r + O'B = 2 + r + r\sqrt{2}$$

$$\Rightarrow r(\sqrt{2} + 1) = 2(\sqrt{2} - 1)$$

$$\Rightarrow r = \frac{2(\sqrt{2} - 1)}{(\sqrt{2} + 1)} = \frac{2(\sqrt{2} - 1)^2}{2 - 1} = 2(2 + 1 - 2\sqrt{2})$$

$$6 - 4\sqrt{2}$$

18. (a) Let $YX = YZ = r$ (Same radii); OYZ is a straight line (Contact of circles)

$YX \perp AB$ (Tangent $\perp$ to radius);

$AX = 9$, $XB = 5$ (Given)

$\Rightarrow AB = 14$, $OB = OZ = 7$ (same radii);

$OX = 7 - 5 = 2$

In triangle OXY , $OY = 7 - r$; $YX = r$, $OX = 2$

$\Rightarrow OY^2 = YX^2 + OX^2$ (Pythagoras' Theorem)

$$(7 - r)^2 = r^2 + 2^2 \Rightarrow 49 - 14r + r^2 = r^2 + 4$$

$$\Rightarrow 14r = 45 \Rightarrow r = 3\frac{3}{14} \text{ cm.}$$

19. (b) Let X, Y, Z be the radii of the circle, centres X, Y, Z respectively YAZ, XYB, XZC are straight lines (Contact of circles)

$$XY = X - Y = 6$$

$$XZ = X - Z = 7$$

$$YZ = Y + Z = 9$$

$$\Rightarrow (1) + (2) + (3)$$

$$2X = 22 \Rightarrow X = 11, Y = 5, Z = 4$$

The radius of the circle, centre X , is 11 cm.

The radius of the circle, centre Y , is 5 cm.

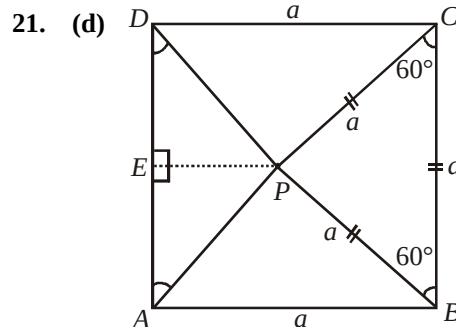
The radius of the circle, centre Z , is 4 cm.

20. (c) $m \angle ABM = 180^\circ - 120^\circ = 60^\circ$
 $\therefore \triangle AMB$ is a $30^\circ - 60^\circ - 90^\circ$ triangle.

$$\therefore AM = \frac{\sqrt{3}}{2} AB = \frac{\sqrt{3}}{2} \times 8 = 4\sqrt{3}$$

$$MB = \frac{1}{2} AB = \frac{1}{2} \times 8 = 4$$

$$(AC)^2 = (AM)^2 + (MC)^2 = (4\sqrt{3})^2 + (4 + 7)^2 = 48 + 121 = 169; AC = \sqrt{169} = 13.$$



$$\angle PBA = \angle ABC - \angle PBC = 90^\circ - 60^\circ = 30^\circ$$

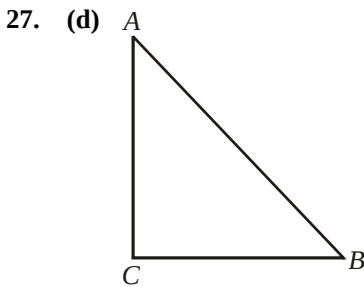
Further in $\triangle ABP$

$$PB = AB = a \Rightarrow \angle BPA = \angle BAP$$

Further $2(\angle BPA) + \angle PBA = 180^\circ$

$$\Rightarrow 2\angle BPA = 180^\circ - 30^\circ = 150^\circ$$

$$\Rightarrow \angle BPA = 75^\circ = \angle BAP$$

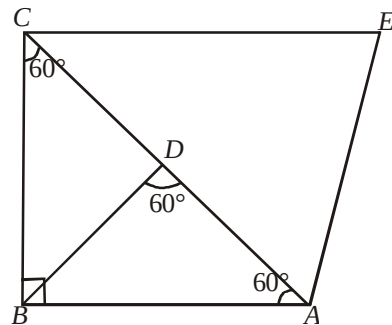


$$\begin{aligned}\angle A + \angle B &= 90^\circ \\ \angle A - \angle B &= 89 - 1 = 88 \\ 89 - 1 &= 88 \\ 88 - 2 &= 86 \\ 87 - 3 &= 84 \\ \dots &\dots \dots \\ 45 - 45 &= 0 \\ 44 - 46 &= -2\end{aligned}$$

$$\begin{aligned}\dots &\dots \dots \\ 1 - 89 &= -88\end{aligned}$$

Thus k can assume total '44 + 1 + 44 = 89' values

28. (b) $\triangle ABC$ is a right angled



and $\angle ABC = 90^\circ$

Lt $AB = x$

then $AB = AD = CD = BD = x$

$\triangle ABD$ is equilateral triangle

$$\therefore \angle CAE = 60^\circ$$

$$\therefore \angle BCA = 30^\circ$$

$$\Rightarrow \angle ACE = 60^\circ$$

$$\therefore \angle CEA = 60^\circ, \text{ also}$$

Hence, $\triangle ACE$ is an equilateral triangle

Thus, $AC = AE = CE = 2x$

$$\text{and } \frac{BC}{AB} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow BC = AB\sqrt{3} = x\sqrt{3}$$

$$\therefore \frac{BC}{AE} = \frac{x\sqrt{3}}{2x} = \frac{\sqrt{3}}{2}$$

29. (c) $\angle ROQ = 180^\circ - 50 = 130^\circ$

$$\left(\because \angle OQP + \angle ORP + \angle QOR + \angle QPR = 360^\circ \right)$$

$$\left(\text{and } \angle OQP = \angle ORP = 90^\circ \right)$$

Now, since $RT = TM$ and $QS = SM$

$\therefore$ also $OR = OM = OQ$

$$\therefore \angle ROT = \angle TOM \text{ and } \angle MOS = \angle SOQ$$

$$\therefore \angle SOT = \frac{1}{2} \angle ROQ$$

$$\therefore \angle SOT = \frac{130}{2} = 65^\circ$$

30. (a) $\angle OCT = 90^\circ, \angle DCT = 45^\circ$

$$\therefore \angle OCB = 45^\circ$$

$\therefore \angle COB = 45^\circ$ ($\triangle BOC$ is a right angled triangle)

$$\therefore \angle AOC = 180^\circ - 45^\circ = 135^\circ$$

Now, $\therefore CD = 10 \Rightarrow BC = 5 \text{ cm} = OB$

$$\Rightarrow OC = 5\sqrt{2} \text{ cm} = OA$$

$$\text{Again, } AC^2 = OA^2 + OC^2 - 2OA \cdot OC \cos 135^\circ$$

$$= 2(OA)^2 - 2(OA)^2 \cdot \cos 135^\circ$$

$$= 2(5\sqrt{2})^2 - 2(5\sqrt{2})^2 \times \left(-\frac{1}{\sqrt{2}}\right)$$

$$= 100 + \frac{100}{\sqrt{2}}$$

$$AC^2 = 170.70$$

$$\Rightarrow AC \approx 13 \text{ cm}$$

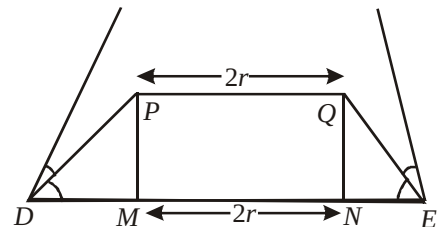
$$\therefore \text{Perimeter of } \triangle OAC = OA + OC + AC$$

$$= 5\sqrt{2} + 5\sqrt{2} + 13 = 27 \text{ cm}$$

31. (c) Let the radius of each circle be r unit then

$$PQ = QR = PR = 2r$$

$$\angle PDM = \angle QEM = 30^\circ$$



$$\therefore \frac{DM}{DP} = \cos 30^\circ$$

$$DM = DP \times \frac{\sqrt{3}}{2} \quad [DP = QE = (r)]$$

$$DM = \frac{r\sqrt{3}}{2}$$

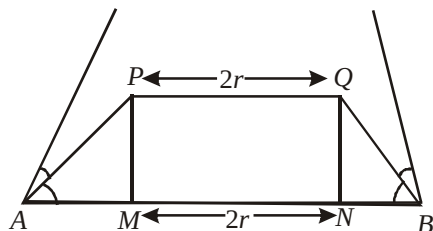
$$\therefore DE = DM + MN + NE$$

$$= \frac{r\sqrt{3}}{2} + 2r + \frac{r\sqrt{3}}{2} = (2 + \sqrt{3})r$$

$$DE = DF = EF = (2 + \sqrt{3})r$$

Again $\angle PAM = \angle QBN = 30^\circ$

$$\frac{r}{AM} = \frac{1}{\sqrt{3}}$$



$$\Rightarrow AM = r\sqrt{3} = BN$$

$$\therefore AB = AM + MN + NB$$

$$= r\sqrt{3} + 2r + r\sqrt{3} = 2r(1 + \sqrt{3})$$

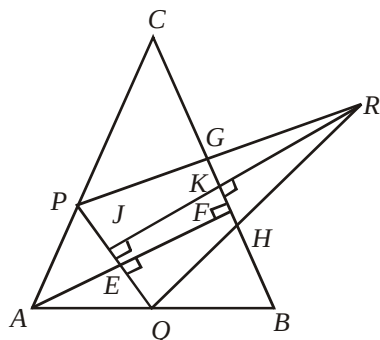
$$AB = BC = AC = 2r(1 + \sqrt{3})$$

Ratio of perimeter of equilateral triangle = ratio of their sides

$\therefore$ Ratio of perimeter of $\triangle ABC : \triangle DEF : \triangle PQR$

$$= 2(1+\sqrt{3}) : (2+\sqrt{3}) : 2$$

32. (a)



$$\triangle APO \sim \triangle ACB, BC = 2PQ$$

and $BC \parallel PQ$

$$\Rightarrow AE = 2AF$$

$$\Rightarrow AE = EF$$

Again $\triangle RGH \sim \triangle RPQ$

and $PO = 2GH$

(By mid-point theorem)

$$\Rightarrow RJ = 2RK$$

$$\Rightarrow RK = JK$$

But since $EF = JK$

$$\therefore AE = EF = JK = RK$$

$$\therefore RJ = RK + JK \text{ and } AF = AE + EF$$

and $RJ = AF = h$ (say),

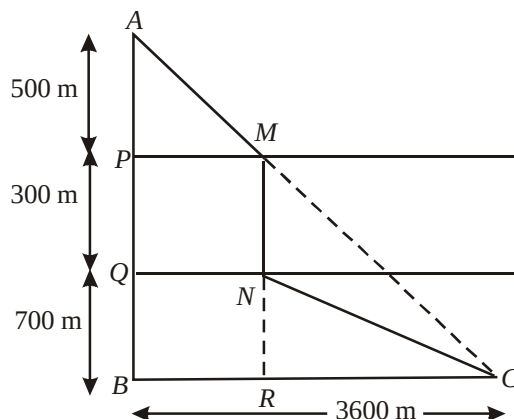
$$\text{then } \frac{\text{Area of } \triangle PQR}{\text{Area of } \triangle ABC} = \frac{\frac{1}{2} \times PQ \times h}{\frac{1}{2} \times BC \times h} = \frac{PQ}{BC} = \frac{1}{2}$$

33. (a) Let MN be the bridge

$$\triangle APM \sim \triangle ABC$$

$$\frac{AP}{PM} = \frac{AB}{BC}$$

$$\frac{500}{PM} = \frac{1500}{3600} \Rightarrow PM = 1200 = QN = BR$$



$$\therefore RC = BC - BR = 2400 \text{ m}$$

and $NR = BQ = 700$ m

$$\therefore NC = \sqrt{NR^2 + RC^2}$$

$$NC = 2500 \text{ m}$$

Also $AM = \sqrt{AP^2 + PM^2}$

$AM = 1300 \text{ m}$

$\therefore$ Total distance to be travelled = $AM + MN + NC$

$$= 1300 + 300 + 2500$$

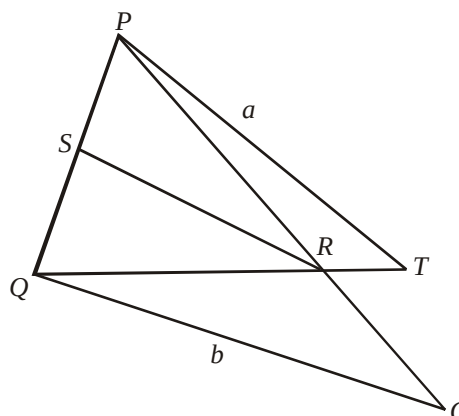
$$= 4100 \text{ m}$$

34. (c) $\Delta PQU \sim PSR$

$$\Rightarrow \frac{PS}{PW} = \frac{SR}{OU} \quad \dots(1)$$

$$\Delta POU \sim SOR$$

$$\Rightarrow \frac{SR}{PO} = \frac{SR}{PT} \quad \dots(2)$$



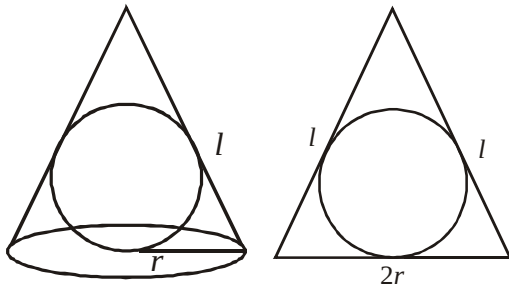
From (1) and (2)

$$PQ \times SR = PS \times QU = SQ \times PT \Rightarrow \frac{SQ}{PS} = \frac{b}{a}$$

Now use componendo and equation (1) to obtain

$$SR = \frac{ab}{a+b}$$

35. (b) Consider the biggest cross-section of the cone as a isosceles triangle therefore the circle inscribed in the triangle will be the biggest cross-section of the sphere.

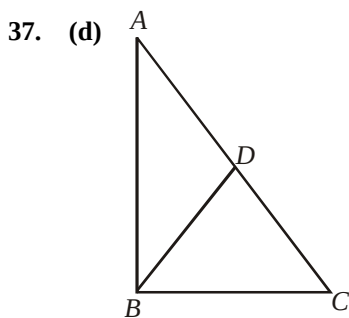


We know that in radius $\times$ semi perimeter = Area of the triangle.

A little calculator will lead to the answer, i.e.,

$$\frac{4}{3}\pi r^3 \left(\frac{l-r}{l+r} \right)^{3/2}$$

36. (a) Angle $XPA = \text{angle } ABP = x$
 Angle $CPX = \text{angle } CDP = x + y$
 Angle CDP is exterior angle of triangle PDB
 So angle $CDP = DBP + DPB$
 $x + y = x + DPB$
 $DPB = y$
 So angle $CPA = DPB$



$$\begin{aligned} AC \cdot AD &= AB^2 \\ AC \cdot AD &= BC^2 \\ \Delta ABC &\sim \Delta ADB \end{aligned}$$

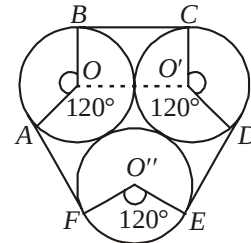
$$\therefore \frac{AC}{AB} = \frac{AB}{AD}$$

(Corresponding sides of similar triangle are proportional)

$$AC \cdot AD = AB^2 \quad \dots(1)$$

Also, $\Delta ABC \sim \Delta BDC \Rightarrow \frac{AC}{BC} = \frac{BC}{CD}$
 $AC \cdot CD = BC^2 \quad \dots(2)$
 Both conditions of option (d) are found, therefore (d) is the answer.

38. (c) $ABCD$ is a cyclic quadrilateral. Therefore
 $\angle DCB = 180^\circ - \angle A = 180^\circ - 60^\circ = 120^\circ$
 $\angle ABC = 80^\circ$; therefore $\angle BCQ = 180^\circ - 120^\circ = 60^\circ$
 And $\angle CBQ = 180^\circ - 80^\circ = 100^\circ$
 (because, sum of angles on a line = 180°)
 Then in ΔBCQ , $\angle Q = 180^\circ - (100^\circ + 60^\circ) = 20^\circ$
 ($\because$ sum of angles of triangle = 180°)
39. (a) $\angle AOB = \angle COD = \angle FO'E = 120^\circ$
 Distance between 2 centres = 2 m
 $\therefore BC = DE = FA = 2m$
 Perimeter of the figure = $BC + DE + FA +$
 circumference of sectors AOB , $CO'D$ and $FO'E$.
 But three equal sectors of $120^\circ = 1$ full circle of same radius.

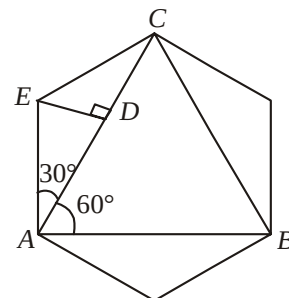


Therefore, perimeter of surface
 $= 2\pi r + BC + DE + FA = (2\pi + 6)m$

40. (a) In square $ADCE$, area = x^2
 So, side $AC = x$
 In square $AHIB$, area = y^2
 So, side $AB = y$
 As $x^2 > y^2$ so, $x > y$
 and in ΔABC ($\angle ABC = 90^\circ$)
 $(AC)^2 - (AB)^2 = (BC)^2$
 Hence area $BCGF = (AC)^2 - (AB)^2 = x^2 - y^2 = (x-y)(x+y)$
41. (d) As, hexagon is regular and $AD = CD$
 So, ΔABC is equilateral triangle with

$$AB = BC = AC = \frac{P}{3}$$

$$AD = \frac{P}{6}$$



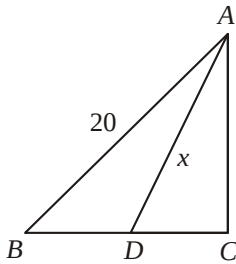
So,

$$\cos 30^\circ = \frac{\sqrt{3}}{2} = \frac{AD}{AE}$$

$$AE = AD \times \frac{2}{\sqrt{3}} = \frac{2p}{3\sqrt{3}}$$

$$\text{So, perimeter of hexagon} = 6 \times \frac{2p}{6\sqrt{3}} = \frac{2p}{\sqrt{3}}$$

42. (a) The solution of the above question is based on the following construction, where $AB = 20$ and $BD = 10$



The question is asking for the exact length AD , of triangle ABD .

Think only of length measuring formulae (Pythagoras theorem is obvious in this case).

If we extend the side BD upto a point C , the length AC will give the Altitude or height of the $\triangle ABD$. Then we will get:

$$\frac{1}{2} b \times h \Rightarrow 80 \Rightarrow \frac{1}{2} \times 10 \times h = 80 \Rightarrow h = 16, \text{ i.e. } AC = 16.$$

And now as $\triangle ABC$ is a right angled triangle, we can easily get the length of DC as 2, based on the triplet 12, 16, 20.

Now, if $AC = 16$, $DC = 2$, we can easily get the exact length of AD using Pythagoras theorem i.e. $AC =$

$$\begin{aligned} & \sqrt{2^2 + 16^2} \\ &= \sqrt{260} \end{aligned}$$

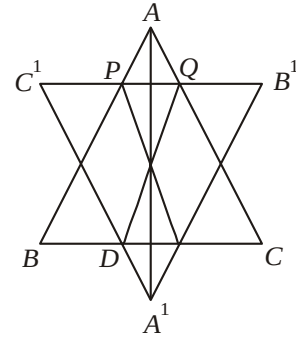
$$43. (c) \frac{\sqrt{3}}{4} a^2 = 300 \text{ cm}^2$$

$$a = \sqrt{\frac{1200}{\sqrt{3}}}$$

So, AP will be equal to $AB/3$

So, the area common to both triangles.

$$\begin{aligned} &= 300 - 3 \times \frac{\sqrt{3}}{4} \times \frac{1}{9} \times \frac{1200}{\sqrt{3}} \\ &= 200 \text{ cm}^2 \end{aligned}$$



$$44. (a) \text{Area}(\triangle AOB) = \frac{1}{2} \times OA \times OB \times \sin(120).$$

$$PR = RQ = \sqrt{2}. \text{Area}(\triangle PRQ) = \frac{1}{2} \times PR \times RQ \times \sin(90).$$

$$\text{So ratio of areas} = \frac{\frac{1}{2} \times 2 \times 1}{\frac{1}{2} \times \frac{\sqrt{3}}{2}} = \frac{4}{\sqrt{3}}$$

$$45. (a) \text{Area of the quarter circle} = \frac{\pi r^2}{4} \Rightarrow 0.25\pi. \text{ Going by}$$

options, we have to see that the area of the inserted circle is less than the area of the quarter circle.

$$\text{option (b)} \frac{(\sqrt{2}+1)}{2} \Rightarrow \text{Area} = (1.5 + \sqrt{2})\pi$$

$$2.9\pi > 0.25\pi. \text{ Hence discarded.}$$

$$\text{Option (c)} \sqrt{2} - \frac{1}{2} \Rightarrow \text{Area} = \pi \left(2 + \frac{1}{4} - \sqrt{2} \right) \Rightarrow 0.75\pi$$

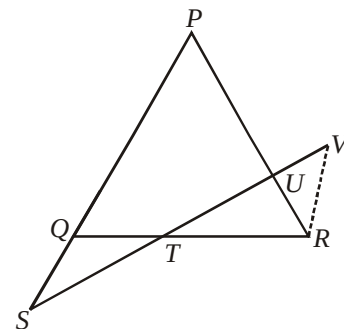
$$> 0.25\pi. \text{ Hence discarded.}$$

$$\text{Option (d)} 1 - \sqrt{2} \Rightarrow \text{Area} = \pi(1 + 8 - \sqrt{2}) \Rightarrow 0.85\pi > 0.25\pi. \text{ Hence discarded.}$$

$$\text{Option (a)} \sqrt{2} - 1 \Rightarrow \text{Area} \pi(2 + 1 - \sqrt{2}) \Rightarrow 0.20\pi < 0.25\pi$$

Hence this option is correct.

$$46. (c) \text{ We have } QT = TR \text{ and } PU = PS \text{ and } UR = 2 \text{ units}$$



Now, in ΔQST and ΔTVR

$$QT = TR$$
$$\therefore QS = VR$$

In $\triangle UVR$, $\angle VUR = \angle RVU$

From (1) and (2)

47. (a) In $\triangle COD$

$$b^2 = 2r^2$$

and $\triangle AOB =$ equilateral triangle

Hence, $a = r$

$$CD = 24 \text{ cm}$$
$$CE = ED = 12 \text{ cm}$$

$$= 12^2 + 5^2 = 169$$

$$OE = 13$$

$$\therefore OO' = OE + EO' = 13 + 13 = 26 \text{ cm}$$

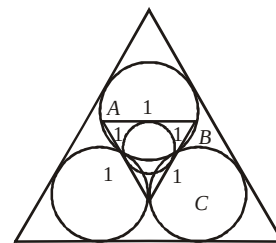
Hence, $OO' = AB = 26$ cm

49. (a) $\triangle ABC$ is equilateral

$$\tan 30 = r/x = x = \sqrt{3r} \quad (\text{where } r = 1)$$

$$AB = 2 + 2x = 2 + 2\sqrt{3} = 2(1 + \sqrt{3})$$

$$\text{Required area of } \triangle ABC = \sqrt{3}(1+\sqrt{3})^2$$



50. (d) Height of $\triangle ABC = \sqrt{\frac{3}{2}}a$

$$\frac{\sqrt{3}}{2} \times 2h = h = \sqrt{3}$$

$\frac{2h}{3}$ is the radius of the circle inscribed in ABC

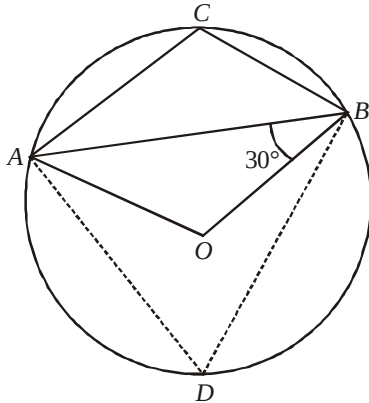
So radius of circumscribed circle

$$= \frac{2\sqrt{3}}{3} + 1 = \left(\frac{2}{\sqrt{3}} + 1 \right)$$

Hence, the required portion = $\pi \left(\frac{2}{\sqrt{3}} + 1 \right)^2 - 3\pi$

Explanation of Test Yourself

1. (b)



$OB = OA = \text{radius of the circle}$
 $\angle AOB = 180 - (30 + 30)$
 {Sum of angles of triangle = 180° }
 $\Rightarrow 120^\circ$

Then $\angle ADB = \frac{120}{2} = 60^\circ$, because the angle subtended by a chord at the centre is twice of what it can subtend at the circumference. Again, $ABCD$ is a cyclic quadrilateral;
 So $\angle ACB = 180^\circ - 60^\circ = 120^\circ$ (because opposite angles of cyclic quadrilateral are supplementary).

2. (a) $\triangle DOC$ and $\triangle AOB$ are similar (by AAA property)

$$\frac{\text{ar } \triangle AOB}{\text{ar } \triangle DOC} = \frac{(AB)^2}{(DC)^2} = \frac{9}{1}$$

So area of AOB : Area of $DOC = (3 : 1)^2 \Rightarrow 9 : 1$

3. (a) In $\triangle QRS$; $QR = RS$, therefore $\angle RQS = \angle RSQ$ (because angles opposite to equal sides are equal).

$$\text{Thus } \angle RQS + \angle RSQ = 180^\circ - 100^\circ = 80^\circ$$

$$\therefore \angle RQS = \angle RSQ = 40^\circ$$

$$\angle PQS = 180^\circ - 40^\circ = 140^\circ$$

(sum of angles on a line = 180°)

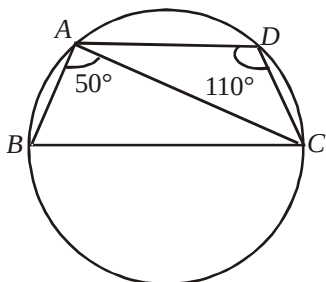
$$\text{Then again } \angle QRS = \angle QSP$$

($\because$ angles opposite to equal sides are equal)

$$\text{Thus } \angle QPS + \angle QSP = 180^\circ - 140^\circ = 40^\circ$$

$$\text{And } \angle QPS = \angle QSP = 20^\circ$$

4. (a) $\angle ABC + \angle ADC = 180^\circ$ (sum of opposites angles of cyclic quadrilateral is 180°)



$$\Rightarrow \angle ABC + 110^\circ = 180^\circ$$

($ABCD$ is a cyclic quadrilateral)

$$\Rightarrow \angle ABC = 180 - 110 \Rightarrow \angle ABC = 70^\circ$$

$$\therefore AD \parallel BC$$

$\therefore \angle ABC + \angle BAD = 180^\circ$ (Sum of the interior angles on the same side of transversal is 180°)

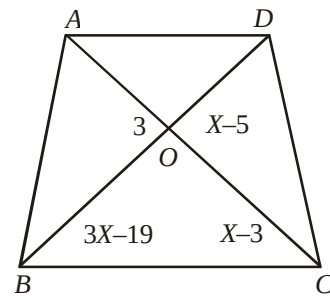
$$70^\circ + \angle BAD = 180^\circ$$

$$\Rightarrow \angle BAD = 180^\circ - 70^\circ = 110^\circ$$

$$\Rightarrow \angle BAC + \angle DAC = 110^\circ \Rightarrow 50^\circ + \angle DAC = 110^\circ$$

$$\Rightarrow \angle DAC = 110^\circ - 50^\circ = 60^\circ$$

5. (a)



$\triangle AOD$ and $\triangle BOC$ are similar (AAA property)

$$\text{Then } \frac{3}{3X-19} = \frac{X-5}{X-3}$$

$$\Rightarrow 3X - 9 = 3X^2 - 15X - 19X + 95$$

$$\Rightarrow 3X^2 - 37X + 104 = 0$$

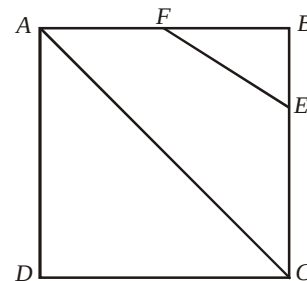
On solving this quadratic equation, we get $x = 8$ or 9 .

6. (b) In the given figure, $\triangle ABD$ is similar to $\triangle ACD$

$$\text{Then } \frac{AB}{BD} = \frac{AC}{DC} \Rightarrow \frac{6}{3} = \frac{5}{DC} \Rightarrow DC = 2.5 \text{ cm}$$

7. (b) Let the side of the square be x , then

$$BE = \frac{x}{3} \text{ and } BF = \frac{x}{2}$$



$$\text{Area of } \triangle FEB = \frac{1}{2} \times \frac{x}{3} \times \frac{x}{2} = \frac{x^2}{12}$$

$$\text{Now, } \frac{x^2}{12} = 108$$

$$\Rightarrow x^2 = 108 \times 12 = 1296$$

In $\triangle ADC$, we have

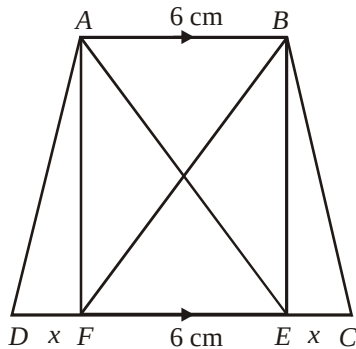
$$AC^2 = AD^2 + DC^2$$

$$= x^2 + x^2 = 2x^2$$

$$= 2 \times 1296 = 2592$$

$$\text{or } AC = \sqrt{2592} = 36\sqrt{2}$$

8. (b)



In the above question:

$$FE = AB = 6 \text{ cm}$$

$$\triangle ADF \cong \triangle BEC; \text{ so } DF = EC$$

$$\text{Let } DF = EC = x$$

Solving through options; e.g. option (b) $1/3$; $x = 6$

Then by Pythagoras triplet $AF = 8$

$$\text{Area of } ABEF = 8 \times 6 = 48 \text{ cm}^2$$

$$\text{Area of } \triangle AFD + \triangle BEC = 2 \times \frac{1}{2} \times 6 \times 8 \Rightarrow 48 \text{ cm}^2$$

$\therefore$ Area of $ABCD = 48 + 48 = 96 \text{ cm}^2$. Hence the condition is proved.

9. (b) As F is the mid-point of AD , CF is the median of the triangle ACD to the side AD .

Hence area of the triangle FCD = area of the triangle ACF .

Similarly area of triangle BCE = area of triangle ACE .

$$\therefore \text{Area of } ABCD = \text{Area of triangles } (CDF + CFA + ACE + BCE)$$

$$= 2 \text{ Area of triangles } (CFA + ACE) = 2 \times 13 = 26 \text{ sq. units.}$$

$$10. (b) m \angle ACD = \frac{1}{2} m(\text{arc } CXD) = m \angle DEC$$

$$\therefore m \angle DEC = x = 40^\circ$$

$$\therefore m \angle ECB = \frac{1}{2} m(\text{arc } EYC) = m \angle EDC$$

$$\therefore m \angle ECB = y = 54^\circ$$

$$54 + x + z = 180^\circ \text{ ..(Sum of all the angles of a triangle)}$$

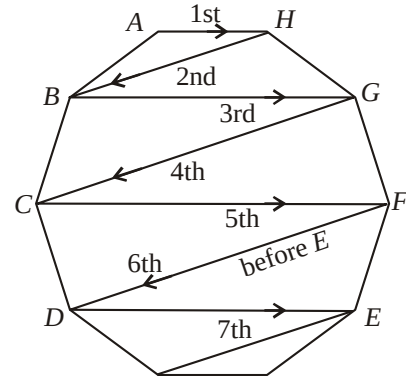
$$54 + 40 + z = 180^\circ \quad \therefore z = 86^\circ.$$

11. (d) As the point 'O' is formed by the $\perp$ bisectors to the three sides of the $\triangle$, so point 'O' is the circumcenter. This means that virtually, points A, B and C are on the circumference of the circle.

Thus $m \angle BOC = 2m \angle BAC$ ($\therefore$ angle subtended by an arc at the centre of the circle is twice the angle subtended at the circumference).

12. (d) The measure of the angle EDF has to be 90° since it should be half of the 180° angle.

13. (c)

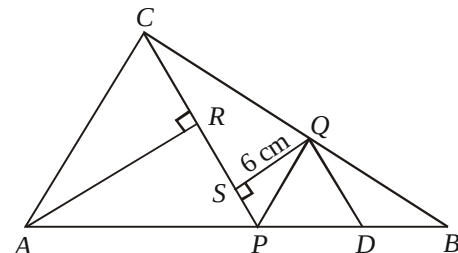


A frog couldn't jump on the vertices E. Therefore, there are 6 other vertices where it jumped.

Thus, there are 6 jumps before reaching E.

14. (c) Join $AC \Rightarrow \angle ACB = 90^\circ \Rightarrow \angle CAB = 55^\circ$. But $\angle BDC = \angle CAB$ as they are subscribed by the same arc.

15. (c) From figure



Given that,

$$PQ \parallel AC,$$

$$\therefore \frac{CQ}{QB} = \frac{AP}{PB} = \frac{4}{3}$$

Again, $QD \parallel CP$,

$$\therefore \frac{PD}{BD} = \frac{CQ}{QB} = \frac{4}{3}$$

$$\text{As } \frac{PD}{DB} = \frac{4}{3} \Rightarrow \frac{PD}{DB + PD} = \frac{4}{3 + 4} \Rightarrow \frac{PD}{PB} = \frac{4}{7} \Rightarrow PD = \frac{4}{7} PB$$

$$\therefore \frac{AP}{PD} = \frac{AP}{\frac{4}{7} PB} = \frac{7}{4} \times \frac{AP}{PB} = \frac{7}{4} \times \frac{4}{3} = 7:3$$

19

MENSURATION

- Introduction
- Basic Conversion of Units
- Plane Figures
- Area of a Triangle
- Area of a Quadrilateral
- Area of a Regular Hexagon
- Area of Irregular Plane Figures

- Paths
- Area Related to a Circle
- Surface Area and Volume of Solids
- Euler's Rule
- Circle Packing in a Square
- Circles Packing in a Circle
- Some Other Important Concepts

INTRODUCTION

In this chapter, we study how to measure perimeter, area and volume of plane and solid figures. In CAT and other equivalent aptitude tests, this chapter contributes almost 4–6 problems in CAT and other equivalent aptitude tests. Level of problems is moderate to tough since CAT exam aims to evaluate a person on a wide variety of skills and logically planning. Therefore it is advised that CAT aspirants must take this chapter seriously for the sake of their score in aptitude test.

BASIC CONVERSION OF UNITS

(i) Length:

- 1 m = 10 dm = 100 cm = 1000 mm
- 1 dm = 10 cm = 100 mm
- 1 cm = 10 mm
- 1 feet (ft) = 12 inches
- 1 inch = 2.54 cm
- 1 yard (y) = 3 feet (ft)
- 1 m = 1.094 yard (y) = 39.37 inches
- 1 yard (y) = 0.914 metre (m)
- 1 km = 1000 m = $\frac{5}{8}$ miles
- 1 mile = 1760 yards (y) = 5280 feet (ft)
- 1 nautical mile (knot) = 6080 feet (ft)

(ii) Surface Area:

Surface areas are measured in square units.

$$1 \text{ square metre} = 1 \text{ m} \times 1 \text{ m} = 100 \text{ cm} \times 100 \text{ cm} = 10000 \text{ cm}^2$$

$$1 \text{ square yard} = 1 \text{ y} \times 1 \text{ y} = 3 \text{ ft} \times 3 \text{ ft} = 9 \text{ ft}^2$$

$$1 \text{ acre} = 4047 \text{ m}^2 \text{ (approx.)}$$

$$1 \text{ hectare} = 10000 \text{ m}^2$$

(iii) Mass:

$$1 \text{ kg} = 1000 \text{ grams (g)} = 2.2 \text{ pounds (approx.)}$$

$$1 \text{ gram} = 10 \text{ miligram (mg)}$$

$$1 \text{ quintal} = 100 \text{ kg}$$

$$1 \text{ tonne} = 10 \text{ quintal} = 1000 \text{ kg}$$

(iv) Volume:

Volumes are measured in cubic units.

$$1 \text{ litre} = 1000 \text{ cm}^3 \text{ or cc}$$

$$1 \text{ m}^3 = 10000 \text{ litres} (= 10^4 \text{ l}) = 10^7 \text{ cm}^3$$

Note that

$$\sqrt{2} = 1.414, \sqrt{3} = 1.732, \sqrt{5} = 2.236,$$

$$\sqrt{6} = 2.45, \pi = \frac{22}{7} \text{ or } 3.14$$

PLANE FIGURES

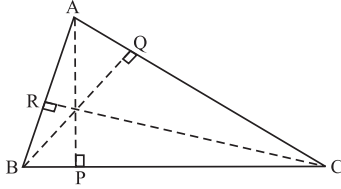
We have already dealt with plane figures (Triangles, Quadrilaterals and Circles) in geometry chapter. In this chapter, we will deal with perimeter and area of plane figures.

Perimeter: The perimeter of a plane geometrical figure is the total length of sides (or boundary) enclosing the figure. Units of measuring perimeter can be cm, m, km, etc.

Area: The area of any figure is the amount of surface enclosed within its bounding lines. Area is always expressed in square units.

AREA OF A TRIANGLE

- If in a triangle, we draw a perpendicular AP from vertex A on opposite side BC then AP is called altitude (or height) of the triangle ABC corresponding to base BC .

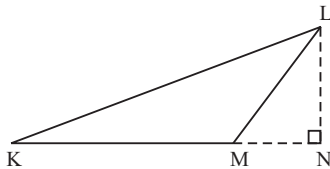


Similarly, BQ and CR are altitude of $\triangle ABC$ corresponding to, bases AC and AB respectively.

$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{corresponding altitude}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times BC \times AP = \frac{1}{2} \times AC \times BQ = \frac{1}{2} \times AB \times CR$$

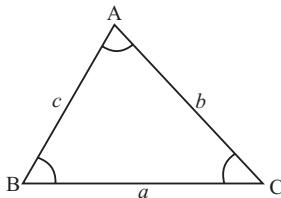
Note that in $\triangle KLM$, LN is the perpendicular on KM produced.



Here, LN is the altitude corresponding to the base KM of $\triangle KLM$.

$$\therefore \text{Area of } \triangle KLM = \frac{1}{2} \times KM \times LN$$

- Let in $\triangle ABC$, $BC = a$, $AC = b$ and $AB = c$; then perimeter of $\triangle ABC = a + b + c$



$$\text{Semi-perimeter of } \triangle ABC, s = \frac{a + b + c}{2}$$

$$\text{Area of } \triangle ABC = \sqrt{s(s-a)(s-b)(s-c)} \quad (\text{Heron's formula})$$

- Area of $\triangle ABC = \frac{1}{2} \times (\text{Product of two sides}) \times (\text{Sine of the included angle})$

$$= \frac{1}{2} ac \sin B \text{ or } \frac{1}{2} ab \sin C \text{ or } \frac{1}{2} bc \sin A$$

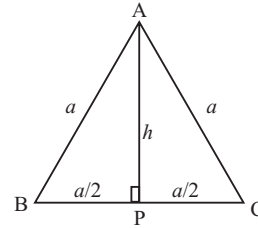
$$\text{Note that } \sin 30^\circ = \frac{1}{2}, \sin 45^\circ = \frac{1}{\sqrt{2}},$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \sin 90^\circ = 1$$

Area of an Equilateral Triangle

Since, $\triangle ABC$ is an equilateral triangle.

$$\therefore AB = BC = CA = a \text{ (say)}$$



From $\triangle APC$,

$$AP^2 = AC^2 - PC^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4}$$

$$AP = \frac{\sqrt{3}}{2} a \Rightarrow h = \frac{\sqrt{3}}{2} a$$

$$\text{Area of an equilateral } \triangle = \frac{1}{2} \times a \times \frac{\sqrt{3}}{2} a = \frac{\sqrt{3}}{4} a^2,$$

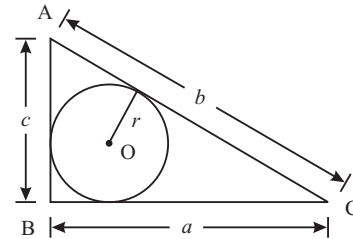
where a is the length of its one side

Note that

- among all the triangles that can be formed with a given perimeter, the equilateral triangle will have the maximum area.
- For a given area of triangle, the perimeter of equilateral triangle is minimum.

Area of Incircle and Circumcircle of a Triangle

- If a circle touches all the three sides of a triangle, then it is called incircle of the triangle.



Area of incircle of a triangle = $r \cdot s$, where r is the radius of the incircle and s is the half of the perimeter of the triangle.

If a, b, c are the length of the sides of $\triangle ABC$, then

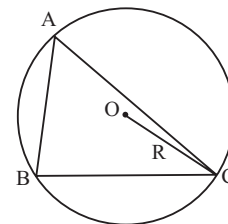
$$s = \frac{a + b + c}{2}$$

For an equilateral triangle,

$$r = \frac{\text{Length of a side of the triangle}}{2\sqrt{3}} = \frac{h}{3},$$

where h is the height of the triangle.

- If a circle passes through the vertices of a triangle, then the circle is called circumcircle of the triangle.



Area of the circumcircle = $\frac{abc}{4R}$, where R is the radius of the circumcircle and a, b, c are the length of sides of the triangle.

For an equilateral triangle,

$$R = \frac{\text{Length of a side of the triangle}}{\sqrt{3}} = \frac{2h}{3},$$

where h is the height or altitude of the equilateral triangle.

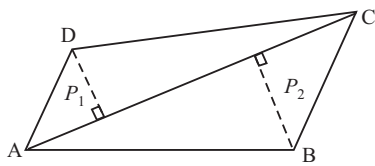
Hence for an equilateral triangle, $R = 2r$.

Note that an equilateral triangle inscribed in a circle will have the maximum area compared to other triangles inscribed in the same circle.

AREA OF A QUADRILATERAL

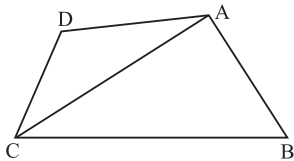
1. Area of quadrilateral $ABCD$

$$= \frac{1}{2} \times (\text{Length of the longest diagonal}) \times (\text{Sum of length of perpendicular to the longest diagonal from its opposite vertices})$$



$$= \frac{1}{2} \times d \times (p_1 + p_2), \text{ where } d = AC \text{ (i.e. longest diagonal)}$$

2. If length of four sides and one of its diagonals of quadrilateral $ABCD$ are given, then

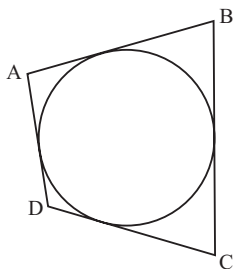


$$\begin{aligned} \text{Area of the quadrilateral } ABCD \\ = \text{Area of } \triangle ABC + \text{Area of } \triangle ADC \end{aligned}$$

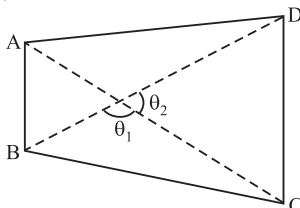
3. Area of circumscribed quadrilateral

$$= \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

where $s = \frac{a+b+c+d}{2}$ and a, b, c, d are length of sides of quadrilateral $ABCD$.



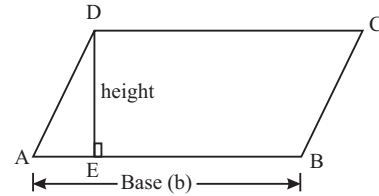
4. If θ_1 and θ_2 are the angles between the diagonals of a quadrilateral, then



$$\text{Area of the quadrilateral} = \frac{1}{2} d_1 d_2 \sin \theta_1 \text{ or } \frac{1}{2} d_1 d_2 \sin \theta_2$$

Here d_1 and d_2 are the length of the diagonals of the quadrilateral.

Area of a Parallelogram



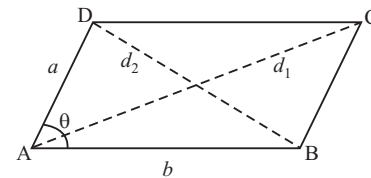
$$\text{Area of parallelogram} = \text{Base} \times \text{Corresponding height}$$

$$A = b \times h$$

Perimeter of a parallelogram = $2(a + b)$, where a and b are length of adjacent sides.

If θ be the angle between any two adjacent sides of a parallelogram whose length are a and b , then

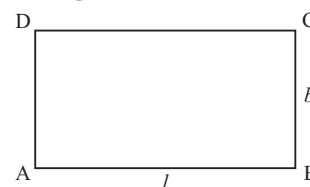
$$\text{Area of parallelogram} = ab \sin \theta$$



Note that in a parallelogram sum of squares of two diagonals = 2 (sum of squares of two adjacent sides)

$$\text{i.e., } d_1^2 + d_2^2 = 2(a^2 + b^2)$$

Area of a Rectangle

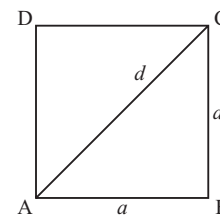


$$\text{Area of a rectangle} = \text{Length} \times \text{Breadth} = l \times b$$

[If any one side and diagonal is given]

$$\text{Perimeter of a rectangle} = 2(l + b)$$

Area of a Square



$$\text{Area of square} = \text{side} \times \text{side} = a \times a = a^2$$

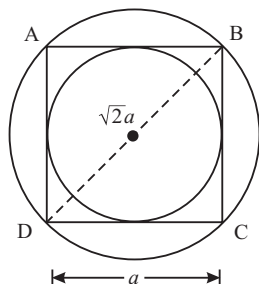
$$\text{Length of diagonal } (d) = a\sqrt{2} \text{ (by Pythagoras theorem)}$$

$$\text{Hence area of the square} = \left(\frac{d}{\sqrt{2}}\right)^2 = \frac{d^2}{2}$$

$$\text{Perimeter of square} = 4 \times \text{side} = 4 \times a$$

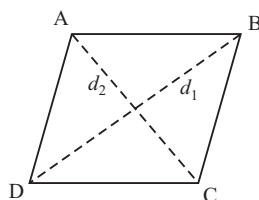
For a given perimeter of a rectangle, a square has maximum area.

Note that the side of a square is the diameter of the inscribed circle and diagonal of the square is the diameter of the circumscribing circle.



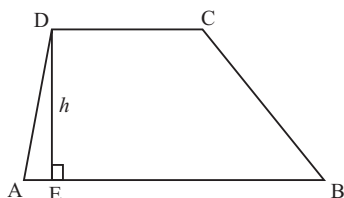
$$\text{Hence inradius} = \frac{a}{2} \text{ and circumradius} = \frac{\sqrt{2}a}{2} = \frac{a}{\sqrt{2}}$$

Area of a Rhombus



$$\begin{aligned} \text{Area of a rhombus} &= \frac{1}{2} \times \text{product of diagonals} \\ &= \frac{1}{2} \times d_1 \times d_2 \end{aligned}$$

Area of a Trapezium



Distance between parallel sides of a trapezium is called height of trapezium.

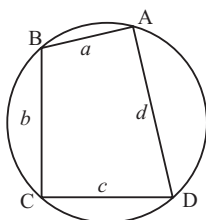
In fig. ABCD is a trapezium, whose sides AB and CD are parallel,

$$\begin{aligned} DE = h &= \text{Height of the trapezium} \\ &= \text{Distance between } \parallel \text{ sides.} \end{aligned}$$

$$\begin{aligned} \text{Area of trapezium} &= \frac{1}{2} (\text{sum of } \parallel \text{ sides}) \times \text{height} \\ &= \frac{1}{2} \times (AB + CD) \times DE \end{aligned}$$

Area of a Cyclic Quadrilateral

For a given quadrilateral ABCD inscribed in a circle with sides measuring a, b, c, and d;



$$\text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

where

$$s = \frac{a+b+c+d}{2}$$

Illustration 1: A rectangular parking space is marked out by painting three of its sides. If the length of the unpainted side is 9 feet, and the sum of the lengths of the painted sides is 37 feet, then what is the area of the parking space in square feet?

- (a) 46 (b) 81
(c) 126 (d) 252

Solution: (c) Clearly, we have : $l = 9$ and $l + 2b = 37$ or $b = 14$.
 $\therefore \text{Area} = (l \times b) = (9 \times 14) \text{ sq. ft.} = 126 \text{ sq. ft.}$

Illustration 2: A square carpet with an area 169 m^2 must have 2 metres cut-off one of its edges in order to be a perfect fit for a rectangular room. What is the area of rectangular room?

- (a) 180 m^2 (b) 164 m^2
(c) 152 m^2 (d) 143 m^2

Solution: (d) Side of square carpet $\sqrt{\text{Area}} = \sqrt{169} = 13 \text{ m}$
After cutting of one side,
Measure of one side = $13 - 2 = 11 \text{ m}$
and other side = 13 m (remain same)
 $\therefore \text{Area of rectangular room} = 13 \times 11 = 143 \text{ m}^2$

Illustration 3: The ratio between the length and the breadth of a rectangular park is 3 : 2. If a man cycling along the boundary of the park at the speed of 12 km/hr completes one round in 8 minutes, then the area of the park (in sq. m) is:

- (a) 15360 (b) 153600
(c) 30720 (d) 307200

Solution: (b) Perimeter = Distance covered in 8 min.

$$= \left(\frac{12000}{6} \times 8 \right) \text{ m} = 1600 \text{ m.}$$

Let length = $3x$ metres and breadth = $2x$ metres.

Then, $2(3x + 2x) = 1600$ or $x = 160$.

$\therefore$ Length = 480 m and Breadth = 320 m .

$\therefore \text{Area} = (480 \times 320) \text{ m}^2 = 153600 \text{ m}^2$.

Illustration 4: The length and breadth of a playground are 36m and 21 m respectively. Poles are required to be fixed all along the boundary at a distance 3m apart. The number of poles required will be

- (a) 39 (b) 38
(c) 37 (d) 40

Solution: (b) Given, playground is rectangular.

Length = 36 m , Breadth = 21 m

Now, perimeter of playground = $2(21 + 36) = 114$

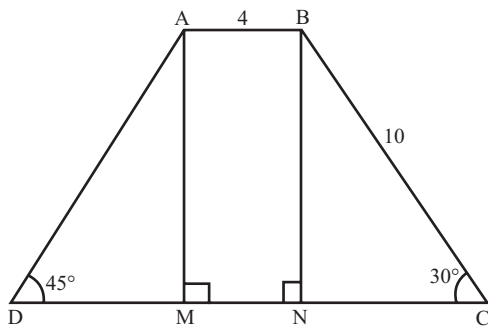
Now, poles are fixed along the boundary at a distance 3 m .

$\therefore$ Required no. of poles = $\frac{114}{3} = 38$.

Illustration 5: Find the area of the trapezium ABCD.

- (a) $\frac{5}{2}(13 + 2\sqrt{3})$ (b) $\frac{5\sqrt{3}(13 + 5\sqrt{3})}{2}$
(c) $13(13 + 2\sqrt{3})$ (d) None of these

Solution: (d)



AB and DC are the parallel sides

Height = $AM = BN$

$AB = MN = 4$

$\triangle BNC$ and $\triangle AMD$ are right angled triangles

In $\triangle BNC \Rightarrow \sin 30 = \frac{BN}{10} \Rightarrow BN = 5$

Using Pythagoras theorem $NC = \sqrt{10^2 - 5^2} = 5\sqrt{3}$

In $\triangle ADM$; $AM = 5$; $\tan 45 = \frac{AM}{DM} = 1 \Rightarrow DM = 5$

$\Rightarrow DM = 5$

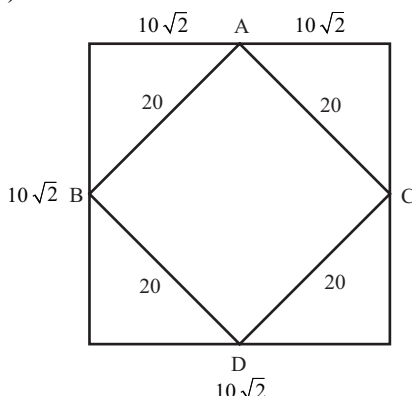
Area of trapezium $\Rightarrow \frac{1}{2} (\text{Sum of parallel sides}) \times \text{height}$

$\Rightarrow \frac{1}{2} (4 + 4 + 5\sqrt{3} + 5) \times 5 = \frac{5(13 + 5\sqrt{3})}{2}$

Illustration 6: Two goats tethered to diagonally opposite vertices of a field formed by joining the mid-points of the adjacent sides of another square field of side $20\sqrt{2}$. What is the total grazing area of the two goats?

- (a) $10\pi\text{m}^2$ (b) $50(\sqrt{2} - 1)\pi\text{m}^2$
(c) $100\pi(3 - 2\sqrt{2})\text{m}^2$ (d) $200\pi(2 - \sqrt{2})\text{m}^2$

Solution: (a)



The length of rope of goat = $10\sqrt{2}$ m

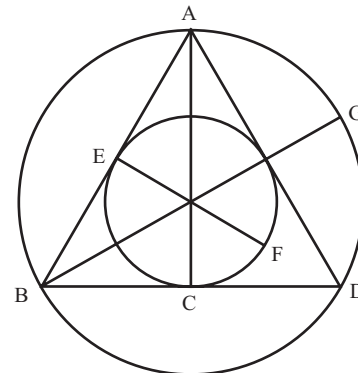
Then the two goats will graze an area = Area of a semicircle with radius $10\sqrt{2}$ m.

So total area grazed = $\frac{\pi r^2}{2} \Rightarrow 100\pi\text{m}^2$

Illustration 7: Find the ratio of the diameter of the circles inscribed and circumscribing an equilateral triangle to its height

- (a) 1 : 2 : 1 (b) 2 : 4 : 3
(c) 1 : 3 : 4 (d) 3 : 2 : 1

Solution: (b)



Let side of equilateral triangle = a

Then height = $\frac{a\sqrt{3}}{2}$

Area = $\frac{\sqrt{3}}{4}a^2$; $S = \frac{a+a+a}{2} = \frac{3a}{2}$

Diameter of inner circle = $\frac{2 \times \text{Area}}{\text{Perimeter of triangle}}$

$= \frac{\sqrt{3}}{4}a^2 \times \frac{2 \times 2}{3a} = \frac{a}{\sqrt{3}}$

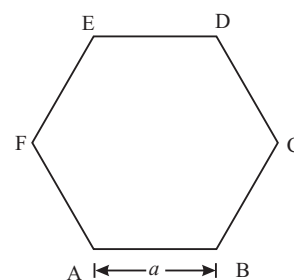
Diameter of outer circle = $\frac{a^3}{2 \times \text{Area}} = \frac{a^3}{2} \times \frac{4}{\sqrt{3}a^2}$

$\Rightarrow \frac{2a}{\sqrt{3}}$

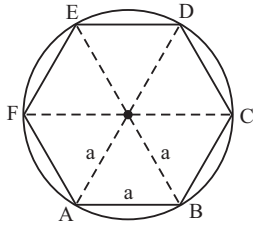
Ratio = $\frac{a}{\sqrt{3}} : \frac{2a}{\sqrt{3}} : \frac{a\sqrt{3}}{2} \Rightarrow \text{Ratio} = 2 : 4 : 3$

AREA OF A REGULAR HEXAGON

Area = $\frac{3\sqrt{3}}{2}a^2$, where ' a ' is the length of each side of the regular hexagon.

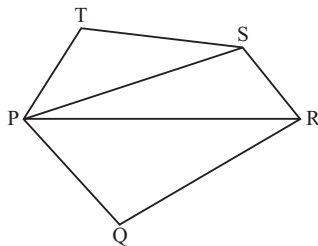


Diagonals of a hexagon divide it into six equilateral triangles. Hence, radius of the circumcircle of the hexagon = Length of a side of the hexagon = a



AREA OF IRREGULAR PLANE FIGURES

1. By drawing all the diagonals from anyone vertex, The polygon divided into several triangles.
Hence area of the polygon = Sum of area of all the triangles

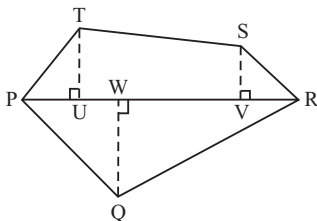


For example,

Area of pentagon $PQRST$

= Area of ΔPQR + Area of ΔPRS + Area of ΔPST .

2. By drawing longest diagonal and perpendicular from all vertices on two sides of the longest diagonal to the longest diagonals, the polygon is divided into several right triangles and trapeziums. By finding the sum of all the triangles and trapeziums, so formed we get the area of the polygon.

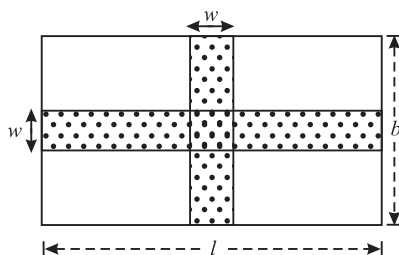


For example,

Area of pentagon $PQRST$ = Area of ΔPTU + Area of trapezium $(TUVS)$ + Area of ΔSVR + Area of ΔRQW + Area of ΔQWS .

PATHS

1. Pathways Running Across the Middle of a Rectangle



$$\begin{aligned}\text{Area of the path} &= l \cdot w + b \cdot w - w \cdot w \\ &= (l + b - w) \cdot w\end{aligned}$$

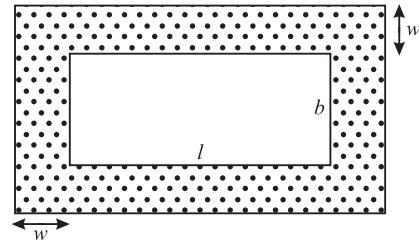
Perimeter of the path

$$= 2l + 2b - 4w$$

$$= 2(l + b - 2w)$$

Here w is the width of the path.

2. Pathways Outside a Rectangle



$$\begin{aligned}\text{Area of path} &= 2(lw) + 2(b \cdot w) + 4(w \cdot w) \\ &= (l + b + 2w)2w\end{aligned}$$

Perimeter of path

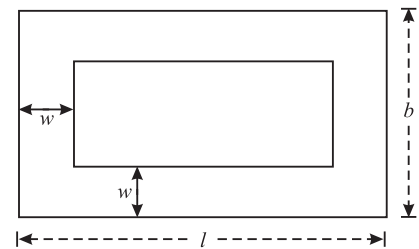
$$= (\text{Internal perimeter}) + (\text{External perimeter})$$

$$= 2(l + b) + 2(l + b + 4w)$$

$$= 4(l + b + 2w)$$

Here w is the width of the path.

3. Pathway Inside a Rectangle



$$\begin{aligned}\text{Area of path} &= 2(l \cdot w) + 2(b \cdot w) - 4(w \cdot w) \\ &= (l + b - 2w) \cdot 2w\end{aligned}$$

Perimeter of path

$$= \text{Length of outer path} + \text{Length of inner path}$$

$$= 2(l + b) + 2(l + b - 4w)$$

$$= 4(l + b - 2w)$$

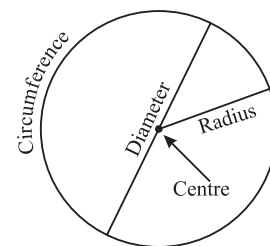
AREA RELATED TO A CIRCLE

Circle

Set of all points in a plane which are at a fixed distance from a fixed point in the same plane is called a circle.

The fixed point is called centre of the circle and the fixed distance is called radius of the circle.

Circumference or perimeter of a circle of radius r is

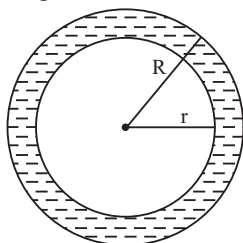


$$c = 2\pi r = \pi d \quad (2r = d = \text{diameter})$$

$$\text{Area of the circle} = \pi r^2 = \frac{\pi d^2}{4} = \frac{c^2}{4\pi} = \frac{1}{2} \times c \times r$$

Circular Ring

Region enclosed between two concentric circles of different radii in a plane is called a ring.



$$\text{Area of the ring} = \pi R^2 - \pi r^2 = \pi (R^2 - r^2)$$

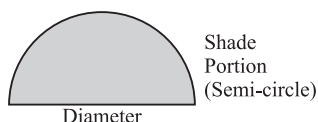
Circumference of the ring

$$= (\text{External circumference}) + (\text{Internal circumference})$$

$$= 2\pi R + 2\pi r = 2\pi(R + r)$$

Semi-circle

A semi-circle is a figure enclosed by a diameter and one half of the circumference of the circle.

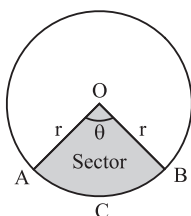


$$\text{Area of the semi-circle} = \frac{\pi r^2}{2}$$

$$\text{Circumference of the semi-circle} = \pi r + 2r = r(\pi + 2)$$

Sector of a Circle

Sector of a circle is the portion of a circle enclosed by two radii and an arc of the circle. $OACB$ is a sector of the circle.



Length of arc ACB (which make angle θ at the centre)

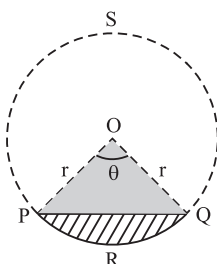
$$= (2\pi r) \times \frac{\theta}{360} = \frac{\pi r \theta}{180}$$

$$\text{Perimeter of the sector } OACB = 2r + \frac{\pi r \theta}{180}$$

$$\text{Area of sector } OACB = (\pi r^2) \times \frac{\theta}{360}$$

Segment of a Circle

A segment of a circle is a region enclosed by a chord and an arc of the circle.



Any chord of a circle which is not a diameter divides the circle into two segments, one of which is the major segment and other is minor segment.

Perimeter of the segment $PRQP$

$$= \text{Length of the arc } PRQ + \text{Length of } PQ$$

$$= \frac{\pi r \theta}{180} + 2r \sin \frac{\theta}{2}$$

Area of (minor) segment PQR

$$= \text{Area of sector } OPRQO - \text{Area of } \triangle OPQ$$

Area of (major) segment PSQ

$$= \text{Area of circle} - \text{Area of segment } PQR$$

Illustration 8: A circular grass lawn of 35 metres in radius has a path 7 metres wide running around it on the outside. Find the area of path.

- (a) 1694 m² (b) 1700 m²
(c) 1598 m² (d) None of these

Solution: (a) Radius of a circular grass lawn (without path) = 35 m

$$\therefore \text{Area} = \pi r^2 = \pi (35)^2$$

Radius of a circular grass lawn (with path)

$$= 35 + 7 = 42 \text{ m}$$

$$\therefore \text{Area} = \pi r^2 = \pi (42)^2$$

$$\therefore \text{Area of path} = \pi (42)^2 - \pi (35)^2$$

$$= \pi (42^2 - 35^2)$$

$$= \pi (42 + 35) (42 - 35)$$

$$= \pi \times 77 \times 7 = \frac{22}{7} \times 77 \times 7 = 1694 \text{ m}^2$$

Illustration 9: A wire can be bent in the form of a circle of radius 56 cm. If it is bent in the form of a square, then its area will be:

- (a) 3520 cm² (b) 6400 cm²
(c) 7744 cm² (d) 8800 cm²

Solution: (c) Length of wire = $2\pi \times R = \left(2 \times \frac{22}{7} \times 56\right) \text{ cm}$
 $= 352 \text{ cm}$

$$\text{Side of the square} = \frac{352}{4} \text{ cm} = 88 \text{ cm}$$

$$\text{Area of the square} = (88 \times 88) \text{ cm}^2 = 7744 \text{ cm}^2$$

Illustration 10: There are two concentric circular tracks of radii 100 m and 102 m, respectively. A runs on the inner track and goes once round on the inner track in 1 min 30 sec, while B runs on the outer track in 1 min 32 sec. Who runs faster?

- (a) Both A and B are equal
(b) A
(c) B
(d) None of these

Solution: (b) Radius of the inner track = 100 m

and time = 1 min 30 sec $\equiv$ 90 sec.

Also, Radius of the outer track = 102 m

and time = 1 min 32 sec $\equiv$ 92 sec.

Now, speed of A who runs on the inner track

$$= \frac{2\pi (100)}{90} = \frac{20\pi}{9} \approx 6.98$$

And speed of B who runs on the outer track

$$= \frac{2\pi(102)}{90} = \frac{51\pi}{23} \quad 6.96$$

Since, speed of $A >$ speed of B

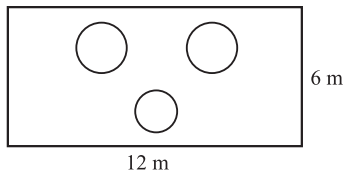
$\therefore$ A runs faster than B .

Illustration 11: A rectangular plate is of 6 m breadth and 12 m length. Two apertures of 2 m diameter each and one apertures of 1 m diameter have been made with the help of a gas cutter. What is the area of the remaining portion of the plate?

- (a) 68.5 sq. m. (b) 62.5 sq m
(c) 64.5 sq. m (d) None of these

Solution: (c) Given, Length = 12 m and Breadth = 6 m

$\therefore$ Area of rectangular plate = $12 \times 6 = 72 \text{ m}^2$



Since, two apertures of 3 m diameter each have been made from this plate.

$$\therefore \text{Area of these two apertures} = \pi(1)^2 + \pi(1)^2 = \pi + \pi = 2\pi$$

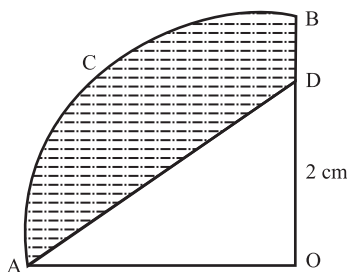
$$\text{Area of 1 aperture of 1m diameter} = \pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$$

$$\therefore \text{Total area of aperture} = 2\pi + \frac{\pi}{4} = \frac{9\pi}{4} = \frac{9}{4} \times \frac{22}{7} = \frac{99}{14}$$

$$\therefore \text{Area of the remaining portion of the plate} = 72 - \frac{99}{14} \text{ sq. m} = \frac{909}{14} \text{ sq. m} \approx 64.5 \text{ sq. m}$$

Illustration 12: In the adjoining figure, $AOBCA$ represents a quadrant of a circle of radius 3.5 cm with centre O . Calculate the area of the shaded portion.

- (a) 35 cm^2 (b) 7.875 cm^2
(c) 9.625 cm^2 (d) 6.125 cm^2

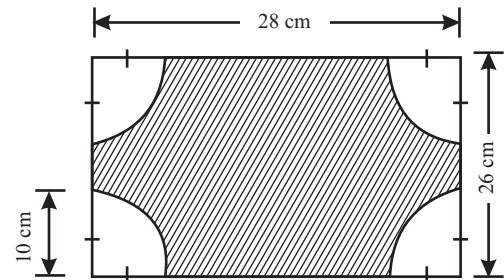


Solution: (d)

Area of shaded portion = Area of quadrant – Area of triangle

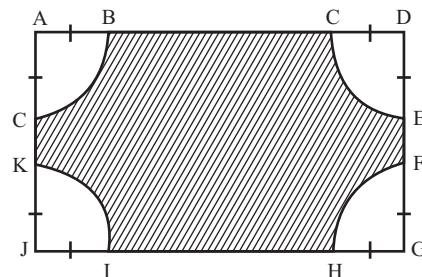
$$\Rightarrow \frac{\pi r^2}{4} - \frac{1}{2} \times 3.5 \times 2 = \frac{3.14 \times (3.5)^2}{4} - 3.5 = 6.125 \text{ cm}^2$$

Illustration 13: Find the perimeter and area of the shaded portion of the adjoining diagram:



- (a) 90.8 cm, 414 cm^2 (b) 181.6 cm, 423.7 cm^2
(c) 90.8 cm, 827.4 cm^2 (d) 181.6 cm, 827.4 cm^2

Solution: (a)



KJ = radius of semicircles = 10 cm

4 quadrants of equal radius = 1 circle of that radius

Area of shaded portion $\Rightarrow$ Area of rectangle – Area of circle
 $(28 \times 26) - (3.14 \times 10^2) \Rightarrow 414 \text{ cm}^2$

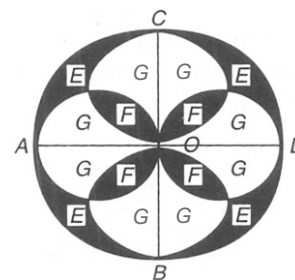
$BC = 28 - (10 + 10) = 8$ and $EF = 26 - (10 + 10) = 6$

Perimeter of shaded portion = $28 \text{ cm} + 2\pi r$

Answer $\Rightarrow 414 \text{ cm}^2 =$ Area and

Perimeter = 90.8

Illustration 14: $ABDC$ is a circle and circles are drawn with AO , CO , DO and OB as diameters. Areas E and F are shaded E/F is equal to



- (a) 1/1 (b) 1/2
(c) 1/2 (d) $\pi/4$

Solution: (a)

$AO = CO = DO = OB =$ radius of bigger circle = r (let)

$$\text{Then area of } (G + F) = \frac{\pi r^2}{2}$$

Area of $2(G + F) = \pi r^2$. Also area of $2G + F + E = \pi r^2$

i.e. $2G + F + F = 2G + F + E \Rightarrow F = E$

So the ratio of areas E and $F = 1 : 1$

SURFACE AREA AND VOLUME OF SOLIDS

Solid

A solid body has three dimensions namely length, breadth (or width) and height (or thickness). The surfaces that bind it are called faces and the lines where faces meet are called edges.

The area of the surface that binds the solid is called its surface area.

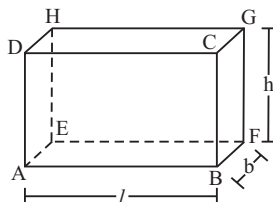
We measure the size of a solid body in terms of its volume.

The amount of space that any solid body occupies is called its volume.

Surface areas are measured in square units and volumes are measured in cubic units.

Cuboid

A cuboid is like a three dimensional box. It is defined by its length (l), breadth (b) and height (h). A cuboid can also be visualised as a room. It has six rectangular faces. It is also called rectangular parallelepiped.



A cuboid is shown in the figure with length ' l ', breadth ' b ' and height ' h '. ' d ' denotes the length of a diagonal (AG , CE , BH or DF) of the cuboid.

Total surface area of a cuboid = $2(lb + bh + hl)$

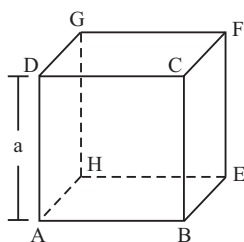
Lateral surface area (i.e., total area excluding area of the base and top) = $2h(l + b)$

Length of a diagonal of a cuboid = $\sqrt{l^2 + b^2 + h^2}$

Volume of a cuboid = Space occupied by cuboid
= Area of base $\times$ height
= $(l \times b) \times h = lbh$

Cube

A cube is a cuboid whose all edges are equal i.e.,
length = breadth = height = a (say)



Area of each face of the cube is a^2 square units.

Total surface area of the cuboid = Area of 6 square faces of the cube
= $6 \times a^2 = 6a^2$

Lateral surface area of cube i.e., total surface area excluding top and bottom faces = $4a^2$

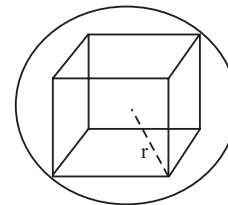
Length of diagonal (d) of the cube

$$= \sqrt{a^2 + a^2 + a^2}$$

$$= \sqrt{3a^2} = a\sqrt{3}$$

Volume of the cube (V) = Base area $\times$ Height
= $a^2 \times a = a^3$

Note that if a cube of the maximum volume is inscribed in a sphere of radius ' r ', then the edge of the cube = $\frac{2r}{\sqrt{3}}$



Cylinder

A cylinder is a solid object with circular ends of equal radius and the line joining their centres perpendicular to them. This line is called axis of the cylinder. The length of axis between centres of two circular ends is called height of the cylinder.

In the figure, a cylinder with circular ends each of radius r and height h is shown.

Curved surface area of a cylinder
= Circumference of base $\times$ height
= $2\pi r \times h = 2\pi rh$

If cylinder is closed at both the ends then total surface area of the cylinder

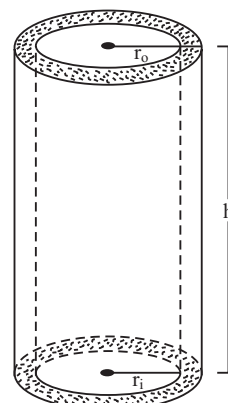
= Curved surface area + Area of circular ends
= $2\pi rh + 2 \times \pi r^2 = 2\pi r(h + r)$

Volume of the cylinder (V) = Base area $\times$ Height
= $\pi r^2 \times h = \pi r^2 h$

- Note that a cylinder can be generated by rotating a rectangle by fixing one of its sides.
- The curved surface of a cylinder is also called lateral surface.

Hollow Cylinder

A hollow cylinder is like a pipe.



Inner radius = r_i and outer radius = r_o
Hence $r_o - r_i$ = thickness of material of the cylinder.

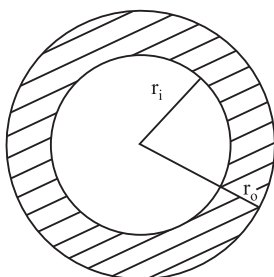
Let length or height of the cylinder = h ,

Curved surface area (C.S.A) of the hollow cylinder

= Outer curved surface area of the cylinder
+ Inner curved surface area of the cylinder

$$= 2\pi r_o h + 2\pi r_i h = 2\pi h(r_o + r_i)$$

Total surface area of hollow cylinder
= C.S.A. of hollow cylinder
+ Area of 2 circular end rings.



(one end of the pipe)

$$= 2\pi h(r_o + r_i) + 2\pi(r_o^2 - r_i^2)$$

$$= 2\pi(r_o + r_i)(h + r_o + r_i)$$

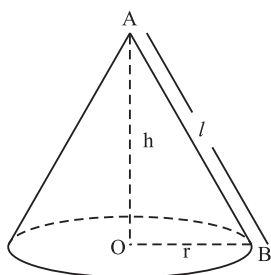
Volume of hollow cylinder = Volume of the material used in making the cylinder

$$= \pi(r_o^2 - r_i^2)h$$

Cone

A cone is a solid obtained by rotating a strip in the shape of a right angled triangle about its height. It has a circular base and a slanting lateral curved surface that converges at a point. Its dimensions are defined by the radius of the base (r), the height (h) and slant height (l).

A structure similar to cone is the ice-cream cone.



Height (AO) of cone is always perpendicular to base radius (OB) of the cone.

$$\text{Slant height } (l) = \sqrt{h^2 + r^2}$$

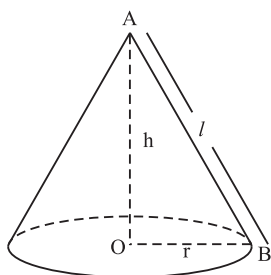
$$\text{Volume of cone} = \frac{1}{3} \times \text{base area} \times \text{height} = \frac{1}{3} \times \pi r^2 \times h$$

$$\text{Curved surface area (C.S.A.)} = \pi r l$$

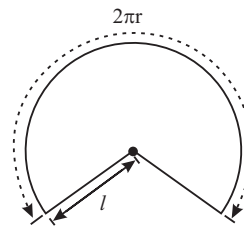
$$\text{Total surface area (T.S.A.)} = \text{C.S.A.} + \text{Base area}$$

$$= \pi r l + \pi r^2 = \pi r(l + r)$$

When a conical cup of paper (hollow cylinder) is unrolled, it forms a sector of a circle



Conical cup of paper



Unrolled conical cup, which is a sector of a circle.

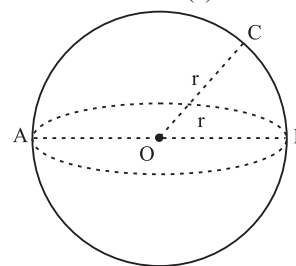
Radius of this sector is equal to slant height of the cone.

Length of curved edge of this sector is equal to the circumference of the base of the cone.

Sphere

A sphere is formed by revolving a semi-circle about its diameter. It has one curved surface which is such that all points on it are equidistant from a fixed point within it, called the centre.

Length of a line segment joining the centre to any point of the curved surface is called the radius (r) of the sphere.



Any line segment passing through the centre and joining two points on the curved surface is called the diameter (d) of the sphere.

Centre = O

Radius = $OC = OA = OB = r$,

Diameter = AB

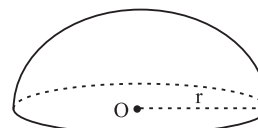
$$= d = 2r$$

Surface area of a sphere = $4\pi r^2$

$$\text{Volume of a sphere } (V) = \frac{4}{3} \pi r^3$$

Hemisphere

A plane through the centre of the sphere cuts the sphere into two equal parts. Each part is called a hemisphere.



$$\text{Volume of a hemisphere} = \frac{2}{3} \pi r^3$$

Curved surface area (C.S.A.) of a hemisphere = $2\pi r^2$

Total surface area (T.S.A.) of a hemisphere

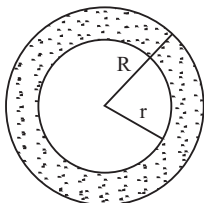
$$= \text{C.S.A.} + \text{Base area}$$

$$= 2\pi r^2 + \pi r^2 = 3\pi r^2$$

Note that if a sphere is inscribed in a cylinder then the volume of the sphere is $\frac{2}{3}$ rd of the volume of the cylinder.

Hollow Sphere or Spherical Shell

A rubber ball is an example of hollow sphere. In the rubber ball air is filled inside it. Thickness of the rubber in the ball is uniform. If outer and inner radii are R and r , then thickness of rubber or material used in hollow sphere = $R - r$.

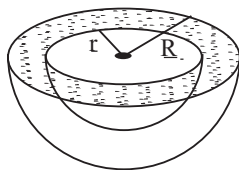


$$\begin{aligned}\text{Volume of the rubber or material used in hollow sphere} &= \text{External volume} - \text{Internal volume} \\ &= \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 \\ &= \frac{4}{3} \pi (R^3 - r^3)\end{aligned}$$

$$\text{External surface area} = 4\pi R^2.$$

Hemispherical Bowl

When a spherical shell is cut off in two equal parts, then each part is called a hemispherical bowl as shown in the figure.



If R and r are external and internal radii of the hemisphere respectively, then

$$\begin{aligned}\text{Volume of the material used in the hemispherical bowl} &= \text{External volume} - \text{Internal volume} \\ &= \frac{2}{3} \pi R^3 - \frac{2}{3} \pi r^3 = \frac{2}{3} \pi (R^3 - r^3)\end{aligned}$$

$$\text{External curved surface area} = 2\pi R^2$$

$$\text{Internal surface area} = 2\pi r^2$$

$$\text{Area of the cross-sectional ring} = \pi R^2 - \pi r^2 = \pi(R^2 - r^2)$$

$$\text{Total surface area}$$

$$\begin{aligned}&= (\text{External curved surface area}) + (\text{Internal curved surface area}) \\ &\quad + (\text{Area of cross-sectional ring}) \\ &= 2\pi R^2 + 2\pi r^2 + \pi(R^2 - r^2) \\ &= \pi(3R^2 + r^2)\end{aligned}$$

Illustration 15: If the radius of a sphere is increased by 2 cm, then its surface area increases by 352 cm². The radius of the sphere before the increase was:

- (a) 3 cm (b) 4 cm
(c) 5 cm (d) 6 cm

Solution: (d) $4\pi (r+2)^2 - 4\pi r^2 = 352$

$$\Rightarrow (r+2)^2 - r^2 = \left(352 \times \frac{7}{22} \times \frac{1}{4}\right) = 28.$$

$$\Rightarrow (r+2+r)(r+2-r) = 28$$

$$\Rightarrow 2r+2 = \frac{28}{2} \Rightarrow 2r+2 = 14 \Rightarrow r = 6 \text{ cm}$$

Illustration 16: A cylindrical bucket of height 36 cm and radius 21 cm is filled with sand. The bucket is emptied on the ground and a conical heap of sand is formed, the height of the heap being 12 cm. The radius of the heap at the base is :

- (a) 63 cm (b) 53 cm
(c) 56 cm (d) 66 cm

Solution: (a) Volume of the bucket = volume of the sand emptied

$$\text{Volume of sand} = \pi (21)^2 \times 36$$

Let r be the radius of the conical heap.

$$\text{Then, } \frac{1}{3} \pi r^2 \times 12 = \pi (21)^2 \times 36$$

$$\text{or } r^2 = (21)^2 \times 9 \text{ or } r = 21 \times 3 = 63 \text{ cm}$$

Illustration 17: The length of the longest rod that can be placed in a room which is 12 m long, 9 m broad and 8 m high is

- (a) 27 m (b) 19 m
(c) 17 m (d) 13 m

Solution: (c) Required length = length of the diagonal

$$= \sqrt{12^2 + 9^2 + 8^2} = \sqrt{144 + 81 + 64} = \sqrt{289} = 17 \text{ m}$$

Illustration 18: The internal measurements of a box with lid are 115 × 75 × 35 cm and the wood of which it is made is 2.5 cm thick. Find the volume of wood.

- (a) 82,125 cm³ (b) 70,054 cm³
(c) 78,514 cm³ (d) None of these

Solution: (a) Internal volume = 115 × 75 × 35 = 3,01,875 cm³

$$\begin{aligned}\text{External volume} &= (115 + 2 \times 2.5) \times (75 + 2 \times 2.5) \times \\ &\quad (35 + 2 \times 2.5)\end{aligned}$$

$$= 120 \times 80 \times 40 = 3,84,000 \text{ cm}^3$$

$$\therefore \text{Volume of wood} = \text{External volume} - \text{Internal volume}$$

$$= 3,84,000 - 3,01,875 = 82,125 \text{ cm}^3$$

Illustration 19: A rectangular tank is 225 m by 162 m at the base. With what speed must water flow into it through an aperture 60 cm by 45 cm that the level may be raised 20 cm in 5 hours ?

- (a) 5000 m/hr (b) 5400 m/hr
(c) 5200 m/hr (d) 5600 m/hr

Solution: (b) Required speed of flow of water

$$= \frac{225 \times 162 \times 20}{5 \times 100} = \frac{60}{100} \times \frac{45}{100} \times h$$

$$\therefore h = 5400$$

Illustration 20: A metallic sheet is of rectangular shape with dimensions 48 cm × 36 cm. From each one of its corners, a square of 8 cm is cut off. An open box is made of the remaining sheet. Find the volume of the box

- (a) 5110 cm³ (b) 5130 cm³
(c) 5120 cm³ (d) 5140 cm³

Solution: (c) Volume of the box made of the remaining sheet

$$= 32 \times 20 \times 8 = 5120 \text{ cm}^3$$

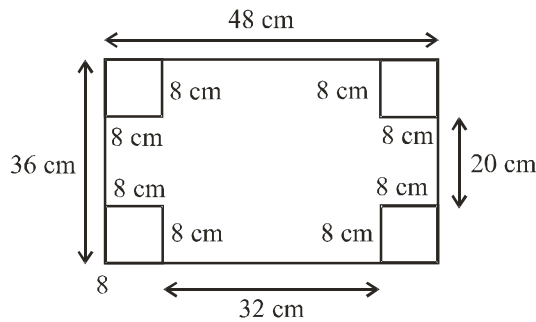


Illustration 21: The capacity of a cylindrical tank is 246.4 litres. If the height is 4 metres, what is the diameter of the base?

- (a) 1.4 m (b) 2.8 m
(c) 14 m (d) None of these

Solution: (d) Volume of the tank = 246.4 litres = 246400 cm³.

Let the radius of the base be r cm. Then,

$$\left(\frac{22}{7} \times r^2 \times 400\right) = 246400$$

$$\Rightarrow r^2 = \left(\frac{246400 \times 7}{22 \times 400}\right) = 196 \Rightarrow r = 14.$$

$\therefore$ Diameter of the base = $2r = 28$ cm = .28 m

Illustration 22: A cone, a hemisphere and a cylinder stand on equal bases and have the same height. What is the ratio of their volumes?

- (a) 2 : 1 : 3 (b) 2.5 : 1 : 3
(c) 1 : 2 : 3 (d) 1.5 : 2 : 3

Solution: (c)

As they stand on the same base so their radius is also same.

$$\text{Then; volume of cone} = \frac{\pi r^2 h}{3}$$

$$\text{Volume of hemisphere} = \frac{2\pi r^2}{3}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$\text{Ratio} = \frac{\pi r^2 h}{3} : \frac{2\pi r^2}{3} : \pi r^2 h$$

$$\Rightarrow \frac{h}{3} : \frac{2r}{3} : h$$

$$\Rightarrow h : 2r : 3h$$

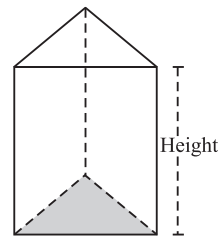
Radius of a hemisphere = Its height

$$\text{So } h : 2h : 3h \Rightarrow 1 : 2 : 3$$

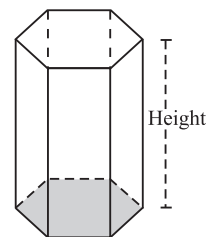
Prism

A 'prism' is a solid having identical and parallel top and bottom (or base) faces. These identical faces are regular polygon of any number of sides. The side faces of a prism are rectangular and are known as lateral faces. Number of lateral faces is equal to the number of sides in the base.

Here are some example of prisms



Triangular base prism



Hexagonal base prism

Lateral surface area of the prism

$$= (\text{Perimeter of the base}) \times (\text{Height})$$

Total surface area of the prism

$$= (\text{Surface area of the top and bottom}) + (\text{Lateral surface area})$$

$$= 2 \times \text{Area of the base} + \text{Perimeter of base} \times \text{Height}$$

$$\text{Volume of the prism} = (\text{Area of base}) \times (\text{Height})$$

The actual formula used to find the surface area and volume will depend upon the number of sides in the base of the prism.

Pyramid

It is a three-dimensional body made up of a regular polygon shaped base and triangular lateral faces that meet at a point called vertex, which is also called the apex of the pyramid.

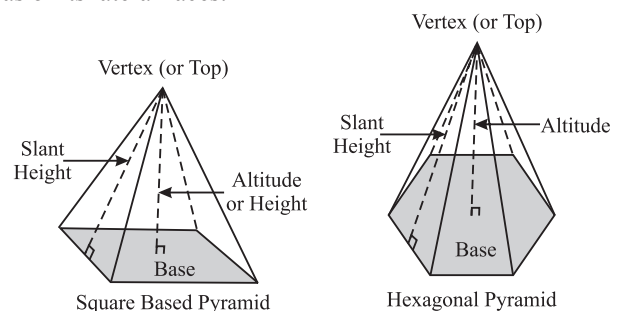
The number of triangular faces is equal to the number of sides in the base.

For example: A pyramid with a square base has four triangular faces, whereas a pyramid with a hexagonal face is made up of six triangular faces, and so on.

Lower face is called the base and the perpendicular distance of the vertex (or top) from the base is called the height or altitude of the pyramid.

The altitude of a lateral face of a pyramid is the slant height, which is the perpendicular distance of the vertex (or top) from the mid-point of any side of the base.

The lateral surface area of a regular pyramid is the sum of the areas of its lateral faces.



Lateral surface area of a pyramid

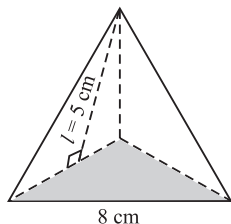
$$= \frac{1}{2} \times (\text{Area of the base}) \times (\text{Slant height})$$

Total surface area of a pyramid

$$= \frac{1}{2} \times (\text{Perimeter of the base}) \times (\text{Slant height}) + (\text{Area of the base})$$

$$\text{Volume of a pyramid} = \frac{1}{3} \times \text{Area of base} \times \text{Height}$$

Illustration 23: Find the lateral surface area of a regular pyramid with triangular base, if each edge of the base measures 8 cm and slant height is 5 cm.

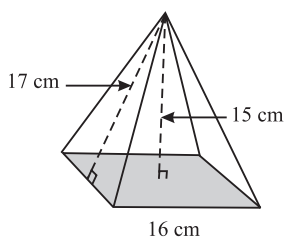


Solution: The perimeter of the base is the sum of the sides,

$$p = 3(8) = 24 \text{ cm}$$

$$\text{L.S.A.} = \frac{1}{2} \times (24) \times (5) = 60 \text{ cm}^2$$

Illustration 24: Find the total surface area of a pyramid with a square base if each side of the base measures 16 cm, the slant height of a side is 17 cm and the altitude is 15 cm.



Solution: The perimeter of the base,

$$p = 4 \times 16 = 64 \text{ cm}$$

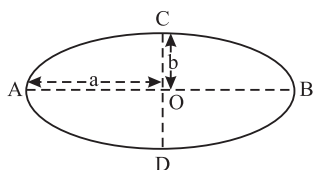
The area of the base

$$= 16^2 = 256 \text{ cm}^2$$

$$\begin{aligned} \text{T.S.A.} &= \frac{1}{2} (64) (17) + 256 \\ &= 544 + 256 = 800 \text{ cm}^2 \end{aligned}$$

Ellipse

Figure of an ellipse is given below.



AB and CD are length of major and minor axis of an ellipse

Length of major axis, $AB = 2a$

and length of the minor axis, $CD = 2b$

Then $AO = a$, $OC = b$

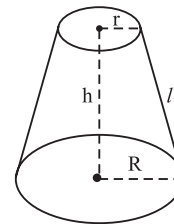
Perimeter of the ellipse $= \pi(a + b)$

Area of the ellipse $= \pi ab$

Frustum of a Cone

When top portion of a cone cut off by a plane parallel to the base of it, the left-over part is called the frustum of the cone.

In the figure, r and R are the radius of two ends, h is the height and l is the slant height of the frustum of cone.



$$\text{Slant height, } l = \sqrt{(R - r)^2 + h^2}$$

$$\text{Curved surface area} = \pi(R + r)l$$

Total surface area

$$= (\text{Curved surface area}) + (\text{Area of two circular ends})$$

$$= \pi(R + r)l + \pi R^2 + \pi r^2$$

$$= \pi(Rl + rl + R^2 + r^2)$$

$$\text{Height of the original cone} = \frac{Rh}{R - r}$$

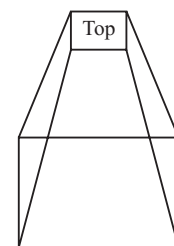
Volume of the frustum of cone

$$= \frac{\pi h}{3} (R^2 + r^2 + Rr)$$

Frustum of a Pyramid

When top portion of a pyramid is cut off by a plane parallel to the base of it, the left-over part is called the frustum of the pyramid.

If A_1, A_2 are of top and bottom face, P_1 and P_2 are the perimeters of top and bottom face, h is the height and l is the slant height of the frustum of the pyramid, then



$$\text{Lateral surface area} = \frac{1}{2} (P_1 + P_2) l$$

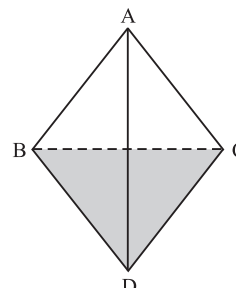
$$\text{Total surface area} = \text{Lateral surface area} + A_1 + A_2$$

$$= \frac{1}{2} (P_1 + P_2) l + A_1 + A_2$$

$$\text{Volume} = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 \cdot A_2})$$

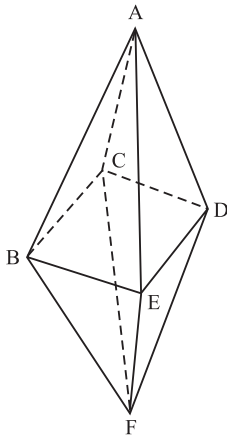
Tetrahedron (Only Shape)

A tetrahedron is a solid object which has 4 faces. All the faces of a tetrahedron are equilateral triangles. A tetrahedron has 4 vertices and 6 edges.



Octahedron (Only Shape)

An octahedron is a solid object which has 8 faces. All the faces of an octahedron are equilateral triangles. An octahedron has 6 vertices and 12 edges.



EULER'S RULE

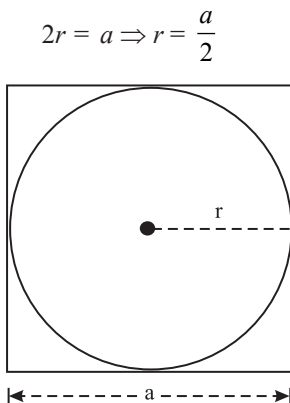
For any regular shape solid (like cuboid, cube, cylinder, etc)

$$\begin{aligned} \text{Number of faces (F)} + \text{Number of vertices (V)} \\ = \text{Number of edges (E)} + 2 \\ \text{i.e., } F + V = E + 2 \end{aligned}$$

CIRCLE PACKING IN A SQUARE

Let 'a' be the length of a side of the square and 'r' be the radius of the circle.

Case- (i): One circle



Case- (ii): Two circles

In the isosceles right angled $\triangle BCD$,

$$BD = \sqrt{2}r$$

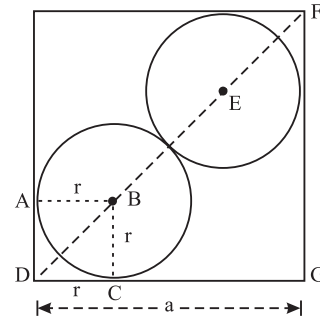
In the isosceles right angled $\triangle DFG$,

$$DF = \sqrt{2}a$$

$$\begin{aligned} \text{Now } DF &= DB + BE + EF \\ &= \sqrt{2}r + 2r + \sqrt{2}r \\ &= 2r + 2\sqrt{2}r = 2(\sqrt{2} + 1)r \end{aligned}$$

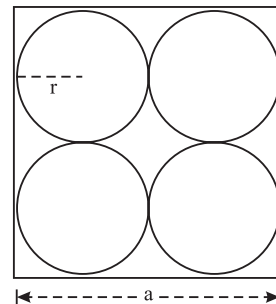
$$\therefore 2(\sqrt{2} + 1)r = \sqrt{2}a$$

$$\Rightarrow r = \frac{a}{\sqrt{2}(\sqrt{2}+1)} = \frac{a}{2+\sqrt{2}}$$



Case- (iii): Four circles

$$4r = a \Rightarrow r = \frac{a}{4}$$



Case- (iv): Five circles

In isosceles right angled triangle ABC ,

$$AC = \sqrt{2}r$$

$$AE = AC + CE = \sqrt{2}r + 2r = (2 + \sqrt{2})r$$

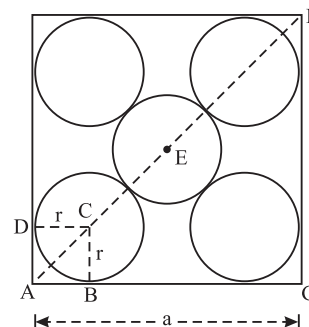
$$AF = 2AE = 2(2 + \sqrt{2})r$$

In isosceles right angle triangle AGF ,

$$AF = \sqrt{2}a$$

$$\therefore 2(2 + \sqrt{2})r = \sqrt{2}a$$

$$\Rightarrow r = \frac{a}{2(\sqrt{2}+1)}$$



Case- (v): Eight circles

In isosceles right $\triangle ABC$,

$$AC = \sqrt{2}r$$

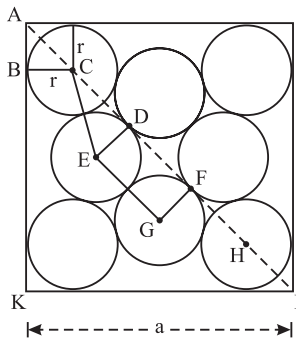
In right $\triangle CDE$,

$$CD = \sqrt{CE^2 - ED^2}$$

$$\Rightarrow CD = \sqrt{4r^2 - r^2} = \sqrt{3}r$$

In rectangle $DEGF$,

$$DF = EG = 2r$$



Now, $AI = AC + CD + DF + FH + HI$

$$= \sqrt{2}r + \sqrt{3}r + 2r + CD + AC \quad [\because FH = CD, HI = AC]$$

$$\Rightarrow AI = \sqrt{2}r + \sqrt{3}r + 2r + \sqrt{3}r + \sqrt{2}r$$

$$= (2\sqrt{2} + 2\sqrt{3} + 2)r$$

$$= 2(\sqrt{3} + \sqrt{2} + 1)r \quad \dots(1)$$

Also in isosceles right triangle AKI ,

$$AI = \sqrt{2}a \quad \dots(2)$$

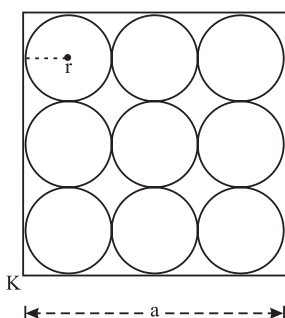
From (1) and (2),

$$2(\sqrt{3} + \sqrt{2} + 1)r = \sqrt{2}a$$

$$\therefore r = \frac{\sqrt{2}a}{2(\sqrt{3} + \sqrt{2} + 1)} = \frac{a}{\sqrt{2}(\sqrt{3} + \sqrt{2} + 1)}$$

Case-(vi): Nine circles

$$6r = a \Rightarrow r = \frac{a}{6}$$

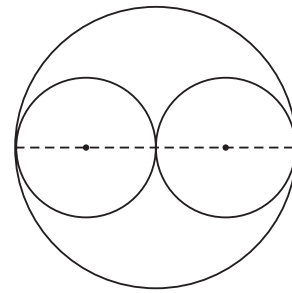


CIRCLES PACKING IN A CIRCLE

Let R be the radius of larger circle and r be the radius of smaller circle.

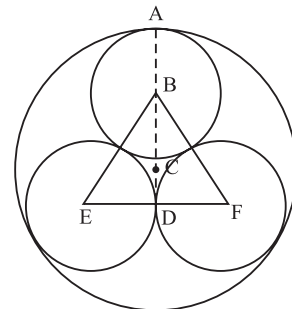
Case-(i): Two circles

$$R = 2r \Rightarrow r = \frac{R}{2}$$



Case-(ii): Three circles

C is the centroid of equilateral $\triangle BEF$



$$\therefore BC : CD = 2 : 1$$

$$\therefore BC = \frac{2}{3} BD \quad \dots(1)$$

In right angled $\triangle BDE$,

$$BD = \sqrt{BE^2 - DE^2}$$

$$BD = \sqrt{4r^2 - r^2} = \sqrt{3}r \quad \dots(2)$$

From (1) and (2),

$$BC = \frac{2}{3} \times \sqrt{3}r = \frac{2}{\sqrt{3}}r$$

Now $AC = AB + BC$

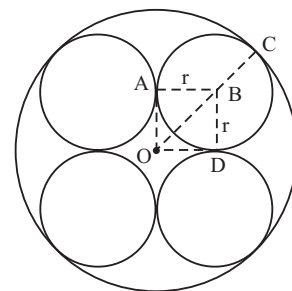
$$= r + \frac{2}{\sqrt{3}}r = \left(\frac{\sqrt{3} + 2}{\sqrt{3}} \right) r$$

Also $AC = R$

$$\therefore \left(\frac{\sqrt{3} + 2}{\sqrt{3}} \right) r = R \Rightarrow r = \frac{\sqrt{3}R}{\sqrt{3} + 2}$$

$$\Rightarrow r = (2\sqrt{3} - 3)R$$

Case- (iii): Four circles



In right $\triangle BDO$,

$$OB = \sqrt{2}r$$

$$OC = OB + BC = \sqrt{2}r + r$$

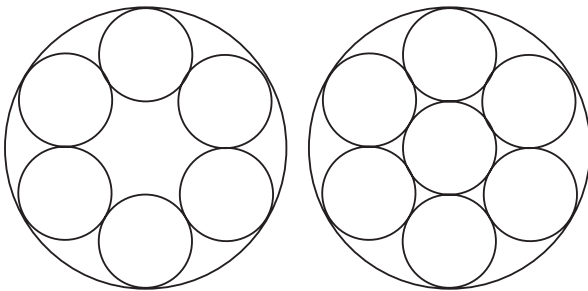
$$OC = (\sqrt{2} + 1)r$$

Also $OC = R$

$$\therefore (\sqrt{2} + 1)r = R$$

$$\Rightarrow r = \frac{R}{(\sqrt{2} + 1)} = (\sqrt{2} - 1)R$$

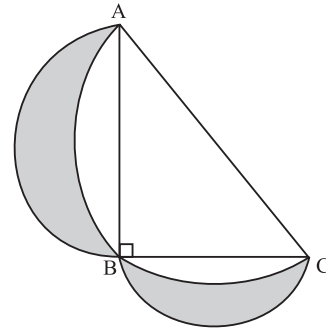
Case- (iv): Six/Seven circles



$$6r = 2R \Rightarrow r = \frac{1}{3}R$$

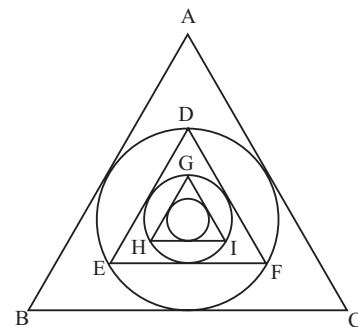
SOME OTHER IMPORTANT CONCEPTS

1. In the figure ABC is a triangle right angled at B . Three semi-circles are drawn taking the three sides AB , BC and CA as diameter. The region enclosed by the three semi-circles is shaded.



Area of the shaded region = Area of the right angled triangle.

2. In the figure given below all triangles are equilateral triangles and circles are inscribed in these triangles. If the side of triangle $ABC = a$, then the side of triangle $DEF = \frac{a}{2}$ and the side of triangle $GHI = \frac{a}{4}$



Thus length of a side of an inner triangle is half the length of immediate outer triangle. Similarly the radius of an inner circle is half the radius of immediate outer circle.

Practice Exercises

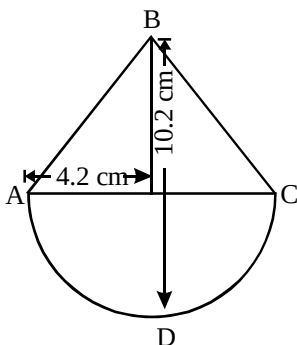


Foundation Level

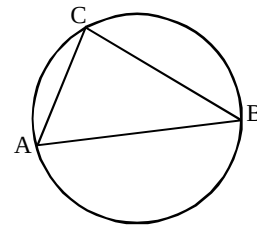
- The side and the height of a rhombus are 13 and 20 cms respectively. Find the area.
(a) 260 cm^2 (b) 275 cm^2
(c) 290 cm^2 (d) None of these
- The circumference of a circle is 44 metres. Find the area of the circle.
(a) 154 m^2 (b) 160 m^2
(c) 175 m^2 (d) 168 m^2
- The length and breadth of a rectangle are in the ratio 9 : 5. If its area is 720 m^2 , find its perimeter.
(a) 112 metre (b) 115 metre
(c) 110 metre (d) 118 metre
- How many squares are there in a 5 inch by 5 inch square grid, if the grid is made up one inch by one inch squares?
(a) 50 (b) 150
(c) 55 (d) 25
- If the ratio of areas of two squares is 9 : 1, the ratio of their perimeter is :
(a) 9 : 1 (b) 3 : 4
(c) 3 : 1 (d) 1 : 3
- A circle and a rectangle have the same perimeter. The sides of the rectangle are 18 cm and 26 cm. What is the area of the circle ?
(a) 88 cm^2 (b) 154 cm^2
(c) 1250 cm^2 (d) 616 cm^2
- If the perimeter and diagonal of a rectangle are 14 and 5 cms respectively, find its area.
(a) 12 cm^2 (b) 16 cm^2
(c) 20 cm^2 (d) 24 cm^2
- When the circumference and area of a circle are numerically equal, then the diameter is numerically equal to
(a) area (b) circumference
(c) 4 (d) 2π
- In a parallelogram, the length of one diagonal and the perpendicular dropped on that diagonal are 30 and 20 metres respectively. Find its area.
(a) 600 m^2 (b) 540 m^2
(c) 680 m^2 (d) 574 m^2
- The area of a triangle is 615 m^2 . If one of its sides is 123 metre, find the length of the perpendicular dropped on that side from opposite vertex.
(a) 15 metres (b) 12 metres
(c) 10 metres (d) None of these
- How many plants will be there in a circular bed whose outer edge measure 30 cms, allowing 4 cm^2 for each plant ?
(a) 18 (b) 750
(c) 24 (d) 120
- A square carpet with an area 169 m^2 must have 2 metres cut-off one of its edges in order to be a perfect fit for a rectangular room. What is the area of rectangular room?
(a) 180 m^2 (b) 164 m^2
(c) 152 m^2 (d) 143 m^2
- If the area of a circle decreases by 36%, then the radius of a circle decreases by
(a) 20% (b) 18%
(c) 36% (d) 64%
- The altitude drawn to the base of an isosceles triangle is 8 cm and the perimeter is 32 cm. The area of the triangle is
(a) 72 cm^2 (b) 60 cm^2
(c) 66 cm^2 (d) None of these
- The area of a square field is 576 km^2 . How long will it take for a horse to run around at the speed of 12 km/h ?
(a) 12 h (b) 10 h
(c) 8 h (d) 6 h
- Four equal circles are described about the four corners of a square so that each touches two of the others. If a side of the square is 14 cm, then the area enclosed between the circumferences of the circles is :
(a) 24 cm^2 (b) 42 cm^2
(c) 154 cm^2 (d) 196 cm^2
- The ratio between the length and the breadth of a rectangular park is 3 : 2. If a man cycling along the boundary of the park at the speed of 12km / hr completes one round in 8 minutes, then the area of the park (in sq. m) is:
(a) 15360 (b) 153600
(c) 30720 (d) 307200
- A wire can be bent in the form of a circle of radius 56 cm. If it is bent in the form of a square, then its area will be:
(a) 3520 cm^2 (b) 6400 cm^2
(c) 7744 cm^2 (d) 8800 cm^2
- The length of a room is double its breadth. The cost of colouring the ceiling at ` 25 per sq. m is ` 5,000 and the cost of painting the four walls at ` 240 per sq. m is ` 64,800. Find the height of the room.
(a) 4.5 m (b) 4 m
(c) 3.5 m (d) 5 m
- A metal cube of edge 12 cm is melted and formed into three smaller cubes. If the edges of two smaller cubes are 6 cm and 8 cm, then find the edge of the third smaller cube.
(a) 10 cm (b) 14 cm
(c) 12 cm (d) 16 cm
- A well 22.5 deep and of diameter 7 m has to be dug out. Find the cost of plastering its inner curved surface at ` 3 per sq. metre.
(a) ` 1465 (b) ` 1485
(c) ` 1475 (d) ` 1495

22. The length, breadth and height of a cuboid are in the ratio 1 : 2 : 3. The length, breadth and height of the cuboid are increased by 100%, 200% and 200%, respectively. Then, the increase in the volume of the cuboid will be
(a) 5 times (b) 6 times
(c) 12 times (d) 17 times
23. The surface area of a cube is 150 m^2 . The length of its diagonal is
(a) $5\sqrt{3} \text{ m}$ (b) 5 m
(c) $\frac{10}{\sqrt{3}} \text{ m}$ (d) 15 m
24. The length of the longest rod that can be placed in a room which is 12 m long, 9 m broad and 8 m high is
(a) 27 m (b) 19 m
(c) 17 m (d) 13 m
25. If the volume of a sphere is divided by its surface area, the result is 27 cms. The radius of the sphere is
(a) 9 cms (b) 27 cms
(c) 81 cms (d) 243 cms
26. The volume of water measured on a rectangular field $500 \text{ m} \times 300 \text{ m}$ is 3000 m^3 . Find the depth (amount) of rain that has fallen.
(a) 2 cms (b) 3 cms
(c) 4 cms (d) 3.5 cms
27. How many spherical bullets can be made out of a lead cylinder 28 cm high and with base radius 6 cm, each bullet being 1.5 cm in diameter?
(a) 1845 (b) 1824
(c) 1792 (d) 1752
28. Water flows out through a circular pipe whose internal diameter is 2 cm, at the rate of 6 metres per second into a cylindrical tank, the radius of whose base is 60 cm. By how much will the level of water rise in 30 minutes?
(a) 2 m (b) 4 m
(c) 3 m (d) 5 m
29. A spherical ball of lead, 3 cm in diameter, is melted and recast into three spherical balls. The diameter of two of these balls are 1.5 cm and 2 cm respectively. The diameter of the third ball is
(a) 2.5 cm (b) 2.66 cm
(c) 3 cm (d) 3.5 cm
30. A cube of 384 cm^2 surface area is melt to make x number of small cubes each of 96 mm^2 surface area. The value of x is
(a) 80,000 (b) 8
(c) 8,000 (d) 800
31. The capacity of a cylindrical tank is 246.4 litres. If the height is 4 metres, what is the diameter of the base?
(a) 1.4 m (b) 2.8 m
(c) 14 m (d) None of these
32. A conical cavity is drilled in a circular cylinder of 15 cm height and 16 cm base diameter. The height and the base diameter of the cone are same as those of the cylinder. Determine the total surface area of the remaining solid.
(a) $440 \pi \text{ cm}^2$ (b) $215 \pi \text{ cm}^2$
(c) $542 \pi \text{ cm}^2$ (d) $376 \pi \text{ cm}^2$
33. If the radius of a sphere is increased by 2 cm, then its surface area increases by 352 cm^2 . The radius of the sphere before the increase was:
(a) 3 cm (b) 4 cm
(c) 5 cm (d) 6 cm
34. A hollow sphere of internal and external diameters 4 cm and 8 cm respectively is melted into a cone of base diameter 8 cm. The height of the cone is:
(a) 12 cm (b) 14 cm
(c) 15 cm (d) 18 cm
35. The length and breadth of a playground are 36m and 21 m respectively. Poles are required to be fixed all along the boundary at a distance 3m apart. The number of poles required will be
(a) 39 (b) 38
(c) 37 (d) 40
36. A rectangular plate is of 6 m breadth and 12 m length. Two apertures of 2 m diameter each and one apertures of 1 m diameter have been made with the help of a gas cutter. What is the area of the remaining portion of the plate?
(a) 68.5 sq. m. (b) 62.5 sq m
(c) 64.5 sq. m (d) None of these
37. Four sheets $50 \text{ cm} \times 5 \text{ cm}$ are arranged without overlapping to form a square having side 55 cm. What is the area of inner square so formed?
(a) 2500 cm^2 (b) 2025 cm^2
(c) 1600 cm^2 (d) None of these
38. A garden is 24 m long and 14 m wide. There is a path 1 m wide outside the garden along its sides. If the path is to be constructed with square marble tiles $20 \text{ cm} \times 20 \text{ cm}$, the number of tiles required to cover the path is
(a) 1800 (b) 200
(c) 2000 (d) 2150
39. The length of a rectangular field is double its width. Inside the field there is a square-shaped pond 8 m long. If the area of the pond is $\frac{1}{8}$ of the area of the field, what is the length of the field?
(a) 32 m (b) 16 m
(c) 64 m (d) 20 m
40. A horse is tethered to one corner of a rectangular grassy field 40 m by 24 m with a rope 14 m long. Over how much area of the field can it graze?
(a) 154 cm^2 (b) 308 m^2
(c) 150 m^2 (d) None of these
41. The length of a cold storage is double its breadth. Its height is 3 metres. The area of its four walls (including the doors) is 108 m^2 . Find its volume.
(a) 215 m^3 (b) 216 m^3
(c) 217 m^3 (d) 218 m^3
42. The cost of the paint is ₹ 36.50 per kg. If 1 kg of paint covers 16 square feet, how much will it cost to paint outside of a cube having 8 feet each side?
(a) ₹ 692 (b) ₹ 768
(c) ₹ 876 (d) ₹ 972

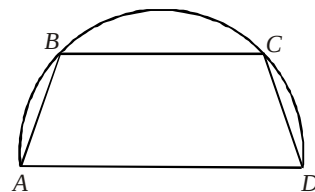
43. A cuboidal block of $6 \text{ cm} \times 9 \text{ cm} \times 12 \text{ cm}$ is cut up into an exact number of equal cubes. The least possible number of cubes will be:
 (a) 6 (b) 9
 (c) 24 (d) 30
44. A semicircular sheet of paper of diameter 28 cm is bent to cover the exterior surface of an open conical ice-cream cup. The depth of the ice-cream cup is
 (a) 10.12 cm (b) 8.12 cm
 (c) 12.12 cm (d) 13.27 cm
45. How many squares are there in a 5 inch by 5 inch square grid, if the grid is made up one inch by one inch squares?
 (a) 50 (b) 150
 (c) 55 (d) 25
46. A conical vessel of base radius 2 cm and height 3 cm is filled with kerosene. This liquid leaks through a hole in the bottom and collects in a cylindrical jar of radius 2 cm. The kerosene level in the jar is
 (a) $\pi \text{ cm}$ (b) 1.5 cm
 (c) 1 cm (d) 3 cm
47. A square contains four times the area of another square. If one side of the larger square be 4 cm greater than that of smaller square, then the perimeter of smaller square will be equal to
 (a) 8 cm (b) 16 cm
 (c) 24 cm (d) 32 cm
48. The length, breadth and height of a room are X, Y and Z feet respectively. The cost of whitewashing the four walls of this room is Rs.2500. Find the cost of preparing another room whose length, breadth and height are double that of the previous room.
 (a) 5000 (b) 12,500
 (c) 10,000 (d) 20,000
49. The altitude of a triangular billboard is one third of the base. If the cost of preparing this billboard is ₹ 11000, find the height of the triangle if the cost per sq dm is ₹ 10.
 (a) 285.5 m (b) 296.4 m
 (c) 270.8 m (d) 264.7 m
50. A solid wooden toy in the shape of a right circular cone is mounted on a hemisphere. If the radius of the hemisphere is 4.2 cm and the total height of the toy is 10.2 cm, find the volume of the wooden toy.



- (a) 104 cm^3 (b) 162 cm^3
 (c) 427 cm^3 (d) 266 cm^3
51. The dimensions of a field are 20 m by 9 m. A pit 10 m long, 4.5 m wide and 3 m deep is dug in one corner of the field and the earth removed has been evenly spread over the remaining area of the field. What will be the rise in the height of field as a result of this operation?
 (a) 1 m (b) 2 m
 (c) 3 m (d) 4 m
52. In a triangle ABC, points P, Q and R are the mid-points of the sides AB, BC and CA respectively. If the area of the triangle ABC is 20 sq. units, find the area of the triangle PQR
 (a) 10 sq. units (b) 5.3 sq. units
 (c) 5 sq. units (d) None of these
53. From a circular sheet of paper with a radius of 20 cm, four circles of radius 5 cm each are cut out. What is the ratio of the areas of uncut to the cut portion?
 (a) 1 : 3 (b) 4 : 1
 (c) 3 : 1 (d) 4 : 3
54. The figure shows a circle of diameter AB and radius 6.5 cm. If chord CA is 5 cm long, find the area of triangle ABC

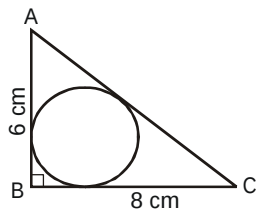


- (a) 60 sq. cm. (b) 30 sq. cm.
 (c) 40 sq. cm. (d) 52 sq. cm.
55. The sides of a triangle are 5, 12 and 13 units respectively. A rectangle is constructed which is equal in area to the triangle and has a width of 10 units. Then the perimeter of the rectangle is
 (a) 30 (b) 26
 (c) 13 (d) None of these
56. On a semicircle with diameter AD, chord BC is parallel to the diameter. Further, each of the chords AB and CD has length 2, while AD has length 8. What is the length of BC?



- (a) 7.5 (b) 7
 (c) 7.75 (d) None of these
57. One diagonal of a rhombus is 24 cm whose side is 13 cm. Find the area of the rhombus.
 (a) 25 sq. cm (b) 312 sq. cm.
 (c) 125 sq. cm. (d) 120 sq. cm.

58. The radius of the incircle in the given diagram will be

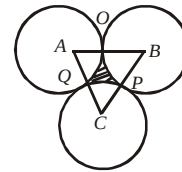


- (a) 1.8 cm (b) 2 cm
(c) 2.5 cm (d) 3.6 cm
59. A beam 9m long, 50 cm wide and 20 cm deep is made of wood which weighs 30 kg per m^3 , find the weight of the beam.
(a) 36 kg (b) 63 kg
(c) 40 kg (d) 39 kg
60. A square field of 2 sq. kilometres is to be divided into two equal parts by a fence which coincides with a diagonal. Find the length of the fence.
(a) 2 km (b) 4 km
(c) 6 km (d) 8 km
61. If a rectangular paper of length 6 cm. and width 3 cm. is rolled to form a cylinder with height equal to the width of the paper, then its base radius is –
(a) $\frac{6}{\pi}$ cm (b) $\frac{3}{\pi}$ cm
(c) $\frac{3}{2\pi}$ cm (d) $\frac{9}{2\pi}$ cm
62. A hollow spherical shell is made of metal of density 4.8 g/cm^3 . If its internal and external radii are 10 cm and 12 cm respectively, find the weight of the shell
(a) 15.24 kg (b) 12.84 kg
(c) 14.64 kg (d) None of these
63. If the sum of the circumferences of two circles with radii R_1 and R_2 is equal to the circumference of a circle of radius R , then
(a) $R_1 + R_2 = R$
(b) $R_1 + R_2 > R$
(c) $R_1 + R_2 < R$
(d) Nothing definite can be said about the relation among R_1, R_2 and R
64. The radius of a circle whose circumference is equal to the sum of the circumferences of the two circles of diameters 36 cm and 20 cm is
(a) 56 cm (b) 42 cm
(c) 28 cm (d) 16 cm
65. The area of the circle that can be inscribed in a square of side 6 cm is
(a) $36\pi \text{ cm}^2$ (b) $18\pi \text{ cm}^2$
(c) $12\pi \text{ cm}^2$ (d) $9\pi \text{ cm}^2$

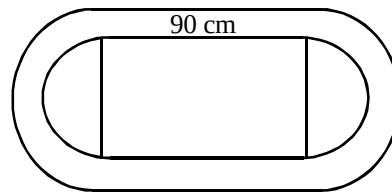
66. Circumference of a sector of angle p° of a circle with radius R is

- (a) $\frac{p}{180} \times 2\pi R$ (b) $\frac{p}{180} \times \pi R^2$
(c) $\frac{p}{360} \times 2\pi R$ (d) $\frac{p}{720} \times 2\pi R^2$

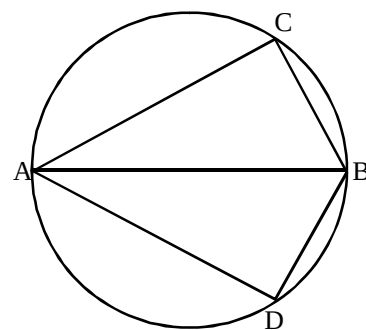
67. Three circles with centres A, B and C and with unit radii touch each other at O, P and Q . Find the area of the shaded region.



- (a) 0.16 sq. units (b) 1.21 sq. units
(c) 0.03 sq. units (d) 0.32 units
68. The inside perimeter of a practice running track with semi-circular ends and straight parallel sides is 312 m. The length of the straight portion of the track is 90 m. If the track has a uniform width of 2 m throughout, find its area.



- (a) 5166 m^2 (b) 5802.57 m^2
(c) 636.57 m^2 (d) 1273.14 m^2
69. AB is the diameter of the given circle, while points C and D lie on the circumference as shown. If AB is 15 cm. AC is 12 cm and BD is 9 cm, find the area of the quadrilateral $ABCD$.

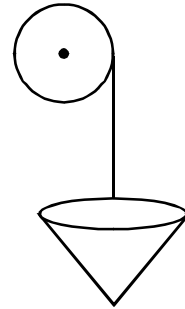


- (a) 54 sq. cm (b) 216 sq. cm
(c) 162 sq. cm (d) None of these
70. What is the greatest area (in sq. units) of a rectangle the sum of whose 3 sides is equal = 100.
(a) 625 (b) 1250
(c) 883.33 (d) 666.66

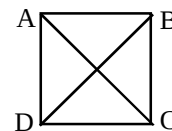
Standard Level

- 2 cm of rain has fallen on a sq. km of land. Assuming that 50% of the raindrops could have been collected and contained in a pool having a $100 \text{ m} \times 10 \text{ m}$ base, by what level would the water level in the pool have increased?
(a) 15 m (b) 20 m
(c) 10 m (d) 25 m
- The area of a right angled isosceles triangle whose hypotenuse is equal to 270 m is
(a) 19000 m^2 (b) 18225 m^2
(c) 17256 m^2 (d) 18325 m^2
- A right circular solid cylinder of base radius 4 cm and vertical height 22.5 cm is melted to form 8 equal solid spheres. If there is a process loss of 20% during such formation, then what is the radius of each of the solid sphere so formed?
(a) 2 cm (b) 3 cm
(c) 2.5 cm (d) 3.5 cm
- The volume of a sphere is changing @ 100 cc/min. The rate at which the surface area of the sphere is changing when the radius of the sphere = 10 cm, is
(a) $30 \text{ cm}^2 / \text{min}$ (b) $20 \text{ cm}^2 / \text{min}$
(c) $20\pi \text{ cm}^2 / \text{min}$ (d) $30\pi \text{ cm}^2 / \text{min}$
- There is an error of + 1.5% while measuring the radius of a sphere. What is the percentage error in calculating the volume of the sphere?
(a) 4.6% (b) 3.2%
(c) 9.5% (d) 4.3%
- If the radius of a circle is diminished by 10%, the area is diminished by
(a) 36% (b) 20%
(c) 19% (d) 10%
- A landowner increased the length and breadth of a rectangular plot by 10% and 20% respectively. Find the percentage change in the cost of the plot.
(a) 35% (b) 33%
(c) 22.22% (d) 32%
- In measuring the side of a square, an error of 5% in excess is made. The error % in the calculated area is,
(a) $10\frac{1}{4}\%$ (b) $10\frac{3}{4}\%$
(c) $1\frac{3}{4}\%$ (d) 25%
- In the given diagram a rope is wound round the outside of a circular drum whose diameter is 70 cm and a bucket is tied

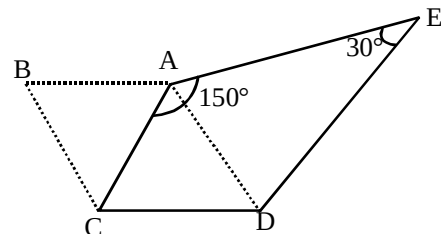
to the other end of the rope. Find the number of revolutions made by the drum if the bucket is raised by 11 m.



- (a) 10 (b) 2.5
(c) 5 (d) 5.5
- The perimeter of a sector of a circle of radius 5.7 m is 27.2 m. Find the area of the sector.
(a) 90.06 cm^2 (b) 135.09 cm^2
(c) 45 cm^2 (d) None of these
- $ABCD$ is a trapezium such that AB, DC is parallel and BC is perpendicular to them. If angle $(DAB) = 45^\circ$, $BC = 2 \text{ cm}$ and $CD = 3 \text{ cm}$, then find the length of AB ?
(a) 6 cm (b) 4 cm
(c) 3 cm (d) 5 cm
- $ABCD$ is a square of area 4, which is divided into four non overlapping triangles as shown in the fig. Then the sum of the perimeters of the triangles is



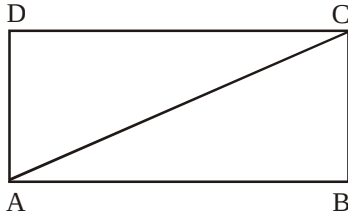
- (a) $8(2 + \sqrt{2})$ (b) $8(1 + \sqrt{2})$
(c) $4(1 + \sqrt{2})$ (d) $4(2 + \sqrt{2})$
- In $\triangle ACD$, $AD = AC$ and $\angle C = 2\angle E$. The distance between parallel lines AB and CD is h .



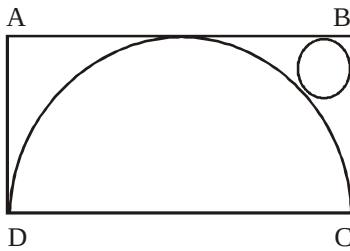
Then

- Area of parallelogram $ABCD$
 - Area of $\triangle ADE$
- (a) $I > II$ (b) $I < II$
(c) $I = II$ (d) Nothing can be said

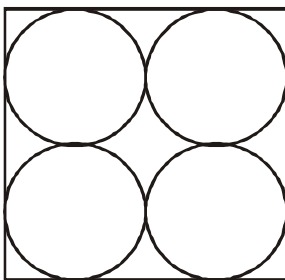
14. In the adjoining figure, $AC + AB = 5AD$ and $AC - AD = 8$. Then the area of the rectangle $ABCD$ is



- (a) 36 (b) 50
(c) 60 (d) Cannot be answered
15. The figure shows a rectangle $ABCD$ with a semi-circle and a circle inscribed inside it as shown. What is the ratio of the area of the circle to that of the semi-circle?



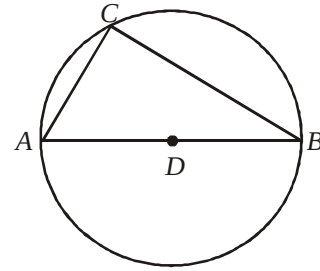
- (a) $(\sqrt{2}-1)^2$ (b) $2(\sqrt{2}-1)^2$
(c) $(\sqrt{2}-1)^2/2$ (d) None of these
16. A cone, a hemisphere and a cylinder stand on equal bases and have the same height, the height being equal to the radius of the circular base. Their total surface areas are in the ratio:
- (a) $(\sqrt{2}+1):3:4$ (b) $(\sqrt{3}+1):3:4$
(c) $\sqrt{2}:3:4$ (d) $\sqrt{3}:7:8$
17. Four identical coins are placed in a square. For each coin, the ratio of area to circumference is same as the ratio of circumference to area.



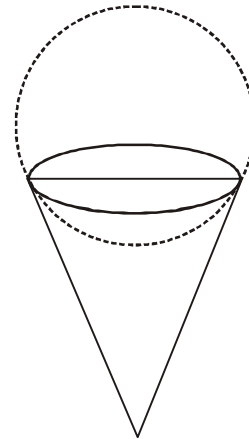
Then, find the area of the square that is not covered by the coins

- (a) $16(\pi-1)$ (b) $16(8-\pi)$
(c) $16(4-\pi)$ (d) $16\left(4-\frac{\pi}{2}\right)$

18. The figure shows a circle of diameter AB and radius 6.5 cm. If chord CA is 5 cm. long, find the area of $\triangle ABC$

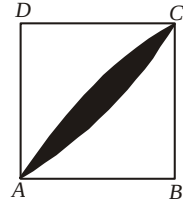


- (a) 60 sq. cm (b) 30 sq. cm
(c) 40 sq. cm (d) 52 sq. cm.
19. The diameter of hollow cone is equal to the diameter of a spherical ball. If the ball is placed at the base of the cone, what portion of the ball will be outside the cone –



- (a) 50% (b) less than 50%
(c) more than 50% (d) 100%
20. A slab of ice 8 inches in length, 11 inches in breadth, and 2 inches thick was melted and resolidified in the form of a rod of 8 inches diameter. The length of such a rod, in inches, is nearest to
- (a) 3 (b) 3.5
(c) 4 (d) 4.5
21. A passage 12 m long, 3m high and 4 m wide has two doors of 2.5 m by 1.5 m and a window of 2 m by 0.60 m. The cost of colouring the walls and ceiling at ` 15 per sq. m is
- (a) ` 1023 (b) ` 432
(c) ` 2029.5 (d) ` 1635
22. What is the side of the largest possible regular octagon that can be cut out of a square of side 1 cm?

- (a) $\sqrt{2}-1$ (b) $\frac{1}{\sqrt{2}+2}$ cm
(c) $\frac{1}{\sqrt{2}-1}$ cm (d) $\frac{\sqrt{2}}{\sqrt{2}+1}$ cm

23. If h be the height and α the semi-vertical angle of a right circular cone, then its volume is given by
- (a) $\frac{1}{3}\pi h^3 \tan^2 \alpha$ (b) $\frac{1}{3}\pi h^2 \tan^2 \alpha$
 (c) $\frac{1}{3}\pi h^2 \tan^3 \alpha$ (d) $\frac{1}{3}\pi h^3 \tan^3 \alpha$
24. If length, breadth and height of a cuboid is increased by $x\%$, $y\%$ and $z\%$ respectively then its volume is increased by
- (a) $\left[x + y + z + \frac{xy + xz + yz}{100} + \frac{xyz}{(100)^2} \right] \%$
 (b) $\left[x + y + z + \frac{xy + xz + yz}{100} \right] \%$
 (c) $\left[x + y + z + \frac{xyz}{(100)^2} \right] \%$
 (d) None of these
25. Consider a cylinder of height n cms and radius $\frac{3}{\pi}$ cms. A string of width h cms, when wound around the cylinder without keeping any space between two turns, covers the lateral surface of the cylinder completely. What is the required length of the string?
- (a) $\frac{6h}{n}$ cms (b) $\frac{12h}{n}$ cms
 (c) $\frac{36h}{n}$ cms (d) $6n$ cms
26. A pipes each of 3 inch diameter are to be replaced by a single pipe discharging the same quantity of water. What should be the diameter of the single pipe, if the speed of water is the same?
- (a) 6 inch (b) 3 inch
 (c) 9 inch (d) 12 inch
27. A sphere is melted and half of the molten liquid is used to form 11 identical cubes, whereas the remaining half is used to form 7 identical smaller spheres. The ratio of the side of the cube to the radius of the new small sphere is
- (a) $(4/3)^{1/3}$ (b) $(8/3)^{1/3}$
 (c) $(3)^{1/3}$ (d) 2
28. The greatest possible sphere is turned from a cubical block of wood. If the volume of the block removed be 35280 cu.in., the diameter of the sphere ($\pi = 22/7$) will be
- (a) 33 in. (b) 27 in.
 (c) 39 in. (d) None of these
29. A regular hexagonal prism has its perimeter of base as 600 cm and height 200 cm. How many litres of milk can it hold? Find the weight of milk if density is 0.8 gm/cc.
- (a) 4210, 4156.9 gm (b) 5196, 4156.9 kg
 (c) 5916, 5261.8 kg (d) 6412, 8296.1 kg
30. Find the area of an isosceles triangle whose equal sides are 8 cm each and the third side is 10 cm?
- (a) 10 cm² (b) 48 cm²
 (c) $5\sqrt{39}$ cm² (d) $10\sqrt{10}$ cm²
31. In the figure given below, $ABCD$ is a square of side 4 cm. Two quadrants of a circle with B and D as centres are draw. The radius of each of the quadrants is 4cm. What is the area of the shaded portion?
- 
- (a) 4.56 sq. cm (b) 9.12 sq. cm
 (c) 13.68 sq. cm (d) 7.76 sq. cm
32. Find the volume and the total surface area of a solid right pyramid of its height is 4 cm, and its square base is of side 6 cm.
- (a) 86 sq. cm. (b) 90 sq. cm.
 (c) 80 sq. cm. (d) 96 sq. cm.
33. What is the area of a regular hexagon inscribed in a circle of radius r ?
- (a) $2\sqrt{3} r^2$ sq. units (b) $\frac{3\sqrt{3}}{2} r^2$ sq. units
 (c) $\frac{2}{\sqrt{3}} r^2$ sq. units (d) $\frac{\sqrt{3}}{2} r^2$ sq. units
34. The radius of the incircle of triangle when sides are 18, 24 and 3 cms is
- (a) 2 cm. (b) 4 cm.
 (c) 6 cm. (d) 9 cm.
35. The central park of the city is 40 metres long and 30 metres wide. The mayor wants to construct two roads of equal width in the park such that the roads intersect each other at right angles and the diagonals of the park are also the diagonals of the small square formed at the intersection of the two roads. Further, the mayor wants that the area of the two roads to be equal to the remaining area of the park. What should be the width of the roads?
- (a) 10 metres (b) 12.5 metres
 (c) 14 metres (d) 15 metres
36. A spherical metal of radius 10 cm is melted and made into 1000 smaller spheres of equal sizes. In this process the surface area of the metal is increased by:
- (a) 1000 times (b) 100 times
 (c) 10 times (d) None of these
37. If the length, breadth and height of a cube are decreased, decreased and increased by 5%, 5% and 20% respectively, then what will be the impact on the surface area of the cube (in percentage terms)?
- (a) 7.25% (b) 5%
 (c) 8.33% (d) 20.75%

38. A shuttle cock used for playing badminton has the shape of a frustum of a cone mounted hemisphere (as shown in the figure). The external diameters of the frustum are 5 cm and 2 cm, the height of the entire shuttle cock is 7 cm. Then its external surface area is

(a) 4.25 cm^2 (b) 74.26 cm^2
(c) 73.26 cm^2 (d) 74.36 cm^2

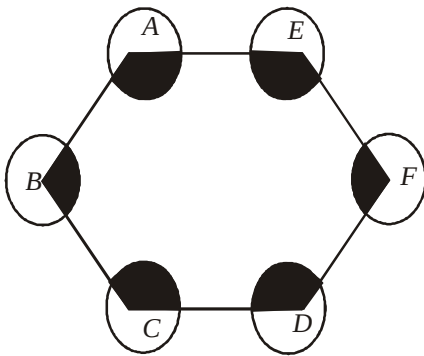
39. The sides of a triangle are 21, 20 and 13 cm. Find the area of the larger triangle into which the given triangle is divided by the perpendicular upon the longest side from the opposite vertex.

(a) 72 cm^2 (b) 96 cm^2
(c) 168 cm^2 (d) 144 cm^2

40. A solid sphere of radius 6 cm is melted into a hollow cylinder of uniform thickness. If the external radius of the base of the cylinder is 5 cm and its height is 32 cm, find the uniform thickness of the cylinder.

(a) 2 cm (b) 3 cm
(c) 1 cm (d) 3.5 cm

41. Find the sum of the areas of the shaded sectors given that $ABCDEF$ is any hexagon and all the circles are of same radius r with different vertices of the hexagon as their centres as shown in the figure.



(a) πr^2 (b) $2\pi r^2$
(c) $5\pi r^2/4$ (d) $3\pi r^2/2$

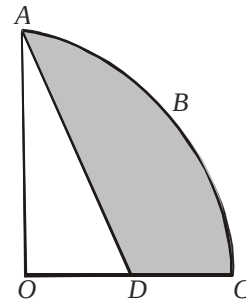
42. A cube is inscribed in a hemisphere of radius R , such that four of its vertices lie on the base of the hemisphere and the other four touch the hemispherical surface of the half-sphere. What is the volume of the cube?

(a) $0.25 R^3$ (b) $0.67 \sqrt{\frac{2}{3}} R^3$
(c) $0.5 \sqrt{\frac{2}{3}} R^3$ (d) $0.67 R^3$

43. Two different sides of a parallelogram are 8 cm and 6 cm and the ratio of the diagonals is 3 : 4. Find the difference between the lengths of the diagonals?

(a) 5 cm (b) 7 cm
(c) 6 cm (d) $\sqrt{8}$ cm

44. In the figure given below, $ABCO$ represents a quadrant of a circle of radius 10.5 cm with centre O . Calculate the area of shaded portion, if $OD = DC$.



(a) 59 cm^2 (b) 69 cm^2
(c) 79 cm^2 (d) 49 cm^2

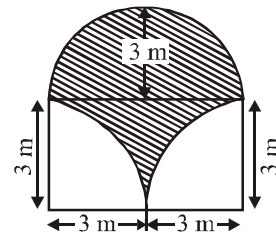
45. The volume of spheres are proportional to the cubes of their radii. Two spheres of the same material weigh 3.6 kg and 2.7 kg and the radius of the smaller one is 2 cm. If the two were melted down and formed into a single sphere, what would be its radius?

(a) 4 cm (b) 4.3 cm
(c) 3 cm (d) 2.6 cm

46. If the sides 50 m and 130 m of the triangular field meet at an angle of 72° , then find the area in which wheat is cultivated. ($\sin 72^\circ = 0.9510$, $\cos 72^\circ = 0.309$).

(a) $100 \pi \text{ m}^2$ (b) $125 \pi \text{ m}^2$
(c) $160 \pi \text{ m}^2$ (d) None of these

47. In the adjoining figure is a park in which shaded area is to be covered by grass. If the rate of covering with grass is ₹ 0.70 per sq. m.



Find the expenditure of covering its field with grass

($\pi = 22/7$)

(a) ₹ 12.60 (b) ₹ 6.30
(c) ₹ 9.30 (d) ₹ 10.30

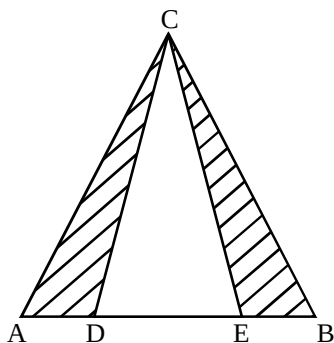
48. $ABCD$ is a quadrilateral. The diagonals of $ABCD$ intersect at the point P . The area of the triangles APD and BPC are 27 and 12, respectively. If the areas of the triangles APB and CPD are equal, then the area of triangle APB is

(a) 12 (b) 18
(c) 15 (d) 16

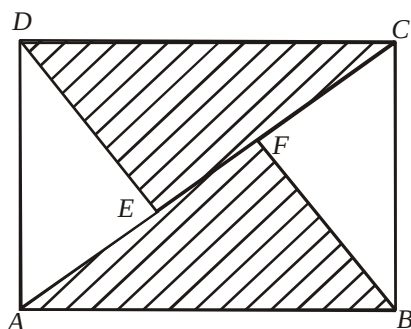
49. Two circles of radius 1 cm touch at point P . A third circle is drawn through the points A , B and C such that PA is the diameter of the first circle and BC perpendicular to AP is the diameter of the second circle. The radius of the third circle in cm.

(a) $\frac{9}{5}$ (b) $\frac{7}{4}$
(c) $\frac{\sqrt{10}}{2}$ (d) 2

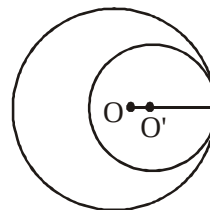
50. A toy is in the shape of a right circular cylinder with a hemisphere on one end and a cone on the other. The height and radius of the cylindrical part are 13 cm and 5 cm respectively. The radii of the hemispherical and conical parts are the same as that of the cylindrical part. Calculate the surface area of the toy if the height of conical part is 12 cm.
- (a) 1440 cm^2 (b) 385 cm^2
(c) 1580 cm^2 (d) 770 cm^2
51. A square hole of cross-sectional area 4 cm^2 is drilled across a cube with its length parallel to a side of the cube. If an edge of the cube measures 5 cm, what is the total surface area of the body so formed?
- (a) 158 cm^2 (b) 190 cm^2
(c) 166 cm^2 (d) 182 cm^2
52. In the equilateral triangle ABC , $AD = DE = BE$, D and E lies on the AB . If each side of the triangle (i.e., AB , BC and AC) be 6 cm, then the area of the shaded region is:



- (a) 9 cm^2 (b) $6\sqrt{3} \text{ cm}^2$
(c) $5\sqrt{3} \text{ cm}^2$ (d) None of these
53. $ABCD$ is a rectangle of dimensions $6 \text{ cm} \times 8 \text{ cm}$. DE and BF are the perpendiculars drawn on the diagonal of the rectangle. What is the ratio of the shaded to that of unshaded region?



- (a) $7 : 3$ (b) $16 : 9$
(c) $4 : 3\sqrt{2}$ (d) Data insufficient
54. Two circles touch internally and their centres are O and O' as shown. The sum of their areas is $180\pi \text{ sq. cm.}$ and the distance between their centres is 6 cm.

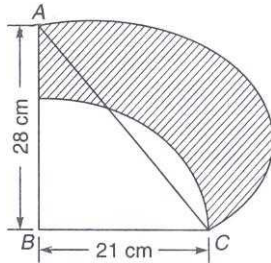


- What is the diameter of the larger circle ?
- (a) 16 cm (b) 12 cm
(c) 18 cm (d) 24 cm

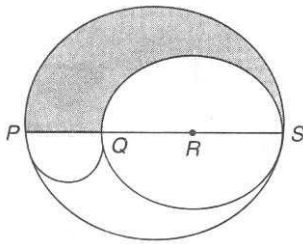
Directions for Qs 55-58 : The given question is followed by three statements labelled I, II and III. You have to study the question and all the three statements given do decide whether any information provided in the statement(s) is/are redundant and can be dispensed with while answering the given question.

55. What is the are of the hall ?
- I. Material cost of flooring per square metre is ` 2.50.
II. Labour cost of flooring the hall is ` 3500
III. Total cost of flooring the hall is ` 14500
- (a) I and II only (b) II and III only
(c) All I, II and III (d) None of these
56. What is the area of a right-angled triangle ?
- I. The perimeter of the triangle is 30 cm.
II. The ratio between the base and the height of the triangle is 5 : 12.
III. The area of the triangle is equal to the area of a rectangle of length 10 cm.
- (a) I and II only (b) II and III only
(c) I, and III only (d) III, and either I or II only
57. What is the area of the given rectangle?
- I. Perimeter of the rectangle is 60 cm.
II. Breadth of the rectangle is 12 cm.
III. Sum of two adjacent sides is 30 cm.
- (a) I only (b) II only
(c) III only (d) I or III only
58. A solid metallic cone is melted and recast into a sphere. What is the radius of the sphere ?
- I. The radius of the base of the cone is 2.1 cm.
II. The height of the cone is four times the radius of its base.
III. The height of the cone is 8.4 cm.
- (a) Only I and II (b) Only II and III
(c) Only I and III (d) Any two of the three
59. In a city, there is a circular park. There are four points of entry into the park, namely P , Q , R and S . Three paths were constructed which connected the points PQ , RS and PS . The length of the path PQ is 10 units and the length of the path RS is 7 units. Later, the municipal corporation extended the paths PQ and RS so that they meet at a point T on the main road outside the park. The path from Q to T measures 8 units and it was found that the angle PTS is 60° . Find the area (in square units) enclosed by the paths PT , TS and PS .
- (a) $36\sqrt{3}$ (b) $54\sqrt{3}$
(c) $72\sqrt{3}$ (d) $90\sqrt{3}$

60. In the figure, ABC is a right angled triangle with $B = 90^\circ$, $BC = 21$ cm and $AB = 28$ cm. With AC as diameter of a semicircle and with BC as radius, a quarter circle is drawn. Find the area of the shaded portion correct to two decimal places

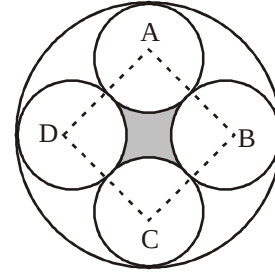


- (a) 428.75 cm^2 (b) 857.50 cm^2
 (c) 214.37 cm^2 (d) 371.56 cm^2
61. PQRS is the diameter of a circle of radius 6 cm. The lengths PQ, QR and RS are equal. Semi-circles are drawn with PQ and QS as diameters as shown in the figure alongside. Find the ratio of the area of the shaded region to that of the unshaded region.

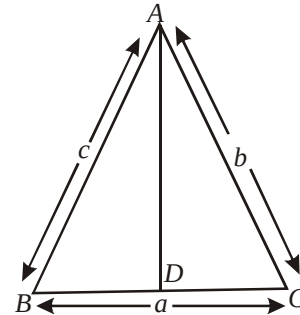


- (a) 1 : 2 (b) 25 : 121
 (c) 5 : 18 (d) 5 : 13
62. In the figure below, the radius of the bigger circle is $(\sqrt{2} + 1)$ cm and the radius of all the smaller circles are equal. Each

of the smaller circles touches two of the other three smaller circles and the larger circle as shown. Find the area (in cm^2) of the shaded portion.



- (a) $2 - \frac{\pi}{2}$ (b) 1
 (c) $\pi/4$ (d) Cannot be determined
63. In a triangle ABC , AD is the angle bisector of $\angle BAC$ and $\angle BAD = 60^\circ$. What is the length of AD ?



- (a) $\frac{b+c}{bc}$ (b) $\frac{bc}{b+c}$
 (c) $\sqrt{b^2 + c^2}$ (d) $\frac{(b+c)^2}{bc}$

Expert Level

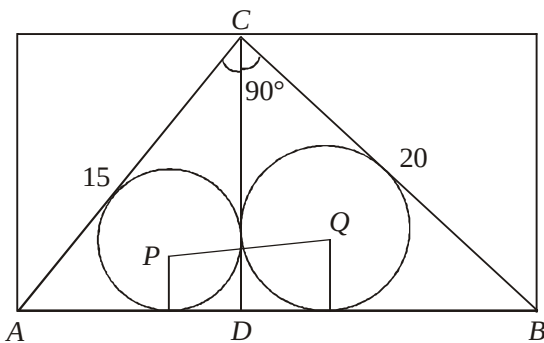
1. A square is inscribed in a circle which is inscribed in an equilateral triangle. If one side of the triangle is 'a', find the area of the square.

(a) $\frac{1^2}{3}$ (b) $\frac{a^2}{6}$
(c) $\frac{3a^2}{8}$ (d) $\frac{\pi a^2}{12}$

2. There is a solid cube with side 10 m. If the largest possible cone is carved out of it, then what is the surface area of the remaining part of the cube?

(a) $600 + 25\sqrt{5}\pi$ (b) $500 + 25\sqrt{5}\pi$
(c) $600 - 25(\sqrt{5} + 1)\pi$ (d) $600 + 25(\sqrt{5} - 1)\pi$

3. In the figure given below, ACB is a right angled triangle. CD is the altitude. Circles are inscribed within the triangles ACD , BCD . P and Q are the centres of the circles. The distance PQ is

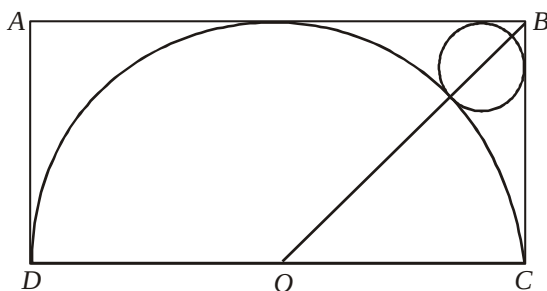


(a) 5 (b) $\sqrt{50}$
(c) 7 (d) 8

4. A right pyramid 15 cm high stands on a square base of side 16 cm. Find its total surface area.

(a) 376 sq. cm (b) 1280 sq. cm
(c) 736 sq. cm (d) 800 sq. cm

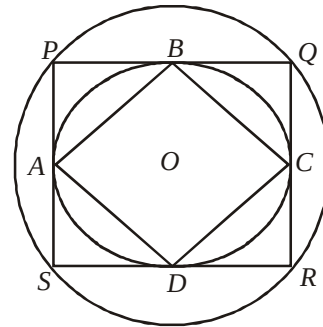
5. The figure shows the rectangle $ABCD$ with a semicircle and a circle inscribed inside it as shown. What is the ratio of the area of the circle to that of the semicircle?



(a) $(\sqrt{2} - 1)^2 : 1$ (b) $2(\sqrt{2} - 1)^2 : 1$
(c) $(\sqrt{2} - 1)^2 : 2$ (d) None of these

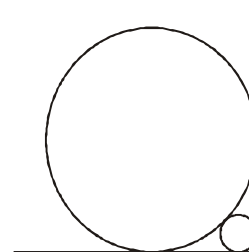
6. The figure shows two concentric circles with centre O . $PQRS$ is a square inscribed in the outer circle. It also circumscribes the inner circle, touching it at point B, C, D and A .

What is the ratio of the perimeter of the outer circle to that of polygon $ABCD$?



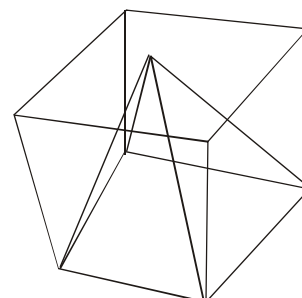
(a) $\frac{\pi}{4}$ (b) $\frac{3\pi}{2}$
(c) $\frac{\pi}{2}$ (d) π

7. A circle with radius 2 is placed against a right angle. Another smaller circle is also placed as shown in the adjoining figure. What is the radius of the smaller circle?



(a) $3 - 2\sqrt{2}$ (b) $4 - 2\sqrt{2}$
(c) $7 - 4\sqrt{2}$ (d) $6 - 4\sqrt{2}$

8. A regular square pyramid is placed in a cube so that the base of the pyramid and that of the cube coincide. The vertex of the pyramid lies on the face of the cube opposite to the base, as shown. An edge of the cube is 7 inches.

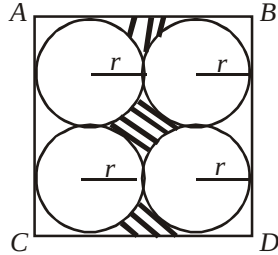


How many square inches (approximately) are in the positive difference between the surface area of the cube and the surface area of the pyramid?

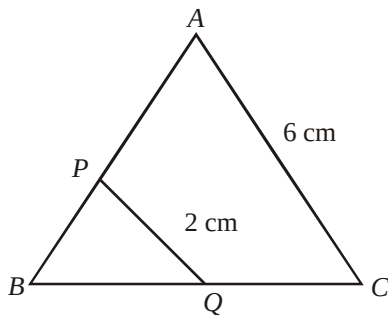
- (a) 134.4 (b) 133.4
(c) 138.4 (d) 135.4

9. In the figure given below $ABCD$ is a square of side 5 cm and all the four circles are of equal radius. What is the area of the shaded region?

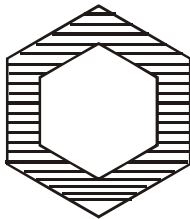
- (a) $\frac{25}{8} (4 - \pi)$ sq. cm.
(b) $\frac{9}{4} (4 - \pi)$ sq. cm.
(c) $\frac{1}{4} (8 - \pi)$ sq. cm.
(d) None of these



10. ABC is an equilateral triangle with side 6 cm. BPQ is a small equilateral triangle of side 2 cm cut out from ABC . How many such small triangles can be cut out from ABC ?

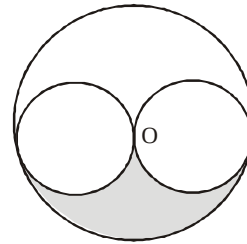


- (a) 8 (b) 9
(c) 12 (d) 16
11. Two concentric regular hexagons are drawn such that the sides of one are parallel to the sides of the other. If the side of the outer hexagon is 6 cms and the shortest distance between one side of one hexagon and the closest parallel



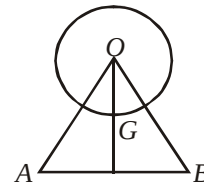
side of the other hexagon is $2\sqrt{3}$ cm. find the shaded area (in sq. cm).

- (a) $36\sqrt{3}$ (b) $81\sqrt{3}/2$
(c) $48\sqrt{3}$ (d) None of these
12. In the following figure, three circles are given. The two smaller circles are equal. The radius of the larger circle is 8 cm. O is the centre of the larger circle.



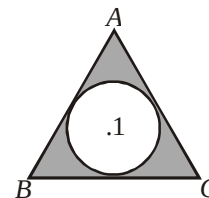
Find area of shaded portion.

- (a) 16π (b) 20π
(c) 4π (d) 8π
13. In the given figure, G , a point on the circle, is the centroid of the triangle OAB .



Find the ratio of the area of the equilateral triangle OAB to that of the circle whose centre is O .

- (a) $7\sqrt{3} : 44$ (b) $21\sqrt{3} : 88$
(c) $7\sqrt{3} : 88$ (d) $21\sqrt{3} : 44$
14. In ΔABC , $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$ and the side $a = 2$, then area of the triangle is
- (a) 1 (b) 2
(c) $\sqrt{3}/2$ (d) $\sqrt{3}$
15. Find the area of the shaded region if ΔABC is an equilateral triangle of side 6 cm.



- (a) 6.15 cm. (b) 5.15 cm.
(c) 4 cm. (d) 3.12 cm.
16. A city has a park shaped as a right angled triangle. The length of the longest side of this park is 80 m. The Mayor of the city wants to construct three paths from the corner point opposite to the longest side such that these three paths divide the longest side into four equal segments. Determine the sum of the squares of the lengths of the three paths.
- (a) 4000 m (b) 4800 m
(c) 5600 m (d) 6400 m
17. A conical tent of given capacity has to be constructed. The ratio of the height to the radius of the base for the minimum area of canvas required for the tent is
- (a) 1 : 2 (b) 2 : 1
(c) $1 : \sqrt{2}$ (d) $\sqrt{2} : 1$

18. A circular tent is cylindrical to a height of 3 metres and conical above it. If its diameter is 105 m and the slant height of the conical portion is 53 m, calculate the length of the canvas 5 m wide to make the required tent.

(a) 3894 (b) 973.5
(c) 1947 m (d) 1800 m

19. All five faces of a regular pyramid with a square base are found to be of the same area. The height of the pyramid is 3 cm. The total area of all its surfaces (in cm^2) is

(a) 8 (b) 10
(c) 12 (d) 16

Directions for Question 20:

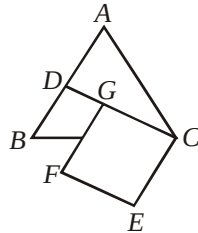
There are 300 coins, each coin having radius 2 cm and height 1 cm. The coins are so kept that each coin touches the other two. The base has 3 coins and the figure is built upon this base.

20. Find the volume of the region enclosed by the coins.

(a) $600(2\sqrt{3} - \pi)\text{cm}^3$ (b) $400(2\sqrt{3} - \pi)\text{cm}^3$
(c) $100(2\sqrt{3} - \pi)\text{cm}^3$ (d) $200(2\sqrt{3} - \pi)\text{cm}^3$

21. Find the ratio of the areas of an equilateral triangle ABC and square $EFGC$, if G is the centroid of the triangle ABC .

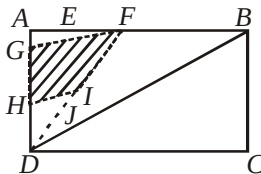
(a) 27 : 16
(b) 1 : 4
(c) $4 : 3\sqrt{3}$
(d) $3\sqrt{3} : 4$



22. In a triangle ABC , the lengths of the sides AB and AC equal 17.5 cm and 9 cm respectively. Let D be a point on the line segment BC such that AD is perpendicular to BC . If $AD = 3$ cm, then what is the radius (in cm) of the circle circumscribing the triangle ABC ?

(a) 17.05 (b) 27.85
(c) 22.45 (d) 26.25

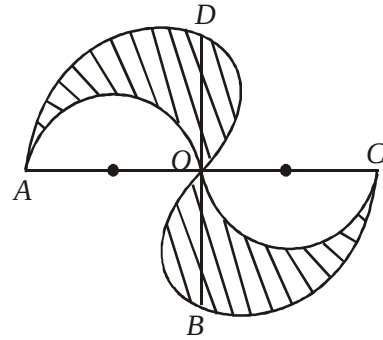
23. In rectangle $ABCD$, E, F and G, H are points of trisection of AB and AD respectively. Also, I and J are points of trisection of line FD .



Find the area (in sq. units) of the shaded region, if the area of the rectangle is 216 sq. units

(a) 32 (b) 24
(c) 40 (d) 35

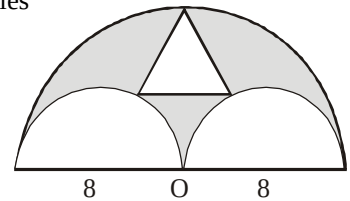
24. In the given figure below, the boundary of the shaded region comprises of four semicircles and two quarter circles. If $OA = OB = OC = OD = 7$ cm and the straight lines AC and BD are perpendicular to each other, find the length of the boundary



(a) 68 cm (b) 49 cm
(c) 66 cm (d) 44 cm

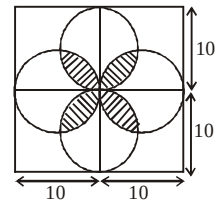
25. Find the area of the shaded region in the diagram below where the given triangle is isosceles with vertices of base lying on axis of the radius perpendicular to the diameters of the two small semicircles

(a) $8 + 16\sqrt{2}$
(b) $16 - \sqrt{2}$
(c) $16(\pi - 1)$
(d) 16π

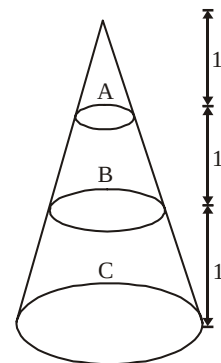


26. Find the area of the shaded region. [All the circles shown in the figure are congruent]

(a) $25\left(\frac{\pi}{2} - 1\right)$
(b) 50π
(c) $100\left(\frac{\pi}{2} - 1\right)$
(d) $200(\pi - 1)$



27. A right circular cone is divided into 3 portions $A < B$ and C by planes parallel to the base as shown in the figure.



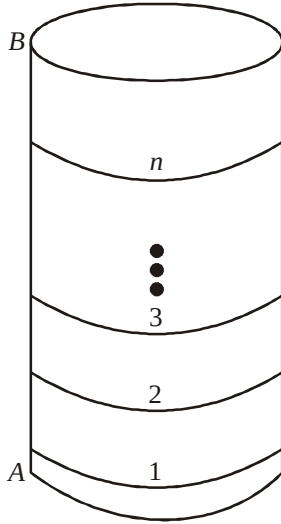
The height of each portion is 1 unit. calculate.

- (1) the ratio of the volume of A to the volume of B .
(2) the ratio of the volume of B to that of C .
(3) the ratio of the area of the curved surface of B to that of C .

(a) 1 : 7, 7 : 19, 5 : 3 (b) 1 : 7, 19 : 7, 5 : 3
(c) 1 : 7, 19 : 7, 5 : 3 (d) 1 : 7, 7 : 19, 3 : 5

Directions for Qs 28–30 : Answer the questions on the basis of the information given

Consider a cylinder of height h cm and radius $r = \frac{2}{\pi}$ cms as shown in the figure (not drawn to scale).

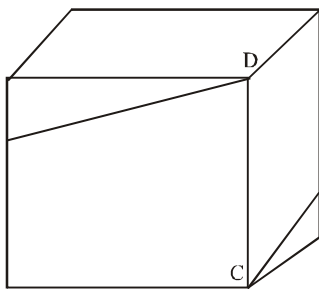


A string of a certain length, when wound on its cylindrical surface, starting at point A and ending at point B , gives a maximum of n turns (in other words, the string's length is the minimum length required to wind n turns).

28. What is the vertical spacing in cms between two consecutive turns ?

- (a) h/n
- (b) $h/\sqrt{n}$
- (c) h/n^2
- (d) Cannot be determined with given information

29. The same string, when wound on the exterior four walls of a cube of side n cms, starting at point C and ending at point D , can give exactly one turn (see figure, not drawn to scale)/ The length of the string, in cms, is



- (a) $\sqrt{2}n$
- (b) $\sqrt{17}n$
- (c) n
- (d) $\sqrt{13}n$

30. In the setup of the previous two questions, how is h related to n

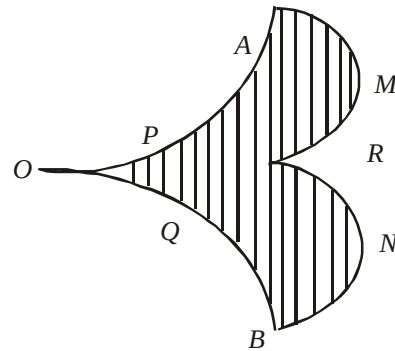
- (a) $h = \sqrt{2}n$
- (b) $h = \sqrt{17}n$
- (c) $h = n$
- (d) $h = \sqrt{13}n$

31. What is the area of the shaded region show, if the radius of each circle is equal to the side of the hexagon, which in turn is equal to 6 cm, and A and B are the centres of the circles?

- (a) $12(2\sqrt{13} - \pi)$ sq. cm
- (b) $86(\sqrt{3} - 2\pi)$ sq. cm
- (c) $6(9\sqrt{3} - 4\pi)$ sq. cm
- (d) $4(8\sqrt{3} - 2\pi)$ sq. cm

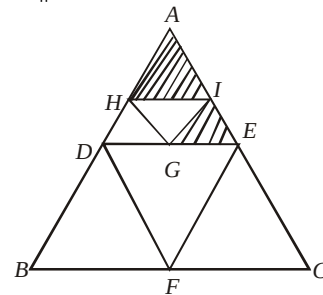


32. If $AB = 10$ cm, what is the area of the shaded portion ? it is given that OPA and OQB are quadrants of a circle and AB is a tangent to them. AMR and RNB are two identical semicircles.



- (a) $25(2 - \pi/4)$
- (b) $25\pi/4$
- (c) $50 - 25\pi/2$
- (d) None of these

33. In the diagram $AD = DB$ and $AH = HD$ Find the ratio of the area of the shaded portion to that of the triangle DEF , if $DE \parallel BC$ and $HG \parallel AE$



- (a) 2 : 3
- (b) 1 : 3
- (c) 1 : 2
- (d) $1 : 1\frac{1}{3}$

34. Carpenter Rajesh has a circular piece of plywood of diameter 30 feet. He has cut out two disks of diameter 20 feet and 10 feet. What is the diameter of the largest disk that can be cut out from the remaining portion of the plywood piece?

- (a) > 8.00 feet and ≤ 8.20 feet
- (b) > 8.21 feet and ≤ 8.40 feet
- (c) > 8.41 feet and ≤ 8.60 feet
- (d) > 8.61 feet and ≤ 8.80 feet

35. Consider a square $ABCD$ of side 60 cm. It contains arcs BD and AC drawn with centres at A and D respectively. A circle is drawn such that it touches side AB , arcs BD and arc AC . What is the radius of the circle?

(a) 9 cm (b) 10 cm
(c) 12 cm (d) 15 cm

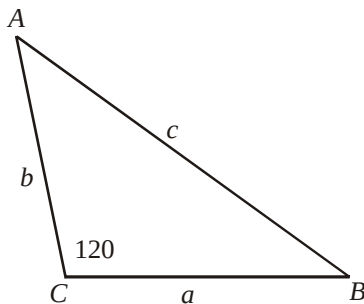
36. Rakhal is looking for a field where he can graze his cow. He finds a local farmer, Gopal, who agrees to rent his 2 field to Rakhal for `1000 a year. Rakhal finds a post in the field and ties his cow to the post with a 25 feet rope. After some months, Gopal tells Rakhal that he will build a shed with four walls on the field with the post as one of the corner posts. The shed would be 15 feet by 10 feet. Rakhal agrees but he realizes that this arrangement would reduce the available area for grazing. What should be the modified rent to compensate for this loss grazing area if Rakhal has to keep the cow tied to the same post with the same rope ?

(a) `800 (b) `880
(c) `888 (d) `930

37. Find the area of the triangle inscribed in a circle circumscribed by a square made by joining the midpoints of the adjacent sides of a square of side a .

(a) $3a^2/16$ (b) $\frac{3\sqrt{3}a^2}{16}$
(c) $3/4 a^2(\pi - 1/2)$ (d) $\frac{3\sqrt{3}a^2}{32}$

38. Two persons start walking on a road that diverge at an angle of 120° . If they walk at the rate of 3 km/h and 2 km/h respectively. Find the distance between them after 4 hours.

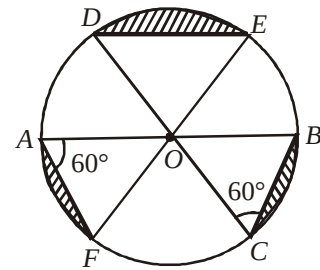


(a) $4\sqrt{19}$ km (b) 5 km
(c) 7 m (d) $8\sqrt{19}$ km

39. A right elliptical cylinder full of petrol has its widest elliptical side 2.4 m and the shortest 1.6 m. Its height is 7 m. Find the time required to empty half the tank through a hole of diameter 4 cm if the rate of flow of petrol is 120 m/min

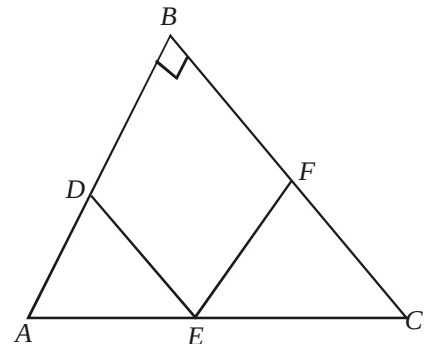
(a) 60 min (b) 90 min
(c) 75 min (d) 70 min

40. In the adjoining figure O is the centre of the circle with radius r . AB , CD and EF are the diameters of the circle. $\angle OAF = \angle OCB = 60^\circ$. What is the area of the shaded region?



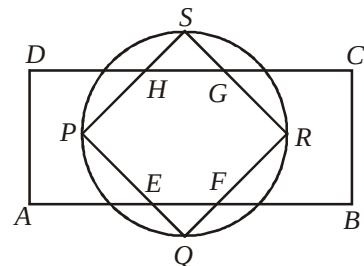
(a) $\frac{r^2}{2} \left(\pi - \frac{3\sqrt{3}}{2} \right)$ (b) $\frac{r^2}{2} \left(\pi - \frac{3\sqrt{3}}{4} \right)$
(c) $\frac{r^2}{3} \left(\pi - \frac{2\sqrt{3}}{3} \right)$ (d) Data insufficient

41. In the adjoining figure ABC is a right angled triangle, $BDEF$ is a square, $AE = 7.5$ cm and $AC = 18$ cm. What is the area of triangle ABC ?



(a) 76.621 cm^2 (b) 70.054 cm^2
(c) 83.25 cm^2 (d) 90.90 cm^2

42. In the adjoining figure $ABCD$ is a rectangle in which length is twice of breadth. H and G divide the line CD into three equal parts. Similarly points E and F trisect the line AB . A circle $PQRS$ is circumscribed by a square $PQRS$ which passes through the points E , F , G and H . What is the ratio of areas of circles to that of rectangle?



(a) $3\pi : 7$ (b) $3\pi : 4$
(c) $25\pi : 72$ (d) $32\pi : 115$

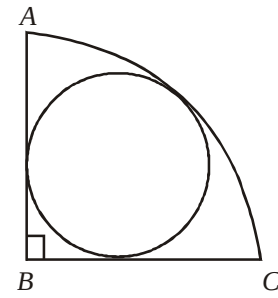
43. In an equilateral D , 3 coins of radii 1 unit each are kept in such a way that they touch each other and also the sides of the triangle. What is the area of the triangle (in sq. units)?

(a) $4 + 5\sqrt{2}$ (b) $6 + 4\sqrt{3}$
(c) $4 + 6\sqrt{3}$ (d) $3 + 8\sqrt{3}$

44. The length of the sides CB and CA of a triangle ABC are given by a and b , and the angle C is $2\pi/3$. The line CD bisects the angle C and meets AB at D . Then the length of CD is

- (a) $\frac{a^2 + b^2}{2(a+b)}$ (b) $\frac{ab}{2(a+b)}$
 (c) $\frac{1}{a+b}$ (d) $\frac{ab}{a+b}$

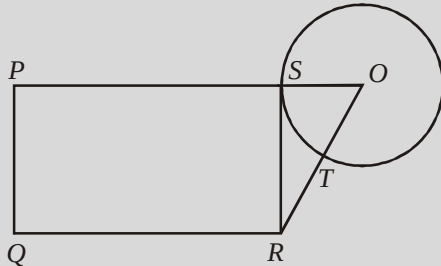
45. If ABC is a quarter circle and a circle is inscribed in it and if $AB = 1$ cm, find radius of the smaller circle.



- (a) $\sqrt{2} - 1$ (b) $(\sqrt{2} + 1)/2$
 (c) $\sqrt{2} - 1/2$ (d) $1 - 2\sqrt{2}$

Test Yourself

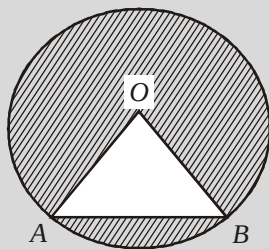
1. PQRS is a square. SR is a tangent (at point S) to the circle with centre O and TR = OS. Then the ratio of area of the circle to the area of the square is



- (a) $\pi/3$ (b) $11/7$
(c) $3/\pi$ (d) $7/11$
2. In the diagram, all triangles are equilateral. If $AB = 16$, then the total area of all the black triangles is



- (a) $25\sqrt{3}$ (b) $27\sqrt{3}$
(c) $35\sqrt{3}$ (d) $37\sqrt{3}$
3. The length and breadth of the floor of the room are 20 feet and 10 feet respectively. Square tiles of 2 feet length of different colours are to be laid on the floor. Black tiles are laid in the first row on all sides. If white tiles are laid in the one-third of the remaining and blue tiles in the rest, how many blue tiles will be there?
- (a) 16 (b) 24
(c) 32 (d) 48
4. O is the centre of a circle of radius 5 cm. The chord AB subtends an angle of 60° at the centre. Find the area of the shaded portion (approximate value).



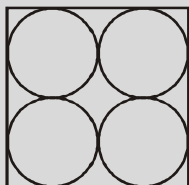
- (a) 50 cm^2 (b) 62.78 cm^2
(c) 49.88 cm^2 (d) 67.67 cm^2

5. The length, breadth and height of a cuboid are in the ratio 1 : 2 : 3. The length, breadth and height of the cuboid are increased by 100%, 200% and 200%, respectively. Then, the increase in the volume of the cuboid will be :
(a) 5 times (b) 6 times
(c) 12 times (d) 17 times
6. A boat having a length 3 m and breadth 2 m is floating on a lake. The boat sinks by 1 cm when a man gets on it. The mass of man is :
(a) 12 kg (b) 60 kg
(c) 72 kg (d) 96 kg
7. An ice-cream company makes a popular brand of ice-cream in rectangular shaped bar 6 cm long, 5 cm wide and 2 cm thick. To cut the cost, the company has decided to reduce the volume of the bar by 20%, the thickness remaining the same, but the length and width will be decreased by the same percentage amount. The new length L will satisfy :
(a) $5.5 < L < 6$ (b) $5 < L < 5.5$
(c) $4.5 < L < 5$ (d) $4 < L < 4.5$
8. A hemispherical bowl is filled to the brim with a beverage. The contents of the bowl are transferred into a cylindrical vessel whose radius is 50% more than its height. If the diameter is same for both the bowl and the cylinder, the volume of the beverage in the cylindrical vessel is:
(a) $66\frac{2}{3}\%$ (b) $78\frac{1}{2}\%$
(c) 100% (d) More than 100%
9. If the curved surface area of a cone is thrice that of another cone and slant height of the second cone is thrice that of the first, find the ratio of the area of their base.
(a) 81 : 1 (b) 9 : 1
(c) 3 : 1 (d) 27 : 1
10. The minute hand of a clock is 10 cm long. Find the area of the face of the clock described by the minute hand between 9 a.m. and 9 : 35 a.m.
(a) 183.3 cm^2 (b) 366.6 cm^2
(c) 244.4 cm^2 (d) 188.39 cm^2
11. Find the length of the string wound on a cylinder of height 48 cm and a base diameter of $5\frac{1}{11}$ cm. The string makes exactly four complete turns round the cylinder while its two ends touch the cylinder's top and bottom.
(a) 192 cm (b) 80 cm
(c) 64 cm (d) Cannot be determined

12. Iron weighs 8 times the weight of oak. Find the diameter of an iron ball whose weight is equal to that of a ball of oak 18 cm in diameter.

(a) 4.5 cm (b) 9 cm
(c) 12 cm (d) 15 cm

13. Four identical coins are placed in a square. For each coin the ratio of area to circumference is same as the ratio of circumference to area. Then find the area of the square that is not covered by the coins

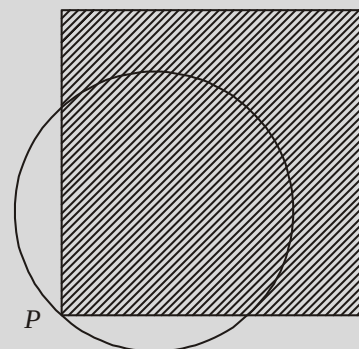


(a) $16(\pi-2)$ (b) $16(8-\pi)$
(c) $16(4-\pi)$ (d) $16(4-\pi/2)$

Directions for Questions 14 & 15 : Answer the questions on the basis of the information given below.

A punching machine is used to punch a circular hole of diameter two units from a square sheet of aluminium of width 2 units, as shown below. The hole is punched such that the circular hole

touches one corner P of the square sheet and the diameter of the hole originating at P is in line with a diagonal of the square.



14. The proportion of the sheet area that remains after punching is:

(a) $(\pi + 2) / 8$ (b) $(6 - \pi) / 8$
(c) $(4 - \pi) / 4$ (d) $(\pi - 2) / 4$

15. Find the area of the part of the circle (round punch) falling outside the square sheet.

(a) $\pi/4$ (b) $(\pi - 1) / 2$
(c) $(\pi - 1) / 4$ (d) $(\pi - 2) / 4$



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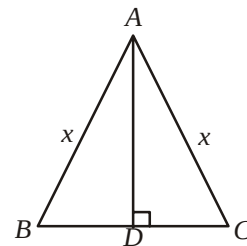
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Hints & Solutions

Foundation Level

1. (a) Area of rhombus = side $\times$ height
 $= 13 \times 20 = 260 \text{ cm}^2$
2. (a) In a circle, circumference = $2\pi r$
Hence, $44 = 2\pi r \quad \therefore r = \frac{44}{2\pi}$
Now, area of circle = $\pi r^2 = \pi \times \frac{44}{2\pi} \times \frac{44}{2\pi} = 154 \text{ m}^2$
3. (a) Let the length and breadth of a rectangle are $9x \text{ m}$ and $5x \text{ m}$ respectively.
In a rectangle, area = length $\times$ breadth
 $\therefore 720 = 9x \times 5x$
or $x^2 = 16 \Rightarrow x = 4$
Thus, length = $9 \times 4 = 36 \text{ m}$
and breadth = $5 \times 4 = 20 \text{ m}$
Therefore, perimeter of rectangle = $2(36 + 20) = 112 \text{ m}$
4. (d) Required no. of squares = $\frac{5^2}{1^2} = 25$
5. (c) Let the area of two squares be $9x$ and x respectively.
So, sides of both squares will be
 $\sqrt{9x}$ and $\sqrt{x}$ respectively. [since, side = $\sqrt{\text{area}}$]
Now, perimeters of both squares will be
 $4 \times \sqrt{9x}$ and $4 \times \sqrt{x}$ respectively.
[since, perimeter = $4 \times$ side]
Thus, ratio of their perimeters $\frac{4\sqrt{9x}}{4\sqrt{x}} = 3 : 1$
6. (d) Perimeter of the circle = $2\pi r = 2(18 + 26)$
 $\Rightarrow 2 \times \frac{22}{7} \times r = 88 \Rightarrow r = 14$
 $\therefore$ Area of the circle
 $= \pi r^2 = \frac{22}{7} \times 14 \times 14 = 616 \text{ cm}^2$
7. (a) In a rectangle,
 $\frac{(\text{perimeter})^2}{4} - (\text{diagonal})^2 + 2 \times \text{area}$
 $\Rightarrow \frac{(14)^2}{4} - 5^2 + 2 \times \text{area}$
 $49 = 25 + 2 \times \text{area}$
 $\therefore \text{Area} = \frac{49 - 25}{2} = \frac{24}{2} = 12 \text{ cm}^2$
8. (c) Circumference of circle = Area of circle
or $\pi d = \pi \left(\frac{d}{2}\right)^2$ [where d = diameter]
 $\therefore d = 4$

9. (a) In a parallelogram.
Area = Diagonal $\times$ length of perpendicular on it.
 $= 30 \times 20 = 600 \text{ m}^2$
10. (c) In a triangle,
Area = $\frac{1}{2} \times$ length of perpendicular $\times$ base
or $615 = \frac{1}{2} \times$ length of perpendicular $\times 123$
 $\therefore$ Length of perpendicular $\frac{615 \times 2}{123} = 10 \text{ m}$.
11. (a) Circumference of circular bed = 30 cm
Area of circular bed $\frac{(30)^2}{4\pi}$
Space for each plant = 4 cm^2
 $\therefore$ Required number of plants
 $\frac{(30)^2}{4\pi} \div 4 = 17.89 = 18$ (Approx)
12. (d) Side of square carpet $\sqrt{\text{Area}} = \sqrt{169} = 13 \text{ m}$
After cutting of one side,
Measure of one side = $13 - 2 = 11 \text{ m}$
and other side = 13 m (remain same)
 $\therefore$ Area of rectangular room = $13 \times 11 = 143 \text{ m}^2$
13. (a) If area of a circle decreased by $x\%$ then the radius of a circle decreases by
 $(100 - 10\sqrt{100 - x})\% = (100 - 10\sqrt{100 - 36})\%$
 $= (100 - 10\sqrt{64})\%$
 $100 - 80 = 20\%$
14. (b) Let ABC be the isosceles triangle and AD be the altitude.
Let $AB = AC = x$. Then, $BC = (32 - 2x)$.



Since, in an isosceles triangle, the altitude bisects the base. So, $BD = DC = (16 - x)$.
In $\triangle ADC$, $AC^2 = AD^2 + DC^2$
 $\Rightarrow x^2 = (8)^2 + (16 - x)^2$
 $\Rightarrow 32x = 320 \Rightarrow x = 10$.
 $\therefore BC = (32 - 2x) = (32 - 20) \text{ cm} = 12 \text{ cm}$.

Hence, required area = $\left(\frac{1}{2} \times BC \times AD\right)$

$$\left(\frac{1}{2} \times 12 \times 10\right) \text{ cm}^2 = 60 \text{ cm}^2.$$

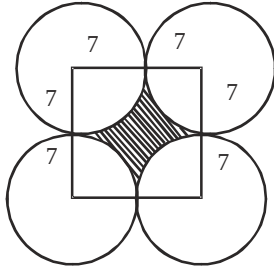
15. (c) Area of field = 576 km². Then,

each side of field = $\sqrt{576}$ = 24 km

Distance covered by the horse
= Perimeter of square field
= 24 × 4 = 96 km

$$\therefore \text{Time taken by horse} = \frac{\text{distance}}{\text{speed}} = \frac{96}{12} = 8 \text{ h}$$

16. (b)



The shaded area gives the required region.

Area of the shaded region

= Area of the square – area of four quadrants of the circles

$$= (14)^2 - 4 \times \frac{1}{4} \pi (7)^2$$

$$196 - \frac{22}{7} \times 49 = 196 - 154 = 42 \text{ cm}^2$$

17. (b) Perimeter = Distance covered in 8 min.

$$\left(\frac{12000}{60} \times 8 \right) \text{ m} = 1600 \text{ m.}$$

Let length = 3x metres and breadth = 2x metres.

Then, 2 (3x + 2x) = 1600 or x = 160.

$\therefore$ Length = 480 m and Breadth = 320 m.

$\therefore$ Area = (480 × 320) m² = 153600 m².

18. (c) Length of wire = $2\pi \times R = \left(2 \times \frac{22}{7} \times 56 \right) \text{ cm} = 352 \text{ cm.}$

$$\text{Side of the square} = \frac{352}{4} \text{ cm} = 88 \text{ cm.}$$

$$\text{Area of the square} = (88 \times 88) \text{ cm}^2 = 7744 \text{ cm}^2.$$

19. (a) Let the length of the room be ℓ m

Then its, breadth = $\ell/2$

$$\text{Therefore, } \ell \times \frac{\ell}{2} = \frac{5000}{25}$$

$$\text{or } \ell^2 = 400$$

$$\text{or } \ell = 20 \text{ m}$$

$$\text{Also, } 2\ell h + 2 \times \frac{\ell}{2} \times h = \frac{64800}{240}$$

$$\Rightarrow 3\ell h = 270$$

$$\text{or } h = \frac{270}{3 \times 20} = \frac{270}{60} = 4.5 \text{ m}$$

20. (a) Let the edge of the third cube be x cm.

$$\text{Then, } x^3 + 6^3 + 8^3 = 12^3$$

$$\Rightarrow x^3 + 216 + 512 = 1728$$

$$\Rightarrow x^3 = 1000 \Rightarrow x = 10.$$

Thus the edge of third cube = 10 cm.

21. (b) Area of the inner curved surface of the well dug

$$= [2\pi \times 3.5 \times 22.5] = 2 \times \frac{22}{7} \times 3.5 \times 22.5$$

$$= 44 \times 0.5 \times 22.5 = 495 \text{ sq. m.}$$

$$\therefore \text{Total cost} = 495 \times 3 = \text{` } 1485.$$

22. (d) Let the length, breadth and height of the cuboid be x, 2x and 3x, respectively.

$$\text{Therefore, volume} = x \times 2x \times 3x = 6x^3$$

New length, breadth and height = 2x, 6x and 9x, respectively.

$$\text{New volume} = 108x^3$$

$$\text{Thus, increase in volume} = (108 - 6)x^3 = 102x^3$$

$$\frac{\text{Increase in volume}}{\text{Original volume}} = \frac{102x^3}{6x^3} = 17$$

23. (a) In a cube,

$$\text{Area} = 6 (\text{side})^2$$

$$\text{or } 150 = 6 (\text{side})^2$$

$$\therefore \text{side} = \sqrt{25} = 5 \text{ m}$$

$$\text{Length of diagonal} = \sqrt{3} \times \text{side} = 5\sqrt{3} \text{ m}$$

24. (c) Required length = length of the diagonal

$$\sqrt{12^2 + 9^2 + 8^2} = \sqrt{144 + 81 + 64} = \sqrt{289} = 17 \text{ m}$$

25. (c) In a sphere, volume = $\frac{4}{3} \pi r^3$

$$\text{and surface area} = 4\pi r^2$$

$$\text{According to question, } \frac{4}{3} \pi r^3 \div 4\pi r^2 = 27$$

$$\text{or } r = 27 \times 3 = 81 \text{ cms}$$

26. (a) Let depth of rain be h metre. Then,
volume of water

$$= \text{area of rectangular field} \times \text{depth of rain}$$

$$\text{or } 3000 = 500 \times 300 \times h$$

$$\therefore h = \frac{3000}{500 \times 300} \text{ m} = \frac{3000 \times 100}{500 \times 300} \text{ cms} = 2 \text{ cms}$$

27. (c) Volume of cylinder = $(\pi \times 6 \times 6 \times 28) \text{ cm}^3$
 $= (36 \times 28) \pi \text{ cm}^3$.

Volume of each bullet = $\left(\frac{4}{3} \pi \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4}\right) \text{ cm}^3$
 $= \frac{9\pi}{16} \text{ cm}^3$.

Number of bullets = $\frac{\text{Volume of cylinder}}{\text{Volume of each bullet}}$
 $= \left[(36 \times 28) \pi \times \frac{16}{9\pi}\right] = 1792$.

28. (c) Let h be the required height then, $\frac{22}{7} \times (60)^2 \times h$

$= 30 \times 60 \times \frac{22}{7} \times (1)^2 \times (600)$

$\Rightarrow 60h = 30 \times 600$

$\Rightarrow h = 300 \text{ cm} = 3 \text{ m}$

29. (a) Let radius of the 3rd spherical ball be R ,

$\therefore \frac{4}{3} \pi \left(\frac{3}{2}\right)^3 = \frac{4}{3} \pi \left(\frac{3}{4}\right)^3 + \frac{4}{3} \pi (1)^3 + \frac{4}{3} \pi R^3$

$\Rightarrow R^3 \left[\left(\frac{3}{2}\right)^3 - \left(\frac{3}{4}\right)^3 - 1\right]$

$\frac{27}{8} - \frac{27}{64} - 1 = \frac{125}{64} = \left(\frac{5}{4}\right)^3 \Rightarrow R = \frac{5}{4} = 1.25$

$\therefore$ Diameter of the third spherical ball
 $= 1.25 \times 2 = 2.5 \text{ cm}$.

30. (c) Let 'A' be the side of bigger cube and 'a' be the side of smaller cube

Surface area of bigger cube = $6A^2$

or $384 = 6A^2$

$\therefore A = 8 \text{ cm}$.

Surface area of smaller cube = $6a^2$

$96 = 6a^2$

$\therefore a = 4 \text{ mm} = 0.4 \text{ cm}$

So, Number of small cube $\frac{\text{Volume of bigger cube}}{\text{Volume of smaller cube}}$

$\frac{(8)^3}{(0.4)^3} = \frac{512}{0.064} = 8,000$

31. (d) Volume of the tank = 246.4 litres = 246400 cm^3 .
 Let the radius of the base be $r \text{ cm}$. Then,

$\left(\frac{22}{7} \times r^2 \times 400\right) = 246400$

$\Rightarrow r^2 = \left(\frac{246400 \times 7}{22 \times 400}\right) = 196 \Rightarrow r = 14$.

$\therefore$ Diameter of the base = $2r = 28 \text{ cm} = .28 \text{ m}$

32. (a) Total surface area of the remaining solid = Curved surface area of the cylinder + Area of the base + Curved surface area of the cone

$= 2\pi rh + \pi r^2 + \pi r \ell$

$= 2\pi \times 8 \times 15 + \pi \times (8)^2 + \pi \times 8 \times 17$

$= 240\pi + 64\pi + 136\pi$

$= 440\pi \text{ cm}^2$

33. (d) $4\pi(r+2)^2 - 4\pi r^2 = 352$

$\Rightarrow (r+2)^2 - r^2 = \left(352 \times \frac{7}{22} \times \frac{1}{4}\right) = 28$.

$\Rightarrow (r+2+r)(r+2-r) = 28$

$\Rightarrow 2r+2 = \frac{28}{2} \Rightarrow 2r+2 = 14 \Rightarrow r = 6 \text{ cm}$

34. (b) Volume of material in the sphere

$= \left[\frac{4}{3} \pi \times \{(4)^3 - (2)^3\}\right] \text{ cm}^3 = \left(\frac{4}{3} \pi \times 56\right) \text{ cm}^3$.

Let the height of the cone be $h \text{ cm}$.

Then, $\frac{1}{3} \pi \times 4 \times 4 \times h = \left(\frac{4}{3} \pi \times 56\right)$

$\Rightarrow h = \left(\frac{4 \times 56}{4 \times 4}\right) = 14 \text{ cm}$.

35. (b) Given, playground is rectangular.

Length = 36 m, Breadth = 21 m

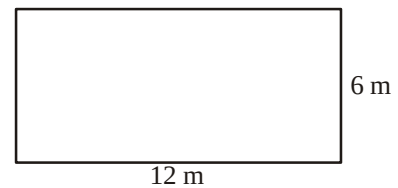
Now, perimeter of playground = $2(21 + 36) = 114$

Now, poles are fixed along the boundary at a distance 3m.

$\therefore$ Required no. of poles = $\frac{114}{3} = 38$.

36. (c) Given, Length = 12 m and Breadth = 6 m

$\therefore$ Area of rectangular plate = $12 \times 6 = 72 \text{ m}^2$



Since, two apertures of 3 m diameter each have been made from this plate.

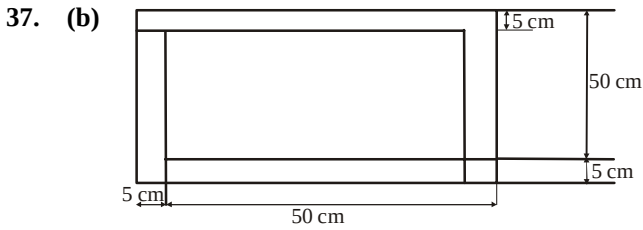
$\therefore$ Area of these two apertures = $\pi(1)^2 + \pi(1)^2$
 $= \pi + \pi = 2\pi$

Area of 1 aperture of 1m diameter = $\pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$

$\therefore$ Total area of aperture = $2\pi + \frac{\pi}{4} = \frac{9\pi}{4} = \frac{9}{4} \times \frac{22}{7} = \frac{99}{14}$

$\therefore$ Area of the remaining portion of the plate

$= 72 - \frac{99}{14} \text{ sq. m} = \frac{909}{14} \text{ sq. m} \approx 64.5 \text{ sq. m}$



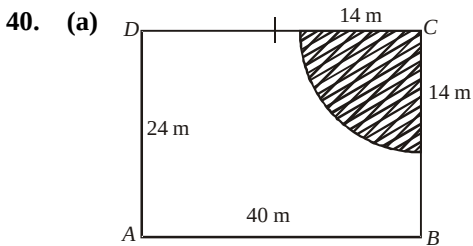
Side of the inner square = $55 - 10 = 45$
 $\therefore$ Area of inner square = $45 \times 45 = 2025$ sq. m.

38. (c) Given, length of garden = 24 m and breadth of garden = 14 m
 $\therefore$ Area of the garden = $24 \times 14 \text{ m}^2 = 336 \text{ m}^2$.
 Since, there is 1 m wide path outside the garden
 $\therefore$ Area of Garden (including path)
 $= (24 + 2) \times (14 + 2) = 26 \times 16 \text{ m}^2 = 416 \text{ m}^2$.
 Now, Area of Path = Area of garden (including path) - Area of Garden
 $= 416 - 336 = 80 \text{ m}^2$.
 Now, Area of Marbles = $20 \times 20 = 400 \text{ cm}^2$

$$\therefore \text{Marbles required} = \frac{\text{Area of Path}}{\text{Area of Marbles}}$$

$$= \frac{80,0000}{400} = 2000$$

39. (a) Let width of the field = b m
 $\therefore$ length = $2b$ m
 Now, area of rectangular field = $2b \times b = 2b^2$
 Area of square shaped pond = $8 \times 8 = 64$
 According to the question,
 $64 \times \frac{1}{8}(2b^2) \Rightarrow b^2 = 64 \times 4 \Rightarrow b = 16$ m
 $\therefore$ length of the field = $16 \times 2 = 32$ m



Area of the shaded portion

$$\frac{1}{4} \times \pi (14)^2 = 154 \text{ m}^2$$

41. (b) Let ℓ be the length and b be the breadth of cold storage.
 $L = 2B$, $H = 3$ metres
 Area of four walls = $2[L \times H + B \times H] = 108$
 $\Rightarrow 6BH = 108 \Rightarrow B = 3$
 $\therefore L = 12$, $B = 6$, $H = 3$
 Volume = $12 \times 6 \times 3 = 216 \text{ m}^3$
42. (c) Surface area of the cube = (6×8^2) sq. ft. = 384 sq. ft.
 Quantity of paint required = $\left(\frac{384}{16}\right)$ kg = 24 kg.
 $\therefore$ Cost of painting = $(36.50 \times 24) = ₹ 876$.

43. (c) Volume of block = $(6 \times 9 \times 12) \text{ cm}^3 = 648 \text{ cm}^3$.
 Side of largest cube = H.C.F. of 6 cm, 9 cm, 12 cm = 3 cm.
 Volume of the cube = $(3 \times 3 \times 3) = 27 \text{ cm}^3$.

$$\therefore \text{Number of cubes} = \left(\frac{648}{27}\right) = 24.$$

44. (d) Circumference of the base of ice-cream cup = Diameter of the sheet = 28 cm
 $2\pi r = 28$

$$r = \frac{14}{\pi} \text{ cm} = 4.45 \text{ cm}$$

Slant height of cone = radius of the sheet = 14 cm

$$\therefore 14^2 = (4.45)^2 + h^2$$

$$\text{or } h^2 = 196 - 19.80 = 176.20$$

$$\therefore h = 13.27 \text{ cm}$$

45. (d) Required no. of squares = $\frac{5^2}{1^2} = 25$

46. (c) Let the kerosene level of cylindrical jar be h .

$$\text{Now, volume of conical vessel} = \frac{1}{3} \pi r^2 h$$

Since, radius (r) = 2 cm and height (h) = 3 cm of conical vessel.

$$\therefore \text{Volume} = \frac{1}{3} \pi \times 4 \times 3 = 4\pi$$

Now, volume of cylindrical jar = $\pi r^2 h = \pi (b)^2 h = 4\pi h$

Now, volume of conical vessel = Volume of cylindrical Jar

$$\Rightarrow 4\pi = 4\pi h$$

$$h = 1 \text{ cm}$$

Hence, kerosene level in jar is 1 cm.

47. (b) Let, side of smaller square = x cm.

$\therefore$ Side of larger square = $(x + 4)$ cm

By the question; $(x + 4)^2 = 4x^2$.

$$\Rightarrow x^2 + 8x + 16 = 4x^2 \Rightarrow 3x^2 - 8x - 16 = 0$$

$$\Rightarrow 3x^2 - 12x + 4x - 16 = 0$$

$$\Rightarrow 3x(x - 4) + 4(x - 4) = 0$$

$$\Rightarrow (x - 4)(3x + 4) = 0, \therefore x = 4 \text{ cm}$$

$$\Rightarrow \text{perimeter of smaller square} = 4 \times 4 = 16 \text{ cm}.$$

48. (c) Surface area of the walls of the first room = $2(XZ + YZ)$

Surface area of the walls of the second room

$$= 2(4XZ + 4YZ) = 8(XZ + YZ) = 4 \text{ times the first area}$$

$$\Rightarrow \text{Cost required} = 4 \times 2500 = ₹ 10,000.$$

49. (c) Area = $\frac{1}{2} \times \text{Base} \times \text{Altitude}$

$$\Rightarrow \text{Area} = \frac{1}{2} (3 \times X) = \frac{3}{2} X^2$$

$$\text{Area painted} = (₹ 11000) / (₹ 10 \text{ per dm}^2)$$

$$\Rightarrow \text{Area} = 1,10,000 \text{ m}^2$$

$$\Rightarrow \text{On equating, we have } \frac{3}{2} X^2 = 1,10,000$$

$$\Rightarrow \text{Height} = X = 270.8 \text{ m}.$$

50. (d) Volume of the cone is given by $= \frac{1}{3} \times \pi r^2 h$
Here, $r = 4.2$ cm, $h = 10.2 - r = 6$ cm
Therefore the volume of the cone $= \frac{1}{3} \pi \times (4.2)^2 \times 6$ cm
 $= 110.88$ cm³

$$\text{Volume of the hemisphere} = \frac{1}{2} \times \frac{4}{3} \pi r^3 = 155.23 \text{ cm}^3$$

$$\text{Total volume} = 110.88 + 155.232 = 266.112$$

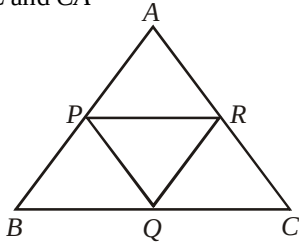
51. (a) Volume of mud dugout $= 10 \times 4.5 \times 3 = 135$ m³

Let the remaining ground rise by $= h$ m

$$\text{Then } \{(20 \times 9) - (10 \times 4.5)\} h = 135$$

$$135 h = 135 \Rightarrow h = 1 \text{ m}$$

52. (c) Consider for an equilateral triangle. Hence $\triangle ABC$ consists of 4 such triangles with end points on mid pts AB , BC and CA



$$\Rightarrow \frac{1}{4} \text{ar}(\triangle ABC) = \text{ar}(\triangle PQR)$$

$$\Rightarrow \text{ar}(\triangle PQR) = 5 \text{ sq. units}$$

53. (c) $\frac{\text{Area of uncut portion}}{\text{Area of cut portion}} = \frac{(\pi \times 20 \times 20) - (100\pi)}{(4 \times \pi \times 5 \times 5)}$

$$= \frac{300\pi}{100\pi} = \frac{3}{1}$$

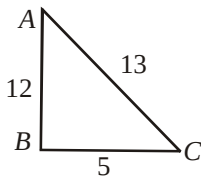
54. (b) In the figure $\angle ACB$ is 90°
(angle subtended by diameter $= 90^\circ$)
 $AC = 5$, $AB = 13$

Using pythagoras theorem,

$$AB^2 = AC^2 + CB^2 \Rightarrow CB = \sqrt{13^2 - 5^2} = 12$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times 5 \times 12 = 30$$

55. (b)



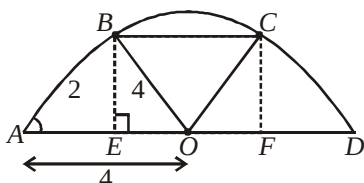
ABC forms a right angled triangle

$$\therefore \text{Area} = \frac{1}{2} \times 12 \times 5 = 30$$

Area of rectangle $= 30 \times \ell \times 10$ or $\ell = 3$ units

$$\therefore \text{Perimeter} = 2(10 + 3) = 26$$

56. (b)



$$BO = \text{radius} = 4 = AO$$

$$AE = 2 \cos A \quad 2 \times \left(\frac{2^2 + 4^2 - 4^2}{2 \times 2 \times 4} \right) \quad \frac{2}{4} \quad \frac{1}{2}$$

$$\therefore BC = AD - AE - FD \quad 8 - \frac{1}{2} - \frac{1}{2} = 7$$

$$(\because AE = FD)$$

57. (d) (side)²

$$= \left(\frac{1}{2} \times \text{onedagonal} \right)^2 + \left(\frac{1}{2} \times \text{other diagonal} \right)^2$$

$$13^2 = \left(\frac{1}{2} \times \text{onedagonal} \right)^2 + \left(\frac{1}{2} \times 24 \right)^2$$

$$169 - 144 = \left(\frac{1}{2} \times \text{diagonal} \right)^2$$

$$25 = \left(\frac{1}{2} \times \text{diagonal} \right)^2$$

$$5 = \frac{1}{2} \times \text{diagonal} \quad \therefore \text{diagonal} = 10$$

$$\therefore \text{Area} = \frac{1}{2} \times 10 \times 24 = 120 \text{ sq. cm.}$$

58. (b) $AC^2 = AB^2 + BC^2 \Rightarrow AC = 10$

$$\text{We have } r = (A/s); A = \frac{1}{2} \times (6 \times 8) = 24$$

$$s = (6 + 8 + 10)/2 = 12$$

$$r = A/s = 24/12 = 2.$$

59. (a) Volume of the beam $= \ell \times b \times h = 9 \times \frac{1}{2} \times \frac{1}{5} = \frac{9}{10}$ m³

$$\text{Weight of the beam} = \frac{9}{10} \times 40 \text{ kg} = 36 \text{ kg}$$

60. (a) Area of square $= 2 \text{ km}^2$

$$\therefore \text{Diagonal} = \sqrt{2 \times 2} \text{ km} = 2 \text{ kilometres}$$

61. (b) Circumference of base $= 2\pi r = 6 \quad \therefore r = \frac{3}{\pi}$

62. (c) Volume of spherical shell

$$= \frac{4\pi}{3} (R^3 - r^3) = \frac{4\pi}{3} (12^3 - 10^3)$$

$$= \frac{4}{3} \times \pi \times (12 - 10) (12^2 + 12 \times 10 + 10^2)$$

$$= \frac{4}{3} \times \pi \times 2 \times 364 \text{ cm}^3$$

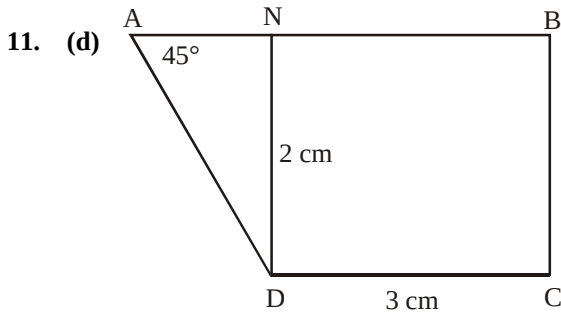
Weight = volume $\times$ density

$$= \frac{4}{3} \times \pi \times 364 \times 4.8 = 14.64 \text{ kg}$$

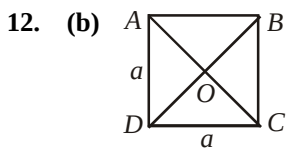
63. (a)
64. (b)
65. (d) Let $ABCD$ be a square with side = 6 cm. Then the radius of the circle touches the square = 3 cm.
Area of circle = $\pi(r)^2 = 9\pi \text{ cm}^2$
66. (c) $\therefore \text{Circumference} = \frac{p}{360} = 2\pi R$
67. (a) Area of shaded region = Area of equilateral $\Delta ABC - 3$ (Area of sector AQO)
 $= \frac{\sqrt{3}}{4} \times (2)^2 = 3 \times \frac{60}{360} \times \frac{22}{7} \times (1)^2$
 $= \sqrt{3} - \frac{11}{7} = 1.73 - 1.57 = 0.16 \text{ sq. units.}$
68. (c) 2 semicircles = 1 circle with equal radius
So $2\pi r = 132 \Rightarrow 2r = \frac{132}{3.14} = 42 \text{ m diameter}$
Area of track = Area within external border - Area within internal border.
 $\Rightarrow \pi (23^2 - 21^2) + 90 \times 46 - 90 \times 4^2$
 $\Rightarrow 88\pi + 360 \Rightarrow 636.3 \text{ m}^2$
69. (d) Since AB is the diameter of the circle, $\angle ACB$ would be right angle. In this triangle, we know $AB = 15$ and $AC = 12$. So, we can find BC . Since $3 - 4 - 5$ forms a triplet, $3 \times (3 - 4 - 5)$ also forms a triplet. So, $9 - 12 - 15$ forms a triplet. Hence, $BC = 9$. Since $BC = BD$, $ADB - ACB$ (similar triangles). Hence, area of $\Delta ABC = \text{Area of } \Delta ABD = \frac{1}{2} AC \times CB = \frac{1}{2} \times 12 \times 9 = 54$
So, area of quadrilateral $ABCD = 2 \times 54 = 108 \text{ sq. cm.}$
70. (b) Given is $2a + b = 100$, and area = ab . Assuming $2a = x$, we get $x + b = 100$, and area = $\frac{1}{2}xb$. Now area will be maximum when $x = b$, or $a = b/2$. Thus $a = 25$, & $b = 50$, maximum area = 1250.
3. (b) (Volume of solid cylinder) $\times 0.8 = 8 \times$ Volume of each solid sphere.
 $\therefore (\pi \times r_1^2 \times h) \times 0.8 = 8 \times (4/3) \times \pi \times r_2^3$
 $\Rightarrow (4^2 \times 22.5 \times 0.8) = 8 \times (4/3) \times r_2^3 \Rightarrow r_2 = 3 \text{ cm.}$
4. (b) $V = \frac{4}{3} \pi r^3 \Rightarrow dV/dt = \frac{4}{3} \pi \cdot 3 r^2 \cdot dr/dt \Rightarrow dr/dt = 1/(4\pi)$
Now surface area $s = 4 \pi r^2 \Rightarrow ds/dt = 8 \pi r \cdot dr/dt$.
Now substitute the value of dr/dt
 $\Rightarrow ds/dt = 20 \text{ cm}^2/\text{min.}$
5. (a) $V = \frac{4}{3}\pi r^3$; $V' = \frac{4}{3}\pi (1.015r)^3 \Rightarrow V'/V = 1.0456$
 $\Rightarrow \text{error} = 4.6\%$.
6. (c) If the radius is diminished by $r\%$, then
Area is diminished by $\left(2r - \frac{r^2}{100}\right)\%$
 $2 \times 10 - \frac{10^2}{100} = 19\%$
7. (d) Increase in Area = $10 + 20 + \frac{10 \times 20}{100} = 30 + \frac{200}{100} = 32\%$
Hence, there will be 32% change in the cost of the plot
8. (a) If side is increased by $a\%$, area increased by $\left(2a + \frac{a^2}{100}\right)\%$
 $2 \times 5 + \frac{5^2}{100} = 10\frac{1}{4}\%$
9. (c) Circumference of the circular face of the cylinder = $2\pi r$
 $\Rightarrow 2 \times \frac{22}{7} \times \frac{35}{100} = 2.2 \text{ m}$
Number of revolutions required to lift the bucket by $11 \text{ m} = \frac{11}{2.2} = 5$
10. (d) Let the angle subtended by the sector at the centre be θ
Then, $5.7 + 5.7 + (2\pi) \times 5.7 \times \frac{8}{360} = 27.2$
 $11.4 + \frac{11.4 \times 3.14 \times \theta}{360} = 27.2$
 $\Rightarrow \frac{\theta}{360} = 0.44$
Area of the sector = $\pi r^2 \frac{\theta}{360} \Rightarrow (22/7) \times (5.7)^2 \times 0.44 = 44.92 \text{ approx.}$

Standard Level

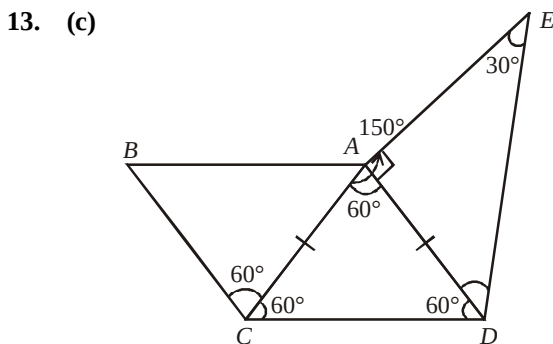
1. (c) Volume of rain that is to be collected in a pool = $2 \times 1 \times 10^{10} \times \frac{1}{2}$
 $= 10^{10} \text{ cm} = 10^4 \text{ meter}$
Volume of pool = $L \times B \times h$
 $10^4 = 100 \times 10 \times h$
 $h = \frac{10^4}{100 \times 10} = 10 \text{ m.}$
2. (b) Hypotenuse = 270 m
 $\Rightarrow \text{Hypotenuse}^2 = \text{Side}^2 + \text{Side}^2 = 2 \text{ Side}^2$
 $\Rightarrow \text{Side}^2 = (270)^2/2 = 72900/2 = 36450$ or side = 190.91m
 $\Rightarrow \text{Required Area} = \frac{1}{2} \times 190.91 \times 190.91 = 36446.6/2 = 18225 \text{ m}^2 \text{ (approx).}$



$\tan(45^\circ) = \frac{DN}{AN}$ or $1 = \frac{DN}{AN}$
 Therefore, $AN = DN = 2$ cm
 $AB = AN + BN = 2$ cm + 3 cm = 5 cm
 Hence, answer option is (d).



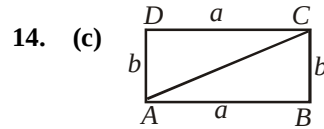
$ABCD$ is square $a^2 = 4 \Rightarrow a = 2$
 $ac = BD = 2\sqrt{2}$
 perimeters of four triangles
 $AB + BC + CD + DA + 2(AC + BD)$
 $8 + 2(2\sqrt{2} + 2\sqrt{2}) = 8(1 + \sqrt{2})$



$\angle A = \angle C = 60$ (alternative angles)
 $\angle C = \angle D = 60^\circ$
 (since $AC = AD$ and $\angle A = 60^\circ$)
 ΔACD is equilateral
 so its area $\frac{x^2\sqrt{3}}{4}$ (where x is side)
 Area of parallelogram $ABCD = 2 \times \frac{x^2\sqrt{3}}{4} = \frac{x^2\sqrt{3}}{2}$
 Area of $\Delta ADE = \frac{1}{2} \times AD \times AE$

$$\frac{1}{2} \times x \times x \tan 60^\circ = \frac{x^2\sqrt{3}}{2}$$

therefore we see,
 Area of parallelogram $ABCD =$ Area of ΔADE



$$AC + AB = 5AD \text{ or } AC + a = 5b \quad \dots(1)$$

$$AC - AD = 8 \text{ or } AC = b + 8 \quad \dots(2)$$

$$\text{Using (1) and (2), } a + b + 8 = 5b \text{ or } a + 8 = 4b \quad \dots(3)$$

Using Pythagoras theorem,

$$a^2 + b^2 = (b + 8)^2 = b^2 + 64 + 16b$$

$$\text{or } a^2 - 16b + 64 = (4b - 8)^2 = 16b^2 + 64 - 64b$$

[From (3)]

$$\Rightarrow 16b^2 - 80b = 0 \text{ or } b = 0 \text{ or } 5$$

$$\text{Putting } b = 5 \text{ in (3), } a = 4b - 8 = 20 - 8 = 12$$

$$\text{Area of rectangle} = 12 \times 5 = 60$$

15. (d) Let the radius of the semi-circle be R and that of the circle be r , then from the given data, it is not possible to express r in terms of R . Thus option (d) is the correct alternative.

$$\begin{aligned} 16. (a) \quad \pi r(r+1) : 3\pi r^2 : 2\pi r(r+h) \\ = \pi \times 1(1+\sqrt{2}) : 3 \times \pi \times 1 : 2 \times \pi \times 1(1+1) \\ = (\sqrt{2}+1) : 3 : 4 \end{aligned}$$

17. (c) Let r be the radius of each circle.
 Then by given condition,

$$\frac{\pi R^2}{2\pi R} = \frac{2\pi R}{\pi R^2} \Rightarrow R^2 = 4 \Rightarrow R = 2$$

$\therefore$ The length of the side of the square = 8

Now the area covered by 4 coins = $4 \times \pi (2)^2 = 16\pi$
 and area of the square = 64

$$\therefore \text{The area which is not covered by the coins} = 64 - 16\pi = 16(4 - \pi)$$

18. (b) $AD = 6.5$
 $\therefore AB = 13$ (diameter)
 Now $\angle ACB = 90^\circ$ (since the diameter of a circle subtends 90° at the circumference)
 So by pythagoras theorem, $CB = 2$ cm.

$$\therefore \text{area of } \Delta ACB = \frac{1}{2} \times 5 \times 2 = 5 \text{ sq. cm}$$

19. (c) Though it is given that diameter of the cone is equal to the diameter of the spherical ball. But the ball will not fit into the cone because of its slant shape. Hence more than 50% of the portion of the ball will be outside the cone.

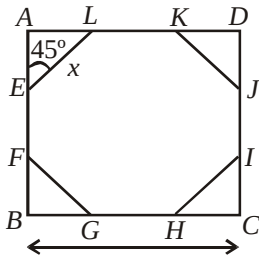
20. (b) Volume of the given ice cuboid = $8 \times 11 \times 2 = 176$
Let the length of the required rod is ℓ .

$$\therefore \pi \ell \frac{8^2}{4} = 176 \quad \therefore \ell = 3.5 \text{ inches}$$

21. (c) Surface area of walls = $2(lh + bh)$
= $2[(12 \times 3) + (4 \times 3)] = 2(36 + 12) = 96 \text{ m}^2$
Area of doors = $2 \times 2.5 \times 1.5 = 7.5 \text{ m}^2$
Area of window = $2 \times 0.60 = 1.2 \text{ m}^2$ and
area of ceiling = $2 \times 4 = \text{m}^2$
Area to be coloured = area of walls + area of ceiling –
area of doors – area of window
= $96 + 48 - 7.5 - 1.2 = 135.3 \text{ m}^2$

Required cost = $135.3 \times 15 = 2029.5$.

22. (a) Since the interior angle of an octagon is 135° , the angles in $\triangle AEL$ are each 45° .
If x is the side of the octagon $LE = x$.
Hence, in $\triangle ALE$.



$$AL = AE = \frac{x}{\sqrt{2}} = KD$$

$$\therefore AL + LK + KD$$

$$= \frac{x}{\sqrt{2}} + x + \frac{x}{\sqrt{2}}$$

But this is the side of the square.

$$\text{Hence, } \frac{2x}{\sqrt{2}} + x = 1 \Rightarrow (1 + \sqrt{2}) = 1$$

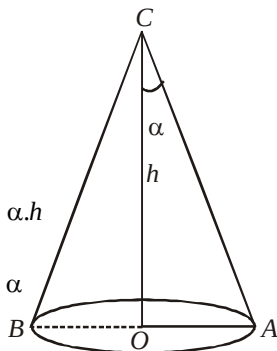
$$\therefore x = \frac{1}{1 + \sqrt{2}} = \sqrt{2} - 1$$

23. (a) $r = h \tan \alpha$

$$\therefore \text{Volume} = \frac{1}{3} \pi r^2 h$$

$$\frac{1}{3} \pi h^2 \tan^2 \alpha \cdot h$$

$$\frac{1}{3} \pi h^3 \tan^2 \alpha$$



24. (a) Let us suppose that each side of the cuboid be 100 units. Then its volume = 100^3 units.
Now sides of the cuboid are : $(100 + x)$, $(100 + y)$ and $(100 + z)$
Then its new volume = $(100 + x)(100 + y)(100 + z)$
= $100^3 + 100^2(x + y + z) + 100(xy + xz + yz) + xyz$

% change in volume

$$= \frac{100^2(x + y + z) + 100(xy + xz + yz) + xyz}{100^3} \times 100$$

$$= \left[x + y + z + \frac{xy + xz + yz}{100} + \frac{xyz}{(100)^2} \right] \%$$

25. (a) If we open the cylinder (along vertical line), its base length = $2\pi r = 2\pi \frac{3}{\pi} = 6\text{cm}$.
Since, width of the string is h cms.

$$\text{Number of turns} = \frac{\text{height of cylinder}}{\text{width of the string}} \frac{n}{h} \text{ and}$$

$$\text{required length of the string} = \text{base length} \times \text{no. of turns}$$

$$= \frac{6h}{n} \text{ cms.}$$

26. (a) Let h be the length of water column discharged in 1 hour or 1 minute.
Volume discharged by the 4 pipe = Volumes discharged by the single pipe
 $4 \times \pi \times (1.5)^2 \times h = \pi \times (r)^2 \times h$
 $\therefore r^2 = 9 \quad \therefore r = 3$
Diameter = 6 inches.
27. (b) As per the given conditions,

$$11a^3 - 7 \times \frac{4}{3} \times \pi \times r^3 \therefore \frac{a}{r} \left(\frac{8}{3} \right)^{1/3}$$

28. (d) Let the edge of the cube measure x in.
Then the diameter of the sphere is x in.
Now volume of wood removed

$$= \text{volume of cube} - \text{volume of sphere} = \left(x^3 - \frac{\pi x^3}{6} \right) \text{ c.}$$

in By hypothesis, this volume = 35280 c. in.

$$\therefore x^3 - \frac{\pi x^3}{6} = 35280 \Rightarrow x^3 \left(1 - \frac{22}{7 \times 6} \right) = 35280$$

$$\Rightarrow x^3 (20/42) = 35280 \Rightarrow x = 42.$$

29. (b) Side of hexagon = $\frac{\text{Perimeter}}{\text{Number of sides}} = \frac{600}{6} = 100\text{cm}$.

Area of regular hexagon

$$= \frac{3\sqrt{3}}{2} \times 100 \times 100 = 25980.8 \text{ sq. cm.}$$

Volume = Base area $\times$ height

$$= 25980.8 \times 200 \text{ c.c.} = 5196160 \text{ c.c. or } 5196 \text{ lit.}$$

Weight of milk = $5196160 \times 0.8 \text{ gm}$

$$= 4156928 \text{ gm} = 4156.9 \text{ kg.}$$

30. (c) Area of isosceles triangle = $\frac{b}{4} (\sqrt{4a^2 - b^2})$
where b is the base and a is any of the equal sides.

$$\text{Area of the required triangle} = \frac{10}{4} (\sqrt{4(8)^2 - (10)^2})$$

$$= \frac{10}{4} \sqrt{156} = 5\sqrt{39} \text{ cm}^2$$

31. (b) Area of the shaded portion = Area of quadrant ABC + Area of quadrant ACD - Area of square ABCD.

$$= \frac{\pi}{2} \times 4^2 - 4^2 = \left(\frac{\pi}{2} - 1\right) 4^2 = (\pi - 2) 8 = 9.12 \text{ sq. cm}$$

32. (d) Volume of pyramid = $\frac{1}{3} \times \text{base area} \times \text{height}$

$$= \frac{1}{3} \times 6 \times 6 \times 4 = 48 \text{ cc}$$

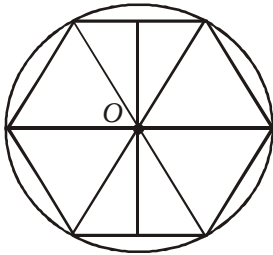
$$\Rightarrow \text{Height of slant face (X)} = \sqrt{4^2 + 3^2} = 5 \text{ cm.}$$

$$\text{Area of each slant face} = \frac{1}{2} \times 5 \times 6 = 15 \text{ sq. cm}$$

$$\text{Area of base} = 6 \times 6 \text{ sq. cm} = 36 \text{ sq. cm.}$$

$$\Rightarrow \text{Total surface area} = 4(15) + 36 \text{ sq. cm.} = 96 \text{ sq. cm.}$$

33. (b) We can divide the regular hexagon into 6 equilateral triangles. Since the hexagon is in a circle the radius r is the side of the equilateral triangle.



$\therefore$ Area of the hexagon

$$= 6 \times \frac{\sqrt{3}}{4} r^2 = \frac{3\sqrt{3}}{2} r^2 \text{ sq. units.}$$

34. (c) $r = \frac{\text{Area of triangle}}{s} = \frac{\Delta}{s}$

$$s = \frac{a+b+c}{2} = \frac{18+24+30}{2} = 36$$

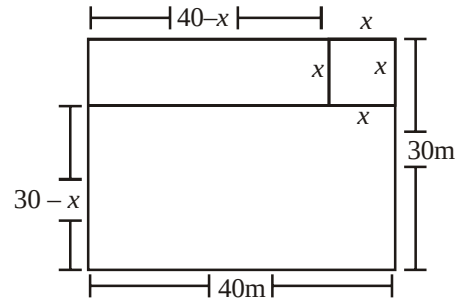
$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\Rightarrow \Delta = \sqrt{36(36-18)(36-24)(36-30)}$$

$$\Delta = \sqrt{36 \times 18 \times 12 \times 6} = 216$$

$$\text{So, radius of incircle} = \frac{216}{36} = 6 \text{ cm.}$$

35. (a)



$$\begin{aligned} \text{Hence, } & (x)(40-x) + (x)(30-x) + x^2 \\ & = 1200 - [(x)(40-x) + (x)(30-x) + x^2] \\ & 2[(x)(40-x) + (x)(30-x) + x^2] = 1200 \\ & 40x - x^2 + 30x - x^2 + x^2 = 600 \\ & -x^2 + 70x - 600 = 0 \\ & x^2 - 70x + 600 = 0 \\ & (x-60)(x-10) = 0 \\ & x = 10 \text{ or } 60. \\ & \text{As } x \text{ must be less than } 30. \\ & \therefore x = 10 \end{aligned}$$

36. (d) $\frac{4}{3} \pi R^3 = 1000. \frac{4}{3} \pi r^3$

$$R = 10r$$

$$\therefore r = \text{radius of smaller sphere} = 1 \text{ cm}$$

$$R = \text{radius of spherical metal}$$

$$\text{Initial surface area of metal} = 4\pi R^2 = 4\pi \times 100 = 400\pi$$

$$\text{Final surface area of 1000 smaller sphere} = 1000 \cdot 4\pi r^2 = 1000 \times 4\pi = 4000\pi$$

$$\therefore \text{Increase in surface area} = 4000\pi - 400\pi = 3600\pi$$

$$\frac{3600\pi}{400\pi} = 9 \text{ times}$$

Hence correct option is (d)

37. (d) Assume the initial surface area as 100 on each side. A total of 6 such surfaces would give a total surface area of 600. Two surface areas would be impacted by the combined effect of length and breadth, two would be affected by length and height. Thus, the respective surface areas would be (110.25 twice, 126 twice and 126 twice) Thus, new surface area = 220.5 + 504 = 724.5. A percentage increase of 20.75%. Option (d) is correct.

38. (b) External surface area

$$= \text{Curved surface area of frustum of the cone} + \text{Surface area of hemisphere}$$

$$= \pi(r_1 + r_2)l + 2\pi r_1^2 \quad [r_1 = 1, r_2 = 2.5, h = 7 - 1 = 6 \text{ cm}]$$

$$= \pi(3.5)(6.2) + 2\pi(1)^2 l = \sqrt{h^2 + (r_1 - r_2)^2} = \sqrt{6^2 + (1-2.5)^2}$$

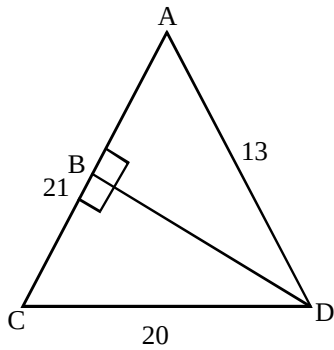
$$= \frac{22}{7} \times \frac{35}{10} \times \frac{618}{100} + 2 \times \frac{22}{7}$$

$$= \sqrt{36 + 2.25} = \sqrt{38.25} = 6.18$$

$$= \frac{11 \times 618}{100} + \frac{44}{7} = 67.98 + 6.28 = 74.26 \text{ cm}^2$$

$\therefore$ (b) holds.

39. (b)



Let the original triangle be $\triangle ACD$
 Longest side $= AC = 21$ cm
 In the right angled $\triangle ABD$, by Pythagorean triplets,
 we get $AB = 5$ cm and $BD = 12$ cm
 Then, $BC = 21 - 5 = 16$
 By Pythagoras theorem,
 $BD^2 = CD^2 - BC^2 \Rightarrow BD = 12$ cm

Area of the larger $\triangle BDC = \frac{1}{2} \times 16 \times 12 = 96$ cm²

40. (c) Let the internal radius of the cylinder $= r$
 Then, the volume of sphere $=$ Volume of hollow cylinder

$$\Rightarrow \frac{4\pi \cdot 6^3}{3} = \pi(5^2 - r^2)$$

$$\Rightarrow \frac{864\pi}{3} = 32\pi(25 - r^2)$$

$$\Rightarrow r^2 = 16 = r = 4$$
 cm

So thickness of the cylinder $= 5 - 4 = 1$ cm

41. (b) Sum of interior angles of a hexagon $= 720^\circ$
 6 sectors with same radius $r = 2$ full circles of same radius.

So area of shaded region $\Rightarrow 2\pi r^2$

42. (b) Let $ABCDEFGH$ be the cube of side a and O be the centre of the hemisphere.

$$AC = \sqrt{2}a$$

$$OD = OC = R$$

Let P be the mid-point of AC

$$OP = a$$

Now in $\triangle AOC$

$$R^2 = a^2 + \frac{a^2}{2} \therefore a = \sqrt{\frac{2}{3}}R$$

$$\therefore \text{Volume} = a^3 = 0.67\sqrt{\frac{2}{3}}R^3$$

43. (d) As for the diagonals X & Y of a parallelogram.
 $X^2 + Y^2 = 2(A^2 + B^2)$; where A & B are the sides of the parallelogram.

Sides of a parallelogram are 8 cm and 6 cm.

And ratio of the diagonals is 3 : 4 such that $X = 4k$ and $Y = 3k$

$$16k^2 + 9k^2 = 2(8^2 + 6^2), \text{ so, } K = \sqrt{8}$$

Therefore, the required difference in $4k - 3k$

$$= k = \sqrt{8} \text{ cm.}$$

44. (a) Since $ABCOA$ is a quadrant of circle of radius 10.5 cm $OA = OC = r = 10.5$ cm and

$$OD = DC = \frac{10.5}{2} = 5.25 \text{ cm}$$

Area of shaded portion $=$ (Area of the quadrant) $-$ (Area of $\triangle AOD$)

$$\text{Area} = \frac{\theta}{360} \pi r^2 - \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{90}{360} \times \frac{22}{7} \times (10.5)^2 - \frac{1}{2} \times 5.25 \times 10.25$$

$$= 86.625 - 27.5625 = 59.06 \text{ cm}^2.$$

45. (d) 1 kg $= 1000$ cm³

$$2700 = k \cdot 2^3$$

$$k = \frac{2700}{8}$$

$$6300 = k \cdot r^3$$

$$r^3 = \frac{6300}{k} = \frac{6300}{\frac{2700}{8}} = \frac{8}{3},$$

$$r^3 = \frac{56}{3}, r = 2.6 \text{ cm}$$

46. (d) Area $= \frac{1}{2} \cdot 50.130 \sin 72^\circ$

$$= \frac{1}{2} \cdot 50.130 \times 0.9510 = 3090.75 \text{ m}^2$$

47. (a) From the fig. the shaded area $=$ (Area of the rectangle $- 2 \times$ quarter of circle) $+ \text{area of rectangle}$

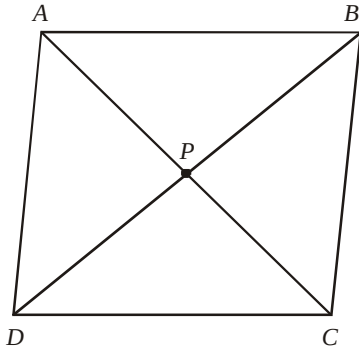
$$= \left[\left(3 \times 6 - 2 \times \frac{\pi}{4} \times 3^2 \right) + \frac{1}{2} \pi \times 3^2 \right] \text{ sq. m}$$

$$= \left[18 - \frac{9\pi}{2} + \frac{9\pi}{2} \right] = 18 \text{ sq. m}$$

$$\therefore \text{Cost of covering with grass} = \frac{18 \times 70}{100}$$

$$= \frac{630 \times 2}{100} = 12.60$$

48. (b) We know ratio of area of triangles divided by diagonals are same.



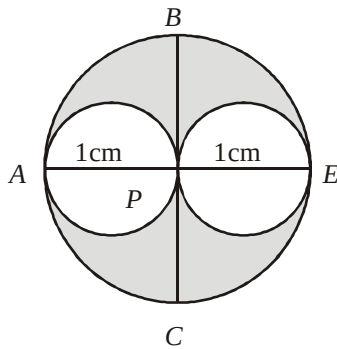
$$\frac{\text{Area of } \triangle APD}{\text{Area of } \triangle APB} = \frac{\text{Area of } \triangle DPC}{\text{Area of } \triangle CPB}$$

$$\Rightarrow \frac{27}{x} = \frac{x}{12}$$

$$x^2 = 27 \times 12 = 3 \times 3 \times 3 \times 3 \times 2 \times 2$$

$$x = 18$$

49. (d)



Since, radius of first two circles is same

$\therefore P$ is mid-point of AE .

$\therefore P$ is centre of third circle.

$\therefore \text{Radius} = AP = 1 + 1 = 2 \text{ cm}$

50. (d) Radius of cylinder, hemisphere and cone = 5 cm

Height of cylinder = 13 cm

Height of cone = 12 cm

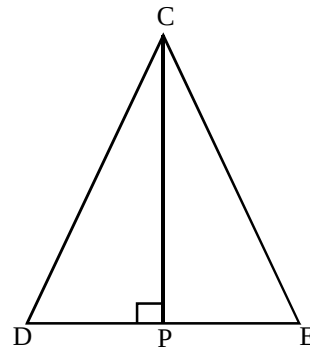
$$\text{Surface area of toy} = 2\pi rh + \frac{4\pi r^2}{2} + \pi rL$$

$$L = \sqrt{h^2 + r^2} = \sqrt{12^2 + 5^2} = 13$$

$$\text{Then } \Rightarrow (2 \times 3.14 \times 5 \times 13) + (2 \times 3.14 \times 25) + (3.14 \times 5 \times 13) \Rightarrow 770 \text{ cm}^2.$$

51. (d) Total surface area of the cube = $6 (\text{Side})^2 = 150$
 New surface area added = $4 \times (2 \times 5)$ [Surfaces $\times$ side of square cross section $\times$ depth]
 Total old surface area to be subtracted = $4 + 4 = 8$
 Hence net surface area = $150 + 40 - 8 = 182 \text{ cm}^2$

52. (b)



Area of equilateral triangle

$$ABC = \frac{\sqrt{3}}{4} \times (6)^2 = 9\sqrt{3} \text{ cm}^2$$

$$\text{Area of } \triangle ADE = \frac{1}{2} \times DE \times CP$$

$$= \frac{1}{2} \times 2 \times \frac{\sqrt{3}}{2} \times 6$$

$$= 3\sqrt{3} \text{ cm}^2$$

$$\therefore \text{Area of shaded region} = 9\sqrt{3} - 3\sqrt{3} = 6\sqrt{3} \text{ cm}^2$$

53. (b) $\frac{\text{Area of } \triangle DAE}{\text{Area of } \triangle DEC} = \frac{\frac{1}{2} \times DE \times AE}{\frac{1}{2} \times DE \times CE}$

$$= \frac{AE}{CE} = \frac{(AD)^2}{(DC)^2} = \left(\frac{6}{8}\right)^2 = \frac{9}{16}$$

Similarly, in $\triangle ABC$,

$$\frac{\text{Area of } \triangle BCF}{\text{Area of } \triangle BFA} = \frac{9}{16}$$

$$\therefore \text{The area of shaded to unshaded region} = \frac{16}{9}$$

54. (d) $\pi r_1^2 + \pi r_2^2 = 180\pi$... (1)

$$\Rightarrow r_1^2 + r_2^2 = 180 \text{ and distance between centers i.e.}$$

$$= r_1 + r_2 = 6 \Rightarrow r_2 = 6 - r_1$$

$$\text{From the eq. (1), } r_1^2 + (6 - r_1)^2 = 180$$

$$\Rightarrow r_1^2 + (r_1^2 - 12r_1 + 36) = 180$$

$$\Rightarrow 2r_1^2 - 12r_1 + 36 = 180$$

$$\Rightarrow 2r_1^2 - 12r_1 - 144 = 0$$

$$\Rightarrow (r_1 - 12)(r_1 + 6) = 0.$$

$$\text{Hence, } r_1 = 12 \text{ cm and } d_1 = 24 \text{ cm.}$$

55. (c) I. material cost = ₹ 2.50 per m².
 II. Labour cost = ₹ 3500
 III. Total cost = ₹ 14500
 Let the area be A sq. metres
 ∴ Material cost = ₹ (14500 - 3500) = ₹ 11000

$$\therefore \frac{5A}{2} = 11000 \Rightarrow A = \left(\frac{11000 \times 2}{5} \right) = 4400 \text{ m}^2$$

Thus, all I, II and III are needed to get the answer.

56. (a) From II, base : height 5 : 12
 Let base = 5x and height = 12x.

$$\text{Then, hypotenuse} = \sqrt{(5x)^2 + (12x)^2} = 13x$$

From I, perimeter of the triangle = 30 cm.

$$\therefore 5x + 12x + 13x = 30 \Rightarrow x = 1$$

So, base 5x = 5 cm., height = 12x = 12 cm.

$$\therefore \text{Area} = \left(\frac{1}{2} \times 5 \times 12 \right) \text{ cm}^2 = 30 \text{ cm}^2$$

Thus, I and II together give the answer.

Clearly III is redundant, since the breadth of the rectangle is not given.

57. (d) From I and II, we can find the length and breadth of the rectangle and therefore the area can be obtained.
 So, III is redundant.
 Also, from II and III, we can find the length and breadth and therefore the area can be obtained.
 So, I is redundant.

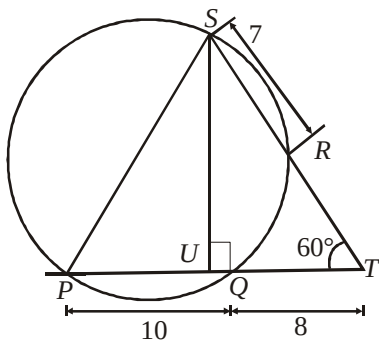
58. (d) $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi r^2 h$

Now r and h can be determined from any two of I, II and III.

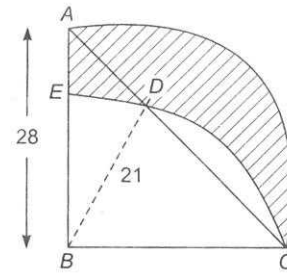
Thus, R can be calculated.

59. (c) Let the circle represent the park.
 Area enclosed by the paths PT, TS and PS

$$= \text{area of } \triangle PTS = \frac{1}{2}(PT)(SU) \\ = \frac{1}{2}(18) \frac{\sqrt{3}}{2} (16) = 72\sqrt{3}$$



60. (a)



Area of shaded portion = Area of ADC - Area of sector DC + Area of $\triangle ADB$ - sector BED

$$\Rightarrow \text{Area of ADC} = \pi \times (17.5)^2 \times \frac{1}{2} = 481 \text{ cm}^2$$

$$\frac{\angle DBC}{\angle ABC} = \frac{21}{28} \Rightarrow \angle DBC = 67.5 \text{ and } \angle DBA = 22.5$$

$$\Rightarrow \text{Area of sector DC} = \left(\pi \times 21^2 \times \frac{67.5}{360} \right) - \left(\frac{1}{2} \times 21^2 \times \sin 67.5 \right) = 56 \text{ cm}^2$$

$$\text{Area of ADE} = \left(\frac{1}{2} \times 28 \times 21 \right)$$

$$- \left(204 + \frac{1}{2} \times 21^2 \times \sin 22.5 \right) = 5.6 \text{ cm}^2$$

$$\text{Thus area of shaded portion} = 480 - 56 + 5.6 = 429 \text{ cm}^2$$

61. (d) $PQ = QR = RS = \frac{12}{3} = 4 \text{ cm}$

$$\text{Area of unshaded region} \Rightarrow \frac{\pi 6^2}{2} + \frac{\pi 4^2}{2}$$

$$\Rightarrow 18\pi + 8\pi \Rightarrow 26\pi$$

$$\text{Area of shaded region} \Rightarrow \frac{\pi 6^2}{2} - \frac{\pi 4^2}{2}$$

$$\Rightarrow 18\pi - 8\pi = 10\pi$$

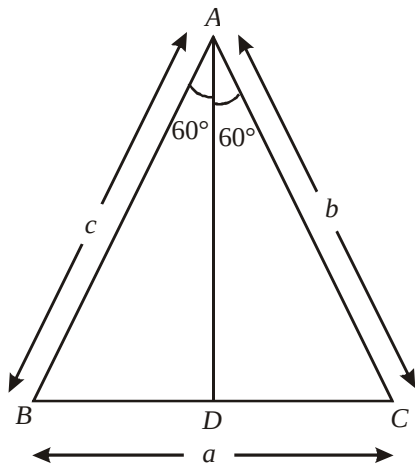
$$\text{Ratio} = \frac{10\pi}{26\pi} \Rightarrow \frac{5}{13} \Rightarrow 5:13$$

62. (d) If the radius of smaller circle is 1 unit, then the radius of the bigger circle is $\sqrt{2} + 1$ units.

So, the answer in this case would be the area of square ABCD - 4 quadrants of the smaller circle.

$$= 4 - \pi$$

63. (b)



Let $AD = h$ (say)

then area of $\triangle ABC$

$$= \frac{1}{2} bc \sin 120^\circ = \frac{\sqrt{3}}{4} bc$$

Area of $\triangle BAD$

$$= \frac{1}{2} ch \sin 60^\circ$$

$$= \frac{\sqrt{3}}{4} ch$$

$$\text{and area of } \triangle CAD = \frac{1}{2} bh \sin 60^\circ = \frac{\sqrt{3}}{4} bh$$

$$\text{Now, } ar(\triangle ACD) = ar(\triangle BAD) + ar(\triangle CAD)$$

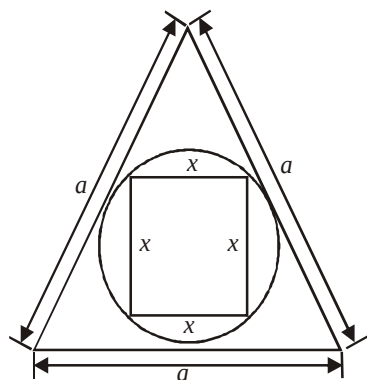
$$\frac{\sqrt{3}}{4} bc = \frac{\sqrt{3}}{4} ch + \frac{\sqrt{3}}{4} bh$$

$$\Rightarrow bh = h(b + c)$$

$$\Rightarrow h = \frac{bc}{b + c}$$

Expert Level

1. (b)



$$\text{Radius of circle } R = \frac{\frac{\sqrt{3}}{4} a}{\frac{3a}{2}} = \frac{\sqrt{3}}{6} a$$

Diagonal of square = diameter of circle

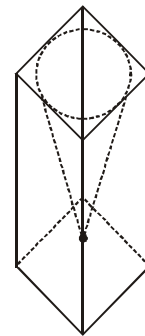
$$= 2 \cdot \frac{\sqrt{3}}{6} a$$

$$= \sqrt{2} x = \frac{\sqrt{3}}{3} a$$

$$x = \frac{\sqrt{3}}{3\sqrt{2}} a = \frac{\sqrt{6}}{6} a$$

$$\text{Area} = x^2 \left[\frac{(\sqrt{6}a)}{6} \right]^2 = \frac{6}{36} a^2 = \frac{1}{6} a^2$$

2. (d) The cone is shown below with its face as a circle inscribed in one of the surfaces of the cube and its vertex on the opposite side.



Area of the cube

$$= 6 \times 100 = 600 \text{ cm}^2.$$

The base of the cone = $25\pi \text{ cm}^2$

Lateral surface of cone

$$= \pi \times 5\sqrt{100+25} = 25\sqrt{5} \pi \text{ cm}^2$$

$\therefore$ New surface area

= Area of cube – area of base of cone + lateral surface

$$\text{area of cone} = 600 + 25(\sqrt{5} - 1) \pi$$

3. (c) Area of $\triangle ABC = \frac{1}{2} \times 15 \times 20 = 150 \text{ sq. cm.}$

$$AB = \sqrt{AC^2 + BC^2} = \sqrt{225 + 400} = \sqrt{625} = 25$$

$$\text{Area of } ABC = \frac{1}{2} \times 25 \times CD = 150 ; CD = 12 \text{ cm.}$$

PQ = Radius of incircle of ACD + Radius of incircle of BCD

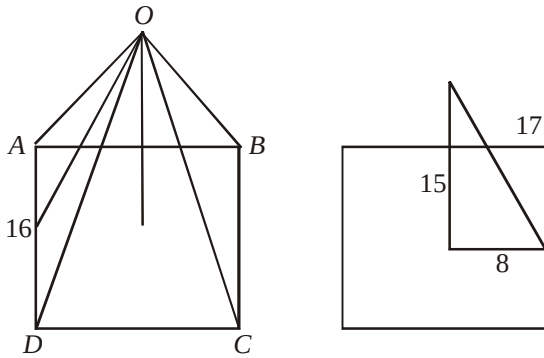
Radius of incircle triangle ADC

$$\text{Area of } ADC = r \times s$$

$$\text{Area of } ADC = \frac{1}{2} \times 12 \times 16 = 96$$

$$96 = r \times \left(\frac{12 + 20 + 16}{2} \right) = 24r \Rightarrow r = 4 \quad \therefore PQ = 7$$

4. (d) $\ell^2 = 15^2 + 8^2 = 225 + 64 = 289 \Rightarrow \ell = \sqrt{289} = 17$ cm.



Total surface area = Area of square base + 4 × Area of triangular side faces

$$= (16)^2 + 4 \times \frac{1}{2} \times 16 \times 17 = 256 + 544 = 800 \text{ sq. cm.}$$

5. (c) Let the radius of the semicircle be R .
Now join O to B

$$OC = OD = R \quad \therefore OB = R\sqrt{2}$$

The diameter of the smaller circle

$$= (R\sqrt{2} - R) = R(\sqrt{2} - 1)$$

$$\text{Area of the semicircle} = \frac{\pi R^2}{2};$$

$$\text{Area of the circle} = \frac{\pi R^2 (\sqrt{2} - 1)^2}{2^2}$$

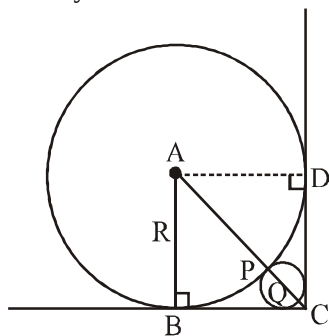
Hence the ratio of the area of the smaller circle to that

$$\text{of the semicircle} = \frac{(\sqrt{2} - 1)^2}{2} \text{ or } (\sqrt{2} - 1)^2 : 2$$

6. (c) Joining B to O and C to O
Let the radius of the outer circle be r
 $\therefore$ Perimeter = $2\pi r$
But $OQ = BC = r$ [diagonals of the square $BQCO$]
 $\therefore$ Perimeter of $ABCD = 4r$.

$$\text{Hence, ratio} = \frac{2\pi r}{4r} = \frac{\pi}{2}$$

7. (d) Let the radii of the bigger and smaller circles be R and r respectively.



$\therefore$ In the figure $AB = AD = R$

As $\angle ADC = 90^\circ$, $\angle ABC = 90^\circ$ and $\angle DCB = 90^\circ$

$$\therefore BC = R \text{ and } AC = \sqrt{2} R \text{ and } AC = AP + PQ + QC \\ = R + r + \sqrt{2} r \text{ (QC = } \sqrt{2} r \text{ can be proved in the same}$$

$$\text{way as we proved } AC = \sqrt{2} R \therefore r = \frac{(\sqrt{2} - 1) R}{\sqrt{2} + 1}$$

Rationalising the denominator, we get

$$r = (3 - 2\sqrt{2}) R$$

$$\text{Given } R = 2, \text{ we get } r = 2(3 - 2\sqrt{2}) = 6 - 4\sqrt{2}$$

8. (d) The surface area of the cube is
cube = $6 \times 7^2 = 294$
The height of one of the four triangular faces of the pyramid is

$$h = \sqrt{7^2 + \frac{7^2}{4}} = \frac{1}{2} \times 7 \times \sqrt{5} = 7.82624$$

The area one of the four triangular faces of the pyramid is

$$t = \frac{1}{2} \times 7.82624 \times 7 = 27.39183$$

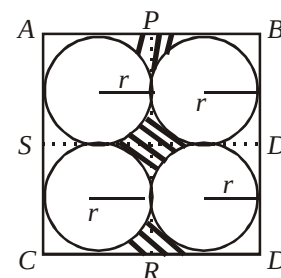
The surface area of the pyramid is

$$\text{pyramid} = 7^2 + 4t = 7^2 + 4 \times 27.39183 = 158.56732$$

The answer is :

$$\text{cube} - \text{pyramid} = 294 - 158.56732 = 135.43268 = 135.4$$

9. (a) The side of the square given is 5 cm, and the radius of the circle is $5/4$ cm. By dividing the square as shown alongside the shaded area of each section is
(area of the square
– area of the circle) $\times 1/2$



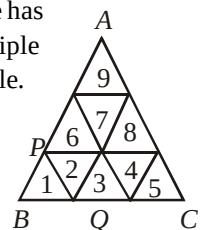
$$= [(5/2)^2 - \pi (5/4)^2] 1/2 \\ = 25/40 [4 - \pi] 1/2 \\ = 25/32 (4 - \pi)$$

Area of the shaded portion in the given

$$\text{Figure is } 4 \times 25/32 (4 - \pi) = 25/8 (4 - \pi) \text{ sq. cm}$$

10. (b) Since larger equilateral triangle has a side which is an integral multiple of the side of the smaller triangle.
 $\therefore$ The number of triangles

$$= \frac{\sqrt{3}/4 \times 6 \times 6}{\sqrt{3}/4 \times 2 \times 2} = 9$$



11. (c) Shortest distance between the parallel sides
= (height of the equilateral triangle in outer hexagon)
– (height of the equilateral triangle in inner hexagon)

$$\sqrt{3}/2 (6) - \sqrt{3}/2 (a) = 2\sqrt{3} \cdot 6 - a = 4, a = 2$$

$$\text{shaded area} = 6(36 - 4) \sqrt{3}/4 = 48\sqrt{3}$$

12. (a) Sum of areas of two smaller circles = $2 \times \pi \times 4^2 = 32\pi$.
Area of larger circle = $\pi \times 8^2 = 64\pi$.

$$\text{Area of shaded portion} = \frac{64\pi - 32\pi}{2} = \frac{32\pi}{2} = 16\pi$$

13. (b) Let the radius of the circle, i.e., OG, be r cm
Area of circle = πr^2 sq. cm.

$$\text{Height of the triangle} = \frac{3r}{2} \text{ cm,}$$

$$\frac{\sqrt{3}}{2} \times s = \frac{3r}{2} \Rightarrow s = \frac{\sqrt{3}r}{2}$$

$$\text{Area of the equilateral triangle} = \frac{h^2}{\sqrt{3}} - \frac{3\sqrt{3}r^2}{4}$$

Ratio of the area of equilateral triangle to the that of circle

$$= \frac{3\sqrt{3}r^2}{4} : \pi r^2 = 21\sqrt{3} : 88.$$

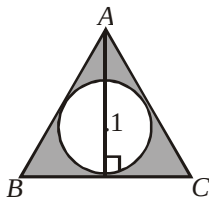
14. (d) $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c} \Rightarrow \frac{\cos A}{k \sin A} = \frac{\cos B}{k \sin B} = \frac{\cos C}{k \sin C}$
 $\Rightarrow \cot A = \cot B = \cot C \Rightarrow A = B = C = 60^\circ$
 $\Rightarrow \Delta ABC$ is equilateral.
 $\therefore \Delta = \frac{\sqrt{3}}{4} a^2 = \sqrt{3}$

15. (a) Height = $AD = \frac{\sqrt{3}}{2} \times \text{side} = \frac{\sqrt{3}}{2} \times 6 = 3\sqrt{3}$ cm
 $\therefore$ Inradius = $\frac{1}{3} \times \text{height} = \frac{1}{3} \times 3\sqrt{3} = \sqrt{3}$ cm.

$$A(\text{shaded region}) = A(\Delta ABC) - A(\text{incircle})$$

$$= \frac{\sqrt{3}}{4} \times (\text{side})^2 = \pi r^2$$

$$= \frac{\sqrt{3}}{4} \times 6 \times 6 - 3.14 \times (\sqrt{3})^2$$



$$= 15.57 - 9.42 = 6.15 \text{ cm.}$$

16. (c) ABC is a city park
AC = 80

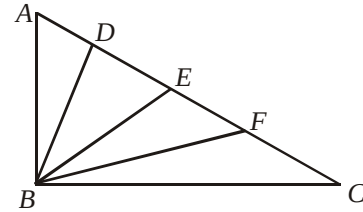
$$AD = DE = EF = FC = 20 \text{ m}$$

$$\text{Let } AB = a, BC = b$$

$$\therefore a^2 + b^2 = 80^2$$

$$\text{Also } BE^2 = AE \cdot EC = 40 \cdot 40$$

$$\therefore BE = 40 \text{ m}$$



Using Apollonius theorem in ΔABE , as $AD = DE$
 $AB^2 + BE^2 = 2(BD^2 + AD^2)$

$$BD^2 + 20^2 = \frac{1}{2}(a^2 + 40^2) \quad \dots (1)$$

Similarly, for ΔBEC , as $EF = FC$
 $BE^2 + BC^2 = 2(BF^2 + FC^2)$

$$BF^2 + 20^2 = \frac{1}{2}(BE^2 + BC^2) = \frac{1}{2}(b^2 + 40^2) \quad \dots (2)$$

Adding (1) and (2)

$$BD^2 + BF^2 + 2 \cdot 20^2 = \frac{1}{2}(a^2 + b^2 + 2 \cdot 40^2)$$

$$= \frac{1}{2}(80^2 + 2 \cdot 40^2)$$

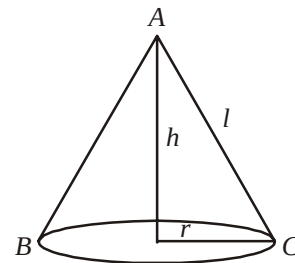
$$\therefore BD^2 + BE^2 + BF^2 = \frac{1}{2}(6400 + 3200)$$

$$= 3200 + 1600 - 800 + 1600$$

$$= 5600$$

17. (d) Let ABC be the conical tent of given capacity = $\frac{1}{3}\pi r^2 h$,

where 'h' be the height and 'r' be the radius of the base.



Let 'l' be the slant height of the conical tent.

$$\text{Now, surface area (S.A)} = \pi r l$$

$$\pi r \sqrt{h^2 + r^2}$$

$$\pi r^2 \sqrt{\left(\frac{h}{r}\right)^2 + 1}$$

Now, to find the ratio of the height to the radius for minimum amount of canvas, we consider options

- (a) $h = 1, r = 2 \Rightarrow S.A = 4\pi\sqrt{5}/4 = 2\sqrt{5}\pi$
 (b) $h = 2, r = 1 \Rightarrow S.A = \pi\sqrt{5}$
 (c) $h = 1, r = \sqrt{2} \Rightarrow S.A = 2\pi\sqrt{3/2} = \sqrt{6}\pi$
 (d) $h = \sqrt{2}, r = 1 \Rightarrow S.A = \pi\sqrt{2+1} = \sqrt{3}\pi$ (min)

Hence, only option (d) is the correct option.

18. (c) Radius = $\frac{105}{2} = 52.5$ cm

Area of the entire canvas, used for the tent
 = Area of cylinder + centre of cone
 = $2\pi rh + \pi r^2$

$$= \pi r (2h + r) = 3.14 \times 52.5 \left(2\sqrt{53^2 - 52.5^2} + 52.5 \right)$$

= $5 \times l$ (because area of canvas = $l \times b$ also)
 = $l \Rightarrow 1947$ m

19. (c) Equate the area of the square ABCD and triangle PDC and find a relation between the slant height and the length of the base of the pyramid.

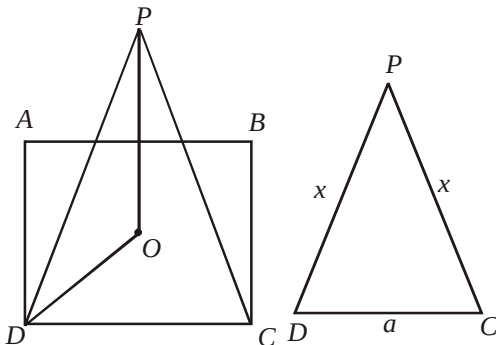
$$a^2 = \frac{a}{4} \sqrt{4x^2 - a^2}$$

$$16a^2 = 4x^2 - a^2$$

$$\Rightarrow 17a^2 = 4x^2$$

$$\frac{a}{x} = \frac{2}{\sqrt{17}}$$

$$\text{Let } a = 2\pi; x = \sqrt{17}p$$



Now in ΔPOD

$$OD = \sqrt{2}P \quad PD = \sqrt{17}P$$

$$AO = \sqrt{15}P \quad \text{But } AO = 3 \Rightarrow P = \frac{3}{\sqrt{15}}$$

$$2P = a = \frac{6}{\sqrt{15}}$$

$$\text{Area of the base} = \frac{36}{15}$$

$$\text{Total surface area} = \frac{36}{15} \times 5 = 12$$

20. (d) Volume of hollow space = area of the triangle $\times$ height
 (100 cm) $- \frac{1}{2}$ volume of the cylinder of base radius 2 cm and height 100 cm

21. (d) Let the side of the equilateral triangle be a cm.

$$\text{Area of the triangle } ABC = \frac{\sqrt{3}a^2}{4}$$

$$\text{Height of the triangle} = \frac{\sqrt{3}a}{2} \text{ cm}$$

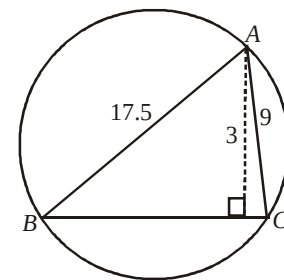
As G is circumcentre

$$\text{Area of the square } EFGC = \left(\frac{a}{\sqrt{3}} \right)^2 = \frac{a^2}{3} \text{ sq. cm.}$$

Ratio area of ΔABC to $EFGC$

$$\frac{\sqrt{3}a^2}{4} : \frac{a^2}{3} \Rightarrow 3\sqrt{3} : 4.$$

22. (d)



Radius of circum circle of a triangle,

$$R = \frac{a \times b \times c}{4\Delta}$$

$$\therefore \Delta = \frac{a \times b \times c}{4R} = \frac{17.5 \times 9 \times BC}{4R}$$

$$\text{Also } \Delta = \frac{1}{2} \times BC \times 3$$

$$\therefore \frac{17.5 \times 9 \times BC}{4R} = \frac{1}{2} \times BC \times 3$$

$$\therefore R = 26.25.$$

23. (a) Area of $\Delta ABD = 1/2 \times$ area of rectangle ABCD

As $AF = 2/3 AB$

Area of $\Delta AFD = 2/3$ area of ΔABD

As $AG = GH = HD = 1/3 AD$,

Area of $GFH = 1/3$ Area of ΔAFD

$$\therefore \text{Area of } \Delta GFH = \frac{1}{3} \times \frac{2}{3} \times \frac{1}{2} \times \text{Area of rectangle ABCD}$$

$$= \frac{1}{9} \times 216 = 24 \text{ sq. units}$$

As $FI = IJ = JD = 1/3 FD$.

Area of ΔFHI

$1/3 \times$ area of ΔFHD

$= 1/3 \times 1/3 \times 2/3 \times 1/2 \times$ Area of rectangle $ABCD$

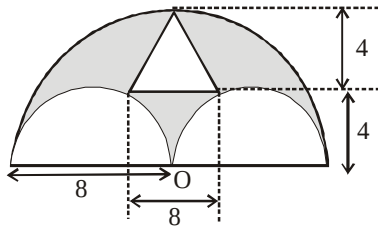
$= 1/27 \times 216 = 8$ sq. units.

$\therefore$ Area of shaded region $= 24 + 8 = 32$ sq. units

24. (c) Boundary of shaded region = Circumference of four semicircles (two circles, $r = 7/2$) + Circumference of two quarter circles (one semi-circles, $r = 7$)

$$\Rightarrow (2 \times 2\pi r) + \pi \times 2r = (4 \times 22/7 \times 7/2) + (22/7 \times 7) \\ = 44 + 22 = 66 \text{ cm.}$$

25. (c)

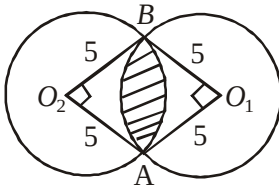


Shaded area = Area of big semicircle – (Area of 2 semi-circles + area of triangle)

$$= \frac{\pi 8^2}{2} - \left(\frac{2\pi 4^2}{2} + \frac{1}{2} \times 8 \times 4 \right) = 32\pi - 16\pi - 16$$

$$= 16(\pi - 1)$$

26. (c) It is clear that any two circles in the figure intersect orthogonally.



Consider the area of shaded region in this figure

= Area (sector O_1AB) + Area

(sector of O_2BA) – area (square O_1AO_2B)

$$= \frac{1}{4}\pi(5)^2 + \frac{1}{4}\pi(5)^2 - (5)^2$$

$$\frac{1}{2}\pi(5)^2 - (5)^2 = 25\left(\frac{\pi}{2} - 1\right)$$

Hence, the required answer = 4 times the area calcu-

$$\text{lated above} = 4 \times 25\left(\frac{\pi}{2} - 1\right) = 100\left(\frac{\pi}{2} - 1\right)$$

27. (d) Let the radii of the three base circle be $r, 2r, 3r$,

(1) Volume $A = 1/3\pi r^2$

Volume $B = 1/3\pi (2r)^2 \times 2 - 1/3\pi r^2 = 7/3\pi r^2$

$\Rightarrow$ Volume A : Volume $B = 1/3\pi r^2 : 7/3\pi r^2 = 1 : 7$.

$$(2) \text{ Volume } C = 1/3\pi (3r)^2 \times 3 - 7/3\pi r^2 - 1/3\pi r^2 \\ = 9\pi r^2 - 7/3\pi r^2 - 1/3\pi r^2 \\ = 19\pi r^2/3 \text{ Volume } B : \text{Volume } C = 7/3\pi r^2 : 19\pi r^2 \\ = 7 : 19$$

$$(3) \text{ Curved surface area of } B : \text{curved surface area of } C \\ [\pi(2r)(21) - \pi(r)(1)] / [\pi(3r)(31) - \pi(2r)(21)] \\ = \pi r(41) / \pi r(91 - 42) = 41/49 \text{ or } 41 : 49$$

28. (a) Let each turn be of length

$$2\pi r = \frac{2\pi \times 2}{\pi} = 4 \text{ cms.}$$

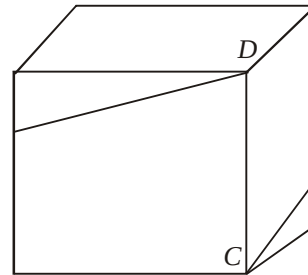
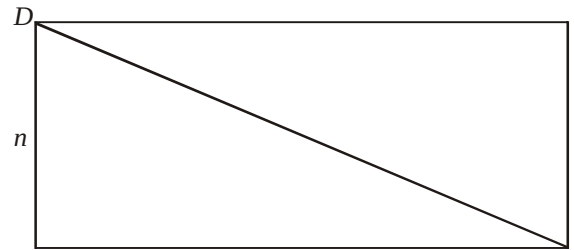
Thus for n turns, length needed will be $4n$ cms.

$$\text{Total surface area} = 2\pi rh \Rightarrow 2\pi \times \frac{2}{\pi} \times h = 4h$$

If turns are equally spaced, then distances between

$$\text{vertical turns is } \frac{4h}{4n} = \frac{h}{n}$$

29. (a) If we cut open the cube, we will get a rectangle with sides $4n$ and n .



So the length of the string is $\sqrt{17}n$

30. (c) Based on the above solution itself, we can say $h = n$.

$$31. (c) \text{ Interior angle of hexagon} = \frac{(2 \times 6 - 4)}{6} \times 90 = 120^\circ$$

$$\text{Area of hexagon} = \frac{1}{2} \times 6 \times 6 \times \frac{\sqrt{3}}{2} \times 6 = 54\sqrt{3} \text{ sq. cm}$$

Area of a hexagon occupied by circles

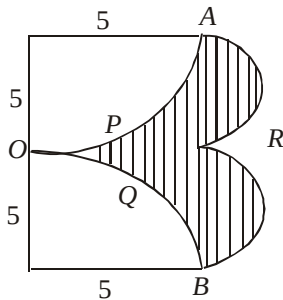
$$= 2 \times \frac{120}{360} \times \pi \times 6^2 = 24\pi \text{ sq. cm.}$$

Area of a shaded region

$$= 54\sqrt{3} - 24\pi = 6(9\sqrt{3} - 4\pi) \text{ sq. cm.}$$

32. (a) Area of two small semicircles

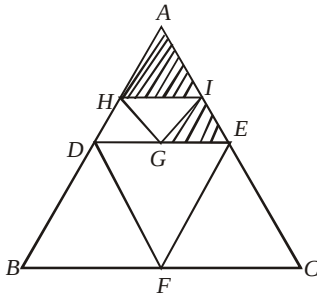
$$= 2 \times \frac{\pi(5/2)^2}{2} = 25\pi/4 \text{ cm}^2$$



Area of the shaded region = Area of the rectangle
 - (Area of the quadrants APQ and BQO)
 $= (5 \times 10) - 2 \times \frac{\pi \times 5^2 \times 1/4}{} = 50 - 25\pi/2$
 $\therefore$ Total area of the shaded region

$$= 50 - \frac{25\pi}{2} + \frac{25\pi}{4} = 25 \left(2 - \frac{\pi}{4} \right)$$

33. (c)



Consider the above diagram, where the area of $\triangle ABC = A$
 The heights of $\triangle AHI$, $\triangle HGI$, $\triangle GEI$ are same as they are
 between the same two parallels.

Area of $\triangle HGI$ + Area of $\triangle AHI$ + Area of $\triangle GEI$

$$= \frac{3}{4} \times \frac{1}{4} A \quad \dots(1)$$

$$\text{Since } HG = \frac{1}{2} AE,$$

$$\triangle HGI = \frac{1}{2} (\text{sum of } \triangle AHI + \triangle HGI) \quad \dots(2)$$

From (1) and (2)

$$\text{Area of the shaded portion} = \frac{2}{3} \times \frac{3}{16} A = \frac{1}{8} A$$

$$\text{Area of } \triangle DEF = \frac{1}{3} \text{ of trapezium } DEBC$$

$$= \frac{1}{3} \times \frac{3}{4} \text{ of } A = \frac{1}{4} A$$

$$\text{Therefore required ratio} = \frac{1}{8} A : \frac{1}{4} A = 1 : 2$$

34. (c) Let C_1, C_2, C_3, C_4 be the circles as shown having centre O, A, B and C respectively. C_2 and C_3 meet at D .

Let r be the radius of circle C_4 having centre C . O, A, B be the centre of circle C_1, C_2 and C_3 respectively.

Now given radius of C_2 and C_3 are 10 cm and 5 cm respectively and radius of C_1 is 15 cm.

$$\therefore AC = 10 + r, BC = 5 + r, OC = 15 - r$$

Let $DC = x$

$\therefore AO = OD$ so we can use Apollonius theorem in $\triangle ADC$

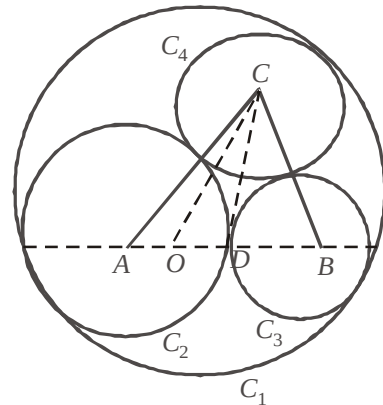
$$AC^2 + DC^2 = 2(OC^2 + OD^2)$$

$$(10 + r)^2 + x^2 = 2[5^2 + (15 - r)^2]$$

Similarly, for $\triangle BCO$ as $BD = OD$

$$OC^2 + BC^2 = 2[CD^2 + BD^2]$$

$$(15 - r)^2 + (5 + r)^2 = 2[x^2 + 5^2]$$



Eliminating x and solving for r

$$r = \frac{60}{14} = \frac{30}{7}$$

$$2r = \text{Diameter} = \frac{60}{7} = 8.57$$

Hence (c) is the correct option

35. (b) Let the side of the square be = 1 unit.

Radius of the circle = $OR = x$.

Also $AO = AP - OP = 1 - x$

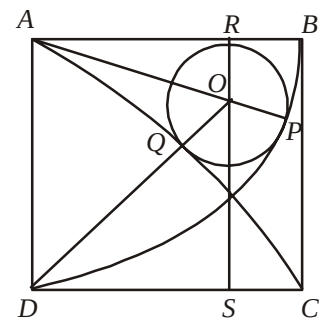
$DO = DQ + OQ = 1 + x$

$OS = 1 - x$

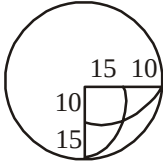
$$\therefore (1 - x)^2 - x^2 = (1 + x)^2 - (1 - x)^2$$

$$\Rightarrow 1 = 6x \Rightarrow x = \frac{1}{6}$$

$$\text{As } AB = 60, OR = \frac{1}{6} (60) = 10$$



36. (b)



For 3 quadrants there would be no difference as all of the field here is available for grazing.

For the IV one area for grazing before shed was made

$$= \frac{\pi \times 25^2}{4}$$

$$\text{after shed is made} = \frac{\pi}{4} (10^2 + 15^2)$$

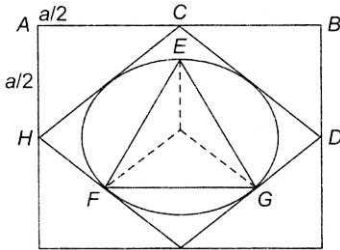
Fraction of the total area over which cow can graze

$$= \frac{10^2 + 15^2}{25^2} = \frac{2^2 + 3^2}{5^2} = \frac{13}{25}$$

$$\therefore \text{Rent for the quadrant would be } \frac{13}{25} (250) = 130$$

$$\therefore \text{The new rent should be } 3 \times \frac{1000}{4} + 130 = 880$$

37. (d)



AB - side of the outermost triangle = a

AC = CB = a/2

$$HC = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \frac{a}{\sqrt{2}}$$

$$\text{Diameter of circle} = \frac{a}{\sqrt{2}}; \text{radius} = \frac{a}{2\sqrt{2}}$$

O is the centre of the circle. Then $\angle EOF = \angle 120^\circ$

$$\text{Then Area of } \triangle EOF = \frac{1}{2} EO \cdot OF \cdot \sin 120^\circ$$

$$\Rightarrow \frac{1}{2} \times \frac{a^2}{8} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}a^2}{32}$$

$$\text{Then area of } \triangle EFG = \frac{3\sqrt{3}a^2}{32}$$

38. (a) Distance after 4 hours = AB = C

$$a = 3 \times 4 = 12; b = 2 \times 4 = 8$$

$$\text{and } \frac{a+b+c}{2} \Rightarrow \frac{12+8+C}{2} \Rightarrow \left(\frac{10+C}{2} \right)$$

$$\text{Area} = \sqrt{S(S-a)(S-b)(S-c)}$$

$$\text{Area} = \frac{1}{2} ab \sin 120^\circ$$

$$\text{Area} \Rightarrow 48 \times \frac{\sqrt{3}}{2} = 24\sqrt{3}$$

As per question:

$$24\sqrt{3} \sqrt{\left(10 + \frac{c}{2}\right)\left(\frac{c}{2} - 2\right)\left(2 + \frac{c}{2}\right)\left(10 - \frac{c}{2}\right)}$$

On solving, we get $c = 4\sqrt{19}$ km

39. (d) Volume of the elliptical cylinder

$$= \pi \times \frac{2.4}{2} \times \frac{1.6}{2} \times 7$$

$$= 3.14 \times 1.2 \times 0.8 \times 7 \Rightarrow 9\text{m}^3$$

Amount of water emptied per minute

$$120 \times 3.14 \times \left(\frac{2}{100}\right)^2$$

Time required to empty half the tank

$$\frac{4.5}{120 \times 3.14 \times 0.02^2} = 70 \text{ min}$$

40. (a) Notice that all the given triangles are equilateral

$$\text{Area of shaded region} = 3 \left[\pi r^2 \frac{60}{360} - \frac{\sqrt{3}}{4} \times r^2 \right]$$

$$= \frac{r^2}{2} \left[\pi - \frac{3\sqrt{3}}{2} \right]$$

$$41. (a) \frac{AE}{EC} = \frac{AB}{BC} = \frac{7.5}{10.5} = \frac{5}{7}$$

$$\text{Now, } AB^2 + BC^2 = AC^2$$

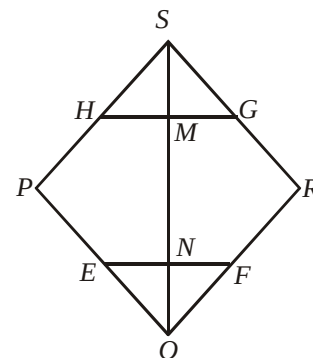
$$(5k)^2 + (7k)^2 = (18)^2$$

$$74k^2 = 324 \Rightarrow k^2 = \frac{324}{74}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} \times AB \times BC = \frac{1}{2} \times 5k \times 7k$$

$$= \frac{35}{2} k^2 = \frac{35}{2} \times \frac{324}{74} = 76.621 \text{ cm}^2$$

42. (c)



Let $AD = 3a$ and $DC = 6a$

$$\therefore DH = HG = GC = \frac{6a}{3} = 2a$$

$$HM = MG = \frac{2a}{a} = a = SM$$

$$\therefore NQ = a$$

$$\therefore SQ = SM + MN + NQ$$

$$= a + 3a + a = 5a$$

Since diagonal of square $SQ = 5a$

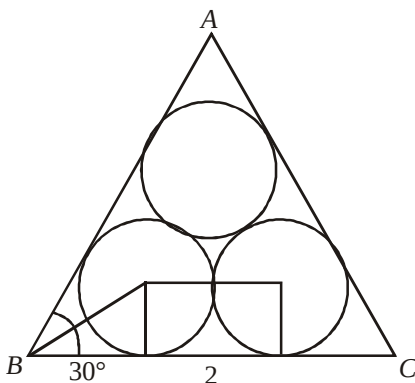
But, diameter of circle $SQ =$ diagonal of square SQ

$$\therefore \text{Radius of the circle} = \frac{5a}{2}$$

$$\Rightarrow \text{Area of the circle} = \pi \times \left(\frac{5a}{2}\right)^2$$

$$\text{Here } \frac{\text{Area of circle}}{\text{Area of rectangle}} = \frac{\frac{25}{4}(a^2\pi)}{3a \times 6a} = \frac{25\pi}{72}$$

43. (b) It can be seen that the side of the triangle is



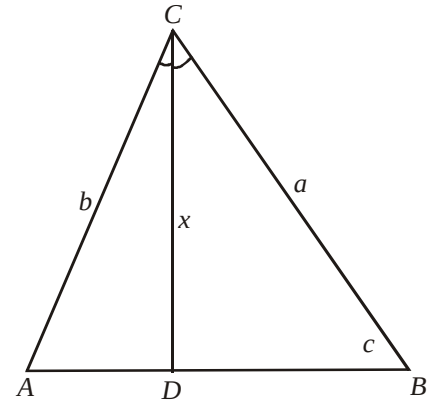
44. (d) Let $CD = x$

$$\text{Area } \triangle ABC = \triangle ACD + \triangle DCB$$

$$\Rightarrow \frac{1}{2}ab \sin 120^\circ = \frac{1}{2}bx \sin 60^\circ + \frac{1}{2}ax \sin 60^\circ$$

Now it can be observed that x

$$x = \frac{ab}{a+b}$$



45. (a) Area of the quarter circle = $\frac{\pi r^2}{4} \Rightarrow 0.25\pi$. Going by options, we have to see that the area of the inserted circle is less than the area of the quarter circle.

$$\text{Option (b)} \frac{(\sqrt{2}+1)}{2} \Rightarrow \text{Area} = (1.5 + \sqrt{2})\pi$$

$$2.9\pi > 0.25\pi. \text{ Hence discarded.}$$

$$\text{Option (c)} \sqrt{2} - \frac{1}{2} \Rightarrow \text{Area} = \pi \left(2 + \frac{1}{4} - \sqrt{2}\right)$$

$$\Rightarrow 0.75\pi > 0.25\pi. \text{ Hence discarded}$$

$$\text{Option (d)} 1 - \sqrt{2} \Rightarrow \text{Area} = \pi(1 + 8 - \sqrt{2})$$

$$\Rightarrow 0.85\pi > 0.25\pi. \text{ Hence discarded.}$$

$$\text{Option (a)} \sqrt{2} - 1 = \text{Area} = \pi(2 + 1 - \sqrt{2})$$

$$\Rightarrow 0.20\pi < 0.25\pi$$

Hence this option is correct.

Explanation of Test Yourself

1. (a) Looking at the options we can easily eliminate option (b) and (d), because in the ratio of the area of the circle to the area of the triangle we cannot eliminate π and hence the answer should contain π .

$\frac{\text{Area of the circle}}{\text{Area of the square}}$ so, π should be in the numerator.

2. (d) In the diagram, there are 27 black triangles. If the entire diagram was divided into the smallest size equilateral triangles, there would be $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 = 64$ equilateral triangles. Thus, $27/64$ of $\triangle ABC$ is coloured black, so $37/64$ is unshaded.

$$\text{Area of triangle } ABC = \left\{ \sqrt{(3)/4} \right\} \times 16 \times 16 = 64\sqrt{3}.$$

Hence, the area of the unshaded portion is $37\sqrt{3}$. Drop a perpendicular from A, meeting BC at D. Since $\triangle ABC$ is equilateral and $AB = 16$, then $BD = DC = 8$.

3. (a) Area left after laying black tiles
 $= [(20 - 4) \times (10 - 4)] \text{ sq. ft.} = 96 \text{ sq. ft.}$

$$\text{Area under white tiles} = \left(\frac{1}{3} \times 96 \right) \text{ sq. ft} = 32 \text{ sq. ft.}$$

$$\text{Area under blue tiles} = (96 - 32) \text{ sq. ft} = 64 \text{ sq. ft.}$$

$$\text{Number of blue tiles} = \frac{64}{(2 \times 2)} = 16.$$

4. (d) Area of the shaded portion = Area of circle – Area of triangle

$$\Rightarrow \text{Area of circle} = \pi r^2 \Rightarrow \frac{22}{7} \times 5 \times 5$$

$$\Rightarrow \frac{22 \times 25}{7} \text{ cm}^2 = 78.50 \text{ cm}^2$$

$$\text{Area of triangle} \Rightarrow \frac{1}{2} \pi r^2 \sin \theta \Rightarrow \frac{1}{2} \times 25 \times \sin \theta$$

$$= \frac{1}{2} \times 25 \times \sin 60^\circ$$

$$\Rightarrow \frac{25\sqrt{3}}{4} \Rightarrow 6.25 \times 1.732 \Rightarrow 10.8 \text{ cm}^2$$

$$\therefore \text{Area of shaded portion} = 78.50 - 10.8 = 67.7 \text{ cm}^2$$

5. (d) Let the length, breadth and height of the cuboid be x , $2x$ and $3x$, respectively.

$$\text{Therefore, volume} = x \times 2x \times 3x = 6x^3$$

New length, breadth and height = $2x$, $6x$ and $9x$, respectively.

$$\text{New volume} = 108x^3$$

$$\text{Thus, increase in volume} = (108 - 6)x^3 = 102x^3$$

$$\frac{\text{Increase in volume}}{\text{Original volume}} = \frac{102x^3}{6x^3} = 17 \text{ times}$$

6. (b) Volume of water displaced = $(3 \times 2 \times 0.01) \text{ m}^3 = 0.06 \text{ m}^3$.

$\therefore$ Mass of man

$$= \text{Volume of water displaced} \times \text{Density of water}$$

$$= (0.06 \times 1000) \text{ kg} = 60 \text{ kg.}$$

7. (b) $L \times B \times 2 = 80\% \text{ of } (6 \times 5 \times 2) = 48$

$$\Rightarrow L \times B = 24$$

$$\text{Now, } 6 - 6 \times 10\% = 5.4,$$

$$5 - 5 \times 10\% = 4.5 \text{ and}$$

$$\text{Therefore, } 5.4 \times 4.5 = 24.3$$

$$\text{Clearly, } 5 < L < 5.5$$

8. (c) Let the height of the vessel be x .

Then, radius of the bowl = radius of the vessel = $x/2$.

$$\text{Volume of the bowl, } V_1 = \frac{2}{3} \pi \left(\frac{x}{2} \right)^3 = \frac{1}{12} \pi x^3.$$

$$\text{Volume of the vessel, } V_2 = \pi \left(\frac{x}{2} \right)^2 x = \frac{1}{4} \pi x^3.$$

Since $V_2 > V_1$, so the vessel can contain 100% of the beverage filled in the bowl.

9. (a) Let the slant height of 1st cone = L

Then the slant height of 2nd cone = $3L$

Let the radius of 1st cone = r_1

And let the radius of 2nd cone = r_2

$$\text{Then, } \pi r_1 L = 3 \times \pi r_2 \times 3L$$

$$\Rightarrow \pi r_1 L = 9\pi r_2 L \Rightarrow r_1 = 9r_2$$

Ratio of area of the base

$$\frac{\pi r_1^2}{\pi r_2^2} \Rightarrow \left(\frac{r_1}{r_2} \right)^2 = \left(\frac{9}{1} \right)^2 \Rightarrow 81 : 1$$

10. (a) The entire dial of the clock = 360°

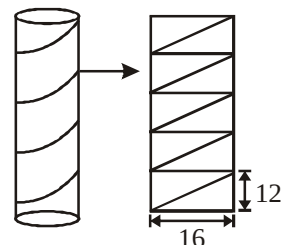
$$\text{Every 5 minutes} = \frac{360^\circ}{12} = 30^\circ$$

$$\text{So 35 minutes} = 30 \times 17 = 210$$

$$\text{Area} = \pi r^2 \times \frac{8}{360} = 3.14 \times 100 \times \frac{210}{360}$$

$$\Rightarrow 183.3 \text{ cm}^2$$

11. (d)



$$\text{The base circumference} = \frac{22}{7} \times \frac{56}{11} = 16 \text{ cm. The}$$

$$\text{length of one complete turn} = \sqrt{16^2 + 12^2} = 20 \text{ cm.}$$

$$\text{Hence, total length} = 80 \text{ cm.}$$

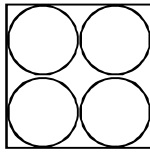
12. (b) Let the radius of iron ball = r_1
Let the radius of oak ball = r_0
Then, as iron weights 8 times more than Oak

$$\therefore \frac{4\pi r_0^3}{3} = \frac{8 \times 4\pi r_1^3}{3} = \frac{r_0}{r_1} = 2 \Rightarrow r_0 = 2r_1$$

So diameter of iron = $\frac{1}{2}$ diameter of oak

$$\Rightarrow \frac{1}{2} \times 18 = 9 \text{ cm}$$

13. (c) Let 'r' be the radius of each circle. Then by given condition

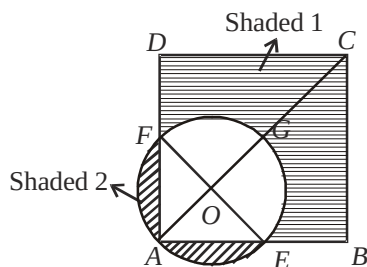


$$\frac{\pi r^2}{2\pi r} = \frac{2\pi r}{\pi r^2} \Rightarrow \pi r^2 = 2\pi r \Rightarrow r = 2 \quad [\text{since } r \neq 0]$$

$\therefore$ Length of the side of the square = 8

The area of square which is not covered by the coins
= $64 - 4\pi(2)^2 = 16(4 - \pi)$

For Qs. 14-15.



In $\triangle FAE$, $\angle FAE = 90^\circ$ as it is an angle of a square. Further EF will be the diameter of the circle as an angle subtended by a diameter on the circumference of a circle = 90° . So, EF will pass through the centre O.

In $\triangle FOA$ and $\triangle AOE$

$OF = OE$ (radius of circle)

AO is common

$\angle FAO = \angle EAO$ (45°)

So $\triangle FOA \sim \triangle AOE \Rightarrow AF = AE$

$$\text{Area of } \triangle AFE = \frac{1}{2}bh = \frac{1}{2}AF^2$$

$$AF^2 + AE^2 = FE^2 = 2^2 = 4$$

(Pythagoras theorem)

$$\text{or } AF^2 = 2 \Rightarrow \text{Area of } \triangle AFE = \frac{1}{2} \times 2 = 1$$

Area of shaded region 2

$$= \frac{\pi r^2}{2} - 1 = \frac{\pi \times 1^2}{2} - 1 = \frac{\pi - 2}{2}$$

Area of the white portion of circle

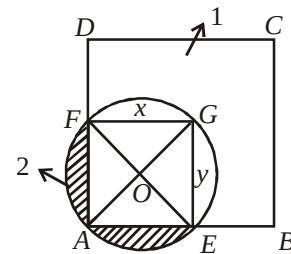
$$= \pi r^2 - \frac{\pi - 2}{2} = \frac{\pi - (\pi - 2)}{2} = \frac{\pi + 2}{2}$$

$$\therefore \text{Area of shaded region 1} = 2^2 - \frac{\pi + 2}{2} = \frac{6 - \pi}{2}$$

Hence required proportion of the sheet

$$= \frac{\left(\frac{6 - \pi}{2}\right)}{4} = \frac{6 - \pi}{8}$$

Alternatively



Complete the square AEGF. Note 2 diagonals of a square make angles of 90° with each other. The area of the shaded region 2 can also found by:

$2(\text{Area of sector } AOE - \text{Area of } \triangle AOE)$

$$= 2\left(\frac{\pi r^2}{4} - \frac{1}{2} \times 1 \times 1\right) \quad [AO = OE = \text{radius} = 1]$$

$$= 2\left(\frac{\pi}{4} - \frac{1}{2}\right) = \frac{\pi - 2}{2}$$

Again, area of region 1 = Area of square ABCD - Area of square AEGF - Area of region 2.

[Note : Area of region 2 = Area of region x + region y]

$$\Rightarrow \text{Area (region 1)} = 2^2 - 2 - \left(\frac{\pi - 2}{2}\right)$$

[Note : Area of square AEGF = AE^2

Further $AE^2 + GE^2 = 2^2 = 2AE^2$ or $AE^2 = 2$]

$$\Rightarrow \text{Area (region 1)} = 2 - \left(\frac{\pi - 2}{2}\right) = \frac{6 - \pi}{2}$$

$$\text{Required proportion} = \frac{(6 - \pi)/2}{4} = \frac{6 - \pi}{8}$$

14. (b)

15. (d)

20

COORDINATE GEOMETRY

- Introduction
- Rectangular Coordinate Axes
- Sign Conventions in the xy -Plane
- Quadrants of xy -Plane and Sign of x and y -Coordinate of a Point in different Quadrants
- Plotting a Point Whose Coordinates are Known
- Distance Formula
- Applications of Distance Formula
- Section Formula
- Coordinates of Some Particular Points
- Area of Triangle and Quadrilateral
- Transformation of Axes
- Image of a Point
- Equation of Straight Line Parallel to An Axis
- Inclination of a Straight Line
- Slope (or Gradient) of a Straight Line
- Equation of Straight Lines
- Different Forms of the Equation of a Straight Line
- Point of Intersection of Two Lines
- Position of a Point Relative to a Line
- Angle Between Two Straight Lines
- Equation of Parallel and Perpendicular Lines
- Distance of a Line from a Point
- Distance Between Two Parallel Lines

INTRODUCTION

Coordinates are a pair of values that show an exact position of a point in a plane. This chapter contributes very few problems to CAT and other equivalent aptitude test. It is advised that CAT aspirants should not leave this chapter as it is very basic and solutions of this problem require formulatic approach.

RECTANGULAR COORDINATE AXES

Let XOX' be a horizontal straight line and YOY' be a vertical straight line drawn through a point O in the plane of the paper. Then

the line XOX' is called x -axis

the line YOY' is called y -axis

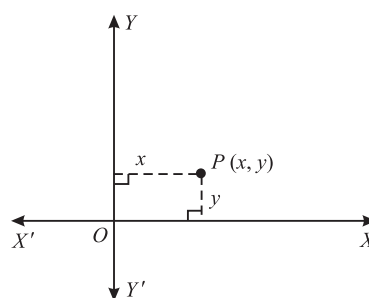
plane of paper is called xy -plane or cartesian plane.

x -axis and y -axis together are called co-ordinate axes or axis of reference.

The point O is called the origin.

Cartesian Coordinates

Position of any point in a cartesian plane can be described by their cartesian coordinates. The ordered pair of perpendicular distances first from y -axis and second from x -axis of a point P is called cartesian coordinates of P .



If the cartesian coordinates of point P are (x, y) , then x is called abscissa or x -coordinate of P and y is called the ordinate or y -coordinate of point P .

SIGN CONVENTIONS IN THE xy -PLANE

- (i) All the distances are measured from origin (o).
- (ii) All the distances measured along or parallel to x -axis but right side of origin are taken as $+ve$.
- (iii) All the distances measured along or parallel to x -axis but left side of origin are taken as $-ve$.
- (iv) All the distances measured along or parallel to y -axis but above the origin are taken as $+ve$.
- (v) All the distances measure along or parallel to y -axis but below the origin are taken as $-ve$.

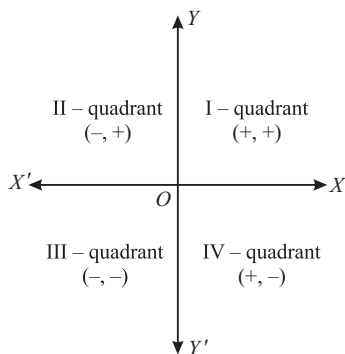
According to the Above Sign Conventions

- Coordinate of origin is $(0, 0)$
- Coordinate of any point on the x -axis but right side of origin is of the form $(x, 0)$, where $x > 0$.
- Coordinate of any point on the x -axis but left side of origin is of the form $(-x, 0)$, where $x > 0$.
- Coordinate of any point on the y -axis but above the origin is of the form $(0, y)$, where $y > 0$.
- Coordinate of any point on the y -axis but below the origin is of the form $(0, -y)$, where $y > 0$.

QUADRANTS OF xy -PLANE AND SIGN OF x AND y -COORDINATE OF A POINT IN DIFFERENT QUADRANTS

x and y -axis divide the xy -plane in four parts. Each part is called a quadrant.

The four quadrants are written as I-quadrant (XOY), II-quadrant (YOX'), III-quadrant ($X'OY'$) and IV-quadrant ($Y'OX$). Each of these quadrants shows the specific quadrant of the xy -plane as shown below:



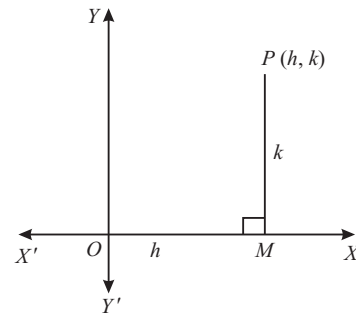
- Any of the four quadrants does not include any part of x or y -axis.
- In the first quadrant both x and y -coordinates of any point are $+ve$.
- In second quadrant x -coordinate of any point is $-ve$ but y -coordinate of any point is $+ve$.
- In third quadrant, both x and y -coordinates of any point are $-ve$.
- In fourth quadrant, x -coordinate of any point is $+ve$ but y -coordinate of any point is $-ve$ as shown in the above diagram.

PLOTTING A POINT WHOSE COORDINATES ARE KNOWN

The point can be plotted by measuring its proper distances from both the axes. Thus, any point P whose coordinates are (h, k) can be plotted as follows:

- Measure OM equal to h (i.e. x -coordinate of point P) along the x -axis.
- Now perpendicular to OM equal to k .

Mark point P above M such that PM is parallel to y -axis and $PM = k$ (i.e. y -coordinate of point P)



In this chapter, now we shall study to find the distance between two given points, section formula, mid-point formula, slope of a line, angles between two straight lines and equation of a line in different forms etc.

DISTANCE FORMULA

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \left| \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \right| \quad \text{or} \quad \left| \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \right|$$

$$\text{Distance of point } P(x, y) \text{ from the origin} = \left| \sqrt{x^2 + y^2} \right|$$

Illustration 1: If distance between the point $(x, 2)$ and $(3, 4)$ is 2, then find the value of x .

Solution:

$$2 = \left| \sqrt{(x - 3)^2 + (2 - 4)^2} \right| \Rightarrow 2 = \left| \sqrt{(x - 3)^2 + 4} \right|$$

Squaring both sides

$$4 = (x - 3)^2 + 4 \Rightarrow x - 3 = 0 \Rightarrow x = 3$$

Illustration 2: Find the distance between each of the following points :

$A(-6, -1)$ and $B(-6, 11)$

Solution: Here the points are $A(-6, -1)$ and $B(-6, 11)$

By using distance formula, we have

$$AB = \sqrt{\{ -6 - (-6) \}^2 + \{ -1 - 11 \}^2} = \sqrt{0^2 + 12^2} = 12$$

Hence, $AB = 12$ units.

APPLICATIONS OF DISTANCE FORMULA

- For given three points A, B, C to decide whether they are collinear or vertices of a particular triangle. After finding the distances AB, BC and CA ; we shall find that the points are
 - collinear, if the sum of any two distances is equal to the third.
 - vertices of an equilateral triangle if $AB = BC = CA$
 - vertices of an isosceles triangle if $AB = BC, BC = CA$ or $CA = AB$
 - vertices of a right angled triangle if $AB^2 + BC^2 = CA^2$ or $BC^2 + CA^2 = AB^2$ or $CA^2 + AB^2 = BC^2$
- For given four points A, B, C, D ;
 - $AB = BC = CD = DA; AC = BD \Rightarrow ABCD$ is a square
 - $AB = BC = CD = DA \Rightarrow ABCD$ is a rhombus
 - $AB = CD, BC = DA; AC = BD \Rightarrow ABCD$ is a rectangle
 - $AB = CD, BC = DA \Rightarrow ABCD$ is a parallelogram

Illustration 3: What kind of triangle is formed by $A(1, 2)$, $B(4, 3)$ and $C(5, 6)$?

Solution: $AB^2 = (4-1)^2 + (3-2)^2 = 9 + 1 = 10$
 $BC^2 = (5-4)^2 + (6-3)^2 = 1 + 9 = 10$
 $CA^2 = (5-1)^2 + (6-2)^2 = 16 + 16 = 32$
 $AB^2 = BC^2 \Rightarrow$ it is isosceles.
 $CA^2 > AB^2 + BC^2$ since $32 > 10 + 10 \Rightarrow \angle C$ is obtuse
Hence, ABC is an obtuse isosceles Δ .

SECTION FORMULA

Co-ordinates of a point which divides the line segment joining two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio $m_1 : m_2$ are :

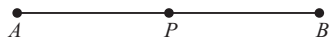
- (i) $\left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$, for internal division.



P divides AB internally in the ratio $m : n$

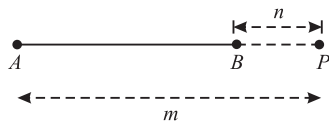
If $m_1 = m_2$, then the point will be the mid point of PQ whose

$$\text{co-ordinates} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$



P is the mid-point of AB

- (ii) $\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2} \right)$, for external division



P divides AB externally in the ratio $m : n$

- (iii) When we need to find the ratio in which a point on a line segment divides it, we suppose the required ratio as $k : 1$ or $m/n : 1$.

Note:

- (i) Co-ordinates of any point on the line segment joining two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ are

$$\left(\frac{x_1 + \lambda x_2}{1 + \lambda}, \frac{y_1 + \lambda y_2}{1 + \lambda} \right), (\lambda \neq -1)$$

- (ii) **Division by axes:** Line segment joining the points (x_1, y_1) and (x_2, y_2) is divided by

(a) x -axis in the ratio $-y_1 / y_2$

(b) y -axis in the ratio $-x / x_2$

If ratio is positive division internally and if ratio is negative division is externally.

- (iii) **Division by a line:** Line $ax + by + c = 0$ divides the line joining the points (x_1, y_1) and (x_2, y_2) in the ratio

$$\left(-\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} \right)$$

Illustration 4: Find the ratio in which the line $3x + 4y = 7$ divides the line segment joining the points $(1, 2)$ and $(-2, 1)$.

Solution: Ratio = $-\frac{3(1) + 4(2) - 7}{3(-2) + 4(1) - 7} = \frac{4}{-9} = \frac{4}{9} \quad 4:9$.

Illustration 5: Find the points of trisection of line joining the points $A(2, 1)$ and $B(5, 3)$.

Solution: $(2, 1) \xrightarrow{\frac{1}{2}} P_1 \xrightarrow{\frac{1}{2}} P_2 \xrightarrow{\frac{1}{2}} B(5, 3)$

$$P_1(x, y) = \left(\frac{1 \times 5 + 2 \times 2}{1 + 2}, \frac{1 \times 3 + 2 \times 1}{1 + 2} \right) = \left(3, \frac{5}{3} \right)$$

$$P_2(x, y) = \left(\frac{2 \times 5 + 1 \times 2}{2 + 1}, \frac{2 \times 3 + 1 \times 1}{2 + 1} \right) = \left(4, \frac{7}{3} \right)$$

Illustration 6: Prove that points $A(1, 1)$, $B(-2, 7)$ and $C(3, -3)$ are collinear.

$$\text{Solution: } AB = \sqrt{(1+2)^2 + (1-7)^2} = \sqrt{9+36} = 3\sqrt{5}$$

$$BC = \sqrt{(-2-3)^2 + (7+3)^2} = \sqrt{25+100} = 5\sqrt{5}$$

$$CA = \sqrt{(3-1)^2 + (-3-1)^2} = \sqrt{4+16} = 2\sqrt{5}$$

Clearly, $BC = AB + AC$. Hence A, B, C are collinear.

Illustration 7: Find the ratio in which the join of $(-4, 3)$ and $(5, -2)$ is divided by (i) x -axis (ii) y -axis.

Solution:

- (i) x -axis divides the join of (x_1, y_1) and (x_2, y_2) in the ratio of $-y_1 : y_2 = -3 : -2 = 3 : 2$.

- (ii) y -axis divides, in the ratio of $-x_1 : x_2 \Rightarrow 4 : 5$.

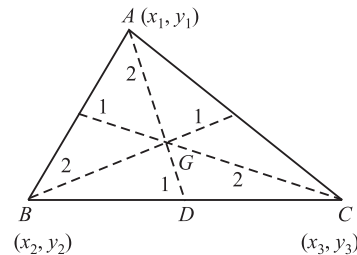
COORDINATES OF SOME PARTICULAR POINTS

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are vertices of any triangle ABC , then

Centroid

Centroid is the point of intersection of the medians of a triangle. Centroid divides each median in the ratio of $2 : 1$.

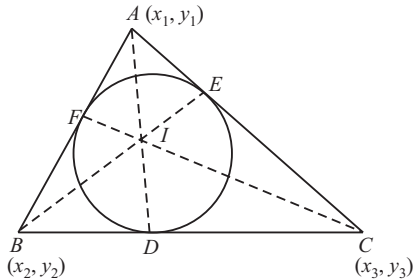
A median is a line segment joining the mid point of a side to its opposite vertex of a triangle.



$$\text{Co-ordinates of centroid, } G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Incentre

Incentre is the point of intersection of internal bisectors of the angles of a triangle. Also incentre is the centre of the circle touching all the sides of a triangle.



Co-ordinates of incentre,

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right),$$

where a, b, c are length of the sides opposite to vertices A, B, C respectively of triangle ABC .

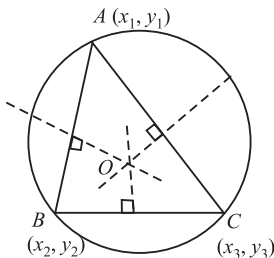
- (i) Angle bisector divides the opposite sides in the ratio of the sides included in the angle. For example

$$\frac{BD}{DC} = \frac{AB}{AC} \quad \frac{c}{b} =$$

- (ii) Incentre divides the angle bisectors AD, BE and CF in the ratio $(b + c) : a, (c + a) : b$ and $(a + b) : c$ respectively.

Circumcentre

It is the point of intersection of perpendicular bisectors of the sides of a triangle. It is also the centre of a circle passing through the vertices of the triangle. Thus if O is circumcentre of any triangle ABC , then $OA^2 = OB^2 = OC^2$.



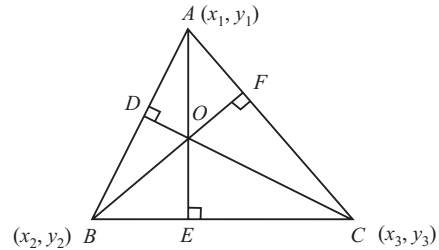
- (i) If a triangle is right angle, then its circumcentre is the mid-point of hypotenuse.
(ii) If $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ are the vertices of a $\triangle ABC$, then coordinates of the circumcentre of the triangle ABC is

$$\left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

Note: $\sin 30^\circ = \frac{1}{2}, \sin 45^\circ = \frac{1}{\sqrt{2}}, \sin 60^\circ = \frac{\sqrt{3}}{2}, \sin 90^\circ = 1, \sin 120^\circ = \frac{\sqrt{3}}{2}$

Orthocentre

It is the point of intersection of perpendiculars drawn from vertices on opposite sides.



Coordinate of the orthocentre of $\triangle ABC$

$$= \left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right)$$

If the triangle is right angled triangle, then orthocentre is the vertex where right angle is formed.

Note:

(i) $\tan 30^\circ = \frac{1}{\sqrt{3}}, \tan 45^\circ = 1, \tan 60^\circ = \sqrt{3}, \tan 90^\circ =$

$\frac{1}{0}$ or not defined

- (ii) If the triangle is equilateral, then the centroid, incentre, orthocentre, and circumcentre coincide.
(iii) Orthocentre, centroid and circumcentre in the ratio 2 : 1.
(iv) In an isosceles triangle centroid, orthocentre, incentre, and circumcentre lie on the same line.

Illustration 8: Find incentre (I) of triangle whose vertices are $A(-36, 7), B(20, 7), C(0, -8)$.

Solution: Using distance formula

$$a = BC = \sqrt{20^2 + (7 + 8)^2} = 25$$

$$b = CA = \sqrt{36^2 + (7 + 8)^2} = 39$$

$$c = AB = \sqrt{(36 + 20)^2 + (7 - 7)^2} = 56$$

$$I = \left(\frac{25(-36) + 39(20) + 56(0)}{25 + 39 + 56}, \frac{25(7) + 39(7) + 56(-8)}{25 + 39 + 56} \right)$$

$$I = (-1, 0).$$

Illustration 9: If $(0, 1), (1, 1)$ and $(1, 0)$ are mid-points of the sides of a triangle then find its incentre.

Solution: Let $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ are vertices of a triangle, then

$$x_1 + x_2 = 0, x_2 + x_3 = 2, x_3 + x_1 = 2$$

$$y_1 + y_2 = 2, y_2 + y_3 = 2, y_3 + y_1 = 0$$

Solving these equations, we get

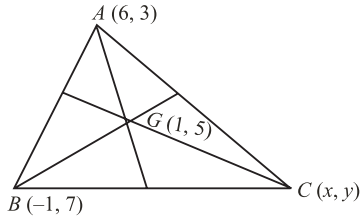
$$A(0, 0), B(0, 2) \text{ and } C(2, 0)$$

Now $a = BC = 2\sqrt{2}$, $b = CA = 2$, $c = AB = 2$

Thus incentre of a $\triangle ABC$ is $(2 - \sqrt{2}, 2 - \sqrt{2})$.

Illustration 10: The two vertices of a triangle are $(6, 3)$ and $(-1, 7)$ and its centroid is $(1, 5)$. Find the third vertex.

Solution: Let ABC be a triangle whose vertices are $A = (6, 3)$, $B = (-1, 7)$, $C = (x, y)$ and centroid $G = (1, 5)$



Then using the formula, for coordinates of centroid

$$1 = \frac{6 + (-1) + x}{3} \text{ and } 5 = \frac{3 + 7 + y}{3}$$

$$\Rightarrow x = -2 \text{ and } y = 5$$

Hence, the third vertex is $C = (-2, 5)$

Solution: Let the third vertex be (x_3, y_3) , area of triangle

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

As, $x_1 = 2, y_1 = 1, x_2 = 3, y_2 = 2$, Area of $\Delta = 5$

$$\Rightarrow 5 = \frac{1}{2} |2(2 - y_3) + 3(y_3 - 1) - x_3(1 - 2)|$$

$$\Rightarrow 10 = |3x_3 + y_3 - 7| \Rightarrow 3x_3 + y_3 = \pm 19$$

Taking positive sign, $3x_3 + y_3 - 7 = 10$

$$\Rightarrow 3x_3 + y_3 = 17 \quad \dots (1)$$

Taking negative sign, $3x_3 + y_3 - 7 = -10$

$$\Rightarrow 3x_3 + y_3 = -3 \quad \dots (2)$$

Given that (x_3, y_3) lies on $y = x + 3$

$$\text{So, } -x_3 + y_3 = 3 \quad \dots (3)$$

$$\text{Solving eqs. (1) and (3), } x_3 = \frac{7}{2}, y_3 = \frac{13}{2}$$

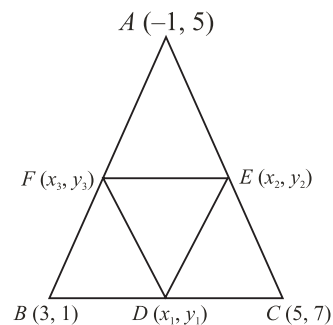
$$\text{Solving eqs. (2) and (3), } x_3 = \frac{-3}{2}, y_3 = \frac{3}{2}$$

So the third vertex are $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(\frac{-3}{2}, \frac{3}{2}\right)$

Illustration 12: The coordinates of A, B and C are $(-1, 5)$, $(3, 1)$ and $(5, 7)$ respectively, D, E and F are the middle points of BC, CA and AB respectively. Calculate the area of the triangle DEF .

Solution: Mid-point $D(x_1, y_1) = \left(\frac{3+5}{2}, \frac{1+7}{2}\right) = (4, 4)$

$$\text{Mid-point } E(x_2, y_2) = \left(\frac{-1+5}{2}, \frac{5+7}{2}\right) = (2, 6)$$



$$\text{Mid-point } F(x_3, y_3) = \left(\frac{-1+3}{2}, \frac{5+1}{2}\right) = (1, 3)$$

Now, using the formula, area of triangle

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$\Rightarrow \text{Area of } \triangle DEF = \frac{1}{2} |4(6-3) + 2(3-4) + 1(4-6)|$$

$$= 4 \text{ square units.}$$

Hence, the area of $\triangle DEF$ is 4 square units.

Illustration 13: $(1, 1), (3, 4), (5, -2)$ and $(4, -7)$ are vertices of a quadrilateral then find its area.

AREA OF TRIANGLE AND QUADRILATERAL

Area of a Triangle

Let $A(x_1, y_1), B(x_2, y_2)$ and $C(x_3, y_3)$ are vertices of a triangle, then area of the triangle ABC .

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

(i) When one vertex is origin i.e. if the vertices are $(0, 0)$, (x_1, y_1) and (x_2, y_2) then area of the triangle,

$$\Delta = \frac{1}{2} |x_1 y_2 - x_2 y_1|$$

(ii) When two vertices are on x -axis say $(a, 0), (b, 0)$ and third vertex is (h, k) then its area $= \frac{1}{2} |(a-b)k|$.

(iii) If the area of a triangle ABC is zero, then three points A, B and C are collinear.

(iv) Three points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are collinear if

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_2}{x_3 - x_2}$$

Area of a Quadrilateral

If $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4) are vertices of a quadrilateral then its area

$$= \frac{1}{2} |(x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + (x_3 y_4 - x_4 y_3) + (x_4 y_1 - x_1 y_4)|$$

Illustration 11: The area of a triangle is 5. Two of its vertices are $(2, 1)$ and $(3, -2)$. The third vertex lies on $y = x + 3$. Find the third vertex.

Solution: Area = $\frac{1}{2} | 1 \times 4 - 3 \times 1 + 3 \times (-2) - 5 \times 4$
 $+ 5(-7) - 4(-2) + 4 \times 1 - 1(-7) |$
 $= \frac{1}{2} | 4 - 3 - 6 - 20 - 35 + 8 + 4 + 7 | = \frac{41}{2} \text{ units}$

TRANSFORMATION OF AXES

Let origin $O(0, 0)$ be shifted to a point (a, b) by moving the x -axis and y -axis parallel to themselves. If the coordinates of point P with reference to old axis are (x_1, y_1) , then coordinate of this point P with respect to new axis will be $(x_1 - a, y_1 - b)$

Illustration 14: If origin $(0, 0)$ shifted to $(-3, 4)$, what will be the coordinates of the point in the new position of axes (i.e., new coordinate system) which is represented by $(4, 2)$ in the old position of axes (i.e., old coordinate system)?

Solution: Coordinates in the new coordinate system
 $= (4 - (-3), 2 - 4) = (4 + 3, -2) = (7, -2)$.

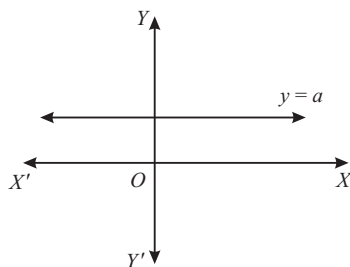
IMAGE OF A POINT

Let (x, y) be any point, then its image with respect to

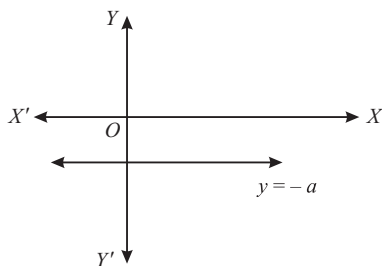
- (i) x -axis is $(x, -y)$ (ii) y -axis is $(-x, y)$
- (iii) origin is $(-x, -y)$ (iv) line $y = x$ is (y, x)

EQUATION OF STRAIGHT LINE PARALLEL TO AN AXIS

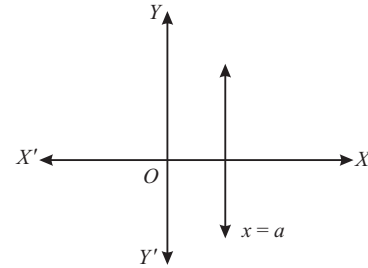
- (i) Equation of x -axis is $y = 0$
- (a) Equation of a line parallel to x -axis (or perpendicular to y -axis) at a distance ' d ' above the y -axis is $y = a$.



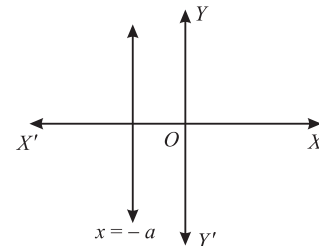
- (b) Equation of a line parallel to x -axis (or perpendicular to y -axis) at a distance ' a ' below the y -axis is $y = -a$.



- (ii) Equation of y -axis is $x = 0$
- (a) Equation of a line parallel to y -axis (or perpendicular to x -axis) at a distance ' a ' towards right side of y -axis is $x = a$.



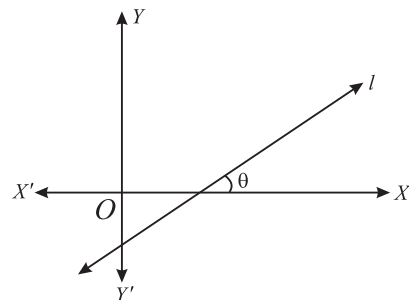
- (b) Equation of a line parallel to y -axis (or perpendicular to x -axis) at a distance ' a ' towards left side of y -axis is $x = -a$.



Example: Equation of a line which is parallel x -axis and at a distance of 4 units below the y -axis is $y = -4$.

INCLINATION OF A STRAIGHT LINE

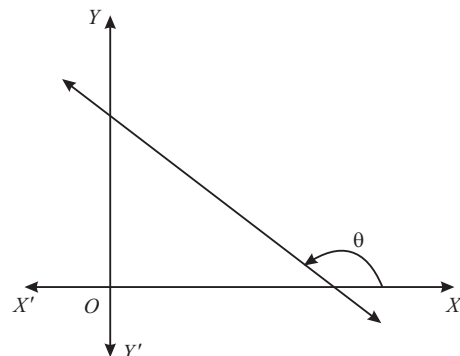
An angle made by a straight line with the positive direction of x -axis in anti-clock wise direction is called the angle of inclination of the straight line.



Here angle of inclination of the line $l = \theta$.

SLOPE (OR GRADIENT) OF A STRAIGHT LINE

- (i) Slope of a straight line is equal to the tangent of the angle which the straight line makes with the positive direction of x -axis in anticlock-wise direction and it is generally denoted by m .



Thus if a line makes an angle θ with the positive direction of x -axis in anticlock-wise direction then its slope, $m = \tan \theta$.

In other words, slope of line is the tangent of the angle of inclination other line.

(ii) Slope (m) of a line joining two points (x_1, y_1) and (x_2, y_2) is given by $m = \frac{y_2 - y_1}{x_2 - x_1}$.

(iii) Slope m of line $ax + by + c = 0$ is $-\frac{a}{b}$ i.e., $m = -\frac{a}{b}$.

Condition of Collinearity of Three Points

Three points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ will be collinear if one of the following three conditions is satisfied.

(i) If sum of any two distances AB , BC and CA is equal to the third distance.

(ii) $\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_2}{x_3 - x_2}$

(iii) Slope of AB = Slope of BC

(iv) Area of $\triangle ABC = 0$

Illustration 15: If three points $(h, 0)$, (a, b) and $(0, k)$ lies on

a line, show that $\frac{a}{h} + \frac{b}{k} = 1$.

Solution: The given points are $A(h, 0)$, $B(a, b)$, $C(0, k)$, they lie on the same plane.

$\therefore$ Slope of AB = Slope of BC

$\therefore$ Slope of $AB = \frac{b-0}{a-h} = \frac{b}{a-h}$, Slope of $BC = \frac{k-b}{0-a} = \frac{k-b}{-a}$

$\therefore \frac{b}{a-h} = \frac{k-b}{-a}$ or by cross multiplication

$-ab = (a-h)(k-b)$ or $-ab = ak - ab - hk + hb$

or $0 = ak - hk + hb$ or $ak + hb = hk$

Dividing by hk , $\frac{ak}{hk} + \frac{hb}{hk} = 1$ or $\frac{a}{h} + \frac{b}{k} = 1$

EQUATION OF STRAIGHT LINES

A linear relation between x and y which is satisfied by coordinates of every point lying on a straight line is called the equation of Straight Line. Every linear equation in two variables x and y always represents a straight line. For example,

$4x + 9y = 40$, $-10x + 6y = 8$, $x + 3y = 0$, etc.

General form of the equation of a straight line is given by $ax + by + c = 0$.

DIFFERENT FORMS OF THE EQUATION OF A STRAIGHT LINE

General Form

$$ax + by = 0,$$

where a, b, c are any real numbers but a and b cannot be zero simultaneously.

Slope of the line $ax + by + c = 0$

$$m = -\frac{\text{coefficient of } x}{\text{coefficient of } y}$$

$a = 0 \Rightarrow by + c = 0$, which is a line parallel to x -axis

$b = 0 \Rightarrow ax + c = 0$, which is a line parallel to y -axis

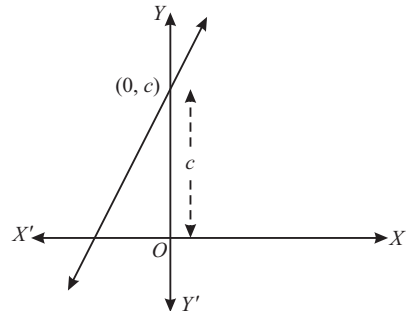
$c = 0 \Rightarrow ax + by = 0$, which is a line passing through the origin

Slope Intercept Form

$$y = mx + c,$$

where m is the slope of the line and c is the y -intercept of the line.

y -intercept of a line is the y -coordinate of the point where the line intersect the y -axis.



Slope Point Form

The equation of a line with slope m and passing through a point (x_1, y_1) is $y - y_1 = m(x - x_1)$.

Two Point Form

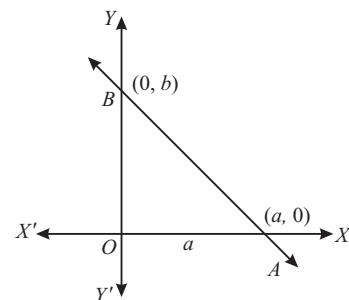
The equation of a line passing through two given points (x_1, y_1) and (x_2, y_2) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

Intercept Form

$$\frac{x}{a} + \frac{y}{b} = 1 = 1$$

where a and b are x and y -intercept respectively of the line.



x -intercept of a line is the x -coordinate of the point where the line intersects the x -axis. y -intercept of a line is the y -coordinate of the point where the line intersects the y -axis.

Illustration 16: Find the equation of a line which passes through $(3, -4)$ and makes an angle of 45° with x -axis.

Solution: $y - (-4) = \tan 45^\circ (x - 3)$

$$\Rightarrow y + 4 = x - 3 \Rightarrow x - y - 7 = 0$$

Illustration 17: Find the equation of a line which makes intercepts 3 and 4 on x -axis and y -axis respectively.

$$\text{Solution: } \frac{x}{3} + \frac{y}{4} = 1 \Rightarrow 4x + 3y = 12.$$

Illustration 18: Find the slope intercept form and intercept form of a line $3x + 4y = 5$.

Solution:

$$\begin{aligned} \text{(i)} \quad 3x + 4y &= 5 \\ \Rightarrow 4y &= -3x + 5 \\ \Rightarrow y &= -\frac{3}{4}x + \frac{5}{4} \end{aligned}$$

This is the slope intercept form.

$$\text{Here } m = -\frac{3}{4}, c = \frac{5}{4}$$

$$\begin{aligned} \text{(ii)} \quad 3x + 4y &= 5 \\ \Rightarrow \frac{3x}{5} + \frac{4y}{5} &= 1 \end{aligned}$$

$$\frac{x}{5/3} + \frac{y}{5/4} = 1$$

This is the intercept form

$$\text{Here } a = 5/3, b = 5/4.$$

Illustration 19: Find the equation of the line passing through the points $(-1, 1)$ and $(2, -4)$.

Solution: The line passes through the points $A(-1, 1)$ $B(2, -4)$

Equation of the line passing through (x_1, y_1) and (x_2, y_2) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$\text{We have } x_1 = -1, y_1 = 1, x_2 = 2, y_2 = -4$$

$$\therefore \text{Equation of } AB \text{ is } y - 1 = \frac{-4 - 1}{2 + 1}(x + 1)$$

$$\Rightarrow y - 1 = \frac{-5}{3}(x + 1) \Rightarrow 3(y - 1) = -5(x + 1)$$

$$3y - 3 = -5x - 5 \Rightarrow 5x + 3y - 3 + 5 = 0 \Rightarrow 5x + 3y + 2 = 0$$

Illustration 20: Find the equation of the line passing through $(-3, 5)$ and perpendicular to the line through the points $(2, 5)$ and $(-3, 6)$.

Solution: $y - 5 = 5(x + 3) \Rightarrow 5x - y + 20 = 0$

Illustration 21: Find the equation of the line, which passes through $(2, -5)$ and cuts off equal intercepts on both the axes.

Solution: Here the intercepts on the both axes are equal, So, $a = b$.

$$\text{Using the intercept form } \frac{x}{a} + \frac{y}{b} = 1$$

The line passes through $(2, -5)$,

$$\Rightarrow \frac{2}{a} + \frac{-5}{a} = 1 \Rightarrow a = -3$$

The required equation is

$$\Rightarrow x + y + 3 = 0$$

Illustration 22: Find the equation of a line passing through point $(5, 1)$ and parallel to the line $7x - 2y + 5 = 0$.

Solution: The required equation of the line is $7x - 2y - 33 = 0$.

POINT OF INTERSECTION OF TWO LINES

Point of intersection of two lines can be obtained by solving the their equations.

Point of intersection of two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ is given by

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} \right)$$

POSITION OF A POINT RELATIVE TO A LINE

- The point (x_1, y_1) lies on the line $ax + by + c = 0$ if, $ax_1 + by_1 + c = 0$
- If $P(x_1, y_1)$ and $Q(x_2, y_2)$ do not lie on the line $ax + by + c = 0$, then they are on the same side of the line if $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of the same sign and they lie on the opposite sides of the line if $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of the opposite sign.

Illustration 23: Point $(3, 4)$ and $(-9, 6)$ lie on which side of line $7x + 5y - 9 = 0$?

Solution: Putting both the points in the left hand side part $(7x + 5y - 9)$ of the given equation, we get

$$7 \times 3 + 5 \times 4 - 9 = 32 \quad \text{and} \quad 7 \times (-9) + 5 \times 6 - 9 = -42.$$

As both 32 and -42 are of opposite sign, so the given points lie opposite side of the given line.

ANGLE BETWEEN TWO STRAIGHT LINES

If θ be the angle between two straight lines $y = m_1x + c_1$ and $y = m_2x + c_2$, then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1m_2} \right|$$

If θ be the angle between two straight lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, then

$$\tan \theta = \left| \frac{a_1b_2 - b_1a_2}{a_1a_2 + b_1b_2} \right|$$

There are two angles between two lines, but generally we consider the acute angle as the angle between them, so in above both formula we take only positive value of $\tan \theta$.

Parallel Lines

Two lines are parallel if their slopes m_1 and m_2 are equal i.e., $m_1 = m_2$.

Lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

Perpendicular Lines

Two lines are perpendicular if product of their slopes m_1 and m_2 is -1 i.e., $m_1 \times m_2 = -1$.

Lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are perpendicular if $a_1a_2 + b_1b_2 = 0$.

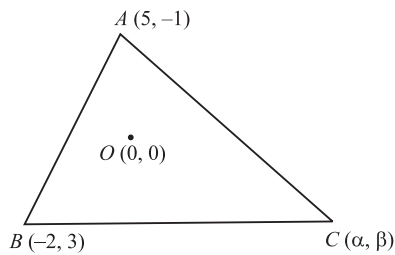
Coincident Lines

Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are coincident

$$\text{if } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

Illustration 24: Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If origin is the orthocentre, then find the third vertex of the triangle.

Solution: Let $C(\alpha, \beta)$ be the third vertex



$$AO \perp BC = \left(\frac{\beta - 3}{\alpha + 2} \right) \left(\frac{-1}{5} \right) = -1$$

$$\Rightarrow 5\alpha - \beta = -13 \quad \dots (1)$$

$$BO \perp AC \Rightarrow \left(\frac{\beta + 1}{\alpha - 5} \right) \left(\frac{3}{-2} \right) = -1$$

$$\Rightarrow 2\alpha - 3\beta = 13 \quad \dots (2)$$

Solving (1) and (2), $(\alpha, \beta) = (-4, -7)$.

Illustration 25: Find the angle between $y = x + 6$ and $y = \sqrt{3}x + 7$.

Solution: Here, $m_1 = 1, m_2 = \sqrt{3}$.

$$\tan \theta = \left| \frac{1 - \sqrt{3}}{1 + \sqrt{3}} \right| \Rightarrow \theta = \tan^{-1} \frac{1 - \sqrt{3}}{1 + \sqrt{3}} = 15^\circ$$

$$\text{Since } \tan 15^\circ = \frac{1 - \sqrt{3}}{1 + \sqrt{3}} \Rightarrow 15^\circ = \tan^{-1} \left(\frac{1 - \sqrt{3}}{1 + \sqrt{3}} \right).$$

Note that if $\tan \theta = x$, then $\theta = \tan^{-1} x$.

Here $\tan^{-1} x$ read as $\tan$ inverse x .

Illustration 26: If $7x + 3y + 9 = 0$ and $y = kx + 7$ are two parallel lines then find the value of k .

Solution: $m_1 = -\frac{7}{3}, m_2 = k$

Two lines are parallel, if $m_1 = m_2 \Rightarrow k = -\frac{7}{3}$.

EQUATION OF PARALLEL AND PERPENDICULAR LINES

- Equation of a line which is parallel to $ax + by + c = 0$ is in the form $ax + by + k = 0$.
- Equation of a line which is perpendicular to $ax + by + c = 0$ is in the form $bx - ay + k = 0$.
The value of k in both cases is obtained with the help of additional information given in the problem.

Illustration 27: Find the equation of a line which passes through $(-3, 2)$ and perpendicular to the $3x + 4y = 5$.

Solution: Let the equation be $4x - 3y + k = 0$, this line passes through $(-3, 2)$

$$\text{Hence } 4(-3) - 3(2) + k = 0 \Rightarrow k = 18$$

Required equation is $4x - 3y + 18 = 0$.

DISTANCE OF A LINE FROM A POINT

- The length P of the perpendicular from the point (x_1, y_1) on the line $ax + by + c = 0$ is given by

$$P = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

- Length of perpendicular from origin to the line $ax + by + c = 0$ is given by

$$\left| \frac{c}{\sqrt{a^2 + b^2}} \right|$$

Illustration 28: Find the points on y -axis whose perpendicular distance from the line $4x - 3y - 12 = 0$ is 3.

Solution: Let the required point be $P(0, \alpha)$. It is given that the length of the perpendicular from $P(0, \alpha)$ to the line $4x - 3y - 12 = 0$ is 3.

$$\therefore \left| \frac{4 \times 0 - 3\alpha - 12}{\sqrt{4^2 + (-3)^2}} \right| = 3$$

$$\Rightarrow |3\alpha + 12| = 15$$

$$\Rightarrow |\alpha + 4| = 5$$

$$\Rightarrow \alpha + 4 = \pm 5$$

$$\Rightarrow \alpha = 1, -9$$

Hence, the required points are $(0, 1)$ and $(0, -9)$.

Illustration 29: Find the coordinates of a point on $x + y + 3 = 0$, whose distance from $x + 2y + 2 = 0$ is $\sqrt{5}$.

Solution: Let the required point be (x_1, y_1) . Since it lies on $x + y + 3 = 0$

$$\therefore x_1 + y_1 + 3 = 0 \quad \dots (1)$$

Now,

Length of the perpendicular from (x_1, y_1) to $x + 2y + 2 = 0$ is $\sqrt{5}$.

$$\Rightarrow \left| \frac{x_1 + 2y_1 + 2}{\sqrt{1^2 + 2^2}} \right| = \sqrt{5} \Rightarrow x_1 + 2y_1 + 2 = \pm 5 \quad \dots (2)$$

Solving equations (1) and (2), we get: $x_1 = -9, y_1 = 6$ and $x_1 = 1, y_1 = -4$

Hence, the required points are $(-9, 6)$ and $(1, -4)$.

DISTANCE BETWEEN TWO PARALLEL LINES

The distance between two parallel lines $ax + by + c_1 = 0$ and $ax + by + c_2 = 0$ is

$$\left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right|$$

Illustration 30: Find the distance between $3x + 2y + 7 = 0$ and $6x + 4y + 3 = 0$.

$$\text{Solution: } \frac{\left| 7 - \frac{3}{2} \right|}{\sqrt{3^2 + 2^2}} = \frac{\frac{11}{2}}{\sqrt{13}} = \frac{11}{2\sqrt{13}}$$

Practice Exercises



Foundation Level

- If distance between the point $(x, 2)$ and $(3, 4)$ is 2, then the value of $x =$
 (a) 0 (b) 2
 (c) 3 (d) 4
- Find the mid-point of the line-segment joining two points $(3, 4)$ and $(5, 12)$.
 (a) $(-4, 8)$ (b) $(0, 8)$
 (c) $(4, 8)$ (d) $(4, 0)$
- The mid-point of the line segment joining the points $(-2, 4)$ and $(6, 10)$ is
 (a) $(2, 5)$ (b) $(2, 7)$
 (c) $(3, 7)$ (d) $(3, 8)$
- For which value of k given below the point $A(-1, 4)$, $B(2, 5)$ and $C(3, k)$ are collinear?
 (a) $16/3$ (b) 16
 (c) 5 (d) -1
- The points $A(-4, -1)$, $B(-2, -4)$, $C(4, 0)$ and $D(2, 3)$ are the vertices of a
 (a) Parallelogram (b) Rectangle
 (c) Rhombus (d) Square
- The line $x + y = 4$ divides the line joining the points $(-1, 1)$ and $(5, 7)$ in the ratio
 (a) 2 : 1 (b) 1 : 2
 (c) 1 : 2 externally (d) None of these
- If $A(3, 5)$, $B(-3, -4)$, $C(7, 10)$ are the vertices of a parallelogram taken in the order, then the co-ordinates of the fourth vertex are
 (a) $(10, 19)$ (b) $(15, 10)$
 (c) $(19, 10)$ (d) $(15, 19)$
- The centroid of a triangle, whose vertices are $(2, 1)$, $(5, 2)$ and $(3, 4)$ is
 (a) $\left(\frac{8}{3}, \frac{7}{3}\right)$ (b) $\left(\frac{10}{3}, \frac{7}{3}\right)$
 (c) $\left(-\frac{10}{3}, \frac{7}{3}\right)$ (d) $\left(\frac{10}{3}, -\frac{7}{3}\right)$
- The incentre of the triangle with vertices $(1, \sqrt{3})$, $(0, 0)$ and $(2, 0)$ is
 (a) $\left(1, \frac{\sqrt{3}}{2}\right)$ (b) $\left(\frac{2}{3}, \frac{1}{\sqrt{3}}\right)$
 (c) $\left(\frac{2}{3}, \frac{\sqrt{3}}{2}\right)$ (d) $\left(1, \frac{1}{\sqrt{3}}\right)$
- If the coordinates of the points A, B, C be $(4, 4)$, $(3, -2)$ and $(3, -16)$ respectively, then the area of the triangle ABC is:
 (a) 27 (b) 15
 (c) 18 (d) 7
- The equation of the line passing through $(2, -4)$ and parallel to $X - 2Y - 5 = 0$, is
 (a) $2X + Y + 3 = 0$ (b) $X - 2Y - 10 = 0$
 (c) $X - 2Y + 8 = 0$ (d) $X - 2Y + 13 = 0$
- The centroid of the triangle whose vertices are $(3, 10)$, $(7, 7)$, $(-2, 1)$ is
 (a) $(8/3, 6)$ (b) $(6, 8/3)$
 (c) $(-4, -7/3)$ (d) None of these
- The coordinates of the centroid G of a triangle with vertices at $(3, 7)$, $(5, 5)$ and $(-3, 2)$ is
 (a) $(10/3, 14/3)$ (b) $(10/3, 10/3)$
 (c) $(5/3, 14/3)$ (d) $(11/3, 10/3)$
- The equation of the line passing through the points $(5, 3)$ and $(3, 5)$ is
 (a) $X - Y + 8 = 0$ (b) $X + Y + 8 = 0$
 (c) $X - Y - 8 = 0$ (d) $X + Y - 8 = 0$
- The slope of the line $a^2X - aY + 1 = 0$, where a is constant, is
 (a) $-a^2$ (b) $-a$
 (c) a (d) None of these
- Equation of a line which makes intercepts 3 and 4 on x axis and y axis respectively is
 (a) $4x + 3y = 12$ (b) $3x + 4y = 12$
 (c) $6x + y = 12$ (d) $4x - 3y = 12$
- The coordinates of a point which divides the join of $(5, -5)$ and $(2, -3)$ in the ratio 4 : 3, externally, are:
 (a) $(3, 4)$ (b) $(-7, 3)$
 (c) $(-7, 9)$ (d) $(8, 3)$
- Distance between $P(x, y)$ and $Q(3, -6)$ is 10 units and x is positive integer, then $x =$
 (a) 3 (b) 9
 (c) 7 (d) 11
- Distance between the points $(a \cos 35^\circ, 0)$ and $B(a, a \cos 65^\circ)$ is
 (a) a^2 (b) $\sqrt{a^2}$
 (c) $-a$ (d) a
- The vertices of a parallelogram in order are $A(1, 2)$, $B(4, y)$, $C(x, 6)$, $D(3, 6)$, then $(x, y) =$
 (a) $(6, 3)$ (b) $(3, 6)$
 (c) $(5, 6)$ (d) $(1, 4)$
- The point which divides the line segment joining the points $(7, -6)$ and $(3, 4)$ in ratio 1 : 2 internally lies in the
 (a) I quadrant (b) II quadrant
 (c) III quadrant (d) IV quadrant

22. How many squares are possible if two of the vertices of a quadrilateral are (1, 0) and (2, 0)?
 (a) 1 (b) 2
 (c) 3 (d) 4
23. In what ratio is the line segment made by the points (7, 3) and (-4, 5) divided by the y-axis?
 (a) 2 : 3 (b) 4 : 7
 (c) 3 : 5 (d) 7 : 4
24. If the coordinates of the mid-point of the line segment joining the points (2, 1) and (1, -3) is (x, y), then the relation between x and y can be best described by
 (a) $3x + 2y = 5$ (b) $6x + y = 8$
 (c) $5x - 2y = 4$ (d) $2x - 5y = 4$
25. Points (4, -1), (6, 0), (7, 2) and (5, 1) are joined to be a vertex of a quadrilateral. What will be the structure?
 (a) Rhombus (b) Parallelogram
 (c) Square (d) Rectangle
26. Which of the following three points represent a straight line?
 (a) $\left(-\frac{1}{2}, 3\right)$, (-5, 6) and (-8, 8)
 (b) $\left(-\frac{1}{2}, 3\right)$, (5, 6) and (-8, 8)
 (c) $\left(\frac{1}{2}, 3\right)$, (-5, 6) and (-8, 8)
 (d) $\left(-\frac{1}{2}, 3\right)$, $\left(\frac{5}{6}\right)$ and (8, 8)
27. Which of the following will be the equation of a straight line parallel to the y-axis at a distance of 9 units to the left?
 (a) $x = -9$ (b) $x = 9$
 (c) $y = 9$ (d) $y = -9$
28. What can be said about the equation of the straight line $x = 7$?
 (a) It is the equation of a straight line at a distance of 7 units towards the right of the y-axis.
 (b) It is the equation of a straight line at a distance of 7 units towards the left of the y-axis.
 (c) It is the equation of a straight line at a distance of 7 units below the x-axis.
 (d) It is the equation of a straight line at a distance of 7 units above the x-axis.
29. Find the third vertex of the triangle whose two vertices are (-3, 1) and (0, -2) and the centroid is the origin.
 (a) (2, 3) (b) $\left(-\frac{4}{3}, \frac{14}{3}\right)$
 (c) (3, 1) (d) (6, 4)
30. Which of the following straight lines passes through the origin?
 (a) $x + y = 4$ (b) $x^2 + y^2 = -6$
 (c) $x + y = 5$ (d) $x = 4y$
31. What will be the reflection of the point (4, 5) in the second quadrant?
 (a) (-4, -5) (b) (-4, 5)
 (c) (4, -5) (d) None of these
32. If the origin gets shifted to (2, 2), then what will be the new coordinates of the point (4, -2)?
 (a) (-2, 4) (b) (2, 4)
 (c) (4, 2) (d) (2, -4)
33. What will be the length of the perpendicular drawn from the point (4, 5) upon the straight line $3x + 4y = 10$?
 (a) $\frac{12}{5}$ (b) $\frac{32}{5}$
 (c) $\frac{22}{5}$ (d) $\frac{42}{5}$
34. The straight line joining (1, 2) and (2, -2) is perpendicular to the line joining (8, 2) and (4, p). What will be the value of p?
 (a) -1 (b) 1
 (c) 3 (d) None of these
35. The orthocentre of the triangle formed by the points (0, 0), (8, 0) and (4, 6) is
 (a) $\left(4, \frac{8}{3}\right)$ (b) (3, 4)
 (c) (4, 3) (d) $\left(3, \frac{5}{2}\right)$
36. What will be the new equation of straight line $3x + 4y = 6$ if the origin gets shifted to (3, -4)?
 (a) $3x + 4y = 5$ (b) $4x - 3y = 4$
 (c) $3x + 4y + 1 = 0$ (d) $3x + 4y - 13 = 0$
37. What will be the value of p if the equation of straight line $2x + 5y = 4$ gets changed to $2x + 5y = p$ after shifting the origin at (3, 3)?
 (a) 16 (b) -17
 (c) 12 (d) 10
38. Which of the following two lines are perpendicular?
 1. $x + 2y = 5$ 2. $2x - 4y = 6$
 3. $2x + 3y = 4$ 4. $2x - y = 4$
 (a) 1 and 2 (b) 2 and 4
 (c) 2 and 3 (d) 1 and 4
39. A line passing through the points (a, 2a) and (-2, 3) is perpendicular to the line $4x + 3y + 5 = 0$. Find the value of a?
 (a) -14/3 (b) 18/5
 (c) 14/3 (d) -18/5
40. If the point R(1, -2) divides externally the line segment joining P(2, 5) and Q in the ratio 3 : 4, what will be the coordinates of Q?
 (a) (-3, 6) (b) (2, -4)
 (c) (3, 6) (d) (1, 2)
41. C is the mid-point of PQ, if P is (4, x), C is (y, -1) and Q is (-2, 4), then x and y respectively are
 (a) -6 and 1 (b) -6 and 2
 (c) 6 and -1 (d) 6 and -2
42. The three vertices of a triangle are given as (0, 1), (0, -5) and (4, -), '-' denotes an integer which has been erased. Which of the following can be the area of the triangle?
 (in sq. units)
 (a) 12 (b) 14
 (c) 16 (d) cannot be determined

43. A quadrilateral has the vertices at the points $(-4, 2)$, $(2, 6)$, $(8, 5)$ and $(9, -7)$. Show that the mid-points of the sides of this quadrilateral are the vertices of a parallelogram.
 (a) Rectangle (b) Square
 (c) Parallelogram (d) Rhombus
44. The coordinates of the points A, B, C, D are $(2, a)$, $(3, 5)$, $(3, 4)$ and $(0, 6)$ respectively. If the lines AC and BD be perpendicular, then $a = ?$
 (a) 7 (b) 1
 (c) -1 (d) -7
45. If the points $(a, 0)$, $(0, b)$ and $(1, 1)$ are collinear, then
 (a) $\frac{1}{a^2} + \frac{1}{b^2} = 1$ (b) $\frac{1}{a^2} - \frac{1}{b^2} = 1$
 (c) $\frac{1}{a} + \frac{1}{b} = 1$ (d) $\frac{1}{a} - \frac{1}{b} = 1$
46. The points $A(a, b+c)$, $B(b, a+c)$, $C(c, a+b)$ are
 (a) vertices of an equilateral triangle.
 (b) collinear
 (c) vertices of an isosceles triangle
 (d) vertices of a right triangle
47. The angle between the lines $y = (2 - \sqrt{3})X + 5$ and $y = (2 + \sqrt{3})X - 7$ is
 (a) 30° (b) 60°
 (c) 45° (d) $\tan^{-1} 3$
48. Find the ratio in which the point $(2, y)$ divides the join of $(-4, 3)$ and $(6, 3)$ and hence find the value of y
 (a) $2:3, y=3$ (b) $3:2, y=4$
 (c) $3:2, y=3$ (d) $3:2, y=2$
49. The area of quadrilateral with vertices $(2, 4)$, $(0, 4)$, $(0, -4)$, $(2, -4)$ is equal to (sq. units)
 (a) 8 (b) 12
 (c) 16 (d) 32
50. If $P\left(\frac{a}{3}, 4\right)$ is the mid-point of the line segment joining the points $Q(-6, 5)$ and $R(-2, 3)$, then the value of a is
 (a) -4 (b) -12
 (c) 12 (d) -6
51. The ratio in which the line $2x + y - 4 = 0$ divides the line segment joining the points $A(2, -2)$ and $B(3, 7)$ is
 (a) $3:7$ (b) $4:7$
 (c) $2:9$ (d) $4:9$
52. Which of the following points is the nearest to the origin?
 (a) $(0, -6)$ (b) $(-8, 0)$
 (c) $(-3, -4)$ (d) $(7, 0)$
53. Find the distance between the two parallel straight lines $y = mx + c$ and $y = mx + d$ [Assume $c > d$]
 (a) $\left(\frac{c-d}{(1+m^2)^{1/2}}\right)$ (b) $\left(\frac{d-c}{(1+m^2)^{1/2}}\right)$
 (c) $\left(\frac{d}{(1+m^2)^{1/2}}\right)$ (d) $\left(\frac{-d}{(1+m^2)^{1/2}}\right)$
54. If the points $(1, 1)$, $(-1, -1)$ and $(-\sqrt{3}, k)$ are vertices of an equilateral triangle then the value of k will be :
 (a) 1 (b) -1
 (c) $\sqrt{3}$ (d) $-\sqrt{3}$
55. The image of the point $(3, 8)$ in the line $x + 3y = 7$ is
 (a) $(4, 7)$ (b) $(2, 3)$
 (c) $(-1, -4)$ (d) None of these
56. The points $(3, 0)$, $(-3, 0)$, $(0, -3\sqrt{3})$ are the vertices of
 (a) equilateral triangle (b) isosceles triangle
 (c) right triangle (d) scalene triangle
57. D is a point on AC of the triangle with vertices $A(2, 3)$, $B(1, -3)$, $C(-4, -7)$ and BD divides ABC into two triangles of equal area. The equation of the line drawn through B at right angles to BD is
 (a) $y - 2x + 5 = 0$ (b) $2y - x + 5 = 0$
 (c) $y + 2x - 5 = 0$ (d) $2y + x - 5 = 0$
58. Ratio in which the line $3x + 4y = 7$ divides the line segment joining the points $(1, 2)$ and $(-2, 1)$ is
 (a) $3:5$ (b) $4:6$
 (c) $4:9$ (d) None of these
59. If the area of a triangle with vertices $(-3, 0)$, $(3, 0)$ and $(0, k)$ is 9 sq unit, then what is the value of k ?
 (a) 3 (b) 6
 (c) 9 (d) 12
60. The line $mx + ny = 1$ passes through the points $(1, 2)$ and $(2, 1)$. What is the value of m ?
 (a) 1 (b) 3
 (c) $\frac{1}{2}$ (d) $\frac{1}{3}$
61. The line $y = 0$ divides the line joining the points $(3, -5)$ and $(-4, 7)$ in the ratio
 (a) $3:4$ (b) $4:5$
 (c) $5:7$ (d) $7:9$
62. For what value of k , the equations $3x - y = 8$ and $9x - ky = 24$ will have infinitely many solutions?
 (a) 6 (b) 5
 (c) 3 (d) 1
63. The equations of two equal sides AB and AC of an isosceles triangle ABC are $x + y = 5$ and $7x - y = 3$ respectively. What will be the equation of the side BC if area of triangle ABC is 5 square units.
 (a) $x + 3y - 1 = 0$ (b) $x - 3y + 1 = 0$
 (c) $2x - y = 5$ (d) $x + 2y = 5$
64. What is the equation of a line parallel to x -axis at a distance of 5 units below x -axis?
 (a) $x = 5$ (b) $x = -5$
 (c) $y = 5$ (d) $y = -5$
65. What are the points on the axis of x whose perpendicular distance from the straight line $x/p + y/q = 1$ is p ?
 (a) $\frac{p}{q}\left[q + \sqrt{(p^2 + q^2)}\right], 0$ (b) $\frac{p}{q}\left[q - \sqrt{(p^2 + q^2)}\right], 0$
 (c) Both (a) and (b) (d) None of these

Standard Level

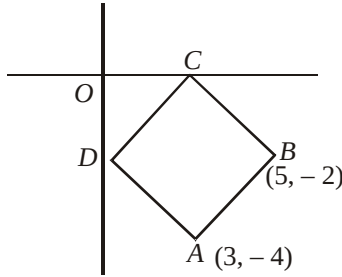
- The fourth vertex of a rectangle whose other vertices are (4, 1) (7, 4) and (13, -2) is
(a) (10, -5) (b) (10, 5)
(c) (-10, 5) (d) (-10, -5)
- P, Q, R are three collinear points. The coordinates of P and R are (3, 4) and (11, 10) respectively, and PQ is equal to 2.5 units. Coordinates of Q are
(a) (5, 11/2) (b) (11, 5/2)
(c) (5, -11/2) (d) (-5, 11/2)
- The coordinates of vertices A and B of an equilateral triangle ABC are (-4, 0) and (4, 0) respectively. Which of the following could be coordinates of C
(a) (0, 2√3) (b) (0, 4)
(c) (0, 4√3) (d) (0, 3)
- The three vertices of a parallelogram are A (3, -4), B (-2, 1) and C (-6, 5). Which of the following cannot be the fourth one
(a) (-1, 0) (b) (7, -8)
(c) (1, -5) (d) All of these
- The intercept made by a line on y-axis is double to the intercept made by it on x-axis. If it passes through (1, 2) then its equation
(a) 2x + y = 4 (b) 2x + y + 4 = 0
(c) 2x - y = 4 (d) 2x - y + 4 = 0
- The mid-points of sides of a triangle are (2, 1), (-1, -3) and (4, 5). Then the coordinates of its vertices are:
(a) (7, 9), (-3, -7), (1, 1) (b) (-3, -7), (1, 1), (2, 3)
(c) (1, 1), (2, 3), (-5, 8) (d) None of these
- The point whose abscissa is equal to its ordinate and which is equidistant from the points (1, 0) and (0, 3) is
(a) (1, 1) (b) 2, 2
(c) (3, 3) (d) (4, 4)
- If the point dividing internally the line segment joining the points (a, b) and (5, 7) in the ratio 2 : 1 be (4, 6), then
(a) a = 1, b = 2 (b) a = 2, b = -4
(c) a = 2, b = 4 (d) a = -2, b = 4
- The distance of point of intersection of 2X - 3Y + 13 = 0 and 3X + 7Y - 15 = 0 from (4, -5), will be
(a) 10 units (b) 12 units
(c) 11 units (d) None of these
- A (-2, 4) and B (-5, -3) are two points. The coordinates of a point P on Y axis such that PA = PB, are
(a) (3, 4) (b) (0, 9)
(c) (9, 0) (d) (0, -1)
- If 7x + 3y + 9 = 0 and y = kx + 7 are two parallel lines then k is
(a) 3/7 (b) -7/3
(c) 3 (d) 7
- A line intersects the straight lines 5x - y - 4 = 0 and 3x - 4y - 4 = 0 at A and B respectively. If a point P (1, 5) on the line AB is such that AP : PB = 2:1 (internally), find the point A.
(a) $\begin{pmatrix} 75 & 304 \\ 17 & 17 \end{pmatrix}$ (b) $\begin{pmatrix} 65 & 304 \\ 17 & 17 \end{pmatrix}$
(c) $\begin{pmatrix} 75 & 104 \\ 17 & 17 \end{pmatrix}$ (d) $\begin{pmatrix} 75 & 180 \\ 17 & 17 \end{pmatrix}$
- The vertices of triangle ABC are A (4, 4), B (6, 3), C (2, -1); then angle ABC is equal to
(a) 45° (b) 90°
(c) 60° (d) None of these
- A line intersects the y-axis and x-axis at the points P and Q respectively. If (2, -5) is the mid-point of PQ, then the coordinates of P and Q are respectively
(a) (0, -5) and (2, 0) (b) (0, 10) and (-4, 0)
(c) (0, 4) and (-10, 0) (d) (0, -10) and (4, 0)
- If the point P (2, 1) lies on the line segment joining points A(4, 2) and B(8, 4), then
(a) AP = $\frac{1}{3}$ AB (b) AP = PB
(c) PB = $\frac{1}{3}$ AB (d) AP = $\frac{1}{2}$ AB
- If area of Δ formed by joining mid-points of the sides of ΔABC is 2sq. units, then area of ΔABC =
(a) 8 (b) 4
(c) 2 (d) 1
- The centroid of a triangle formed by (7, p), (q, -6), (9, 10) is (6, 3). Then p + q
(a) 6 (b) 5
(c) 7 (d) 8
- If the points A(1, 2), B(2, 4) and C(3, a) are collinear, what is the length BC?
(a) √2 unit (b) √3 unit
(c) √5 unit (d) 5 unit
- If (-5, 4) divides the line segment between the coordinate axes in the ratio 1: 2, then what is its equation?
(a) 8x + 5y + 20 = 0 (b) 5x + 8y - 7 = 0
(c) 8x - 5y + 60 = 0 (d) 5x - 8y + 57 = 0
- What is the slope of the line perpendicular to the line $\frac{x}{4} + \frac{y}{3} = 1$?
(a) $\frac{3}{4}$ (b) $-\frac{3}{4}$
(c) $-\frac{4}{3}$ (d) $\frac{4}{3}$
- Two sides of a square lie on the lines x + y = 1 and x + y + 2 = 0. What is its area?
(a) $\frac{11}{2}$ (b) $\frac{9}{2}$
(c) 5 (d) 4
- If the three vertices of a rectangle taken in order are the points (2, -2), (8, 4) and (5, 7). The coordinates of the fourth vertex is
(a) (1, 1) (b) (1, -1)
(c) (-1, 1) (d) None of these

23. If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram $PQRS$, then
 (a) $a = 2, b = 4$ (b) $a = 3, b = 4$
 (c) $a = 2, b = 3$ (d) $a = 3, b = 5$
24. Angle between $y = x + 6$ and $y = \sqrt{3}x + 7$ is
 (a) 75° (b) 45°
 (c) 15° (d) 30°
25. If (a, b) , (c, d) and $(a - c, b - d)$ are collinear, then which one of the following is correct?
 (a) $bc - ad = 0$ (b) $ab - cd = 0$
 (c) $bc + ad = 0$ (d) $ab + cd = 0$
26. Find the coordinates of the points that trisect the line segment joining $(1, -2)$ and $(-3, 4)$
 (a) $\left(-\frac{1}{3}, 0\right)$ (b) $\left(-\frac{5}{3}, 2\right)$
 (c) Both (a) and (b) (d) None of these
27. The distance between the lines $4x + 3y = 11$ and $8x + 6y = 15$ is
 (a) 4 (b) $\frac{7}{10}$
 (c) $\frac{5}{7}$ (d) 26
28. If the mid-point of the line joining $(3, 4)$ and $(p, 7)$ is (x, y) and $2x + 2y + 1 = 0$, then what will be the value of p ?
 (a) 15 (b) $-\frac{17}{2}$
 (c) -15 (d) $\frac{17}{2}$
29. The number of lines that are parallel to $2x + 6y + 7 = 0$ and have an intercept of length 10 between the coordinate axes is
 (a) 0 (b) 1
 (c) 2 (d) Infinite
30. Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If the orthocentre of the triangle is the origin, what will be the coordinates of the third point?
 (a) $(4, 7)$ (b) $(-4, 7)$
 (c) $(-4, -7)$ (d) $(4, -7)$
31. One side of a rectangle lies along the line $4x + 7y + 5 = 0$. Two of its vertices are $(-3, 1)$ and $(1, 1)$. Which of the following may be an equation which represents any of the other three straight lines?
 (a) $7x - 4y = 3$ (b) $7x - 4y + 3 = 0$
 (c) $y + 1 = 0$ (d) $4x + 7y = 3$
32. The point A divides the join the points $(-5, 1)$ and $(3, 5)$ in the ratio $k : 1$ and coordinates of points B and C are $(1, 5)$ and $(7, -2)$ respectively. If the area of $\triangle ABC$ be 2 units, then k equals
 (a) 7, 9 (b) 6, 7
 (c) 7, 31/9 (d) 9, 31/9
33. The distance between the lines $X + 2Y - 3 = 0$ and $11X + 22Y + 88 = 0$, will be
 (a) $7/5$ (b) $49/\sqrt{5}$
 (c) $7/\sqrt{5}$ (d) None of these
34. A point P is equidistant from A $(3, 1)$ and B $(5, 3)$ and its abscissa is twice its ordinate, then its co-ordinates are.
 (a) $(2, 1)$ (b) $(1, 2)$
 (c) $(4, 2)$ (d) $(2, 4)$
35. If $(-1, -1)$ and $(3, -1)$ are two opposite corners of a square, the other two corners are
 (a) $(2, 0)$, $(-2, 2)$ (b) $(2, -2)$, $(0, 2)$
 (c) $(3, 0)$, $(4, -2)$ (d) None of these
36. Two points on a line are such that their Y co-ordinates is $(3 + \sqrt{2})$ times the X co-ordinate. Then the line
 (a) has negative slope (b) passes through the origin
 (c) is parallel to the X-axis (d) is parallel to the Y-axis
37. Equation of a line which is passing through $(3, -4)$ and making an angle of 45° with x-axis is
 (a) $x - y - 7 = 0$ (b) $x + y + 7 = 0$
 (c) $x - y + 7 = 0$ (d) $x + y - 7 = 0$
38. The diagonals AC and BD of a rhombus intersect at $(5, 6)$. If $A = (3, 2)$ then equation of diagonal BD is
 (a) $y - x = 1$ (b) $2y - x = 17$
 (c) $y - 2x + 4 = 0$ (d) $2y + x = 17$
39. A line passes through the point $(3, 4)$ and cuts off intercepts, from the coordinates axes such that their sum is 14. The equation of the line is :
 (a) $4x - 3y = 24$ (b) $4x + 3y = 24$
 (c) $3x - 4y = 24$ (d) $3x + 4y = 24$
40. Two straight lines $x - 3y - 2 = 0$ and $2x - 6y - 6 = 0$
 (a) never intersect
 (b) intersect at a single point
 (c) intersect at infinite number of points
 (d) intersect at more than one point (but finite number of points)
41. If $(a, 0)$, $(0, b)$ and $(1, 1)$ are collinear, what is $(a + b - ab)$ equal to?
 (a) 2 (b) 1
 (c) 0 (d) -1
42. The equation of a straight line which makes an angle 45° with the x-axis with y-intercept 101 units is
 (a) $10x + 101y = 1$ (b) $101x + y = 1$
 (c) $x + y - 101 = 0$ (d) $x - y + 101 = 0$
43. The area of triangle formed by the points $(p, 2-2p)$, $(1-p, 2p)$ and $(-4-p, 6-2p)$ is 70 unit. How many integral values of p are possible?
 (a) 2 (b) 3
 (c) 4 (d) None of these
44. What is the perimeter of the triangle with vertices $A(-4, 2)$, $B(0, -1)$ and $C(3, 3)$?
 (a) $7 + 3\sqrt{2}$ (b) $10 + 5\sqrt{2}$
 (c) $11 + 6\sqrt{2}$ (d) $5 + \sqrt{2}$
45. $P(3, 1)$, $Q(6, 5)$ and $R(x, y)$ are three points such that the angle PRQ is a right angle and the area of $\triangle PRQ$ is 7. The number of such points R that are possible is
 (a) 1 (b) 2
 (c) 3 (d) 4
46. The coordinates of the mid-points of the sides of a triangle are $(4, 2)$, $(3, 3)$ and $(2, 2)$. What will be the coordinates of the centroid of the triangle?
 (a) $\left(3, \frac{7}{3}\right)$ (b) $\left(-3, \frac{-7}{3}\right)$
 (c) $\left(3, \frac{-7}{3}\right)$ (d) $\left(-3, \frac{7}{3}\right)$

Expert Level

- The equation of the line which bisects the obtuse angle between the lines $x - 2y + 4 = 0$ and $4x - 3y + 2 = 0$, is
 - $(4 - \sqrt{5})x - (3 - 2\sqrt{5})y + (2 - 4\sqrt{5}) = 0$
 - $(4 + \sqrt{5})x + (3 + 2\sqrt{5})y + (2 + 4\sqrt{5}) = 0$
 - $(4 + \sqrt{5})x - (3 + 2\sqrt{5})y + (2 + 4\sqrt{5}) = 0$
 - None of these
- Which of the following points will be collinear with the points $(-3, 4)$ and $(2, -5)$?
 - $(0, 0)$
 - $(7, -14)$
 - $(0, -1)$
 - $(3, 1)$
- If the lines $\ell x + my + n = 0$, $mx + ny + \ell = 0$ and $nx + \ell y + m = 0$ are concurrent then
 - $\ell + m + n = 0$
 - $\ell - m - n = 0$
 - $\ell + m - n = 0$
 - $m + n - \ell = 0$
- If the middle points of the sides of a triangle be $(-2, 3)$, $(4, -3)$ and $(4, 5)$, then the centroid of the triangle is:
 - $(5/3, 2)$
 - $(5/6, 1)$
 - $(2, 5/3)$
 - $(1, 5/6)$
- What is the inclination of the line $\sqrt{3}x - y - 1 = 0$?
 - 30°
 - 60°
 - 135°
 - 150°
- What will be the equation of the straight line that passes through the intersection of the straight lines $2x - 3y + 4 = 0$ and $3x + 4y - 5 = 0$ and is perpendicular to the straight line $3x - 4y = 5$?
 - $8x + 6y = \frac{32}{7}$
 - $4x + 3y = \frac{84}{17}$
 - $4x + 3y = \frac{62}{17}$
 - $8x + 6y = \frac{58}{17}$
- Find the equation of the straight line passing through the origin and the point of intersection of the lines $x/a + y/b = 1$ and $x/b + y/a = 1$.
 - $y = x$
 - $y = -x$
 - $y = 2x$
 - $y = -2x$
- Orthocentre of the triangle whose sides are given by $4x - 7y + 10 = 0$, $x + y - 5 = 0$ & $7x + 4y - 15 = 0$ is
 - $(-1, -2)$
 - $(1, -2)$
 - $(-1, 2)$
 - $(1, 2)$
- If a line passes through the point $P(1, 2)$ makes an angle of 45° with the x axis and meets the line $x + 2y - 7 = 0$ in Q , then PQ equals
 - $\frac{2\sqrt{2}}{3}$
 - $\frac{3\sqrt{2}}{2}$
 - $\sqrt{3}$
 - $\sqrt{2}$
- If $(4, P)$ lies on the line which passes through $(2, 3)$ and which is parallel to $4X + 3Y - 6 = 0$, the value of P is
 - 3
 - 1
 - $1/3$
 - $1/2$
- If one vertex of equilateral Δ is at $A(3, 4)$ and the base BC is $x + y - 5 = 0$, then the length of each side of the Δ is
 - $3\sqrt{3}$
 - $4\sqrt{3}$
 - $\frac{2\sqrt{2}}{\sqrt{3}}$
 - $2\sqrt{2}$
- The perimeter of the triangle whose vertices are $(-1, 4)$, $(-4, -2)$, $(3, -4)$, will be
 - 38
 - 16
 - 42
 - None of these
- The area of a triangle is 5. Two of its vertices are $(2, 1)$ and $(3, -2)$. The third vertex is (x, y) where $y = x + 3$. Then the co-ordinate of the third vertex is
 - $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$
 - $\left(-\frac{7}{2}, \frac{13}{2}\right)$ or $\left(\frac{3}{2}, -\frac{3}{2}\right)$
 - $\left(\frac{1}{2}, \frac{3}{2}\right)$
 - $\left(\frac{3}{2}, \frac{1}{2}\right)$
- The middle point of $A(1, 2)$ and $B(x, y)$ is $C(2, 4)$. If BD is perpendicular to AB such that $CD = 3$ unit, then what is the length BD ?
 - $2\sqrt{2}$ unit
 - 2 unit
 - 3 unit
 - $3\sqrt{2}$ unit
- The points $(p - 1, p + 2)$, $(p, p + 1)$, $(p + 1, p)$ are collinear for
 - $p = 0$
 - $p = 1$
 - $p = -1/2$
 - Any value of p
- One side of a rectangle lies along the line $4x + 7y + 5 = 0$. Two of its vertices are $(-3, 1)$ and $(1, 1)$. Which of the following is not an equation of the other three straight lines?
 - $14x - 8y = 6$
 - $7x - 4y = -25$
 - $4x + 7y = 11$
 - $14x - 8y = 20$
- A triangle ABC is given by $A(2, 5)$, $B(-1, -1)$, $C(3, 1)$. The equation of median drawn on BC from A , is:
 - $2X + Y - 9 = 0$
 - $5X - Y - 5 = 0$
 - $3X + Y = -9$
 - None of these

18. $(3, -4)$ and $(5, -2)$ are two consecutive vertices of a square in which $(2, -2)$ is an interior point. The centre of the square is at



- (a) $(1, 2)$ (b) $(3, -2)$
 (c) $(0, 2)$ (d) $(4, -2)$
19. The coordinates of P and Q are $(-3, 4)$ and $(2, 1)$, respectively. If PQ is extended to R such that $PR = 2QR$, then what are the coordinates of R ?
- (a) $(3, 7)$ (b) $(2, 4)$
 (c) $\left(-\frac{1}{2}, \frac{5}{2}\right)$ (d) $(7, -2)$
20. Which one of the following points on the line $2x - 3y = 5$ is equidistant from $(1, 2)$ and $(3, 4)$?
- (a) $(7, 3)$ (b) $(4, 1)$
 (c) $(1, -1)$ (d) $(-2, -3)$
21. If the straight lines $x - 2y = 0$ and $kx + y = 1$ intersect at the point $\left(1, \frac{1}{2}\right)$, then what is the value of k ?
- (a) 1 (b) 2
 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$
22. If the sum of the squares of the distances of the point (x, y) from the points $(a, 0)$ and $(-a, 0)$ is $2b^2$, then which one of the following is correct ?
- (a) $x^2 + a^2 = b^2 + y^2$ (b) $x^2 + a^2 = 2b^2 - y^2$
 (c) $x^2 - a^2 = b^2 + y^2$ (d) $x^2 + a^2 = b^2 - y^2$

23. For what value of k , are the lines $x + 2y - 9 = 0$ and $kx + 4y + 5 = 0$ parallel?
- (a) 2 (b) -1
 (c) 1 (d) 0
24. Let p, q, r, s be the distances from origin of the points $(2, 6)$, $(3, 4)$, $(4, 5)$ and $(-2, 5)$ respectively. Which one of the following is a whole number?
- (a) p (b) q
 (c) r (d) s
25. If the medians PT and RS of a triangle with vertices $P(0, b)$, $Q(0, 0)$ and $R(a, 0)$ are perpendicular to each other, which of the following satisfies the relationship between a and b ?
- (a) $4b^2 = a^2$ (b) $2b^2 = a^2$
 (c) $a = -2b$ (d) $a^2 + b^2 = 0$
26. If P and Q are two points on the line $3x + 4y = -15$, such that $OP = OQ = 9$ units, the area of the triangle POQ will be
- (a) $18\sqrt{2}$ sq units (b) $3\sqrt{2}$ sq units
 (c) $6\sqrt{2}$ sq units (d) $15\sqrt{2}$ sq units
27. If p is the length of the perpendicular from the origin to the line $\frac{x}{a} + \frac{y}{b} = 1$, then which of the following is true?
- (a) $\frac{1}{p^2} = \frac{1}{b^2} - \frac{1}{a^2}$ (b) $\frac{1}{p^2} = \frac{1}{a^2} - \frac{1}{b^2}$
 (c) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$ (d) None of these
28. What will be the area of the rhombus $ax \pm by \pm c = 0$?
- (a) $\frac{3c^2}{ab}$ (b) $\frac{4c^2}{ab}$
 (c) $\frac{2c^2}{ab}$ (d) $\frac{c^2}{ab}$

Test Yourself

- Find the orthocentre of the triangle whose sides have the equations $y = 15$, $3x = 4y$, and $5x + 12y = 0$.
 (a) $(0, 0)$ (b) $(0, -33)$
 (c) $(-33, 33)$ (d) $(-33, 0)$
- The ratio in which the point $(2, y)$ divides the join of $(-4, 3)$ and $(6, 3)$ and hence the value of y is
 (a) $2:3, y=3$ (b) $3:2, y=4$
 (c) $3:2, y=3$ (d) $3:2, y=2$
- The area of the triangle formed by the line $5x - 3y + 15 = 0$ with coordinate axes is
 (a) 15 sq. units (b) 5 sq. units
 (c) 8 sq. units (d) $\frac{15}{2}$ sq. units
- Find the in-centre of the right angled isosceles triangle having one vertex at the origin and having the other two vertices at $(6, 0)$ and $(0, 6)$.
 (a) $3(2 + \sqrt{2}), 3(2 + \sqrt{2})$
 (b) $3(2 - \sqrt{2}), 3(2 - \sqrt{2})$
 (c) $2 + \sqrt{2}, 2 + \sqrt{2}$
 (d) $2 - \sqrt{2}, 2 - \sqrt{2}$
- What is the point of intersection of the lines $2x + 3y = 5$ and $3x - 4y = 10$?
 (a) $\frac{50}{17}, \frac{-5}{17}$ (b) $\frac{-50}{17}, \frac{5}{17}$
 (c) $\frac{-10}{17}, \frac{35}{17}$ (d) $\frac{10}{17}, \frac{-35}{17}$
- Find the equation of the straight line which passes through $(3, 4)$ and the sum of whose X and Y intercepts is 14.
 (a) $4x + 3y = 24$ (b) $x + y = 7$
 (c) Both (a) and (b) (d) None of these
- The three vertices of a rhombus, taken in order are $(2, -1)$, $(3, 4)$ and $(-2, 3)$. The fourth vertex is
 (a) $(-2, -3)$ (b) $(-1, -4)$
 (c) $(-3, -4)$ (d) $(-3, -2)$
- Find the ratio in which the point $(2, y)$ divides the join of $(-4, 3)$ and $(6, 3)$ and hence find the value of y .
 (a) $3:2; 1$ (b) $3:2; 3$
 (c) $2:3; 1$ (d) $2:3; 3$
- The equation of two equal sides of an isosceles triangle are $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side is passes through the point $(1, -10)$. The equation of the third side is
 (a) $x - 3y - 31 = 0$ but not $3x + y + 7 = 0$
 (b) neither $3x + y + 7 = 0$ nor $x - 3y - 31 = 0$
 (c) $3x + y + 7 = 0$ or $x - 3y - 31 = 0$
 (d) $3x + y + 7 = 0$ but not $x - 3y - 31 = 0$
- $P(x, y)$ moves such that the area of the triangle with vertices at $P(x, y)$, $(1, -2)$, $(-1, 3)$ is equal to the area of the triangle with vertices at $P(x, y)$, $(2, -1)$, $(3, 1)$. The locus of P is the pair of lines
 (a) $3x + 3y + 4 = 0 = 7x + y - 6$
 (b) $x + y - 2 = 0 = 7x + y + 4$
 (c) $x + y - 6 = 0 = x + 7y + 4$
 (d) $7x + 3y + 4 = 0 = 3x + y - 6$
- The incentre of the triangle with vertices $(1, \sqrt{3})$, $(0, 0)$ and $(2, 0)$ is
 (a) $\left(1, \frac{\sqrt{3}}{2}\right)$ (b) $\left(\frac{2}{3}, \frac{1}{\sqrt{3}}\right)$
 (c) $\left(\frac{2}{3}, \frac{\sqrt{3}}{2}\right)$ (d) $\left(1, \frac{1}{\sqrt{3}}\right)$
- Find the equation of a straight line passing through $(2, -3)$ and having a slope of 1 unit.
 (a) $y - x + 5 = 0$ (b) $x + y + 5 = 0$
 (c) $-x - y = -5$ (d) $x - y - 5 = 0$
- What will be the circumcentre of a triangle whose sides are $3x - y + 3 = 0$, $3x + 4y + 3 = 0$ and $x + 3y + 11 = 0$?
 (a) $(3, 0)$ (b) $(-3, 0)$
 (c) $(3, -3)$ (d) $(-3, 3)$
- Point of intersection of the diagonals of square is at origin and coordinate axis are drawn along the diagonals. If the side is of length a , then one which is not the vertex of square is :
 (a) $(a/\sqrt{2}, 0)$ (b) $(0, a/\sqrt{2})$
 (c) $(a\sqrt{2}, 0)$ (d) $(-a/\sqrt{2}, 0)$
- Find the equation of a straight line perpendicular to the straight line $3x + 4y = 7$ and passing through the point $(3, -3)$.
 (a) $x - 3y = 21$ (b) $4x = 3y$
 (c) $4x - 3y = 21$ (d) $3x - 4y = 21$

Hints & Solutions

Foundation Level

1. (c) $2 = \sqrt{(x-3)^2 + (2-4)^2} \Rightarrow 2 = \sqrt{(x-3)^2 + 4}$

Squaring both sides

$$4 = (x-3)^2 + 4 \Rightarrow x-3 = 0 \Rightarrow x = 3$$

2. (c) Let $A(3, 4)$ and $B(5, 12)$ be the given points.
Let $C(x, y)$ be the mid-point of AB . Using mid-point

$$\text{formula, we have, } x = \frac{3+5}{2} = 4 \text{ and } y = \frac{4+12}{2} = 8$$

$\therefore C(4, 8)$ are the co-ordinates of the mid-point of the line segment joining two points $(3, 4)$ and $(5, 12)$.

3. (b)

4. (a)

5. (b)

6. (b) Ratio $-\left(\frac{-1+1-4}{5+7-4}\right) = \frac{1}{2}$

7. (d) Mid point of $A(3, 5)$ and $C(7, 10) = M\left(5, \frac{15}{2}\right)$

$$\therefore \text{Mid points of } BD = M\left(5, \frac{15}{2}\right)$$

$B(-5, -4)$ and $D(x, y)$

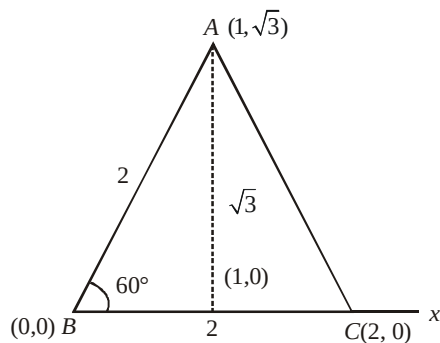
$$\therefore \frac{-5+x}{2} = 5, \quad x = 10 + 5 = 15$$

$$\frac{-4+y}{2} = \frac{15}{2}, \quad y = 15 + 4 = 19$$

Co-ordinates of fourth vertex $D = (15, 19)$

8. (b) $x = \frac{2+5+3}{3} = \frac{10}{3}$ and $y = \frac{1+2+4}{3} = \frac{7}{3}$

9. (d) Clearly, the triangle is equilateral.



So, the incentre is the same as the centroid.

$$\therefore \text{Incentre} = \left(\frac{1+0+2}{3}, \frac{\sqrt{3}+0+0}{3}\right) = \left(1, \frac{1}{\sqrt{3}}\right)$$

10. (d) $\frac{1}{2}[4 - (2+16) + 3(-16-4) + 3(4+2)]$

$$= \frac{1}{2}[56 - 60 + 18] = 7$$

11. (b) Line parallel to $X - 2Y - 5 = 0$ will be $X - 2Y + K = 0$.
Put the point in this equation we have
 $2 + 8 + K = 0$
or $K = -10$ or line is $X - 2Y - 10 = 0$

12. (a) Centroid $= \left(\frac{3+7-2}{3}, \frac{10+7+1}{3}\right) = \left(\frac{8}{3}, 6\right)$

13. (c) Let G be (X, Y) , then $X = \{3 + 5 + (-3)\} / 3 = 5/3$
 $Y = \{7 + 5 + 2\} / 3 = 14/3 \Rightarrow G$ is $(5/3, 14/3)$

14. (d) The equation will be

$$Y - 3 = \frac{(5-3)}{(3-5)} \times (X - 5)$$

$$\Rightarrow Y - 3 = 5 - X \text{ or } X + Y - 8 = 0$$

15. (c) Slope $= -\frac{\text{Coefficient of } X}{\text{Coefficient of } Y} = \frac{-a^2}{-a} = a$

16. (a) $\frac{x}{3} + \frac{y}{4} = 1 \Rightarrow 4x + 3y = 12$

17. (b) Let the ratio be $4 : 3$ or $4/3 : 1$.

$$\text{Now } X = \frac{\frac{4}{3} \times 2 - 5}{\frac{4}{3} - 1} = \frac{\frac{8}{3} - 5}{\frac{4}{3} - 1} = \frac{-\frac{7}{3}}{\frac{1}{3}} = -7$$

$$Y = \frac{\frac{4}{3} \times 3 + 5}{\frac{4}{3} - 1} = \frac{4 + 5}{\frac{4}{3} - 1} = \frac{9}{\frac{1}{3}} = 3. \text{ Hence } (-7, 3)$$

18. (b)

19. (b)

20. (a) Mid-point of AC is $\left(\frac{1+x}{2}, \frac{2+6}{2}\right)$ i.e., $\left(\frac{1+x}{2}, 4\right)$;

$$\text{Mid-point of } BD \text{ is } \left(\frac{4+3}{2}, \frac{y+5}{2}\right)$$

Since for a || gm, diagonals bisect each other

$$\therefore \frac{1+x}{2} = \frac{7}{2} \text{ and } \frac{y+5}{2} = 4 \Rightarrow x = 6, y = 3$$

21. (d)

22. (c)

23. (d)

24. (b)

25. (a)

26. (a)
27. (a)
28. (a)
29. (c)
30. (d)
31. (b)
32. (d)
33. (c)
34. (b) Form the equation of the straight lines and then use the options.
35. (a) Orthocentre is the point of intersection of altitudes of a triangle and centroid divides the straight line formed by joining circumcentre and the orthocentre in the ratio 2 : 1.
Let the vertices of the triangle be $O(0, 0)$, $A(8, 0)$ and $B(4, 6)$.
The equation of an altitude through O and perpendicular to AB is $y = 2/3x$ and similarly the equation of an altitude through A and perpendicular to OB is $2x + 3y = 16$. Now find the point of intersection of these two straight lines.
36. (c)
37. (b) If the origin gets changed to (h, k) from $(0, 0)$, then
Old x co-ordinate = New x co-ordinate + h
Old y co-ordinate = New y co-ordinate + k
38. (d)
39. (b) Equation of any straight line perpendicular to the line $4x + 3y + 5 = 0$ will be of the form of $3x - 4y = k$, where k is any constant.
Now form the equation of the straight line with the given two points and then equate.
40. (c)
41. (a)
42. (a)
43. (c)
44. (b) Slope $AC = \frac{4-a}{3-2} = 4-a$
Slope $BD = \frac{6-5}{0-3} = -\frac{1}{3}$
For perpendicular $(4-a)(-1/3) = -1 \Rightarrow a = 1$
45. (c) $\frac{1}{2} \begin{vmatrix} 1 & 0 & 1 \\ 0 & b & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0 \Rightarrow a(b-1) + 0 + 1(-b) = 0$
 $\Rightarrow ab - a - b = 0 \Rightarrow \frac{1}{a} + \frac{1}{b} = 1$
46. (b) Slope of $AB = \frac{a+c-b-c}{b-a} = \frac{a-b}{b-a} = -1$
Slope of $BC = \frac{a+b-a-c}{c-b} = \frac{b-c}{c-b} = -1$
Hence, collinear.

47. (b) $\theta = \tan^{-1} \left(\frac{2-\sqrt{3}-2-\sqrt{3}}{1+4-3} \right) \tan^{-1}(-\sqrt{3}) = 120^\circ$

Considering smaller angle $\theta = 60^\circ$

48. (c) Let the required ratio be $k : 1$

Then, $2 \frac{6k-4 \times 1}{k+1} \Rightarrow k = \frac{3}{2}$

$\therefore$ The required ratio is $\frac{3}{2} :: 1 \Rightarrow 3 : 2$

Also, $y = \frac{3 \times 3 + 2 \times 3}{3+2} = 3$

49. (c) Let A, B, C, D be the four vertices.

Then area of quadrilateral = Area of two triangles ABD and BCD .

Now area of triangle ABD with $A(2, 4)$, $B(0, 4)$, $D(2, -4)$

$= \frac{1}{2} [2 \times 4 + 0 \times -4 + 2 \times 4 - 4 \times 0 - 4 \times 2 - (-4) \times 2]$

$= \frac{1}{2} [8 + 8 - 8 + 8] = 8$ square units.

Area of triangle BCD with $B(0, 4)$, $C(0, -4)$, $D(2, -4)$ will be

$= \frac{1}{2} [0 \times -4 + 0 \times -4 + 2 \times 4 - 4 \times 0 - 4 \times 2 - 2 \times 0]$

$= \frac{1}{2} [8 + 8] = 8$ square units.

$\Rightarrow$ Total area = $8 + 8 = 16$ sq. units.

50. (d)

51. (d)

52. (c)

53. (a) See theory of this chapter.

Point of intersection of $y = mx + c$ with x -axis is $(-c/m, 0)$.

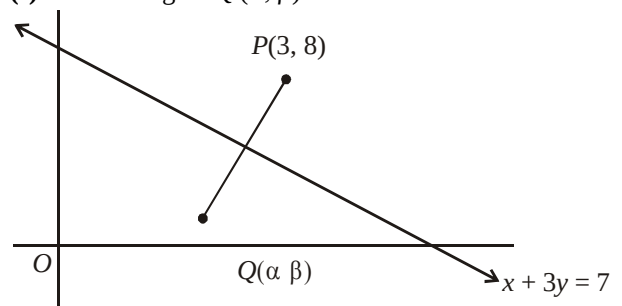
Now use the formula for the distance of a point to a straight line.

54. (c) The equilateral Δ has its sides equal.

Hence the distance between the vertices should be equal.

$a \sqrt{2^2 + 2^2} = \sqrt{(\sqrt{3} + 1)^2 + k(k-1)^2} \Rightarrow k = \sqrt{3}$

55. (c) If the image is $Q(\alpha, \beta)$ then



slope of $PQ \times \left(-\frac{1}{3}\right) = -1 \Rightarrow \frac{\beta-8}{\alpha-3} \times \left(-\frac{1}{3}\right) = -1$

$$\begin{aligned}\beta - 8 &= 3\alpha - 9 \\ \Rightarrow 2\alpha - \beta &= 1 \quad \dots(1) \\ \text{Mid point of } PQ \text{ lies on } x + 3y &= 7\end{aligned}$$

$$\text{then } \left(\frac{\alpha+3}{2}\right) + 3\left(\frac{\beta+8}{2}\right) = 7 \Rightarrow \alpha + 3 + 3\beta + 24 = 14$$

$$\therefore \alpha + 3\beta = -13 \quad \dots\dots(2)$$

From (1) & (2) we get $\alpha = -1, \beta = -4$

56. (a) Find the three lengths separately

$$AB = 6, BC = \sqrt{3^2 + (3\sqrt{3})^2} = 6,$$

$$AC = \sqrt{3^2 + (3\sqrt{3})^2} = 6$$

Hence, the point are the vertices of equilateral triangle.
57. (a) Since BD divides the triangle into two of equal area, BD is a median and $D \equiv (-1, -2)$.

Slope of $BD = -\frac{1}{2}$. The equation of the required line is
 $y + 3 = 2(x - 1)$ i.e., $y - 2x + 5 = 0$.

58. (c) $-\frac{3(1)+4(2)-7}{3(-2)+4(1)-7} = -\frac{4}{-9} = \frac{4}{9}$

59. (a) Let the vertices of the ΔABC be $A(-3,0), B(3,0)$ and $C(0,k)$.

Given, area is 9

$$\Rightarrow 9 = \frac{1}{2} \{-3(-k) + 1(3k)\}$$

$$\Rightarrow 18 = 3k + 3k$$

$$\Rightarrow k = \frac{18}{6} = 3$$

60. (d) Since line $mx + ny = 1$ passes through $(1, 2)$ and $(2, 1)$ therefore they satisfied the equation.

$$\Rightarrow m + 2n = 1 \quad \dots(i)$$

$$\text{and } 2m + n = 1 \quad \dots(ii)$$

From eqs. (i) and (ii), we get $m = n = \frac{1}{3}$

61. (c) Let $P(x, y)$ be the point of division that divides the line joining $(3, -5)$ and $(-4, 7)$ in the ratio of $k : 1$

$$\text{Now, } y = \frac{7k - 5}{k + 1} \quad \dots(i)$$

Since, P lies on $y = 0$ or x -axis then, from eq. (i)

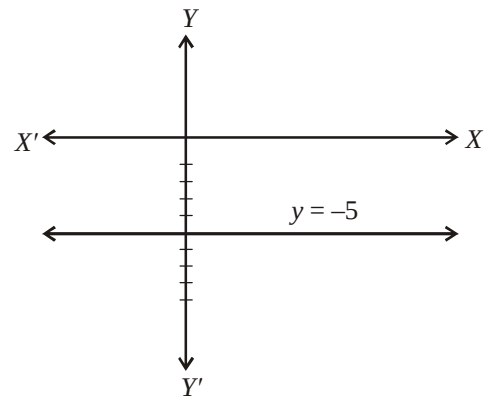
$$0 = \frac{7k - 5}{k + 1} \Rightarrow 7k = 5 \Rightarrow k = \frac{5}{7}$$

62. (c) For infinite solution $\frac{3}{9} = \frac{-1}{-k} = \frac{8}{24}$

$$\Rightarrow \frac{1}{3} = \frac{1}{k} \Rightarrow k = 3$$

63. (d) Draw the points and then check with the options.
Alternative: Find out the point of intersection with the help of options and then use the formula for area of triangle.

64. (d) Equation of a line parallel to x -axis at a distance of 5 units below x -axis is $y = -5$



65. (c) Use the formula of distance of a point from the straight line using the options.

Standard Level

1. (a)
2. (a)
3. (c)
4. (d)
5. (a)

6. (a) $\frac{X_1 + X_2}{2} = 2, \frac{X_2 + X_3}{2} = -1, \frac{X_3 + X_1}{2} = 4$

$$\Rightarrow X_1 = 7, X_2 = -3, X_3 = 1$$

Similarly, y_1, y_2, y_3 can be found

7. (b) Let the point be (X, X) , so according to the condition $(X - 1)^2 + (X - 0)^2 = (X - 0)^2 + (X - 3)^2$

$$\Rightarrow 2X + 1 = -6X + 9 \Rightarrow X = 2$$

Hence the point is $(2, 2)$

8. (c) $\frac{2 \times 5 + 1(a)}{2 + 1} = 4 \Rightarrow a = 2$

$$\text{and } \frac{2 \times 7 + 1(b)}{2 + 1} = 6 \Rightarrow b = 4$$

9. (b) The point of intersection will be obtained by simultaneously solving the two equations and then by the distance formula, distance can be found.

10. (d) Take points P one by one and see which one $(0, -1)$ satisfies.

11. (b) $m_1 = -7/3, m_2 = k$
Two lines are parallel if $m_1 = m_2$
 $k = -7/3$

12. (a) Any point A on the first line is $(t, 5t - 4)$. Any point B on the second line is $\left(r, \frac{3r - 4}{4}\right)$.

$$\text{Hence, } 1 = \frac{2r + t}{3} \text{ and } 5 = \frac{\frac{3r - 4}{4} + 5t - 4}{3}$$

$$\Rightarrow 2r + t = 3 \text{ and } 3r + 10t = 42.$$

$$\text{on solving, we get } t = \frac{75}{17}. \text{ Hence } A \text{ is } \left(\frac{75}{17}, \frac{304}{17}\right)$$

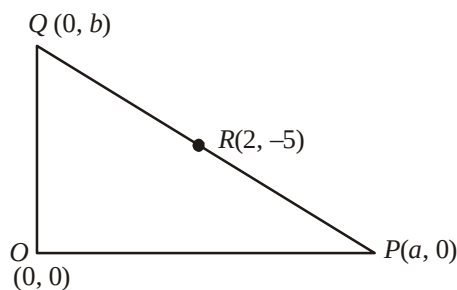
13. (d) Angle ABC is the angle between the lines AB and BC .

$$\text{Now slope of line } AB = m_1 = \frac{3-4}{6-4} = -\frac{1}{2}$$

$$\text{and slope of line } BC = m_2 = \frac{-1-3}{2-3} = \frac{-4}{-4} = 1$$

$$\text{Now, } \tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2} = -3$$

14. (d) Let the line $\frac{x}{a} + \frac{y}{b} = 1$ meet x -axis at $P(a, 0)$ and y -axis at $Q(0, b)$. Since P is mid-point at PQ .



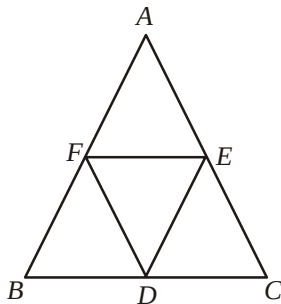
$$\therefore \frac{a+0}{2} = 2, \frac{0+b}{2} = -5$$

$$\therefore a = 4, b = -10$$

$$\therefore P \text{ is } (4, 0), Q \text{ is } (0, -10)$$

15. (d)

16. (a) Area of $\triangle ABC = 4$ area of $\triangle DEF$



where D, E, F are the mid-points of BC, CA, AB respectively.

Reqd. area = $4(2) = 8$ sq. units

17. (c) By the given condition $\frac{7+q+9}{3} = 6$

$$\text{and } \frac{p-6+10}{3} = 3$$

$$\Rightarrow q = 2 \text{ and } p = 5 \therefore p + q = 5 + 2 = 7$$

18. (c) Since the points are collinear, therefore $\Rightarrow (4-a) - 2(2-3) + 1(2a - 12) = 0$
 $\Rightarrow 4 - a + 2 + 2a - 12 = 0$

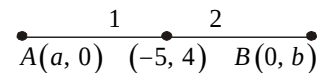
$$\Rightarrow a - 6 = 0$$

$$\Rightarrow a = 6$$

Thus, coordinates of C are $(3, 6)$.

$$\text{Thus, } BC = \sqrt{(3-2)^2 + (6-4)^2} \\ = \sqrt{1+4} = \sqrt{5} \text{ unit}$$

19. (c) Let $A(a, 0)$ and $B(0, b)$ be two points on x -axis and y -axis respectively



Given $(-5, 4)$ divides line AB in the ratio $1 : 2$.

By section formula we have

$$\therefore -5 = \frac{1 \times 0 + 2 \times a}{3}$$

$$\Rightarrow a = \frac{-15}{2} \text{ and } 4 = \frac{1 \times b + 2 \times 0}{3}$$

$$\Rightarrow b = 12$$

$$\text{Thus, } A = \left(\frac{-15}{2}, 0 \right) \text{ and } B = (0, 12)$$

Hence, equation of line joining $\left(\frac{-15}{2}, 0 \right)$ and $(0, 12)$ is

$$(y-0) = \frac{12-0}{0+\frac{15}{2}} \cdot \left(x + \frac{15}{2} \right)$$

$$\Rightarrow y = \frac{2}{5}x + 15$$

$$\Rightarrow 5y = (2x + 60) \Rightarrow 8x - 5y + 60 = 0$$

20. (d) Given equation of the line $\frac{x}{4} + \frac{y}{3} = 1$

can be written as

$$3x + 4y = 12 \Rightarrow 4y = -3x + 12$$

$$\Rightarrow y = \frac{-3}{4}x + \frac{12}{4}$$

The slope of the line $\frac{x}{4} + \frac{y}{3} = 1$ is $\frac{-3}{4}$

$\therefore$ Slope of the line perpendicular to this line $\left(\frac{-1}{3/4} \right) = \frac{4}{3}$

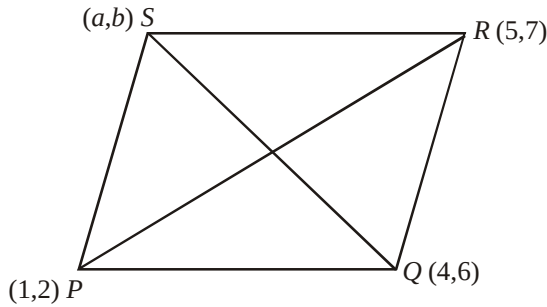
21. (b) Length of the square can be find out using the method of finding out the distance between two parallel lines.

$$\frac{x+8}{2} \quad \frac{2+5}{2}$$

22. (c) Let fourth vertex be (x, y) , then

$$\text{and } \frac{y+4}{2} = \frac{-2+7}{2} \Rightarrow x = -1, y = 1$$

23. (c) Diagonals cut each other at middle points.



$$\text{Hence, } \frac{a+4}{2} = \frac{1+5}{2} \Rightarrow a = 2$$

$$\frac{b+6}{2} = \frac{2+7}{2} \Rightarrow b = 3$$

24. (c) $\tan \theta = \left| \frac{1-\sqrt{3}}{1+\sqrt{3}} \right| \Rightarrow \theta = \tan^{-1} \frac{1-\sqrt{3}}{1+\sqrt{3}} = 15^\circ$

25. (a) Let A, B and C having co-ordinates (a, b), (c, d) and {(a - c), (b - d)} respectively be the points
If these points are collinear then
 $a(d - b + d) + C(b - d - b) + (a - c)(b - d) = 0$
 $\Rightarrow ad - ab - bc + ab = 0$
 $\Rightarrow bc - ad = 0$

26. (c)

27. (b)

28. (c)

29. (c)

30. (c)

31. (a)

32. (c) $A = \left(\frac{3k-5}{k+1}, \frac{5k+1}{k+1} \right)$, Area of $\Delta ABC = 2$ units

$$\Rightarrow \frac{1}{2} \left[\frac{3k-5}{k+1}(5+2) + 1 \left(-2 - \frac{5k+1}{k+1} \right) + 7 \left(\frac{5k+1}{k+1} - 5 \right) \right] = \pm 2$$

$$\Rightarrow 14k - 66 = \pm 4(k+1) \Rightarrow k = 7 \text{ or } 31/9$$

33. (d) We have the lines as $X + 2Y - 3 = 0$
and $X + 2Y + 8 = 0$

$$\text{Now, the distance} = \frac{C_1 - C_2}{\sqrt{a^2 + b^2}} = 11\sqrt{5}$$

34. (c) Let the point be P(2X, X). The choices we are left with are (1, 2) and (2, 4).

$$AP = \sqrt{(3-2X)^2 + (1-X)^2},$$

$$BP = \sqrt{(5-2X)^2 + (3-X)^2}$$

$$AP = BP. \text{ (only } (4, 2) \text{ satisfies)}$$

35. (d) We have the mid-point of diagonal = (1, -1) which should be the mid point of the other two points as well and which is not satisfied by any given alternative.

36. (b) If the y-coordinate bears a constant ratio in the x-coordinate, the equation of the line is $y = mx$ and it passes through the origin.

37. (a) $y - (-4) = \tan 45^\circ (x - 3)$
 $\Rightarrow y + 4 = x - 3 \Rightarrow x - y - 7 = 0$

38. (d) $m_{AC} = \frac{6-2}{5-3} = 2 \Rightarrow m_{BD} = -\frac{1}{2}$

Thus equation of BD is

$$(y - 6) = -\frac{1}{2}(x - 5) \text{ i.e., } 2y + x - 17 = 0.$$

39. (b) Given $a + b = 14 \Rightarrow a = 14 - b$

$$\text{Hence, the equation of straight line is } \frac{x}{14-b} + \frac{y}{b} = 1$$

$$\text{Also it passes through } (3, 4) \Rightarrow \frac{3}{14-b} + \frac{4}{b} = 1$$

$$\Rightarrow b = 8 \text{ or } 7$$

$$\text{Therefore equations are } 4x + 3y = 24 \text{ and } x + y = 7$$

40. (a) Given equation of straight lines are $x - 3y - 2 = 0$ and $2x - 6y - 6 = 0$

$$\text{Here, } a_1 = 1, a_2 = 2, b_1 = -3, b_2 = -6, \\ c_1 = -2, c_2 = -6$$

$$\text{Now, } \frac{a_1}{a_2} = \frac{1}{2}, \frac{b_1}{b_2} = \frac{1}{2}, \frac{c_1}{c_2} = \frac{1}{3}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

$\therefore$ Both straight lines never intersect.

41. (c) Since, (a, 0), (0, b) and (1, 1) are collinear.

$$a(b - 1) + 1(0 - b) = 0$$

$$\Rightarrow ab - a - b = 0$$

$$\Rightarrow a + b - ab = 0$$

42. (d) The equation of the required line is,

$$y = mx + c \quad \dots (1)$$

$$\text{where } m = \tan 45^\circ = 1$$

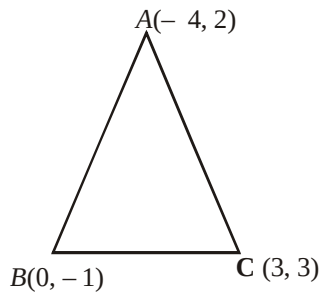
$$\text{and } c = y - \text{intercept} = 101 \text{ units}$$

$$\therefore \text{ from (1)}$$

$$y = x + 101 \Rightarrow x - y + 101 = 0$$

43. (d) Use the formula of area of a triangle which will lead to a quadratic equation. Now solve the quadratic equation to see the number of integral solutions it can have.

44. (b) By using distance formula,



$$\text{We have, } AB = \sqrt{(0+4)^2 + (-1-2)^2} = \sqrt{16+9} = 5$$

$$BC = \sqrt{9+16} = 5$$

$$CA = \sqrt{49+(1)^2} = \sqrt{50} = 5\sqrt{2}$$

$$\text{Hence, required perimeter} = AB + BC + CA \\ = 10 + 5\sqrt{2}$$

45. (b)

46. (a)

Expert Level

1. (a)

2. (b) The line passing through $(-3, 4)$ and $(2, -5)$ is

$$y - 4 = \frac{-5-4}{2+3}(x+3) \text{ or } y - 4 = \frac{-9}{5}(x+3)$$

$$\text{or } 5y - 20 = -9x - 27$$

$$\text{or } 9x + 5y = -7$$

Point (2) satisfies above equation.

3. (a) Since the lines are concurrent, so

$$3\ell m n - \ell^3 - m^3 - n^3 = 0$$

$$\Rightarrow (\ell + m + n)(\ell^2 + m^2 + n^2 - \ell m - mn - n\ell) = 0$$

$$\Rightarrow \ell + m + n = 0 \quad [\because \ell^2 + m^2 + n^2 > \ell m + mn + n\ell]$$

4. (c) Let the vertices of the triangle are

$A(X_1, Y_1)$, $B(X_2, Y_2)$ and $C(X_3, Y_3)$, then

$$X_1 + X_2 = 8 \quad \dots\dots\dots (1)$$

$$Y_1 + Y_2 = 10 \quad \dots\dots\dots (2)$$

$$X_2 + X_3 = -4 \quad \dots\dots\dots (3)$$

$$Y_2 + Y_3 = 6 \quad \dots\dots\dots (4)$$

$$X_3 + X_1 = 8 \quad \dots\dots\dots (5)$$

$$Y_3 + Y_1 = -6 \quad \dots\dots\dots (6)$$

On solving these equations, we get

$$X_1 = 10, X_2 = -2, X_3 = -2, Y_1 = -1, Y_2 = 11 \text{ and } Y_3 = -5.$$

Hence, the centroid is $(2, 5/3)$

Examination method : As we know that the centroid of the triangle ABC and that of the triangle formed by joining the middle points of the sides of triangle BC is same.

$$\frac{4+4-2}{3}, \frac{5-3+3}{3} = \left(2, \frac{5}{3}\right)$$

5. (b) Given equation can be written as

$$y = \sqrt{3}x - 1 \text{ on comparing with } y = mx + c$$

$$\text{We get } \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ$$

6. (c) Find the point of intersection of the lines and then put the coordinate of this point into the equation $4x + 3y = K$, which is perpendicular to the equation of straight line $3x - 4y = 5$, which is perpendicular to the equation of straight line $3x - 4y = 5$, to find out K .

7. (a) Find the point of intersection of the lines by solving the simultaneous equations and then use the two point formula of a straight line.

Alternative: After finding out the point of intersection, use options to check.

8. (d)

9. (a)

10. (c) Line parallel to $4X + 3Y - 6 = 0$ will be $4X + 3Y + K = 0$

$$\text{Put } (2, 3) \Rightarrow 8 + 9 + K = 0 \text{ or } K = -17,$$

$$\text{hence line is } 4X + 3Y - 17 = 0$$

$$\text{Now } (4, P) \text{ lies on this line so, } 4 \times 4 + 3 \times P - 17 = 0$$

$$\text{or } 3P = 1 \text{ or } P = 1/3.$$

11. (c) If AD the altitude from A ,

$$AD = \frac{|3+4-5|}{\sqrt{2}} = \sqrt{2}$$

$$\therefore \text{side of the } \Delta = AD \operatorname{cosec} 60^\circ = \sqrt{2} \times \frac{2}{\sqrt{3}} = \frac{2\sqrt{2}}{\sqrt{3}}$$

12. (d) The three length AB, BC, AC will be

$$AB = \sqrt{[(-1+4)^2 + (4+2)^2]} = \sqrt{45}$$

$$BC = \sqrt{[(-4-3)^2 + (-2+2)^2]} = \sqrt{7^2 + 2^2} = \sqrt{53}$$

$$AC = \sqrt{4^2 + 8^2} = \sqrt{80}$$

$$\text{Perimeter} = AB + BC + AC$$

13. (a)

14. (b) Given that mid point of $A(1, 2)$ and $B(x, y)$ is $C(2, 4)$,

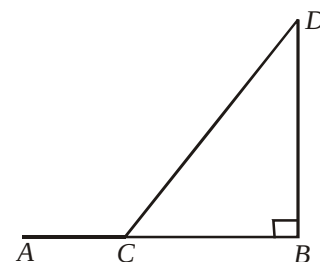
$$\frac{1+x}{2} = 2 \text{ and } \frac{2+y}{2} = 4$$

$$\Rightarrow x = 3 \text{ and } y = 6$$

This given coordinates of B are $(3, 6)$.

Given that

$$BD \perp AB \text{ and } CD = 3 \text{ unit}$$



$$BC = \sqrt{(2-3)^2 + (4-6)^2} = \sqrt{1+4} = \sqrt{5}$$

$$\text{In right angled } \Delta BCD, CD^2 = BC^2 + BD^2$$

$$\Rightarrow 9 = 5 + BD^2 \Rightarrow BD^2 = 4 \Rightarrow BD = 2 \text{ unit}$$

15. (d) For 3 points to be collinear,
(i) Either the slope of any two of the 3 points should be equal to the slope of any other two points. Or
(ii) The area of the triangle formed by the three points should be equal to zero.

16. (d) Use the options. $(-3, 1)$ and $(1, 1)$ satisfy the all lines
 $\therefore$ (d) is incorrect.

17. (b) Median will be the line through A and mid-point of BC i.e., $(1, 0)$.
Hence, equation through $(2, 5)$, $(1, 0)$ will be

$$Y - 5 = \frac{(0 - 5)}{(1 - 2)} \times (X - 2) \text{ or } 5X - Y - 5 = 0$$

18. (b) Slope of $AB = 1$ and hence AB is inclined at 45° to the x -axis. $AB = 2\sqrt{2}$
 $\therefore$ diagonal $AC = 4$ and AC is inclined at 90° to the x -axis. Hence, the centre of the square is at E , 2 units above A and its coordinates are $(3, -2)$. A second square possible is ABC_1D_1 on the side of AB opposite to CD but $(2, -2)$ will not be an interior point.

19. (d) As given :
Coordinates of P and Q are $(-3, 4)$ and $(2, 1)$ respectively.
Let coordinates of R be (x, y) .
As given : $PR = 2 QR$
 $\Rightarrow PR - QR = QR \Rightarrow PQ = QR$
So, Q is the mid point of P and R

$$\Rightarrow 2 = \frac{-3 + x}{2} \text{ and } \frac{4 + y}{2}$$

$$\Rightarrow x = 7 \text{ and } y = -2$$

$$\therefore \text{Coordinates of } R = (7, -2).$$

20. (b) Let point $P(x_1, y_1)$ be equidistant from point $A(1, 2)$ and $B(3, 4)$.

$$\therefore PA = PB$$

$$\Rightarrow PA^2 = PB^2$$

$$\Rightarrow (1 - x_1)^2 + (2 - y_1)^2 = (3 - x_1)^2 + (4 - y_1)^2$$

$$\Rightarrow 1 + x_1^2 - 2x_1 + 4 + y_1^2 - 4y_1$$

$$9 + x_1^2 - 6x_1 + 16 + y_1^2 - 8y_1$$

$$\Rightarrow x_1 + y_1 = 5 \quad \dots(1)$$

$$\text{As } P(x_1, y_1) \text{ lies on } 2x - 3y = 5$$

$$\therefore 2x_1 - 3y_1 = 5 \quad \dots(2)$$

On solving Eqs. (1) and (2), we get

$$x_1 = 4 \text{ and } y_1 = 1$$

$$\therefore \text{Coordinates of } P \text{ are } (4, 1).$$

21. (c) Since, the straight lines $x - 2y = 0$ and $kx + y = 1$ intersect at the point $\left(1, \frac{1}{2}\right)$.

$$\therefore \text{The point } \left(1, \frac{1}{2}\right) \text{ satisfies the equation } kx + y = 1$$

$$\therefore \text{Put } x = 1, \text{ and } y = \frac{1}{2} \text{ in eqn } kx + y = 1,$$

$$\text{we get } k.1 + \frac{1}{2} = 1 \Rightarrow k = \frac{1}{2}$$

22. (d) Let $P(x, y)$ be a point and $A = (a, 0)$, $B = (-a, 0)$.

$$\text{Now, } PA^2 = (x - a)^2 + y^2$$

$$PB^2 = (x + a)^2 + y^2$$

Since the sum of the distances of the point $P(x, y)$ from the points $A(a, 0)$ and $B(-a, 0)$ is $2b^2$.

$$\therefore PA^2 + PB^2 = 2b^2$$

$$(x - a)^2 + (y - 0)^2 + (x + a)^2 + (y - 0)^2 = 2b^2$$

$$\Rightarrow x^2 + a^2 - 2ax + y^2 + x^2 + a^2 + 2ax + y^2 = 2b^2$$

$$\Rightarrow x^2 + a^2 + y^2 = b^2$$

$$\Rightarrow x^2 + a^2 = b^2 - y^2$$

23. (a) Given equation of lines are

$$x + 2y - 9 = 0 \Rightarrow 2y = -x + 9$$

$$\Rightarrow y = -\frac{1}{2}x + \frac{9}{2} \quad \dots(1)$$

$$\text{and } kx + 4y + 5 = 0 \Rightarrow 4y = -kx - 5$$

$$\Rightarrow y = -\frac{k}{4}x - \frac{5}{4} \quad \dots(2)$$

Since line (1) and line (2) are parallel therefore their slopes are equal.

$$\frac{-1}{2} = \frac{-k}{4} \Rightarrow k = 2$$

24. (b) Let $A(2, 6)$; $B(3, 4)$; $C(4, 5)$ and $D(-2, 5)$ are the given points. Let O be the origin, i.e., $O(0, 0)$

$$OA = \sqrt{(2 - 0)^2 + (6 - 0)^2} = \sqrt{40} = 2\sqrt{10} \text{ units}$$

$$OB = \sqrt{(3 - 0)^2 + (4 - 0)^2} = \sqrt{9 + 16} = 5 \text{ units}$$

$$OC = \sqrt{(4 - 0)^2 + (5 - 0)^2} = \sqrt{16 + 25} = \sqrt{41} \text{ units}$$

$$OD = \sqrt{(-2 - 0)^2 + (5 - 0)^2} = \sqrt{4 + 25} = \sqrt{29} \text{ units}$$

So, $q = OB = 5$ units is the correct answer.

25. (b)

26. (a)

27. (c) Use the formula (perpendicular distance of a point from a straight line).

28. (c)

Explanation of Test Yourself

1. (b) Let ABC be the triangle whose sides BC , CA and AB have the equations $y = 15$, $3x = 4y$, and $5x + 12y = 0$ respectively.

Solving these equations pairwise, we get coordinates of A , B and C as $(0, 0)$, $(-36, 15)$ and $(20, 15)$ respectively. AD is a line passing through $A(0, 0)$ and perpendicular to $y = 15$.

So, equation of AD is $x = 0$.

The equation of any line perpendicular to $3x - 4y = 0$ is represented by $4x + 3y + k = 0$.

This line will pass through $(-36, 15)$ if $-144 + 45 + k = 0 \Rightarrow k = 99$.

So the equation of BE is $4x + 3y + 99 = 0$.

Solving the equations of AD and BE we get $x = 0, y = -33$. Hence, the coordinates of the orthocentre are $(0, -33)$.

2. (c) Let the required ratio be $k : 1$

$$\text{Then, } 2 = \frac{6k - 4(1)}{k + 1} \text{ or } k = \frac{3}{2}$$

$$\therefore \text{ The required ratio is } \frac{3}{2} \text{ or } 3 : 2$$

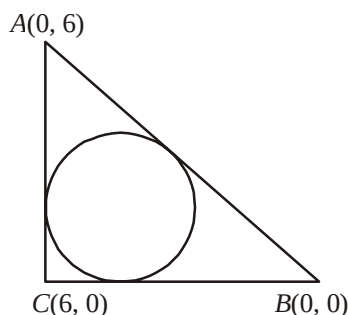
$$\text{Also, } y = \frac{3(3) + 2(3)}{3 + 2} = 3$$

3. (d) $\frac{5x}{-15} + \frac{3y}{15} = 1 \Rightarrow \frac{x}{-3} + \frac{y}{5} = 1$

$$\therefore \text{ area of } \Delta = \frac{1}{2} \times 3 \times 5 = \frac{15}{2}$$

4. (b) Obviously, the length of the two sides AB and BC of the triangle is 6 units and the length of the third side is $(6^2 + 6^2)^{1/2}$.

$$\text{Hence, } a = c = 6, b = 6\sqrt{2}$$



$$\begin{aligned} & \frac{(6.0 + 6\sqrt{2}.0 + 6.6)}{(6 + 6 + 6\sqrt{2})}, \frac{(6.6 + 6\sqrt{2}.0 + 6.0)}{(6 + 6 + 6\sqrt{2})} \\ &= \frac{36}{12 + 6\sqrt{2}}, \frac{36}{12 + 6\sqrt{2}} \Rightarrow (2 - \sqrt{2}), 3(2 - \sqrt{2}) \end{aligned}$$

5. (a) To find out the point of intersection, we just need to solve the simultaneous equations.

$$2x + 3y = 5 \quad \dots\dots(1)$$

$$3x - 4y = 10 \quad \dots\dots(2)$$

Multiplying equation (1) by 3 and equation (2) by 2.

$2x + 3y = 5$ becomes $6x + 9y = 15$ (equation 3) And

$3x - 4y = 10$ becomes $6x - 8y = 20$ (equation 4).

Doing equation (3) - equation (4) given us:

$$\begin{array}{r} 6x + 9y = 15 \\ 6x - 8y = 20 \\ \hline 17y = -5 \end{array}$$

Hence, $y = -5/17$. This gives us $x = 50/17$

$$\text{So point of intersection} = \left(\frac{50}{17}, \frac{-5}{17} \right)$$

6. (c) Let the intercepts made by the line x -axis and y -axis be α and $(14 - \alpha)$ respectively.

$$\text{Then, its equation is } \frac{X}{\alpha} + \frac{Y}{14 - \alpha} = 1 \quad \dots(1)$$

Since it passes through $(3, 4)$, we have:

$$\frac{3}{\alpha} + \frac{4}{14 - \alpha} = 1 \Rightarrow \alpha^2 - 13\alpha + 42 = 0 \Rightarrow (\alpha - 6)$$

$$(\alpha - 7) = 0.$$

$$\alpha = 6 \text{ and } \alpha = 7.$$

$$\text{So, the required equation is: } \frac{x}{6} + \frac{y}{8} = 1 \text{ or } \frac{x}{7} + \frac{y}{7} = 1 \text{ i.e.}$$

$$4x + 3y = 24 \text{ or } x + y = 7.$$

7. (d) Let the fourth vertex be $D(x, y)$
 $\therefore$ Mid-point of Diagonal BD and mid-point of diagonal AC are equal

$$\therefore \left(\frac{x+3}{2}, \frac{y+4}{2} \right) = \left(\frac{-2+2}{2}, \frac{3-1}{2} \right)$$

$$\text{Hence, } D(-3, -2)$$

8. (b) Let the required ratio be $k : 1$

$$\text{Then, } 2 = \frac{6k - 4 \times 1}{k + 1} \Rightarrow k = \frac{3}{2}$$

$$\therefore \text{ The required ratio is } \frac{3}{2} : 1, \text{ i.e., } 3 : 2.$$

$$\text{Also, } y = \frac{3 \times 3 + 2 \times 3}{3 + 2} = 3$$

9. (c) Third side passes through $(1, -10)$ so let its equation be $y + 10 = m(x - 1)$

If it makes equal angle, say θ with given two sides, then

$$\tan \theta = \frac{m - 7}{1 + 7m} = \frac{m - (-1)}{1 + m(-1)} \Rightarrow m = -3 \text{ or } 1/3$$

Hence possible equations of third side are

$$y + 10 = -3(x - 1) \text{ and } y + 10 = \frac{1}{3}(x - 1)$$

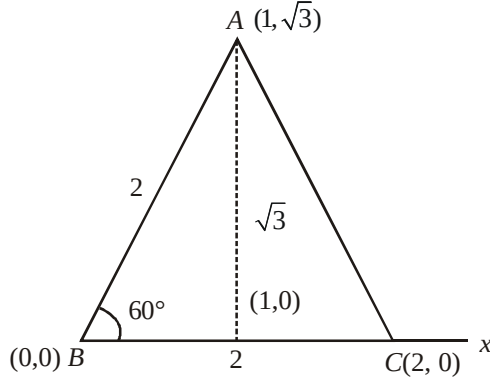
$$\text{or } 3x + y + 7 = 0 \text{ and } x - 3y - 31 = 0$$

10. (a) Locus of P is given by

$$\frac{1}{2} \begin{vmatrix} 1 & x & y \\ 1 & 1 & -2 \\ 1 & -1 & 3 \end{vmatrix} = \pm \frac{1}{2} \begin{vmatrix} 1 & x & y \\ 1 & 2 & -1 \\ 1 & 3 & 1 \end{vmatrix}$$

i.e., $1 - 5x - 2y = \pm (5 - 2x + y)$ giving the two lines $3x + 3y + 4 = 0$ and $7x + y - 6 = 0$.

11. (d) Clearly, the triangle is equilateral.



So, the incentre is the same as the centroid.

$$\therefore \text{Incentre} = \left(\frac{1+0+2}{3}, \frac{\sqrt{3}+0+0}{3} \right) = \left(1, \frac{1}{\sqrt{3}} \right)$$

12. (a) Here slope = 1

And point given is $(2, -3)$.

So, we will use point-slope formula for finding the equation of straight line. This formula is given by:

$$(y - y_1) = m(x - x_1)$$

So, equation of the line will be $y - (-3) = 1(x - 2)$

$$\Rightarrow y + 3 = x - 2$$

$$\Rightarrow y - x + 5 = 0$$

13. (c) Let ABC be the triangle whose sides AB , BC and CA have the equations $3x - y + 3 = 0$, $3x + 4y + 3 = 0$ and $x + 3y + 11 = 0$ respectively.

Solving the equations, we get the points A , B and C as $(-2, -3)$, $(-1, 0)$ and $(7, -6)$ respectively

The equation of a line perpendicular to BC is $4x - 3y + k = 0$. [For students unaware of this formula, read the section on straight lines later in the chapter.]

This will pass through $(3, -3)$, the mid-point of BC , if $12 + 9 + k = 0 \Rightarrow k = -21$

Putting $k_1 = -21$ in $4x - 3y + k = 0$, we get $4x - 3y - 21 = 0$... (1)

as the equation of the perpendicular bisector of BC .

Again, the equation of a line perpendicular to CA is $3x - y + k_1 = 0$.

This will pass through $(5/2, -9/2)$, the mid-point of AC if $5/2 + 9/2 + k_1 = 0 \Rightarrow k_1 = -12$

Putting $k_1 = -12$ in $3x - y + k_1 = 0$, we get

$$3x - y - 12 = 0 \quad \dots (2)$$

as the perpendicular bisector of AC .

Solving (1) and (2), we get $x = 3, y = -3$.

Hence, the coordinates of the circumcentre of $\triangle ABC$ are $(3, -3)$.

14. (a) Obviously from right angled triangle BOA

$$OA = OB = a / \sqrt{2}$$

Hence the vertex $(a / \sqrt{2}, 0)$ is not the vertex of square.

15. (c) Equation of the line perpendicular to $3x + 4y = 7$ will be of the form $4x - 3y = K$.

This line passes through $(3, -3)$, so this point will satisfy the equation of straight line $4x - 3y = K$. So, $4(3) - 3(-3) \Rightarrow K = 21$.

Hence equation of required straight line will be $4x - 3y = 21$



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UNIT-V

Counting

Principles

Chapter 21	Permutations and Combinations
Chapter 22	Probability

21

PERMUTATIONS AND COMBINATIONS

- Introduction
- Fundamental Principle of Counting
- Factorials
- Meaning of Permutation and Combination
- Counting Formula for Linear Permutations
- Number of Linear Permutations Under Certain Conditions
- Circular Permutations
- Counting Formula for Combination
- Division and Distribution of Objects
- Dearrangement Theorem
- Important Results about Points
- Finding the Rank of a Word

INTRODUCTION

Permutation is the different arrangement of a given number of things by taking some or all at a time and on the other hand combinations are the selection which can be done by taking some or all from a number of objects.

This chapter is considered a tricky one by many CAT aspirants and hence, a clear understanding of concept is required. It is very important topic for CAT and other MBA entrance exams, usually 2–3 questions are asked in CAT exam either directly or indirectly as application in probability.

FUNDAMENTAL PRINCIPLE OF COUNTING

Multiplication Principle

If an operation can be performed in ‘ m ’ different ways; followed by a second operation performed in ‘ n ’ different ways, then the two operations in succession can be performed in $m \times n$ ways. This can be extended to any finite number of operations.

Illustration 1: A person wants to go from station P to station R via station Q . There are 4 routes from P to Q and 5 routes from Q to R . In how many ways can he travel from P to R ?

Solution: He can go from P to Q in 4 ways and Q to R in 5 ways.

So number of ways of travel from P to R is $4 \times 5 = 20$.

Illustration 2: A college offers 6 courses in the morning and 4 in the evening. Find the possible number of choices with the student if he wants to study one course in the morning and one in the evening.

Solution: The college has 6 courses in the morning out of which the student can select one course in 6 ways.

In the evening the college has 4 courses out of which the student can select one in 4 ways.

Hence the required number of ways = $6 \times 4 = 24$.

Illustration 3: In how many ways can 5 prizes be distributed among 4 boys when every boy can take one or more prizes?

Solution: First prize may be given to any one of the 4 boys, hence first prize can be distributed in 4 ways.

Similarly every one of second, third, fourth and fifth prizes can also be given in 4 ways.

$\therefore$ The number of ways of their distribution
= $4 \times 4 \times 4 \times 4 \times 4 = 4^5 = 1024$

Addition Principle

If an operation can be performed in ‘ m ’ different ways and another operation, which is independent of the first operation, can be performed in ‘ n ’ different ways. Then either of the two operations can be performed in $(m + n)$ ways. This can be extended to any finite number of independent operations.

Illustration 4: A college offers 6 courses in the morning and 4 in the evening. Find the number of ways a student can select exactly one course, either in the morning or in the evening.

Solution: The college has 6 courses in the morning out of which the student can select one course in 6 ways.

In the evening the college has 4 courses out of which the student can select one in 4 ways.

Hence the required number of ways = $6 + 4 = 10$.

Illustration 5: A person wants to leave station Q . There are 4 routes from station Q to P and 5 routes from Q to R . In how many ways can he travel from the station Q ?

Solution: He can go from Q to P in 4 ways and Q to R in 5 ways. To go from Q to P and Q to R are independent to each other. Hence the person can leave station Q in $4 + 5 = 9$ ways.

FACTORIALS

If n is a natural number then the product of all natural numbers upto n is called factorial n and it is denoted by $n!$ or $\underline{n}$

Thus, $n! = n(n-1)(n-2) \dots 3.2.1$

Note that $0! = 1 = 1!$

$$n! = n(n-1)!$$

$$= n(n-1)(n-2)!$$

$$= n(n-1)(n-2)(n-3)!, \text{ etc.}$$

For example $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1$

But $4! = 4 \times 3 \times 2 \times 1$

$$\therefore 6! = 6 \times 5 \times 4! \text{ or } 6 \times 5 \times 4 \times 3!$$

Remember that

$$0! = 1, 1! = 1, 2! = 2, 3! = 6, 4! = 24, 5! = 120, 6! = 720, \text{ etc.}$$

MEANING OF PERMUTATION AND COMBINATION

Each of the different arrangements which can be made by taking some or all of a number of things is called a permutation. Note that in an arrangement, the order in which the things arranged is considerable i.e., arrangement AB and BA of two letters A and B are different because in AB , A is at the first place and B is at the second place from left whereas in BA , B is at the first place and A is at the second place.

The all different arrangements of three letters A , B and C are ABC , ACB , BCA , BAC , CAB and CBA .

Here each of the different arrangements ABC , ACB , BCA , BAC , CAB and CBA is a permutation and number of different arrangement i.e. 6 is the number of permutations.

ABC , ACB , BCA , BAC , CAB and CBA are different arrangements of three letters A , B and C , because in each arrangement, order in which the letters arranged, is considered. But if the order in which the things are arranged is not considered; then ABC , ACB , BCA , BAC , CAB and CBA are not different but the same. Similarly AB and BA are not different but the same.

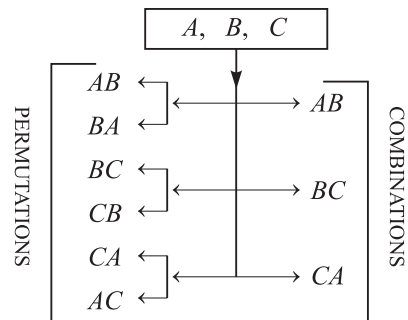
Each of the different selections or groups which can be made by some or all of a number of given things without reference to the order of things in any selection or group is called a combination.

As in selection order in which things are selected is not considered; hence, selections of two letters AB and BA out of three letters A , B and C are the same. Similarly selections of BC and CB are the same.

Also selections of CA and AC are the same.

Hence selection of two letters out of the three letters A , B and C can be made as AB , BC and CA only.

As in arrangements, order in which things are arranged is considered. Hence all arrangements of two letters out of the three letters A , B and C are AB , BA , BC , CB , CA and AC .



Number of permutations (or arrangements) of two letters out of three letters A , B and $C = 6$.

Number of combinations (or groups) of two letters out of three letters A , B and $C = 3$.

Permutations of three different letters A , B and C taken two at a time is also understood as selections of any two different letters AB , BC or CA out of A , B and C , then the selected two letters arranged in two ways as

AB, BA ; BC, CB or CA, AC

Hence using multiplication principle, number of permutations of three different letters A , B and C taken two at a time

$= (\text{Number of ways to select any two different letters out of the three given letters}) \times (\text{Number of arrangements of two selected letters})$

$$= 3 \times 2 = 6$$

Thus permutations means selection of some or all of the given things at a time and then arrangements of selected things. In most of the problems, it is mentioned that the problem is of permutation or combination but in some problems it is not mentioned. In the case where it is not mentioned that problem given is of permutation or combination, you can easily identify the given problem is of permutation or combination using the following classifications of problems:

Problems of Permutations

- (i) Problems based on arrangements
- (ii) Problems based on standing in a line
- (iii) Problems based on seated in a row
- (iv) Problems based on digits
- (v) Problems based on arrangement letters of a word
- (vi) Problems based on rank of a word (in a dictionary)

Problems of Combinations

- (i) Problems based on selections or choose
- (ii) Problems based on groups or committee
- (iii) Problems based on geometry

If in any problem, it is neither mentioned that the problem is of permutation or combination nor does the problem fall in the categories mentioned above for the problems of permutations or problems of combinations, then do you think whether arrangement (i.e. order) is meaningful or not? If arrangement (i.e., order) is considerable in the given problem, then the problem is of permutation otherwise it is of combination. This will be more clear through the following illustrations:

Suppose you have to select three batsmen out of four batsmen B_1, B_2, B_3 and B_4 , you can select three batsmen $B_1 B_2 B_3, B_2 B_3 B_4, B_3 B_4 B_1$ or $B_4 B_1 B_2$.

Here order of selections of three batsmen in any group of three batsmen is not considerable because it does not make any difference in the match.

Hence in the selection process; $B_2 B_3 B_4, B_2 B_4 B_3, B_3 B_2 B_4, B_3 B_4 B_2, B_4 B_2 B_3$ and $B_4 B_3 B_2$ all are the same.

But for batting, the order of batting is important.

Therefore for batting; $B_2 B_3 B_4, B_2 B_4 B_3, B_3 B_2 B_4, B_3 B_4 B_2, B_4 B_2 B_3$ and $B_4 B_3 B_2$, are different because $B_2 B_3 B_4$ means batsman B_2 batting first then batsman B_3 and then batsman B_4 whereas $B_2 B_4 B_3$ means batsman B_2 batting first then batsman B_4 and then batsman B_3 .

COUNTING FORMULA FOR LINEAR PERMUTATIONS

Without Repetition

1. Number of permutations of n different things, taking r at a time is denoted by ${}^n P_r$ or $P(n, r)$, which is given by

$${}^n P_r = \frac{n!}{(n-r)!} \quad (0 \leq r \leq n)$$

$$= n(n-1)(n-2) \dots (n-r+1),$$

where n is a natural number and r is a whole number.

2. Number of arrangements of n different objects taken all at a time is ${}^n P_n = n!$

Note:

$${}^n P_1 = n, \quad {}^n P_r = n \cdot {}^{n-1} P_{r-1}, \quad {}^n P_r = (n-r+1) \cdot {}^n P_{r-1},$$

$${}^n P_n = {}^n P_{n-1}$$

Illustration 6: Find the number of ways in which four persons can sit on six chairs.

Solution: ${}^6 P_4 = 6 \cdot 5 \cdot 4 \cdot 3 = 360$

With Repetition

1. Number of permutations of n things taken all at a time, if out of n things p are alike of one kind, q are alike of second kind, r are alike of a third kind and the rest $n - (p + q + r)$ are all different is

$$\frac{n!}{p!q!r!}$$

2. Number of permutations of n different things taken r at a time when each thing may be repeated any number of times is n^r .

Illustration 7: Find the number of words that can be formed out of the letters of the word COMMITTEE taken all at a time.

Solution: There are 9 letters in the given word in which two T's, two M's and two E's are identical. Hence the required number of

$$\text{words} = \frac{9!}{2!2!2!} = \frac{9!}{(2!)^3} = \frac{9!}{8} = 45360$$

NUMBER OF LINEAR PERMUTATIONS UNDER CERTAIN CONDITIONS

1. Number of permutations of n different things taken all together when r particular things are to be placed at some r given places $= {}^{n-r} P_{n-r} = (n-r)!$
2. Number of permutations of n different things taken r at a time when m particular things are to be placed at m given places $= {}^{n-m} P_{r-m}$
3. Number of permutations of n different things, taken r at a time, when a particular thing is to be always included in each arrangement, is $r \cdot {}^{n-1} P_{r-1}$.
4. Number of permutation of n different things, taken r at a time, when m particular thing is never taken in each arrangement is ${}^{n-m} P_r$.
5. Number of permutations of n different things, taken all at a time, when m specified things always come together is $m! \times (n-m+1)!$
6. Number of permutations of n different things, taken all at a time, when m specific things never come together is $n! - m! \times (n-m+1)!$

Illustration 8: How many different words can be formed with the letters of the word 'JAIPUR' which start with 'A' and end with 'I'?

Solution: After putting A and I at their respective places (only in one way) we shall arrange the remaining 4 different letters at 4 places in $4!$ ways. Hence the required number $= 1 \times 4! = 24$.

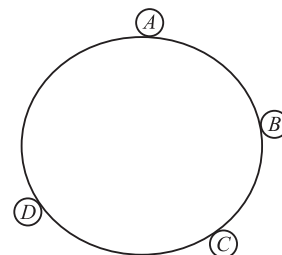
Illustration 9: How many different 3 letter words can be formed with the letters of word 'JAIPUR' when A and I are always to be excluded?

Solution: After leaving A and I , we are remained with 4 different letters which are to be used for forming 3 letters words. Hence the required number $= {}^4 P_3 = 4 \times 3 \times 2 = 24$.

CIRCULAR PERMUTATIONS

1. Arrangement Around a Circular Table

In circular arrangements, there is no concept of starting point (i.e. starting point is not defined). Hence number of circular permutations of n different things taken all at a time is $(n-1)!$ if clockwise and anti-clockwise order are taken as different.



In the case of four persons A, B, C and D sitting around a circular table, then the two arrangements $ABCD$ (in clock-

wise direction) and $ADCB$ (the same order but in anti-clockwise direction) are different.

Hence the number of arrangements (or ways) in which four different persons can sit around a circular table $= (4 - 1)! = 3! = 6$.

2. Arrangement of Beads or Flowers (All Different) Around a Circular Necklace or Garland

The number of circular permutations of n different things taken all at a time is $\frac{(n-1)!}{2}$, if clockwise and anti-clockwise order are taken as the same.

If we consider the circular arrangement, if necklace made of four precious stones A, B, C and D ; the two arrangements $ABCD$ (in clockwise direction) and $ADCB$ (the same but in anti-clockwise direction) are the same because when we take one arrangement $ABCD$ (in clockwise direction) and then turn the necklace around (front to back), then we get the arrangement $ADCB$ (the same but in anti-clockwise direction). Hence the two arrangements will be considered as one arrangement because the order of the stones is not changing with the change in the side of observation. So in this case, there is no difference between the clockwise and anti-clockwise arrangements.

Therefore number of arrangements of four different stones in the necklace $= \frac{(n-1)!}{2}$.

3. Number of Circular Permutations of n Different Things Taken r at a Time

Case I: If clockwise and anti-clockwise orders are taken as different, then the required number of circular permutations

$$= \frac{{}^nP_r}{r}.$$

Case II: If clockwise and anti-clockwise orders are taken as same, then the required number of circular permutations

$$= \frac{{}^nP_r}{2r}.$$

4. Restricted Circular Permutations

When there is a restriction in a circular permutation then first of all we shall perform the restricted part of the operation and then perform the remaining part treating it similar to a linear permutation.

Illustration 10: In how many ways can 5 boys and 5 girls be seated at a round table so that no two girls may be together?

Solution: Leaving one seat vacant between two boys, 5 boys may be seated in $4!$ ways. Then at remaining 5 seats, 5 girls can sit in $5!$ ways. Hence the required number $= 4! \times 5!$

Illustration 11: In how many ways can 4 beads out of 6 different beads be strung into a ring?

Solution: In this case a clockwise and corresponding anticlockwise order will give the same circular permutation. So the required

$$\text{number} = \frac{{}^6P_4}{4.2} = \frac{6.5.4.3}{4.2} = 45.$$

Illustration 12: Find the number of ways in which 10 persons can sit round a circular table so that none of them has the same neighbours in any two arrangements.

Solution: 10 persons can sit round a circular table in $9!$ ways. But here clockwise and anti-clockwise orders will give the same neighbours. Hence the required number of ways $= \frac{1}{2}9!$.

COUNTING FORMULA FOR COMBINATION

1. Selection of Objects Without Repetition

The number of combinations or selections of n different things taken r at a time is denoted by nC_r or $C(n, r)$ or

$$C \left(\begin{matrix} n \\ r \end{matrix} \right)$$

$$\text{where } {}^nC_r = \frac{n!}{r!(n-r)!}; (0 \leq r \leq n)$$

$$= \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 2.1};$$

where n is a natural number and r is a whole number.

Some Important Results

- (i) ${}^nC_n = 1, {}^nC_0 = 1$
- (ii) ${}^nC_r = \frac{{}^nP_r}{r!}$
- (iii) ${}^nC_r = {}^nC_{n-r}$
- (iv) ${}^nC_x = {}^nC_y \Rightarrow x + y = n$
- (v) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
- (vi) ${}^nC_r = \frac{n}{r} \cdot {}^{n-1}C_{r-1}$
- (vii) ${}^nC_1 = {}^nC_{n-1} = n$

Illustration 13: If ${}^{20}C_r = {}^{20}C_{r-10}$, then find the value of ${}^{18}C_r$

Solution: ${}^{20}C_r = {}^{20}C_{r-10} \Rightarrow r + (r-10) = 20 \Rightarrow r = 15$

$$\therefore {}^{18}C_r = {}^{18}C_{15} = {}^{18}C_3 = \frac{18.17.16}{1.2.3} = 816$$

Illustration 14: How many different 4-letter words can be formed with the letters of the word 'JAIPUR' when A and I are always to be included?

Solution: Since A and I are always to be included, so first we select 2 letters from the remaining 4, which can be done in ${}^4C_2 = 6$ ways. Now these 4 letters can be arranged in $4! = 24$ ways, so the required number $= 6 \times 24 = 144$.

Illustration 15: How many combinations of 4 letters can be made of the letters of the word 'JAIPUR'?

Solution: Here 4 things are to be selected out of 6 different things.

$$\text{So the number of combinations} = {}^6C_4 = \frac{6.5.4.3}{4.3.2.1} = 15$$

2. Selection of Objects With Repetition

The total number of selections of r things from n different things when each thing may be repeated any number of times is ${}^{n+r-1}C_r$.

3. Restricted Selection

- (i) Number of combinations of n different things taken r at a time when k particular things always occur is ${}^{n-k}C_{r-k}$.

- (ii) Number of combinations of n different things taken r at a time when k particular things never occur is ${}^{n-k}C_r$.

4. Selection From Distinct Objects

Number of ways of selecting at least one thing from n different things is

$${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1.$$

This can also be stated as the total number of combination of n different things is $2^n - 1$.

Illustration 16: Ramesh has 6 friends. In how many ways can he invite one or more of them at a dinner ?

Solution: He can invite one, two, three, four, five or six friends at the dinner. So total number of ways of his invitation
 $= {}^6C_1 + {}^6C_2 + {}^6C_4 + {}^6C_5 + {}^6C_6 = 2^6 - 1 = 63$

5. Selection From Identical Objects

- The number of combination of n identical things taking r ($r \leq n$) at a time is 1.
- The number of ways of selecting any number r ($0 \leq r \leq n$) of things out of n identical things is $n + 1$.
- The number of ways to select one or more things out of $(p + q + r)$ things; where p are alike of first kind, q are alike of second kind and r are alike of third kind
 $= (p + 1)(q + 1)(r + 1) - 1$.

Illustration 17: There are n different books and p copies of each in a library. Find the number of ways in which one or more than one books can be selected.

Solution: Required number of ways
 $= (p + 1)(p + 1) \dots n \text{ terms} - 1 = (p + 1)^n - 1$

Illustration 18: A bag contains 3 one ₹ coins, 4 five ₹ coins and 5 ten ₹ coins. How many selection of coins can be formed by taking atleast one coin from the bag?

Solution: There are 3 things of first kind, 4 things of second kind and 5 things of third kind, so the total number of selections
 $= (3 + 1)(4 + 1)(5 + 1) - 1 = 119$

6. Selection When Both Identical and Distinct Objects are Present

If out of $(p + q + r + t)$ things; p are alike one kind, q are alike of second kind, r are alike of third kind and t are different, then the total number of combinations is

$$(p + 1)(q + 1)(r + 1)2^t - 1$$

7. Number of ways in which it is possible to make a selection of r things from $m + n + p = N$ things, where m are alike of one kind, n alike of second kind and p alike of third kind taken r at a time is given by coefficient of x^r in the expansion of

$$(1 + x + x^2 + \dots + x^m)(1 + x + x^2 + \dots + x^n)(1 + x + x^2 + \dots + x^p).$$

For example the number of ways in which a selection of four letters can be made from the letters of the word

PROPORTION is given by coefficient of x^4 in
 $(1 + x + x^2 + x^3)(1 + x + x^2)(1 + x + x^2)(1 + x)(1 + x)(1 + x).$

DIVISION AND DISTRIBUTION OF OBJECTS

1. The number of ways in which $(m + n)$ different things can be divided into two groups which contain m and n things respectively is

$${}^{m+n}C_m \cdot {}^nC_n = \frac{(m + n)!}{m!n!}, m \neq n$$

Particular case:

When $m = n$, then total number of ways is

$$\frac{(2m)!}{(m!)^2}, \text{ when order of groups is considered and}$$

$$\frac{(2m)!}{2!(m!)^2}, \text{ when order of groups is not considered.}$$

2. The number of ways in which $(m + n + p)$ different things can be divided into three groups which contain m , n and p things respectively is

$${}^{m+n+p}C_m \cdot {}^{n+p}C_p \cdot {}^nC_p = \frac{(m + n + p)!}{m!n!p!}, m \neq n \neq p$$

Particular case:

When $m = n = p$, then total number of ways is

$$\frac{(3m)!}{(m!)^3}, \text{ when order of groups is considered and}$$

$$\frac{(3m)!}{3!(m!)^3}, \text{ when order of groups is not considered.}$$

3. (i) Total number of ways to divide n identical things among r person is ${}^{n+r-1}C_{r-1}$
 (ii) Also total number of ways to divide n identical things among r persons so that each gets atleast one is ${}^{n-1}C_{r-1}$.

Illustration 19: In how many ways 20 identical mangoes may be divided among 4 persons if each person is to be given at least one mango?

Solution: If each person is to be given at least one mango, then number of ways will be ${}^{20-1}C_{4-1} = {}^{19}C_3 = 969$.

Illustration 20: In how many ways can a pack of 52 cards be divided in 4 sets, three of them having 17 cards each and fourth just one card?

Solution: Since the cards are to be divided into 4 sets, 3 of them having 17 cards each and 4th just one card, so number of ways
 $= \frac{52!}{1!51!} \cdot \frac{51!}{(17!)^3 3!} = \frac{52!}{(17!)^3 3!}.$

DEARRANGEMENT THEOREM

Any change in the given order of the thing is called a Dearrangement.

1. If n items are arranged in a row, then the number of ways in which they can be dearranged so that no one of them occupies the place assigned to it is

$$n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!} \right]$$

2. If n things are arranged at n places then the number of ways to dearrange such that exactly r things remain their original places is

$$\frac{n!}{r!} \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} + \dots + (-1)^{n-r} \frac{1}{(n-r)!} \right]$$

Illustration 21: There are 3 letters and 3 envelopes. Find the number of ways in which all letters are put in the wrong envelopes

Solution: The required number of ways

$$= 3! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} \right] = 3 - 1 = 2$$

Illustration 22: There are 4 balls of different colour and 4 boxes of colours the same as those of the balls. Find the number of ways to put one ball in each box so that only two balls are in boxes with respect to their colour.

Solution: The required number of ways

$$= \frac{4!}{2!} \left[1 - \frac{1}{1!} + \frac{1}{2!} \right] = 4 \times 3 \left[1 - 1 + \frac{1}{2} \right] = 6$$

IMPORTANT RESULTS ABOUT POINTS

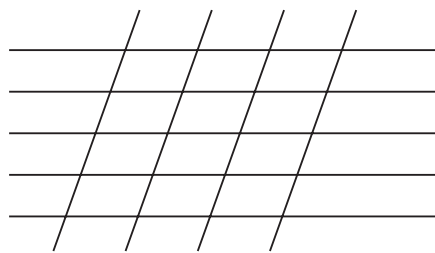
- If there are n points in a plane of which m ($< n$) are collinear, then
 - Total number of different straight lines obtained by joining these n points is ${}^nC_2 - {}^mC_2 + 1$.
 - Total number of different triangles formed by joining these n points is ${}^nC_3 - {}^mC_3$
- Number of diagonals of a polygon of n sides is ${}^nC_2 - n$ i.e., $\frac{n(n-3)}{2}$.
- If m parallel lines in a plane are intersected by a family of other n parallel lines, then total number of parallelograms so formed is ${}^mC_2 \times {}^nC_2$ i.e., $\frac{mn(m-1)(n-1)}{4}$.
- Given n points on the circumference of a circle, then
 - Number of straight lines obtained by joining these n points $= {}^nC_2$
 - Number of triangles obtained by joining these n points $= {}^nC_3$
 - Number of quadrilaterals obtained by joining these n points $= {}^nC_4$

Illustration 23: There are 10 points in a plane and 4 of them are collinear. Find the number of straight lines joining any two of them.

Solution: Total number of lines $= {}^{10}C_2 - {}^4C_2 + 1 = 40$.

Illustration 24: If 5 parallel straight lines are intersected by 4 parallel straight lines, then find the number of parallelograms thus formed.

Solution:



Number of parallelograms $= {}^5C_2 \times {}^4C_2 = 60$.

FINDING THE RANK OF A WORD

We can find the rank of a word out of all the words with or without meaning formed by arranging all the letters of a given word in all possible ways when these words are listed as in a dictionary. You can easily understand the method to find the above mentioned rank by the following illustrations.

Illustration 25: If the letters of the word RACHIT are arranged in all possible ways and these words (with or without meaning) are written as in a dictionary, then find the rank of this word RACHIT.

Solution: The order of the alphabet of RACHIT is A, C, H, I, R, T.

The number of words beginning with A (i.e. the number of words in which A comes at first place) is ${}^5P_5 = 5!$.

Similarly, number of words beginning with C is 5!, beginning with H is 5! and beginning with I is also 5!.

So before R, four letters A, C, H, I can occur in $4 \times (5!) = 480$ ways.

Now the word RACHIT happens to be the first word beginning with R. Therefore the rank of this word RACHIT $= 480 + 1 = 481$.

Illustration 26: The letters of the word MODESTY are written in all possible orders and these words (with or without meaning) are listed as in a dictionary then find the rank of the word MODESTY.

Solution:

The order of the alphabet of MODESTY is D, E, M, O, S, T, Y.

Number of words beginning with D is ${}^6P_6 = 6!$

Number of words beginning with E is ${}^6P_6 = 6!$

Number of words beginning with MD is ${}^5P_5 = 5!$

Number of words beginning with ME is ${}^5P_5 = 5!$

Now the first word start with MO is MODESTY.

Hence rank of the word MODESTY

$$\begin{aligned} &= 6! + 6! + 5! + 5! + 5! + 1 \\ &= 720 + 720 + 120 + 120 + 1 \\ &= 1681. \end{aligned}$$

Practice Exercises



Foundation Level

- The sum of all the four digit even numbers which can be formed by using the digits 0, 1, 2, 3, 4 and 5 if repetition of digits is allowed is
(a) 1765980 (b) 1756980
(c) 1769580 (d) 1759680
- How many words beginning with vowels can be formed with the letters of the word EQUATION?
(a) 25200 (b) 15200
(c) 25300 (d) 35200
- The number of words that can be formed out of the letters of the word COMMITTEE is
(a) $\frac{9!}{(2!)^3}$ (b) $\frac{9!}{(2!)^2}$
(c) $\frac{9!}{2!}$ (d) 9!
- If ${}^{10}P_r = 720$, then r is equal to
(a) 4 (b) 2
(c) 3 (d) 1
- The number of ways of selecting exactly 4 fruits out of 4 apples, 5 mangoes, 6 oranges is
(a) 10 (b) 15
(c) 20 (d) 25
- Number of ways in which 12 different balls can be divided into groups of 5, 4 and 3 balls are
(a) $\frac{12!}{5!4!}$ (b) $\frac{12!}{5!4!3!}$
(c) $\frac{12!}{5!4!3!3!}$ (d) None of these
- How many different letter arrangements can be made from the letter of the word EXTRA in such a way that the vowels are always together?
(a) 48 (b) 60
(c) 40 (d) 30
- In how many ways can a committee of 5 made out 6 men and 4 women containing atleast one woman?
(a) 246 (b) 222
(c) 186 (d) None of these
- How many integers greater than 5000 can be formed with the digit 7, 6, 5, 4 and 3, using each digit at most once?
(a) 72 (b) 144
(c) 84 (d) 192
- Every body in a room shakes hands with every else. If total number of hand-shaken is 66, then number of persons in the room is
(a) 11 (b) 12
(c) 13 (d) 14
- The number of words from the letters of the words BHARAT in which B and H will never come together, is
(a) 360 (b) 240
(c) 120 (d) None of these
- A bag contains 3 black, 4 white and 2 red balls, all the balls being different. The number of at most 6 balls containing balls of all the colours is
(a) $42(4!)$ (b) $2^6 \times 4!$
(c) $(2^6 - 1)(4!)$ (d) None of these
- How many different ways are possible to arrange the letters of the word "MACHINE" so that the vowels may occupy only the odd positions?
(a) 800 (b) 125
(c) 348 (d) 576
- If ${}^nP_r = {}^nP_{r+1}$ and ${}^nC_r = {}^nC_{r-1}$, then the values of n and r are
(a) 4, 3 (b) 3, 2
(c) 4, 2 (d) None of these
- If ${}^nP_r = 720$ nC_r , then r is equal to
(a) 3 (b) 7
(c) 6 (d) 4
- In how many ways a hockey team of eleven can be elected from 16 players?
(a) 4368 (b) 4267
(c) 5368 (d) 4166
- The number of values of r satisfying the equation ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$ is
(a) 1 (b) 2
(c) 3 (d) 4
- The total number of all proper factors of 75600 is
(a) 120 (b) 119
(c) 118 (d) None of these

19. In how many ways can six different rings be worn on four fingers of one hand?
(a) 10 (b) 12
(c) 15 (d) 16
20. Find the number of ways in which 8064 can be resolved as the product of two factors?
(a) 20 (b) 21
(c) 22 (d) 24
21. In how many ways can twelve girls be arranged in a row if two particular girls must occupy the end places?
(a) $\frac{10!}{2!}$ (b) $12!$
(c) $10! \times 2!$ (d) $\frac{12!}{2!}$
22. To fill a number of vacancies, an employer must hire 3 programmers from among 6 applicants, and 2 managers from among 4 applicants. What is the total number of ways in which she can make her selection?
(a) 1,490 (b) 132
(c) 120 (d) 60
23. A father has 2 apples and 3 pears. Each weekday (Monday through Friday) he gives one of the fruits to his daughter. In how many ways can this be done?
(a) 120 (b) 10
(c) 24 (d) 12
24. If a secretary and a joint secretary are to be selected from a committee of 11 members, then in how many ways can they be selected?
(a) 110 (b) 55
(c) 22 (d) 11
25. On a railway route there are 20 stations. What is the number of different tickets required in order that it may be possible to travel from every station to every other station?
(a) 40 (b) 380
(c) 400 (d) 420
26. If $P(32, 6) = kC(32, 6)$, then what is the value of k ?
(a) 6 (b) 32
(c) 120 (d) 720
27. How many times does the digit 3 appear while writing the integers from 1 to 1000?
(a) 269 (b) 308
(c) 300 (d) None of these
28. A person X has four notes of rupee 1, 2, 5 and 10 denomination. The number of different sums of money she can form from them is
(a) 16 (b) 15
(c) 12 (d) 8
29. There are 4 qualifying examinations to enter into Oxford University: RAT, BAT, SAT, and PAT. An Engineer cannot go to Oxford University through BAT or SAT. A CA on the other hand can go to the Oxford University through the RAT, BAT & PAT but not through SAT. Further there are 3 ways to become a CA (viz., Foundation, Inter & Final). Find the ratio of number of ways in which an Engineer can make it to Oxford University to the number of ways a CA can make it to Oxford University.
(a) 3 : 2 (b) 2 : 3
(c) 2 : 9 (d) 9 : 2
30. How many straight lines can be formed from 8 non-collinear points on the X - Y plane?
(a) 28 (b) 56
(c) 18 (d) 19860
31. A man has 3 shirts, 4 trousers and 6 ties. What are the number of ways in which he can dress himself with a combination of all the three?
(a) 13 (b) 72
(c) $13! \cdot 3! \cdot 4! \cdot 6!$ (d) $3! \cdot 4! \cdot 6!$
32. If $({}^{28}C_{2r} : {}^{24}C_{2r-4}) = 225 : 11$. Find the value of r .
(a) 10 (b) 11
(c) 7 (d) 9
33. There is a question paper consisting of 15 questions. Each question has an internal choice of 2 options. In how many ways can a student attempt one or more questions of the given fifteen questions in the paper?
(a) 3^7 (b) 3^8
(c) 3^{15} (d) $3^{15} - 1$
34. How many numbers can be formed with the digits 1, 6, 7, 8, 6, 1 so that the odd digits always occupy the odd places.
(a) 15 (b) 12
(c) 18 (d) 20
35. There are 6 boxes numbered 1, 2, ... 6. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is
(a) 5 (b) 21
(c) 33 (d) 60
36. In how many ways five chocolates can be chosen from an unlimited number of Cadbury, Five-star, and Perk chocolates?
(a) 81 (b) 243
(c) 21 (d) 31
37. There are 20 people among whom two are sisters. Find the number of ways in which we can arrange them around a circle so that there is exactly one person between the two sisters.
(a) $18!$ (b) $2!19!$
(c) $19!$ (d) None of these
38. In a company, each employee gives a gift to every other employee. If the number of gifts is 61, then the number of employees in the company is :
(a) 11 (b) 13
(c) 12 (d) 8

39. In how many ways can Ram choose a vowel and a constant from the letters of the word ALLAHABAD?
(a) 4 (b) 6
(c) 9 (d) 5
40. There are three rooms in a hotel: one single, one double and one for four persons. How many ways are there to house seven persons in these rooms?
(a) $7!/1!2!4!$ (b) $7!$
(c) $7!/3$ (d) $7!/3!$
41. The digits, from 0 to 9 are written on 10 slips of paper (one digit on each slip) and placed in a box. If three of the slips are drawn and arranged, then the number of possible different arrangements is
(a) 1000 (b) 720
(c) 810 (d) None of these
42. The number of ways in which 7 different books can be given to 5 students if each can receive none, one or more books is
(a) 5^7 (b) 7^5
(c) ${}^{11}C_5$ (d) $12!$
43. In how many ways can 13 different alphabets ($a, b, c, \dots, m$) be arranged so that the alphabets f and g never come together?
(a) $13! - 12!$ (b) $13! - 12! / 2!$
(c) $13! - 2 \times 12!$ (d) None of these
44. Number of ways in which the letters of word GARDEN can be arranged with vowels in alphabetical order, is
(a) 360 (b) 240
(c) 120 (d) 480
45. The number of ways in which a mixed double tennis game can be arranged from amongst 9 married couple if no husband and wife plays in the same game is
(a) 756 (b) 3024
(c) 1512 (d) 6048
46. In how many ways can 21 identical white balls and 19 identical black balls be arranged in a row so that no 2 black balls are together?
(a) 1540 (b) 1640
(c) 1240 (d) 1440
47. If 5 parallel straight lines are intersected by 4 parallel straight, then the number of parallelograms thus formed is
(a) 20 (b) 60
(c) 101 (d) 126
48. The number of ways in which a couple can sit around a table with 6 guests if the couple take consecutive seat is
(a) 1440 (b) 720
(c) 5040 (d) None of these
49. How many different words beginning with O and ending with E can be formed with the letters of the word ORDINATE, so that the words are beginning with O and ending with E?
(a) 8! (b) 6!
(c) 7! (d) $7!/2!$
50. How many 6 digit number can be formed from the digits 1, 2, 3, 4, 5, 6 which are divisible by 4 and digits are not repeated?
(a) 192 (b) 122
(c) 140 (d) 242
51. There are 5 candidates in an election and 3 of them are to be elected. A voter can cast any number of votes but not more than three. The number of ways in which he can cast his vote is
(a) 5 (b) 15
(c) 20 (d) 25
52. If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3 : 5$, the possible value of n will be :
(a) 3 (b) 5
(c) 4 (d) 2
53. All possible two factors products are formed from the numbers 1, 2, 3, 4,, 200. The number of factors out of total obtained which are multiples of 5 is
(a) 5040 (b) 7180
(c) 8150 (d) None of these
54. A set of 15 different words are given. In how many ways is it possible to choose a subset of not more than 5 words?
(a) 4944 (b) 4^{15}
(c) 15^4 (d) 4943
55. In an examination, there are 3 multi-choice questions and each question has 4 alternatives. If a student is declared pass only when he attempts all question correctly, then number of ways in which he can fail is
(a) 1 (b) 12
(c) 27 (d) 63
56. Seven nouns, five verbs, and two adjectives are written on a blackboard. We can form a sentence by choosing one word of each type, and we do not care about how much sense the sentence makes. How many ways are there to do this?
(a) $7^2 \times 5^2 \times 2^2$ (b) $7^1 \times 5^1 \times 2^1 \times 3!$
(c) $7! \times 5! \times 2!$ (d) $2^7 \times 2^5 \times 2^2$
57. In how many ways can the eight directors, the vice-chairman and the chairman of a firm be seated at a round-table, if the chairman has to sit between the vice-chairman and the director?
(a) $9! \times 2$ (b) $2 \times 8!$
(c) $2 \times 7!$ (d) None of these
58. How many 4 digit numbers divisible by 5 can be formed with the digits 0, 1, 2, 3, 4, 5, 6 and 6?
(a) 220 (b) 249
(c) 432 (d) 288
59. The number of circles that can be drawn out of 10 points of which 7 are collinear is
(a) 130 (b) 85
(c) 45 (d) Cannot be determined

Standard Level

1. The number of ways of choosing a committee of 2 women and 3 men from 5 women and 6 men, if Mr. A refuses to serve on the committee if Mr. B is a member and Mr. B can only serve, if Miss C is the member of the committee, is
(a) 60 (b) 84
(c) 124 (d) None of these
2. 5 men and 6 women have to be seated in a straight row so that no two women are together. Find the number of ways this can be done.
(a) 48400 (b) 39600
(c) 9900 (d) 86400
3. The total number of ways in which 8 men and 6 women can be arranged in a line so that no 2 women are together is
(a) 48 (b) ${}^8P_8 \cdot {}^9P_6$
(c) $8! (84)$ (d) ${}^8C_8 \cdot {}^9C_6$
4. How many words can be formed with the letters of the word INTERNATIONAL?
(a) 129729600 (b) 129729500
(c) 29729600 (d) 127829600
5. Numbers of ways in which atleast three fruits be selected out of 20 fruits in which 10-mangoes, 5-apples, 2-oranges and rest are different, are
(a) 1583 (b) 1577
(c) 1559 (d) None of these
6. The number of different ways in which 8 persons can stand in a row so that between two particular person A and B there are always two person, is
(a) $60 (5!)$ (b) $15(4!) \times (5!)$
(c) $4! \times 5!$ (d) None of these
7. The total number of eight digit numbers in which all digits are different, is
(a) $9!9$ (b) $9!9/2$
(c) $9!$ (d) None of these
8. In how many ways can the letters of the word "VALEDICTORY" be arranged so that the vowels are never separated?
(a) 883490 (b) 967680
(c) 563680 (d) 483840
9. From 6 boys and 7 girls a committee of 5 is to be formed so as to include atleast one girl. The number of ways this can be done is
(a) ${}^{13}C_4$ (b) ${}^6C_4 \cdot {}^7C_1$
(c) $7 \cdot {}^6C_4$ (d) ${}^{13}C_5 - {}^6C_1$
10. The number of all possible selections of one or more questions from 10 given questions, each question having one alternative is
(a) 3^{10} (b) $2^{10} - 1$
(c) $3^{10} - 1$ (d) 2^{10}
11. The number of ways in which 13 gold coins can be distributed among three persons such that each one gets at least two gold coins is
(a) 36 (b) 24
(c) 42 (d) 6
12. The number of ways of distributing 8 identical balls in 3 distinct boxes so that none of the boxes is empty is
(a) 8C_3 (b) 21
(c) 3^8 (d) 5
13. How many different nine digit numbers can be formed from the number 223355888 by rearranging its digits so that the odd digits occupy even positions?
(a) 16 (b) 36
(c) 60 (d) 180
14. If two dices are tossed simultaneously, the number of elements in the resulting sample space is
(a) 6 (b) 8
(c) 36 (d) 24
15. A boy has 3 library cards and 8 books of his interest in the library. Of these 8, he does not want to borrow Chemistry part II unless Chemistry part I is also borrowed. In how many ways can he choose the three books to be borrowed?
(a) 56 (b) 27
(c) 26 (d) 41
16. By stringing together 9 different coloured beads, how many different bracelets can be made?
(a) 20160 (b) 40320
(c) 80640 (d) 10080
17. How many ways are there to place a set of chess pieces on the first row of chessboard. The set consists of a king, a queen, two identical rooks, knights & bishops?
(a) $8!$ (b) 8^8
(c) 5040 (d) 4280
18. In how many ways can 7 persons stand in the form of a ring?
(a) $P(7, 2)$ (b) $7!$
(c) $6!$ (d) $\frac{7!}{2}$
19. What is $\frac{(n+2)! + (n+1)(n-1)!}{(n+1)(n-1)!}$ equal to?
(a) 1
(b) Always an odd integer
(c) A perfect square
(d) None of the above
20. In a football championship 153 matches were played. Every team played one match with each other team. How many teams participated in the championship?
(a) 21 (b) 18
(c) 17 (d) 15
21. If $P(77, 31) = x$ and $C(77, 31) = y$, then which one of the following is correct?
(a) $x = y$ (b) $2x = y$
(c) $77x = 31y$ (d) $x > y$

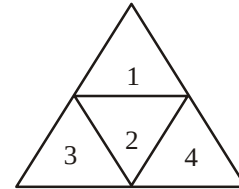
22. 6 equidistant vertical lines are drawn on a board. 6 equidistant horizontal lines are also drawn on the board cutting the 6 vertical lines, and the distance between any two consecutive horizontal lines is equal to that between any two consecutive vertical lines. What is the maximum number of squares thus formed?
(a) 37 (b) 55
(c) 91 (d) 225
23. In how many ways can 12 papers be arranged if the best and the worst paper never come together?
(a) $12!/2!$ (b) $12! - 11!$
(c) $(12! - 11!)/2$ (d) $12! - 2 \cdot 11!$
24. In the Suniti building in Mumbai there are 12 floors plus the ground floor. 9 people get into the lift of the building on the ground floor. The lift does not stop on the first floor. If 2, 3 and 4 people alight from the lift on its upward journey, then in how many ways can they do so?
(Assume they alight on different floors.)
(a) ${}^{11}C_3 \times {}^3P_3$ (b) ${}^{11}P_3 \times {}^9C_4 \times {}^5C_3$
(c) ${}^{10}P_3 \times {}^9C_4 \times {}^5C_3$ (d) ${}^{12}C_3$
25. In how many ways can a selection be made of 5 letters out of 5As, 4Bs, 3Cs, 2Ds and 1E?
(a) 70 (b) 71
(c) ${}^{15}C_5$ (d) None of these
26. If a team of four persons is to be selected from 8 males and 8 females, then in how many ways can the selections be made to include at least one male.
(a) 1550 (b) 1675
(c) 1725 (d) 1750
27. Letters of the word DIRECTOR are arranged in such a way that all the vowels come together. Find out the total number of ways for making such arrangement.
(a) 4320 (b) 2720
(c) 2160 (d) 1120
28. 4 boys and 2 girls are to be seated in a row in such a way that the two girls are always together. In how many different ways can they be seated?
(a) 1200 (b) 7200
(c) 148 (d) 240
29. In how many ways can 7 Englishmen and 7 Americans sit down at a round table, no 2 Americans being in consecutive positions?
(a) 3628800 (b) 2628800
(c) 3628000 (d) 3328800
30. In a jet there are 3 seats in front and 3 in the back. Number of different ways can six persons of different heights be seated in the jeep, so that every one in front is shorter than the person directly behind is
(a) 90 (b) 60
(c) 54 (d) 15
31. The total number of integral solutions for (x, y, z) such that $xyz = 24$ is
(a) 36 (b) 90
(c) 120 (d) None of these
32. A shopkeeper has 10 copies of each of nine different books, then number of ways in which atleast one book can be selected is
(a) $9^{11} - 1$ (b) $10^{10} - 1$
(c) $11^9 - 1$ (d) 10^9
33. How many numbers greater than one million can be formed with 2, 3, 0, 3, 4, 2, 3? (repetitions not allowed)
(a) 720 (b) 360
(c) 120 (d) 240
34. 5 Indian and 5 American couples meet at a party & shake hands. If no wife shakes hands with her husband and no Indian wife shakes hands with a male, then the number of hand shakes that takes place in the party is
(a) 95 (b) 110
(c) 135 (d) 150
35. In a college examination, a candidate is required to answer 6 out of 10 question which are divided into two section each containing 5 questions. Further the candidates is not permitted to attempt more than 4 questions from either of the section. The number of ways in which he can make up a choice of 6 question is
(a) 200 (b) 150
(c) 100 (d) 50
36. The total number of ways in which letters of the word ACCOST can be arranged so that the two C's never come together will be
(a) 120 (b) 360
(c) 240 (d) $6! - 2!$
37. In how many ways can a term of 11 cricketers be chosen from 6 bowlers, 4 wicket keepers and 11 batsmen to give a majority of bastemen if at least 4 bowlers are to be included and there is one wicket keeper?
(a) 27730 (b) 27720
(c) 17720 (d) 26720
38. There are 4 boys and 4 girls. In how many ways can they be seated ina row so that all the girls do not sit together?
(a) 17440 (b) 37440
(c) 37340 (d) 37450
39. Rajdhani express going from Bombay to Delhi stops at 5 intermediate stations. 10 passengers enter the train during the journey with ten different ticket of two classes. The number of different sets of tickets they may have is
(a) ${}^{15}C_{10}$ (b) ${}^{20}C_{10}$
(c) ${}^{30}C_{10}$ (d) None of these
40. Find the minimum possible number of boxes that Rahul must have, given that the total number of coins in all the boxes put together is between 3235 and 3256 (both values inclusive).
(a) 110 (b) 111
(c) 117 (d) 118
41. Three dice are rolled. The number of possible outcomes in which at least one die shows 5 is
(a) 215 (b) 36
(c) 125 (d) 91

42. The sides AB , BC , CA of a triangle ABC have 3, 4 and 5 interior points respectively on them. The total number of triangles that can be constructed by using these points as vertices is
(a) 220 (b) 204
(c) 205 (d) 195
43. If all permutations of the letters of the word $AGAIN$ are arranged as in dictionary, then fiftieth word is
(a) $NAAGI$ (b) $NAGAI$
(c) $NAAIG$ (d) $NAIAG$
44. In a chess tournament, where the participants were to play one game with another, two chess players fell ill, having played 3 games each. If the total number of games played is 84, the number of participants at the beginning was
(a) 15 (b) 16
(c) 20 (d) 21
45. All the words that can be formed using alphabets A, H, L, U and R are written as in a dictionary (no alphabet is repeated). Rank of the word $RAHUL$ is
(a) 71 (b) 72
(c) 73 (d) 74
46. In how many different ways can a cube be painted if each face has to be painted either red or blue?
(a) 20 (b) 16
(c) 12 (d) 10
47. How many new words are possible from the letters of the word $PERMUTATION$?
(a) $11!/2!$ (b) $(11!/2!) - 1$
(c) $11! - 1$ (d) None of these
48. There are five boys and three girls who are sitting together to discuss a management problem at a round table. In how many ways can they sit around the table so that no two girls are together?
(a) 1220 (b) 1400
(c) 1420 (d) 1440
49. Seven different objects must be divided among three people. In how many ways can this be done if at least one of them gets exactly 1 object?
(a) 2484 (b) 1218
(c) 729 (d) None of these
50. How many 6-digit numbers have all three digits either all odd or all even?
(a) 31,250 (b) 28,125
(c) 15,625 (d) None of these
51. There are three piles of identical red, blue and green balls and each pile contains at least 10 balls. The number of ways of selecting 10 balls if twice as many red balls as green balls are to be selected, is
(a) 3 (b) 4
(c) 6 (d) 8
52. Out of 10 consonants and four vowels, the number of words that can be formed using six consonants and three vowels is
(a) $^{10}P_6 \times ^6P_3$ (b) $^{10}C_6 \times ^6C_3$
(c) $^{10}C_6 \times ^4C_3 \times 9!$ (d) $^{10}P_6 \times ^4P_3$
53. The number of 5 digit numbers that can be made using the digits 1 and 2 and in which at least one digit is different, is
(a) 30 (b) 31
(c) 32 (d) None of these
54. In how many ways can 10 books on English and 8 books on physics be placed in a row on a shelf so that two books on physics may not be together?
(a) 160 (b) 165
(c) 170 (d) 180
55. In how many ways can 2310 be expressed as a product of 3 factors?
(a) 40 (b) 41
(c) 42 (d) 43
- Directions for Qs. 56–58:**
Different words are formed with the help of letters of the word $SIGNATURE$. Find the number of words in which
56. vowels always occupy even places.
(a) $^4C_4 \times 4! \times 5!$ (b) $^5C_4 \cdot (5!)^2$
(c) $^6C_4 \cdot (4!)^2$ (d) None of these
57. letters S, G and N are always together
(a) $8! \times 2!$ (b) $7! \times 3!$
(c) $6! \times 4!$ (d) None of these
58. no two letters from N, T and R come together
(a) $^7C_4 \times 3! \times 4!$ (b) $^7C_2 \times 2! \times 5!$
(c) $^7C_6 \times 3! \times 6!$ (d) None of these
59. The number of ways in which ten candidates $A_1, A_2, \dots, A_{10}$ can be ranked so that A_1 is always above A_2 is
(a) $\frac{10!}{2}$ (b) $10!$
(c) $9!$ (d) $\frac{8!}{2}$
60. A class photograph has to be taken. The front row consists of 6 girls who are sitting. 20 boys are standing behind. The two corner positions are reserved for the 2 tallest boys. In how many ways can the students be arranged?
(a) $18! \times 1440$ (b) $6! \times 1440$
(c) $18! \times 2! \times 1440$ (d) None of these

61. A, B, C and D are four towns any three of which are non-collinear. Then the number of ways to construct three roads each joining a pair of towns so that the roads do not form a triangle is
- (a) 7 (b) 8
(c) 9 (d) More than 9
62. There are 10 points on a line and 11 points on another line, which are parallel to each other. How many triangles can be drawn taking the vertices on any of the line?
- (a) 1,050 (b) 2,550
(c) 150 (d) 1,045
63. How many motor vehicle registration number plates can be formed with the digits 1, 2, 3, 4, 5 (No digits being repeated) if it is given that registration number can have 1 to 5 digits?
- (a) 100 (b) 120
(c) 325 (d) 205
64. Find the number of 6-digit numbers that can be found using the digits 1, 2, 3, 4, 5, 6 once such that the 6-digit number is divisible by its unit digit. (The unit digit is not 1).
- (a) 620 (b) 456
(c) 520 (d) 528
65. How many different 9-digit numbers can be formed from the number 223355888 by rearranging its digits so that the odd digits occupy even positions?
- (a) 120 (b) $9!(2!)^3 \cdot 3!$
(c) $(4!)(2!)^3 \cdot (3!)$ (d) None of these
66. Boxes numbered 1, 2, 3, 4 and 5 are kept in a row and they are to be filled with either a red or a blue ball, such that no two adjacent boxes can be filled with blue balls. Then how many different arrangements are possible, given that all balls of a given colour are exactly identical in all respects?
- (a) 8 (b) 10
(c) 15 (d) 22
67. Five persons A, B, C, D and E along with their wives are seated around a round table such that no two men are adjacent to each other. The wives are three places away from their husbands. Mrs. C is on the left of Mr. A , Mrs. E is two places to the right of Mrs. B . Then, who is on the right hand side of Mr. A ?
- (a) Mrs. B (b) Mrs. D
(c) Mrs. E (d) Either Mrs. B or Mrs. D
68. N persons stand on the circumference of a circle at distinct points. Each possible pair of persons, not standing next to each other, sings a two-minute song one pair after the other. If the total time taken for singing is 28 minutes, what is N ?

- (a) 5 (b) 7
(c) 9 (d) None of these

69. In the given figure, what is the maximum number of different ways in which 8 identical balls can be placed in the small triangles 1, 2, 3 and 4 such that each triangle contains at least one ball?



- (a) 32 (b) 35
(c) 44 (d) 56
70. There are 5 different Jeffrey Archer books, 3 different Sidney Sheldon books and 6 different John Grisham books. The number of ways in which at least one book can be given away is
- (a) $2^{10} - 1$ (b) $2^{11} - 1$
(c) $2^{12} - 1$ (d) $2^{14} - 1$
71. The number of natural numbers of two or more than two digits in which digits from left to right are in increasing order is
- (a) 127 (b) 128
(c) 502 (d) 512
72. How many natural numbers not more than 4300 can be formed with the digits 0, 1, 2, 3, 4 (if repetitions are allowed)?
- (a) 574 (b) 570
(c) 575 (d) 569
73. The sides of a triangle have 4, 5 and 6 interior points marked on them respectively. The total number of triangles that can be formed using any of these points
- (a) 371 (b) 415
(c) 286 (d) 421
74. Total number of ways in which six '+' and four '-' signs can be arranged in a line such that no two '-' signs occur together, is
- (a) 35 (b) 18
(c) 15 (d) 42
75. Between two junction stations A and B , there are 12 intermediate stations. The number of ways in which a train can be made to stop at 4 of these stations so that no two of these halting stations are consecutive, is
- (a) 8C_4 (b) 9C_4
(c) ${}^{12}C_4 - 4$ (d) None of these

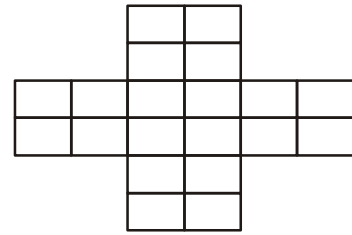
76. In a unique hockey series between India & Pakistan, they decide to play till a team wins 5 matches . The number of ways in which the series can be won by India, if no match ends in a draw is
 (a) 126 (b) 252
 (c) 225 (d) None of those
77. The different letters of the alphabet are given, Out of which five letter words are formed. Then the numbers of words in which at least one letter is repeated is
 (a) 50400 (b) 840
 (c) 30240 (d) 69760
78. With 17 consonants and 5 vowels the number of words of four letters that can be formed having two different vowels in the middle and one consonant, repeated or different at each end is
 (a) 5780 (b) 2890
 (c) 5440 (d) 2720
79. In how many ways can 5 prizes be distributed among 4 boys when every boy can take one or more prizes?
 (a) 1024 (b) 625
 (c) 120 (d) 600
80. A man invites 4 men and 4 women to a party. In how many ways can they sit at a round table so that no two men are together?
 (a) 24 (b) 6
 (c) 144 (d) 120
81. Three dice are rolled. The number of possible outcomes in which at least one die shows 5 is
 (a) 215 (b) 36
 (c) 125 (d) 91
82. There are 10 points in a plane out of which 5 are collinear. The number of triangles that can be drawn will be
 (a) 120 (b) 110
 (c) 100 (d) 78

Expert Level

- How many possible values of n will make ${}^{13}C_n < {}^{13}C_{n+2}$?
(a) 4 (b) 3
(c) 6 (d) 2
 - The sum of 5 digit numbers in which only odd digits occur without any repetition is
(a) 277775 (b) 555550
(c) 1111100 (d) None of these
- Directions for Qs. 3–5 :** Refer to the following information to answer the questions that follow.
A number of 4 different digits is formed by using the digits 1, 2, 3, 4, 5, 6, 7 in all possible ways without repetition.
- How many of them are greater than 3400?
(a) 840 (b) 560
(c) 480 (d) 120
 - How many of them are exactly divisible by 25?
(a) 20 (b) 35
(c) 40 (d) 50
 - How many of them are exactly divisible by 4?
(a) 150 (b) 160
(c) 120 (d) 200
 - A person has 12 friends out of which 7 are relatives. In how many ways can he invite 6 friends such that at least 4 of them are relatives?
(a) 462 (b) 562
(c) 450 (d) 400
 - Messages are conveyed by arranging 4 white, 1 blue and 3 red flags on a pole. Flags of the same colour are alike. If a message is transmitted by the order in which the colours are arranged then the total number of messages that can be transmitted if exactly 6 flags are used is
(a) 45 (b) 65
(c) 125 (d) 185
 - If the letters of the word 'PARKAR' are written down in all possible manner as they are in a dictionary, then the rank of the word 'PARKAR' is
(a) 98 (b) 99
(c) 100 (d) 101
 - The number of integers satisfying the inequality ${}^{n+1}C_3 - {}^{n+1}C_2 \leq 100$ is
(a) nine (b) eight
(c) five (d) None of these
 - The number of triangles whose vertices are at the vertices of an octagon but none of whose sides happen to come from the sides of the octagon is
(a) 24 (b) 52
(c) 48 (d) 16
 - The number of 5 digit numbers of the form $xyzyz$ in which $x < y$ is
(a) 350 (b) 360
(c) 380 (d) 390
 - Three are n points in a plane, no three being collinear except m of them which are collinear. The number of triangles that can be drawn with their vertices at three of the given points is
(a) ${}^{n-m}C_3$ (b) ${}^nC_3 - {}^mC_3$
(c) ${}^nC_3 - m$ (d) None of these
 - The number of arrangements of the letters of the word BANANA is which the two 'N's do not appear adjacently is
(a) 40 (b) 60
(c) 80 (d) 100
 - Number of integers greater than 7000 and divisible by 5 that can be formed using only the digits 3, 6, 7, 8 and 9, no digit being repeated, is
(a) 46 (b) 48
(c) 72 (d) 42
 - There are 10 points in a plane out of which 5 are collinear. The number of straight lines that can be drawn by joining these points will be
(a) 35 (b) 36
(c) 45 (d) 24
 - The streets of a city are arranged like the lines of a chess board. There are m streets running North to South and ' n ' streets running East to West. The number of ways in which a man can travel from NW to SE corner going the shortest possible distance is
(a) $\sqrt{m^2 + n^2}$ (b) $\sqrt{(m-1)^2 + (n-1)^2}$
(c) $\frac{(m+n)!}{m! \cdot n!}$ (d) $\frac{(m+n-2)!}{(m-1)! \cdot (n-1)!}$
 - In a conference 10 speakers are present. If S_1 wants to speak before S_2 and S_2 wants to speak after S_3 , then the number of ways all the 10 speakers can give their speeches with the above restriction if the remaining seven speakers have no objection to speak at any number is
(a) ${}^{10}C_3$ (b) ${}^{10}P_8$
(c) ${}^{10}P_3$ (d) $10!/3$
 - Six persons A, B, C, D, E and F are to be seated at a circular table. The number of ways this can be done if A must have either B or C on his right and B must have either C or D on his right is
(a) 36 (b) 12
(c) 24 (d) 18

19. To fill up 12 vacancies, there are 25 candidates of which 5 are from SC. If 3 of these vacancies are reserved for SC candidates while the remaining are open to all then the number of ways in which the selection can be made is
(a) ${}^5C_3 \times {}^{15}C_9$ (b) ${}^5C_3 \times {}^{22}C_9$
(c) ${}^5C_3 \times {}^{20}C_9$ (d) None of these
20. The number of non negative integral solution of the equation, $x + y + 3z = 33$ is
(a) 120 (b) 135
(c) 210 (d) 520
21. On a plane there are 37 straight lines, of which 13 pass through the point A and 11 pass through the point B. Besides, no three lines pass through one point, no lines passes through both points A and B, and no two are parallel, then the number of intersection points the lines have is equal to
(a) 535 (b) 601
(c) 728 (d) 963
22. How many numbers lying between 3000 and 4000 and which are divisible by 5 can be made with the digits 3, 4, 5, 6, 7 and 8? (Digits are not to be repeated in any number.)
(a) 11 (b) 12
(c) 13 (d) 14
23. In how many ways can n women be seated in a row so that a particular women will not be next to each other?
(a) $(n-2) \times (n-1)!$ (b) $(n-2) \times (n-2)!$
(c) $(n-1) \times (n-1)!$ (d) None of these
24. The number of ways in which n distinct objects can be put into two different boxes so that no box remains empty is
(a) $2^n - 1$ (b) $n^2 - 1$
(c) $2^n - 2$ (d) $n^2 - 2$
25. The number of words of four letters containing equal number of vowels and consonants, repetition being allowed, is
(a) 105^2 (b) 210×243
(c) 105×243 (d) None of these
26. Let S be the set of five-digit numbers formed by the digits 1, 2, 3, 4 and 5, using each digit exactly once such that exactly two odd positions are occupied by odd digits. What is the sum of the digits in the rightmost position of the numbers in S?
(a) 228 (b) 216
(c) 294 (d) 192
27. m distinct animals of a circus have to be placed in m cages, one in each cage. If n ($n < m$) cages are too small to accommodate p ($n < p < m$) animals, then the number of ways of putting the animals into cages are
(a) $({}^{m-n}P_p) {}^{m-p}P_{m-p}$ (b) ${}^{m-n}C_p$
(c) $({}^{m-n}C_p) ({}^{m-p}C_{m-p})$ (d) None of these
28. Two series of a question booklets for an aptitude test are to be given to twelve students. In how many ways can the students be placed in two rows of six each so that there should be no identical series side by side and that the students sitting one behind the other should have the same series?
(a) $2 \times {}^{12}C_6 \times (6!)^2$ (b) $6! \times 6!$
(c) $7! \times 7!$ (d) None of these
29. A, B, C D,X, Y, Z are the players who participated in a tournament. Everyone played with every other player exactly once. A win scores 2 points, a draw scores 1 point and a loss scores 0 points. None of the matches ended in a draw. No two players scored the same score. At the end of the tournament, the ranking list is published which is in accordance with the alphabetical order. Then
(a) M wins over N
(b) N wins over M
(c) M does not play with N
(d) None of these
30. Out of $2n+1$ students, n students have to be given the scholarships. The number of ways in which at least one student can be given the scholarship is 63. What is the number of students receiveing the scholarship?
(a) 5 (b) 7
(c) 3 (d) 9
31. A graph may be defined as a set of points connected by lines called edges. Every edge connects a pair of points. Thus, a triangle is a graph with 3 edges and 3 points. The degree of a point is the number of edges connected to it. For example, a triangle is a graph with three points of degree 2 each. Consider a graph with 12 points. It is possible to reach any point from any other point through a sequence of edges. The number of edges, e, in the graph must satisfy the condition
(a) $11 \leq e \leq 66$ (b) $10 \leq e \leq 66$
(c) $11 \leq e \leq 65$ (d) $0 \leq e \leq 11$
32. There are three coplanar parallel lines. If any p points are taken on each of the lines, then find the maximum number of triangles with the vertices of these points.
(a) $p^2(4p-3)$ (b) $p^3(4p-3)$
(c) $p(4p-3)$ (d) p^3
33. There are three books on table A which has to be moved to table B. The order of the book on Table A was 1, 2, 3, with book 1 at the bottom. The order of the book on table B should be with book 2 on top and book 1 on bottom. Note that you can pick up the books in the order they have been arranged. You can't remove the books from the middle of the stack. In how many minimum steps can we place the books on table B in the required order?
(a) 1 (b) 2
(c) 3 (d) 4
34. In how many ways is it possible to choose a white square and a black square on a chess board so that the squares must not lie in the same row or column?
(a) 56 (b) 896
(c) 60 (d) 768

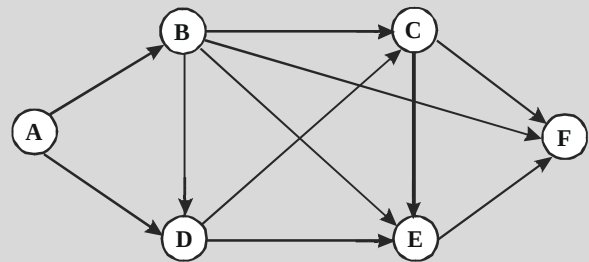
35. There are 12 towns grouped into four zones with three towns per zone. It is intended to connect the towns with telephone lines such that every two towns are connected with three direct lines if they belong to the same zone, and with only one direct line otherwise. How many direct telephone lines are required?
- (a) 72 (b) 90
(c) 96 (d) 144
36. There are 6 tasks and 6 persons. Task I cannot be assigned either to person 1 or to person 2; task 2 must be assigned to either person 3 or person 4. Every person is to be assigned one task. In how many ways can the assignment be done?
- (a) 144 (b) 180
(c) 192 (d) 360
37. The letters of the word ALLAHABAD are rearranged to form new words and put in a dictionary. If the dictionary has only these words and one word on every page in alphabetical order then what is the page number on which the word LABADALAH comes?
- (a) 6089 (b) 6088
(c) 6087 (d) 6086
38. How many natural numbers smaller than 10,000 are there in the decimal notation of which all the digits are different?
- (a) 2682 (b) 4474
(c) 5274 (d) 1448
39. Sameer has to make a telephone call to his friend Harish. Unfortunately he does not remember the 7-digit phone number. But he remembers that the first 3 digits are 635 or 674, the number is odd and there is exactly one 9 in the number. The minimum number of trials that Sameer has to make to be successful is
- (a) 10,000 (b) 3,402
(c) 3,200 (d) 5,000
40. 10 straight lines, no two of which are parallel and no three of which pass through any common point, are drawn on a plane. The total number of regions (including finite and infinite regions) into which the plane would be divided by the lines is
- (a) 56 (b) 255
(c) 1024 (d) not unique
41. Suppose you have a currency, named Miso, in three denominations: 1 Miso, 10 Misos and 50 Misos. In how many ways can you pay a bill of 107 Misos?
- (a) 19 (b) 17
(c) 16 (d) 18
42. Each of 8 identical balls is to be placed in the squares shown in the figure given in a horizontal direction such that one horizontal row contains 6 balls and the other horizontal row contains 2 balls. In how many maximum different ways can this be done?



- (a) 38 (b) 28
(c) 16 (d) 14
43. Two variants of the CAT paper are to be given to twelve students. In how many ways can the students be placed in two rows of six each so that there should be no identical variants side by side and that the students sitting one behind the other should have the same variant?
- (a) $2 \times {}^{12}C_6 \times (6!)^2$ (b) $6! \times 6!$
(c) $7! \times 7!$ (d) None of these
44. The straight lines S_1, S_2, S_3 are in a parallel and lie in the same plane. A total number of A points on S_1 ; B points on S_2 and C points on S_3 are used to produce triangles. What is the maximum number of triangles formed?
- (a) $A+B+C C_3 - A C_3 - B C_3 - C C_3 + 1$
(b) $A+B+C C_3$
(c) $A+B+C C_3 + 1$
(d) $(A+B+C C_3 - A C_3 - B C_3 - C C_3)$
45. If x, y and z are whole numbers such that x, y, z , then how many solutions are possible for the equation $x + y + z = 36$?
- (a) 361 (b) 323
(c) 382 (d) 342
46. The HCF of three natural numbers x, y and z is 13. If the sum of x, y and z is 117, then how many ordered triplets (x, y, z) exist?
- (a) 28 (b) 27
(c) 54 (d) 55
47. The crew of an 8-member rowing team is to be chosen from 12 men ($M_1, M_2, \dots, M_{12}$) and 8 women ($W_1, W_2, \dots, W_8$). There have to be 4 people on each side with at least one woman on each side. Further it is also known that on the right side of the boat (while going forward) W_1 and M_1 must be selected while on the left side of the boat M_2, M_3 and M_{10} must be selected. What is the number of ways in which the rowing team can be arranged?
- (a) $1368 \times 4! \times 4!$ (b) $1200 \times 4! \times 4!$
(c) $1120 \times 4! \times 4!$ (d) $728 \times 4! \times 4!$
48. A boy plays a mathematical game where he tries to write the number 1998 into the sum of 2 or more consecutive positive even numbers (e.g., $1998 = 998 + 1000$). In how many different ways can he do so?
- (a) 5 (b) 6
(c) 7 (d) 8

Test Yourself

- How many three-digit number can be generated from 1, 2, 3, 4, 5, 6, 7, 8, 9, such that the digits are in ascending order?
(a) 80 (b) 81
(c) 83 (d) 84
- How many ways are there to arrange the letters in the word GARDEN with vowels in alphabetical order
(a) 480 (b) 240
(c) 360 (d) 120
- There are 5 historical moments, 6 gardens and 7 shopping malls in the city. In how many ways a tourist can visit the city, if he visits at least one shopping mall?
(a) $2^5 \cdot 2^6 \cdot (2^7 - 1)$ (b) $2^4 \cdot 2^6 (2^7 - 1)$
(c) $2^5 \cdot 2^6 (2^6 - 1)$ (d) None of these
- In how many ways 7 men and 7 women can sit on a round table such that no two women sit together?
(a) $(7!)^2$ (b) $7! \times 6!$
(c) $(6!)^2$ (d) $7!$
- On a triangle ABC, on the side AB, 5 points are marked, 6 points are marked on the side BC and 3 points are marked on the side AC (none of the points being the vertex of the triangle). How many triangles can be made by using these points?
(a) 90 (b) 333
(c) 328 (d) None of these
- How many 6-digit numbers have at least 1 even digit?
(a) 884375 (b) 3600
(c) 880775 (d) 15624
- There is a 7-digit telephone number with all different digits. If the digit at extreme right and extreme left are 5 and 6 respectively, find how many such telephone numbers are possible.
(a) 120 (c) 1,00,000
(c) 6720 (d) None of these
- The numbers of ways in which the letters of the word 'VOWEL' can be arranged so that the letters O, E occupy only even places is
(a) 12 (b) 24
(c) 18 (d) 36
- A bouquet has to be formed from 18 different flowers so that it should contain not less than three flowers. How many ways are there of doing this in?
(a) 5,24,288 (b) 2,62,144
(c) 2,61,972 (d) None of these
- There are 10 points on a straight line AB and 8 on another straight line, AC none of them being A. How many triangles can be formed with these points as vertices?
(a) 720 (b) 640
(c) 816 (d) None of these
- How many numbers can be formed from 1,2,3,4 and 5 (without repetition), when the digit at the units place must be greater than that in the tenth place?
(a) 54 (b) 60
(c) $\frac{5!}{3}$ (d) $2 \times 4!$
- The figure below shows the network connecting cities A, B, C, D, E and F. The arrows indicate permissible direction of travel. What is the number of distinct paths from A to F?
(a) 9 (b) 10
(c) 11 (d) None of these



Hints & Solutions

Foundation Level

1. (c)

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Last place can be filled by 0, 2, 4

So total sum = $5 \times 6 \times 6 (0 + 2 + 4) + 5 \times 6 \times 3 \times 10 (0 + 1 + 2 + 3 + 4 + 5) + 5 \times 6 \times 3 \times 100 (0 + 1 + 2 + 3 + 4 + 5) + 6 \times 6 \times 3 \times 1000 (0 + 1 + 2 + 3 + 4 + 5)$
 $= 180 \times 6 + 900 \times 15 + 9000 \times 15 + 10800 \times 15$
 $= 1080 + 13500 + 135000 + 162000 = 1769580$

2. (a) There are 8 letters in the word EQUATION.

A/E/I/O/U							
5 in ways	${}^7P_7 = 7! = 5040$						

$\therefore$ Reqd. no. = $5 \times 5040 = 25200$

3. (a) There are 9 letters in the given word in which two T's, two M's and two E's are identical. Hence the required

$$\text{number of words} = \frac{9!}{2!2!2!} = \frac{9!}{(2!)^3}$$

4. (c) Given, ${}^{10}P_r = 720$

$$\therefore \frac{10!}{(10-r)!} = 720$$

$$\therefore 10 \times 9 \times 8 \times \dots \text{to } r \text{ factors} = 720 = 10 \times 9 \times 8$$

$$\therefore r = 3$$

5. (b) 4 apples, 5 mangoes and 6 oranges
 coeff. of x^4 in $(1 + x + x^2 + x^3 + x^4)^3$
 $=$ coeff. of x^4 in $(1 - x)^{-3} = {}^6C_2 = 15$

6. (b) $\frac{12!}{5!4!3!}$

7. (a) Considering the two vowels E and A as one letter, the total no. of letters in the word 'EXTRA' is 4 which can be arranged in 4P_4 , i.e. 4! ways and the two vowels can be arranged among themselves in 2! ways.

$$\therefore \text{reqd. no.} = 4! \times 2! = 4 \times 3 \times 2 \times 1 \times 2 \times 1 = 48$$

8. (a) A committee of 5 out of 6 + 4 = 10 can be made in ${}^{10}C_5 = 252$ ways.

If no woman is to be included, then number of ways = ${}^5C_5 = 6$

$$\therefore \text{the required number} = 252 - 6 = 246$$

9. (d) 4 digit number

3	4	3	2
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 = 72,

5 digit number = 120

Total = 192

10. (b) If number of persons be n , then total number of hand-shaken = ${}^nC_2 = 66$

$$\Rightarrow n(n-1) = 132 \Rightarrow (n+1)(n-12) = 0$$

$$\therefore n = 12 \quad (\because n \neq -11)$$

11. (b) There are 6 letters in the word BHARAT, 2 of them are identical.

Hence total number of words with these letter = 360

Also the number of words in which B and H come together = 120

$$\therefore \text{The required number of words} = 360 - 120 = 240$$

12. (a) The required number of selections
 $= {}^3C_1 \times {}^4C_1 \times {}^2C_1 ({}^6C_3 + {}^6C_2 + {}^6C_0) = 42 \times 4!$

13. (d) MACHINE has 4 consonants and 3 vowels.

The vowels can be placed in position no. 1, 3, 5, 7

$$\Rightarrow \text{Total number of ways possible} = {}^4P_3 = 24.$$

For each of these 24 ways the 4 consonants can occupy the other 4 places in 4P_4 ways

$$\Rightarrow \text{Total} = 24 \times 24 = 576$$

14. (b) We have, ${}^nP_r = {}^nP_{r+1}$

$$\Rightarrow \frac{n!}{(n-r)!} = \frac{n!}{(n-r-1)!} \Rightarrow \frac{1}{(n-r)} = 1$$

$$\text{or } n - r = 1$$

...(1)

$$\text{Also, } {}^nC_r = {}^nC_{r-1} \Rightarrow r + r - 1 = n$$

$$\Rightarrow 2r - n = 1$$

...(2)

Solving (1) and (2), we get $r = 2$ and $n = 3$

15. (c) ${}^nP_r = 720 {}^nC_r$

$$\text{or } \frac{n!}{(n-r)!} = \frac{720(n!)}{(n-r)!r!}$$

$$\Rightarrow r! = 720 = 1 \times 2 \times 3 \times 4 \times 5 \times 6!$$

$$\text{or } r = 6$$

16. (a) Total number of ways = ${}^{16}C_{11} = \frac{16!}{11! \times 5!} = 4368.$

$$= \frac{16 \times 15 \times 14 \times 13 \times 12}{5 \times 4 \times 3 \times 2 \times 1} = 4368.$$

17. (b) ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$

$$\Rightarrow {}^{39}C_{3r-1} + {}^{39}C_{3r} = {}^{39}C_{r^2-1} + {}^{39}C_{r^2}$$

$$\Rightarrow {}^{40}C_{3r} = {}^{40}C_{r^2}$$

$$\Rightarrow r^2 = 3r \text{ or } r^2 = 40 - 3r$$

$$\Rightarrow r = 0, 3 \text{ or } -8, 5$$

3 and 5 are the values as the given equation is not defined by $r = 0$ and $r = -8$. Hence, the number of values of r is 2.

18. (c) We have, $75600 = 2^4 \cdot 3^3 \cdot 5^2 \cdot 7$
The total number of ways of selecting some or all out of four 2's, three 3's, two 5's and one 7's is
 $(4 + 1)(3 + 1)(2 + 1)(1 + 1) - 1 = 119$
But this includes the given number itself. Therefore, the required number of proper factors is 118.
19. (c) Required number of ways = ways of selecting 4 objects out of 6 given objects
 $= {}^6C_4 \cdot \frac{6 \times 5}{2} = 15$
20. (d) First of all we will prime factorize 8064.
 $8064 = 2 \times 4032 = 2^2 \times 2016 = 2^3 \times 1008 = 2^4 \times 504$
 $= 2^5 \times 252 = 2^6 \times 126 = 2^7 \times 63$
 $= 2^7 \times 3^2 \times 7^1$
Required no. of ways = $(7 + 1)(2 + 1) \cdot 1$
 $= 8 \times 3 = 24$
21. (c) Two particular girls can be arranged in $2!$ ways and remaining 10 girls can be arranged in $10!$ ways.
Required no. of ways = $2! \times 10!$
22. (c) Required no. of the ways = ${}^6C_3 \times {}^4C_2 = 20 \times 6 = 120$
23. (b) Required number of ways = $\frac{5!}{2!3!} \cdot 10$
24. (b) Selection of 2 members out of 11 has ${}^{11}C_2$ number of ways
 ${}^{11}C_2 = 55$
25. (b) From each railway station, there are 19 different tickets to be issued. There are 20 railway station
So, total number of tickets = $20 \times 19 = 380$.
26. (d) Since ${}^{32}P_6 = k \cdot {}^{32}C_6$
 $\Rightarrow \frac{32!}{(32-6)!} = k \cdot \frac{32!}{6!(32-6)!}$
 $\Rightarrow k = 6! = 720$
27. (c) Before 1000 there are one digit, two digits and three digits numbers.
Numbers of times 3 appear in one digit number = 20×9
Number of times 3 appear in two digit numbers = 11×9
Number of times 3 appear in three digit numbers = 21
Hence total number of times the digit 3 appear while writing the integers from 1 to 1000
 $= 180 + 99 + 21 = 300$
28. (b) $2^4 - 1 = 15$ sums of money can be formed.
29. (b) An IITian can make it to IIMs in 2 ways, while a CA can make it through in 3 ways. Required ratio is 2 : 3.
30. (a) For a straight line we just need to select 2 points out of the 8 points available. 8C_2 would be the number of ways of doing this.
31. (b) ${}^3C_1 \times {}^4C_1 \times {}^6C_1 = 72$
32. (c) At $r = 7$, the value becomes
 $(28!/14! \times 14!)/(24!/10! \times 14!) \rightarrow 225 : 11$
33. (d) For each question we have 3 choices of answering the question (2 internal choices + 1 non-attempt).
Thus, there are a total of 3^{15} ways of answering the question paper. Out of this there is exactly one way in which the student does not answer any question.
Thus there are a total of $3^{15} - 1$ ways in which at least one question is answered.
34. (c) The digits are 1, 6, 7, 8, 7, 6, 1. In this seven-digit no. there are four odd places and three even places
OEOEOEO. The four odd digits 1, 7, 7, 1 can be arranged in four odd places in $[4!/2! \times 2] = 6$ ways [as 1 and 7 are both occurring twice].
The even digits 6, 8, 6 can be arranged in three even places in $3!/2! = 3$ ways.
Total no. of ways = $6 \times 3 = 18$
35. (b) With one green ball there would be six ways of doing this. With 2 green balls 5 ways, with 3 green balls 4 ways, with 4 green balls 3 ways, with 5 green balls 2 ways and with 6 green balls 1 way. So a total of $1 + 2 + 3 + 4 + 5 + 6 = 21$ way.
36. (b) For each selection there are 3 ways of doing it. Thus, there are a total of $3 \times 3 \times 3 \times 3 \times 3 = 243$.
37. (d) First arrange the two sisters around a circle in such a way that there will be one seat vacant between them. [This can be done in $2!$ ways since the arrangement of the sisters is not circular.]
Then, the other 18 people can be arranged on 18 seats in $18!$ ways.
38. (c) Let the total number of employees in the company be n .
Total number of gifts = ${}^nC_2 = \frac{n(n-1)}{2} = 61$
 $\Rightarrow n^2 - n - 132 = 0$ or $(n+11)(n-12) = 0$
or $n = 12$ [- 11 is rejected]
39. (a) In the letters of the word ALLAHABAD there is only 1 vowel available for selection (A). Note that the fact that A is available 4 times has no impact on this fact. Also, there are 4 consonants available - viz. L, H, B and D. Thus, the number of ways of selecting a vowel and a consonant would be $1 \times {}^4C_1 = 4$.
40. (a) Choose 1 person for the single room & from the remaining choose 2 people for the double room & from the remaining choose 4 people for the 4 persons room
 $\rightarrow {}^7C_1 \times {}^6C_2 \times {}^4C_4$.
41. (b) ${}^{10}P_3 = 720$
42. (a) Ist book can be given to any of the five students. Similarly other six books also have 5 choices. Hence the total number of ways is 5^7 .
43. (c) Total possible arrangements = ${}^{13}P_{13} = 13!$
Total number in which f and g are together
 $= 2 \times {}^{12}P_{12} = 2 \times 12!$
44. (a) Order of vowels is fixed
 $\therefore$ required number of ways are $\frac{6!}{2!}$

45. (c) Selection of two husbands = 9C_2
 Selection of two wives whose husbands are not chosen yet = 7C_2
 Total number of ways to form two teams
 = ${}^9C_2 \cdot {}^7C_2 \cdot 2! = 1512$
46. (a) First arrange 21 white balls in a row. This can be done in 1 way (Since they are identical). Now there are 22 place for the 19 black balls and so the place can be filled in
- $${}^{22}C_{19} \text{ ways} = \frac{22!}{3!9!} \text{ ways}$$
- $$\text{or } {}^{22}C_3 = \frac{22 \times 21 \times 20}{2 \times 3} = 1540$$
47. (b) Number of parallelograms = ${}^5C_2 \times {}^4C_2 = 60$.
48. (a) A couple and 6 guests can be arranged in $(7-1)$ ways. But in two people forming the couple can be arranged among themselves in $2!$ ways.
 $\therefore$ the required number of ways = $6! \times 2! = 1440$
49. (b) $6!$ ways, O fixed 1st and E fixed in last.
50. (a) For the number to be divisible by 4, the last two digits must be any of 12, 24, 16, 64, 32, 36, 56 and 52. The last two digit places can be filled in 8 ways. Remaining 3 places in 4P_3 ways. Hence no. of 5 digit nos. which are divisible by 4 are $24 \times 8 = 192$.
51. (d) The voter can cast one or two or three votes. So total number of ways in which he can cast his vote
 = ${}^5C_1 + {}^5C_2 + {}^5C_3 = 5 + 10 + 10 = 25$
52. (c) Given : ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3/5$
 $\Rightarrow \frac{(2n+1)!}{(2n+1-n+1)!} \div \frac{(2n-1)!}{(n+2)(n+1)!} = \frac{3}{5}$
 $\Rightarrow \frac{(2n+1) \times 2n \times (2n-1)!}{(n+2)(n+1)n(n-1)!} \times \frac{(n-1)!}{(2n-1)!} = \frac{3}{5}$
 $\Rightarrow \frac{2(2n+1)}{(n+2)(n+1)} \times \frac{3}{5} \Rightarrow 5(4n+2) = 3(n^2+3n+2)$
 $\Rightarrow 20n+10 = 3n^2+9n+6 \Rightarrow 3n^2-11n-4=0$
 $\Rightarrow (3n+1)(n-4)=0 \Rightarrow n=4$
53. (b) The total number of two factor products = ${}^{200}C_2$. The number of numbers from 1 to 200 which are not multiples of 5 is 160. Therefore the total number of two factor products which are not multiple of 5 is ${}^{160}C_2$. Hence, the required number of factors which are multiples of 5 = ${}^{200}C_2 - {}^{160}C_2 = 7180$.
54. (a) ${}^{15}C_0 + {}^{15}C_1 + {}^{15}C_2 + {}^{15}C_3 + {}^{15}C_4 + {}^{15}C_5$
 = $1 + 15 + 105 + 455 + 1365 + 3003 = 4944$
55. (d) Since every question can be attempted in 4 ways and each question has only one correct answer, hence number of required ways = $(4 \times 4 \times 4) = 1 = 63$

56. (b) Total number of ways
 = $7^1 \times 5^1 \times 2^1 \times 3!$
57. (b) Let the vice-chairman and the chairman from 1 unit along with the eight directors, we now have to arrange 9 different units in a circle.
 This can be done in $8!$ ways.
 At the same time, the vice-chairman & the chairman can be arranged in two different ways. Therefore, the total number of ways = $2 \times 8!$.
58. (b) We need to think of this as: Number with two sixes or numbers with one six or number with no six.
 0, 1, 2, 3, 4, 5, 6
 Number with 2 sixes:
 Number ending n zero ${}^5C_1 \times 3!/2! = 15$
 Numbers ending in 5 and
 (a) Starting with 6 ${}^5C_1 \times 2! = 10$
 (b) Not starting with 6 4C_1 (as zero is not allowed) = 4
 Number with 1 six or no sixes.
 Number ending in 0 ${}^6C_3 \times 3! = 120$
 Number ending in 5 ${}^5C_1 \times {}^5C_2 \times 2! = 100$
 Thus a total of 249 numbers.
59. (b) For drawing a circle we need 3 non collinear points.
 This can be done in:
 ${}^3C_3 + {}^3C_2 \times {}^7C_1 + {}^3C_1 \times {}^7C_2 = 1 + 21 + 63 = 85$.

Standard Level

1. (c) (i) Miss C is taken
 (1) B included $\Rightarrow$ A excluded $\Rightarrow {}^4C_1 \cdot {}^4C_2 = 24$
 (2) B excluded $\Rightarrow {}^4C_1 \cdot {}^5C_3 = 40$
 (ii) Miss C is not taken
 $\Rightarrow$ B does not comes ; ${}^4C_2 \cdot {}^5C_3 = 60 \Rightarrow$ Total = 124
2. (d) Total seats = $5 + 6 = 11$.
 Arrangement will be : W M W M W M W M W M W
 $\Rightarrow$ Total possible arrangements will be :
 ${}^6P_6 \times {}^5P_5 = 86400$.
3. (b) 8 men can sit in a row in 8P_8 ways. Then for the 6 women, there are 9 seats to sit
 $\therefore$ the women can sit in 9P_6 ways
 $\therefore$ total number of ways = ${}^8P_8 \cdot {}^9P_6$
4. (a) There are 13 letters in the word INTERNATIONAL, of which N occurs thrice, each of I, T and A occurs twice and the rest are different.
 $\therefore$ Req'd. no. = $\frac{13!}{3!2!2!2!}$
 $= \frac{13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2}{6 \times 2 \times 2 \times 2}$
 $= 13 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 3 \times 2 = 129729600$

5. (c) Total number of ways of selecting any number of fruits
 $= 11 \times 6 \times 3 \times 2 \times 2 \times 2 = 1584$
 Number of ways in which no fruit is selected = 1
 Number of ways in which only one fruit is selected = 6
 Number of ways in which two fruit are selected
 $= {}^6C_2 + 3 = 18$
 $\therefore$ Number of ways in which at least three fruits are selected = $1584 - (1 + 6 + 18) = 1559$.
6. (a) The number of 4 persons including A, B = 6C_2
 Considering these four as a group, number of arrangements with the other four = 5!
 But in each group the number of arrangements = $2! \times 2!$
 $\therefore$ The required number of ways = ${}^6C_2 \times 5! \times 2! \times 2!$
7. (b) There are ten digits 0, 1, 2,, 9. Permutations of these digits taken eight at a time = ${}^{10}P_8$ which include permutations having 0 at the first. When 0 is fixed at the first place, then number of such permutations
 $= {}^9P_7 - {}^9P_7 = \frac{10!}{2} - \frac{9!}{2} = \frac{9!}{2}$
8. (b) Consider the vowels to be one entity
 $\Rightarrow$ (a, e, i, o), v, l, d, c, t, r, y have to be permuted and the 4 vowels can also permute in the set
 $\Rightarrow$ Total number of arrangements possible
 $= {}^8P_8 \times {}^4P_4 = 967680$
9. (d) From total 13 members 5 can be select as ${}^{13}C_5$
 For at least one girl in the committee number of ways are ${}^{13}C_5 - {}^6C_1$
10. (c) Since each question can be selected in 3 ways, by selecting it or by selecting its alternative or by rejecting it. Thus, the total number of ways of dealing with 10 given questions is 3^{10} including a way in which we reject all the questions.
 Hence, the number of all possible selections is $3^{10} - 1$.
11. (a) No. of ways = coeff. of x^{13} in $(x^2 + x^3 + x^4 + \dots + x^9)^3$
 [Max coin one can get is 9]
 $=$ coeff. of x^{13} in $x^6 (1 + x + x^2 + \dots + x^7)^3$
 $=$ coeff. of x^7 in $\left(\frac{1-x^8}{1-x}\right)^3 =$ coeff. of x^7 in $(1-x)^{-3}$
 $=$ coeff. of x^7 in $(1 + {}^3C_1 x + {}^4C_2 x^2 + \dots) = {}^9C_7 = 36$
12. (b) Required number of ways
 $=$ coefficient of x^2 in $(x + x^2 + \dots x^6)^3$
 $[\because \text{each box can receive minimum 1 and maximum 6 balls}]$

$$= \text{coeff of } x^8 \text{ in } x^2(1 + x + x^2 + \dots + x^5)^3$$

$$\text{coeff of } x^5 \text{ in } \left(\frac{1-x^6}{1-x}\right)^3$$

$$= \text{coeff of } x^5 \text{ in } (1-x)^{-3}$$

$$= \text{coeff of } x^5 \text{ in } (1 + {}^3C_1 x + {}^4C_2 x^2 + \dots) = {}^7C_5 = 21$$

13. (c) X - X - X - X - X. The four digits 3, 3, 5, 5 can be arranged at (-) places in $\frac{4!}{2!2!} = 6$ ways.

The five digits 2, 2, 8, 8, 8 can be arranged at (X) places

$$\text{in } \frac{5!}{2!3!} \text{ ways} = 10 \text{ ways}$$

$$\text{Total no. of arrangements} = 6 \times 10 = 60 \text{ ways}$$

14. (c) Number of elements in the sample space
 $= 6 \times 6 = 36$

The sample space is given by

(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6)

(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6)

.....

.....

.....

(6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)

15. (d) Two possibilities are there :
 (i) Chemistry part I is available in 8 books with Chemistry part II.

or

- (ii) Chemistry part II is available in 8 books but Chemistry part I is not available.

Total No. of ways

$$= 1 \times {}^6C_1 + {}^7C_3$$

$$= 6 + \frac{7 \times 6 \times 5}{3 \times 2} = 6 + 35 = 41$$

16. (a) There are 8! arrangements of the beads on the bracelet, but half of these can be obtained from the other half simply by turning the bracelet over.

$$\text{Hence there are } \frac{1}{2} (8!) = 20160 \text{ different bracelets.}$$

17. (c) We have 8 pieces and 8 squares on the chessboard so we can distribute them in 8! ways but 2 pieces are identical in three cases so total ways

$$= \frac{8!}{2! \times 2! \times 2!} = 5040.$$

18. (c) Number of ways in which 7 persons can stand in the form of a ring = $(7 - 1)! = 6!$

19. (c) Given expression is :

$$\frac{(n+2)! + (n+1)!(n-1)!}{(n+1)!(n-1)!} \times (\text{let})$$

$$\Rightarrow x \frac{(n+2)(n+1)n(n-1)! + (n+1)(n-1)!}{(n+1)(n-1)!}$$

$$= (n+2)n + 1 = n^2 + 2n + 1 = (n+1)^2$$

Which is a perfect square.

20. (b) Let total no. of team participated in a championship be n .

Since, every team played one match with each other team.

$$\therefore {}^nC_2 = 153 \Rightarrow \frac{n!}{2!(n-2)!} = 153$$

$$\Rightarrow \frac{n(n-1)(n-2)!}{2!(n-2)!} = 153 \Rightarrow \frac{n(n-1)}{2} = 153$$

$$\Rightarrow n(n-1) = 306$$

$$\Rightarrow n^2 - n - 306 = 0$$

$$\Rightarrow n^2 - 18n + 17n - 306 = 0$$

$$\Rightarrow n(n-18) + 17(n-18) = 0$$

$$\Rightarrow n = 18, -17$$

n cannot be negative

$$\therefore n \neq -17$$

$$\Rightarrow n = 18$$

21. (d) As we know

$$P(n, r) = r! \cdot C(n, r)$$

$\therefore$ From the question, we have

$$x = r! \cdot (y)$$

Here $r = 31$

$$\therefore x = (31)! \cdot y.$$

22. (c) The number of squares would be

$$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 = 91.$$

23. (d) All arrangements – Arrangements with best and worst paper together = $12! - 2! \times 11!$.

24. (b) We just need to select the floors and the people who get down at each floor.

The floors selection can be done in ${}^{11}C_3$ ways.

The people selection is ${}^9C_4 \times {}^5C_3$.

Also, the floors need to be arranged using $3!$

Thus, ${}^{11}C_3 \times {}^9C_4 \times {}^5C_3 \times 3!$ or ${}^{11}P_3 \times {}^9C_4 \times {}^5C_3$

25. (b) Number of ways of selecting 5 different letters = ${}^5C_5 = 1$

Number of ways of selecting 2 similar and 3 different letters = ${}^4C_1 \times {}^4C_3 = 16$

Number of ways of selecting 2 similar letters and 2 more similar letters and 1 different letter = ${}^4C_2 \times {}^3C_1 = 18$

Number of ways of selecting 3 similar letters and 2 different letters = ${}^3C_1 \times {}^4C_1 = 18$

Number of ways of selecting 3 similar letters and another 2 other similar letters = ${}^3C_1 \times {}^3C_1 = 9$

Number of ways of selecting 4 similar letters and 1 different letter = ${}^2C_1 \times {}^4C_1 = 8$

Number of ways of selecting 5 similar letters = ${}^1C_1 = 1$

Total number of ways = $1 + 16 + 18 + 18 + 9 + 8 + 1 = 71$.

26. (d) $1m + 3f = {}^8C_1 \times {}^8C_3 = 8 \times 56 = 448$

$$2m + 2f = {}^8C_2 \times {}^8C_2 = 28 \times 28 = 784$$

$$3m + 1f = {}^8C_3 \times {}^8C_1 = 56 \times 8 = 448$$

$$4m + 0f = {}^8C_4 \times {}^8C_0 = 70 \times 1 = 70$$

Total = 1750

27. (c) Taking all vowels (IEO) as a single letter (since they come together) there are six letters among which there are two R.

$$\text{Hence no. of arrangements} = \frac{6!}{2!} \times 3! = 2160$$

There vowels can be arranged in $3!$ ways among themselves, hence multiplied with $3!$.

28. (d) Assume the 2 given students to be together (i.e. one). Now these are five students.

Possible ways of arranging them are = $5! = 120$

Now they (two girls) can arrange themselves in $2!$ ways.

Hence total ways = $120 \times 2 = 240$

29. (a) Putting 1 Englishman in a fixed position, the remaining 6 can be arranged in $6!$ 720 ways, For each such arrangement, there are 7 positions for the 7 Americans and they can be arranged in $7!$ ways.

Total number of arrangements = $7! \times 6! = 3628800$

30. (a) Group 6 persons can be divided into 3 equal groups

$$\text{in } \frac{6!}{2!2!2!} \text{ ways} \quad \begin{matrix} P_1 & P_2 & P_3 \\ P_4 & P_6 & P_6 \end{matrix}$$

say $P_1P_4; P_2P_5; P_3P_6$

Now each elements of a group can be arranged in $3!$ ways.

$$\text{Total ways} = \frac{6!3!}{2!2!2!3!} = \frac{720}{8} = 90$$

31. (c) $24 = 2.3.4, 2.2.6.4, 1.3.8, 1.2.12, 1.1.24$ (as product of three positive integers)
 $\therefore$ the total number of positive integral solution of

$$xyz = 24 \text{ is equal to } 3! + \frac{3!}{2!} + 3! + 3! + \frac{3!}{2!}, \text{ i.e., } 30$$

Any two of the factors in each factorization may be negative

$$\therefore \text{The total number of integral solution} \\ = 30 + 3 \times 30 = 120$$

32. (c) $\underbrace{B_1}_{10} \underbrace{B_2}_{10} \underbrace{B_3}_{10} \dots \dots \dots \underbrace{B_9}_{10}$

Selection of atleast one book

$$\underbrace{(10+1)(10+1)\dots(10+1)}_{9 \text{ times}} - 1 = 10^9 - 1$$

33. (b) Required number is greater than 1 million (7 digits).
 From given digits, total numbers which can be formed = $7!$

Number starting from zero = $6!$

$$\Rightarrow \text{Required number} = 7! - 6!$$

$\therefore$ Repetition not allowed, so required answer

$$= \frac{7! - 6!}{2!3!} = 360$$

34. (c) Total number of hand shakes = ${}^{20}C_2$ of those no Indian female shakes hand with male

$$\Rightarrow 5 \times 10 = 50 \text{ hand shakes}$$

No American wife shakes hand with her husband

$$= 5 \times 1 = 5 \text{ hand shakes}$$

$$\Rightarrow \text{total number of hand shakes occurred}$$

$$= {}^{20}C_2 - (50 + 5) = 190 - 55 = 135$$

35. (a) The required number of ways
 $= {}^5C_4 \cdot {}^5C_2 + {}^5C_3 \cdot {}^5C_3 + {}^5C_2 \cdot {}^5C_4 = 50 + 100 + 50 = 200$

36. (c) Total number of ways to permute 6 alphabets 2 of which are common = $6! / 2! = 360$.

(1) Treat the two C's as one

$$\Rightarrow \text{Number of possible ways} = {}^5P_5 = 120$$

(b) Number of ways = Total arrangements – Number of arrangements in which they always come together
 $= 360 - 120 = 240$.

37. (b) 1 wicket keeper from 4 can be selected in

$${}^4C_1 = \frac{4!}{3!1!} = 4 \text{ ways}$$

If 4 bowlers are chosen then remaining 6 batsmen - can be chosen in ${}^{11}C_6$.

$${}^6C_4 \cdot {}^{11}C_6 = \frac{6!}{4!2!} \times \frac{11!}{3!1!} = \frac{5 \times 6}{2} \times \frac{11 \times 10 \times 9 \times 8 \times 7}{5 \times 4 \times 3 \times 2} \\ = 15 \times 14 \times 33 = 6930$$

If we choose 5 bowlers then we have to choose 5 batsmen

$\therefore$ there is no majority.

$$\therefore \text{Total number of ways} = 4 \times 6930 = 27720.$$

38. (b) Total no. of persons = $4 + 4 = 8$

When there is no restriction they can be seated in a row in $8!$ ways. But when all the 4 girls sit together, we can consider the group of 4 girls as one person. Therefore, we have only 4 (no. of boys) + 1 = 5 persons, who can be arranged in a row in $5!$ ways.

But the 4 girls can be arranged among themselves in ${}^4P_4 = 4!$ ways.

$$\therefore \text{No. of ways when all the 4 girls are together} \\ = 5! \times 4!$$

$$\therefore \text{Reqd. no. of ways in which all the 4 girls do not sit together} = 8! - 5! \times 4! = 8 \times 7 \times 6 \times 5! - 5! \times 24 = 5! (336 - 24) = 120 \times 312 = 37440$$

39. (c) For a particular class total number of different tickets from first intermediate station = 5

Similarly number of different tickets from second intermediate station = 4

So total number of different tickets

$$= 5 + 4 + 3 + 2 + 1 = 15$$

And same number of tickets for another class

$$\Rightarrow \text{total number of different tickets} = 30 \text{ and number of selection} = {}^{30}C_{10}$$

40. (d) To minimise the number of boxes we have to fill the maximum number of boxes possible with the maximum number of coins (in the descending order) till the desired number of coins is reached

(i.e., at least 3235)

$\therefore$ 10 boxes of 33 coins each + 10 boxes of 32 coins each till 10 boxes of 21 coins each

$$= 10 \times [33 + 32 + \dots + 21] = 10 \times 13 \times \frac{[32 + 21]}{2}$$

$$= 10 \times 13 \times 27 \times 10 = 3510 \rightarrow 130 \text{ boxes}$$

Which is more than 3235

$$\therefore 3510 - 21 \times 10 = 3300 \rightarrow 120 \text{ boxes}$$

$$3300 - 22 \times 3 = 3234 \rightarrow 117 \text{ boxes.}$$

$$117 + 1 \text{ more box} = 118 \text{ boxes}$$

41. (d) Required number of possible outcomes

= Total number of possible outcomes –

Number of possible outcomes in which 5 does

not appear on any dice. (hence 5 possibilities in each throw)

$$= 6^3 - 5^3 = 216 - 125 = 91$$

42. (c) We have in all 12 points. Since, 3 points are used to form a triangle, therefore the total number of triangles including the triangles formed by collinear points on AB , BC and CA is ${}^{12}C_3 = 220$. But this includes the following :

The number of triangles formed by 3 points on

$$AB = {}^3C_3 = 1$$

The number of triangles formed by 4 points on

$$BC = {}^4C_3 = 4.$$

The number of triangles formed by 5 points on

$$CA = {}^5C_3 = 10.$$

Hence, required number of triangles

$$= 220 - (10 + 4 + 1) = 205.$$

43. (c) Starting with the letter A , and arranging the other four letters, there are $4! = 24$ words. These are the first 24 words. Then starting with G , and arranging A, A, I , and

N in different ways, there are $\frac{4!}{2!1!1!} \cdot \frac{24}{2} = 12$ words.

Hence, total 36 words.

Next, the 37th word starts with I . There are 12 words starting with I . This accounts up to the 48th word. The 49th word is $NAAGI$. The 50th word is $NAAIG$.

44. (a) Let there be n participants in the beginning. Then the number of games played by $(n - 2)$ players $= {}^{n-2}C_2$

$$\therefore {}^{n-2}C_2 + 6 = 84$$

(Two players played three games each)

$$\Rightarrow {}^{n-2}C_2 = 78 \Rightarrow (n-2)(n-3) = 156$$

$$\Rightarrow n^2 - 5n - 150 = 0 \Rightarrow n = 15.$$

45. (d) No. of words starting with A are $4! = 24$

No. of words starting with H are $4! = 24$

No. of words starting with L are $4! = 24$

These account for 72 words

Next word is $RAHLU$ and the 74th word $RAHUL$.

46. (d) Let " x R, y B" denote x Red and y Blue faces such that $x + y = 6$.

I. " 6 R, 0 B" and " 0 R, 6 B":

Only 1 such case is possible for each.

Hence, total 2 possibilities.

II. " 5 R, 1 B" and " 1 R, 5 B":

Only 1 such case is possible for each.

Hence, total 2 possibilities.

III. " 4 R, 2 B" and " 2 R, 4 B":

In " 4 R, 2 B", the two blue faces are either adjacent or not. So 2 such cases are possible for each. Hence, total 4 possibilities.

IV. " 3 R, 3 B":

One case is when the three blue faces are adjacent to one another. The other case is when one blue face is adjacent to the other two but the other two are not adjacent to each other. Hence, total 2 possibilities.

Total possibilities $= 2 + 2 + 4 + 2 = 10$.

47. (b) Number of 11 letter words formed from the letter P, E, R, M, U, T, A, I, O, N $= 11!/2!$.

Number of new words formed = total words

$$- 1 = 11!/2! - 1.$$

48. (d) We have no girls together, let us first arrange the 5 boys and after that we can arrange the girls in the space between the boys.

Number of ways of arranging the boys around a circle

$$= [5 - 1]! = 24.$$

Number of ways of arranging the girls would be by placing them in the 5 spaces that are formed between the boys. This can be done in 5P_3 ways = 60 ways.

Total arrangements $= 24 \times 60 = 1440$.

49. (b) If only one get 1 object

The remaining can be distributed as:

$$(6, 0), (4, 2), (3, 3).$$

$$({}^7C_1 \times {}^6C_6 \times 3! + {}^7C_1 \times {}^6C_4 \times 3! + {}^7C_1 \times {}^6C_3 \times 3!/2!) = 42 + 630 + 420 = 1092.$$

If 2 people get 1 object each:

$${}^7C_1 \times {}^6C_1 \times {}^5C_5 \times 3!/2! = 126.$$

Thus, a total of 1218.

50. (b) When all digits are odd

$$5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^6$$

When all digits are even

$$4 \times 5 \times 5 \times 5 \times 5 \times 5 = 4 \times 5^5$$

$$5^6 + 4 \times 5^5 = 28125$$

51. (b) Let the number of green balls be x . Then the number of red balls is $2x$. Let the number of blue balls be y . Then,

$$x + 2x + y = 10 \Rightarrow 3x + y = 10 \Rightarrow y = 10 - 3x$$

Clearly, x can take values 0, 1, 2, 3. The corresponding values of y are 10, 7, 4 and 1. Thus, the possibilities are (0, 10, 0), (2, 7, 1), (4, 4, 2) and (6, 1, 3), where (r , b , g) denotes the number of red, blue and green balls.

52. (c) Six consonants and three vowels can be selected from 10 consonants and 4 vowels in ${}^{10}C_6 \times {}^4C_3$ ways. Now, these 9 letters can be arranged in $9!$ ways. So, required number of words $= {}^{10}C_6 \times {}^4C_3 \times 9!$.

53. (a) Total number of numbers without restriction $= 2^5$

Two numbers have all the digits equal. So, the required numbers $= 2^5 - 2 = 30$.

54. (b) In order that two books on physics are never together we place the books as follows
 $X E X E X E X E X E X E X E X E X E$
 Where E denotes the position of English books
 X denotes the vacant places in between English books, where the books on physics is to be placed
 So, there are 11 x marked placed for 8 books on physics.

Hence, the required no. of ways = ${}^{11}C_8$

$$= \frac{11 \times 10 \times 9}{3 \times 2} = 165$$

55. (b) $2310 = 2 \times 3 \times 5 \times 7 \times 11 = \text{say } a \times b \times c$
 if $a = 1$ then $b \times c$ can be chosen in

$$\frac{(1+1)(1+1)(1+1)}{2} = 16 \text{ ways}$$

If $a = 2$ then $b \times c$ can be chosen in

$$\frac{(1+1)(1+1)(1+1)}{2} = 8 \text{ ways of which one way has}$$

already been included in the case $a = 1$

$\therefore$ totally 7 ways.

Similarly if $a = 3, 5, 7$ and 11 then number of ways will be $(8-2), (8-3), (8-4)$, and $(8-5)$

Respectively and all possible combinations would have been covered.

$\therefore 16 + 7 + 6 + 5 + 4 + 3 = 41$ ways.

Alternate method :

$$2340 = 2 \times 3 \times 5 \times 7 \times 11$$

When a number can be expressed as a product of n distinct prime numbers, then it can be expressed as a product of 3 numbers in

$$\frac{3^{n+1} + 1}{2} = \frac{3^4 + 1}{2} = \frac{82}{2} = 41 \text{ ways.}$$

(56–58)

SIGNATURE

A, E, G, I, N, R, S, T, U, i.e. all the letters are different.

56. (a)

1	2	3	4	5	6	7	8	9
---	---	---	---	---	---	---	---	---

 ↓ ↓ ↓ ↓

Vowel are A, E, I, U

There are 4 vowels and 4 even places.

$\therefore$ Required number of words in which vowels always occupy the even places = ${}^4C_4 \times 4! \times 5!$

57. (b) **By string method :**

$$\textcircled{\text{SGN}} + 6 = 7$$

$\therefore$ Required number of works = $7! \times 3!$

58. (d) **By gap method :**

$$\times \textcircled{1} \times \textcircled{2} \times \textcircled{3} \times \textcircled{4} \times \textcircled{5} \times \textcircled{6} \times$$

Required no. of words

$${}^7C_3 \times 3! \times 6!$$

59. (a) Ten candidates can be ranked in $10!$ ways. In half of these ways A_1 is above A_2 and in another half A_2 is above A_1 . So, required number of ways is $\frac{10!}{2}$.

60. (a) Two tallest boys can be arranged in $2!$ ways. Rest 18 can be arranged in $18!$ ways.

Girls can be arranged in $6!$ ways.

$$\text{Total number of ways of arrangement} = 2! \times 18! \times 6! = 18! \times 2 \times 720 = 18! \times 1440$$

61. (d) To construct 2 roads, three towns can be selected out of 4 in $4 \times 3 \times 2 = 24$ ways.

Now if the third road goes from the third town to the first town, a triangle is formed, and if it goes to the fourth town, a triangle is not formed. So, there are 24 ways to form a triangle and 24 ways of avoiding a triangle.

62. (d) For a triangle, two points on one line and one on the other has to be chosen.

$$\text{No. of ways} = {}^{10}C_2 \times {}^{11}C_1 + {}^{11}C_2 \times {}^{10}C_1 = 1,045.$$

63. (c) Single digit numbers = 5

$$\text{Two digit numbers} = 5 \times 4 = 20$$

$$\text{Three digit numbers} = 5 \times 4 \times 3 = 60$$

$$\text{Four digit numbers} = 5 \times 4 \times 3 \times 2 = 120$$

$$\text{Five digit numbers} = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$\text{Total} = 5 + 20 + 60 + 120 + 120 = 325$$

64. (d) The unit digit can either be 2, 3, 4, 5 or 6.

When the unit digit is 2, the number would be even and hence will be divisible by 2. Hence all numbers with unit digit 2 will be included which is equal to 5! or 120.

When the unit digit is 3, then in every case the sum of the digits of the number would be 21 which is a multiple of 3. Hence all numbers with unit digit 3 will be divisible by 3 and hence will be included.

Total number of such numbers is 5! or 120.

Similarly for unit digit 5 and 6, the number of required numbers is 120 each.

When the unit digit is 4, the number would be divisible by 4 only if the ten's digit is 2 or 6. Total number of such numbers is $2! \times 4!$ or 48.

$$\text{Hence, Total numbers} = 120 + 120 + 120 + 120 + 48 = 528$$

65. (d) The odd digits have to occupy even positions. This

can be done in $\frac{4!}{2!2!} = 6$ ways

The other digits have to occupy the other positions.

This can be done in $\frac{5!}{3!2!} = 10$ ways

Hence total number of rearrangements possible
 $= 6 \times 10 = 60$.

66. (d) Each box can be filled in 2 ways.

Hence, total no. of ways $= 2^5 = 32$

Blue balls cannot be filled in adjacent boxes

Total no. of such cases in which blue ball is filled in 2 adjacent boxes is

2 blue + 3 blue + 4 blue + 5 blue

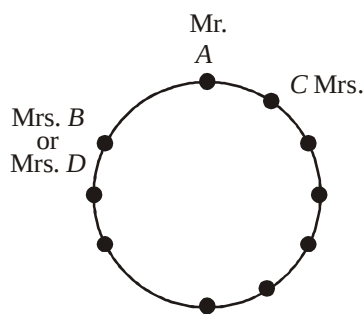
$= 4$ ways (12, 23, 34, 45) + 3 ways (123, 234, 345)

+ 2 ways (1234, 2345) + 1 way

$= 10$ ways

Hence, total cases in which blue balls can not be filled in adjacent boxes $= 32 - 10 = 22$

67. (d)



Since no two men are adjacent to each other, therefore no male is on the right of Mr. A.

Since wives are three places away from their husbands, therefore Mrs. A cannot be on the right of Mr. A.

Mrs. E cannot be on the right of Mr. A, since Mrs. B cannot be on left of Mr. A. Hence, either Mrs. B or Mrs. D can be on the right of Mr. A.

68. (b) There are 28 minutes, hence total no. of songs are 14. Since each pair sings one song. Hence, total number of pairs is 14.

Since, in each possible pair persons are not standing next to each other.

$$\therefore {}^nC_2 - n = 14 \Rightarrow n = 7$$

Hence, total number of people $= 7$.

69. (b) The ways of placing the balls would be 5, 1, 1, 1 ($\frac{4!}{3!} = 4$ ways);

4, 2, 1, 1 ($\frac{4!}{2!} = 12$ ways);

3, 3, 1, 1 ($\frac{4!}{2!} \times 2! = 6$ ways);

3, 2, 2, 1 ($\frac{4!}{2!} = 12$ ways) and 2, 2, 2, 2 (1 way).

Total number of ways $= 4 + 12 + 6 + 12 + 1 = 35$ ways.

70. (d) For each book we have two options, give or not give. Thus, we have a total of 2^{14} ways in which the 14 books can be decided upon. Out of this, there would be 1 way in which no book would be given. Thus, the number of ways is $2^{14} - 1$.

71. (c) We cannot take '0' since the smallest digit must be placed at the left most place. We have only 9 digits from which to select the numbers. First select any number of digits. Then for any selection there is only one possible arrangement where the required condition is met. This can be done in ${}^9C_1 + {}^9C_2 + {}^9C_3 + \dots + {}^9C_9$ ways $2^9 - 1 = 511$ ways.

But we can't take numbers which have only one digit, hence the required number is $511 - 9 = 502$

72. (c) The condition is that we have to count the number of natural numbers not more than 4300.

The total possible numbers with the given digits
 $= 5 \times 5 \times 5 \times 5 = 625 - 1 = 624$.

Subtract from this the number of natural number greater than 4300 which can be formed from the given digits
 $= 1 \times 2 \times 5 \times 5 - 1 = 49$.

Hence, the required number of numbers $= 624 - 49$.

73. (d) You can form triangles by taking 1 point from each side, or by taking 2 points from any 1 side and the third point from either of the other two sides.

This can be done in: $4 \times 5 \times 6 = {}^4C_2 \times {}^{11}C_1 + {}^5C_2 \times {}^{10}C_1 + {}^6C_2 \times {}^9C_1 = 120 + 66 + 100 + 135 = 421$

74. (a) First we write six '+' sings at alternate places i.e., by leaving one place vacant between two successive '+' sings. Now there are 5 places vacant between these sings and these are two places vacant at the ends. If we write 4 '-' sings these 7 places then no two '-' will come together. Hence total number of ways ${}^7C_4 = 35$

75. (b) Let x_1 be the number of stations before the first halting station, x_2 between first and second, x_3 between second and third, x_4 between third and fourth and x_5 on the

right of 4th stations. Then $x_1 \geq 0, x_5 \geq 0, x_2, x_3, x_4 \geq 1$ satisfying $x_1 + x_2 + x_3 + x_4 + x_5 = 8$... (1)

The total number of ways is the number of solution of the above equation

Let $y_2 = x_2 - 1, y_3 = x_3 - 1, y_4 = x_4 - 1$.

Then (1) reduces to $x_1 + y_2 + y_3 + y_4 + x_5 = 5$,

where $y_2, y_3, y_4 \geq 0$.

The number of solution of this equation is

$${}^{5+5-1}C_{5-1} = {}^9C_4.$$

76. (a) India wins exactly in 5 matches $\Rightarrow$ loses in none $\Rightarrow {}^5C_0$ ways
India wins exactly in 6 matches $\Rightarrow$ wins the 6th and loses anyone in the 1st five $\Rightarrow {}^5C_1$ ways and so on.
Total number of ways = ${}^5C_0 + {}^5C_1 + {}^6C_2 + {}^7C_3 + {}^8C_4$
= 126
77. (d) Number of words in which no letter is repeated
= $10 \times 9 \times 8 \times 7 \times 6 = 30240$
Hence, the number of words in which at least one letter is repeated = $10^5 - 30240 = 69760$
78. (a) The two letters, the first and the last of the four lettered word can be chosen in $(17)^2$ ways, as repetition is allowed for consonants. The two vowels in the middle are distinct so that the number of ways of filling up the two places is ${}^5P_2 = 20$. The number of different words = $(17)^2 \cdot 20 = 5780$.
79. (a) First prize may be given to any one of the 4 boys, hence first prize can be distributed in 4 ways.
Similarly every one of second, third, fourth and fifth prizes can also be given in 4 ways.
 $\therefore$ the number of ways of their distribution
= $4 \times 4 \times 4 \times 4 \times 4 = 4^5 = 1024$.
80. (c) (W) – – fixed
M M Fix the position of one woman
 $\Rightarrow$ Remaining women can sit in 3P_3 ways.
W W
M M
W
Remaining men can sit in 4P_4 ways
 $\Rightarrow$ Total ${}^3P_3 \times {}^4P_4 = 144$ ways
81. (d) Required number of possible outcomes
= Total number of possible outcomes – Number of possible outcomes in which 5 does not appear on any dice
= $6^3 - 5^3 = 91$.
82. (b) A triangle requires 3 non collinear points, ${}^{10}C_3$ combinations. But 5 points give us straight line. Hence number of triangles = ${}^{10}C_3 - {}^5C_3 = 120 - 10 = 110$.

Expert Level

1. (c) ${}^{13}C_n < {}^{13}C_{n+2} \Rightarrow \frac{13!}{(13-n)!n!} < \frac{13!}{(13-n-2)!(n+2)!}$
 $\Rightarrow \frac{13!}{(13-n)!n!} < \frac{13!}{(11-n)!(n+2)!}$
 $\Rightarrow \frac{1!}{(13-n)(12-n)(11-n)!} < \frac{1!}{(11-n)!(n+2)(n+1)n!}$

$$\Rightarrow \frac{1}{(13-n)(12-n)} < \frac{1}{(n+2)(n+1)}$$

$$\Rightarrow (n+2)(n+1) < (13-n)(12-n)$$

$$\Rightarrow n^2 + 3n + 2 < n^2 - 25n + 156$$

$$\Rightarrow 3n + 2 < -25n + 156$$

$$\Rightarrow 28n < 154 \Rightarrow n < 154 / 28 \text{ or } n < 5.5$$

Since we need integral values, so n can take any value from 0 to 5.

2. (d) The digits that make the numbers are 1, 3, 5, 7 and 9. The number of numbers with one of these in the first place = 4!
Every one of the digits appear in all the 5 places – digital, 10th, 100th, etc. in one or the other 5-digit number. The required sum of all the numbers is, therefore,
 $= 25(10^4 + 10^3 + 10^2 + 10 + 1) \times 4! = 600 \times \frac{10^5 - 1}{10 - 1}$
 $= 600 \times 11111 = 6666600$.
3. (b) Numbers greater than 3400 will have, 4 or 5 or 6 or 7 in the first place. Having filled the first place say by 4, we have to choose 3 digits out of the remaining 6 and the number will be ${}^6P_3 = 6!/3! = 6 \times 5 \times 4 = 120$.
Therefore total of such numbers will be
 $4 \times 120 = 480$ (1)
Numbers greater than 3400 can be those which have 34, 35, 36, 37 in the first two places. Having filled up 34 in the first two places we will have to choose 2 more out of remaining 5 and the number will be
 ${}^5P_2 = 5!/2! = 5 \times 4 = 20$
Therefore, total as above will be $20 \times 4 = 80$ (2)
Hence all the numbers greater than 3400 will be
 $480 + 80 = 560$
Alternate method : Numbers less than 3400 will have 1 or 2 in 1st place or 31, 32 in the first two position.
 ${}^6P_3 + {}^6P_3 = 120 + 120 = 240$
 ${}^5P_2 + {}^5P_2 = 20 + 20 = 40$.
Total numbers which are less than 3400
 $= 240 + 40 = 280$.
Also total number of numbers formed is ${}^7P_4 = 840$.
Hence numbers greater than 3400 is $840 - 280 = 560$
4. (c) A number will be divisible by 24 if the last two digits are divisible by 25 and this can be done in two ways for either 25 or 75 can be three and remaining two places out of 5 digits can be filled in 5P_2 ways.
Hence the required number = $2 \times {}^5P_2 = 2 \times 20 = 40$
5. (d) A number is divisible by 4 if the last two digits are divisible by 4 which can be done in 10 ways (12, 16, 24, 32, 36, 52, 56, 64, 72, 76).
Hence number = $10 \times {}^5P_2 = 10 \times 20 = 200$

6. (a) No. of non-relative friends = $12 - 7 = 5$.
He may invite 6 friends in follows ways :
I : 4 relatives + 2 non-relatives $\Rightarrow {}^7C_4 \times {}^5C_2$
II : 5 relatives + 1 non-relatives $\Rightarrow {}^7C_5 \times {}^5C_1$
III : 6 relatives + 0 non-relatives $\Rightarrow {}^7C_6$
 $\therefore$ Req'd. no. of ways = ${}^7C_4 \times {}^5C_2 + {}^7C_5 \times {}^5C_1 + {}^7C_6$
 $= 35 \times 10 + 21 \times 5 + 7 = 462$

7. (d) We will consider the following cases,
I case : 4 alike and 2 others alike
II case : 4 alike and 2 different
III case : 3 alike and 3 others alike
IV case : 3 alike and 2 other alike and 1 different
Total ways

$$= 1 \times \frac{6!}{4! 2!} + 1 \times \frac{6!}{4!} + 1 \times \frac{6!}{3! 3!} + {}^2C_1 \times \frac{6!}{3! 2!}$$

$$= 15 + 30 + 20 + 120 = 185.$$

8. (b) Letters of the word PARKAR written in alphabetical order are A A K P R R

Number of words starting with A is	= 60
Number of words starting with K is	= 30
Number of words starting with PAA is	= 3
Number of words starting with PAK is	= 3
Number of words starting with PARA is	= 20
Number of words starting with PARKAR is	= 1

$$\therefore \text{Rank of word PARKAR is } 99$$

9. (b) The inequality is ${}^{n+1}C_3 - {}^{n+1}C_2 \leq 100$

We must have $n+1 \geq 3$ and $n+1 \geq 2$

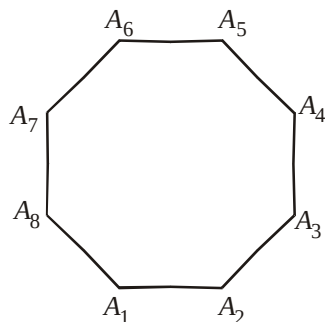
$$\Rightarrow n \geq 2 \text{ and } n \geq 1$$

$$\Rightarrow n \geq 2 \text{ and also } \frac{(n+1)n(n-1)}{6} - \frac{(n+1)n}{2} \leq 100$$

$$\Rightarrow (n+1)n(n-4) \leq 600$$

By trial the values of n satisfying this are 2, 3, 4, 5, 6, 7, 8, 9 which are eight in number.

10. (d) Number of all possible triangles
= Number of selections of 3 points from 8 vertices
 $= {}^8C_3 = 56$



Number of triangle with one side common with octagon
 $= 8 \times 4 = 32$

(Consider side A_1A_2 . Since two points A_3, A_8 are adjacent, 3rd point should be chosen from remaining 4 points.)

Number of triangles having two sides common with octagon : All such triangles have three consecutive vertices, viz., $A_1A_2A_3, A_2A_3A_4, \dots, A_8A_1A_2$.

Number of such triangles = 8

$\therefore$ Number of triangles with no side common
 $= 56 - 32 - 8 = 16$.

11. (b) The first digit x can be any one of 1 to 8 whereas z can be any one of 0 to 9.

When x is 1, y can assume the values 2 to 9 ;

when x is 2, y can assume the values 3 to 9 and so on.

Thus the total number = $(8 + 7 + \dots + 1) \times 10$

$$= \frac{8 \cdot 9}{2} \cdot 10 = 360.$$

12. (b) Given n points the number of triangles that can be drawn by joining any three non-collinear points nC_3 out of this mC_3 is to be subtracted as m points are collinear and no triangle is possible within the m points.

13. (a) Required number = $\frac{6!}{2!3!} - \frac{5!}{3!} = 40$

14. (d) Four digit numbers = $3 \cdot 3 \cdot 2 \cdot 1 = 18$
Five digit numbers = $4 \cdot 3 \cdot 2 \cdot 1 \cdot 1 = 24$
 $\therefore$ Total number of numbers = 42

15. (b) To draw a straight line, we need two points. Hence ${}^{10}C_2$ lines are possible. But 5 points are collinear, hence we subtract 5C_2 but these 5 points give 1 straight line
 $\Rightarrow$ Number of straight lines possible
 $= {}^{10}C_2 - {}^5C_2 + 1 = 45 - 10 + 1 = 36$

16. (d) Number of ways = Arrangement of (m - 1) things of one kind and (n - 1) things of the other kind

$$= \frac{(m+n-2)!}{(m-1)!(n-1)!}$$

17. (d) $S_1 S_3 S_2$ or $S_3 S_1 S_2$
Three places can be chosen in ${}^{10}C_3$ ways and there are only 2 possible arrangements.

Now remaining seven can be arranges in 7! ways

$$\text{Total ways} = {}^{10}C_3 \times 2 \times 7!$$

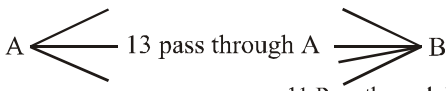
18. (d) When A has B or C to his right we have AB or AC
when B has C or D to his right we have BC or BD
Thus $\Rightarrow$ we must have ABC or ABD or AC and BD
for ABC D, E, F on a circle number of ways = $3! = 6$
for ABD C, E, F on a circle number of ways = $3! = 6$
for AC, BD E, F the number of ways = $3! = 6$
 $\Rightarrow$ Total = 18

19. (b) 3 vacancies for SC candidates can filled up from 5 candidates in 5C_3 ways.
After this for remaining $12 - 3 = 9$ vacancies, there will be $25 - 3 = 32$ candidates.

These vacancies can be filled up in ${}^{22}C_9$ ways.
Hence required number of ways = ${}^5C_3 \times {}^{22}C_9$

20. (c) Consider cases when $z = 0, 1, 2, \dots, 11$
 $\Rightarrow x + y = 33, 33, 27, \dots$
Total number of solution of $x + y = 33, 30, 27, \dots$
 $= 34 + 31 + 28 + \dots + 1$ (12 times)

$$= \frac{12}{2} (1 + 34) = 210$$

21. (a) 
13 pass through A
11 Pass through B

$\therefore$ number of intersection points

$${}^{37}C_2 - {}^{13}C_2 - {}^{11}C_2 + 2$$

($\because$ two points A & B) = 535.

22. (b) Every number between 3000 and 4000, which is divisible by five and which can be formed by the given digits, must contains 5 in unit's place and 3 in thousand's place. Thus we are left with four digits out of which we are to place two digits between 3 and 5, which can be done in ${}^4P_2 = 12$ ways. Hence 12 numbers can be formed.

23. (a) With no restrictions, n women may be seated in a row in nP_n ways.

If 2 of the n women must always sit next to each other, the number of arrangements = $2!({}^{n-1}P_{n-1})$.

Hence the number of ways n women can be seated in a row if 2 particular women may never sit together

$$= {}^nP_n - 2({}^{n-1}P_{n-1}) = n! - 2(n-1)!$$

$$= n(n-1)! - 2(n-1)!$$

$$= (n-2) \times (n-1)!$$

24. (c) Each object can be put either in box B_1 (say) or in box B_2 (say). So, there are two choices for each of the n objects. Therefore the number of choices for n distinct

$$\text{objects is } \underbrace{2 \times 2 \times \dots \times 2}_{n\text{-times}} = 2^n.$$

One of these choices correspond to either the first or the second box being empty.

Thus, there are $2^n - 2$ ways in which neither box is empty.

25. (b) The number of selections of 1 pair of vowels and 1 pair of consonants = ${}^5C_1 \times {}^{21}C_2$

The number of selections of 2 different vowels and 2 different consonants = ${}^5C_2 \times {}^{21}C_2$

$\therefore$ the required number of words

$$= {}^5C_1 \times {}^{21}C_1 \times \frac{4!}{2!2!} + {}^5C_2 \times {}^{21}C_2 \times 4!$$

26. (b) There can be 2 possibilities – last digit is odd or even.

Case I : Last digit is odd. Fixing one out of 1, 3 & 5 in the last position. Then only one odd number can occupy odd position which can be chosen in 2C_1 ways = 2.

One of the two odd digits can be selected for this position in again, 2C_1 ways = 2.

The other odd number can be put in either of the two even places in 2 ways.

Finally the two even numbers can be arranged in 2! ways.

Hence sum of last digit of these nos.

$$= (2 \times 2 \times 2 \times 2) (1 + 3 + 5) = 144 \text{ ways}$$

Case II : Last digit is even. Then 2 odd nos. out of 3 can be arranged in ${}^3P_2 = 3!$ ways.

Again the even nos. can be arranged in 2! ways

$$\therefore \text{Sum} = (3! \times 2) (2 + 4) = 72 \text{ ways.}$$

$$\text{Total ways} = 144 + 72 = 216.$$

27. (a) Number of ways of arranging p big animals into $m - n$ big cages = ${}^{m-n}P_p$.

Now remaining animals can be arranged in any cage in ${}^{m-p}P_{m-p}$ ways

$$\therefore \text{Desired number of ways} = {}^{m-n}P_p \times {}^{m-p}P_{m-p}$$

28. (b) Six students can be arranged in a row is 6! ways. Another six students can be further arranged in 6! ways.

Hence, total number of ways = $6! \times 6!$

Note: Do not get confused with the two type of booklets. The booklets can be distributed in 2 ways.

Both these arrangements will be part of the permutation of students arrangement.

1	2	1	2	1	2
1	2	1	2	1	2

2	1	2	1	2	1
2	1	2	1	2	1

29. (a) Each one of the 26 players played 25 matches and none of the matches ended in a draw Hence all the scores must be even. Also each one of them scored different from the other. The maximum score possible is 50 and minimum score is 0. There are exactly 26 possible scores, 50, 48, 46, ..., 0.

The ranking is in an alphabetical order means A scored 50, B – 48, Z – 0.

This is possible if A wins all the matches B loses only to A win against all others etc.

In final rank, every player win only with all players who are below in final ranking. Since $M > N$ hence M wins over N.

30. (c) The no. of ways are

$${}^{2n+1}C_1 + {}^{2n+1}C_2 + \dots + {}^{2n+1}C_n = 63$$

By option elimination, $2n + 1 = 7$. So $n = 3$.

31. (d) There are 12 points. Since they can be reached from any other point, the edges will be $^{12}C_2 = 66$. Also the number of edges will be maximum 11.

32. (a) The number of triangles with vertices on different lines ${}^pC_1 \times {}^pC_1 \times {}^pC_1 = p^3$

The number of triangles with 2 vertices on one line and the third vertex on any one of the other two lines

$$= {}^3C_1 ({}^pC_2 \times {}^{2p}C_1) = 6p \cdot \frac{p(p-1)}{2} = 3p^2(p-1)$$

$\therefore$ the required number of triangles

$$= p^3 + 3p^2(p-1) = 4p^3 - 3p^2 = p^2(4p-3)$$

(The work "maximum" shows that no selection of points from each of the three lines are collinear).

33. (d) First step — take book 3 to the table B and, second step — put the book 2 on top of 3. Third step — Transfer the arrangement and keep it over book 1 on table A. The last step is transfer the whole arrangement to the table B which is the fourth step to take. Thus total 4 steps are required.

34. (d) There are 32 black and 32 white squares on a chess board. Then no. of ways in choosing one white and one black square on the chess

$$= {}^{32}C_1 \times {}^{32}C_1 = 32 \times 32 = 1024$$

There are 8 rows and 8 columns on a chess board.

In each row or column, there are 4 white and 4 black squares.

Therefore number of ways to choose a white and a black square from the same row

$$= {}^4C_1 \times {}^4C_1 \times 8 = 128$$

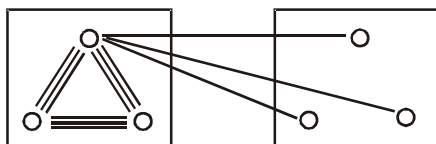
No. of ways to choose a white and a black square from the same column

$${}^4C_1 \times {}^4C_1 \times 8 = 128$$

Total ways in which a white and a black squares lie on the same row or same column = $128 + 128 = 256$

Hence, required no. of ways = $1024 - 256 = 768$

35. (b)



Consider zone 1

No. of lines for internal connections in each zone = 9

Total number of lines for internal connections in four zones = $9 \times 4 = 36$

No. of lines for external connections between any two zones = $3 \times 3 = 9$ (as shown in figure)

$\therefore$ Total no. of lines required for connecting towns of

$$\text{different zones} = {}^4P_2 \times 9 = 6 \times 9 = 54$$

$$\therefore \text{Total no. of lines in all} = 54 + 36 = 90$$

36. (a) Task 2 can be assigned to 3 or 4

So, there are only 2 options for task 2.

Now, task 1 can not be assigned to 1 or 2 i.e. there are 3 options.

So required no. of ways

$$= (2 \text{ options for task 2}) \times (3 \text{ options for task 1}) \\ \times (4 \text{ options for task 3}) \times (3 \text{ options for task 4}) \\ \times (2 \text{ options for task 5}) \times (1 \text{ option for task 6}). \\ = 2 \times 3 \times 4 \times 3 \times 2 \times 1 = 144.$$

37. (a) No. of words starting with A = $8!/2!3! = 3360$.

$$\text{No. of words starting with B} = 8!/2!4! = 840.$$

$$\text{No. of words starting with C} = 8!/2!4! = 840$$

$$\text{No. of words starting with D} = 8!/2!4! = 840$$

$$\text{No. of words starting with H} = 8!/2!4! = 840$$

$$\text{Now of words starting with LAA} = 6!/2! = 180$$

Now LAB starts and first word starts with LABA.

$$\text{No. of words starting with LABAA} = 4! = 24.$$

After this the next words will be LABADAHL, LABADAALH, LABADAHAL, LABADAHLA and hence, Option (a) is the answer.

38. (c) No. of 1 digit nos = 9

$$\text{No. of 2 digit nos} = 81$$

$$\text{No. of 3 digit nos} = 9 \times 9 \times 8 = 648$$

$$\text{No. of 4 digit nos} = 9 \times 9 \times 8 \times 7 = 4536$$

$$\text{Total nos} = 9 + 81 + 648 + 4536 = 5274$$

39. (b) There are two ways of selecting 635 or 674. If last digit is 9, then there are 9 ways of filling each of the remaining 3 digits.

$$\text{Thus total no. of this type of numbers} = 2 \times 9^3 = 1,458.$$

$$\text{When last digit is not 9, total no. of this type of numbers} = 2 \times 3 \times 4 \times 9^2 = 1944.$$

[9 can be selected at any of the 4th, 5th or 6th place in 3 ways. Also at the unit place 4 odd nos. except 9 can be used.]

$$\text{Thus required no.} = 1,944 + 1,458 = 3,402$$

40. (a) For 2 such lines, no. of regions formed are 4

$$\text{For 3 lines no. of regions formed are } 7 (= 4 + 3)$$

$$\text{For 4 lines, no. of regions formed are } 11 (= 7 + 4)$$

$$\text{For 5 lines no. of regions formed are } 16 (= 11 + 5)$$

Similarly for 6, 7, 8, 9 and 10 lines, no. of regions are

$$16 + 6 = 22$$

$$22 + 7 = 29$$

$$29 + 8 = 37$$

$$37 + 9 = 46$$

$$46 + 10 = 56$$

$$\therefore \text{For 10 lines no. of regions} = {}^{10}C_2 + 10 + 1 \\ = 45 + 11 = 56$$

41. (d) **Case (I):** When number of 50 misos is 0, The

No. of 10 misos	No. of 1 misos
10	7
9	17
8	27
⋮	⋮
1	97
0	107

Number of ways to pay the bill = 11

Case (II): When number of 50 misos is 1, then

No. of 10 misos	No. of 1 misos
5	7
4	17
3	27
2	37
1	47
0	57

Number of ways to pay the bill = 6

Case (III) : When number of 50 misos is 2, then

No. of 10 misos	No. of 1 misos
0	7

Number of ways to pay the bill = 1. Hence, from all the three cases, we got total numbers of ways to pay a bill of 107 misos = $11 + 6 + 1 = 18$.

42. (a) The 6 balls must be on either of the middle rows. This can be done in 2 ways. Once, we put the 6 balls in their single horizontal row — it becomes evident that for placing the 2 remaining balls on a straight line there are 2 principal options:
1. Placing the two balls in one of the four rows with two squares. In this case the numbers of ways of placing the balls in any particular row would be 1 way (since once you were to choose one of the 4 rows, the balls would automatically get placed as there are only two squares in each row.) Thus the total number of ways would be $2 \times 4 \times 1 = 8$ ways.
 2. Placing the two balls in the other row with six squares. In this case the number of ways of placing the 2 balls in that row would be 6C_2 . This would give us ${}^2C_1 \times 1 \times {}^6C_2 = 30$ ways. Total is $30 + 8 = 38$ ways.
43. (a) First select six people out of 12 for the first row. The other six automatically get selected for the second row. Arrange the two rows of people amongst themselves.

Besides, the papers can be given in the pattern of 121212 or 212121. Hence the answer is

$$2 \times ({}^{12}C_6 \times 6! \times 6!)$$

44. (d) For the maximum possibility assume that no three points other than given in the question are in a straight line.
45. (a) The total number of solutions for $x + y + z = 36$, if x, y and z are whole numbers is given by ${}^{36+3-1}C_{3-1} = {}^{38}C_2 = 703$.
The number of solutions where $x = y$ will be 19 {from $(x, y) = (0, 0)$ to $(18, 18)$ }.
The number of solutions where x is not equal to $y = 703 - 19 = 684$
Among these 684 solutions, half will have $x > y$ and the rest will have $y > x$.
Hence, the total number of solutions where $x \geq y$
- $$19 + \frac{684}{2} = 361$$
46. (b) Let the three numbers be $13a, 13b$ and $13c$, where a, b and c are coprime.
 $\therefore 13a + 13b + 13c = 117$
 $\Rightarrow 13(a + b + c) = 13 \times 9$
 $\Rightarrow a + b + c = 9$
The number of positive integer solutions of $a + b + c = 9$ is ${}^{9-1}C_{3-1}$ i.e., ${}^8C_2 = 28$.
However, there is a case, $a = b = c = 3$, where a, b and c are not coprime.
So the answer = $28 - 1 = 27$
47. (a) In this question you will first have to complete the selection of 4 people for either side and then arrange the rowers on each side (which would be done by using 4!)
The solution would depend on the following structure the structure would very based on whether you select 2 more men for the right side or you select 1 man and 1 woman for the right side or you select 2 women for the right side.
The solution would be given by:
 ${}^{12}C_2 \times 4! \times {}^8C_1 \times 4! + {}^{12}C_1 \times {}^8C_1 \times 4! \times {}^7C_1 \times 4! + {}^8C_2 \times 4! \times {}^6C_1 \times 4! = 1368 \times 4! \times 4!$
48. (c) For answering this question we need to plan the use of the factors of 1998.
 $1998 = 2 \times 3^3 \times 37 \rightarrow 16$ factors viz. $1 \times 1998, 2 \times 999, 3 \times 666, 9 \times 222, 18 \times 111, 27 \times 74, 54 \times 37$.
Thus we could form 7 APs as follows:
(1) An AP with 2 terms and average 999
(2) An AP with 3 terms and average 666 and so on $\rightarrow 7$ ways.

Explanation of Test Yourself

1. (d) Numbers starting with 12 – 7 numbers
Numbers starting with 13 – 6 numbers; 14 – 5, 15 – 4, 16 – 3, 17 – 2, 18 – 1. Thus total number of numbers starting from 1 is given by the sum of 1 to 7 = 28.
Number of numbers starting from 2 – would be given by the sum of 1 to 6 = 21
Number of numbers starting from 3 – sum of 1 to 5 = 15
Number of numbers starting from 4 – sum of 1 to 4 = 10
Number of numbers starting from 5 – sum of 1 to 3 = 6
Number of numbers starting from 6 = 1 + 2 = 3
Number of numbers starting from 7 = 1
Thus a total of: $28 + 21 + 15 + 10 + 6 + 3 + 1 = 84$ such numbers.
2. (c) Total number of arrangements of letters in the word GARDEN = $6! = 720$ there are two vowels A and E, in half of the arrangements A precedes E and other half A follows E.

So, vowels in alphabetical order in $\frac{1}{2} \times 720 = 360$
3. (a) For each historical monument, there are two possibilities either he visit or does not visit.
Total number of ways = $2^5 \cdot 2^6 (2^7 - 1)$
4. (b) n items can be arranged in a circle or circular arrangement of n items can be made in $(n - 1)!$ ways.
The 7 men can be seated around round table in $6!$ ways such that 7 places are left empty between them. In those 7 places 7 women can be arranged in $7!$ ways.
 $\therefore$ Required number of ways = $6! \cdot 7!$
5. (b) There will be two types of triangles
The first type will have its vertices on the three sides of the $\triangle ABC$.
The second type will have two of its vertices on the same side and the third vertex on any of the other two sides.
Hence, the required number of triangles
 $= 6 \times 5 \times 3 \cdot {}^6C_2 \times 8 + {}^5C_2 \times 9 + {}^3C_2 \times 11$
 $= 90 + 120 + 90 + 33$
 $= 333$
6. (a) All six digit numbers – Six digit numbers with only odd digits.
 $= 900000 - 5 \times 5 \times 5 \times 5 \times 5 \times 5 = 884375$.
7. (c) The number would be of the form $\underline{6} \quad \underline{\quad} \quad \underline{5}$
The 5 missing digits have to be formed using the digits 0, 1, 2, 3, 4, 7, 8, 9 without repetition.
Thus, ${}^8C_5 \times 5! = 6720$
8. (a)

V	O		W	E	L
$\textcircled{X}$					$\textcircled{X}$
1	2		3	4	5

O and E occupying only even places [$\textcircled{X}$ marked places] can be arranged in 2P_2 ways = 2
Remaining letters can be arranged in $(3!)$ ways = 6
Required number of words = $2 \times 6 = 12$
9. (d) ${}^{18}C_4 + {}^{18}C_5 + {}^{18}C_6 + \dots + {}^{18}C_{17} + {}^{18}C_{18}$
 $= [{}^{18}C_0 + {}^{18}C_4 + \dots + {}^{18}C_{18}] - [{}^{18}C_0 + {}^{18}C_1 + {}^{18}C_2 + {}^{18}C_3]$
 $= 2^{18} - [1 + 18 + 153 + 816]$
 $= 261158$
10. (b) ${}^{10}C_2 \times {}^8C_1 + {}^{10}C_1 \times {}^8C_2 = 360 + 280 = 640$
11. (b) The numbers should be formed from 1, 2, 3, 4 and 5 (without repetition), such that the digit at the units place must be greater than in the tenth place. Tenth place has five options.
If 5 is at the tenth place then the digit at the unit's place cannot be filled by the digit greater than that at the tenth place.
If 4 is at the tenth place, then the unit's place has only option of 5, while the three places can be filled up in $3!$ Ways.
If 3 is at the tenth place, then the unit's place can be filled up by 4 or 5, i.e. in 2 ways. While other three places can be filled up in $3!$ ways.
If 2 is at the tenth place, then the unit's place can be filled up by 3, 4 or 5 i.e. in 3 ways. While other three places can be filled up in $3!$ Ways.
If 1 is at the tenth place, then any other four places can be filled up in $4!$ Ways.
Thus the total number of numbers satisfying the given conditions is
 $0 + 3! + 2(3!) + 3(3!) + 4! = 60$.
12. (b) Paths from A to F are
ABCF, ADEF, ABEF, ADCF, ABDEF, ABDCE, ABDCEF, ABCE, ADCEF and ABF
Hence, 10 possible distinct roots.

13. (c) Three numbers can be selected and arranged out of 10 numbers in ${}^{10}P_3$ ways

$$\frac{10!}{7!} = 10 \times 9 \times 8.$$

Now this arrangement is restricted to a given condition that first number is always less than the second number and second number is always less than the third number. Hence three numbers can be arranged among themselves in ${}_3P_3$ ways.

Hence, required number of arrangements

$$= \frac{10 \times 9 \times 8}{3 \times 2} = 120 \text{ ways}$$

14. (c) A and B can occupy the first and the ninth places, the second and the tenth places, the third and the eleventh place and so on... This can be done in 18 ways.

A and B can be arranged in 2 ways.

All the other 24 alphabates can be arranged in $24!$ ways.

Hence the required answer

$$= 2 \times 18 \times 24! = 36 \times 24!$$

15. (a) Let number of girls = x and the number of boys = y
45 games in which both the players were girls

$$\Rightarrow {}^xC_2 = 45$$

$$\Rightarrow \frac{x!}{2!(x-2)!} \quad x(x-1) = 90 \Rightarrow x = 10$$

190 games, where both the players were boys.

$${}^yC_2 = 190 \Rightarrow y(y-1) = 380 \quad \therefore y = 20$$

Hence, the total number of games in which one player was a boy and the other was a girl = $10 \times 20 = 200$

22

PROBABILITY

- Introduction
- Concept of Probability
- Basic Terms
- Mathematical Definition of Probability
- Odds against and Odds in Favour of an Event
- Addition Theorem
- Independent Events
- Conditional Probability
- Geometrical Applications
- Probability Regarding n Letters and Their Envelopes
- Expectation

INTRODUCTION

Probability is most important concept that we use in our day to day life. A mathematically measure of uncertainty is known as probability.

Probability is one of the topics that is considered by CAT aspirants to be important. Analysis of the past CAT question papers can also show that a number of problems have been asked from this chapter every year.

Number of questions based on probability are increasing; probability question tends to be bundled among the difficult questions hence good CAT aspirants must encounter them.

In probability, concept of permutation and combination are used, therefore students would need to master the concepts of combination and permutation in order to crack the CAT and other aptitude tests.

CONCEPT OF PROBABILITY

If you go to buy 10 kg of sugar at ₹ 40 per kg, you can easily find the exact price of your purchase is ₹ 400. On the other hand, the shopkeeper may have a good estimate of the number of kg of sugar that will be sold during the day, but it is impossible to predict the exact amount, because the number of kg of sugar that the consumers will purchase during a day is random.

There are various phenomenon in nature, leading to an outcome, which cannot be predicted in advance. For example, we cannot exactly predict that (i) a head will occur on tossing a coin, (ii) a student will clear the CAT, (iii) India will win the cricket match against Pakistan, etc. But we can measure the amount of

certainty of occurrence of an outcome of a phenomenon. This amount of certainty of occurrence of an outcome of a phenomenon is called probability. For example, on tossing a coin certainty of occurrence of each of a head and a tail are the same. Hence amount certainty of occurrence of each of a head and a tail is 50% i.e., $\frac{50}{100} = \frac{1}{2}$. Therefore $\frac{1}{2}$ is the amount of certainty of occurrence of a head (or a tail) on tossing a coin and hence $\frac{1}{2}$ is the probability of occurrence of a head (or a tail) on tossing a coin. On throwing a dice (a dice is a cuboid having one of the numbers 1, 2, 3, 4, 5 and 6 on each of its six faces) certainty of occurrence of each of the numbers 1, 2, 3, 4, 5 and 6 on its top face are the same.

Therefore certainty of occurrence of each of the numbers 1, 2, 3, 4, 5 and 6 is $\frac{1}{6}$.

Therefore $\frac{1}{6}$ is the amount of certainty of occurrence of each of the numbers 1, 2, 3, 4, 5 or 6 on the top face of the dice on throwing the dice and hence $\frac{1}{6}$ is the probability of occurrence of each of the numbers 1, 2, 3, 4, 5, or 6 on the top face of the dice on tossing a dice is $\frac{1}{6}$.

BASIC TERMS

1. **An Experiment:** An action or operation resulting in two or more outcomes is called an experiment. For examples

- (i) Tossing of a coin is an experiment because there are two possible outcomes head and tail.
- (ii) Drawing a card from a pack of 52 cards is an experiment because there are 52 possible outcomes.
2. **Sample Space:** The set of all possible outcomes of an experiment is called the sample space, denoted by S . An element of S is called a sample point. For examples
- (i) In the experiment of tossing a coin, the sample space has two points corresponding to head (H) and Tail (T) i.e., $S\{H, T\}$.
- (ii) When we throw a dice then any one of the numbers 1, 2, 3, 4, 5 and 6 will come up. So the sample space, $S = \{1, 2, 3, 4, 5, 6\}$
3. **An Event:** Any subset of a sample space is an event. For example,
If we throw a dice then $S = \{1, 2, 3, 4, 5, 6\}$
Then $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$, the null set ϕ and S itself are some events of S , because they all are subsets of set S .
4. **Impossible Event:** The null set ϕ is called the impossible event or null event. For example,
Getting 7 when a dice is thrown is an impossible or a null event.
5. **Sure Event:** The entire sample space is called sure or certain event. For example,
Here the event:
Getting an odd or even number on throwing a dice is a sure event, because the event $= S$.
6. **Complement of an Event:** The complement of an event A is denoted by $\bar{A}$, A' or A^c , is the set of all sample points of the sample space other than the sample points in A . For example, in the experiment of tossing a fair dice, $S = \{1, 2, 3, 4, 5, 6\}$ If $A = \{1, 3, 5, 6\}$, then $A^c = \{2, 4\}$
Note that $A \cup A^c = S$, $A \cap A^c = \phi$.
7. **Simple (or Elementary) Event:** An event is called a simple event if it is a singleton subset of the sample space S . The singleton subset means the subset having only one element. For example,
- (i) When a coin is tossed, sample space $S = \{H, T\}$
Let $A = \{H\}$ = the event of occurrence of head and $B = \{T\}$ = the event of occurrence of tail.
Here A and B are simple events.
- (ii) When a dice is thrown then sample space, $S = \{1, 2, 3, 4, 5, 6\}$
Let $A = \{5\}$ = the event of occurrence of 5
 $B = \{2\}$ = the event of occurrence of 2
Here A and B are simple events.
8. **Compound Event:** It is the joint occurrence of two or more simple events. For example,
The event of at least one head appears when two fair coins are tossed is a compound event,
 $A = \{HT, TH, HH\}$
9. **Equally Likely Events:** A number of simple events are said to be equally likely if there is no reason for one event to occur in preference to any other event. For example,

In drawing a card from a well shuffled pack of 52 cards, there are 52 outcomes and hence 52 simple events which are equally likely because there is no reason for one event to occur in preference to any other event.

10. **Exhaustive Events:** Consider the experiment of throwing a dice. We have $S = \{1, 2, 3, 4, 5, 6\}$. Let us define the following events
 A : a number less than 4 appears.
 B : a number greater than 2 but less than 5 appears and
 C : a number greater than 4 appears.
Then, $A = \{1, 2, 3\}$, $B = \{3, 4\}$ and $C = \{5, 6\}$. We observe that $A \cup B \cup C = \{1, 2, 3\} \cup \{3, 4\} \cup \{5, 6\} = \{1, 2, 3, 4, 5, 6\} = S$. Such events A , B and C are called exhaustive events. In general, if $E_1, E_2, \dots, E_n$ are n events of a sample space S . Such that $E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n = S$, then $E_1, E_2, \dots, E_n$ are called exhaustive events.
11. **Mutually Exclusive Events:** If two events cannot occur simultaneously then they are mutually exclusive. If A and B are mutually exclusive events, then $A \cap B = \phi$.
Ex. In drawing a card from a well shuffled pack of 52 cards, consider the following events:
 A = the card is a spade, and
 B = the card is a heart
The two events A and B are mutually exclusive.
Following two events are not mutually exclusive.
(a) The card is a heart (b) The card is a king
Because the card can be king of heart.
12. **Mutually Exclusive and Exhaustive Events:** Let S be the sample space associated with a random experiment and $A_1, A_2, \dots, A_n$ be the subsets of S , such that
(i) $A_i \cap A_j = \phi$ for $i \neq j$ and (ii) $A_1 \cup A_2 \cup \dots \cup A_n = S$
Then the collection of events $A_1, A_2, \dots, A_n$ is said to form a mutually exclusive and exhaustive system of events.
If $E_1, E_2, \dots, E_n$ are elementary events associated with a random experiment, then
(i) $E_i \cap E_j = \phi$ for $i \neq j$ and (ii) $E_1 \cup E_2 \cup \dots \cup E_n = S$
So, the collection of all elementary events associated with a random experiment always form a system of mutually exclusive and exhaustive events.

MATHEMATICAL DEFINITION OF PROBABILITY

If an event A consists of m sample points of a sample space S having n elements ($0 \leq m \leq n$), then the probability of occurrence of event A , denoted by $P(A)$ is defined to be $\frac{m}{n}$ i.e., $P(A) = \frac{m}{n}$

$$\therefore 0 \leq m \leq n \Rightarrow 0 \leq \frac{m}{n} \leq 1 \Rightarrow 0 \leq P(A) \leq 1$$

If the event A has m elements, then A' has $(n - m)$ elements.

$$\therefore P(A') = \frac{n-m}{n} = 1 - \frac{m}{n} = 1 - P(A)$$

Let $S = \{a_1, a_2, \dots, a_n\}$ be the sample space

$$P(S) = \frac{n}{n} = 1, \text{ corresponding to the certain event.}$$

$P(\phi) = \frac{0}{n} = 0$, corresponding to the null event ϕ
(or impossible event)

If $A_i = \{a_i\}$, $i = 1, 2, \dots$ or n ; then A_i is the event corresponding to a single sample point a_i , then $P(A_i) = \frac{1}{n}$.

If $E_1, E_2, E_3, \dots, E_n$ are exhaustive events, then

$$P(E_1 \cup E_2 \cup E_3 \dots \cup E_n) = P(S) = 1.$$

Illustration 1: Two dice are thrown at a time. Find the probability of the followings:

(i) the numbers shown are equal

(ii) the difference of numbers shown is 1

Solution: The sample space in a throw of two dice

$$S = \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), (2, 2), \dots, (2, 6), (3, 1), \dots, (3, 6), (4, 1), \dots, (4, 6), (5, 1), \dots, (5, 6), (6, 1), \dots, (6, 6)\}$$

$\therefore$ total no. of outcomes, $n(S) = 36$

(i) Here E_1 = the event of showing equal number on both dice

$$= \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

$$\therefore n(E_1) = 6, \Rightarrow P(E_1) = \frac{n(E_1)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

(ii) Here E_2 = the event of showing numbers whose difference is 1.

$$= \{(1, 2), (2, 1), (2, 3), (3, 2), (3, 4), (4, 3), (4, 5), (5, 4), (5, 6), (6, 5)\}$$

$$\therefore n(E_2) = 10, \Rightarrow P(E_2) = \frac{n(E_2)}{n(S)} = \frac{10}{36} = \frac{5}{18}$$

Illustration 2: If three cards are drawn from a pack of 52 cards, what is the chance that all will be queen?

Solution: If the sample space be S , then $n(S)$ = the total number of ways of drawing 3 cards out of 52 cards = ${}^{52}C_3$

Now, if A = the event of drawing three queens, then

$$n(A) = {}^4C_3$$

$$\therefore P(E) = \frac{n(A)}{n(S)} = \frac{{}^4C_3}{{}^{52}C_3} = \frac{4}{\frac{52 \times 51 \times 50}{3 \times 2}} = \frac{1}{5525}$$

Note that in a pack of playing cards,

Total number of cards: 52(26 red, 26 black)

Four suits: Heart, Diamond, Spade, Club-13 cards of each suit

Court number of cards: 12(4 kings, 4 queens, 4 jacks)

Face number of cards: 16(4 aces, 4 kings, 4 queens, 4 jacks)

Illustration 3: Words are formed with the letters of the word PEACE. Find the probability that 2 E's come together.

Solution: Total number of words which can be formed with the letters of the word P E A C E = $\frac{5!}{2!} = 60$

Number of words in which 2 E's come together = $4! = 24$

$$\therefore \text{Required prob.} = \frac{24}{60} = \frac{2}{5}$$

Illustration 4: A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both of them win a prize. The probability that they will not win a prize in a single trial is

(a) $\frac{1}{25}$

(b) $\frac{24}{25}$

(c) $\frac{2}{25}$

(d) None of these

Solution: (b) Total number of possibilities = 25×25

Favourable cases for their winning = 25

$$\therefore P(\text{they win a prize}) = \frac{25}{25 \times 25} = \frac{1}{25}$$

$$\therefore P(\text{they will not win a prize}) = 1 - \frac{1}{25} = \frac{24}{25}$$

ODDS AGAINST AND ODDS IN FAVOUR OF AN EVENT

Let there be $(m + n)$ equally likely outcomes of an experiment and an event of this experiment has m elements. Then by definition of probability of occurrence of event A , $P(A) = \frac{m}{m + n}$.

The probability of non-occurrence of event A , $P(A') = \frac{n}{m + n}$

$$\therefore P(A) : P(A') = m : n$$

Odd in favour of occurrence of the event A is defined by $m : n$ i.e., $P(A) : P(A')$; and the odds against the occurrence of the event A is defined by $n : m$ i.e., $P(A') : P(A)$.

Illustration 5: In a single toss of two dice, find the odds against drawing 7 as the sum of numbers on two dice.

Solution: $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$

$$n(S) = 36, \quad n(E) = 6$$

$$\therefore P(E) = \frac{6}{36} = \frac{1}{6}$$

So, the odds against drawing 7 as the sum of numbers on two dice

$$= \frac{P(\bar{E})}{P(E)} = \frac{1 - \frac{1}{6}}{\frac{1}{6}} = \frac{5}{1} = 5 : 1.$$

Illustration 6: Find the odds in favours of getting a king when a card is drawn from a well shuffled pack of 52 cards.

Solution:

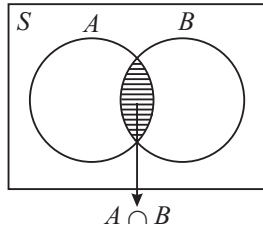
$$\text{Required probability} = \frac{{}^4C_1 / {}^{52}C_1}{{}^{48}C_1 / {}^{52}C_1} = \frac{{}^4C_1}{{}^{48}C_1} = \frac{4}{48} = \frac{1}{12}$$

ADDITION THEOREM

If A and B are any events in S , then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{i.e., } P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$



For three events A , B and C in S , we have

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C).$$

Special Addition Rule

If A , B , and C are mutually exclusive, then $P(A \cap B)$, $P(B \cap C)$, $P(C \cap A)$, $P(A \cap B \cap C) = 0$, hence $P(A \cup B) = P(A) + P(B)$ and $P(A \cup B \cup C) = P(A) + P(B) + P(C)$

Illustration 7: A bag contains 6 white, 5 black and 4 red balls. Find the probability of getting either a white or a black ball in a single draw.

Solution: Let A = Event that we get a black ball
Two events A and B are mutually exclusive.

$$P(A) = \frac{{}^6C_1}{{}^{15}C_1} = \frac{6}{15}, P(B) = \frac{{}^5C_1}{{}^{15}C_1} = \frac{5}{15} =$$

$$\text{So, } P(A \cup B) = P(A) + P(B) = \frac{6}{15} + \frac{5}{15} = \frac{11}{15}.$$

Illustration 8: One digit is selected from first 20 positive integers. What is the probability that it is divisible by 3 or 4.

Solution:

Let A = Event that the selected number is divisible by 3
 B = Event that the selected number is divisible by 4

Here, the events A and B are not mutually exclusive because 12 is divisible by both 3 and 4.

$$P(A) = \frac{6}{20}, P(B) = \frac{5}{20}, P(A \cap B) = \frac{1}{20} \\ \therefore P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ = \frac{6}{20} + \frac{5}{20} - \frac{1}{20} = \frac{10}{20} = \frac{1}{2}.$$

Illustration 9: The probability that at least one of the events A and B occurs is 0.7 and they occur simultaneously with probability 0.2. Then $P(\bar{A}) + P(\bar{B}) =$

- (a) 1.8 (b) 0.6
(c) 1.1 (d) 0.4

Solution: (c) We have $P(A \cup B) = 0.7$ and $P(A \cap B) = 0.2$

$$\text{Now, } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow P(A) + P(B) = 0.9 \Rightarrow 1 - P(\bar{A}) + 1 - P(\bar{B}) = 0.9$$

$$\Rightarrow P(\bar{A}) + P(\bar{B}) = 1.1$$

INDEPENDENT EVENTS

Two or more events are said to be independent if occurrence or non-occurrence of any of them does not influence the probability of occurrence or non-occurrence of other events.

For example, when two cards are drawn from a pack of 52 playing cards with replacement (i.e., the first card drawn is put back in the pack and then the second card is drawn), then the event of occurrence of a king in the first draw and the event of occurrence of a king in the second draw are independent events because the occurrence or non-occurrence of a king in first draw does not influence the probability of occurrence or non-occurrence of the king in second draw. You can also see that the probability of drawing a king in the second draw is $\frac{4}{52}$ whether a king is

drawn in the first draw or not. But if the two cards are drawn without replacement, then the two events are not independent, because in this case probability of drawing a king in the second draw depends on whether a king is drawn in first draw or not. If a king is drawn in first draw, then probability of drawing a king in second draw will be $\frac{3}{51}$ but if a king is not drawn in first draw, then the probability of drawing a king in second draw will be $\frac{4}{51}$.

Illustration 10: A fair coin is tossed repeatedly. If the tail appears on first four tosses, then the probability of the head appearing on the fifth toss equals

- (a) $\frac{1}{2}$ (b) $\frac{1}{32}$
(c) $\frac{31}{32}$ (d) $\frac{1}{5}$

Solution: (a) The event that the fifth toss results a head is independent of the event that the first four tosses results tails.
 $\therefore$ Probability of the required event = $1/2$.

CONDITIONAL PROBABILITY

Let A and B be two events associated with a random experiment. Then, the probability of occurrence of A under the condition that B has already occurred and $P(B) \neq 0$, is called the conditional probability of occurrence of A when B has already occurred and it is denoted by $P(A/B)$.

Thus, $P(A/B)$ = Probability of occurrence of A , if B has already occurred and $P(B) \neq 0$

$$= \frac{P(A \cap B)}{P(B)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}} = \frac{n(A \cap B)}{n(B)}$$

Similarly, $P(B/A)$ = Probability of occurrence of B , if A has already occurred and $P(A) \neq 0$

$$= \frac{P(A \cap B)}{P(A)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(A)}{n(S)}} = \frac{n(A \cap B)}{n(A)}$$

1. Multiplication Theorem on Probability

If A and B are two events associated with a random experiment, then

$$P(A \cap B) = P(A) \cdot P(B/A), \text{ if } P(A) \neq 0$$

$$\text{or } P(A \cap B) = P(B) \cdot P(A/B), \text{ if } P(B) \neq 0.$$

2. Multiplication Theorem for Independent Events

If A and B are independent events associated with a random experiment, then $P(A/B) = P(A)$ and $P(B/A) = P(B)$

$$\therefore P(A \cap B) = P(A) \cdot P(B/A) = P(A) \cdot P(B)$$

i.e., the probability of simultaneous occurrence of two independent events is equal to the product of probability of their individual occurrence.

Extension of multiplication theorem for independent events

If $A_1, A_2, \dots, A_n$ are independent events associated with a random experiment, then

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = P(A_1) P(A_2) \dots P(A_n).$$

3. Probability of Occurrence of at Least One of the n Independent Events

If $p_1, p_2, p_3, \dots, p_n$ be the probabilities of occurrence of n independent events $A_1, A_2, A_3, \dots, A_n$ respectively, then

(i) Probability of happening none of them

$$\begin{aligned} &= P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \dots \cap \bar{A}_n) \\ &= P(\bar{A}_1) P(\bar{A}_2) \cdot P(\bar{A}_3) \dots P(\bar{A}_n) \\ &= (1 - p_1)(1 - p_2)(1 - p_3) \dots (1 - p_n) \end{aligned}$$

(ii) Probability of happening at least one of them

$$\begin{aligned} &= P(A_1 \cup A_2 \cup A_3 \dots \cup A_n) \\ &= 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \bar{A}_3 \dots \cap \bar{A}_n) \\ &= 1 - P(\bar{A}_1) P(\bar{A}_2) P(\bar{A}_3) \dots P(\bar{A}_n) \\ &= 1 - (1 - p_1)(1 - p_2)(1 - p_3) \dots (1 - p_n) \end{aligned}$$

Illustration 11: A man and his wife appear for an interview

for two posts. The probability of the husband's selection is $\frac{1}{7}$

and that of the wife's selection is $\frac{1}{5}$. The probability that only one of them will be selected is

- (a) $\frac{6}{7}$ (b) $\frac{4}{35}$
(c) $\frac{6}{35}$ (d) $\frac{2}{7}$

Solution: (d) Probability that only husband is selected

$$= P(H) P(\bar{W}) = \frac{1}{7} \left(1 - \frac{1}{5}\right) = \frac{1}{7} \times \frac{4}{5} = \frac{4}{35}$$

Probability that only wife is selected

$$= P(\bar{H}) P(W) = \left(1 - \frac{1}{7}\right) \left(\frac{1}{5}\right) = \frac{6}{7} \times \frac{1}{5} = \frac{6}{35}$$

$\therefore$ Probability that only one of them is selected

$$= \frac{4}{35} + \frac{6}{35} = \frac{10}{35} = \frac{2}{7}$$

Illustration 12: A bag contains 4 red and 4 blue balls. Four balls are drawn one by one from the bag, then find the probability that the drawn balls are in alternate colour.

Solution:

E_1 : Event that first drawn ball is red, second is blue and so on.

E_2 : Event that first drawn ball is blue, second is red and so on.

$$\therefore P(E_1) = \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5} \quad \text{and} \quad P(E_2) = \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{3}{5}$$

$$P(E) = P(E_1) + P(E_2) = 2 \times \frac{4}{8} \cdot \frac{4}{7} \cdot \frac{3}{6} \cdot \frac{3}{5} = \frac{6}{35}$$

Illustration 13: A bag contains 5 red and 4 green balls. Four balls are drawn at random then find the probability that two balls are of red colour and two balls are of green.

Solution:

$n(S)$ = The total number of ways of drawing 4 balls out of total 9 balls = 9C_4 .

If A_1 = The event of drawing 2 red balls out of 5 red balls then $n(A_1) = {}^5C_2$.

A_2 = The event of drawing 2 green balls out of 4 greens balls then $n(A_2) = {}^4C_2$.

Let A = The event of drawing 2 balls are of red colour and 2 balls are of green colour.

$$\therefore n(A) = n(A_1) \cdot n(A_2) = {}^5C_2 \times {}^4C_2$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{{}^5C_2 \times {}^4C_2}{{}^9C_4} = \frac{5 \times 4 \times 4 \times 3}{2 \times 2 \times 9 \times 8 \times 7 \times 6} = \frac{10}{21}$$

Illustration 14: Let A, B, C be 3 independent events such that $P(A) = \frac{1}{3}, P(B) = \frac{1}{2}, P(C) = \frac{1}{4}$. Then find the probability of exactly 2 events occurring out of 3 events.

Solution: P (exactly two of A, B, C occur)

$$\begin{aligned} &= P(A \cap B) + P(B \cap C) + P(C \cap A) - 3P(A \cap B \cap C) \\ &= P(A) \cdot P(B) + P(B) \cdot P(C) + P(C) \cdot P(A) - 3P(A) \cdot P(B) \cdot P(C) \\ &= \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{1}{3} - 3 \cdot \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{4} \end{aligned}$$

Illustration 15: A bag contains 3 red, 6 white and 7 blue balls. Two balls are drawn one by one. What is the probability that first ball is white and second ball is blue when first drawn ball is not replaced in the bag?

Solution:

Let A = Event of drawing a white ball in first draw

and B = Event of drawing a blue ball in second draw

Here A and B are dependent events.

$$P(A) = \frac{6}{16}, P\left(\frac{B}{A}\right) = \frac{7}{15}$$

$$P(A \cap B) = P(A) \cdot P\left(\frac{B}{A}\right) = \frac{6}{16} \times \frac{7}{15} = \frac{7}{40}$$

Illustration 16: Three coins are tossed together. What is the probability that first shows head, second shows tail and third shows head?

Solution: Let A = The event first coin shows head

B = The event that second coin shows tail

C = The event that third coin shows head

These three events are mutually independent.

$$\text{So, } P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Illustration 17: A problem of mathematics is given to three students A, B, and C; whose chances of solving it are 1/2, 1/3, 1/4 respectively. Then find the probability that the problem will be solved.

Solution: Obviously the events of solving the problem by A, B and C are independent.

The problem will be solved if at least one of the three students will solve the problem.

Therefore required probability

$$= 1 - \left[\left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{4}\right) \right] = 1 - \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{3}{4}$$

Illustration 18: Two dice are thrown simultaneously. Find the probability that the sum of the number appeared on two dice is 8, if it is known that the second dice always exhibits 4.

Solution: Let A be the event of occurrence of 4 always on the second

dice = {(1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4)}, $\therefore n(A) = 6$ and B be the event of occurrences of such numbers on both dice whose sum is 8 = {(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)}

Thus, $A \cap B = \{(4, 4)\}$

$\therefore n(A \cap B) = 1$

$$\therefore P\left(\frac{B}{A}\right) = \frac{n(A \cap B)}{n(A)} = \frac{1}{6}.$$

Illustration 19: A coin is tossed thrice. If E be the event of showing at least two heads and F be the event of showing head in the first throw, then find $P\left(\frac{E}{F}\right)$.

Solution:

$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

$E = \{HHH, HHT, HTH, THH\}$

$F = \{HHH, HHT, HTH, HTT\}$

$E \cap F = \{HHH, HHT, HTH\}$

$n(E \cap F) = 3, n(F) = 4$

$$\therefore \text{Reqd prob.} = P\left(\frac{E}{F}\right) = \frac{n(E \cap F)}{n(F)} = \frac{3}{4}.$$

GEOMETRICAL APPLICATIONS

The following statements are axiomatic:

1. If a point is taken at random on a given line segment AB, the probability that it falls on a particular segment PQ of the line segment is PQ/AB

$$\text{i.e. probability} = \frac{\text{favourable area}}{\text{total area}}$$

2. If a point is taken at random on the area S which includes an area σ , the probability that the point falls on σ is σ/S .

$$\text{i.e. probability} = \frac{\text{favourable area}}{\text{total area}}$$

PROBABILITY REGARDING n LETTERS AND THEIR ENVELOPES

If n letters corresponding to n envelopes are placed in the envelopes at random, then

$$\text{Probability that all letters are in right envelopes} = \frac{1}{n!}$$

$$\text{Probability that all letters are not in right envelopes} = 1 - \frac{1}{n!}$$

Probability that no letters is in right envelopes

$$= \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^n \frac{1}{n!}$$

Probability that exactly r letters are in right envelopes

$$= \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \dots + (-1)^{n-r} \frac{1}{(n-r)!} \right]$$

Illustration 20: There are four letters and four envelopes, the letters are placed into the envelopes at random, find the probability that all letters are placed in the wrong envelopes.

Solution: Since all letters are to be placed in wrong envelopes, hence required probability

$$= \left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right] = \frac{1}{2} - \frac{1}{6} + \frac{1}{24} = \frac{3}{8}.$$

EXPECTATION

If there are n possibilities $A_1, A_2, \dots, A_n$ in an experiment having the probabilities $p_1, p_2, \dots, p_n$ respectively. If value $M_1, M_2, \dots, M_n$ are associated with the respective possibility, then the expected value of the experiment is given by

$$\sum_{i=1}^n p_i \cdot M_i$$

Practice Exercises



Foundation Level

1. Two dice are thrown simultaneously. The probability of obtaining a total score of seven is
 - (a) $\frac{1}{6}$
 - (b) $\frac{1}{3}$
 - (c) $\frac{2}{7}$
 - (d) $\frac{5}{6}$
2. Four balls are drawn at random from a bag containing 5 white, 4 green and 3 black balls. The probability that exactly two of them are white is
 - (a) $\frac{14}{33}$
 - (b) $\frac{7}{16}$
 - (c) $\frac{18}{33}$
 - (d) $\frac{9}{16}$
3. Two dice are tossed. The probability that the total score is a prime number is :
 - (a) $\frac{1}{6}$
 - (b) $\frac{5}{12}$
 - (c) $\frac{1}{2}$
 - (d) $\frac{7}{9}$
4. Anil can kill a bird once in 3 shots. On the assumption that he fires 3 shots, find the probability that the bird is killed.
 - (a) $\frac{1}{3}$
 - (b) $\left(\frac{1}{3}\right)^3$
 - (c) $\frac{19}{27}$
 - (d) $\frac{8}{9}$
5. If A and B are two independent events with $P(A) = 0.6$, $P(B) = 0.3$, then $P(A' \cap B')$ is equal to :
 - (a) 0.18
 - (b) 0.28
 - (c) 0.82
 - (d) 0.72
6. The probabilities that A and B will die with in a year are p and q respectively, then the probability that only one of them will be alive at the end of the year is -
 - (a) $p + q$
 - (b) $p + q - pq$
 - (c) $p + q + pq$
 - (d) $p + q - 2pq$
7. A pair of dice is thrown thrice. The probability of throwing doublets at least once is
 - (a) $\frac{1}{36}$
 - (b) $\frac{25}{216}$
 - (c) $\frac{125}{216}$
 - (d) None of these
8. The probability of getting number 5 in throwing a dice is
 - (a) 1
 - (b) $\frac{1}{3}$
 - (c) $\frac{1}{6}$
 - (d) $\frac{5}{6}$
9. The probability of getting head and tail alternately in three throws of a coin (or a throw of three coins), is
 - (a) $\frac{1}{8}$
 - (b) $\frac{1}{4}$
 - (c) $\frac{1}{3}$
 - (d) $\frac{3}{8}$
10. A die is thrown once. What is the probability of occurrence of an odd number on the upper face?
 - (a) $\frac{2}{3}$
 - (b) $\frac{1}{2}$
 - (c) $\frac{1}{4}$
 - (d) $\frac{1}{8}$
11. A die is thrown once. Find the probability that 3 or greater than 3 turns up.
 - (a) $\frac{1}{2}$
 - (b) $\frac{1}{3}$
 - (c) $\frac{1}{4}$
 - (d) $\frac{2}{3}$
12. Find the probability of getting a multiple of 2 in the throw of a die.
 - (a) $\frac{1}{2}$
 - (b) $\frac{1}{4}$
 - (c) $\frac{1}{3}$
 - (d) $\frac{1}{6}$
13. India and Pakistan play a 5 match test series of hockey, the probability that India wins at least three matches is
 - (a) $\frac{1}{2}$
 - (b) $\frac{3}{5}$
 - (c) $\frac{4}{5}$
 - (d) None of these
14. The probability that a man can hit a target is $\frac{3}{4}$. He tries 5 times. The probability that he will hit the target at least three times is
 - (a) $\frac{291}{364}$
 - (b) $\frac{371}{461}$
 - (c) $\frac{471}{502}$
 - (d) $\frac{459}{512}$
15. From eighty cards numbered 1 to 80, two cards are selected randomly. The probability that both the cards have the numbers divisible by 4 is given by
 - (a) $\frac{21}{316}$
 - (b) $\frac{19}{316}$
 - (c) $\frac{1}{4}$
 - (d) None of these

16. The probability of getting sum more than 7 when a pair of dice are thrown is
 (a) $\frac{7}{36}$ (b) $\frac{5}{12}$
 (c) $\frac{7}{12}$ (d) None of these
17. Two dice are thrown simultaneously then the probability of obtaining a total score of 5 is
 (a) $\frac{1}{18}$ (b) $\frac{1}{12}$
 (c) $\frac{1}{9}$ (d) None of these
18. The probability that the two digit number formed by digits 1, 2, 3, 4, 5 is divisible by 4 is
 (a) $\frac{1}{30}$ (b) $\frac{1}{20}$
 (c) $\frac{1}{5}$ (d) None of these
19. Probability of throwing 16 in one throw with three dice is
 (a) $\frac{1}{36}$ (b) $\frac{1}{18}$
 (c) $\frac{1}{72}$ (d) $\frac{1}{9}$
20. Of a total of 600 bolts, 20% are too large and 10% are too small. The remainder are considered to be suitable. If a bolt is selected at random, the probability that it will be suitable is
 (a) $\left(\frac{1}{5}\right)$ (b) $\left(\frac{7}{10}\right)$
 (c) $\left(\frac{1}{10}\right)$ (d) $\left(\frac{3}{10}\right)$
21. The probability that in the toss of two dice we obtain the sum 7 or 11 is
 (a) $\frac{1}{6}$ (b) $\frac{1}{18}$
 (c) $\frac{2}{9}$ (d) $\frac{23}{108}$
22. A card is drawn at random from a pack of 100 cards numbered 1 to 100. The probability of drawing a number which is a square, is
 (a) $\frac{1}{10}$ (b) $\frac{1}{100}$
 (c) $\frac{9}{10}$ (d) $\frac{90}{100}$
23. The alphabets of word ALLAHABAD are arranged at random. The probability that in the words so formed, all identical alphabets are found together, is
 (a) $\frac{1}{63}$ (b) $\frac{16}{17}$
 (c) $\frac{5!}{9!}$ (d) None of these
24. 3 integers are chosen at random from the set of first 20 natural numbers. The chance that their product is a multiple of 3, is.
 (a) $\frac{194}{285}$ (b) $\frac{1}{57}$
 (c) $\frac{13}{19}$ (d) $\frac{3}{4}$
25. The probability that Krishna will be alive 10 years hence, is $\frac{7}{15}$ and that Hari will be alive is $\frac{7}{10}$. What is the probability that both Krishna and Hari will be dead 10 years hence ?
 (a) $\frac{21}{150}$ (b) $\frac{24}{150}$
 (c) $\frac{49}{150}$ (d) $\frac{56}{150}$
26. The probability that in the random arrangement of the letters of the word 'UNIVERSITY', the two I's does not come together is
 (a) $\frac{4}{5}$ (b) $\frac{1}{5}$
 (c) $\frac{1}{10}$ (d) $\frac{9}{10}$
27. Among 15 players, 8 are batsmen and 7 are bowlers. Find the probability that a team is chosen of 6 batsmen and 5 bowlers:
 (a) $\frac{{}^8C_6 \times {}^7C_5}{{}^{15}C_{11}}$ (b) $\frac{28}{15}$
 (c) $\frac{15}{28}$ (d) None of these
28. A four digit number is formed by the digits 1, 2, 3, 4 with no repetition. The probability that the number is odd is
 (a) zero (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) None of these
29. X speaks truth in 60% and Y in 50% of the cases. The probability that they contradict each other narrating the same incident is
 (a) $\frac{1}{4}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{2}$ (d) $\frac{2}{3}$
30. An integer is chosen at random from the numbers 1, 2, ..., 25. The probability that the chosen number is divisible by 3 or 4, is
 (a) $\frac{2}{25}$ (b) $\frac{11}{25}$
 (c) $\frac{12}{25}$ (d) $\frac{14}{25}$

31. The probability that a leap year will have 53 Friday or 53 Saturday, is
- (a) $\frac{2}{7}$ (b) $\frac{3}{7}$
(c) $\frac{4}{7}$ (d) $\frac{1}{7}$
32. An experiment yields 3 mutually exclusive and exhaustive events A, B, C . If $P(A) = 2P(B) = 3P(C)$, then $P(A)$ is equal to
- (a) $\frac{1}{11}$ (b) $\frac{2}{11}$
(c) $\frac{3}{11}$ (d) $\frac{6}{11}$
33. If $P(A \cup B) = 0.8$ and $P(A \cap B) = 0.3$, then $P(A') + P(B')$ equals to
- (a) 0.3 (b) 0.5
(c) 0.7 (d) 0.9
34. Five coins whose faces are marked 2, 3 are thrown. What is the probability of obtaining a total of 12?
- (a) $\frac{1}{16}$ (b) $\frac{3}{16}$
(c) $\frac{5}{16}$ (d) $\frac{7}{16}$
35. An aircraft has three engines A, B and C . The aircraft crashes if all the three engines fail. The probabilities of failure are 0.03, 0.02 and 0.05 for engines A, B and C respectively. What is the probability that the aircraft will not crash?
- (a) 0.00003 (b) 0.90
(c) 0.99997 (d) 0.90307
36. A coin is tossed three times. What is the probability of getting head and tail (HTH) or tail and head (THT) alternatively?
- (a) $\frac{1}{4}$ (b) $\frac{1}{5}$
(c) $\frac{1}{6}$ (d) $\frac{1}{8}$
37. The probability that a student passes in mathematics is $\frac{4}{9}$ and that he passes in physics is $\frac{2}{5}$. Assuming that passing in mathematics and physics are independent of each other, what is the probability that he passes in mathematics but fails in physics?
- (a) $\frac{4}{15}$ (b) $\frac{8}{45}$
(c) $\frac{26}{45}$ (d) $\frac{19}{45}$
38. From a pack of 52 cards, two cards are drawn, the first being replaced before the second is drawn. What is the probability that the first is a diamond and the second is a king?
- (a) $\frac{1}{4}$ (b) $\frac{4}{13}$
(c) $\frac{1}{52}$ (d) $\frac{4}{15}$
39. In throwing of two dice, what is the number of exhaustive events?
- (a) 6 (b) 12
(c) 36 (d) 18
40. In a lottery, 16 tickets are sold and 4 prizes are awarded. If a person buys 4 tickets, what is the probability of his winning a prize?
- (a) $\frac{4}{16^4}$ (b) $\frac{175}{256}$
(c) $\frac{1}{4}$ (d) $\frac{81}{256}$
41. A card is drawn from a pack of 52 cards and a gambler bets that it is a spade or an ace. Which one of the following are the odds against his winning this bet?
- (a) 13 to 4 (b) 4 to 13
(c) 9 to 4 (d) 4 to 9
42. Each of A and B tosses two coins. What is the probability that they get equal number of heads?
- (a) $\frac{3}{16}$ (b) $\frac{5}{16}$
(c) $\frac{4}{16}$ (d) $\frac{6}{16}$
43. What is the probability that in a family of 4 children there will be at least one boy?
- (a) $\frac{15}{16}$ (b) $\frac{3}{8}$
(c) $\frac{1}{16}$ (d) $\frac{7}{8}$
44. The chance of winning the race of the horse A is $\frac{1}{5}$ and that of horse B is $\frac{1}{6}$. What is the probability that the race will be won by A or B ?
- (a) $\frac{1}{30}$ (b) $\frac{1}{3}$
(c) $\frac{11}{30}$ (d) $\frac{1}{15}$
45. What is the probability of two persons being born on the same day (ignoring date)?
- (a) $\frac{1}{49}$ (b) $\frac{1}{365}$
(c) $\frac{1}{7}$ (d) $\frac{2}{7}$
46. If A and B are two mutually exclusive and exhaustive events with $P(B) = 3P(A)$, then what is the value of $P(\bar{B})$?
- (a) $\frac{3}{4}$ (b) $\frac{1}{4}$
(c) $\frac{1}{3}$ (d) $\frac{2}{3}$
47. The probabilities of two events A and B are given as $P(A) = 0.8$ and $P(B) = 0.7$. What is the minimum value of $P(A \cap B)$?
- (a) 0 (b) 0.1
(c) 0.5 (d) 1

48. In tossing three coins at a time, what is the probability of getting at most one head?
- (a) $\frac{3}{8}$ (b) $\frac{7}{8}$
(c) $\frac{1}{2}$ (d) $\frac{1}{8}$
49. Two balls are selected from a box containing 2 blue and 7 red balls. What is the probability that at least one ball is blue?
- (a) $\frac{2}{9}$ (b) $\frac{7}{9}$
(c) $\frac{5}{12}$ (d) $\frac{7}{12}$
50. The probability of guessing a correct answer is $\frac{x}{12}$. If the probability of not guessing the correct answer is $\frac{2}{3}$, then what is x equal to?
- (a) 2 (b) 3
(c) 4 (d) 6
51. If A and B are two mutually exclusive events, then what is $P(AB)$ equal to?
- (a) 0 (b) $P(A) + P(B)$
(c) $P(A)P(B)$ (d) $P(A)P\left(\frac{B}{A}\right)$
52. If $P(E)$ denotes the probability of an event E , then E is called certain event if :
- (a) $P(E) = 0$ (b) $P(E) = 1$
(c) $P(E)$ is either 0 or 1 (d) $P(E) = 1/2$
53. A programmer noted the results of attempting to run 20 programs. The results showed that 2 programs ran correctly in the first attempt, 7 ran correctly in the second attempt, 5 ran correctly in the third attempt, 4 ran correctly in the fourth attempt and 2 ran correctly in the fifth attempt. What is the probability that his next programme will run correctly on the third run ?
- (a) $\frac{1}{4}$ (b) $\frac{1}{3}$
(c) $\frac{1}{6}$ (d) $\frac{1}{5}$
54. The digits 1, 2, 3, 4, 5, 6, 7, 8, 9 are written in random order to form a nine digit number. Find the probability that this number is divisible by 4:
- (a) $\frac{4}{9}$ (b) $\frac{2}{9}$
(c) $\frac{17}{81}$ (d) None of these
55. Two cards are drawn from a pack of 52 cards. The probability that either both are red or both are kings is
- (a) $\frac{1}{2}$ (b) $\frac{1}{321}$
(c) $\frac{325}{1326}$ (d) None of these
56. A man and his wife appear for an interview for two posts. The probability of the husband's selection is $\frac{1}{7}$ and that of the wife's selection is $\frac{1}{5}$. The probability that only one of them will be selected is
- (a) $\frac{6}{7}$ (b) $\frac{4}{35}$
(c) $\frac{6}{35}$ (d) $\frac{2}{7}$
57. The probability that a person will hit a target in shooting practice is 0.3. If he shoots 10 times, the probability that he hits the target is
- (a) 1 (b) $1 - (0.7)^{10}$
(c) $(0.7)^{10}$ (d) $(0.3)^{10}$
58. Suppose six coins are tossed simultaneously. Then the probability of getting at least one tail is
- (a) $\frac{71}{72}$ (b) $\frac{53}{54}$
(c) $\frac{63}{64}$ (d) $\frac{1}{12}$
59. In a given race the odds in favour of three horses A, B, C are 1 : 3; 1 : 4; 1 : 5 respectively. Assuming that dead head is impossible the probability that one of them wins is
- (a) $\frac{7}{60}$ (b) $\frac{37}{60}$
(c) $\frac{1}{5}$ (d) $\frac{1}{8}$
60. The probability that the 13th day of a randomly chosen month is a Friday, is
- (a) $\frac{1}{12}$ (b) $\frac{1}{7}$
(c) $\frac{1}{84}$ (d) $\frac{1}{13}$
61. In a single throw with four dice, the probability of throwing seven is
- (a) $\frac{4}{6^4}$ (b) $\frac{8}{6^4}$
(c) $\frac{16}{6^4}$ (d) $\frac{20}{6^4}$
62. Six dice are thrown. The probability that different number will turn up is
- (a) $\frac{129}{1296}$ (b) $\frac{1}{54}$
(c) $\frac{5}{324}$ (d) $\frac{5}{54}$

63. If A and B are events such that $P(A|B) = P(B|A)$, then
 (a) $A \subset B$ but $A \neq B$ (b) $A = B$
 (c) $A \cap B = \phi$ (d) $P(A) = P(B)$
64. If two dice are tossed, find the probability of throwing a total of ten or more.
 (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) $\frac{2}{3}$
65. From a pack of 52 cards two are drawn with replacement. The probability, that the first is a diamond and the second is a king, is
 (a) $1/26$ (b) $17/2704$
 (c) $1/52$ (d) None of these
66. Two cards are selected at random from a deck of 52 playing cards. The probability that both the cards are greater than 2 but less than 9 is
 (a) $\frac{46}{221}$ (b) $\frac{63}{221}$
 (c) $\frac{81}{221}$ (d) $\frac{93}{221}$
67. If A and B are two independent events such that $P(a) = 1/6$ and $P(b) = 1/2$, then
 (a) $P(AB) = 1/24$ (b) $P(A \cup B) = 1/12$
 (c) $P(A^c B) = 1/6$ (d) $P(A^c B^c) = 5/12$
68. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{5}$, then which is not true?
 (a) $P(A \cup B) = \frac{3}{5}$ (b) $P\left(\frac{A}{B}\right) = \frac{1}{4}$
 (c) $P(A / A \cup B) = \frac{5}{6}$ (d) $P(A \cap B / \bar{A} \cup \bar{B}) = 0$
69. A card is drawn from a pack of 52 cards. A gambler bets that it is a spade or an ace. What are the odds against his winning this bet?
 (a) 17 : 52 (b) 52 : 17
 (c) 9 : 4 (d) 4 : 9
70. The probability that a man will live 10 more years is $\frac{1}{4}$ and the probability that his wife will live 10 more years is $\frac{1}{3}$. Then the probability that neither will be alive in 10 years is
 (a) $\frac{5}{12}$ (b) $\frac{7}{12}$
 (c) $\frac{1}{2}$ (d) $\frac{11}{12}$
71. A and B play a game where each is asked to select a number from 1 to 25. If the two numbers match, both of them win a prize. The probability that they will not win a prize in a single trial is
 (a) $\frac{1}{25}$ (b) $\frac{24}{25}$
 (c) $\frac{2}{25}$ (d) None of these
72. In a box containing 100 bulbs, 10 are defective. The probability that out of a sample of 5 bulbs, none is defective is
 (a) 10^{-1} (b) $\left(\frac{1}{2}\right)^5$
 (c) $\left(\frac{9}{10}\right)^5$ (d) $\frac{9}{10}$
73. The probability of happening an event A in one trial is 0.4. The probability that the event A happens at least once in three independent trials is –
 (a) 0.936 (b) 0.216
 (c) 0.904 (d) 0.784
74. Find the probability of drawing a jack or an ace from a pack of playing cards.
 (a) $\frac{1}{8}$ (b) $\frac{1}{6}$
 (c) $\frac{1}{3}$ (d) $\frac{2}{13}$
75. When two dice are thrown, the probability that the difference of the numbers on the dice is 2 or 3 is
 (a) $\frac{7}{18}$ (b) $\frac{3}{11}$
 (c) $\frac{5}{18}$ (d) $\frac{1}{2}$
76. In shuffling a pack of cards three are accidentally dropped. The probability that the missing cards are of distinct colours is
 (a) $\frac{169}{425}$ (b) $\frac{165}{429}$
 (c) $\frac{162}{459}$ (d) $\frac{164}{529}$
77. A quadratic equation is chosen from the set of all quadratic equations which are unchanged by squaring their roots. The chance that the chosen equation has equal roots is
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{1}{4}$ (d) None of these

78. Four persons are selected at random out of 3 men, 2 women and 4 children. The probability that there exactly 2 children in the selection is

(a) $\frac{11}{21}$ (b) $\frac{9}{21}$
(c) $\frac{10}{21}$ (d) None of these

79. It is given that the events A and B are such that

$$P(A) = \frac{1}{4}, P(A|B) = \frac{1}{2} \text{ and } P(B|A) = \frac{2}{3}. \text{ Then } P(B) \text{ is}$$

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{2}$

80. A coin is tossed and a dice is rolled. The probability that the coin shows the head and the dice shows 6 is

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$
(c) $\frac{1}{12}$ (d) $\frac{1}{24}$

Directions for Questions 81 and 82:

An urn contains one black ball and one green ball. A second urn contains one white and one green ball. One ball is drawn at random from each urn.

81. What is the probability that both balls are of same colour ?

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

82. What is the probability of getting at least one green ball ?

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

83. The probability that a student will pass in Mathematics is $\frac{3}{5}$ and the probability that he will pass in English is $\frac{1}{3}$. If the probability that he will pass in both Mathematics and English is $\frac{1}{8}$, what is the probability that he will pass in at least one subject?

(a) $\frac{97}{120}$ (b) $\frac{87}{120}$
(c) $\frac{53}{120}$ (d) $\frac{120}{297}$

Standard Level

- A bag contains 6 red and 4 green balls. A fair dice is rolled and a number of balls equal to that appearing on the dice is chosen from the urn at random. The probability that all the balls selected are red is
 - $\frac{1}{3}$
 - $\frac{3}{10}$
 - $\frac{1}{8}$
 - $\frac{1}{5}$
- If $P(A) = 0.8$, $P(B) = 0.9$, $P(AB) = p$, which one of the following is correct?
 - $0.72 \leq p \leq 0.8$
 - $0.7 \leq p \leq 0.8$
 - $0.72 < p < 0.8$
 - $0.7 < p < 0.8$
- A, B, C are three mutually exclusive event associated with a random experiment. Find $P(A)$ if it is given that $P(B) = \frac{3}{2} P(A)$ and $P(C) = \frac{1}{2} P(B)$.
 - $\frac{4}{13}$
 - $\frac{2}{3}$
 - $\frac{12}{13}$
 - $\frac{1}{13}$
- Two cards are drawn at random from a well-shuffled pack of 52 cards. What is the probability that either both are red or both are queens?
 - $\frac{17}{112}$
 - $\frac{55}{221}$
 - $\frac{55}{221}$
 - $\frac{33}{221}$
- Four cards are drawn at a time from a pack of 52 playing cards. Find the probability of getting all the four cards of the same suit.
 - $\frac{5}{13}$
 - $\frac{12}{65}$
 - $\frac{44}{4165}$
 - $\frac{44}{169}$
- Two persons A and B throw a die alternatively till one of them gets a three and wins the game. Find their respective probabilities of winning.
 - $\frac{6}{11}, \frac{5}{11}$
 - $\frac{5}{11}, \frac{8}{11}$
 - $\frac{3}{11}, \frac{7}{11}$
 - $\frac{8}{11}, \frac{3}{11}$
- The probability that A can solve a problem is $\frac{2}{3}$ and B can solve it is $\frac{3}{4}$. If both attempt the problem, what is the probability that the problem gets solved?
 - $\frac{11}{12}$
 - $\frac{7}{12}$
 - $\frac{5}{12}$
 - $\frac{9}{12}$
- A speaks the truth in 70 percent cases and B in 80 percent. The probability that they will contradict each other when describing a single event is
 - 0.36
 - 0.38
 - 0.4
 - 0.42
- If A and B are two independent events and $P(C) = 0$, then A, B, C are :
 - independent
 - dependent
 - not pairwise independent
 - None of these
- A dice is thrown 6 times. If 'getting an odd number' is a 'success', the probability of 5 successes is
 - $\frac{1}{10}$
 - $\frac{3}{32}$
 - $\frac{5}{6}$
 - $\frac{25}{26}$
- A bag contains 5 white and 3 black balls, and 4 are successively drawn out and not replaced. What's the chance of getting different colours alternatively?
 - $\frac{1}{6}$
 - $\frac{1}{5}$
 - $\frac{1}{4}$
 - $\frac{1}{7}$
- A bag contains 5 white and 7 black balls and a man draws 4 balls at random. The odds against these being all black is
 - 7 : 92
 - 92 : 7
 - 92 : 99
 - 99 : 92
- The letters of the word SOCIETY are placed at random in a row. The probability that the three vowels come together is
 - $\frac{1}{6}$
 - $\frac{1}{7}$
 - $\frac{2}{7}$
 - $\frac{5}{6}$

14. Course materials are sent to students by a distance education institution. The probability that they will send a wrong programme's study material is $\frac{1}{5}$. There is a probability of $\frac{3}{4}$ that the package is damaged in transit, and there is a probability of $\frac{1}{3}$ that there is a short shipment. What is the probability that the complete material for the course arrives without any damage in transit ?
- (a) $\frac{4}{5}$ (b) $\frac{8}{60}$
(c) $\frac{8}{15}$ (d) $\frac{4}{20}$
15. A coin is tossed 5 times. What is the probability that head appears an odd number of times?
- (a) $\frac{2}{5}$ (b) $\frac{1}{5}$
(c) $\frac{1}{2}$ (d) $\frac{4}{25}$
16. Two dice are tossed. The probability that the total score is a prime number is
- (a) $\frac{1}{6}$ (b) $\frac{5}{12}$
(c) $\frac{1}{2}$ (d) $\frac{7}{9}$
17. The probability that the sum of the square of the two numbers, which show up when two fair dice are thrown, is even is
- (a) $\frac{3}{7}$ (b) $\frac{4}{7}$
(c) $\frac{5}{7}$ (d) None of these
18. There are 5 pairs of shoes in a cupboard from which 4 shoes are picked at random. The probability that there is at least one pair is
- (a) $\frac{8}{21}$ (b) $\frac{11}{21}$
(c) $\frac{13}{21}$ (d) $\frac{12}{31}$
19. A die is rolled three times, find the probability of getting a larger number than the previous number each time.
- (a) $\frac{5}{24}$ (b) $\frac{1}{24}$
(c) $\frac{5}{54}$ (d) $\frac{1}{8}$
20. The fair dice are thrown. The probability that the number appear are not all distinct is
- (a) $\frac{5}{9}$ (b) $\frac{4}{9}$
(c) $\frac{1}{6}$ (d) $\frac{5}{6}$
21. Two dice are thrown simultaneously. What is the probability of obtaining a multiple of 2 on one of them and a multiple of 3 on the other
- (a) $\frac{5}{36}$ (b) $\frac{11}{36}$
(c) $\frac{1}{6}$ (d) $\frac{1}{3}$
22. Two dice are thrown at a time, find the probability that the sums of the numbers on the upper faces of the dice are equal to 7.
- (a) $\frac{1}{8}$ (b) $\frac{1}{4}$
(c) $\frac{1}{3}$ (d) $\frac{1}{6}$
23. If 4 whole numbers are taken at random, and multiplied 27 together, the chance that the last digit in the product is 1, 3, 7 or 9 will be
- (a) $\frac{13}{976}$ (b) $\frac{17}{529}$
(c) $\frac{16}{625}$ (d) $\frac{13}{625}$
24. One card is drawn from a well-shuffled pack of 52 cards. What is the probability, that it is not the ace of hearts ?
- (a) $\frac{51}{52}$ (b) $\frac{1}{52}$
(c) $\frac{1}{12}$ (d) $\frac{1}{2}$
25. A dice is thrown twice. The probability of getting 4, 5 or 6 in the first throw and 1, 2, 3 or 4 in the second throw is
- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$
26. Ram and Shyam appear for an interview for two vacancies in an organisation for the same post. The probabilities of their selection are $\frac{1}{6}$ and $\frac{2}{5}$ respectively. What is the probability that none of them will be selected?
- (a) $\frac{5}{6}$ (b) $\frac{1}{5}$
(c) $\frac{1}{2}$ (d) $\frac{3}{5}$
27. Three groups of children contain respectively 3 girls and 1 boy, 2 girls and 2 boys and 1 girl and 3 boys. One child is selected at random from each group. The probability that the three selected consists of 1 girl and 2 boys is
- (a) $\frac{13}{32}$ (b) $\frac{12}{32}$
(c) $\frac{15}{32}$ (d) $\frac{11}{32}$
28. A class consists of 80 students, 25 of them are girls and 55 are boys. If 10 of them are rich and the remaining poor and also 20 of them are intelligent then the probability of selecting an intelligent rich girl is
- (a) $\frac{5}{128}$ (b) $\frac{25}{128}$
(c) $\frac{5}{512}$ (d) None of these

29. If the probability of A to fail in an examination is 0.2 and that for B is 0.3, then probability that either A or B is fail, is :
- (a) 0.5 (b) 0.44
(c) 0.8 (d) 0.25
30. The probability of choosing at random a number that is divisible by 6 or 8 from among 1 to 90 is equal to
- (a) $\frac{1}{6}$ (b) $\frac{1}{30}$
(c) $\frac{11}{80}$ (d) $\frac{23}{90}$
31. In single cast with two dice the odds against drawing 7 is
- (a) 5 (b) $\frac{1}{5}$
(c) 6 (d) $\frac{1}{6}$
32. From a group of 7 men and 4 women a committee of 6 persons is formed. What is the probability that the committee will consist of exactly 2 women?
- (a) $\frac{5}{11}$ (b) $\frac{3}{11}$
(c) $\frac{4}{11}$ (d) $\frac{2}{11}$
33. A natural number is chosen at random from the first 100 natural numbers. What is the probability that the number chosen is a multiple of 2 or 3 or 5?
- (a) $\frac{30}{100}$ (b) $\frac{1}{33}$
(c) $\frac{74}{100}$ (d) $\frac{7}{10}$
34. Two numbers a and b are chosen at random from the set of first 30 natural numbers. The probability that $a^2 - b^2$ is divisible by 3 is:
- (a) $\frac{37}{87}$ (b) $\frac{47}{87}$
(c) $\frac{17}{29}$ (d) None of these
35. A problem is given to three students whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved ?
- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) $\frac{7}{12}$
36. An article manufactured by a company consists of two parts X and Y. In the process of manufacture of the part X, 9 out of 100 parts may be defective. Similarly, 5 out of 100 are likely to be defective in the manufacture of the part Y. Calculate the probability that the assembled product will not be defective.
- (a) 0.6485 (b) 0.6565
(c) 0.8645 (d) None of these
37. If $P(A) = 3/7$, $P(B) = 1/2$ and $P(A' \cap B') = 1/14$, then are A and B mutually exclusive events?
- (a) No (b) Yes
(c) Either yes or no (d) Cannot be determined
38. Michael Jordan's probability of hitting any basketball shot in three times than mine, which never exceeds a third. To beat him in a game, I need to hit a shot myself and have Jordan miss the same shot. If I pick my shot optimally, what is the maximum probability of winning which I can attain?
- (a) $\frac{1}{16}$ (b) $\frac{1}{12}$
(c) $\frac{5}{6}$ (d) $1/14$
39. Triangles are formed by joining vertices of an octagon. Any one of those triangle is selected at random. What is the probability that the selected triangle has no side common with the octagon?
- (a) $3/7$ (b) $2/7$
(c) $5/7$ (d) $1/7$
40. One bag contains 4 white balls and 2 black balls. Another bag contains 3 white balls and 5 black balls. If one ball is drawn from each bag, determine the probability that one ball is white and another is black.
- (a) $6/24$ (b) $5/24$
(c) $7/24$ (d) $13/24$
41. A woman goes to visit the house of some friend whom she has not seen for many years. She knew that besides the two married adults in the household, there are two children of different ages. But she does not know their genders. When she knocks on the door of the house, a boy answers. What is the probability that the younger child is a boy?
- (a) $\frac{2}{3}$ (b) $\frac{1}{2}$
(c) $\frac{1}{3}$ (d) $\frac{1}{4}$
42. The odds against P solving a problem are 8 : 6 and odds in favour of Q solving the same problem are 14 : 10. The probability of the problem being solved, if both of them try it, is
- (a) $\frac{5}{21}$ (b) $\frac{16}{21}$
(c) $\frac{5}{12}$ (d) $\frac{5}{7}$

43. The probability that A can solve a problem is $\frac{2}{3}$ and B can solve it is $\frac{3}{4}$. If both attempt the problem, what is the probability that the problem gets solved?
- (a) $\frac{11}{12}$ (b) $\frac{7}{12}$
(c) $\frac{5}{12}$ (d) $\frac{9}{12}$
44. Atul can hit a target 3 times in 6 shots, Bhola can hit the target 2 times in 6 shots and Chandra can hit the 4 times in 4 shots. What is the probability that at least 2 shots (out of 1 shot taken by each one of them) hit the target?
- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) $\frac{1}{3}$ (d) $\frac{5}{6}$
45. Suppose six coins are tossed simultaneously. Then the probability of getting at least one tail is :
- (a) $\frac{71}{72}$ (b) $\frac{53}{54}$
(c) $\frac{63}{64}$ (d) $\frac{1}{12}$
46. A and B are events such that $P(A \cup B) = \frac{3}{4}$, $P(A \cap B) = \frac{1}{4}$, $P(\bar{A}) = \frac{2}{3}$ then $P(\bar{A} \cap B)$ is
- (a) $\frac{5}{12}$ (b) $\frac{3}{8}$
(c) $\frac{5}{8}$ (d) $\frac{1}{4}$
47. Seven digits from the numbers 1, 2, 3, 4, 5, 6, 7, 8, 9 are written in a random order. The probability that this seven digit number is divisible by 9 is
- (a) $\frac{2}{9}$ (b) $\frac{7}{36}$
(c) $\frac{1}{9}$ (d) $\frac{7}{12}$
48. A committee of 5 Students is to be chosen from 6 boys and 4 girls. Find the probability that the committee contains exactly 2 girls.
- (a) $\frac{10}{21}$ (b) $\frac{11}{21}$
(c) $\frac{12}{21}$ (d) $\frac{13}{21}$
49. There are 10 envelopes and 10 letters to go inside them. Each letter is meant for a specified envelope only. What is the probability that exactly 9 of them are in the right envelopes?
- (a) $1/10!$ (b) 1
(c) 0 (d) None of these
50. A three digit number is written down by random choice of the digits 1 to 9 with replacements. The probability that atleast one of the digits chosen is a perfect square is
- (a) $\frac{8}{27}$ (b) $\frac{4}{9}$
(c) $\frac{2}{9}$ (d) None of these
51. 4 gentlemen and 4 ladies take seats at random round a table. The probability that they are sitting alternately is
- (a) $4/35$ (b) $1/70$
(c) $2/35$ (d) $1/35$
52. Two cards are drawn one by one from a pack of cards. The probability of getting first card an ace and second a coloured one is (before drawing second card, first card is not placed again in the pack) :
- (a) $1/26$ (b) $5/52$
(c) $5/221$ (d) $4/13$
53. Seven people seat themselves indiscriminately at round table. The probability that two distinguished persons will be next to each other is
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{1}{4}$ (d) $\frac{2}{3}$
54. The odds against A solving a certain problem are 3 to 2 and the odds in favour of B solving the same problem are 2 to 1. The probability that the problem will be solved if they both try, is
- (a) $\frac{2}{5}$ (b) $\frac{11}{15}$
(c) $\frac{4}{5}$ (d) $\frac{2}{3}$
55. If two events A and B are such that $P(\bar{A}) = 0.3$, $P(B) = 0.4$ and $P(A \cap \bar{B}) = 0.5$ then $P\left(\frac{B}{A \cup \bar{B}}\right)$
- (a) 0.9 (b) 0.5
(c) 0.6 (d) 0.25
56. If A and B are two events such that $P(A) \neq 0$ and $P(B) \neq 1$, then $P\left(\frac{\bar{A}}{\bar{B}}\right)$
- (a) $1 - P\left(\frac{A}{B}\right)$ (b) $1 - P\left(\frac{\bar{A}}{B}\right)$
(c) $\frac{1 - P(A \cup B)}{P(\bar{B})}$ (d) $\frac{P(\bar{A})}{P(\bar{B})}$

57. Let $0 < P(A) < 1$, $0 < P(B) < 1$ and $P(A \cup B) = P(A) + P(B) - P(A)P(B)$, then :
- (a) $P(B/A) = P(B) - P(A)$
 (b) $P(A' \cup B') = P(A') + P(B')$
 (c) $P(A \cap B) = P(A')P(B')$
 (d) None of these
58. A group of investigators took a fair sample of 1972 children from the general population and found that there are 1000 boys and 972 girls. If the investigators claim that their research is so accurate that the sex of a new born child can be predicted based on the ratio of the sample of the population, then what is the expectation in terms of the probability that a new child born will be a girl?
- (a) $\frac{243}{250}$ (b) $\frac{250}{257}$
 (c) $\frac{9}{10}$ (d) $\frac{243}{493}$
59. A life insurance company insured 25,000 young boys, 14,000 young girls and 16,000 young adults. The probability of death within 10 years of a young boy, young girl and a young adult are 0.02, 0.03 and 0.15 respectively. One of the insured persons dies. What is the probability that the dead person is a young boy?
- (a) $\frac{36}{165}$ (b) $\frac{25}{166}$
 (c) $\frac{26}{165}$ (d) $\frac{32}{165}$
60. Eleven books, consisting of five Engineering books, four Mathematics books and two Physics books, are arranged in a shelf at random. What is the probability that the books of each kind are all together?
- (a) $\frac{5}{1155}$ (b) $\frac{2}{1155}$
 (c) $\frac{3}{1155}$ (d) $\frac{1}{1155}$
61. 12 persons are seated around a round table. What is the probability that two particular persons sit together?
- (a) $\frac{2}{11}$ (b) $\frac{1}{6}$
 (c) $\frac{3}{11}$ (d) $\frac{3}{15}$
62. The probability of a bomb hitting a bridge is $\frac{1}{2}$ and two direct hits are needed to destroy it. The least number of bombs required so that the probability of the bridge being destroyed is greater than 0.9 is:
- (a) 7 bombs (b) 3 bombs
 (c) 8 bombs (d) 9 bombs
63. A number is chosen at random from the numbers 10 to 99. By seeing the number, a man will sing if the product of the digits is 12. If he chooses three numbers with replacement, then the probability that he will sing at least once is:
- (a) $1 - \left(\frac{43}{45}\right)^3$ (b) $\left(\frac{43}{45}\right)^3$
 (c) $1 - \frac{48 \times 86}{90^3}$ (d) None of these
64. Probabilities that Rajesh passes in Maths, Physics and Chemistry are m , p and c respectively. Of these subjects, Rajesh has a 75% chance of passing in at least one, 50% chance of passing in at least two and 40% chance of passing in exactly two. Find which of the following is true.
- (a) $p + m + c = \frac{19}{20}$ (b) $p + m + c = \frac{27}{20}$
 (c) $pmc = \frac{1}{20}$ (d) $pmc = \frac{1}{8}$
65. Two small squares on a chess board are chosen at random. Find the probability that they have a common side:
- (a) $\frac{1}{12}$ (b) $\frac{1}{18}$
 (c) $\frac{2}{15}$ (d) $\frac{3}{14}$

Expert Level

- There are 6 positive and 8 negative numbers. Four numbers are chosen at random and multiplied. The probability that the product is a positive number is
 (a) $\frac{500}{1001}$ (b) $\frac{503}{1001}$
 (c) $\frac{505}{1001}$ (d) $\frac{101}{1001}$
- The probability of getting 10 in a single throw of three fair dice is
 (a) $\frac{1}{6}$ (b) $\frac{1}{8}$
 (c) $\frac{1}{9}$ (d) None of these
- In an examination, there are 500 students, 150 passed the first paper and 350 passed the second paper. 50 students passed both the papers. Find the probability that a student selected at random has failed in both the papers.
 (a) $\frac{1}{5}$ (b) $\frac{1}{10}$
 (c) $\frac{3}{10}$ (d) $\frac{3}{5}$
- A, B and C shoot to hit a target. If A hits the target 4 times in 5 trials, B hits it 3 times in 4 trials and C hits it 2 times in 3 trials. What is the probability that the target is hit by at least 2 persons?
 (a) $\frac{5}{6}$ (b) $\frac{3}{4}$
 (c) $\frac{4}{5}$ (d) $\frac{1}{9}$
- A bag contains 5 red and 4 green balls and another bag contains 3 red and 7 black balls. If a ball is drawn from each bag. Find the probability that both are of different colours.
 (a) $\frac{47}{90}$ (b) $\frac{5}{6}$
 (c) $\frac{7}{18}$ (d) $\frac{2}{15}$
- A MNC has two Grids – Grid I and Grid II. Out of 5 Directors and 4 General Managers of Grid I, one person is transferred to Grid II having already 3 Directors and 7 General Managers. If, one person superannuates from Grid II, then the probability that this person was a director is
 (a) $\frac{32}{99}$ (b) $\frac{4}{45}$
 (c) $\frac{20}{99}$ (d) $\frac{3}{10}$
- Two dice are thrown n times in succession. The probability of obtaining double – six at least once is
 (a) $\frac{1}{36}$ (b) $1 - \left(\frac{35}{36}\right)^n$
 (c) $\left(\frac{1}{36}\right)^n$ (d) $\left(\frac{35}{36}\right)^n$
- Two cards are drawn from a well shuffled deck of 52 cards. The probability that one is a red card and the other is a queen is
 (a) $\frac{101}{1326}$ (b) $\frac{4}{51}$
 (c) $\frac{16}{221}$ (d) None of these
- Each of two persons tosses three fair coins. The probability that they obtain the same number of heads is
 (a) $\frac{1}{2}$ (b) $\frac{5}{16}$
 (c) $\frac{7}{16}$ (d) None of these
- There are three events E_1 , E_2 and E_3 one of which must, and only one can happen. The odds are 7 to 4 against E_1 and 5 to 3 against E_2 . The odds against E_3 is
 (a) 4 : 11 (b) 3 : 8
 (c) 23 : 88 (d) 65 : 23
- If A and B are two events, the probability that at most one of these events occurs is :
 (a) $P(A') + P(B') - P(A' \cap B')$
 (b) $P(A') + P(B') + P(A \cup B) - 1$
 (c) $P(A \cap B') + P(A' \cap B) + P(A' \cap B')$
 (d) All above are correct.
- A and B are two independent events. The probability that both A and B occur is $\frac{1}{6}$ and the probability that neither of them occurs is $\frac{1}{3}$. The probability of occurrence of A is.
 (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
 (c) $\frac{5}{6}$ (d) $\frac{1}{6}$

13. In each of a set of games it is 2 to 1 in favour of the winner of the previous game. The chance that the player who wins the first game shall win three at least of the next four is
- (a) $\frac{8}{27}$ (b) $\frac{4}{81}$
(c) $\frac{4}{9}$ (d) None
14. Given two bags A and B as follows : Bag A contains 3 red and 2 white balls and bag B contains 2 red and 5 white balls. A bag is selected at random, a ball is drawn and put into the other bag, then a ball is drawn from the second bag. The probability that both balls drawn are of the same colour is
- (a) $\frac{187}{1680}$ (b) $\frac{901}{1680}$
(c) $\frac{439}{1680}$ (d) None of these
15. A positive integer N is selected such that $100 < N < 200$. The probability that it is divisible by either 4 or 7 is :
- (a) $\frac{38}{99}$ (b) $\frac{24}{99}$
(c) $\frac{34}{99}$ (d) $\frac{14}{99}$
16. If three vertices of a regular hexagon are chosen at random, then the chance that they form an equilateral triangle is :
- (a) $\frac{1}{3}$ (b) $\frac{1}{5}$
(c) $\frac{1}{10}$ (d) $\frac{1}{2}$
17. Two dice are thrown. The probability that the sum of the numbers coming up on them is 9, if it is known that the number 5 always occurs on the first die, is
- (a) $\frac{1}{6}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{2}$
18. If $\frac{1+4p}{4}$, $\frac{1-p}{2}$ and $\frac{1-2p}{2}$ are the probabilities of three mutually exclusive events, then value of p is
- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{2}{3}$
19. The probability of getting 10 in a single throw of three fair dice is
- (a) $\frac{1}{6}$ (b) $\frac{1}{8}$
(c) $\frac{1}{9}$ (d) $\frac{1}{5}$
20. The probability that when 12 balls are distributed among three boxes, the first will contain three balls is,
- (a) $\frac{2^9}{3^{12}}$ (b) $\frac{{}^{12}C_3 \cdot 2^9}{3^{12}}$
(c) $\frac{{}^{12}C_3 \cdot 2^{12}}{3^{12}}$ (d) None of these
21. If n integers taken at random are multiplied together, then the probability that the last digit of the product is 1, 3, 7 or 9 is
- (a) $\frac{2^n}{5^n}$ (b) $\frac{4^n - 2^n}{5^n}$
(c) $\frac{4^n}{5^n}$ (d) None of these
22. The probabilities of four cricketers A, B, C and D scoring more than 50 runs in a match are $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{10}$. It is known that exactly two of the players scored more than 50 runs in a particular match. The probability that these players were A and B is
- (a) $\frac{27}{65}$ (b) $\frac{5}{6}$
(c) $\frac{1}{6}$ (d) None of these
23. From a bag containing 4 white and 5 black balls a man draws 3 at random; what are the odds against these being all black?
- (a) 37 : 5 (b) 5 : 37
(c) 23 : 19 (d) 19 : 23
24. One hundred cards are numbered from 1 to 100. Find the probability that a card chosen at random has the digit 5.
- (a) $\frac{19}{100}$ (b) $\frac{11}{100}$
(c) $\frac{12}{100}$ (d) $\frac{1}{100}$
25. Two sisters A and B appeared for an audition. The probability of selection of A is $\frac{1}{5}$ and that of B is $\frac{2}{7}$. Find the probability that both of them are selected.
- (a) $\frac{2}{35}$ (b) $\frac{1}{35}$
(c) $\frac{4}{35}$ (d) $\frac{7}{35}$
26. A 5 digit number is formed by using the digits 0, 1, 2, 3, 5 and 5 without repetition. The probability that the number is divisible by 6 is
- (a) 0.08 (b) 0.17
(c) 0.18 (d) 0.36

27. If M and N are any two events. The probability, that exactly one of them occurs, is
- $P(M) + P(N) - P(M \cap N)$
 - $P(M) + P(N) + P(M \cap N)$
 - $P(M) + P(N)$
 - $P(M) + P(N) - 2P(M \cap N)$
28. It is known that at noon at a certain place the sun is hidden by clouds on an average two days out of every three. The chance that the sun will be shining at noon on at least four out of five specified future days is
- $\frac{1}{81}$
 - $\frac{16}{243}$
 - $\frac{11}{243}$
 - $\frac{7}{20}$
29. A, B, C in order, cut a pack of cards, replacing them after each cut, on condition that the first who cuts a spade shall win a prize. Then A 's chance of winning is
- $\frac{16}{37}$
 - $\frac{12}{37}$
 - $\frac{9}{37}$
 - $\frac{14}{37}$
30. What is the probability that any two different cards of a well-shuffled deck of 52 cards will be together in the deck if their suit is not considered?
- $\frac{9}{11}$
 - $\frac{10}{3}$
 - $\frac{8}{13}$
 - $\frac{10}{11}$
31. Three integers are chosen at random from the first 20 integers. The probability that their product is even, is
- $\frac{2}{19}$
 - $\frac{3}{29}$
 - $\frac{17}{19}$
 - $\frac{4}{29}$
32. The probability that two integers chosen at random and their product will have the same last digit is :
- $\frac{3}{10}$
 - $\frac{1}{25}$
 - $\frac{4}{15}$
 - $\frac{7}{15}$
33. The probability that the birth days of six different persons will fall in exactly two calendar months is
- $\frac{1}{6}$
 - ${}^{12}C_2 \times \frac{2^6}{12^6}$
 - ${}^{12}C_2 \times \frac{2^6 - 1}{12^6}$
 - $\frac{341}{12^5}$
34. A book has 999 pages. If a page is opened at random the probability that the sum of the digits in its number is 9, is
- $\frac{55}{999}$
 - $\frac{22}{333}$
 - $\frac{117}{999}$
 - $\frac{4}{37}$
35. Two integers x and y are chosen with replacement out of the set $\{0, 1, 2, 3, \dots, 10\}$. Then the probability that $|x - y| > 5$ is
- $\frac{81}{121}$
 - $\frac{30}{121}$
 - $\frac{25}{121}$
 - $\frac{20}{121}$
36. The coefficients a and b in the quadratic equation $x^2 + ax + b = 0$ are the numbers that appear when a pair of fair dice is tossed. The probability that the roots are real is-
- $\frac{15}{36}$
 - $\frac{17}{36}$
 - $\frac{19}{36}$
 - $\frac{21}{36}$
37. Four whole numbers taken at random are multiplied together. What is the chance that the last digit in the product is 1, 3, 7 or 9?
- $\frac{16}{625}$
 - $\frac{1}{210}$
 - $\frac{8}{125}$
 - $\frac{4}{25}$
38. A die is thrown 7 times. The chance that an odd number turns up at least 4 times, is
- $\frac{1}{4}$
 - $\frac{1}{2}$
 - $\frac{1}{8}$
 - None of these
39. A letter is taken from the word ASSISTANT and another from the word STATISTICS. What is the probability that both the letters are the same?
- $\frac{1}{45}$
 - $\frac{17}{90}$
 - $\frac{19}{90}$
 - $\frac{13}{90}$
40. A box contains 3 white and 2 red balls. If first drawing ball is not replaced then the probability that the second drawing ball will be red is
- $\frac{8}{25}$
 - $\frac{2}{5}$
 - $\frac{3}{5}$
 - $\frac{21}{35}$
41. An event X can happen with probability P , and event Y can happen with probability P' . What is the probability that exactly one of them happens?
- $P + P' - 2PP'$
 - $2PP' - P' + PP'$
 - $P - P' + 2PP'$
 - $2P'P - P' + P$

42. In a bombing of the Nathula pass, the Indian troops have to destroy a bridge on the pass. The bridge is such that it is destroyed when exactly 2 bombs hit it. A MIG-27 is dispatched in order to do the bombing. Flt Lt. Rakesh Sharma needs to ensure that there is at least 97% probability for the bridge to be destroyed. He knows that when he drops a bomb on the bridge the probability of the bomb hitting the bridge is 90%. Weather conditions and visibility being poor he is unable to see the bridge from his plane. How many bombs does he need to drop to be 95% sure that the bridge will be destroyed?
- (a) 3 (b) 4
(c) 5 (d) 6
43. 20 girls, among whom are A and B sit down at a round table. The probability that there are 4 girls between A and B is:
- (a) $\frac{17}{19}$ (b) $\frac{2}{19}$
(c) $\frac{13}{19}$ (d) $\frac{6}{19}$
44. Two integers x and y are chosen with replacement out of the set $\{0, 1, 2, 3, \dots, 10\}$. Then the probability that $|x - y| > 5$ is:
- (a) $\frac{7}{11}$ (b) $\frac{40}{121}$
(c) $\frac{35}{121}$ (d) $\frac{30}{121}$
45. A consignment of 15 wristwatches contain 4 defectives. The wristwatches are selected at random, one by one and examined. The ones examined are not put back. What is the probability that ninth one examined is the last defective?
- (a) $\frac{11}{195}$ (b) $\frac{17}{195}$
(c) $\frac{8}{195}$ (d) $\frac{16}{195}$
46. A letter is taken out at random from 'ASSISTANT' and another is taken out from 'STATISTICS'. The probability that they are the same letters is:
- (a) $\frac{35}{87}$ (b) $\frac{19}{90}$
(c) $\frac{19}{96}$ (d) None of these

Test Yourself

- In four schools B_1, B_2, B_3, B_4 the percentage of girls students is 12, 20, 13, 17 respectively. From a school selected at random, one student is picked up at random and it is found that the student is a girl. The probability that the school selected is B_2 , is
 (a) $\frac{6}{31}$ (b) $\frac{10}{31}$
 (c) $\frac{13}{62}$ (d) $\frac{17}{62}$
- The probability that a leap year will contain 53 Sunday is
 (a) $\frac{1}{7}$ (b) $\frac{2}{7}$
 (c) $\frac{4}{7}$ (d) $\frac{5}{7}$
- A bag contains 3 red and 7 black balls, two balls are taken out at random, without replacement, If the first ball taken out is red, then the probability that the second taken out red ball is
 (a) $\frac{1}{10}$ (b) $\frac{1}{15}$
 (c) $\frac{3}{10}$ (d) $\frac{2}{21}$
- The probabilities of a problem being solved by two students A and B are $\frac{1}{2}, \frac{1}{3}$ respectively. Then the probability of the problem being solved is :
 (a) $\frac{2}{3}$ (b) $\frac{4}{3}$
 (c) $\frac{1}{3}$ (d) 1
- The ratio of number of officers and ladies in the Scorpion Squadron and in the Gunners Squadron are 3 : 1 and 2 : 5 respectively. An individual is selected to be the chairperson of their association. The chance that this individual is selected from the Scorpions is $\frac{2}{3}$. Find the probability that the chairperson will be an officer.
 (a) $\frac{25}{42}$ (b) $\frac{13}{43}$
 (c) $\frac{11}{43}$ (d) $\frac{7}{42}$
- The probability that A speaks truth is $\frac{4}{5}$, while the probability for B is $\frac{3}{4}$. The probability that they contradict each other when asked to speak on a fact is
 (a) $\frac{4}{5}$ (b) $\frac{1}{5}$
 (c) $\frac{7}{20}$ (d) $\frac{3}{20}$
- A bag contains 5 black and 3 red balls. A ball is taken out from the bag and is not returned to it. If this process is repeated three times, then what is the probability of drawing a black ball in the next draw of a ball?
 (a) 0.7 (b) 0.625
 (c) 0.1 (d) None of these
- If two squares are chosen at random on a chess board, the probability that they have a side in common is
 (a) $\frac{1}{9}$ (b) $\frac{2}{7}$
 (c) $\frac{1}{18}$ (d) $\frac{2}{9}$
- In a horse race the odds in favour of three horses are 1 : 2, 1 : 3 and 1 : 4. The probability that one of the horse will win the race is
 (a) $\frac{37}{60}$ (b) $\frac{47}{60}$
 (c) $\frac{1}{4}$ (d) $\frac{3}{4}$
- The probability that a contractor will get a plumbing contract is $\frac{2}{3}$ and the probability that he will get an electric contract is $\frac{5}{9}$. If the probability of getting at least one contract is $\frac{4}{5}$, what is the probability that he will get both the contracts?
 (a) $\frac{19}{45}$ (b) $\frac{13}{45}$
 (c) $\frac{12}{35}$ (d) $\frac{11}{23}$
- One hundred identical coins each with probability P of showing up Heads are tossed once. If $0 < P < 1$ and the probability of Heads showing on 50 coins is equal to that of Heads showing on 51 coins, then value of P is
 (a) $\frac{1}{21}$ (b) $\frac{49}{101}$
 (c) $\frac{50}{101}$ (d) $\frac{51}{101}$

12. A bag contains 15 tickets numbered 1 to 15. A ticket is drawn and replaced. Then one more ticket is drawn and replaced. The probability that first number drawn is even and second is odd is
- (a) $\frac{56}{225}$ (b) $\frac{26}{578}$
- (c) $\frac{57}{289}$ (d) None of these
13. A and B are two independent events. The probability that both A and B occur is $\frac{1}{6}$ and the probability that neither of them occurs is $\frac{1}{3}$. The probability of occurrence of A is :
- (a) $\frac{5}{6}$ (b) $\frac{1}{6}$
- (c) $\frac{1}{5}$ (d) None of these
14. A pair of fair dice are rolled together till a sum of either 5 or 7 is obtained. The probability that 5 comes before 7 is
- (a) 0.45 (b) 0.4
- (c) 0.5 (d) 0.7
15. A book contains 1000 pages numbered consecutively. The probability that the sum of the digits of the number of a page is 9, is :
- (a) zero (b) $\frac{55}{1000}$
- (c) $\frac{33}{1000}$ (d) $\frac{44}{1000}$

Hints & Solutions

Foundation Level

1. (a) When two are thrown then there are 6×6 exhaustive cases $\therefore n = 36$. Let A denote the event "total score of 7" when 2 dice are thrown then $A = [(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)]$.
Thus there are 6 favourable cases.

$$\therefore m = 6 \quad \text{By definition } P(A) = \frac{m}{n}$$

$$\therefore P(A) = \frac{6}{36} = \frac{1}{6}$$

2. (a) $\therefore$ No of ways of drawing 2 white balls from 5 white balls = 5C_2 .

Also, No of ways of drawing 2 other from remaining 7

$$\text{balls} = {}^7C_2$$

Total number of balls = 12

$$\text{Hence, required probability} = \frac{{}^5C_2 \times {}^7C_2}{{}^{12}C_4} = \frac{14}{33}$$

3. (b) Total no. of outcomes when two dice are thrown = $n(S) = 36$ and the possible cases for the event that the sum of numbers on two dice is a prime number, are (1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 1), (5, 6), (6, 1), (6, 5)
Number of outcomes favouring the event = $n(A) = 15$

$$\text{Required probability} = \frac{n(A)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$

4. (c) $P(A) = \frac{1}{3} \quad P(\bar{A}) = \frac{2}{3}$

$$P(\text{bird killed}) = 1 - P(\text{none of 3 shots hit})$$

$$1 - \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} = \frac{19}{27}$$

5. (b) Since, A and B are independent events
 $\therefore A'$ and B' are also independent events
 $\Rightarrow P(A' \cap B') = P(A') \cdot P(B')$
 $(0.4)(0.7) = 0.28$

$$[\because P(A') = 1 - P(A), P(B') = 1 - P(B)]$$

6. (d)

7. (d) Doublets occur when the numbers thrown are (1, 1), (2, 2), (6, 6). Therefore the probability of a

$$\text{doublet occurring in single throw} = \frac{6}{36} = \frac{1}{6}$$

The probability of a doublet not occurring at all in three

$$\text{throws} = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$$

$$\text{Required probability} = 1 - \frac{125}{216} = \frac{91}{216}$$

8. (c) Required probability = $1/6$.

9. (b) Total probable ways = 8

Favourable number of ways = [HTH = THT]

$$\text{Hence required probability} = \frac{2}{8} = \frac{1}{4}$$

10. (b) Any of the six numbers 1, 2, 3, 4, 5, 6 may appear on the upper face $\therefore n = 6$

Number of odd numbers = 3, since the odd numbers are

1, 3, 5. $\therefore m = 3$.

$\therefore$ The required probability

$$= \frac{\text{number of favourable cases}}{\text{number of all cases}} = \frac{m}{n} = \frac{3}{6} = \frac{1}{2}$$

11. (d) n = Number of all cases = 6

m = Number of favourable cases = 4 (since the numbers that appear are 3, 4, 5, 6)

$$\therefore \text{The required probability} = P = \frac{m}{n} = \frac{4}{6} = \frac{2}{3}$$

12. (a) $S = (1, 2, 3, 4, 5, 6) \therefore n(S) = 6$

Let A be the event that the die shows a multiple of 2.

$$A = \{2, 4, 6\} \quad \therefore n(A) = 3$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

13. (a) India win atleast three matches

$$= {}^5C_3 \left(\frac{1}{2}\right)^5 + {}^5C_4 \left(\frac{1}{2}\right)^5 + {}^5C_5 \left(\frac{1}{2}\right)^5 = \left(\frac{1}{2}\right)^5$$

$$(16) = \left(\frac{1}{2}\right)$$

14. (d) $P = \frac{3}{4}, q = \frac{1}{4}, n = 5$

Required probability

$$= {}^5C_3 \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^2 + {}^5C_4 \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right) + {}^5C_5 \left(\frac{3}{4}\right)^5 = \frac{459}{512}$$

15. (b) Total no. of divisible by 4 between 1 to 80
 $80 = 4 + (n - 1)4$
 $80 = 4n$
 $\Rightarrow n = 20$

$$\therefore \text{Required probability} = \frac{{}^{20}C_2}{{}^{80}C_2} = \frac{19}{316}$$

16. (b) Here $n(S) = 6^2 = 36$
 Let E be the event "getting sum more than 7" i.e. sum of pair of dice = 8, 9, 10, 11, 12

$$\text{i.e., } E = \left\{ \begin{array}{ccccc} (2, 6) & (3, 5) & (4, 4) & (5, 3) & (6, 2) \\ (3, 6) & (4, 5) & (5, 4) & (6, 3) & \\ (4, 6) & (5, 5) & (6, 4) & & \\ (5, 6) & (6, 5) & (6, 6) & & \end{array} \right\}$$

$$\therefore n(E) = 15$$

$$\therefore \text{Required prob} \quad \frac{n(E)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$

17. (c) Number of sample points on throwing two dice
 $= 6 \times 6 = 36$
 The possible outcomes are
 (1, 4), (2, 3), (3, 2), (4, 1)
 The probability of obtaining a total score of 5 is
 $= \frac{4}{6 \times 6} = \frac{1}{9}$

18. (c) Given digits are 1, 2, 3, 4, 5
 Total no. of 2 digits numbers formed = $(5)^2 = 25$
 Favourable cases are 12, 24, 32, 44, 52
 No. of favourable cases = 5

$$\therefore \text{Required Probability} = \frac{5}{25} = \frac{1}{5}$$

19. (a) Total no. of cases = $6^3 = 216$
 16 can appear on three dice in following ways
 (6, 6, 4), (6, 5, 5), (6, 4, 6), (4, 6, 6), (5, 5, 6),
 (5, 6, 5).
 $\therefore$ No. of favourable cases = 6

$$\text{Hence, the required probability} \quad \frac{6}{6^3} = \frac{1}{36}$$

20. (b) Total number of bolts = 600
 Number of too large bolts = 20% of 600
 $\frac{20 \times 600}{100} = 120$
 Number of too small bolts = 10% of 600 = 60
 Number of suitable bolts = $600 - 120 - 60 = 420$

$$\text{Thus required probability} = \frac{420}{600} = \frac{7}{10}$$

21. (c) Total possible outcomes = 36
 E = Event of getting sum 7
 $= \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$
 F = Event of getting sum 11
 $= \{(6, 5), (5, 6)\}$
 Total no. of favourable cases = $6 + 2 = 8$
 Now required probability

$$\frac{\text{Total favourable cases}}{\text{Total outcomes}} = \frac{8}{36} = \frac{2}{9}$$

22. (a) $n(S) = 100$
 E = square of terms lies between 1 to 100.
 $= 1, 4, 9, 16, 25, 36, 49, 64, 81, 100$
 $n(E) = 10$

$$\therefore \text{Required probability} \quad \frac{n(E)}{n(S)} = \frac{10}{100} = \frac{1}{10}$$

23. (a) (AAAA), (LL), HBD

$$P = \frac{5!}{9!} = \frac{5! \times 4! \times 2!}{9!}$$

$$= \frac{24 \times 2}{9 \times 8 \times 7 \times 6} = \frac{1}{63}$$

24. (a) Total number of ways of selecting 3 integers from 20 natural numbers = ${}^{20}C_3 = 1140$
 Their product is a multiple of 3 means, at least one number is divisible by 3.
 The numbers which are divisible by 3 are 3, 6, 9, 12, 15, 18 and the number of ways of selecting atleast one of them is ${}^6C_1 \times {}^{14}C_2 + {}^6C_2 \times {}^{14}C_1 + {}^6C_3 = 776$

$$\text{Required Probability} = \frac{776}{1140} = \frac{194}{285}$$

25. (b) The probability that Krishna will be alive 10 years hence, is $\frac{7}{15}$
 So, probability that Krishna will be dead 10 years hence, the

$$= 1 - \frac{7}{15} = \frac{8}{15}$$

Also, probability that Hari will be alive 10 years hence is $\frac{7}{10}$

So, the probability that Hari will be dead 10 years

$$\text{hence,} = 1 - \frac{7}{10} = \frac{3}{10}$$

So, the probability that both Krishna and Hari will be dead 10 years hence

$$\frac{8}{15} \times \frac{3}{10} = \frac{24}{150}$$

26. (a) Total no. of arrangements of the letters of the word

$$\text{UNIVERSITY is } \frac{10!}{2!}.$$

No. of arrangements when both I's are together = 9!
So, the no. of ways in which 2 I's do not together

$$= \frac{10!}{2!} - 9!$$

∴ Required probability

$$= \frac{\frac{10!}{2!} - 9!}{\frac{10!}{2!}} = \frac{10! - 9! \cdot 2!}{10!}$$

27. (a) Total no. of players = 15
Total no. of batsmen = 8
Total no. of bowlers = 7
Total no. of players in the team = 11

∴ No. of ways to choose a team = ${}^{15}C_{11}$

∴ No. of way to choose 6 batsmen and 5 bowler

$$= {}^8C_6 \times {}^7C_5$$

$$\therefore \text{Required Probability} = \frac{{}^8C_6 \times {}^7C_5}{{}^{15}C_{11}}$$

28. (d) Total number of numbers = $4! = 24$
For odd nos. 1 or 3 has to be at unit's place
If 1 is at unit place, then total number of numbers
= $3! = 6$
And if 3 is at units place, then total number of numbers
= $3! = 6$
∴ Total number of odd number = $6 + 6 = 12$
∴ Required probability = $\frac{12}{24} = \frac{1}{2}$

29. (c) Required Probability = $P(X) \cdot P(\bar{Y}) + P(\bar{X}) \cdot P(Y)$

$$= \frac{60}{100} \times \frac{50}{100} + \frac{40}{100} \times \frac{50}{100} = \frac{1}{2}$$

30. (c) $P(3 \cup 4) = P(c) + P(d) - P(3 \cap 4)$

$$= \frac{8}{25} + \frac{6}{25} - \frac{2}{25} = \frac{12}{25}$$

31. (b) In a leap year there are 366 days in which 52 weeks and two days. The combination of 2 days may be : Sun-Mon, Mon-Tue, Tue-Wed, Wed-Thu, Thu-Fri, Fri-Sat, Sat-Sun.

$$P(53 \text{ Fri}) = \frac{2}{7}; P(53 \text{ Sat}) = \frac{2}{7}$$

$$\text{and } P(53 \text{ Fri and } 53 \text{ Sat}) = \frac{1}{7}$$

$$\therefore P(53 \text{ Fri or Sat}) = P(53 \text{ Fri}) + P(53 \text{ Sat}) - P(53 \text{ Fri and Sat})$$

$$= \frac{2}{7} + \frac{2}{7} - \frac{1}{7} = \frac{3}{7}$$

32. (d) Clearly, $P(A \cup B \cup C) = 1$

$$\Rightarrow P(A) + P(B) + P(C) = 1$$

$$\Rightarrow P(A) + \frac{1}{2}P(A) + \frac{1}{3}P(A) = 1$$

$$\Rightarrow \frac{11}{6}P(A) = 1$$

$$\Rightarrow P(A) = \frac{6}{11}$$

33. (d) Now, $P(A' \cap B') = P(A' \cup B')$
= $1 - P(A \cup B) = 1 - 0.8 = 0.2$
and $P(A' \cup B') = 1 - P(A \cap B) = 1 - 0.3 = 0.7$
But $P(A' \cup B') = P(A') + P(B') - P(A' \cap B')$
 $\Rightarrow 0.7 = P(A') + P(B') - 0.2$
 $\Rightarrow P(A') + P(B') = 0.9$

34. (c) Let E be the event of total of 12.

$E = (2, 2, 2, 3, 3), (2, 2, 3, 3, 2), (2, 3, 3, 2, 2),$
 $(3, 3, 2, 2, 2), (3, 2, 3, 2, 2), (3, 2, 2, 3, 2),$
 $(3, 2, 2, 2, 3), (2, 3, 2, 3, 2), (2, 3, 2, 2, 3),$
 $(2, 2, 3, 2, 3)$

$$n(E) = 10$$

Sample sapce contain total possibility = $2^5 = 32$

Hence, $n(s) = 32$

$$\text{So, } P(E) = \frac{n(E)}{n(S)} = \frac{10}{32} = \frac{5}{16}$$

35. (c) Since, probabilities of failure for engines A, B and C $P(A)$, $P(B)$ and $P(C)$ are 0.03, 0.02 and 0.05 respectively.

The aircraft will crash only when all the three engine fail. So, probability that it crashes = $P(A) \cdot P(B) \cdot P(C)$
= $0.03 \times 0.02 \times 0.05$

$$= 0.00003$$

Hence, the probability that the aircraft will not crash,
= $1 - 0.00003$

$$= 0.99997$$

36. (a) Total possible outcomes, $S = \{HHH, HHT, HTH, THT, TTH, THH, TTT, HTT\}$ and desired outcomes $E = \{HTH, THT\}$

$$\Rightarrow n(E) = 2 \text{ and } n(S) = 8$$

$$\text{Hence, required probability} = P(E) = \frac{n(E)}{n(S)} = \frac{2}{8} = \frac{1}{4}$$

37. (a) Probability of passing in mathematics $\frac{4}{9}$
 Probability of passing in physics $\frac{2}{5}$
 Probability of failure in physics $= 1 - \frac{2}{5} = \frac{3}{5}$
 Given that both the events are independent.
 Required probability $= \frac{4}{9} \times \frac{3}{5} = \frac{4}{15}$
38. (c) Probability of getting a diamond, $P(D) = \frac{13}{52}$ $\frac{1}{4}$
 and probability to king, $P(K) = \frac{4}{52} = \frac{1}{13}$
 So, required probability $= P(D) \cdot P(K)$
 $\frac{1}{4} \times \frac{1}{13} = \frac{1}{52}$
39. (c) A dice has six faces. So, in throwing of two dice, the number of exhaustive events is $6 \times 6 = 36$.
40. (c) 16 tickets are sold and 4 prizes are awarded. A person buys 4 tickets, then required probability $= \frac{4}{16} = \frac{1}{4}$
41. (c) Probability of getting a spade $= \frac{13}{52}$
 Probability of an ace $= \frac{4}{52}$
 and probability of getting a spade ace $= \frac{1}{52}$
 $\therefore$ Required probability $= \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$
 Odds against his winning $= \frac{1 - \frac{4}{13}}{\frac{4}{13}} = \frac{\frac{9}{13}}{\frac{4}{13}} = \frac{9}{4}$
42. (b) If both get one head then it is $\frac{1}{4} \times \frac{1}{4}$
 and if both get two heads then it is $\frac{1}{2} \times \frac{1}{2}$
 $\Rightarrow$ Prob (getting same number of heads) $= \frac{1}{4} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{2}$
 $\frac{1}{16} + \frac{1}{4} = \frac{5}{16}$
43. (a) Total possibility of 4 children, either girl or boy is $2^4 = 16$. Out of these there is one possibility in which

there will be no boy and only girls. So, total possibility of at least one boy is $16 - 1 = 15$

$$\Rightarrow P(\text{at least one boy}) = \frac{15}{16}$$

44. (c) Let $P(A)$ be the probability that the race will be won by A and $P(B)$ be the probability that the race will be won by B.

$$\therefore P(A) = \frac{1}{5} \quad \text{and} \quad P(B) = \frac{1}{6}$$

$\therefore$ Probability that the race will be won by

$$A \text{ or } B = P(A) + P(B) = \frac{1}{5} + \frac{1}{6} = \frac{11}{30}$$

45. (b) Required probability $= \frac{365}{365} \times \frac{1}{365} = \frac{1}{365}$

46. (b) A and B are mutually exclusive and exhaustive events with

$$P(A \cap B) = 0, P(A \cup B) = 1$$

we know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 1 = P(A) + 3P(A)$$

$$\Rightarrow P(A) = \frac{1}{4} \quad \therefore P(B) = \frac{3}{4}$$

$$\text{Hence, } P(\bar{B}) = 1 - P(B) = 1 - \frac{3}{4} = \frac{1}{4}$$

47. (c) As we know $P(A \cup B) \leq 1$

$$\therefore P(A) + P(B) - P(A \cap B) \leq 1$$

$$\Rightarrow 0.8 + 0.7 - P(A \cap B) \leq 1$$

$$\Rightarrow P(A \cap B) \geq 1.5 - 1$$

$$\Rightarrow P(A \cap B) \geq 0.5$$

Hence, the minimum value of $P(A \cap B)$ is 0.5.

48. (c) Possible samples are as follows

$$\{HHH, HTH, HHT, THH, TTH, THT, HTT, TTT\}$$

Let A be the event of getting one head.

Let B be the event of getting no head.

Favourable outcome for

$$A = \{TTH, THT, HTT\}$$

Favourable outcome for

$$B = \{TTT\}$$

Total no. of outcomes = 8

$$\therefore P(A) = \frac{3}{8}, P(B) = \frac{1}{8}$$

$\therefore$ Required probability = Probability of getting one head + Probability of getting no head

$$= P(A) + P(B) = \frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$$

49. (a) No. of blue balls = 2
No. of red balls = 7
Total no. of balls = 9
Required probability
= $P(\text{one ball is blue}) + P(\text{both ball is blue})$
$$\frac{2}{9} \times \frac{7}{8} + \frac{2}{9} \times \frac{1}{8} = \frac{14}{72} + \frac{2}{72} = \frac{16}{72} = \frac{2}{9}$$
50. (c) Given probability of guessing a correct answer $\frac{x}{12}$
and probability of not guessing the correct answer $\frac{2}{3}$
As we know
 $P(\text{occurrence of an event}) + P(\text{non-occurrence of an event}) = 1$
$$\therefore \frac{x}{12} + \frac{2}{3} = 1 \Rightarrow \frac{x+8}{12} = 1 \Rightarrow x = 12 - 8 = 4$$
51. (a) Since, A and B are mutually exclusive events.
 $\therefore P(AB) = P(A \cap B) = 0$
52. (b) Prob. (certain event) = 1
 $\Rightarrow P(E) = 1$
53. (a) Total number of attempts = 20
Favourable no. of attempts = 5
Required probability (running the program correctly in the third run) = $\frac{5}{20} = \frac{1}{4}$
54. (b) Total possible nine digit number = 9!
Out of these 9! numbers only those numbers are divisible by 4 which have their last digits as even natural number and the numbers formed by their last two digits are divisible by 4. The possible numbers of last two digits are
12, 32, 52, 72, 92, 24, 64, 84, 16, 36, 56, 76, 96, 28, 48, 68.
Thus there are 16 ways of choosing the last two digits. Corresponding to each of these ways the remaining 7 digits can be arranged in 7! ways. Therefore, the total number of 9 digits numbers divisible by 4 is $16 \times 7!$.
Hence, required probability = $\frac{16 \times 7!}{9!} = \frac{2}{9}$
55. (d) 2 cards can be drawn from the pack in ${}^{52}C_2$ ways.
Let A be the event "Two cards are red" and B be the event "Two cards drawn are kings".
The required probability is $P(A \cup B)$.
From addition theorem, we have
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$... (1)
Now, $P(A)$ = Probability of drawing two red cards
$$= \frac{{}^{26}C_2}{{}^{52}C_2} \quad [\because \text{There are total 26 red cards}]$$

$$P(B) = \text{Probability of drawing two king cards} \\ = \frac{{}^4C_2}{{}^{52}C_2} \quad [\because \text{There are 4 king cards}]$$

$$P(A \cap B) = \text{Probability of drawing 2 red king cards} \\ = \frac{{}^2C_2}{{}^{52}C_2} \quad [\because \text{There are just 2 red kings}]$$

Substituting the values in (a), we get

$$P(A \cup B) = \frac{{}^{26}C_2}{{}^{52}C_2} + \frac{{}^4C_2}{{}^{52}C_2} - \frac{{}^2C_2}{{}^{52}C_2} = \frac{325}{1326} + \frac{6}{1326} - \frac{1}{1326} \\ = \frac{55}{221}$$

56. (d) Probability that only husband is selected

$$P(H)P(\bar{W}) = \frac{1}{7} \left(1 - \frac{1}{5}\right) = \frac{1}{7} \times \frac{4}{5} = \frac{4}{35}$$

Probability that only wife is selected

$$= P(\bar{H})P(W) = \left(1 - \frac{1}{7}\right) \left(\frac{1}{5}\right) = \frac{6}{7} \times \frac{1}{5} = \frac{6}{35}$$

$\therefore$ Probability that only one of them is selected

$$\frac{4}{35} + \frac{6}{35} = \frac{10}{35} = \frac{2}{7}$$

57. (b) The probability that the person hits the target = 0.3
 $\therefore$ The probability that he does not hit the target in a trial = $1 - 0.3 = 0.7$
 $\therefore$ The probability that he does not hit the target in any of the ten trials = $(0.7)^{10}$
 $\therefore$ Probability that he hits the target
= Probability that at least one of the trials succeeds
= $1 - (0.7)^{10}$.
58. (c) If six coins are tossed, then the total no. of outcomes = $(2)^6 = 64$

Now, probability of getting no tail $\frac{1}{64}$

Probability of getting at least one tail

$$1 - \frac{1}{64} = \frac{63}{64}$$

59. (b) Suppose E_1 , E_2 and E_3 are the events of winning the race by the horses A, B and C respectively

$$\therefore P(E_1) = \frac{1}{1+3} = \frac{1}{4}, \quad P(E_2) = \frac{1}{1+4} = \frac{1}{5}$$

$$P(E_3) = \frac{1}{1+5} = \frac{1}{6}$$

$\therefore$ Probability of winning the race by one of the horses A, B and C

$$P(E_1 \text{ or } E_2 \text{ or } E_3) = P(E_1) + P(E_2) + P(E_3)$$

$$\frac{1}{4} + \frac{1}{5} + \frac{1}{6} = \frac{37}{60}$$

60. (c) Probability of selecting a month = $\frac{1}{12}$.
13th day of the month is Friday if its first day is Sunday
and the probability of this = $\frac{1}{7}$.
 $\therefore$ Required probability = $\frac{1}{12} \cdot \frac{1}{7} = \frac{1}{84}$.
61. (d) Total of seven can be obtained in the following ways
1, 1, 1, 4 in $\frac{4!}{3!} = 4$ ways
[there are four objects, three repeated]
Similarly,
1, 1, 2, 3 in $\frac{4!}{2!} = 12$ ways
1, 2, 2, 2 in $\frac{4!}{3!} = 4$ ways
Hence, required probability = $\frac{4+12+4}{6^4} = \frac{20}{6^4}$
[$\therefore$ Exhaustive no. of cases = $6 \times 6 \times 6 \times 6 = 6^4$]
62. (c) The number of ways of getting the different number 1, 2, ..., 6 in six dice = $6!$.
Total number of ways = 6^6
Hence, required probability = $\frac{6!}{6^6}$
$$\frac{1 \times 2 \times 3 \times 4 \times 5 \times 6}{6^6} = \frac{5}{324}$$
63. (d)
64. (a) Here the number of favourable cases, consists of throwing 10, 11 or 12 with the two dice. The number of ways in which a sum of 10 can be thrown are (4, 6), (5, 5), (6, 4) i.e. 3 ways. The number of ways in which a total of 11 can be thrown are (5, 6), (6, 5) i.e. 2 ways. The number of ways in which a total of 12 can be thrown in (6, 6) i.e. 1 way.
 m = number of favourable cases = $3 + 2 + 1 = 6$
 n = Total number of cases = $6 \times 6 = 36$
 $\therefore$ Probability = $P = \frac{m}{n} = \frac{6}{36} = \frac{1}{6}$
65. (c) Required probability
= $P(\text{Diamond}) \cdot P(\text{King}) = \frac{13}{52} \cdot \frac{4}{52} = \frac{1}{52}$
66. (a) The cards are of four colours and the number of cards of given description is 24.
The probability = $\frac{24}{52} \cdot \frac{23}{51} = \frac{46}{221}$.
67. (d) $P(1) = \frac{1}{6}$, $P(2) = \frac{1}{2}$, A and B are independent events,
that means $P(AB) = P(1) P(2) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}$
and $P(A^c B^c) = 1 - P(A \cup B)$
 $P(A^c B^c) = 1 - P(a) - P(b) + P(AB)$
$$= 1 - \frac{1}{6} - \frac{1}{2} + \frac{1}{12} = \frac{12-2-6+1}{12} = \frac{5}{12}$$
68. (b) Since A and B are independent
 $\therefore P(A \cap B) = P(A) \cdot P(B)$
and $P(A/B) = P(A)$
Thus, $P(A/B) = \frac{1}{2}$
Hence, option (b) is not true.
69. (c) Probability of the card being a spade or an ace
= $\frac{16}{52} = \frac{4}{13}$. Hence odds in favour is 4 : 9.
So, the odds against his winning is 9 : 4.
70. (c) The probability that a man will not live 10 more years = $3/4$ and the probability that his wife will not live 10 more years = $2/3$. Then the probability that neither will be alive in 10 years = $3/4 \times 2/3 = 1/2$
71. (b) Total number of possibilities = 25×25
Favourable cases for their winning = 25
 $\therefore P(\text{they win a prize}) = \frac{25}{25 \times 25} = \frac{1}{25}$
 $\therefore P(\text{they will not win a prize}) = 1 - \frac{1}{25}$
72. (c)
73. (d) Here $P(A) = 0.4$ and $P(\bar{A}) = 0.6$
Probability that A does not happen at all = $(0.6)^3$
Thus required probability = $1 - (0.6)^3 = 0.784$
74. (d) As there are four jacks and four aces, the number of favourable cases = 8
 $\therefore$ The required probability $P = \frac{8}{52} = \frac{2}{13}$
75. (a) The favourable cases are (1, 3), (2, 4), (3, 5), (4, 6) and (1, 4), (2, 5), (3, 6) and their reversed cases like (3, 1),
Total number of favourable cases = 2×7
 $\therefore P = \frac{14}{36} = \frac{7}{18}$
76. (a) The first card can be one of the 4 colours, the second can be one of the three and the third can be one of the two. The required probability is therefore
$$4 \times \frac{13}{52} \times 3 \times \frac{13}{51} \times 2 \times \frac{13}{50} = \frac{169}{425}$$

77. (a) Quadratic equations which are unchanged by squaring their roots = 4
whose roots are (0, 0), (0, 1), (1, 1) and (ω , ω^2)
If which two equations have equal roots

$$\therefore \text{Probability} = \frac{2}{4} \times \frac{1}{2}$$

78. (c) Total number of ways in which 4 persons can be selected out of 3 + 2 + 4 = 9 persons = ${}^9C_4 = 126$
Number of ways in which a selection of 4 contains exactly 2 children = ${}^4C_2 \times {}^5C_2 = 60$

$$\therefore \text{reqd. prob.} = \frac{60}{126} \times \frac{10}{21}$$

79. (b) $P(A) = 1/4$, $P(A/B) = \frac{1}{2}$, $P(B/A) = 2/3$

By conditional probability,

$$P(A \cap B) = P(A)P(B/A) = P(B)P(A/B)$$

$$\Rightarrow \frac{1}{4} \times \frac{2}{3} = P(B) \times \frac{1}{2} \Rightarrow P(B) = \frac{1}{3}$$

80. (c) Probability of getting a head on tossing a coin (P_1) = $\frac{1}{2}$.

Probability of getting a six on rolling a dice (P_2) = $\frac{1}{6}$.

These two events are independent.

So the probability that the coin shows the head and the dice shows 6 is given by

$$P = P_1 \times P_2 = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}.$$

Solutions for 81 and 82

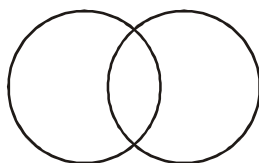
Total number of balls in urn – I = 1 Black + 1 Green = 2 Balls

Total number of balls in urn – II = 1 White + 1 Green = 2 Balls

81. (c) Required prob = $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

82. (d) Required prob

$$= \left(\frac{1}{2} \times \frac{1}{2}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$



83. (a)

We have: $\frac{19}{40} + \frac{1}{8} + \frac{5}{24} = \frac{97}{120}$

Standard Level

1. (d) If the number on the dice is 1, then the probability that 1 red ball is

$$\text{chosen} = \frac{{}^6C_1}{{}^{10}C_1}$$

If number on the dice is 2, then the } $\frac{{}^6C_2}{{}^{10}C_2}$
that two red balls can be chosen

and so on, till the number on the dice is six. Now, each number on the dice can appear with a probability of 1/6.

$\therefore$ Required probability

$$\frac{1}{6} \left[\frac{{}^6C_1}{{}^{10}C_1} + \frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^6C_3}{{}^{10}C_3} + \frac{{}^6C_4}{{}^{10}C_4} + \frac{{}^6C_5}{{}^{10}C_5} + \frac{{}^6C_6}{{}^{10}C_6} \right]$$

$$\frac{1}{6} \left[\frac{6}{10} + \frac{15}{45} + \frac{20}{120} + \frac{15}{210} + \frac{6}{252} + \frac{1}{210} \right] = \frac{1}{5}.$$

2. (b) We know,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow 0.8 + 0.9 - p \leq 1$$

$$\Rightarrow 1.7 - p \leq 1$$

$$\Rightarrow 0.7 \leq p$$

$$\text{Now, } P(A) < P(B)$$

$$\therefore P(A \cap B) \leq P(A)$$

$$\Rightarrow p \leq 0.8$$

$$\text{Hence, } 0.7 \leq p \leq 0.8$$

3. (a) $P(A) + P(B) + P(C) = 1 \rightarrow 2P(B)/3 + P(B) + P(B)/2 = 1$
 $\rightarrow 13P(B)/6 = 1 \rightarrow P(B) = 6/13$. Hence, $P(A) = 4/13$

4. (c) $n(s) = {}^{52}C_2 = 1326$

Let A = event of getting both red cards

and B = event of getting both queens

then $A \cap B$ = event of getting two red queens

$$n(A) = {}^{26}C_2 = 325, n(B) = {}^4C_2 = 6$$

$$n(A \cap B) = {}^2C_2 = 1$$

$$\therefore P(A) = \frac{325}{1326}, P(B) = \frac{6}{1326} \times \frac{1}{221}$$

$$P(A \cap B) = \frac{1}{1326}$$

$$\therefore P(\text{both red or both queens}) = P(A \cap B)$$

$$= P(A) + P(B) - P(A \cap B)$$

$$= \frac{325}{1326} + \frac{1}{221} - \frac{1}{1326} = \frac{55}{221}$$

5. (c) $n(S) = {}^{52}C_4$

Let E_1, E_2, E_3, E_4 be the events of getting all spades, all clubs, all hearts and all diamonds respectively.

Then $n(E_1) = {}^{13}C_4$

$n(E_2) = {}^{13}C_4$

$n(E_3) = {}^{13}C_4$

$n(E_4) = {}^{13}C_4$

$\therefore P(E_1) = \frac{{}^{13}C_4}{{}^{52}C_4}, P(E_2) = \frac{{}^{13}C_4}{{}^{52}C_4},$

$P(E_3) = \frac{{}^{13}C_4}{{}^{52}C_4}, P(E_4) = \frac{{}^{13}C_4}{{}^{52}C_4},$

Since E_1, E_2, E_3 and E_4 are mutually exclusive events.

$\therefore P(\text{getting all the 4 cards of the same suit})$

$P(E_1 \text{ or } E_2 \text{ or } E_3 \text{ or } E_4) = P(E_1) + P(E_2) + P(E_3) + P(E_4)$

$= 4 \times \left(\frac{{}^{13}C_4}{{}^{52}C_4} \right) = \frac{44}{4165}$

6. (a) Let E = the event that A gets a three and F the event that B gets a three

Then, $P(E) = \frac{1}{6}, P(F) = \frac{1}{6}$

$P(\bar{E}) = \frac{5}{6}, P(\bar{F}) = \frac{5}{6}$

Suppose A wins then, he gets a three in 1st or 3rd or 5th... throw etc.

$\therefore P(A \text{ wins}) = P[E \text{ or } (\bar{E} \bar{F} E) \text{ or } (\bar{E} \bar{F} \bar{E} \bar{F} E) \text{ or } \dots]$

$= P[E \text{ or } (\bar{E} \text{ and } \bar{F} \text{ and } E) \text{ or } (\bar{E} \text{ and } \bar{F} \text{ and } \bar{E} \text{ and } \bar{F} \text{ and } E) \text{ or } \dots]$

$= P(E) + P(\bar{E} \text{ and } \bar{F} \text{ and } E) + P(\bar{E} \text{ and } \bar{F} \text{ and } \bar{E} \text{ and } \bar{F} \text{ and } E) + \dots$

$= P(E) + P(\bar{E}) P(\bar{F}) P(E) + P(\bar{E}) P(\bar{F}) P(E) + \dots$

$= \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \dots$

$= \frac{1}{6} + \frac{1}{6} \left(\frac{5}{6} \right)^2 + \frac{1}{6} \left(\frac{5}{6} \right)^4 + \dots$

$= \frac{1}{6} \left[1 + \left(\frac{5}{6} \right)^2 + \left(\frac{5}{6} \right)^4 + \dots \right]$

$= \frac{1}{6} \cdot \frac{1}{1 - \left(\frac{5}{6} \right)^2} = \left(\frac{1}{6} \cdot \frac{36}{11} \right) = \frac{6}{11}$

Thus, $P(A \text{ wins}) = \frac{6}{11}$ and $P(B \text{ wins}) = \frac{5}{11}$

7. (a) The probability that A cannot solve the problem

$1 - \frac{2}{3} = \frac{1}{3}$

The probability that B cannot solve the problem

$1 - \frac{3}{4} = \frac{1}{4}$

The probability that both A and B cannot solve the

problem $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$

$\therefore$ The probability that at least one of A and B can solve

the problem $1 - \frac{1}{12} = \frac{11}{12}$

$\therefore$ The probability that the problem is solved $= \frac{11}{12}$

8. (b) A and B will contradict each other if one of the events $A \cap B'$ or $A' \cap B$ occurs. The probability of this happening is

$P[(A \cap B') \cup (A' \cap B)] = P(A \cap B') + P(A' \cap B)$
 $= P(A)P(B') + P(A')P(B),$

because A and B are independent. Therefore, putting $P(A) = 0.7$ and $P(B) = 0.8$ the required probability is $(0.7)(0.2) + (0.3)(0.8) = 0.38$.

9. (a) Since, A and B are independent events.

$\therefore P(A \cap B) = P(A)P(B)$

Further since, $A \cap C, B \cap C, A \cap B \cap C$ are subsets of C , we have

$P(A \cap C) \leq P(C) = 0$

$P(B \cap C) \leq P(C) = 0$

and $P(A \cap B \cap C) \leq P(C) = 0$

$\Rightarrow P(A \cap C) = 0 = P(A)P(C)$

$P(B \cap C) = 0 = P(B)P(C)$

$P(A \cap B \cap C) = 0 = P(A)P(B)P(C).$

Clearly A, B, C are pairwise independent as well as mutually independent. Thus, A, B, C are independent events.

10. (b) Let A be the event of getting an odd number.

Here, $n(S) = 6$ and

$n(A) = 3$

Probability of getting an odd number $\frac{3}{6} = \frac{1}{2}$

Hence, probability of not getting an odd number

$$1 - \frac{1}{2} = \frac{1}{2}$$

Required probability of 5 successes

$${}^6C_5 \times \left(\frac{1}{2}\right)^5 \times \frac{1}{2} = \frac{3}{32}$$

11. (d) Total number of balls = 8. Let the first drawn ball is

white, so required probability = $\frac{5}{8} \times \frac{3}{7} \times \frac{4}{6} \times \frac{2}{5} = \frac{1}{14}$.

But here we had started with a white ball. When we start with a black ball, the required probability

$$= \frac{3}{8} \times \frac{5}{7} \times \frac{2}{6} \times \frac{4}{5} = \frac{1}{14}.$$

Since these two cases are mutually exclusive.

$$\text{Total probability} = \frac{1}{14} + \frac{1}{14} = \frac{2}{14} = \frac{1}{7}.$$

12. (b) There are $7 + 5 = 12$ balls in the bag and the number of ways in which 4 balls can be drawn is ${}^{12}C_4$ and the number of ways of drawing 4 black balls (out of seven) is 7C_4 .

Hence, $P(4 \text{ black balls})$

$$\frac{{}^7C_4}{{}^{12}C_4} = \frac{7 \cdot 6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3 \cdot 4} \times \frac{1 \cdot 2 \cdot 3 \cdot 4}{12 \cdot 11 \cdot 10 \cdot 9} = \frac{7}{99}$$

Thus the odds against the event 'all black balls' are

$$\left(1 - \frac{7}{99}\right) : \frac{7}{99} \text{ i.e., } \frac{92}{99} : \frac{7}{99} \text{ or } 92 : 7.$$

13. (b) The word 'SOCIETY' contains seven distinct letters and they can be arranged at random in a row in 7P_7 ways, i.e., in $7! = 5040$ ways.

Let us now consider those arrangements in which all the three vowels come together. So in this case we have to arrange four letters. S, C, T, Y and a pack of three vowels in a row which can be done in 5P_5 i.e. $5! = 120$ ways.

Also, the three vowels in their pack can be arranged in 3P_3 i.e. $3! = 6$ ways.

Hence, the number of arrangements in which the three vowels come together is $120 \times 6 = 720$

$\therefore$ The probability that the vowels come together

$$= \frac{720}{5040} = \frac{1}{7}$$

14. (b) Probability (sending a correct programme)

$$1 - \frac{1}{5} = \frac{4}{5}$$

Probability (the packet is not damaged) $1 - \frac{3}{4} = \frac{1}{4}$

Probability (there is no short shipment) $1 - \frac{1}{3} = \frac{2}{3}$

Required probability $\frac{4}{5} \times \frac{1}{4} \times \frac{2}{3} = \frac{2}{15} = \frac{8}{60}$

15. (c) Probability of occurrence of head in a toss of a coin is $1/2$.

Required probability = Prob [Head appears once] + Prob. [Head appears thrice] + Prob. [Head appears five times]

$$= {}^5C_1 \left(\frac{1}{2}\right)^5 + {}^5C_3 \left(\frac{1}{2}\right)^5 + {}^5C_5 \left(\frac{1}{2}\right)^5$$

$$= \left(\frac{1}{2}\right)^5 [5 + 10 + 1] = \frac{16}{32} = \frac{1}{2}$$

16. (b) Total no. of outcomes when two dice are thrown = $n(S) = 36$ and the possible cases for the event that the sum of numbers on two dice is a prime number, are (1, 1), (1, 2), (1, 4), (1, 6), (2, 1), (2, 3), (2, 5), (3, 2), (3, 4), (4, 1), (4, 3), (5, 1), (5, 6), (6, 1), (6, 5)

Number of outcomes favouring the event = $n(A) = 15$

$$\text{Required probability} = \frac{n(A)}{n(S)} = \frac{15}{36} = \frac{5}{12}$$

17. (d) Out of 36 possible outcomes the ones which are favourable for the event are

(i) When the numbers are both even and

(ii) When the numbers are both odd. There are six doublets and the pairs. (1, 3), (1, 5), (2, 4), (2, 6) etc. Which make a total of $6 \times 3 = 18$. The required probability is $1/2$.

18. (c) There are 5 pairs of shoes and 4 shoes can be picked in $10 \times 9 \times 8 \times 7$ ways. Number of ways in which 4 shoes can be picked such that no two are alike = $10 \times 8 \times 6 \times 4$.

$$\text{The required probability} = 1 - \frac{10 \times 8 \times 6 \times 4}{10 \times 9 \times 8 \times 7} = \frac{13}{21}.$$

19. (c) The total number of ways = $6 \times 6 \times 6 = 216$. Now we count the number of favourable ways. Clearly the second number has to be greater than 1.

If the second number is i ($i > 1$)

Thus the number of favourable ways

$$\sum_{i=1}^5 (i-1)(6-i) = 1 \times 4 + 2 \times 3 + 3 \times 2 + 4 \times 1 = 20$$

$$\therefore \text{Required probability} = \frac{20}{216} = \frac{5}{54}$$

20. (b) Out of the 6^3 possible outcomes 6.5.5 outcomes will have all distinct numbers.
- The probability = $1 - \frac{6.5.4}{6^3} = \frac{4}{9}$.
21. (b) Favourable cases for one are there i.e., 2, 4 and 6 and for other are two i.e., 3, 6.
- Hence required probability = $\left[\left(\frac{3 \times 2}{36} \right)^2 - \frac{1}{36} \right] = \frac{11}{36}$
- [As same way happen when dice changes numbers among themselves]
22. (d) If a die is thrown, there are 6 equally likely and mutually exclusive cases. Since two dice are thrown, the total number of ways = $6 \times 6 = 36$. If a sum of 7 is to be obtained from the numbers appearing on the two upper faces, the numbers in the two dice can be (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1), which are six in number.
- $\therefore$ Number of favourable cases = $m = 6$
- Total number of cases = 36
- $\therefore$ The required probability = $p = \frac{m}{n} = \frac{6}{36} = \frac{1}{6}$
23. (c) If the last digit in the product is not 1, 3, 5, 7 or 9, it must be 0, 5, or even.
- Hence none of the four numbers must end in 0, 2, 4, 5, 6, 8.
- Hence out of ten numbers in which any of the four whole numbers may end, four are favourable to the event that none of the four numbers must end in 0, 2, 4, 5, 6, 8.
- Hence the probability that each of the four numbers must not end in any of these = $\frac{4}{10} \times \frac{2}{5}$
- $\Rightarrow$ Required probability = $\left(\frac{2}{5} \right)^4 = \frac{16}{625}$.
24. (a) The ace of hearts can be drawn in only 1 way ($\therefore$ in a pack of cards there is only one ace of heart)
- $P(A)$ = Probability of drawing the ace of hearts = $\frac{1}{52}$.
- Hence the probability of not drawing an ace of hearts
- $= P(\bar{A}) = 1 - P(A) = 1 - \frac{1}{52} = \frac{51}{52}$
25. (a) Let $P(A)$ and $P(B)$ be the probability of the events of getting 4, 5 or 6 in the first throw and 1, 2, 3 or 4 in the second throw respectively, then
- $P(A \text{ and } B) = P(A).P(B) = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$
26. (c) Required probability = $\left(1 - \frac{1}{6} \right) \times \left(1 - \frac{2}{5} \right) = \frac{5}{6} \times \frac{3}{5} = \frac{1}{2}$
27. (b) The event definition is
- A girl is selected from the first group and one boy each are selected from the second and third groups. OR A girl is selected from the second group and one boy each are selected from the first and third groups. OR A girl is selected from the third group and one boy each are selected from the first and second groups.
28. (c) Total 80, Girls = 25, Boys = 55
10 R, 70 P, 20 I
- $\frac{1}{4} \times \frac{1}{8} \times \frac{25}{80} = \frac{5}{512}$
29. (b) Given $P(A_f) = 0.2$ and $P(B_f) = 0.3$
- Since, A and B are independent events
- $\therefore P(A \cap B) = P(A).P(B)$
- $(0.2) \times (0.3) = 0.06$
- $\therefore$ Required prob = $P(A \cup B)$
- $P(A) + P(B) - P(A \cap B)$
- $= 0.2 + 0.3 - 0.06 = 0.44$
30. (d) Nos. divisible by 6 are 6, 12, 18,, 90.
Nos. divisible by 8 are 8, 16, 24,, 88.
Now, total no. divisible by 6 = 15
and total no. divisible by 8 = 11
Now, the no. divisible by both 6 and 8 are 24, 48, 72.
So, total no. divisible by both 6 and 8 = 3
- $\therefore$ Probability (number divisible by 6 or 8)
- $= \frac{15 + 11 - 3}{90} = \frac{23}{90}$
31. (a) Let $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$
- $\therefore P(E) = \frac{6}{6 \times 6} = \frac{1}{6}$
- So, odds against drawing 7
- $\frac{P(\bar{E})}{P(E)} = \frac{1 - \frac{1}{6}}{\frac{1}{6}} = \frac{5}{1}$
32. (a) ${}^6C_2 \times [(7/11) \times (6/10) \times (5/9) \times (4/8) \times (4/7) \times (3/6)] = 5/11$.
33. (c) $n(S) = 100$
- Let A be the event of getting a number divisible by 2 and B be the event of getting a number divisible by 3 and C be the event of getting of number divisible by 5.
- $\therefore (A \cap B)$ be the event of getting a number divisible by both 2 and 3.
- $(B \cap C)$ be the event of getting a number divisible by both 3 and 5.
- $(A \cap C)$ be the event of getting a number divisible by both 2 and 5.
- $(A \cap B \cap C)$ be the event of getting a number divisible by A, B and C.

Now, $n(A) = 50$, $n(B) = 33$, $n(C) = 20$, $n(A \cap B) = 16$,
 $n(B \cap C) = 6$, $n(A \cap C) = 10$, $n(A \cap B \cap C) = 3$

$$\therefore P(A) = \frac{50}{100} = \frac{1}{2}, \quad P(B) = \frac{33}{100},$$

$$P(C) = \frac{20}{100} = \frac{1}{5}, \quad n(A \cap B) = \frac{16}{100}$$

$$n(B \cap C) = \frac{6}{100}, \quad n(A \cap C) = \frac{10}{100},$$

$$n(A \cap B \cap C) = \frac{3}{100}$$

Required probability = $P(A \cup B \cup C)$

$$= P(A) + P(B) + P(C) - [P(A \cap B) + P(B \cap C) + P(A \cap C)] + P(A \cap B \cap C)$$

$$= \frac{50}{100} + \frac{33}{100} + \frac{20}{100} - \left(\frac{16}{100} + \frac{6}{100} + \frac{10}{100} \right) = \frac{3}{100}$$

$$= \frac{74}{100}$$

34. (b) Out of 30 numbers 2 numbers can be chosen in ${}^{30}C_2$ ways. So, exhaustive number of cases = ${}^{30}C_2 = 435$
 Since $a^2 - b^2$ is divisible by 3 if either a and b are divisible by numbers, of cases = ${}^{10}C_2 + {}^{20}C_2 = 235$

$$\text{Hence, required probability} = \frac{235}{435} = \frac{47}{87}$$

35. (c) Let A, B, C be the respective events of solving the problem and $\bar{A}, \bar{B}, \bar{C}$ be the respective events of not solving the problem. Then A, B, C are independent events

$\therefore \bar{A}, \bar{B}, \bar{C}$ are independent events

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{3} \text{ and } P(C) = \frac{1}{4}$$

$$P(\bar{A}) = \frac{1}{2}, P(\bar{B}) = \frac{2}{3} \text{ and } P(\bar{C}) = \frac{3}{4}$$

$$\therefore P(\text{none solves the problem}) = P(\text{not } A) \text{ and } (\text{not } B) \text{ and } (\text{not } C)]$$

$$= P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= P(\bar{A}) P(\bar{B}) P(\bar{C}) \quad (\because \bar{A}, \bar{B} \text{ and } \bar{C} \text{ are independent})$$

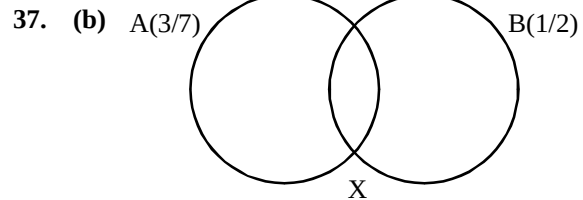
$$= \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{1}{4}$$

Hence, $P(\text{the problem will be solved})$

$$= 1 - P(\text{none solves the problem})$$

$$= 1 - \frac{1}{4} = \frac{3}{4}$$

36. (c) Required probability
 $= P(X \text{ not defective and } Y \text{ not defective})$
 $= P(\bar{X})P(\bar{Y})$
 $= \{1 - P(X)\}\{1 - P(Y)\}$
 $= \frac{91}{100} \times \frac{95}{100} = \frac{8645}{10000} = 0.8645$



$$\text{Also } P(A \cup B) = 1 - P(A' \cap B')$$

$$(3/7) + (1/2) - x = 13/14 \rightarrow x = 0$$

Thus, there is no interference between A and B as $P(A \cup B) = x = 0$. Hence, A and B are mutually exclusive.

38. (b) If I choose a shot that I will make probability p (where p is between 0 to $1/3$), then Michael Jordan will make the same shot with probability $3p$. Hence, the probability that I make a shot that Jordan subsequently misses is $p(1 - 3p)$. The graph of this function is a

parabola which equals zero when $p = 0$ and $p = \frac{1}{3}$.

By symmetry the vertex (maximum) is midway between the two, at $p = \frac{1}{6}$.

Hence, the best chance I have of winning the game is $\frac{1}{12}$.

39. (b) Total number of triangles formed = ${}^8C_3 = 56$
 Triangles having two sides common = 8
 Triangles having one side common = ${}^8C_1 \times {}^4C_1 = 32$
 Triangle having three sides common = 0
 Triangles having no side common = $56 - 40 = 16$
 So, probability = $16/56 = 2/7$

40. (d) Probability that first ball is white and second black
 $= (4/6) \times (5/8) = 5/12$
 Probability that first ball is black and second white
 $= (2/6) \times (3/8) = 1/8$
 These are mutually exclusive events hence the required probability

$$P = \frac{5}{12} + \frac{1}{8} = \frac{13}{24}$$

41. (a) The total possible pairs of children (B, B), (B, G), (G, B). Now the one child is boy, is confirmed, but we don't know whether he is youngest or elder one. So the three ordered pairs could be the one describing the children in this family. So the probability of the younger children to be boy = $\frac{2}{3}$.

42. (b) The odd against P solving a problem = 8 : 6.

$$\therefore \text{Probability of } P \text{ not solving the problem } \frac{8}{14} = \frac{4}{7}$$

The odds in favour of Q solving problem = 14 : 10

$$\therefore \text{Probability of } Q \text{ not solving the problem } \frac{10}{24} = \frac{5}{12}$$

Hence, the probability of P and Q not solving the problem

$$\frac{4}{7} \times \frac{5}{12} = \frac{5}{21}$$

$\therefore$ Probability of the problem being solved
= 1 - probability of the problem not being solved

$$1 - \frac{5}{21} = \frac{16}{21}$$

43. (a) The probability that A cannot solve the problem

$$1 - \frac{2}{3} = \frac{1}{3}$$

The probability that B cannot solve the problem

$$1 - \frac{3}{4} = \frac{1}{4}$$

The probability that both A and B cannot solve the

$$\text{problem } \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$$

$\therefore$ The probability that at least one of A and B can solve

$$\text{the problem } 1 - \frac{1}{12} = \frac{11}{12}$$

$$\therefore \text{The probability that the problem is solved} = \frac{11}{12}$$

44. (b) Chandra hits the target 4 times in 4 shots. Hence, he hits the target definitely.

The required probability, therefore, is given by.

$P(\text{both Atul and Bhola hit}) + P(\text{Atul hits, Bhola does not hit}) + P(\text{Atul does not hit, Bhola hits})$

$$= \frac{3}{6} \times \frac{2}{6} + \frac{3}{6} \times \frac{4}{6} + \frac{3}{6} \times \frac{2}{6}$$

$$\frac{1}{6} + \frac{1}{3} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$$

45. (c) If six coins are tossed, then the total no. of outcomes = $(2)^6 = 64$

Now, probability of getting no tail $\frac{1}{64}$

Probability of getting at least one tail

$$1 - \frac{1}{64} = \frac{63}{64}$$

46. (a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$;

$$\Rightarrow \frac{3}{4} = 1 - P(\bar{A}) + P(B) - \frac{1}{4}$$

$$\Rightarrow 1 = 1 - \frac{2}{3} + P(B) \Rightarrow P(B) = \frac{2}{3}$$

$$\text{Now, } P(\bar{A} \cap B) = P(B) - P(A \cap B) = \frac{2}{3} - \frac{1}{4} = \frac{5}{12}$$

47. (c) A number is divisible by 9, if the sum of its digits is divisible by 9. Here $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 45$ is divisible by 9.

$\therefore$ the two numbers to be removed should be such that their sum is 9.

$\therefore$ they can be any one of the following pairs

(1, 8), (2, 7), (3, 6), (4, 5).

Hence the number of favourable cases = 4

Total number of cases of removing two numbers = 9C_2

$$\therefore \text{Required probability} = \frac{4}{{}^9C_2} = \frac{4}{36} = \frac{1}{9}$$

48. (a) 5 Students can be selected from 10 in ${}^{10}C_5$ ways.

$$\therefore n(S) = {}^{10}C_5 = \frac{10!}{5!5!} = \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2} = 252$$

Let A be the event that the committee includes exactly 2 girls and 3 boys. The two girls can be selected in 4C_2 ways and the 3 boys can be selected in 6C_3 ways.

$$\therefore n(A) = {}^4C_2 \times {}^6C_3 = 6 \times 20 = 120$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{120}{252} = \frac{10}{21}$$

49. (c) If 9 letters are in right envelope, then the 10th automatically has to go to the right envelope.

Hence the probability of such occurrence is 0.

50. (d) The probability that none of the digits chosen turn out

to be a perfect square = $\left(\frac{6}{9}\right)^3 = \frac{8}{27}$, as 1, 4, 9 are the

ones which are perfect squares, the required probability

$$= \frac{19}{27}$$

51. (d) $n(S) = 7!$, $n(E) = (3!) \times (4!)$

$$\therefore P(E) = \frac{(3!) \times (4!)}{7!} = \frac{6}{7 \times 6 \times 5} = \frac{1}{15} = \frac{1}{35}$$

52. (c) Let A be the event of getting first card an ace and B be the event of getting second a coloured one. Since, both the events associated with a random experiment.

(i.e. condition of probability)

Therefore, the probability of getting first card an ace

$$P(A) = \frac{4}{52} = \frac{1}{13}$$

and probability of drawing a coloured one in second draw

$$P(B/A) = \frac{15}{51} = \frac{5}{17}$$

(since one card has already been drawn)

Hence, by conditional probability,

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

$$\Rightarrow \frac{5}{17} = \frac{P(A \cap B)}{\frac{1}{13}}$$

$$\Rightarrow P(A \cap B) = \frac{5}{17} \times \frac{1}{13} = \frac{5}{221}$$

53. (a) Seven people can seat themselves at a round table in $6!$ ways. The number of ways in which two distinguished persons will be next to each other = $2(5)!$, Hence, the required probability

$$\frac{2(5)!}{6!} = \frac{1}{3}$$

54. (c) Probability of A solving the problem = $\frac{2}{5}$.

$$\text{Probability of } B \text{ solving the problem} = \frac{2}{3}$$

As A 's solving the problem and B 's solving the problem are two independent events, required probability

$$= \frac{2}{5} + \frac{2}{3} - \frac{2}{5} \times \frac{2}{3} = \frac{4}{5}$$

55. (d) $P\left(\frac{B}{A \cup \bar{B}}\right) = \frac{P[B \cap (A \cup \bar{B})]}{P(A \cup \bar{B})}$

$$= \frac{P[B \cap A] \cup (B \cap \bar{B})}{P(A) + P(\bar{B}) - P(A \cap \bar{B})}$$

Given

$$\Rightarrow P(A \cap \bar{B}) = 0.5$$

$$\Rightarrow P(A) - P(A \cap B) = 0.5$$

$$\Rightarrow P(A \cap B) = P(A) - P(A \cap \bar{B}) = 0.7 - 0.5 = 0.2$$

$$P\left(\frac{B}{A \cup \bar{B}}\right) = \frac{P(A \cap B)}{P(A) + P(\bar{B}) - P(A \cap \bar{B})}$$

$$\frac{0.2}{0.7 + 0.6 - 0.5} = \frac{0.2}{0.8} = \frac{1}{4} = 0.25$$

56. (c) $P\left(\frac{\bar{A}}{\bar{B}}\right) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{P(\overline{A \cup B})}{P(\bar{B})} = \frac{1 - P(A \cup B)}{P(\bar{B})}$

57. (d) Given $P(A) + P(B) - P(A \cap B) = P(A \cup B)$

Comparing with

$$P(A) + P(B) - P(A \cap B) = P(A \cup B)$$

we get $P(A \cap B) = P(A) \cdot P(B)$

$\therefore A$ and B independent events.

58. (d) $972/1972 = 243/493$.

59. (b) The required probability will be given by the expression:

$$\frac{\text{The number of young boys who will die}}{\text{The total number of people who will die}}$$

60. (d) $\frac{(5! \times 4! \times 2! \times 3!)}{11!} = \frac{24 \times 2 \times 6}{11 \times 10 \times 9 \times 8 \times 7 \times 6} = 1/1155$.

61. (a)

$$P = \frac{\text{Total no. of ways in which two people sit together}}{\text{Total no. of ways}}$$

$$= (10! \times 2!)/11!$$

62. (a) Try to find the number of ways in which 0 or 1 bomb hits the bridge if n bombs are thrown.

The required value of the number of bombs will be such that the probability of 0 or 1 bomb hitting the bridge should be less than 0.1.

63. (a) The number of events for the condition that he will sing = 4 [34, 43, 26, 62]

The number of events in the sample = 90.

Probability that he will sing at least once

= $1 - \text{Probability that he will not sing}$.

64. (b) At least one means (exactly one + exactly two + exactly three)

At least two means (exactly two + exactly three)

The problem gives the probabilities for passing in at least one, at least two and exactly two.

65. (b) The common side could be horizontal or vertical. Accordingly, the number of ways the event can occur is.

$$n(E) = 8 \times 7 + 8 \times 7 = 112$$

$$n(S) = {}^{64}C_2$$

$$\text{Required probability} = \frac{2 \times 8 \times 7 \times 2}{64 \times 63} = \frac{1}{18}$$

Expert Level

1. (c) Total no. of numbers = 6 positive + 8 negative = 14
 $n(S) = {}^{14}C_4$

The product of four numbers could be positive when,

- (a) all the four numbers chosen are positive or
 (b) all the four numbers chosen are negative or
 (c) two of the chosen numbers are positive and two are negative.

$$\begin{aligned} \text{Required Prob.} &= \frac{{}^6C_4}{{}^{14}C_4} + \frac{{}^8C_4}{{}^{14}C_4} + \frac{{}^6C_2 \times {}^8C_2}{{}^{14}C_4} \\ &= \frac{505}{1001} \end{aligned}$$

2. (b) With three dice the cases favourable for the event of getting 10 are

$$\left. \begin{array}{l} 1+3+6 \\ 1+4+5 \\ 2+3+5 \end{array} \right\} \times 3! \quad \left. \begin{array}{l} 2+2+6 \\ 2+4+4 \\ 3+3+4 \end{array} \right\} \times \frac{3!}{2!} \Rightarrow \text{Total of 27.}$$

$$\text{Required probability} = \frac{27}{6^3} = \frac{1}{8}$$

3. (b) Total number students = Number passing in I paper only + Number passing II paper only + Number passing in both + Number failed in both
 $\Rightarrow 500 = (150 - 50) + (350 - 50) + X$

$$\Rightarrow X = 50 \Rightarrow \text{Required probability} = \frac{50}{500} = \frac{1}{10}$$

4. (a) Let A, B, C be the events that A hits the target, B hits the target and C hits the target respectively.

$$\text{Then, } P(A) = \frac{4}{5}, P(B) = \frac{3}{4}, P(C) = \frac{2}{3}$$

$$P(\bar{A}) = \frac{1}{5}, P(\bar{B}) = \frac{1}{4}, P(\bar{C}) = \frac{1}{3}$$

Case I. $P(A, B, \text{ and } C, \text{ all hit the target}) = P(A \cap B \cap C)$

$$= P(A) P(B) P(C) = \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} = \frac{2}{5}$$

Case II. $P(A \text{ and } B \text{ hit but not } C) = P[A \cap B \cap \bar{C}]$

$$= P(A) P(B) P(\bar{C}) = \frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{1}{5}$$

Case III. $P(A \text{ and } C \text{ hit but not } B) = P[A \cap C \cap \bar{B}]$

$$= P(A) P(C) P(\bar{B}) = \frac{4}{5} \times \frac{2}{3} \times \frac{1}{4} = \frac{2}{15}$$

Case IV. $P(B \text{ and } C \text{ hit but not } A) = P[B \cap C \cap \bar{A}]$

$$= P(B) P(C) P(\bar{A}) = \frac{3}{4} \times \frac{2}{3} \times \frac{1}{5} = \frac{1}{10}$$

All the above cases being mutually exclusive, we have the required probability

$$= \frac{2}{5} + \frac{1}{5} + \frac{2}{15} + \frac{1}{10} = \frac{5}{6}$$

5. (b)

5 Red + 4 Green	3 Red + 7 Black
-----------------	-----------------

The no. of ways in which two different coloured balls can be drawn are $G + R, R + B, G + B$.

$\therefore P$ (two different balls) = P (1st red and 2nd black) + P (1st green and 2nd red) + P (1st green and 2nd Black)

$$\left(\frac{5}{9} \times \frac{7}{10} \right) + \left(\frac{4}{9} \times \frac{3}{10} \right) + \left(\frac{4}{9} \times \frac{7}{10} \right)$$

$$\frac{35+12+28}{90} = \frac{75}{90} = \frac{5}{6}$$

6. (a) Two cases are possible here

- (1) The person transferred to Grid-II is a director
 (2) The person transferred to Grid-II is a General Manager.

$\therefore P$ (director super annuated) = P (director transferred) $\times P$ (director super annuated) + P (General Manager transferred) $\times P$ (director super annuated)

$$= \left(\frac{5}{4+56} \right) \left(\frac{3+1}{3+7+1} \right) + \left(\frac{4}{5+4} \right) \left(\frac{3}{3+7+1} \right)$$

$$\frac{5}{9} \cdot \frac{4}{11} + \frac{4}{9} \cdot \frac{3}{11} = \frac{32}{99}$$

7. (b) P (at least once) = $1 - P$ (none double six), for one trial

$$P \text{ (double six)} = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

$$P \text{ (no double six)} = 1 - \frac{1}{36} = \frac{35}{36}$$

$$P \text{ (atleast once)} = 1 - \left(\frac{35}{36} \right)^n, \text{ for } n \text{ trials.}$$

8. (a) One red card and one queen can be drawn in the following mutually exclusive ways;

- (1) by drawing one red card out of 24 red cards (excluding 2 red queens) and one red queen out of 2 red queens. Let the event be A .
 (2) by drawing one red card out of 26 red cards (including 2 red queens and one queen out of 2 black queens. Let this event be B .

- (3) by drawing two red queens of red. Let this event be C

Required probability = $P(A \cup B \cup C) = P(A) + P(B) + P(C)$

$$= \frac{{}^{24}C_1 \cdot {}^2C_1}{{}^{52}C_2} + \frac{{}^{26}C_1 \cdot {}^2C_1}{{}^{52}C_2} + \frac{{}^2C_2}{{}^{52}C_2} = \frac{101}{1326}$$

9. (b) Since, they obtain equal number of heads, so number of heads obtained by them must be any of 0, 1, 2 and 3.

No. of heads	Out comes	Probability
0	TTT 1/8	
1	TTH, HTT, THT	3/8
2	THH, HTH, HHT	3/8
3	HHH 1/8	

So, the required probability

$$\left(\frac{1}{8}\right)^2 + \left(\frac{3}{8}\right)^2 + \left(\frac{3}{8}\right)^2 + \left(\frac{1}{8}\right)^2$$

$$\frac{1}{64} + \frac{9}{64} + \frac{9}{64} + \frac{1}{64} = \frac{20}{64} = \frac{5}{16}$$

10. (d) Since, one and only one of the three events E_1, E_2 and E_3 can happen, therefore

$$P(E_1) + P(E_2) + P(E_3) = 1 \quad \dots(1)$$

$\therefore$ Odds against E_1 are 7 : 4

$$\Rightarrow P(E_1) = \frac{4}{4+7} = \frac{4}{11} \quad \dots(2)$$

$\therefore$ Odds against E_2 are 5 : 3

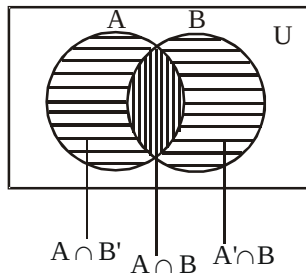
$$\Rightarrow P(E_2) = \frac{3}{3+5} = \frac{3}{8} \quad \dots(3)$$

From (1), (2) and (3), we have, $\frac{4}{11} + \frac{3}{8} + P(E_3) = 1$.

$$\text{i.e. } P(E_3) = 1 - \frac{4}{11} - \frac{3}{8} = \frac{88-32-33}{88} = \frac{23}{88} = \frac{23}{23+65}$$

Hence odds against E_3 is 65 : 23.

11. (d)



At most one of two events occurs if the event $A' \cup B'$ occurs.

$$P(A' \cup B') = 1 - P(A \cap B)$$

$$\text{Now, } P(A' \cup B') = P(A') + P(B') - P(A' \cap B')$$

$$P(A') + P(B') - [1 - P(A \cup B)]$$

$$P(A') + P(B') + P(A \cup B) - 1.$$

Finally, since

$$P(A' \cup B') = P[(A')' \cap B] + P[A' \cap (B')'] + P(A' \cap B')$$

$$= P(A \cap B) + P(A' \cap B) + P(A' \cap B')$$

$$[\because P(A \cup B) = P(A' \cap B) + P(A \cap B') + P(A \cap B)]$$

[See the Venn diagram].

12. (a) Let $P(A) = a$ and $P(B) = b$ Then $P(A \cap B) = \frac{1}{6}$

$$\Rightarrow P(A)P(B) = \frac{1}{6}, \text{ because } A \text{ and } B \text{ are independent.}$$

$$\therefore ab = \frac{1}{6} \quad \dots(1)$$

$$\text{Also } P(\bar{A} \cap \bar{B}) = [1 - P(A)][1 - P(B)];$$

$$\therefore [1 - a][1 - b] = \frac{1}{3} \Rightarrow 1 - a - b + ab = \frac{1}{3} \quad \dots(2)$$

$$\text{From (1) and (2) we have } a + b = \frac{5}{6} \quad \dots(3)$$

Solving (1) and (3) we get,

$$a = \frac{1}{2}, b = \frac{1}{3}, \therefore P(A) = \frac{1}{2}.$$

13. (c) Let W stand for the winning of a game and L for losing it. Then there are 4 mutually exclusive possibilities

(i) W, W, W (ii) W, W, L, W

(iii) W, L, W, W (iv) L, W, W, W.

[Note that case (i) includes both the cases whether he losses or wins the fourth game.]

By the given conditions of the question, the probabilities for (i), (ii), (iii) and (iv) respectively are

$$\frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3}; \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3}; \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} \text{ and } \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{2}{3}.$$

Hence the required probability

$$= \frac{8}{27} + \frac{4}{81} + \frac{4}{81} + \frac{4}{81} = \frac{36}{81} = \frac{4}{9}.$$

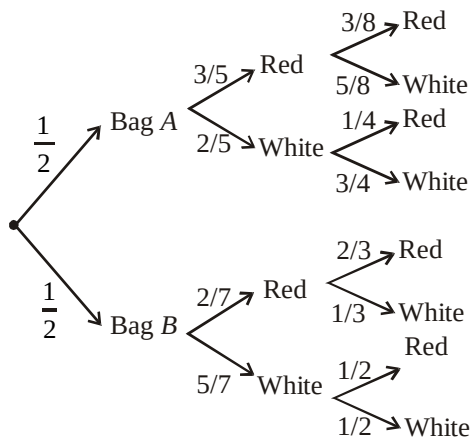
$\therefore$ The probability of winning the game if previous

game was also won is $\frac{2}{1+2} = \frac{2}{3}$ and the probability

of winning the game if previous game was a loss is

$$\frac{1}{1+2} = \frac{1}{3}]$$

14. (b) The whole event consists of the following mutually exclusive ways.
- (1) Selecting the bag A, drawing a red ball from A and putting it into bag B and then drawing a red ball from B.
 - (2) Selecting the bag A, drawing a white ball from A and putting it into bag B and then drawing a white ball from B.
 - (3) Selecting the bag B, drawing a red ball from B and putting it into A and then drawing a red ball from A.
 - (4) Selecting the bag B, drawing a white ball from B and putting it into A and then drawing a white ball from A.
- The tree diagram of the above processes are shown below, with respective probability of each step



The required probability is

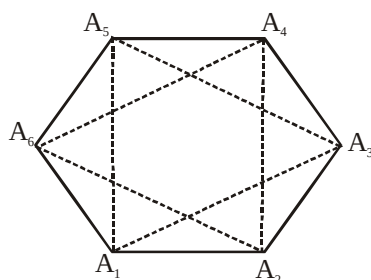
$$\frac{1}{2} \times \frac{3}{5} \times \frac{3}{8} + \frac{1}{2} \times \frac{2}{5} \times \frac{5}{8} + \frac{1}{2} \times \frac{2}{7} \times \frac{2}{3} + \frac{1}{2} \times \frac{5}{7} \times \frac{1}{2}$$

$$\frac{9}{80} + \frac{3}{20} + \frac{2}{21} + \frac{5}{28} = \frac{901}{1680}$$

15. (c) Numbers divisible by 4 are 104, 108..., 196; 24 in number. Numbers divisible by 7 are 105, 112, ..., 196; 14 in number. Numbers divisible by both, i.e. divisible by 28 are 112, 140, 168, 196; 4 in number. Hence, required probability

$$\frac{24}{99} + \frac{14}{99} - \frac{4}{99} = \frac{34}{99}$$

16. (c) Three vertices can be selected in 6C_3 ways.



The only equilateral triangles possible are $A_1A_3A_5$ and $A_2A_4A_6$

$$p = \frac{2}{{}^6C_3} = \frac{2}{20} = \frac{1}{10}$$

17. (a) $\therefore S = \{1, 2, 3, 4, 5, 6\} \times \{1, 2, 3, 4, 5, 6\}$

$$\therefore n(S) = 36$$

& Let $E_1 \equiv$ the event that the sum of the numbers coming up is 9.

& $E_2 \equiv$ the event of occurrence of 5 on the first die.

$$E_1 \equiv \{(3, 6), (6, 3), (4, 5), (5, 4)\}$$

$$\therefore n(E_1) = 4 \text{ and}$$

$$E_2 = \{(5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\}$$

$$\therefore n(E_2) = 6$$

$$E_1 \cap E_2 = \{(5, 4)\} \therefore n(E_1 \cap E_2) = 1$$

$$\text{Now, } P(E_1 \cap E_2) = \frac{n(E_1 \cap E_2)}{n(S)} = \frac{1}{36}$$

$$\text{and } P(E_2) = \frac{n(E_2)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

$\therefore$ Required probability

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\frac{1}{36}}{\frac{1}{6}} = \frac{1}{6}$$

18. (a) $\frac{1+4p}{4}, \frac{1-p}{2}, \frac{1-2p}{2}$ are probabilities of the three mutually exclusive events, then

$$0 \leq \frac{1+4p}{4} \leq 1, 0 \leq \frac{1-p}{2} \leq 1, 0 \leq \frac{1-2p}{2} \leq 1$$

$$\text{and } 0 \leq \frac{1+4p}{4} + \frac{1-p}{2} + \frac{1-2p}{2} \leq 1$$

$$\therefore -\frac{1}{4} \leq p \leq \frac{3}{4}, -1 \leq p \leq 1, -\frac{1}{2} \leq p \leq \frac{1}{2}, \frac{1}{2} \leq p \leq \frac{5}{2}$$

$$\therefore \frac{1}{2} \leq p \leq \frac{1}{2}$$

[The intersection of above four intervals]

$$\therefore p = \frac{1}{2}$$

19. (b) Exhaustive no. of cases = 6^3

10 can appear on three dice either as distinct number as following (1, 3, 6); (1, 4, 5); (2, 3, 5) and each can occur in $3!$ ways. Or 10 can appear on three dice as repeated digits as following (2, 2, 6), (2, 4, 4), (3, 3, 4)

and each can occur in $\frac{3!}{2!}$ ways.

∴ No. of favourable cases $3 \times 3! + 3 \times \frac{3!}{2!} = 27$

Hence, the required probability $\frac{27}{6^3} = \frac{1}{8}$

20. (b) Since each ball can be put into any one of the three boxes. So, the total number of ways in which 12 balls can be put into three boxes is 3^{12} .

Out of 12 balls, 3 balls can be chosen in ${}^{12}C_3$ ways. Now, remaining 9 balls can be put in the remaining 2 boxes in 2^9 ways. So, the total number of ways in which 3 balls are put in the first box and the remaining in other two boxes is ${}^{12}C_3 \times 2^9$.

Hence, required probability $\frac{{}^{12}C_3 \cdot 2^9}{3^{12}}$

21. (a) In any number the last digit can be one of 0, 1, 2, 8, 9. Therefore, the last digit of each number can be chosen in 10 ways. Thus, exhaustive number of ways = 10^n . If the last digit be 1, 3, 7 or 9 none of the numbers can be even or end in 0 or 5. Thus, we have a choice of 4 digits viz. 1, 3, 7 or 9 with which each of n number should end. So, favourable number of ways = 4^n .

$$\frac{4^n}{10^n} \left(\frac{2}{5} \right)^n$$

22. (a) Let E_1 be the event that exactly two players scored

more than 50 runs then $P(E_1) = \frac{1}{2} \times \frac{1}{3} \times \frac{3}{4} \times \frac{9}{10}$

$$+ \frac{1}{2} \times \frac{2}{3} \times \frac{1}{4} \times \frac{9}{10} + \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{1}{10} + \frac{1}{2} \times \frac{1}{3} \times \frac{1}{4} \times \frac{9}{10}$$

$$+ \frac{1}{2} \times \frac{1}{3} \times \frac{3}{4} \times \frac{1}{10} + \frac{1}{2} \times \frac{2}{3} \times \frac{1}{4} \times \frac{1}{10} = \frac{65}{240}$$

Let E_2 be the event that A and B scored more than 50

runs, then $P(E_1 \cap E_2) = \frac{1}{2} \times \frac{1}{3} \times \frac{3}{4} \times \frac{9}{10} = \frac{27}{240}$

∴ Desired probability

$$= P(E_2 / E_1) = \frac{P(E_1 \cap E_2)}{P(E_1)} = \frac{27}{65}$$

23. (a) The total number of ways in which 3 balls can be drawn is 9C_3 and the number of ways of drawing 3 black balls is 5C_3 ; therefore the chance of drawing 3 black balls

$$= \frac{{}^5C_3}{{}^9C_3} = \frac{5 \times 4 \times 3}{9 \times 8 \times 7} = \frac{5}{42}$$

Thus the odds against the event are 37 to 5.

24. (a) Total number of cases = $n = 100$

All the numbers from 50 to 59 have the digit 5. They are 10 in number. Besides these numbers the numbers 5, 15, 25, 35, 45, 65, 75, 85, 95 have the digit 5. These

are 9 in number. Number of favourable cases = $m =$ Number of numbers which have the digit 5 = $10 + 9 = 19$

$$\therefore \text{Probability} = P = \frac{m}{n} = \frac{19}{100}$$

25. (a) Let A be the event that the A is selected and B be the event that B is selected

$$\therefore P(A) = \frac{1}{5} \text{ and } P(B) = \frac{2}{7}$$

Let C be the event that both are selected.

$$\therefore C = A \cap B : \therefore P(C) = P(A \cap B)$$

∴ $P(C) = P(A) \cdot P(B)$ as A and B are independent events

$$= \frac{1}{5} \times \frac{2}{7} = \frac{2}{35}$$

26. (c) $n(S) =$ Total number of numbers = $5 \times {}^5C_4 \times 4! = 5(5!)$

Five digit numbers divisible by 6 are formed by using the numbers 0, 1, 2, 4 and 5 or 1, 2, 3, 4, and 5.

∴ number of such numbers

$$= n(E) = 2(4!) + 2 \times 3 \times 3! + 4 \times 3! = 108$$

27. (d) Probability of exactly M occurs = $P(M \cap \bar{N})$

and probability of exactly N occurs = $P(\bar{M} \cap N)$

∴ The probability that exactly one of them occurs is

$$P(M \cap \bar{N}) + P(\bar{M} \cap N)$$

$$P(M) - P(M \cap N) + P(N) - P(M \cap N)$$

$$P(M) + P(N) - 2P(M \cap N)$$

28. (c) Here the probability that the sun is hidden = $2/3$.

∴ The probability that it is out = $1/3$.

Now at least 4 days shining $\Rightarrow$ 4 day out and 1 day hidden + 5 days out =

$$5 \left(\frac{2}{3} \right) \times \left(\frac{1}{3} \right)^4 + \left(\frac{1}{3} \right)^5 = \frac{5 \times 2}{243} + \frac{1}{243} = \frac{11}{243}$$

29. (a) There are 13 cards of spade in a pack of 52 cards. So the chance that any of them will cut a spade is $13/52 = 1/4$

$\Rightarrow$ Probability that it is not a spade

$$= 1 - 1/4 = 3/4.$$

Consider the following mutually exclusive ways in which A may win : A wins in the first cut, OR A, B, C have failed and then A wins, OR A, B, C, A, B, C have failed and then A wins and so on upto infinity

$\Rightarrow$ Respective chances of these events are

$$= 1/4, [(3/4) \times (3/4) \times (3/4) \times (1/4)],$$

$$[(3/4) \times (3/4) \times (3/4) \times (3/4) \times (3/4) \times (1/4)] \dots \text{to infinity}$$

$\Rightarrow$ A's chance of succeeding

$$= (1/4) + (3/4)^3 \times (1/4) + (3/4)^6 \times 1/4 + \dots$$

$$= 1/4 [1/(1 - 27/64)] = 16/37.$$

30. (c) Consider the probability that, for example, an ace and a king are together. There are 4 aces and 4 kings in a deck.

Hence an ace can be chosen in 4 ways, and when that is done a king can be chosen in 4 ways.

Thus an ace and then a king can be selected in $4 \times 4 = 16$ ways.

Similarly, a king and then an ace can be selected in 16 ways.

Then an ace and a king can be together in $2 \times 16 = 32$ ways.

For every one way the combination (ace, king) occurs, the remaining 50 cards and the (ace, king) combination can be permuted in $51!$ ways. The number of favourable arrangements is thus $32(51!)$.

Since the total number of arrangements of all the cards in the deck is $52!$, the required probability is

$$\frac{32(51!)}{52!} = \frac{32}{52} = \frac{8}{13}.$$

31. (c) The total number of ways in which 3 integers can be chosen from first 20 integers is ${}^{20}C_3$.

The product of three integers will be even if at least one of the integers is even. Therefore, the required probability = $1 - \text{Prob. that none of the three integers is even}$

$$= 1 - \frac{{}^{10}C_3}{{}^{20}C_3} = 1 - \frac{2}{19} = \frac{17}{19}.$$

[Three odd integers can be chosen in ${}^{10}C_3$ ways as there are 10 even and 10 odd integers.]

32. (b) The condition implies that the last digit in both the integers should be 0, 1, 5 or 6 and the probability

$$4 \left(\frac{1}{10} \right)^2 = \frac{4}{100} = \frac{1}{25}$$

[$\therefore$ The squares of numbers ending in 0 or 1 or 5 or 6 also 0 or 1 or 5 or 6 respectively]

33. (d) Exhaustive number of cases = 12

Favourable cases = ${}^{12}C_2 (2^6 - 2)$

$$\therefore \text{Probability} = \frac{{}^{12}C_2 (2^6 - 2)}{12^6} = \frac{341}{12^5}$$

34. (a) Total no. of cases = 999

Favourable cases.

The page number can be with all identical digits,

i.e., 333 $\longrightarrow$ 1

The identical and one different like 009, 117, 225 and

$$441 \longrightarrow 4 \times \frac{3!}{2!} = 12$$

All different digits like

$$018, 027, 036, 045, 126, 135, 234 \longrightarrow 7 \times 3! = 42$$

So favourable cases = 55

$$\text{Desired probability} = \frac{55}{999}$$

35. (b) Total number of selections = 11×11

Now $|x - y| > 5$ clearly $x \neq 5$

If $x = 0$, then $y > 5 \Rightarrow 5$ favourable cases

If $x = 1$, then $y > 6 \Rightarrow 4$ favourable cases

If $x = 2$, then $y > 7 \Rightarrow 3$ favourable cases

If $x = 3$, then $y > 8 \Rightarrow 2$ favourable cases

If $x = 4$, then $y > 9 \Rightarrow 1$ favourable cases

Symmetrical cases will be obtained for $x = 6, 7, 8, 9, 10$

$\therefore$ Favourable cases = 30

$$\therefore \text{Desired probability} = \frac{30}{121}$$

36. (c) The condition for real is $a^2 - 4b \geq 0$

$a = 1$: for no value of b the condition is satisfied

2 : value of b can be 1

3 : 1, 2

4 : 1, 2, 3, 4

5 : 1, 2, 3, 4, 5, 6

6 : 1, 2, 3, 4, 5, 6

Thus (a, b) can be one of the 19 pairs and the probability

$$= \frac{19}{36}$$

37. (a) If the product of the four numbers ends in one of the digits 1, 3, 7, or 9, each number should have the last digit as one of these 4 digits.

$\therefore$ the number of favourable cases = 4^4

Total number of all possible cases = 10^4

$$\text{Hence the required probability} = \frac{4^4}{10^4} = \frac{2^4}{5^4} = \frac{16}{625}.$$

38. (b) For at least 4 successes, required probability

$$= {}^7C_4 \left(\frac{1}{2} \right)^4 \left(\frac{1}{2} \right)^3 + {}^7C_5 \left(\frac{1}{2} \right)^5 \left(\frac{1}{2} \right)^2$$

$$+ {}^7C_6 \left(\frac{1}{2} \right)^6 \left(\frac{1}{2} \right)^1 + {}^7C_7 \left(\frac{1}{2} \right)^7 = \frac{1}{2}.$$

39. (c) The word ASSISTANT contains two A's, one I, one N, three S's and two T's whereas the word STATISTICS contains one A, one C, two I's, three S's and three T's.
 $\therefore$ total number of ways of choosing one letter from each word = ${}^9C_1 \cdot {}^{10}C_1 = 90$.
 Common letters are A, I, S, T.
 $\therefore$ the number of favourable cases
 $= {}^2C_1 \cdot {}^1C_1 + {}^1C_1 \cdot {}^2C_1 + {}^3C_1 \cdot {}^3C_1 + {}^2C_1 \cdot {}^3C_1$
 $= 2 + 2 + 9 + 6 = 19$
 Hence the required probability = $\frac{19}{90}$.
40. (b) Let A $\equiv$ the event that drawing ball is white.
 B $\equiv$ the event that drawing ball is red.
 There are two mutually exclusive cases of the required event : WR and RR
 Now $P(WR) = P(W) P\left(\frac{R}{W}\right) = \frac{2}{5} \cdot \frac{2}{4} = \frac{6}{20}$
 $P(RR) = P(R) P\left(\frac{R}{R}\right) = \frac{3}{5} \cdot \frac{1}{4} = \frac{2}{20}$
 $\therefore$ reqd. Prob. = $P(WR + RR) = P(WR) + P(RR)$
 $= \frac{6}{20} + \frac{2}{20} = \frac{8}{20} = \frac{2}{5}$
41. (a) The event definition will be: Event X happens and Y doesn't happen or Y happens and X does not happen.
42. (a) The probability if he drops 3 bombs will be given by: Hit and Hit OR Miss and Hit and Hit OR Hit and Miss and Miss = $0.9 \times 0.9 + 0.1 \times 0.9 \times 0.9 \times 0.1 \times 0.9$
 $= 0.81 + 0.081 + 0.081 = 0.972 > 0.97$.
 Hence, 3 bombs would give him a probability of higher than 97% for the bridge to be destroyed.
43. (b) 20 girls can be seated around a round table in 19! ways.
 So, exhaustive number of cases = 19!
 Excluding A and B, out of remaining 18 girls, 4 girls can be selected ${}^{18}C_4$ ways which can be arranged in 4! ways.
 Remaining $20 - (4 + 2) = 14$ girls can be arranged in 14! ways. Also A and B mutually can be arranged in 2! ways.
 $\therefore$ Required number of arrangement = ${}^{18}C_4 \times 4! \times 2! \times 14!$
 $= 18! \times 2$
 Required probability = $\frac{18! \times 2}{19!} = \frac{2}{19}$
44. (d) Since x and y can take values from 0 to 10. So, the total number of ways of selecting x and y is $11 \times 11 = 121$

Now, $|x - y| > 5 \Rightarrow x - y < -5$ or $x - y > 5$

There are 30 pairs of values of x and y satisfying these two inequalities, so favourable number of ways = 30

Hence, required probability = $\frac{30}{121}$

45. (c) Let A be the event of getting exactly 3 defectives in the examination of 8 wristwatches.
 A and B be the event of getting ninth wristwatch defective
 Then

Required probability = $P(A \cap B) = P(A)P\left(\frac{B}{A}\right)$

Now, $P(A) = \frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8}$

And $P\left(\frac{B}{A}\right)$ = Probability that the ninth examined wristwatch is defective given that there were 3 defectives in the first 8 pieces examined = $\frac{1}{7}$

Hence, required probability = $\frac{{}^4C_3 \times {}^{11}C_5}{{}^{15}C_8} \times \frac{1}{7} = \frac{8}{195}$

46. (b) ASSISTANT $\rightarrow$ AA I N SSS TT
 STATISTICS $\rightarrow$ A II C SSS TTT
 Hence N and C are not common and same letters can be A, I, S, T, Therefore

Probability of choosing A = $\frac{{}^2C_1}{{}^9C_1} \times \frac{{}^1C_1}{{}^{10}C_1} = \frac{1}{45}$

Probability of choosing I = $\frac{1}{{}^9C_1} \times \frac{{}^2C_1}{{}^{10}C_1} = \frac{1}{45}$

Probability of choosing S = $\frac{{}^3C_1}{{}^9C_1} \times \frac{{}^3C_1}{{}^{10}C_1} = \frac{1}{10}$

Probability of choosing T = $\frac{{}^2C_1}{{}^9C_1} \times \frac{{}^3C_1}{{}^{10}C_1} = \frac{1}{15}$

Hence, required probability

$= \frac{1}{45} + \frac{1}{45} + \frac{1}{10} + \frac{1}{15} = \frac{19}{90}$

Explanation of Test Yourself

1. (b) Total no. of students in four schools
 $= 12 + 20 + 13 + 17 = 62$.
 Now, one student is selected at random.
 $\therefore$ Total outcomes $= 62_{C_1}$
 Now, no. of students in school $B_2 = 20$.
 No. of ways to select a student from $B_2 = 20_{C_1}$.

$$\therefore \text{Required probability} = \frac{20_{C_1}}{62_{C_1}} = \frac{20}{62} = \frac{10}{31}$$

2. (b) Total no. of days in a leap year $= 366$
i.e., 52 weeks and 2 days.
 Thus, leap year has 52 Sunday and remaining two days can be :

- | | |
|-----------------|-----------------|
| (1) Sun and Mon | (2) Mon and Tue |
| (3) Tue and Wed | (4) Wed and Thu |
| (5) Thu and Fri | (6) Fri and Sat |
| (7) Sat and Sun | |

Total no. of events $= 7$

Let A be the event that leap year has 53 Sundays.

$\therefore$ There is two ways that remaining two days contain Sunday is

- | | |
|-----------------|------------------|
| (1) Sun and Mon | (2) Sat and Sun. |
|-----------------|------------------|

$\therefore$ Favourable no. of cases $= 2$

$$\therefore P(\text{leap year contain 53 Sunday}) = \frac{2}{7}$$

$$\text{i.e. } P(A) = \frac{2}{7}.$$

3. (b) Total no. of balls $= 10$
 $P(\text{drawing one red ball}) = \frac{3}{10}$ ($\because$ red ball $= 3$)
 $P(\text{drawing second red ball without replacement}) = \frac{2}{9}$
 Required probability $\frac{3}{10} \times \frac{2}{9} = \frac{1}{15}$

4. (a) The probability of the problem being not solved by A , $(P_1) = 1 - \frac{1}{2} = \frac{1}{2}$
 Similarly, the probability of the problem being not solved by B , $(P_2) = 1 - \frac{1}{3} = \frac{2}{3}$
 $\therefore$ The probability that the problem being solved by neither A nor B , $(P_3) = P_1 \times P_2 = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$

Hence the required probability

$$P = 1 - P_3 = 1 - \frac{1}{3} = \frac{2}{3}.$$

5. (a) $(2/3) \times (3/4) + (1/3) \times (2/7) = (1/2) + (2/21) = (25/42)$
 6. (c) A and B will contradict each other if one speaks truth and other false. So, the

$$\text{Required probability} = \frac{4}{5} \left(1 - \frac{3}{4}\right) + \left(1 - \frac{4}{5}\right) \frac{3}{4}$$

$$\frac{4}{5} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4} = \frac{7}{20}$$

7. (a) The three balls that are taken out can be either 3 black balls or 2 black and 1 red ball or 1 black and 2 red ball or 3 red balls.

Each of these will give their own probabilities of drawing a black ball.

8. (c) Two squares out of 64 can be selected in

$${}^{64}C_2 = \frac{64 \times 63}{2} = 32 \times 63 \text{ ways}$$

The number of ways of selecting those pairs which have a side in common

$$= \left(\frac{1}{2}\right) (4 \times 2 + 24 \times 3 + 36 \times 4) = 112$$

[Since each of the corner squares has two neighbours each of 24 squares in border rows, other than corner ones has three neighbours and each of the remaining 36 squares have four neighbours and in this computation, each pair of squares has been considered twice].

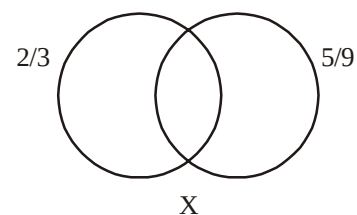
$$\text{Hence required probability} = \frac{112}{32 \times 63} = \frac{1}{18}.$$

9. (b) Probabilities of winning the race by three horses are $\frac{1}{3}$, $\frac{1}{4}$ and $\frac{1}{5}$.

Hence required probability

$$= \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{47}{60}$$

10. (a)



From the venn diagram we get:

$$(2/3) + (5/9) - x = 4/5 \rightarrow x = (2/3) + (5/9) - (4/5) = 19/45$$

11. (d) P of heads showing on 50 coins
 $= {}^{100}C_{50} \times P^{50}(1-P)^{50}$

P of heads showing on 51 coins $= {}^{100}C_{51} \times P^{51}(1-P)^{49}$

Both are equal

$${}^{100}C_{50} \times P^{50}(1-P)^{50} = {}^{100}C_{51} \times P^{51}(1-P)^{49}$$

$$\text{or } \frac{100 \times 49 \times \dots \times 52 \times 51}{1 \times 2 \times \dots \times 49 \times 50} \times (1-P)$$

$$= \frac{100 \times 99 \times \dots \times 53 \times 52}{1 \times 2 \times \dots \times 48 \times 49} \times P$$

$$\text{or } \frac{51}{50} \times (1-P) = P$$

$$\text{or } 51 \times (1-P) = 50P$$

$$\text{or } 51 - 51P = 50P$$

$$\text{or } 51 = 101P$$

$$\therefore P = \frac{51}{101}$$

12. (a) In the first draw, we have 7 even tickets out of 15 and in the second we have 8 odd tickets out of 15.
 Thus, $(7/15) \times (8/15) = 56/225$.

13. (d) Since A and B are two independent events

$$\therefore P(A \cap B) = P(A)P(B)$$

$$\Rightarrow P(A)P(B) = \frac{1}{6} \quad (\text{given}) \dots (i)$$

$$\text{and } P(\text{neither of } A \text{ nor } B) = P(\overline{A \cup B}) = \frac{1}{3}$$

$$\Rightarrow P(A \cup B) = 1 - P(\overline{A \cup B}) = 1 - \frac{1}{3} = \frac{2}{3}$$

We know that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{2}{3} = P(A) + P(B) - \frac{1}{6}$$

$$\Rightarrow P(A) + P(B) = \frac{2}{3} + \frac{1}{6} = \frac{5}{6}$$

$$\Rightarrow P(A) + \frac{1}{6P(A)} = \frac{5}{6} \quad [\text{from (i)}]$$

$$\Rightarrow 6[P(A)]^2 - 5P(A) + 1 = 0$$

$$\Rightarrow (2P(A) - 1)(3P(A) - 1) = 0 \Rightarrow P(A) = \frac{1}{2}, \frac{1}{3}$$

14. (b) We do not have to consider any sum other than 5 or 7 occurring.

A sum of 5 can be obtained by any of

$[4 + 1, 3 + 2, 2 + 3, 1 + 4]$

Similarly a sum of 7 can be obtained by any of

$[6 + 1, 5 + 2, 4 + 3, 3 + 4, 2 + 5, 1 + 6]$

For 5: $n(E) = 4, n(S) = 6 + 4$

$$P = 0.4$$

For 7: $n(E) = 6$

$$n(S) = 6 + 4$$

$$P = 0.6$$

15. (b) Total no. of pages = 1000

No. of one digit nos. summing 9 = 1

No. of two digit nos. summing 9

$(1 \& 8, 2 \& 7, 3 \& 6, 4 \& 5, 90) = 2! \times 4 + 1 = 9$

No. of three digit nos. summing 9 = nos. in which last digit is 0 (180, 270, 360, 450, 900)

+ nos. with all 3 different digits (126, 135, 234)

+ nos. with 2 same digit (144, 171, 252)

+ nos. with all 3 digits (333)

$$17 + 3 \times 3! + \frac{3!}{2!} \times 3 + 1 = 45$$

[$\therefore$ each number out of Nos. 180, 270, 360 and 450 can be arranged in 4 ways.]

$\therefore$ no. of ways = $4(4) = 16$

Total favourable cases = $1 + 9 + 45 = 55$

Therefore, required probability

$$\frac{\text{Total favourable cases}}{\text{Total outcomes}} = \frac{55}{1000}$$

Mock Tests

- ✦ **Mock Test-1**
- ✦ **Mock Test-2**
- ✦ **Mock Test-3**
- ✦ **Mock Test-4**
- ✦ **Mock Test-5**

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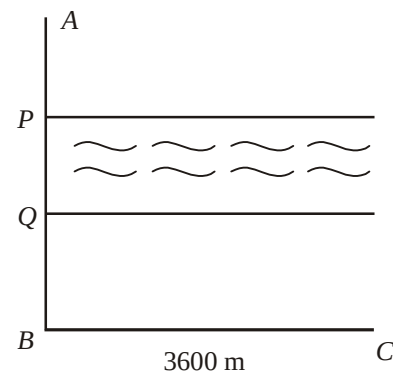
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Mock Test-1

- Define a number K such that it is the sum of the squares of the first M natural numbers. (i.e. $K = 1^2 + 2^2 + \dots + M^2$) where $M < 55$. How many values of M exist such that K is divisible by 4?
(a) 10 (b) 11
(c) 12 (d) None of these
- The sum of the first three terms of the arithmetic progression is 30 and the sum of the squares of the first term and the second term of the same progression is 116. Find the seventh term of the progression if its fifth term is known to be exactly divisible by 14.
(a) 36 (b) 40
(c) 43 (d) 22
- In a management entrance test, a student scores 2 marks for every correct answer and loses 0.5 marks for every wrong answer. A student attempts all the 100 questions and scores 120 marks. The number of questions he answered correctly was
(a) 50 (b) 45
(c) 60 (d) 68
- A vessel is full of a mixture of kerosene and petrol in which there is 18% kerosene. Eight litres are drawn off and then the vessel is filled with petrol. If the kerosene is now 15%, how much does the vessel hold?
(a) 40 litres (b) 32 litres
(c) 36 litres (d) 48 litres
- In an election of 3 candidates A, B and C , A gets 50% more votes than B . A also beats C by 18,000 votes. If it is known that B gets 5 percentage point more votes than C , find the number of voters on the voting list (given 90% of the voters on the voting list voted and no votes were illegal)
(a) 72,000 (b) 81,000
(c) 90,000 (d) 1,00,000
- Due to a price hike of 20%, 4 kgs less tea is available for ₹ 120. What is the original price of tea?
(a) ₹ 4/kg (b) ₹ 5/kg
(c) ₹ 6/kg (d) ₹ 4.5/kg
- Two alloys made up of copper and tin. The ratio of copper and tin in the first alloy is 1 : 3 and in the second alloy it is 2 : 5. In what ratio the two alloys should be mixed to obtain a new alloy in which the ratio of tin and copper be 8 : 3?
(a) 3 : 5 (b) 4 : 7
(c) 3 : 8 (d) 5 : 11
- Mohit Anand borrows a certain sum of money from the AMS Bank at 10% per annum at compound interest. The entire debt is discharged in full by Mohit Anand on payment of two equal amounts of ₹ 1000 each, one at the end of the first year and the other at the end of the second year. What is the approximate value of the amount borrowed by him?
(a) ₹ 1852 (b) ₹ 1736
(c) ₹ 1694 (d) ₹ 1792
- Two typists of varying skills can do a job in 6 minutes if they work together. If the first typist typed alone for 4

minutes and then the second typist typed alone for 6 minutes, they would be left with $\frac{1}{5}$ of the whole work. How many minutes would it take the slower typist to complete the typing job working alone?

- (a) 10 minutes (b) 15 minutes
(c) 12 minutes (d) 20 minutes
- Ravi, who lives in the countryside, caught a train for home earlier than usual yesterday. His wife normally drives to the station to meet him. But yesterday he set out on foot from the station to meet his wife on the way. He reached home 12 minutes earlier than he would have done had he waited at the station for his wife. The car travels at a uniform speed, which is 5 times Ravi's speed on foot. Ravi reached home at exactly 6 O'clock. At what time would he have reached home if his wife, forewarned of his plan, had met him at the station?
(a) 5 : 48 (b) 5 : 24
(c) 5 : 00 (d) 5 : 36
 - In the given diagram, river PQ is just perpendicular to the national highway AB . At a point B highway just turns at right angle and reaches to C . $PA = 500$ m and $BQ = 700$ m and width of the uniformly wide river (i.e., PQ) is 300m. Also $BC = 3600$ m. A bridge has to be constructed across the river perpendicular to its stream in such a way that a person can reach from A to C via bridge covering least possible distance. PQ is the widthness of the river, then what is the minimum possible required distance from A to C including the length of bridge?



- (a) 4100 m (b) 3900 m
(c) $3000\sqrt{2}$ m (d) None of these
- If the string is wound on the exterior four walls of a cube of side a cm starting at point C and ending at point D exactly above C , making equally spaced 4 turns. The side of the cube is :
(a) $a = \frac{2n}{\sqrt{255}}$ (b) $a = \frac{(n)^2}{16}$
(c) $a = \frac{8n}{\sqrt{257}}$ (d) $a = 2\sqrt{15}n$

13. The area of a triangle is 5. Two of its vertices are (2, 1) and (3, -2). The third vertex lies on $y = x + 3$. Find the third vertex :
- (a) $\left(2, \frac{13}{5}\right)$ (b) $\left(\frac{7}{2}, \frac{13}{2}\right)$
 (c) $\left(\frac{9}{2}, \frac{13}{2}\right)$ (d) $\left(\frac{7}{2}, \frac{13}{2}\right)$ or $\left(-\frac{3}{2}, \frac{3}{2}\right)$
14. The function $y = 1/x$ shifted 1 unit down and 1 unit right is given by
 (a) $y - 1 = 1/(x + 1)$ (b) $y - 1 = 1/(x - 1)$
 (c) $y + 1 = 1/(x - 1)$ (d) $y + 1 = 1/(x + 1)$
15. If the sum of the roots of the quadratic equations $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then $\frac{b^2}{ac} + \frac{bc}{a^2} =$
 (a) 0 (b) -1
 (c) 1 (d) 2
- Directions for Qs 16 and 17:** A survey shows that 41%, 35% and 60% of the people watch "Maine Pyaar Kiya" "Maine Pyaar Kyun Kiya" and "Pyaar to Hona Hi Tha" respectively. 27% people watch exactly two of the three movies and 3% watch none.
16. What percentage of people watch all the three movies?
 (a) 40% (b) 6%
 (c) 9% (d) 12%
17. If another survey indicates that 16% of the people watch Maine Pyaar Kiya and Pyaar to Hona Hi Tha, and 14% watch Maine Pyaar Kyun Kiya and Pyaar to Hona Hi Tha, then what percentage of the people watch only Maine Pyaar Kyun Kiya? (Use the data from the previous question, if necessary)
 (a) 10% (b) 8%
 (c) 12% (d) 15%
18. If $\log_{30} 3 = x$ and $\log_{30} 5 = y$, then $\log_8 30$ is equal to
 (a) $3(1 - x - y)$ (b) $\frac{1}{3(1 - x - y)}$
 (c) $\frac{3}{(1 - x - y)}$ (d) $\frac{(1 - x - y)}{3}$
19. A tea party is arranged for 16 people along the two sides of a long table with 8 chairs on each side. Four men wish to sit on one particular side and two on the other side. In how many ways can they be seated?
 (a) ${}^{10}C_4 \times 8!$ (b) ${}^{10}P_4 \times (8!)^2$
 (c) ${}^{10}C_4 \times (8!)^2$ (d) $4! \times 2! \times (8!)^2$
20. What is the probability that four S come consecutively in the word MISSISSIPPI?
 (a) $\frac{4}{165}$ (b) $\frac{2}{165}$
 (c) $\frac{3}{165}$ (d) $\frac{1}{165}$

Mock Test-2

1. 16 students were writing a test in a class. Rahul made 14 mistakes in the paper, which was the highest number of mistakes made by any student. Which of the following statements is definitely true?

(a) At least two students made the same number of mistakes.
 (b) Exactly two students made the same number of mistakes.
 (c) At most two students made the same number of mistakes.
 (d) All students made different number of mistakes.

Directions for Q. 2: Answer the question based on the following information.

There are 250 integers $a_1, a_2, \dots, a_{250}$, not all of them necessarily different. Let the greatest integer of these 250 integers be referred to as max, and the smallest integer be referred to as min. The integers a_1 through a_{124} form sequence A, and the rest form sequence B. Each member of A is less than or equal to each member of B.

2. Elements of A are in ascending order, and those of B are in descending order. a_{124} and a_{125} are interchanged. Then which of the following statements is true?
 (a) A continues to be in ascending order.
 (b) B continues to be in descending order.
 (c) A continues to be in ascending order and B in descending order.
 (d) None of the above
3. The average age of a group of persons going for a picnic is 16.75 years. 20 new persons with an average age of 13.25 years join the group on the spot due to which the average of the group becomes 15 years. Find the number of persons initially going for the picnic.
 (a) 24 (b) 20
 (c) 15 (d) 18
4. A mixture worth ₹ 3.25 a kg is formed by mixing two types of flour, one costing ₹ 3.10 per kg while the other ₹ 3.60 per kg. In what proportion must they have been mixed?
 (a) 3 : 7 (b) 7 : 10
 (c) 10 : 3 (d) 7 : 3
5. To pass an examination, 40% marks are essential. A obtains 10% marks less than the pass marks and B obtains 11.11% marks less than A. What percent less than the sum of A's and B's marks should C obtain to pass the exam?
 (a) 40% (b) $41\frac{3}{17}\%$
 (c) 28% (d) Any of these
6. The RBI lends a certain amount to the SBI on simple interest for two years at 20%. The SBI gives this entire amount to Bharti Telecom on compound interest for two years at the same rate annually. Find the percentage earning of the SBI at the end of two years on the entire amount.

(a) 4% (b) $3\frac{1}{7}\%$
 (c) $3\frac{2}{7}\%$ (d) $3\frac{6}{7}\%$

7. A milkman professes to sell milk at its CP only. But still he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture?

(a) 80% (b) 83.33%
 (c) 75% (d) 66.66%

8. Arvind Singh purchased a 40 seater bus. He started his services on route no. 2 (from Terhipuliya to Charbagh with route length of 50 km). His profit (P) from the bus depends upon the no. of passengers over a certain minimum number of passengers 'n' and upon the distance travelled by bus. His profit is ₹ 3600 with 29 passengers in the bus for a journey of 36 km and ₹ 6300 when there are 36 passengers travelled for 42 km. What is the minimum no. of passengers are required so that he will not suffer any loss?

(a) 12 (b) 20
 (c) 18 (d) 15

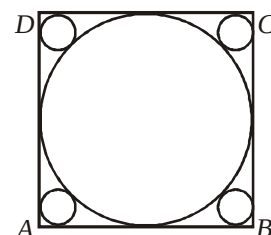
9. A tank of 3600 cu m capacity is being filled with water. The delivery of the pump discharging the tank is 20% more than the delivery of the pump filling the same tank. As a result, twelve minutes more time is needed to fill the tank than to discharge it. Determine the delivery of the pump discharging the tank.

(a) 40 m³/min (b) 50 m³/min
 (c) 60 m³/min (d) 80 m³/min

10. An urgent message had to be delivered from the house of the Peshwas in Pune to Shivaji who was camping in Bangalore. A horse rider travels on horse back from Pune to Bangalore at a constant speed. If the horse increased its speed by 6 km/h, it would take the rider 4 hours less to cover that distance. And travelling with a speed 6 km/h lower than the initial speed, it would take him 10 hours more than the time he would have taken had he travelled at a speed 6 kmph higher than the initial speed. Find the distance between Pune and Bangalore.

(a) 120 km (b) 600 km
 (c) 720 km (d) 750 km

11. ABCD is a square, in which a circle is inscribed touching all the sides of square. In the four corners of square 4 smaller circles of equal radii are drawn, containing maximum possible area.



What is the ratio of the area of larger circle to that of sum of the areas of four smaller circles?

- (a) $1:(68-48\sqrt{2})$ (b) $1:17\sqrt{2}$
(c) $3:(34-12\sqrt{2})$ (d) None of these

12. The radius of a cone is $\sqrt{2}$ times the height of the cone. A cube of maximum possible volume is cut from the same cone. What is the ratio of the volume of the cone to the volume of the cube?

- (a) 3.18π (b) 2.25π
(c) 2.35π (d) can't be determined

13. A straight line intersects the x-axis at A and the y-axis at B. AB is divided internally at C(8, 10) in the ratio 5 : 4. Find the equation of AB

- (a) $x + y = 18$ (b) $x + y + 2 = 0$
(c) $x + y - 2 = 0$ (d) None of these

14. The values of x, for which the functions $f(x) = x$,

$g(x) = (\sqrt{x})^2$ and $h(x) = x^2/x$ are identical, is

- (a) $0 \leq x$ (b) $0 < x$
(c) All real values (d) All real values except 0

15. If α, β, γ be the roots of the equation

$x(1+x^2) + x^2(6+x) + 2 = 0$, then the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ is

- (a) -3 (b) $\frac{1}{2}$
(c) $-\frac{1}{2}$ (d) None of these

Directions for Questions 16 and 17: Answer the questions based on the following information.

The following data was observed from a study of car complaints received from 180 respondents at Colonel Verma's car care

workshop, viz., engine problem, transmission problem or mileage problem. Of those surveyed, there was no one who faced exactly two of these problems. There were 90 respondents who faced engine problems, 120 who faced transmission problems and 150 who faced mileage problems.

16. How many of them faced all the three problems?

- (a) 45 (b) 60
(c) 90 (d) 20

17. How many of them faced either transmission problems or engine problems?

- (a) 30 (b) 60
(c) 90 (d) 40

18. If $\log x : \log y : \log z = (y-z) : (z-x) : (x-y)$, then

- (a) $X^y \cdot Y^z \cdot Z^x = 1$ (b) $X^x Y^y Z^z = 1$

- (c) $x\sqrt{x}, y\sqrt{y}, z\sqrt{z}$ 1 (d) None of these

19. Sam has forgotten his friend's seven-digit telephone number. He remembers the following: the first three digits are either 635 or 674, the number is odd, and the number nine appears once. If Sam were to use a trial and error process to reach his friend, what is the minimum number of trials he has to make before he can be certain to succeed?

- (a) 1000 (b) 2430
(c) 3402 (d) 3006

20. The probability that a man will live 10 more years is $\frac{1}{4}$ and

the probability that his wife will live 10 more years is $\frac{1}{3}$.

Then the probability that neither will be alive in 10 years is

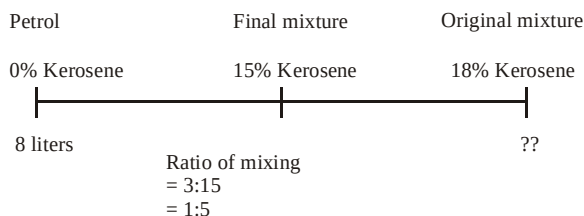
- (a) $\frac{5}{12}$ (b) $\frac{7}{12}$
(c) $\frac{1}{2}$ (d) $\frac{11}{12}$

Mock Test-3

- If highest power of 8 in $N!$ is 19 find highest power of 8 in $(N + 1)!$.
(a) 19 or 20 (b) 21 or 22
(c) 19 or 21 (d) None of these
- Find the value of x such that $(x + 3)(3x - 2)^5(7 - x)^3(5x + 8)^2 \geq 0$
(a) $(-\infty, -3] \cup [2/3, 7]$
(b) $(-\infty, -3] \cup [2/3, 7] \cup \{-8/5\}$
(c) $(-\infty, -3] \cup [2/3 - 8/5]$
(d) None of these
- If $a^2 + b^2 = 7ab$, then find the value of $\log_2 \left(\frac{a+b}{3} \right)$
(a) $1/2(\log_2 a + \log_2 b)$
(b) $1/2(\log_2 a - \log_2 b)$
(c) $(\log_2 a + \log_2 b)$
(d) None of these
- Find the value of $(x^{\log y - \log z})(y^{\log z - \log x})(z^{\log x - \log y}) = 1$
(a) xyz (b) $x + y + z$
(c) 1 (d) 0
- Convert $(332.22)_4$ to base 5.
(a) $(222.\overline{30})_5$ (b) $(222.30)_5$
(c) $(222.33)_5$ (d) None of these
- Find the value of x such that $\log_{0.3}(x - 1) = \log_{0.09}(x - 1)$.
(a) $[2, \infty)$ (b) $(2, \infty)$
(c) $(3, \infty)$ (d) None of these
- If a and b are two negative integers whose difference is 3 and sum of their squares is 89, then find the value of $a + b$.
(a) -13 (b) -12
(c) -14 (d) -15
- The total number of ways in which 5 balls of different colours can be distributed among 3 persons so that each person gets at least one ball is
(a) 75 (b) 150
(c) 210 (d) 243
- Find the ratio of circum radius (R) to inradius (r) of a triangle whose sides are 4, 6 and 8.
(a) 2 : 3 (b) 16 : 5
(c) 5 : 16 (d) 4 : 7
- For a Δ with sides a, b and c , which one of the following statement is correct?
(i) If $a^2 + b^2 < c^2$, then $\angle A$ is obtuse angle.
(ii) If $a^2 + b^2 < c^2$, then $\angle B$ is acute angle.
(iii) If $a^2 + b^2 > c^2$, then ΔABC must be an acute angled.
(a) Only (i) and (ii)
(b) Only (ii) and (iii)
(c) Only (iii)
(d) None of these
- If α and β are roots of $ax^2 + bx + c = 0$, then find the value of $a \left(\frac{\alpha^2 + \beta^2}{\beta \alpha} \right) + b \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right)$
(a) b (b) a
(c) ab (d) a/b
- If $(p^2 + 2)x^2 + 2p^2x + (p^2 - 4) = 0$ has roots of opposite sign, then find the range of p .
(a) $(-2, 2]$ (b) $[-2, 2]$
(c) $(-2, 3)$ (d) $(-2, 2)$
- Solve : $|x^2 + 3x| + x^2 - 2 \geq 0$
(a) $(-\infty, -2/3]$
(b) $(-\infty, -2/3] \cup [1, \infty)$
(c) $(-\infty, -2/3] \cup [1/2, \infty)$
(d) None of these
- If $A + B = 45^\circ$, then find the value of $(1 + \tan A)(1 + \tan B)$
(a) 0 (b) 1
(c) 2 (d) None of these
- How many values of $[x]$ exists such that $4\{x\} = x + [x]$
(a) 2 (b) 1
(c) 0 (d) None of these
- In how many ways can two squares be chosen on a 8×8 chessboard such that they have only one corner in common?
(a) 98 (b) 94
(c) 108 (d) None of these
- Price of two types of rice is ` 20 and ` 16/kg. A shopkeeper mixed them in the ratio of $n : (n + 1)$ where n is an integer and sold it at ` 17.8/kg. What is the minimum value of n so that he is in loss?
(a) 3 (b) 7
(c) 6 (d) 5
- If A can complete a project in 3 days, B can complete the same project in 4 days while C can complete it in 5 days. All of them stated the project together but after 3 days C left, after 4 days B left and after 5 days A left, then how many projects they have finished in 5 days?
(a) 2 (b) 1
(c) 3 (d) 4
- If $N = 10! + 20! + 30! + \dots + 100!$. Then find the highest power of 10 in N^N
(a) N (b) $2N$
(c) $4N$ (d) 124
- Solve for x if $(x)^2 + (x + 1)^2 = 25$, where (x) denotes the least integer function
(a) $[-5, -4] \cup [2, 3]$
(b) $[-5, -4] \cup [2, 3] \cup [5, 7]$
(c) $[-5, -4] \cup [2, 3] \cup \{1\}$

Hints and Solutions of Mock Test-1

1. (c) The sum of squares of the first n natural numbers is given by $n(n+1)(2n+1)/6$.
For this number to be divisible by 4, the product of $n(n+1)(2n+1)$ should be a multiple of 8. Out of n , $(n+1)$ and $(2n+1)$ only one of n or $(n+1)$ can be even and $(2n+1)$ would always be odd.
Thus, either n or $(n+1)$ should be a multiple of 8.
This happens if we use $n = 7, 8, 15, 16, 23, 24, 31, 32, 39, 40, 47, 48$. Hence, 12 such numbers.
2. (b) Since the sum of the first three terms of the AP is 30, the average of the AP till 3 terms would be $30/3 = 10$. The value of the second term would be equal to this average and hence the second term is 10. Using the information about the sum of squares of the first and second terms being 116, we have that the first term must be 4. Thus, the AP has a first term of 4 and a common difference of 6. The seventh term would be 40. Thus, option (b) is correct.
3. (d) If the number of questions correct is N , then the number of wrong answers is $100 - N$. Using this we get:
 $N \times 2 - (100 - N) \times 0.5 = 120 \rightarrow 2.5N = 170 \rightarrow N = 68$.
4. (d) The following visualization would help:



From the figure, we can see that the original mixture would be 40 liters and the petrol being mixed is 8 litres. Thus, the vessel capacity is 48 litres.

5. (d) The only values that fit this situation are C 25%, B 30% and A 45%. These are the percentage of votes polled. (Note: these values can be got either through trial and error or through solving $c + c + 5 + 1.5(c + 5) = 100\%$
Then, 20% is 18000 (the difference between A & C.)
Hence, 90000 people must have voted and 100000 people must have been on the voter's list.
6. (b) Due to price hike of 20%, quantity of less tea available

$$= \frac{1}{6}x = 4 \quad x = 24 \text{ (where } x = \text{initial quantity)}$$

$$\text{Original price} = \frac{120}{24} = ₹ 5$$

7. (b)

(A_1)	(A_2)
C T	C T
1 3	2 5

$$\text{Copper} \rightarrow \frac{1}{4} \quad \text{Copper} \rightarrow \frac{2}{7}$$

$$\text{Required copper} \rightarrow \frac{3}{11}$$

So, the required ratio is 4 : 7

Since it is clear from the above values
($1 + 2 \rightarrow 3$ and $4 + 7 \rightarrow 11$)

Alternatively : By Alligation

$$\begin{array}{ccc} \frac{1}{4} & \searrow & \frac{2}{7} \\ & 3 & \\ \left(\frac{2}{7} - \frac{3}{11}\right) & \nearrow & \left(\frac{3}{11} - \frac{1}{4}\right) \end{array}$$

$$\frac{1}{77} : \frac{1}{44}$$

$$\Rightarrow \frac{1}{7} : \frac{1}{4} \\ 4 : 7$$

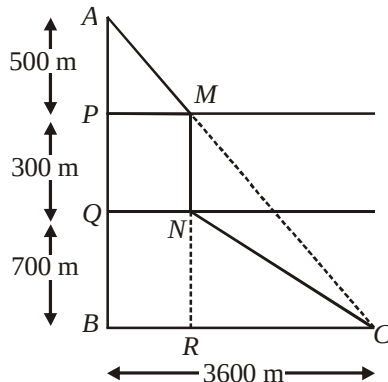
8. (b) $P + 2$ years interest on $P = 1000 + 1$ years interest on $1000 + 1000$
 $\rightarrow 1.21P = 2100 \rightarrow P = 1736$ (approx).
9. (b) Since the first typist types for 4 minutes and the second typist types for exactly 6 minutes, the work left (which is given as $1/5$ of the total work) would be the work the first typist can do in 2 minutes. Thus, the time taken by the first typist to do the work would be 10 minutes and his rate of work would be 10% per minute. Also, since both the typists can do the work together in 6 minutes, their combined rate of work would be $100/6 = 16.66\%$ per minute.
Thus, the second typist's rate of work would be $16.66 - 10 = 6.66\%$ per minute.
He would take $100/6.66 = 15$ minutes to complete the task alone.

10. (d) The wife drives for 12 minutes less than her driving on normal days.
Thus, she would have saved 6 minutes each way.
Hence, Ravi would have walked for 30 minutes (since his speed is $1/5^{\text{th}}$ of the car's speed).
In effect, Ravi spends 24 minutes extra on the walking (rather than if he had traveled the same distance by car.)
Thus, if Ravi had got the car at the station only, he would have saved 24 minutes more and reached at 5 : 36.

11. (a) Let MN be the bridge.
 $\triangle APM \sim \triangle ABC$

$$\therefore \frac{AP}{PM} = \frac{AB}{BC}$$

$$\frac{500}{PM} = \frac{1500}{3600} \Rightarrow PM = 1200 = QN = BR$$



$$\therefore RC = BC - BR = 2400 \text{ m}$$

$$\text{and } NR = BQ = 700 \text{ m}$$

$$\therefore NC = \sqrt{NR^2 + RC^2}$$

$$NC = 2500 \text{ m}$$

$$\text{Also } AM = \sqrt{AP^2 + PM^2}$$

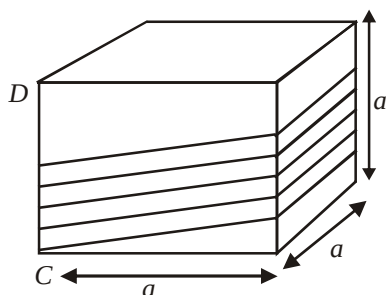
$$AM = 1300 \text{ m}$$

$$\therefore \text{Total distance to be travelled} = AM + MN + NC$$

$$= 1300 + 300 + 2500$$

$$= 4100 \text{ m}$$

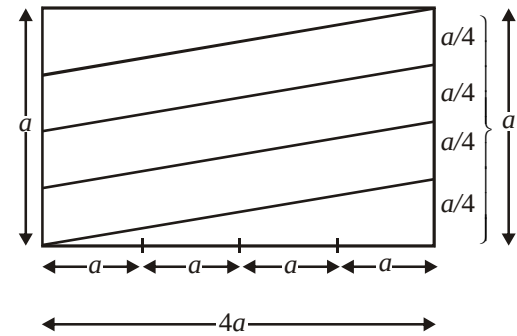
12. (c) Total length of string = $8n$ cm
Since, total length of string
= number of turns $\times$ perimeter of cylinder
= $8 \times n = 8n$ cm



Length of string required for 1 turn (or round)

$$= \frac{8n}{4} = 2n$$

$$\text{but } 2n = \sqrt{\left(\frac{a}{4}\right)^2 + (4a)^2}$$



$$\Rightarrow a = \frac{8n}{\sqrt{257}}$$

where a is the side of cube.

13. (d) Let (x_1, y_1) be the third vertex, then $y_1 = x_1 + 3 \dots (1)$
The area of the triangle formed by the points $(2, 1)$, $(3, -2)$ and (x_1, y_1)

$$= \frac{1}{2} \{-4 - 3 + 3y_1 + 2x_1 + x_1 - 2y_1\}$$

$$= \frac{1}{2} (3x_1 + y_1 - 7)$$

By the given condition,

$$\pm \frac{1}{2} (3x_1 + y_1 - 7) = 5$$

$$\Rightarrow 3x_1 + y_1 - 7 = \pm 10$$

$$\Rightarrow 3x_1 + y_1 = 17$$

$$\text{and } 3x_1 + y_1 = -3$$

...(2)

...(3)

$$\text{Solving (1) and (2), we get } x_1 = \frac{7}{2}, y_1 = \frac{13}{2}$$

$$\text{Solving (1) and (3), we get } x_1 = -\frac{3}{2} \text{ and } y_1 = \frac{3}{2}$$

Hence, the coordinates of the third vertex is either

$$\left(\frac{7}{2}, \frac{13}{2}\right) \text{ or } \left(-\frac{3}{2}, \frac{3}{2}\right)$$

14. (c) Looking at the options, one unit right means x is replaced by $(x - 1)$. Also, 1 unit down means -1 on the RHS.

$$\text{Thus, } (y + 1) = 1/(x - 1)$$

15. (d) Let the roots be α and β .

$$\therefore \alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$$

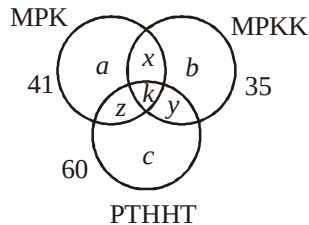
$$\Rightarrow \frac{-b}{a} \frac{\alpha^2 + \beta^2}{\alpha^2 \beta^2} \Rightarrow \frac{-b}{a} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha^2 \beta^2}$$

$$\Rightarrow \frac{-b}{a} = \frac{(b/a)^2 - \frac{2c}{a}}{c^2/a^2}$$

$$\Rightarrow \frac{-b}{a} \frac{b^2 - 2ca}{c^2} \Rightarrow b^2 a + bc^2 = 2ca^2$$

$$\Rightarrow \frac{b^2}{ac} + \frac{bc}{a^2} = 2$$

16. (b) $\alpha + \beta + \gamma = 97\%$



$$\alpha + 2\beta + 3\gamma = 41 + 35 + 60 = 136\%$$

$$\text{But } \beta = (x + y + z) = 27\%$$

$$\therefore (\alpha + 2\beta + 3\gamma) - (\alpha + \beta + \gamma) = \beta + 2\gamma = 39\%$$

$$\therefore (\beta + 2\gamma) - \beta = 2\gamma = 39 - 27 = 12\%$$

$$\therefore \gamma = 6\% = (k)$$

$$\therefore 6\% \text{ people watch all the three movies}$$

17. (c) $z + k = 16 \Rightarrow z = 10$ $\{\because \gamma = k = 6\}$

$$y + k = 14 \Rightarrow y = 8$$

$$\therefore x = 9 \quad (\because x + y + z = 27)$$

$$\therefore b = 35 - (x + k + y) = 35 - (9 + 6 + 8) = 12\%$$

18. (b) $\log_{30} 3 = x, \log_{30} 5 = y$

$$\therefore x + y = \log_{30} 15$$

$$\Rightarrow x + y = \log_{30} (30/2)$$

$$\Rightarrow (x + y) = 1 - \log_{30} 2 \Rightarrow \log_{30} 2 = 1 - x - y$$

$$\Rightarrow 3 \log_{30} 2 = 3(1 - x - y) \Rightarrow \log_{30} 8 = 3(1 - x - y)$$

$$\therefore \log_8 30 = \frac{1}{3(1 - x - y)}$$

19. (c) Four persons have chosen to sit on one particular side (assume side A) and two persons on the other side (assume side B). So, we are supposed to select four persons for side A from the remaining 10 persons and remaining six persons will be sitting on side B.

Number of ways 4 persons can be selected from 10 persons = $^{10}C_4$

Number of ways 6 persons can be selected from the remaining 6 persons = 6C_6

Number of ways 8 persons can be arranged on side A = $8!$

Number of ways 8 persons can be arranged on side B = $8!$

$$\text{Total number of ways} = ^{10}C_4 \times ^6C_6 \times 8! \times 8!$$

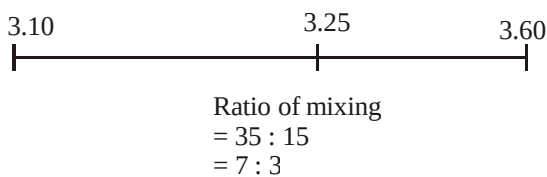
20. (a) $P = \frac{\text{No. of arrangements with four S together}}{\text{Total no. of arrangements}}$

$$= \frac{[8! / (4! \times 2!)]}{[11! / (4! \times 4! \times 2!)]}$$

$$= 8! \times 4! / 11! = 24/990 = 4/165.$$

Hints and Solutions of Mock Test-2

1. (a) The number of mistakes made by all the students will be between 0 and 14, i.e., students are having a total of 15 options to make mistakes. Since the number of students = 16, at least two students will have the same number of mistakes (that can be zero also, i.e., two students are making no mistakes). Hence, option (a) is the answer.
2. (a) If elements of A are in ascending order a_{124} would be the largest value in A . Also a_{125} would be the largest value in B . On interchanging a_{124} and a_{125} , A continues to be in ascending order, but B would lose its descending order arrangement since a_{124} would be the least value in B . Hence, option (a) is correct.
3. (b) Solve using alligation. Since 15 is the mid-point of 13.25 and 16.75, the ratio is 1:1 and hence there are 20 people who were going for the picnic initially.
4. (d) The required ratio would be 7 : 3 as seen in the following figure.



5. (d) Let the exam be of 100 marks. A obtains 36 marks (10% or $1/10^{\text{th}}$ less than the pass marks) while B obtains 32 marks (11.11% or $1/9^{\text{th}}$ less than A). The sum of A and B 's marks are $36 + 32 = 68$. To pass C can obtain 28 marks less than 68 – which is percentage of $41(3/17)\%$. If C obtains 28% less marks than 68 or if C obtains 40% less marks than 68 he would still pass. Thus, option (d) is correct.
6. (a) SBI would be paying 40% on the capital as interest over two years and it would be getting 44% of the capital as interest from Bharti Telecom. Hence, it earns 4%.
7. (b) CP of 100% pure milk = ₹ 100

$$\text{CP of the mixture} = \left(100 - 100 \times \frac{1}{6} \right) = ₹ 83.33$$

Hence, 83.33% milk is present 1 litre of mixture.
 Alternatively, if he has initially 10 L of milk, he must have made it 12 L of mixture to get a profit of 20% (since SP per litre = CP per litre). Hence out of 12 litres of mixture, 10 litres is milk and 2 litres is water.
8. (d) The minimum number of passengers n , at which there is no loss and number of passengers travelling = m and let the distance travelled is d , then

$$p \propto (m - n)d$$
 or
$$p = k(m - n)d; k \text{ is constant.}$$

When $P = 3600$, $m = 29$ and $d = 36$, then

$$3600 = k(29 - n) \times 36 \quad \dots(1)$$

Again, when $p = 6300$, $m = 36$, $d = 42$ km

$$6300 = k(36 - n) \times 42 \quad \dots(2)$$

Dividing equation (2) by (1)

$$\frac{6300}{3600} = \frac{k(36 - n) \times 42}{k(29 - n) \times 36}$$

$$\Rightarrow \frac{(36 - n)}{(29 - n)} = \frac{9}{6}$$

$$\Rightarrow 3n = 45 \Rightarrow n = 15$$

Hence to avoid loss, minimum number of 15 passengers are required.

9. (c) Use options for this question as follows:
 If discharging delivery is 40, filling delivery will be 16.66% less (this will give a decimal value right at the start and is unlikely to be the answer. Hence, put this option aside for the time being.)

Option (c) gives good values. If discharging delivery is 60, filling delivery will be 50. Also, time taken for discharge of 3600 cu m will be 60 minutes and the time taken for delivery will be 72 minutes (12 minutes more – which is the basic condition of the problem).

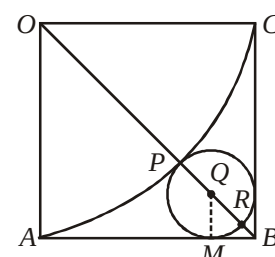
10. (c) The questions structure (and solving) have to be done on the basis of integers. The following equations emerge:

$$\frac{d}{s} - \frac{d}{(s + 6)} = 4 \text{ and } \frac{d}{(s - 6)} - \frac{d}{(s + 6)} = 10$$

Solving these expressions through normal solving methods is close to impossible (at the very least it would take a huge amount of time.) Instead this question has to be solved using the logic that integral difference in ratios in such a situation can only occur in all the three ratios (d/s , $d/(s + 6)$ and $d/(s - 6)$) are integers. Hence, d should have three divisors which are 6 units apart from each other.

11. (a) $OA = AB = BC = OC = OP$

Let $OA = R$ (radius of the larger circle), then $OB = R\sqrt{2}$
 Similarly, $PQ = MQ = QR = r$ (radius of the smaller circle),



then $BQ = r\sqrt{2}$

$$\therefore BP = r + r\sqrt{2}$$

and $BP = OB - OP = R\sqrt{2} - R$

$$R\sqrt{2} - R = r + r\sqrt{2}$$

$$R(\sqrt{2} - 1) = r(\sqrt{2} + 1)$$

$$\Rightarrow r = R(\sqrt{2} - 1)^2$$

$$r = R(3 - 2\sqrt{2})$$

$$\begin{aligned} \therefore \frac{\text{Area of larger circle}}{\text{Area of smaller circle}} &= \frac{\pi R^2}{4\pi r^2} \\ &= \frac{R^2}{4(3 - 2\sqrt{2})^2 R^2} \times \frac{1}{4(17 - 12\sqrt{2})} \end{aligned}$$

12. (b) Let the each side of cube be a , then

$$CD = \sqrt{2}a$$

$$\therefore CQ = \frac{a}{\sqrt{2}}$$

Let the radius of cone be r and height be h , then

$$r = h\sqrt{2}$$

$\therefore$ In $\triangle APO$ and $\triangle CQO$
(Similar triangles)

$$\frac{AP}{PO} = \frac{CQ}{OQ} = \frac{r}{h} = \frac{\frac{a}{\sqrt{2}}}{(h-a)}$$

$$\Rightarrow \frac{\frac{a}{\sqrt{2}}}{(h-a)} = \sqrt{2}$$

$$\Rightarrow a = 2(h-a)$$

$$\Rightarrow h = \frac{3a}{2}$$

$$\therefore r = \frac{3a}{2} \times \sqrt{2}$$

$$\text{and } h = \frac{3a}{2}$$

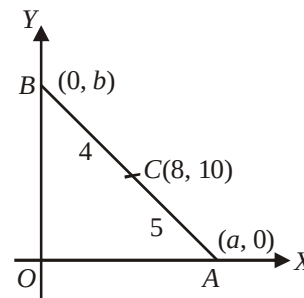
$$\begin{aligned} \therefore \text{Volume of cone} &= \frac{1}{3} \pi \times \left(\frac{3a\sqrt{2}}{2} \right)^2 \times \frac{3a}{2} \\ &= \frac{9}{4} a^3 \pi \end{aligned}$$

and Volume of cube $= a^3$

$$\therefore \text{Required ratio} = \frac{\frac{9}{4} \pi a^3}{a^3} = \frac{9}{4} \pi = 2.25\pi$$

13. (a) Let the equation of the line AB be $\frac{x}{a} + \frac{y}{b} = 1$... (1)

Then the coordinates of A and B are respectively $(a, 0)$ and $(0, b)$.



Since $C(8, 10)$ divides AB in the ratio $5 : 4$, we have

$$\frac{5 \times 0 + 4 \times a}{5 + 4} = 8$$

$$\text{and } \frac{5 \times b + 4 \times 0}{5 + 4} = 10$$

or $a = 18$ and $b = 18$

Hence from (1), the required equation of the line AB is

$$\frac{x}{18} + \frac{y}{18} = 1 \text{ or } x + y = 18$$

14. (b) All three functions would give the same values for $x > 0$. As $g(x)$ is not defined for negative x , and $h(x)$ is not defined for $x = 0$.

15. (c) $x(1 + x^2) + x^2(6 + x) + 2 = 0$
 $\Rightarrow 2x^3 + 6x^2 + x + 2 = 0$

For roots: α, β, γ

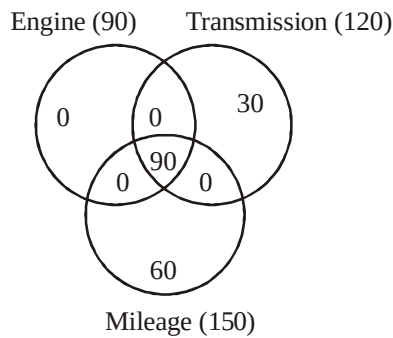
$$\alpha + \beta + \gamma = -\left(\frac{6}{2}\right) = -3$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{1}{2}$$

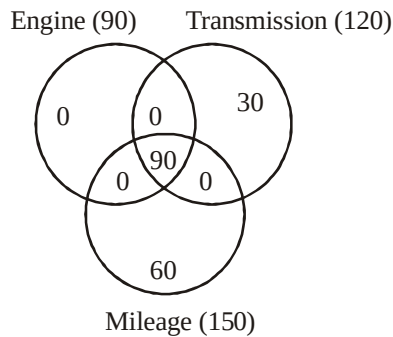
$$\alpha\beta\gamma = -\left(\frac{2}{2}\right) = -1$$

$$\therefore \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{1/2}{-1} = -1/2$$

16. (c)



17. (b) There are 30 such people. Option (b) is correct.



18. (b) $\log x : \log y : \log z = (y - z) : (z - x) : (x - y)$

$$\therefore \frac{\log x}{y - z} = \frac{\log y}{z - x} = \frac{\log z}{x - y} = k \text{ (say)}$$

$$\therefore \log x = k(y - z)$$

$$\Rightarrow x \log x = kx(y - z)$$

$$\text{Similarly, } y \log y = ky(z - x)$$

$$z \log z = kz(x - y)$$

Adding all 3 equations

$$x \log x + y \log y + z \log z = 0$$

$$\Rightarrow \log X^x \cdot Y^y \cdot Z^z = 0$$

$$\therefore X^x \cdot Y^y \cdot Z^z = 1$$

19. (c) The possible numbers are:

635__9	9 in the units place	$9 \times 9 \times 9 = 729$ numbers
635___	9 used before the units place	$3 \times 9 \times 9 \times 4 = 972$ numbers
674__9	9 in the units place	$9 \times 9 \times 9 = 729$ numbers
674___	9 used before the units place	$3 \times 9 \times 9 \times 4 = 972$ numbers
Total		3402 numbers

20. (c) The probability that a man will not live 10 more years = $3/4$ and the probability that his wife will not live 10 more years = $2/3$. Then the probability that neither will be alive in 10 years = $3/4 \times 2/3$.

Hints and Solutions of Mock Test-3

1. (c) Since highest power of 8 in $N!$ is 19 hence that of 2 in $N!$ is 57, 58 or 59.

Since the highest power of 2 in $62!$ and $63!$ is 57 and that in $64!$ is 63, and hence $N = 62$ or 63 .

If $N = 62$, then $N + 1 = 63$ and the highest power of 2 is 57 and that of 8 is 19.

If $N = 63$, then $N + 1 = 64$ and the highest power of 2 is 63 and that of 8 is 21.

2. (b) Given that

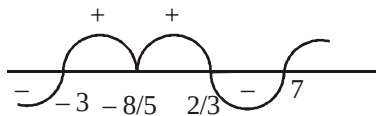
$$(x+3)(3x-2)^5(7-x)^3(5x+8)^2 \geq 0$$

$$\Rightarrow (x+3)(3x-2)^5(x-7)^3(5x+8)^2 \leq 0$$

(Taking coefficient of x , +ve in all brackets)

$$\Rightarrow (x+3)(x-2/3)^5(x-7)^3(x+8/5)^2 \leq 0$$

The curve of function is as follows



$\therefore$ The critical points are $-3, -8/5, 2/3, 7$

$$\therefore x \in (-\infty, -3] \cup [2/3, 7] \cup \{-8/5\}$$

3. (a) $a^2 + b^2 = 7ab \Rightarrow a^2 + b^2 + 2ab = 9ab \Rightarrow (a+b)^2 = 9ab$

$$\Rightarrow \left(\frac{a+b}{3}\right)^2 = ab \Rightarrow 2 \log_2 \left(\frac{a+b}{3}\right) = (\log_2 a + \log_2 b)$$

$$\Rightarrow \log_2 \left(\frac{a+b}{3}\right) = \frac{1}{2}(\log_2 a + \log_2 b)$$

4. (c) Let $K = x^{\log y - \log z} \cdot y^{\log z - \log x} \cdot z^{\log x - \log y}$

$$\Rightarrow \log K = (\log y - \log z) \cdot \log x + (\log z - \log x) \cdot \log y$$

$$+ (\log x - \log y) \cdot \log z$$

$$= \log x \log y - \log z \log x + \log y \log z - \log x \log y$$

$$+ \log z \log x - \log y \log z = 0$$

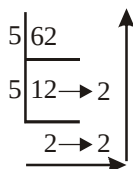
$$\Rightarrow K = 1$$

$$\text{So, } x^{\log y - \log z} \cdot y^{\log z - \log x} \cdot z^{\log x - \log y} = 1$$

5. (a) 1st we will convert $(332.22)_4$ to base 10

$$(332.22)_4 = (3 \times 4^2) + (3 \times 4^1) + (2 \times 4^0) + (2 \times 4^{-1}) + (2 \times 4^{-2}) = 48 + 12 + 2 + 0.5 + 0.125 = (62.625)_{10}$$

Now in next step we will convert $(62.625)_{10}$ to base 5 for this 1st we will convert the fractional part.



$$\text{Hence } (62)_{10} = (222)_5$$

Then in next step we have to convert $(0.625)_{10}$ to base 5

$$0.625 \times 5 = 3.125 \quad [\text{Take out the integral part i.e., 3 in this case}]$$

$$0.125 \times 5 = 0.625 \quad [\text{Take out the integral part i.e., 0 in this case}]$$

$$0.625 \times 5 = 3.125 \quad [\text{Take out the integral part i.e., 3 in this case}]$$

$$0.125 \times 5 = 0.625 \quad [\text{Take out the integral part i.e., 0 in this case}]$$

Hence final result is recurring number $(222.303030\dots)_5$

$$\text{So } (332.22)_4 = (62.625)_{10} = (222.\overline{30})_5$$

6. (b) log function is defined if $(x-1) > 0 \Rightarrow x > 1 \dots (1)$

$$\log_{0.3}(x-1) < \log_{(0.3)^2}(x-1)$$

$$\log_{0.3}(x-1) < 1/2 \log_{(0.3)}(x-1)$$

$$\log_{0.3}(x-1)^2 < 1/2 \log_{(0.3)}(x-1)$$

$$\text{or, } (x-1)^2 < (x-1)$$

$$\text{or, } (x-1)(x-1-1) > 0$$

$$\text{or, } (x-1)(x-2) > 0$$

$$\Rightarrow x < 1 \text{ or } x > 2 \dots (2)$$

Therefore from (1) and (2), $x \in (2, \infty)$

7. (a) Let the integers be a and $b = a + 3$

$$\text{Therefore, } (a)^2 + (a+3)^2 = 89$$

$$\Rightarrow a^2 + 3a - 40 = 0$$

From which on solving, we get

$$A = -8, 5$$

Since, a is negative

$$A = -8$$

Hence numbers are -8 and 5

8. (b) Total number of ways is given by

$$3^5 - {}^3C_1 \times (3-1)^5 + {}^3C_2 (3-2)^5 = 243 - 3 \times 32 + 3 = 150$$

Alternately:

Case (1): If distribution is done such that $5 = (1, 2, 2)$,

$$\text{then number of ways is } \left\{ \frac{5!}{(2!)(2!)} \right\} \left\{ \frac{3!}{2!} \right\} = 90$$

Case (2): If distribution is done such that $5 = (1, 1, 3)$,

$$\text{then number of ways is } \left\{ \frac{5!}{(3!)} \right\} \left\{ \frac{3!}{2!} \right\} = 60$$

So total number of ways is $90 + 60 = 150$

9. (b) Area of triangle with sides 4, 6 and 8 is

$$\sqrt{9(9-4)(9-6)(9-8)} = \sqrt{9 \times 5 \times 3 \times 1} = 3\sqrt{15}$$

We know that area of Δ is $\frac{abc}{4R} = r.s$

$$\text{Hence } \frac{abc}{4R} = 3\sqrt{15} \Rightarrow \frac{4 \times 6 \times 8}{4R} = 3\sqrt{15} \Rightarrow \frac{16}{\sqrt{15}}$$

Similarly, $r.s = 3\sqrt{15}$

$$r.9 = 3\sqrt{15} \Rightarrow r = \frac{\sqrt{15}}{3}$$

$$\text{Hence } \frac{R}{r} = \frac{16}{\sqrt{15}} \times \frac{3}{\sqrt{15}} = \frac{16 \times 3}{15} = \frac{16}{5}$$

10. (d) Consider each statement one by one

- (i) If $a^2 + b^2 < c^2$, then $\angle C$ is obtuse angle and the remaining two angles $\angle A$ and $\angle B$ must be acute angle. Hence this statement is wrong.
 (ii) Similar to explanation of statement (i) $\angle B$ is acute angle. Hence this statement is true.
 (iii) If $a^2 + b^2 > c^2$, then ΔABC is acute angled triangle if side c is the longest one, hence this statement is wrong. Only statement (ii) is right

11. (a) $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$

$$\begin{aligned} \therefore a \left(\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} \right) + b \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right) &= \frac{a(\alpha^3 + \beta^3) + b(\alpha^2 + \beta^2)}{\alpha\beta} \\ &= \frac{a[(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)] + [(\alpha + \beta)^2 - 2\alpha\beta]}{\alpha\beta} \\ &= \frac{a \left[\left(-\frac{b}{a} \right)^3 - 3 \frac{c}{a} \left(-\frac{b}{a} \right) \right] + b \left[\left(-\frac{b}{a} \right)^2 - \frac{2c}{a} \right]}{\frac{c}{a}} = \frac{bc}{\frac{c}{a}} = b \end{aligned}$$

12. (d) Since, the roots are real

$$D = 4p^4 - 4(p^2 + 2)(p^2 - 4) > 0$$

$$= p^2 > -4, \text{ which is true for all values of } p.$$

Now, product < 0 (since roots are of opposite sign)

$$\Rightarrow \frac{p^2 - 4}{p^2 + 2} < 0$$

$$\Rightarrow p^2 - 4 < 0 \text{ (As } p^2 + 2 \text{ is always positive)}$$

$$\Rightarrow -2 < p < 2$$

Therefore $p \in (-2, 2)$

13. (c) **Case -1:**

$$x^2 + 3x \geq 0 \Rightarrow x \in (-\infty, -3] \cup [0, \infty)$$

$$\Rightarrow x^2 + 3x + x^2 - 2 \geq 0$$

$$\Rightarrow 2x^2 + 3x - 2 \geq 0$$

$$\Rightarrow (2x - 1)(x + 2) \geq 0$$

$$\Rightarrow x \in (-\infty, -2] \cup \left[\frac{1}{2}, \infty \right)$$

$$\text{Hence } x \in (-\infty, -3] \cup \left[\frac{1}{2}, \infty \right)$$

Case -2:

$$x^2 + 3x < 0 \Rightarrow x \in (-3, 0)$$

$$\Rightarrow -x^2 - 3x + x^2 - 2 \geq 0$$

$$\Rightarrow -3x - 2 \geq 0$$

$$\Rightarrow x \leq -2/3$$

$$\text{Solution set of } x \in \left(-3, -\frac{2}{3} \right]$$

$$\text{Therefore } x \in \left(-\infty, -\frac{2}{3} \right] \cup \left[\frac{1}{2}, \infty \right)$$

14. (c) $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$$1 = \frac{\tan A + \tan B}{1 - \tan A \tan B} \text{ [as } A + B = 45^\circ, \tan(A + B) = 1]$$

$$\therefore \tan A + \tan B + \tan A \tan B = 1$$

$$\text{or, } 1 + \tan A + \tan B + \tan A \tan B = 1 + 1$$

$$\Rightarrow (1 + \tan A) + \tan B(1 + \tan A) = 2$$

$$\Rightarrow (1 + \tan A)(1 + \tan B) = 2$$

15. (a) We know $x = [x] + \{x\}$

$$\Rightarrow 4\{x\} = 2[x] + \{x\}$$

$$\Rightarrow \{x\} = \frac{2}{3}[x]$$

$$\Rightarrow 0 \leq \frac{2}{3}[x] < 1$$

$$\Rightarrow 0 \leq [x] < \frac{3}{2}$$

Here $[x] = 0, 1$

16. (a) Since the given condition is that corner of two selected squares is common hence they must be on two adjacent columns.

Now 1st let us assume that 2 squares that are chosen from the first 2 columns.

When the first square is chosen either from the top row or the bottom row, only one square can be chosen from the 2nd column. Thus from the first two columns, $2 \times 1 + 6 \times 2$ or 14 pairs of squares can be chosen. In 8×8 chessboards, 7 pairs of adjacent columns can be chosen.

Therefore, the total number of ways of choosing the squares = $7 \times 14 = 98$.

17. (d) Price of two varieties are ₹ 20 and ₹ 16 and they mixed in the ratio of $n : (n + 1)$, then cost of the mixture is $[20n + 16(n + 1)] / [n + (n + 1)] = (36n + 16) / (2n + 1)$. Since shopkeeper is in loss hence this resultant price should be more than the selling price hence $(36n + 16) / (2n + 1) > 17.8$ or $n > 9/2$ so minimum value of n is 5.
18. (c) Since A, B and C can complete the job in 3, 4 and 5 days so we will assume total work as LCM of 3, 4 and 5 that is 60. So let us assume project is making 60 dolls, then A, B and C can make 20, 15 and 12 dolls per day. Since A worked for 5 days, B worked for 4 days, and C worked for 3 days, so total number of dolls made by them is $20 \times 5 + 15 \times 4 + 12 \times 3 = 196$ dolls.

Since making 60 dolls is equal to one project so making 196 dolls = $196/60 = 3.26$ projects, so they will complete 3 projects.

19. (b) Highest power of 10 in $10!$ is 2, hence the required highest power is 2 N.

20. (a) **Case 1:** $x \in I \Rightarrow (x) = x$
 $\Rightarrow x^2 + x^2 + 2x + 1 = 25$
 $\Rightarrow x = 3, -4$

Case 2:

$$\begin{aligned} x \notin I &\Rightarrow (x) = [x] + 1 \\ \Rightarrow [x]^2 + [x] + 1 + [x]^2 + 4[x] + 4 &= 25 \\ \Rightarrow [x]^2 + 3[x] - 10 &= 0 \\ \therefore [x] &= 2, -5 \end{aligned}$$

Therefore $x = [-5, -4] \cup [2, 3]$

Hence solution set of x is $[-5, -4] \cup [2, 3]$

Mock Test-4

- The product of all integers from 1 to 100 will have the following number of zeros at the end:
(a) 20 (b) 21
(c) 19 (d) 24
 - The equation $(x^x)^x = 2$ is satisfied when x is equal to :
(a) infinity (b) 2
(c) $\sqrt[4]{2}$ (d) $\sqrt{2}$
 - a, b, c and d are consecutive integers. If $d^2 + b^2 - c^2 - a^2 = 22$, then the numbers are
(a) 5, 6, 7, 8 (b) 4, 5, 6, 7
(c) 3, 4, 5, 6 (d) Can't say
 - Ashutosh has six bags full of blue and white balls which he wishes to sell. Bag 1 has 29 balls, bag 2 has 14 balls, bag 3 has 23 balls, bag 4 has 6 balls, bag 5 has 5 balls and bag 6 has 12 balls. Ashutosh now tells his friends : "If I sell all the balls of a particular bag, I will have twice as many blue balls left as white balls. "If we know that a bag contains balls of only one colour, can you tell the bag he was talking about?
(a) bag 1 (b) bag 2
(c) bag 4 (d) bag 5
 - The last time Raman was buying Deepawali cards, he found four types of cards that he liked priced at Rs. 2.00, Rs.3.50, Rs.4.50 and Rs.5.00 each. As Raman wanted 30 cards he took five each of two kinds and ten each of other two putting down the exact number of ten rupee notes on the counter for payment. How many notes did Raman pay?
(a) 8 (b) 9
(c) 10 (d) 11
 - If $k + 4$ is an odd integer, which of the following is necessarily odd? ($k > 10$)
(a) $\frac{k+5}{2}$ (b) $(k-6)(k+6)$
(c) $(k+1)(k-1)$ (d) $(k+3)(k+4)$
 - $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \dots + \log_{10}(\log_{1023} 1024)$ equals
(a) 10 (b) e
(c) 1 (d) 0
 - We define a function f on the integers $f(x) = x/10$, if x is divisible by 10, $f(x) = x + 1$ if x is not divisible by 10. If $A_0 = 1994$ and $A_{n+1} = f(A_n)$.
What is the smallest n such that $A_n = 2$?
(a) 9 (b) 18
(c) 128 (d) 1993
 - If $x^3 + 2x^2 + ax + b$ is exactly divisible by $x^2 - 1$, then the value of a and b are respectively
(a) 1 and 2 (b) 1 and 0
(c) -1 and -2 (d) 0 and 1
 - Let $f(n) = \left[\frac{1}{2} + \frac{n}{100} \right]$ where [n] denotes the integral part of n. Then the value of $\sum_{n=1}^{100} f(n)$ is
(a) 50 (b) 51
(c) 1 (d) 0
 - A textile manufacturing firm employs 50 looms. It makes fabrics for a branded company. The aggregate sales value of the output of the 50 looms is Rs. 5,00,000 and the monthly manufacturing expenses is Rs. 1,50,000. Assume that each loom contributes equally to the sales and the manufacturing expenses are evenly spread over the number of looms. Monthly establishment charges are Rs. 75,000. If one loom breaks down and remains idle for one month, the decrease in profit is
(a) Rs. 13000 (b) Rs. 10000
(c) Rs. 7000 (d) Rs. 5500
- DIRECTIONS (Qs. 12 & 13) : Read the information carefully and answer the questions that follows:**
- A truck travelled from town A to town B over several days. During the first day, it covered $1/p$ of the total distance, where p is a natural number. During the second day, it travelled $1/q$ of the remaining distance, where q is a natural number. During the third day, it travelled $1/p$ of the distance remaining after the second day, and during the fourth day, $1/q$ of the distance remaining after third day. By the end of the fourth day the truck had travelled $3/4$ of the distance between A and B.
- The value of $p + q$ is
(a) 4 (b) 5
(c) 6 (d) 7
 - If the total distance is 100 kilometres, the minimum distance that can be covered on day 1 is kilometres.
(a) 25 (b) 30
(c) 33 (d) 35
 - Shyam went from Delhi to Simla via Chandigarh by car. The distance from Delhi to Chandigarh is $3/4$ times the distance from Chandigarh to Simla. The average speed from Delhi to Chandigarh was half as much again as that from Chandigarh to Simla. If the average speed for the entire journey was 49 kmph. What was the average speed from Chandigarh to Simla?
(a) 39.2 kmph (b) 63 kmph
(c) 42 kmph (d) 21 kmph
 - Water drops fall at a uniform rate from a kalash (pot) over the Shiv Ling in the Shiva temple. If 9 drops fall in 10 seconds, how many drops fall in 15 seconds?
(a) 14 (b) 12
(c) 13 (d) 13.5

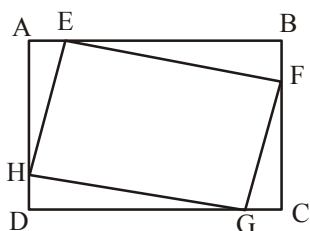
16. (17:8) as (25:7) :: (32:5) as (? : ?)
 (a) 39:12 (b) 27:9
 (c) 37:10 (d) 29:11

DIRECTIONS (Qs. 17-19): *In our camp there are 10 soldiers, four of them run at the speed of 7.2 km/hr., 3 at 5.4 km/hr, and another three at 3.6 km/hr. Seeing the liquid missile the soldiers are asked to vacate the camp. Because of the hilly area, the soldiers have to essentially stop for 2 seconds for breathing, after every 30 seconds of run. The missile can kill all life upto exactly 420 m from the camp. Assuming that the missile will exactly hit our camp, answer the following:*

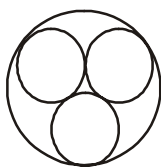
17. How many soldiers will survive, if the missile hits in exactly five minutes?
 (a) All 10 (b) Four
 (c) Seven (d) None
18. Assuming that the missile will exactly hit our camp, answer the following:
 How many soldiers will survive if the missile hits 15 seconds late?
 (a) All 10 (b) Four
 (c) Seven (d) None
19. Assuming that the missile will exactly hit our camp, answer the following:
 What is the minimum time required for all the soldiers to come out safely from the missile range?
 (a) 420 seconds (b) 450 seconds
 (c) 446 seconds (d) 444 seconds
20. On a certain pasture the grass grow at an even rate. It is known that 40 cows can graze on it for 40 days before the grass is exhausted, but 30 cows can graze there for as long as 60 days. How many days would the pasture last if 20 cows were to graze on it ?
 (a) 90 days (b) 80 days
 (c) 100 days (d) 120 days
21. A person buys some apples and mangoes from the market at a rate such that a mango is two times costlier than an apple and he sells them such that a mango is thrice the price of an apple. By selling the apple at twice its cost price, he makes the total of 150 % profit. Find the proportion of mangoes to the apples.
 (a) 3 : 5 (b) 3 : 4
 (c) 1 : 2 (d) 1 : 2
22. Shyama and Vyom walk up an escalator (moving stairway). The escalator moves at a constant speed. Shyama takes three steps for every two of Vyom's steps. Shyama gets to the top of the escalator after having taken 25 steps, while Vyom (because his slower pace lets the escalator do a little more of the work) takes only 20 steps to reach the top. If the escalator were turned off, how many steps would they have to take to walk up?
 (a) 40 (b) 50
 (c) 60 (d) 80

23. A player rolls a die and receives the same number of rupees as the number of dots on the face that turns up. What should the player pay for each roll if he wants to make a profit of one rupee per throw of the die in the long run ? Assume that the die has six faces carrying numbers 1 through 6.
 (a) Rs.2.50 (b) Rs.2
 (c) Rs.3.50 (d) Rs.4
24. Consider all digits from 1 to 9. Suppose that three digits are selected in strictly increasing order or strictly descending order. How many such selections would have such property that all three digits form an arithmetic progression
 (a) 24 (b) 30
 (c) 32 (d) 48
25. A number lock can be unlocked by a 3 digit code; each digit of the code could be any one of 0, 1 ...,9. What is the maximum number of trials required to unlock, if the following are known (a) the last digit (b) the sum of the first two digits is always less than or equal to the third?
 (a) 9 (b) 45
 (c) 55 (d) 36
26. If $\sin \theta = \frac{1}{2}$ and θ is acute, then $(3\cos\theta - 4\cos^3\theta)$ is equal to
 (a) 0 (b) $\frac{1}{2}$
 (c) $\frac{1}{6}$ (d) -1
27. An aeroplane flying horizontally 1 km. above the ground is observed at an elevation of 60° and after 10 seconds the elevation is observed to be 30° . The uniform speed of the aeroplane in km/h is
 (a) 240 (b) $240\sqrt{3}$
 (c) $60\sqrt{3}$ (d) None of these
28. A and B start walking along a circle from the same point, in opposite directions. They meet 30 m. from the starting point. While B has still not completed his circle, they meet again after having travelled at the same speed as before, at a distance of 50 m. from the starting point, but measured in the other direction as before. What is the circumference of the circle?
 (a) 80 m (b) 110 m
 (c) 130 m (d) 160 m
29. A, B, C, D, are four towns at the vertices of a rectangle. W is a well inside the rectangle. If the shortest distance to the well from the towns, A, B, C is 4 km, 5 km and 6 km respectively, find the shortest distance to the well from town D.
 (a) 7 km (b) $4\sqrt{2}$ km
 (c) 9 km (d) $3\sqrt{3}$ km

30. ABCD is a square, EFGH is another square. $AE = FD = GC = HB = a$ and $AF = DG = CH = BE = b$, then the ratio of the perimeters of ABCD to EFGH is

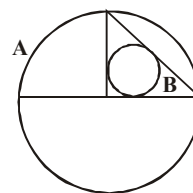


- (a) $a : b$ (b) $(a + b) : (a - b)$
 (c) $(a + b) : \sqrt{a^2 + b^2}$ (d) $2a^2 : b^2$
31. Three circles of radius 1 cm are circumscribed by a circle of radius r , as shown in the figure. Find the value of r ?



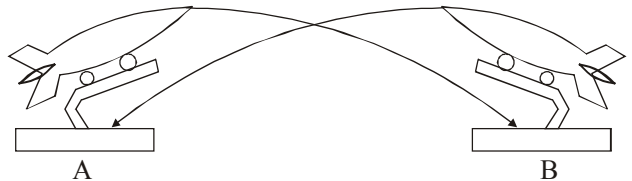
- (a) $\sqrt{3} + 1$ (b) $\frac{2 + \sqrt{3}}{\sqrt{3}}$
 (c) $\frac{\sqrt{3} + 2}{\sqrt{2}}$ (d) $2 + \sqrt{3}$

32. What is the ratio of the area of circle A to that of circle B ?

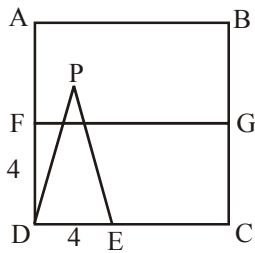


- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
 (c) $\frac{2}{(\sqrt{2} + 1)^2}$ (d) $\frac{2}{(\sqrt{2} - 1)^2}$
33. A cylinder 84 cm high has a circumference of 16 cm. A string makes exactly 7 complete turns round the cylinder while its two ends touch the cylinder's top and bottom. How long is the string in cm?
- (a) 160 (b) 190
 (c) 140 (d) 180
34. A conical cap just covers two spheres placed one above the other on a table. If the radii of the sphere are 1" and 2¼" find the height of the cone
- (a) 5.3" (b) 7.8"
 (c) 8.1" (d) 6.5"

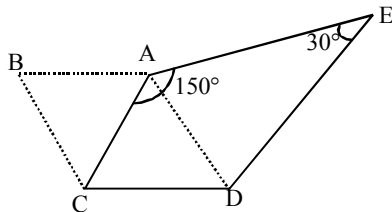
Mock Test-5

- The last two digits of $25^{63} \times 63^{25}$ are :
(a) 35 (b) 75
(c) 55 (d) 45
- When $4^{101} + 6^{101}$ is divided by 25, the remainder is
(a) 20 (b) 10
(c) 5 (d) 0
- The number of solutions of $|[x] - 2x| = 4$, where $[x]$ is the greatest integer $\leq x$ is
(a) 2 (b) 4
(c) 1 (d) infinite
- If $\log_{12} 27 = a$, then $\log_6 16$ is :
(a) $(3-a)/4(3+a)$ (b) $(3+a)/4(3-a)$
(c) $4(3+a)/(3-a)$ (d) $4(3-a)/(3+a)$
- Rohan is asked to figure out the marks scored by Sunil in three different subjects with the help of certain clues. He is told that the product of the marks obtained by Sunil is 72 and the sum of the marks obtained by Sunil is equal to the Rohan's current age (in completed years). Rohan could not answer the question with this information. When he was also told that Sunil got the highest marks in Physics among the three subjects, he immediately answered the question correctly. What is the sum of the marks scored by Sunil in the two subjects other than Physics?
(a) 6 (b) 8
(c) 10 (d) Cannot be determined
- How many divisors of 25200 can be expressed in the form $4n + 3$, where n is a whole number?
(a) 6 (b) 8
(c) 9 (d) None of these
- If the difference between the roots of the equations $x^2 + ax + b = 0$ is equal to the difference between the roots of the equation $x^2 + bx + a = 0$ ($a \neq b$), then
(a) $a + b = 4$ (b) $a + b = -4$
(c) $a - b = 4$ (d) $a - b = -4$
- Let $f(x) = |x-2| + |2.5-x| + |3.6-x|$, where x is a real number, attains a minimum at
(a) $x = 2.3$ (b) $x = 2.5$
(c) $x = 2.7$ (d) $x = 1.6$
- Given that $\frac{4p+9q}{p} = \frac{5q}{p-q}$ and 'p' and 'q' are both positive, calculate $\frac{4p}{5q}$?
(a) $\frac{5}{6}$ (b) $\frac{12}{11}$
(c) $\frac{16}{25}$ (d) $\frac{6}{5}$
- Two persons are awarded pensions in proportion to the square root of the number of years that they have served. First has 9 years longer service than the second. If the length of service of the first had exceeded that of the second by 4.25 years their pension would have been in the ratio of 9:8. How long had they served?
(a) 43, 34 (b) 25, 16
(c) 5, 4 (d) 34, 25
- Seth Bajaj, Khetan and Agarwal are playing a game of chance. Seth Bajaj borrows Rs. 10,000 from Khetan and lends 25 % of it to Agarwal who, in turn has already borrowed Rs. 6000 from Khetan. However, lady luck has her say and Seth Agarwal wins the entire pooled money (discussed above and belonging to all three) after round 1. So he decides to lend half his money equally to Bajaj and Khetan. Find the money left with Agarwal finally. Assume that Khetan initially had Rs.20,000 which has been used in the game. Starting balance of Bajaj and Agarwal can be taken as zero.
(a) Rs.5000 (b) Rs. 20,000
(c) Rs.7500 (d) Rs10,000
- The wordlists for MBA entrance (provided by a particular tutorials) contain an average of 30 words per list and there exist 100 such lists. A student starts cramming one list per day but unfortunately forgets 10% of it after every 25 days. Find the total number of words she can recall just when she finishes the last list.
(a) 2625 (b) 2550
(c) 2085 (d) 2750
- Three satellites A,B and C move around the earth in circular orbits and complete a rotation in 90 minutes, 120 minutes and 150 minutes respectively. If during their rotation they are all in line with the earth as shown at some instant, after how long will they be in similar position?
(a) 25hrs (b) 30hrs
(c) 10hrs (d) 20hrs
- Two rockets are launched simultaneously from two different positions.

Rocket A will land at the same spot from which Rocket B was launched, the Rocket B will land at the same spot where Rocket A was launched, allowing a small distance to the left or right to avoid a midair collision.

- The rockets are launched from the same angle, and therefore travel the same distance both vertically and horizontally. If the rockets reach their destinations in one and nine hours, respectively, after passing each other, how much faster is one rocket than the other?
- (a) 2 times (b) $2\frac{1}{2}$ times
(c) $1\frac{1}{2}$ times (d) 3 times
15. My watch is correct whereas Abhijit's watch is not. In Abhijit's watch the time for the minute hand overtaking the hour hand is 65 minutes, according to my watch. If both the watches show the same time at 6:30 a.m., find the time shown by Abhijit's watch at 6:25 p.m.
- (a) 6:21 p.m. (b) 6:34 p.m.
(c) 6:30 p.m. (d) 6:20 p.m.
16. A trader has a faulty balance, the left pan of which weighs 100 gm more than the right. The trader has a measure of 1 kg only. While buying goods, he keeps the weight measure in the left pan, and while selling keeps the weight measure in the right pan. If he sells at the cost price, find his gain percentage.
- (a) 10% (b) $(200/9)\%$
(c) $(110/9)\%$ (d) $(200/11)\%$
17. If $8a765 + 8765a$ is divisible by 9 then what is 'a' (where 'a' is an integer between 09) ?
- (a) 1 (b) 3
(c) 5 (d) 7
18. In the same amount of time, a new production assembly robot can assemble 8 times as many transmissions as an old assembly line. If the new robot can assemble X transmissions per hour, how many transmissions can the new robot and the old assembly line produce together in 5 days of round the clock production?
- (a) 15X (b) $45X/8$
(c) $135X/8$ (d) 135X
19. A is twice as efficient as B and B is four times as efficient as C. A, B and C take turns constructing a wall starting with A on day 1 followed by B then C and so on. If C alone can construct the wall in 120 days, what percentage of the wall is constructed by A?
- (a) 45 (b) 52.5
(c) 60 (d) 62.5
20. In a war field, the All- Baida group has the dangerous Stinger liquid missile with two inlets for fuel intake. One inlet can fill the missile in 5 minutes while the other in 6 minutes. However, since the missile is quite old and the residual of a previous war, there is a hole in the missile from which the fuel recedes and it takes 12 minutes to fully drain the missile. All three inlets/outlets are open now. In what minimum time from now can we expect the missile hitting our camp? (The missile is launched as soon as it is full and it would take three seconds for it to reach our camp after launch)
- (a) 2.55 minutes (b) 3.58 minutes
(c) 5.55 minutes (d) 6.47 minutes
21. Meena and Radhika have 200 and 150 marbles each. They decide to play a simple game : Meena will give 5 % of her marbles to Radhika who will then give 5% of her marbles to Meena. The game will continue till any one of them has a number of marbles 10 % different than the number held at the beginning. Rules of the game dictate that marbles to be exchanged must be integral in number (rounded off to next higher integer). How many times will the exchange take place before the game stops ?
- (a) 10 (b) 29
(c) 14 (d) 16
22. There are two swimmers who start swimming towards each other from opposite banks of the river. First time they meet at a point 900 ft from one shore. They cross each other, touch the opposite bank and return. They meet each other again, this time 300 ft from the other shore. What is the width of the river?
- (a) 2,400 ft (b) 1,800 ft
(c) 2,700 ft (d) 3,600 ft
23. A management institute has six senior professors and four junior professors. Three professors are selected at random for a Government project. The probability that at least one of the junior professors would get selected is:
- (a) $5/6$ (b) $2/3$
(c) $1/5$ (d) $1/6$
24. India plays two matches each with West Indies and Australia. In any match the probabilities of India getting points 0, 1 and 2 are 0.45, 0.05 and 0.50 respectively. Assuming that the outcomes are independent, the probability of India getting at least 7 points is
- (a) 0.8750 (b) 0.0875
(c) 0.0625 (d) 0.0250
25. The number of ways in which a team of eleven players can be selected from 22 players, always including 2 of them and excluding 4 of them, is
- (a) ${}^{16}C_{11}$ (b) ${}^{16}C_5$
(c) ${}^{16}C_9$ (d) ${}^{20}C_9$.
26. $\frac{\cos 70^\circ}{\sin 20^\circ} + \frac{\cos 59^\circ}{\sin 31^\circ} - 8\sin^2 30^\circ$ is equal to
- (a) 1 (b) -1
(c) 0 (d) 2
27. The angle of elevation of the top of a tower at point on the ground is 30° . If on walking 20 metres toward the tower, the angle of elevation become 60° , then the height of the tower is
- (a) 10 metre (b) $\frac{10}{\sqrt{3}}$ metre
(c) $10\sqrt{3}$ metre (d) None of these
28. The area of the square ABCD is 64. Let E, F and G be midpoints of DC, AD and BC respectively. If P is any point inside the rectangle ABGF and if X is the area of the triangle DPE, then which one of the following is necessarily true?



- (a) $8 < x < 16$ (b) $8 < x < 32$
 (c) $16 < x < 32$ (d) $16 < x < 64$
29. ABC is right-angled triangle in which $\angle B = 90^\circ$ and $BC = a$. If n points $L_1, L_2, \dots, L_n$ on AB are such that AB is divided in $n + 1$ equal parts and $L_1M_1, L_2M_2, \dots, L_nM_n$ are line segments parallel to BC and $M_1, M_2, \dots, M_n$ are on AC then the sum of the lengths of $L_1M_1, L_2M_2, \dots, L_nM_n$ is
- (a) $\frac{a(n+1)}{2}$ (b) $\frac{a(n-1)}{2}$
 (c) $\frac{an}{2}$ (d) $\frac{a(n)(n+1)}{2}$
30. In $\triangle ACD$, $AD = AC$ and $\angle C = 2\angle E$. The distance between parallel lines AB and CD is h .



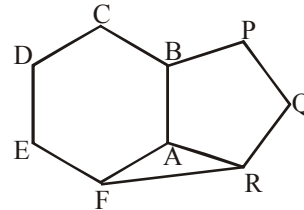
Then

I. Area of parallelogram ABCD

II. Area of $\triangle ADE$

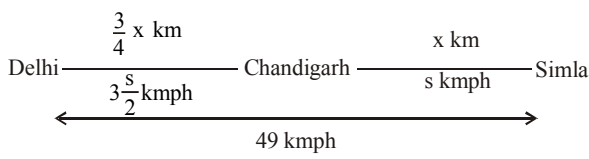
- (a) $I > II$ (b) $I < II$
 (c) $I = II$ (d) $I = 2 II$

31. In the figure, ABCDEF is a regular hexagon and ABPQR is a regular pentagon. The side AB is common. Calculate $\angle AFR$.



- (a) 24° (b) 48°
 (c) 12° (d) 40°
32. Which of the following has the smallest volume ?
- (a) A cube with side 5 cm.
 (b) A sphere with radius 3 cm.
 (c) A cuboid with sides 2cm, 5cm, 10cm
 (d) A cylinder with radius 2cm and height 10cm.
33. Ramesh has a special cylinder such that a sphere and a cone (right circular) can be perfectly inscribed inside it. Thus, the ratio of volume of the cylinder to the sum of volumes of the inscribed sphere and cone will be :
- (a) Equal to one (b) more than 1
 (c) less than 1 (d) equal to 2
34. Spherical balls of radius r are arranged in a lattice structure as an imaginary cube. There is a sphere at each vertex of the cube and one sphere at the centre of the cube. If this imaginary cube is to be as small as possible, what is the distance between the centres of the spheres on two adjacent vertices?
- (a) r (b) $2r/\sqrt{3}$
 (c) $r\sqrt{2}$ (d) $4r/\sqrt{3}$

Hints and Solutions of Mock Test-4

1. (d) The greatest power of a prime number a in $n!$ is $1[n/a] + 1[n/a^2] + 1[n/a^3] + \dots$. One 2 and one 5 give 10 on multiplication with 1 zero in the units place. The greatest power of 5 in $100! = 1[100/5] + 1[100/5^2] + \dots = 20 + 4 = 24$.
 $\therefore 100!$ contains 5^{24} . There are more than 24 powers of 2. Taking any 24 of them; we have $(2^{24} \times 5^{24}) = 10^{24}$ is contained in $100!$. So there are 24 zeros in $100!$.
2. (d) $(x^x)^x = x^{x \cdot x}$ Try all the alternatives.
3. (b) $d^2 - c^2 + b^2 - a^2 = (d + c)(d - c) + (b + a)(b - a)$
 $= d + c + b + a$ as $(d - c) = 1$ & $(b - a) = 1$
 $= 4a + 6 = 22$ (Given)
 $\therefore a = 4, b = 5, c = 6$, and $d = 7$.
4. (a) Question can be solved by suitable assumptions. Bags 3, 5 and 6 had blue balls. Bags 1, 2, and 4 had white balls. When he sells white balls of bag 1, he is left with $14 + 6 = 20$ white balls. No. of blue balls left $= 23 + 5 + 12 = 40$.
5. (d) Since total price is a multiple of 10, number of cards priced at Rs 3.5 and Rs 4.5 should be either 5 or 10 each. If these are 5 each and the others are 10 each, Total payment
 $= (10 \times 2 + 5 \times 3.5 + 5 \times 4.5 + 10 \times 5) = \text{Rs } 110$.
 This is a multiple of 10 ($11 \times 10 = 110$). However, in the other case, Total payment works out to Rs 115 $\Rightarrow$ Infeasible.
6. (b) 1. $k + 5$ is even, $(k + 5)/2$ may be even or odd depending on k .
 2. $k - 6$ is odd, $k + 6$ is odd.
 $\therefore (k - 6)(k + 6)$ is definitely odd.
 3. Both $(k + 1)$ and $(k - 1)$ are even.
 $\therefore$ Product is even.
 4. $k + 3$ is even. $\therefore (k + 3)(k + 4)$ is even.
7. (c) $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \dots + \log_{10}(\log_{1023} 1024)$
 $= \log_{10} \left[\log_2 4 \times \log_3 2 \times \log_4 3 \times \dots \times \log_{1023} 1024 \right]$
 $= \log_{10} [\log_2 1024] = \log_{10} [\log_2 (2^{10})]$
 $= \log_{10} 10 = 1$
8. (a) $A_0 = 1994$
 $A_1 = f(A_0) = 1994 + 1 = 1995$
 $A_2 = f(A_1) = 1995 + 1 = 1996$
 $A_3 = f(A_2) = 1996 + 1 = 1997$
 $A_4 = f(A_3) = 1997 + 1 = 1998$
 $A_5 = f(A_4) = 1998 + 1 = 1999$
 $A_6 = f(A_5) = 1999 + 1 = 2000$
 $A_7 = f(A_6) = \frac{2000}{10} = 200$
9. (c) Putting $x = 1$ and $x = -1$ in the given expression we get
 $3 + a + b = 0$ and $1 - a + b = 0$
 $b = -2, a = -1$
10. (b) $f(1) = \left[\frac{1}{2} + \frac{1}{100} \right] = 0, \dots, \dots$
 $f(49) = \left[\frac{1}{2} + \frac{49}{100} \right] = \left[\frac{99}{100} \right] = 0$
 $f(50) = \left[\frac{1}{2} + \frac{50}{100} \right] = 1, f(51) = \left[\frac{1}{2} + \frac{51}{100} \right] = 1, \dots$
 $f(100) = \left[\frac{1}{2} + \frac{100}{100} \right] = 1$.
 $\Rightarrow \sum_{n=1}^{100} f(n) = f(1) + f(2) + \dots + f(100) = 51$
 [As each of $f(1), \dots, f(49) = 0$ and each of $f(50), \dots, f(100) = 1$]
11. (c) Total profit of 50 looms $= 500000 - 150000 - 75000 = 275000$
 Total profit of 49 looms
 $= (500000 - 150000) \times \frac{49}{50} - 75000$
 $= 7000 \times 49 - 75000 = 268000$.
 Decrease in profit = Rs. 7000.
12. (d) By the end of the 4th day, the truck had travelled $3/4$ of the distance between A and B.
 As p and q are natural numbers, the distance has to be a multiple of $4 + 3 = 7$.
 So, the value of $p + q = 7$.
13. (a) Minimum distance traveled on day
 $1 = \frac{1}{4} \times 100 = 25 \text{ kms}$
14. (c) 
 Let average speed from Chandigarh to Simla

$$\text{Time taken from Delhi to Chandigarh} = \frac{3x \times 2}{4 \times 3s} = \frac{x}{2s}$$

$$\text{Time taken from Chandigarh to Simla} = \frac{x}{s}$$

$$\text{Total time} = \frac{x}{2s} + \frac{x}{s} = \frac{3x}{2s}$$

$$\text{also total time} = \frac{(3/4)x + x}{49} = \frac{7x}{4 \times 49}$$

$$\therefore \text{Total time would be same } \frac{3x}{2s} = \frac{7x}{4 \times 49}$$

$$\Rightarrow s = 42 \text{ km/h}$$

15. (c) There are 8 intervals of time between 9 drops. If 10 seconds can accommodate 8 intervals, 15 seconds can accommodate 12 intervals, or 13 drops.
16. (c) The first two digits enclosed within any parentheses are added together to get the second number contained within each parentheses. To get the first two digits of any following parentheses, add the number found in the preceding parentheses together. In this case, that is 37 : 10.
17. (c) Speed of 4 soldiers = 2m/sec = 60 m / 32 sec.
Now, distance covered by soldier in 5 minutes
$$= \frac{60}{32} \times 5 \times 60 = 562.5 \text{ m}$$

Speed of 3 soldiers = 1.5 m/sec = 45 m / 32 sec.
Therefore distance covered by soldier in 5 minutes = 421.87 m
Speed of 3 soldiers = 1 m/sec. = 30 m / 32 sec.
Therefore distance covered by soldier in 5 minutes = 281.25 m
If missile hits in 5 minutes 7 soldiers would survive.
18. (c) If missile hits 15 seconds late i.e. if the missile hits after 315 seconds.

$$\text{Distance covered by 4 soldiers} = \frac{60}{32} \times 315 \text{ m} \approx 590.6 \text{ m}$$

$$\text{Distance covered by 3 soldiers} = \frac{45}{32} \times 315 \text{ m} \approx 443 \text{ m}$$

$$\text{Distance covered by 3 soldiers} = \frac{30}{32} \times 315 \text{ m} \approx 295.3 \text{ m}$$

19. (c) The minimum time required for all the soldiers to come out safely from the missile range would be the time required by the slowest soldier = $420 \times \frac{32}{30} = 448 \text{ sec.}$
Therefore required time = 448 - 2 = 446 sec.
20. (d) In this problem, remember that grass is growing at an even rate. In 40 days the pasture provides 1600 feeds and in 60 days 1800 feeds. In 20 days growth in the pasture allows for 200 extra feeds, or enough grass grows every day to feed 10 cows. Thus the grass on the pasture before the cows start grazing is enough for (1600 - 40 × 10) i.e. 1200 feeds.

Thus the pasture lasts 120 days if 20 cows were to graze on it.

21. (c) Let 'a' be the number of apples, C_a its cost price and P_a its selling price. Let 'm' be the number of mangoes, C_m its cost price and P_m its selling price

$$\therefore \frac{mP_m + aP_a}{mC_m + aC_a} = 2.5 \text{ OR}$$

$$\frac{aP_a}{aC_a} \left[\left(\frac{m}{a} \right) \left(\frac{P_m}{P_a} \right) + 1 \right] = 2.5 \text{ OR } 2 \left[\frac{k \times 3 + 1}{k \times 2 + 1} \right] = 2.5$$

$$\Rightarrow 6k + 2 = 5k + 2.5 \text{ OR } k = 0.5 \text{ or } m/a = 1/2$$

22. (b) If Shyam takes 1 minutes for every 3 steps, then he takes 1/3 minutes for every step. For 25 steps, he takes 25/3 minutes = 8.33 minutes. So, Vyom takes $\frac{1}{2}$ minutes for every step. For 20 steps, he takes 10 minutes.
Difference between their times = 1.66 minutes
Escalator takes 5 steps in 1.66 minutes.
Speed of escalator is 1 step for 0.33 minutes = 3 steps per minutes.
If escalator is moving, then Shyam takes 25 steps and escalator also takes 25 steps.
Hence, total number of steps = 50.
23. (a) The probability of any of the six faces turning up = 1/6
Thus in the long run, on the average, the number turning on the face is $(1+2+3+4+5+6)/6 = 21/6 = 3.5$
Thus he receives Rs 3.5 per roll in the long run. Hence he must pay (3.5 - 1) = Rs 2.5 per roll to make a profit of Re 1 on each throw in the long run.
24. (c) Selections required by the question are (1, 2, 3), (2, 3, 4), (3, 4, 5), (4, 5, 6), (5, 6, 7), (6, 7, 8), (7, 8, 9), (2, 4, 6), (4, 6, 8), (1, 3, 5), (3, 5, 7), (5, 7, 9), (3, 6, 9), (1, 4, 7), (2, 5, 8), (1, 5, 9).
These are all in ascending order and we can find in descending order by just reversing these selections. Hence there are a total of $16 \times 2 = 32$ selections possible.
25. (c) The number could be between 000 to 999. It is required to find the number of 3-digit numbers between 000 and 999 satisfying the given conditions.
- | Possibilities | Number of possibilities |
|---------------|-------------------------|
| 900 - 999 | 909 |
| 800 - 899 | 809, 819 |
| 700 - 799 | 709, 719, 729 |
| : | : |
| : | : |
| : | : |
| 000 - 099 | 009, 019, 029,099 |

$$\therefore \text{Total number} = 1 + 2 + \dots + 10 = \frac{10 \times 11}{2} = 55$$

26. (a) We have, $\sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$
Now, $3\cos\theta - 4\cos^3\theta$

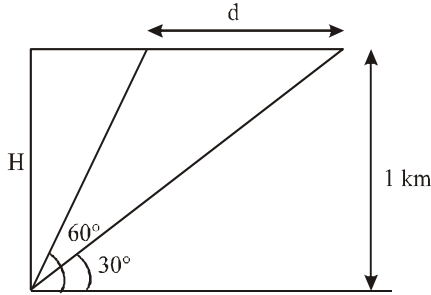
$$\Rightarrow 3 \cos 30^\circ - 4 \cos^3 30^\circ$$

$$\Rightarrow 3 \times \frac{\sqrt{3}}{2} - 4 \times \left(\frac{\sqrt{3}}{2}\right)^3 \Rightarrow \frac{3\sqrt{3}}{2} - \frac{4 \times 3\sqrt{3}}{8}$$

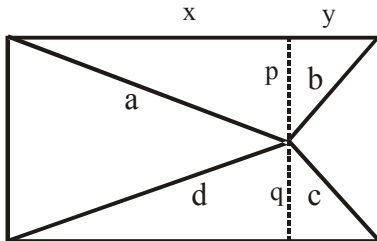
$$\Rightarrow \frac{3\sqrt{3}}{2} - \frac{3\sqrt{3}}{2} = 0$$

27. (b) $d = H \cot 30^\circ - H \cot 60^\circ$
Time taken = 10 second

$$\text{speed} = \frac{\cot 30^\circ - \cot 60^\circ}{10} \times 60 \times 60 = 240\sqrt{3}$$



28. (b) Let A have travelled x m. when A and B meet. x is less than the circumference by 30, or circumference $= (x + 30)$. When they meet for the second time, A has travelled $2x$ m., which is 50 m. more than the circumference, or circumference $= (2x - 50)$. Equating, $(x + 30) = (2x - 50)$, or $x = 80$, i.e. Circumference = 110 m.
29. (d) Here, $a^2 = p^2 + x^2$, $b^2 = p^2 + y^2$, $c^2 = q^2 + y^2$, $d^2 = q^2 + x^2$



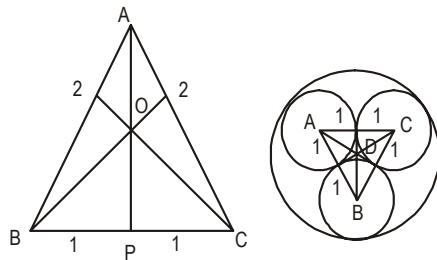
Hence, it follows that
 $a^2 + c^2 = b^2 + d^2$

$$\text{Therefore, } 4^2 + 6^2 = 5^2 + d^2, \text{ or } d = 3\sqrt{3}$$

30. (c) $FE = FG = GH = HE = \sqrt{a^2 + b^2}$;

$$\text{Ratio} = \frac{4(a+b)}{4\sqrt{a^2+b^2}} = \frac{a+b}{\sqrt{a^2+b^2}}$$

31. (b) $AP = \sqrt{3}$, $AO = \frac{2}{3}\sqrt{3} = \frac{2}{\sqrt{3}}$,



$$r = AO + 1 = \frac{2}{\sqrt{3}} + 1 = \frac{2 + \sqrt{3}}{\sqrt{3}}$$

32. (d) $CB = \frac{1}{2}\sqrt{2}r = \frac{r}{\sqrt{2}}$ $CQ = a$, $DQ = a$, $DB = r - a$

$$BQ^2 = a^2 + \left(\frac{r}{\sqrt{2}}\right)^2 = a^2 + (r-a)^2$$

$$\therefore a^2 + \frac{r^2}{2} = a^2 + r^2 + a^2 - 2ar$$

$$a^2 + \frac{r^2}{2} - 2ar = 0$$

$$\text{i.e. } a^2 - 2ra + \frac{r^2}{2} = 0$$

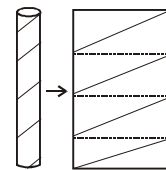
$$a = \frac{2r \pm \sqrt{4r^2 - 2r^2}}{2} = \frac{2 \pm \sqrt{2}}{2} r = \left(1 \pm \frac{1}{\sqrt{2}}\right) r$$

$$\text{since } a < r \therefore a = \left(1 - \frac{1}{\sqrt{2}}\right) r.$$

$$\text{Hence, } \frac{\pi r^2}{\pi a^2} = \frac{1}{\left(1 - \frac{1}{\sqrt{2}}\right)^2} = \frac{2}{(\sqrt{2} - 1)^2}$$

33. (c) For the given problem, the cylinder can be unrolled into a rectangle with dimensions 84 cm by 16 cm. The string thus creates 7 diagonal lines whose measure can be found out by considering the 7 right triangles they can form:

- the base of the triangle is 16 cm and the height is $\frac{1}{7}$ of 84 or 12 cm.
- from these, we can get the measure of one diagonal line as 20 cm [i.e., $(16^2 + 12^2)^{1/2}$].
- Therefore, the whole string measures 140 cm (i.e., 7×20 cm).



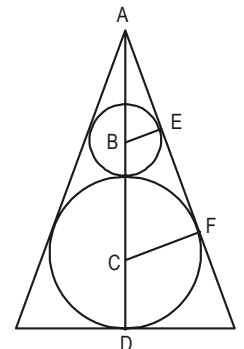
34. (c) Let $AB = h$; $\triangle ABE \sim \triangle ACF$

$$\frac{AB}{BE} = \frac{AC}{CF}; \frac{h}{1} = \frac{h+1+2\frac{1}{4}}{2\frac{1}{4}}$$

$$\therefore 2\frac{1}{4}h = h + 3\frac{1}{4}; \quad 1\frac{1}{4}h = 3\frac{1}{4}$$

$$\therefore \frac{5h}{4} = \frac{13}{4}; \therefore 5h = 13; \quad h = \frac{13}{5}$$

$$\text{Total height} = \frac{13}{5} + 1 + 2\frac{1}{4} + 2\frac{1}{4} = 8\frac{1}{10} = 8.1$$



Hints and Solutions of Mock Test-5

1. (b) $5, 5^2, 5^3, 5^4, 5^5$
 $5, 25, 125, 625, 3125$
 $\therefore$ Last two digits for any power of 5 = 25
 $\therefore$ last two digits in $25^{63} = 25$
 $3^1, 3^2, 3^3, 3^4, 3^5, 3^6, 3^7, 3^8$
 Last digit 3, 9, 7, 1, 3, 9, 7, 1
 $\therefore$ Last digit in $63^{25} = (63)^{6 \times 4 + 1}$
 Last digit in $63^{25} = 3$
 $\therefore$ Last two digits of $25^{63} \times 63^{25} = 75$.
2. (c) $4^2 + 6^2 = 52$, Remainder = 2, when divided by 25.
 $4^3 + 6^3 = 280$, Remainder = 5, when divided by 25.
 $4^4 + 6^4 = 1552$, Remainder = 2, when divided by 25.
 The series continues. For odd powers, remainder is 5 while for even powers, remainder is 2.
3. (b) $|[x] - 2x| = 4 \Rightarrow [x] - 2x = \pm 4 \Rightarrow [x] - 2[x] - 2\{x\} = \pm 4$
 $\Rightarrow -[x] - 2\{x\} = \pm 4 \Rightarrow \{x\} = \frac{-[x] \pm 4}{2}$
 If $\{x\} = \frac{-[x] - 4}{2}$, then
 $0 \leq \{x\} < 1 \Rightarrow 0 \leq -[x] - 4 < 2 \Rightarrow -6 < [x] \leq -4$
 $\therefore [x] = -5$ or $-4 \Rightarrow \{x\} = \frac{1}{2}$ or $0 \therefore x = -5 + \frac{1}{2}$
 or $-4 + 0 \Rightarrow x = -\frac{9}{2}$ or -4
 If $\{x\} = \frac{-[x] + 4}{2}$, then $0 \leq \{x\} < 1 \Rightarrow 0 \leq -[x] + 4 < 2$
 $\Rightarrow 2 < [x] \leq 4 \therefore [x] = 3$ or $4 \Rightarrow \{x\} = \frac{1}{2}$ or 0
 $\therefore x = 3 + \frac{1}{2}$ or $4 + 0 \Rightarrow x = \frac{7}{2}$ or 4
 $\therefore$ Solution set is $x \in \left\{-\frac{9}{2}, -4, \frac{7}{2}, 4\right\}$
4. (d) $\log_{12} 27 = a$
 $\Rightarrow \log_{12} 3^3 = a \Rightarrow \log_{12} 3 = \frac{a}{3}$
 $\Rightarrow \log_3 12 = \frac{3}{a} \Rightarrow \log_3 4 + \log_3 3 = \frac{3}{a}$
 $\Rightarrow 2 \log_3 2 = \frac{3-a}{a}$

$$\Rightarrow \log_3 2 = \frac{3-a}{2a} \quad \dots(i)$$

$$\text{Now, } \log_6 16 = 4 \log_6 2 = A \text{ (say)}$$

$$\Rightarrow \log_6 2 = \frac{A}{4} \Rightarrow \log_2 6 = \frac{4}{A} \Rightarrow \log_2 3 = \frac{4-A}{A}$$

$$\Rightarrow \log_3 2 = \frac{A}{4-A} \quad \dots(ii)$$

From (i) and (ii)

$$\frac{3-a}{2a} = \frac{A}{4-A} \text{ or } \frac{2a}{3-a} = \frac{4-A}{A}$$

$$\Rightarrow \frac{2a}{3-a} + 1 = \frac{4-A}{A} + 1 \Rightarrow \frac{2a+3-a}{3-a} = \frac{4-A+A}{A}$$

$$\Rightarrow \frac{a+3}{3-a} = \frac{4}{A} \Rightarrow A = \frac{4(3-a)}{3+a}$$

5. (a) The product of the marks obtained = 72
 As Rohan was not able to figure out the marks obtained by Sunil initially, there must be at least two possible ways of getting that same sum. The two possible cases are 2, 6, 6 and 3, 3, 8 (Sum = 14).
 When Rohan got to know that Sunil got the highest in Physics among the three subjects, he could answer correctly as this is possible only with 3, 3 and 8.
 Therefore, the sum of the marks obtained by Sunil in the other two subjects is $3 + 3$ i.e. 6.
6. (c) $25200 = 2^4 \times 3^2 \times 5^2 \times 7^1$
 As the required divisors when divided by 4 leave remainder 3, the power of 2 in the divisors has to be 0. Therefore, any such divisor is of the form $3^a \times 5^b \times 7^c$, which when divided by 4 leaves the remainder $(-1)^a \times 1^b \times (-1)^c$.
 For the remainder to be 3 i.e. -1, one of 'a' or 'c' must be even/0 and the other should be odd. Also, 'b' can take all the three possible values without making a difference to the remainder.
 The nine possibilities are listed below:
 $a = 0, b = 0, c = 1$
 $a = 2, b = 0, c = 1$
 $a = 1, b = 0, c = 0$
 $a = 0, b = 1, c = 1$
 $a = 2, b = 1, c = 1$
 $a = 1, b = 1, c = 0$
 $a = 0, b = 2, c = 1$
 $a = 2, b = 2, c = 1$
 $a = 1, b = 2, c = 0$

7. (b) $x^2 + ax + b = 0$

Let the roots of this equation be α and β .

$$\Rightarrow \alpha + \beta = -a$$

$$\alpha \cdot \beta = b$$

$$x^2 + bx + a = 0$$

Let the roots be γ and $\delta \Rightarrow \gamma + \delta = -b$

$$\gamma\delta = a$$

$$\text{Given, } |\alpha - \beta| = |\gamma - \delta|$$

$$\text{On squaring, } (\alpha - \beta)^2 = (\gamma - \delta)^2$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = (\gamma + \delta)^2 - 4\gamma\delta$$

$$\text{or } a^2 - 4b = b^2 - 4a \text{ or } a^2 - b^2 + 4(a - b) = 0$$

$$\text{or } (a - b)(a + b + 4) = 0$$

$$\Rightarrow a + b = -4 \quad \therefore a \neq b$$

8. (c) $f(x) = |x - 2| + |2.5 - x| + |3.6 - x|$ can attain minimum value when either of the terms = 0.

$$\text{Case I : When } |x - 2| = 0 \Rightarrow x = 2, \text{ value of } f(x) = 0.5 + 1.6 = 2.1.$$

$$\text{Case II : When } |2.5 - x| = 0 \Rightarrow x = 2.5 \text{ value of } f(x) = 0.5 + 0 + 1.1 = 1.6.$$

$$\text{Case III : When } |3.6 - x| = 0 \Rightarrow x = 3.6$$

$$\Rightarrow f(x) = 1.6 + 1.1 + 0 = 2.7$$

Hence the minimum value of $f(x)$ is 1.6 at $x = 2.5$.

9. (d) $\frac{4p+9q}{p} = \frac{5q}{p-q} \Rightarrow (4p+9q)(p-q) = 5pq$

$$\Rightarrow 4p^2 + 9pq - 4pq - 9p^2 = 5pq$$

$$\Rightarrow 4p^2 = 9q^2$$

$$\Rightarrow 2p = 3q \Rightarrow \frac{p}{q} = \frac{3}{2}, \text{ Now } \frac{4}{5} \frac{p}{q} = \frac{4}{5} \times \frac{3}{2} = \frac{6}{5}$$

10. (b) Since $\sqrt{\frac{T_2 + 4.25}{T_2}} = \frac{9}{8} \Rightarrow 81T_2 = 64(T_2 + 4.25)$

$$\Rightarrow T_2 = 16 \text{ years and } \therefore T_1 = 25 \text{ years}$$

11. (d) Stage 0: $A = 0, B = 0, K = \text{Rs. } 20,000$.

$$\text{Stage 1: } K = 20,000 - 10,000 - 6000 = 4000.$$

$$B = 10,000 - 2500 = 7500. A = 2500 + 6000 = 8500$$

$$\text{Stage 2 : } A = 8500 + 7500 + 4000 = 20,000 \text{ (starting balance)}$$

$$\Rightarrow \text{Finally, } A = 10,000, B = 5,000, K = 5,000.$$

12. (b) **Case I :** Wordlist 1 to 25 - the student will forget 10% of 30 = 3 words every 25 days, i.e. 3 times (words of 1st list are forgot on 26, 51 and 76 days). Hence words remembered in 1st list = $30 - 3 - 3 - 3 = 21$.

Total words remembered in 1-25 days = 25×21

Case II : Word list 26 to 50 - he forgets 10% of words twice. Hence words remembered in any one list = $30 - 3 - 3 = 24$.

Total words remembered = 25×24

Case III : Word list 51-75

Words remembered in any one list = $30 - 3 = 27$

Total words remembered = 25×27

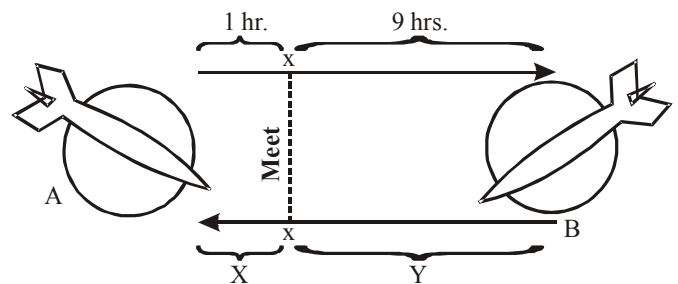
Case IV : Words list 76 to 100 - do not forget any word.

Total words remembered = 25×30

$\therefore$ Total words remembered

$$= 25(21 + 24 + 27 + 30) = 2550.$$

13. (b) The required. time should be a common multiple of the time periods of three satellites.
The LCM of 90, 120 and 150 = 1800 min. = 30 hrs.
14. (d) Here's how you figure it out.



$X + Y = \text{total distance}$

$V_f = \text{velocity of faster rocket}$

$V_s = \text{velocity of slower rocket}$

$T_b = \text{time before meeting}$

$Y = \text{distance covered by faster rocket before meeting}$

$$\Rightarrow Y = V_f \times T_b$$

$X = \text{distance covered by slower rocket before meeting}$

$$\Rightarrow X = V_s \times T_b$$

$$\text{Therefore, } \frac{X}{Y} = \frac{V_s}{V_f} \quad \dots(1)$$

Now after the rockets meet,

$Y = \text{distance covered by slower rocket.}$

$$\Rightarrow Y = 9V_s$$

and $X = \text{distance covered by faster rocket} = 1 \cdot V_f$

$$\text{Thus, } \frac{X}{Y} = \frac{V_f}{9V_s}$$

From (1) and (2), we get

$$\frac{V_s}{V_f} = \frac{V_f}{9V_s} \Rightarrow V_f^2 = 9V_s^2$$

$$\Rightarrow \sqrt{V_f^2} = \sqrt{9V_s^2} \Rightarrow V_f = 3V_s$$

15. (c) In a correct watch, a minute hand overtakes the hour hand in $60(60/55) = 720/11$ min. In Abhijit's clock, this occurs in 65 min. of correct time, i.e. $715/11$ min. Therefore Gain in 65 min is $5/11$ min. Therefore Gain in 11 hr. 55 min. (= 715 min.) = $(715/65)(5/11) = 5$ min. Therefore time shown = 6.30p.m.
16. (b) The trader buys 1100 gm goods for goods worth 1000 gm. He sells only 900 gm goods, but charges the customer for goods worth 1100 gm. So he makes a profit of $[(1100 - 900) / 900] \times (100) = 200 / 9\%$.
17. (a) Now any number is divisible by 9, if the sum of digits is divisible by 9
 $\therefore$ here the sum of digits is $(8+a+7+6+5) + (8+7+6+5+a) = 52+2a$
 $\therefore 2a = 2 \therefore a = 1$
 Now by observation it is obvious that $a = 1$
 $\therefore$ total has to be 54 for the number to be divisible by 9.
18. (d) Number of transmission made by the new and the old robots are X and X/8 respectively per hour each
 $\Rightarrow$ Total per hour = $X + \frac{X}{8} = \frac{9X}{8}$
 $\Rightarrow$ Total in 5 days (=120 hrs) = $\left(\frac{9X}{8}\right) \times 120 = 135X$
19. (d) 'C' takes 120 days
 $\therefore$ 'B' will take $120/4 = 30$ days and 'A' will take $30/2 = 15$ days as time $\propto \frac{1}{\text{efficiency}}$
 $\therefore$ Day 1 'A' Does $(1/15)$ th work.
 Day 2 'B' does $(1/30)$ th work.
 Day 3 'C' does $(1/120)$ th work.
 Total = $\frac{1}{15} + \frac{1}{30} + \frac{1}{120} = \left(\frac{13}{130}\right)$ th work in 3 days
 $\therefore$ In 27 days $\frac{9 \times 13}{130} = \frac{117}{120}$ th work is finished.
 On the 28th day $(3/120)$ th work is remaining $< (1/15)$
 $\therefore$ On the 28th day 'A' Does $(3/120)$ th of work and work gets completed.
 $\therefore$ A does = $9 \times \frac{1}{15} + \frac{3}{120} = \frac{3}{5} + \frac{1}{40} = \frac{25}{40}$ th work
 i.e. A does $\frac{25}{40} \times 100 = 62.5\%$ work
20. (b) Speed of inlet 1 = $\frac{1}{5}$ /min.
 Speed of inlet 2 = $\frac{1}{6}$ /min.
 Speed of outlet = $\frac{1}{12}$ /min.

Therefore the missile will be filled at the speed

$$\frac{1}{5} + \frac{1}{6} - \frac{1}{12} \text{ / min} = \frac{17}{60} \text{ / sec.}$$

Therefore the missile will be filled in

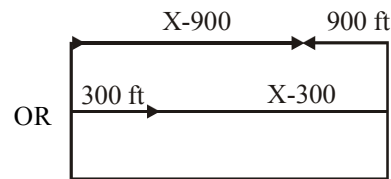
$$\frac{60}{17} \text{ minutes} = 3.53 \text{ min.}$$

$$\text{Hence, Missile hits the camp} = 3.53 + \frac{3}{60} = 3.58 \text{ min}$$

21. (c) Step 1: M = 200 - 10 = 190, R = 150 + 10 = 160;
 $M = 190 + 8 = 198$, $R = 160 - 8 = 152$
 Step 2: M = 198 - 10 = 188, R = 152 + 10 = 162;
 $M = 188 + 9 = 197$, $R = 162 - 9 = 153$
 Step 3: M = 197 - 10 = 187, R = 153 + 10 = 163;
 $M = 187 + 9 = 196$, $R = 163 - 9 = 154$

Clearly a cycle is formed. Since Radhika started off with lesser number, concentrate on her, 10% of 150 = 15. Thus the game will stop when Radhika has 165 marbles. This will happen after 14 steps.

22. (a) **Short cut :** $3(900) - 300 = 2,400$ ft.



Assume width = X

Ratio of distances travelled when they first meet

$$= \frac{900}{X - 900}$$

Ratio of distances travelled when they meet for the second time, after they cross

$$= \frac{(X - 900) + 300}{900 + (X - 300)} = \frac{X - 600}{X + 600}$$

Two ratio have to be equal

$$\therefore \frac{900}{X - 900} = \frac{X - 600}{X + 600} \text{ Hence, } [1].$$

(Do not solve, look at the alternatives. It is obvious that $X = 2,400$).

23. (a) Probability that atleast one of the junior professors is selected
 $= 1 - \text{Probability that none of the junior professors is selected } (P_1)$

$$P_1 = \frac{{}^6C_3}{{}^{10}C_3} = \frac{6 \times 5 \times 4}{10 \times 9 \times 8} = \frac{1}{6} \Rightarrow P = 1 - \frac{1}{6} = \frac{5}{6}$$

24. (b) Probability of getting at least seven points
 $= \text{Probability of getting 7 points or 8 points}$
 $= \text{Prob. of getting 7 points} + \text{Prob. of getting 8 points.}$
 Seven points in four matches can be obtained in the following four different ways.
 2, 2, 2, 1; 2, 2, 1, 2; 2, 1, 2, 2; 1, 2, 2, 2

The probability of each of these ways = $(0.50)^3 (0.05)$
(by multiplication theorem for independent events)
= 0.00625

Hence, probability of getting 7 points
= $4 (0.00625)$ (by add. Theorem)
= 0.0250

Eight points in four matches can be obtained only in one way i.e. 2, 2, 2, 2. Hence

Prob. of getting 8 points = $(0.50)^4 = 0.0625$

Thus, the required prob. = $0.0250 + 0.0625 = 0.0875$

25. (c) Leaving 4 from 22, we have to select 9 from the remaining $22 - 4 - 2 = 16$ Players.

$\therefore$ No. of selection = ${}^{16}C_9$ which is given in (c).

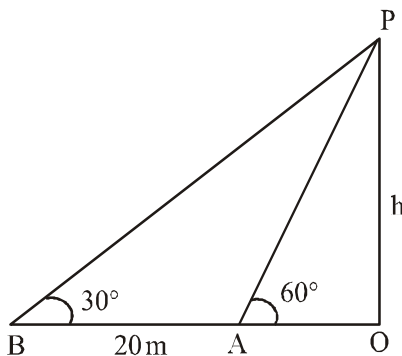
26. (c) We have, $\frac{\cos 70^\circ}{\sin 20^\circ} + \frac{\cos 59^\circ}{\sin 31^\circ} - 8 \sin^2 30^\circ$

$$\Rightarrow \frac{\sin 20^\circ}{\sin 20^\circ} + \frac{\sin 31^\circ}{\sin 31^\circ} - 8 \left(\frac{1}{2} \right)^2$$

$[\because \cos \theta = \sin (90^\circ - \theta), \sin 30^\circ = 1/2]$

$$\Rightarrow 1 + 1 - 8 \times \frac{1}{4} \Rightarrow 2 - 2 = 0$$

27. (c)



$$OA = h \cot 60^\circ, OB = h \cot 30^\circ$$

$$OB - OA = 20 = h (\cot 30^\circ - \cot 60^\circ)$$

$$\Rightarrow h = \frac{20}{\left(\sqrt{3} - \frac{1}{\sqrt{3}} \right)} = \frac{20\sqrt{3}}{2} = 10\sqrt{3}$$

28. (a) Area of a $\Delta = \frac{1}{2} \times b \times h$

$$x = \frac{1}{2} \times 4 \times (8 > h > 4)$$

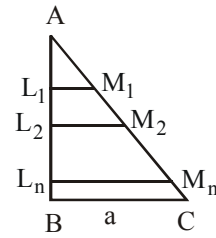
$$x = 2 (8 > h > 4)$$

$$\therefore 8 < x < 16$$

29. (c) $\frac{AL_1}{AB} = \frac{L_1M_1}{BC};$

$$\therefore \frac{1}{n+1} = \frac{L_1M_1}{a};$$

$$\therefore L_1M_1 = \frac{a}{n+1};$$



$$\frac{AL_2}{AB} = \frac{L_2M_2}{BC};$$

$$\therefore \frac{2}{n+1} = \frac{L_2M_2}{a};$$

$$\therefore L_2M_2 = \frac{2a}{n+1}, \text{ etc}$$

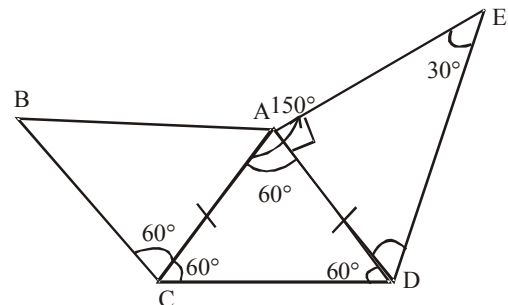
$\therefore$ the required sum

$$= \frac{a}{n+1} + \frac{2a}{n+1} + \frac{3a}{n+1} + \dots + \frac{na}{n+1}$$

$$\frac{a}{n+1} (1 + 2 + 3 + \dots + n) = \frac{a}{n+1} \cdot \frac{n(n+1)}{2}$$

$$= \frac{an}{2}.$$

30. (c)



$\angle CAD = \angle ACD = 60^\circ$ (alternative angles)

$\angle ACD = \angle ADC = 60^\circ$ (since $AC = AD$ and $\angle A = 60^\circ$)

ΔACD is equilateral

Let each side of ΔADC be x

$$\text{So, its area} = \frac{x^2 \sqrt{3}}{4} \text{ (where } x \text{ is side)}$$

$$\text{Area of parallelogram } ABCD = 2 \times \frac{x^2 \sqrt{3}}{4} = \frac{x^2 \sqrt{3}}{2}$$

$$\text{Area of } \Delta ADE = \frac{1}{2} \times AD \times AE$$

$$= \frac{1}{2} \times x \times x \tan 60^\circ = \frac{x^2 \sqrt{3}}{2}$$

Therefore we see, area of parallelogram $ABCD$
= area of ΔADE

$$31. (a) \angle BAF = (6 - 2) \frac{180^\circ}{6} = 120^\circ,$$

$$\angle BAR = (5 - 2) \frac{180^\circ}{5} = 108^\circ$$

$$\angle FAR + \angle BAF + \angle BAR = 360^\circ \quad (\text{angles at a point})$$

$$\Rightarrow \angle FAR = 360^\circ - 120^\circ - 108^\circ = 132^\circ$$

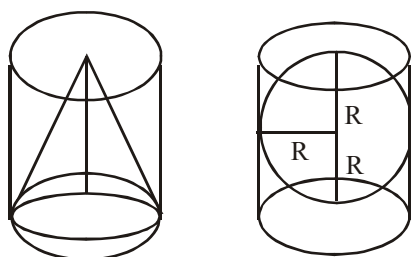
$$\Rightarrow \angle AFR = \frac{(180^\circ - 132^\circ)}{2} = 24^\circ$$

$$32. (c) \text{ The volumes are : } 5^3 = 125,$$

$$\frac{4}{3}\pi 3^3 = 36\pi, 100, \pi \times 2^2 \times 10 = 40\pi$$

$$33. (a) \text{ Radius of cylinder} = R = \text{Radius of sphere} = \text{Radius of cone's base.}$$

$$\text{Height of cylinder} = \text{Height of cone} = 2R = \text{Diameter of sphere.}$$



$$\text{Volume of cylinder} = \pi R^2 \times 2R = 2\pi R^3,$$

$$\text{volume of sphere} = \frac{4}{3}\pi R^3,$$

$$\text{volume of cone} = \frac{1}{3}\pi R^2(2R) = \frac{2}{3}\pi R^3$$

$$\Rightarrow \text{Ratio} = 2\pi R^3 : \left[\frac{4}{3}\pi R^3 + \frac{2}{3}\pi R^3 \right] = 1 : 1$$

34. (d) If the side of the imaginary cube is a , then the longest diagonal is $a\sqrt{3}$. If the cube is the smallest possible, then the spheres at two opposite corners, and the centre sphere would be touching each other.

Therefore $(r + 2r + r) = 4r = a\sqrt{3}$. Since ' a ' is the distance between the centres of the spheres on two

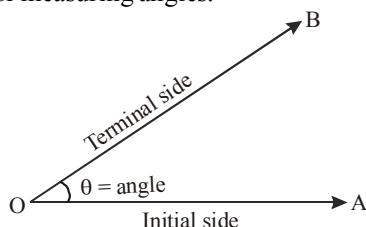
$$\text{adjacent corners, } a = \frac{4r}{\sqrt{3}}.$$

23

TRIGONOMETRY AND ITS APPLICATIONS

ANGLE

Consider a ray $\overrightarrow{OA}$. If this ray rotates about its end point O and takes the position OB , then the angle $\angle AOB$ has been generated. An angle is considered as the figure obtained by rotating a given ray about its end-point. The initial position OA is called the initial side and the final position OB is called terminal side of the angle. The end point O about which the ray rotates is called the vertex of the angle. The measure of an angle is the amount of rotation performed to get the terminal side from initial side. There are several units for measuring angles.

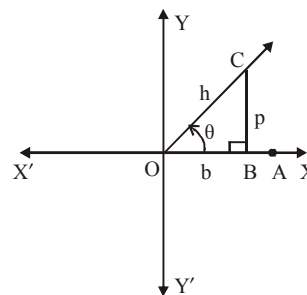


TRIGONOMETRIC RATIOS

Let XOX' and YOY' be horizontal and vertical axes of respectively. Let A be a point on OX . Let the ray OA start rotating in the plane XY in an anti-clockwise direction from the initial position OA about the point O till it reaches its final position OC after some interval of time. (See Fig.). Thus, an angle COA is formed with x -axis. Let $\angle COA = \theta$. (θ is a Greek letter, and we read it as “theta”). Draw $CB \perp OX$. Now clearly ΔCBO is a right angled triangle.

In right ΔCBO , OC is the hypotenuse. For angle $\theta = \angle COA$, BC and OB are called side opposite to angle θ and adjacent side of angle θ respectively.

Let $CB = p$, $OB = b$ and $OC = h$. We define the different ratios between hypotenuse, side opposite to angle θ and adjacent side of angle θ as trigonometric ratios for angle θ . Horizontal axis $X'OX$ is called X -axis and vertical axis YOY' is called Y -axis.



These trigonometrical ratios are :

$$\text{Sine of } \theta = \frac{\text{Side opposite to angle } \theta}{\text{Hypotenuse}} = \frac{CB}{OC} = \frac{p}{h};$$

$$\text{Cosine of } \theta = \frac{\text{Adjacent side to angle } \theta}{\text{Hypotenuse}} = \frac{OB}{OC} = \frac{b}{h}$$

$$\text{Tangent of } \theta = \frac{\text{Side opposite to angle } \theta}{\text{Adjacent side to angle } \theta} = \frac{CB}{OB} = \frac{p}{b};$$

$$\text{Cotangent of } \theta = \frac{\text{Adjacent side to angle } \theta}{\text{Side opposite to angle } \theta} = \frac{OB}{CB} = \frac{b}{p}$$

$$\text{Secant of } \theta = \frac{\text{Hypotenuse}}{\text{Adjacent side to angle } \theta} = \frac{OC}{OB} = \frac{h}{b};$$

$$\text{Cosecant of } \theta = \frac{\text{Hypotenuse}}{\text{Side opposite to angle } \theta} = \frac{OC}{CB} = \frac{h}{p}$$

Sine of θ is abbreviated as $\sin \theta$, Cosine of θ is abbreviated as $\cos \theta$, Tangent of θ is abbreviated as $\tan \theta$

Cotangent of θ is abbreviated as $\cot \theta$, Secant of θ is abbreviated as $\sec \theta$ and Cosecant of θ is abbreviated as $\csc \theta$

(i) Throughout the study of trigonometry we shall be using only abbreviated form of these trigonometric ratios.

Thus,

$$\sin \theta = \frac{p}{h}, \cos \theta = \frac{b}{h}; \tan \theta = \frac{p}{b}, \cot \theta = \frac{b}{p};$$

$$\sec \theta = \frac{h}{b}, \operatorname{cosec} \theta = \frac{h}{p}$$

(ii) $\sin \theta$ is an abbreviation for "sine of angle θ " and not the product of $\sin$ and θ .

VALUE OF TRIGONOMETRIC RATIOS FOR SOME SPECIFIC ANGLES

The values of trigonometric ratios for angles 0° , 30° , 45° , 60° and 90° are quite often used in solving problems in our day-to-day life. Thus the following table is very useful.

IMPORTANT TABLE

Trigonometrical ratio (θ)	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined
$\cot \theta$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\operatorname{cosec} \theta$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	not defined

BASIC FORMULAE OR TRIGONOMETRIC IDENTITY

$$(i) \sin \theta \cdot \operatorname{cosec} \theta = 1 \text{ or } \sin \theta = \frac{1}{\operatorname{cosec} \theta} \text{ or } \operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$(ii) \cos \theta \cdot \sec \theta = 1 \text{ or } \cos \theta = \frac{1}{\sec \theta} \text{ or } \sec \theta = \frac{1}{\cos \theta}$$

$$(iii) \tan \theta \cdot \cot \theta = 1 \text{ or } \cot \theta = \frac{1}{\tan \theta} \text{ or } \tan \theta = \frac{1}{\cot \theta}$$

$$(iv) \sin^2 \theta + \cos^2 \theta = 1$$

$$\text{or } \cos^2 \theta = 1 - \sin^2 \theta \text{ or } \sin^2 \theta = 1 - \cos^2 \theta$$

$$(v) \sec^2 \theta - \tan^2 \theta = 1$$

$$\text{or } \sec^2 \theta = 1 + \tan^2 \theta \text{ or } \tan^2 \theta = \sec^2 \theta - 1$$

$$(vi) \operatorname{cosec}^2 \theta - \cot^2 \theta = 1 \text{ or } \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$$

$$\text{or } \cot^2 \theta = \operatorname{cosec}^2 \theta - 1$$

$$(vii) \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$(viii) \cot \theta = \frac{\cos \theta}{\sin \theta}$$

TRIGONOMETRIC RATIOS FOR COMPLEMENTARY ANGLES

$$\sin (90^\circ - \theta) = \frac{OM}{OP} = \cos \theta, \cos (90^\circ - \theta) = \frac{PM}{OP} = \sin \theta,$$

$$\tan (90^\circ - \theta) = \frac{OM}{PM} = \cot \theta,$$

$$\cot (90^\circ - \theta) = \frac{PM}{OM} = \tan \theta, \operatorname{cosec} (90^\circ - \theta) = \frac{OP}{OM} = \sec \theta$$

$$\text{and } \sec (90^\circ - \theta) = \frac{OP}{PM} = \operatorname{cosec} \theta$$

Illustration 1: Evaluate $\frac{\cos 43^\circ}{\cos 47^\circ} + \frac{\sec 32^\circ}{\operatorname{cosec} 58^\circ}$

Sol. We know that $\cos (90^\circ - \theta) = \sin \theta$

$$\sin 47^\circ = \sin (90^\circ - \theta) = \cos 43^\circ$$

$$\text{Also, } \operatorname{cosec} 58^\circ = \operatorname{cosec} (90^\circ - 32^\circ) = \sec 32^\circ$$

$$\therefore \frac{\cos 43^\circ}{\cos 47^\circ} + \frac{\sec 32^\circ}{\operatorname{cosec} 58^\circ} = \frac{\cos 43^\circ}{\cos 43^\circ} + \frac{\sec 32^\circ}{\sec 32^\circ} = 1 + 1 = 2$$

Illustration 2: Evaluate $\frac{\sec^2 54^\circ - \cot^2 36^\circ}{\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ}$

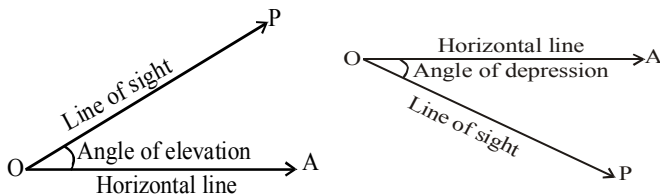
$$+ 2\sin^2 38^\circ \sec^2 52^\circ - \sin^2 45^\circ + \frac{2}{\sqrt{3}} \tan 17^\circ \tan 60^\circ \tan 73^\circ.$$

Sol. The given expression is

$$\begin{aligned} & \frac{\sec^2 54^\circ - \cot^2 36^\circ}{\operatorname{cosec}^2 57^\circ - \tan^2 33^\circ} + 2\sin^2 38^\circ \sec^2 52^\circ - \sin^2 45^\circ \\ & + \frac{2}{\sqrt{3}} \tan 17^\circ \tan 60^\circ \tan 73^\circ \\ & = \frac{\sec^2 (90^\circ - 36^\circ) - \cot^2 36^\circ}{\operatorname{cosec}^2 (90^\circ - 33^\circ) - \tan^2 33^\circ} + 2\sin^2 38^\circ \sec^2 \\ & (90^\circ - 38^\circ) - \sin^2 45^\circ + \frac{2}{\sqrt{3}} \tan (90^\circ - 73^\circ) \tan 73^\circ \tan 60^\circ \\ & = \frac{1}{1} + 2\sin^2 38^\circ \times \frac{1}{\sin^2 38^\circ} - \frac{1}{2} + \frac{2}{\sqrt{3}} \times \frac{1}{\tan 73^\circ} \times \tan 73^\circ \times \sqrt{3} \\ & [\because \operatorname{cosec}^2 \theta - \cot^2 \theta = 1, \sec^2 \theta - \tan^2 \theta = 1] \\ & = 1 + 2 - \frac{1}{2} + 2 = 5 - \frac{1}{2} = \frac{9}{2} \end{aligned}$$

ANGLE OF ELEVATION AND ANGLE OF DEPRESSION

Let an observer at the point O is observing an object at the point P. The line OP is called the LINE OF SIGHT of the point P. Let OA be the horizontal line passing through O. O, A and P should be in the same vertical plane. If object P be above the line of sight OA, then the acute angle AOP, between the line of sight and the horizontal line is known as ANGLE OF ELEVATION of object P. If the object P is below the horizontal line OA then the angle AOP, between the line of sight and horizontal line is known as ANGLE OF DEPRESSION of object P.



TO FIND THE HEIGHT AND THE DISTANCE OF AN INACCESSIBLE TOWER STANDING ON A HORIZONTAL PLANE

Let AB be a tower and B be its foot. On the horizontal line through B, take two points P and Q. Measure the length PQ.

Let PQ = a.

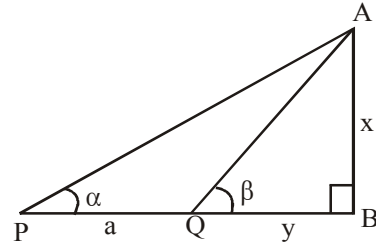
Let the angles of elevation of the top A of the tower as seen from P and Q be respectively α and β ($\beta > \alpha$) then

$\angle APB = \alpha$, $\angle AQB = \beta$. Let $AB = x$, $BQ = y$.

From right angled $\triangle ABP$, $\tan \alpha = \frac{AB}{PB} = \frac{x}{a+y}$

$$\therefore a+y = x \cot \alpha.$$

.....(i)



From right angled $\triangle ABQ$, $\tan \beta = \frac{AB}{BQ} = \frac{x}{y}$

$$\therefore y = x \cot \beta$$

.....(ii)

From equations (i) and (ii),

$$\therefore a = x \cot \alpha - x \cot \beta.$$

$$\Rightarrow x = \frac{a}{\cot \alpha - \cot \beta}$$

$$\text{Also } y = x \cot \alpha - a \Rightarrow y = \frac{a \cot \alpha}{\cot \alpha - \cot \beta} - a$$

$$\Rightarrow y = \frac{a \cot \alpha - a(\cot \alpha - \cot \beta)}{\cot \alpha - \cot \beta} \Rightarrow y = \frac{a \cot \beta}{\cot \alpha - \cot \beta}$$

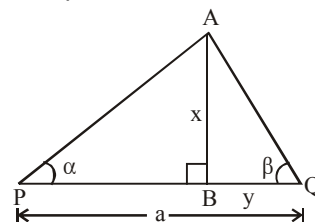
In the above case, P and Q are on the same side of the tower. If the two points are on the opposite sides of the tower then from the adjoining figure, we get

$$\tan \alpha = \frac{x}{PB} \text{ or } PB = x \cot \alpha$$

$$\text{and } \tan \beta = \frac{x}{BQ} \text{ or } BQ = x \cot \beta.$$

$$\therefore a = PB + BQ = x(\cot \alpha + \cot \beta)$$

$$\therefore x = \frac{a}{\cot \alpha + \cot \beta}$$



$$\text{and } y = BQ = x \cot \beta$$

NOTE : Here, all the lines AP, AQ, AB are in the same plane.

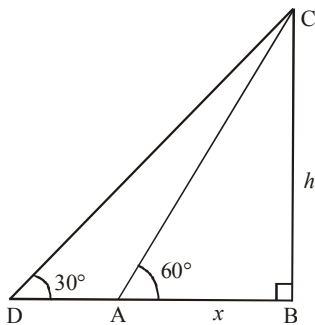
Illustration 3: A person, standing on the bank of a river, observes that the angle subtended by a tree on the opposite bank is 60° ; when he retreats 20m from the bank, he finds the angle to be 30° . Find the height of the tree and the breadth of the river.

Sol. Let AB be the width of the river and BC be the tree which makes an angle of 60° at a point A on the opposite bank. Let D be the position of the person after retreating 20 m from the bank.

Let $AB = x$ metres and $BC = h$ metres.

From right angled triangles ABC and DBC ,

$$\text{we have } \tan 60^\circ = \frac{BC}{AB} \text{ and } \tan 30^\circ \Rightarrow \sqrt{3} = \frac{h}{x}$$



$$\text{and } \frac{1}{\sqrt{3}} = \frac{h}{x+20} \Rightarrow h = x\sqrt{3}$$

$$\text{and } h = \frac{x+20}{\sqrt{3}} \Rightarrow x\sqrt{3} = \frac{x+20}{\sqrt{3}}$$

$$\Rightarrow 3x = x+20 \Rightarrow x = 10 \text{ m}$$

Putting $x = 10$ in $h = \sqrt{3}x$, we get

$$h = 10\sqrt{3} = 17.32 \text{ m}$$

Hence, height of the tree = 17.32 m and the breadth of the river = 10 m.

Illustration 4: A man is standing on the deck of a ship, which is 8m above water level. He observes the angle of elevation of the top of a hill as 60° and the angle of depression of the base of the hill as 30° . Calculate the distance of the hill from the ship and the height of the hill.

Sol. Let x be the distance of hill from man and $h+8$ be height of hill which is required.

In rt. $\triangle ACB$,

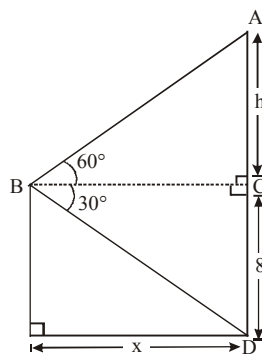
$$\tan 60^\circ = \frac{AC}{BC} = \frac{h}{x}$$

$$\Rightarrow \sqrt{3} = \frac{h}{x}$$

In rt. $\triangle BCD$,

$$\tan 30^\circ = \frac{CD}{BC} = \frac{8}{x}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{8}{x} \Rightarrow x = 8\sqrt{3}$$



$$\therefore \text{Height of hill} = h+8 = \sqrt{3}x+8 = (\sqrt{3})(8\sqrt{3})+8 = 32 \text{ m}$$

$$\text{Distance of ship from hill} = x = 8\sqrt{3} \text{ m}$$

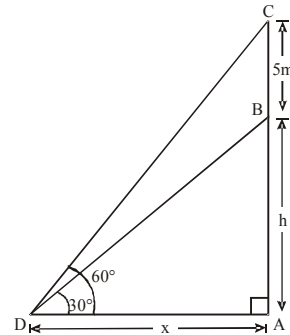
Illustration 5: A vertical to stands on a horizontal plane and is surmounted by a vertical flag staff of height 6 meters. At

point on the plane, the angle of elevation of the bottom and the top of the flag staff are respectively 30° and 60° . Find the height of tower.

Sol. Let AB be the tower of height h meter and BC be the height of flag staff surmounted on the tower.

Let the point of the plane be D at a distance m meter from the foot of the tower.

In $\triangle ABD$,



$$\tan 30^\circ = \frac{AB}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x} \Rightarrow x = \sqrt{3}h \quad \dots\dots (i)$$

In $\triangle ADC$,

$$\tan 60^\circ = \frac{AC}{AD} \Rightarrow \sqrt{3} = \frac{5+h}{x} \Rightarrow x = \frac{5+h}{\sqrt{3}} \quad \dots\dots (ii)$$

$$\text{From (i) and (ii), } \sqrt{3}h = \frac{5+h}{\sqrt{3}} \Rightarrow 3h = 5+h \Rightarrow 2h = 5$$

$$\Rightarrow h = \frac{5}{2} = 2.5 \text{ m}$$

So, the height of tower = 2.5 m

Illustration 6: The angles of depressions of the top and bottom of 8m tall building from the top of a multistoried building are 30° and 45° respectively. Find the height of multistoried building and the distance between the two buildings.

Sol. Let AB be the multistoried building of height h and let the distance between two buildings be x meters.

$$\angle XAC = \angle ACB = 45^\circ \quad (\text{Alternate angles})$$

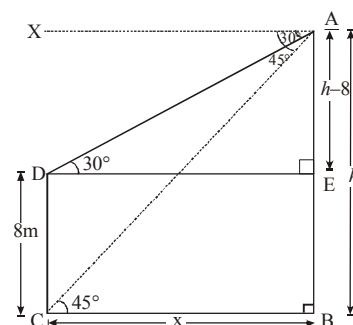
$$\angle XAD = \angle ADE = 30^\circ \quad (\text{Alternate angles})$$

$$\text{In } \triangle ADE, \tan 30^\circ = \frac{AE}{ED} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h-8}{x} \quad [\because CB = DE = x]$$

$$\Rightarrow x = \sqrt{3}(h-8) \quad \dots\dots (i)$$

In $\triangle ACB$,

$$\tan 45^\circ = \frac{h}{x} \Rightarrow 1 = \frac{h}{x} \Rightarrow h = x \quad \dots\dots (ii)$$



From (i) and (ii),

$$\sqrt{3}(h-8) = h \Rightarrow \sqrt{3}h - 8\sqrt{3} = h$$

$$\Rightarrow \sqrt{3}h - h = 8\sqrt{3} \Rightarrow h(\sqrt{3}-1) = 8\sqrt{3}$$

$$\Rightarrow h = \frac{8\sqrt{3}}{\sqrt{3}-1} \times \frac{(\sqrt{3}+1)}{(\sqrt{3}+1)} \Rightarrow h = \frac{8\sqrt{3}(\sqrt{3}+1)}{2}$$

$$\Rightarrow h = 4\sqrt{3}(\sqrt{3}+1) \Rightarrow h = 4(3+\sqrt{3}) \text{ metres}$$

From (ii), $x = h$

So, $x = 4(3+\sqrt{3})$ metres

Hence, height of multistoried building = $4(3+\sqrt{3})$ metres

distance between two building = $4(3+\sqrt{3})$ metres.

Illustration 7: The angle of elevation of an aeroplane from a point on the ground is 45° . After a flight of 15 sec, the elevation changes to 30° . If the aeroplane is flying at a height of 3000 metres, find the speed of the aeroplane.

Sol. Let the point on the ground is E which is y metres from point B and let after 15 sec. flight it covers x metres distance

$$\text{In } \triangle AEB, \tan 45^\circ = \frac{AB}{EB}$$

$$\Rightarrow 1 = \frac{3000}{y} \Rightarrow y = 3000\text{m} \quad \dots\dots\dots (i)$$

$$\text{In } \triangle CED, \tan 30^\circ = \frac{CD}{ED} \Rightarrow \frac{1}{\sqrt{3}} = \frac{3000}{x+y}$$

($\because AB = CD$)

$$\Rightarrow x + y = 3000\sqrt{3} \quad \dots\dots\dots (ii)$$

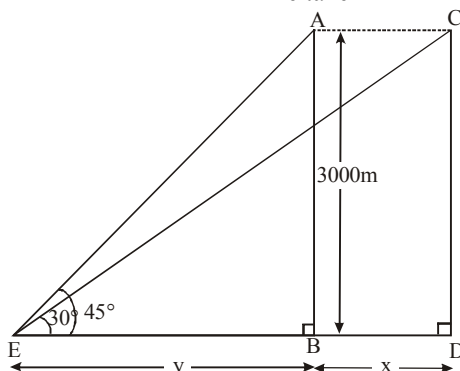
From eqs. (i) and (ii)

$$x + 3000 = 3000\sqrt{3} \Rightarrow x = 3000\sqrt{3} - 3000$$

$$\Rightarrow x = 3000(\sqrt{3}-1) \Rightarrow x = 3000 \times (1.732-1)$$

$$\Rightarrow x = 3000 \times 0.732 \Rightarrow x = 2196\text{m}$$

$$\text{Speed of aeroplane} = \frac{\text{Distance covered}}{\text{Time taken}}$$



$$= \frac{2196}{15} \text{ m/sec} = 146.4 \text{ m/sec}$$

$$= \frac{2196}{15} \times \frac{18}{5} \text{ km/hr} = 527.04 \text{ km/hr}$$

Hence, the speed of aeroplane is 527.04 km/hr.

Illustration 8: A boy is standing on the ground and flying a kite with 100m of string at an elevation of 30° . Another boy is standing on the roof of a 10m high building and is flying his at an elevation of 45° . Both the boys are on opposite sides of both the kites. Find the length of the string that the second boy must have so that the two kites meet.

Sol. Let the length of second string be x m.

In $\triangle ABC$,

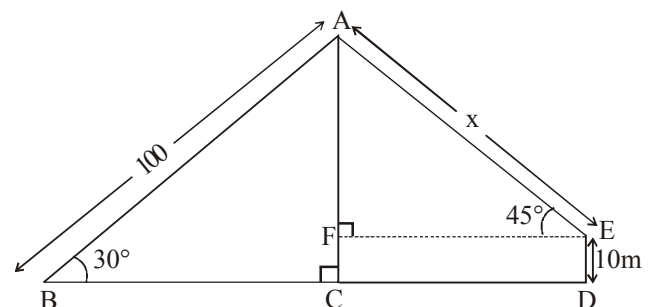
$$\sin 30^\circ = \frac{AC}{AB} \quad \text{or} \quad \frac{1}{2} = \frac{AC}{100} \Rightarrow AC = 50\text{m}$$

In $\triangle AEF$,

$$\sin 45^\circ = \frac{AF}{AE} \Rightarrow \frac{1}{\sqrt{2}} = \frac{AF-FC}{x}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{50-10}{x} \quad [\because AC = 50\text{m}, FC = ED = 10\text{m}]$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{40}{x} \Rightarrow x = 40\sqrt{2}\text{m}$$



So the length of string that the second boy must have so that the two kites meet = $40\sqrt{2}\text{m}$

Practice Exercise

Foundation Level

1. $\cos 1^\circ \cdot \cos 2^\circ \cdot \cos 3^\circ \dots \cos 179^\circ$ is equal to
 (a) -1 (b) 0
 (c) 1 (d) $1/\sqrt{2}$

2. $\sin^2 \theta + \operatorname{cosec}^2 \theta$ is always
 (a) greater than 1
 (b) less than 1
 (c) greater than or equal to 2
 (d) equal to 2

3. If $\sin \theta + \cos \theta = a$ and $\frac{\sin \theta + \cos \theta}{\sin \theta \cos \theta} = b$, then
 (a) $b = \frac{2a}{a^2 - 1}$ (b) $a = \frac{2b}{b^2 - 1}$
 (c) $ab = b^2 - 1$ (d) $a + b = 1$

4. The value of $(\sin^2 7\frac{1}{2}^\circ + \cos^2 7\frac{1}{2}^\circ) - (\sin^2 30^\circ + \cos^2 30^\circ) + (\sin^2 7^\circ + \sin^2 83^\circ)$ is equal to
 (a) 3 (b) $3\frac{1}{2}$
 (c) 2 (d) 1

5. If $\tan 15^\circ = 2 - \sqrt{3}$, then the value of $\cot^2 75^\circ$ is
 (a) $7 + \sqrt{3}$ (b) $7 - 2\sqrt{3}$
 (c) $7 - 4\sqrt{3}$ (d) $7 + 4\sqrt{3}$

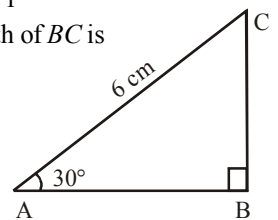
6. If $x = p \sec \theta$ and $y = q \tan \theta$ then
 (a) $x^2 - y^2 = p^2 q^2$ (b) $x^2 q^2 - y^2 p^2 = pq$
 (c) $x^2 q^2 - y^2 p^2 = \frac{1}{p^2 q^2}$ (d) $x^2 q^2 - y^2 p^2 = p^2 q^2$

7. If $b \tan \theta = a$, the value of $\frac{a \sin \theta - b \cos \theta}{a \sin \theta + b \cos \theta}$
 (a) $\frac{a - b}{a^2 + b^2}$ (b) $\frac{a + b}{a^2 + b^2}$
 (c) $\frac{a^2 + b^2}{a^2 - b^2}$ (d) $\frac{a^2 - b^2}{a^2 + b^2}$

8. If $\tan \theta + \sin \theta = m$ and $\tan \theta - \sin \theta = n$, then the value of $m^2 - n^2$ is equal to
 (a) $4mn$ (b) $2\sqrt{mn}$
 (c) $4\sqrt{mn}$ (d) $2\sqrt{m/n}$

9. $\tan 9^\circ \times \tan 27^\circ \times \tan 63^\circ \times \tan 81^\circ =$
 (a) 4 (b) 3
 (c) 2 (d) 1

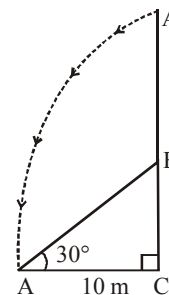
10. In the adjoining figure, the length of BC is
 (a) $2\sqrt{3}$ cm
 (b) $3\sqrt{3}$ cm
 (c) $4\sqrt{3}$ cm
 (d) 3 cm



11. If the angle of depression of an object from a 75 m high tower is 30° , then the distance of the object from the tower is
 (a) $25\sqrt{3}$ m (b) $50\sqrt{3}$ m
 (c) $75\sqrt{3}$ m (d) 150 m

12. The angle of elevation of the top of a tower at point on the ground is 30° . If on walking 20 metres toward the tower, the angle of elevation become 60° , then the height of the tower is
 (a) 10 metre (b) $\frac{10}{\sqrt{3}}$ metre
 (c) $10\sqrt{3}$ metre (d) None of these

13. The top of a broken tree has its top touching the ground (shown in the adjoining figure) at a distance of 10 m from the bottom. If the angle made by the broken part with ground is 30° , then the length of the broken part is
 (a) $10\sqrt{3}$ cm (b) $\frac{20}{\sqrt{3}}$ m
 (c) 20 cm (d) $20\sqrt{3}$ m



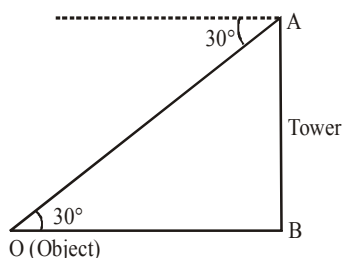
14. An aeroplane flying horizontally 1 km. above the ground is observed at an elevation of 60° and after 10 seconds the elevation is observed to be 30° . The uniform speed of the aeroplane in km/h is

- (a) 240 (b) $240\sqrt{3}$
 (c) $60\sqrt{3}$ (d) None of these
15. A 25 m ladder is placed against a vertical wall of a building. The foot of the ladder is 7 m from the base of the building. If the top of the ladder slips 4m, then the foot of the ladder will slide
 (a) 5 m (b) 8 m
 (c) 9 m (d) 15 m
16. If the length of the shadow of a tower is $\sqrt{3}$ times that of its height, then the angle of elevation of the sun is
 (a) 15° (b) 30°
 (c) 45° (d) 60°
17. The angles of elevation of the top of a tower from two points at distances m and n metres are complementary. If the two points and the base of the tower are on the same straight line, then the height of the tower is
 (a) $\sqrt{mn}$ (b) mn
 (c) $\frac{m}{n}$ (d) None of these
18. An aeroplane at a height of 600 m passes vertically above another aeroplane at an instant when their angles of elevation at the same observing point are 60° and 45° respectively. How many metres higher is the one from the other?
 (a) 286.53 m (b) 274.53 m
 (c) 253.58 m (d) 263.83 m

Hints & Solutions

1. (b) 2. (c) 3. (a)
 4. (d) $\sin 83^\circ = \cos 7^\circ$
 $\therefore$ the given expression is $1 - 1 + 1 = 1$
5. (c) $\cot^2 75^\circ = (2 - \sqrt{3})^3 = 7 - 4\sqrt{3}$
6. (d) We know $\sec^2 \theta - \tan^2 \theta = 1$ and $\sec \theta = \frac{x}{p}$, $\tan \theta = \frac{y}{q}$
 $\therefore x^2 q^2 - p^2 y^2 = p^2 q^2$
7. (d) $\tan \theta = \frac{a}{b}$

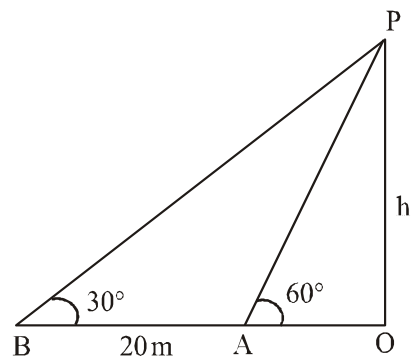
$$\frac{a \sin \theta - b \cos \theta}{a \sin \theta + b \cos \theta} = \frac{a \tan \theta - b}{a \tan \theta + b} = \frac{a^2 - b^2}{a^2 + b^2}$$
8. (c) 9. (d)
10. (d) **Hint:** $\sin 30^\circ = \frac{BC}{AC} \Rightarrow \frac{1}{2} = \frac{BC}{6 \text{ cm}} \Rightarrow BC = 3 \text{ cm}$.
11. (c) **Hint:** $\tan 30^\circ = \frac{AB}{OB}$



$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{75 \text{ m}}{OB}$$

$$\Rightarrow OB = 75\sqrt{3} \text{ m}$$

12. (c)



$$OA = h \cot 60^\circ, OB = h \cot 30^\circ$$

$$OB - OA = 20 = h (\cot 30^\circ - \cot 60^\circ)$$

$$\Rightarrow h = \frac{20}{\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)} = \frac{20\sqrt{3}}{2} = 10\sqrt{3}$$

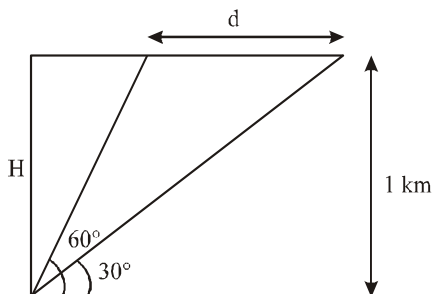
13. (b) **Hint:** $\cos 30^\circ = \frac{AC}{AB} \Rightarrow \frac{\sqrt{3}}{2} = \frac{10 \text{ m}}{AB}$

$$\Rightarrow AB = \frac{20}{\sqrt{3}} \text{ m}$$

14. (b) $d = H \cot 30^\circ - H \cot 60^\circ$

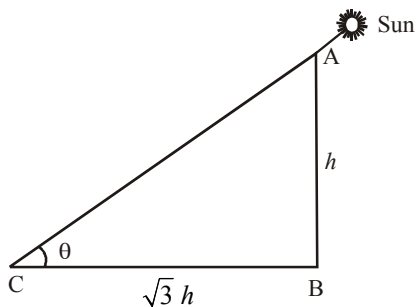
Time taken = 10 second

$$\text{speed} = \frac{\cot 30^\circ - \cot 60^\circ}{10} \times 60 \times 60 = 240\sqrt{3}$$



15. (b)

16. (b) **Hint:** Let height of tower (AB) be h metres, then length of its shadow (BC) = $\sqrt{3} h$ metres.



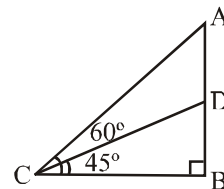
Let angle of elevation be θ ,

$$\text{then } \tan \theta = \frac{h}{\sqrt{3}h} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = 30^\circ$$

17. (a)

18. (c) Let the aeroplanes are at point A and D respectively. Aeroplane A is flying 600m above the ground.



So, $AB = 600$.

$\angle ACB = 60^\circ$, $\angle DCB = 45^\circ$

$$\text{From } \triangle ABC, \frac{AB}{BC} = \tan 60^\circ \Rightarrow BC = \frac{600}{\sqrt{3}} = 200\sqrt{3}.$$

$$\text{From } \triangle DCB, \frac{DB}{BC} = \tan 45^\circ \Rightarrow DB = 200\sqrt{3}.$$

$$\text{So, the distance } AD = AB - BD = 600 - 200\sqrt{3} \\ = 200(3 - \sqrt{3}) = 200(3 - 1.7321) = 253.58\text{m}.$$



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